Contents

Polyfold Theory, verified: the machine-checked edition 1

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A companion to the expanded exposition of Part I, of Chapters 7–11 of Part II and of Part III (in progress), of H. Hofer, K. Wysocki and E. Zehnder's Polyfold and Fredholm Theory *(Springer 2021). Statement numbers and pages are those of the Springer edition, followed in parentheses by those of the text the formalization was first checked against: for Part I the 2014 preprint, arXiv 1407.3185 ("preprint"), for Parts II and III, which have no preprint, the book's arXiv version, 1707.08941 ("2017"); a page that could not be located in the Springer text is given in the old numbering only, marked as such. The V- and G-numbers are the project's own, and the Lean names carry the preprint's (Part I) and the 2017 (Part II) numbers;* `Exposition/BookConcordance.md` *gives the correspondence). That exposition is written to be read; this edition is written to be checked. It records, chapter by chapter, exactly which statements of the text have been verified by the Lean 4 proof assistant against the Mathlib library, what each verified statement says when read back in ordinary language, which hypotheses the text left silent, where the formal statement or proof departs from the text's, and what remains unverified and why. Nothing here needs to be read in Lean; every checked statement is restated in prose, and the Lean names are given only so that the claims can be audited.*

Where to start. The project has three parts:

- the expanded exposition (`Exposition/Chapter1.md`, …), the book for reading;
- *Polyfold Theory, Checked*, the Lean formalization (`Polyfold/`) and the reports on it;
- the sign-off page, the list of statements a reader is asked to confirm.

This edition is the way into the second part. Its reports each answer one question:

- **What is proved?** This edition: every checked result in plain language, with all its hypotheses, and the list of statements to sign off.
- **How is it proved?** *Polyfold Theory, proof by proof* (`Exposition/ProofByProof.md`): the formal proofs retold as mathematics, with a trace tag on each step naming the Lean declaration that carries it.
- **How does it differ from the text?** *Polyfold Theory beside Lean*, the comparison (`Exposition/polyfold-beside-lean.html`): each result of the book beside the checked statement, with a verdict (the same, restated, hypotheses made explicit, stronger, narrower) and the differences.

A result has the same number, V1.5 say, in this edition, the comparison and the sign-off page.

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0. Summary

Chapter	HWZ	Content	Status	Lean lines
Chapter 1	Ch. 1, §§1.1–1.4	sc-Banach spaces, sc-operators and sc-Fredholm operators, sc-differentiability and its relation to classical differentiability, the chain rule, the degeneracy index	**verified**	10,705
Chapter 2	Ch. 2	sc-smooth retractions and retracts, M-polyfolds, the degeneracy index, tame M-polyfolds, strong bundles	**verified; Remark 2.4.8's (preprint 2.28) negative claim is false**	27,046
Chapter 3	Ch. 3	sc-Fredholm sections and the implicit function theorem	**partly verified**	69,061
Chapter 4	Ch. 4, §§4.1–4.4	manifolds with boundary with corners: sc^+-retracts and the characterization, smooth finite-dimensional submanifolds, families and Sard's theorem (vendored)	**verified except two remarks**	34,173

Chapter	HWZ	Content	Status	Lean lines
Chapter 5	Ch. 5, §§5.1–5.5 (preprint §§5.1–5.4)	the Fredholm package for M-polyfolds: auxiliary norms and reflexive auxiliary norms, compactness, sc^+-sections and bump functions, transversality, general and good position, homotopies, the ∂-extension theorem, partitions of unity	**verified except Theorem 5.27 (preprint; not in the 2021 edition) and the Sobolev sc-structure on general bounded domains**	29,629
Chapter 6	Ch. 6, §§6.2–6.6, 6.8 (preprint §§6.1–6.5, 6.7)	linearizations and orientations: determinant lines and exact sequences, the determinant of a Fredholm operator, good left-projections, stabilization, the determinant line bundle, the propagation of orientations along paths, orientable and oriented sc-Fredholm sections, the orientation of the solution set	**verified except the global parts of Theorem 6.6.18 (preprint 6.41)**	48,976

Chapter	HWZ	Content	Status	Lean lines
Chapter 7	Part II, Ch. 7, §§7.1–7.3, 7.5–7.6 (2017 §§7.1–7.5)	ep-groupoids: groupoids of M-polyfolds with local sc-diffeomorphisms as source and target and a local properness, the orbit space, the natural representation of the isotropy groups, local uniformizers, ep-subgroupoids, effective and reduced ep-groupoids, the topology of the orbit space, index lifts, invariant sc-smooth partitions of unity, proper subsets	**verified except the tangent $T\varphi$ of Definition 7.1.4 (2017 Definition 7.4), the remark after Definition 7.1.7 (2017 Definition 7.7) in its 2017 form, and the Hausdorff property of $\mathbf{X}^R$ in the 2017 Proposition 7.9 and Proposition 7.10 (rewritten in 2021 as Proposition 7.2.10 and Theorem 7.2.14, which assume it)**	23,439

Chapter	HWZ	Content	Status	Lean lines
Chapter 8	Part II, Ch. 8, §§8.1–8.3, 8.5–8.6 (2017 §§8.1–8.5)	bundles and covering functors: the tangent ep-groupoid and the tangent functor, natural transformations, the orbit space of the tangent, strong bundles over ep-groupoids and their functorial picture, strong bundle maps and section functors, proper covering functors, their local structure and geometric lifts, proper strong bundle covering functors, the counting of the fibres	**8.3) for nonempty D and the degeneracy remark after Definition 8.5.1 for tame X, Y**	15,943

Chapter	HWZ	Content	Status	Lean lines
Chapter 9	Part II, Ch. 9, §§9.1–9.5 (2017 Chapter 9)	branched ep⁺-subgroupoids: local branching structures, the tangent and the boundary, good points, orientations, the geometry of local branching structures, sets of measure zero and the canonical σ-algebras (integration and Stokes not formalized)	**verified for §§9.1–9.4 and §9.6.1 with the readings and hypotheses recorded below, unconditionally for ep-groupoids whose object M-polyfold is $\mathbb{R}^m$, except the global part of Proposition 9.1.11, Lemma 9.4.3 and, in Figure 9.1, the orientations of the faces of the faces; for §9.5 the sets of measure zero, the σ-algebras and the isotropy factors are verified, branched integration of sc-forms over sheets covered by countably many graph charts, with "oriented, tame" read as a covering by oriented tame structures, the boundary measures and Stokes' theorem as printed for every $n \geq 1$, and Remark 9.5.20 in part (its M-polyfold, the**	59,408

Chapter	HWZ	Content	Status	Lean lines
Chapter 10	Part II, Ch. 10, §§10.1–10.6 (2017 Chapter 10)	equivalences and localization: equivalences of ep-groupoids, natural equivalences, the weak fibred product, localization at the equivalences and generalized maps, strong bundle equivalences, localization in the strong bundle case	**verified for equivalences, natural equivalences, the weak fibred product, the localization $\mathcal{EP}(E^{-1})$ with its universal property, strong bundle equivalences and generalized strong bundle maps, Remark 10.1.8 with Zhou's example, and Propositions 10.4.6, 10.4.7 and 10.5.6 with Definition 8.4.2 read for local strong bundle isomorphisms, with the readings recorded below; Theorems 10.3.19 and 10.5.7 do not hold as we read them, and Theorem 10.3.19 holds for an effective target**	17,543

Chapter	HWZ	Content	Status	Lean lines
Chapter 11	Part II, Ch. 11, §§11.1–11.6 (2017 Chapter 11)	geometry up to equivalences: the tangent functor and faces up to equivalence, branched ep$^+$-subgroupoids under generalized isomorphisms, strong bundles up to equivalence, coverings and equivalences (sc-differential forms and integration not formalized)	**verified for the tangent functor and faces up to equivalence, sc-differential forms up to equivalence, push-forwards and pull-backs of branched ep$^+$-subgroupoids with their tangents, boundaries and orientations, auxiliary norms and multisection functors under generalized strong bundle isomorphisms, proper coverings and proper strong bundle coverings up to equivalence with their localizations, and the canonical measures and boundary measures under generalized isomorphisms, with the readings recorded below (the intersection clause of Proposition 11.1.6 does not hold as we read it), except the boundary**	20,313

Chapter	HWZ	Content	Status	Lean lines
Chapter 12	Part III, Ch. 12, §§12.1–12.5 (2017 Chapter 12)	Fredholm theory in ep-groupoids: sc-Fredholm section functors, auxiliary norms, the good neighbourhood theorem, continuous partitions of unity, the extension of auxiliary norms from the boundary, reflexive auxiliary norms, bump functions and sc^+-section functors, extension from the boundary, compactness and properness, the orientation bundle and its naturality, the reflexive local extension, Theorem 12.2.13 and Theorem 12.5.7 (waves 1–7)	**begun 2026-11-13 (Part III wave 1, continued in waves 2–7), verified for sc-Fredholm section functors, auxiliary norms and their existence, continuous partitions of unity, the good neighbourhood theorem, the extension from the boundary (Theorem 12.2.7 as printed), reflexive auxiliary norms (Theorem 12.2.13 with the para-compactness of $\lvert X\rvert$ named; Part III wave 3), bump functions and sc^+-section functors (Theorems 12.3.2 (2) and 12.3.3 with the pointwise reading of bump functions), the local and equivariant extension from the boundary and Theorem 12.3.9, the compactness notions of §12.4 with the**	13,293

Chapter	HWZ	Content	Status	Lean lines
Chapter 13	Part III, Ch. 13, §§13.1–13.5 (2017 Chapter 13)	sc$^+$-multisections: good systems of neighbourhoods, the natural bi-action, existence and extension of good systems; support, pointwise norm, symmetric structures, decorations, sums, scalar multiples and locally finite sums; structurable and atomic sc$^+$-multisections; equivalences, coverings and structurability, commensurable sums, the constructions of §13.5, Proposition 13.4.2 (waves 1–7)	**begun 2026-11-13 (Part III wave 1, continued in waves 2–5), verified for good systems of neighbourhoods (Theorem 13.1.7 as printed), the natural bi-action, the existence of good systems and Proposition 13.1.9; for the support, pointwise norm, symmetric structures and the decorated algebra of sc$^+$-multisections; for structurable sc$^+$-multisections with Proposition 13.3.19; for the pull-back and push-forward along equivalences and coverings (Theorems 13.4.1, 13.4.3, 13.4.4) and commensurable sums; and for atomic sc$^+$-multisections, Theorem 13.5.3 given a good system, Theorem 13.5.7, and Proposition**	12,388

Chapter	HWZ	Content	Status	Lean lines
Chapter 14	Part III, Ch. 14, §§14.1–14.6 (2017 Chapter 14)	extension of sc$^+$-multisections from the boundary: good neighbourhoods and structured multisections over ∂X, the extension of correspondences, z-loops and coherent neighbourhood systems, local extensions, the size control; Proposition 14.5.4, Theorem 14.1.9 with nonempty index sets, Proposition 14.1.8, the assembly over faces (waves 3–7)	**begun 2026-11-13 (Part III waves 3–7), verified for good neighbourhoods and good systems for ∂X (Theorem 14.1.6 with the chapter's standing assumptions), the tameness of the faces $\bar{\theta}_j$, sc$^+$-multisections and structured sc$^+$-multisections over ∂X, the local extension result, Proposition 14.1.8 in a stronger form, the good structured version, the extension of the correspondences, z-loops, the relations $\sim_z$ and $\approx_z$ with Proposition 14.4.6 as printed, coherent neighbourhood systems and the local extensions of Proposition 14.4.14 for the objects of §14.3, Lemma 14.5.1 with (14.53), Proposition 14.5.4, the**	13,255

Chapter	HWZ	Content	Status	Lean lines
Chapter 15	Part III, Ch. 15, §§15.1–15.4 (2017 Chapter 15)	transversality and invariants: the linearization T(f, Λ), its well-definedness, the separation lemmas; transversality, good position and local solution sets; compactness control, admissible perturbations and general position; orientations and invariants (waves 1–7)	**begun 2026-11-13 (Part III wave 1, continued in waves 2–7), verified for the linearization $T(f,\Lambda)$, its well-definedness and functor property at level-1 points, the sc-Fredholm property of its summands in the chart reading and the separation lemmas (Proposition 15.1.3 for a non-isolated point), for transversality, good position and the local solution sets of §15.2 with Part I's tame germ data, Theorem 15.2.5 also under general position, for the compactness control, admissible perturbations, general position, Remark 15.3.6 and the auxiliary norm on $[0,1]\times X$ of §15.3, and for Theorems 15.3.7, 15.3.9, Corollary 15.3.10, Theorem**	17,485

Chapter	HWZ	Content	Status	Lean lines
Chapter 16	Part III, Ch. 16, §§16.1–16.6 (2017 Chapter 16)	polyfolds: polyfold structures and their equivalence, bump functions and partitions of unity, the degeneracy index, the boundary, tameness; the tangent polyfold and sc-differential forms; strong polyfold bundles; branched suborbifolds and Stokes' theorem; sc$^+$-multisections and Fredholm theory over polyfolds (waves 1–7)	**begun 2026-11-13 (Part III wave 1, continued in waves 2–7), verified for polyfold structures and their metrizability, their equivalence and morphisms, polyfolds and sc-smooth maps, the invariance of bump functions, partitions of unity and tameness, the degeneracy index and the boundary; the tangent polyfold, the raised polyfolds, the tangent functor on PS and on effective polyfolds and the de Rham complex with pull-backs; strong polyfold bundles, auxiliary and reflexive auxiliary norms, independent of the model, and sections, the zero section included; branched suborbifolds**	9,770

Total: 422,427 lines of Lean in 959 files under `Polyfold/` (`wc -l`, excluding unfinished `WIP_` files). Toolchain: Lean v4.33.1, Mathlib v4.33.1. To reproduce: `lake build`, then `lake env lean -DmaxSynthPendingDepth=3 Polyfold/Test.lean`, which prints the axioms of every main theorem. Edition of 13 November 2026.

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1. How to read a verified statement

The statements are the contract. A machine-checked proof guarantees that the *stated* theorem follows from Lean's axioms. It guarantees nothing about whether the statement is the one the authors meant. The reader's job, and the only mathematical judgement this document asks for, is to read each prose restatement and confirm that it says what the text means. Where the restatement is weaker or stronger than the text, or uses a different but equivalent definition, this is flagged as a **Deviation**.

The setting. An sc-Banach space is formalized as an indexed family of real Banach spaces E_m, $m \geq 0$, with injective compact continuous linear inclusions $E_{m+1} \to E_m$ and the smooth points dense in every level; the text's subspace picture $E_0 \supset E_1 \supset \cdots$ is recovered by identifying E_m with its image in E_0, and "$x \in E_m$" for $x \in E_0$ means that x is in that image. A map $f : U \to F$ on a subset $U \subseteq E_0$ is a total function $E_0 \to F_0$ whose values off U play no role; its level maps $f_m : U_m \to F_m$ are determined by the injectivity of the inclusions. "Relatively open in a partial quadrant" enters most theorems only through its consequences on each level, that U_m is the intersection of an open and a convex set and has unique differentiability at each of its points (an *sc-domain*); whenever the text's hypothesis is replaced by this weaker one the item says so.

Classical differentiability. C^k on a relatively open subset of a partial quadrant means Mathlib's C^k *within the set*, with derivatives within the set; on these sets derivatives are unique, so the notion agrees with the exposition's Definition 3.C.

Axioms. Every verified theorem depends only on the three standard axioms of Lean's Mathlib (`propext`, `Classical.choice`, `Quot.sound`), as `Polyfold/Test.lean` prints; there is no `sorry`.

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2. Chapter 1 (HWZ §§1.1–1.4): verified

Chapter 1 of Part I (pp. 5–20 (preprint pp. 6–17)) sets up the sc-calculus: sc-Banach spaces, sc-operators, sc-subspaces and quotients (Definitions 1.1.1–1.1.6 (preprint 1.1–1.5), Propositions 1.1.7–1.1.8 (preprint 1.6–1.7)), sc-Fredholm operators and their stability under sc$^+$-perturbations (Definition 1.1.9 (preprint 1.8), Propositions 1.1.10–1.1.11 (preprint 1.9–1.10)), partial quadrants, sc^0, sc^1, sc^k and sc-smooth maps (Definitions 1.1.12–1.1.15 and the unnumbered definition after them (preprint Definitions 1.11–1.15)), the relation between sc-differentiability and classical differentiability on the levels (Propositions 1.2.1–1.2.4 (preprint 1.16–1.19), Corollary 1.2.5 (preprint 1.20)), the chain rule (Theorem 1.3.1 (preprint 1.21)), the degeneracy index of a partial quadrant (Definition 1.3.3 (preprint 1.23), Lemma 1.3.4 (preprint 1.24)) and boundary recognition (Theorem 1.3.5 (preprint 1.25)). Every numbered statement is verified, Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the text's own hypothesis (C^{l+1} from U_{m+l} to F_m for $l \leq k$), by the 2010 paper's induction (S_k) on elementary maps (V1.37–V1.39). One is verified in a stronger form: Theorem 1.3.5 (preprint 1.25), which the text quotes from another paper, is proved here for sc^1 germs. Example 1.1.3 (preprint 1.2), the weighted Sobolev spaces, is verified since (V1.42; see the addendum below). The remarks of the text that the exposition turns into lemmas are verified as well (V1.28–V1.36): the invariance of the index under sc$^+$-perturbations and on every level, the uniqueness of the sc-structure in finite dimensions, sc-splittings as sc-projections, the smooth points of a quotient, the compact-open continuity of Df, sc^1 = C^1 in finite dimensions on quadrants with boundary, and the level shift and tangent of a tuple. An addendum (2026-11-12) adds Example 1.1.3 (preprint 1.2): the weighted Sobolev spaces $H^{m,\delta_m}(\mathbb{R} \times S^1)$ form an sc-structure on $L^2(\mathbb{R} \times S^1)$, with Rellich's theorem on bounded cylinders proved by Fourier series (V1.42, closing gap G.1), and the content of Remark 1.1.17 (preprint 1.15): relatively open subsets of closed convex partial cones are sc-domains, so §§1.2–1.3 apply to them (V1.43, closing gap G.2). The setting throughout: an sc-Banach space is an indexed family of real Banach spaces E_m with injective compact continuous linear inclusions $E_{m+1} \to E_m$ and dense smooth points; $\iota_m : E_m \to E_0$ is the composite inclusion, and "$x \in E_m$" for $x \in E_0$ means $x \in \iota_m(E_m)$. A map $f : U \to F$, $U \subseteq E_0$, is a total function $E_0 \to F_0$ whose values off U do not matter, with level maps $f_m : U_m \to F_m$ determined by the injectivity of ι_m. The text's hypothesis "U relatively open in a partial quadrant C of E" is, in most theorems, weakened to the *sc-domain* condition it implies: each U_m is the intersection of an open and a convex subset of E_m and has unique differentiability at each of its points. Where a theorem is stated for sc-domains the text's hypothesis is a special case. The infrastructure that Mathlib lacked and that is built here, the Riesz–Schauder theorem and the stability of Fredholm operators under compact perturbations (used without citation in §1.4.1), the compact-open lemma behind §1.2, and sc-domains, has its own items (V1.24–V1.27); so has the machinery of the proof of Proposition 1.2.4 (preprint 1.19) under the text's hypothesis, the derivative tower, the elementary maps with the induction (S_k), and the sum rule for sc^k maps (V1.37–V1.39).

V1.1 (sc-Banach spaces, sums and shifts; `Polyfold.ScScale`, `Polyfold.ScScale.toBase`, `Polyfold.ScScale.smooth`, `Polyfold.ScScale.smoothLvl`, `Polyfold.ScBanach`, `Polyfold.ScBanach.denseRange_inclK`, `Polyfold.ScBanach.const`, `Polyfold.ScBanach.prod`, `Polyfold.ScBanach.prod_smoothLvl`, `Polyfold.ScBanach.shift`, `Polyfold.ScBanach.shift_smoothLvl`, `Polyfold.ScBanach.shiftK`, `Polyfold.ScBanach.lvlSet`). A *scale* consists of real Banach spaces E_m, $m \in \mathbb{N}$, and injective compact continuous linear maps $E_{m+1} \to E_m$; $\iota_m : E_m \to E_0$ is the composite, $E_\infty \subseteq E_0$ the subspace of points that lie in $\iota_m(E_m)$ for every m, and $E_\infty \cap E_m := \iota_m^{-1}(E_\infty)$. An *sc-Banach space* is a scale in which $E_\infty \cap E_m$ is dense in E_m for every m. Consequences: the image of E_{m+k} in E_m is dense. Constructions: for a finite-dimensional normed space V the constant family $E_m = V$, with identity inclusions, is an sc-Banach space all of whose points are smooth; for sc-Banach spaces E, F, the family $E_m \times F_m$ with the product inclusions is an sc-Banach space $E \oplus F$ whose smooth points at level m are the pairs of smooth points; the family E_{m+1} with the inclusions of E is an sc-Banach space E^1 whose smooth points at level m are those of E at level $m+1$; E^k is the k-fold iterate. For $U \subseteq E_0$, the level set $U_m := \iota_m^{-1}(U) \subseteq E_m$. *Book:* Definition 1.1.1 (preprint 1.1), p. 5 (preprint p. 6): a decreasing sequence of Banach spaces $E = E_0 \supset E_1 \supset \cdots$ with compact inclusions and $E_\infty = \bigcap E_m$ dense in every E_m; the constant structure on a finite-dimensional space; $(E \oplus F)_m = E_m \oplus F_m$; the filtration $A_m = A \cap E_m$ of a subset and the shift A^k. *Hidden hypotheses:* the inclusions are injective (implicit in "$\subset$") and continuous (part of compactness). *Deviation (restated):* the E_m are separate Banach spaces, not subsets of one space; the two pictures are equivalent and nothing in Chapter 1 uses more than the family. The text's remarks that a finite-dimensional space has *only* the constant structure and that the constant family on an infinite-dimensional space is not an sc-structure are V1.31.

V1.2 (sc-operators and sc-isomorphisms; `Polyfold.ScOperator`, `Polyfold.ScOperator.ext_base`, `Polyfold.IsScOperator`, `Polyfold.ScOperator.isScOperator`, `Polyfold.IsScOperator.toScOperator`, `Polyfold.ScOperator.comp`, `Polyfold.ScOperator.mapsTo_smoothLvl`, `Polyfold.ScIso`, `Polyfold.ScIso.symm`, `Polyfold.ScIso.trans`, `Polyfold.isScOperator_of_mapsTo`, `Polyfold.ScIso.ofBijective`). An *sc-operator* $T : E \to F$ between sc-Banach spaces is a family of continuous linear maps $T_m : E_m \to F_m$ with $\iota^F_m T_{m+1} = T_m \iota^E_m$; the family is determined by T_0, and a continuous linear $T_0 : E_0 \to F_0$ *is an sc-operator* exactly when for each m some continuous linear $T_m : E_m \to F_m$ satisfies $\iota_m T_m = T_0 \iota_m$ (the two formulations correspond). Sc-operators compose, add, and map smooth points to smooth points. An *sc-isomorphism* is a family of continuous linear equivalences $E_m \simeq F_m$ commuting with the inclusions; inverses and composites are sc-isomorphisms. A continuous linear $T_0 : E_0 \to F_0$ with $T_0(E_m) \subseteq F_m$ for every m is an sc-operator (the continuity of $T_0 : E_m \to F_m$ is the closed graph theorem), and an sc-operator bijective on every level is an sc-isomorphism (the open mapping theorem). *Book:* Definition 1.1.4 (preprint 1.3), p. 6 (preprint p. 7). *Hidden hypotheses:* none. *Deviation (restated):* the level maps are data, with the text's predicate on the base map recovered as `Polyfold.IsScOperator`; the continuity of the inverse of a bijective sc-operator on each level, which the text's definition of an sc-isomorphism demands, is built into the formal notion, and derived from the open mapping theorem separately (`Polyfold.ScIso.ofBijective`); the continuity on the levels asked by Definition 1.1.4 (preprint 1.3) is automatic for a bounded T_0 that maps E_m into F_m (Findings, Chapter 3, item 233).

V1.3 (sc$^+$-operators; `Polyfold.ScPlus`, `Polyfold.ScBanach.shiftIncl`, `Polyfold.ScPlus.toScOperator`, `Polyfold.ScPlus.isCompactOperator`, `Polyfold.ScPlus.toScOperator_mem_range_toBase_succ`). An sc$^+$-*operator* $E \to F$ is an sc-operator $E \to F^1$, i.e. continuous linear $S_m : E_m \to F_{m+1}$ commuting with the inclusions. Composed with the inclusion $F^1 \to F$ (an sc-operator) it is an sc-operator $E \to F$ whose level-m map $E_m \to F_m$ is compact for every m, and which raises the level: $x \in \iota_m(E_m)$ implies $S_0 x \in \iota_{m+1}(F_{m+1})$. *Book:* Definition 1.1.5 (preprint 1.4), p. 7 (preprint p. 7), and the remark that follows it: an sc-operator S with $S(E_m) \subset F_{m+1}$ and $S : E \to F^1$ an sc-operator is sc-compact. *Hidden hypotheses:* none; the condition $S(E_m) \subset F_{m+1}$ is contained in "$S : E \to F^1$ is an sc-operator" and is dropped.

V1.4 (sc-subspaces and sc-splittings; `Polyfold.ScBanach.levelSub`, `Polyfold.`

`ScBanach.isClosed_levelSub`, `Polyfold.ScBanach.IsScSubspace`, `Polyfold.ScBanach.ofSubspaceScale`, `Polyfold.ScBanach.ofSubspace`, `Polyfold.ScBanach.IsScSplitting`, `Polyfold.ScBanach.IsScSplitting.closedComplemented`, `Polyfold.ScBanach.IsScSplitting.symm`). For a subspace $A \subseteq E_0$ let $A_m := \iota_m^{-1}(A) \subseteq E_m$, closed if A is. A is an *sc-subspace* if A is closed and A_m is contained in the closure of $A_m \cap E_\infty$ for every m. A closed A with the levels A_m and the restricted inclusions is a scale (the restricted inclusions are compact), and an sc-subspace is in this way an sc-Banach space. $E = F \oplus G$ is an *sc-splitting* if F and G are sc-subspaces and F_m, G_m are complementary subspaces of E_m for every m; then F_m is closed complemented in E_m, and the splitting is symmetric in F and G. *Book:* Definition 1.1.6 (preprint 1.5), p. 7 (preprint p. 7): F closed with $(F \cap E_m)$ an sc-structure; an sc-complement G with $(G \cap E_m)$ an sc-structure and $E_m = F_m \oplus G_m$ a topological direct sum on every level. *Hidden hypotheses:* none. *Deviation (restated):* the closedness of F_m in E_m and the compactness of $F_{m+1} \to F_m$ are automatic and are not conditions; "topological direct sum" is read as complementary closed subspaces, the continuity of the projections following from the closed graph theorem (`Polyfold.ScBanach.IsScSplitting.closedComplemented`).

V1.5 (Proposition 1.1.7 (preprint 1.6); `Polyfold.ScBanach.proposition_1_1_7_iff`, `Polyfold.ScBanach.proposition_1_1_7`). Let E be an sc-Banach space and $F \subseteq E_0$ a finite-dimensional subspace. F is an sc-subspace if and only if $F \leq E_\infty$; and if $F \leq E_\infty$ there is a subspace $G \subseteq E_0$ such that $E = F \oplus G$ is an sc-splitting. *Book:* preprint p. 7, the same. *Hidden hypotheses:* none. *Proof:* as in the text, with a continuous projection $E_0 \to F$ from Mathlib (Hahn–Banach), lifted to each level through $F_m \cong F$.

V1.6 (Proposition 1.1.8 (preprint 1.7); `Polyfold.ScBanach.quotientScale`, `Polyfold.ScBanach.quotient`, `Polyfold.ScBanach.quotientMap`, `Polyfold.ScBanach.proposition_1_1_8`). Let E be an sc-Banach space and $A \subseteq E_0$ a closed subspace. The quotient Banach spaces E_m/A_m, with the inclusions $E_{m+1}/A_{m+1} \to E_m/A_m$ induced by ι (injective and compact), and with the smooth points dense on every level, form an sc-Banach space E/A, and the quotient maps $E_m \to E_m/A_m$ form an sc-operator $E \to E/A$. As a statement: there exist an sc-Banach space Q and an sc-operator $q : E \to Q$ with every q_m surjective and $\ker q_m = A_m$. *Book:* preprint p. 8, for an sc-subspace A: E/A with the filtration E_m/A_m is an sc-Banach space, where E_m/A_m is identified with $\{(x + A) \cap E_m : x \in E\}$. *Hidden hypotheses:* none. *Deviation (stronger):* only closedness of A is used, not that A is an sc-subspace. The text's identification of E_m/A_m with a subset of E/A is the injectivity of the induced inclusion, which holds because A_m is the full preimage of A. The text's description of $(E/A)_\infty$ as the classes of points of E_∞ is V1.33; only the trivial inclusion is used here.

V1.7 (sc-Fredholm operators and the index; `Polyfold.ScFredholmSplitting`, `Polyfold.IsScFredholm`, `Polyfold.ScFredholmSplitting.index`, `Polyfold.scIndex`, `Polyfold.ScFredholmSplitting.index_eq`, `Polyfold.ScFredholmSplitting.K_le_smooth`, `Polyfold.ScFredholmSplitting.ker_lvl`, `Polyfold.ScFredholmSplitting.range_lvl`, `Polyfold.ScFredholmSplitting.isFredholm`). An *sc-Fredholm splitting* of an sc-operator $T : E \to F$ is given by subspaces $K, X \subseteq E_0$ and $C, Y \subseteq F_0$ such that $E = K \oplus X$ and $F = C \oplus Y$

are sc-splittings, $K = \ker T_0$, $Y = \operatorname{range} T_0$, K and C are finite-dimensional, and for every m the map T_m is a bijection of X_m onto Y_m. T is *sc-Fredholm* if a splitting exists. The index $\dim K - \dim C$ of a splitting equals $\dim \ker T_0 - \dim(F_0/\operatorname{range} T_0)$, hence does not depend on the splitting. Moreover $K \leq E_\infty$; $\ker T_m = K_m$ and $\operatorname{range} T_m = Y_m$; and every $T_m : E_m \to F_m$ is a Fredholm operator in Mathlib's sense (finite-dimensional kernel, closed range of finite codimension, strict, with closed complemented kernel). *Book:* Definition 1.1.9 (preprint 1.8), p. 8 (preprint p. 9), with the remarks that $K \subset E_\infty$, $T(X_m) = Y_m$, $E_m = K \oplus X_m$ and "$F = C \oplus T(E_m)$" (read F_m), and $\operatorname{ind}(T) = \dim K - \dim C$. *Hidden hypotheses:* none. *Deviation (restated):* condition (3), "$T : X \to Y$ is an sc-isomorphism", is taken as the bijectivity of $T_m : X_m \to Y_m$ for every m; the continuity of the inverses is automatic by the open mapping theorem on the closed subspaces X_m, Y_m. The independence of the index from the splitting and the level-wise Fredholm property are not stated in the text.

V1.8 (Proposition 1.1.10 (preprint 1.9); `Polyfold.proposition_1_1_10`, `Polyfold.ScOperator.IsRegularizing`, `Polyfold.IsScFredholm.isRegularizing`, `Polyfold.isScFredholm_iff`). Let $T : E \to F$ be an sc-operator with an sc-Fredholm splitting, $e \in E_0$ and $m \in \mathbb{N}$. If $T_0 e \in \iota_m(F_m)$ then $e \in \iota_m(E_m)$. As a property of sc-operators (`Polyfold.ScOperator.IsRegularizing`): every sc-Fredholm operator is regularizing. Conversely, an sc-operator T with T_0 Fredholm that is regularizing is sc-Fredholm (V1.9); so T is sc-Fredholm if and only if $T_0 : E_0 \to F_0$ is Fredholm and T is regularizing. *Book:* preprint p. 9, the same ("an sc-Fredholm operator is regularizing"). *Hidden hypotheses:* none; the step "$c = 0$" uses $C \cap Y = 0$ on level 0, as in the text. The converse is not stated in the text; it is the tool for the text's unargued "is sc-Fredholm" for operators obtained by conjugation and restriction in Chapter 3 (V3.8, V3.46, V3.96; Findings, Chapter 3, item 233).

V1.9 (Proposition 1.1.11 (preprint 1.10); `Polyfold.proposition_1_1_11_preprint`, `Polyfold.ScPlus.regularizing_add`, `Polyfold.isScFredholm_of_isRegularizing`, `Polyfold.isScFredholm_of_isRegularizing_index`). Let E, F be sc-Banach spaces, $T : E \to F$ an sc-Fredholm operator and $S : E \to F$ an sc$^+$-operator. Then $T + S$ (with S read as an sc-operator $E \to F$) is sc-Fredholm. Along the way, $T + S$ is regularizing in the sense of V1.8. *Book:* preprint p. 10, the same; the proof is in §1.4.1, pp. 14–16 (preprint pp. 14–15). *Hidden hypotheses:* none. *Proof:* the text's, with three additions. The opening claim that $T + S : E_m \to F_m$ is Fredholm is the classical compact-perturbation theorem, proved here since Mathlib lacks it (V1.24–V1.25). The step "$y = (T + S)e$, so by Proposition 1.9 $e \in E_m$" needs a bootstrap, since Se is only one level up: $Te = y - Se \in F_1$ gives $e \in E_1$, then $Se \in F_2$, and so on m times; the same bootstrap gives $K_m = K_{m+1}$. The complement C is the range of a linear right inverse of $F_0 \to F_0/Y$ with values in F_∞ rather than the span of a lifted basis, and condition (3) of Definition 1.1.9 (preprint 1.8) for $T + S$, which the text does not verify, is proved as the bijectivity of $(T + S)_m : X_m \to Y_m$. The text does not assert that the index is preserved; it is, and this is V1.30. The construction uses only two properties of $T + S$, that it is Fredholm on level 0 and regularizing; with these as hypotheses it is proved as a criterion: an sc-operator that is Fredholm on level 0 and regularizing is sc-Fredholm, and every level map is then Fredholm of the level-0 index (not stated in the text).

V1.10 (partial quadrants and tuples; `Polyfold.quadrant`, `Polyfold.quadrant_convex`, `Polyfold.quadrant_isClosed`, `Polyfold.lvlSet_quadrant`, `Polyfold.IsPartialQuadrant`, `Polyfold.IsPartialQuadrant.convex`, `Polyfold.IsPartialQuadrant.isClosed`, `Polyfold.IsPartialQuadrant.uniqueDiffWithinAt_lvlSet`, `Polyfold.isPartialQuadrant_univ`, `Polyfold.IsRelOpenInQuadrant`). In the sc-Banach space $\mathbb{R}^n \oplus W$ (with $\mathbb{R}^n$ constant) the *quadrant* on level m is $\{(a, w) : a_i \geq 0 \text{ for all } i\}$; it is closed and convex with non-empty interior, and the level sets of the level-0 quadrant are the level quadrants. $C \subseteq E_0$ is a *partial quadrant* if there are n, an sc-Banach space W and an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ with $C = T_0^{-1}(\text{quadrant})$. Then every C_m is closed, convex, with non-empty interior in E_m, and C_m has unique differentiability (in Mathlib's sense) at each of its points; E_0 itself is a partial quadrant ($n = 0$). (U, C, E) is a *tuple*, i.e. U is a *relatively open subset of the partial quadrant* C, if C is a partial quadrant and $U = O \cap C$ with $O \subseteq E_0$ open. *Book:* Definition 1.1.12 (preprint 1.11), p. 9 (preprint p. 10), and the sentence after it. *Hidden hypotheses:* none; "closed convex" is derived from the model. *Deviation:* none in the definition. The text's tacit use of C_k as a partial quadrant in E^k is V1.36 (for $k = 1$, and by iteration).

V1.11 (sc⁰ maps; `Polyfold.levelMap`, `Polyfold.IsSc0`, `Polyfold.IsSc0.levelMap_mapsTo`, `Polyfold.IsSc0.continuousOn_base`, `Polyfold.IsSc0.congr`, `Polyfold.IsSc0.mono`, `Polyfold.IsSc0.comp`, `Polyfold.isSc0_id`, `Polyfold.ScOperator.isSc0`). For $f : E_0 \to F_0$ the *level map* $f_m : E_m \to F_m$ sends x to the unique $y \in F_m$ with $\iota_m y = f(\iota_m x)$, when such y exists (and to a junk value otherwise). f is sc^0 *on* $U \subseteq E_0$ if for every m and $x \in U_m$ the point $f(\iota_m x)$ lies in $\iota_m(F_m)$, and f_m is continuous on U_m. Then f is continuous on U; if $f(U) \subseteq V$ then $f_m(U_m) \subseteq V_m$; the property depends only on the values of f on U and passes to subsets of U; the composition of an sc^0 map on U with values in V and an sc^0 map on V is sc^0 on U; the identity and every sc-operator are sc^0, the latter with level maps T_m. *Book:* Definition 1.1.13 (preprint 1.12), p. 9 (preprint p. 10): $f(U_m) \subset V_m$ (meaning U'_m) and $f : U_m \to U'_m$ continuous for all m. *Hidden hypotheses:* none. *Deviation (restated):* maps are total functions on E_0, and the target tuple enters only through $f(U) \subseteq U'$.

V1.12 (the tangent; `Polyfold.ScBanach.tangent`, `Polyfold.ScBanach.tangentSet`, `Polyfold.ScBanach.lvlSet_tangentSet`, `Polyfold.ScBanach.tangentSet_mono`, `Polyfold.IsScDomain.tangentSet`, `Polyfold.ScBanach.shift_tangentSet`). The *tangent* of an sc-Banach space E is $TE := E^1 \oplus E$, with levels $E_{m+1} \times E_m$. For $U \subseteq E_0$, $TU \subseteq (TE)_0 = E_1 \times E_0$ is $\{(x, h) : x \in U_1\}$, with level-m part $U_{m+1} \times E_m$; $U \subseteq V$ implies $TU \subseteq TV$. If U is an sc-domain (V1.27), TU is an sc-domain in TE; and the level-1 part of TU is $T(U_1)$ in $T(E^1) = (TE)^1$. *Book:* Definition 1.1.14 (preprint 1.13), p. 9 (preprint p. 10): $T(U, C, E) = (U^1 \oplus E, C^1 \oplus E, E^1 \oplus E)$, "again a tuple". *Hidden hypotheses:* none. *Deviation:* none; that TC is a partial quadrant in TE, so that $T(U, C, E)$ is again a tuple, is V1.36, while the proofs of §§1.2–1.3 propagate the sc-domain property (V1.27) instead.

V1.13 (sc¹ maps and the tangent map; `Polyfold.tangentMapWith`, `Polyfold.IsSc1With`, `Polyfold.IsSc1`, `Polyfold.scDeriv`, `Polyfold.tangentMap`, `Polyfold.IsSc1With.eq_of_uniqueDiffWithinAt`, `Polyfold.IsSc1.scDeriv_eqOn`, `Polyfold.IsSc1.tangentMap_`

`eqOn`, `Polyfold.IsSc1.congr`, `Polyfold.IsSc1With.continuousOn_deriv`, `Polyfold.IsSc1With.exists_delta_deriv`, `Polyfold.exists_isSc1_not_continuousOn_deriv`, `Polyfold.ScOneExample.isSc1With`, `Polyfold.ScOneExample.not_continuousAt_D`). Let $f : E_0 \to F_0$, $U \subseteq E_0$ and $D : E_1 \to \mathcal{L}(E_0, F_0)$ a family of bounded operators. The tangent map $T_D f : (TE)_0 \to (TF)_0$ is $(x, h) \mapsto (f_1(x), D(x)h)$. f is sc^1 *on U with derivatives D* if (1) f is sc^0 on U; (2) for every $x \in U_1$, the map $f \circ \iota_1 : E_1 \to F_0$ has derivative $D(x) \circ \iota_1$ at x within U_1; (3) $T_D f$ is sc^0 on TU. f is sc^1 *on U* if some D makes it so; `Polyfold.scDeriv` is a chosen such D (junk if none exists) and `Polyfold.tangentMap` the tangent map built from it. If D and D' both satisfy (2) at a point $x \in U_1$ where U_1 has unique differentiability, then $D(x) = D'(x)$, because E_1 has dense image in E_0; hence on an sc-domain the chosen derivative agrees on U_1 with any admissible one, and the tangent map on TU. The property depends only on the values of f on U. Level 0 of condition (3) is the continuity of $(x, h) \mapsto D(x)h$ on $U_1 \times E_0$, and it implies: for $x \in U_1$, a compact $K \subseteq E_0$ and $c > 0$ there is $\delta > 0$ with $\|D(x')w - D(x)w\| \le c$ for all $x' \in U_1$ with $\|x' - x\|_1 < \delta$ and all $w \in K$ (continuity of $x \mapsto D(x)$ in the compact-open topology). It need not be norm continuous: on $E = \mathbb{R} \oplus X$, with X the sc-Banach space whose level m consists of the real sequences with $(k+1)^m x_k \to 0$, normed by $\sup_k (k+1)^m |x_k|$ (every level a copy of c_0, the inclusions diag$(1/(k+1))$), the map $f(t, x)_k = \cos(kt)x_k$ is sc^1 with $Df(t, x)(\tau, \xi)_k = -k \sin(kt) x_k \tau + \cos(kt)\xi_k$, and for every derivative family D of f the map $x \mapsto D(x)$ is not continuous on E_1 into $\mathcal{L}(E_0, X_0)$ with the operator norm: $(\pi/(n+1), 0) \to 0$ in E_1, but $\|Df(\pi/(n+1), 0) - Df(0, 0)\| \ge 2$. *Book:* Definition 1.1.15 (preprint 1.14), pp. 10 ff. (preprint pp. 10–11), with conditions (1)–(3) as stated, and the remarks that $x \mapsto Df(x)$ is continuous for the compact-open but not in general for the norm topology, and that sc¹ maps between finite-dimensional spaces are the C^1 maps. *Hidden hypotheses:* the uniqueness of $Df(x)$, without which "the tangent map" is undefined; it holds because E_1 is dense in E_0 and U_1 has unique differentiability (true for relatively open subsets of partial quadrants, V1.10). *Deviation (hypotheses explicit):* the derivative family is an explicit parameter of `Polyfold.IsSc1With`, with `Polyfold.IsSc1` the text's notion. The compact-open remark is V1.34, its negative half the example above, which the text does not give (Findings, Chapter 1, follow-up item 1), and the finite-dimensional remark is V1.35 (on relatively open subsets of quadrants, with boundary; the open case is also in `Polyfold/Test.lean`, `isSc1_iff_contDiffOn_of_finiteDimensional`).

V1.14 (sc^k and sc-smooth maps; `Polyfold.IsSck`, `Polyfold.IsScSmooth`, `Polyfold.IsSck.of_succ`, `Polyfold.IsSck.of_le`, `Polyfold.isScSmooth_iff`, `Polyfold.IsSck.congr`). By recursion: f is sc^0 on U in the sense of V1.11; f is sc^{k+1} on U if it is sc^1 on U and its tangent map (built from the chosen derivative) is sc^k on $TU \subseteq (TE)_0$ as a map into $(TF)_0$. f is *sc-smooth* on U if it is sc^k for every k. Then sc^{k+1} implies sc^k, and sc^l for $l \le k$; f is sc-smooth iff it is sc^1 with sc-smooth tangent map; on an sc-domain, sc^k depends only on the values on U. *Book:* preprint p. 11, the inductive definition after Definition 1.1.15 (preprint 1.14) (numbered 1.15 here, as in the Lean files; Remark 1.1.17 (preprint 1.15) on partial cones is not formalized). *Hidden hypotheses:* none beyond V1.13.

V1.15 (Proposition 1.2.1 (preprint 1.16); `Polyfold.IsSc1Levels`, `Polyfold.IsSc1Levels.continuousOn_clm`, `Polyfold.IsSc1Levels.contDiffOn`, `Polyfold.IsSc1Levels.toBase_`

`apply`, `Polyfold.IsSc1Levels.isSc1With`, `Polyfold.IsSc1With.isSc1Levels`, `Polyfold.proposition_1_2_1`, `Polyfold.IsSc1With.contDiffOn_base`, `Polyfold.IsSc1With.scOperator_deriv`). Let U be an sc-domain in E and f sc^0 on U. Then f is sc^1 on U if and only if there is a family $D_m(x) \in \mathcal{L}(E_m, F_m)$, $m \in \mathbb{N}$, $x \in U_{m+1}$, such that for every m: (a) $f_m \circ \iota : U_{m+1} \to F_m$ has derivative $D_m(x) \circ \iota$ at every $x \in U_{m+1}$ within U_{m+1}; (b) $(x, h) \mapsto D_m(x)h$ is continuous on $U_{m+1} \times E_m$. For any such family, $x \mapsto D_m(x) \circ \iota$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$ in the operator norm, so $f : U_{m+1} \to F_m$ is C^1 within U_{m+1}; and $\iota_m D_m(x) = D_0(\iota x)\iota_m$, so all $D_m(x)$ are restrictions of the single operator $Df(x) := D_0(x)$. In the direction "sc^1 □ levels" the family is the level operators of V1.26; in particular $f \circ \iota_1 : U_1 \to F_0$ is C^1. If f is sc^1 with derivatives Df and $x \in U_1$ is a smooth point, $Df(x) : E_0 \to F_0$ is an sc-operator. *Book:* preprint p. 12: an sc^0 map on a relatively open subset of a partial quadrant is sc^1 iff (1) for $m \geq 1$, $f : U_m \to F_{m-1}$ is C^1 with $df : U_m \to \mathcal{L}(E_m, F_{m-1})$ continuous and (2) $df(x)$ extends to a bounded $Df(x) : E_{m-1} \to F_{m-1}$ with $(x, h) \mapsto Df(x)h$ continuous on $U_m \oplus E_{m-1}$; in particular $Df(x)$ is an sc-operator at smooth points. *Hidden hypotheses:* the compatibility of the extensions $Df(x)$ across levels, which the text's notation assumes and the direction (□) needs (Lemma 3.H of the exposition); it is proved. *Deviation (restated):* the extensions form an explicit family; the C^1 property and the norm continuity of df are consequences of (a)–(b) rather than part of the condition; the hypothesis on U is the weaker sc-domain condition.

V1.16 (Proposition 1.2.2 (preprint 1.17); `Polyfold.IsSc0.shift`, `Polyfold.IsSc1With.shift`, `Polyfold.IsSck.shift`, `Polyfold.IsScSmooth.shift`, `Polyfold.IsScDomain.shift`). Let U be an sc-domain in E and f sc^k on U. Then the level-1 map $f_1 : E_1 \to F_1$ is sc^k on U_1 as a map between the shifted sc-Banach spaces E^1 and F^1 (whose level sets of U_1 are the U_{m+1}). For $k = 0$ no hypothesis on U is needed; for an sc^1 map with derivatives Df, the derivatives of f_1 are the level-1 operators $Df(x) : E_1 \to F_1$, $x \in U_2$; sc-smooth maps lift to sc-smooth maps; and U_1 is an sc-domain in E^1. *Book:* preprint p. 12: if $f : U \to V$ is sc^k between relatively open subsets of partial quadrants then $f : U^1 \to V^1$ is sc^k. *Hidden hypotheses:* none. *Deviation:* none; the target V plays no role and the hypothesis on U is weaker.

V1.17 (Proposition 1.2.3 (preprint 1.18); `Polyfold.IsSck.contDiffOn_levels'`, `Polyfold.IsSck.contDiffOn_levels`, `Polyfold.IsSck.contDiffOn_levels_of_le`, `Polyfold.IsSck.contDiffOn_base`, `Polyfold.IsScSmooth.contDiffOn_levels`). Let U be an sc-domain in E and f sc^k on U. Then for every m, the map $f_m \circ \iota : U_{m+k} \to F_m$ is C^k within U_{m+k}; for $l \leq k$, $f_m \circ \iota : U_{m+l} \to F_m$ is C^l; in particular $f \circ \iota_k : U_k \to F_0$ is C^k. If f is sc-smooth, $f : U_{m+k} \to F_m$ is C^k for all m and k. *Book:* preprint p. 13, the same for $f : U \to V$ between relatively open subsets of partial quadrants. *Hidden hypotheses:* the meaning of "C^k" on a relatively open subset of a quadrant, never defined in the text: here Mathlib's C^k within the set, with unique derivatives on these sets. *Deviation:* none in the statement. *Proof:* a direct induction on k over all sc-domains, with a compatible level-lowering operator $\rho : E_{m+k} \to E_m$ as parameter, and the compact-open lemma (V1.26) applied to the second component of the level tangent map; the 2010 paper's multilinear maps $\Gamma(x)$ and its differentiation under the integral sign are not used.

V1.18 (Proposition 1.2.4 (preprint 1.19); `Polyfold.proposition_1_2_4_text`, `Polyfold.`

`IsScSmooth.of_contDiffOn_levels_text`, `Polyfold.proposition_1_2_4_zero`, `Polyfold.proposition_1_2_4`, `Polyfold.IsScSmooth.of_contDiffOn_levels`). Let U be an sc-domain in E and f sc^0 on U. If for every $m \geq 0$ and $0 \leq l \leq k$ the map $f_m \circ \iota : U_{m+l} \to F_m$ (the level-m map restricted along the inclusion $E_{m+l} \to E_m$) is C^{l+1} within U_{m+l}, then f is sc^{k+1} on U; if this holds for every l, then f is sc-smooth. Also proved, under the stronger hypothesis that every level map $f_m : U_m \to F_m$ is C^{k+1} within U_m: f is sc^{k+1}; for $k = 0$, where the two hypotheses agree, f is sc^1 with $Df(x)$ the derivative of $f : U_0 \to F_0$ within U at $\iota_1 x$; and if f maps U_m into F_m for every m and every f_m is C^∞, then f is sc-smooth. *Book:* preprint p. 13: if $f : U_{m+l} \to F_m$ is C^{l+1} for all $m \geq 0$ and $0 \leq l \leq k$, then f is sc^{k+1}. *Hidden hypotheses:* none ("f induces a map $U_{m+l} \to F_m$" is the sc^0 hypothesis). *Deviation:* none; the sc-smooth statement is a direct consequence. *Proof:* the 2010 paper's (Proposition 2.4): f is the elementary map with no vectors, and every elementary map with j vectors and $j + s \leq k + 1$ is sc^s (V1.38), with the derivatives $d^j f(x)$ supplied by the derivative tower of f (V1.37). The induction cannot be run on the hypothesis itself: if f satisfies it with k, then Tf in general does not satisfy it with $k-1$, since its second component $(x, h) \mapsto Df(x)h : U_{m+l+1} \times E_{m+l} \to F_m$ is only C^l. Under the stronger hypothesis the level maps of Tf are C^k and the induction is immediate.

V1.19 (Corollary 1.2.5 (preprint 1.20); `Polyfold.corollary_1_2_5_text`, `Polyfold.IsScSmooth.of_contDiffOn_levels_finiteDimensional_text`, `Polyfold.IsScSmooth.of_contDiffOn_finiteDimensional`, `Polyfold.corollary_1_2_5_zero`, `Polyfold.corollary_1_2_5`, `Polyfold.corollary_1_2_5_base`). Let U be an sc-domain in E, V a finite-dimensional normed space with its constant sc-structure and $f : E_0 \to V$. If for some k and all $0 \leq l \leq k$ the map $f \circ \iota_l : U_l \to V$ is C^{l+1} within U_l, then f is sc^{k+1} on U. If $f \circ \iota_l$ is C^{l+1} within U_l for every l, then f is sc-smooth; if f is C^∞ within U, then f is sc-smooth. Also proved, under the stronger hypothesis: if $f \circ \iota_m$ is C^{k+1} within U_m for every m, in particular if f is C^{k+1} within U, then f is sc^{k+1}; for $k = 0$ this is the text's hypothesis. *Book:* preprint p. 13: if $f : U_l \to \mathbb{R}^N$ is C^{l+1} for $0 \leq l \leq k$ then f is sc^{k+1}. *Hidden hypotheses:* none; no sc^0 hypothesis is needed, since $f \circ \iota_m$ is the continuous $f \circ \iota_0$ composed with an inclusion. *Deviation:* none ($\mathbb{R}^N$ is any finite-dimensional space); the two sc-smooth statements are the exposition's additions. *Proof:* Proposition 1.2.4 (preprint 1.19) (V1.18) for the constant structure, the hypothesis on $U_{m+l} \to V$ being the restriction of the one on U_l.

V1.20 (Theorem 1.3.1 (preprint 1.21), the chain rule; `Polyfold.IsSc1With.hasFDerivWithinAt_comp`, `Polyfold.IsSc1With.comp`, `Polyfold.theorem_1_3_1'`, `Polyfold.theorem_1_3_1`, `Polyfold.IsSck.comp`, `Polyfold.IsSck.comp_of_isRelOpenInQuadrant`, `Polyfold.IsScSmooth.comp`). Let $U \subseteq C \subseteq E$ and $V \subseteq D \subseteq F$ be relatively open subsets of partial quadrants, G an sc-Banach space, $f : E_0 \to F_0$ sc^1 on U with $f(U) \subseteq V$, and $g : F_0 \to G_0$ sc^1 on V. Then $g \circ f$ is sc^1 on U, and $T(g \circ f) = Tg \circ Tf$ on TU. More precisely: if g is sc^1 on V with derivatives Dg, f is sc^1 on U with derivatives Df, $f(U) \subseteq V$, and V_1 is the intersection of an open and a convex set, then $g \circ f$ is sc^1 on U with derivatives $x \mapsto Dg(f_1(x)) \circ Df(x)$, with no hypothesis on U; and $g \circ f \circ \iota_1$ has this derivative at every $x \in U_1$ within U_1. Iterated: if U, V are sc-domains (in particular relatively open subsets of partial quadrants), f sc^k on U with $f(U) \subseteq V$ and g sc^k on V, then $g \circ f$ is sc^k on U; sc-smooth maps compose to sc-smooth maps. *Book:* preprint p. 13, with a third tuple $W \subset Q \subset G$ for the target of g; the proof is §1.4.2, pp. 15–16 (preprint pp. 15–16). *Hidden hypotheses:* the segment from $f(x)$ to $f(x+h)$ lies in V_1 for $|h|_1$ small, which uses the convexity of D_1, the openness of V_1 in D_1 and the continuity of $f : U_1 \to F_1$; the uniqueness of $D(g \circ f)(x)$ behind "$T(g \circ f) = Tg \circ Tf$". *Deviation:* none in the statement; $W \subset Q \subset G$ plays no role. *Proof:* the text's two integrals are replaced by the mean value inequality for $t \mapsto g(y_t) - t\,Dg(f(x))(f(x+h) - f(x))$ on $[0,1]$; the first integral is not needed since $Dg(f(x))$ is bounded on F_0; the uniform convergence on the compact closure of $\{Df(x)h/|h|_1\}$ is Mathlib's `IsCompact.eventually_forall_of_forall_eventually` rather than a subsequence argument. The text's appeal to Proposition 1.2.3 (preprint 1.18) is not used.

V1.21 (Definition 1.3.3 (preprint 1.23); `Polyfold.quadrantDeg`, `Polyfold.degIndexWith`, `Polyfold.degIndex`, `Polyfold.degIndex_eq`, `Polyfold.degIndex_quadrant`). For $p = (a, w) \in \mathbb{R}^n \oplus W$ let $\#(p)$ be the number of i with $a_i = 0$; for an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ presenting a partial quadrant C let $d_T(x) := \#(T_0 x)$. The *degeneracy index* $d_C : E_0 \to \mathbb{N}$ is d_T for a chosen presentation T of C (and 0 if C is not a partial quadrant); for every presentation T and every $x \in C$, $d_C(x) = d_T(x)$. On the quadrant itself $d(p) = \#(p)$. *Book:* preprint p. 14: $d_C(x) = \#\{i : a_i = 0\}$ where $T(x) = (a_1, \dots, a_k, w)$; interior points, boundary points and corners. *Hidden hypotheses:* none. *Deviation:* none; d_C is total on E_0 with junk values off C, and the classification into interior, boundary and corner points is terminology.

V1.22 (Lemma 1.3.4 (preprint 1.24); `Polyfold.quadrantDeg_eq_of_equiv`, `Polyfold.lemma_1_3_4`, `Polyfold.lemma_1_3_4_dim`). Let $T : E \to \mathbb{R}^n \oplus W$ and $T' : E \to \mathbb{R}^{n'} \oplus W'$ be sc-isomorphisms with $C = T_0^{-1}(Q_n) = T_0^{\prime -1}(Q_{n'})$. Then $d_T(x) = d_{T'}(x)$ for every $x \in C$, and $n = n'$. Core: if $S : \mathbb{R}^n \times W_0 \to \mathbb{R}^{n'} \times W_0'$ is a continuous linear equivalence with $q \in Q_n \iff Sq \in Q_{n'}$, then $\#(p) = \#(Sp)$ for every $p \in Q_n$. *Book:* p. 14 (preprint p. 14), "d_C does not depend on the choice of T", proof on preprint p. 17. *Hidden hypotheses:* none; the proof uses only linear algebra on level 0. *Proof:* the text's, except that $k = k'$ is not proved first but derived from the lemma at $x = 0$; the codimension of E_x is $\dim(V/E_x) = \#I(x)$ with $E_x = \ker((b, v) \mapsto (b_i)_{i \in I(x)})$, and "$S(E_x) \subset E_y'$ hence equality" is completed by the same inclusion for S^{-1}.

V1.23 (Theorem 1.3.5 (preprint 1.25), boundary recognition; `Polyfold.theorem_1_3_5`, `Polyfold.theorem_1_3_5_lvl1`, `Polyfold.degIndexWith_le_of_germ`, `Polyfold.eventually_degIndexWith_le`, `Polyfold.exists_tendsto_lvl1_degIndexWith`, `Polyfold.IsSc1With.mapsTo_minimalSubspace`). Let $U \subseteq C \subseteq E$ and $U' \subseteq C' \subseteq E'$ be relatively open subsets of partial quadrants, $f : E_0 \to E_0'$ sc^1 on U with $f(U) \subseteq U'$, $g : E_0' \to E_0$ sc^1 on U' with $g(U') \subseteq U$, $g(f(x)) = x$ for $x \in U$ and $f(g(y)) = y$ for $y \in U'$. Then $d_C(x) = d_{C'}(f(x))$ for every $x \in U$. Ingredients: $d_{C'}$ is upper semicontinuous ($d_{C'}(y') \le d_{C'}(y)$ for y' near y); every $x \in C$ is a limit of points of C on level 1 with the same index; and for $x \in U_1$ the derivative $Df(x)$ maps $E_x := \{v : (T_0 v)_i = 0 \text{ whenever } (T_0 x)_i = 0\}$ into $E'_{f(x)}$, from which $d_C(x) \le d_{C'}(f(x))$ follows by comparing codimensions through the chain rule. *Book:* preprint p. 14: for tuples (U, C, E), (U', C', E') and a germ $f : (U, x) \to (U', x')$ of a local sc-diffeomorphism, $d_C(x) = d_{C'}(f(x))$; no proof, a reference to "[20], Theorem 1.19". *Hidden hypotheses:* the notion "germ of a local sc-diffeomorphism", undefined in Chapter 1, read as an

sc^1 map $f : U \to U'$ with sc^1 inverse $g : U' \to U$ on relatively open neighbourhoods U, U' of x, $f(x)$ (which may be shrunk at will). *Deviation (stronger):* only sc^1 is required of f and g, where the text has sc-smooth. *Proof:* the argument of §2.4 (preprint §2.3) (Lemma 2.4.9 (preprint 2.29), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)) for the identity retraction, with level-1 points in place of smooth points; the Lean file is `Polyfold/Chapter2/DegeneracyIndex.lean`.

V1.24 (Fredholm operators between Banach spaces, Riesz–Schauder; `Polyfold.isStrictMap_of_isClosed_range`, `Polyfold.isFredholm_of_banach`, `Polyfold.isCompactOperator_of_finiteDimensional_range`, `Polyfold.finiteDimensional_ker_one_add`, `Polyfold.isClosed_range_one_add`, `Polyfold.coFG_range_one_add`, `Polyfold.isFredholm_one_add`). Let E, F be real Banach spaces. A continuous linear $u : E \to F$ with closed range is a strict map; if moreover $\ker u$ is finite-dimensional and range u has finite codimension, u is a Fredholm operator in Mathlib's sense. A continuous linear map of finite rank is a compact operator. For a compact operator $c : E \to E$: $\ker(1 + c)$ is finite-dimensional, $\mathrm{range}(1 + c)$ is closed and of finite codimension, and $1 + c$ is Fredholm. *Book:* not stated; used in §1.4.1, preprint p. 14 ("$T + S : E_m \to F_m$ is Fredholm", "its image Y is closed"). *Deviation (not in the book):* infrastructure Mathlib v4.33.1 lacks.

V1.25 (compact perturbations of Fredholm operators; `Polyfold.isFredholm_add_isCompactOperator`). Let E, F be real Banach spaces, $u : E \to F$ Fredholm and $k : E \to F$ compact. Then $u + k$ is Fredholm. *Book:* not stated; the first line of §1.4.1. *Deviation (not in the book):* infrastructure; the invariance of the index, which the exposition records (Kato IV.5.26) and the text does not, is V1.28.

V1.26 (the compact-open lemma and the level operators; `Polyfold.exists_delta_of_continuousOn`, `Polyfold.continuousOn_clm_comp_of_isCompactOperator`, `Polyfold.contDiffOn_clm_comp_of_isCompactOperator`, `Polyfold.levelDeriv`, `Polyfold.IsSc1With.exists_levelDeriv`, `Polyfold.IsSc1With.toBase_levelDeriv`, `Polyfold.IsSc1With.levelMap_tangentMapWith`, `Polyfold.IsSc1With.continuousOn_levelDeriv`, `Polyfold.IsSc1With.continuousOn_levelDeriv_clm`, `Polyfold.IsSc1With.hasFDerivWithinAt_level`, `Polyfold.IsSc1With.contDiffOn_level`, `Polyfold.IsSc1With.levelDeriv_eq`). *Compact-open lemma.* Let X, Y, Y', Z be real normed spaces, $\iota : Y' \to Y$ a compact operator and $A : X \to \mathcal{L}(Y, Z)$. If $(x, y) \mapsto A(x)y$ is continuous on $s \times Y$, then $x \mapsto A(x) \circ \iota$ is continuous on s into $\mathcal{L}(Y', Z)$ with the operator norm; if s is the intersection of an open and a convex set with unique differentiability at every point and $(x, y) \mapsto A(x)y$ is C^k within $s \times Y$, then $x \mapsto A(x) \circ \iota$ is C^k within s. *Level operators.* Let f be sc^1 on U with derivatives Df, $m \in \mathbb{N}$ and $x \in U_{m+1}$. There is a unique continuous linear $Df(x) : E_m \to F_m$ (written `levelDeriv`) with $\iota_m \circ Df(x) = Df(\iota x) \circ \iota_m$, namely the second component of the level-m tangent map, which is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ on $U_{m+1} \times E_m$; $(x, h) \mapsto Df(x)h$ is continuous on $U_{m+1} \times E_m$; $x \mapsto Df(x) \circ \iota$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$. If U is an sc-domain, $f_m \circ \iota : U_{m+1} \to F_m$ has derivative $Df(x) \circ \iota$ at x within U_{m+1}, hence is C^1 within U_{m+1}; and if $f_m : U_m \to F_m$ has derivative L at ιx within U_m, then $L = Df(x)$. *Book:* not stated as such; the compactness argument of the 2010 paper's Propositions 2.1–2.3 (quoted in §1.2) and its integral identity $f(x+h) - f(x) = \int_0^1 Df(x+\tau h)h\,d\tau$. *Deviation (not in the book):* the C^k case of the lemma, proved by induction with the mean value inequality, replaces the multilinear maps $\Gamma(x)$ and the differentiation under the integral sign of the 2010 paper; the integral identity is formed as a Bochner integral in F_m and identified with $f(x+h) - f(x)$ by the fundamental theorem of calculus for the C^1 map $f : U_1 \to F_0$.

V1.27 (sc-domains; `Polyfold.IsScDomain`, `Polyfold.IsRelOpenInQuadrant.isScDomain`, `Polyfold.IsScDomain.tangentSet`, `Polyfold.IsScDomain.shift`). $U \subseteq E_0$ is an *sc-domain* if for every m the level set U_m is the intersection of an open and a convex subset of E_m, and U_m has unique differentiability at each of its points. A relatively open subset of a partial quadrant is an sc-domain; if U is an sc-domain, so are TU in TE and U_1 in E^1. *Book:* not stated; the text works with tuples (U, C, E) and asserts that $T(U, C, E)$ is a tuple. *Deviation (not in the book):* the level-wise regularity that the proofs of §§1.2–1.3 use, stable under the tangent construction and the level shift, stated in place of the tuple.

V1.28 (the Fredholm index and its invariance; `Polyfold.linIndex`, `Polyfold.HasFiniteIndex`, `Polyfold.linIndex_comp`, `Polyfold.linIndex_eq_zero_of_finiteRange_sub_id`, `Polyfold.HasFiniteIndex.of_bijective_add_and_linIndex`, `Polyfold.HasFiniteIndex.of_comp_of_comp`, `Polyfold.fredholmIndex`, `Polyfold.IsFredholm.hasFiniteIndex`, `Polyfold.fredholmIndex_add_of_norm_lt`, `Polyfold.fredholmIndex_add_isCompactOperator`). For a linear map $u : E \to F$ over a field let $\operatorname{ind}(u) := \dim\ker u - \dim(F/\operatorname{range} u) \in \mathbb{Z}$, meaningful when both dimensions are finite ("finite index"). If u and $v : F \to G$ have finite index, so has $v \circ u$, and $\operatorname{ind}(v \circ u) = \operatorname{ind}(u) + \operatorname{ind}(v)$. If a is bijective and r has finite rank, $a + r$ has finite index 0 (in particular $1 + r$). If $v \circ u$ and $u \circ v$ have finite index, so has u. For real Banach spaces E, F: a Fredholm operator $u : E \to F$ has finite index, and $\operatorname{ind}(u)$ is its *Fredholm index*. If $v : F \to E$ is a quasi-inverse of u ($vu - 1$ and $uv - 1$ have finite rank) and $\|v\|\,\|s\| < 1$, then $u + s$ has finite index and $\operatorname{ind}(u + s) = \operatorname{ind}(u)$. If u is Fredholm and $k : E \to F$ is compact, then $\operatorname{ind}(u + k) = \operatorname{ind}(u)$. *Book:* not stated; the classical fact behind the exposition's (F2) (Kato IV.5.26), which the text's §1.4.1 would need to conclude that the index is preserved. *Deviation (not in the book):* infrastructure Mathlib v4.33.1 lacks (it has Fredholm operators but no index). *Proof:* additivity by rank–nullity along the exact sequence $0 \to \ker u \to \ker vu \to \operatorname{range} u \cap \ker v \to F/\operatorname{range} u \to G/\operatorname{range} vu \to G/\operatorname{range} v \to 0$; small perturbations through $v(u + s) = (1 + vs) + (vu - 1)$ and $(u + s)v = (1 + sv) + (uv - 1)$, where $1 + vs$, $1 + sv$ are invertible by the Neumann series; compact perturbations by the local constancy of $t \mapsto \operatorname{ind}(u + tk)$ on $\mathbb{R}$ (each $u + t_0 k$ is Fredholm by V1.25), not by Kato's argument.

V1.29 (Lemma 2.D, the level operators of an sc-Fredholm operator; `Polyfold.ScFredholmSplitting.C_le_smooth`, `Polyfold.ScBanach.finrank_levelSub_of_le_smooth`, `Polyfold.ScFredholmSplitting.fredholmIndex_lvl`, `Polyfold.IsScFredholm.fredholmIndex_lvl`, `Polyfold.scIndex_eq_fredholmIndex`). Let $T : E \to F$ be an sc-operator with an sc-Fredholm splitting (K, X, C, Y) (V1.7). Then $C \le F_\infty$ (and $K \le E_\infty$, V1.7); a subspace $A \le E_\infty$ has $\dim A_m = \dim A$ for every m; and for every m the level map $T_m : E_m \to F_m$

is a Fredholm operator with $\operatorname{ind}(T_m) = \dim K - \dim C$. Hence for an sc-Fredholm T, $\operatorname{ind}(T_m) = \operatorname{ind}(T) := \dim \ker T_0 - \dim(F_0/\operatorname{range} T_0)$ for every m; the latter is the Fredholm index of T_0. *Book:* preprint p. 9, after Definition 1.1.9 (preprint 1.8): $K \subset E_\infty$, $T(X_m) = Y_m$, $E_m = K \oplus X_m$, "$F = C \oplus T(E_m)$"; the level-wise index is not stated. *Hidden hypotheses:* none. *Deviation (stronger):* the Fredholm property and the index on every level are added (the exposition's Lemma 2.D). *Proof:* $\ker T_m = K_m \cong K$ and $F_m/\operatorname{range} T_m = F_m/Y_m \cong C_m \cong C$, the isomorphisms $A_m \cong A$ coming from $A \leq E_\infty$.

V1.30 (Proposition 1.1.11 (preprint 1.10), the index; `Polyfold.proposition_1_1_11_index`, `Polyfold.proposition_1_1_11_index_lvl`). Let $T : E \to F$ be sc-Fredholm and $S : E \to F$ an sc$^+$-operator. Then $\operatorname{ind}(T + S) = \operatorname{ind}(T)$, the index being $\dim \ker(\cdot)_0 - \dim \operatorname{coker}(\cdot)_0$; and for every m the level map $(T+S)_m : E_m \to F_m$ is Fredholm of index $\operatorname{ind}(T)$. *Book:* p. 9 (preprint p. 10), Proposition 1.1.11 (preprint 1.10) states only that $T + S$ is sc-Fredholm (V1.9). *Hidden hypotheses:* none. *Deviation (stronger):* the index statement is the exposition's addition. *Proof:* on level 0, T_0 is Fredholm (V1.7) and S_0 compact (V1.3), so V1.28 applies; on level m, V1.29 for the sc-Fredholm operator $T + S$ (V1.9).

V1.31 (finite-dimensional spaces; `Polyfold.ScBanach.smooth_eq_top_of_finiteDimensional`, `Polyfold.ScBanach.toBase_bijective_of_finiteDimensional`, `Polyfold.ScBanach.isoConstOfFiniteDimensional`, `Polyfold.ScScale.finiteDimensional_of_surjective_incl`). Let E be an sc-Banach space with E_0 finite-dimensional. Then every point of E_0 is smooth, every $\iota_m : E_m \to E_0$ is bijective, and the maps ι_m form an sc-isomorphism of E onto E_0 with its constant structure. For a scale E (no density assumed) and any m: if the inclusion $E_{m+1} \to E_m$ is surjective, then E_m is finite-dimensional. *Book:* preprint p. 6, after Definition 1.1.1 (preprint 1.1): a finite-dimensional Banach space has precisely one sc-structure, the constant one; on an infinite-dimensional space the constant structure is not an sc-structure, because the inclusions are not compact. *Hidden hypotheses:* none. *Deviation (restated):* "precisely one" is read, for the indexed families of V1.1, as "sc-isomorphic to the constant structure by the inclusions" (so the level norms are all equivalent), and "the constant structure fails (1)" as "an onto inclusion forces finite dimension". *Proof:* E_∞ is a dense, hence (finite dimension) closed, hence full subspace of E_0; an onto compact inclusion is an isomorphism (open mapping theorem), so the identity of E_m is compact.

V1.32 (sc-splittings are sc-projections; `Polyfold.ScBanach.IsScSplitting.isTopCompl`, `Polyfold.ScBanach.IsScSplitting.scProj`, `Polyfold.ScBanach.IsScSplitting.scProj_idem`, `Polyfold.ScBanach.IsScSplitting.range_scProj`, `Polyfold.ScBanach.IsScSplitting.ker_scProj`, `Polyfold.ScOperator.idem_lvl`, `Polyfold.ScOperator.levelSub_range_eq`, `Polyfold.ScBanach.isScSplitting_of_idem`, `Polyfold.ScBanach.isScSplitting_iff_exists_scProj`). Let $A, B \subseteq E_0$. If $E = A \oplus B$ is an sc-splitting, then on every level A_m, B_m are topological complements, and the continuous projections of E_m onto A_m along B_m form an sc-operator $P : E \to E$ with $P_m^2 = P_m$, $P_m(E_m) = A_m$ and $\ker P_m = B_m$. Conversely, if $P : E \to E$ is an sc-operator with $P_0^2 = P_0$, then $P_m^2 = P_m$ and $(P_0(E_0))_m = P_m(E_m)$ for every m, and $E = P_0(E_0) \oplus \ker P_0$ is an sc-splitting. Hence $E = A \oplus B$ is an sc-splitting iff some sc-operator P with $P_0^2 = P_0$ has range A and kernel B. *Book:* pp. 6–8 (preprint pp. 7–8): the definition of an sc-splitting, and in the proof of Proposition 1.1.7 (preprint 1.6) the step "since $P \circ P = P$, it is a sc-projection" with $G := (1 - P)(E)$. *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* the equivalence the text uses is stated and proved (the exposition's Lemma 2.B); $\ker P_0$ is the text's $(1 - P)(E)$. The formal Proposition 1.1.7 (preprint 1.6) (V1.5) does not depend on it.

V1.33 (the smooth points of a quotient; `Polyfold.ScBanach.exists_smooth_mem_coset`, `Polyfold.ScBanach.mk_mem_quotient_smooth_iff`). Let A be an sc-subspace of E and $x \in E_0$. If for every j there is $y_j \in E_j$ with $\iota_j y_j - x \in A$, then there is $y \in E_\infty$ with $y - x \in A$. Equivalently: the class of x is a smooth point of the quotient E/A (V1.6) if and only if $x + A$ contains a smooth point of E. *Book:* preprint p. 9, in the proof of Proposition 1.1.8 (preprint 1.7): "$(E/A)_\infty$ consists of all elements of the form $x + A_\infty$ with $x \in E_\infty$". *Hidden hypotheses:* that A is an sc-subspace (the density of A_∞ in every A_j) is used; the text has it. *Deviation:* none. *Proof:* a Mittag-Leffler construction (the exposition's Lemma 2.C), with $\prod_{i<j} \max(1, \|E_{i+1} \to E_i\|)$ bounding all the inclusions $E_j \to E_i$, $i \leq j$.

V1.34 (continuity of Df for the compact-open topology; `Polyfold.continuousOn_apply_iff_uniformOnCompact`, `Polyfold.IsSc1With.deriv_uniformOnCompact`, `Polyfold.exists_isSc1_not_continuousOn_deriv`). Let X be a metric space, Y, Z real normed spaces, $s \subseteq X$ and $A : X \to \mathcal{L}(Y, Z)$. Then $(x, h) \mapsto A(x)h$ is continuous on $s \times Y$ if and only if for every $x \in s$, every compact $K \subseteq Y$ and every $\varepsilon > 0$ there is a neighbourhood of x in s on which $\|A(x')w - A(x)w\| < \varepsilon$ for all $w \in K$. Applied to an sc^1 map with derivatives Df: $x \mapsto Df(x)$ is continuous on U_1 for the topology of uniform convergence on compact subsets of E_0. There is an sc^1 map on a relatively open subset of a partial quadrant for which no derivative family is continuous on U_1 into $\mathcal{L}(E_0, F_0)$ with the operator norm (the example of V1.13). *Book:* preprint p. 11: $x \mapsto Df(x)$ is in general not continuous for the operator norm, but it is continuous for the compact-open topology. *Hidden hypotheses:* none. *Deviation (restated):* the compact-open topology on $\mathcal{L}(Y, Z)$ is written out as uniform convergence on compact sets, and the remark becomes an equivalence with level 0 of condition (3) of Definition 1.1.15 (preprint 1.14) (the exposition's Lemma 3.F). The negative half of the remark is proved by a sequence-space example (V1.13), not by the shift map of the exposition's §6, which needs weighted Sobolev spaces.

V1.35 (finite dimensions: sc¹ is C^1, also at the boundary; `Polyfold.ScBanach.lvlSet_const`, `Polyfold.isSc1_iff_contDiffOn_const`, `Polyfold.isSc1_iff_fderivWithin_const`, `Polyfold.isScSmooth_iff_contDiffOn_const`, `Polyfold.isSc1_iff_contDiffOn_of_isRelOpenInQuadrant`, `Polyfold.isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant`). Let V, W be finite-dimensional real normed spaces with their constant sc-structures, $f : V \to W$ and $U \subseteq V$ an sc-domain (V1.27), for instance a relatively open subset of a partial quadrant of V. Then f is sc^1 on U if and only if f is C^1 within U; equivalently, f is differentiable within U at every point of U and the derivative within U, $x \mapsto$ fderivWithin $f\,U\,x$, is continuous on U. And f is sc-smooth on U if and only if it is C^∞ within U. *Book:* preprint p. 11: "the sc^1-maps between finite dimensional Banach spaces are the familiar C^1-maps". *Hidden hypotheses:* the meaning of C^1 at boundary points of a quadrant, never defined in the text; here the derivative within U, which at a boundary point is the one-sided derivative along U and is unique because the quadrant has non-empty interior. *Deviation (hypotheses explicit):* none beyond this; by V1.31 every sc-structure on a finite-dimensional space is the constant one up to sc-isomorphism. *Proof:* Propositions 1.2.1 and 1.2.4 (preprint 1.16 and 1.19) at $k = 0$ (V1.15, V1.18) for the levels of the constant structure, which all equal U.

V1.36 (level shifts and tangents of tuples; `Polyfold.ScIso.shift`, `Polyfold.IsPartialQuadrant.shift`, `Polyfold.IsRelOpenInQuadrant.shift`, `Polyfold.tangentQuadrantIso`, `Polyfold.IsPartialQuadrant.tangentSet`, `Polyfold.IsRelOpenInQuadrant.tangentSet`, `Polyfold.ScBanach.shift_tangentSet`). Let C be a partial quadrant in E, presented by an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$. Then C_1 is a partial quadrant in E^1, presented by the level shift $T^1 : E^1 \to \mathbb{R}^n \oplus W^1$; and $TC = \{(x,h) \in E_1 \times E_0 : x \in C_1\}$ is a partial quadrant in TE, presented by $(x,h) \mapsto (a,(w,T_0 h))$ where $T_1 x = (a,w)$, an sc-isomorphism $TE \to \mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$. If U is relatively open in C, then U_1 is relatively open in C_1 and TU in TC; and the level-1 part of TU is $T(U_1)$. *Book:* preprint p. 10, after Definition 1.1.14 (preprint 1.13): "$T(U,C,E)$ is a tuple consisting again of a relatively open subset TU in the partial quadrant TC"; the corresponding fact for $C^1 \subset E^1$ is used without being stated. *Hidden hypotheses:* none. *Deviation:* none; C^k for $k \geq 2$ is the iterate. These are the exposition's Lemmas 3.A(2) and 3.B, proved in the Chapter 2 files (`Polyfold/Chapter2/LevelPolyfold.lean`, `Polyfold/Chapter2/TangentBundle.lean`).

V1.37 (the derivative tower; `Polyfold.IsDerivTower`, `Polyfold.levelDerivTower`, `Polyfold.baseDerivTower`, `Polyfold.contDiffOn_levels_min`, `Polyfold.levelDerivTower_toBase`, `Polyfold.baseDerivTower_toBase`, `Polyfold.isDerivTower_of_contDiffOn_levels`, `Polyfold.IsDerivTower.eq_of_toBase`). A derivative tower of order $k+1$ for $f : E_0 \to F_0$ on U consists of maps $\Phi_i : E_0 \times E_0^i \to F_0$ with $\Phi_0(x) = f(x)$ and, for $x \in U_{m+l}$, continuous i-linear maps $\Psi_{i,m,l}(x) : E_{m+l}^i \to F_m$, for $i \leq l+1$ and $i \leq k+1$, such that (a) $\iota_m \Psi_{i,m,l}(x)(h_1, \dots, h_i) = \Phi_i(\iota_{m+l}x)(\iota_{m+l}h_1, \dots, \iota_{m+l}h_i)$; (b) $\Psi_{i,m,l}$ is continuous on U_{m+l} into the space of continuous i-linear maps; (c) for $i \leq l$ and $i \leq k$, $\Psi_{i,m,l}$ is differentiable within U_{m+l} at every point, with derivative $\xi \mapsto \Psi_{i+1,m,l}(x)(\xi, \cdot)$. If U is an sc-domain, f is sc^0 on U and $f_m \circ \iota : U_{m+l} \to F_m$ is C^{l+1} within U_{m+l} for all m and $l \leq k$, then the iterated derivatives within the level sets, $\Psi_{i,m,l} := d^i(f_m \circ \iota)$ on U_{m+l} and Φ_i the i-th derivative of $f : U_{i-1} \to F_0$ at lifts to E_{i-1}, form a derivative tower of order $k+1$. In a tower, two values $\Psi_{i,m,l}(x)(h)$, $\Psi_{i,m,l'}(x')(h')$ with the same images in E_0 are equal. *Book:* not stated; the 2010 paper's proof of Proposition 2.4 (pp. 51–53) writes one symbol $d^j f(x)$ for the derivatives on all level pairs. *Hidden hypotheses:* the compatibility (a) is the identification the text makes tacitly. *Deviation (not in the book):* a tool of the formalization. *Proof:* (a) from the chain rule for iterated derivatives within a set composed on the left and on the right with continuous linear maps (the inclusions), and the uniqueness of derivatives on the level sets; for $l > k$ the hypothesis is used with order $k+1$, restricted from U_{m+k}.

V1.38 (elementary maps and the induction (S_k); `Polyfold.ElemData`, `Polyfold.ElemData.Good`,

`Polyfold.ElemData.dom`, `Polyfold.ElemData.map`, `Polyfold.ElemData.map_toBase`, `Polyfold.ElemData.levelMap_eq`, `Polyfold.ElemData.isSc0`, `Polyfold.ElemData.deriv`, `Polyfold.ElemData.hasFDerivWithinAt_level`, `Polyfold.ElemData.continuousOn_deriv`, `Polyfold.ElemData.isSc1Levels`, `Polyfold.ElemData.isSc1With`, `Polyfold.ElemData.shiftData`, `Polyfold.ElemData.derivData`, `Polyfold.ElemData.slotData`, `Polyfold.ElemData.tangentMap_eqOn`, `Polyfold.ElemData.isSck`, `Polyfold.ElemData.base`, `Polyfold.ScBanach.shiftAdd`, `Polyfold.ScBanach.shiftAddZeroIso`, `Polyfold.ScBanach.shiftAddShiftIso`). Fix a derivative tower of order $k+1$ for f on U (V1.37), and write E^p for the sc-Banach space with levels $(E^p)_l = E_{p+l}$ ($E \cong E^0$ and $(E^p)^1 \cong E^{p+1}$ as sc-Banach spaces). Elementary data on an sc-Banach space X are an sc-operator $\pi_0 : X \to E^p$, sc-operators $\pi_a : X \to E^{n_a}$ ($1 \le a \le i$) and a number m; the elementary map is $g(z) = \Phi_i(\pi_0 z)(\pi_1 z, \dots, \pi_i z)$, a map $X \to F^m$, on $W := \pi_0^{-1}(U_p)$. Assume $m + i \le p$, $m + i \le n_a + 1$ for all a, and $i \le k$. Then g is sc^0 on W, its level-l map being Ψ_i on level $(m+l, i-1)$ at the slots. If W is an sc-domain, g is sc^1 with

$$Dg(z)\zeta = d^{i+1} f(\pi_0 z)(\pi_0 \zeta, \pi_1 z, \dots, \pi_i z) + \sum_{a=1}^{i} d^i f(\pi_0 z)(\pi_1 z, \dots, \pi_a \zeta, \dots, \pi_i z),$$

and, if moreover $i + 1 \le k$, its tangent map on TW is $(g', g_0 + \sum_a g_a)$, through $(F^m)^1 \cong F^{m+1}$ in the first component, with g' the elementary map of the data (π_0, π_a) taken on the first factor of $TX = X^1 \oplus X$ (levels $p+1$, $n_a + 1$, $m+1$), g_0 the one with $i+1$ vectors (π_0 on the second factor, π_a on the first) and target shift m, and g_a the one with π_a moved to the second factor; all satisfy the three conditions. Finally, if W is an sc-domain and $i + s \le k + 1$, then g is sc^s on W. *Book:* 2010 paper, Proposition 2.4, pp. 51–53, the inductive statement (S$_k$): the components of $T^{k+1} f$ are linear combinations of $d^j f(x)(h_1, \dots, h_j)$ on $U^{k+1} \oplus E^{n_1} \oplus \dots \oplus E^{n_j} \to F^l$ with $j + l - 1 \le n_i \le k$, each sc^1 with the derivative (11), and those of $T^{k+2} f$ are again of this form. *Hidden hypotheses:* the base-point condition $j + l \le k + 1$, which the text derives from its bounds. *Deviation (not in the book):* the induction runs on s over all elementary maps on all X, not on the description of $T^{k+1} f$; the upper bound $n_i \le k$ is not needed, and its consequences $j + l \le p$ and $j \le k$ are conditions of their own. *Proof:* sc^1 by the level criterion of Proposition 1.2.1 (preprint 1.16) (V1.15): Dg exists one level down because $d^{i+1} f$ and $d^i f$ do, by the derivative of the multilinear evaluation (V1.39); sc^{s+1} from the tangent map, the pairing, the sum rule (V1.39) and the induction hypothesis.

V1.39 (sums of sc^k maps and the derivative of multilinear evaluation; `Polyfold.IsSck.zero`, `Polyfold.IsSck.add`, `Polyfold.IsSck.sum`, `Polyfold.hasFDerivAt_cmm_eval`). Let W be an sc-domain in X. The constant map 0 is sc^k on W; if g_1, g_2 are sc^k on W, so is $g_1 + g_2$; a finite sum of sc^k maps is sc^k. For continuous multilinear maps $A : \prod_a M_a \to Z$ between normed spaces, the evaluation $(A, h) \mapsto A(h)$ is differentiable at every (A, h) with derivative $(B, \eta) \mapsto B(h) + \sum_a A(h_1, \dots, \eta_a, \dots)$. *Book:* not stated; the 2010 paper's proof of Proposition 2.4 uses both (the terms of $\pi \circ T^{k+1} f$ treated one at a time, and the derivative (11)). *Hidden hypotheses:* none. *Deviation (not in the book):* tools; only sums, not scalar multiples, are needed. *Proof:* $g_1 + g_2$ is the sc-smooth addition $Y \oplus Y \to Y$ composed with the pairing (g_1, g_2) (`Polyfold.IsSck.scPair`), by the chain rule (V1.20); the sum by induction.

V1.40 (Remark 1.1.2, Banachable spaces and equivalent norms; `Polyfold.remark_1_1_2`, `Polyfold.remark_1_1_2_openMapping`). Let E_1, E_2 be real normed spaces and $e : E_1 \to E_2$ a linear equivalence (one vector space carrying two norms). (1) If e and e^{-1} are continuous (the two norms induce the same topology), there is $c > 0$ with $\|ex\| \leq c\|x\|$ and $\|x\| \leq c\|ex\|$ for all x. (2) If both spaces are complete and e is continuous, then e^{-1} is continuous, and the two norms are equivalent as in (1). *Book:* Remark 1.1.2, p. 5, new in the 2021 edition: "Banach space" means a Banachable space, a vector space with a topology induced by a complete norm, and all norms inducing that topology are equivalent "in view of the open mapping theorem". *Hidden hypotheses:* none. *Deviation (restated):* the remark is a convention, not a statement; the Lean records its mathematical content in two readings. Part (1) needs neither completeness nor the open mapping theorem (a continuous linear map between normed spaces is bounded); the open mapping theorem is what gives part (2), two complete norms one of which dominates the other. *Proof:* (1) by the operator norms of e and e^{-1}; (2) by Mathlib's open mapping theorem (`ContinuousLinearEquiv.ofBijective`), then (1).

V1.41 (Proposition 1.1.11 (preprint 1.10), the 2021 statement with the index; `Polyfold.proposition_1_1_11`). Let $T : E \to F$ be sc-Fredholm and $S : E \to F$ an sc$^+$-operator. Then $T+S$ is sc-Fredholm, $\text{ind}(T+S) = \text{ind}(T)$, and for every m the level map $(T+S)_m : E_m \to F_m$ is Fredholm of index $\text{ind}(T)$. *Book:* Proposition 1.1.11, p. 9 (preprint 1.10, p. 10); the 2021 statement adds that $T + S$ "has the same index". The proof in Appendix 1.4.1, pp. 15–16, is the one of the earlier texts and does not argue the index. *Hidden hypotheses:* none. *Deviation (stronger):* the level statement is the formalization's addition; the index statement, the exposition's addition in V1.30, is now the book's own. *Proof:* the conjunction of V1.9 (`Polyfold.proposition_1_1_11_preprint`) and V1.30 (`Polyfold.proposition_1_1_11_index`, `Polyfold.proposition_1_1_11_index_lvl`); the index rests on the invariance of the Fredholm index under compact perturbations (V1.28).

V1.42 (Example 1.1.3 (preprint 1.2): the weighted Sobolev spaces $H^{m,\delta_m}(\mathbb{R} \times S^1)$ form an sc-structure; `Polyfold.GapP1Sobolev.example_1_1_3`, `Polyfold.GapP1Sobolev.example_1_1_3_lvl`, `Polyfold.GapP1Sobolev.inclH`, `Polyfold.GapP1Sobolev.valCLM_inclH`, `Polyfold.GapP1Sobolev.completeSpace_H`, `Polyfold.GapP1Sobolev.isClosed_range_ΦH`, `Polyfold.GapP1Sobolev.isCompactOperator_inclH`, `Polyfold.GapP1Sobolev.rellich`, `Polyfold.GapP1Sobolev.hasSum_bc`, `Polyfold.GapP1Sobolev.tail_le`, `Polyfold.GapP1Sobolev.dense_test`, `Polyfold.GapP1Sobolev.IsTest.mem_Hsub`, `Polyfold.GapP1Sobolev.SmPer.hasWeakDeriv`, `Polyfold.GapP1Sobolev.tendsto_cut`, `Polyfold.GapP1Sobolev.leibniz`, `Polyfold.GapP1Sobolev.approx_supp`, `Polyfold.GapP1Sobolev.level0_isometry`, `Polyfold.GapP1Sobolev.level0_surjective`, `Polyfold.GapP1Sobolev.HasWeakDeriv`, `Polyfold.GapP1Sobolev.IsTest`, `Polyfold.GapP1Sobolev.H`, `Polyfold.GapP1Sobolev.nonneg_of_strictMono`). `example_1_1_3 h0 hmono : ScBanach`, for every $\delta : \mathbb{N} \to \mathbb{R}$ with $\delta_0 = 0$ and δ strictly increasing, has the levels $E_m = H^{m,\delta_m}(\mathbb{R} \times S^1)$ (`example_1_1_3_lvl`, by definition), with $E_{m+1} \to E_m$ the identity on the underlying L^2 function (`inclH`, `valCLM_inclH`). The ingredients of the text's proof: each $H^{m,\delta}$ is a Banach space (`completeSpace_H`, an isometric embedding into a finite product of L^2 spaces with closed range); the inclusion is compact for $\delta < \delta'$ (`isCompactOperator_inclH`), by Rellich on bounded cylinders (`rellich`, with Fourier series on boxes, Parseval `hasSum_bc`) and the weight tail estimate (`tail_le`); test functions lie in every level and are dense in it (`IsTest.mem_Hsub`, `dense_test`, by cut-off `tendsto_cut` with the Leibniz rule `leibniz`, and Fourier truncation `approx_supp`); level 0 is $L^2(\mathbb{R} \times S^1)$ (`level0_isometry`, `level0_surjective`). *Book:* §1.1, Example 1.1.3 (preprint 1.2), pp. 6 ff. (preprint pp. 6–7), With $0 = \delta_0 < \delta_1 < \cdots$, the spaces $E_m = H^{m,\delta_m}(\mathbb{R} \times S^1)$ of L^2 functions whose weak derivatives up to order m, weighted by $e^{\delta_m |s|}$, are in L^2 form an sc-structure on $L^2(\mathbb{R} \times S^1)$, "using Sobolev's compact embedding theorem for bounded domains and the assumption that (δ_m) is strictly increasing". *Deviation (restated):* $\mathbb{R} \times S^1$ is the strip $\mathbb{R} \times (0, 1]$ with Lebesgue measure; $S^1 = \mathbb{R}/\mathbb{Z}$ enters through the test functions (`IsTest`: smooth, 1-periodic in t, zero for $|s|$ large), against which weak derivatives are defined (`HasWeakDeriv`); this is the distributional derivative on the cylinder. The norm is $\|u\|^2_{m,\delta} = \sum_{a+b\le m} \|e^{\delta|s|}\partial^a_s \partial^b_t u\|^2_{L^2}$. The text fixes no norm; the usual choices are equivalent, and an sc-structure depends only on the topologies. Derivatives are taken in the order $\partial^a_s \partial^b_t$, which for test functions does not matter. The hypotheses are the text's: $\delta_0 = 0$ and (δ_m) strictly increasing; $\delta_m \ge 0$ is derived (`nonneg_of_strictMono`). The proof follows the text's outline (compact embedding on bounded domains plus the growth of the weights); the exposition's Proposition 1.C is the paper proof.

V1.43 (Remark 1.1.17 (preprint 1.15): partial cones in place of partial quadrants; `Polyfold.IsPartialCone`, `Polyfold.IsPartialCone.isScDomain`, `Polyfold.IsPartialCone.isScDomain_of_convex`, `Polyfold.IsPartialCone.tangentSet`, `Polyfold.IsPartialQuadrant.isPartialCone`, `Polyfold.remark_1_1_17_chainRule`, `Polyfold.interior_lvlSet_nonempty_of_interior`). `IsPartialCone E P`: P closed, convex, $\lambda P \subseteq P$ for $\lambda \ge 0$, with nonempty interior. For $U = O \cap P$ with O open, U is an sc-domain (V1.27; `IsPartialCone.isScDomain`; convexity and nonempty interior suffice, `IsPartialCone.isScDomain_of_convex`, since every level set P_m has nonempty interior, `interior_lvlSet_nonempty_of_interior`). $TP = P^1 \oplus E$ is again a closed convex partial cone with nonempty interior (`IsPartialCone.tangentSet`). Partial quadrants are partial cones (`IsPartialQuadrant.isPartialCone`), and the chain rule holds for relatively open subsets of partial cones (`remark_1_1_17_chainRule`). *Book:* §1.1, Remark 1.1.17 (preprint 1.15), p. 10 (preprint p. 11), Instead of a partial quadrant C one may take a closed convex partial cone P ($\lambda P \subset P$ for $\lambda \ge 0$) with non-empty interior; for U open in P, sc^1 is defined as in Definition 1.1.15 (preprint 1.14), with the tangent map on $TP = P^1 \oplus E$, again a closed cone with non-empty interior. Many results should carry over, "but one should check carefully which arguments carry over". *Deviation (restated):* The remark says that many results should carry over and that one should check which arguments do. Here: sc^1 (`IsSc1`) is defined for every U, and the results of §§1.2–1.3 are stated for sc-domains, so they hold for relatively open subsets of partial cones; the chain rule is recorded as an instance. The degeneracy index, tameness and faces (Chapters 2 ff.) use the quadrant structure and are not generalized, as in the text.

What is not verified in this chapter

- pp. 6 ff. (preprint pp. 6–7), Example 1.1.3 (preprint 1.2): the weighted Sobolev spaces $H^{m,\delta_m}(\mathbb{R} \times S^1)$ as an sc-structure on L^2. Mathlib has no weighted Sobolev spaces and no Rellich–Kondrachov theorem; the exposition proves it (Proposition 1.C). Planned, with the shift map of the 2010 paper (exposition §6), as the project's step H2. Verified since: the spaces form an sc-Banach space with level 0 equal to $L^2(\mathbb{R} \times S^1)$, by Rellich on bounded cylinders (Fourier series), the weight tail estimate and the density of test functions (V1.42) (gap G.1, closed); the shift map on them is not formalized.
- p. 10 (preprint p. 11), Remark 1.1.17 (preprint 1.15): partial cones. Not formalized, as in the exposition. Verified since: relatively open subsets of closed convex partial cones with nonempty interior are sc-domains, TP is again such a cone, and the chain rule holds there (V1.43) (gap G.2, closed); the later chapters stay with quadrants, as in the text.
- preprint p. 11, after Definition 1.1.15 (preprint 1.14): the text's own instance of a derivative that is not norm continuous, the shift map of the exposition's §6, needs the weighted Sobolev spaces of Example 1.1.3 (preprint 1.2), which are formalized since (V1.42); the shift map on them is not. The remark itself is verified, the negative half by a sequence-space example (V1.13, V1.34), the positive half by V1.34.

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3. Chapter 2 (HWZ §§2.1, 2.3–2.7 (preprint §§2.1–2.6)): verified; Remark 2.4.8's (preprint 2.28) negative claim is false

Chapter 2 of Part I (pp. 21–88 (preprint pp. 18–60)) builds differential geometry on the images of sc-smooth retractions. §2.1 defines retractions and retracts, proves that the tangent $TO = Tr(TU)$ does not depend on the retraction (Proposition 2.1.11 (preprint 2.4)), defines sc-smooth maps between retracts and proves the chain rule (Definitions 2.1.1–2.1.14 (preprint 2.1–2.7), Proposition 2.1.13 (preprint 2.6), Theorem 2.1.15 (preprint 2.8)). §2.3 (preprint §2.2) glues retracts into M-polyfolds, gives them a filtration X_m and a tangent bundle TX, proves metrizability (Theorem 2.3.10 (preprint 2.16), Proposition 2.3.15 (preprint 2.19)) and treats sub-M-polyfolds (Definitions 2.3.2–2.3.18 (preprint 2.11–2.20), Proposition 2.3.19 (preprint 2.21)). §2.4 (preprint §2.3) transports the degeneracy index to M-polyfolds and compares it with the tangent space (Definitions 2.4.1–2.4.7 (preprint 2.22–2.27), Lemmas 2.4.3, 2.4.6, 2.4.9 (preprint 2.23, 2.26, 2.29), Propositions 2.4.4, 2.4.10 (preprint 2.24, 2.30), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)). §2.5 (preprint §2.4) introduces tame retracts and the faces of tame M-polyfolds (Definitions 2.5.1–2.5.10 (preprint 2.33–2.39), Propositions 2.5.3–2.5.20 (preprint 2.35–2.46), with the products of M-polyfolds that Proposition 2.5.20 (preprint 2.46) uses), §2.6 (preprint §2.5) strong bundles (Definitions 2.6.1–2.6.7 (preprint 2.47–2.53), Proposition 2.6.8 (preprint 2.54)), and the appendix §2.7 (preprint §2.6) proves Proposition 2.1.2 (preprint 2.2), Theorem 2.3.10 (preprint 2.16) and Proposition 2.3.15 (preprint 2.19) with Smirnov's theorem (Theorem 2.7.1 (preprint 2.55)), Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57). Verified: every definition except 2.9 and 2.38 (which are given only in the forms the theory uses; 2.9, the 2021 Definition 2.1.16, is formalized since, apart from its categories, V2.92; 2.38, the 2021 Definition 2.5.8, is formalized since, V2.111) and every numbered result of §§2.1, 2.3, 2.4, 2.6 (preprint §§2.1–2.3 and §2.5), with these exceptions: Proposition 2.1.2 (preprint 2.2) (the classical submanifold statement, motivation only; verified since, V2.108), Lemma 2.4.9 (preprint 2.29) in its path formulation (its content, the sign arguments, is verified and Proposition 2.4.10 (preprint 2.30) is proved from it; verified since, V2.109), the openness of $p : TX \to X_1$ in Proposition 2.3.15(2) (preprint 2.19). The second statement of Theorem 2.4.12 (preprint 2.31) (equality makes C_xO a partial quadrant) is verified without its hypothesis, and **Remark 2.4.8's (preprint 2.28) claim that C_xO is in general not a partial quadrant in T_xO is false**: it is one at every smooth point of every sc-smooth retract (V2.33), by Flor's theorem on nonnegative idempotent matrices, proved in the needed case. Every numbered result of §2.5 (preprint §2.4) is verified: Propositions 2.5.3, 2.5.6 (preprint 2.35, 2.36) (with "C_xO is a partial quadrant", which holds for every retract, V2.33), Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41), Propositions 2.5.14, 2.5.15 (preprint 2.42, 2.43; the retraction onto a face, and the face as a tame sub-M-polyfold with $d_F = d_O - 1$), 2.5.16 (preprint 2.44; both sentences, the intersections of local faces included), Theorem 2.5.17 (preprint 2.45) and Proposition 2.5.20 (preprint 2.46) (fibred products, with the products of M-polyfolds it uses). In §2.6 (preprint §2.5) the M-polyfold $Y[1]$, the sc-smoothness of $P[1]$ and the chart-independence of $Y_{m,k}$ are verified, and the chart-wise definition of sc$^+$-sections is shown to be the text's; equivalence of atlases, of M-polyfolds and of strong bundles, is an equivalence relation (V2.11, V2.57), strong bundle charts restrict to open subsets of the base (V2.57), and the charts of a strong bundle depend only on its class of atlases (V2.58). **Theorem 2.3.10 (preprint 2.16) holds with the text's hypotheses**: the formal proof adds none, avoids Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57) (which need a regularity of X_m that the text does not check) and proves Smirnov's theorem, which Mathlib lacks, in a pull-back form. Several statements are proved in a stronger form: Propositions 2.4.10, 2.5.3 (preprint 2.30, 2.35) and Theorem 2.4.12 (preprint 2.31) at points on level 1 and for sc^1 germs, Proposition 2.5.3 (preprint 2.35) from condition (2) of tameness alone, Corollary 2.4.13 (preprint 2.32) for every relatively open subset of a partial quadrant, and Proposition 2.5.6's (preprint 2.36) $d_O = d_C$ in the form "a tame chart realizes d_X" for every M-polyfold. An addendum (2026-11-12) adds Proposition 2.1.2 (preprint 2.2), proved by conjugating the retraction to the projection $Dr(x)$ (V2.108, closing gap G.4), Lemma 2.4.9 (preprint 2.29) in its path formulation with $T_xO = C_xO - C_xO$ (V2.109, closing gap G.10), the extended faces of a partial quadrant from C and d_C alone (V2.110, closing gap G.12), Definition 2.5.8 (preprint 2.38) in T_xX, with C_xX a partial quadrant, $d_X(x) = d_{C_xX}(0_x)$ and the transition derivatives as sc-isomorphisms (V2.111, closing gap G.13), and Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57) as printed (V2.112, closing gap G.15).

The setting. Sc-Banach spaces and maps are those of Chapter 1. A *retraction* is a total map $r : E_0 \to E_0$ with an explicit domain U; an *M-polyfold* is a Hausdorff paracompact space with one sc-smooth atlas representing its class, and every notion is defined through the maximal atlas of all charts compatible with it. Sc-smoothness between retracts quantifies over all retractions onto the source (equivalent to the text's "one"). Where the text computes in the model $C = [0, \infty)^k \oplus W$, the formal statements use a presentation $T : E \cong \mathbb{R}^n \oplus W$ of C and the coordinate functionals $\lambda_i = (T\,\cdot)_i$; the subspaces E_x, the cones C_x and tameness are defined through T. The tangent bundle TX

is a quotient of tuples (x, φ, h) with its topology and charts; its fibres $T_x X$ are sets, identified with $T_{\varphi(x)} O$ by the charts, with no linear structure of their own. Theorem 1.3.5 (preprint 1.25) of Chapter 1 (V1.23) is proved in `Polyfold/Chapter2/DegeneracyIndex.lean` by the argument of §2.4 (preprint §2.3).

V2.1 (sc-smooth retractions; `Polyfold.ScRetraction`, `Polyfold.ScRetraction.retract`, `Polyfold.ScRetraction.mem_retract_iff`, `Polyfold.ScRetraction.apply_eq_self_of_mem_retract`, `Polyfold.ScRetraction.lvlSet_retract_subset`, `Polyfold.ScRetraction.scDeriv_comp_self`). Let C be a partial quadrant in E. An *sc-smooth retraction* on C consists of a set $U \subseteq E_0$ relatively open in C and a map $r : E_0 \to E_0$ with $r(U) \subseteq U$, r sc-smooth on U and $r(r(x)) = r(x)$ for $x \in U$. Its *retract* is $O = r(U)$; $O \subseteq U$, $x \in O$ iff $x \in U$ and $r(x) = x$, and $O_m \subseteq U_m$. For $x \in O_1$, $Dr(x) \circ Dr(x) = Dr(x)$. *Book:* Definition 2.1.1 (preprint 2.1), p. 21 (preprint p. 18), and the remark after Definition 2.1.3 (preprint 2.3), p. 26 (preprint p. 19). *Hidden hypotheses:* none. *Deviation:* none; the values of r off U play no role.

V2.2 (sc-smooth retracts; `Polyfold.IsScRetract`, `Polyfold.ScRetraction.isScRetract`, `Polyfold.ScRetraction.id`, `Polyfold.IsRelOpenInQuadrant.isScRetract`, `Polyfold.LocalModel`). (O, C, E) is an *sc-smooth retract* if $O = r(U)$ for an sc-smooth retraction on C. A relatively open $U \subseteq C$ is a retract ($r = \mathrm{id}$). A *local model* is a triple (E, C, O) with C a partial quadrant and O a retract in C. *Book:* Definition 2.1.3 (preprint 2.3), p. 22 (preprint p. 18). *Hidden hypotheses:* none.

V2.3 (Proposition 2.1.11 (preprint 2.4); `Polyfold.proposition_2_1_11`, `Polyfold.ScRetraction.comp_eq_self_of_retract_subset`, `Polyfold.ScRetraction.tangent_subset_of_retract_subset`, `Polyfold.ScRetraction.tangent`, `Polyfold.ScRetraction.mem_tangent_iff`). For sc-smooth retractions r on U and r' on U', both relatively open in C, with $r(U) = r'(U')$: $Tr(TU) = Tr'(TU')$. If only $r(U) \subseteq r'(U')$, then $r' \circ r = r$ on U and $Tr(TU) \subseteq Tr'(TU')$. A point p lies in $Tr(TU)$ iff $p \in TU$ and $Tr(p) = p$. *Book:* Proposition 2.1.11 (preprint 2.4), p. 26 (preprint p. 19). *Hidden hypotheses:* the composition $r' \circ r$ is defined because $O \subseteq U'$. *Proof:* as in the text, by Theorem 1.3.1 (preprint 1.21) applied to $r' \circ r = r$.

V2.4 (Definition 2.1.12 (preprint 2.5), the tangent of a retract; `Polyfold.retractTangent`, `Polyfold.retractTangent_eq`, `Polyfold.ScRetraction.mem_tangent_iff_exists`, `Polyfold.ScRetraction.mem_tangent_iff_mem_range`, `Polyfold.tangentSpace`, `Polyfold.tangentSpace_eq_range`, `Polyfold.tangentSpace_eq_coe_submodule`). For a retract O, $TO \subseteq E_1 \times E_0$ is $Tr(TU)$ for any retraction r onto O (V2.3); $TO = \{(r(x), Dr(x)h) : x \in U_1, h \in E_0\}$, and for $x \in O_1$, $(x, h) \in TO$ iff h is in the range of $Dr(x)$. The tangent space $T_x O = \{h : (x, h) \in TO\}$ is this range, a closed subspace of E_0. *Book:* Definition 2.1.12 (preprint 2.5), p. 26 (preprint p. 19), and Definition 2.3.12 (preprint 2.17), p. 38 (preprint p. 25) (the fibre $T_x O$). *Hidden hypotheses:* none. *Deviation (restated):* TO is defined for every subset O by a choice of retraction, and is empty if O is not a retract.

V2.5 (Proposition 2.1.13 (preprint 2.6); `Polyfold.proposition_2_1_13_one`, `Polyfold.proposition_2_1_13_two`, `Polyfold.proposition_2_1_13_three`, `Polyfold.ScRetraction.`

`comp_comp_eqOn`). Let r on U and s on V be retractions onto the same O and $f : E_0 \to E'_0$. (1) If $f \circ r$ is sc-smooth on U, then $f \circ s$ is sc-smooth on V. (2) Then $T(f \circ r) = T(f \circ s)$ on $Tr(TU)$. (3) If $f \circ r$ is sc-smooth and f maps O into the retract $O' = r'(U')$, then $T(f \circ r)$ maps $Tr(TU)$ into $Tr'(TU')$. *Book:* Proposition 2.1.13 (preprint 2.6), p. 27 (preprint p. 20). *Hidden hypotheses:* in (3), $f(O) \subseteq O'$ (contained in "$f : O \to O'$"). *Proof:* as in the text, with $f \circ r \circ s = f \circ s$ and $\rho \circ f \circ r = f \circ r$ and Theorem 1.3.1 (preprint 1.21).

V2.6 (Definition 2.1.14 (preprint 2.7), sc-smooth maps between retracts; `Polyfold.IsScSmoothRetractMap`, `Polyfold.isScSmoothRetractMap_iff_exists`, `Polyfold.retractTangentMap`, `Polyfold.retractTangentMap_eqOn`, `Polyfold.IsScSmoothRetractMap.tangentMap_mapsTo`, `Polyfold.retractTangentMap_apply`, `Polyfold.IsScSmoothRetractMap.continuousOn`, `Polyfold.IsScSmoothRetractMap.mem_range_toBase`). f is *sc-smooth from* (O, C, E) *to* (O', C', E') if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth on U for every sc-smooth retraction r on C with $r(U) = O$; for a retract O this holds iff $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth for one such r. The tangent map Tf is $T(f \circ r)$ for a chosen r; on TO it equals $T(f \circ r)$ for every r onto O, maps TO into TO' when O, O' are retracts, and $Tf(x, h) = (f(x), D(f \circ r)(x)h)$. Such an f is continuous on O and maps points of O on level m to points on level m. *Book:* Definition 2.1.14 (preprint 2.7), pp. 27–28 (preprint pp. 20–21). *Hidden hypotheses:* none. *Deviation (restated):* the definition quantifies over all retractions and contains $f(O) \subseteq O'$; the text's form is `Polyfold.isScSmoothRetractMap_iff_exists`.

V2.7 (Theorem 2.1.15 (preprint 2.8), the chain rule for retracts; `Polyfold.theorem_2_1_15`, `Polyfold.IsScSmoothRetractMap.comp`, `Polyfold.retractTangentMap_comp`, `Polyfold.isScSmoothRetractMap_id`, `Polyfold.ScRetraction.isScSmoothRetractMap`, `Polyfold.IsScSmoothRetractMap.congr`). Let (O, C, E) and (O', C', E') be retracts, f sc-smooth from O to O' and g sc-smooth from O' to (O'', C'', E''). Then $g \circ f$ is sc-smooth from O to O'' and $T(g \circ f) = Tg \circ Tf$ on TO. The identity is sc-smooth, a retraction is sc-smooth as a map $(U, C, E) \to (O, C, E)$, and sc-smoothness depends only on the values on O. *Book:* Theorem 2.1.15 (preprint 2.8), p. 28 (preprint p. 21), which writes (O'', C', E'') for (O'', C'', E''). *Hidden hypotheses:* $r' \circ f \circ r = f \circ r$, which needs $f(O) \subseteq O'$. *Proof:* the text gives none ("follows from Theorem 1.21"); $(g \circ f) \circ r = (g \circ r') \circ (f \circ r)$ on U and Theorem 1.3.1 (preprint 1.21).

V2.8 (the tangent functor on retracts; `Polyfold.IsPartialQuadrant.tangentSet`, `Polyfold.IsRelOpenInQuadrant.tangentSet`, `Polyfold.ScRetraction.tangentRetraction`, `Polyfold.ScRetraction.tangentRetraction_retract`, `Polyfold.IsScRetract.tangentSet`, `Polyfold.LocalModel.tangent`, `Polyfold.IsScSmoothRetractMap.tangent`, `Polyfold.retractTangentMap_id`, `Polyfold.retractTangent_inter_isOpen`, `Polyfold.retractTangentMap_inter_isOpen`). TC is a partial quadrant in TE and TU is relatively open in TC; Tr is an sc-smooth retraction on TU with retract TO, so (TO, TC, TE) is a retract. If f is sc-smooth between the retracts O and O', then Tf is sc-smooth from (TO, TC, TE) to (TO', TC', TE'); T of a map equal to the identity on O is the identity on TO. For $W \subseteq E_0$ open: $T(O \cap W) = TO \cap (W_1 \times E_0)$, and Tf computed on $(O \cap W, C, E)$ equals Tf computed on (O, C, E) there. *Book:* preprint p. 21, after Theorem 2.1.15 (preprint 2.8) (T a functor $\mathscr{R} \to \mathscr{R}$), and preprint p. 22, after Definition 2.1.16 (preprint 2.9) (Tr a bundle retraction). *Hidden hypotheses:* the locality of Tf under restriction to open subsets, used in Definitions 2.3.12–2.3.14 (preprint 2.17–2.18). *Deviation (restated):* the bundle structure is V2.92 (Tr a bundle retraction covering r^1, Tf a bundle map; gap G.5, formalized since); the category $\mathcal{BR}$ is not formalized.

V2.9 (Proposition 2.3.1 (preprint 2.10); `Polyfold.proposition_2_3_1_one`, `Polyfold.proposition_2_3_1_two`, `Polyfold.ScRetraction.restrict`, `Polyfold.ScRetraction.restrict_retract`, `Polyfold.IsScSmoothRetractMap.restrict`, `Polyfold.IsScSmoothRetractMap.of_locally`, `Polyfold.LocalModel.restrict`). (1) If (O, C, E) is a retract and $W \subseteq E_0$ is open, $(O \cap W, C, E)$ is a retract (with r restricted to $U \cap r^{-1}(W)$). (2) If (V, C, E) is a retract and s is sc-smooth from (V, C, E) to itself with $s \circ s = s$ on V, then $(s(V), C, E)$ is a retract, with the retraction $s \circ r$. Sc-smooth maps between retracts restrict to $O \cap W$, and are sc-smooth as soon as every point of O has an open neighbourhood W with f sc-smooth on $O \cap W$. *Book:* Proposition 2.3.1 (preprint 2.10), p. 33 (preprint p. 23). *Hidden hypotheses:* the restriction of an sc-smooth map to a smaller relatively open set is sc-smooth (V2.63). *Deviation (restated):* an open subset of O is written $O \cap W$.

V2.10 (Definition 2.3.2 (preprint 2.11), charts; `Polyfold.MChart`, `Polyfold.MChart.overlap`, `Polyfold.MChart.isScRetract_overlap`, `Polyfold.MChart.transition`, `Polyfold.MChart.Compatible`, `Polyfold.MChart.compatible_self`, `Polyfold.MChart.Compatible.restrict`, `Polyfold.MChart.compatible_of_locally`). A *chart* on a space X is a local model (O, C, E) and a partial equivalence φ between X and E_0 with open source V and target O, continuous on V, with inverse continuous on O. The overlap $\varphi(V \cap V')$ is a retract. Two charts are *compatible* if $\psi \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V \cap V'), C, E)$ to $(\psi(V \cap V'), C', E')$ and $\varphi \circ \psi^{-1}$ back. A chart is compatible with itself, compatibility passes to restrictions, and two charts each compatible with every chart of a family covering $V \cap V'$ are compatible. *Book:* Definition 2.3.2 (preprint 2.11), pp. 34 ff. (preprint pp. 23–24). *Hidden hypotheses:* none. *Deviation (restated):* the chart is a partial equivalence; the last statement, the locality of compatibility, is not in the text and is what makes the maximal atlas (V2.12) work.

V2.11 (Definition 2.3.3 (preprint 2.12), atlases; `Polyfold.MAtlas`, `Polyfold.MAtlas.Equiv`, `Polyfold.MAtlas.equiv_iff`, `Polyfold.MAtlas.Equiv.refl`, `Polyfold.MAtlas.Equiv.symm`, `Polyfold.MAtlas.Equiv.trans`, `Polyfold.MAtlas.equivalence`). An *atlas* is a set of pairwise compatible charts whose sources cover X; two atlases are *equivalent* if their union is an atlas, equivalently if every chart of one is compatible with every chart of the other; the relation is an equivalence relation. *Book:* Definition 2.3.3 (preprint 2.12), p. 35 (preprint p. 24). *Hidden hypotheses:* none; the transitivity of the equivalence, which "an equivalence class of atlases" (Definition 2.3.4 (preprint 2.13)) presupposes, is not discussed in the text and is proved from the locality of compatibility (Findings, Chapter 2, item 41).

V2.12 (Definition 2.3.4 (preprint 2.13), M-polyfolds; `Polyfold.MPolyfold`, `Polyfold.MPolyfold.IsChart`, `Polyfold.MPolyfold.IsChart.compatible`, `Polyfold.MPolyfold.IsChart.restrict`, `Polyfold.MPolyfold.isChart_congr`, `Polyfold.MPolyfold.exists_isChart_mem_source`, `Polyfold.LocalModel.toMPolyfold`). An *M-polyfold* structure on a

Hausdorff paracompact space X is an atlas. The *charts of the structure* are all charts compatible with every chart of this atlas; any two are compatible, restrictions of charts are charts, every point lies in the source of one, and equivalent atlases give the same charts. A nonempty retract O is an M-polyfold with the chart id. *Book:* Definition 2.3.4 (preprint 2.13), p. 35 (preprint p. 24). *Hidden hypotheses:* charts compatible with an atlas are compatible with each other (the text does not discuss it; it needs V2.10). *Deviation (restated):* one atlas represents its equivalence class.

V2.13 (Definition 2.3.8 (preprint 2.14), sc-smooth maps between M-polyfolds; `Polyfold.MPolyfold.IsScSmoothOn`, `Polyfold.MPolyfold.IsScSmoothMap`, `Polyfold.MPolyfold.IsScSmoothOn.of_exists_charts`, `Polyfold.MPolyfold.isScSmoothMap_id`, `Polyfold.MPolyfold.IsScSmoothOn.comp`, `Polyfold.MPolyfold.IsScSmoothMap.comp`). $f : X \to Y$ is *sc-smooth on the open set* U if it is continuous on U and, for all charts φ of X, ψ of Y and every open $V \subseteq U \cap \operatorname{dom} \varphi$ with $f(V) \subseteq \operatorname{dom} \psi$, the map $\psi \circ f \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V), C, E)$ to (O', C', E'); f is sc-smooth if it is sc-smooth on X. It suffices that every $x \in U$ has one such pair of charts and one such $V \ni x$. The identity is sc-smooth; if f is sc-smooth on U with $f(U) \subseteq U'$ and g on U', then $g \circ f$ is sc-smooth on U. *Book:* Definition 2.3.8 (preprint 2.14), p. 37 (preprint p. 24), and "the definition does not depend on the choice of charts". *Hidden hypotheses:* f continuous (for charts with $f(V) \subseteq V'$ to exist around every point). *Deviation (hypotheses explicit):* continuity is part of the definition; maps on open subsets are included.

V2.14 (the filtration X_m; `Polyfold.MPolyfold.levelSet`, `Polyfold.MPolyfold.mem_levelSet_iff`, `Polyfold.MPolyfold.levelSet_zero`, `Polyfold.MPolyfold.levelSet_succ_subset`, `Polyfold.MPolyfold.smoothSet`, `Polyfold.MPolyfold.levelTopology`, `Polyfold.MPolyfold.Level`, `Polyfold.MPolyfold.continuous_levelVal`, `Polyfold.MPolyfold.isEmbedding_levelLift`). X_m is the set of points x with $\varphi(x) \in \iota_m(E_m)$ for some chart φ of the structure around x; then this holds for every chart around x. $X_0 = X$, $X_{m+1} \subseteq X_m$, $X_\infty = \bigcap_m X_m$. X_m carries the topology generated by the sets $\{x \in V : \varphi(x) \in W\}$, φ a chart, $W \subseteq E_m$ open (in terms of the lift of $\varphi(x)$ to E_m). The inclusion $X_m \to X$ is continuous, X_m is Hausdorff (`Polyfold.MPolyfold.instT2SpaceLevel`), and on $V_m = V \cap X_m$ the lift of φ to E_m is a topological embedding. *Book:* preprint p. 25, the paragraph after Definition 2.3.8 (preprint 2.14). *Hidden hypotheses:* that the level is independent of the chart needs that transition maps are sc^0 on the overlap retracts (the text: "of course"). *Deviation (restated):* the topology is the one generated by the text's family; that the family is a basis is not proved.

V2.15 (Lemma 2.3.9 (preprint 2.15); `Polyfold.lemma_2_3_9`, `Polyfold.MPolyfold.levelIncl`). The inclusion $X_{m+1} \to X_m$ is continuous for every m. *Book:* Lemma 2.3.9 (preprint 2.15), p. 37 (preprint p. 25). *Hidden hypotheses:* none.

V2.16 (the M-polyfold X^1; `Polyfold.ScRetraction.shift`, `Polyfold.ScRetraction.shift_retract`, `Polyfold.IsScRetract.shift`, `Polyfold.LocalModel.shift`, `Polyfold.IsScSmoothRetractMap.shift`, `Polyfold.MPolyfold.levelChart`, `Polyfold.MPolyfold.levelChart_compatible`, `Polyfold.MPolyfold.levelAtlas`, `Polyfold.MPolyfold.shift`, `Polyfold.MPolyfold.shift_isChart`, `Polyfold.MPolyfold.mem_shift_levelSet_iff`). A retraction r on U induces an sc-smooth retraction on $U_1 \subseteq C_1 \subseteq E^1$ with retract O_1, so (O_1, C_1, E^1)

is a retract; an sc-smooth map between retracts induces an sc-smooth map between their level-1 parts (Proposition 1.2.2 (preprint 1.17) for retracts). For every chart φ of X, $(V_1, \varphi, (O_1, C_1, E^1))$ is a chart of X_1; any two are compatible, they form an atlas, and X_1 with it is an M-polyfold X^1. *Book:* preprint p. 25 ("$(V_m, \varphi, (O_m, C_m, E_m))$ is a chart on X_m … the collection of such charts is an atlas on X_m"). *Hidden hypotheses:* the paracompactness of X_1 (Theorem 2.3.10 (preprint 2.16)), an instance argument supplied by V2.17. The levels of X^1 are those of X shifted by one, as sets: $(X^1)_m = X_{m+1}$. *Deviation (narrower):* $m = 1$ here; for $m \geq 2$ the iterate $(X^1)^1 \cdots$ has the points of X_m, and its topology and charts are those of X_m by V2.93 (gap G.6, closed; the set identity was added in a fourth addendum, Findings, Chapter 5, item 142).

V2.17 (Theorem 2.3.10 (preprint 2.16); `Polyfold.theorem_2_3_10`, `Polyfold.MPolyfold.metrizableSpace`, `Polyfold.MPolyfold.metrizableSpace_level`, `Polyfold.MPolyfold.paracompactSpace_level`, `Polyfold.MPolyfold.SmoothLevel`, `Polyfold.MPolyfold.smoothLevel_topology_eq`, `Polyfold.MPolyfold.metrizableSpace_smoothLevel`, `Polyfold.MPolyfold.paracompactSpace_subtype`). Let X be an M-polyfold. Then X is metrizable; for every m, the level X_m with the topology of V2.14 is metrizable and paracompact; and X_∞, with the initial topology of the inclusions $X_\infty \to X_m$ ($m \geq 0$), is metrizable. Every subset of X is paracompact. *Book:* Theorem 2.3.10 (preprint 2.16), p. 37 (preprint p. 25), proof pp. 74–76 (preprint pp. 57–58). *Hidden hypotheses:* the topology of X_∞, which the text does not define. *Deviation (hypotheses explicit):* **no hypothesis is added.** *Proof:* not the text's. The text needs X_m regular for Proposition 2.7.4 (preprint 2.57) and does not check it. Here a partition of unity of the paracompact Hausdorff X subordinate to the chart domains is pulled back to X_m along the continuous inclusion; with the metrizable chart domains $V_m \cong O_m$ it gives an explicit metric (V2.62). Paracompactness of X_m follows by Stone's theorem; X_∞ embeds in $\prod_m X_m$.

V2.18 (Definition 2.3.12 (preprint 2.17), tangent vectors in charts; `Polyfold.MPolyfold.TangentPre`, `Polyfold.MPolyfold.TangentPre.transport`, `Polyfold.MPolyfold.TangentPre.transport_self`, `Polyfold.MPolyfold.TangentPre.transport_mem`, `Polyfold.MPolyfold.TangentPre.transport_transport`, `Polyfold.MPolyfold.TangentPre.Rel`, `Polyfold.MPolyfold.TangentPre.Rel.symm`, `Polyfold.MPolyfold.TangentPre.Rel.trans`, `Polyfold.MPolyfold.TangentPre.setoid`). A tangent tuple is (x, φ, h) with $x \in X_1$, φ a chart of the structure around x and $(\varphi(x), h) \in TO$. Its transport to a chart ψ around x is $T(\psi \circ \varphi^{-1})(\varphi(x), h)$, computed on the overlap retract; it lies in TO' over $\psi(x)$, transport to φ is the identity, and transport to ψ followed by transport to χ is transport to χ. Two tuples are equivalent if their points agree and the transport of the first to the chart of the second is the second; this is an equivalence relation. *Book:* Definition 2.3.12 (preprint 2.17), pp. 38–39 (preprint pp. 25–26). *Hidden hypotheses:* that the relation is an equivalence relation (the text does not check it); it needs Theorem 2.1.15 (preprint 2.8) on the triple overlap and the locality of Tf (V2.8). *Deviation (restated):* none in substance.

V2.19 (Definition 2.3.14 (preprint 2.18), the tangent space TX; `Polyfold.MPolyfold.TangentSpace`, `Polyfold.MPolyfold.tangentProj`, `Polyfold.MPolyfold.tangentFiber`, `Polyfold.MPolyfold.tangentChartSet`, `Polyfold.MPolyfold.tangentChartMap`, `Polyfold.`

`MPolyfold.tangentChartMap_mk_self`, `Polyfold.MPolyfold.tangentChartMap_injOn`, `Polyfold.MPolyfold.tangentChartMap_mk_ofVec`, `Polyfold.MPolyfold.tangentChartMap_mk_ofVec_of_mem`, `Polyfold.MPolyfold.tangentBasis`). TX is the set of equivalence classes of tangent tuples, with $p : TX \to X_1$, $[(x, \varphi, h)] \mapsto x$, and $T_xX = p^{-1}(x)$. For a chart φ, $TV = p^{-1}(V_1)$ and $T\varphi : TV \to TO$ sends a class to its transport to φ; $T\varphi[(x, \varphi, h)] = (\varphi(x), h)$, $T\varphi$ is injective on TV with inverse $a \mapsto [(\varphi^{-1}(a_1), \varphi, a_2)]$ on TO, and $T\psi \circ (T\varphi)^{-1} = T(\psi \circ \varphi^{-1})$ on $T(\varphi(V \cap V'))$. TX carries the topology generated by the sets $\{\alpha \in TV : T\varphi(\alpha) \in W\}$, W open. *Book:* Definition 2.3.14 (preprint 2.18) and the paragraphs after it, pp. 38–40 (preprint pp. 26–27). *Hidden hypotheses:* none. *Deviation (narrower):* the vector space, Banach and sc-Banach structures of T_xX are not formalized.

V2.20 (Proposition 2.3.15 (preprint 2.19); `Polyfold.MPolyfold.continuous_tangentProj`, `Polyfold.MPolyfold.continuous_levelVal_tangentProj`, `Polyfold.MPolyfold.isOpen_tangentChartSet`, `Polyfold.MPolyfold.metrizableSpace_tangentSpace`, `Polyfold.MPolyfold.paracompactSpace_tangentSpace`, `Polyfold.proposition_2_3_15_three`). (1) TX is Hausdorff (the instance `Polyfold.MPolyfold.instT2SpaceTangentSpace`). (2) $p : TX \to X_1$ is continuous, and so is $TX \to X$; each TV is open. (3) TX is metrizable and paracompact. *Book:* Proposition 2.3.15 (preprint 2.19), p. 39 (preprint p. 27), proof pp. 75–78 (preprint pp. 58–60). *Hidden hypotheses:* none for (3). *Deviation (narrower):* the openness of p in (2) is not proved; the basis property in (1) is not needed. *Proof:* (3) as for Theorem 2.3.10 (preprint 2.16), pulling a partition of unity of X back along $TX \to X_1 \to X$; the charts $TV \cong TO$ are metrizable.

V2.21 (TX as an M-polyfold; `Polyfold.MPolyfold.tangentChart`, `Polyfold.MPolyfold.tangentChart_overlap`, `Polyfold.MPolyfold.tangentChart_transition`, `Polyfold.MPolyfold.tangentChart_compatible`, `Polyfold.MPolyfold.tangentAtlas`, `Polyfold.MPolyfold.tangentPolyfold`, `Polyfold.MPolyfold.isScSmoothRetractMap_tangent_fst`, `Polyfold.MPolyfold.tangentProj_isScSmooth`). For a chart φ of X, $(TV, T\varphi, (TO, TC, TE))$ is a chart of TX, with $T\varphi(TV \cap TV') = T(\varphi(V \cap V'))$ and transition map $T(\varphi' \circ \varphi^{-1})$ (formula (5)); these charts are pairwise compatible, form an atlas, and make TX an M-polyfold. The projection $TO \to O_1$ is sc-smooth from (TO, TC, TE) to (O_1, C_1, E^1), and $p : TX \to X^1$ is sc-smooth. *Book:* preprint p. 27, after Proposition 2.3.15 (preprint 2.19). *Hidden hypotheses:* the paracompactness of TX and X_1 (instance arguments, supplied by V2.20 and V2.17). *Deviation (narrower):* “sc-smooth M-polyfold bundle” is read as “p is sc-smooth”, and through the bundle models and the transition maps in V2.94; the fibrewise linear structure is not formalized.

V2.22 (tangent maps of sc-smooth maps between M-polyfolds; `Polyfold.MPolyfold.tangentMap`, `Polyfold.MPolyfold.tangentMap_mk`, `Polyfold.MPolyfold.levelVal_tangentProj_tangentMap`, `Polyfold.MPolyfold.tangentMap_mapsTo_fiber`, `Polyfold.MPolyfold.tangentChartMap_tangentMap`, `Polyfold.MPolyfold.tangentMap_id`, `Polyfold.MPolyfold.tangentMap_comp`, `Polyfold.MPolyfold.continuous_tangentMap`, `Polyfold.MPolyfold.tangentMap_isScSmooth`, `Polyfold.MPolyfold.IsScSmoothMap.mapsTo_levelSet`). An sc-smooth $f : X \to Y$ maps X_m into Y_m. Its tangent map $Tf : TX \to TY$, $[(x, \varphi, h)] \mapsto [(f(x), \psi, T(\psi \circ f \circ \varphi^{-1})(\varphi(x), h)_2)]$, is independent of the chart ψ around $f(x)$

and of the representative; $p \circ Tf = f \circ p$, $Tf(T_x X) \subseteq T_{f(x)} Y$, $T\psi \circ Tf = T(\psi \circ f \circ \varphi^{-1}) \circ T\varphi$ in charts, $T(\mathrm{id}) = \mathrm{id}$, $T(g \circ f) = Tg \circ Tf$, and Tf is continuous and sc-smooth $TX \to TY$. *Book:* not stated; the text uses Tf for M-polyfolds from §2.4 (preprint §2.3) on (Proposition 2.3.19(4) (preprint 2.21), Definition 2.5.8 (preprint 2.38)). *Deviation (not in the book):* the construction the text leaves implicit.

V2.23 (Definition 2.3.18 (preprint 2.20), sub-M-polyfolds; `Polyfold.MPolyfold.IsSubMPolyfold`, `Polyfold.MPolyfold.isSubMPolyfold_univ`). $A \subseteq X$ is a *sub-M-polyfold* if every $a \in A$ has an open $V \ni a$ and a map r with $r(V) \subseteq V$, r sc-smooth on V, $r \circ r = r$ on V and $r(V) = A \cap V$. *Book:* Definition 2.3.18 (preprint 2.20), p. 42 (preprint p. 28). *Hidden hypotheses:* none.

V2.24 (Proposition 2.3.19 (1), (2) (preprint 2.21); `Polyfold.MPolyfold.SubData`, `Polyfold.MPolyfold.SubData.shrink`, `Polyfold.MPolyfold.SubData.image_shrink`, `Polyfold.MPolyfold.SubData.rho_isScSmooth`, `Polyfold.MPolyfold.SubData.isScRetract_image`, `Polyfold.MPolyfold.subChart`, `Polyfold.MPolyfold.subChart_compatible`, `Polyfold.MPolyfold.subAtlas`, `Polyfold.MPolyfold.subPolyfold`, `Polyfold.MPolyfold.subPolyfold_inclusion_isScSmooth`, `Polyfold.MPolyfold.subPolyfold_retraction_isScSmooth`, `Polyfold.IsScSmoothRetractMap.mono_source`, `Polyfold.MPolyfold.paracompactSpace_of_isClosed`). Let A be a sub-M-polyfold with the subspace topology. For data (V, r) at a point and a chart φ, the set $W = (V \cap \operatorname{dom} \varphi) \cap r^{-1}(V \cap \operatorname{dom} \varphi)$ is open with $r(W) \subseteq W$ and $r(W) = W \cap A$; $\rho = \varphi \circ r \circ \varphi^{-1}$ is sc-smooth and idempotent on $\varphi(W)$, so $\varphi(W \cap A)$ is a retract and φ restricted to $W \cap A$ is a chart of A with model $(\varphi(W \cap A), C, E)$. Any two such charts are compatible; they form an atlas, and A is an M-polyfold. (1) The inclusion $A \to X$ is sc-smooth. (2) For every data (V, r), the map $i^{-1} \circ r$ is sc-smooth on V. A closed subset of X is paracompact. *Book:* Proposition 2.3.19 (1), (2) (preprint 2.21), pp. 42–44 (preprint pp. 28–30). *Hidden hypotheses:* the topology of A (the subspace topology; then (1)'s homeomorphism is by construction) and the paracompactness of A, an instance argument of the structure, supplied for every A by `Polyfold.MPolyfold.paracompactSpace_subtype` (V2.17). *Deviation (hypotheses explicit):* none otherwise.

V2.25 (Proposition 2.3.19 (3), (4) (preprint 2.21), in a chart; `Polyfold.ScRetraction.compRetraction`, `Polyfold.ScRetraction.compRetraction_retract`, `Polyfold.proposition_2_3_19_three`, `Polyfold.proposition_2_3_19_four`). Let r be a retraction onto O, s sc-smooth from (O, C, E) to itself with $s \circ s = s$, $t = s \circ r$ (a retraction onto $s(O)$) and $a \in s(O)_1$. If a is smooth: $T_a O = Dr(a)E_0$, $T_a s(O) = Dt(a)E_0 \subseteq T_a O$, the subspaces $Dt(a)E_0$ and $(Dr(a) - Dt(a))E_0$ are sc-subspaces of E, and on every level m their level-m parts are disjoint with sum the level-m part of $T_a O$. For every $a \in s(O)_1$: s is sc-smooth from O to $s(O)$ and Ts maps $T_a O$ onto $T_a s(O)$. *Book:* Proposition 2.3.19 (3), (4) (preprint 2.21), preprint p. 30, proved in coordinates with "$T_a O = Ds(a)Dr(a)E \oplus (I - Ds(a))Dr(a)E$". *Hidden hypotheses:* $Ds(a)$ means $Dt(a)$; the splitting is an sc-splitting because $Dr(a)$, $Dt(a)$ are commuting sc-projections with $Dr(a)Dt(a) = Dt(a) = Dt(a)Dr(a)$. *Deviation (narrower):* in a chart only, as the text proves it; the transport to $T_a X$ is not formalized (no sc-structure on the fibres of TX).

V2.26 (Definition 2.4.1 (preprint 2.22), the degeneracy index; `Polyfold.MPolyfold.`

`chartDegIndex`, `Polyfold.MPolyfold.degIndex`, `Polyfold.MPolyfold.degIndex_le`, `Polyfold.MPolyfold.exists_isChart_degIndex_eq`, `Polyfold.MPolyfold.eventually_chartDegIndex_le`**).** For a chart φ around x, $d(x,\varphi) = d_C(\varphi(x))$; $d_X(x)$ is the least $d(x,\varphi)$ over the charts of the structure around x. The minimum is attained, $d_X(x) \leq d(x,\varphi)$ for every such chart, and $d(\cdot,\varphi)$ is upper semicontinuous on the source of φ. *Book:* Definition 2.4.1 (preprint 2.22), p. 44 (preprint p. 31) ("smooth charts"). *Hidden hypotheses:* none. *Deviation (restated):* "smooth charts" are the charts of the maximal atlas.

V2.27 (Lemma 2.4.3 (preprint 2.23); `Polyfold.lemma_2_4_3`, `Polyfold.eventually_degIndexWith_le`, `Polyfold.MPolyfold.isOpen_setOf_degIndex_lt`, `Polyfold.MPolyfold.isClosed_setOf_le_degIndex`**).** For every $x \in X$, $d_X(y) \leq d_X(x)$ for all y near x; hence $\{d_X < k\}$ is open and $\{d_X \geq k\}$ closed. The degeneracy index d_C of a partial quadrant is upper semicontinuous. *Book:* Lemma 2.4.3 (preprint 2.23), p. 45 (preprint p. 31) ("evident"). *Hidden hypotheses:* none.

V2.28 (Proposition 2.4.4 (preprint 2.24); `Polyfold.MPolyfold.IsScDiffeoOn`, `Polyfold.proposition_2_4_4`, `Polyfold.MPolyfold.transportChart`, `Polyfold.MPolyfold.transportChart_isChart`, `Polyfold.MPolyfold.degIndex_le_of_isScDiffeoOn`**).** Let $U \subseteq X$, $V \subseteq Y$ be open, f sc-smooth on U with $f(U) \subseteq V$ and g sc-smooth on V with $g(V) \subseteq U$, inverse to each other. Then $d_X(x) = d_Y(f(x))$ for every $x \in U$. For a chart ψ of Y, the map $\psi \circ f$ on $U \cap f^{-1}(\operatorname{dom}\psi)$ is a chart of the structure of X. *Book:* Proposition 2.4.4 (preprint 2.24), p. 45 (preprint p. 31) ("from the definitions one deduces immediately"). *Hidden hypotheses:* none. *Deviation (restated):* a germ is represented by an sc-diffeomorphism of open sets.

V2.29 (Definition 2.4.5 (preprint 2.25), the boundary; `Polyfold.MPolyfold.boundary`, `Polyfold.MPolyfold.IsWithoutBoundary`, `Polyfold.MPolyfold.isWithoutBoundary_iff`, `Polyfold.MPolyfold.isClosed_boundary`, `Polyfold.MPolyfold.isOpen_interior`, `Polyfold.MPolyfold.corners`, `Polyfold.MPolyfold.isClosed_corners`, `Polyfold.MPolyfold.IsScDiffeoOn.mem_boundary_iff`**).** $\partial X = \{d_X \geq 1\}$; X is without boundary iff $d_X \equiv 0$ iff $\partial X = \emptyset$. ∂X and $\{d_X \geq 2\}$ are closed, $\{d_X = 0\}$ is open, and sc-diffeomorphisms of open sets preserve ∂X. *Book:* Definition 2.4.5 (preprint 2.25), p. 45 (preprint p. 31). *Hidden hypotheses:* none.

V2.30 (Lemma 2.4.6 (preprint 2.26); `Polyfold.lemma_2_4_6`**).** For a sub-M-polyfold A with the structure of V2.24, $d_A(a) \leq d_X(a)$ for every $a \in A$. *Book:* Lemma 2.4.6 (preprint 2.26), pp. 45–46 (preprint pp. 31–32). *Hidden hypotheses:* the paracompactness of A, as in V2.24.

V2.31 (the subspaces E_x and the cones C_x; `Polyfold.minimalSubspaceWith`, `Polyfold.mem_minimalSubspaceWith`, `Polyfold.isClosed_minimalSubspaceWith`, `Polyfold.finrank_quotient_minimalSubspaceWith`, `Polyfold.partialQuadrantAtWith`, `Polyfold.minimalSubspaceWith_subset_partialQuadrantAtWith`, `Polyfold.isClosed_partialQuadrantAtWith`, `Polyfold.exists_tendsto_lvl1_degIndexWith`, `Polyfold.exists_tendsto_smooth_degIndexWith`, `Polyfold.ScBanach.subset_closure_ker_inter_range_toBase`, `Polyfold.partialQuadrantAtWith_subset_closure`**).** Let $T : E \cong \mathbb{R}^n \oplus W$ present C. $E_x = \{v : (Tv)_i = 0 \text{ whenever } (Tx)_i = 0\}$ is a closed subspace of E_0 with

$\dim(E_0/E_x) = d_C(x)$; $C_x = \{v : (Tv)_i \geq 0 \text{ whenever } (Tx)_i = 0\}$ is closed and convex and contains E_x. Every $x \in C$ is a limit of points of level 1 (and of smooth points) with the same index and in C. The level-1 points of the kernel of any continuous linear map into a finite-dimensional space are dense in it, and the level-1 points of C_x are dense in C_x. *Book:* preprint p. 32 (E_i, E_I, $I(x)$, E_x, H_i, C_x in the model $\mathbb{R}^k \oplus W$). *Hidden hypotheses:* the densities, which the text uses without proof ("the closure of $(T_x^R O)_\infty$ is $T_x^R O$"). *Deviation (restated):* defined through a presentation of C.

V2.32 (Definition 2.4.7 (preprint 2.27); `Polyfold.reducedTangentSpaceWith`, `Polyfold.partialConeWith`, `Polyfold.quotDim`, `Polyfold.quotDim_inf_minimalSubspace_le`, `Polyfold.quotDim_reducedTangentSpaceWith_le`). For a retraction r, a presentation T and $x \in O_1$: $T_x^R O = Dr(x)E_0 \cap E_x$ and $C_x O = Dr(x)E_0 \cap C_x$. For subspaces A, B, $\dim(A/(A \cap B))$ is the natural number $\operatorname{quotDim}(A, B)$; $\dim(T_x O / T_x^R O) \leq d_C(x)$. *Book:* Definition 2.4.7 (preprint 2.27), p. 47 (preprint p. 33). *Hidden hypotheses:* none. *Deviation (stronger):* defined at every $x \in O_1$, not only at smooth points.

V2.33 (Remark 2.4.8 (preprint 2.28); `Polyfold.reducedTangentSpaceWith_subset_partialConeWith`, `Polyfold.remark_2_4_8_isPartialQuadrant`, `Polyfold.remark_2_4_8_partialConeWith`, `Polyfold.flor_nonnegRange_simplicial`, `Polyfold.remark_2_4_8_quotient_simplicial`, `Polyfold.ScRetraction.isScSubspace_reducedTangentSpaceWith`, `Polyfold.partialConeWith_eq_image_scDeriv`). $T_x^R O \subseteq C_x O \subseteq T_x O$. Moreover, for every sc-smooth retraction r in $C = T^{-1}([0,\infty)^n \oplus W)$ and every smooth $x \in O$ (no tameness), $T_x O$ and $T_x^R O$ are sc-subspaces of E and $C_x O$ is a partial quadrant in the sc-Banach space $T_x O$ (with the sc-structure induced from E), of dimension $r = \operatorname{rank} M = \dim(T_x O / T_x^R O) \leq d_C(x)$, where M is the matrix of $Dr(x)$ on $E/E_x \cong \mathbb{R}^{I(x)}$. The core is Flor's theorem in the needed case: for a linear M on $\mathbb{R}^\iota$ with $M^2 = M$ mapping the orthant into itself, the cone range $M \cap [0,\infty)^\iota$ is the set of nonnegative combinations of linearly independent columns of M. *Book:* Remark 2.4.8 (preprint 2.28), p. 47 (preprint p. 33): the inclusions, and "naively one might expect that $C_x O$ is a partial quadrant in $T_x O$. However, this is in general not the case", without an example. *Hidden hypotheses:* none. *Deviation (stronger):* **the remark's negative claim does not hold in our formal reading** (checked in Lean); the remark may mean the narrower point below. *Proof:* $Dr(x)$ maps C_x and E_x into themselves (the sign arguments of Lemma 2.4.9 (preprint 2.29)), so $C_x O = Dr(x)(C_x)$; modulo $T_x^R O$ it is the column cone of the nonnegative idempotent matrix M, simplicial by Flor's theorem (minimal column supports, generation by descent on the support, independence through a private index of each minimal support); the sc-isomorphism $T_x O \cong \mathbb{R}^r \oplus T_x^R O$ of the proof of Theorem 2.4.12 (preprint 2.31) (V2.42) carries $C_x O$ onto $[0,\infty)^r \oplus T_x^R O$. What can fail for a non-tame retract is only $r = d_C(x)$ (Theorem 2.4.12 (preprint 2.31)), which the text proves for tame retracts (Proposition 2.5.6 (preprint 2.36), V2.42) (Findings, Chapter 2, item 50; round-trip item V2.33).

V2.34 (the sign arguments; `Polyfold.IsSc1With.deriv_mem_minimalSubspace_of_lvl1`, `Polyfold.IsSc1With.mapsTo_minimalSubspace`, `Polyfold.IsSc1With.deriv_mem_partialQuadrant_of_lvl1`, `Polyfold.IsSc1With.mapsTo_partialQuadrant`, `Polyfold.`

`eventually_add_smul_mem_lvlSet`, `Polyfold.eventually_add_smul_mem_lvlSet_of_cone`, `Polyfold.ScRetraction.scDeriv_mapsTo_minimalSubspace`). Let f be sc^1 on U, relatively open in C (presented by T), with values in C' (presented by T'), and $x \in U_1$. For $v \in E_1$ with $\iota_1 v \in E_x$: $x + tv \in U_1$ for $|t|$ small and $Df(x)v \in E'_{f(x)}$; hence $Df(x)$ maps E_x into $E'_{f(x)}$. For $v \in E_1$ with $\iota_1 v \in C_x$: $x + tv \in U_1$ for small $t \geq 0$ and $Df(x)v \in C'_{f(x)}$; hence $Df(x)$ maps C_x into $C'_{f(x)}$. For a retraction r and $x \in O_1$, $Dr(x)$ maps E_x into E_x. *Book:* the proofs of Lemma 2.4.9 (preprint 2.29), pp. 48–49 (preprint pp. 33–35), and Proposition 2.5.3 (preprint 2.35), p. 54 (preprint p. 40). *Deviation (not in the book):* the derivative-level content of Lemma 2.4.9 (preprint 2.29), isolated; Lemma 2.4.9 (preprint 2.29) itself, about sc-smooth paths, is not formalized. *Proof:* the function $t \mapsto (T'f(x+tv))_j$ is ≥ 0 near 0 and vanishes at 0; its derivative at 0 vanishes (two-sided) or is ≥ 0 (one-sided).

V2.35 (Proposition 2.4.10 (preprint 2.30); `Polyfold.ScRetraction.IsDiffeoGerm`, `Polyfold.IsScSmoothRetractMap.isDiffeoGerm`, `Polyfold.ScRetraction.IsDiffeoGerm.deriv_inv_deriv`, `Polyfold.ScRetraction.IsDiffeoGerm.deriv_deriv_inv`, `Polyfold.proposition_2_4_10_reduced`, `Polyfold.proposition_2_4_10_cone`). A *germ of sc-diffeomorphisms* between the retracts $O = r(U)$ and $O' = r'(U')$ is a pair F, G with $F \circ r$, $G \circ r'$ sc^1, $F(O) \subseteq O'$, $G(O') \subseteq O$, $G \circ F = \mathrm{id}$ on O and $F \circ G = \mathrm{id}$ on O'; an sc-smooth map between retracts with an sc-smooth inverse is one. For $x \in O_1$ and $y = (F \circ r)_1(x)$: $D(G \circ r')(y) \circ D(F \circ r)(x) = Dr(x)$, $D(F \circ r)(x) \circ D(G \circ r')(y) = Dr'(y)$, and, with $Tf(x) = D(F \circ r)(x)$, $Tf(x)(T^R_x O) = T^R_y O'$ and $Tf(x)(C_x O) = C_y O'$. *Book:* Proposition 2.4.10 (preprint 2.30), p. 49 (preprint p. 35) (smooth x, sc-smooth germs; deduced from Lemma 2.4.9 (preprint 2.29)). *Hidden hypotheses:* none. *Deviation (stronger):* level-1 points and sc^1 germs. *Proof:* from the sign arguments (V2.34) and the two derivative identities, not from Lemma 2.4.9 (preprint 2.29).

V2.36 (a retract as an M-polyfold; `Polyfold.LocalModel.selfChart`, `Polyfold.LocalModel.toMPolyfold`, `Polyfold.IsRelOpenInQuadrant.toMPolyfold`, `Polyfold.IsRelOpenInQuadrant.selfChart_isChart`, `Polyfold.MChart.Compatible.isDiffeoGerm`). A nonempty retract O, and a nonempty relatively open $U \subseteq C$, is an M-polyfold on the subtype with the single chart id. Two compatible charts give a germ of sc-diffeomorphisms (in the sense of V2.35) between the overlap retracts, with the restricted retractions. *Book:* preprint p. 35, before Theorem 2.4.12 (preprint 2.31). *Deviation (restated):* the text's $d_O(x) = \min d_{C'}(\varphi(x))$ over germs into retracts is Definition 2.4.1 (preprint 2.22) for this structure.

V2.37 (Theorem 2.4.12 (preprint 2.31), first statement; `Polyfold.theorem_2_4_12`, `Polyfold.theorem_2_4_12_retract`, `Polyfold.ScRetraction.IsDiffeoGerm.quotDim_le`, `Polyfold.MPolyfold.quotDim_le_chartDegIndex`, `Polyfold.ScRetraction.scDeriv_restrict`, `Polyfold.degIndexWith_le_of_germ`, `Polyfold.remark_2_4_8_isPartialQuadrant`). Let X be an M-polyfold, φ a chart of the structure around x with $\varphi(x) = \iota_1 z$ on level 1, r a retraction onto the model O and T a presentation of C. Then $\dim(T_z O / T^R_z O) \leq d_X(x)$. For a retract O itself as an M-polyfold and $x \in O_1$: $\dim(T_x O / T^R_x O) \leq d_O(x)$. Along a germ of sc-diffeomorphisms F between retracts, $\dim(T_x O / T^R_x O) \leq d_{C'}(F(x))$. *Book:* Theorem 2.4.12 (preprint 2.31), p. 50 (preprint p. 35) (a retract, a smooth x). *Hidden hypotheses:* none. *Deviation* *(stronger):* every M-polyfold, every chart, points on level 1. The second statement of the theorem (equality makes C_xO a partial quadrant) holds without its hypothesis: C_xO is a partial quadrant at every smooth point of every retract (V2.33). *Proof:* for every chart ψ around x, $D(\psi \circ \varphi^{-1})$ induces $T_zO/T_z^RO \cong T_yO'/T_y^RO'$, which embeds in E'/E'_y of dimension $d_{C'}(\psi(x))$; the text instead reduces to a chart realizing $d_O(x)$.

V2.38 (Corollary 2.4.13 (preprint 2.32); `Polyfold.corollary_2_4_13`, `Polyfold.IsRelOpenInQuadrant.degIndex_le`, `Polyfold.IsRelOpenInQuadrant.degIndex_le_degIndex_of_lvl1`, `Polyfold.degIndexWith_le_of_germ`). For a nonempty relatively open subset U of a partial quadrant C, viewed as the M-polyfold X_U, $d_{X_U}(x) = d_C(x)$ for every $x \in U$. *Book:* Corollary 2.4.13 (preprint 2.32), p. 51 (preprint p. 37) ($U = C$). *Hidden hypotheses:* the semicontinuity of $d_{C'}$ and Proposition 2.4.4 (preprint 2.24) at the approximating points. *Deviation (stronger):* every relatively open U; level-1 approximants suffice. Theorem 1.3.5 (preprint 1.25) (V1.23, `Polyfold.theorem_1_3_5`) is proved in the same file by the same argument.

V2.39 (Definition 2.5.1 (preprint 2.33); `Polyfold.partialQuadrantAtWith`, `Polyfold.minimalSubspaceWith`, `Polyfold.minimalSubspaceWith_subset_partialQuadrantAtWith`, `Polyfold.convex_partialQuadrantAtWith`, `Polyfold.finrank_quotient_minimalSubspaceWith`, `Polyfold.minimalSubspaceWith_le`, `Polyfold.minimalSubspaceWith_eq`). For a presentation T of C and $e \in C$: the partial quadrant C_e and the minimal subspace E_e of V2.31, with $E_e \subseteq C_e$, C_e convex and $\dim(E_0/E_e) = d_C(e)$. For two presentations T, T' of the same C, the two subspaces E_e agree. *Book:* Definition 2.5.1 (preprint 2.33), p. 53 (preprint p. 39), through the extended faces. *Hidden hypotheses:* none. *Deviation (restated):* through a presentation, of which E_e is independent (the argument of Lemma 1.3.4 (preprint 1.24)); the intrinsic description by extended faces (preprint p. 39) is not formalized.

V2.40 (Definition 2.5.2 (preprint 2.34), tame retractions; `Polyfold.ScRetraction.IsTameWith`, `Polyfold.ScRetraction.IsTame`, `Polyfold.IsTameRetract`, `Polyfold.IsTameRetract.isScRetract`, `Polyfold.ScRetraction.id_isTame`, `Polyfold.ScRetraction.IsTameWith.isTame`, `Polyfold.ScRetraction.IsTameWith.restrict`). A retraction r is *tame for T* if (1) $d_C(r(x)) = d_C(x)$ for $x \in U$ and (2) for every smooth $x \in O_1$ there is a subspace $A \leq E_x$ such that $E = Dr(x)E_0 \oplus A$ is an sc-splitting. r is *tame* if it is tame for every presentation of C; tameness for one presentation implies it. A *tame retract* is the retract of a tame retraction. Identities of relatively open subsets are tame, and the restriction of a tame retraction to $U \cap r^{-1}(V)$, V open, is tame. *Book:* Definition 2.5.2 (preprint 2.34), pp. 53–54 (preprint pp. 39–40). *Hidden hypotheses:* that the definition does not depend on the presentation, which the text uses tacitly. *Deviation (restated):* none in substance.

V2.41 (Proposition 2.5.3 (preprint 2.35); `Polyfold.proposition_2_5_3`, `Polyfold.ScRetraction.scDeriv_mapsTo_minimalSubspace`, `Polyfold.ScRetraction.isScOperator_scDeriv`, `Polyfold.ScRetraction.isScSplitting_range_scDeriv`, `Polyfold.coord_scDeriv_eq`). Let r be tame for T and $x \in O_1$ smooth. Then $(1 - Dr(x))E_0 \leq E_x$, and $E = Dr(x)E_0 \oplus (1 - Dr(x))E_0$ is an sc-splitting; equivalently $\lambda_j(Dr(x)v) = \lambda_j(v)$ for $j \in I(x)$ and all v. *Book:* Proposition 2.5.3 (preprint 2.35), p. 54 (preprint p. 41), with the discussion on p. 54

(preprint p. 40). *Hidden hypotheses:* none. *Deviation (stronger):* only condition (2) of tameness is used; $Dr(x)E_x \subseteq E_x$ holds for every retraction (V2.34), where the text uses condition (1).

V2.42 (Proposition 2.5.6 (preprint 2.36); `Polyfold.proposition_2_5_6`, `Polyfold.proposition_2_5_6_degIndex`, `Polyfold.quotDim_reducedTangentSpaceWith_eq`, `Polyfold.MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, `Polyfold.MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract`, `Polyfold.LocalModel.degIndex_toMPolyfold_of_isTameWith`, `Polyfold.quotDim_eq_finrank_quotient_of_isCompl`, `Polyfold.proposition_2_5_6_partialCone`, `Polyfold.splitIso`, `Polyfold.isPartialQuadrant_ofSubspace`). Let r be tame for T. (13) At a smooth $x \in O_1$, $\dim(T_xO/T_x^RO) = d_O(x) = d_C(x)$, where d_O is the index of the M-polyfold O; moreover T_xO and T_x^RO are sc-subspaces of E, and $C_xO = T_xO \cap C_x$ is a partial quadrant in the sc-Banach space T_xO (with the sc-structure induced from E). (14) $d_O(x) = d_C(x)$ for every $x \in O$. More generally, a chart φ of any M-polyfold whose model is a tame retract satisfies $d_X(y) = d_C(\varphi(y))$ for every y in its source. For sc-subspaces $B \le A$ with $B = A \cap \bigcap_j \ker \ell_j$ and smooth $e_j \in A$ with $\ell_i(e_j) = \delta_{ij}$, the map $v \mapsto ((\ell_j v)_j, v - \sum_j \ell_j(v)e_j)$ is an sc-isomorphism $A \cong \mathbb{R}^d \oplus B$, and $\{v \in A : \ell_j(v) \ge 0 \ \forall j\}$ is a partial quadrant in A. *Book:* Proposition 2.5.6 (preprint 2.36), p. 56 (preprint p. 41). *Hidden hypotheses:* none. *Deviation (stronger):* the chart form, which Proposition 2.5.14 (preprint 2.42) needs; T_x^RO is an sc-subspace, where the text says a Banach space. *Proof:* (13), (14) as in the text: $E_x = T_x^RO \oplus A$ gives $\dim(T_xO/T_x^RO) = d_C(x)$, Theorem 2.4.12 (preprint 2.31) gives $\le d_X$; a general point is approximated by smooth points $r(T^{-1}(a, w_j))$ of the same index, with Lemma 2.4.3 (preprint 2.23). The partial quadrant not from the equality case of Theorem 2.4.12 (preprint 2.31), as in the text, but directly (it holds without tameness, V2.33; what tameness adds is the dimension $d_O(x)$ of the quadrant): the smooth vectors $\hat{e}_j = Dr(x)T^{-1}(e_{i_j}, 0)$ satisfy $\lambda_{i_k}(\hat{e}_j) = \delta_{jk}$ by Proposition 2.5.3 (preprint 2.35), and the splitting applies to $A = T_xO$, $B = T_x^RO$; T_x^RO is an sc-subspace because the smooth points $Dr(x)T^{-1}(a, w')$, $w' \in W_\infty$, lie in it ($Dr(x)E_x \subseteq E_x$) and approximate u, $Tu = (a, w)$, on every level.

V2.43 (Definition 2.5.7 (preprint 2.37); `Polyfold.MPolyfold.IsTame`, `Polyfold.MPolyfold.isChart_of_mem_equiv`, `Polyfold.MPolyfold.IsTame.exists_chart`). X is *tame* if some atlas equivalent to its atlas consists of charts with tame models; then every point lies in a chart of the structure whose model is a tame retract. *Book:* Definition 2.5.7 (preprint 2.37), p. 57 (preprint p. 42). *Hidden hypotheses:* none.

V2.44 (Definition 2.5.10 (preprint 2.39); `Polyfold.MPolyfold.IsFace`, `Polyfold.MPolyfold.IsFaceStructured`, `Polyfold.MPolyfold.IsFace.isClosed`, `Polyfold.MPolyfold.IsFace.subset_boundary`). A *face* is the closure of the connected component in $\{d_X = 1\}$ of a point with $d_X = 1$; X is *face-structured* if every x lies in exactly $d_X(x)$ faces. Faces are closed and contained in ∂X. *Book:* Definition 2.5.10 (preprint 2.39), p. 57 (preprint p. 43). *Hidden hypotheses:* none.

V2.45 (the local geometry of faces; `Polyfold.subset_coordHyper_of_isPreconnected`, `Polyfold.IsSc1With.apply_deriv_eq_zero_of_ray`, `Polyfold.ray_zeroSet`, `Polyfold.local_face_structure`, `Polyfold.eventually_zeroSet_subset`, `Polyfold.convex_ball_`

`inter_zeroSet_eq`, `Polyfold.ScRetraction.faceStratum`, `Polyfold.coordHyper`). With $\lambda_i = (T\cdot)_i$, $I(z) = \{i : \lambda_i(z) = 0\}$ and $f_i = \{\lambda_i = 0\}$: (a) a preconnected subset of $\{z \in C : d_C(z) = 1\}$ containing a point of f_i lies in f_i; (b) if r is tame for T, $x \in O_1$ smooth, $i \in I(x)$, $\lambda_i(y) = 0$ and $\lambda_l(y) > 0$ for the other $l \in I(x)$, then for all small $\tau > 0$, $x + \tau y \in U$, $I(x + \tau y) = \{i\}$ and $I(r(x + \tau y)) = \{i\}$; (c) for $e \in O$ with $\lambda_i(e) = 0$ there are $\varepsilon > 0$ and $p \in S = \{z \in O : d_C(z) = 1\}$ with $\lambda_i(p) = 0$ such that $B = B_\varepsilon(e) \cap C \subseteq U$, $I(z) \subseteq I(e)$ and $I(r(z)) \subseteq I(e)$ on B, r maps the points of B with $I = \{i\}$ into the component of S through p, and maps $f_i \cap B$ into its closure. *Book:* the proof of Lemma 2.5.12 (preprint 2.40), p. 59 (preprint p. 43) (a sketch). *Deviation (not in the book):* the exposition's Lemmas 4.A–4.C; (b) uses tameness condition (2) through Proposition 2.5.3 (preprint 2.35).

V2.46 (Lemma 2.5.12 (preprint 2.40); `Polyfold.ScRetraction.IsRetractFace`, `Polyfold.lemma_2_5_12`, `Polyfold.lemma_2_5_12_injective`, `Polyfold.closure_connectedComponentIn_faceStratum`, `Polyfold.connectedComponentIn_coordHyper_eq`, `Polyfold.LocalModel.isFace_toMPolyfold_iff`, `Polyfold.subset_of_relClopen_of_isPreconnected`). Let r be tame for T. A *face of the retract* is $\widehat{F} \cap O$ for a connected component $\widehat{F}$ of S. Then G is a face of the retract iff G is the connected component of $f_i \cap O$ through some $e \in O$ with $\lambda_i(e) = 0$; and for $x \in O$ with $\lambda_i(x) = 0$, equal components of $f_i \cap O$ and $f_j \cap O$ through x force $i = j$. The faces of the M-polyfold O are exactly the faces of the retract. *Book:* Lemma 2.5.12 (preprint 2.40), p. 59 (preprint p. 43). *Hidden hypotheses:* the face preservation of V2.45, which the text's sketch does not address. *Deviation (hypotheses explicit):* extended faces are the coordinate hyperplanes of a presentation.

V2.47 (Corollary 2.5.13 (preprint 2.41); `Polyfold.corollary_2_5_13`, `Polyfold.corollary_2_5_13_retract`, `Polyfold.LocalModel.isFaceStructured_toMPolyfold`). For a nonempty tame retract O, the M-polyfold O is face-structured: every $x \in O$ lies in exactly $d_O(x)$ faces. *Book:* Corollary 2.5.13 (preprint 2.41), p. 59 (preprint p. 44). *Hidden hypotheses:* none.

V2.48 (Proposition 2.5.14 (preprint 2.42); `Polyfold.proposition_2_5_14`, `Polyfold.MPolyfold.faces_encard_le`, `Polyfold.MPolyfold.IsTame.exists_chart`). Let X be tame and $x \in X$. There is a chart $\varphi : V \to O$ of the structure around x such that for all $y \in V$, $d_X(y) = d_C(\varphi(y))$ and $d_O(\varphi(y)) = d_X(y)$, and the M-polyfold O is face-structured. Globally, x lies in at most $d_X(x)$ faces of X. *Book:* Proposition 2.5.14 (preprint 2.42), p. 60 (preprint p. 44). *Hidden hypotheses:* "$d_V = d_X$" needs that a tame chart realizes d_X (V2.42). *Deviation (restated):* the neighbourhood V is represented by the M-polyfold O.

V2.49 (Proposition 2.5.15 (preprint 2.43); `Polyfold.proposition_2_5_15`, `Polyfold.killCoord`, `Polyfold.killCoord_mem`, `Polyfold.killCoord_eq_self`, `Polyfold.LocalModel.isSubMPolyfold_of_isFace`, `Polyfold.faceIncl`, `Polyfold.faceProj`, `Polyfold.degIndexWith_eq_quadrantDeg_faceProj_add_one`, `Polyfold.proposition_2_5_15_tameModel`, `Polyfold.MPolyfold.faceChart`, `Polyfold.MPolyfold.faceChart_compatible_subChart`, `Polyfold.LocalModel.proposition_2_5_15_isTame`). Let r be tame for T and F a face of the retract. (a) $F \subseteq f_i$ for some i, and there is an sc-smooth retraction s on a set U^* with $F \subseteq U^* \subseteq U$, U^* relatively open in C, with retract F and $s = r \circ P_i$ on U^*, where

P_i is the sc-projection onto f_i killing the i-th coordinate; every face of the M-polyfold O is a sub-M-polyfold of O. (b) With the sc-operators $\iota : \mathbb{R}^k \oplus W \to E$ (insert a zero i-th coordinate, onto f_i) and $\pi : E \to \mathbb{R}^k \oplus W$ (delete it), $\pi \circ \iota = 1$, $\iota \circ \pi = 1$ on f_i and $d_C(y) = d_{C'}(\pi y) + 1$ on f_i ($C' = [0,\infty)^k \oplus W$); $r' = \pi \circ r \circ \iota$ is a tame retraction of C' onto $\pi(F)$, and $d_{C'}(\pi y) + 1 = d_C(y)$ for $y \in F$. (c) For a face F of a nonempty tame retract O, with its sub-M-polyfold structure, F is a tame M-polyfold and $d_F(x) + 1 = d_O(x)$ for $x \in F$; the chart $\pi : F \to \pi(F)$ is compatible with the sub-charts of Proposition 2.3.19 (preprint 2.21). *Book:* Proposition 2.5.15 (preprint 2.43), pp. 61–62 (preprint pp. 44–45). *Hidden hypotheses:* the face preservation of V2.45; the paracompactness of F (faces are closed). *Deviation (restated):* the tame model lives on $\mathbb{R}^k \oplus W$ through ι, π instead of the sc-subspace f_1; tameness (2) of r' is proved with the complement $(1 - Dr'(x))E'$ by Proposition 2.5.3 (preprint 2.35), not by intersecting with f_1. U^* is a union of balls intersected with their P_i-preimages, since P_i need not be norm-decreasing. *No hypothesis is added.*

V2.50 (Proposition 2.5.16 (preprint 2.44); `Polyfold.MPolyfold.proposition_2_5_16`, `Polyfold.proposition_2_5_14`, `Polyfold.LocalModel.proposition_2_5_15_isTame`, `Polyfold.MPolyfold.proposition_2_5_16_corner`, `Polyfold.corner_retraction`, `Polyfold.corner_tameModel`, `Polyfold.cornerIncl`, `Polyfold.cornerProj`, `Polyfold.cornerKill`, `Polyfold.degIndexWith_eq_quadrantDeg_cornerProj_add_card`). Let X be tame and $x \in X$. There is a chart $\varphi : V \to O$ of the structure around x, with a tame retraction r onto O (tame for a presentation T), such that $\varphi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O (the local faces), and every face F of O is a sub-M-polyfold of O which, with the induced structure, is tame with $d_F(\varphi(x)) + 1 = d_X(x)$ when $\varphi(x) \in F$. For every $\sigma \subseteq I(\varphi(x))$ (a set of $\#\sigma$ face germs at x) there are pairwise distinct faces $F_i \ni \varphi(x)$ of the retract, $F_i \subseteq f_i$ ($i \in \sigma$), and an open $B \ni \varphi(x)$ such that $A = \{y \in V : \varphi(y) \in B,\ \varphi(y) \in F_i\ \forall i \in \sigma\}$ is a sub-M-polyfold of X, tame with its induced structure, with $d_A(y) + \#\sigma = d_X(y)$ for all $y \in A$. The ingredients: the corner $f_\sigma = \bigcap_{i\in\sigma} f_i$ is modelled on $\mathbb{R}^{n-\#\sigma} \oplus W$ through sc-operators ι_σ, π_σ, with $d_C(y) = d_{C'}(\pi_\sigma y) + \#\sigma$ on f_σ; near a point $o \in O$ with $\sigma \subseteq I(o)$, $r \circ P_\sigma$ ($P_\sigma = \iota_\sigma \pi_\sigma$) is an sc-smooth retraction onto $O \cap V \cap f_\sigma = O \cap V \cap \bigcap_{i\in\sigma} F_i$; and $\pi_\sigma \circ r \circ \iota_\sigma$ is a tame retraction of $[0,\infty)^{n-\#\sigma} \oplus W$ onto its image under π_σ. *Book:* Proposition 2.5.16 (preprint 2.44), p. 62 (preprint p. 46) (stated without proof). *Hidden hypotheses:* none. *Deviation (restated):* local faces are the faces of the M-polyfold $O \cong V$, face germs at x are the faces F_i, $i \in I(\varphi(x))$, "the germ F^σ_x" is represented by the set A, and "codimension $\#\sigma$" is the number of quadrant coordinates removed in the tame model; the index identity holds on all of A. *Proof:* not by the exposition's induction on $\#\sigma$ but at once: Lemma 4.C for each $i \in \sigma$ shows that r maps points of $C \cap f_\sigma$ near $\varphi(x)$ into f_σ; then as in Proposition 2.5.15 (preprint 2.43) and Theorem 2.5.17 (preprint 2.45).

V2.51 (Theorem 2.5.17 (preprint 2.45); `Polyfold.MPolyfold.theorem_2_5_17_interior`, `Polyfold.MPolyfold.isSubMPolyfold_of_local`, `Polyfold.MPolyfold.theorem_2_5_17_isSubMPolyfold_interior`, `Polyfold.MPolyfold.exists_localFace_of_isFaceStructured`, `Polyfold.MPolyfold.theorem_2_5_17_isSubMPolyfold_face`, `Polyfold.MPolyfold.theorem_2_5_17_isTame_face`, `Polyfold.MPolyfold.isTame_of_forall_exists_chart`). Let X be tame, G the connected component of $\{d_X = 1\}$ through a point and $F = \overline{G}$ the face. (a)

$F \cap \{d_X = 1\} = G$, and G is a sub-M-polyfold of X. (b) If X is face-structured, every face F agrees near each of its points, in a tame chart, with a face of the model retract; F is a sub-M-polyfold of X, tame with its induced structure, and $d_X(x) = d_F(x) + 1$ for $x \in F$. (A subset A is a sub-M-polyfold as soon as near each point $\varphi(A)$ agrees on an open set with an sc-smooth retract contained in O; X is tame as soon as every point lies in a chart of the structure with a tame model.) *Book:* Theorem 2.5.17 (preprint 2.45), p. 62 (preprint p. 46) (stated without proof). *Hidden hypotheses:* the meaning of "the interior of a face", here $F \cap \{d_X = 1\}$. *Deviation (hypotheses explicit):* the counting argument behind "each global face through x contains exactly one local face" (onto by Lemma 4.C, injective by face-structuredness) is made explicit. The theorem's face counts are V2.48 and the definition of face-structured (V2.44).

V2.52 (products of M-polyfolds, exposition Lemma 4.D; `Polyfold.IsPartialQuadrant.prod`, `Polyfold.prod_eq_prodPresentation`, `Polyfold.degIndex_prod`, `Polyfold.minimalSubspaceWith_prodPresentation`, `Polyfold.ScRetraction.prod`, `Polyfold.IsScRetract.prod`, `Polyfold.ScRetraction.IsTameWith.prod`, `Polyfold.MChart.prod`, `Polyfold.MChart.Compatible.prod`, `Polyfold.MPolyfold.paracompactSpace_prod`, `Polyfold.MPolyfold.prod`, `Polyfold.MPolyfold.degIndex_prod_le`, `Polyfold.MPolyfold.isTame_prod`, `Polyfold.MPolyfold.isScSmoothMap_fst`, `Polyfold.MPolyfold.isScSmoothMap_snd`, `Polyfold.MPolyfold.IsScSmoothOn.prodMk`). For partial quadrants $C \subseteq E$, $C' \subseteq E'$, $C \times C'$ is a partial quadrant of $E \oplus E'$, presented by $E \oplus E' \cong \mathbb{R}^{n+n'} \oplus (W \oplus W')$, with $d_{C\times C'}(x, x') = d_C(x) + d_{C'}(x')$ and minimal subspace $E_x \oplus E'_{x'}$; products of sc-smooth (tame) retractions are sc-smooth (tame) retractions. For M-polyfolds X, Y the space $X \times Y$ is paracompact and the product charts are an atlas of an M-polyfold structure; $d_{X\times Y}(x, y) \leq d_X(x) + d_Y(y)$, and if X, Y are tame, $X \times Y$ is tame with equality. The projections are sc-smooth and the pairing of sc-smooth maps is sc-smooth. *Book:* used without statement in the proof of Proposition 2.5.20 (preprint 2.46), preprint p. 47 ("$d_{X\times Y} = d_X + d_Y$"). *Hidden hypotheses:* the additivity needs tameness; in general only the inequality holds. Paracompactness of $X \times Y$ is Theorem 2.3.10 (preprint 2.16). *Deviation (extra):* not stated in the text; the exposition's Lemma 4.D.

V2.53 (Proposition 2.5.20 (preprint 2.46); `Polyfold.MPolyfold.IsLocalScDiffeo`, `Polyfold.MPolyfold.fibredProduct`, `Polyfold.MPolyfold.fibredSubDataL`, `Polyfold.MPolyfold.isSubMPolyfold_fibredProduct`, `Polyfold.MPolyfold.isScDiffeoOn_snd_fibredProduct`, `Polyfold.MPolyfold.isScDiffeoOn_fst_fibredProduct`, `Polyfold.MPolyfold.degIndex_fibredProduct_eq_snd`, `Polyfold.MPolyfold.degIndex_fibredProduct_eq_fst`, `Polyfold.MPolyfold.isTame_fibredProduct`, `Polyfold.MPolyfold.proposition_2_5_20`, `Polyfold.MPolyfold.proposition_2_5_20_of_isLocalScDiffeo`). Let $f : X \to Z$ be a local sc-diffeomorphism (every point has an open neighbourhood that f maps sc-diffeomorphically onto an open subset of Z) and $g : Y \to Z$ sc-smooth. Then $X_f \times_g Y = \{(x, y) : f(x) = g(y)\}$ is closed in $X \times Y$, hence paracompact, and a sub-M-polyfold of $X \times Y$, near each point the retract of $R(x', y') = ((f|_U)^{-1}(g(y')), y')$; $d_{X_f\times_g Y}(x, y) = d_Y(y)$; if Y is tame, so is the fibred product. If g is also a local sc-diffeomorphism, then $d_{X_f\times_g Y}(x, y) = d_X(x) = d_Y(y)$, and if moreover X and Y are tame, $2\, d_{X_f\times_g Y}(x, y) = d_{X\times Y}(x, y)$. *Book:* Proposition 2.5.20 (preprint 2.46), pp. 63–64 (preprint pp. 46–47). *Hidden hypotheses:* the meaning of "local sc-diffeomorphism";

the paracompactness of the fibred product. The text's $\pi_2 : X_f \times_g Y \to X$ has target Y. *Deviation (stronger):* $d = d_Y$, and $d = d_X = d_Y$ in the last statement, hold without tameness (Proposition 2.4.4 (preprint 2.24) along π_2, π_1); tameness is used only for $d_{X\times Y} = d_X + d_Y$ (V2.52). The inverse of π_2 near (x, y) is $y' \mapsto ((f|_U)^{-1}(g(y')), y')$; the exposition's factorization through the constant-in-x map $y' \mapsto (x, y')$ is not sc-smooth unless x is a smooth point. *No hypothesis is added.*

V2.54 (Definition 2.6.1 (preprint 2.47); `Polyfold.doubleLevel`, `Polyfold.shiftFiber`, `Polyfold.IsStrongBundleMap`, `Polyfold.IsStrongBundleRetraction`, `Polyfold.IsPartialQuadrant.prod_univ`, `Polyfold.IsRelOpenInQuadrant.prod_univ`, `Polyfold.IsStrongBundleRetraction.baseRetraction`, `Polyfold.IsStrongBundleRetraction.fiber_idem`, `Polyfold.IsStrongBundleRetraction.toScRetraction`, `Polyfold.IsStrongBundleRetraction.toScRetractionOne`). $C \times F_0$ is a partial quadrant in $E \oplus F$ and $U \times F_0$ is relatively open in it. A *strong bundle map* on $U \lhd F$ is $\Phi : E_0 \times F_0 \to E'_0 \times F'_0$ mapping $U \times F_0$ into $U' \times F'_0$, whose first component does not depend on h and second is linear in h for $u \in U$, mapping $U_m \oplus F_k$ into $U'_m \oplus F'_k$ for $k \leq m + 1$, sc-smooth on $U \times F_0$ for $E \oplus F$, and whose induced map on $U \times F_1$ (fibre component lifted to F'_1) is sc-smooth for $E \oplus F^1$. A *strong bundle retraction* is one with $R \circ R = R$ on $U \times F_0$. Then $r(u) = R(u, 0)_1$ is an sc-smooth retraction on U, $\Gamma(u, \cdot)$ is a projection for $u \in O$, and R is an sc-smooth retraction on $(U \lhd F)[0]$ and $[1]$. *Book:* Definition 2.6.1 (preprint 2.47) and the remarks after it, pp. 65–66 (preprint pp. 48–49). *Hidden hypotheses:* $U \lhd F$ relatively open in a partial quadrant (for "sc-smooth" to make sense). *Deviation (restated):* strong bundle isomorphisms are not defined separately.

V2.55 (Definition 2.6.2 (preprint 2.48) and $K[i]$; `Polyfold.IsStrongBundleRetract`, `Polyfold.strongLevel`, `Polyfold.strongLevelShift`, `Polyfold.strongShiftOne`, `Polyfold.IsStrongBundleRetraction.mem_image_iff`, `Polyfold.IsStrongBundleRetraction.toScRetractionOne_retract`, `Polyfold.IsStrongBundleRetract.isScRetract`, `Polyfold.IsStrongBundleRetract.isScRetract_one`, `Polyfold.IsStrongBundleRetract.isScSmoothRetractMap_fst`, `Polyfold.IsStrongBundleRetract.isScSmoothRetractMap_fst_one`, `Polyfold.IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl`). K is a *local strong bundle retract* over C if $K = R(U \times F_0)$ for a strong bundle retraction on a relatively open $U \subseteq C$; then $K = \{(u, h) : u \in O, \Gamma(u, h) = h\}$. $K_{m,k}$ and $K[1] = \{(u, h) \in E_0 \times F_1 : (u, \iota_1 h) \in K\}$ are defined level by level; K is a retract in $(C \times F_0, E \oplus F)$, $K[1]$ (the image of R on $(U \lhd F)[1]$) a retract in $(C \times F_0, E \oplus F^1)$, and the projections $K[i] \to O$ and the inclusion $K[1] \to K[0]$ are sc-smooth. *Book:* Definition 2.6.2 (preprint 2.48) and pp. 65–67 (preprint pp. 49–50). *Hidden hypotheses:* that R preserves the double filtration at $(0, 1)$, for $K[1] = R((U \lhd F)[1])$. *Deviation (restated):* none in substance.

V2.56 (Definition 2.6.3 (preprint 2.49); `Polyfold.IsScSmoothSection`, `Polyfold.IsScPlusSection`, `Polyfold.IsScPlusSection.mem`, `Polyfold.IsScPlusSection.isScSmoothSection`). An *sc-smooth section* of $p : K \to O$ is f with $f(x)_1 = x$ on O, sc-smooth from (O, C, E) to $(K, C \times F_0, E \oplus F)$; an *sc$^+$-section* is $f : E_0 \to E_0 \times F_1$ with $f(x)_1 = x$ on O, sc-smooth into $(K[1], C \times F_0, E \oplus F^1)$. An sc$^+$-section is an sc-smooth section. *Book:* Definition 2.6.3 (preprint 2.49), p. 66 (preprint p. 50) ($K(0)$, $K(1)$ for $K[0]$, $K[1]$). *Deviation* *(restated):* the category $\mathcal{SBR}$ is not formalized.

V2.57 (Definition 2.6.4 (preprint 2.50); **`Polyfold.IsBanachable`, `Polyfold.StrongBundleChart`, `Polyfold.StrongBundleChart.K`, `Polyfold.StrongBundleChart.continuous_fiber`, `Polyfold.StrongBundleChart.Compatible`, `Polyfold.StrongBundleChart.Compatible.symm`, `Polyfold.StrongBundleAtlas`, `Polyfold.StrongBundleAtlas.Equiv`, `Polyfold.StrongBundleChart.compatible_of_locally`, `Polyfold.StrongBundleAtlas.equiv_iff`, `Polyfold.StrongBundleAtlas.Equiv.trans`, `Polyfold.StrongBundleAtlas.equivalence`, `Polyfold.IsStrongBundleRetraction.restrict`, `Polyfold.StrongBundleChart.restrict`, `Polyfold.StrongBundleChart.Compatible.restrict`).** For $P : Y \to X$ with linear structures on the fibres: a *strong bundle chart* is a chart φ of the M-polyfold X, an sc-Banach space F, a strong bundle retraction R on a relatively open $U \subseteq C$ with base retract O, and a homeomorphism $\Phi : P^{-1}(V) \to K = R(U \times F_0)$ with $\Phi(y)_1 = \varphi(P(y))$ and linear on each fibre (hence continuous on fibres). Two charts are *compatible* if $\Phi' \circ \Phi^{-1}$ and $\Phi \circ \Phi'^{-1}$ are sc-smooth between the overlaps on [0] and, through the lift to F_1, on [1], and carry fibre components in F_1 to fibre components in F_1'. Atlases and their equivalence as in V2.11; two atlases are equivalent iff every chart of one is compatible with every chart of the other, and the relation is an equivalence relation (it rests on the locality of compatibility: near a point of the overlap a transition factors through a chart of the middle atlas, on [0] and on [1]). A strong bundle chart restricts to every open $s \subseteq V$ (retraction domain $U \cap r^{-1}(W)$ for $\varphi(s) = O \cap W$, $K_s = \{(u, h) \in K \mid u \in \varphi(s)\}$, Φ unchanged), and restrictions of compatible charts are compatible. A space is *Banachable* if it is linearly homeomorphic to a Banach space. *Book:* Definition 2.6.4 (preprint 2.50), pp. 67–68 (preprint pp. 51–52). *Hidden hypotheses:* φ a chart of X (the text: a homeomorphism onto O); the [1]-transition condition. *Deviation (hypotheses explicit):* both stated. The transitivity of the equivalence, which "an equivalence class of strong bundle atlases" (Definition 2.6.5 (preprint 2.51)) presupposes, and the restriction of charts, which the text uses silently (Lemma 5.1.2 (preprint 5.2)), are proved (Findings, Chapter 2, items 40, 42); a restriction whose retraction domain lies inside a given open $W \ni 0$ of E is not, since $U \cap W$ need not be invariant under r.

V2.58 (Definition 2.6.5 (preprint 2.51), strong bundles, $Y[0]$, $Y[1]$, $P[0]$, $P[1]$; **`Polyfold.StrongBundle`, `Polyfold.StrongBundleChart.toMChart`, `Polyfold.StrongBundleChart.Compatible.toMChart`, `Polyfold.StrongBundle.atlas0`, `Polyfold.StrongBundle.toMPolyfold0`, `Polyfold.StrongBundle.isScSmoothMap_proj`, `Polyfold.StrongBundle.doubleLevel`, `Polyfold.StrongBundle.mem_doubleLevel_iff`, `Polyfold.StrongBundle.oneChart`, `Polyfold.StrongBundle.oneChart_compatible`, `Polyfold.StrongBundle.atlas1`, `Polyfold.StrongBundle.toMPolyfold1`, `Polyfold.StrongBundle.isScSmoothMap_proj1`, `Polyfold.StrongBundle.maximalAtlas`, `Polyfold.StrongBundle.maximalAtlas_equiv`, `Polyfold.StrongBundle.isChart_iff_exists_equiv`, `Polyfold.StrongBundle.isChart_congr`, `Polyfold.StrongBundle.IsChart.compatible`, `Polyfold.StrongBundle.IsChart.restrict`).** A *strong bundle* is a continuous surjection $P : Y \to X$, Y Hausdorff paracompact, with Banachable fibres and a strong bundle atlas. The charts Φ form an sc-smooth atlas of Y (the M-polyfold $Y[0]$), and $P : Y[0] \to X$ is sc-smooth. $Y_{m,k}$ ($k \leq m + 1$) is the set of y with $\Phi(y) \in E_m \times F_k$ in the charts of the atlas around y; one chart around y suffices. The charts

$\Phi[1] : y \mapsto (\Phi(y)_1, \Phi(y)_2$ lifted to $F_1)$, defined on $Y_{0,1}$ with the topology they generate and onto the retracts $K[1]$, are pairwise compatible, their transition maps being the [1]-transitions of Definition 2.6.4 (preprint 2.50), and make $Y[1]$ an M-polyfold; $P[1] : Y[1] \to X$ is sc-smooth. The charts of the strong bundle (compatible with the atlas) are those of the maximal atlas, which is equivalent to the given one: a chart is one iff it belongs to some atlas of the class, any two are compatible, restrictions of charts are charts, and equivalent atlases have the same charts. *Book:* Definition 2.6.5 (preprint 2.51) and the remarks after it, preprint p. 52. *Hidden hypotheses:* the paracompactness of $Y[1]$ (proved, V2.64); for the chart-independence of $Y_{m,m+1}$ and of $Y_{0,1}$, the F_1-condition on the [1]-transition (V2.57). *Deviation (restated):* $Y[1]$ carries the topology generated by the charts $\Phi[1]$, as X_m in §2.3 (preprint §2.2); the charts are those induced by the given atlas. The notions defined through the charts compatible with the atlas depend only on the class; those defined with the charts of the given atlas itself ($Y_{m,k}$, $Y[1]$ and its atlas, sc⁺-sections) are not shown invariant (Findings, Chapter 2, item 43).

V2.59 (Definition 2.6.6 (preprint 2.52); `Polyfold.IsSection`, `Polyfold.StrongBundle.IsScSmoothSection`, `Polyfold.StrongBundle.localRepOne`, `Polyfold.StrongBundle.IsScPlusSection`). A *section* is f with $P \circ f = \mathrm{id}$; it is *sc-smooth* if it is sc-smooth $X \to Y[0]$, and an *sc⁺-section* if in every chart of the atlas its fibre values lie in F_1 and its local representative, lifted to $E_0 \times F_1$, is sc-smooth from O into $K[1]$. *Book:* Definition 2.6.6 (preprint 2.52), p. 68 (preprint p. 53). *Deviation (restated):* sc⁺ is defined chart by chart; this is equivalent to the text's definition: f is an sc⁺-section iff $P \circ f = \mathrm{id}$ and $f = \iota \circ g$ for an sc-smooth $g : X \to Y[1]$, $\iota : Y[1] \to Y$ the inclusion (`Polyfold.StrongBundle.isScPlusSection_iff`).

V2.60 (Definition 2.6.7 (preprint 2.53); `Polyfold.pullbackSpace`, `Polyfold.pullbackProj`, `Polyfold.pullbackProj'`, `Polyfold.pullback_comm`, `Polyfold.continuous_pullbackProj`, `Polyfold.isClosed_pullbackSpace`). $f^*Y = \{(z, y) : P(y) = f(z)\}$ with the projections P_f, P'; $P \circ P' = f \circ P_f$; P_f is continuous, and f^*Y is closed in $Z \times Y$ when X is Hausdorff and P, f are continuous. *Book:* Definition 2.6.7 (preprint 2.53), p. 69 (preprint p. 53). *Hidden hypotheses:* none.

V2.61 (Proposition 2.6.8 (preprint 2.54); `Polyfold.proposition_2_6_8`, `Polyfold.pullbackAtlas`, `Polyfold.PullbackChartData`, `Polyfold.PullbackChartData.toChart`, `Polyfold.PullbackChartData.transition_eqOn`, `Polyfold.PullbackChartData.transition_isScSmooth`, `Polyfold.PullbackChartData.transitionOne_isScSmooth`, `Polyfold.PullbackChartData.compatible_toChart`, `Polyfold.paracompactSpace_pullbackSpace`, `Polyfold.pullbackFiberCLE`, `Polyfold.pullbackRetraction`, `Polyfold.isStrongBundleRetraction_pullback`, `Polyfold.pullback_image`, `Polyfold.isStrongBundleRetract_pullback`). Let $P : Y \to X$ be a strong bundle and $f : Z \to X$ sc-smooth. Then f^*Y is paracompact, and $P_f : f^*Y \to Z$ is a strong bundle whose atlas consists of the pulled-back charts: for a strong bundle chart Φ over φ and a chart ψ of Z with $f(\mathrm{dom}\,\psi) \subseteq \mathrm{dom}\,\varphi$ and $g = \varphi \circ f \circ \psi^{-1}$, the chart $\Phi_f(z, y) = (\psi(z), \mathrm{pr}_F \Phi(y))$ onto $K_g = \{(w, h) : w \in O_Z, (g(w), h) \in K\}$, the image of the strong bundle retraction $R_g(w, h) = (r_Z(w), \Gamma(g(r_Z(w)), h))$. Its fibres are $P_f^{-1}(z) \cong P^{-1}(f(z))$ (a linear homeomorphism). Transition maps of pulled-back charts are $(w, h) \mapsto (\psi_2\psi_1^{-1}(w), \mathrm{pr}_F(\Phi_2\Phi_1^{-1})(g_1(w), h))$

and are sc-smooth on $[0]$ and $[1]$. *Book:* Proposition 2.6.8 (preprint 2.54), p. 69 (preprint p. 53) ("the easy proof is left to the reader"). *Hidden hypotheses:* f^*Y paracompact (required by Definition 2.6.5 (preprint 2.51), not mentioned); proved from Theorem 2.3.10 (preprint 2.16) for Z and $Y[0]$. *Deviation (hypotheses explicit):* "natural structure" is the atlas of all pulled-back charts. That the projection $P' : f^*Y \to Y$ is a strong bundle map (sc-smooth on $[0]$ and $[1]$), which "natural" presumably also says, is not formalized (Findings, Chapter 2, item 51).

V2.62 (Theorem 2.7.1 (preprint 2.55), Smirnov's theorem; `Polyfold.metrizableSpace_of_locally`, `Polyfold.metrizableSpace_of_partition`, `Polyfold.metrizableSpace_of_pullback`, `Polyfold.cone_triangle`, `Polyfold.truncDist`, `Polyfold.paracompactSpace_of_metrizableSpace`). A paracompact Hausdorff space in which every point has a metrizable open neighbourhood is metrizable. More generally, a space Y with open metrizable subspaces V_i and a locally finite family of continuous $\rho_i \geq 0$ with $\operatorname{supp} \rho_i \subseteq V_i$, not all vanishing at any point, is metrizable, with the metric $\sum_i |\rho_i(x) - \rho_i(y)| + \min(\rho_i(x), \rho_i(y)) \min(d_i(x, y), 1)$; and if $\pi : Y \to X$ is continuous into a paracompact Hausdorff X whose points have open neighbourhoods U with $\pi^{-1}(U)$ metrizable, Y is metrizable. Metrizable spaces are paracompact. *Book:* Theorem 2.7.1 (preprint 2.55), p. 72 (preprint p. 56), quoted from Smirnov. *Deviation (stronger):* proved, in the pull-back form used for X_m and TX.

V2.63 (restriction and locality of sc^k maps; `Polyfold.IsSck.of_subset`, `Polyfold.IsScSmooth.of_subset`, `Polyfold.IsSc1.tangentMap_eqOn_of_subset`, `Polyfold.IsSc1.of_locally`, `Polyfold.IsSck.of_locally`, `Polyfold.IsScSmooth.of_locally`, `Polyfold.IsScSmooth.congr`). On sc-domains: an sc^k map restricts to sc^k on every smaller sc-domain, with the same tangent map there; a map that is sc^1 (resp. sc^k, sc-smooth) on $U \cap W$ for an open W around each point of U is sc^1 (resp. sc^k, sc-smooth) on U; sc-smoothness depends only on the values on U. *Book:* used without statement in §§2.1, 2.3 (preprint §2.1–§2.2) (Propositions 2.1.11, 2.3.1 (preprint 2.4, 2.10), Definitions 2.3.4–2.3.8 (preprint 2.13–2.14)). *Deviation (not in the book):* infrastructure the text uses silently.

V2.64 (the space $Y[1]$ and the inclusion $Y[1] \to Y[0]$; `Polyfold.StrongBundle.oneSet`, `Polyfold.StrongBundle.mem_oneSet_iff`, `Polyfold.StrongBundle.One`, `Polyfold.StrongBundle.continuous_oneVal`, `Polyfold.StrongBundle.metrizableSpace_one`, `Polyfold.StrongBundle.paracompactSpace_one`, `Polyfold.StrongBundle.isScSmoothMap_oneVal`). For a strong bundle $P : Y \to X$, $Y_{0,1}$ is the set of points whose fibre component lies in F_1 in every chart of the atlas around them, and one such chart suffices. With the topology generated by the charts $\Phi[1]$ (V2.58), the inclusion $Y[1] \to Y$ is continuous, so $Y[1]$ is Hausdorff; $Y[1]$ is metrizable, hence paracompact; and the inclusion $Y[1] \to Y[0]$ is sc-smooth (in the charts $\Phi[1]$, Φ it is $(u, h) \mapsto (u, \iota h)$). *Book:* not stated; the text takes $Y[1]$ to be an M-polyfold on $Y_{0,1}$ (preprint p. 52). *Hidden hypotheses:* paracompactness of $Y[1]$, required by Definition 2.3.4 (preprint 2.13). *Deviation (not in the book):* proved; metrizability by the pull-back form of Smirnov's theorem over the paracompact base (V2.62), as in Theorem 2.3.10 (preprint 2.16).

V2.65 (Proposition 2.1.13 (preprint 2.6), the 2021 statement; `Polyfold.proposition_2_1_13`). Let $r : U \to U$ and $s : V \to V$ be sc-smooth retractions onto the same retract O, ρ an sc-smooth retraction onto O', and f a map with $f(O) \subseteq O'$. Then (1) $f \circ r$ is sc-smooth on U iff $f \circ s$ is sc-smooth on V; (2) if $f \circ r$ is sc-smooth, $T(f \circ r) = T(f \circ s)$ on $TO = Tr(TU)$; (3) if $f \circ r$ is sc-smooth, $T(f \circ r)$ maps TO into TO'. *Book:* Proposition 2.1.13, p. 27 (preprint 2.6, p. 20); the 2021 statement of (3) has the hypothesis "$f \circ r$ sc-smooth", which the preprint and 2017 statements lack and their proofs use. *Hidden hypotheses:* $f(O) \subseteq O'$, contained in "$f : O \to O'$". *Deviation:* none. The Lean statement of (3) in V2.5 already carried the 2021 hypothesis, so the 2021 change confirms the formalization's reading. *Proof:* the conjunction of `Polyfold.proposition_2_1_13_one` (both ways), `Polyfold.proposition_2_1_13_two` and `Polyfold.proposition_2_1_13_three` (V2.5).

V2.66 (Proposition 2.1.4, complete closed neighbourhoods in a retract; `Polyfold.ScRetraction.exists_isClosed_retract_inter_closedBall`, `Polyfold.IsScRetract.exists_isComplete_closure`, `Polyfold.proposition_2_1_4`). Let O be an sc-smooth retract in E, with the metric $d_O(x, y) = |x - y|_0$ of E_0. Every $x \in O$ has open neighbourhoods $V \subseteq W$ in O with $\mathrm{cl}_O(V) \subseteq W$ and $\mathrm{cl}_O(V)$ complete for d_O; W can be chosen inside any given neighbourhood of x. For a retraction r onto O, $O \cap \bar{B}_\varepsilon(x)$ is closed in E_0 for all small $\varepsilon > 0$. *Book:* Proposition 2.1.4, p. 22, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* W arbitrarily small. *Proof:* with $O \cap \bar{B}_\varepsilon(x)$ in place of the text's V' with $r(V') \subseteq V'$: for $\bar{B}_\varepsilon(x) \cap C \subseteq U$, a limit of points of $O \cap \bar{B}_\varepsilon(x)$ lies in U and is fixed by r; a closed subset of the Banach space E_0 is complete.

V2.67 (Definition 2.1.5, locally complete spaces; `Polyfold.AdmitsCompleteMetricOn`, `Polyfold.admitsCompleteMetricOn_of_isComplete`, `Polyfold.AdmitsCompleteMetricOn.metrizableSpace`, `Polyfold.AdmitsCompleteMetricOn.of_isEmbedding`, `Polyfold.AdmitsCompleteMetricOn.mono`, `Polyfold.isComplete_inter_of_isClosed`, `Polyfold.IsLocallyComplete`, `Polyfold.IsLocallyComplete.exists_subset`, `Polyfold.IsLocallyComplete.regularSpace`, `Polyfold.IsLocallyComplete.t1Space`, `Polyfold.IsLocallyComplete.t2Space`, `Polyfold.isLocallyComplete_of_completeSpace`, `Polyfold.isLocallyComplete_real`, `Polyfold.isLocallyComplete_of_metric`). A topological space X is locally complete if every x has open $V \ni x$ and W with $\mathrm{cl}_X(V) \subseteq W$ such that W carries a metric d inducing its subspace topology for which $\mathrm{cl}_X(V)$ is complete. Then W can be chosen inside any neighbourhood of x, and X is regular, T_1 and Hausdorff. A complete metric space is locally complete ($\mathbb{R}$ in particular), and so is a metric space in which every point has open $V \subseteq W$, $\mathrm{cl}(V) \subseteq W$, with $\mathrm{cl}(V)$ complete. *Book:* Definition 2.1.5, p. 23, new in the 2021 edition, for a Hausdorff space X, with (1) the metric on $W(x)$ and (2) $V(x) \subseteq W(x)$, $\mathrm{cl}_X(V(x)) \subseteq W(x)$, $\mathrm{cl}_X(V(x))$ complete. *Hidden hypotheses:* none. *Deviation (stronger):* the Hausdorff assumption is not part of the definition; it follows (T_1 from the metrizable W, regularity from the global condition $\mathrm{cl}_X(V) \subseteq W$), so V2.70 and V2.71 need no Hausdorff hypothesis except for the conclusion $T_{3\frac{1}{2}}$. $V \subseteq W$ is a consequence, not a separate condition.

V2.68 (Remark 2.1.6; `Polyfold.remark_2_1_6`, `Polyfold.IsLocallyCompleteRel`, `Polyfold.IsLocallyComplete.isLocallyCompleteRel`). If $\mathrm{cl}_X(V) \subseteq W$, the closure of V in the subspace W is $\mathrm{cl}_X(V)$. The modified notion, in which $\mathrm{cl}_X(V) \subseteq W$ is replaced by $V \subseteq W$ and completeness is asked of $\mathrm{cl}_W(V)$, is implied by local completeness. *Book:* Remark 2.1.6, p. 23, new in the 2021 edition: the condition $\mathrm{cl}_X(V(x)) \subseteq W(x)$ is global, and the modified definition is weaker.

Hidden hypotheses: none. *Deviation (narrower):* "weaker" is proved here as the implication; that the modified notion is strictly weaker is V2.91, an example (gap G.76, closed).

V2.69 (Corollary 2.1.7; `Polyfold.corollary_2_1_7`, `Polyfold.isLocallyComplete_univ_retract`). An sc-smooth retract O, with the topology of E_0, is locally complete; so is E_0 itself. *Book:* Corollary 2.1.7, p. 23, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* V2.66, with the metric d_O restricted to W (`Polyfold.isLocallyComplete_of_metric`).

V2.70 (Proposition 2.1.8; `Polyfold.IsLocallyComplete.of_isEmbedding`, `Polyfold.proposition_2_1_8_1`, `Polyfold.proposition_2_1_8_2`, `Polyfold.proposition_2_1_8_3`, `Polyfold.proposition_2_1_8_4`). Let X be locally complete. (1) A locally closed subspace A (open $\cap$ closed) is locally complete; (2) a space homeomorphic to X is locally complete; (3) an open subspace is locally complete. (4) A space Y with an open cover (Y_i) by locally complete subspaces such that every $z \in Y_i$ has an open $U \ni z$, $U \subseteq Y_i$, with $\mathrm{cl}_Y(U) \subseteq Y_i$, is locally complete. More generally, a space embedded in X with locally closed image is locally complete. *Book:* Proposition 2.1.8, p. 24, new in the 2021 edition, for Hausdorff spaces. *Hidden hypotheses:* none. *Deviation (stronger):* no Hausdorff hypothesis (V2.67). "Locally closed" is Mathlib's notion, equivalent to the text's use ("$U \cap A$ closed in U for an open $U \ni a$"). *Proof:* (1)–(3) from the embedding statement; (4) by transporting the metric of $W \subseteq Y_i$ to $W \subseteq Y$.

V2.71 (Lemma 2.1.9 and Proposition 2.1.10; `Polyfold.lemma_2_1_9`, `Polyfold.IsLocallyComplete.completelyRegularSpace`, `Polyfold.proposition_2_1_10`). A locally complete space is first countable and completely regular; if it is Hausdorff it is $T_{3\frac{1}{2}}$ (Tychonoff). *Book:* Lemma 2.1.9 (countable neighbourhood bases) and Proposition 2.1.10 (Hausdorff locally complete spaces are completely regular), p. 25, both new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* no Hausdorff hypothesis for complete regularity. *Proof:* without sequences: the neighbourhood filter of x is the image of that of the metrizable open W, and the bump function is built in W and extended by a constant outside $\mathrm{cl}_X(V)$. Mathlib's complete regularity separates with $f(x) = 0$ and $f = 1$ on the closed set (the text's β is $1 - f$).

V2.72 (Lemma 2.2.1, the levels of a connected relatively open set; `Polyfold.ScBanach.lvlSetInfty`, `Polyfold.ScBanach.smoothSeq`, `Polyfold.ScBanach.toBase_smoothSeq`, `Polyfold.ScBanach.smoothSeq_mem`, `Polyfold.ScBanach.smoothSeq_zero`, `Polyfold.ScBanach.image_lvlSetInfty`, `Polyfold.ScBanach.joinedIn_lvlSetInfty_of_segment`, `Polyfold.IsPartialQuadrant.exists_smooth_near`, `Polyfold.IsRelOpenInQuadrant.exists_ball_inter_subset`, `Polyfold.lemma_2_2_1`, `Polyfold.lemma_2_2_1_infty`). Let U be relatively open in a partial quadrant C and connected. For every $k \in \mathbb{N}$ the level $U_k = U \cap E_k$, with the topology of E_k, is path-connected; and U_∞, represented as the compatible sequences $(v_m) \in \prod_m E_m$ with $v_0 \in U$ (with the product topology, i.e. the topology of all the norms $|\cdot|_m$), is path-connected. Smooth points are dense in every level C_m, and the image of this U_∞ in E_0 is $U \cap E_\infty$. *Book:* Lemma 2.2.1, p. 30, new in the 2021 edition, for $k \in \mathbb{N} \cup \{\infty\}$. *Hidden hypotheses:* the topology of E_∞, which the text does not specify. *Deviation (explicit):* U_∞ carries the inverse-limit topology. *Proof:* not the text's polygon along a path σ in U_0: the set of points of U near which every point of the level is joined to a fixed one is open and closed in the connected

U; segments in convex pieces $B_\varepsilon \cap C$ join smooth points, and smooth points are dense.

V2.73 (Lemma 2.2.2, Definition 2.2.3 and Theorem 2.2.4; `Polyfold.lemma_2_2_2`, `Polyfold.definition_2_2_3`, `Polyfold.theorem_2_2_4`, `Polyfold.IsScRetract.locallyPathConnectedSpace`). For a retraction $r : U \to U$ onto O, $x \in O$ and $V = V' \cap U$ with $V' \ni x$ open in E_0, the set $W = V \cap r^{-1}(V)$ is relatively open in C, contains x, and satisfies $W \cap O = V \cap O$, $r(W) \subseteq W$, $r(W) = O \cap V$. Mathlib's local path-connectedness is the text's (every open neighbourhood of z contains an open path-connected neighbourhood). A retract is locally path-connected. *Book:* Lemma 2.2.2, p. 31, Definition 2.2.3 and Theorem 2.2.4, p. 32, all new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* Theorem 2.2.4 was already proved for Chapter 7 (`Polyfold.IsScRetract.locallyPathConnectedSpace`), joining nearby points by $\tau \mapsto r(o + \tau(p - o))$ instead of through Lemma 2.2.2 and the components of $W(x)$.

V2.74 (Definition 2.2.5 and Proposition 2.2.6; `Polyfold.IsAlmostLocallyContractible`, `Polyfold.proposition_2_2_6`, `Polyfold.isAlmostLocallyContractible_univ`). A space Z is almost locally contractible if for every z and open $V \ni z$ there are an open W, $z \in W \subseteq V$, and a continuous $\Phi : [0,1] \times W \to Z$ with values in V, $\Phi(0,y) = y$, $\Phi(t,z) = z$ and $\Phi(1,y) = z$. An sc-smooth retract is almost locally contractible; so is E_0. *Book:* Definition 2.2.5 and Proposition 2.2.6, p. 33, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* the text's $\Phi(t,y) = r((1-t)y + tx)$ on $W = B_\varepsilon(x) \cap O$.

V2.75 (the footnote to Definition 2.3.4 (preprint 2.13), paracompactness, and the convention of Appendix 2.7.2; `Polyfold.IsLocallyFiniteSetFamily`, `Polyfold.IsParacompactStar`, `Polyfold.isParacompactStar_iff`, `Polyfold.definition_2_3_4`, `Polyfold.isParacompactStar_real`, `Polyfold.IsParacompactBySubcover`, `Polyfold.not_isParacompactBySubcover_real`). A space is paracompact* if every open cover has a locally finite open cover refining it, with no separation axiom; this is Mathlib's paracompactness, so "Hausdorff paracompact*" *is exactly the hypothesis of the M-polyfold structure (V2.12); $\mathbb{R}$ is paracompact*. Under the reading"every open cover has a locally finite open **sub**cover", $\mathbb{R}$ is not paracompact (the cover by the intervals $(-n-1, n+1)$). *Book:* Definition 2.3.4, p. 35 (preprint 2.13, p. 23), with the footnote, new in the 2021 edition, that paracompactness does not include the Hausdorff property; Appendix 2.7.2, p. 72 (and 2017 Appendix 2.6.2, p. 51), states the convention with a locally finite open subcovering. *Hidden hypotheses:* none. *Deviation (restated):* the footnote changes nothing for M-polyfolds, whose Hausdorff property is required separately. The subcover wording of Appendix 2.7.2 does not hold as we read it for $\mathbb{R}$; it is read as "refinement" (possible misprint), as in the footnote.

V2.76 (Lemma 2.3.5 and Lemma 2.7.2; `Polyfold.lemma_2_7_2`, `Polyfold.exists_isOpen_closure_subset_of_paracompact`, `Polyfold.lemma_2_3_5`). In a regular space every open $U \ni x$ contains an open $V \ni x$ with $\mathrm{cl}(V) \subseteq U$. In a Hausdorff paracompact space, and so in an M-polyfold, every open $W \ni x$ contains such a V inside any given neighbourhood of x. *Book:* Lemma 2.3.5, p. 35, and Lemma 2.7.2, p. 72, new in the 2021 edition; Lemma 2.7.2 is stated for regular Hausdorff spaces. *Hidden hypotheses:* none. *Deviation (stronger):* Lemma 2.7.2 needs only regularity. *Proof:* Mathlib (regular spaces; Hausdorff paracompact spaces are normal).

V2.77 (Propositions 2.3.6 and 2.3.7; `Polyfold.MChart.isLocallyComplete_source`, `Polyfold.MPolyfold.isLocallyComplete`, `Polyfold.proposition_2_3_6`, `Polyfold.proposition_2_3_7`, `Polyfold.MPolyfold.locallyPathConnectedSpace`). A chart domain of an M-polyfold is locally complete. An M-polyfold X is locally complete, with $W(x)$ inside any given neighbourhood of x: there are open $V \subseteq W$ around x with $\mathrm{cl}_X(V) \subseteq W$ and a metric on W inducing its topology for which $\mathrm{cl}_X(V)$ is complete. An M-polyfold is locally path-connected. *Book:* Propositions 2.3.6 and 2.3.7, p. 36, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* Proposition 2.3.6 by the open-cover part of V2.70, the cover by chart domains ($V \cong O$, V2.69 and V2.70 (2)) with condition (b) from V2.76, instead of the text's direct argument in a chart; Proposition 2.3.7 was already proved for Chapter 7 (`Polyfold.MPolyfold.locallyPathConnectedSpace`).

V2.78 (Definitions 2.5.18 and 2.5.19, tame sub-M-polyfolds; `Polyfold.MPolyfold.HasTameInducedStructure`, `Polyfold.MPolyfold.HasReducedComplementAt`, `Polyfold.MPolyfold.hasReducedComplementAt_self`, `Polyfold.MPolyfold.TameSubData`, `Polyfold.MPolyfold.IsTameSubMPolyfold`, `Polyfold.MPolyfold.IsTameSubMPolyfold.isSubMPolyfold`, `Polyfold.MPolyfold.hasTameInducedStructure_univ`, `Polyfold.MPolyfold.isTameSubMPolyfold_univ`). For a sub-M-polyfold Y of X, the induced structure is tame if the M-polyfold Y (V2.24) is tame. For a tame X, Y is a tame sub-M-polyfold if every $y \in Y$ lies in an open V with an sc-smooth retraction r of the sub-M-polyfold data ($r(V) = Y \cap V$) such that $d_X(r(x)) = d_X(x)$ on V and, at every smooth $y \in Y \cap V$, in some chart around y with a presentation of the partial quadrant, the tangent space of the chart image Q of Y has an sc-complement A in T_zO inside the reduced tangent space $T_zO \cap E_z$ ($(T_zO)_m = (T_zQ)_m \oplus A_m$ on every level). A tame sub-M-polyfold is a sub-M-polyfold; in a tame X, $Y = X$ is a tame sub-M-polyfold with tame induced structure. *Book:* Definitions 2.5.18, p. 62, and 2.5.19, p. 63, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* condition (3) is stated in a chart, for some chart around y (T_yX has no linear structure here, gap G.7); its independence of the chart is V2.99, the remark that a tame sub-M-polyfold is tame V2.100, and the examples in $[0, \infty)^2 \times \mathbb{R}$ V2.101 (gap G.77, closed).

V2.79 (faces of open sets and the corner axioms; `Polyfold.TameBoundary.faceAt`, `Polyfold.TameBoundary.IsFaceIn`, `Polyfold.TameBoundary.facesThrough`, `Polyfold.TameBoundary.LocalPieces`, `Polyfold.TameBoundary.IsCornerIndex`, `Polyfold.TameBoundary.faceAt_inter_eq`, `Polyfold.TameBoundary.faceExt`, `Polyfold.TameBoundary.faceExt_unique`, `Polyfold.TameBoundary.exists_faceExt_eq`, `Polyfold.TameBoundary.isCornerIndex_zero`, `Polyfold.MPolyfold.exists_localPieces_chart`, `Polyfold.MPolyfold.isCornerIndex_degIndex`, `Polyfold.MPolyfold.isFaceIn_univ_iff`). For a map $d : X \to \mathbb{N}$ on a space X and an open V, a face of V is the closure in V of a connected component of $V \cap \{d = 1\}$; the faces of X itself are those of V2.44. The corner axioms ask that d be upper semicontinuous and that every x have an open D such that every $y \in D$ has, in arbitrarily small neighbourhoods, $d(y)$ preconnected pieces of $\{d = 1\}$ accumulating at y and covering $\{d = 1\}$ near y, no preconnected subset of $D \cap \{d = 1\}$ meeting two of them. For open $U \subseteq V$, a face θ of U lies in a unique face θ' of V, the one containing it. The degeneracy index of a tame M-polyfold satisfies the corner axioms; so does $d \equiv 0$. *Book:* not stated; §2.7.4, pp. 78–82, new in the 2021 edition, uses the faces of open subsets V (Definition 2.5.10 for V with its induced structure, whose degeneracy index is $d_X|V$ by Proposition 2.4.4) and the local structure of a tame chart. *Hidden hypotheses:* none. *Deviation (not in the book):* all of §2.7.4 is proved from the corner axioms, so that it applies to the object spaces of tame ep-groupoids (2021 §7.6.5); the local pieces in a tame chart come from the local face structure of V2.45.

V2.80 (Proposition 2.7.5; `Polyfold.TameBoundary.exists_nbhd_faces_mem`, `Polyfold.TameBoundary.proposition_2_7_5`, `Polyfold.MPolyfold.proposition_2_7_5`). In a tame M-polyfold (more generally, for d with the corner axioms) every x has an open $V \ni x$ with (1) $d(y) \leq d(x)$ on V, (2) every $y \in V$ in exactly $d(y)$ faces of V, (3) an open $U(x)$, $x \in U(x) \subseteq V$, such that every face of V meeting $U(x)$ contains x; (3) holds for every open $V \ni x$. *Book:* Proposition 2.7.5, p. 78, new in the 2021 edition. *Hidden hypotheses:* none.

V2.81 (Definition 2.7.6 and Proposition 2.7.7; `Polyfold.TameBoundary.IsFaceStructuredOpen`, `Polyfold.TameBoundary.faceStructuredOpens`, `Polyfold.TameBoundary.faceStructuredNbhds`, `Polyfold.TameBoundary.faceStructuredNbhdsAt`, `Polyfold.TameBoundary.faceStructuredNbhds_eq_iUnion`, `Polyfold.TameBoundary.faceStructuredNbhdsAt_nonempty`, `Polyfold.TameBoundary.sUnion_faceStructuredNbhds`, `Polyfold.TameBoundary.mem_faceStructuredNbhds_of_subset`, `Polyfold.TameBoundary.inter_mem_faceStructuredNbhds`, `Polyfold.TameBoundary.inter_mem_faceStructuredNbhdsAt`, `Polyfold.MPolyfold.proposition_2_7_7`, `Polyfold.TameBoundaryExample.isFaceStructuredOpen_quadrant`). An open V is face-structured if every $y \in V$ lies in exactly $d(y)$ faces of V; N_X is the set of nonempty face-structured V with $d|V$ bounded, $N_{X,x}$ the set of $V \in N_X$ with $x \in V$ and $d \leq d(x)$ on V, and $N_X = \bigcup_x N_{X,x}$. For a tame X: every $N_{X,x}$ is nonempty (so N_X covers X); a nonempty open subset of a member of N_X is in N_X; nonempty intersections of members of N_X are in N_X, and $N_{X,x}$ is closed under intersections. Every open subset of $[0, \infty)^n \oplus W$ is face-structured. *Book:* Definition 2.7.6, p. 79, and Proposition 2.7.7, p. 80, new in the 2021 edition. *Hidden hypotheses:* none.

V2.82 (Definitions 2.7.8 and 2.7.9, the category L_X and the boundary structure functor; `Polyfold.TameBoundary.IsMor`, `Polyfold.TameBoundary.IsMor.trans`, `Polyfold.TameBoundary.LObj`, `Polyfold.TameBoundary.boundaryFunctor`, `Polyfold.TameBoundary.boundaryFunctor_map_injective`, `Polyfold.TameBoundary.boundaryFunctor_map_bijective`, `Polyfold.MPolyfold.LocalBoundaryCategory`, `Polyfold.MPolyfold.boundaryStructureFunctor`). The objects of L_X are the pairs (x, V) with $d(x) \geq 1$ and $V \in N_{X,x}$; there is a morphism $(x, V) \to (y, W)$, and then only one, iff $V \subseteq W$, x lies in $d(x)$ faces of W and each of them contains y; morphisms compose. The functor F_X sends (x, V) to the set of faces of V through x and a morphism to $\theta \mapsto \theta'$, the face of W containing θ; $F_X(\varphi)$ is injective, and bijective for a morphism $(x, V) \to (x, W)$. *Book:* Definitions 2.7.8, p. 80, and 2.7.9, p. 81, new in the 2021 edition, with the text's argument for the composition and the injection (2.43). *Hidden hypotheses:* none. *Deviation (restated):* L_X is a preorder (a thin category) and F_X takes values in types; θ' is defined as the union of the faces of W containing θ, which is the text's "the face of W through x and y with $\theta \cap \theta'$ open in θ" (`Polyfold.TameBoundary.faceExt_unique`).

V2.83 (Definition 2.7.10, the set bX and the map B; `Polyfold.TameBoundary.BTriple`, `Polyfold.TameBoundary.BRel`, `Polyfold.TameBoundary.BRel.refl`, `Polyfold.TameBoundary.BRel.symm`, `Polyfold.TameBoundary.BRel.trans`, `Polyfold.TameBoundary.bSetoid`, `Polyfold.TameBoundary.BSpace`, `Polyfold.TameBoundary.bMap`, `Polyfold.TameBoundary.range_bMap`, `Polyfold.TameBoundary.encard_bMap_preimage`, `Polyfold.MPolyfold.TameBoundarySpace`, `Polyfold.MPolyfold.tameBoundaryMap`, `Polyfold.MPolyfold.range_tameBoundaryMap`, `Polyfold.MPolyfold.encard_tameBoundaryMap_preimage`). An admissible triple is (x, V, θ) with (x, V) an object of L_X and θ a face of V through x; $(x, V, \theta) \sim (x', V', \theta')$ iff $x = x'$ and some $V'' \in N_{X,x}$, $V'' \subseteq V \cap V'$, has a face θ'' through x sent to θ and to θ'. This is an equivalence relation; bX is the set of classes and $B[x, V, \theta] = x$. The image of B is $\partial X = \{d \geq 1\}$, and $B^{-1}(x)$ has exactly $d(x)$ points. *Book:* Definition 2.7.10, p. 81, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the fibre count, the set-level content of Theorem 2.7.13 (4).

V2.84 (Definition 2.7.11 and (2.44), the sets $U(x, V, \theta)$ and the maps Φ; `Polyfold.TameBoundary.uSet`, `Polyfold.TameBoundary.phi`, `Polyfold.TameBoundary.phi_injective`, `Polyfold.TameBoundary.reach`, `Polyfold.TameBoundary.IsCenteredAt`, `Polyfold.TameBoundary.isOpen_reach`, `Polyfold.TameBoundary.bMap_image_uSet`, `Polyfold.TameBoundary.phi_bijective_of_isCenteredAt`, `Polyfold.TameBoundary.isCenteredAt_reach`, `Polyfold.TameBoundary.reach_mem_faceStructuredNbhdsAt`). $U(x, V, \theta)$ is the set of classes with a representative (y, V', θ') such that $y \in \theta$, there is a morphism $(y, V') \to (x, V)$, and it sends θ' to θ; $\Phi_{(x,V,\theta)} : U(x, V, \theta) \to \theta$ is $[y, V', \theta'] \mapsto y$. Φ is injective, and its image is $\theta \cap V^x$, where $V^x = \{y \in V :$ every face of V through y contains $x\}$ is open. If every face of V contains x (V centred at x), Φ is a bijection onto θ; V^x is centred at x, and $V^x \in N_{X,x}$ for $V \in N_{X,x}$. *Book:* Definition 2.7.11 and (2.44), p. 82, new in the 2021 edition: $\Phi_{(x,V,\theta)}$ is a canonical bijection onto θ. *Hidden hypotheses:* that V be small enough; the text says, for the topology of bX (p. 85), that the main point is to take V sufficiently small. *Deviation (narrower):* the bijection is proved for V centred at x; for every $V \in N_{X,x}$ it does not hold as we read it (V2.85). The centred neighbourhoods are cofinal in $N_{X,x}$, which is what "V sufficiently small" provides.

V2.85 ((2.44) for every $V \in N_{X,x}$: an example in $[0, \infty)^3$; `Polyfold.TameBoundaryExample.dq`, `Polyfold.TameBoundaryExample.isCornerIndex_dq`, `Polyfold.TameBoundaryExample.dq_eq`, `Polyfold.TameBoundaryExample.part`, `Polyfold.TameBoundaryExample.dq_eq_one_iff`, `Polyfold.TameBoundaryExample.subset_part`, `Polyfold.TameBoundaryExample.faceAt_subset_coord_eq_zero`, `Polyfold.TameBoundaryExample.exists_part_mem_faceAt`, `Polyfold.TameBoundaryExample.slab`, `Polyfold.TameBoundaryExample.slab_mem_faceStructuredNbhdsAt`, `Polyfold.TameBoundaryExample.slabFace`, `Polyfold.TameBoundaryExample.counterTriple`, `Polyfold.TameBoundaryExample.exists_face_not_mem`, `Polyfold.TameBoundaryExample.pt_010_not_mem_image`, `Polyfold.TameBoundaryExample.phi_not_surjective`, `Polyfold.TameBoundaryExample.not_forall_bijective_phi`). In the tame M-polyfold $X = [0, \infty)^3 \oplus W$ (W any sc-Banach space), let $x = (0, 0, 1)$, $V = \{p_0 < 1/2,\ 1/2 < p_1 + p_2 < 3/2\}$ (a convex open slab around the segment from x to $(0, 1, 0)$), and θ the face of V through that segment. Then $V \in N_{X,x}$, $d_X(x) = 2$, (x, V, θ) is an admissible triple, $(0, 1, 0) \in \theta$ also lies on the face of V in $\{p_2 = 0\}$, which misses x, and so $(0,1,0)$ is not in the image of $\Phi_{(x,V,\theta)}$: Φ is not onto θ, and not every $\Phi_{(x,V,\theta)}$ is a bijection. *Book:* Definition 2.7.11, (2.44), p. 82, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (not in the book):* the example shows that (2.44) does not hold as we read it for every $V \in N_{X,x}$, and that some restriction such as "V centred at x" (V2.84) is needed. *Proof:* $d_X(p)$ is the number of vanishing half-line coordinates, a face of an open V through a point with only $p_k = 0$ lies in $\{p_k = 0\}$.

V2.86 (Lemma 2.7.15, the topology of bX; `Polyfold.TameBoundary.IsBOpen`, `Polyfold.TameBoundary.BTriple.exists_shrink`, `Polyfold.TameBoundary.BRel.exists_common`, `Polyfold.TameBoundary.eq_of_mem_uSet`, `Polyfold.TameBoundary.isBOpen_iff`, `Polyfold.TameBoundary.bTopology`, `Polyfold.TameBoundary.isOpen_iff_isBOpen`, `Polyfold.TameBoundary.isOpen_iff_forall_uSet`, `Polyfold.TameBoundary.lemma_2_7_15`, `Polyfold.MPolyfold.lemma_2_7_15`). A set $\mathcal{U} \subseteq bX$ is in T_{bX} if every class $p \in \mathcal{U}$ has a representative (x,V,θ) and a set $O \ni p$, $O \subseteq \mathcal{U} \cap U(x,V,\theta)$, with $\Phi_{(x,V,\theta)}(O)$ open in θ; equivalently, if every $p \in \mathcal{U}$ has a representative with $U(x,V,\theta) \subseteq \mathcal{U}$. These sets are the open sets of a topology. *Book:* the description of T_{bX} in §2.7.4.3 and Lemma 2.7.15, p. 85, new in the 2021 edition. *Hidden hypotheses:* $p \in O$, without which $O = \emptyset$ would make every set open. *Deviation (explicit):* the first description is read with O containing the class. *Proof:* from the second description and a common refinement of equivalent triples, without the sc-smoothness of the transition maps that the text's proof of the intersection axiom uses.

V2.87 (Proposition 2.7.12, topological part; `Polyfold.TameBoundary.isOpen_uSet`, `Polyfold.TameBoundary.isTopologicalBasis_uSet`, `Polyfold.TameBoundary.continuous_bMap`, `Polyfold.TameBoundary.exists_nbhd_faceExt_inter_subset`, `Polyfold.TameBoundary.exists_isOpen_bMap_image`, `Polyfold.TameBoundary.isOpenEmbedding_phi`, `Polyfold.TameBoundary.phiHomeomorph`, `Polyfold.TameBoundary.proposition_2_7_12`, `Polyfold.MPolyfold.proposition_2_7_12`). The sets $U(x,V,\theta)$ are open and form a basis of T_{bX}; B is continuous; every $\Phi_{(x,V,\theta)}$ is an open embedding into θ, and a homeomorphism onto θ for V centred at x. For two triples, the image of $U(x,V,\theta) \cap U(x',V',\theta')$ is open in θ and in θ', and the transition map $\Phi_{(x',V',\theta')} \circ \Phi^{-1}_{(x,V,\theta)}$ is the identity on underlying points of X. *Book:* Proposition 2.7.12, p. 83, new in the 2021 edition: the images are open and the transition map is an sc-diffeomorphism. *Hidden hypotheses:* none. *Deviation (narrower):* the sc-diffeomorphism statement is not part of this item: for representatives with V centred at x it is proved with the M-polyfold structure of bX (V2.103), and for every representative in V2.104 (gap G.78, closed).

V2.88 (Proposition 2.7.16; `Polyfold.TameBoundary.tripleOf`, `Polyfold.TameBoundary.exists_mem_uSet_tripleOf`, `Polyfold.TameBoundary.metrizableSpace_preimage`, `Polyfold.TameBoundary.proposition_2_7_16`, `Polyfold.MPolyfold.proposition_2_7_16`). For d with the corner axioms on a Hausdorff X, bX is Hausdorff; on a metrizable X, bX is metrizable. For a tame M-polyfold, T_{bX} is Hausdorff and metrizable. *Book:* Proposition 2.7.16, pp. 85–87, new in the 2021 edition. *Hidden hypotheses:* none. *Proof:* not the text's paracompactness argument (a locally finite refinement built from one of the cover (V_x) of ∂X): the pull-back form of Smirnov's theorem (V2.62) along B, each point having an open neighbourhood $B^{-1}(W) \cap \bigcup_\theta U(x,W,\theta)$

embedded into the metrizable $X \times \mathbb{R}^{\text{faces}}$ by B and the indicators of the clopen sets $U(x, W, \theta)$.

V2.89 (Theorem 2.7.13, topological part, and Definition 2.7.14; `Polyfold.TameBoundary.eq_of_isOpen_uSet`, `Polyfold.TameBoundary.theorem_2_7_13_topology`, `Polyfold.MPolyfold.theorem_2_7_13`, `Polyfold.MPolyfold.tameBoundary`). For a tame M-polyfold X: (1) bX carries the metrizable topology T_{bX}; every $U(x, V, \theta)$ is open and every $\Phi_{(x,V,\theta)}$ is an open embedding into θ; T_{bX} is the only topology in which every $U(x, V, \theta)$ is open and every Φ maps open sets to relatively open subsets of θ; $B : bX \to X$ is continuous, with image ∂X, and every x has exactly $d_X(x)$ preimages. The tame boundary is B as a continuous map. *Book:* Theorem 2.7.13, pp. 83–84, and Definition 2.7.14, p. 84, new in the 2021 edition: (1) the metrizable topology, (2) a natural tame M-polyfold structure on bX, (3) B sc-smooth, (4) $d_X(B(p)) - 1 = d_{bX}(p)$, characterized by: every $U(x, V, \theta)$ open and every Φ a homeomorphism and an sc-diffeomorphism onto θ. *Hidden hypotheses:* none. *Deviation (narrower):* this item is (1) and the topological half of the characterization; "homeomorphism onto θ" is proved for V centred at x (V2.84, V2.87), the open embedding for every V. Parts (2)–(4) beyond the fibre count are V2.102 and V2.106; the sc-diffeomorphism half of the characterization is V2.103 for centred representatives and V2.104 for every representative, with image $\theta \cap V^x$ (V2.105) (gap G.78, closed).

V2.90 (the tame quadrant as a witness; `Polyfold.MPolyfold.quadrantPolyfold`, `Polyfold.MPolyfold.quadrantPolyfold_isTame`, `Polyfold.MPolyfold.quadrantPolyfold_encard_preimage`, `Polyfold.MPolyfold.quadrantPolyfold_tameBoundarySpace_nonempty`, `Polyfold.MPolyfold.quadrantPolyfold_univ`). The quadrant $[0, \infty)^n \oplus W$ is a tame M-polyfold; its corner 0 has exactly n preimages under B, so $bX \neq \emptyset$ for $n \geq 1$; and $Y = X$ has a tame induced structure and is a tame sub-M-polyfold. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book):* satisfiability of the hypotheses of V2.78–V2.89.

V2.91 (Remark 2.1.6, strictness: the K-topology on $\mathbb{R}$; `Polyfold.remark_2_1_6_strict`, `Polyfold.isLocallyCompleteRel_of_forall_isCompletelyMetrizable`, `Polyfold.isCompletelyMetrizableSpace_of_isEmbedding_compl_range`, `Polyfold.kSeq`, `Polyfold.kTop`, `Polyfold.RealKTop`, `Polyfold.RealKTop.not_regularSpace`, `Polyfold.RealKTop.not_isLocallyComplete`, `Polyfold.RealKTop.isLocallyCompleteRel`). A space in which every point has an open, completely metrizable neighbourhood satisfies the modified definition of Remark 2.1.6 (with $V = W$). A space embedded in $\mathbb{R}$ onto the complement of a sequence is completely metrizable. Let $\mathbb{R}_K$ be $\mathbb{R}$ with the K-topology, generated by the usual open sets and $\mathbb{R} \setminus K$, $K = \{1/(n+1) : n \in \mathbb{N}\}$. It is Hausdorff; it satisfies the modified definition, the points outside K lying in $\mathbb{R} \setminus K$ and those of K in $\mathbb{R} \setminus \{0\}$, both with the usual subspace topology; and it is not regular, so not locally complete (V2.67). So some Hausdorff space satisfies the modified definition and not Definition 2.1.5. *Book:* Remark 2.1.6, p. 23, new in the 2021 edition: the modified definition would be weaker. *Hidden hypotheses:* none. *Deviation (not in the book):* the example; with V2.68 the remark holds as we read it, and "weaker" is strict (gap G.76, closed). *Proof:* every closed neighbourhood of 0 in $\mathbb{R}_K$ meets K, through irrational points near $1/(n+1)$. The complement of a sequence (a_n) embeds as a closed set into $\mathbb{R} \times \mathbb{R}^{\mathbb{N}}$ by $y \mapsto (y, ((y - a_n)^{-1})_n)$, in place of the theorem that G_δ subsets of complete metric spaces are completely metrizable, which Mathlib lacks.

V2.92 (Definition 2.1.16 (preprint 2.9), bundle retractions and bundle maps; `Polyfold.ScRetraction.IsBundleRetractionOver`, `Polyfold.ScRetraction.tangentRetraction_isBundleRetractionOver`, `Polyfold.IsBundleMapOn`, `Polyfold.IsBundleMapOn.fst_eq`, `Polyfold.IsScSmoothRetractMap.retractTangentMap_isBundleMapOn`). A retraction R of $E \oplus F$ (V2.1) is a bundle retraction covering the retraction r of E if its domain is $U \oplus F$, U the domain of r, and $R(u,v) = (r(u), \rho(u)v)$ there, with $\rho(u)$ a continuous linear map of F_0. A map Φ from $E \oplus F$ to $E' \oplus F'$ is a bundle map on B covering a if $\Phi(u,v) = (a(u), \phi(u)v)$ on B with $\phi(u)$ continuous linear; then $p' \circ \Phi = a \circ p$ on B. The tangent retraction Tr is a bundle retraction covering the level-1 retraction r^1, and for an sc-smooth f between retracts Tf is a bundle map on TO covering the level-1 map f^1. *Book:* Definition 2.1.16, p. 29 (preprint 2.9, p. 22). *Hidden hypotheses:* none. *Deviation (restated):* the definition fixes the shape of the maps; sc-smoothness and $R \circ R = R$ are part of the data (a retraction of V2.1, an sc-smooth map of V2.6), and $\rho(u)$ is continuous on level 0, which the text's linear $\rho(u, \cdot)$ is as a restriction of the continuous R. "$p' \circ \Phi = p$" is read as $p' \circ \Phi = a \circ p$ (possible misprint; Findings, Chapter 2, item 76). The categories $\mathcal{R}$, $\mathcal{BR}$ and the forgetful functor are not formalized; they name these objects and maps and state nothing further (gap G.5, formalized since apart from them).

V2.93 (the levels of X^1, gap G.6; `Polyfold.MPolyfold.levelLift_transition`, `Polyfold.MPolyfold.levelLift_levelChart`, `Polyfold.MPolyfold.shiftLevelMap`, `Polyfold.MPolyfold.shiftLevelInv`, `Polyfold.MPolyfold.continuous_shiftLevelMap`, `Polyfold.MPolyfold.continuous_shiftLevelInv`, `Polyfold.MPolyfold.shiftLevelHomeomorph`, `Polyfold.MPolyfold.shift_level_eq`). For every m, the identity of the underlying points of X is a homeomorphism $(X^1)_m \cong X_{m+1}$; in a chart φ of X, the level-m lift of the level chart φ^1 of X^1 is the level-$(m+1)$ lift of φ ($(E^1)_m = E_{m+1}$), and level lifts in two charts differ by the level map of the transition map. So the iterate $X^m = (X^1)^1 \cdots$ is, for each m, the space X_m with its topology and the charts $(V_m, \varphi, (O_m, C_m, E_m))$, by composing these homeomorphisms. *Book:* the paragraph after Definition 2.3.8, p. 37 (preprint after Definition 2.14, p. 25): these charts form an atlas of X_m, which with it is the M-polyfold X^m. *Hidden hypotheses:* none. *Deviation (restated):* X^m is the iterate, identified with X_m for each fixed m; one homeomorphism for the family indexed by m is not stated. This closes gap G.6 (V2.16).

V2.94 (Theorem 2.3.16 as a bundle statement; `Polyfold.theorem_2_3_16`, `Polyfold.tangentMap_levelMap_one`, `Polyfold.toBase_tangentMap_levelMap_one`, `Polyfold.retractTangent_shift_of_isScRetract`, `Polyfold.toBase_retractTangentMap_shift`). In a local model, for an sc¹ map f on U, the tangent map of $f^1 : U^1 \to F^1$ is the level-1 map of Tf on $T(U^1) = (TU)^1$; $T(O^1) = (TO)^1$ for a retract O; and $T(f^1)$ is the level-1 map of Tf for an sc-smooth map between retracts. For an M-polyfold X: the tangent charts $T\varphi$ belong to the structure of TX; $p : TX \to X^1$ is sc-smooth; in a tangent chart, p is the first projection onto the level chart φ^1 of X^1; for every retraction r onto a model O, the model (TO, TC, TE) is a bundle retract (Tr covers r^1, V2.92); and the transition map of two tangent charts is a bundle map on $T(\varphi(V \cap V'))$ covering the level-1 map of the transition map of the charts. *Book:* Theorem

2.3.16, p. 40 (2017: running text after Proposition 2.5; preprint: after Proposition 2.19, p. 27): the construction $X \mapsto (p : TX \to X^1)$; the text before it says that p is locally built on the bundle retractions $TO \to O^1$ and that the transition maps are sc-smooth bundle maps. *Hidden hypotheses:* the paracompactness of TX and X_1 (instance arguments, supplied by V2.20 and V2.17). *Deviation (restated):* "sc-smooth M-polyfold bundle" is read through the models and the transition maps, with TX the M-polyfold of V2.21; the fibres T_xX carry no linear structure of their own (gap G.7).

V2.95 (Proposition 2.3.17 for $i = 1$, $T(X^1) \equiv (TX)^1$; `Polyfold.MPolyfold.someChart`, `Polyfold.MPolyfold.TangentPre.lower`, `Polyfold.MPolyfold.TangentPre.transport_lower`, `Polyfold.MPolyfold.tangentLower`, `Polyfold.MPolyfold.tangentShiftMap`, `Polyfold.MPolyfold.tangentShiftEquiv`, `Polyfold.MPolyfold.tangentShiftHomeomorph`, `Polyfold.MPolyfold.tangentProjShift`, `Polyfold.MPolyfold.tangentShiftEquiv_isScDiffeoOn`, `Polyfold.MPolyfold.tangentChartMap_tangentLower`, `Polyfold.MPolyfold.levelChart_tangentShiftMap`, `Polyfold.proposition_2_3_17`). The inclusion $T(X^1) \to TX$, defined in a chosen chart and independent of it, takes values in $(TX)_1$ and gives a bijection $\Phi : T(X^1) \to (TX)^1$ with: (1) Φ is an sc-diffeomorphism of the M-polyfolds $T(X^1)$ and $(TX)^1$, in particular a homeomorphism; (2) $p^1 \circ \Phi = p_{X^1}$, where $p^1 : (TX)^1 \to (X^1)^1$ is the level-1 map of $p : TX \to X^1$; (3) p^1 is sc-smooth; (4) in every chart ψ of X around the base point, $T\psi(\Phi(\alpha))$ is the inclusion of $T(\psi^1)(\alpha)$, so Φ does not depend on the chart used to define it; (5) in the charts $T(\varphi^1)$ of $T(X^1)$ and $(T\varphi)^1$ of $(TX)^1$, Φ is the identity of $T(O^1) = (TO)^1$, in particular linear on the fibres. *Book:* Proposition 2.3.17, p. 42, new in the 2021 edition, with pp. 40–42 ($(TO)^i = T(O^i)$ in a local model, compatible with the tangent charts): natural identifications of $(TX)^i$ with $T(X^i)$ and of the bundles $(TX)^i \to (X^1)^i$ and $T(X^i) \to (X^i)^1$, covering $(X^1)^i \equiv (X^i)^1$. *Hidden hypotheses:* none. *Deviation (restated):* "naturally identified, also as bundles" is read as (1)–(5); for $i = 1$ the identification $(X^1)^1 \equiv (X^1)^1$ is the identity.

V2.96 (Proposition 2.3.17 for $i = 2$, and raising the index of an sc-diffeomorphism; `Polyfold.MPolyfold.levelOneEquiv`, `Polyfold.MPolyfold.levelOneEquiv_isScDiffeoOn`, `Polyfold.MPolyfold.tangentShiftEquivTwo`, `Polyfold.MPolyfold.tangentProjShiftTwo`, `Polyfold.proposition_2_3_17_two`). A bijection $e : Y \to Z$ of M-polyfolds that is sc-smooth in both directions induces an sc-diffeomorphism $e^1 : Y^1 \to Z^1$. Composing Φ of V2.95 for X^1 with the index raise of Φ^{-1} for X gives an sc-diffeomorphism $\Phi_2 : T(X^2) \to (TX)^2$, $X^2 = (X^1)^1$, $(TX)^2 = ((TX)^1)^1$, covering the identity of $(X^1)^2 = (X^2)^1$: $p^2 \circ \Phi_2 = p_{X^2}$. *Book:* Proposition 2.3.17, p. 42, for $i = 2$. *Hidden hypotheses:* none. *Deviation (narrower):* $i = 2$; every i is V2.97.

V2.97 (Proposition 2.3.17 for every i; `Polyfold.MPolyfold.ScEquiv`, `Polyfold.MPolyfold.ScEquiv.shift`, `Polyfold.MPolyfold.ScMap`, `Polyfold.MPolyfold.tangentShiftScEquiv`, `Polyfold.MPolyfold.tangentProjScMap`, `Polyfold.MPolyfold.tangentLower_tangentMap`, `Polyfold.BundledMPolyfold`, `Polyfold.BundledMPolyfold.nonempty`, `Polyfold.BundledMPolyfold.lift`, `Polyfold.BundledMPolyfold.tangent`, `Polyfold.BundledMPolyfold.tangentLiftEquiv`, `Polyfold.BundledMPolyfold.shiftLiftEquiv`, `Polyfold.BundledMPolyfold.tangentProjLift`,`Polyfold.BundledMPolyfold.incl`,`Polyfold.BundledMPolyfold.tangentIncl`, `Polyfold.BundledMPolyfold.tangentLiftEquiv_one`,

`Polyfold.BundledMPolyfold.tangentLiftEquiv_unique`, `Polyfold.BundledMPolyfold.tangentLiftEquiv_natural`, `Polyfold.proposition_2_3_17_all`). Bundle an M-polyfold X with its space, so that the lifts $X^0 = X$, $X^{i+1} = (X^i)^1$ form a family indexed by i. For every i: (1) $\Phi_i : T(X^i) \to (TX)^i$, with $\Phi_0 = \text{id}$ and $\Phi_{i+1} = (\Phi_i)^1 \circ \Phi$, and $J_i : (X^1)^i \to (X^i)^1$ are sc-diffeomorphisms; (2) $p^i : (TX)^i \to (X^1)^i$, the i-th index raise of p, and $p : T(X^i) \to (X^i)^1$ are sc-smooth; (3) $J_i \circ p^i \circ \Phi_i = p$; (4) Φ_i commutes with the inclusions $T(X^i) \to TX$ and $(TX)^i \to TX$, (5) and with the inclusions $T(X^{i+1}) \to T(X^i)$ and $(TX)^{i+1} \to (TX)^i$ of the two towers of the text; (6) J_i is the identity on the points of X; (7) Φ_i is the only map with (4). Φ_1 is the Φ of V2.95, and $\Phi_i \circ T(f^i) = (Tf)^i \circ \Phi_i$ for every sc-smooth $f : X \to Y$, from the naturality $\iota \circ T(f^1) = Tf \circ \iota$ of the inclusion $\iota : T(X^1) \to TX$. A bundled M-polyfold exists (the half-line). *Book:* Proposition 2.3.17, p. 42, with the two towers of p. 41. *Hidden hypotheses:* none. *Deviation (restated):* "natural identification" is read as (1)–(7) and the naturality in X. Linearity on the fibres is not part of the statement, since T_xX has no linear structure (gap G.7); Φ_i is a composite of index raises of maps that are the identity in charts (V2.95 (5)).

V2.98 (Remark 2.4.2, the diagonal half-line in $[0,\infty)^3$; `Polyfold.ScOperator.ofConst`, `Polyfold.diagonalAverage`, `Polyfold.diagonalRetraction`, `Polyfold.mem_diagonalRetraction_retract_iff`, `Polyfold.diagonalModel`, `Polyfold.diagonalPolyfold`, `Polyfold.diagonalEmbed`, `Polyfold.diagonalProject`, `Polyfold.diagonalEmbed_isScDiffeoOn`, `Polyfold.degIndex_diagonal_quadrant`, `Polyfold.diagonalPolyfold_degIndex`, `Polyfold.diagonalHalfLineChart`, `Polyfold.diagonalHalfLineChart_spec`, `Polyfold.remark_2_4_2`). In $C = [0,\infty)^3 \oplus W$, W any sc-Banach space, replacing the three coordinates by their average is an sc-smooth retraction onto $\Delta \oplus W$, $\Delta = \{(s,s,s) : s \geq 0\}$. With the M-polyfold structure of this retract: (1) $(s,w) \mapsto ((s,s,s),w)$ is an sc-diffeomorphism from the half-line M-polyfold $[0,\infty) \oplus W$ onto $\Delta \oplus W$; (2) $d_C((s,s,s),w)$ is 3 for $s = 0$ and 0 otherwise, while d_Δ is 1 for $s = 0$ and 0 otherwise, and in the defining chart (the identity of the retract) the chart index is d_C; (3) one chart, defined on all of $\Delta \oplus W$, has as model the relatively open subset $[0,\infty) \oplus W$ of a partial quadrant, and in it the chart index is d_Δ. *Book:* Remark 2.4.2, pp. 44–45, new in the 2021 edition, for $\Delta \subseteq [0,\infty)^3 \subseteq \mathbb{R}^3$. *Hidden hypotheses:* none. *Deviation (stronger):* any W ($W = 0$ is the text's case); "classically smooth" is read as sc-smooth for the constant sc-structure, and the retraction, which the text does not give, is the orthogonal projection onto the diagonal. *Proof:* d_Δ by Proposition 2.4.4 (preprint 2.24) (V2.28) applied to the sc-diffeomorphism (1).

V2.99 (Definition 2.5.19 (3) does not depend on the chart; `Polyfold.MPolyfold.mem_smoothSet_iff_of_isChart`, `Polyfold.MPolyfold.TameSubData.hasReducedComplementAt_of_isChart`, `Polyfold.MPolyfold.TameSubData.complement_forall_charts`). A point of an M-polyfold is smooth iff its image in one chart is a smooth point of the model. Let Y have tame sub-M-polyfold data (V, r) (V2.78), condition (3) being asked in one chart around each smooth point, for one presentation of its partial quadrant. Then at every smooth $y \in Y \cap V$ condition (3) holds in every chart $\varphi \ni y$, for every presentation T of the partial quadrant of φ and every level-1 representative z of $\varphi(y)$, for the chart image $\varphi(W' \cap Y)$ of any sub-M-polyfold retraction r' onto Y near y, in particular for $r' = r$. *Book:* Definition 2.5.19 (3), p. 63, new in the 2021 edition, stated on T_yX.

Hidden hypotheses: none. *Deviation (restated):* (3) remains a statement in charts, since T_yX has no linear structure here (gap G.7); the chart, the presentation and the retraction do not matter (gap G.77, closed). *Proof:* by Chapter 4's chart change for the general-position condition (`Polyfold.MPolyfold.isGeneralPositionRetractAt_chart_change`, V4.45), which is (3) word for word.

V2.100 (a tame sub-M-polyfold is tame; `Polyfold.ScBanach.levelSub_sup_of_le_of_isScSplitting`, `Polyfold.ScBanach.isScSplitting_sup_of_complementIn`, `Polyfold.MPolyfold.TameSubData.isTameRetract_image`, `Polyfold.MPolyfold.IsTameSubMPolyfold.hasTameInducedStructure`). If $E = M \oplus B$ is an sc-splitting and $N, A \subseteq M$ are sc-subspaces with $M_m = N_m \oplus A_m$ on every level, then $E = N \oplus (A \oplus B)$ is an sc-splitting. For tame sub-M-polyfold data (V, r) and a chart φ of X with a tame model, the chart image $\varphi(W \cap Y)$ of Y is a tame retract, of the retraction $(\varphi r \varphi^{-1}) \circ t$ with t the tame retraction of the model. Hence a tame sub-M-polyfold Y of a tame X has a tame induced structure (Definition 2.5.18): the sub-charts built on tame charts of X have tame models. *Book:* the text after Definition 2.5.19, p. 63, new in the 2021 edition ("not difficult to see"). *Hidden hypotheses:* none. *Proof:* the first condition of tameness from (2) of Definition 2.5.19 and $d_X = d_C \circ \varphi$ in a tame chart; the sc-complement of the tangent space of $\varphi(W \cap Y)$ inside E_x is $A \oplus B$, with A from (3) in the chart φ (V2.99) and B the complement of T_xO given by the tameness of t.

V2.101 (the examples A and B in $[0,\infty)^2 \times \mathbb{R}$; `Polyfold.TameSubExample.fixSet`, `Polyfold.TameSubExample.isSubMPolyfold_fixSet`, `Polyfold.TameSubExample.isTameSubMPolyfold_fixSet`, `Polyfold.TameSubExample.setA`, `Polyfold.TameSubExample.setB`, `Polyfold.TameSubExample.isTameSubMPolyfold_setA`, `Polyfold.TameSubExample.hasTameInducedStructure_setA`, `Polyfold.TameSubExample.isSubMPolyfold_setB`, `Polyfold.TameSubExample.not_mem_tameSubData_setB`, `Polyfold.TameSubExample.not_isTameSubMPolyfold_setB`, `Polyfold.TameSubExample.halfLineToB_isScDiffeoOn`, `Polyfold.TameSubExample.hasTameInducedStructure_setB`, `Polyfold.TameSubExample.examples_A_B`). In $X = [0,\infty)^n \oplus W$, W any sc-Banach space, the fixed set of an idempotent sc-operator L mapping the quadrant into itself is a sub-M-polyfold, and a tame sub-M-polyfold if L does not change the $\mathbb{R}^n$-component. For $n = 2$: $A = \{(x, y, 0)\}$ (with $L(a, w) = (a, 0)$) is a tame sub-M-polyfold with tame induced structure; $B = \{(x, x, 0)\}$ is a sub-M-polyfold, but no retraction onto B with conditions (1) and (2) of Definition 2.5.19 is defined near the corner 0, so B is not a tame sub-M-polyfold; $t \mapsto (t, t, 0)$ is an sc-diffeomorphism from the half-line $[0, \infty)$ onto B with its induced structure, which is tame. For $W = \mathbb{R}$ these are the text's statements. *Book:* the text after Definition 2.5.19, p. 63, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* any W. *Proof:* for small $t > 0$ the point $(t, 0, 0)$ of V has $d_X = 1$, while d_X takes only the values 0 and 2 on B; (3) is not needed.

V2.102 (Theorem 2.7.13 (2)–(4), the tame M-polyfold bX; `Polyfold.MPolyfold.exists_tame_chart_subset`, `Polyfold.MPolyfold.exists_localFace_of_faceIn`, `Polyfold.MPolyfold.BChartData`, `Polyfold.MPolyfold.BChartData.chart`, `Polyfold.MPolyfold.BChartData.chart_transition_isScSmooth`, `Polyfold.MPolyfold.BChartData.chart_compatible`, `Polyfold.MPolyfold.exists_bChartData`, `Polyfold.MPolyfold.`

`paracompactSpace_tameBoundarySpace`, `Polyfold.MPolyfold.bAtlas`, `Polyfold.MPolyfold.tameBoundaryPolyfold`, `Polyfold.MPolyfold.tameBoundaryPolyfold_isTame`, `Polyfold.MPolyfold.tameBoundaryMap_isScSmooth`, `Polyfold.MPolyfold.degIndex_tameBoundaryMap`, `Polyfold.MPolyfold.degIndex_tameBoundaryMap_sub_one`**).** Let X be tame. A chart datum of bX consists of a triple (x, V, θ) with V centred at x (V2.84), a chart φ of X, an open $V_1 \subseteq \operatorname{dom} \varphi$, an sc-isomorphism T of the model space of φ with $\mathbb{R}^{k+1} \oplus W$ and a coordinate i vanishing on $T\varphi(\theta \cap V_1)$, such that the image of $\varphi(\theta \cap V_1)$ under the face projection π (Proposition 2.5.15 (preprint 2.43)) is a tame retract of $[0, \infty)^k \oplus W$ and $d_X(y) = d(\pi\varphi(y)) + 1$ on $\theta \cap V_1$. Its chart is $p \mapsto \pi(\varphi(B(p)))$ on $U(x, V, \theta) \cap B^{-1}(V_1)$; two such charts are compatible, with transition map $\pi' \circ (\varphi' \circ \varphi^{-1}) \circ \iota$ (ι the face inclusion); every point of θ lies in the V_1 of a chart datum with the triple (x, V, θ), so the charts cover bX. With the topology T_{bX}, paracompact by V2.88, they make bX a tame M-polyfold (2); $B : bX \to X$, which is ι in these charts, is sc-smooth (3); and $d_X(B(p)) = d_{bX}(p) + 1$, hence $d_X(B(p)) - 1 = d_{bX}(p)$ (4). *Book:* Theorem 2.7.13 (2)–(4), pp. 83–84, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation:* none. *Proof:* the charts are the charts $\pi \circ \varphi$ of the face θ composed with $\Phi_{(x,V,\theta)}$; the local faces of θ in a tame chart inside V are found as in Theorem 2.5.17 (preprint 2.45).

V2.103 (the characterization after Theorem 2.7.13, and Proposition 2.7.12, for centred representatives; `Polyfold.TameBoundary.BSpace.exists_centred_rep`, `Polyfold.MPolyfold.isSubMPolyfold_of_isFaceIn`, `Polyfold.MPolyfold.facePolyfold`, `Polyfold.MPolyfold.phiTot`, `Polyfold.MPolyfold.phi_isScDiffeoOn`, `Polyfold.MPolyfold.tameBoundaryPolyfold_unique`, `Polyfold.MPolyfold.tameBoundaryPolyfold_satisfies`, `Polyfold.MPolyfold.proposition_2_7_12_scDiffeo`, `Polyfold.MPolyfold.theorem_2_7_13_structure`**).** Let X be tame. Every class of bX has a representative (x, V, θ) with V centred at x. A face θ of a face-structured open V is a sub-M-polyfold of X and carries the induced structure. For V centred at x, $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism of the open set $U(x, V, \theta)$ onto θ, with inverse Φ^{-1}; an M-polyfold structure on bX (with the topology T_{bX}) for which all these Φ are sc-diffeomorphisms is equivalent to that of V2.102, which has the property. For two triples with centred neighbourhoods, the transition map $\Phi_{(x',V',\theta')} \circ \Phi^{-1}_{(x,V,\theta)}$ is an sc-diffeomorphism between the open subsets $\Phi_{(x,V,\theta)}(U \cap U')$ of θ and $\Phi_{(x',V',\theta')}(U \cap U')$ of θ', and the identity on points of X. One statement collects (2)–(4) with this characterization and the uniqueness. *Book:* Proposition 2.7.12, p. 83, and the characterization after Theorem 2.7.13, p. 84, new in the 2021 edition: for every class and every representative, $U(x, V, \theta)$ is open and $\Phi_{(x,V,\theta)}$ a homeomorphism and an sc-diffeomorphism onto θ, with the structure induced from V. *Hidden hypotheses:* none. *Deviation (narrower):* centred representatives; every representative is V2.104. "The structure induced from V" is read as the sub-M-polyfold structure of θ in X, and "uniquely characterized" as: every structure with the property has an equivalent atlas.

V2.104 (Proposition 2.7.12 and the characterization for every representative; `Polyfold.TameBoundary.BTriple.exists_centred_subset`, `Polyfold.TameBoundary.phiInv`, `Polyfold.TameBoundary.phiInv_spec`, `Polyfold.MPolyfold.reachSet`, `Polyfold.MPolyfold.isOpen_reachSet`, `Polyfold.MPolyfold.reachSet_eq_univ_of_isCenteredAt`, `Polyfold.MPolyfold.phiTot_isScSmoothOn`, `Polyfold.MPolyfold.phiInv_isScSmoothOn`, `Polyfold.MPolyfold.phi_ `isScDiffeoOn_nonCentred`, `Polyfold.MPolyfold.tameBoundaryPolyfold_unique_allReps`, `Polyfold.MPolyfold.tameBoundaryPolyfold_satisfies_allReps`, `Polyfold.MPolyfold.proposition_2_7_12_scDiffeo_nonCentred`**).** Every triple s has an equivalent triple s' centred at its point with $U(s') \subseteq U(s)$. For every representative (x, V, θ), $\Phi_{(x,V,\theta)}$ is sc-smooth on $U(x, V, \theta)$, its inverse is sc-smooth on the image $\theta \cap V^x$, and Φ is an sc-diffeomorphism of $U(x, V, \theta)$ onto the open subset $\theta \cap V^x$ of θ, which is θ when V is centred at x. An M-polyfold structure on bX with this property for every representative is equivalent to that of V2.102, which has it. For every pair of representatives, $\Phi_{(x,V,\theta)}(U \cap U')$ and $\Phi_{(x',V',\theta')}(U \cap U')$ are open in θ and θ', and the transition map between them is an sc-diffeomorphism, with inverse $\Phi_{(x,V,\theta)} \circ \Phi^{-1}_{(x',V',\theta')}$, the identity on points of X. *Book:* as V2.103. *Hidden hypotheses:* none. *Deviation (narrower):* Proposition 2.7.12 holds as printed for all representatives; in the characterization the image of Φ is $\theta \cap V^x$, and "onto θ" does not hold as we read it for every representative (V2.105) (gap G.78, closed). *Proof:* near each point, Φ^{-1} is $\Phi^{-1}_{s'}$ for a centred refinement s', composed with the inclusion of an open piece of θ into the face of s'; the sub-M-polyfold calculus (a map into a sub-M-polyfold is sc-smooth if it is sc-smooth into X) replaces the text's open sc-embeddings of faces.

V2.105 ("Φ is an sc-diffeomorphism onto θ" for every representative: the slab of V2.85; `Polyfold.MPolyfold.not_surjOn_phiTot`**).** For the triple (x, V, θ) of V2.85 in $[0, \infty)^3 \oplus W$, $\Phi_{(x,V,\theta)}$ does not map $U(x, V, \theta)$ onto θ. *Book:* the characterization after Theorem 2.7.13, p. 84, for "any representative". *Hidden hypotheses:* none. *Deviation (not in the book):* read for every representative, the characterization does not hold as we read it; the strongest version that holds is V2.104, onto $\theta \cap V^x$, which for centred V is the printed statement (V2.103). *Proof:* V2.85.

V2.106 (Definition 2.7.14 and the iterated boundaries (2.45)–(2.47); `Polyfold.MPolyfold.TameMPolyfoldBundle`, `Polyfold.MPolyfold.TameMPolyfoldBundle.b`, `Polyfold.MPolyfold.TameMPolyfoldBundle.iter`, `Polyfold.MPolyfold.TameMPolyfoldBundle.iterMap`, `Polyfold.MPolyfold.TameMPolyfoldBundle.iterMap_isScSmooth`, `Polyfold.MPolyfold.TameMPolyfoldBundle.degIndex_iterMap`, `Polyfold.MPolyfold.TameMPolyfoldBundle.range_iterMap`, `Polyfold.MPolyfold.definition_2_7_14`**).** The tame boundary of a tame X is bX with the tame structure of V2.102 and the sc-smooth B, whose image is ∂X. For a tame M-polyfold bundled with its space, $b^0X = X$ and $b^kX = b(b^{k-1}X)$ are tame M-polyfolds; $B^0 = \mathrm{id}$ and $B^{k+1} = B_X \circ B^k_{bX}$, so $B^k = B_X \circ B_{bX} \circ \cdots \circ B_{b^{k-1}X}$; $B^k : b^kX \to X$ is sc-smooth, $d_X(B^k(p)) = d_{b^kX}(p) + k$, and the image of B^k is $\{d_X \geq k\}$. *Book:* Definition 2.7.14 and (2.45)–(2.47), p. 84, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the index formula, Theorem 2.7.13 (4) iterated. The bundling makes b a map that can be iterated.

V2.107 (witnesses for Theorem 2.7.13 (2)–(4); `Polyfold.MPolyfold.quadrantPolyfold_bChartData_nonempty`, `Polyfold.MPolyfold.quadrantPolyfold_degIndex_tameBoundary`**).** For the tame quadrant $[0, \infty)^n \oplus W$, $n \geq 1$, chart data of bX exist, and the n points of bX over the corner 0 have $d_{bX} = n - 1$. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book):* the hypotheses of V2.102–V2.106 are satisfiable, and (4) is not vacuous.

V2.108 (Proposition 2.1.2 (preprint 2.2): the image of a classical smooth retraction is

a submanifold; `Polyfold.GapP1Prop212.proposition_2_1_2`, `Polyfold.GapP1Prop212.isIdempotentElem_fderiv`, `Polyfold.GapP1Prop212.retractChart`, `Polyfold.GapP1Prop212.retractChart_comp`, `Polyfold.GapP1Prop212.hasFDerivAt_retractChart`). `proposition_2_1_2`: for a real Banach space E, $U \subseteq E$ open, r C^∞ on U with $r(U) \subseteq U$ and $r \circ r = r$ on U, and $x \in r(U)$, there are open $V \ni x$ with $V \subseteq U$, open $W \ni 0$, closed complementary subspaces $R = \operatorname{range} Dr(x)$ and $N = \ker Dr(x)$, and a C^∞ diffeomorphism $\psi : V \to W$ (ψ and ψ^{-1} C^∞) with $\psi(x) = 0$ and $\psi(r(U) \cap V) = R \cap W$. *Book:* §2.1, Proposition 2.1.2 (preprint 2.2), p. 22 (preprint p. 18) (proof §2.7.1 (preprint §2.6.1), preprint pp. 54–55), If $r : U \to U$ is a C^∞ retraction of an open subset of a Banach space E, then $O = r(U)$ is a C^∞ submanifold: near each $x \in O$ there are a splitting $E = R \oplus N$ and a diffeomorphism $\psi : V \to W$ with $\psi(O \cap V) = R \cap W$ (the text prints $\psi(0) = x$ for $\psi(x) = 0$). *Deviation (restated):* The printed normalization $\psi(0) = x$, for $\psi : V \to W$ with $x \in V$ and $0 \in W$, is read as $\psi(x) = 0$. The proof is not the text's (the estimates of §2.7.1): $P = Dr(x)$ is idempotent by the chain rule (`isIdempotentElem_fderiv`); $\varphi(y) = P(r(y) - x) + (1 - P)(y - r(y))$ (`retractChart`) has $D\varphi(x) = 1$ (`hasFDerivAt_retractChart`) and conjugates r to P, $\varphi \circ r = P \circ \varphi$ (`retractChart_comp`); Mathlib's inverse function theorem gives ψ. C^∞ is `ContDiffOn` ∞ (smooth, not analytic).

V2.109 (Lemma 2.4.9 (preprint 2.29): $T^R_x O$ and $C_x O$ through sc-smooth paths; $T_x O = C_x O - C_x O$; `Polyfold.GapP1Lemma249.lemma_2_4_9_reduced`, `Polyfold.GapP1Lemma249.lemma_2_4_9_cone`, `Polyfold.GapP1Lemma249.lemma_2_4_9_tangent`, `Polyfold.GapP1Lemma249.pathDeriv`, `Polyfold.GapP1Lemma249.twoSidedPathDerivs`, `Polyfold.GapP1Lemma249.oneSidedPathDerivs`, `Polyfold.GapP1Lemma249.hasDerivWithinAt_pathDeriv`, `Polyfold.GapP1Lemma249.scDeriv_pathDeriv`, `Polyfold.GapP1Lemma249.pathDeriv_mem_minimalSubspaceWith`, `Polyfold.GapP1Lemma249.pathDeriv_mem_partialQuadrantAtWith`, `Polyfold.GapP1Lemma249.path_retract_segment`, `Polyfold.GapP1Lemma249.partialConeWith_subset_closure`, `Polyfold.GapP1Lemma249.isScDomain_Ioo`, `Polyfold.GapP1Lemma249.isScDomain_Ico`). For a retraction r with retract O and a presentation of C, paths are maps $\alpha : \mathbb{R} \to E_0$, sc-smooth on $(-\varepsilon, \varepsilon)$ or on $[0, \varepsilon)$ for the constant sc-structure on $\mathbb{R}$ (both sc-domains: `isScDomain_Ioo`, `isScDomain_Ico`), with values in O and $\alpha(0) = x$; $\dot\alpha(0)$ is `pathDeriv`, $D\alpha(0)1$, the classical two-sided or one-sided derivative (`hasDerivWithinAt_pathDeriv`). At a smooth $x \in O$: (1) $T^R_x O$ is the closure of the set of $\dot\alpha(0)$ for two-sided paths (`lemma_2_4_9_reduced`); (2) $C_x O$ is the closure of the set of $\dot\alpha(0)$ for one-sided paths (`lemma_2_4_9_cone`); (3) $T_x O = C_x O - C_x O$, for every $x \in O_1$ (`lemma_2_4_9_tangent`). The proof is the text's: $Dr(x)\dot\alpha(0) = \dot\alpha(0)$ (`scDeriv_pathDeriv`), the sign arguments (`pathDeriv_mem_minimalSubspaceWith`, `pathDeriv_mem_partialQuadrantAtWith`), the paths $r(x + tv)$ (`path_retract_segment`) and the density of smooth points (`partialConeWith_subset_closure`). *Book:* §2.4 (preprint §2.3), Lemma 2.4.9 (preprint 2.29), pp. 48–49 (preprint pp. 33–35), For a retract and a smooth $x \in O$: (1) $T^R_x O$ is the closure of the derivatives $\dot\alpha(0)$ of sc-smooth paths $\alpha : (-\varepsilon, \varepsilon) \to O$ through x; (2) $C_x O$ the closure of the one-sided derivatives of sc-smooth paths $\alpha : [0, \varepsilon) \to O$; (3) $T_x O = C_x O - C_x O$. *Deviation (same):* none.

V2.110 (The extended faces of a partial quadrant, intrinsically (preprint p. 39); `Polyfold.GapP1ExtendedFaces.stratumOne`, `Polyfold.GapP1ExtendedFaces.faceComponents`,

`Polyfold.GapP1ExtendedFaces.extendedFaces`, `Polyfold.GapP1ExtendedFaces.facesAt`, `Polyfold.GapP1ExtendedFaces.faceComponents_eq`, `Polyfold.GapP1ExtendedFaces.ncard_faceComponents`, `Polyfold.GapP1ExtendedFaces.span_faceOpen`, `Polyfold.GapP1ExtendedFaces.extendedFaces_eq`, `Polyfold.GapP1ExtendedFaces.ncard_extendedFaces`, `Polyfold.GapP1ExtendedFaces.add_coordFace`, `Polyfold.GapP1ExtendedFaces.isClosed_add_coordFace`, `Polyfold.GapP1ExtendedFaces.subset_add_coordFace`, `Polyfold.GapP1ExtendedFaces.ncard_facesAt`, `Polyfold.GapP1ExtendedFaces.iInter_facesAt`, `Polyfold.GapP1ExtendedFaces.connectedComponentIn_eq`). The definitions use only C and d_C: `stratumOne` $= \{e \in C \mid d_C(e) = 1\}$, `faceComponents` its connected components, `extendedFaces` the spans of the components (the smallest linear subspaces containing them), `facesAt e` $= \mathscr{F}(e) = \{f \mid e \in f\}$. For any presentation T (used only in the proofs): the components are $A_i = \{(Te)_i = 0,\ (Te)_j > 0\ (j \neq i)\}$ (`faceComponents_eq`) and there are exactly k of them (`ncard_faceComponents`); $\operatorname{span} A_i = f_i = \{(Te)_i = 0\}$ (`span_faceOpen`, `extendedFaces_eq`), so there are exactly k extended faces (`ncard_extendedFaces`); $C + f_i = \{(Te)_i \geq 0\}$, closed and containing C (`add_coordFace`, `isClosed_add_coordFace`, `subset_add_coordFace`); $d_C(e) = \#\mathscr{F}(e)$ (`ncard_facesAt`); and $\bigcap_{f \in \mathscr{F}(e)}(C + f) = C_e$ (`iInter_facesAt`, Definition 2.5.1). *Book:* §2.5 (preprint §2.4), preprint p. 39, $\{e \in C : d_C(e) = 1\}$ has exactly k connected components A_i; each lies in a smallest subspace f_i (an extended face), with a half-space $H_{f_i} \supset C$; $\mathscr{F}(e) = \{f : e \in f\}$ and $d_C(e) = \#\mathscr{F}(e)$. *Deviation (restated):* The text names a half-space $H_{f_i} \supset C$ without a formula; it is read as $C + f_i$, which is the half-space H_i of (2.14) (preprint numbering) for every presentation. The presentation-based forms that the proofs of §2.5 use remain V2.39 and V2.45.

V2.111 (Definition 2.5.8 (preprint 2.38): $T_x^R X$ and $C_x X$ of an M-polyfold; $C_x X$ a partial quadrant, $d_X(x) = d_{C_x X}(0_x)$; `Polyfold.MPolyfold.reducedTangentFiber`, `Polyfold.MPolyfold.partialConeFiber`, `Polyfold.MPolyfold.reducedTangentFiber_eq`, `Polyfold.MPolyfold.partialConeFiber_eq`, `Polyfold.MPolyfold.finrank_quotient_reducedTangentFiber_eq_degIndex`, `Polyfold.degIndex_zero_eq_finrank_quotient`, `Polyfold.mem_minimalSubspaceWith_iff_lineality`, `Polyfold.MPolyfold.mem_reducedTangentFiber_iff_lineality`, `Polyfold.MPolyfold.partialConeFiberModel`, `Polyfold.MPolyfold.mem_partialConeFiberModel_iff`, `Polyfold.MPolyfold.isPartialQuadrant_partialConeFiberModel`, `Polyfold.MPolyfold.degIndex_partialConeFiberModel_eq_finrank`, `Polyfold.MPolyfold.definition_2_5_8_degIndex`, `Polyfold.ScBanach.IsScOperator.toBase_toScOperator`, `Polyfold.ScBanach.restrLvl`, `Polyfold.ScBanach.subspaceIso`, `Polyfold.MPolyfold.fiberCoord_eq_transitionCLM`, `Polyfold.MPolyfold.transitionCLM_transitionCLM`, `Polyfold.MPolyfold.tangentFiberScIso`, `Polyfold.MPolyfold.fiberCoordSc`, `Polyfold.MPolyfold.tangentFiberScIso_comp`, `Polyfold.MPolyfold.mem_partialConeFiberModel_fiberCoordSc`, `Polyfold.MPolyfold.tangentFiberScIso_image_partialConeFiberModel`, `Polyfold.MPolyfold.definition_2_5_8_scStructure`). With the linear structure of $T_x X$ (G.7), $T_x^R X$ and $C_x X$ are the preimages under $T\varphi(x)$ of $T^R_{\varphi(x)} O$ and $C_{\varphi(x)} O$ (`reducedTangentFiber`, `partialConeFiber`) and do not depend on the chart or the presentation (`reducedTangentFiber_eq`, `partialConeFiber_eq`, by

Proposition 2.4.10); $T^R_x X = C_x X \cap (-C_x X)$ (`mem_reducedTangentFiber_iff_lineality`). At a smooth x, $T_x X$ carries the sc-structure of $T_{\varphi(x)} O$ transported by $T\varphi(x)$; in it $C_x X$ is a partial quadrant (any chart, `isPartialQuadrant_partialConeFiberModel`), and in a tame chart $d_X(x) = \dim(T_x X / T^R_x X) = d_{C_x X}(0_x)$ (`definition_2_5_8_degIndex`). For two charts the transition derivative $T_{\varphi(x)} O \to T_{\psi(x)} O$ is an sc-isomorphism (`tangentFiberScIso`, from `ScBanach.subspaceIso`) that carries φ-coordinates to ψ-coordinates (`tangentFiberScIso_comp`) and the model of $C_x X$ onto the model (`tangentFiberScIso_image_partialConeFiberModel`); `definition_2_5_8_scStructure` collects the definition. $d_C(0) = \dim(F/(C \cap -C))$ for every partial quadrant (`degIndex_zero_eq_finrank_quotient`). *Book:* §2.5 (preprint §2.4), Definition 2.5.8 (preprint 2.38), p. 57 (preprint p. 42), For a smooth $x \in X$, $T^R_x X$ and $C_x X$ are the preimages of $T^R_o O$ and $C_o O$ under $T\psi(x)$ for any chart ψ; well defined by Proposition 2.4.10 (preprint 2.30). For tame X, $C_x X$ is a partial quadrant and $d_X(x) = \dim(T_x X / T^R_x X) = d_{C_x X}(0_x)$. *Deviation (restated):* The sc-structure of $T_x X$ is that of $T_{\varphi(x)} O$, well defined up to the sc-isomorphisms `tangentFiberScIso` between charts; this is what the text uses. $d_{C_x X}(0_x)$ does not depend on the choice (`degIndex_zero_eq_finrank_quotient`, `mem_reducedTangentFiber_iff_lineality`). The local forms are V2.32, V2.35; in a chart, "partial quadrant" is V2.42 (`proposition_2_5_6_partialCone`).

V2.112 (Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57): paracompactness through closed locally finite refinements; `Polyfold.GapP1Michael.lemma_2_7_3`, `Polyfold.GapP1Michael.proposition_2_7_4`, `Polyfold.GapP1Michael.paracompactSpace_of_closed_refinement`, `Polyfold.GapP1Michael.exists_closed_refinement`). `lemma_2_7_3 [T3Space X]`: X is paracompact (`ParacompactSpace`) iff every open cover has a closed locally finite refinement; the converse direction holds for every topological space (`paracompactSpace_of_closed_refinement`), the direct one for regular paracompact spaces (`exists_closed_refinement`). `proposition_2_7_4 [T3Space Y]`: a locally finite cover of Y by closed sets, each paracompact as a subspace, makes Y paracompact. *Book:* §2.7.2 (preprint §2.6.2), Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57), p. 74 (preprint p. 56), Lemma 2.7.3 (preprint 2.56) (Michael): a regular Hausdorff space is paracompact iff every open cover has a closed locally finite refinement. Proposition 2.7.4 (preprint 2.57): a regular space with a locally finite cover by closed paracompact subspaces is paracompact. *Deviation (same):* none.

What is not verified in this chapter

- p. 22 (preprint p. 18), Proposition 2.1.2 (preprint 2.2) (classical C^∞ retractions have submanifold images): motivation only; not attempted (the exposition proves it, §8.1). Verified since, with $\psi(x) = 0$ for the printed $\psi(0) = x$ (V2.108) (gap G.4, closed).
- p. 29 (preprint p. 22), Definition 2.1.16 (preprint 2.9) (sc-smooth bundle retractions, the categories $\mathcal{R}$, $\mathcal{BR}$): not formalized as a definition; its instances (Tr, strong bundle retractions) are. Formalized since as a definition: bundle retractions and bundle maps, with Tr and Tf as instances (V2.92); the categories $\mathcal{R}$, $\mathcal{BR}$ and their forgetful functors are still not formalized (gap G.5, formalized since apart from the categories; Findings, item 76).

- preprint p. 25, X_m as an M-polyfold for $m \geq 2$: only X^1 is constructed (V2.16); the iterate $(X^1)^1 \cdots$ has the points of X_m, its topology is not compared with that of X_m; Theorem 2.3.10 (preprint 2.16) is proved for every m. Verified since: $(X^1)_m \cong X_{m+1}$ through the identity, with the charts of X_{m+1}, for every m, so each iterate is X_m with its topology and charts (V2.93) (gap G.6, closed; Findings, item 77).
- pp. 38–44 (preprint pp. 26–30), the vector space, Banach and sc-Banach structures of T_xX, and Proposition 2.3.19 (3),
  (4) (preprint 2.21) in T_aX rather than in a chart: not formalized.
- p. 39 (preprint p. 27), Proposition 2.3.15 (2) (preprint 2.19), openness of $p : TX \to X_1$: not proved (not used).
- pp. 48–49 (preprint pp. 33–35), Lemma 2.4.9 (preprint 2.29) in its path formulation, and $T_xO = C_xO - C_xO$: not formalized; the sign arguments are V2.34. Verified since, as printed (V2.109) (gap G.10, closed).
- preprint p. 39, the intrinsic description of extended faces (k components of $\{d_C = 1\}$, minimal subspaces, independence of the presentation): the formalization works in a presentation. Verified since, from C and d_C alone, with H_f read as $C + f$ (V2.110) (gap G.12, closed).
- p. 57 (preprint p. 42), Definition 2.5.8 (preprint 2.38) ($T_x^R X$, C_xX in T_xX): only the local form, its invariance and the partial quadrant C_oO (in a tame chart V2.42, in any chart V2.33); not the transport to T_xX or $d_{C_xX}(0_x) = d_X(x)$. Verified since: C_xX is a partial quadrant in the sc-structure transported from a chart, $d_X(x) = d_{C_xX}(0_x)$ in a tame chart, and the transition derivatives are sc-isomorphisms matching coordinates and cones (V2.111) (gap G.13, closed).
- pp. 65–67 (preprint pp. 48–51), strong bundle isomorphisms, the category $\mathcal{SBR}$ and the functors forget$[i]$: not formalized.
- p. 69 (preprint p. 53), Proposition 2.6.8 (preprint 2.54): that the projection $P' : f^*Y \to Y$ of the pull-back is a strong bundle map; the pulled-back atlas is V2.61.
- p. 72 (preprint p. 56), Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57): not needed, since the proofs of Theorem 2.3.10 (preprint 2.16) and Proposition 2.3.15 (3) (preprint 2.19) avoid them. Verified since, as printed (V2.112) (gap G.15, closed); the proofs still avoid them.
- p. 23, Remark 2.1.6: that the modified notion of local completeness is strictly weaker; only the implication is proved (V2.68). Gap G.76. Verified since: $\mathbb{R}$ with the K-topology is Hausdorff, satisfies the modified notion and is not locally complete (V2.91) (gap G.76, closed; Findings, item 75).
- p. 30, Remark 2.1.17: it fixes the words "tangent" and "tangent bundle" of a retract and has no further content.
- pp. 62–63, Definition 2.5.19 (3): its independence of the chart, the remark that a tame sub-M-polyfold is tame, and the examples in $[0, \infty)^2 \times \mathbb{R}$ (V2.78). Gap G.77. Verified since, (3) still in charts (gap G.7) (V2.99, V2.100, V2.101) (gap G.77, closed; Findings, items 81, 82).
- pp. 83–84, Proposition 2.7.12 and the characterization after Theorem 2.7.13 for representatives (x, V, θ) whose V is not centred at x: $\Phi_{(x,V,\theta)}$ is an open embedding there (V2.87), but the sc-smoothness of the transition maps is not formalized; every class has a centred representative.

Gap G.78. Verified since: Proposition 2.7.12 for every representative, and the characterization with Φ an sc-diffeomorphism onto the open set $\theta \cap V^x$, which is θ for centred V (V2.104); onto θ it does not hold as we read it for every representative (V2.105) (gap G.78, closed; Findings, items 83, 84).

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4. Chapter 3 (HWZ §§3.1–3.8): partly verified

Chapter 3 of Part I (pp. 89–174 (preprint pp. 61–146)) develops sc-Fredholm theory. §3.1 defines germs, sections of strong bundles, fillers, basic germs, sc-Fredholm germs and sections, proves local compactness (Theorem 3.1.13 (preprint 3.12)) and the stability under sc⁺-perturbations (Theorem 3.1.18 (preprint 3.15)), and states the implicit function theorems (Theorems 3.1.22, 3.1.26 (preprint 3.19, 3.23)), with good position (Definition 3.1.24 (preprint 3.21), Proposition 3.1.25 (preprint 3.22)). §3.2 introduces the tangent germ property and shows that it makes a subset a sub-M-polyfold (Definition 3.2.1 (preprint 3.25), Proposition 3.2.2 (preprint 3.26), Theorem 3.2.3 (preprint 3.27)). §3.3 proves the germ implicit function theorem (Definition 3.3.1 (preprint 3.28), Theorems 3.3.2–3.3.7 (preprint 3.29–3.34) and Theorem 3.35 (preprint; not in the 2021 edition)). §3.4 proves that basic germs are stable under sc⁺-perturbations (Theorem 3.4.2 (preprint 3.38)). §3.5 proves local regularity and compactness (Theorem 3.5.1 (preprint 3.40)) and local solvability (Corollary 3.5.3 (preprint 3.41), Theorem 3.5.6 (preprint 3.43), Theorem 3.5.11 (preprint Corollary 3.47)). §3.6 states and proves the implicit function theorem for sc-Fredholm sections (Proposition 3.6.5 (preprint 3.52), Definition 3.6.6 (preprint 3.53), Theorem 3.6.8 (preprint 3.54), Lemma 3.6.9 (preprint 3.55)) and its global forms (Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)). §3.7 gives a criterion for being conjugate to a basic germ (Theorem 3.7.1 (preprint 3.58), Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61)). The appendix §3.8 proves Proposition 3.1.25 (preprint 3.22), Lemma 3.5.10 (preprint 3.46), Lemma 3.6.9 (preprint 3.55), and the inverse and implicit function theorems in partial quadrants (Theorems 3.8.15, 3.8.22 (preprint 3.72, 3.78)).

What is verified. **The germ implicit function theorem (Theorem 3.3.6 (preprint 3.33)) holds with the text's hypotheses**, and so do Corollary 3.3.7 (preprint 3.34) and Theorem 3.35 (preprint; not in the 2021 edition): the solution germ is sc^k on the germ of neighbourhoods shrunk by $2^k - 1$ levels, and sc-smooth on one diagonal germ, which shrinks fast with the level (V3.31, V3.32). **The implicit function theorems also hold with the text's hypotheses, local and global**: Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) (V3.94, V3.95), Theorem 3.6.11 (preprint 3.57), Corollary 3.1.27 (preprint 3.24) and Theorem 3.6.10 (preprint 3.56) (V3.87, V3.70, V3.86), once three steps of the text's proof are replaced. The surjectivity of $f'(y)$ at the zeros y near a given zero, which the text derives from Corollary 3.5.3 (preprint 3.41) and which does not follow from it (V3.82; the same step stands in the book's arXiv version), is proved by a new argument, an implicit function theorem at the nearby zero for a perturbed equation (V3.93); the regularity of the solution germ away from the centre is proved by re-centring (V3.73, V3.74); and the relative openness of the set Σ in the proof of Theorem 3.2.3 (preprint 3.27) by the choice of the good complement (V3.75, V3.77). With them: the dimension of the solution set is the Fredholm index (V3.96, V3.97), the index is locally constant along the solution set (V3.98; also without good position, with the text's hypotheses: tame germ data and no bump functions), and so, with the text's hypotheses, is the index of Definition 6.2.1 (preprint 6.1) (read with the text's locally defined sc⁺-sections) along sc-smooth paths $\mathbb{R} \to X$, the text's Proposition 6.2.2 (preprint 6.2) for paths on $\mathbb{R}$ (V3.98), ker $f'(y)$ is the tangent space of the solution set in a chart (V3.99), and good position is the surjectivity of $f'(y)$ at zeros with $d_X(y) = 0$, so that Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) hold with the text's hypothesis "$f'(x)$ surjective" (V3.100, V3.101). The forms first proved under an added filler condition at the nearby zeros (V3.68, V3.69, V3.82, V3.85) are kept; that condition follows from good position (V3.102), and with Definition 3.1.4 (preprint 3.4) as stated the text's argument gives the kernel half of Theorem 3.6.8 (1) (preprint 3.54) (V3.81) but not the surjectivity. All these statements are the repaired ones. They were first proved under the hypothesis that the base retraction of every strong bundle chart compatible with the atlas is tame, which no strong bundle over a nonempty base satisfies (V3.104, where only the half of the argument that concerns retractions is machine-checked), so those first statements were vacuous; they now assume what the text's Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) ask, germ data and good position in a chart whose base retraction is tame, and conclude good position in such charts (V3.103, V3.107). Also verified: every definition of §§3.1–3.6 (Definitions 3.1.1–3.1.4, 3.1.6, 3.1.7, 3.1.11, 3.1.16, 3.1.24, 3.2.1, 3.3.1, 3.4.1, 3.6.1, 3.6.3, 3.6.4, 3.6.6 (preprint 3.1–3.4, 3.6, 3.7, 3.11, 3.14, 3.21, 3.25, 3.28, 3.37, 3.48, 3.50, 3.51, 3.53); since the switch of Definition 3.1.11 (preprint 3.11) to the text's germ reading, the fillings, the local sc⁺-sections and every notion built on them are germs whose levels may shrink with m, as in the text, V3.7, V3.9, V3.13, and so, since a fourth addendum, is Definition 3.1.3 (preprint 3.3), V3.5); Proposition 3.1.5 (preprint 3.5), also in sc form (V3.8); Lemma 3.1.9 (preprint 3.9) as stated (V3.11); Proposition 3.1.10 (preprint 3.10), both statements, the second on every level (V3.12); Theorems 3.1.13 (preprint 3.12; with fewer hypotheses), 3.1.18 (preprint 3.15; without tameness), Corollary 3.1.15 (preprint 3.13); Theorem 3.1.21 (preprint 3.18) (without tameness), with the linearization of the perturbed section (V3.91, V3.92); Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17); Proposition 3.1.25 (preprint 3.22) (without the extreme-ray lemmas); Proposition 3.2.2 (preprint 3.26); Theorem 3.2.3 (preprint 3.27) under two added regularity conditions, each with a counterexample to the step of the text's proof it replaces (for solution sets both are theorems); Theorems 3.3.2, 3.3.3 (preprint 3.29, 3.30), Lemmas 3.3.4, 3.3.5 (preprint 3.31, 3.32) (as one induction), Theorem 3.4.2 (preprint 3.38) (with a linear coordinate change), Corollary 3.4.3 (preprint 3.39), Theorem 3.5.1 (preprint 3.40) (with compact closures), Corollary 3.5.3 (preprint 3.41), Lemmas 3.5.5, 3.5.8–3.5.10 (preprint 3.42, 3.44–3.46); Theorem 3.5.6 (preprint 3.43) and Theorem 3.5.11 (preprint Corollary 3.47) (their clause (1), "the solution map is an sc-smooth germ", in the level-wise sense: C^k into level k on a neighbourhood depending on k); Remark 3.6.2 (preprint 3.49); Proposition 3.6.5 (preprint 3.52) in charts, as an sc-Fredholm operator with the index $n-N$ of the conjugating basic germ on every level, in every chart (V3.46, V3.96); Lemma 3.8.25 (preprint 3.81) as stated (V3.66); Lemma 3.6.9 (preprint 3.55) (with the added hypothesis that the good complement is closed); Theorem 3.6.8 (preprint 3.54) as described above, with its graph description; Theorem 3.7.1 (preprint 3.58) for a general sc-splitting (since the sixth addendum; first in the coordinates of its proof); Lemmas 3.7.2 and 3.7.3 (preprint 3.59 and 3.60) (in an abstract form, with their hypotheses derived in Theorem 3.7.1 (preprint 3.58)) and Lemma 3.7.4 (preprint 3.61); and the appendix, including, since the sixth addendum, Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11, 3.8.20 (preprint 3.62–3.64, 3.66–3.68, 3.76), which the formal proofs avoid (Lemma 3.8.1 for $P \neq \{0\}$; at $P = \{0\}$ it does not hold as we read it). Not verified: Proposition 3.6.5 (preprint 3.52) on T_xX itself, which carries no linear structure here; Proposition 6.2.2 (preprint 6.2) for paths sc-smooth on $[0, 1]$ only (the project has no M-polyfold $[0, 1]$); the passage from an atlas of tame charts to tame germ data (Definition 3.1.11 (preprint 3.11) ⇒ Definition 3.6.3 (preprint 3.50)), which is open. Remarks 3.1.8 and 3.1.23 (preprint 3.8 and 3.20) compare the definitions with the earlier paper [21] and make no claim to verify. Items V3.68–V3.90 were added after the first edition of this section (the Lean formalized after it); the earlier items whose content changed with them (V3.13, V3.23, V3.24, V3.34, V3.38, V3.42, V3.46–V3.54, V3.59, V3.62) are updated in place. Items V3.91–V3.102 were added in a second addendum (Theorem 3.1.21 (preprint 3.18), Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, the index of the solution set, the kernel as tangent space, good position at interior zeros); the items whose content changed with them (V3.46, V3.47, V3.68, V3.69, V3.81–V3.87) are updated in place. Items V3.103–V3.107 were added after the repair of the tameness hypothesis (Findings, Chapter 3, items 190–199: the tame germ notions, the non-tame retraction behind the vacuity, Theorems 3.1.18 and 3.1.21 (preprint 3.15 and 3.18) for tame germ data, good position in any chart of the germ data); the items whose statements changed with it (V3.45, V3.47, V3.48, V3.68–V3.70, V3.80–V3.82, V3.84–V3.87, V3.94, V3.95, V3.97–V3.102) are updated in place. A third addendum (sc-Fredholm theory on all levels, the one-germ form of Theorem 3.35 (preprint), Lemmas 3.1.9 and 3.8.25 (preprint 3.9 and 3.81) as stated, good position with the text's sc-complement, the index without good position, and the difference between the text's germs and the formal definitions on the whole chart) updated V3.5, V3.7–V3.9, V3.11–V3.13, V3.16, V3.19, V3.20, V3.22, V3.28, V3.31, V3.32, V3.43, V3.45–V3.47, V3.66, V3.68, V3.70, V3.86, V3.87, V3.94–V3.98 and V3.103 in place. The switch of Definition 3.1.11 (preprint 3.11) and of everything built on it to the text's germ reading (phase 2: the canonical notions and theorems are the germ reading, the project's earlier whole-chart ones are kept as `…WholeChart`, `…_wholeChart`; Findings, Chapter 3, items 330–334) updated V3.5, V3.7–V3.9, V3.13, V3.16, V3.17, V3.19, V3.43, V3.45–V3.47, V3.68, V3.78, V3.79, V3.81, V3.82, V3.91, V3.93, V3.96, V3.98, V3.102, V3.103, V3.105 and V3.107 in place. A fourth addendum (2026-11-02: Definition 3.1.3 (preprint 3.3) in the germ reading, Definition 6.2.1 (preprint 6.1) and Proposition 6.2.2 (preprint 6.2), germ data and the charts of a tame atlas; Findings, Chapter 3, items 340–355) updated V3.5, V3.13, V3.98 and V3.103 in place. A fifth addendum (2026-11-02: the restriction of a strong bundle to an open set, the index along the zero set and Proposition 6.2.2 (preprint 6.2) without bump functions, Definition 6.2.1 (preprint 6.1) with the text's local sections; Findings, Chapter 3, items 349, 360–367) updated V3.68 and V3.98 in place. A sixth addendum (2026-11-12; Findings, Chapter 3, items 386, 387) adds Remark 3.3.8 (preprint 3.36) in quantitative form (V3.122, closing gap G.19), Lemmas 3.8.1–3.8.3 (preprint 3.62–3.64) (V3.123, V3.124, closing gap G.25; Lemma 3.8.1 is proved for $P \neq \{0\}$ and does not hold as we read it at $P = \{0\}$), Lemmas 3.8.9–3.8.11 (preprint 3.66–3.68) with the text's notions (V3.125, gap G.26), Definition 3.8.14 (preprint 3.71) and its agreement with Mathlib's C^k and with sc^1 (V3.126, gap G.27), Lemma 3.8.20 (preprint 3.76) (V3.127, gap G.28), Lemma 3.8.13 (preprint 3.70) for any presentation of C, with $d_M = d_X$ near the centre in Theorem 3.2.3 (preprint 3.27) (V3.128, gap

G.29), and an sc-smooth family of projections whose rank jumps (V3.129); it updates in place V3.62 (Lemma 3.8.21 (preprint 3.77) as printed, with f only C^1), V3.68 (ker $f'(y) = T_yS(f,U)$ in T_yX, with an intrinsic linearization at the zeros) and V3.89 (Theorem 3.7.1 (preprint 3.58) for a general sc-splitting). It also settles gap G.79 (Findings, Chapter 3, item 388): with the 2021 germ data as printed, in any strong bundle chart, Theorem 3.6.8 (2) does not hold as we read it, by a tame strong bundle over $\mathbb{R}$ with an sc-Fredholm section whose solution set is a closed half-line (V3.130), and such a germ need not have tame germ data (V3.131); V3.85, V3.94, V3.103, V3.110 and V3.115 are updated in place.

The setting. Germs are fixed sequences of sets (Definition 3.1.1 (preprint 3.1)), and the calculus of germs — restriction, chain rule, pairing, Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs — is proved, not assumed. The finite-dimensional parameter spaces $\mathbb{R}^n$, $\mathbb{R}^N$ are arbitrary finite-dimensional normed spaces P, Q, and the parameter set $\widetilde{C}$ a convex set containing 0 with nonempty interior, which is closed or a partial quadrant where a theorem needs it. A "strong bundle chart" in Definitions 3.1.3, 3.1.6, 3.1.11 (preprint 3.3, 3.6, 3.11) is a chart compatible with the atlas. The fillings and the local sc⁺-sections of Definition 3.1.11 (preprint 3.11) are, as in the text, germs on sc-germs of neighbourhoods whose levels may shrink with m (the text's germ reading, since phase 2; V3.7, V3.9, V3.13); the project's earlier notions, with these data on the whole U of the chart (levels $U \cap E_m$), are stricter and are kept under the names …`WholeChart`. The sc-smooth section germs of Definition 3.1.3 (preprint 3.3), which no theorem assumes, are germs in the same sense since a fourth addendum (V3.5; Findings, Chapter 3, items 220–221, 330). Tameness of the strong bundle is nowhere used in §3.1. It enters §3.6 as the text's Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) put it: the germ data of f at a point, and its good position, lie in a strong bundle chart whose base retraction is tame (V3.103; "tame germ data" and "good position in a tame chart" below). It is used, for the chart at the zero x_0, through Proposition 2.5.3 (preprint 2.35), $(1 - Dr(0))E \subseteq E_0$, and, for the tangent germ property of the solution set, through the tameness of the model of the chart $\psi \circ \varphi$ (V3.71). The reading first formalized, that the base retractions of all charts compatible with the atlas are tame, cannot be met (V3.104, the retraction-level half machine-checked). Classical regularity in the appendix is Mathlib's C^k within the set, which on relatively open subsets of partial quadrants is the text's Definition 3.8.14 (preprint 3.71). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 3".

V3.1 (Definition 3.1.1 (preprint 3.1), sc-germs of neighbourhoods; `Polyfold.ScNbhdGerm`, `Polyfold.ScNbhdGerm.tangent`, `Polyfold.ScNbhdGerm.ofSet`, `Polyfold.IsQuadrantLike`, `Polyfold.IsPartialQuadrant.isQuadrantLike`). An sc-germ of neighbourhoods around 0 in $C \subseteq E_0$ is a family $U_m \subseteq E_m$, each of the form $O \cap C_m$ with O open in E_m, containing 0, with $\iota U_{m+1} \subseteq U_m$. Its tangent germ has levels $U_{m+1} \times E_m$. A relatively open $U \ni 0$ in C gives the germ $U_m = U \cap E_m$. C is quadrant-like if $0 \in C$ and every C_m is convex with nonempty interior; partial quadrants are. *Book:* Definition 3.1.1 (preprint 3.1), p. 89 (preprint p. 62). *Hidden hypotheses:* none. *Deviation (restated):* a germ is a fixed sequence, not a class up to shrinking; the text's "the size of the sets U_m does not matter" is not built in, and the theorems say on which germ they hold. $(U_{m+1})_m$ is a germ only when all levels coincide (finite-dimensional E).

V3.2 (Definition 3.1.2 (preprint 3.2), sc^k-germs; `Polyfold.IsSc0Germ`, `Polyfold.IsSc1GermWith`, `Polyfold.IsSc1Germ`, `Polyfold.IsSckGerm`, `Polyfold.IsScSmoothGerm`, `Polyfold.IsSck.isSckGerm`). f is an sc⁰-germ on (U_m) if $f(U_m) \subseteq F_m$ and $f : U_m \to F_m$ is continuous for every m; sc¹ if moreover, for a family of bounded $Df(x) : E_0 \to F_0$, $x \in U_1$, f is differentiable within U_1 at x (increments in E_1, remainder in F_0) with derivative $Df(x)$, and the tangent map $(x, h) \mapsto (f(x), Df(x)h)$ is an sc⁰-germ on the tangent germ; sc^{k+1} if sc¹ with sc^k tangent map; sc-smooth if sc^k for all k. An sc^k map on a relatively open $U \ni 0$ is an sc^k-germ on $(U \cap E_m)$. *Book:* Definition 3.1.2 (preprint 3.2), pp. 90–91 (preprint pp. 62–63). *Hidden hypotheses:* for quadrant-like C the derivative is unique on U_1. *Deviation:* none.

V3.3 (the calculus of germs; `Polyfold.IsSckGerm.congr`, `Polyfold.IsSckGerm.mono`, `Polyfold.ScNbhdGerm.MapsToGerm`, `Polyfold.IsSckGerm.comp`, `Polyfold.IsSc1Germ.comp`, `Polyfold.ScOperator.isScSmoothGerm`, `Polyfold.IsSckGerm.pair`, `Polyfold.ScNbhdGerm.preimage`). For quadrant-like sets: sc^k-germs depend only on their values on U_0 and restrict to smaller germs; if f is an sc^k-germ on (U_m), g an sc^k-germ on (V_m) and $f(U_m) \subseteq V_m$ for every m, then $g \circ f$ is an sc^k-germ, with $T(g \circ f) = Tg \circ Tf$ on the tangent germ (k = 1); sc-operators are sc-smooth germs on every germ; pairings of sc^k-germs are sc^k-germs; for an sc⁰-germ f with $f(0) = 0$ and a germ G' around 0 there is a germ inside (U_m) on which f maps into G' level by level. *Book:* not stated; used throughout §§3.3–3.6. *Hidden hypotheses:* the level-wise inclusion $f(U_m) \subseteq V_m$ for compositions. *Deviation (not in the book).*

V3.4 (Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs; `Polyfold.IsSc1GermWith.exists_levelDeriv`, `Polyfold.IsSc1GermWith.continuousOn_levelDeriv_clm`, `Polyfold.IsSc1GermWith.hasFDerivWithinAt_level`, `Polyfold.ScNbhdGerm.shift`, `Polyfold.IsSckGerm.shift`, `Polyfold.IsSckGerm.contDiffOn_levels`). For an sc¹-germ with derivative family D and $x \in U_{m+1}$, $D(x)$ restricts to bounded operators $E_m \to F_m$, continuous in x in the operator norm on U_{m+1}, and $f : U_{m+1} \to F_m$ is differentiable within U_{m+1} with these derivatives. $(U_{m+1})_m$ is a germ in E^1 around $0 \in C^1$, and an sc^k-germ on (U_m) is an sc^k-germ $(U_{m+1})_m \to F^1$. An sc^k-germ is C^k from U_{m+k} to F_m. *Book:* not stated for germs; the text applies Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) to germs (Lemma 3.1.9 (preprint 3.9): "recalling Proposition 1.17", which is 1.16; Lemma 3.3.4 (preprint 3.31); Corollary 3.3.7 (preprint 3.34)). *Hidden hypotheses:* none. *Deviation (not in the book):* the germ versions, with the proofs of Chapter 1.

V3.5 (Definition 3.1.3 (preprint 3.3), sc-smooth section germs; `Polyfold.StrongBundle.IsScSmoothSectionGerm`, `Polyfold.StrongBundle.IsChart`, `Polyfold.StrongBundle.IsScSmoothSection.isScSmoothSectionGerm`, `Polyfold.StrongBundle.isChart_congr`). (f, x_0) is an sc-smooth section germ of the strong bundle $P : Y \to X$ if x_0 is smooth and in some strong bundle chart compatible with the atlas, centred at x_0, the principal part **f** of f satisfies: $\mathbf{f} \circ r$ is an sc-smooth germ on some sc-germ of neighbourhoods G of 0 with $G_0 \subseteq U$. An sc-smooth section is an sc-smooth section germ at every smooth point that has a centred chart (with the germ $U \cap E_m$). *Book:* Definition 3.1.3 (preprint 3.3), p. 91 (preprint p. 63). *Hidden hypotheses:* "a strong bundle chart" means a chart compatible with the atlas (a chart of the atlas need not be centred);

the notion does not depend on the atlas within its class (`Polyfold.StrongBundle.isChart_congr`, V2.58). *Deviation (hypotheses explicit):* the chart, as just said. Since a fourth addendum the germ is the text's: $\mathbf{f} \circ r$ is an sc-smooth germ on some sc-germ of neighbourhoods G with $G_0 \subseteq U$, whose levels may shrink with m, as for Definitions 3.1.4, 3.1.6 and 3.1.11 (preprint 3.4, 3.6 and 3.11) since phase 2. Until then it was required on the whole U of the chart, which is stricter (a function C^k on $(-1/k, 1/k)$ for every k is an sc-smooth germ in the constant scale $\mathbb{R}$ and sc-smooth on no neighbourhood of 0, `Polyfold.exists_isScSmoothGerm_not_isScSmooth`), and the item was "narrower". No theorem assumes the definition (Findings, Chapter 3, items 220–221, 260).

V3.6 (the section algebra and the sum rule; `Polyfold.IsScSmooth.add`, `Polyfold.sectionAdd`, `Polyfold.StrongBundleChart.principal_sectionAdd`, `Polyfold.StrongBundleChart.isZeroAt_iff`, `Polyfold.StrongBundleChart.add_mem_K`, `Polyfold.StrongBundle.IsScSmoothSection.isScSmoothRetractMap`, `Polyfold.StrongBundle.IsScPlusSection.isScSmoothRetractMap_one`). Sums, differences and negatives of sc-smooth maps on relatively open subsets of partial quadrants are sc-smooth. The fibrewise sum $f + s$ of sections is a section, with principal part $\mathbf{f} + \mathbf{s}$ in every chart; $f(x) = 0$ iff $\mathbf{f}(\varphi(x)) = 0$; the fibres of K over O are linear. In every strong bundle chart compatible with the atlas, the representative $z \mapsto (z, \mathbf{f}(z))$ of an sc-smooth section is sc-smooth $O \to K$, and the representative $z \mapsto (z, \mathbf{s}_1(z))$ of an sc$^+$-section is sc-smooth $O \to K[1]$. *Book:* not stated; used in Theorems 3.1.13, 3.1.18 (preprint 3.12, 3.15). *Hidden hypotheses:* the compatibility on $[1]$ of Definition 2.6.4 (preprint 2.50) (its first use). *Deviation (not in the book).*

V3.7 (Definition 3.1.4 (preprint 3.4), fillers; `Polyfold.IsFillingGerm`, `Polyfold.fillDefect`, `Polyfold.fillDefect_isScSmoothGerm`, `Polyfold.IsFillerOf`, `Polyfold.StrongRetr.r`, `Polyfold.StrongRetr.`, `Polyfold.StrongRetr. CLM`). For a strong bundle retraction $R(u, h) = (r(u), \rho(u)h)$ on $U \lhd F$ with $O = r(U)$, $0 \in U$: g is a filling of f if g is an sc-smooth germ on an sc-germ of neighbourhoods G with $G_0 \subseteq U$ and (1) $f = g$ on O near 0; (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in O; (3) $y \mapsto [1 - \rho(r(y))]g(y)$ is an sc^1-germ on G with a derivative family Dk, and $Dk(0)$ restricts to a topological linear isomorphism $\ker Dr(0) \to \ker \rho(0)$. ($\rho(u)$ is a bounded operator, $\rho(x)$ a projection for $x \in O$.) *Book:* Definition 3.1.4 (preprint 3.4), p. 91 (preprint p. 64). *Hidden hypotheses:* the map in (3) is differentiable at 0 (true for every sc-smooth germ g, and $Dk(0)$ is unique). *Deviation (hypotheses explicit):* stated for the principal parts; tameness unused. Since the switch of Definition 3.1.11 (preprint 3.11) to the text's germ reading (phase 2) g and the map in (3) are germs whose levels may shrink with m; the stricter whole-chart filler (`Polyfold.IsFillerOf`, the map in (3) sc^1 on all of U, the sc-smoothness of g on U added in Definition 3.1.6 (preprint 3.6)) is kept for the whole-chart notions (Findings, Chapter 3, items 220–221, 261, 330). Formerly "narrower", for the whole-chart reading.

V3.8 (Proposition 3.1.5 (preprint 3.5) and the matrix of $Dg(0)$; `Polyfold.IsFillerOf.proposition_3_1_5`, `Polyfold.proposition_3_1_5_preprint`, `Polyfold.fillerRestrict`, `Polyfold.fillerRestrict_kerEquiv`, `Polyfold.fillerRestrict_cokerEquiv`, `Polyfold.StrongRetr.scDeriv_fiber_zero`, `Polyfold.IsFillerOf.deriv_comp_r_eq_zero`, `Polyfold.IsFillerOf.scDeriv_section_comp_r`, `Polyfold.IsScFredholm.restrict`, `Polyfold.`

`IsFillerOf.proposition_3_1_5_sc`, `Polyfold.IsFillerOf.filler_condition_three_sc`, `Polyfold.IsFillerOf.proposition_3_1_5_sc_converse`, `Polyfold.IsFillerOf.proposition_3_1_5_sc_iff`). Let g be sc-smooth on U and fill the section f of $K \to O$, with $g(0) = 0$. Then $D(\rho(r(\cdot))g)(0) = \rho(0)Dg(0)$, $(1 - \rho(0))Dg(0)Dr(0) = 0$, $D(f \circ r)(0) = Dg(0)Dr(0)$; and $Tf(0) := Dg(0)$ restricted to $Dr(0)E \to \rho(0)F$ is surjective iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, and has the same index ($\dim\ker - \dim\operatorname{coker}$). The linear algebra holds for any idempotents p, q and L with $(1-q)Lp = 0$ and $(1-q)L : \ker p \to \ker q$ a topological isomorphism. In sc form: if $Dg(0)$ is an sc-Fredholm sc-operator, then $Dr(0)E$, $\rho(0)F$, $\ker Dr(0)$ and $\ker \rho(0)$ are sc-subspaces, $Tf(0) : Dr(0)E \to \rho(0)F$ is sc-Fredholm with the index of $Dg(0)$ on every level, and $(1 - \rho(0))Dg(0) : \ker Dr(0) \to \ker \rho(0)$ is an sc-isomorphism (condition (3) on every level); for $Dg(0)$ an sc-operator, $Dg(0)$ is sc-Fredholm iff $Tf(0)$ is sc-Fredholm and condition (3) holds on every level. *Book:* the paragraph after Definition 3.1.4 (preprint 3.4) and Proposition 3.1.5 (preprint 3.5), pp. 91–93 (preprint pp. 64–65). *Hidden hypotheses:* $(u,h) \mapsto \rho(u)h$ differentiable at $(0,0)$; $g(0) = 0$. *Deviation (stronger):* the index equality without any finiteness assumption, via isomorphisms of kernels and cokernels instead of the text's factorization. The text's target "$(1 - D\rho(0))F$" is $(1 - \rho(0))F$. The sc form needs condition (3) only on level 0, as the text states it (the higher levels of $Tf(0)$ come from the regularity of $Dg(0)$); its converse needs condition (3) on every level and fails with level 0 alone (an example with the shift on a weighted ℓ^2, not formalized; Findings, Chapter 3, items 234, 237, 238). The same holds for a filling germ (V3.7), with $Dg(0)$ the value at 0 of any derivative family of g (`Polyfold.IsFillingGerm.proposition_3_1_5`, `Polyfold.IsFillingGerm.proposition_3_1_5_sc`, `Polyfold.IsFillingGerm.proposition_3_1_5_sc_iff`; Findings, Chapter 3, items 272, 289, 314).

V3.9 (Definition 3.1.6 (preprint 3.6), filled versions; `Polyfold.IsFillingGerm`, `Polyfold.IsFilledVersion`, `Polyfold.IsFilledVersion.isFillingGerm`, `Polyfold.IsFillingGerm.isFilledVersion`). A filling of the principal part **f** (V3.7) is an sc-smooth germ g on an sc-germ of neighbourhoods inside U, whose levels may shrink with m, filling **f** near 0; the centred chart is that of Definition 3.1.11 (preprint 3.11) (V3.13). *Book:* Definition 3.1.6 (preprint 3.6), p. 93 (preprint p. 66). *Hidden hypotheses:* none. *Deviation (restated):* the datum in a given chart, for the principal parts; the chart is quantified in Definition 3.1.11 (preprint 3.11). The project's earlier filled version (`Polyfold.IsFilledVersion`), g sc-smooth on all of U, is stricter for germs whose levels shrink with m (round-trip items V3.9, V3.43); it is exactly a filling sc-smooth on all of U (`IsFilledVersion.isFillingGerm`, `IsFillingGerm.isFilledVersion`; Findings, Chapter 3, items 220–221, 260–262, 330). Formerly "narrower", for the whole-chart reading.

V3.10 (Definition 3.1.7 (preprint 3.7), basic germs; `Polyfold.IsBasicGerm`, `Polyfold.IsSc0ContractionGerm`). For a finite-dimensional P, a set $\widetilde{C} \subseteq P$, a finite-dimensional Q and a germ G around 0 in $\widetilde{C} \oplus W$: f is a basic germ with contraction map B if f is an sc-smooth germ on G with $f(0) = 0$, the W-component of $f(a,w)$ is $w - B(a,w)$ on U_0, and B is an sc^0-contraction germ (V3.25). *Book:* Definition 3.1.7 (preprint 3.7), p. 94 (preprint p. 66) (Remark 3.1.8 (preprint 3.8): historical). *Hidden hypotheses:* the points of the contraction estimate lie in U_m. *Deviation (restated):* $\mathbb{R}^n$, $\mathbb{R}^N$ replaced by finite-dimensional spaces.

V3.11 (Lemma 3.1.9 (preprint 3.9); `Polyfold.ContractionGerm.norm_levelDeriv_snd_le`, `Polyfold.IsSc0ContractionGerm.deriv_zero_snd`, `Polyfold.IsSc0ContractionGerm.exists_norm_deriv_snd_le`, `Polyfold.IsSc0ContractionGerm.exists_norm_levelDeriv_snd_le`). For an sc^0-contraction germ B that is an sc^1-germ: (a) in the working form, $\|D_2B(a,w)\|_{L(W_m)} \leq \frac{1}{2}$ at points $(a,w) \in U_{m+1}$ with $a \in N(m+1) \cap \widetilde{C}$, $w \in O(m+1)$, $\iota w \in O(m)$; (b) $D_2B(0,0) = 0$; (c) for every $\varepsilon > 0$ there is $r > 0$ with $|D_2B(x)\zeta|_0 \leq \varepsilon|\zeta|_0$ for $x \in U_1$, $|x|_1 < r$, $\zeta \in W_0$; (d) as stated, for $\widetilde{C}$ quadrant-like: for every $\varepsilon > 0$ and m there is $r > 0$ with $|D_2B(x)\zeta|_m \leq \varepsilon|\zeta|_m$ for $\zeta \in W_m$ at every $x \in U_{m+1}$ with $|x|_m < r$. *Book:* Lemma 3.1.9 (preprint 3.9), p. 94 (preprint p. 67) (every ε, every level, $(a,w) \in E_{m+1}$ small in E_m; B sc-smooth). *Hidden hypotheses:* differentiability of $B : U_{m+1} \to W_m$ (Proposition 1.2.1 (preprint 1.16) for germs, V3.4). *Deviation (stronger):* sc^1 suffices, and the general statement is proved as stated (d); smallness on level $m+1$ is not needed, since the contraction estimate of Definition 3.3.1 (preprint 3.28) is applied at points small on level 0 (Findings, Chapter 3, item 215). Formerly "narrower", when only (a)–(c) were proved.

V3.12 (Proposition 3.1.10 (preprint 3.10); `Polyfold.proposition_3_1_10`, `Polyfold.proposition_3_1_10_scPlus`, `Polyfold.IsSc1GermWith.isScOperator_deriv_zero`, `Polyfold.ScBanach.scIndex_projWOp`, `Polyfold.StabilityData.exists_isFredholm_Dfs'`, `Polyfold.proposition_3_1_10_level`, `Polyfold.proposition_3_1_10_full`, `Polyfold.IsBasicGerm.levelDeriv_add_eq`, `Polyfold.IsBasicGerm.snd_levelDeriv_add`). Let $\widetilde{C}$ be a partial quadrant, h a basic germ, s an sc^1-germ into $(Q \oplus W)^1$ and $f = h + \iota s$ an sc^1-germ with derivative family Df. Then $Df(0)$ is an sc-operator, sc-Fredholm of index $\dim P - \dim Q$: it is $T_0 + S$ with $T_0(\delta a, \delta w) = (0, \delta w)$ and S an sc$^+$-operator. For the data of Theorem 3.4.2 (preprint 3.38), $D(h+s)(x) : E_0 \to F_0$ is Fredholm of index $\dim P - \dim Q$ for every $x \in U_1$ small on level 1. On every level (the second statement): for every m there is $\delta > 0$ such that at every $x \in U_{m+1}$ with $|x|_m < \delta$ the level-m operator $Df(x)_m : P \oplus W_m \to Q \oplus W_m$ is Fredholm of index $\dim P - \dim Q$ ($\widetilde{C}$ quadrant-like); $f(0) = 0$ is not used. *Book:* Proposition 3.1.10 (preprint 3.10), p. 95 (preprint p. 68). *Hidden hypotheses:* the finite-rank terms are sc$^+$ because $Dh(0)$ is an sc-operator at the smooth point 0. *Deviation (stronger):* both statements as in the text, with s and f only sc^1 and $f(0) = 0$ unused. The second statement by the text's proof on level m: $Df(x)_m = T_0 + S(x) + K(x)$ with $\|S(x)\| \leq \varepsilon$ by Lemma 3.1.9 (preprint 3.9) on level m (V3.11) and $K(x)$ compact; the W-component of $Df(x)_m$, which the text uses silently, follows from the uniqueness of derivatives within U_1 (Findings, Chapter 3, item 230). Formerly "narrower", the second statement on level 0 only.

V3.13 (Definition 3.1.11 (preprint 3.11), sc-Fredholm germs; `Polyfold.StrongBundle.IsScFredholmGerm`, `Polyfold.IsFillingGerm`, `Polyfold.IsScPlusGerm`, `Polyfold.IsConjugateToBasicGerm`, `Polyfold.IsStrongBundleIso`, `Polyfold.StrongBundle.IsScFredholmGermWholeChart`, `Polyfold.StrongBundle.IsScFredholmGermWholeChart.isScFredholmGerm`, `Polyfold.StrongBundle.isScFredholmGerm_iff_germG_wholeChart`). (f, x_0) is an sc-Fredholm germ if x_0 is smooth and in a strong bundle chart of P centred at x_0 the principal part has a filling g (V3.9: an sc-smooth germ on a germ of neighbourhoods inside

U), there is an sc⁺-germ s (`IsScPlusGerm`: $s : E_0 \to F_1$ an sc-smooth germ into F^1 on a germ inside U) with $\iota s(0) = g(0)$, and $k = g - \iota s$ is conjugated to a basic germ: strong bundle maps $\Psi : U \lhd F \to U' \lhd F'$ and Ψ' back, inverse to each other, with base map ψ, $\psi(0) = 0$, U' relatively open in $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant of a finite-dimensional space), a basic germ h on a germ inside U', and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0. *Book:* Definition 3.1.11 (preprint 3.11), p. 96 (preprint p. 69). *Hidden hypotheses:* none. *Deviation (restated):* the identity is required for $u \in U$; tameness unused; $C' = T\psi(0)C$ not imposed (it is proved, V3.48). The witnesses of the conjugation are also available as data (`Polyfold.ConjugationData`, V3.46). Since the switch to the text's germ reading (phase 2) the filling and the sc⁺-section are germs whose levels may shrink with m; Ψ on all of U is no restriction (on a relatively open $U_0 \subseteq U$ it is the same after restricting the chart on level 0). The project's earlier notion, with g, s sc-smooth on the whole U (`IsScFredholmGermWholeChart`), implies this one and is exactly this one with g, s sc-smooth on U; whether the two agree for sections sc-smooth on X is open, and no theorem depends on it (Findings, Chapter 3, items 220–221, 261–264, 330). The existential over charts is essential: a filling in a chart gives a topological isomorphism ker $Dr(0) \cong$ ker $\rho(0)$, a condition on the chart (`Polyfold.IsFillingGerm.nonempty_kerEquiv`), so a filling in one chart gives none in another unless these normal spaces match; a compatible tame chart of the trivial bundle carries no filling of any section (`Polyfold.TrivialBundle.not_isFillingGerm_evenBundleChart`; Findings, Chapter 3, items 350–352).

V3.14 (Theorem 3.1.13 (preprint 3.12); `Polyfold.theorem_3_1_13_preprint`, `Polyfold.theorem_3_1_13_local_subset`, `Polyfold.MPolyfold.continuousOn_levelChartSymm`). Let f be a section of a strong bundle $P : Y \to X$ ($P \circ f = \mathrm{id}$) with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. There are open sets $O(0) \supseteq O(1) \supseteq \cdots$ of X containing x_0 such that for every i the closure in X of $\{x \in O(i) \mid f(x) = 0\}$ is the image of a compact subset of X_i, with the level topology, under the inclusion $X_i \to X$. *Book:* Theorem 3.1.13 (preprint 3.12), p. 97 (preprint p. 70) (sc-smooth section of a tame strong bundle). *Hidden hypotheses:* none beyond those used. *Deviation (stronger):* only $P \circ f = \mathrm{id}$ of the section is used; no tameness. *Proof:* the local form (Theorem 3.5.1 (preprint 3.40) for any subset of the solution set), transported through chart, filler and conjugation, with the inverse chart $O_i \to X_i$ continuous for the level topology.

V3.15 (Corollary 3.1.15 (preprint 3.13); `Polyfold.corollary_3_1_15_preprint`, `Polyfold.corollary_3_1_15_section`, `Polyfold.tendsto_of_isCompact_of_tendsto_map`). (a) Under the hypotheses of V3.14, if $f(x_k) = 0$ and $x_k \to x_0$ in X, then for every m there are $y_k \in X_m$ with $\iota y_k = x_k$ for k large and $y_k \to y_0$ in X_m, $\iota y_0 = x_0$. (b) If f is an sc-Fredholm section and $x_k \to x$ are zeros, then x is a zero and a smooth point, and the same holds at x. *Book:* Corollary 3.1.15 (preprint 3.13), p. 97 (preprint p. 70). *Hidden hypotheses:* the germ property at the limit. *Deviation (restated):* the text's arbitrary limit x is covered for $x = x_0$ and, for sc-Fredholm sections, for every x.

V3.16 (Definition 3.1.16 (preprint 3.14), sc-Fredholm sections; `Polyfold.StrongBundle.IsScFredholmSection`, `Polyfold.StrongBundle.IsScFredholmSectionWholeChart`, `Polyfold.StrongBundle.IsScFredholmSectionWholeChart.isScFredholmSection`). f is an sc-Fredholm section of the strong bundle P if it is sc-smooth, regularizing ($x \in X_m$, $f(x) \in Y_{m,m+1}$ imply $x \in X_{m+1}$) and an sc-Fredholm germ (V3.13) at every smooth point. *Book:* Definition 3.1.16 (preprint 3.14), p. 97 (preprint p. 70). *Hidden hypotheses:* none. *Deviation:* none; the bundle need not be tame. Clause (3) is the text's germ reading since phase 2; the project's earlier whole-chart notion (`IsScFredholmSectionWholeChart`), stricter, implies it (Findings, Chapter 3, items 220–221, 330). Formerly "restated", for the whole-chart reading.

V3.17 (Theorem 3.1.18 (preprint 3.15); `Polyfold.theorem_3_1_18_preprint`, `Polyfold.StrongBundle.IsScSmoothSection.sectionAdd`, `Polyfold.StrongBundle.regularizing_sectionAdd`, `Polyfold.StrongBundle.IsScFredholmGerm.sectionAdd`). If f is an sc-Fredholm section and s an sc⁺-section of a strong bundle, the fibrewise sum $f + s$ is an sc-Fredholm section. *Book:* Theorem 3.1.18 (preprint 3.15), p. 98 (preprint p. 71) (tame bundle). *Hidden hypotheses:* the sum rule; the [1]-compatibility of charts (V3.6). *Deviation (stronger):* no tameness. In the germ reading of Definition 3.1.11 (preprint 3.11) (V3.13), with the text's proof: $\widetilde{t} = \widehat{s} \circ r$ is sc-smooth on the whole U, and a germ plus a whole-chart map is a germ on the same germ (V3.19; Findings, Chapter 3, item 300); the whole-chart form is `Polyfold.theorem_3_1_18_wholeChart`.

V3.18 (Lemma 3.1.19 (preprint 3.16); `Polyfold.lemma_3_1_19`). For a section s of $K \to O$: $s(r(x)) = s(x)$ for $x \in O$, and $\rho(r(u))s(r(u)) = s(r(u))$, $R(r(u), s(r(u))) = (r(u), s(r(u)))$ for $u \in U$. *Book:* Lemma 3.1.19 (preprint 3.16), p. 99 (preprint p. 72). *Hidden hypotheses:* s takes values in K. *Deviation (restated):* the sc⁺-regularity of $s \circ r$ is proved inside Theorem 3.1.18 (preprint 3.15) (V3.17).

V3.19 (Lemma 3.1.20 (preprint 3.17); `Polyfold.IsFillingGerm.add_section_germ`, `Polyfold.IsFillingGerm.add_section`, `Polyfold.lemma_3_1_20`). If g is a filling of $\mathbf{f}$ (V3.7) and s a section of $K \to O$, then $g + s \circ r$ is a filling of $\mathbf{f} + s$, with the same derivative family in (3): for s sc-smooth on U, on the same germ; for s an sc-smooth germ on G_s, on $G \cap r^{-1}(G_s)$. *Book:* Lemma 3.1.20 (preprint 3.17), p. 99 (preprint p. 72). *Hidden hypotheses:* none. *Deviation (stronger):* the regularity of $g + s \circ r$ is part of the conclusion, and of s only the sc-smoothness is used, not that it is sc⁺. This is the text's statement for fillings that are germs, proved in the project since the switch to the germ reading (phase 2); the whole-chart `Polyfold.lemma_3_1_20` (fillers in the sense of `IsFillerOf`, any section s, the regularity left to Theorem 3.1.18 (preprint 3.15)) neither implies it nor follows from it (round-trip item V3.19; Findings, Chapter 3, items 220–221, 273). Formerly "restated".

V3.20 (Definition 3.1.24 (preprint 3.21), good position; `Polyfold.IsGoodComplement`, `Polyfold.IsInGoodPosition`, `Polyfold.IsScGoodComplement`, `Polyfold.IsInGoodPositionSc`, `Polyfold.IsInGoodPosition.isInGoodPositionSc`, `Polyfold.isScGoodComplement_iff`, `Polyfold.isInGoodPositionSc_iff`). A subspace P is a good complement of N with respect to C if $E_0 = N \oplus P$ (algebraically) and for some $\varepsilon > 0$, $n \in N$, $p \in P$, $|p|_0 \leq \varepsilon |n|_0$ imply ($n \in C \iff n + p \in C$). N is in good position to C if $N \cap C$ has nonempty interior in N and N has a good complement. Definition 3.1.24 (preprint 3.21) as written (`IsScGoodComplement`: $E = N \oplus P$ an sc-splitting with the constant ε; `IsInGoodPositionSc`) implies this; on a partial quadrant a smooth finite-dimensional N in good position is in good position with an sc-complement, also inside any sc-subspace $T \supseteq N$, and for smooth N the text's good complements are the project's closed ones. *Book:* Definition 3.1.24 (preprint 3.21), pp. 102–103 (preprint pp. 76–77) (an sc-complement, or a topological one). *Hidden hypotheses:* none. *Deviation (stronger):* only the algebraic splitting is required, as a hypothesis. As a conclusion the algebraic form is weaker (an algebraic good complement need not be closed, round-trip item V3.20), so the text's form is formalized as well and proved where the theorems conclude good position (V3.47, V3.94; Findings, Chapter 3, items 210, 211).

V3.21 (Proposition 3.1.25 (preprint 3.22); `Polyfold.proposition_3_1_25`, `Polyfold.exists_quadrantNormalForm`). If the finite-dimensional subspace N is in good position to the partial quadrant C, then $N \cap C$ is a partial quadrant in N. More precisely, for $C = \{\ell_i \geq 0\}$ with continuous functionals jointly onto $\mathbb{R}^n$, there is a linear isomorphism $e : N \to \mathbb{R}^k \times N_0$ carrying $N \cap C$ onto $[0, \infty)^k \times N_0$, with $k = \dim N - \dim(N \cap W)$ and the degeneracy counts preserved off $N \cap W$. *Book:* Proposition 3.1.25 (preprint 3.22), p. 103 (preprint p. 77), proof pp. 143–153 (preprint pp. 120–126). *Hidden hypotheses:* none. *Deviation:* none in the statement. *Proof:* a segment argument and Lemma 3.8.8 (preprint 3.65) instead of the extreme-ray Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11 (preprint 3.62–3.64, 3.66–3.68).

V3.22 (Definition 3.2.1 (preprint 3.25), the tangent germ property; `Polyfold.MPolyfold.TangentGermChart`, `Polyfold.MPolyfold.HasTangentGermProperty`, `Polyfold.IsScInftyGermAt`). $M \subseteq X$ has the n-dimensional tangent germ property if $M \subseteq X_\infty$ and every $x \in M$ has: a chart $\varphi : (U, x) \to (O, 0)$ of X onto a tame retract (O, C, E); an n-dimensional smooth $N \subseteq E$ in good position to C with a good complement Y that is **closed**; a relatively open $V \ni 0$ in $N \cap C$; and a continuous $\delta : V \to E$ with values in Y, such that $\varphi(M \cap U) = \{v + \delta(v) \mid v \in V\}$, $\delta(0) = 0$, $D\delta(0) = 0$, and for every m, k there is a relatively open $V_{m,k} \ni 0$ in $N \cap C$ with $\delta(V_{m,k}) \subseteq E_m$ and $\delta : V_{m,k} \to E_m$ of class C^k. *Book:* Definition 3.2.1 (preprint 3.25), p. 104 (preprint p. 79). *Hidden hypotheses:* Y closed (the text's sc-complement is). *Deviation (narrower):* the added closedness; "sc^∞-germ" read with k-dependent neighbourhoods, which the fixed-germ and the set-sense readings imply (`Polyfold.IsScSmoothGerm.isScInftyGermAt`, `Polyfold.IsScSmooth.isScInftyGermAt`). The germ implicit function theorem gives the fixed-germ reading as well (V3.31, V3.32); Theorem 3.5.6 (1) (preprint 3.43) is proved in the k-dependent one (V3.38).

V3.23 (Proposition 3.2.2 (preprint 3.26); `Polyfold.proposition_3_2_2`, `Polyfold.proposition_3_2_2_image`, `Polyfold.MPolyfold.TangentGermChart.transport`). Let $\psi : X \to X'$ be an sc-diffeomorphism of M-polyfolds (sc-smooth on all of X, with an sc-smooth inverse ψ' on all of X') and $\psi(M) = M'$. Then M' has the n-dimensional tangent germ property (V3.22) if and only if M has it; the data at $\psi(x)$ are the chart $\varphi \circ \psi^{-1}$ with the same N, Y, V, δ. *Book:* Proposition 3.2.2 (preprint 3.26), p. 105 (preprint p. 80). *Hidden hypotheses:* none. *Deviation:* none (the tameness of X is not used). The text's "$X' = \psi(X)$ is tame" is not needed here; that tameness is carried by sc-diffeomorphisms is V3.71.

V3.24 (Theorem 3.2.3 (preprint 3.27); `Polyfold.theorem_3_2_3_of_isRegular`, `Polyfold.theorem_3_2_3_isSubMPolyfold`, `Polyfold.MPolyfold.TangentGermChart.IsRegular`, `Polyfold.MPolyfold.theorem_3_2_3_isChart`, `Polyfold.MPolyfold.theorem_3_2_3_`

`isTame`, `Polyfold.MPolyfold.theorem_3_2_3_isManifoldWithCorners`, `Polyfold.MPolyfold.theorem_3_2_3_degIndex`). Let $M \neq \emptyset$ be a subset of an M-polyfold X such that every point of M has data of Definition 3.2.1 (preprint 3.25) (V3.22) that are *regular*: (a) δ is sc-smooth on V as a map on a set, and (b) the projection π along Y maps O into C. Then M is a sub-M-polyfold of X; its induced M-polyfold structure (Proposition 2.3.19 (preprint 2.21)) is tame and has an equivalent atlas of charts on relatively open subsets of n-dimensional partial quadrants; each map $y \mapsto \pi\varphi(y)$ on $M \cap \varphi^{-1}(\Sigma \cap O)$, modelled on $(V, N \cap C, N)$, is a chart of the induced structure containing x; and $d_M(y) = d_{N \cap C}(\pi\varphi(y))$ there. *Book:* Theorem 3.2.3 (preprint 3.27), p. 106 (preprint p. 81), proof pp. 106–109 (preprint pp. 81–84). *Hidden hypotheses:* (a) and (b); the text's proof asserts both and each fails in an example (a smooth curve with a vertical tangent written over the horizontal axis; $C = [0, \infty)^3$, $N = \mathbb{R}(1, 1, 1)$, $Y = \{x + y - z = 0\}$). *Deviation (narrower):* the two added conditions; X need not be tame; the relation $d_{N \cap C} = d_X$ via Lemma 3.8.13 (preprint 3.70) is not transported to a general presentation of C. For the solution set of an sc-Fredholm section both conditions are theorems (V3.74, V3.77, V3.85); for any set with the tangent germ property, (a) follows from (b) after shrinking the data (V3.74). With a dimension that may vary from point to point: V3.72.

V3.25 (Definition 3.3.1 (preprint 3.28), sc⁰-contraction germs; `Polyfold.IsSc0ContractionGerm`, `Polyfold.ContractionGerm`, `Polyfold.IsSc0ContractionGerm.toContractionGerm`). For a germ G around 0 in $\widetilde{C} \oplus W$: B is an sc⁰-contraction germ if B is an sc⁰-germ, $B(0, 0) = 0$, and for every $\varepsilon > 0$ and m there is $r > 0$ such that the points (a, w) with $a \in \widetilde{C}$, $|a| < r$, $|w|_m < r$ lie in U_m and $|B(a, w) - B(a, w')|_m \leq \varepsilon |w - w'|_m$ for such (a, w), (a, w'). Every such B has a working form: open $N(0) \supseteq N(1) \supseteq \cdots \ni 0$ in P and $0 \in K(m) \subseteq O(m) \subseteq W_m$ (K closed, O open, $\iota K(m + 1) \subseteq K(m)$) with $\{a\} \times O(m) \subseteq U_m$, $B_m(a, \cdot)$ a $\frac{1}{2}$-contraction on $O(m)$ mapping $K(m)$ into itself, for $a \in N(m) \cap \widetilde{C}$. *Book:* Definition 3.3.1 (preprint 3.28), p. 109 (preprint p. 85). *Hidden hypotheses:* the points lie in U_m. *Deviation (restated):* explicit regions, needed for Lemma 3.3.4 (preprint 3.31).

V3.26 (Theorem 3.3.2 (preprint 3.29); `Polyfold.theorem_3_3_2`, `Polyfold.theorem_3_3_2_of_isSc0ContractionGerm`, `Polyfold.exists_fixedPoint_continuousOn`). For a contraction germ in working form with quadrant-like $\widetilde{C}$: the level-0 fixed point $\delta(a)$ of $B(a, \cdot)$ is an sc⁰-germ on $V_m = N(m) \cap \widetilde{C}$ with $\delta(0) = 0$, $(a, \delta_m(a)) \in U_m$ for $a \in V_m$, $B(a, \delta(a)) = \delta(a)$ on V_0, and every $w \in O(0)$ with $B(a, w) = w$, $a \in V_0$, equals $\delta(a)$. From Definition 3.3.1 (preprint 3.28) directly: the same, with uniqueness among small solutions. The parametrized Banach fixed point theorem: for $\frac{1}{2}$-contractions g_l of a nonempty closed K in a complete metric space, continuous in l pointwise, the fixed point is continuous in l and unique in K. *Book:* Theorem 3.3.2 (preprint 3.29), p. 110 (preprint p. 86). *Hidden hypotheses:* the uniqueness region. *Deviation (hypotheses explicit).*

V3.27 (Theorem 3.3.3 (preprint 3.30); `Polyfold.theorem_3_3_3`, `Polyfold.ContractionGerm.isSc1GermWith_sol`). If B is an sc¹-germ, the solution germ δ is a solution germ and an sc¹-germ on $V_m = N(m + 1) \cap \widetilde{C}$ (the germ shifted by one level), with $D\delta(a) = (1 - D_2 B(a, \delta(a)))^{-1} D_1 B(a, \delta(a))$. *Book:* Theorem 3.3.3 (preprint 3.30), p. 110 (preprint p. 86), quoted from [21] without proof. *Hidden hypotheses:* none. *Deviation (restated):* sc¹ on the shifted germ; a germ only because $\widetilde{C}$ is finite-dimensional. *Proof:* supplied; it needs $(a_0, \delta(a_0)) \in U_2$, hence the shift.

V3.28 (Lemma 3.3.4 (preprint 3.31); `Polyfold.ContractionGerm.lemma_3_3_4`, `Polyfold.ContractionGerm.next`, `Polyfold.ContractionGerm.isSc0ContractionGerm_next`). If B is sc^{k+1} with derivative family D and δ is sc^j on $(N(m+s) \cap \widetilde{C})_m$, then $f^{(1)}(v, b, u, w) = (u - B(v, u), w - DB(v, \delta(v))(b, w))$, in working form with parameter set $\widetilde{C} \times P$, fibre TW and regions $N(m+2+s) \times P$, $O(m+1) \times W_m$, $K(m+1) \times W_m$, is a contraction germ of class sc^{min(k, j)} whose solution germ is $(v, b) \mapsto (\delta(v), D\delta(v)b)$ on $(N(2+s) \cap \widetilde{C}) \times P$. If the working form also satisfies Definition 3.3.1 (preprint 3.28) and B is sc^1, then $f^{(1)}$ satisfies Definition 3.3.1 (preprint 3.28) on its germ, for every s (an ε-contraction near 0 for every $\varepsilon > 0$ on every level), by Lemma 3.1.9 (preprint 3.9) on level m (V3.11). *Book:* Lemma 3.3.4 (preprint 3.31), pp. 111 ff. (preprint pp. 86–87). *Hidden hypotheses:* uniqueness of the fixed point only in a region unbounded in the tangent directions; the shifted solution germ (Proposition 1.2.2 (preprint 1.17) for germs). *Deviation (restated):* the working form records a $\frac{1}{2}$-contraction on explicit regions, all the proof of Theorem 3.3.6 (preprint 3.33) uses; the text's "$f^{(1)}$ is an sc^0-contraction germ" in the sense of Definition 3.3.1 (preprint 3.28) is proved separately (Findings, Chapter 3, item 252).

V3.29 (Lemma 3.3.5 (preprint 3.32), as an induction; `Polyfold.ContractionGerm.sol_isSckGerm`, `Polyfold.ContractionGerm.isSckGerm_succ_of_next`). For every contraction germ over any finite-dimensional parameter space: if B is sc^k and $j \le k$, then δ is an sc^j-germ on $(N(m+s) \cap \widetilde{C})_m$ for some s; the step from j to $j+1$ turns shifts s (for δ) and s' (for $T\delta$) into $s + s' + 1$. *Book:* Lemma 3.3.5 (preprint 3.32), p. 111 (preprint p. 87) (the iterated germs $f^{(j)}$ with solution $T^j\delta$). *Hidden hypotheses:* none. *Deviation (restated):* the $f^{(j)}$ are not constructed; one induction quantified over all contraction germs.

V3.30 (Theorem 3.3.6 (preprint 3.33), the germ implicit function theorem; `Polyfold.theorem_3_3_6`, `Polyfold.theorem_3_3_6_smooth`, `Polyfold.theorem_3_3_6_of_isSc0ContractionGerm`). If B is an sc^k-germ, there is s such that the solution germ δ is a solution germ and an sc^k-germ on $(N(m+s) \cap \widetilde{C})_m$; the proof gives $s = 2^k - 1$. If B is sc-smooth, this holds for every k (with s depending on k). From Definition 3.3.1 (preprint 3.28) with balls: there is an sc^k solution germ. *Book:* Theorem 3.3.6 (preprint 3.33), p. 112 (preprint p. 88). *Hidden hypotheses:* **none; the theorem holds with the text's hypotheses**. *Deviation (restated):* the germ on which δ is sc^k shrinks with k; in the text's up-to-shrinking reading this is the theorem.

V3.31 (Corollary 3.3.7 (preprint 3.34); `Polyfold.corollary_3_3_7`, `Polyfold.corollary_3_3_7_isScSmoothGerm`, `Polyfold.isSckGerm_of_levels`). If B is an sc-smooth germ, then for all m, k there is a relatively open $V \ni 0$ in $\widetilde{C}$ with $\delta(V) \subseteq W_m$ and $\delta : V \to W_m$ of class C^k. The last sentence on one germ: δ is an sc-smooth germ (sc^k for every k on the same sequence of sets) on a diagonal germ $V_m = N(m + t_m) \cap \widetilde{C}$, $t_m = \sum_{j \le m} s_j$, and a solution germ there. *Book:* Corollary 3.3.7 (preprint 3.34), p. 113 (preprint p. 88). *Hidden hypotheses:* Proposition 1.2.3 (preprint 1.18) for germs. *Deviation:* none; its last sentence ("δ is sc-smooth at 0") holds on one germ: sc^j involves only the levels $\ge j$ of a germ (`Polyfold.isSckGerm_of_levels`), so the k-dependent germs of

Theorem 3.3.6 (preprint 3.33) can be replaced by one diagonal germ, which shrinks fast with the level (Findings, Chapter 3, item 250).

V3.32 (Theorem 3.35 (preprint); `Polyfold.theorem_3_35_preprint`, `Polyfold.theorem_3_35_smooth_preprint`, `Polyfold.theorem_3_35_isScSmoothGerm_preprint`, `Polyfold.theorem_3_35_of_isSc0ContractionGerm_preprint`, `Polyfold.isSckGerm_contractionMap_iff`). For $\widetilde{C}$ a partial quadrant in a finite-dimensional P: if B is sc^k, δ is sc^k on $(N(m+s) \cap \widetilde{C})_m$ for some s; if B is sc-smooth, then for every k there is such an s, and δ is an sc-smooth germ on one germ $N(m+t_m) \cap \widetilde{C}$. In the text's form: for an sc^0-contraction germ $f = w - B(a,w)$ of Definition 3.3.1 (preprint 3.28) on a quadrant-like $\widetilde{C}$ and any representative (V, δ) of its solution germ, if f is sc^k then δ is sc^k on a germ $V' \subseteq V$, and if f is sc-smooth then δ is an sc-smooth germ on one germ $V' \subseteq V$. *Book:* Theorem 3.35 (preprint), preprint p. 89. *Hidden hypotheses:* none. *Deviation (restated):* the conclusion holds on a smaller germ $V' \subseteq V$, as for Theorem 3.3.6 (preprint 3.33) (V3.30); the hypotheses are on f (f is sc^k iff B is); $\mathbb{R}^n$ is generalized to a quadrant-like $\widetilde{C}$ in a finite-dimensional P (Findings, Chapter 3, items 250, 251). Before the third addendum "sc-smooth" held only for every k on a germ depending on k.

V3.33 (Definition 3.4.1 (preprint 3.37), the basic class; `Polyfold.IsBasicGerm`). As V3.10. *Book:* Definition 3.4.1 (preprint 3.37), p. 114 (preprint p. 90). *Hidden hypotheses:* none. *Deviation:* none.

V3.34 (Theorem 3.4.2 (preprint 3.38); `Polyfold.theorem_3_4_2`, `Polyfold.StabilityData`, `Polyfold.StabilityData.exists_of_scPlus`, `Polyfold.StabilityData.pushMap`, `Polyfold.StabilityData.pushB_isSc0ContractionGerm`, `Polyfold.scPlusRemainder_contraction`, `Polyfold.StabilityData.pushMap_eq_zero_iff`, `Polyfold.StabilityData.finrank_Cs_eq_finrank_Zs`). Let f be a basic germ on G with contraction B, s an sc-smooth germ into W^1 with $s(0) = 0$ (the W-component of an sc$^+$-germ), and $W = C \oplus X = R \oplus Z$ an sc-Fredholm splitting of $1 + D_2 s(0)$ (it exists, Proposition 1.1.11 (preprint 1.10)). With $P_1 : W \to X$, $P_2 : W \to R$ the projections and $L = (1 + D_2 s(0))|_X$, the linear maps $\psi(a,w) = ((a,(1-P_1)w), P_1 w)$ and $\Psi(q,w) = ((q,(1-P_2)w), L^{-1}P_2 w)$ carry $f + \iota s$ to the germ $\Psi \circ (f + \iota s) \circ \psi^{-1}$ on the index-raised spaces $(P \oplus C) \oplus X^1 \to (Q \oplus Z) \oplus X^1$, with germ $\psi(U_{m+1})$, which is a basic germ with contraction map $L^{-1}P_2(B - S)$, $S = s - D_2 s(0)$; its zeros correspond to those of $f + \iota s$; and $\dim C = \dim Z$. *Book:* Theorem 3.4.2 (preprint 3.38), pp. 114–116 (preprint pp. 90–93). *Hidden hypotheses:* the operator-norm continuity of the level derivatives of s (for "S is an arbitrarily small contraction"). *Deviation (restated):* a linear coordinate change; only the W-component of s; the text's $\tau : Z \to C$ not applied. The coordinate change is a strong bundle isomorphism of the index-raised structures (`Polyfold.StabilityData.linIso_isStrongBundleIso`, V3.76).

V3.35 (Theorem 3.5.1 (preprint 3.40); `Polyfold.theorem_3_5_1_compactness`, `Polyfold.theorem_3_5_1_regularity`, `Polyfold.IsSc0ContractionGerm.exists_lift`, `Polyfold.theorem_3_1_13_local`). Let B be an sc^0-contraction germ on G around 0 in $\widetilde{C} \oplus W$ with $\widetilde{C}$ **closed**, $\mathbf{s}$ an sc^0-germ into W^1 with $\mathbf{s}(0) = 0$, and S any set of solutions in U_0 of $w = B(a,w) - \iota\mathbf{s}(a,w)$. There are open $O(0) \supseteq O(1) \supseteq \cdots \ni 0$ in E_0 and compact $K_i \subseteq E_i$ with $\mathrm{cl}_{E_0}(S \cap O(i)) = \iota_i(K_i)$; in particular these closures lie in E_i. For a basic germ h and an sc^0-germ s into $(Q \oplus W)^1$ with $s(0) = 0$, the closures of $\{h + \iota s = 0\} \cap O(i)$ are compact subsets of E_i. *Book:* Theorem 3.5.1 (preprint 3.40), p. 118 (preprint p. 94), proof pp. 118–123 (preprint pp. 95–98). *Hidden hypotheses:* $\widetilde{C}$ closed (automatic for a partial quadrant). *Deviation (stronger):* compact closures; B only sc^0 with the contraction estimate, s only sc^0 into W^1.

V3.36 (Corollary 3.5.3 (preprint 3.41); `Polyfold.StabilityData.corollary_3_5_3`, `Polyfold.StabilityData.exists_isFredholm_Dfs'`, `Polyfold.StabilityData.exists_ker_Dfs'_inf_eq_bot`, `Polyfold.isFredholm_add_of_norm_lt`, `Polyfold.HasFiniteIndex.surjective_of_finrank_ker_le`). For the data of V3.34 with $D(f + \iota s)(0)$ surjective: there is $r > 0$ such that every zero x of $f + \iota s$ in U_0 with $|x|_0 < r$ is $\iota(x_1)$ for some $x_1 \in U_1$, and $D(f + \iota s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$. *Book:* Corollary 3.5.3 (preprint 3.41), p. 122 (preprint p. 98). *Hidden hypotheses:* none. *Deviation (restated):* the $\mathbb{R}^N$-component of the sc$^+$-map is absorbed into the basic germ. *Proof:* the Fredholm property holds at every $x \in U_1$ small on level 1 (small plus compact perturbation of $D(f + \iota s)(0)$); surjectivity from $\dim\ker \leq \operatorname{ind}$.

V3.37 (Lemma 3.5.5 (preprint 3.42); `Polyfold.StabilityData.lemma_3_5_5`). For the push-forward $g = \Phi_*(f + \iota s)$ of V3.34 (on level 0): for every $\varepsilon > 0$ there is $r > 0$ such that every zero x' of g in $\psi(U_0)$ with $|x'|_0 < r$ has its X-component $\iota(w_1)$ with $w_1 \in X_1$, $|w_1|_1 < \varepsilon$. *Book:* Lemma 3.5.5 (preprint 3.42), p. 126 (preprint p. 101). *Hidden hypotheses:* none. *Deviation (restated):* the form Theorem 3.5.6 (preprint 3.43) uses; $w' = \delta(a')$ is derived there.

V3.38 (Theorem 3.5.6 (preprint 3.43); `Polyfold.theorem_3_5_6_sc_smooth`, `Polyfold.theorem_3_5_6`, `Polyfold.ShiftedBasicData`, `Polyfold.ShiftedBasicData.Sol.exists_contDiffOn_level`, `Polyfold.exists_proj_mem_of_eq_zero`). Let $\widetilde{C}$ be a partial quadrant, g an sc-smooth germ on G with $g(0) = 0$ whose index-raised germ is a basic germ, with $Dg(0)$ surjective, $K' = \ker Dg(0)$ in good position to $\widetilde{C} \oplus W$, and such that for every $\varepsilon > 0$ there is $r > 0$ with every zero x of g in U_0, $|x|_0 < r$, having W-component $\iota(w_1)$, $|w_1|_1 < \varepsilon$. Then there are a good complement Y of K', a relatively open $V \ni 0$ in $K' \cap (\widetilde{C} \oplus W)$ and $\tau : K' \to P \times W_0$ with values in Y and W-components on level 1, C^1 on V, $\tau(0) = 0$, $D\tau(0) = 0$, such that $y + \tau(y) \in U_0$ and $g(y + \tau(y)) = 0$ for $y \in V$; for every $k \geq 1$ a relatively open $V_k \subseteq V$ containing 0 and a C^k map $\tau_k : V_k \to E_k$ with $\iota_k \tau_k = \tau$ on V_k; and an open $U'' \ni 0$ in which every zero of g in U_0 is $y + \tau(y)$ with $y \in V$. *Book:* Theorem 3.5.6 (preprint 3.43), p. 126 (preprint p. 102). *Hidden hypotheses:* the regularity of Lemma 3.5.5 (preprint 3.42) (made a hypothesis); the projection of a solution to $\widetilde{K}$ lies in $\widetilde{C}$ (a gap in the text, proved); that the solutions given by Theorem 3.8.22 (preprint 3.78) at the different regularities k coincide near 0 (uniqueness clause). *Deviation (restated):* clause (1), "τ is an sc-smooth germ", in the level-wise sense with k-dependent neighbourhoods. With Y closed and its projection onto K' mapping $\widetilde{C} \oplus W$ into itself: V3.77.

V3.39 (Lemma 3.5.8 (preprint 3.44); `Polyfold.ShiftedBasicData.Deriv.ker_Dg_eq`). In the setting of V3.38, $\ker Dg(0) = \{(\alpha, D\delta(0)\alpha) \mid \alpha \in \ker DH(0)\}$, $H(a) = \operatorname{pr}_Q g(a, \delta(a))$. *Book:* Lemma 3.5.8 (preprint 3.44), p. 127 (preprint p. 103). *Hidden hypotheses:* $D_2 B(0) = 0$ on level 0, by density from level 1. *Deviation:* none.

V3.40 (Lemma 3.5.9 (preprint 3.45); `Polyfold.IsGoodComplement.map`, `Polyfold.IsInGoodPosition.map`). A continuous linear isomorphism $S : E_0 \to E'_0$ with

$x \in C \iff Sx \in C'$ carries a good complement P of N to a good complement $S(P)$ of $S(N)$ (constant $\varepsilon/(\|S\|\|S^{-1}\| + 1)$), and good position to good position. *Book:* Lemma 3.5.9 (preprint 3.45), p. 128 (preprint p. 104) (for $T\varphi(0)$). *Hidden hypotheses:* none. *Deviation (stronger):* any such S.

V3.41 (Lemma 3.5.10 (preprint 3.46); `Polyfold.lemma_3_5_10`, `Polyfold.lemma_3_5_10_isInGoodPosition`). If $K' = \{(\alpha, L\alpha) \mid \alpha \in \widetilde{K}\}$ has a good complement with respect to $\widetilde{C} \oplus W$, then $\widetilde{K}$ has a good complement Z with respect to $\widetilde{C}$ such that $Z \times W$ is a good complement of K'; good position of K' gives good position of $\widetilde{K}$. *Book:* Lemma 3.5.10 (preprint 3.46), p. 129 (preprint p. 104), proof pp. 155–161 (preprint pp. 129–132). *Hidden hypotheses:* none. *Deviation:* none (the proof avoids the intermediate complement Y_0).

V3.42 (Theorem 3.5.11 (preprint Corollary 3.47); `Polyfold.theorem_3_5_11_sc_smooth`, `Polyfold.theorem_3_5_11`). For $\widetilde{C}$ a partial quadrant, a basic germ f and the data of V3.34 with $D(f + \iota s)(0)$ surjective and kernel K in good position to $\widetilde{C} \oplus W$: there are a good complement Y of K, a relatively open $V \ni 0$ in $K \cap (\widetilde{C} \oplus W)$, and σ with values in Y and W-components on level 1, C^1 on V, $\sigma(0) = 0$, $D\sigma(0) = 0$, with $v + \sigma(v) \in U_0$ a zero for $v \in V$; for every $k \geq 1$ a C^k lift $\sigma_k : V_k \to E_k$ of σ on a relatively open $V_k \subseteq V$ containing 0; and an open $U'' \ni 0$ in which every zero in U_0 is of this form. *Book:* Theorem 3.5.11 (preprint Corollary 3.47), p. 131 (preprint p. 106). *Hidden hypotheses:* none. *Deviation (restated):* (1) level-wise, as in V3.38. *Proof:* Theorem 3.5.6 (preprint 3.43) for the push-forward, transported back linearly; Lemma 3.8.13 (preprint 3.70) and Theorem 3.8.15 (preprint 3.72) are not needed. With Y closed and its projection onto K mapping $\widetilde{C} \oplus W$ into itself: V3.77.

V3.43 (Definition 3.6.1 (preprint 3.48); `Polyfold.IsFillingGerm`, `Polyfold.IsFillerOf`). As V3.7. *Book:* Definition 3.6.1 (preprint 3.48), p. 134 (preprint p. 109). *Hidden hypotheses:* none. *Deviation:* none. Since the switch to the text's germ reading (phase 2) the filling is the germ notion of V3.7; the stricter whole-chart filler (`IsFillerOf`, with `IsFilledVersion`), for which "filled version = filling" on a given U is false (round-trip item V3.43), is kept (Findings, Chapter 3, items 220–221, 330). Formerly "narrower", for the whole-chart reading.

V3.44 (Remark 3.6.2 (preprint 3.49); `Polyfold.IsFillerOf.remark_3_6_2`). For a filler there is an open $W \ni 0$ with $f = g$ on $O \cap W$ and, for $y \in U \cap W$, $g(y) = \rho(r(y))g(y)$ only if $y \in O$. *Book:* Remark 3.6.2 (preprint 3.49), p. 135 (preprint p. 109). *Hidden hypotheses:* none. *Deviation:* none.

V3.45 (Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51); `Polyfold.StrongBundle.IsScFredholmGerm`, `Polyfold.StrongBundle.IsScFredholmSection`, `Polyfold.StrongBundle.IsScFredholmGermTame`, `Polyfold.StrongBundle.IsScFredholmSectionTame`, `Polyfold.StrongBundle.IsScFredholmGermTameWholeChart`). As V3.13 and V3.16, with the chart of the germ data tame: V3.103. *Book:* Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51), p. 135 (preprint p. 110) (for a tame strong bundle; the chart of Definition 3.6.3 (preprint 3.50) has a tame local strong bundle $K \to O$). *Hidden hypotheses:* none. *Deviation:* none. The forms with a tame chart (V3.103) are the ones the theorems of §3.6 use; since the switch to the text's germ reading (phase 2) they take the filling and the sc^+-section as germs, as the text does, and the earlier whole-chart forms (…`WholeChart`), which imply them (round-trip item V3.45), are kept (Findings, Chapter 3, items 220–221, 330). The forms of V3.13 and V3.16 drop the tameness, which §3.1 does not need. Formerly "narrower", for the whole-chart reading.

V3.46 (Proposition 3.6.5 (preprint 3.52); `Polyfold.proposition_3_6_5_index`, `Polyfold.proposition_3_6_5_local_index`, `Polyfold.ConjugationData`, `Polyfold.ConjugationData.scDeriv_isFredholm_index`, `Polyfold.ConjugationData.index_eq`, `Polyfold.proposition_3_6_5_preprint`, `Polyfold.proposition_3_6_5_local`, `Polyfold.proposition_3_6_5_sc`, `Polyfold.proposition_3_6_5_local_sc`, `Polyfold.ConjugationData.scDeriv_isScFredholm`, `Polyfold.StrongBundle.linAt_isScFredholm`). Let f be a section of a strong bundle with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. In the strong bundle chart of Definition 3.1.11 (preprint 3.11) centred at x_0, with the filling germ g (a derivative family Dg on a germ inside U) and the local sc⁺-germ s for which $g - s$ is conjugated to a basic germ, $Dg(0) : E_0 \to F_0$ and $Tf(x_0) := Dg(0)|_{Dr(0)E_0} : Dr(0)E_0 \to \rho(0)F_0$ are Fredholm operators, and for **every** conjugation of $g - s$ to a basic germ $h : O(\widetilde{C} \oplus W, 0) \to \mathbb{R}^N \oplus W$, $\widetilde{C} \subseteq \mathbb{R}^n$, both have index $n - N$ (`ConjugationData`: the witnesses of a conjugation, Definition 3.1.11 (preprint 3.11), as data). In particular $n - N$ is the same for all conjugations of one germ: for k sc-smooth with $k(0) = 0$ and any conjugation of k to a basic germ, $Dk(0)$ is Fredholm of index $n - N$. They are sc-Fredholm operators: $Dg(0) : E \to F$ is an sc-operator, $T_0O = Dr(0)E$ and $K_0 = \rho(0)F$ are sc-subspaces, and $Tf(x_0) : T_0O \to K_0$ is sc-Fredholm, both of index $n - N$ on every level; for an sc-Fredholm section the same holds in every strong bundle chart around a zero y, at the point z over y, for $\mathrm{Lin}(z) : T_zO \to K_z$. *Book:* Proposition 3.6.5 (preprint 3.52), p. 135 (preprint p. 110) ($Tf(x) : T_xX \to Y_x$ is sc-Fredholm); the index $n - N$ is used in the proof of Theorem 3.6.8 (preprint 3.54). *Hidden hypotheses:* none. *Deviation (restated):* in charts (T_xX carries no linear structure here, Chapter 2). The hypothesis is the text's since the switch to the germ reading (phase 2): the filling and the sc⁺-section are germs (V3.13), and $Dg(0)$ is the value at 0 of any derivative family of the filling germ; the forms for germ data on the whole chart are the `_wholeChart` declarations of the same names (Findings, Chapter 3, items 220–221, 289, 310, 331). Formerly "narrower", for that hypothesis. The sc-Fredholm property, first proved on level 0 only, follows the text's argument with two steps it leaves out: $D(g - s)(0) = \Gamma(0)^{-1}Dh(0)D\psi(0)$ is sc-Fredholm by the criterion of V1.8, and T_0O, K_0 are sc-subspaces (Definition 2.6.1 (preprint 2.47) at the smooth point 0, the closed graph theorem); in other charts the levels are transported by the chart change at smooth points (Findings, Chapter 3, items 235, 239). The first version (`Polyfold.proposition_3_6_5_preprint`, kept for its consumers) states the index as "$n - N$ for some natural numbers n, N", which carries no information; `proposition_3_6_5_index` identifies n and N with the dimensions of the conjugating basic germ (Findings, Chapter 3, item 120). In every other strong bundle chart around x_0 the linearization is Fredholm of the same index (V3.96).

V3.47 (Definition 3.6.6 (preprint 3.53); `Polyfold.IsGoodPositionAtGerm`, `Polyfold.IsGoodPositionAt`, `Polyfold.IsInGoodPositionIn`, `Polyfold.StrongBundle.IsGoodPositionGerm`, `Polyfold.StrongBundle.IsGoodPositionGerm.forall_isGoodPositionAtGerm`, `Polyfold.IsScGoodComplementIn`, `Polyfold.IsInGoodPositionScIn`,

`Polyfold.IsGoodPositionAtScGerm`, `Polyfold.IsGoodPositionAtSc`, `Polyfold.StrongBundle.IsGoodPositionGermSc`, `Polyfold.StrongBundle.IsGoodPositionGermTame.isGoodPositionGermSc`, `Polyfold.StrongBundle.isGoodPositionGermSc_iff`). In a chart centred at the zero, with the filling germ g and $Dg(0)$ the value at 0 of a derivative family of g (the same for all): the germ is in good position if $Tf(0) = Dg(0)|_{T_0O} : T_0O \to K_0$ is onto and $\ker Dg(0)$ is in good position to $T_0O \cap C$ inside T_0O (interior of $\ker Dg(0) \cap C$ in $\ker Dg(0)$ nonempty, a good complement inside T_0O). For a section, (f, x_0) is in good position if it has the data of Definition 3.1.11 (preprint 3.11) at x_0 with the germ in good position in that chart. With the text's sc-complement (`IsGoodPositionAtSc`; for a section `IsGoodPositionGermSc`, in a tame chart) the kernel has a good sc-complement inside T_0O ($T_m = N_m \oplus P_m$ on every level); at a zero of an sc-Fredholm section with tame germ data the two forms agree. *Book:* Definition 3.6.6 (preprint 3.53), p. 136 (preprint p. 111). *Hidden hypotheses:* none. *Deviation (restated):* in a chart rather than in T_xX; no case distinction $d_X(x) \geq 1$, which is recovered as a theorem for tame germ data: at a zero with $d_X = 0$ good position in a tame chart is the surjectivity of the linearization (V3.100). At a zero the linearization and the two conditions do not depend on the chart (V3.78); Definition 3.6.6 (preprint 3.53) at a zero from a chart not centred there is V3.79, and in any chart of the germ data V3.107. The form in a tame chart, which the theorems of §3.6 assume and conclude, is V3.103. The germ data are the text's since the switch to the germ reading (phase 2, V3.13); the whole-chart forms are `IsGoodPositionGermWholeChart`, `IsGoodPositionGermScWholeChart` (Findings, Chapter 3, items 220–221, 280, 320, 330); that "an sc-splitting of T_0O", written in E, is an sc-splitting of the sc-Banach space T_0O is not formalized (item 214).

V3.48 (Theorem 3.6.8 (preprint 3.54), the graph description of the solution set; `Polyfold.theorem_3_6_8_graph`, `Polyfold.theorem_3_6_8_linear`, `Polyfold.theorem_3_6_8_pushforward`, `Polyfold.scDeriv_mem_iff_of_inverse`). Let f be a section of a strong bundle, with $f(x_0) = 0$ and (f, x_0) in good position in a tame chart (V3.103). Then there are: a tame strong bundle chart φ centred at x_0 (compatible with the atlas, with tame base retraction); an sc-smooth ψ on its U with $\psi(0) = 0$ into a relatively open U' of $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant), with an sc-smooth inverse ψ' on U'; the data of Theorem 3.4.2 (preprint 3.38) and a basic germ h' with $D(h' + \iota t)(0)$ surjective and kernel K in good position, K a smooth subspace; a closed good complement Y of K; a relatively open $V' \ni 0$ in $K \cap (\widetilde{C} \oplus W)$; σ with values in Y and W-components on level 1, C^1 on V', $\sigma(0) = 0$, $D\sigma(0) = 0$, such that every $v + \sigma(v)$, $v \in V'$, is a zero of $h' + \iota t$ in its germ, and for every $k \geq 1$ a C^k-lift $\sigma_k : V_k \to E'_k$ of σ on a relatively open $V_k \ni 0$, $V_k \subseteq V'$ (Theorem 3.5.11 (1) (preprint Corollary 3.47)); and an open $V_x \ni x_0$ in the chart domain on which $\psi\varphi$ is injective, $f(y) = 0 \iff (h' + \iota t)(\psi\varphi(y)) = 0$, and $\psi\varphi(\{y \in V_x \mid f(y) = 0\}) = \{v + \sigma(v) \mid v \in V'\}$. On the way: in the chart, $Dg(0)$ is surjective with kernel in good position to C in E, $x \in C \iff D\psi(0)x \in C'$, and $D(\Psi_* g)(0) = \Gamma(0) Dg(0) D\psi(0)^{-1}$ is surjective Fredholm with kernel in good position. *Book:* Theorem 3.6.8 (preprint 3.54), p. 136 (preprint p. 111), the step (3) of its proof, pp. 137–139 (preprint pp. 113–114). *Hidden hypotheses:* the tameness of the chart at x_0, used only as $(1 - Dr(0))E \subseteq E_0$; the neighbourhood V made explicit; the kernel is smooth (the text: without comment). *Deviation (narrower):* only the graph description (3) of the proof; conclusions (1) and (2) of the theorem are

V3.82 and V3.85, the choice of Y with the projection property V3.77.

V3.49 (Lemma 3.6.9 (preprint 3.55); `Polyfold.lemma_3_6_9_preprint`, `Polyfold.lemma_3_6_9_core`). In $E = \mathbb{R}^n \oplus W$, $C = [0, \infty)^n \oplus W$: let N be finite-dimensional in good position with good complement Y, Y **closed**, and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$. There is $\rho > 0$ such that for $v \in N \cap C$ with $|v| < \rho$: $v + \tau(v) \in C$, and Y is a good complement of $N_v = \{n + D\tau(v)n \mid n \in N\}$ with respect to the partial quadrant $C_{v+\tau(v)}$ (so $E_0 = N_v \oplus Y$, both closed, with the good-complement constant independent of v). The core holds for any C cut out by coordinate functionals. *Book:* Lemma 3.6.9 (preprint 3.55), p. 139 (preprint p. 114), proof pp. 160–164 (preprint pp. 132–135). *Hidden hypotheses:* Y closed (needed for $D\tau(v)n \in Y$). *Deviation (narrower):* the added closedness; τ given on all of $N \cap C$. The form used in Theorem 3.6.8 (preprint 3.54) — for a relatively open $V \subseteq N \cap C$ and any presentation of C, with the good position of N_v — is `Polyfold.lemma_3_6_9_relOpen` (V3.80).

V3.50 (Theorem 3.6.10 (preprint 3.56) from the tangent germ property; `Polyfold.theorem_3_6_10_of_tangentGerm`). If X has no boundary and the nonempty zero set S of f has at every point regular data of Definition 3.2.1 (preprint 3.25) (V3.24), then S is a sub-M-polyfold whose induced structure is an n-dimensional manifold with corners without boundary. *Book:* Theorem 3.6.10 (preprint 3.56), p. 140 (preprint p. 115). *Hidden hypotheses:* the regularity conditions of V3.24. *Deviation (narrower):* assumes the tangent germ property of S, which the text derives from Theorem 3.6.8 (preprint 3.54). The theorem with the text's hypotheses is V3.86.

V3.51 (Theorem 3.6.11 (preprint 3.57) from the tangent germ property; `Polyfold.theorem_3_6_11_of_tangentGerm`). If the nonempty zero set S of f has at every point regular data of Definition 3.2.1 (preprint 3.25), then S is a sub-M-polyfold whose induced structure is tame and an n-dimensional manifold with boundary with corners. *Book:* Theorem 3.6.11 (preprint 3.57), p. 140 (preprint p. 115). *Hidden hypotheses:* as V3.50. *Deviation (narrower):* as V3.50. The theorem with the text's hypotheses is V3.87.

V3.52 (Lemma 3.7.2 (preprint 3.59), abstract form; `Polyfold.lemma_3_7_2`, `Polyfold.FamilyCompactness`). Let $b \mapsto L(b)$ be a family of sc-operators $X \to Y$, b in a finite-dimensional space, with $(b, h) \mapsto L(b)h$ continuous on $O \times X_0$ for a set $O \ni 0$, $L(0)$ injective on X_0, and the compactness property on level 0: if $b_j \to b$ in O, (h_j) is bounded in X_0 and $L(b_j)h_j \to 0$, then (h_j) has a convergent subsequence. Then $L(b)$ is injective on X_0 for $b \in O$ near 0. *Book:* Lemma 3.7.2 (preprint 3.59), p. 141 (preprint p. 117) (for $L(b) = P_Y \circ Df(b)|_X$ in the setting of Theorem 3.7.1 (preprint 3.58)). *Hidden hypotheses:* none. *Deviation (restated):* for an abstract family; in the proof of Theorem 3.7.1 (preprint 3.58) its hypotheses are derived from those of the theorem for $L(b) = P_Y \circ Df(b, 0)|_X$ (`Polyfold.familyCompactness_derivFamOp`, `Polyfold.continuousOn_derivFamOp`, V3.89).

V3.53 (Lemma 3.7.3 (preprint 3.60), abstract form; `Polyfold.lemma_3_7_3`). For a compact set K of parameters, a family of sc-operators $L(b)$ with $(b, h) \mapsto L(b)_m h$ continuous on $K \times X_m$, every $L(b)_m$, $b \in K$, injective, and the compactness property on level m: there is $c_m > 0$ with $c_m |h|_m \leq |L(b)h|_m$ for all $b \in K$, $h \in X_m$. *Book:* Lemma 3.7.3 (preprint 3.60), p. 142 (preprint

p. 118). *Hidden hypotheses:* none. *Deviation (restated):* abstract form; the injectivity on level m is a hypothesis, which in Theorem 3.7.1 (preprint 3.58) comes from level 0 (`Polyfold.ScOperator.injective_lvl`, V3.89).

V3.54 (Lemma 3.7.4 (preprint 3.61); `Polyfold.lemma_3_7_4`). Let V be a germ of relatively open neighbourhoods of 0 in a partial quadrant $\widetilde{C}$ of a finite-dimensional G, and $L(v) : E \to F$ sc-isomorphisms such that $(v, h) \mapsto L(v)h$ is an sc-smooth germ on $V \oplus E$ (levels $V_m \times E_m$) and for every m there is $c_m > 0$ with $c_m|h|_m \leq |L(v)h|_m$ for $v \in V_0$, $h \in E_m$. Then $(v, k) \mapsto L(v)^{-1}k$ is an sc-smooth germ on $V \oplus F$. *Book:* Lemma 3.7.4 (preprint 3.61), p. 142 (preprint p. 118), quoted from [25] without proof. *Hidden hypotheses:* none (no operator-norm continuity of $v \mapsto L(v)$). *Deviation (restated):* for germs. *Proof:* supplied; the differentiability uses the compactness of $F_1 \to F_0$. Used in Theorem 3.7.1 (preprint 3.58) in set form (`Polyfold.famInv_isScSmooth_of`, V3.88).

V3.55 (Lemma 3.8.8 (preprint 3.65); `Polyfold.lemma_3_8_8`). For functionals ℓ_i cutting out C, a good complement P of N (constant ε) satisfies $\ell_i(p) = 0$ for $p \in P$ and every i vanishing at some nonzero point of $N \cap C$. *Book:* Lemma 3.8.8 (preprint 3.65), p. 150 (preprint p. 123). *Hidden hypotheses:* none. *Deviation:* none.

V3.56 (Proposition 3.8.12 (preprint 3.69); `Polyfold.proposition_3_8_12`). For N finite-dimensional in good position to $C = [0, \infty)^n \oplus W$ in $\mathbb{R}^n \oplus W$ and $x \in N \cap C$: $d_{N \cap C}(x) = d_C(x)$ if $x \notin N \cap W$, and $d_{N \cap C}(x) + \dim(N \cap W) = \dim N$ if $x \in N \cap W$. *Book:* Proposition 3.8.12 (preprint 3.69), p. 153 (preprint p. 126). *Hidden hypotheses:* none. *Deviation (restated):* (2) additively.

V3.57 (Lemma 3.8.13 (preprint 3.70); `Polyfold.lemma_3_8_13`). For N in good position to $C = [0, \infty)^n \oplus W$ with good complement Y and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$, and $d_{N \cap C}(v) = d_C(v + \tau(v))$ if $v \notin N \cap W$. *Book:* Lemma 3.8.13 (preprint 3.70), p. 154 (preprint p. 127). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* "close to 0" made explicit.

V3.58 (Theorem 3.8.15 (preprint 3.72); `Polyfold.theorem_3_8_15`). For C a partial quadrant in a finite-dimensional V, U a relatively open neighbourhood of 0 in C and f of class C^j, $j \geq 1$, on U with $f(U) \subseteq C$, $f(0) = 0$, $Df(0)C = C$ and $d_C(x) = d_C(f(x))$ on U: there are relatively open $U' \subseteq U$ and V' containing 0 with $f(U') = V'$, f injective on U', and an inverse g of class C^j on V'. *Book:* Theorem 3.8.15 (preprint 3.72), p. 165 (preprint p. 136) (smooth). *Hidden hypotheses:* none. *Deviation (stronger):* C^j; the degeneracy hypothesis used only as "interior iff interior".

V3.59 (Proposition 3.8.17 (preprint 3.73); `Polyfold.proposition_3_8_17`, `Polyfold.InvData`). For U relatively open and convex in C, $f : U \to C$ of class C^j, $j \geq 1$, $\|Df - 1\| \leq \frac{1}{2}$ on U, and x interior iff $f(x)$ interior: $f(U)$ is relatively open in C, f is injective on U, with an inverse of class C^j on $f(U)$. *Book:* Proposition 3.8.17 (preprint 3.73), p. 165 (preprint p. 137). *Hidden hypotheses:* none. *Deviation (stronger):* $0 \in U$, $f(0) = 0$ not needed; (3) weakened. With the hypotheses its proof uses, without the partial quadrant: V3.90.

V3.60 (Lemma 3.8.18 (preprint 3.74); `Polyfold.InvData.injOn`, `Polyfold.InvData.ball_inter_subset_image`, `Polyfold.InvData.image_relOpen`). Under the hypotheses of V3.59: $\frac{1}{2}|x - x'| \leq$

$|f(x)-f(x')| \le \frac{3}{2}|x-x'|$, f is injective, $B(x', 3r) \cap C \subseteq U$ implies $B(f(x'), r) \cap C \subseteq f(U)$, and $f(U)$ is relatively open in C. *Book:* Lemma 3.8.18 (preprint 3.74), p. 165 (preprint p. 137). *Hidden hypotheses:* the interior starting point with $f(x_0) \in B(y', r)$. *Deviation:* none in the statement (openness of f on subsets not stated separately).

V3.61 (Corollary 3.8.19 (preprint 3.75); `Polyfold.corollary_3_8_19`). Under the hypotheses of V3.59, $f : U \to f(U)$ is a homeomorphism. *Book:* Corollary 3.8.19 (preprint 3.75), p. 167 (preprint p. 139). *Hidden hypotheses:* none. *Deviation:* none.

V3.62 (Lemma 3.8.21 (preprint 3.77); `Polyfold.InvData.hasFDerivWithinAt_inv`). Under the hypotheses of V3.59 the inverse is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$. *Book:* Lemma 3.8.21 (preprint 3.77), p. 168 (preprint p. 140) (for any homeomorphism $f : U \to V$ with V relatively open and Df invertible). *Hidden hypotheses:* none. *Deviation (narrower):* under the hypotheses of Proposition 3.8.17 (preprint 3.73), which imply the lemma's. With the hypotheses its proof uses, without the partial quadrant: V3.90. Since the sixth addendum (2026-11-12) the lemma is proved as printed, with f only C^1, and $f \in C^j$ gives $g \in C^j$ (`Polyfold.lemma_3_8_21`, `Polyfold.lemma_3_8_21_contDiffOn`, `Polyfold.lemma_3_8_21_smooth`; row V3.62, now stronger); the form above is `Polyfold.lemma_3_8_21_printed`.

V3.63 (Theorem 3.8.22 (preprint 3.78); `Polyfold.theorem_3_8_22`, `Polyfold.ContractionData.exists_solution`). For a partial quadrant C in a finite-dimensional V, U a relatively open neighbourhood of 0 in C, $f : U \to F$ of class C^j, $j \ge 1$, $f(0) = 0$, $Df(0)$ onto, $K = \ker Df(0)$ in good position with good complement Y: there are a relatively open $V' \ni 0$ in $K \cap C$, $\tau : K \to Y$ of class C^j on V', $\varepsilon, \sigma > 0$ with (1) $\tau(0) = 0$, $D\tau(0) = 0$, $k + \tau(k) \in U$ and $f(k + \tau(k)) = 0$ for $k \in V'$; (2) $|\tau(k)| \le \sigma$ for $k \in V'$, $|k| \le \varepsilon$; (3) $k \in V'$, $|k| \le \varepsilon$, $y \in Y$, $|y| \le \sigma$, $k + y \in U$, $f(k+y) = 0$ imply $y = \tau(k)$. *Book:* Theorem 3.8.22 (preprint 3.78), p. 169 (preprint p. 141). *Hidden hypotheses:* $f(k + \tau(k)) = 0$ and $k + y \in U$, omitted in the text. *Deviation (stronger):* the two clauses; F any normed space; only convexity and interior of C used.

V3.64 (Lemma 3.8.23 (preprint 3.79); `Polyfold.ContractionData.exists_radii`, `Polyfold.Radii.eq_`). In the normalized data there are $\varepsilon, a, b > 0$, $\varepsilon a \le b$, with the region $R = \{|k| \le a, |y| \le b\} \cap C \subseteq U$ and $\|DB\| \le \varepsilon/(2(1+\varepsilon))$ on R; a fixed point y of $B(k, \cdot)$ with $(k, y) \in R$ is $\tau(k)$. *Book:* Lemma 3.8.23 (preprint 3.79), p. 170 (preprint p. 142). *Hidden hypotheses:* none. *Deviation:* none.

V3.65 (Lemma 3.8.24 (preprint 3.80); `Polyfold.Radii.mapsTo_ball`). $B(k, \cdot)$ maps $\{|y| \le \varepsilon|k|\}$ into itself for $k \in K \cap C$, $|k| < a$, and has a fixed point there. *Book:* Lemma 3.8.24 (preprint 3.80), p. 170 (preprint p. 143). *Hidden hypotheses:* the maximum is over $B \cap C$. *Deviation (hypotheses explicit).*

V3.66 (Lemma 3.8.25 (preprint 3.81); `Polyfold.Radii.norm_ _sub_le`, `Polyfold.Radii.norm_ _sub_le_global`, `Polyfold.Radii.norm_ _sub_le_eps`, `Polyfold.Radii.lipschitzOnWith_`). $|\tau(k)| \le \varepsilon|k|/2$, τ is continuous, and $|\tau(k) - \tau(k_0)| \le \frac{\varepsilon}{1+\varepsilon}|k - k_0| \le \varepsilon|k - k_0|$ for all $k, k_0 \in V_0$ (an older form with the constant $1 + 2\varepsilon$ is kept for the proofs that use it). *Book:* Lemma 3.8.25 (preprint 3.81), p. 171 (preprint p. 144) (constant ε). *Hidden hypotheses:* none. *Deviation (stronger):*

the sharper constant $\varepsilon/(1+\varepsilon)$. The text's argument uses the point $(k, \tau(k_0))$, which lies in the region of the contraction when $|k_0| \le 2|k|$; otherwise $|k| < 2|k_0|$ and the same argument with k and k_0 exchanged applies, a case distinction the text omits (Findings, Chapter 3, item 216). Formerly "narrower", with the constant $1+2\varepsilon$.

V3.67 (Lemma 3.8.26 (preprint 3.82); `Polyfold.Radii.contDiffOn_`). τ is differentiable on $V_0 = \{k \in K \cap C : |k| < a\}$ with $D\tau(k) = (1 - D_2 B(k, \tau(k)))^{-1} D_1 B(k, \tau(k))$ and of class C^j on V_0. *Book:* Lemma 3.8.26 (preprint 3.82), p. 172 (preprint p. 144) ("on a possibly smaller neighbourhood"). *Hidden hypotheses:* none. *Deviation (stronger):* no shrinking.

V3.68 (Theorem 3.1.22 (preprint 3.19), the implicit function theorem, interior case, first form; `Polyfold.theorem_3_1_22`, `Polyfold.theorem_3_1_22_linearization`, `Polyfold.MPolyfold.IsWithoutBoundary`, `Polyfold.theorem_3_1_22_linearization_sc`, `Polyfold.StrongBundle.linAt_isScFredholm`). Let X be an M-polyfold with $d_X \equiv 0$, f an sc-Fredholm section (V3.16) of a strong bundle $P : Y \to X$ with tame germ data (V3.103), $f(x_0) = 0$, and let (f, x_0) be in good position in a tame chart (V3.47, V3.103) **with a filler that satisfies condition (3) of Definition 3.1.4 (preprint 3.4) at the zeros of the filler near** 0 (V3.82). Then there are an open $U \ni x_0$ and $d \in \mathbb{N}$ such that every zero $y \in U$ is a smooth point with (f, y) in good position in a tame chart (in particular $f'(y)$ is onto), and $S(f, U) = \{y \in U \mid f(y) = 0\}$ is a sub-M-polyfold of X whose induced structure (Proposition 2.3.19 (preprint 2.21)) is equivalent to that of a d-dimensional manifold with boundary with corners and has degeneracy index 0 at every point. The first clause: for an sc-Fredholm section and a zero x_0 in good position, in the chart of Definition 3.1.11 (preprint 3.11) centred at x_0, $Dg(0)$ and $f'(x_0) = Tf(x_0) = Dg(0)|_{T_0 O} : T_0 O \to K_0$ are sc-Fredholm operators with the same index $n - N$ on every level, and $Tf(x_0)$ is onto on every level (`Polyfold.theorem_3_1_22_linearization_sc`; Proposition 3.6.5 (preprint 3.52) in sc form, V3.46); with no chart singled out, in every strong bundle chart around x_0 the linearization $\mathrm{Lin}(z) : T_z O \to K_z$ is an sc-Fredholm operator with the same index on every level (`Polyfold.StrongBundle.linAt_isScFredholm`, V3.96). *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74) ("$f'(x_0)$ surjective"; the local solution set is a manifold of dimension the Fredholm index, $f'(y)$ onto and $\ker f'(y) = T_y S(f, U)$). *Hidden hypotheses:* tameness of the bundle, read as Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) read it: germ data and good position in charts with tame base retraction (V3.103; the reading "every chart compatible with the atlas has a tame base retraction", under which this item was first proved, cannot be met, V3.104). *Deviation (narrower):* (i) the filler condition at the nearby zeros, through Theorem 3.6.8 (1) (preprint 3.54) (V3.82), which was an added hypothesis when this form was proved and is now a theorem: it holds whenever (f, x_0) is in good position in a tame chart (V3.102), so this form has the text's hypotheses after all, and `Polyfold.theorem_3_1_22'` states it without the condition (V3.95); (ii) the hypothesis "$f'(x_0)$ surjective" is read as good position (Definition 3.6.6 (preprint 3.53) in the chart form), which at $d_X = 0$ is the same (V3.100; the form with "$f'(x_0)$ surjective" is V3.101); (iii) d is existentially quantified here; "the dimension agrees with the Fredholm index" is V3.97; (iv) "$\ker f'(y) = T_y S$" in the chart is V3.99, and since the sixth addendum (2026-11-12) it holds in $T_y X$ at every point of $S(f, U)$ on level 1, for the intrinsic $f'(y) = \Phi_y^{-1} \circ \mathrm{Lin}(\varphi(y)) \circ T\varphi(y) : T_y X \to Y_y$, the same for any two strong bundle charts, with $T_y S$ read through Ti (`Polyfold.theorem_`

`3_1_22_book_ker`, `Polyfold.StrongBundle.linIntr`, `Polyfold.StrongBundle.linIntr_chart_independent`, `Polyfold.StrongBundle.ker_linIntr_eq_subTangentRange`; row V3.68, now restated); (v) $f'(x_0)$ in charts (V3.46); its sc-Fredholm property, first proved only on level 0, holds on every level. *Proof:* Theorem 3.1.26 (preprint 3.23) (V3.69) and Lemma 2.4.6 (preprint 2.26) ($d_S \leq d_X = 0$).

V3.69 (Theorem 3.1.26 (preprint 3.23), the implicit function theorem, boundary case; `Polyfold.theorem_3_1_26_of_nearby`, `Polyfold.theorem_3_1_26`, `Polyfold.MPolyfold.subPolyfold`, `Polyfold.MPolyfold.IsManifoldWithCorners`). Let f be an sc-Fredholm section of a strong bundle, $f(x) = 0$, (f, x) in good position in a tame chart (V3.47, V3.103), and let every zero in an open $V_{x_0} \ni x$ be in good position in a tame chart. Then there are an open U with $x \in U \subseteq V_{x_0}$ and $d \in \mathbb{N}$ such that every zero in U is a smooth point and $S(f, U) = \{y \in U \mid f(y) = 0\}$ is a sub-M-polyfold of X whose induced structure (Proposition 2.3.19 (preprint 2.21)) is tame and equivalent to that of a d-dimensional manifold with boundary with corners (an equivalent atlas of charts on relatively open subsets of d-dimensional partial quadrants). For f with tame germ data (V3.103) and the filler condition of V3.82 at x in place of the nearby good position (`Polyfold.theorem_3_1_26`): the same, and every zero in U is in good position in a tame chart. *Book:* Theorem 3.1.26 (preprint 3.23), p. 103 (preprint p. 77) ($f'(x)$ surjective with kernel in good position; $S(f, U)$ consists of smooth points and is a tame sub-M-polyfold with a uniquely determined structure of a manifold with boundary with corners, the induced one). *Hidden hypotheses:* tameness as in V3.68. *Deviation (narrower):* the good position of the nearby zeros (Theorem 3.6.8 (1) (preprint 3.54)) is a hypothesis of the first form, the filler condition at the nearby zeros (V3.82) of the second; both are now theorems (V3.94, V3.102), and the form with the text's hypotheses is V3.95, with the dimension as the Fredholm index V3.97; d is existentially quantified here; "uniquely determined" is the equivalence of atlases, not stated separately. *Proof:* regular data of Definition 3.2.1 (preprint 3.25) at every point of $S(f, U)$, of the same dimension by the local constancy (V3.85, V3.74), then Theorem 3.2.3 (preprint 3.27) (V3.24).

V3.70 (Corollary 3.1.27 (preprint 3.24); `Polyfold.corollary_3_1_27`, `Polyfold.corollary_3_1_27_sc`, `Polyfold.forall_isGoodPositionGermSc_iff`). Let f be an sc-Fredholm section of a strong bundle such that (f, x) is in good position in a tame chart (V3.103) at every zero x, and let $S = \{x \in X \mid f(x) = 0\}$ be nonempty. Then S is a sub-M-polyfold of X whose induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension (V3.72). The same with good position in the text's sense (sc-complement, V3.47) at every zero, which for an sc-Fredholm section with tame germ data is the same hypothesis (`corollary_3_1_27_sc`). *Book:* Corollary 3.1.27 (preprint 3.24), p. 104 (preprint p. 78). *Hidden hypotheses:* tameness as in V3.68; $S \neq \emptyset$ (for $S = \emptyset$ the formal charts have no values; the statement is then empty). *Deviation (hypotheses explicit):* **no hypothesis added**; the text names no dimension, the formal one is locally constant. It is Theorem 3.6.11 (preprint 3.57) (V3.87) without its last clause.

V3.71 (tameness is carried by sc-diffeomorphisms and by open subsets; `Polyfold.IsScDiffeoQuadrant`, `Polyfold.IsScDiffeoQuadrant.transport`, `Polyfold.IsScDiffeoQuadrant.scDeriv_transport`, `Polyfold.IsScSubspace.map`, `Polyfold.`

`IsScSplitting.map`, `Polyfold.IsScDiffeoQuadrant.isTame_transport`, `Polyfold.IsScDiffeoQuadrant.isTameRetract_image`, `Polyfold.ScRetraction.restrict_isTame`, `Polyfold.IsTameRetract.inter_isOpen`, `Polyfold.MChart.isTameRetract_image`). Let $\psi : U \to U'$ be an sc-diffeomorphism between relatively open subsets of partial quadrants (ψ sc-smooth on U, ψ' sc-smooth on U', mapping U and U' into each other, inverse to each other) and $r : U \to U$ a tame sc-smooth retraction onto O whose domain is U. Then $r' = \psi \circ r \circ \psi'$ is a tame retraction of U' onto $\psi(O)$, so $(\psi(O), C', E')$ is a tame retract; on the way $d_{C'}(\psi(x)) = d_C(x)$ (Theorem 1.3.5 (preprint 1.25)), $Dr'(y) = D\psi(x)Dr(x)D\psi'(y)$ at the level-1 points $y = \psi(x)$ with $x \in O$, and an sc-operator with an sc-operator inverse carries sc-subspaces and sc-splittings. The restriction of a tame retraction to $U \cap r^{-1}(W)$, W open, is tame; an open subset $O \cap W$ of a tame retract is a tame retract; the model $\varphi(s)$ of a chart restricted to an open s of its domain is tame if the model of φ is. *Book:* used without statement in the proof of Proposition 3.2.2 (preprint 3.26) ("$X' = \psi(X)$ is tame", preprint p. 80) and in the proof of Theorem 3.6.8 (preprint 3.54) (the chart $\psi \circ \varphi$ of a tame M-polyfold, preprint p. 114). *Hidden hypotheses:* condition (2) of Definition 2.5.2 (preprint 2.34) is transported at smooth points, where $D\psi(x)$ is an sc-isomorphism, with $D\psi(x)E_x \subseteq E'_{\psi(x)}$ (the one-sided sign argument). *Deviation (not in the book).*

V3.72 (Theorem 3.2.3 (preprint 3.27) with locally constant dimension; `Polyfold.theorem_3_2_3_loc_of_isRegular`, `Polyfold.MPolyfold.IsManifoldWithCornersLoc`, `Polyfold.MPolyfold.theorem_3_2_3_loc_isSubMPolyfold`, `Polyfold.MPolyfold.theorem_3_2_3_loc_isTame`, `Polyfold.MPolyfold.theorem_3_2_3_loc_isManifoldWithCornersLoc`, `Polyfold.MPolyfold.theorem_3_2_3_loc_atlas_equiv`). Let $M \neq \emptyset$ be a subset of an M-polyfold X such that every point of M has regular data of Definition 3.2.1 (preprint 3.25) (V3.22, conditions (a), (b) of V3.24) of some dimension, which may depend on the point. Then M is a sub-M-polyfold of X, its induced structure is tame, and the charts $y \mapsto \pi\varphi(y)$ of all these data form an atlas equivalent to the induced one, consisting of charts on relatively open subsets of partial quadrants of finite-dimensional spaces (`IsManifoldWithCornersLoc`: a manifold with boundary with corners of locally constant dimension); each of these charts is a chart of the induced structure containing its point. *Book:* Theorem 3.2.3 (preprint 3.27), p. 106 (preprint p. 81) (for a fixed dimension n). *Hidden hypotheses:* (a), (b) of V3.24. *Deviation (narrower):* the conditions (a), (b); the dimension may vary from point to point (same proof as V3.24).

V3.73 (the re-centring lemma; `Polyfold.recentre_contDiffOn_levels`, `Polyfold.finrank_le_of_interior_image_nonempty`, `Polyfold.exists_contDiffOn_inverse`). Let χ_1, χ_2 be charts of an M-polyfold X and $S \subseteq X$. Suppose that $\chi_1(S \cap V_{x_1}) = \{v + \sigma_1(v) \mid v \in V_1\}$ for an open V_{x_1} in the domain of χ_1, where K_1 is a finite-dimensional smooth subspace in good position to C_1 with a closed complement Y_1, V_1 is relatively open in $K_1 \cap C_1$, and σ_1 has values in Y_1 and is of class C^1 on all of V_1; that χ_2 is centred at a point $y \in V_{x_1} \cap V_{x_2}$ and $\chi_2(S \cap V_{x_2})$ is the graph of σ_2 over a relatively open $V_2 \ni 0$ in $K_2 \cap C_2$ (the same conditions on K_2, Y_2), with σ_2 continuous, $\sigma_2(0) = 0$, $D\sigma_2(0) = 0$ and C^k-lifts $\sigma_{2,k} : V_{2,k} \to E_{2,k}$ on relatively open neighbourhoods of 0 for every $k \geq 1$; and that $\chi_1(y) = v_0 + \sigma_1(v_0)$, $v_0 \in V_1$. Then $\dim K_1 = \dim K_2$, and for all m, k there is a relatively open $B' \ni v_0$ in $K_1 \cap C_1$, $B' \subseteq V_1$, on which σ_1 takes values in $E_{1,m}$ and is of class C^k into $E_{1,m}$. On the way: a map differentiable on a set whose image has nonempty interior does not lower the dimension (Sard's lemma in fixed dimension); and a C^k map between relatively open subsets of convex sets with nonempty interior in finite-dimensional spaces, with a continuous left inverse on its relatively open image and an invertible derivative at a point, has a C^k inverse near the image of that point. *Book:* the step "δ is an sc$^\infty$-germ near every point $v_0 \in V$" of the proof of Theorem 3.2.3 (preprint 3.27), preprint p. 84. *Hidden hypotheses:* σ_1 differentiable at v_0 (here C^1 on V_1; without it the step fails, V3.24); Y_1, Y_2 closed; the lifts at 0 (Theorem 3.5.11 (1) (preprint Corollary 3.47), V3.42). *Deviation (narrower):* for two graph descriptions of one set, with σ_1 of class C^1 on V_1. *Proof:* the linearization of $\beta = \pi_1 \circ \chi_1\chi_2^{-1} \circ (1 + \sigma_2)$ at 0 is injective by the chain rule for the transition and its inverse on level 1 and the tangency of the graph of σ_2 to the retract; it is onto by the dimension count, not by the symmetric chain rule (which would need σ_1 differentiable into level 1 at v_0); the inverse is C^k by V3.90, not by Theorem 3.8.15 (preprint 3.72) (which would need $d_C(x) = d_C(\beta(x))$).

V3.74 (condition (a) of `IsRegular` by re-centring; `Polyfold.MPolyfold.TangentGermChart.exists_isRegular_of_locally`, `Polyfold.MPolyfold.TangentGermChart.shrink`, `Polyfold.MPolyfold.TangentGermChart.exists_shrinkData`, `Polyfold.MPolyfold.TangentGermChart.shrink_isRegular`, `Polyfold.MPolyfold.TangentGermChart.exists_lift`). Let D be data of Definition 3.2.1 (preprint 3.25) (V3.22) of dimension n for $M \subseteq X$ at x with $\pi(O) \subseteq C$ (condition (b) of V3.24), and let every point of M in an open $U \ni x$ have data of Definition 3.2.1 (preprint 3.25) of some dimension. Then M has at x regular data (conditions (a) and (b) of V3.24) of dimension n whose chart domain lies in U, and every datum of Definition 3.2.1 (preprint 3.25) for M at a point of M in that domain has dimension n. (The proof shrinks D — the chart restricted to an open subset, V to a relatively open V' inside the neighbourhood of 0 on which δ is C^1, the same N, Y, δ — and applies V3.73 at every $v_0 \in V'$; the statement exports only the existence of the regular data.) *Book:* the proof of Theorem 3.2.3 (preprint 3.27), preprint p. 84; the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("proceeding as above"). *Hidden hypotheses:* the data at the nearby points. *Deviation (narrower):* condition (b) is a hypothesis. Read off the statement (not a separate Lean theorem): every point of a set with the n-dimensional tangent germ property has data, so *for such a set, data at x that satisfy (b) can be shrunk to regular data*; condition (a) of V3.24 follows from (b). For the solution set of an sc-Fredholm section (b) holds by the choice of the good complement (V3.75, V3.84).

V3.75 (good complements whose projection maps the partial quadrant into itself; `Polyfold.ProjectsInto`, `Polyfold.exists_goodComplement_projectsInto`, `Polyfold.exists_proj_goodComplement`, `Polyfold.exists_proj_quadCone`, `Polyfold.ProjectsInto.mapL`). `ProjectsInto C N Y` means: $z \in C$, $k \in N$, $z - k \in Y$ imply $k \in C$; for $E_0 = N \oplus Y$ the projection onto N along Y maps C into $N \cap C$. If the finite-dimensional subspace N is in good position to the partial quadrant C of the sc-Banach space E, there is a continuous projection $\pi : E_0 \to N$ with $\pi(C) \subseteq N \cap C$ whose kernel Y is a good complement of N; Y is closed and `ProjectsInto C N Y` holds. The property is carried by continuous linear isomorphisms S with $x \in C \iff Sx \in C'$. *Book:* not stated; the proof of Theorem 3.2.3 (preprint 3.27) (preprint p. 83) asserts that $\Sigma = \{v + y \in C \mid v \in V, y \in Y\}$ is relatively open in C, which fails for an arbitrary good complement (V3.24) and holds when $\pi(O) \subseteq C$. *Hidden hypotheses:* none. *Deviation (not in* *the book):* the construction keeps the coordinates of a set S of indices; when no coordinate vanishes at a nonzero point of $N \cap C$ and $N \cap C \neq N$, one coordinate is kept (the natural choice $S = \Sigma$ fails there: for $C = [0, \infty)^2$, $N = \mathbb{R}(1, 1)$, the projection along the good complement $\mathbb{R}(1, 2)$ maps $(0, 1)$ to $-(1, 1)$).

V3.76 (Corollary 3.4.3 (preprint 3.39); `Polyfold.corollary_3_4_3`, `Polyfold.corollary_3_4_3_isConjugate`, `Polyfold.ConjugationData.compIso_eq_strongShift`, `Polyfold.IsStrongBundleIso.shift`, `Polyfold.IsStrongBundleIso.comp`, `Polyfold.IsStrongBundleIso.pushforward_comp`, `Polyfold.StabilityData.linIso_isStrongBundleIso`, `Polyfold.ScOperator.isStrongBundleIso_prodMap`). Let U be relatively open in a partial quadrant of E with $0 \in U$, $g : U \to F$ sc-smooth and conjugated to a basic germ (V3.13), and $s : U \to F_1$ sc-smooth into F^1 with $s(0) = 0$. Then the level-1 map $(g + \iota s)^1 : U_1 \to F_1$ is conjugated to a basic germ on the shifted structures $E^1 \lhd F^1$. Explicitly, for a conjugation Φ of g to the basic germ h, with $t = \Phi[1]_* s$ ($t(0) = 0$) and $h' = h + (t_{\mathbb{R}^N}, 0)$, there are the data of Theorem 3.4.2 (preprint 3.38) (V3.34) for t_W such that: h' is a basic germ and $\Phi_*(g + s) = h + t$ near 0; the composite $\Gamma = \Lambda \circ \Phi^1$ of the index raise of Φ and the linear coordinate change Λ of Theorem 3.4.2 (preprint 3.38) is a strong bundle isomorphism of the shifted structures onto a relatively open subset of $(\widetilde{C} \oplus C) \oplus X^1$, with base map vanishing at 0 and $\Gamma_* = \Lambda_* \circ \Phi^1_*$; and $(\Gamma(u, (g+s)^1(u)))_2 = (\Lambda_*(h+t))^1(\gamma(u))$ for $u \in U_1$ near 0, a basic germ. The text's form "$\Gamma = \Psi \circ \Phi$, then raise the index" is the same map (`compIso_eq_strongShift`). On the way: a pair of sc-isomorphisms, the index raise of a strong bundle isomorphism and a composition of strong bundle isomorphisms are strong bundle isomorphisms, and push-forwards compose. *Book:* Corollary 3.4.3 (preprint 3.39), p. 116 (preprint p. 93). *Hidden hypotheses:* the index raise of a strong bundle isomorphism is one (used silently). *Deviation (restated):* "sc-germ" read as sc-smooth on U, s as an sc$^+$-map on all of U with $s(0) = 0$; the conclusion is a conjugation of $(g + s)^1$ on the shifted structures; the text's "strong bundle map Ψ" is an isomorphism; the $\mathbb{R}^N$-component of t is absorbed into h; the text's $\tau : Z \to C$ is not applied.

V3.77 (Theorems 3.5.6, 3.5.11 and 3.6.8 (3) (preprint 3.43, 3.47 and 3.54) with a closed good complement whose projection maps the quadrant into itself; `Polyfold.theorem_3_5_6_closed`, `Polyfold.theorem_3_5_6_proj`, `Polyfold.theorem_3_5_11_proj`, `Polyfold.theorem_3_5_11_closed`, `Polyfold.theorem_3_6_8_graph_proj`, `Polyfold.IsGoodComplement.prod_top_graph`, `Polyfold.ShiftedBasicData.exists_sol_of_goodComplement`). The conclusions of V3.38, V3.42 and V3.48 hold with a good complement Y that is closed and satisfies `ProjectsInto` (V3.75) for the partial quadrant ($\widetilde{C} \oplus W$, resp. $C' = \widetilde{C} \oplus W$): the projection onto the kernel along Y maps the partial quadrant into the kernel's cone. In Theorem 3.5.6 (preprint 3.43), $Y = Z \oplus W$ for a good complement Z of $\widetilde{K}$ chosen by V3.75: every good complement Z of $\widetilde{K}$ gives the good complement $Z \oplus W$ of $K' = \{(\alpha, L\alpha)\}$ (constant $\varepsilon/(\|L\| + 1)$), the data of the proof exist for a prescribed Z, and the projection property passes from $(\widetilde{K}, Z)$ to $(K', Z \oplus W)$; in Theorem 3.5.11 (preprint Corollary 3.47) both properties are carried by the linear homeomorphism of Theorem 3.4.2 (preprint 3.38); in Theorem 3.6.8 (3) (preprint 3.54) the shrinking of V' leaves Y unchanged. *Book:* Theorem 3.5.6 (preprint 3.43), p. 126 (preprint p. 102); Theorem 3.5.11 (preprint Corollary 3.47), p. 131 (preprint p. 106); Theorem 3.6.8 (preprint 3.54), step (3) of the proof, preprint p. 114. *Hidden hypotheses:*

none. *Deviation (stronger):* two additional conclusions; the closedness is the text's (its Y' is an sc-complement), the projection property is not in the text and is what makes condition (b) of Theorem 3.2.3 (preprint 3.27) hold for the solution set (V3.84).

V3.78 (the linearization at a zero in charts, and its chart independence; `Polyfold.linAt`, `Polyfold.rho_scDeriv_comp_r`, `Polyfold.linAt_eq_scDeriv`, `Polyfold.IsFilledVersion.fillerRestrict_eq_linAt`, `Polyfold.linAt_eq_germDeriv`, `Polyfold.IsFillingGerm.fillerRestrict_eq_linAt`, `Polyfold.linAt_chart_change`, `Polyfold.StrongBundle.linAt_chart_independence`, `Polyfold.IsInGoodPositionIn.transport`). In a strong bundle chart (retraction r on U onto O, fibre projections ρ, principal part $\mathbf{f}$), Lin$(z) = D(\mathbf{f} \circ r)(z)|_{T_zO}$, $T_zO = Dr(z)E$, at every level-1 point z of the retract. At a zero its values lie in $K_z = \rho(z)F$; it equals $Dg(z)$ on T_zO for every g sc^1 on U agreeing with $\mathbf{f}$ on O near z, and for every sc^1-germ g agreeing with $\mathbf{f}$ near z with z in its germ ($Dg(z)$ the value of its derivative family; Findings, Chapter 3, item 284); at a centred zero it is the $Tf(0)$ of Proposition 3.1.5 (preprint 3.5) (V3.8), also for a filling germ. For strong bundle charts c, c' whose transition is sc-smooth on $[0]$, a section with sc-smooth principal parts in both, and a zero y in both domains with level-1 points z, z' over $\varphi(y)$, $\varphi'(y)$: there are topological linear isomorphisms $A : T_zO \to T_{z'}O'$ and $\Gamma : K_z \to K'_{z'}$ with $\Gamma \circ \text{Lin}(z) = \text{Lin}'(z') \circ A$ and, on T_zO, $x \in C_z \iff Ax \in C'_{z'}$ (the partial quadrants at the two points). Hence for two strong bundle charts of P and an sc-smooth section, Lin(z) and Lin$'(z')$ are simultaneously onto, simultaneously Fredholm with the same index, and their kernels are simultaneously in good position to the partial quadrants inside T_zO, $T_{z'}O'$ (good position inside a subspace is carried by isomorphisms respecting the cones). *Book:* Definition 3.6.6 (preprint 3.53) (p. 136 (preprint p. 111)) presupposes it; Definition 2.5.8 (preprint 2.38) (p. 57 (preprint p. 42)) asserts the chart independence of T_x^RX, C_xX by Proposition 2.4.10 (preprint 2.30); nothing is said about $f'(x)$. *Hidden hypotheses:* the point is a zero (away from zeros the linearization is not chart independent). *Deviation (not in the book):* the chart form of what gap G.13 leaves open; of the compatibility of Definition 2.6.4 (preprint 2.50) only the sc-smoothness of the transition on $[0]$ is used; level-1 points suffice; no tameness.

V3.79 (Definition 3.6.6 (preprint 3.53) at a zero, from a chart not centred there; `Polyfold.StrongBundle.isGoodPositionGermTame_of_chart_dense`, `Polyfold.StrongBundle.isGoodPositionGerm_of_chart`, `Polyfold.StrongBundle.isGoodPositionGerm_of_linAt_surj`, `Polyfold.StrongBundle.isGoodPositionAtGerm_of_chart_dense`). Let f be an sc-smooth section, y a zero with tame germ data (V3.103), c a strong bundle chart of P with y in its domain (not necessarily centred), z a level-1 point over $\varphi(y)$, and g an sc^1-germ on a germ G of c with derivative family Dg and $z \in G_1$, agreeing with $\mathbf{f}$ on O near $\varphi(y)$. If the range of $Dg(z)|_{T_zO}$ contains $\rho(z)k$ for every smooth $k \in F_\infty$, and ker $Dg(z) \cap T_zO$ is in good position to $C_{\varphi(y)}$ inside T_zO, then (f, y) is in good position in a tame chart (V3.47, V3.103). Variants: with all of K_z in the range, for an sc-Fredholm germ, in the chart centred at y of Definition 3.1.11 (preprint 3.11); with the hypotheses on Lin(z) (V3.78). *Book:* the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("proceeding as above", at a nearby zero). *Hidden hypotheses:* the kernel hypothesis is on ker $Dg(z) \cap T_zO$, not on ker $Dg(z)$ (condition (3) of the filler holds at its centre only); z lies in the germ of g. *Deviation (not in the book):* a dense part of K_z in the range suffices, since the range of the linearization in the chart of the germ data is closed (Proposition 3.6.5 (preprint 3.52), V3.46), which is why the dense form is stated for tame germ data. In the germ reading since phase 2; the whole-chart forms are `Polyfold.StrongBundle.isGoodPositionGerm_of_chart_dense`, `Polyfold.StrongBundle.isGoodPositionGerm_of_linAt` and the `_wholeChart` declarations (Findings, Chapter 3, items 285, 301, 313).

V3.80 (property (4) of the proof of Theorem 3.6.8 (preprint 3.54) in the chart at x_0: Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) on the graph; `Polyfold.theorem_3_6_8_nearby`, `Polyfold.lemma_3_6_9_relOpen`, `Polyfold.StabilityData.exists_surjective_Dfs'_graph`). In the data of V3.48, for a section f of a strong bundle with $f(x_0) = 0$ and (f, x_0) in good position in a tame chart (V3.103), there are a presentation T of $C' = \widetilde{C} \oplus W$ and an open $V_{x,1} \ni x_0$ such that for every zero $y \in V_{x,1}$, with $\psi\varphi(y) = v + \sigma(v)$, $v \in V'$: $v + \sigma(v)$ lies on level 1 and $D(h+t)$ there is a surjective Fredholm operator of index $n - N$ ($n = \dim \mathbb{R}^n$, $N = \dim \mathbb{R}^N$ of the basic germ); $v + \sigma(v) \in C'$; and Y is a good complement of $N_v = \{k + D\sigma(v)k \mid k \in K\}$ with respect to the partial quadrant $C'_{v+\sigma(v)}$, and N_v is in good position to it. Lemma 3.6.9 (preprint 3.55) is proved in the form used: for N in good position to C (any presentation) with closed good complement Y, V relatively open in $N \cap C$, $\sigma : V \to Y$ of class C^1 on V with $\sigma(0) = 0$, $D\sigma(0) = 0$, for $v \in V$ near 0: $v + \sigma(v) \in C$, Y is a good complement of N_v with respect to $C_{v+\sigma(v)}$, and N_v is in good position to it. *Book:* the proof of Theorem 3.6.8 (preprint 3.54), property (4), preprint p. 114; Lemma 3.6.9 (preprint 3.55), p. 139 (preprint p. 114). *Hidden hypotheses:* Y closed (V3.49); that $N_v \cap C_z$ has interior in N_v, which Lemma 3.6.9 (preprint 3.55) does not state. *Deviation (narrower):* in the chart at x_0, for the tangent space N_v of the graph; its identification with the kernel at y is V3.81.

V3.81 (Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros, the kernel; `Polyfold.theorem_3_6_8_nearby_kernel`, `Polyfold.theorem_3_6_8_chart_kernel`, `Polyfold.StabilityData.graphSpace_eq_ker_Dfs'`, `Polyfold.IsStrongBundleIso.ker_scDeriv_eq_map`, `Polyfold.scDeriv_mem_partialQuadrantAtWith_iff`, `Polyfold.IsInGoodPosition.isInGoodPositionIn_of_le`, `Polyfold.IsStrongBundleIso.ker_deriv_eq_map_germ`). Let f be a section of a strong bundle, $f(x_0) = 0$, (f, x_0) in good position in a tame chart (V3.103). In a tame chart centred at x_0 (filling germ g with a derivative family Dg, conjugation covering ψ with inverse ψ', the germ $h + t$ with kernel K, closed good complement, graph $\sigma : V' \to Y$) there is an open $V_{x,2} \ni x_0$ such that for every zero $y \in V_{x,2}$, with $z_0 = \varphi(y)$, $\psi(z_0) = v + \sigma(v)$ and level-1 points $z_{0,1}$ over z_0, which lies in the germ of g, x_1 over $v + \sigma(v)$: $\mathbf{f} = g$ on O near z_0 and $g(z_0) = 0$; $D(h+t)(x_1)$ is a surjective Fredholm operator of index $n - N$; $N_v = \ker D(h+t)(x_1)$; $D(\Psi_* g)(x_1) = D(h+t)(x_1)$ and $\ker Dg(z_{0,1}) = D\psi'(x_1)(N_v)$; $Dg(z_{0,1})$ is onto; $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$; and $\ker Dg(z_{0,1})$ is in good position to C_{z_0} inside $T_{z_0}O$. *Book:* Theorem 3.6.8 (1) (preprint 3.54), the kernel half, “from Corollary 3.41 and Lemma 3.55”, preprint p. 114. *Hidden hypotheses:* the C^1-lift of σ into level 1 (Theorem 3.5.11 (1) (preprint Corollary 3.47)) for $N_v \subseteq \ker D(h+t)$; $\ker Dg(z_0) \subseteq T_{z_0}O$ comes from the tangency of the solution set to O, not from the filler (whose condition (3) holds at 0 only). *Deviation (narrower):* in the chart at x_0; **no added hypothesis**; the surjectivity of $f'(y)$ is not part of it (V3.93, V3.94). In the germ reading $z_{0,1}$ lies in the germ of the filling because the basic germ is taken on the points that ψ' maps into the germs of g and s (Findings, Chapter 3, items 276–278, 295).

V3.82 (Theorem 3.6.8 (1) (preprint 3.54) under the filler condition at the nearby zeros; `Polyfold.theorem_3_6_8_one`, `Polyfold.StrongBundle.IsGoodPositionGermFilled`, `Polyfold.exists_range_deriv_eq_ _of_normal_injective_germ`). `IsGoodPositionGermFilled` at x_0: x_0 is smooth and there are the data of Definition 3.6.6 (preprint 3.53) at x_0 in a tame chart (V3.47, V3.103: a centred chart with tame base retraction, a filling germ g with a derivative family Dg on a germ G, a local sc$^+$-germ s with $g - s$ conjugated to a basic germ, the germ in good position) and an open $N \ni 0$ such that at every level-1 point $z_1 \in G_1$ with $z_1 \in O \cap N$ and $g(z_1) = 0$, $Dr(z_1)b = 0$ and $\rho(z_1)Dg(z_1)b = Dg(z_1)b$ imply $b = 0$: condition (3) of Definition 3.1.4 (preprint 3.4), the injectivity of $(1-\rho)Dg$ on ker Dr, at the zeros of g near 0. Theorem: if f is an sc-Fredholm section with tame germ data (V3.103), $f(x_0) = 0$, and (f, x_0) is in good position in this strengthened sense, there is an open $V_x \ni x_0$ such that (f, y) is in good position in a tame chart (V3.47, V3.103) at every zero $y \in V_x$. The step it needs: at a zero z of a filler with $Dg(z)$ onto and condition (3) at z, $\rho(z)F \subseteq Dg(z)(T_zO)$ (Proposition 3.1.5 (1) (preprint 3.5) at z). *Book:* Theorem 3.6.8 (1) (preprint 3.54), p. 136 (preprint p. 111), proof preprint p. 114. *Hidden hypotheses:* the filler condition at the nearby zeros, added when this form was proved; it is a consequence of good position (V3.102), so the form has the text's hypotheses after all. *Deviation (narrower):* the condition is a hypothesis of the statement. Definition 3.1.4 (preprint 3.4) imposes (3) at 0 only (so does the 2017 version, Definition 3.13 there), and then the text's argument does not give the surjectivity of $f'(y)$: with respect to $E = T_{z_0}O \oplus \ker Dr(z_0)$, $F = K_{z_0} \oplus \ker \rho(z_0)$, $Dg(z_0)$ is block triangular and $Tf(z_0)$ is onto iff the block $(1-\rho(z_0))Dg(z_0)|_{\ker Dr(z_0)}$ is injective, which is the local constancy of the index along the solution set (Findings, Chapter 3, items 100, 157); both directions are formalized (V3.102). The surjectivity holds with Definition 3.1.4 (preprint 3.4) as stated, by an implicit function theorem at the nearby zero (V3.93), and Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses is V3.94.

V3.83 (the parameter extension of the proof of Proposition 6.2.2 (preprint 6.2): the obstruction and the tilted kernel; `Polyfold.mem_lineality_of_goodComplement_prod`, `Polyfold.sup_lineality_eq_top_of_goodComplement_prod`, `Polyfold.isInGoodPosition_extendedKernel`). (a) Let $C \subseteq V$ be closed under nonnegative scaling with $0 \in C$ and $N \subseteq V$ a subspace. For every good complement of $\mathbb{R} \times N$ with respect to $\mathbb{R} \times C$, the V-components of its vectors lie in the lineality space $C \cap (-C)$; so $\mathbb{R} \times N$ is in good position only if $N + (C \cap -C) = V$. (b) Let K, Y' be subspaces of $\mathbb{R}^n \oplus W$ and $j : \Lambda \to \mathbb{R}^n \oplus W$ linear, Λ finite-dimensional, with $K \cap j(\Lambda) = 0$, $(K + j(\Lambda)) \oplus Y' = \mathbb{R}^n \oplus W$, Y' in the lineality space of $\widetilde{C} \oplus W$ ($v + y \in \widetilde{C} \oplus W \iff v \in \widetilde{C} \oplus W$ for $y \in Y'$), and $K + j(\Lambda)$ meeting the interior of $\widetilde{C} \oplus W$. Then $\{((\lambda, a), w) \mid (a, w) + j(\lambda) \in K\}$ is in good position to $(\Lambda \times \widetilde{C}) \oplus W$. *Book:* not in Chapter 3; the linear algebra of the perturbation argument of the text's Proposition 6.2.2 (preprint 6.2), tried for the surjectivity of V3.82. *Hidden hypotheses:* none. *Deviation (not in the book):* the rest of the argument is not formalized; it would give the surjectivity only on a neighbourhood depending on Λ (Findings, Chapter 3, item 106). The surjectivity is proved by another argument, with one perturbation direction at a time and the implicit function theorem at the nearby zero (V3.93).

V3.84 (Theorem 3.6.8 (2) (preprint 3.54) at the centre: the data of Definition 3.2.1 (preprint 3.25) for the solution set at x_0; `Polyfold.theorem_3_6_8_tangentGermChart_proj`, `Polyfold.theorem_3_6_8_tangentGermChart`, `Polyfold.MChart.compDiffeo`, `Polyfold.MChart.compDiffeo_isChart`, `Polyfold.isScInftyGermAt_of_lifts`, `Polyfold.StabilityData.ker_Dfs_le_smooth`, `Polyfold.MPolyfold.TangentGermChart._mem_of_projectsInto`). Let f be a section of a strong bundle, $f(x_0) = 0$, (f, x_0) in good position in a tame chart (V3.103). Then there are an open $V_x \ni x_0$ and $d \in \mathbb{N}$ such that $\{y \in V_x \mid f(y) = 0\}$ has at x_0 data of Definition 3.2.1 (preprint 3.25) of dimension d (V3.22) with $\pi(O) \subseteq C$ (condition (b) of V3.24). The data: the chart $\psi \circ \varphi$ onto $\psi(O)$ (for a chart φ of X and an sc-diffeomorphism ψ between relatively open subsets of partial quadrants whose domain contains the retract O of φ, $\psi \circ \varphi$ is a chart of X when $\psi(O)$ is a retract), whose model is a tame retract by V3.71; $N = \ker D(h + t)(0)$, a smooth subspace; the closed good complement with the projection property and the map σ of V3.77; σ is an sc$^\infty$-germ in the sense of V3.22 because it has C^k-lifts into level k for every k. *Book:* the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("S possesses the d-dimensional tangent germ property", $d = \dim \ker f'(x)$). *Hidden hypotheses:* the kernel is a smooth subspace (the text takes it so without comment); $\psi(O)$ is tame (V3.71). *Deviation (narrower):* at x_0 only (the other points: V3.85); d is existentially quantified here (the proof takes $d = \dim \ker D(h + t)(0)$); that d is the Fredholm index of $f'(x_0)$ is V3.97.

V3.85 (Theorem 3.6.8 (2) (preprint 3.54), and Theorem 3.6.8 (preprint 3.54); `Polyfold.theorem_3_6_8_two`, `Polyfold.theorem_3_6_8`, `Polyfold.theorem_3_6_8_tangentGermChart_solutionSet`). Let f be a section of a strong bundle, $f(x_0) = 0$, (f, x_0) in good position in a tame chart (V3.103). (i) The solution set $S = \{f = 0\}$ has at x_0 data of Definition 3.2.1 (preprint 3.25) with $\pi(O) \subseteq C$. (ii) If moreover every zero in an open $V_{x_0} \ni x_0$ is in good position in a tame chart, there are $d \in \mathbb{N}$ and regular data (V3.24) of dimension d for S at x_0 whose chart domain lies in V_{x_0}, and every datum of Definition 3.2.1 (preprint 3.25) for S at a point of S in that domain has dimension d. (iii) Theorem 3.6.8 (preprint 3.54), first form: if f is an sc-Fredholm section with tame germ data and (f, x_0) is in good position with the filler condition of V3.82 (in a tame chart), there is an open $V_x \ni x_0$ such that every zero in V_x is in good position in a tame chart, and (ii) holds with $V_{x_0} = V_x$. *Book:* Theorem 3.6.8 (1), (2) (preprint 3.54), p. 136 (preprint p. 111). *Hidden hypotheses:* tameness as in V3.68. *Deviation (narrower):* (2) is proved from (1), with the nearby good position as a hypothesis; (1) under the filler condition of V3.82, which follows from good position (V3.102). With the text's hypotheses: V3.94; with the dimension as the index and its local constancy: V3.97, V3.98. "S is a sub-M-polyfold with the structure of a manifold with boundary with corners" follows from (ii) by Theorem 3.2.3 (preprint 3.27) (V3.69). The regularity conditions (a), (b) of Theorem 3.2.3 (preprint 3.27), hypotheses for general sets (V3.24), are theorems for the solution set (V3.74, V3.77).

V3.86 (Theorem 3.6.10 (preprint 3.56); `Polyfold.theorem_3_6_10`, `Polyfold.theorem_3_6_10_sc`). Let X be an M-polyfold with $d_X \equiv 0$, f an sc-Fredholm section of a strong bundle, with (f, x) in good position in a tame chart (V3.103) at every zero x, and $S = \{f = 0\}$ nonempty. Then S is a sub-M-polyfold of X whose induced structure is equivalent to that of a manifold with boundary with corners of locally constant dimension and has degeneracy index 0 at every point (a manifold without boundary). The same with good position in the text's sense (sc-complement, V3.47) at every zero, which for an sc-Fredholm section with tame germ data is the same hypothesis (`theorem_3_6_10_sc`). *Book:* Theorem 3.6.10 (preprint 3.56), p. 140 (preprint p. 115) ($f'(x)$ surjective at every zero). *Hidden hypotheses:* as V3.70. *Deviation (restated):* the hypothesis "$f'(x)$ surjective" is read as good position, which by the text's Definition 3.6.6 (preprint 3.53) is surjectivity at points with $d_X(x) = 0$; in the chart form the kernel condition is stated for the chart's partial quadrant, and that it is void when $d_X(x) = 0$ is V3.100; the form with "$f'(x)$ surjective" is V3.101. No other hypothesis is added. The tameness of the induced structure is in V3.87 and V3.101.

V3.87 (Theorem 3.6.11 (preprint 3.57); `Polyfold.theorem_3_6_11`, `Polyfold.theorem_3_6_11_dim`, `Polyfold.solutionSetDim`, `Polyfold.theorem_3_6_11_sc`). Let f be an sc-Fredholm section of a strong bundle, with (f, x) in good position in a tame chart (V3.103) at every zero x. If $S = \{f = 0\}$ is nonempty, S is a sub-M-polyfold of X, its induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension, and at every $x \in S$ there are regular data of Definition 3.2.1 (preprint 3.25) of a dimension $d(x)$ such that every datum at a point of S in their chart domain has dimension $d(x)$. Moreover S is the disjoint union of the sets S_n of points at which S has data of dimension n, each S_n is open and closed in S, and each nonempty S_n is a sub-M-polyfold whose induced structure is tame and equivalent to that of an n-dimensional manifold with boundary with corners. The same with good position in the text's sense (sc-complement, V3.47) at every zero, which for an sc-Fredholm section with tame germ data is the same hypothesis (`theorem_3_6_11_sc`). *Book:* Theorem 3.6.11 (preprint 3.57), p. 140 (preprint p. 115). *Hidden hypotheses:* as V3.70. *Deviation (stronger):* **the text's hypotheses**; the local constancy of the dimension and the parts S_n are not in the text. Two steps of the text's proof are replaced: the regularity of the solution germ away from the centre (V3.73, V3.74) and the relative openness of Σ (V3.75, V3.77). The dimension $d(x)$ is the Fredholm index of $f'(y)$ at the zeros near x (V3.97).

V3.88 (Theorem 3.7.1 (preprint 3.58), given the family of sc-isomorphisms; `Polyfold.theorem_3_7_1_of_family`, `Polyfold.DerivCondition`, `Polyfold.contrMap358_contraction`, `Polyfold.norm_sub_sub_le_of_hasFDerivWithinAt_segment`, `Polyfold.fiberIsoMap_isStrongBundleIso`, `Polyfold.famInv_isScSmooth_of`). In the coordinates of the text's proof — $E = \mathbb{R}^n \oplus X$ ($\mathbb{R}^n$ any finite-dimensional space with a partial quadrant $\widetilde{C}$), $C = \widetilde{C} \oplus X$, $F = \mathbb{R}^N \oplus Y$ — let U be relatively open in C with $0 \in U$ and $f : U \to F$ sc-smooth; let $V_o \ni 0$ be open in $\mathbb{R}^n$ with $(b, 0) \in U$ for $b \in V_o \cap \widetilde{C}$, and $L(b) : X \to Y$ sc-isomorphisms with $L(b)h = (Df(b, 0)(0, h))_Y$ on level 0 and lower bounds $c_m|h|_m \leq |L(b)h|_m$ for $b \in V_o \cap \widetilde{C}$, $h \in X_m$, every m; and let condition (3) hold (`DerivCondition`: for all m and $\varepsilon > 0$ there is $r > 0$ with $|[D_2 f(b, 0) - D_2 f(b, x)]h|_m \leq \varepsilon|h|_m$ for $b \in \widetilde{C}$, $|b| < r$, $x \in X_{m+1}$, $|x|_m < r$, $(b, 0), (b, x) \in U$, $h \in X_m$). Then on $U' = U \cap (V_o \oplus X)$: $k(b, x) = ((f(b, x) - f(0))_{\mathbb{R}^N}, L(b)^{-1}(f(b, x) - f(0))_Y)$ is a basic germ with contraction map $H(b, x) = x - L(b)^{-1}(f(b, x) - f(0))_Y$; $\Phi((b, x), (c, y)) = ((b, x), (c, L(b)^{-1}y))$ is a strong bundle isomorphism $U' \triangleleft (\mathbb{R}^N \oplus Y) \to U' \triangleleft (\mathbb{R}^N \oplus X)$ covering the identity; and $f - f(0)$ is conjugated to a basic germ. *Book:* the second half of the proof of Theorem 3.7.1 (preprint 3.58), pp. 142–145 (preprint pp. 118–120). *Hidden hypotheses:* the mean value inequality along a segment with the derivative bound in the weaker norm of X_m (the text's integral; Findings, Chapter 3, item

73). *Deviation (restated):* in the coordinates of the proof; the sc-smoothness of $(b, y) \mapsto L(b)^{-1}y$ is Lemma 3.7.4 (preprint 3.61) (V3.54) in set form.

V3.89 (Theorem 3.7.1 (preprint 3.58), the basic germ criterion; `Polyfold.theorem_3_7_1`, `Polyfold.CompactnessCondition`, `Polyfold.familyCompactness_derivFamOp`, `Polyfold.fredholmIndex_block`, `Polyfold.surjective_of_fredholmIndex_block`, `Polyfold.derivFamOp`). In the coordinates of V3.88, let U be relatively open in $\widetilde{C} \oplus X$ with $0 \in U$ and $f : U \to \mathbb{R}^N \oplus Y$ sc-smooth with: (1) at every $(b, 0) \in U$, $b \in \widetilde{C}$, $Df(b, 0)$ is sc-Fredholm with the sc-index of $Df(0)$; (2) $h \mapsto (Df(0)(0, h))_Y$ is bijective on level 0, and the compactness condition holds (`CompactnessCondition`: there is $r > 0$ such that for every m, if $b_j \to b_0$ in $\widetilde{C}$ with $(b_j, 0) \in U$ and $|b_0| < r$, (η_j) is bounded in X_m and $Df(b_j, 0)(0, \eta_j) = y_j + z_j$ with $y_j \to 0$ in F_m and (z_j) bounded in F_{m+1}, then (η_j) has a convergent subsequence in X_m); (3) `DerivCondition` (V3.88). Then there are an open $V_o \ni 0$ in $\mathbb{R}^n$ and sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b, 0)(0, h))_Y$ for $b \in V_o \cap \widetilde{C}$, for which the conclusions of V3.88 hold; in particular $f - f(0)$ is conjugated near 0 to a basic germ. On the way: Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) for this family, with their hypotheses derived from (2) (V3.52, V3.53); the index of a block operator, $\operatorname{ind} T = \dim \mathbb{R}^n + \operatorname{ind} L - \dim \mathbb{R}^N$ for $T : \mathbb{R}^n \oplus X \to \mathbb{R}^N \oplus Y$ Fredholm with block L of finite index, and an injective block is onto when $\operatorname{ind} T = \dim \mathbb{R}^n - \dim \mathbb{R}^N$. *Book:* Theorem 3.7.1 (preprint 3.58), p. 140 (preprint p. 116), proof pp. 140–145 (preprint pp. 116–120). *Hidden hypotheses:* the smallness of b in (2) uniform in the level m. *Deviation (narrower):* in the coordinates of the proof; the reduction of a general sc-splitting $E = B \oplus X$, $F = A \oplus Y$ to them is not formalized; (1) is needed only at the points $(b, 0)$; the text's "$B \supseteq \ker Df(0)$ and $Y = Df(0)X$" is replaced by the bijectivity it implies. Since the sixth addendum (2026-11-12) the theorem holds for a general sc-splitting $E = B \oplus X$ with the text's conditions (1)–(3), the target splitting $F = Y \oplus A$ being constructed as in the proof (`Polyfold.theorem_3_7_1_general`, `Polyfold.ScBanach.splitScIso`, `Polyfold.splitScIso_preimage`, `Polyfold.IsConjugateToBasicGerm.of_scIso`; row V3.89, now restated): (3) is read for $h \in X_m$, the radius in (2) is one for all levels, and "smooth point" is a point of $U \cap E_\infty$.

V3.90 (Proposition 3.8.17 (preprint 3.73) and Lemma 3.8.21 (preprint 3.77) with the hypotheses their proofs use; `Polyfold.InvDataW`, `Polyfold.InvDataW.injOn`, `Polyfold.InvDataW.hasFDerivWithinAt_inv`, `Polyfold.InvDataW.contDiffOn_inv`). Let U be a convex subset of a finite-dimensional normed space V with unique differentiability at its points, $f : V \to V$ of class C^j ($j \geq 1$) on U with $\|Df(x) - 1\| \leq \frac{1}{2}$ for $x \in U$, and let $f(U)$ have unique differentiability. Then f is injective on U ($\frac{1}{2}|x - x'| \leq |f(x) - f(x')|$), its inverse g is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$, and g is of class C^j on $f(U)$. *Book:* Proposition 3.8.17 (preprint 3.73), p. 165 (preprint p. 137); Lemma 3.8.21 (preprint 3.77), p. 168 (preprint p. 140). *Hidden hypotheses:* none. *Deviation (restated):* no partial quadrant; the hypothesis "x interior iff $f(x)$ interior" of V3.59 is replaced by the unique differentiability of $f(U)$, which is what the bootstrapping uses; the relative openness of $f(U)$ is not concluded. Used in the re-centring (V3.73), where $f(U)$ is relatively open in a cone for another reason.

V3.91 (Theorem 3.1.21 (preprint 3.18), parametrized perturbations; `Polyfold.theorem_`

`3_1_21_preprint`, `Polyfold.theorem_3_1_21'`, `Polyfold.paramSection`, `Polyfold.paramSection_eq_sectionAdd`, `Polyfold.StrongBundle.pullback`, `Polyfold.StrongBundle.IsScFredholmSection.param`, `Polyfold.StrongBundle.isScPlusSection_paramComb`, `Polyfold.IsBasicGerm.paramExtend`, `Polyfold.IsConjugateToBasicGerm.paramExtend`, `Polyfold.rnPolyfold`, `Polyfold.prod_rn_isTame`). Let $P : Y \to X$ be a strong bundle over an M-polyfold (X, Y Hausdorff and paracompact, as throughout), f an sc-Fredholm section (V3.16) and $s_1, \dots, s_n$ sc⁺-sections of P (Definition 2.6.6 (preprint 2.52)). Let $\mathbb{R}^n$ carry its M-polyfold structure with the constant sc-structure and the charts $r \mapsto r - r_0$, $\mathbb{R}^n \times X$ the product M-polyfold, $\pi(r, x) = x$, and $\pi^* P : \pi^* Y \to \mathbb{R}^n \times X$ the pull-back strong bundle with the natural structure of Proposition 2.6.8 (preprint 2.54) (the atlas of all charts pulled back from charts of P through charts of $\mathbb{R}^n \times X$). Then $F(r, x) = f(x) + \sum_i r_i s_i(x)$, computed in the fibre $P^{-1}(x)$ and read in the fibre of $\pi^* Y$ over (r, x), is an sc-Fredholm section of $\pi^* P$. The paracompactness of $\mathbb{R}^n \times X$ and of $\pi^* Y$, instance arguments of `theorem_3_1_21_preprint`, is supplied by Theorem 2.3.10 (preprint 2.16) in `theorem_3_1_21'`. On the way: $\tilde{f}(r, x) = f(x)$ is an sc-Fredholm section of $\pi^* P$ — in the product chart $(r \mapsto r - r_0) \times \varphi$, centred at (r_0, x), the filling germ is $(a, u) \mapsto g(u)$ on $\mathbb{R}^n \times G$ (filler conditions (1)–(3) from those of g), the sc⁺-germ $(a, u) \mapsto t_0(u)$, and the conjugation is the product of the given one with the identity of $\mathbb{R}^n$, onto the $\mathbb{R}^n$-extension $((a, \lambda), w) \mapsto h(a, w)$ of the basic germ, again a basic germ; $\tilde{s}(r, x) = \sum_i r_i s_i(x)$ is an sc⁺-section (the product rule with a scalar factor, from Proposition 1.2.4 (preprint 1.19)); $F = \tilde{f} + \tilde{s}$ fibrewise, and Theorem 3.1.18 (preprint 3.15) (V3.17) applies; $\mathbb{R}^n \times X$ is tame when X is. *Book:* Theorem 3.1.21 (preprint 3.18), p. 100 (preprint p. 74) (for a tame strong bundle; "$\tilde{f}$ … is a sc-Fredholm section as is readily verified", preprint p. 73). *Hidden hypotheses:* the translation charts of $\mathbb{R}^n$ (a chart centred at (r_0, x) is needed for the germ there), the strong bundle structure of $\pi^* P$, the scalar product rule. *Deviation (stronger):* tameness is not used, as for Theorem 3.1.18 (preprint 3.15).

V3.92 (the linearization of the parametrized section; `Polyfold.linAt_paramSection_zero`, `Polyfold.linAt_paramSection`, `Polyfold.scDeriv_paramSection`, `Polyfold.PullbackChartData.ofProd`). In the setting of V3.91, let c be a strong bundle chart of P compatible with the atlas, with 0 in its retract O, and consider the product chart of $\pi^* P$ through $(r \mapsto r - r_0) \times c$, with the retraction $1 \times r$. At its centre: $D(\hat{F} \circ (1 \times r))(0)(\delta a, \delta u) = D(\hat{G} \circ r)(0)\delta u + \sum_i \delta a_i \hat{s}_i(0)$, $\hat{G} = \hat{f} + \sum_i r_{0,i} \hat{s}_i$; in the chart form of the linearization (V3.78), on $T_0(\mathbb{R}^n \times O) \subseteq \mathbb{R}^n \times T_0 O$: $\mathrm{Lin}_F(0)(\delta a, \delta u) = \mathrm{Lin}_{f + \sum r_{0,i} s_i}(0)\delta u + \sum_i \delta a_i \hat{s}_i(0)$; and at $r_0 = 0$: $\mathrm{Lin}_F(0)(\delta r, \xi) = \mathrm{Lin}_f(0)\xi + \sum_i \delta r_i \hat{s}_i(0)$. *Book:* after Theorem 3.1.22 (preprint 3.19), p. 101 (preprint p. 75) ("the linearization of $\tilde{f}$ at $(0, x_0)$ is $(h, u) \mapsto f'(x_0)u + \sum h_i e_i$"). *Hidden hypotheses:* none (the formula does not use $F = 0$ at the point). *Deviation (restated):* in the chart form (Definition 3.6.6 (preprint 3.53) is formalized in charts); at (r_0, x) with $r_0 \neq 0$ the first term is the linearization of $f + \sum r_{0,i} s_i$; the text's "sc¹-sections s_i" are sc⁺ here, as Theorem 3.1.21 (preprint 3.18) needs.

V3.93 (Theorem 3.6.8 (1) (preprint 3.54), the surjectivity at the nearby zeros, in the chart at x_0; `Polyfold.theorem_3_6_8_chart_onto`, `Polyfold.StrongRetr.mem_O_of_perturbed_zero`, `Polyfold.exists_inward_of_isInGoodPosition`, `Polyfold.inward_`

`add_smul`, `Polyfold.exists_curve_of_inward`, `Polyfold.ShiftedBasicData.exists_radii`, `Polyfold.ShiftedBasicData.perturbContraction`, `Polyfold.ShiftedBasicData.perturbContraction_isSc1GermWith`, `Polyfold.xiMap`, `Polyfold.exists_xi_radii_bounds`, `Polyfold.ShiftedBasicData.surjective_DH_and_ker_eq`, `Polyfold.ShiftedBasicData.exists_perturbed_curve`). In the data of V3.48 at a zero x_0 in good position — the strong bundle chart centred at x_0 with retraction (r, ρ) on U onto O, the filling germ g, the conjugation Ψ covering ψ of $g - s$ to a basic germ, $h' + \iota t$ with kernel $K = \ker D(h' + \iota t)(0)$ in good position, the closed good complement, the C^1 map σ on V' with its lifts, and an open $N_2 \ni 0$ on which ψ maps the zeros of g in O onto the graph of σ — assume condition (2) of Definition 3.1.4 (preprint 3.4) on N_2 and $K \neq 0$. Then there is an open $N_4 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_4$ of g, every level-1 point over z_0 and every smooth $k \in F_\infty$, $\rho(z_0)k = Dg(z_0)x$ for some $x \in T_{z_0}O = Dr(z_0)E$. On the way: (a) a zero of $g + \mu\rho(r(\cdot))k$ at which condition (2) holds lies in O; (b) a nonzero finite-dimensional subspace in good position to the partial quadrant C_z at $z \in C$ contains a vector v with $z + t(v + \zeta) \in C$ for $t \in [0, t_0]$, $|\zeta| < \eta$, and such inward vectors stay inward after adding any vector and rescaling; (c) for $\widetilde{C}$ convex and H of class C^1 on $\Omega \cap (\mathbb{R} \times \widetilde{C})$ with $H(0, a_0) = 0$, $D_aH(0, a_0)$ onto and an inward α_0 with $DH(0, a_0)(1, \alpha_0) = 0$, there is a curve $a(\mu)$, $0 \le \mu < \mu_0$, $a(0) = a_0$, with right derivative α_0 at 0 and $H(\mu, a(\mu)) = 0$; (d) for the normal form of Theorem 3.4.2 (preprint 3.38) (a basic germ $(G_1, w - B)$ on the shifted structure) there are $\frac{1}{4}$-contraction regions of B below any prescribed radii, and $B - \mu P\hat{e}$ is a contraction germ in the working form on $(-\mu_m, \mu_m) \times \{|a| < \nu_m\}$ with the same regions in a for every perturbation $\hat{e}$ with given bounds, its solution germ sc¹ (Theorem 3.3.3 (preprint 3.30)); (e) the push-forward $\hat{e}_k = \Xi(\cdot, k)$ of $u \mapsto \rho(r(u))k$ to the normal form, with Ξ sc-smooth into the index-raised target and linear in k, is bounded with its Lipschitz constant on the contraction regions by $c_m = 2|k|_{m+2}(M_m + 1)/\delta_m$, δ_m, M_m independent of k; (f) at a zero (a_0, w_0) of the normal form with $|a_0| < \nu_3$, $|w_0|_1 < \rho_0$ and a kernel of dimension $n - N$, $D_aH(0, a_0)$ is onto for H the $\mathbb{R}^N$-component of $(G + \mu\hat{e})(a, \delta_\mu(\mu, a))$, the kernel is the graph of $D_a\delta_\mu(0, a_0)$ over $\ker D_aH(0, a_0)$, and there is a curve of zeros of $G + \mu\hat{e}$ through it, right differentiable at 0 into level 2. *Book:* the proof of Theorem 3.6.8 (preprint 3.54), property (4), preprint p. 114 (from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55)). *Hidden hypotheses:* none added. *Deviation (not in the book):* a new argument (Findings, Chapter 3, items 180–187): the text's argument gives the surjectivity of the filled $Dg(z_0)$ only (V3.82). The neighbourhood N_4 is determined by g, r, ρ, Ψ and the normal form, not by k; Corollary 3.4.3 (preprint 3.39) is not used. In the germ reading the curve of zeros lies in the germ of the filling, so g is differentiated within its germ (Findings, Chapter 3, items 276–278, 298).

V3.94 (Theorem 3.6.8 (1) (preprint 3.54), and Theorem 3.6.8 (preprint 3.54), with the text's hypotheses; `Polyfold.theorem_3_6_8_one'`, `Polyfold.theorem_3_6_8'`, `Polyfold.theorem_3_6_8_one_sc_of_tame`, `Polyfold.theorem_3_6_8_one_sc`, `Polyfold.theorem_3_6_8_sc`). Let f be an sc-Fredholm section of a strong bundle with tame germ data (V3.103), $f(x_0) = 0$, and (f, x_0) in good position in a tame chart (V3.47, V3.103). Then there is an open $V_x \ni x_0$ such that (f, y) is in good position in a tame chart at every zero $y \in V_x$; in particular $f'(y)$ is onto (`theorem_3_6_8_one'`). Moreover (`theorem_3_6_8'`) there are $d \in \mathbb{N}$ and regular data of Definition 3.2.1

(preprint 3.25) (V3.24) of dimension d for $S = \{f = 0\}$ at x_0 whose chart domain lies in V_x, and every datum of Definition 3.2.1 (preprint 3.25) for S at a point of S in that domain has dimension d. With the text's notion of good position (sc-complement, V3.47) in the conclusion, from either form at x_0: every zero $y \in V_x$ is in good position with an sc-complement. *Book:* Theorem 3.6.8 (1), (2) (preprint 3.54), p. 136 (preprint p. 111). *Hidden hypotheses:* tameness as in V3.68. *Deviation (restated):* no hypothesis added; (2) as regular tangent germ data with locally constant dimension (the manifold structure of the local solution set is V3.95); the proof of the surjectivity in (1) is V3.93, through Definition 3.6.6 (preprint 3.53) from a chart not centred at y (V3.79), read in the tame chart of the germ data at y (V3.107); for dim $K = 0$ the only zero near x_0 is x_0. The project's good position asks only for an algebraic good complement (V3.20), less than the text's as a conclusion; the sc form follows at each zero (Findings, Chapter 3, items 210, 213).

V3.95 (Theorems 3.1.26 and 3.1.22 (preprint 3.23 and 3.19) with the text's hypotheses; `Polyfold.theorem_3_1_26'`, `Polyfold.theorem_3_1_22'`, `Polyfold.theorem_3_1_26_sc`). Let f be an sc-Fredholm section of a strong bundle with tame germ data (V3.103), $f(x) = 0$ and (f, x) in good position in a tame chart. Then there are an open $U \ni x$ and $d \in \mathbb{N}$ such that every zero in U is in good position in a tame chart and a smooth point, and $S(f, U) = \{y \in U \mid f(y) = 0\}$ is a sub-M-polyfold whose induced structure is tame and equivalent to that of a d-dimensional manifold with boundary with corners (`theorem_3_1_26'`). If moreover $d_X \equiv 0$, the induced structure also has degeneracy index 0 everywhere (`theorem_3_1_22'`). Theorem 3.1.26 (preprint 3.23) with the text's notion of good position (sc-complement, V3.47) at x and at every zero in U: `theorem_3_1_26_sc`. *Book:* Theorem 3.1.26 (preprint 3.23), p. 103 (preprint p. 77); Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74). *Hidden hypotheses:* tameness as in V3.68. *Deviation (hypotheses explicit):* no hypothesis added. For Theorem 3.1.22 (preprint 3.19) the hypothesis "$f'(x_0)$ surjective" is read as good position, the same at $d_X = 0$ (V3.100, V3.101); "the dimension agrees with the Fredholm index" is V3.97, "ker $f'(y) = T_yS$" V3.99; "uniquely determined" is the equivalence of atlases.

V3.96 (the index of the linearization at a zero, in every chart; `Polyfold.StrongBundle.IsSurjLinOfIndex`, `Polyfold.StrongBundle.linAt_index_of_conjugationData`, `Polyfold.StrongBundle.isSurjLinOfIndex_of_conjugationData`, `Polyfold.StrongBundle.IsGoodPositionGerm.exists_isSurjLinOfIndex`, `Polyfold.theorem_3_1_22_linearization_index`, `Polyfold.StrongBundle.linAt_isScFredholm`, `Polyfold.StrongBundle.linAt_isScFredholm_of_isSurjLinOfIndex`, `Polyfold.proposition_3_6_5_sc`, `Polyfold.theorem_3_1_22_linearization_sc`). `IsSurjLinOfIndex f y d`: in every strong bundle chart compatible with the atlas whose domain contains y, at the level-1 point z over $\varphi(y)$, the linearization $\mathrm{Lin}(z) : T_zO \to K_z$ (V3.78) is surjective and Fredholm with $\operatorname{ind}\mathrm{Lin}(z) = \dim\ker\mathrm{Lin}(z) = d$. Theorem: let f be an sc-smooth section, x_0 a zero, c the strong bundle chart of Definition 3.1.11 (preprint 3.11) centred at x_0 with filling germ g, sc$^+$-germ s, $s(0) = g(0)$, and conjugation data of $g - s$ to a basic germ $h : O(\widetilde{C} \oplus W, 0) \to \mathbb{R}^N \oplus W$, $\widetilde{C} \subseteq \mathbb{R}^n$. Then in every strong bundle chart around x_0 the linearization at the point over x_0 is Fredholm of index $n - N$, and it is onto if the germ is in good position in c. Hence at a zero in good position `IsSurjLinOfIndex f x_0 d` holds with $d = n - N$ for every conjugation in the chart of Definition 3.6.6 (preprint 3.53). For an sc-Fredholm section and a zero x_0 in good position, in the chart of Definition

3.6.6 (preprint 3.53), $Dg(0)$ and $Tf(x_0) = Dg(0)|_{T_0O}$ are Fredholm, $Tf(x_0)$ is onto, and for every conjugation $\operatorname{ind} Dg(0) = \operatorname{ind} Tf(x_0) = \dim\ker Tf(x_0) = n - N$. In sc form: for an sc-Fredholm section, a zero y and any strong bundle chart around y, at the point z over $\varphi(y)$ (a smooth point), $T_zO = Dr(z)E$ and $K_z = \rho(z)F$ are sc-subspaces and $\mathrm{Lin}(z) : T_zO \to K_z$ is an sc-Fredholm operator of index $n - N$ on every level; under `IsSurjLinOfIndex f y d` it is onto on every level, of index d on every level; in the chart of Definition 3.6.6 (preprint 3.53), $Dg(0)$ and $f'(x_0)$ are sc-Fredholm and $f'(x_0)$ is onto on every level. *Book:* Proposition 3.6.5 (preprint 3.52), p. 135 (preprint p. 110), and the first clause of Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74). *Hidden hypotheses:* none. *Deviation (stronger):* in charts (T_xX carries no linear structure here, Chapter 2); in every chart by the chart independence at zeros (V3.78), which the text does not state, realised on all levels by the chart change at smooth points (Findings, Chapter 3, items 236, 239); the index on every level, and the surjectivity on every level at a zero in good position, go beyond the text's statement. The hypothesis is the text's since the switch to the germ reading (phase 2): Proposition 3.6.5 (preprint 3.52) enters at the centre of the chart of the germ datum, where $\mathrm{Lin}(0) = Dg(0)$ on T_0O for the filling germ (Findings, Chapter 3, items 284, 310, 314, 330). `theorem_3_1_22_linearization_sc` gives the conclusions in the chart at x_0, and `linAt_isScFredholm_of_isSurjLinOfIndex` with `IsGoodPositionGerm.exists_isSurjLinOfIndex` in every chart. The sc-Fredholm property was first proved on level 0 only; formerly "narrower", for the whole-chart hypothesis.

V3.97 (the dimension of the solution set is the Fredholm index; `Polyfold.theorem_3_6_8_tangentGermChart_index`, `Polyfold.theorem_3_6_8_two_index`, `Polyfold.theorem_3_1_26_index`, `Polyfold.theorem_3_1_22_index`, `Polyfold.theorem_3_6_11_index`, `Polyfold.theorem_3_6_8_graph_proj_conj`, `Polyfold.theorem_3_6_8_tangentGermChart_proj_conj`, `Polyfold.StrongBundle.linAt_isScFredholm_of_isSurjLinOfIndex`). Let f be an sc-smooth section of a strong bundle, $f(x_0) = 0$, (f, x_0) in good position in a tame chart (V3.103). (i) The data of Definition 3.2.1 (preprint 3.25) at x_0 of V3.84 have a dimension d with `IsSurjLinOfIndex f x_0 d` (V3.96): $d = n - N$ for the conjugation in the chart from which $\psi\varphi$ is built. (ii) The regular data of V3.85 (ii) have such a dimension. (iii) For f sc-Fredholm with tame germ data there are an open $U \ni x_0$ and d such that every zero $y \in U$ is in good position in a tame chart with `IsSurjLinOfIndex f y d`, the zeros in U are smooth points, and $S(f, U)$ is a sub-M-polyfold whose induced structure is tame and equivalent to that of a d-dimensional manifold with boundary with corners (`theorem_3_1_26_index`); if $d_X \equiv 0$, also of degeneracy index 0 everywhere (`theorem_3_1_22_index`). (iv) Under the hypotheses of V3.87 (good position in a tame chart at every zero, $S \neq \emptyset$): S is a sub-M-polyfold with tame induced structure, a manifold with boundary with corners of locally constant dimension, and at every $x \in S$ there are regular data of a dimension d such that every datum at a point of S in their chart domain has dimension d and `IsSurjLinOfIndex f y d` holds at every zero y of that domain (`theorem_3_6_11_index`). For an sc-Fredholm section, at a zero with `IsSurjLinOfIndex f y d`, $\mathrm{Lin}(z)$ is moreover sc-Fredholm and onto on every level, of index d on every level, in every chart (V3.96). *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74) ("whose dimension agrees with the Fredholm index"); the proof of Theorem 3.6.8 (preprint 3.54), preprint p. 115 ($d = \dim\ker f'(x)$).

Hidden hypotheses: tameness as in V3.68. *Deviation (stronger):* the index at every zero near the point and in every chart; the text states no dimension in Theorems 3.1.26, 3.6.11 (preprint 3.23, 3.57).

V3.98 (the Fredholm index along the solution set and along paths, Definition 6.2.1 (preprint 6.1), Proposition 6.2.2 (preprint 6.2); `Polyfold.theorem_3_6_8_index_locally_constant`, `Polyfold.theorem_3_6_8_index`, `Polyfold.isSurjLinOfIndex_of_forall_dim`, `Polyfold.theorem_3_6_8_index_locally_constant_sc`, `Polyfold.StrongBundle.linAt_isScFredholm`, `Polyfold.index_locally_constant_tame`, `Polyfold.theorem_3_6_8_index_locally_constant_general_tame`, `Polyfold.index_locally_constant_of_admitsBumpFunctions`, `Polyfold.theorem_3_6_8_index_locally_constant_general`, `Polyfold.StrongBundle.IsLinOfIndex`, `Polyfold.StrongBundle.restrictOpen`, `Polyfold.StrongBundle.IsIndexAtLocal`, `Polyfold.StrongBundle.IsIndexAtLocal.unique`, `Polyfold.StrongBundle.exists_isIndexAtLocal`, `Polyfold.StrongBundle.IsIndexAt`, `Polyfold.StrongBundle.IsIndexAt.unique`, `Polyfold.StrongBundle.exists_isIndexAt`, `Polyfold.StrongBundle.isIndexAtLocal_iff_isIndexAt`, `Polyfold.exists_isScPlusSection_pullback_of_path`, `Polyfold.StrongBundle.isIndexAt_of_pullback_isLinOfIndex`, `Polyfold.proposition_6_2_2_local`, `Polyfold.proposition_6_2_2_local'`, `Polyfold.proposition_6_2_2`). Let f be an sc-Fredholm section of a strong bundle with tame germ data (V3.103), $f(x_0) = 0$, (f, x_0) in good position in a tame chart. Then there are an open $V_x \ni x_0$ and d such that `IsSurjLinOfIndex f y d` holds at every zero $y \in V_x$: $f'(y)$ is onto with $\operatorname{ind} f'(y) = \dim\ker f'(y) = d = \operatorname{ind} f'(x_0)$ (`theorem_3_6_8_index_locally_constant`). Theorem 3.6.8 (preprint 3.54) with the index (`theorem_3_6_8_index`): moreover every zero in V_x is in good position in a tame chart, and S has at x_0 regular data of dimension d whose chart domain is V_x, every datum at a point of S in V_x having dimension d. On the way: at a zero in good position in a tame chart at which every datum of Definition 3.2.1 (preprint 3.25) for S has dimension d, `IsSurjLinOfIndex f y d` (f sc-smooth). In sc form, at every zero in V_x and in every chart, $\operatorname{Lin}(z)$ is sc-Fredholm, onto on every level, of index d on every level (`theorem_3_6_8_index_locally_constant_sc`); at any zero of an sc-Fredholm section, in every chart, $\operatorname{Lin}(z)$ is sc-Fredholm with the same index on every level (`linAt_isScFredholm`). Without good position, with the text's hypotheses: if f is an sc-Fredholm section with tame germ data and $f(x_0) = 0$, there are an open $V \ni x_0$ and $i \in \mathbb{Z}$ such that at every zero $y \in V$, in every strong bundle chart, $f'(y)$ is Fredholm of index i (`IsLinOfIndex f y i`; `index_locally_constant_tame`) and $\operatorname{Lin}(z)$ is sc-Fredholm of index i on every level (`theorem_3_6_8_index_locally_constant_general_tame`); no bump functions are assumed (the first forms, for X admitting sc-smooth bump functions, Definition 5.3.1 (preprint 5.9): `index_locally_constant_of_admitsBumpFunctions`, `theorem_3_6_8_index_locally_constant_general`). **Definition 6.2.1 (preprint 6.1)**, with the text's locally defined section: an sc-Fredholm section f has index i at a smooth point x (`IsIndexAtLocal f x i`) if for some open $V \ni x$ and some sc⁺-section s of the restriction $P|V$ of P to V (`StrongBundle.restrictOpen`: the strong bundle over the M-polyfold V whose atlas consists of the charts of P with base domain inside V), $(f|_V + s)(x) = 0$ and $(f|_V + s)'(x)$ is Fredholm of index i in every strong bundle chart of $P|V$; the index is unique, at a zero it is the index of $f'(x)$, and it exists at every smooth point. With an sc⁺-section on all of X in place of s (`IsIndexAt f x i`) the index is unique as well and implies the local one; if X admits sc-smooth bump functions it exists at every smooth point and the two readings agree (`isIndexAtLocal_iff_isIndexAt`). **Proposition 6.2.2 (preprint 6.2)**: if f is an sc-Fredholm section with tame germ data, then for every sc-smooth path $\varphi : \mathbb{R} \to X$ the index $\mathrm{ind}(f, \varphi(t))$ (local reading) is the same for all t (`proposition_6_2_2_local`), and any two values of the index at $\varphi(0)$ and $\varphi(1)$ agree (`proposition_6_2_2_local'`); if X admits sc-smooth bump functions, the same holds for the global reading (`proposition_6_2_2`). *Book:* Definition 6.2.1 (preprint 6.1), pp. 237–240 (preprint pp. 200–201); Proposition 6.2.2 (preprint 6.2), p. 240 (preprint p. 201) (over a tame M-polyfold, the index is constant along sc-smooth paths $[0, 1] \to X$ of smooth points; the proof: "we do not need a sc-smooth partition of unity", preprint p. 202), whose proof uses the step of V3.82; in Chapter 3 the statement along the solution set is implicit in Theorem 3.6.8 (1) (preprint 3.54). *Hidden hypotheses:* tameness as in V3.68; for Definition 6.2.1 (preprint 6.1), the text's local sc⁺-section s with $s(x) = f(x)$ is a section of the restriction of the bundle to an open neighbourhood of x ($-s$ here), a construction the text uses without comment. *Deviation (narrower):* the path is defined on $\mathbb{R}$, not on $[0, 1]$ ($\mathbb{R} \times X$ is the product of Theorem 3.1.21 (preprint 3.18) and has no boundary, and the text's tame M-polyfold $[0, 1] \times X$ is not used): a path on $[0, 1]$ that extends sc-smoothly to $\mathbb{R}$ is covered, a path sc-smooth on $[0, 1]$ only is not, and the project has no M-polyfold $[0, 1]$. This is the only narrowing left; the hypotheses on f are the text's, in the reading of tameness of V3.103 (as in V3.94), and no bump functions are assumed. *Proof:* Proposition 6.2.2 (preprint 6.2) is proved by the text's device, run on the restriction $P|V$ to the domain V of a chart around $\varphi(t_0)$, where the text's sections need no cut-off — for each t_0 an sc⁺-section S of the pull-back to $\mathbb{R} \times V$ with $\tilde{f} + S = 0$ along the graph of the path near t_0 (the text's $R(x, f(\varphi(t)))$ in the chart), the local constancy of the index of $\tilde{f} + S$ along its zero set, the count $\mathrm{ind}(\tilde{f} + S)'(t, x) = \mathrm{ind}(f, x) + 1$ through the slice $S(t, \cdot)$, and the connectedness of $\mathbb{R}$; the earlier form for M-polyfolds with bump functions cuts the sections off instead (Findings, Chapter 3, items 340–349, 360–367). The local constancy along the zero set without good position is the same device: sc⁺-sections $R(\cdot, e_i)$ on $P|V$ put the parametrized section of Theorem 3.1.21 (preprint 3.18) into general position at $(0, x_0)$, its index is constant along its solution set, at $(0, y)$ it is $k + \mathrm{ind}\, f'(y)$, and the index at a zero is the same for P and $P|V$, through the germ datum (items 240–242, 360–363). Definition 6.2.1 (preprint 6.1) is well defined because two local sections are compared on the intersection of their domains; its independence of s is proved from the germ data and Theorem 3.1.18 (preprint 3.15), not by the text's appeal to Proposition 1.1.11 (preprint 1.10); and it exists at every smooth point through the section $-R(\cdot, f(x))$ on a chart domain (items 340, 364). Under bump functions a local section is cut off to a global one with the same index (item 367). The candidate counterexample of items 188 and 243 is excluded by Theorem 3.6.8 (1) (preprint 3.54). Without tame germ data the local constancy is open: it amounts to the index 0 of the block $(1 - \rho(z))Dg(z)|_{\ker Dr(z)}$ of the filling at the nearby zeros, which condition (3) of Definition 3.1.4 (preprint 3.4) gives at $z = 0$ only (items 243, 343).

V3.99 ($\ker f'(y) = T_yS$ **in a chart; `Polyfold.theorem_3_6_8_ker_eq_tangent`).** Let f be an sc-smooth section of a strong bundle, $f(x_0) = 0$, (f, x_0) in good position in a tame chart (V3.103). Then there are a tame strong bundle chart c centred at x_0 (as in Definition 3.6.6 (preprint 3.53)), the conjugating sc-diffeomorphism $\psi : U \to U' \subseteq C'$ (inverse ψ', $\psi(0) = 0$), the finite-dimensional $K = \ker D(h+t)(0)$, the solution graph $\sigma : V' \to E'$ over a relatively open $V' \ni 0$ of $K \cap C'$ (C^1,

$\sigma(0) = 0$, $D\sigma(0) = 0$), an open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x \mid f(y) = 0\}) = \{v + \sigma(v) \mid v \in V'\}$, and an open $V_{x,2} \subseteq V_x$ containing x_0, such that for every zero $y \in V_{x,2}$, with $\psi(\varphi(y)) = v + \sigma(v)$, $v \in V'$, and every level-1 point z over $\varphi(y)$: $D\psi(z)$ maps the kernel of $\mathrm{Lin}(z) : T_zO \to K_z$ onto $N_v = \{n + D\sigma(v)n \mid n \in K\}$, the tangent space at v of the graph. *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74) ("ker $f'(y) = T_yS(f, U)$ is the tangent space"). *Hidden hypotheses:* tameness as in V3.68. *Deviation (restated):* in the chart $\psi\varphi$ of the tangent germ data, the kernel of the linearization in the chart c; T_yS as a subspace of T_yX is not formalized (Chapter 2, gap G.13).

V3.100 (good position is surjectivity at a zero with $d_X = 0$; `Polyfold.StrongBundle.IsLinSurjAt`, `Polyfold.isInGoodPositionIn_of_subset`, `Polyfold.StrongBundle.isGoodPositionGerm_of_linAt_surj_of_degIndex_eq_zero`, `Polyfold.StrongBundle.isGoodPositionGerm_of_isLinSurjAt`, `Polyfold.StrongBundle.IsGoodPositionGerm.isLinSurjAt`, `Polyfold.StrongBundle.isGoodPositionGerm_iff_isLinSurjAt`, `Polyfold.StrongBundle.IsLinSurjAt.linAt_surjective`). `IsLinSurjAt f y` ("$f'(y)$ is surjective"): in some strong bundle chart compatible with the atlas whose domain contains y, at a level-1 point z over $\varphi(y)$, $\mathrm{Lin}(z)$ maps onto the fibre $K_z = \{k \mid (\varphi(y), k) \in K\}$; then, at a zero, it does so in every such chart and at every level-1 point over $\varphi(y)$ (V3.78). Let f be an sc-Fredholm section of a strong bundle with tame germ data (V3.103) and y a zero with $d_X(y) = 0$. Then (f, y) is in good position in a tame chart if and only if `IsLinSurjAt f y`. On the way: a subspace $N \subseteq T$ is in good position inside T to every cone containing T; in a chart with tame base retraction around a zero y with $d_X(y) = 0$ at which (f, y) is an sc-Fredholm germ, if the linearization is onto there, (f, y) is in good position, since $d_C(\varphi(y)) = d_X(y) = 0$ makes the partial quadrant at $\varphi(y)$ the whole space; the surjectivity is carried to the tame chart of the germ data at y, and the good position read there (V3.107). *Book:* the remark after Theorem 3.6.8 (preprint 3.54), p. 136 (preprint p. 111) (for $\partial X = \emptyset$ good position just means surjectivity), and Corollary 3.1.27 (preprint 3.24) ("an empty condition if $d_X(x) = 0$", p. 104 (preprint p. 79)). *Hidden hypotheses:* the chart is tame, for $d_C = d_X$ (in a non-tame chart $d_C(\varphi(y))$ can exceed $d_X(y)$). *Deviation (hypotheses explicit).*

V3.101 (Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) with "$f'(x)$ surjective", and Theorem 3.6.10 (preprint 3.56) with tameness; `Polyfold.theorem_3_1_22_of_surjective`, `Polyfold.theorem_3_6_10_of_surjective`, `Polyfold.theorem_3_6_10_tame`). Let X be an M-polyfold with $d_X \equiv 0$ and f an sc-Fredholm section of a strong bundle. (i) If f has tame germ data (V3.103), $f(x_0) = 0$ and `IsLinSurjAt f x_0` (V3.100), the conclusions of `theorem_3_1_22_index` hold (V3.97 (iii)): an open $U \ni x_0$ and d with every zero $y \in U$ in good position in a tame chart, `IsSurjLinOfIndex f y d`, the zeros in U smooth, and $S(f, U)$ a sub-M-polyfold whose induced structure is tame, a d-dimensional manifold with degeneracy index 0 everywhere. (ii) If (f, x) is in good position in a tame chart at every zero (`theorem_3_6_10_tame`), or f has tame germ data and `IsLinSurjAt f x` at every zero (`theorem_3_6_10_of_surjective`), and $S \neq \emptyset$, then S is a sub-M-polyfold whose induced structure is tame, a manifold of locally constant dimension with degeneracy index 0 everywhere, with regular data of Definition 3.2.1 (preprint 3.25) at every zero of a dimension d constant near it and equal to the Fredholm index at the zeros of the chart domain. *Book:* Theorem 3.1.22 (preprint 3.19), p. 100 (preprint p. 74); Theorem 3.6.10 (preprint 3.56), p. 140 (preprint p. 115). *Hidden hypotheses:* tameness as in V3.68; $S \neq \emptyset$ as in V3.70. *Deviation (stronger):*

the text's hypotheses; the tameness of the induced structure and the index are added conclusions.

V3.102 (the filler condition at the nearby zeros follows from good position; `Polyfold.normal_injective_of_exists_range_deriv_eq__germ`, `Polyfold.exists_range_deriv_eq__iff_normal_injective_germ`, `Polyfold.isGoodPositionGermFilled_of_isGoodPositionGerm`). (a) Let U be relatively open in a partial quadrant, R a strong bundle retraction on $U \lhd F$, g an sc^1-germ with derivative family Dg, and z a level-1 point in its germ over a point z_0 of the retract O with $\ker Dg(z) \subseteq T_{z_0}O$. If for every $k \in F$ there is $x \in T_{z_0}O$ with $Dg(z)x = \rho(z_0)k$, then $Dr(z)b = 0$ and $\rho(z_0)Dg(z)b = Dg(z)b$ imply $b = 0$: condition (3) of Definition 3.1.4 (preprint 3.4) at z_0. (b) If moreover $\mathbf{f} \circ r$ is sc^1, $\mathbf{f}$ is a section of K on O with $\mathbf{f}(z_0) = 0$, $\mathbf{f} = g$ on O near z_0, and $Dg(z)$ is onto, the two conditions are equivalent. (c) For an sc-Fredholm section with tame germ data (V3.103) and a zero x_0 in good position in a tame chart, (f, x_0) is in good position in a tame chart with the filler condition at the nearby zeros (`IsGoodPositionGermFilled`, V3.82). *Book:* not in the text (Definition 3.1.4 (3) (preprint 3.4), p. 91 (preprint p. 64), at 0 only). *Hidden hypotheses:* none. *Deviation (not in the book):* (c) uses V3.94 and the tangency $\ker Dg(z_0) \subseteq T_{z_0}O$ of V3.81; it shows that the added hypothesis of V3.68, V3.69, V3.82, V3.85 is satisfied whenever the text's hypotheses are.

V3.103 (Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) in a tame chart; `Polyfold.StrongBundle.IsTameChart`, `Polyfold.StrongBundle.IsScFredholmGermTame`, `Polyfold.StrongBundle.IsGoodPositionGermTame`, `Polyfold.StrongBundle.IsScFredholmSectionTame`, `Polyfold.StrongBundle.IsScFredholmGermTame.isScFredholmGerm`, `Polyfold.StrongBundle.IsGoodPositionGermTame.isGoodPositionGerm`, `Polyfold.StrongBundle.IsGoodPositionGermTame.isScFredholmGermTame`, `Polyfold.StrongBundle.IsGoodPositionGermFilled.isGoodPositionGermTame`, `Polyfold.StrongBundle.isTameChart_of_eqOn_id`, `Polyfold.ScRetraction.IsTame.congr`, `Polyfold.IsFillingGerm.nonempty_kerEquiv`, `Polyfold.TrivialBundle.tame_atlas_no_germData_in_atlas_chart`). A strong bundle chart of P is *tame* (`IsTameChart`) if it is compatible with the atlas and its base retraction $r : U \to U$ is tame (Definition 2.5.2 (preprint 2.34)). `IsScFredholmGermTame f x` (Definition 3.6.3 (preprint 3.50), "tame germ data at x"): x is a smooth point and there are the data of Definition 3.1.11 (preprint 3.11) (V3.13: a strong bundle chart centred at x, a filling germ g of the principal part, a local sc$^+$-germ s with $s(0) = g(0)$, and $g - s$ conjugated near 0 to a basic germ; the text's germ reading) in a tame chart. `IsGoodPositionGermTame f x` ("good position in a tame chart"): the same data with the germ in good position in that chart (Definition 3.6.6 (preprint 3.53) in the chart form of V3.47). `IsScFredholmSectionTame f` (Definition 3.6.4 (preprint 3.51), "tame germ data"): f is an sc-Fredholm section (V3.16) with tame germ data at every smooth point. Each of these implies the form without tameness (V3.13, V3.47), good position in a tame chart gives tame germ data, and the strengthened good position of V3.82 is stated in a tame chart and implies good position in it. A retraction agreeing on its domain with a tame retraction is tame; hence a compatible chart whose base retraction is the identity of U (the retract $O = U$ relatively open in the partial quadrant) is tame, so the hypotheses can be met. *Book:* Definition 3.6.3 (preprint 3.50), p. 135 (preprint p. 110) (a strong bundle chart around x "where $K \to O$ is a tame strong local bundle", as in Definition 3.1.6 (preprint 3.6), p. 93 (preprint

p. 66)); Definition 3.6.4 (preprint 3.51), p. 135 (preprint p. 110), and Definition 3.6.6 (preprint 3.53), p. 136 (preprint p. 111) (for a tame strong bundle); a strong bundle retraction is tame when its r is (preprint p. 49); the proof of Proposition 3.6.5 (preprint 3.52) starts from "a tame strong bundle chart" (p. 135 (preprint p. 110)). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* these are the hypotheses of the implicit function theorems of §3.6 (V3.48, V3.68–V3.70, V3.80–V3.82, V3.84–V3.87, V3.94–V3.102), in place of the reading of "tame strong bundle" as the tameness of the base retraction of every compatible chart, under which they were first proved and which cannot be met (V3.104). Whether germ data in some chart (V3.13) can be moved to some compatible tame chart when the atlas consists of tame charts is open; it cannot be done by re-centring a chart of the tame atlas, since whether a filling exists in a chart depends on the chart: the trivial bundle with a one-chart tame atlas has sc-Fredholm sections with tame germ data in compatible charts and no germ data in the chart of the atlas (Findings, Chapter 3, items 350–353). The text builds the tame chart into Definition 3.6.3 (preprint 3.50), and so do the theorems. Since the switch to the text's germ reading (phase 2) the three notions carry the text's germ data (V3.13, V3.45); the earlier whole-chart forms are `IsScFredholmGermTameWholeChart`, `IsGoodPositionGermTameWholeChart`, `IsScFredholmSectionTameWholeChart`, and the non-vacuity witnesses satisfy both readings (Findings, Chapter 3, items 220–221, 304, 330). For Definition 3.6.6 (preprint 3.53) the comparison is that of V3.47. Formerly "restated", for the whole-chart reading (round-trip item V3.103).

V3.104 (a retraction onto a product with a point that is not tame: the earlier tameness hypothesis is unsatisfiable; `Polyfold.ScRetraction.halfLineConst_prod_not_isTame`, `Polyfold.ScRetraction.halfLineConst`, `Polyfold.ScRetraction.halfLineConst_retract`, `Polyfold.halfLineSc`). Let W be an sc-Banach space and c the constant retraction $(a, w) \mapsto (1, 0)$ of the partial quadrant $[0, \infty) \oplus W$ of $\mathbb{R} \oplus W$ (sc-smooth, $(1, 0)$ being a smooth point), with retract $\{(1, 0)\}$. For every sc-smooth retraction r of a nonempty relatively open subset U of a partial quadrant C, the product retraction $c \times r$ of $([0, \infty) \oplus W) \times U$, whose retract is $\{(1, 0)\} \times r(U)$, is not tame: at $((0, 0), u)$ with $u \in r(U)$ the degeneracy index is $1 + d_C(u)$, at its image $((1, 0), u)$ it is $d_C(u)$, against condition (1) of Definition 2.5.2 (preprint 2.34). *Book:* not in the text (Definition 2.5.2 (preprint 2.34), pp. 53–54 (preprint pp. 39–40); Definition 2.6.4 (preprint 2.50), pp. 67–68 (preprint pp. 51–52)). *Hidden hypotheses:* none. *Deviation (not in the book):* the retraction-level half of the argument (Findings, Chapter 3, item 190) that the hypothesis under which the theorems of §3.6 were first proved, "the base retraction of every strong bundle chart compatible with the atlas is tame", fails for every strong bundle over a nonempty base: from a compatible chart (φ, U, R, Φ), the chart $(1, \varphi)$ onto $\{1\} \times O$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ (the retraction above, with $W = 0$) has the transition maps of the first chart composed with affine maps, hence is compatible with every chart the first one is. That chart and its compatibility are not formalized (see "What is not verified"); so the earlier statements, vacuous by this argument, are replaced by those with tame germ data (V3.103).

V3.105 (Theorem 3.1.18 (preprint 3.15) for tame germ data; `Polyfold.theorem_3_1_18_tame`, `Polyfold.StrongBundle.IsScFredholmGermTame.sectionAdd`). If f is an sc-Fredholm section with tame germ data (V3.103) and s an sc$^+$-section, then $f + s$ is an sc-Fredholm section with tame germ data: at every smooth point the germ data of $f + s$ are taken in the tame chart of those of f

(filling germ $g + \hat{s} \circ r$, sc$^+$-germ $\hat{s}_1 \circ r + t_0$, the same conjugation, as in V3.17). *Book:* Theorem 3.1.18 (preprint 3.15), p. 98 (preprint p. 71), recalled after Definition 3.6.4 (preprint 3.51), p. 135 (preprint p. 110), for the tame strong bundle of §3.6. *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* Theorem 3.1.18 (preprint 3.15) for the tame reading of Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51); the proof of V3.17 does not change the chart.

V3.106 (Theorem 3.1.21 (preprint 3.18) for tame germ data; `Polyfold.theorem_3_1_21_tame`, `Polyfold.PullbackChartData.ofProd_isTame`, `Polyfold.StrongBundle.IsScFredholmGermTame.param`, `Polyfold.StrongBundle.IsScFredholmSectionTame.param`). In the setting of V3.91, if f is an sc-Fredholm section with tame germ data (V3.103) and $s_1, \dots, s_n$ are sc$^+$-sections, then $F(r, x) = f(x) + \sum_i r_i s_i(x)$ is an sc-Fredholm section of $\pi^* P$ with tame germ data. On the way: the product chart $(r \mapsto r - r_0) \times c$ of a strong bundle chart c of P (V3.92) has the base retraction $1 \times r$ on $\mathbb{R}^n \times U$, which is tame when r is ($\mathbb{R}^n$ carries the identity retraction of the whole space, so $d(a, u) = d_C(u)$, and the minimal subspaces and tangent spaces are products); so $\tilde{f}(r, x) = f(x)$ has germ data at every smooth (r_0, x) in the tame product chart, and V3.105 adds $\tilde{s}$. *Book:* Theorem 3.1.21 (preprint 3.18), p. 100 (preprint p. 74) (for a tame strong bundle). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* the tame form of V3.91, which is what the implicit function theorems of §3.6 need when applied to F; the tameness of the base $\mathbb{R}^n \times X$ (`prod_rn_isTame`, V3.91) is a different statement.

V3.107 (good position in any chart of the germ data; `Polyfold.StrongBundle.isGoodPositionAtGerm_of_chart`, `Polyfold.StrongBundle.isGoodPositionAtGerm_of_chart_dense`, `Polyfold.StrongBundle.IsGoodPositionGerm.toTame`). (a) Let f be an sc-smooth section, y a zero, c a strong bundle chart of P around y (not necessarily centred), z a level-1 point over $\varphi(y)$, and g an sc^1-germ agreeing with **f** on O near $\varphi(y)$, with z in its germ, such that $Dg(z)$ maps $T_z O$ onto K_z and ker $Dg(z) \cap T_z O$ is in good position to $C_{\varphi(y)}$ inside $T_z O$. Then for **every** datum of Definition 3.1.11 (preprint 3.11) at y (a strong bundle chart c' compatible with the atlas and centred at y, a filling germ g' with a derivative family Dg') the germ is in good position in c' at $Dg'(0)$; with only $\rho(z)k$ in the range for every smooth $k \in F_\infty$ (the hypotheses of V3.79), the same when $Dg'(0)$ is Fredholm. (b) If (f, y), $f(y) = 0$, is in good position in some chart (V3.47) and has tame germ data at y (V3.103), it is in good position in that tame chart. *Book:* Definition 3.6.6 (preprint 3.53), p. 136 (preprint p. 111), presupposes that good position does not depend on the chart; the proof of Theorem 3.6.8 (preprint 3.54), p. 139 (preprint p. 115) ("proceeding as above"). *Hidden hypotheses:* none. *Deviation (not in the book):* V3.79 gives the good position in the chart that Definition 3.1.11 (preprint 3.11) provides at y; (a) gives it in every chart carrying germ data at y. (b) applies (a) at $z = 0$ in the chart of the given good position, centred at y, where 0 lies in every germ, $Tf(0)$ maps onto K_0, ker $Dg(0) \subseteq T_0 O$ and $C_0 = C$; in the germ reading no Fredholm input is needed (Findings, Chapter 3, item 301). This is how the good position of the nearby zeros in Theorem 3.6.8 (1) (preprint 3.54) (V3.94) and the good position from surjectivity (V3.100) are obtained in tame charts. The whole-chart form of (a) is `Polyfold.StrongBundle.isGoodPositionAt_of_chart_dense`.

V3.108 (Definitions 3.1.4 and 3.6.1 (preprint 3.4 and 3.48), the 2021 filling; `Polyfold.`

`IsFilling2021`, `Polyfold.HasFilling2021`, `Polyfold.IsFilling2017`, `Polyfold.isFilling2021_iff`, `Polyfold.IsFilling2021.isFillingGerm`, `Polyfold.IsFillingGerm.isFilling2021`, `Polyfold.IsFilledVersion.isFilling2021`, `Polyfold.IsFilling2017.isFilling2021`, `Polyfold.isFilling2017_iff`, `Polyfold.isFilling2017_id`, `Polyfold.isFilling2017_rn`). Let $K = R(U \lhd F) \to O$ be a strong local bundle, U relatively open in C with $0 \in U$, and $R(u,h) = (r(u), \rho(u)h)$ covering the sc-smooth retraction $r : U \to U$ onto $O = r(U)$. A germ g, sc-smooth on an sc-germ of neighbourhoods of 0 in C inside U, is a 2021 filling of the section with principal part $\hat{f}$ if $r(0) = 0$ and (1) $\hat{f} = g$ on O near 0; (2) every $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in O; (3) the derivative at 0 of $k(y) = [1 - \rho(r(y))]g(y)$, restricted to ker $Dr(0)$, is a topological linear isomorphism onto ker $\rho(0)$. The 2017 filling is the same with r tame. A 2021 filling is exactly a filling of V3.7 with $r(0) = 0$, and a filling of V3.7 is a 2021 filling as soon as $0 \in O$; a 2017 filling is a 2021 filling; with the identity strong bundle retraction (tame) every sc-smooth g fills itself. *Book:* Definition 3.1.4, p. 91 (preprint 3.4, p. 64), and Definition 3.6.1, p. 134 (preprint 3.48, p. 109): the 2021 text asks r to be an sc-smooth retraction, where the 2017 text and the preprint ask for a tame one, and assumes $r(0) = 0$. *Hidden hypotheses:* none. *Deviation (restated):* the germ reading of V3.7, which never asked for tameness; the 2021 change brings the text to the formalization's definition. $r(0) = 0$ is a condition of the definition.

V3.109 (a 2021 filling over a retraction that is not tame; `Polyfold.CornerFilling.Rc`, `Polyfold.CornerFilling.gc`, `Polyfold.CornerFilling.isFilledVersion_gc`, `Polyfold.CornerFilling.isFilling2021_gc`, `Polyfold.CornerFilling.not_isTame_r`, `Polyfold.CornerFilling.not_isFilling2017_gc`, `Polyfold.exists_isFilling2021_not_isFilling2017`). In $E = \mathbb{R} \oplus \mathbb{R}$, $C = U = [0, \infty) \times \mathbb{R}$, $F = \mathbb{R}$, let $R((a,b),h) = ((0,b),0)$, so that $r(a,b) = (0,b)$, $O = \{0\} \times \mathbb{R}$ and $\rho \equiv 0$. Then $g(a,b) = a$ is a 2021 filling of the zero section, and r is not tame ($d_C(1,0) = 0$, $d_C(r(1,0)) = 1$), so it is not a 2017 filling: the 2021 filling is strictly more general than the 2017 one. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book):* compares Definition 3.1.4 with its 2017 form.

V3.110 (Definitions 3.1.11, 3.1.16 = 3.6.3, 3.6.4 (preprint 3.11, 3.14 = 3.50, 3.51) with the 2021 filling; `Polyfold.StrongBundle.IsScFredholmGerm2021`, `Polyfold.StrongBundle.IsScFredholmGerm2017`, `Polyfold.StrongBundle.IsScFredholmSection2021`, `Polyfold.StrongBundle.zero_mem_O_of_chart`, `Polyfold.StrongBundle.isScFredholmGerm2021_iff`, `Polyfold.StrongBundle.isScFredholmGerm2017_iff`, `Polyfold.StrongBundle.IsScFredholmGerm2017.isScFredholmGerm2021`, `Polyfold.StrongBundle.isScFredholmSection2021_iff`, `Polyfold.isScFredholmSection2021_graphSection`, `Polyfold.isScFredholmGerm2017_graphSection`). (f, x_0) is a 2021 sc-Fredholm germ if x_0 is smooth and some compatible strong bundle chart centred at x_0 carries a 2021 filling g of the principal part and an sc$^+$-germ s with $s(0) = g(0)$ such that $g - s$ is conjugate near 0 to a basic germ; the 2017 germ uses the 2017 filling. f is a 2021 sc-Fredholm section if it is sc-smooth, regularizing, and a 2021 sc-Fredholm germ at every smooth point. The 2021 germ is the project's sc-Fredholm germ (V3.13, V3.45), the 2021 section the project's sc-Fredholm section (V3.16), and the 2017 germ the project's germ with tame germ data (V3.103); 2017 implies 2021. The graph section of a C^∞ map $\mathbb{R}^n \to \mathbb{R}^k$ is a 2021 sc-Fredholm section and a 2017 sc-Fredholm germ at every point. *Book:* Definitions 3.1.11, p. 96 (preprint 3.11, p. 69), 3.1.16, p. 97 (preprint 3.14, p. 70), 3.6.3 and 3.6.4, p. 135 (preprint 3.50 and 3.51, p. 110). *Hidden hypotheses:* none. *Deviation (restated):* as V3.13; $r(0) = 0$ holds in a chart centred at x_0. Whether every 2021 germ need not be a 2017 germ (V3.131; gap G.79, settled 2026-11-12).

V3.111 (Proposition 3.1.5 (preprint 3.5) for the 2021 filling; `Polyfold.proposition_3_1_5`). If g is a 2021 filling of $\hat{f}$ with $\hat{f}(0) = 0$ and Dg its derivative germ, then $Dg(0)$ maps $T_0O = \text{range}\, Dr(0)$ into $K_0 = \text{range}\, \rho(0)$, so that its restriction $T\hat{f}(0) : T_0O \to K_0$ is defined, and (1) $T\hat{f}(0)$ is onto iff $Dg(0)$ is, (2) one is Fredholm iff the other is, with the same index. *Book:* Proposition 3.1.5, p. 93 (preprint 3.5, p. 65). *Hidden hypotheses:* as V3.8. *Deviation (restated):* V3.8 for the 2021 definition; $0 \in O$ from $r(0) = 0$.

V3.112 (Theorem 3.1.13 (preprint 3.12) and Corollary 3.1.15 (preprint 3.13) for the 2021 definitions; `Polyfold.theorem_3_1_13`, `Polyfold.corollary_3_1_15`). For a 2021 sc-Fredholm germ (f, x_0) at a zero: nested open neighbourhoods $O(i)$ of x_0 such that the closure of the zeros in $O(i)$ is compact on level i; and a sequence of zeros converging to x_0 converges on every level. *Book:* Theorem 3.1.13 and Corollary 3.1.15, p. 97 (preprint 3.12 and 3.13, p. 70). *Hidden hypotheses:* none. *Deviation (restated):* V3.14 and V3.15 through V3.110; their proofs do not use tameness.

V3.113 (Theorems 3.1.18 (preprint 3.15) and 3.1.21 (preprint 3.18) for the 2021 definitions; `Polyfold.theorem_3_1_18`, `Polyfold.theorem_3_1_21`). If f is a 2021 sc-Fredholm section and s an sc$^+$-section, $f + s$ is a 2021 sc-Fredholm section; for sc$^+$-sections $s_1, \ldots, s_n$, $(\lambda, x) \mapsto f(x) + \sum_i \lambda_i s_i(x)$ is a 2021 sc-Fredholm section of the pull-back bundle over $\mathbb{R}^n \times X$. *Book:* Theorem 3.1.18, p. 98 (preprint 3.15, p. 71), and Theorem 3.1.21, p. 100 (preprint 3.18, p. 74). *Hidden hypotheses:* none. *Deviation (restated):* V3.17 and V3.91 through V3.110.

V3.114 (Proposition 3.6.5 (preprint 3.52) for the 2021 definitions; `Polyfold.proposition_3_6_5`). At a zero x_0 of a 2021 sc-Fredholm germ, in a chart centred at x_0 with a filling g, $Dg(0)$ and the restriction $T_0O \to K_0$ are Fredholm of index $n - N$. *Book:* Proposition 3.6.5, p. 135 (preprint 3.52, p. 110). *Hidden hypotheses:* none. *Deviation (restated):* V3.46 through V3.110.

V3.115 (Theorem 3.6.8 (preprint 3.54), the proof step on p. 138 without a tame chart; `Polyfold.CornerFilling.isGoodPositionAtGerm_gc`, `Polyfold.CornerFilling.not_isInGoodPosition_gc`, `Polyfold.CornerFilling.not_range_le_minimalSubspace`, `Polyfold.theorem_3_6_8_tame_step`). In the example of V3.109: g is a 2021 filling of $\hat{f} = 0$, r is not tame, $0 \in O$, $\hat{f}(0) = 0$, g is sc¹ with $Dg(0)$ onto, and $\ker T\hat{f}(0)$ is in good position to the partial cone in T_0O (V3.47); but the range of $1 - Dr(0)$ does not lie in the subspace E_0 of C at 0, and $\ker Dg(0) = 0 \oplus \mathbb{R}$ is not in good position to C in E. *Book:* Theorem 3.6.8, p. 136 (preprint 3.54, p. 111), proof pp. 137–138: the passage from good position in T_0O to good position of $\ker Dg(0)$ in E uses the sc-complement $\ker Dr(0) \subseteq E_0$ of T_0O, given by Proposition 2.5.3 (preprint 2.35) for a tame retract; the 2021 text calls the retract tame "by assumption". *Hidden hypotheses:* a tame chart. *Deviation (not in the book):* with the untamed 2021 definitions this step does not hold as we read it. In this example the solution set $O \cong \mathbb{R}$ is still a manifold; the counterexample to the conclusion itself is V3.130 (a tame strong bundle over $\mathbb{R}$ whose solution set is a closed half-line). Theorem

3.6.8 and what rests on it are verified with tame germ data (V3.85, V3.94, V3.103) (gap G.79).

V3.116 (Theorem 3.5.1 (preprint 3.40) with the 2021 sequential conclusion; `Polyfold.IsBasicGerm.solves`, `Polyfold.theorem_3_5_1`). Let h be a basic germ on an sc-germ of neighbourhoods G of 0 in $C = \widetilde{C} \times W$ ($\widetilde{C}$ closed), $U = G_0$, s an sc$^+$-germ on G with $s(0) = 0$, and $S = \{x \in U : h(x) + s(x) = 0\}$. Then there are nested neighbourhoods $O(0) \supseteq O(1) \supseteq \cdots$ of 0, relatively open in C and inside U, with $\mathrm{cl}(S \cap O(m)) \subseteq C_m$; and if $x_k \in S$ converges to 0 in E_0, then for every m there are $y_k \in E_m$ with $y_k = x_k$ for k large and $y_k \to 0$ in E_m. *Book:* Theorem 3.5.1, p. 118 (preprint 3.40, p. 94); the sequential conclusion is new in the 2021 edition. *Hidden hypotheses:* as V3.35. *Deviation (restated):* the closure is taken in E_0 (it lies in C, which is closed); "converges on level m" is stated with level-m lifts. *Proof:* from V3.14 and V3.15 in the local form (`Polyfold.theorem_3_1_13_local`, `Polyfold.corollary_3_1_15_local`).

V3.117 (closed convex cones and extreme rays; `Polyfold.IsClosedConvexCone`, `Polyfold.IsExtremeRayGen`, `Polyfold.IsClosedConvexCone.eq_zero`, `Polyfold.isClosedConvexCone_Ici`, `Polyfold.isExtremeRayGen_one`, `Polyfold.isExtremeRayGen_symm_single`, `Polyfold.exists_sub_single_mem`, `Polyfold.exists_quadrantEquiv_of_pointed`). A closed convex cone K is a closed convex set with $K \cap (-K) = \{0\}$ and $tK \subseteq K$ for $t \geq 0$; a generates an extreme ray if $a \in K \setminus \{0\}$ and $y \in K$, $a - y \in K$ imply $y \in \mathbb{R}_+ a$. $[0, \infty)$ is a closed convex cone generated by the extreme ray of 1. If N is in good position to $C = T^{-1}([0, \infty)^n \oplus W)$ and $N \cap C$ is pointed, a linear isomorphism $e : N \cong \mathbb{R}^{\dim N}$ carries $N \cap C$ onto $[0, \infty)^{\dim N}$, and the $e^{-1}(e_j)$ generate extreme rays. *Book:* p. 145, new in the 2021 edition, for finite-dimensional spaces. *Hidden hypotheses:* none. *Deviation (stronger):* the quadrant normal form, from V3.21, is added; it replaces the Krein–Milman-type Lemmas 3.8.1–3.8.3 in the proofs of V3.118–V3.121.

V3.118 (Lemma 3.8.4; `Polyfold.lemma_3_8_4`, `Polyfold.lemma_3_8_8`). Let N be in good position to $C = T^{-1}([0, \infty)^n \oplus W)$ with good complement $N^\perp$, $N \cap C$ a closed convex cone, σ_a the set of i with $a_i = 0$, and $\Gamma = \bigcup_{0 \neq a \in N \cap C} \sigma_a$. Then (1) $y_i = 0$ for $y \in N^\perp$ and $i \in \sigma_a$, a a generator of an extreme ray; (2) Γ is the union of the σ_a over generators of extreme rays; (3) $y_i = 0$ for $y \in N^\perp$ and $i \in \Gamma$. *Book:* Lemma 3.8.4, p. 147, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* (1) and (3) hold without the cone hypothesis (Lemma 3.8.8, V3.55); (2) uses only that $N \cap C$ is pointed. *Proof:* (2) not by writing a as a sum of extreme-ray generators (Krein–Milman, Lemma 3.8.1): for $0 \neq a \in N \cap C$ with $a_i = 0$ and $e(a)_j > 0$, $a - e(a)_j e^{-1}(e_j) \in N \cap C$ forces $e^{-1}(e_j)_i = 0$ (V3.117).

V3.119 (Lemma 3.8.5; `Polyfold.lemma_3_8_5`, `Polyfold.lemma_3_8_5_not_literal`). For N in good position to C with $N \cap C$ a closed convex cone, either $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$, or $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, $\dim N = 1$ and $\Gamma = \emptyset$. In $E = \mathbb{R}$, $C = [0, \infty)$, the subspace $N = 0$ is in good position, $N \cap C = \{0\}$ is a closed convex cone and $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, but $\dim N = 0$. *Book:* Lemma 3.8.5, p. 147, new in the 2021 edition: either $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, in which case $\dim N = 1$ and $\Gamma = \emptyset$, or $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$. *Hidden hypotheses:* $N \neq 0$ for the first case. *Deviation (restated):* stated as the dichotomy that the proof shows (it assumes $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) \neq \{0\}$); read as "$N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$ implies $\dim N = 1$" it does not hold as we read it for $N = 0$. *Proof:* if $\Gamma \neq \emptyset$, a point of N with $x_i = 0$ for $i \in \Gamma$ lies in $N \cap C$ with $-x$; if $\Gamma = \emptyset$ and $\dim N \geq 2$, two generators with all coordinates positive give $b_0 - tb_1 \in N \cap C$ for small $t > 0$, against V3.117.

V3.120 (Lemma 3.8.6; `Polyfold.lemma_3_8_6`, `Polyfold.lemma_3_8_6_two`, `Polyfold.halfLine_witness`). Let N be in good position to C and $P : N \to \mathbb{R}^{\Gamma}$ the coordinate projection. (1) If $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$, P is a linear isomorphism carrying $N \cap C$ onto $[0, \infty)^{\Gamma}$. (2) If $N \cap C$ is a closed convex cone, $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$ and $N \neq 0$, there is a linear isomorphism $N \cong \mathbb{R}$ carrying $N \cap C$ onto $[0, \infty)$. The hypotheses of (2) are satisfiable ($N = E = \mathbb{R}$, $C = [0, \infty)$). *Book:* Lemma 3.8.6, p. 148, new in the 2021 edition, both parts under the cone hypothesis. *Hidden hypotheses:* $N \neq 0$ in (2). *Deviation (narrower):* (2) assumes $N \neq 0$; as printed it does not hold as we read it for $N = 0$ (V3.119). (1) holds without the cone hypothesis.

V3.121 (Corollary 3.8.7; `Polyfold.corollary_3_8_7`). If N is in good position to C and $N \cap C$ is a closed convex cone, then $N \cap C$ is a partial quadrant in N, and a linear isomorphism $N \cong \mathbb{R}^{\dim N}$ carries $N \cap C$ onto $[0, \infty)^{\dim N}$. *Book:* Corollary 3.8.7, p. 149, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the quadrant form, which the text also derives on p. 149; the partial quadrant is V3.21 (Proposition 3.1.25 (preprint 3.22)) without the cone hypothesis.

V3.122 (Remark 3.3.8 (preprint 3.36): the quantitative solution germ; `Polyfold.remark_3_3_8`, `Polyfold.ContractionGerm.exists_radius`, `Polyfold.ContractionGerm.fixLvl_zero_eq`, `Polyfold.exists_antitone_le`). `remark_3_3_8`: for an sc^0-contraction germ $f(a, w) = w - B(a, w)$ that is sc-smooth, on a quadrant-like $\widetilde{C}$, there are δ and decreasing positive sequences (ε_i), (τ_i) such that $\delta(0) = 0$; on $\{|a|_0 \leq \varepsilon_0\}$, δ is continuous with $|\delta(a)|_0 \leq \tau_0$ and $f(a, \delta(a)) = 0$; a solution w with $|w|_0 \leq \tau_0$ is $\delta(a)$; $\delta(a) \in W_i$ and $|\delta(a)|_i \leq \tau_i$ for $|a|_0 \leq \varepsilon_i$; and $\delta : \{|a|_0 \leq \varepsilon_i\} \to W_i$ is of class C^i. *Book:* §3.3, Remark 3.3.8 (preprint 3.36), p. 113 (preprint p. 89), Decreasing sequences (ε_i), (τ_i) with: δ on $\{|a|_0 \leq \varepsilon_0\}$ a continuous solution, unique for $|w|_0 \leq \tau_0$; $\delta(a) \in W_i$, $|\delta(a)|_i \leq \tau_i$ for $|a|_0 \leq \varepsilon_i$; $\delta : \{|a|_0 \leq \varepsilon_i\} \to W_i$ of class C^i. *Deviation (same):* Obtained from Theorem 3.3.2 (preprint 3.29) and Corollary 3.3.7 (preprint 3.34) (V3.26, V3.31) by choosing ε_i and τ_i as running minima of the radii on each level; "decreasing" is read as non-increasing.

V3.123 (Lemma 3.8.1 (preprint 3.62): a closed convex cone is the closed convex hull of its extreme rays; `Polyfold.lemma_3_8_1`, `Polyfold.not_lemma_3_8_1_zero`, `Polyfold.lemma_3_8_1_zero_union`, `Polyfold.rayOf`, `Polyfold.IsExtremeRay`, `Polyfold.extremeRays`, `Polyfold.IsClosedConvexCone.isExtremeRay_rayOf_iff`, `Polyfold.IsClosedConvexCone.exists_pos_functional`). `lemma_3_8_1`: for a closed convex cone $P \neq \{0\}$ in a finite-dimensional space, P is the closure of the convex hull of the union of its extreme rays (`extremeRays`; an extreme ray is a set, `IsExtremeRay`, equivalent to the project's generator form, `IsClosedConvexCone.isExtremeRay_rayOf_iff`). `not_lemma_3_8_1_zero`: the statement for every closed convex cone does not hold as we read it, at $P = \{0\}$. `lemma_3_8_1_zero_union`: with 0 adjoined to the extreme rays it holds for every closed convex cone. *Book:* §3.8.1, Lemma 3.8.1 (preprint 3.62), p. 145 (preprint p. 121), A closed convex cone P in a finite-dimensional vector space N is the closed convex hull of its extreme rays (a version of the Krein–Milman theorem, cited from the literature). *Deviation (narrower):* As printed the lemma does not hold as we read it for $P = \{0\}$: $\{0\}$ is a closed convex cone without extreme rays, and the closed convex hull of the empty family is empty (`not_lemma_3_8_1_zero`; Findings, Chapter 3, item 386). It is proved for $P \neq \{0\}$, by a functional positive on $P \setminus 0$

and Mathlib's Krein–Milman theorem on the compact base, and with 0 adjoined for every cone. Nothing in the text depends on the case $P = \{0\}$: in Lemma 3.8.11 (preprint 3.68) the cone has nonempty interior in $\widetilde{N} \neq 0$ (V3.125).

V3.124 (Lemmas 3.8.2, 3.8.3 (preprint 3.63, 3.64): quadrants and extreme rays; $\dim N - 1 \leq \#\sigma_a$; `Polyfold.lemma_3_8_2`, `Polyfold.IsQuadrantIn`, `Polyfold.extremeRays_nonnegOrthant`, `Polyfold.encard_extremeRays_nonnegOrthant`, `Polyfold.extremeRays_image`, `Polyfold.lemma_3_8_3`, `Polyfold.lemma_3_8_3_goodPosition`, `Polyfold.zeroIdxProj`, `Polyfold.ker_zeroIdxProj_le`). `lemma_3_8_2`: for a closed convex cone P with nonempty interior in a finite-dimensional N, P is a quadrant (`IsQuadrantIn`: a linear isomorphism $N \cong \mathbb{R}^{\dim N}$ carrying P onto $[0,\infty)^{\dim N}$) iff its set of extreme rays has `encard` $\dim N$. `lemma_3_8_3`, `lemma_3_8_3_goodPosition`: in the text's model, if $N \cap C$ is a closed convex cone and a generates an extreme ray of it, $\dim N - 1 \leq \#\sigma_a$ (`quadrantDeg`), with equality when N is in good position; the map (3.34) is `zeroIdxProj`, (3.33) is `ker_zeroIdxProj_le`. *Book:* §3.8.1, Lemmas 3.8.2, 3.8.3 (preprint 3.63, 3.64), pp. 145–146 (preprint pp. 121–122), A closed convex cone with nonempty interior in N is a quadrant iff it has $\dim N$ extreme rays (3.8.2, cited from the literature). In $E = \mathbb{R}^n \oplus W$, $C = [0,\infty)^n \oplus W$, N finite-dimensional and smooth with $N \cap C$ a closed convex cone: a generator a of an extreme ray of $N \cap C$ has $\dim N - 1 \leq \#\sigma_a$, with equality if N is in good position (3.8.3). *Deviation (same):* "$\dim N$-many extreme rays" is read as `encard`. Lemma 3.8.2 is cited by the text from the literature; here it is proved. The formal proof of Proposition 3.1.25 (preprint 3.22) still does not use these lemmas (V3.21; Findings, Chapter 3, item 18).

V3.125 (Lemmas 3.8.9–3.8.11 (preprint 3.66–3.68): the complement $\widetilde{N}$; `Polyfold.lemma_3_8_9_sc`, `Polyfold.lemma_3_8_9`, `Polyfold.lemma_3_8_9_of_isClosed`, `Polyfold.lemma_3_8_10`, `Polyfold.isClosedConvexCone_of_disjoint`, `Polyfold.lemma_3_8_11`, `Polyfold.exists_eq_smul_of_isExtremeRayGen`, `Polyfold.wPart`, `Polyfold.bigSigma`, `Polyfold.sigmaCompl`, `Polyfold.ScBanach.isScSubspace_sup_of_le_smooth`, `Polyfold.ScBanach.isCompl_levelSub_of_isScSplitting`). In the text's model $E = \mathbb{R}^n \oplus W$, $C = [0,\infty)^n \oplus W$, with N in good position (`IsInGoodPositionSc`), $N^{\perp}$ a good sc-complement (`IsScGoodComplement`) and $\widetilde{N}$ an algebraic complement of $N \cap W$ in N: `lemma_3_8_9_sc`: $\widetilde{N}$ is in good position and $(N \cap W) \oplus N^{\perp}$ is a good sc-complement of it (an sc-subspace by `ScBanach.isScSubspace_sup_of_le_smooth`; the algebraic core is `lemma_3_8_9`, which needs only $N^{\perp}$ closed). `lemma_3_8_10`: $C \cap \widetilde{N}$ is a closed convex cone in $\widetilde{N}$, and a generator a of an extreme ray has $\dim \widetilde{N} - 1 = \#\sigma_a = d_C(a)$. `lemma_3_8_11`: $\widetilde{N} \cap (\mathbb{R}^{\Sigma^c} \oplus W) = 0$, or $\widetilde{N} \subseteq \mathbb{R}^{\Sigma^c} \oplus W$ with $\dim \widetilde{N} = 1$ and $\Sigma = \emptyset$. *Book:* §3.8.1, Lemmas 3.8.9–3.8.11 (preprint 3.66–3.68), pp. 150–152 (preprint pp. 123–125), $\widetilde{N}$ is in good position with good complement $(N \cap W) \oplus N^{\perp}$ (3.66); $C \cap \widetilde{N}$ is a closed convex cone whose extreme rays satisfy $\dim \widetilde{N} - 1 = \#\sigma_a = d_C(a)$ (3.67); either $\widetilde{N} \cap (\mathbb{R}^{\Sigma^c} \oplus W) = \{0\}$ or $\widetilde{N} \subseteq \mathbb{R}^{\Sigma^c} \oplus W$ with $\dim \widetilde{N} = 1$, $\Sigma = \emptyset$ (3.68). *Deviation (same):* The proofs are the text's; Lemma 3.8.11 (preprint 3.68) uses Lemma 3.8.1 (preprint 3.62) with 0 adjoined (V3.123). The formal proof of Proposition 3.1.25 (preprint 3.22) still treats its cases directly (V3.21).

V3.126 (Definition 3.8.14 (preprint 3.71): class C^k on partial quadrants; `Polyfold.IsClassCOn`,

`Polyfold.IsClassC1On`, `Polyfold.IsClassSmoothOn`, `Polyfold.bookTangentMap`, `Polyfold.isClassC1On_iff_tendsto`, `Polyfold.isClassC1On_iff_contDiffOn`, `Polyfold.isClassCOn_iff_contDiffOn`, `Polyfold.isClassSmoothOn_iff_contDiffOn`, `Polyfold.isClassCOn_iff_contDiffOn_of_isRelOpenInQuadrant`, `Polyfold.isSc1_iff_isClassC1On`, `Polyfold.isScSmooth_iff_isClassSmoothOn`). `IsClassC1On U f`: there is Df with f differentiable within U at every $x \in U$ with derivative $Df(x)$ (one-sided derivatives in the directions into U; the limit form is `isClassC1On_iff_tendsto`) and Df continuous on U. `IsClassCOn k U f` by recursion through the text's tangent map $Tf(x,h) = (f(x), Df(x)h)$ (`bookTangentMap`): C^0 is continuity, C^{k+2} means f and Tf on $U \times V$ are C^{k+1}; `IsClassSmoothOn` is C^k for all k. On every set with unique derivatives it agrees with Mathlib's `ContDiffOn` (`isClassCOn_iff_contDiffOn`), in particular on relatively open subsets of partial quadrants, where it also agrees with sc¹ and sc$^\infty$ (`isSc1_iff_isClassC1On`, `isScSmooth_iff_isClassSmoothOn`). *Book:* §3.8.5 (preprint §3.8.4), Definition 3.8.14 (preprint 3.71), p. 164 (preprint p. 135), $f : U \to \mathbb{R}^m$ on a relatively open subset of a partial quadrant is C^1 if it has one-sided derivatives depending continuously on x; C^k if f and Tf are C^{k-1}. *Deviation (restated):* The one-sided derivatives are derivatives within U (`HasFDerivWithinAt`), which on these sets are unique. The theorems of §§3.8.5–3.8.6 (preprint §§3.8.4–3.8.5) keep Mathlib's `ContDiffOn`, which is this class by `isClassCOn_iff_contDiffOn_of_isRelOpenInQuadrant`.

V3.127 (Lemma 3.8.20 (preprint 3.76): iterated tangent maps; `Polyfold.lemma_3_8_20`, `Polyfold.lemma_3_8_20_printed`, `Polyfold.QuadMapData`, `Polyfold.QuadMapData.IsSmoothHomeo`, `Polyfold.QuadMapData.tangent`, `Polyfold.QuadMapData.IsSmoothHomeo.tangent`, `Polyfold.QuadMapData.iterTangent`, `Polyfold.constProdIso`). `lemma_3_8_20`: let f be a smooth homeomorphism between relatively open subsets U, $f(U)$ of a partial quadrant C in a finite-dimensional space, with $Df(x)$ invertible (`QuadMapData.IsSmoothHomeo`). Then for every k the iterated tangent map $T^k f$ (`QuadMapData.iterTangent`; $TU = U \times V$ in $C \times V$) is again a smooth homeomorphism of this kind, and its inverse is C^∞ (by Lemma 3.8.21 (preprint 3.77) in its general form, V3.62). `lemma_3_8_20_printed` is the statement under the hypotheses of Proposition 3.8.17 (preprint 3.73). *Book:* §3.8.5 (preprint §3.8.4), Lemma 3.8.20 (preprint 3.76), p. 167 (preprint p. 139), Under the assumptions of Proposition 3.8.17 (preprint 3.73) the iterated tangent maps $T^k f : T^k U \to T^k V$ are smooth homeomorphisms. *Deviation (stronger):* Proved for every smooth homeomorphism with invertible derivative, of which the hypotheses of Proposition 3.8.17 (preprint 3.73) are a case. The proof is the text's: Lemma 3.8.21 applied to $T^k f$, which needs the general form of Lemma 3.8.21, since $T^k f$ does not satisfy the hypotheses of Proposition 3.8.17. The regularity of f^{-1} in Proposition 3.8.17 is still proved by bootstrapping (V3.59).

V3.128 (Lemma 3.8.13 (preprint 3.70) and Lemma 3.6.9 (preprint 3.55) in a general presentation of C; Theorem 3.2.3 (preprint 3.27) and d_X; `Polyfold.lemma_3_8_13_general`, `Polyfold.lemma_3_8_13_local`, `Polyfold.proposition_3_8_12_general`, `Polyfold.mem_lineality_iff`, `Polyfold.coordCLMOf`, `Polyfold.mem_nullSpace_coordFun_iff`, `Polyfold.MPolyfold.theorem_3_2_3_degIndex_eq`). `lemma_3_8_13_general`: for any partial quadrant C of an sc-Banach space, N in good position with a good complement Y and $\tau : N \to Y$ of class C^1 on $N \cap C$ with $\tau(0) = 0$, $D\tau(0) = 0$, there is $\rho > 0$ such that for $v \in N \cap C$ with $|v| < \rho$,

$v + \tau(v) \in C$, and $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $-v \notin C$ (in the text's model: $v \notin N \cap W$). `lemma_3_8_13_local`: the same with τ given only on a relatively open $V \ni 0$. `proposition_3_8_12_general`: Proposition 3.8.12 (preprint 3.69) for any presentation of C. `MPolyfold.theorem_3_2_3_degIndex_eq`: in Theorem 3.2.3 (preprint 3.27), near the centre of a regular datum and off the lineality space, $d_M(y) = d_X(y)$. Lemma 3.6.9 (preprint 3.55) for any presentation is V3.80. *Book:* §3.8.4 (preprint §3.8.3), Lemma 3.8.13 (preprint 3.70), p. 154 (preprint p. 127); §3.2, proof of Theorem 3.2.3 (preprint 3.27), In $E = \mathbb{R}^n \oplus W$, $C = [0,\infty)^n \oplus W$: for N smooth, finite-dimensional, in good position with good complement Y, and $\tau : N \cap C \to Y_1$ of class C^1 with $\tau(0) = 0$, $D\tau(0) = 0$: $d_{N\cap C}(v) = d_C(v + \tau(v))$ for $v \in N \cap C \setminus N \cap W$ near 0. The proof of Theorem 3.2.3 (preprint 3.27) uses it to compute $d_{N\cap C}(v) = d_X(y)$; Lemma 3.6.9 (preprint 3.55) is stated for any partial quadrant $C \subseteq E$. *Deviation (stronger):* The text states Lemma 3.8.13 in the model $\mathbb{R}^n \oplus W$; it holds for every presentation of C, with "$v \notin N \cap W$" as "$-v \notin C$", which is the same in the model. The text does not print $d_M = d_X$ as part of Theorem 3.2.3 (preprint 3.27); the proved form holds near the centre of a datum, off the lineality space, and some restriction is needed: for the diagonal M in $[0,\infty)^2$, $d_M(0) = 1$ while $d_X(0) = 2$ (an example in the docstring, not in Lean; Findings, Chapter 3, item 387). The 2021 use of Lemma 3.8.13 is in the proof of Theorem 3.5.11, (3.30), p. 132.

V3.129 (An sc-smooth family of projections whose rank jumps, and a strong bundle retraction with jumping fibre; `Polyfold.Jump.Family`, `Polyfold.Jump.Family.isScSmooth_map`, `Polyfold.Jump.family`, `Polyfold.Jump.exists_proj_eq`, `Polyfold.Jump.proj_proj`, `Polyfold.Jump.proj_of_nonpos`, `Polyfold.UntamedCorner.isStrongBundleRetraction_R`, `Polyfold.UntamedCorner.not_isTame_baseRetr`). `Jump.Family.isScSmooth_map`: a family of operators $A_m(t)$ on the levels, compatible with the inclusions, continuous in (t,h), norm-differentiable in t on V_{m+1} with derivative $A'_m(t)$, and with $(s,t,h) \mapsto sA'(t)h$ sc-smooth, gives an sc-smooth map $(t,h) \mapsto A(t)h : \mathbb{R} \oplus V \to V$ (`Jump.Family`). On a weighted sequence space there is such a family of projections π_t with $\pi_t = 0$ for $t \leq 0$ and of rank one for $0 < t < 1$ (`Jump.family`, `Jump.proj_of_nonpos`, `Jump.exists_proj_eq`, `Jump.proj_proj`); it is not C^1 on any one level, so Proposition 1.2.4 does not produce it. $(((a,t),w),(x,h)) \mapsto (((0,t),0),(0,\pi_t h))$ is a strong bundle retraction whose fibre jumps (`UntamedCorner.isStrongBundleRetraction_R`), over a base retraction that is not tame (`UntamedCorner.not_isTame_baseRetr`). *Book:* no counterpart in the text (fillings over sc-smooth retractions, Definition 3.1.4, p. 91; retracts of locally varying dimension, after Theorem 2.3.16, p. 40), Not stated. After Theorem 2.3.16 the text refers to the literature for retracts of locally varying dimension. *Deviation (extra):* A tool built for the formalization (gap G.79).

V3.130 (Theorem 3.6.8 with the 2021 germ data as printed: a tame strong bundle over $\mathbb{R}$ whose solution set is a half-line; `Polyfold.UntamedCorner.theorem_3_6_8_untamed_not`, `Polyfold.UntamedCorner.theorem_3_1_22_untamed_not`, `Polyfold.UntamedCorner.theorem_3_6_8_untamed_example`, `Polyfold.UntamedCorner.not_theorem_3_6_8_conclusion`, `Polyfold.StrongBundle.IsTameBundle`, `Polyfold.UntamedCorner.isTameBundle_bundle`, `Polyfold.UntamedCorner.bundle`, `Polyfold.UntamedCorner.chart0`, `Polyfold.UntamedCorner.sec`, `Polyfold.UntamedCorner.isScSmoothSection_sec`, `Polyfold.UntamedCorner.solutionSet_`

`sec`, `Polyfold.UntamedCorner.isScFredholmGerm2021_sec_zero`, `Polyfold.UntamedCorner.isGoodPositionGerm_sec_zero`, `Polyfold.UntamedCorner.subsingleton_fiber_zero`, `Polyfold.UntamedCorner.not_isSubMPolyfold_zeroSet`, `Polyfold.UntamedCorner.isEmpty_tangentGermChart`, `Polyfold.UntamedCorner.Frame`, `Polyfold.UntamedCorner.Frame.chart`, `Polyfold.UntamedCorner.Frame.compatible_chart0`, `Polyfold.UntamedCorner.Frame.isScFredholmGermInterior_sec`, `Polyfold.UntamedCorner.frame0`, `Polyfold.UntamedCorner.frame1`, `Polyfold.UntamedCorner.isScFredholmGermTame_sec_of_ne`, `Polyfold.UntamedCorner.isScFredholmSection2021_sec`, `Polyfold.UntamedCorner.chartTame`, `Polyfold.UntamedCorner.isStrongBundleRetraction_RT`, `Polyfold.UntamedCorner.compatible_chart0_chartTame`, `Polyfold.UntamedCorner.isTameChart_chartTame`, `Polyfold.UntamedCorner.exists_isTameChart`, `Polyfold.UntamedCorner.theorem_3_6_8_untamed_local`, `Polyfold.UntamedCorner.not_isTame_baseRetr`). With the 2021 definitions as printed the statement does not hold as we read it. `bundle : K → O` is a strong bundle over the tame M-polyfold without boundary $O \cong \mathbb{R}$, with an atlas of tame strong bundle charts (`isTameBundle_bundle`, `chartTame`, `exists_isTameChart`); the fibre is 0 for $t \leq 0$ and a line for $0 < t < 1$. The section `sec` is sc-Fredholm in the sense of 2021 Definitions 3.1.16 = 3.6.4 (`isScFredholmSection2021_sec`): at x_0 ($t = 0$) its germ data lie in the atlas chart `chart0`, whose base retraction is not tame (`isScFredholmGerm2021_sec_zero`), at the other points in tame charts (`isScFredholmGermTame_sec_of_ne`, from the frames `frame0`, `frame1`). x_0 is a zero in good position (`isGoodPositionGerm_sec_zero`; $d_X(x_0) = 0$, fibre 0, `subsingleton_fiber_zero`). The solution set is the closed half-line $\{t \leq 0\}$ (`solutionSet_sec`): it has no tangent germ data at x_0 (`isEmpty_tangentGermChart`) and is not a sub-M-polyfold near x_0 (`not_isSubMPolyfold_zeroSet`). Collected in `theorem_3_6_8_untamed_example`; the universal statements are `theorem_3_6_8_untamed_not` (Theorem 3.6.8, conclusion (1) weakened to good position, so only (2) fails) and `theorem_3_1_22_untamed_not` (Theorems 3.1.22, 3.1.26). With tame germ data the theorem holds (V3.85, V3.94, V3.103). *Book:* §3.1 and §3.6, Definitions 3.1.4, 3.6.1 and Theorem 3.6.8, pp. 91, 134–138; with Theorems 3.1.22, 3.1.26, Corollary 3.1.27, Theorems 3.6.10, 3.6.11, Theorem 3.6.8 (p. 136) for an sc-Fredholm section of a tame strong bundle, with the 2021 Definition 3.6.3 (p. 135): the germ data are taken in "a strong bundle chart around x", with a filling over an sc-smooth retraction (Definitions 3.1.4, 3.6.1, pp. 91, 134), not asked to be tame. Conclusion (2): near a zero in good position the solution set is a sub-M-polyfold with the structure of a manifold with boundary with corners. Theorems 3.1.22 and 3.1.26 give the same over an M-polyfold without boundary. *Deviation (narrower):* Every printed hypothesis holds in the example (p1-U-untamed2, §0): the bundle is a strong bundle with a strong bundle atlas over a tame M-polyfold, and it is a "tame strong bundle" in both readings of that undefined phrase (the base is tame, and every point lies in a tame strong bundle chart of the structure, `IsTameBundle`); the germ data at x_0 are in a chart of the atlas, as Definition 3.6.3 asks; the filling, the sc⁺-section $s = 0$ and the basic germ are those of Definitions 3.1.4, 3.1.7, 3.1.11; good position holds at x_0 in every reading ($d_X \equiv 0$, $f'(x_0)$ onto the zero fibre). Only the proof (p. 137) takes the chart of the germ data tame; the preprint and the 2017 text put the tame retraction into the filling. With that reading the theorem is verified (V3.85, V3.94, V3.103). Mechanism: the fibre jumps from 0 to a line at x_0 through the sc-smooth family of projections of V3.129, which is not C^1 on any level (Findings, Chapter 3, item 388; gap G.79).

V3.131 (A 2021 sc-Fredholm germ with no tame germ data, and not a 2017 germ; `Polyfold.UntamedCorner.not_isScFredholmGermTame_sec_zero`, `Polyfold.UntamedCorner.isScFredholmGerm2021_not_tame`, `Polyfold.UntamedCorner.isScFredholmGerm2021_not_2017`, `Polyfold.UntamedCorner.isScFredholmGerm2021_sec_zero`). There are a tame strong bundle and an sc-Fredholm section in the sense of 2021 Definitions 3.1.16 = 3.6.4 with a point at which the germ has germ data (Definition 3.6.3) in a strong bundle chart of the structure but none in a tame one: `not_isScFredholmGermTame_sec_zero` (with tame data at x_0, Theorem 3.6.8 in its tame form would give tangent germ data, which do not exist), `isScFredholmGerm2021_not_tame` (the universal form), `isScFredholmGerm2021_not_2017` (a 2021 germ that is not a 2017 germ). *Book:* §3.6, Definition 3.6.3, p. 135; Definitions 3.1.4, 3.6.1, pp. 91, 134 (no counterpart), Not stated. Definition 3.6.3 takes the germ data in "a strong bundle chart around x"; the 2017 text and the preprint take the filling over a tame retraction. *Deviation (extra):* Answers in the negative whether germ data in a compatible chart can be moved to a tame chart (Findings, Chapter 3, items 353, 388; gap G.79).

What is not verified in this chapter

- pp. 136–139 (preprint pp. 111–114), Theorem 3.6.8 (1) (preprint 3.54): nothing of the statement; the proof is not the text's (V3.93), and the first forms under the filler condition at the nearby zeros (V3.68, V3.69, V3.82, V3.85) are kept beside the forms with the text's hypotheses (V3.94, V3.95). Of the perturbation argument of the text's Proposition 6.2.2 (preprint 6.2), used for the surjectivity at the nearby zeros, only the linear algebra is formalized (V3.83); for the index it is formalized (V3.98).
- preprint p. 201 (Chapter 6), Proposition 6.2.2 (preprint 6.2) for paths sc-smooth on $[0, 1]$ only: it is verified with the text's hypotheses (tame germ data, no bump functions, Definition 6.2.1 (preprint 6.1) with local sections) for paths on $\mathbb{R}$ (V3.98); a path on $[0, 1]$ that does not extend sc-smoothly to $\mathbb{R}$ is not covered, the project having no M-polyfold $[0, 1]$. Without tame germ data the local constancy of the index at the zeros is open (Findings, Chapter 3, items 243, 343).
- p. 135 (preprint p. 110), Proposition 3.6.5 (preprint 3.52) on $T_xX \to Y_x$ itself: verified in charts, as an sc-Fredholm operator $T_zO \to K_z$ in every chart (V3.46, V3.96); T_xX carries no linear structure here (Chapter 2).
- Theorem 3.2.3 (preprint 3.27) for a set whose data do not satisfy condition (b) of V3.24 (it would need a re-graphing along another complement); for solution sets (b) holds by the choice of the complement (V3.77, V3.84).
- Good position in T_xX (Definition 2.5.8 (preprint 2.38), gap G.13 of Chapter 2) and the linearization $f'(x) : T_xX \to Y_x$ away from the zeros: Definition 3.6.6 (preprint 3.53) is verified in charts, with its chart independence at zeros (V3.78); at the zeros on level 1, $f'(x)$ is defined on T_xX and does not depend on the chart (`Polyfold.StrongBundle.linIntr`, V3.68).
- The chart-level half of the argument that the earlier hypothesis "the base retraction of every compatible strong bundle chart is tame" cannot be met (Findings, Chapter 3, items 190, 199):

that the strong bundle chart $(1, \varphi)$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ is a chart compatible with every chart (φ, U, R, Φ) is (sc-calculus for affine maps, not done). Only the non-tameness of its base retraction is verified (V3.104); the theorems no longer use the hypothesis.

- Definition 3.1.11 (preprint 3.11) ⇒ Definition 3.6.3 (preprint 3.50) (Findings, Chapter 3, items 199, 353): whether in a strong bundle whose atlas consists of tame charts every sc-Fredholm germ has germ data in some compatible tame chart is open. Re-centring a tame chart of the atlas does not do it: a filling needs ker $Dr(0) \cong$ ker $\rho(0)$ in its chart, and a one-chart tame atlas of the trivial bundle carries no germ data at all (V3.103). The text builds the tame chart into Definition 3.6.3 (preprint 3.50), and the theorems of §3.6 assume tame germ data.
- §3.1: Definitions 3.1.4, 3.1.6, 3.1.11, 3.1.16 (preprint 3.4, 3.6, 3.11, 3.14) (and 3.48, 3.50, 3.51, 3.53) are in the text's germ reading since phase 2 (V3.7, V3.9, V3.13, V3.16, V3.43, V3.45, V3.47), and Definition 3.1.3 (preprint 3.3) since the fourth addendum (V3.5). Whether the text's germ reading and the project's earlier whole-chart reading of Definition 3.1.11 (preprint 3.11) agree for sections sc-smooth on X is open; no theorem depends on it (Findings, Chapter 3, items 220–221, 264, 330).
- pp. 91, 134–138, the 2021 (untamed) definitions: settled since the sixth addendum (V3.130, V3.131, gap G.79; Findings, Chapter 3, item 388). With germ data in any strong bundle chart, as the 2021 Definition 3.6.3 is printed, Theorem 3.6.8 (2) and the sub-M-polyfold conclusion of Theorems 3.1.22, 3.1.26 do not hold as we read it, and a 2021 sc-Fredholm germ need not have tame germ data; the theorems are verified with tame germ data (V3.85, V3.94, V3.103). Not verified: Corollary 3.1.27 and Theorems 3.6.10, 3.6.11 for untamed germ data, beyond what the counterexample shows.

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5. Chapter 4 (HWZ §§4.1–4.4): verified except two remarks

Chapter 4 of Part I (pp. 175–204 (preprint pp. 147–168)) asks when an M-polyfold is an ordinary manifold with boundary with corners. §4.1 defines a compatible manifold structure with corners (Definition 4.1.1 (preprint 4.1)), states the characterization (Theorem 4.1.2 (preprint 4.2): $X_0 = X_\infty$, $1_0^1 : X \to X^1$ sc-smooth, and centred tame charts whose tangent space at the centre is finite-dimensional and in good position), and prepares its proof: Remark 4.1.3 (preprint 4.3), sc⁺-retractions and sc⁺-retracts (Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7), Proposition 4.1.8 (preprint 4.8)), and the description of an sc⁺-retract near a point as a graph over its tangent space (Proposition 4.1.9 (preprint 4.9), with Lemmas 4.1.11–4.1.13 (preprint 4.10–4.12)). §4.2 defines smooth finite-dimensional submanifolds (Definition 4.2.1 (preprint 4.13), Remark 4.2.2 (preprint 4.14), Proposition 4.2.3 (preprint 4.15), an open question), good position and tameness of submanifolds (Definitions 4.2.5, 4.2.6 (preprint 4.16, 4.17)), proves that a submanifold in good position is a tame manifold with corners (Theorem 4.2.7 (preprint 4.18), Lemma 4.2.8 (preprint 4.19), Remark 4.2.9 (preprint 4.20)), defines general position (Definition 4.2.10 (preprint 4.21)) and describes a submanifold near a point of general position (Theorem 4.2.11 (preprint 4.22)).

§4.3 proves, with Sard's theorem, that the fibres of a family $A \subseteq \mathbb{R}^m \times X$ over almost every parameter are compact manifolds with corners in general position (Theorem 4.3.1 (preprint 4.23), Remark 4.3.2 (preprint 4.24)). §4.4 sketches sc-differential forms, de Rham cohomology and Stokes' theorem (Definitions 4.4.1, 4.4.6 (preprint 4.25, 4.27), Proposition 4.4.5 (preprint 4.26), Theorem 4.4.7 (preprint 4.28)).

What is verified. **Theorems 4.1.2, 4.2.7, 4.2.11 and 4.3.1 (preprint 4.2, 4.18, 4.22 and 4.23) hold with the text's hypotheses**, and Theorem 4.1.2 (preprint 4.2) holds without its tameness hypothesis (V4.5). Every numbered statement of §§4.1–4.3 is verified, including Lemma 4.1.11 (preprint 4.10), which is no longer used (V4.94; eighth addendum): Definition 4.1.1 (preprint 4.1) (V4.1–V4.4), Theorem 4.1.2 (preprint 4.2) (V4.5, V4.27, V4.28), Remark 4.1.3 (preprint 4.3), with an example for "$X^1 \neq X$ is possible", which the text does not give (V4.6), Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6) and Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7) (V4.9–V4.12), Proposition 4.1.8 (preprint 4.8) with its chart form (V4.13, V4.14), Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12) (V4.20–V4.25), Definition 4.2.1 (preprint 4.13) (V4.29), Remark 4.2.2 (preprint 4.14) in a chart (V4.31, V4.32), Proposition 4.2.3 (preprint 4.15) (V4.33–V4.36), Definitions 4.2.5, 4.2.6, 4.2.10 (preprint 4.16, 4.17, 4.21) in charts with their chart independence (V4.38–V4.40, V4.44, V4.45), Theorem 4.2.7 (preprint 4.18), Lemma 4.2.8 (preprint 4.19), Remark 4.2.9 (preprint 4.20) (V4.41–V4.43), Theorem 4.2.11 (1)–(4) (preprint 4.22) (V4.49–V4.52), Theorem 4.3.1 (preprint 4.23) (V4.54, V4.63, V4.64) and Remark 4.3.2 (preprint 4.24) (V4.55). Four places of the text needed more than a reading. (a) Proposition 4.1.9 (preprint 4.9) is proved with two hypotheses the text leaves implicit — the good complement Y is closed, and p is the projection along Y, not $Dt(0)$ — and by a direct argument, because the text's route through Lemma 4.1.11 (preprint 4.10) and Theorem 3.5.11 (preprint Corollary 3.47) yields the graph map only as a germ over a complement of its own choosing (V4.20); where Proposition 4.1.9 (preprint 4.9) is applied the closed complement is chosen, so Theorems 4.1.2, 4.2.7, 4.2.11, 4.3.1 (preprint 4.2, 4.18, 4.22, 4.23) need nothing extra. (b) One step of Lemma 4.2.8 (preprint 4.19) is false ("$z \in Z \subset C$" does not give $n + y \in C$) and is repaired (V4.42). (c) Theorem 4.2.11 (4) (preprint 4.22) has no proof in the text and is proved (V4.51, V4.52). (d) In Theorem 4.3.1 (preprint 4.23) "regular value" is the stratified notion of the proof — every stratified regular value is a regular value, and with ordinary regular values the conclusion fails at the value — and the finite cover of the proof does not exist in general; a countable one is used (V4.53, V4.54, V4.64). Proved although the text does not construct or state it: charts and strong bundle charts centred at a smooth point (V4.15–V4.18), the sc⁺-retraction onto the fibre A_λ and its general position (V4.61, V4.62), Theorem 3.8.22 (preprint 3.78) in C^∞ form (V4.58), and Sard's theorem in all dimensions (V4.56, vendored from our Milnor project, `math_faithful`). Not verified: parts of §4.4 (see below) and the open question after Proposition 4.2.3 (preprint 4.15), whose one settled sentence is verified (see the list at the end of this section). A fourth addendum (2026-11-02; Findings, items 140–146) added the example of Remark 4.1.3 (preprint 4.3) to V4.6. A fifth addendum (2026-11-08; Findings, items 150–159) covers the Springer 2021 edition: the 2021 forms of Theorems 4.2.11 and 4.3.1 (V4.65, V4.66), the sc-smooth vector fields of §4.4 (Definition 4.4.3, Lemma 4.4.4) and their Lie bracket (Proposition 4.4.5, Appendix 4.5) in a local model (V4.67–V4.72), Definition 4.4.3 on an M-polyfold (V4.68), and an example in which the bracket at a point of O_2 does not lie in E_1 (V4.73).

It continues (Findings, items 160–163) with Lemma 4.5.1 in a local model and the independence of the chart (V4.74, V4.75), and on an M-polyfold Proposition 4.4.5 (V4.76, closing gap G.38), Lemma 4.5.1 for every sc-smooth map (V4.77), Lemma 4.4.4 for $i = 1$ (V4.78) and for every i (V4.79). A sixth addendum (2026-11-09; Findings, items 167–172) covers the sc-differential forms of §4.4: Definitions 4.4.1–4.4.2 in the charts of the Whitney sum (V4.80–V4.82, closing gap G.37), the formula for $d\omega$ on normed spaces, where the printed sign does not hold as we read it and $(-1)^{\ell+j+1}$ does (V4.83), d on an M-polyfold, $d \circ d = 0$, d on $\Omega^*_\infty(X)$ and Definition 4.4.6 (V4.84–V4.88), and Stokes' theorem for classical forms on compact oriented Mathlib manifolds with corners (gap G.40 in part). A seventh addendum (2026-11-10; Findings, items 173–174) adds the pull-back f^* with $f^* d_Y = d_X f^*$ (V4.89), forms on ∂X read through the tame boundary, with j^* and $dj^* = j^* d$ (V4.90), the relative complex and $H^*_{dR}(X, \partial X)$ (V4.91, closing gap G.39), and Theorem 4.4.7 for sc-forms on a submanifold with a corner structure (V4.92). Left open then in §4.4: the wedge product and the Poincaré lemmas (gap G.80), the faces of N as tame M-polyfolds (gap G.40), and, for forms on ∂X, whether the agreement of τ implies that of $d\tau$ (V4.90). An eighth addendum (2026-11-12; Findings, items 175–177) verifies the rest of §§4.1–4.3 and most of §4.4: Lemma 4.1.11 (preprint 4.10) as printed (V4.94, closing gap G.32); the 2021 wording of Remark 4.2.2, near $d_A(a) = 0$ and $d_A(a) = 1$, without tameness (V4.32, now "same", closing gap G.33); Theorem 4.2.11 (1) for the faces of U and (4) for the germs of their intersections (V4.50, closing gap G.35); the chart independence of stratified regularity (V4.93, closing gap G.36); the settled sentence of the question after Proposition 4.2.3 (gap G.34, still a question); Lemma 4.5.1 between open subsets and the pull-back bundle $TX|X^2 \to X^2$ with $[A, B]$ as a section (V4.76, V4.77, closing gap G.92); the wedge product and the graded differential algebra $\Omega^*_\infty(X)$ (V4.87, V4.89); the agreement of $d\tau$ for agreeing forms on ∂X (V4.90, now "same"); Theorem 4.4.7 with the sum over the faces of a face-structured N (V4.92, closing gap G.40); and the Poincaré Lemmas 4.4.8 and 4.5.2 (V4.95, V4.96). Good position in Theorem 4.1.2 (2) is automatic in tame M-polyfolds (`hasCornerStructure_of_isTame`; item 175). Not verified in §4.4: the wedge product on $\Omega^*_\infty(X, \partial X)$ and the pull-back of forms on ∂X along a general sc-smooth map (gap G.80). A ninth addendum (2026-11-13; Findings, item 178) verifies the wedge product on $\Omega^*_\infty(X, \partial X)$, which the text calls a differential algebra without giving a product, with the product of the mapping cone of j^* (our choice): d is a derivation of degree $+1$ and the product is associative, with a unit on the left only (V4.91; gap G.80 closed). The 2021 text defines no pull-back of forms on ∂X along a general sc-smooth map; the one it uses, along p_1 on the faces of the solution set in the proof of Theorem 6.7.1, is formalized (`Polyfold.MPolyfold.faceAlongData`).

The setting. An sc⁺-map is a map whose values lie in F_1 and whose lift into F^1 is sc-smooth (V4.7); the text uses the word for maps and defines it only for retractions. $T_x X$ carries no linear structure in the formalization (gap G.7 of Chapter 2), so every condition the text states in $T_a X$ — the tangent space of a submanifold, good and general position, stratified regularity — is stated in a chart and, except for stratified regularity, proved independent of the chart. A chart "with $\psi(x) = 0$" is a chart re-centred at the smooth point x, modelled in the partial quadrant $C_{\varphi(x)}$ of Definition 2.5.1 (preprint 2.33) (V4.17). The map $1^1_0 : X \to X^1$ is defined on X_1, with an arbitrary value off X_1 that plays no role. Tameness of X enters §4.2 only through Lemma 4.2.8 (preprint 4.19) (Theorem 4.2.7

(preprint 4.18), Remark 4.2.9 (preprint 4.20)), the tame centred chart of Theorem 4.2.11 (preprint 4.22) and Proposition 2.5.6 (preprint 2.36) (general position implies good position, V4.48). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 4".

V4.1 (Definition 4.1.1 (preprint 4.1), corner charts and corner structures; `Polyfold.MChart.IsCornerChart`, `Polyfold.MPolyfold.HasCornerStructure`, `Polyfold.MPolyfold.hasCornerStructure_iff`, `Polyfold.MPolyfold.HasCornerStructure.isTame`, `Polyfold.MChart.IsCornerChart.isTameRetract`, `Polyfold.MChart.IsCornerChart.isScPlusRetract`). A chart of X is a *corner chart* if its local model is (U, C, V) with V a finite-dimensional real normed space carrying the constant sc-structure, C a partial quadrant in V, U relatively open in C, and the identity retraction. X has a *corner structure* if it has an atlas of corner charts equivalent to its atlas; equivalently, every point of X lies in a corner chart that is a chart of the M-polyfold structure. The dimension of V may vary from chart to chart. The model of a corner chart is a tame retract and an sc$^+$-retract, and an M-polyfold with a corner structure is tame. *Book:* Definition 4.1.1 (preprint 4.1), p. 175 (preprint p. 147). *Hidden hypotheses:* "sc-smoothly compatible charts" with "φ an sc-diffeomorphism" is read as compatibility with the atlas of X (equivalence of atlases, as for tameness). *Deviation (restated):* the classical manifold structure with corners is represented by the atlas of corner charts; local constancy of the dimension is neither stated by the text nor formalized.

V4.2 (Definition 4.1.1 (preprint 4.1), the transition maps are C^∞; `Polyfold.MChart.IsCornerChart.contDiffOn_transition`, `Polyfold.MChart.isRelOpenInQuadrant_overlap`). For compatible corner charts χ, χ' the overlap $\chi(V \cap V')$ is relatively open in the partial quadrant of χ, and the transition map $\chi' \circ \chi^{-1}$ is of class C^∞ on it (derivatives within the set, one-sided at the boundary). *Book:* Definition 4.1.1 (preprint 4.1), p. 175 (preprint p. 147), the paragraph after the definition. *Hidden hypotheses:* none. *Deviation:* "the atlas defines the structure of a smooth manifold with boundary with corners" is not formalized beyond this (Mathlib's manifolds with corners have a single model space).

V4.3 (finite-dimensional sc-Banach spaces; `Polyfold.ScBanach.finiteDimensional_lvl`, `Polyfold.ScBanach.toBase_surjective`, `Polyfold.ScBanach.scIsoConst`, `Polyfold.ScBanach.mem_smooth_of_finiteDimensional`). If E_0 is finite-dimensional, every level E_m is finite-dimensional, every inclusion $E_m \to E_0$ is onto, E is sc-isomorphic to E_0 with the constant structure (the level maps being the inclusions), and every point of E_0 is smooth. *Book:* §1.1, p. 6 (preprint p. 6), after Definition 1.1.1 (preprint 1.1) ("a finite-dimensional Banach space has precisely one sc-structure"), used in Definition 4.1.1 (preprint 4.1). *Hidden hypotheses:* the density of E_∞ in every level (Definition 1.1.1 (2) (preprint 1.1)) is what makes the inclusions onto; compact injective inclusions alone do not ($E_0 = \mathbb{R}$, $E_m = 0$ for $m \geq 1$). *Deviation (restated):* "one sc-structure" as an sc-isomorphism onto the constant structure.

**V4.4 (Definition 4.1.1 (preprint 4.1) with arbitrary finite-dimensional models, and the manifolds with corners of Chapter 3; `Polyfold.MChart.exists_isCornerChart_of_finiteDimensional`, `Polyfold.IsRelOpenInQuadrant.isScDiffeoQuadrant_toConst`, `Polyfold.MPolyfold.hasCornerStructure_iff_finiteDimensional`, `Polyfold.MPolyfold.isManifoldWithCorners_iff`, `Polyfold.MPolyfold.IsManifoldWithCorners.`

`hasCornerStructure`). A chart of X onto a relatively open subset of a partial quadrant in an sc-Banach space with finite-dimensional E_0 is, with the same map and domain, a corner chart for the constant structure on E_0, of the same dimension. Hence X has a corner structure iff it has an equivalent atlas of charts onto relatively open subsets of partial quadrants in sc-Banach spaces with finite-dimensional E_0 (the manifolds with corners of locally constant dimension of Theorem 3.2.3 (preprint 3.27)). The manifolds with corners of dimension n of Theorem 3.2.3 (preprint 3.27) are exactly the corner structures all of whose corner charts have dimension n. *Book:* Definition 4.1.1 (preprint 4.1), p. 175 (preprint p. 147) ("or in a finite-dimensional vector space"); the relation to Chapter 3 is not stated. *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.5 (Theorem 4.1.2 (preprint 4.2), the characterization; `Polyfold.MPolyfold.theorem_4_1_2`, `Polyfold.MPolyfold.theorem_4_1_2_general`). Let X be an M-polyfold and $x_0 \in X_1$ the junk value of 1^1_0 off X_1. The following are equivalent. (i) X has a corner structure (V4.1). (ii) Every point of X is smooth, $1^1_0 : X \to X^1$ is sc-smooth, and every $x \in X$ lies in a chart ψ of X with $\psi(x) = 0$ whose model (O, C, E) is a tame retract and whose tangent space $N = T_0 O$ (Definition 2.3.12 (preprint 2.17)) is finite-dimensional and in good position to C (Definition 3.1.24 (preprint 3.21)). `theorem_4_1_2` is the text's form, with the hypothesis "X tame". *Book:* Theorem 4.1.2 (preprint 4.2), p. 176 (preprint p. 147); proof pp. 183–186 (preprint pp. 156–157). *Hidden hypotheses:* the chart with $\psi(x) = 0$ is a re-centred chart (V4.17). *Deviation (stronger):* the tameness of X is not used, and (i) implies it (V4.1). *Proof:* V4.27, V4.28.

V4.6 (Remark 4.1.3 (preprint 4.3); `Polyfold.ScBanach.liftOne`, `Polyfold.MPolyfold.toLevelOne`, `Polyfold.MPolyfold.remark_4_1_3_levelVal`, `Polyfold.MPolyfold.remark_4_1_3_isScDiffeo`, `Polyfold.remark_4_1_3_const`, `Polyfold.remark_4_1_3_example`, `Polyfold.remark_4_1_3_retract_example`, `Polyfold.IsLossFamily.isScSmooth`, `Polyfold.exists_retract_smooth_not_scPlus_of_data`, `Polyfold.MPolyfold.mem_shift_levelSet_iff`). The inclusion $1^0_1 : X^1 \to X$ is always sc-smooth; in the charts $(V_1, \varphi, (O_1, C_1, E^1))$ and φ it is the sc-operator $\iota : E^1 \to E$. If every point of X lies in X_1 and $1^1_0 : X \to X^1$ is sc-smooth, then 1^1_0 is an sc-diffeomorphism with inverse 1^0_1. For a finite-dimensional V with the constant structure, $V^1 = V$ as sc-Banach spaces, the lift $V \to V^1$ is the identity, and every sc-smooth retract in V is an sc⁺-retract. The levels of X^1 are those of X shifted by one, as sets: $(X^1)_m = X_{m+1}$. **The example:** there is an M-polyfold all of whose points are smooth ($X_0 = X_\infty$) for which $1^1_0 : X \to X^1$ is not sc-smooth, so $X^1 \neq X$ as M-polyfolds. It is a retract $O \subseteq \mathbb{R} \oplus X$, X the weighted sequence space of V1.13 ($|x|_m = \sup_k (k+1)^m |x_k|$), with a single chart: the image of $r(t, z) = (t, L(t)z)$, $L(t)$ the projection onto $\cos\theta\, e_{n_k} + \sin\theta\, e_{n_{k+1}}$ ($n_k = 2^k$), the angle θ turning from 0 to $\pi/2$ with flat ends while t runs through $[1/(k+2), 1/(k+1)]$, and $L(t) = 0$ for $t \leq 0$. Its points are smooth, and the lift $O \to E_1$ is discontinuous at 0, so O is not an sc⁺-retract (Proposition 4.1.8 (preprint 4.8)). The sc-smoothness of r comes from a criterion proved for it: for level operators $L(p)_m : Z_m \to W_m$ depending on a finite-dimensional parameter p, compatible with the inclusions, with $(p, z) \mapsto L(p)_m z$ continuous and $p \mapsto L(p)_m$ of class C^∞ as a map into the operators $Z_{m+1} \to W_m$ (one level of loss), $(p, z) \mapsto L(p)z$ is sc-smooth. *Book:* Remark 4.1.3 (preprint 4.3), p. 176 (preprint p. 148). *Hidden hypotheses:* 1^1_0 is defined on X_1 (the text: when $X_0 = X_\infty$). *Deviation (stronger):* the sc-diffeomorphism clause needs only $X = X_1$, not

$X_0 = X_\infty$; "$X^1 \neq X$ is possible" has an example, which the text does not give, and in it X and X^1 even have different topologies on the same set. "$X^1 \neq X$" is read as "1_0^1 is not sc-smooth", equivalently "the identity is not an sc-diffeomorphism". An example needs a retraction that loses a level in its first derivative, as the splicings of HWZ do (a retraction with C^1 level maps gives none: an argument, not formalized), so its sc-smoothness does not follow from Proposition 1.2.4 (preprint 1.19). "$(X^1)_m = X_{m+1} = X_i$" misprints X_0; the topologies of the iterated levels agree (`Polyfold.MPolyfold.shiftLevelHomeomorph`; gap G.6, closed) (Findings, items 5, 125, 140–146). Formerly "narrower", for the missing example (gap G.31, now closed).

V4.7 (sc⁺-maps; `Polyfold.IsScPlus`, `Polyfold.IsScPlus.mem_range_succ`, `Polyfold.IsScPlus.iff_exists`, `Polyfold.IsScPlus.isScSmooth`, `Polyfold.IsScPlus.comp_isScSmooth`, `Polyfold.IsScSmooth.comp_isScPlus`, `Polyfold.ScPlus.isScPlus`). A map f on $U \subseteq E_0$ is *sc⁺* if $f(U) \subseteq F_1$ and the lift $U \to F^1$, $x \mapsto f(x) \in F_1$, is sc-smooth. Then $f(U_m) \subseteq F_{m+1}$ for every m. On an sc-domain U: f is sc⁺ iff some sc-smooth $g : U \to F^1$ has $\iota \circ g = f$ on U; an sc⁺-map is sc-smooth; $g \circ f$ is sc⁺ if g is sc⁺ on an sc-domain V and f is sc-smooth with $f(U) \subseteq V$; $g \circ f$ is sc⁺ if f is sc⁺ with $f(U) \subseteq V$ and g is sc-smooth on V (through Proposition 1.2.2 (preprint 1.17) for g); an sc⁺-operator is an sc⁺-map. *Book:* not defined (Definition 4.1.4 (preprint 4.4) is for retractions; the proof of Lemma 4.1.7 (preprint 4.7) says "r is a sc⁺-map"). *Hidden hypotheses:* U, V sc-domains for the composition rules. *Deviation (not in the book).*

V4.8 (the derivative of an sc⁺-map; `Polyfold.IsScPlus.exists_scPlus_deriv`, `Polyfold.IsScPlus.isCompactOperator_deriv`, `Polyfold.finiteDimensional_range_of_isCompactOperator_of_idem`). Let f be sc⁺ on an sc-domain U and $x \in U_1$ a smooth point. There is an sc⁺-operator $S : E \to F$ whose level-0 operator is $Df(x)$; $Df(x) : E_0 \to F_0$ is compact, and so is every level operator of S. A compact continuous linear projection $P = P \circ P$ of a normed space has finite-dimensional range. *Book:* proof of Lemma 4.1.7 (ii) (preprint 4.7), p. 177 (preprint p. 149) ("a sc⁺-operator, since r is a sc⁺-map … must be finite-dimensional"). *Hidden hypotheses:* x smooth; the range of a compact projection is finite-dimensional by Riesz's theorem. *Deviation (hypotheses explicit).*

V4.9 (Definition 4.1.4 (preprint 4.4), sc⁺-retractions; `Polyfold.ScRetraction.IsScPlus`, `Polyfold.ScRetraction.IsScPlus.mem_lvlSet_succ`, `Polyfold.MPolyfold.IsScPlusRetractionOn`). An sc-smooth retraction $r : U \to U$ is an *sc⁺-retraction* if it is an sc⁺-map on U (V4.7); then $r(U_m) \subseteq U_{m+1}$ for every m. On an open V of an M-polyfold: s is an sc⁺-retraction on V if s is sc-smooth on V, $s(V) \subseteq V$, $s \circ s = s$ on V, $s(V) \subseteq X_1$, and $1_0^1 \circ s : V \to X^1$ is sc-smooth (for every junk value). *Book:* Definition 4.1.4 (preprint 4.4), p. 176 (preprint p. 148). *Hidden hypotheses:* in the M-polyfold form, $s(V) \subseteq X_1$, so that "$r : V \to V^1$" makes sense. *Deviation (restated):* $r(U_m) \subseteq U_{m+1}$ is derived, not assumed.

V4.10 (Lemma 4.1.5 (preprint 4.5); `Polyfold.lemma_4_1_5`). If t is an sc⁺-retraction and s an sc-smooth retraction (possibly of another relatively open subset of C) with the same retract, then s is an sc⁺-retraction. *Book:* Lemma 4.1.5 (preprint 4.5), p. 177 (preprint p. 148). *Hidden hypotheses:* none. *Proof:* the text's, $s = t \circ s$ and sc⁺ □ sc-smooth is sc⁺.

V4.11 (Definition 4.1.6 (preprint 4.6), sc⁺-retracts; `Polyfold.IsScPlusRetract`, `Polyfold.IsScPlusRetract.isScPlus`, `Polyfold.isScPlusRetract_iff_forall`). (O, C, E) is an *sc⁺-retract* if O is the retract of an sc⁺-retraction of a relatively open $U \subseteq C$; then every sc-smooth retraction onto O is sc⁺, and O is an sc⁺-retract iff it is a retract all of whose retractions are sc⁺. *Book:* Definition 4.1.6 (preprint 4.6) and the remark after it, pp. 176–178 (preprint pp. 148–149). *Hidden hypotheses:* none.

V4.12 (Lemma 4.1.7 (preprint 4.7); `Polyfold.lemma_4_1_7_i`, `Polyfold.lemma_4_1_7_ii`, `Polyfold.lemma_4_1_7_ii_tangentSpace`). Let r be an sc⁺-retraction onto O. (i) $O \subseteq E_\infty$. (ii) For every level-1 point o with $\iota(o) \in O$ the range of $Dr(o)$ is finite-dimensional; every $o \in O$ has a smooth lift $o_1 \in E_1$, and $T_{o_1}O = Dr(o_1)E_0$ is a finite-dimensional linear subspace. *Book:* Lemma 4.1.7 (preprint 4.7), p. 177 (preprint p. 149). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* the tangent space of Definition 2.3.12 (preprint 2.17) is indexed by level-1 points; the step "$Dr(o)$ is an sc⁺-operator" is V4.8.

V4.13 (Proposition 4.1.8 (preprint 4.8); `Polyfold.proposition_4_1_8`). An sc-smooth retract (O, C, E) is an sc⁺-retract iff $O \subseteq E_\infty$ and the lift 1_0^1 is sc-smooth from (O, C, E) to (O_1, C_1, E^1) in the sense of Definition 2.1.14 (preprint 2.7). *Book:* Proposition 4.1.8 (preprint 4.8), p. 177 (preprint p. 149). *Hidden hypotheses:* none. *Deviation:* none (the proof ends "the lemma is proved").

V4.14 (Proposition 4.1.8 (preprint 4.8) in charts; `Polyfold.MPolyfold.levelChart_toLevelOne`, `Polyfold.MPolyfold.proposition_4_1_8_chart`, `Polyfold.MPolyfold.isScSmoothMap_toLevelOne_iff`, `Polyfold.MPolyfold.isScSmoothMap_toLevelOne_of_cover`). In the charts φ of X and $(V_1, \varphi, (O_1, C_1, E^1))$ of X^1 the map 1_0^1 is the lift $O \to O_1$. For a chart $\varphi : V \to O$ of X: the model is an sc⁺-retract iff $V \subseteq X_\infty$ and 1_0^1 is sc-smooth on V. Hence "every point smooth and 1_0^1 sc-smooth" holds iff every chart of X is modelled on an sc⁺-retract, iff every point lies in some chart so modelled. *Book:* not stated; used in the proofs of Theorem 4.1.2 (preprint 4.2) and Proposition 4.2.3 (4) (preprint 4.15). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.15 (the partial quadrant C_e and the neighbourhood $G(e)$; `Polyfold.recentreNbhd`, `Polyfold.recentreIso`, `Polyfold.isPartialQuadrant_partialQuadrantAtWith`, `Polyfold.image_sub_partialQuadrantAtWith`, `Polyfold.recentreNbhd_inter_eq`, `Polyfold.degIndex_sub_recentre`, `Polyfold.minimalSubspaceWith_recentreIso`, `Polyfold.partialQuadrantAtWith_recentreIso`). Let T present C with coordinates $\lambda_1, \dots, \lambda_n$ and let $e \in E_0$. $G(e) = \{v : \lambda_i(v) > 0 \text{ whenever } \lambda_i(e) \neq 0\}$ is open and, for $e \in C$, contains e. C_e is a partial quadrant, presented by reordering the coordinates as $\mathbb{R}^{I(e)} \oplus (\mathbb{R}^{\text{rest}} \oplus W)$; $C_e - e = C_e$; $G(e) \cap C = G(e) \cap C_e$; and for $v \in G(e)$ the vanishing coordinates of $v - e$ for this presentation are those of v, so that $d_{C_e}(v - e) = d_C(v)$ $(v \in C)$ and the minimal subspace and the partial quadrant of $v - e$ are E_v and C_v. *Book:* Definition 2.5.1 (preprint 2.33), p. 53 (preprint p. 39), asserts without proof that C_e is a partial quadrant; the rest is not in the text. *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.16 (translations and re-centred retracts; `Polyfold.isScSmooth_add_const`, `Polyfold.isScSmooth_sub_const`, `Polyfold.relOpen_translate`, `Polyfold.isScDiffeoQuadrant_`

`translate`, `Polyfold.ScRetraction.restrictIn`, `Polyfold.ScRetraction.restrictIn_isTame`, `Polyfold.isScRetract_recentre`, `Polyfold.isTameRetract_recentre`). Translation by a smooth vector is sc-smooth on every sc-domain. For e smooth and U relatively open in C with $U \subseteq G(e)$, the translate $U - e$ is relatively open in C_e and $v \mapsto v - e$ is an sc-diffeomorphism $U \to U - e$. The restriction of a retraction r to $U \cap r^{-1}(W) \cap W$ (W open) is a retraction onto $r(U) \cap W$, tame if r is. For a retract (tame retract) (O, C, E) and a smooth $e \in E_0$, $((O \cap G(e)) - e, C_e, E)$ is a retract (tame retract), with the retraction $u \mapsto r(u + e) - e$. *Book:* not in the text. *Hidden hypotheses:* e smooth — translation by e preserves every level only then. *Deviation (not in the book).*

V4.17 (charts centred at a smooth point; `Polyfold.MChart.recentre`, `Polyfold.MChart.recentre_isChart`, `Polyfold.MChart.recentre_apply_self`, `Polyfold.MChart.isTameRetract_recentre`, `Polyfold.MChart.chartDegIndex_recentre`, `Polyfold.MPolyfold.exists_centredChart`, `Polyfold.MPolyfold.IsTame.exists_centredChart`). For a chart $\varphi : V \to O \subseteq C$ of X, a presentation of C and a smooth e, the chart $\varphi_e = \varphi - e$ on $V \cap \varphi^{-1}(G(e))$, modelled on $((O \cap G(e)) - e, C_e, E)$, is a chart of X, with $\varphi_e(x) = 0$ when $\varphi(x) = e$, the same degeneracy index as φ on its domain, and a tame model when φ has one. So every smooth point x of the domain of a chart φ lies in a chart χ of X with domain inside that of φ and $\chi(x) = 0$; in a tame M-polyfold every smooth point lies in a tame chart centred at it. *Book:* used without construction in Theorem 4.1.2 (ii) (preprint 4.2), the proofs of Theorems 4.2.7 and 4.2.11 (preprint 4.18 and 4.22), and Lemma 5.3.3 (preprint 5.11). *Hidden hypotheses:* x smooth; the model lies in $C_{\varphi(x)}$, not in C. *Deviation (not in the book).*

V4.18 (strong bundle charts centred at a smooth point; `Polyfold.StrongBundleChart.recentre`, `Polyfold.StrongBundleChart.Compatible.recentre`, `Polyfold.StrongBundle.IsChart.recentre`, `Polyfold.StrongBundle.IsChart.exists_centred`). For a strong bundle chart c over φ, a presentation and a smooth e, the chart c_e over φ_e with bundle retraction $R_e(u, h) = (r(u + e) - e, \Gamma(u + e, h))$ and $\Phi_e(y) = (\Phi(y)_1 - e, \Phi(y)_2)$ is compatible with every chart c is compatible with (the projection being continuous), and is a chart of the strong bundle. Every smooth point of the base in the domain of a strong bundle chart lies in a strong bundle chart centred at it, with smaller domains. *Book:* used in Lemma 5.3.3 (preprint 5.11). *Hidden hypotheses:* e smooth; $(e, 0)$ is then smooth in $E \oplus F$ and $E \oplus F^1$. *Deviation (not in the book).*

V4.19 (the level data of an sc$^+$-retraction and the mean value estimate; `Polyfold.ScRetraction.lvlMap`, `Polyfold.ScRetraction.lvlDeriv`, `Polyfold.ScRetraction.IsScPlus.hasFDerivWithinAt_lvlMap`, `Polyfold.ScRetraction.IsScPlus.continuousOn_lvlDeriv`, `Polyfold.ScRetraction.IsScPlus.exists_mvt`, `Polyfold.exists_norm_sub_sub_le_of_hasFDerivWithinAt`). Let t be an sc$^+$-retraction and $k \geq 0$. Then $t : U_{k+1} \to E_{k+1}$ is differentiable within U_{k+1}, with derivative $Dt(x) \in \mathcal{L}(E_{k+1})$ continuous in x in the operator norm. For $u_0 \in U_{k+1}$ and $\theta > 0$ there is $r > 0$ with $C_{k+1} \cap B(u_0, r) \subseteq U_{k+1}$, $\|Dt(z) - Dt(u_0)\| \leq \theta$ there, and $|t(z') - t(z) - Dt(u_0)(z' - z)|_{k+1} \leq \theta |z' - z|_{k+1}$ for $z, z' \in C_{k+1} \cap B(u_0, r)$. *Book:* (76)–(77) and the estimates in the proofs of Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), pp. 181–183 (preprint pp. 154–155). *Hidden hypotheses:* the mean value inequality is taken on the convex set

$C_{k+1} \cap B(u_0, r)$, since U_{k+1} need not be convex. *Deviation (restated):* the text's integrals are the mean value inequality.

V4.20 (Proposition 4.1.9 (preprint 4.9); `Polyfold.proposition_4_1_9`, `Polyfold.finiteDimensional_tangentSpace_zero`, `Polyfold.GraphData`). Let $t : U \to U$ be an sc$^+$-retraction onto $O \ni 0$, $N = T_0 O$ (finite-dimensional by Lemma 4.1.7 (preprint 4.7)), in good position to C, Y a good complement of N, Y **closed**, and $p : E_0 \to N$ continuous linear with $p = 1$ on N and $p = 0$ on Y (**the projection along** Y). There are an open $W \ni 0$ in E_0, a relatively open $V \ni 0$ in the partial quadrant $N \cap C$ of N (constant structure) and $\Gamma : N \to E_0$, sc-smooth on V, with $p \circ \Gamma = 1$ on V, $\Gamma(a) - a \in Y$, $\Gamma(V) = O \cap W$, $\Gamma \circ p = 1$ on $O \cap W$, $\Gamma(0) = 0$, and p, Γ sc-smooth between the retracts $(O \cap W, C, E)$ and $(V, N \cap C, N)$: $p : O \cap W \to V$ is an sc-diffeomorphism with inverse Γ. *Book:* Proposition 4.1.9 (preprint 4.9), p. 178 (preprint p. 150); proof pp. 178–184 (preprint pp. 150–156). *Hidden hypotheses:* Y closed (Definition 3.1.24 (preprint 3.21) asks for an sc-complement, which is closed; the formal good complement records only the splitting), and p the projection along Y: the text writes $p = Dt(0)$, a projection along $\ker Dt(0)$, which nothing relates to Y, while Lemma 4.1.13 (preprint 4.12) needs the projection along Y (Findings, items 20, 21). *Deviation (hypotheses explicit):* the conclusion is stated on fixed sets, with Γ sc-smooth on the fixed V. *Proof:* not the text's. The text applies Theorem 3.5.11 (preprint Corollary 3.47) to the section $\mathbf{f}(u) = (1 - p)(u - t(u))$ (Lemma 4.1.11 (preprint 4.10)), which gives a graph over a complement of its own choosing, sc-smooth only as a germ (Findings, item 22); the formal proof is direct (V4.21–V4.25).

V4.21 (Lemma 4.1.12 (preprint 4.11), uniqueness; `Polyfold.GraphData.qL_lvlDeriv_zero`, `Polyfold.GraphData.lvlDeriv_zero_of_mem_N`, `Polyfold.GraphData.exists_contraction_radius`, `Polyfold.GraphData.eq_of_p_eq`). In the data of V4.20, on every level $k + 1$: $(1 - p)Dt(0) = 0$ and $Dt(0) = 1$ on N; for $\theta > 0$ there is $r > 0$ for which the estimates of V4.19 hold at $u_0 = 0$ and $|(1 - p)(u' - u)|_{k+1} \le \|1 - p\|\theta|u' - u|_{k+1}$ for $u, u' \in O_{k+1}$ with $|u|_{k+1}, |u'|_{k+1} < r$; and two points u, u' with $p(u) = p(u')$ and $|(1 - p)(u' - u)| \le c|u' - u|$, $c < 1$, coincide. *Book:* Lemma 4.1.12 (preprint 4.11), p. 181 (preprint p. 154): small solutions $a + y$ of $\mathbf{f} = 0$ have $y = \delta(a)$. *Hidden hypotheses:* none. *Deviation (restated):* uniqueness among points of O, without the δ of Theorem 3.5.11 (preprint Corollary 3.47).

V4.22 (Lemma 4.1.13 (preprint 4.12), existence; `Polyfold.GraphData.eps'`, `Polyfold.GraphData.exists_graph_point`). Let ε be the constant of the good complement Y and $\varepsilon' = \varepsilon / (\|\iota_1\|\|\mathrm{lift}_N\| + 1)$. If the estimates of V4.21 hold on $C_1 \cap B(0, r)$ with $\theta < 1$, $\|1 - p\|\theta < 1$ and $\|1 - p\|\theta(1 + \varepsilon') \le \varepsilon'$, there is $\rho > 0$ such that every $a \in N \cap C$ with $|a|_1 < \rho$ has a $u \in O_1$ with $p(u) = a$, $u - a \in Y$, $|u - a|_1 \le \varepsilon'|a|_1$ and $|u|_1 < r$. *Book:* Lemma 4.1.13 (preprint 4.12), pp. 182–183 (preprint pp. 154–155): $a + \delta(a) \in O$ for small a. *Hidden hypotheses:* none. *Deviation (restated):* the point is constructed as the fixed point of $y \mapsto (1 - p)t(a + y)$ on $\{y \in Y_1 : |y|_1 \le \varepsilon'|a|_1\}$, where $a + y \in C$ by the good complement property; then the text's argument $a_1 = a$ shows $t(u) = u$.

V4.23 ($p(O \cap W) \subseteq C$ and the neighbourhoods; `Polyfold.GraphData.p_mem_C`, `Polyfold.GraphData.Nbhd`, `Polyfold.GraphData.exists_nbhd`, `Polyfold.GraphData.Nbhd.image_`

V, `Polyfold.GraphData.Nbhd.injOn_p`). (a) If $u \in O_1$ satisfies $|(1-p)u|_1 \le \kappa|u|_1$ with $0 \le \kappa \le 1/2$ and $2\|\iota_1\|\kappa\|\mathrm{lift}_N\| \le \varepsilon$, then $p(u) \in C$. (b) For every $\theta_0 > 0$ there are $r > 0$, an open $W = B_0(0,\delta_0) \cap p^{-1}(V')$, an open $V' = \{a : |a|_1 < \rho'\}$ of N and Γ on $V = V' \cap C$ with $\Gamma(V) \subseteq O \cap W$, $p \circ \Gamma = 1$, $\Gamma(a) - a \in Y$, $|\Gamma(a)|_1 < r$, $p(O \cap W) \subseteq V$, $\Gamma \circ p = 1$ on $O \cap W$, $\|Dt(z) - Dt(0)\|_{\mathcal{L}(E_1)} \le \theta_0$ on $C_1 \cap B(0,r)$, and $|(1-p)(u'-u)|_1 \le \frac{1}{2}|u'-u|_1$ on $O_1 \cap B_1(0,r)$; then $\Gamma(V) = O \cap W$ and p is injective on $O \cap W$. *Book:* the last paragraph of the proof of Proposition 4.1.9 (preprint 4.9), preprint p. 156 ("$V = \{a \in T_0O \cap C : |a|_0 < \varepsilon\}$"). *Hidden hypotheses:* $p(u) \in C$ for $u \in O$ near 0, which the text never checks (Findings, item 25). *Deviation (not in the book):* the neighbourhoods are open on level 0 and controlled on level 1 through the continuity of $t : U \to E_1$ at 0.

V4.24 (the regularity of Γ: continuity and the frame formula; `Polyfold.GraphData.Nbhd.norm_Γ_lvl_one_sub_le`, `Polyfold.GraphData.Nbhd.continuousOn_Γ_lvl`, `Polyfold.norm_sub_le_of_projection`, `Polyfold.hasFDerivWithinAt_of_projection`, `Polyfold.GraphData.A_injective`, `Polyfold.GraphData.eq_zero_of_mem_Y_of_scDeriv_eq`, `Polyfold.GraphData.scDeriv_sub_frame`, `Polyfold.GraphData.Nbhd.hasFDerivWithinAt_Γ_lvl`). With the data of V4.23: $|\Gamma(a') - \Gamma(a)|_1 \le 2\|\mathrm{lift}_N\||a'-a|$, and Γ is continuous into every level E_m on V. If $\|Dt(u) - Dt(0)\|_{\mathcal{L}(E_1)} \le \theta_0$ with θ_0 small (`FrameSmall`), then $A(u) = p \circ Dt(u)|_N$ is invertible, $T_uO \cap Y = 0$ and $(1 - B_up)Dt(u) = 0$ for the frame $B_u = Dt(u)|_N \circ A(u)^{-1}$; and Γ is differentiable into every level E_{k+1} on V with $D\Gamma(a) = Dt(\Gamma(a))|_N \circ (p \circ Dt(\Gamma(a))|_N)^{-1}$. The abstract step: if s satisfies the mean value estimate at u_0 on a set S, $q \circ B = 1$, $(1 - Bq)Ds(u_0) = 0$ and $\theta\|1 - Bq\| \le 1/2$, then fixed points u, u' of s in S satisfy $|u'-u| \le 2\|B\||q(u'-u)|$, and every continuous right inverse of q with values in the fixed points has derivative B at $q(u_0)$. *Book:* not in the text (Theorem 3.5.11 (preprint Corollary 3.47) gives δ as an sc-smooth germ). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.25 (the C^k-bootstrap and the sc-diffeomorphism; `Polyfold.GraphData.contDiffOn_lvlDeriv_apply`, `Polyfold.GraphData.Nbhd.contDiffOn_Γ_lvl`, `Polyfold.GraphData.Nbhd.isScSmooth_Γ`, `Polyfold.GraphData.Nbhd.V_isRelOpenInQuadrant`, `Polyfold.GraphData.Nbhd.p_isScSmoothRetractMap`, `Polyfold.GraphData.Nbhd.Γ_isScSmoothRetractMap`). $(x, w) \mapsto Dt_{j+1}(x)w$ is C^k on $U_{j+k+1} \times E_{j+k+1}$ (Proposition 1.2.3 (preprint 1.18) for the tangent map of the lift); Γ is C^k into every E_{j+1} on V for every k, hence C^∞ into every level and sc-smooth on V (Proposition 1.2.4 (preprint 1.19)); $(V, N \cap C, N)$ is a local model; p and Γ are sc-smooth between the retracts $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *Book:* the last sentence of the proof of Proposition 4.1.9 (preprint 4.9), preprint p. 156 ("having the sc-smooth map $a \mapsto a + \delta(a)$ as its inverse"). *Hidden hypotheses:* none. *Deviation (not in the book):* derivatives of t are applied only to vectors of the smooth finite-dimensional N, so the level loss of Tt does not accumulate.

V4.26 (charts composed with an sc-diffeomorphism between retracts; `Polyfold.MChart.compRetract`, `Polyfold.MChart.compRetract_isChart`). For a chart φ onto (O, C, E) and mutually inverse f, g that are sc-smooth between (O, C, E) and a retract (O', C', E') (Definition 2.1.14 (preprint 2.7)), $f \circ \varphi$ is a chart onto (O', C', E'), and a chart of the M-polyfold whenever φ is. *Book:* not stated; "we obtain a smoothly compatible atlas", preprint p. 157. *Hidden hypotheses:*

none. *Deviation (not in the book).*

V4.27 (Theorem 4.1.2 (preprint 4.2), (i) □ (ii); `Polyfold.MChart.IsCornerChart.mem_smooth`, `Polyfold.MChart.IsCornerChart.exists_centred`, `Polyfold.MChart.IsCornerChart.exists_centred_tame`, `Polyfold.tangentSpace_relOpen`, `Polyfold.isInGoodPosition_top`). Every point of a corner chart is smooth for its model. A point x in the domain of a corner chart χ of X lies in a chart ψ of X with $\psi(x) = 0$, modelled on a relatively open subset of the partial quadrant $C_{\chi(x)}$ in the same finite-dimensional space; this model is a tame retract (identity retraction), its tangent space at 0 is the whole space, and the whole space is in good position to any partial quadrant (complement 0). Together with V4.14 and V4.1, (i) implies (ii). *Book:* proof of Theorem 4.1.2 (preprint 4.2), preprint p. 156. *Hidden hypotheses:* the centred chart, the tangent space at its centre and its good position, which the text's proof does not address (Findings, item 53). *Deviation (hypotheses explicit).*

V4.28 (Theorem 4.1.2 (preprint 4.2), (ii) □ (i); `Polyfold.exists_proj_goodComplement`, `Polyfold.CornerChartData`, `Polyfold.exists_cornerChartData`, `Polyfold.CornerChartData.chart`, `Polyfold.CornerChartData.chart_isCornerChart`, `Polyfold.CornerChartData.chart_isChart`, `Polyfold.MPolyfold.theorem_4_1_2_chart`, `Polyfold.MPolyfold.hasCornerStructure_of_centred`). If every point is smooth and 1_0^1 is sc-smooth, then for every chart ψ of X with $\psi(x) = 0$ whose tangent space $N = T_0O$ is finite-dimensional and in good position to C there are a closed good complement Y of N, the projection p along it, and W, V, Γ of Proposition 4.1.9 (preprint 4.9); the chart $p \circ \psi$ on $\psi^{-1}(W)$, modelled on $(V, N \cap C, N)$, is a corner chart and a chart of X, with x in its domain and $(p \circ \psi)(x) = 0$. If every point has such a chart ψ (tame or not), X has a corner structure. *Book:* proof of Theorem 4.1.2 (preprint 4.2), pp. 183–186 (preprint pp. 156–157). *Hidden hypotheses:* none; the closed good complement is chosen by the normal form of Proposition 3.1.25 (preprint 3.22), and the compatibility of the charts $p \circ \psi$, not argued in the text, is V4.26 (Findings, items 54, 55). *Deviation (hypotheses explicit).*

V4.29 (Definition 4.2.1 (preprint 4.13), smooth finite-dimensional submanifolds; `Polyfold.MPolyfold.IsSmoothSubmanifold`, `Polyfold.MPolyfold.SubmanifoldData`, `Polyfold.MPolyfold.isSmoothSubmanifold_iff`, `Polyfold.MPolyfold.SubmanifoldData.toSubData`, `Polyfold.MPolyfold.IsSmoothSubmanifold.isSubMPolyfold`). $A \subseteq X$ is a *smooth finite-dimensional submanifold* if every $a \in A$ has an open $V \ni a$ and an sc⁺-retraction s on V (V4.9) with $s(V) = A \cap V$. Such an A is a sub-M-polyfold (Definition 2.3.18 (preprint 2.20)) and carries the induced structure of Proposition 2.3.19 (preprint 2.21). *Book:* Definition 4.2.1 (preprint 4.13), p. 184 (preprint p. 157). *Hidden hypotheses:* none.

V4.30 (the sc⁺-retraction in a chart; `Polyfold.MPolyfold.SubmanifoldData.isScSmoothRetractMap_lift`, `Polyfold.MPolyfold.SubmanifoldData.exists_chartRetraction`, `Polyfold.MPolyfold.SubmanifoldData.isScPlusRetract_image`, `Polyfold.MPolyfold.SubmanifoldData.apply_mem_smooth`). Let (V, s) be data of Definition 4.2.1 (preprint 4.13), φ a chart of X and $W = V \cap \operatorname{dom}\varphi \cap s^{-1}(V \cap \operatorname{dom}\varphi)$. The lift of $\varphi \circ s \circ \varphi^{-1}$ to E^1 is sc-smooth from $(\varphi(W), C, E)$ to (O_1, C_1, E^1). For every retraction r onto O there is an open G with $\varphi(W) = O \cap G$ and an sc⁺-retraction t on $U \cap r^{-1}(G)$, equal to $\varphi \circ s \circ \varphi^{-1} \circ r$ there, onto

$\varphi(W \cap A)$. So $\varphi(W \cap A)$ is an sc$^+$-retract, and $\varphi(a)$ is smooth for $a \in A \cap V \cap \operatorname{dom}\varphi$. *Book:* proof of Theorem 4.2.7 (preprint 4.18), pp. 188–190 (preprint pp. 160–161) ("taking V and U small, we may assume $V = \varphi^{-1}(O)$ … it is a sc$^+$-retraction since s is"). *Hidden hypotheses:* the set W of Proposition 2.3.19 (preprint 2.21) in place of "$V = \varphi^{-1}(O)$", and the sc$^+$ condition read in the charts of X and X^1 (Findings, item 60). *Deviation (hypotheses explicit).*

V4.31 (Remark 4.2.2 (preprint 4.14) in a chart; `Polyfold.MPolyfold.SubmanifoldData.remark_4_2_2`). At $a \in A \cap V \cap \operatorname{dom}\varphi$ and the level-1 point z over $\varphi(a)$: z is smooth; the tangent map of $\rho = \varphi \circ s \circ \varphi^{-1}$ at z maps T_zO onto the tangent space $T_z(\varphi(W \cap A))$; this tangent space lies in T_zO; and it is a finite-dimensional linear subspace. *Book:* Remark 4.2.2 (preprint 4.14), p. 184 (preprint p. 157) ("$T_aA = Ts(a)(T_aX)$ … finite-dimensional"). *Hidden hypotheses:* T_aX is read in a chart (gap G.7). *Deviation (restated):* the sc$^+$-operator is $Dt(z)$ for the chart retraction t of V4.30, not $Ts(a)$.

V4.32 (Remark 4.2.2 (preprint 4.14) at points with $d_X = 0$; `Polyfold.isInGoodPosition_univ`, `Polyfold.partialQuadrantAtWith_eq_univ_of_degIndex_eq_zero`, `Polyfold.MPolyfold.isGoodPositionAt_of_degIndex_eq_zero`, `Polyfold.MPolyfold.hasCornerStructure_of_degIndex_eq_zero`). Every subspace is in good position to the whole space; at a point $e \in C$ with $d_C(e) = 0$, C_e is the whole space. If $a \in A$ and $d_X(a) = 0$, A is in good position at a (Definition 4.2.5 (preprint 4.16), V4.38). If $d_X = 0$ on all of A, the induced structure on A has a corner structure; X need not be tame. (2026-11-12) The 2021 wording, without tameness: the open piece $\{d_A = 0\} \subseteq A$ is a manifold without boundary (`Polyfold.MPolyfold.remark_4_2_2_dA_zero`), and the open piece $\{d_A \le 1\}$ is a manifold with boundary (`Polyfold.MPolyfold.remark_4_2_2_dA_one`). *Book:* Remark 4.2.2 (preprint 4.14), p. 184 (preprint p. 157) ("near points $a \in A$ satisfying $d_X(a) = 0$, necessarily smooth finite-dimensional manifolds in the classical sense"; 2021: d_A, and a manifold with boundary near $d_A(a) = 1$). *Hidden hypotheses:* none. *Deviation (same):* the local form near $d_X(a) = 0$ (`remark_4_2_2_interior`, `remark_4_2_2_near`) and the 2021 forms are proved; at $d_A(a) = 1$ the case where the boundary coordinate vanishes on T_aA is excluded by the graph argument of Lemma 9.2.13 (gap G.33, closed; Findings, items 165, 176).

V4.33 (Proposition 4.2.3 (1) (preprint 4.15); `Polyfold.MPolyfold.IsSmoothSubmanifold.subset_smoothSet`, `Polyfold.MPolyfold.IsSmoothSubmanifold.mem_levelSet_subPolyfold_iff`, `Polyfold.MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_one`). $A \subseteq X_\infty$; the levels of the induced structure are the traces of the levels of X; every point of A is smooth for the induced structure. *Book:* Proposition 4.2.3 (1) (preprint 4.15), p. 186 (preprint p. 158). *Hidden hypotheses:* none. *Deviation (hypotheses explicit):* "$A \subseteq X_\infty$" and "$A = A_\infty$ for the induced structure" are separated; the proof goes through the chart retraction of V4.30 and Lemma 4.1.7 (i) (preprint 4.7).

V4.34 (Proposition 4.2.3 (2) (preprint 4.15); `Polyfold.MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_two`). A is locally closed in X: the intersection of an open and a closed set (equivalently, every $a \in A$ has an open V with $V \cap \overline{A} \subseteq A$). *Book:* Proposition 4.2.3 (2) (preprint 4.15), p. 186 (preprint p. 158), with a garbled definition of "locally closed" (the letter U used twice). *Hidden hypotheses:* none. *Deviation (restated):* Mathlib's notion; a filter replaces the text's sequence, so no metrizability is needed.

V4.35 (Proposition 4.2.3 (3) (preprint 4.15); `Polyfold.MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_three`). The boundary $\{d_A \geq 1\}$ of A with its induced structure lies in $\partial X = \{d_X \geq 1\}$. *Book:* Proposition 4.2.3 (3) (preprint 4.15), p. 186 (preprint p. 158). *Hidden hypotheses:* none. *Proof:* $d_A \leq d_X$ (Lemma 2.4.6 (preprint 2.26)).

V4.36 (Proposition 4.2.3 (4) (preprint 4.15); `Polyfold.MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_four`). For the induced structure on A, every point is smooth and $1_0^1 : A \to A^1$ is sc-smooth (for any junk value). *Book:* Proposition 4.2.3 (4) (preprint 4.15), p. 186 (preprint p. 158). *Hidden hypotheses:* none. *Proof:* every point of A lies in an induced chart whose model $\varphi(W \cap A)$ is an sc⁺-retract (V4.30), and V4.14 applies; the text says only that the sc⁺-retraction "is the identity on the retract A".

V4.37 (tangent spaces of sub-retracts; `Polyfold.tangentSpace_inter_isOpen`, `Polyfold.tangentSpace_subset_of_subset`, `Polyfold.scDeriv_mem_tangentSpace_of_subset`). For a retract Q and an open K, $T_z(Q \cap K) = T_zQ$ at points of $Q \cap K$. For retracts $Q \subseteq O = r(U)$, $T_zQ \subseteq T_zO$. If $F \circ r$ is sc-smooth and F maps the retract $Q \subseteq O$ into a retract Q', then $D(F \circ r)(z)$ maps T_zQ into $T_{F(z)}Q'$. *Book:* not stated. *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.38 (Definition 4.2.5 (preprint 4.16) in a chart; `Polyfold.IsGoodPositionRetractAt`, `Polyfold.MPolyfold.SubmanifoldData.IsGoodPositionChart`, `Polyfold.MPolyfold.IsGoodPositionAt`, `Polyfold.MPolyfold.SubmanifoldData.isGoodPositionChart_iff`). For retracts $Q \subseteq O$, a presentation T and a level-1 point z: T_zQ is in good position to C_z inside T_zO ($T_zQ \subseteq T_zO$, $T_zQ \cap C_z$ has interior points in T_zQ, and T_zQ has a complement $P \subseteq T_zO$ in T_zO with a constant ε as in Definition 3.1.24 (preprint 3.21)). A is *in good position at $a \in A$* if this holds for $Q = \varphi(W \cap A)$ and the model O of some chart φ at a, some data of Definition 4.2.1 (preprint 4.13) at a, some presentation and the level-1 point over $\varphi(a)$. For retractions r onto O and t onto $\varphi(W \cap A)$ the condition says that $Dt(z)E$ is in good position to C_z inside $Dr(z)E$. *Book:* Definition 4.2.5 (preprint 4.16), p. 188 (preprint p. 159). *Hidden hypotheses:* the chart form (gap G.7); the chart form needs no tameness. *Deviation (restated).*

V4.39 (the chart independence of Definition 4.2.5 (preprint 4.16); `Polyfold.IsGoodPositionRetractAt.transport`, `Polyfold.MPolyfold.chart_change_of_transport`, `Polyfold.MPolyfold.isGoodPositionRetractAt_chart_change`, `Polyfold.MPolyfold.IsGoodPositionAt.isGoodPositionChart`, `Polyfold.MPolyfold.isGoodPositionAt_iff_forall`). Good position of a sub-retract is carried by a germ of sc-diffeomorphisms between retracts that maps the sub-retracts into each other. Hence if A is in good position at a, the chart condition holds for every data of Definition 4.2.1 (preprint 4.13) at a, every chart at a, every presentation and the level-1 point over the image of a; and good position at a is equivalent to the condition for all of them. *Book:* implicit in Definition 4.2.5 (preprint 4.16). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.40 (Definition 4.2.6 (preprint 4.17); `Polyfold.MPolyfold.IsTameSubmanifold`). A is a *tame* submanifold if it is a smooth finite-dimensional submanifold whose induced M-polyfold structure is tame. *Book:* Definition 4.2.6 (preprint 4.17), p. 188 (preprint p. 159). *Hidden hypotheses:* none.

V4.41 (Theorem 4.2.7 (preprint 4.18); `Polyfold.MPolyfold.theorem_4_2_7`, `Polyfold.MPolyfold.hasCornerStructure_of_centred_isInGoodPosition`, `Polyfold.MPolyfold.IsGoodPositionAt.isInGoodPosition_of_tame`). Let X be tame and A a smooth finite-dimensional submanifold in good position at every point. Then the induced structure on A has a corner structure (Definition 4.1.1 (preprint 4.1)) and A is tame. More generally, the induced structure has a corner structure as soon as every $a \in A$ lies in a chart ψ of X with $\psi(a) = 0$ in which $T_0(\psi(W \cap A))$ is in good position to C in E; and good position at a gives this in every tame chart centred at a. *Book:* Theorem 4.2.7 (preprint 4.18), p. 188 (preprint p. 159); proof pp. 188–190 (preprint pp. 160–161). *Hidden hypotheses:* none. *Proof:* the text's route through Theorem 4.1.2 (preprint 4.2), with the steps it skips: the centred chart (V4.17), the chart retraction (V4.30), the closed complement (V4.28) and the compatibility of the charts $p \circ \psi$ (V4.26); tameness of X enters only through Lemma 4.2.8 (preprint 4.19).

V4.42 (Lemma 4.2.8 (preprint 4.19); `Polyfold.isGoodComplement_sup_range`, `Polyfold.lemma_4_2_8`, `Polyfold.IsInGoodPositionIn.isInGoodPosition`). Let T present C, π a bounded idempotent with $(1-\pi)E_0 \subseteq E_0^{\min}$, the minimal subspace at 0 ($\{\lambda = 0\}$), and N in good position to C inside πE_0. Then N is in good position to C in E_0, and for every good complement Y of N inside πE_0 with constant ε, $Y \oplus (1-\pi)E_0$ is a good complement of N in E_0, with constant $\varepsilon/(\|\pi\|+1)$. *Book:* Lemma 4.2.8 (preprint 4.19), p. 188 (preprint p. 160). *Hidden hypotheses:* $(1-\pi)E_0$ lies in the minimal subspace at the corner 0 of the centred chart (Proposition 2.5.3 (preprint 2.35)), not merely in C. *Deviation (hypotheses explicit):* the text concludes $n + y \in C$ from $n + y + z \in C$ "since $z \in Z \subset C$", which is false ($C = [0, \infty)$, $z = 1$, $n + y = -\frac{1}{2}$); it holds because $Z \subseteq C \cap (-C)$. The text's norm $|y|_0 + |z|_0$ is replaced by the constant $\varepsilon/(\|\pi\|+1)$ (Findings, item 69).

V4.43 (Remark 4.2.9 (preprint 4.20); `Polyfold.tangentSpace_eq_range_graph`, `Polyfold.lemma_3_6_9_relOpen`, `Polyfold.IsInGoodPosition.isInGoodPositionIn_of_le`, `Polyfold.MPolyfold.remark_4_2_9`). If the retract Q is the image of an sc-diffeomorphism Γ from $(V, N \cap C, N)$ with inverse p, then at $v \in V$ and the level-1 point y over $\Gamma(v)$, $T_yQ = D\Gamma(v)N$ and $p \circ D\Gamma(v) = 1$. For X tame and A in good position at a, there is an open $U \ni a$ such that A is in good position at every point of $U \cap A$. *Book:* Remark 4.2.9 (preprint 4.20), p. 189 (preprint p. 161), without "tame". *Hidden hypotheses:* X tame (Definition 4.2.5 (preprint 4.16) is stated for tame X; Lemma 4.2.8 (preprint 4.19) is used). *Deviation (hypotheses explicit):* "the tangent spaces move only slowly" is $D\sigma(0) = 0$ for $\Gamma = 1 + \sigma$, and Lemma 3.6.9 (preprint 3.55) is applied with the closed complement of the normal form.

V4.44 (Definition 4.2.10 (preprint 4.21) in a chart; `Polyfold.IsGeneralPositionRetractAt`, `Polyfold.MPolyfold.SubmanifoldData.IsGeneralPositionChart`, `Polyfold.MPolyfold.IsGeneralPositionAt`). For retracts $Q \subseteq O$, a presentation T and a level-1 point z: there is an sc-subspace $B \subseteq T_zO \cap E_z$ with $(T_zO)_m = (T_zQ)_m \oplus B_m$ on every level m ($T_zO \cap E_z = T_z^RO$). A is *in general position at* $a \in A$ if this holds for $Q = \varphi(W \cap A)$ and the model of some chart at a, some data, presentation and lift. *Book:* Definition 4.2.10 (preprint 4.21), p. 190 (preprint p. 161). *Hidden hypotheses:* the sc-structure of T_zO is the one induced from E at the smooth point; the chart form (gap G.7). *Deviation (restated).*

V4.45 (the chart independence of Definition 4.2.10 (preprint 4.21); `Polyfold.IsGeneralPositionRetractAt.transport`, `Polyfold.MPolyfold.isGeneralPositionRetractAt_chart_change`, `Polyfold.MPolyfold.IsGeneralPositionAt.isGeneralPositionChart`, `Polyfold.MPolyfold.isGeneralPositionAt_iff_forall`). General position of a sub-retract is carried by a germ of sc-diffeomorphisms at a smooth point; hence general position at a holds in every chart, for every data, presentation and lift, and is equivalent to the condition for all of them. *Book:* implicit in Definition 4.2.10 (preprint 4.21). *Hidden hypotheses:* the point is smooth (true on A). *Deviation (not in the book).*

V4.46 (linear algebra at the corner of a centred chart; `Polyfold.mem_minimalSubspaceWith_zero`, `Polyfold.minimalSubspaceWith_zero_le`, `Polyfold.add_mem_iff_of_mem_zero`, `Polyfold.degIndexWith_add_of_mem_zero`, `Polyfold.coordsOn_surjective`, `Polyfold.exists_isClosed_isCompl_le`, `Polyfold.exists_goodComplement_le_zero`, `Polyfold.degIndex_slice`). For $C = T^{-1}([0,\infty)^n \oplus W)$: the minimal subspace at 0 is $E_0^{\min} = \{\lambda = 0\}$ and lies in every E_x; adding a vector of $E_0^{\min}$ changes neither membership in C nor the degeneracy index. For a finite-dimensional N with $N + E_0^{\min} = E_0$: $\lambda|_N$ is onto $\mathbb{R}^n$, N is in good position to C with a closed good complement $Y \subseteq E_0^{\min}$, and $d_{N\cap C}(v) = d_C(v)$ for $v \in N \cap C$. A finite-dimensional N and a closed K with $N + K$ the whole space give a closed complement of N inside K. *Book:* proof of Theorem 4.2.11 (preprint 4.22), pp. 190–191 (preprint pp. 162–163) ("T_0A has in E an sc-complement Y contained in W"). *Hidden hypotheses:* Y closed. *Deviation (restated):* the text's splitting $T_0A = N \oplus (T_0A \cap W)$ is replaced by the coordinates $\lambda|_{T_0A}$ (Findings, items 82, 83).

V4.47 (Lemma 7.A of the exposition, and general position from $T_yQ + E_0^{\min} = E_0$; `Polyfold.exists_scComplement_le`, `Polyfold.isGeneralPositionRetractAt_of_sup_eq_top`). (a) For $N \subseteq E_\infty$ finite-dimensional, an sc-subspace R and $M = N + R$, there is an sc-subspace $B \subseteq R$ with $M_m = N_m \oplus B_m$ on every level. (b) For r tame, a retract $Q \subseteq O$, a smooth $y \in Q_1$ with T_yQ finite-dimensional, contained in E_∞ and $T_yQ + E_0^{\min} = E_0$, the retract Q is in general position at y in O. *Book:* proof of Theorem 4.2.11 (3) (preprint 4.22), preprint p. 164 ("$T_aO = T_aA + T_a^RO$, implying that T_aA has in T_aO a sc-complement contained in T_a^RO"). *Hidden hypotheses:* T_y^RO *an sc-subspace (Proposition 2.5.6 (preprint 2.36), for tame retracts). Deviation (hypotheses explicit).*

V4.48 (general position implies good position; `Polyfold.IsGeneralPositionRetractAt.isGoodPositionRetractAt`). For r tame and a smooth $z \in O_1$, if Q is in general position at z in O, it is in good position at z. *Book:* not stated; needed for the tameness in Theorem 4.2.11 (4) (preprint 4.22) and for Theorem 4.3.1 (preprint 4.23). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.49 (Theorem 4.2.11 (preprint 4.22) in a tame chart centred at the point; `Polyfold.sup_minimalSubspaceWith_eq_top`, `Polyfold.sup_minimalSubspaceWith_eq_top_graph`, `Polyfold.theorem_4_2_11_chart`). Let r be tame onto $O \ni 0$, t an sc$^+$-retraction onto $Q \subseteq O$ with $0 \in Q$, and $N = T_0Q$, with Q in general position at 0. Then $N + E_0^{\min} = E_0$, N is in good position to C, N has a closed good complement $Y \subseteq E_0^{\min}$ with a projection p whose kernel is exactly Y, and the data W, V, Γ of Proposition 4.1.9 (preprint 4.9) exist for this Y, with (2) $d_{N\cap C}(v) = d_C(\Gamma(v))$ on V and (3) general position of Q at every level-1 point over $Q \cap W$. *Book:* proof of Theorem 4.2.11 (preprint 4.22), pp. 190–193 (preprint pp. 162–164). *Hidden hypotheses:* none. *Deviation (restated):* $T_yQ + E_0^{\min} = E_0$ along the graph comes from $p \circ D\Gamma(v) = 1$ and $\ker p = Y$ (the text: "by means of the particular form of the map α").

V4.50 (Theorem 4.2.11 (preprint 4.22), local structure in general position; `Polyfold.MPolyfold.theorem_4_2_11_preprint`). Let X be tame, A a smooth finite-dimensional submanifold, in general position at $x \in A$. There are a chart ψ of X with $\psi(x) = 0$, a tame retraction r onto its model for a presentation T with n half-line coordinates, and an open $U \ni x$ inside the domain of ψ with: (1) $d_X(x) = n$, and $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O; (2) $d_X(a) = d_A(a)$ for $a \in A \cap U$ (d_A for the induced structure); (3) A is in general position at every point of $A \cap U$; (4) for every $\sigma \subseteq \{1, \dots, n\}$ there are faces $F_i \ni 0$ of the retract ($i \in \sigma$, F_i in the coordinate hyperplane $\lambda_i = 0$) and an open $G \ni 0$ such that $F_\sigma = \psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ contains x, lies in U, and is a tame sub-M-polyfold with $d_{F_\sigma} + \#\sigma = d_X$ on it, and $A \cap F_\sigma$ is a smooth finite-dimensional submanifold of F_σ, in general position in F_σ at each of its points, and tame. *Book:* Theorem 4.2.11 (preprint 4.22), p. 190 (preprint p. 162); proof pp. 190–193 (preprint pp. 162–164). *Hidden hypotheses:* the chart is re-centred at x; "smooth submanifold" is finite-dimensional. *Deviation (restated):* (1) is a statement about the chart model here; with the text's local faces, the faces of a face-structured U, it is `Polyfold.MPolyfold.theorem_4_2_11_faces` (2026-11-12; gap G.35, closed), and (4) holds for the germs of their intersections; F_σ is a germ, taken inside U; (4), without proof in the text, is V4.51, V4.52. (1) and (4) differ from the text's in content, not only in form: the text's local faces are faces of the neighbourhood U itself (the closures in U of the components of $U \cap \{d_X = 1\}$, all through x), the formal ones faces of the model retract of the chart; (2) and (3) are the text's. Identifying the two is the local face theorem for small neighbourhoods, proved since (`Polyfold.MPolyfold.exists_faces_chart`: near x each face of U is a chart face $\psi^{-1}\{\lambda_i = 0\}$) (Findings, Chapter 4, items 90, 130; round-trip item V4.50).

V4.51 (Theorem 4.2.11 (4) (preprint 4.22) in the chart; `Polyfold.killOn`, `Polyfold.exists_scPlus_retraction_corner`, `Polyfold.exists_cornerModel_retraction`, `Polyfold.theorem_4_2_11_four_chart`). In the situation of V4.49, for $\sigma \subseteq \{1, \dots, n\}$ and an open $B \ni 0$: with $\kappa_\sigma(v) = v - s(\lambda(v)1_\sigma)$ (s a linear section of $\lambda|_N$) there are an open $G \ni 0$, $G \subseteq W \cap B$, and an sc⁺-retraction $t_\sigma = \Gamma \circ \kappa_\sigma \circ p \circ t$ onto $Q \cap G \cap f_\sigma$, $f_\sigma = \{\lambda_i = 0, i \in \sigma\}$, whose tangent spaces contain $T_yQ \cap f_\sigma$. A retraction onto a subset of f_σ has a corner model on $[0,\infty)^{n-\#\sigma} \oplus W$, sc⁺ if the retraction is. For every open $B_0 \ni 0$ there are faces $F_i \ni 0$ ($i \in \sigma$) and an open $G \ni 0$, $G \subseteq W \cap B_0$, such that $O \cap G \cap f_\sigma$ is the retract of a retraction whose corner model is tame with $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$, $Q \cap G \cap f_\sigma$ and its corner model are sc⁺-retracts, and the corner model of $Q \cap G \cap f_\sigma$ is in general position at each of its level-1 points. *Book:* not in the text ("the previous arguments, but now applied to $A \cap F_\sigma$, conclude the proof"). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.52 (sc⁺-retractions in a chart, and Theorem 4.2.11 (4) (preprint 4.22) for one σ; `Polyfold.MPolyfold.isScPlusRetractionOn_of_chart`, `Polyfold.MChart.cornerPreimage`, `Polyfold.`

`MPolyfold.theorem_4_2_11_four_of_chart`). For a chart χ of X and an sc$^+$-retraction τ of its model with $\tau(U) \subseteq O$, $\chi^{-1} \circ \tau \circ \chi$ is an sc$^+$-retraction on the open set $\chi^{-1}(\tau.U)$ with image $\chi^{-1}(\tau(U))$. In a tame chart ψ centred at x with the data of V4.49 for the chart retraction of A, for every σ there are faces $F_i \ni 0$ and an open $G \ni 0$ such that $\psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ contains x, lies in the open set W of the submanifold chart, and has the properties (4) of V4.50. *Book:* not in the text. *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.53 (the family $\mathbb{R}^m \times X$ and stratified regular points; `Polyfold.prod_rn_isTame`, `Polyfold.fibreSet`, `Polyfold.rnPres`, `Polyfold.IsStratRegularPoint`, `Polyfold.IsStratRegularValue`). For tame X, $\mathbb{R}^m \times X$ with the product charts is tame. The fibre of $A \subseteq \mathbb{R}^m \times X$ over λ is $A_\lambda = \{x : (\lambda, x) \in A\}$. A point $z \in A$ is a *stratified regular point* if, for some data of Definition 4.2.1 (preprint 4.13) at z, some $a \in \mathbb{R}^m$, some chart φ of X at z_2, some presentation T of its partial quadrant and the level-1 point y over the image of z in the product chart $(r \mapsto r - a) \times \varphi$, every $\mu \in \mathbb{R}^m$ is the first component of a tangent vector v of the chart model of A at y with $v_2 \in E_{\varphi(z_2)}$ (the projection maps $T_zA \cap E_z$, the tangent space of the stratum through z, onto $\mathbb{R}^m$); λ is a *stratified regular value* if every $z \in A$ over λ is a stratified regular point. *Book:* §4.3, preprint p. 164, and the proof of Theorem 4.3.1 (preprint 4.23), p. 194 (preprint p. 166) ("regular values of p_σ"). *Hidden hypotheses:* the chart form (gap G.7). *Deviation (restated):* the chart independence is proved since (V4.93, gap G.36, closed).

V4.54 (Theorem 4.3.1 (preprint 4.23); `Polyfold.theorem_4_3_1_preprint`). Let X be tame and A a smooth finite-dimensional submanifold of $\mathbb{R}^m \times X$ with tame induced structure and compact closure, $\varepsilon > 0$, and suppose (1) for $|\lambda| \leq \varepsilon$ ($|\lambda|^2 = \sum \lambda_i^2$) the set $A \cap \{z_1 = \lambda\}$ is compact and non-empty, (2) A is in general position at every point. Then there is $\Sigma \subseteq \{|\lambda| < \varepsilon\}$ whose complement in $\{|\lambda| < \varepsilon\}$ is Lebesgue-null, consisting of stratified regular values, such that for $\lambda \in \Sigma$ the fibre A_λ is compact, a smooth finite-dimensional submanifold of X, in general position at each of its points, and its induced structure has a corner structure and is tame. *Book:* Theorem 4.3.1 (preprint 4.23), p. 193 (preprint p. 165); proof pp. 193–194 (preprint pp. 165–166). *Hidden hypotheses:* hypothesis (2), misprinted in the text ("there exists a sc-complement T_zA in T_zZ"), is Definition 4.2.10 (preprint 4.21). *Deviation (stronger):* Σ consists of stratified regular values, which are regular values; the tameness of A, the non-emptiness in (1) and $\varepsilon > 0$ are not used, and A_λ is moreover tame. With ordinary regular values the conclusion fails at the value: in $\mathbb{R} \times [0, \infty)$, $A = \{(x, x) : 0 \leq x < 1\} \cup (-1, 1) \times \{5\}$ satisfies the hypotheses, Dp is onto everywhere, and $A_0 = \{0, 5\}$ meets ∂X (Findings, item 102).

V4.55 (Remark 4.3.2 (preprint 4.24); `Polyfold.quotDim_le_finrank_of_eq_sup`, `Polyfold.remark_4_3_2_chart`, `Polyfold.MPolyfold.remark_4_3_2_codim`, `Polyfold.MPolyfold.remark_4_3_2`, `Polyfold.MPolyfold.remark_4_3_2_not_mem_boundary`). For tame X and smooth x, in a tame chart at x the reduced tangent space has codimension $d_X(x)$ in the tangent space. If A is in general position at a, then $d_X(a) \leq \dim T_aA$ (in any tame chart at a); in particular a submanifold in general position with $T_aA = 0$ does not meet ∂X at a. *Book:* Remark 4.3.2 (preprint 4.24), p. 193 (preprint p. 165). *Hidden hypotheses:* X tame and x smooth for the codimension. *Deviation (hypotheses explicit):* stated for any submanifold in general position, not only for the A_λ of Theorem 4.3.1 (preprint 4.23) (to which it applies by V4.54).

V4.56 (Sard's theorem, vendored; `Polyfold.MilnorSard.Sard.critSet`, `Polyfold.MilnorSard.Sard.sardStatement`, `Polyfold.MilnorSard.sard`, `Polyfold.MilnorSard.dense_compl_image_critSet`). Let E, F be finite-dimensional real normed spaces (in the same universe), F with its Borel σ-algebra and a Haar measure μ, $U \subseteq E$ open and f of class C^∞ on U. Then $\mu(f(\text{Crit})) = 0$, where Crit is the set of points of U at which Df is not onto, and the complement of $f(\text{Crit})$ is dense. *Book:* "Sard's theorem", proof of Theorem 4.3.1 (preprint 4.23), p. 194 (preprint p. 166), without statement. *Hidden hypotheses:* Mathlib has only the cases dim $E \leq$ dim F; the general case is Milnor's induction, copied from the Milnor project (`math_faithful`) with only the namespace changed (Findings, item 100). *Deviation (not in the book).*

V4.57 (the strata and Sard's theorem on them; `Polyfold.quadC`, `Polyfold.stratSub`, `Polyfold.zeroSetOf`, `Polyfold.finrank_stratSub`, `Polyfold.stratCritSet`, `Polyfold.measure_image_stratCritSet`). Let V be finite-dimensional with linear coordinates $\ell : V \to \mathbb{R}^n$ and $C = \{\ell \geq 0\}$. The stratum subspace of σ is $\{\ell_i = 0, i \in \sigma\}$, of dimension $\dim V - \#\sigma$ when ℓ is onto; $\sigma(v) = \{i : \ell_i(v) = 0\}$. For f of class C^∞ on $U = O \cap C$ (O open) into a finite-dimensional F, the stratified critical points are the $v \in U$ at which $Df(v)$ restricted to the stratum subspace of $\sigma(v)$ is not onto, and their image is null for every Haar measure. *Book:* proof of Theorem 4.3.1 (preprint 4.23), pp. 193–194 (preprint pp. 165–166) ("$A \cap F_\sigma^\circ$ is a smooth manifold … of dimension $\dim(A) - \#\sigma$"; "Using Sard's theorem we find a subset Σ_σ … having full measure"). *Hidden hypotheses:* the dimension count, which Theorem 4.2.11 (preprint 4.22) does not state (the text cites "Theorem 4.23" for it). *Deviation (restated):* in the model space T_0A of the chart.

V4.58 (Theorem 3.8.22 (preprint 3.78) in C^∞ form; `Polyfold.ContractionData.exists_solution_smooth`, `Polyfold.theorem_3_8_22_smooth`). Under the hypotheses of Theorem 3.8.22 (preprint 3.78) (V3.63) with f of class C^∞, one solution τ is C^∞ on V', with the other conclusions of V3.63 (including the uniqueness clause). *Book:* Theorem 3.8.22 (preprint 3.78), p. 169 (preprint p. 141), states one j; the proof of Theorem 4.3.1 (preprint 4.23) needs C^∞. *Hidden hypotheses:* none. *Deviation (stronger):* the contraction radii of Lemmas 3.8.23, 3.8.24 (preprint 3.79, 3.80) and the fixed point do not depend on j.

V4.59 (the kernel at a stratified regular point is in good position; `Polyfold.quadAt`, `Polyfold.isPartialQuadrant_quadAt`, `Polyfold.exists_goodComplement_stratSub`). For ℓ onto, $\{\ell_i \geq 0, i \in \sigma\}$ is a partial quadrant. If $K + \{\ell_i = 0, i \in \sigma\} = V$, then K is in good position to $\{\ell_i \geq 0, i \in \sigma\}$ with a closed good complement inside the stratum subspace. *Book:* not in the text here; the argument of Lemma 5.26 (preprint; not in the 2021 edition). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.60 (a retraction onto the fibre over a stratified regular point; `Polyfold.exists_fibre_retraction`). For ℓ onto, f of class C^∞ on $O \cap C$ and $v_0 \in O \cap C$ not stratified-critical, there are an open $O' \ni v_0$, $O' \subseteq O$, and a map ρ, C^∞ on $O' \cap C$, mapping $O' \cap C$ into itself and into $f^{-1}(f(v_0))$, equal to the identity on $O' \cap C \cap f^{-1}(f(v_0))$, with $D\rho(v_0)$ of range ker $Df(v_0)$ and equal to the identity there. *Book:* "for $\lambda \in \Sigma$ … A_λ is a smooth compact manifold with boundary with corners”, p. 194 (preprint p. 166), in the model. *Hidden hypotheses:* none. *Deviation (not in the book):* the implicit function theorem at corner points (V4.58 with V4.59).

V4.61 (the sc$^+$-retraction onto A_λ; `Polyfold.isScSmooth_slice`, `Polyfold.scDeriv_slice`, `Polyfold.exists_slice_retraction`). In the model $\mathbb{R}^m \oplus E$ of a product chart, let t be an sc$^+$-retraction described near a point as the graph $\Gamma : V \to t(U) \cap W$ with inverse p (Proposition 4.1.9 (preprint 4.9)), and ρ a C^∞ retraction of $O' \cap (\mathbb{R}^m \times C)$ onto the points v with $\Gamma(v)_1 = c$. Then $\tau(e) = (\Gamma(\rho(p(t(c,e)))))_2$ is an sc$^+$-retraction of E (partial quadrant C) onto $\{e : (c,e) \in t(U) \cap W, p(c,e) \in O'\}$; if $D\rho(v_0) = 1$ on $\{h : (D\Gamma(v_0)h)_1 = 0\}$, the tangent space of its retract at the point over $\Gamma(v_0)_2$ contains $(D\Gamma(v_0)h)_2$ for all such h. The slice $e \mapsto (c,e)$ is sc-smooth with derivative $h \mapsto (0,h)$. *Book:* not constructed (Definition 4.2.1 (preprint 4.13) requires it). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.62 (general position of A_λ; `Polyfold.isGeneralPositionRetractAt_slice`). In a product chart with r_X tame: if the chart model of A is in general position at $z = (c,y)$, the projection maps the tangent space of the stratum $T_zA \cap E_z$ onto $\mathbb{R}^m$, and τ is an sc$^+$-retraction onto $Q \subseteq O$ whose tangent space at the smooth y contains the second components of the tangent vectors of A at z with first component 0, then Q is in general position at y. *Book:* proof of Theorem 4.3.1 (preprint 4.23), p. 194 (preprint p. 166) (“in view of the assumption (2) … $T_{z'}(\{\lambda\} \times A_\lambda)$ has a sc-complement in $T^R_{z'}X$”). *Hidden hypotheses:* stratified regularity at z itself. *Deviation (restated).*

V4.63 (the local step of Theorem 4.3.1 (preprint 4.23); `Polyfold.FibreLocal`, `Polyfold.exists_local_family`). Let X be tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold in general position at every point. Every $z_0 \in A$ has an open neighbourhood U and a Lebesgue-null CV $\subseteq \mathbb{R}^m$ such that at every $z \in A \cap U$ with $z_1 \notin$ CV: z is a stratified regular point, A_{z_1} is near z_2 the image of an sc$^+$-retraction on an open subset of X, and A_{z_1} is in general position and in good position at z_2. *Book:* proof of Theorem 4.3.1 (preprint 4.23), pp. 193–194 (preprint pp. 165–166), in one $U(z)$. *Hidden hypotheses:* none. *Deviation (restated):* in the product chart of a tame chart of X centred at $z_{0,2}$, through V4.49, V4.57, V4.60–V4.62.

V4.64 (the countable cover: Theorem 4.3.1 (preprint 4.23) for any parameter set; `Polyfold.theorem_4_3_1_of_subset`). For X tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with compact closure, in general position at every point, and any $S \subseteq \mathbb{R}^m$, there is $\Sigma \subseteq S$ with $S \setminus \Sigma$ Lebesgue-null such that every $\lambda \in \Sigma$ is a stratified regular value and the conclusions of V4.63 hold at every point of A_λ. *Book:* proof of Theorem 4.3.1 (preprint 4.23), p. 193 (preprint p. 165) (“We cover the set $\{z \in A : |p(z)| \leq \varepsilon\}$ with the finitely many such neighborhoods”). *Hidden hypotheses:* none. *Deviation (stronger):* any S; the text’s set need not be compact and need not have a finite subcover (in $\mathbb{R} \times \mathbb{R}$, $A = (-2\varepsilon, 2\varepsilon) \times \{2\} \cup \{(\varepsilon(1-x), x) : 0 < x < 1\}$); A is second countable, its closure being compact in the metrizable $\mathbb{R}^m \times X$, and countably many neighbourhoods cover it (Findings, item 103).

V4.65 (Theorem 4.2.11 (preprint 4.22) in the 2021 form of (1); `Polyfold.MPolyfold.theorem_4_2_11`). Let X be a tame M-polyfold, $A \subseteq X$ a smooth finite-dimensional submanifold and $x \in A$ a point at which A is in general position. There are a tame chart ψ of X with $\psi(x) = 0$, with a tame retraction R onto its model O and a presentation $T : E \cong \mathbb{R}^n \oplus W$ of its partial quadrant, and an open $U \ni x$ inside the chart domain, such that: (1) $d_X(x) = n$, $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O, and for every $i < n$ there are a face $F_i \ni 0$ of the retract inside the i-th coordinate hyperplane and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap F_i) \subseteq U$; (2)–(4) are those of V4.50. *Book:* Theorem 4.2.11 (preprint 4.22), p. 190 (preprint p. 162); the 2021 edition asks in (1) that the d local faces in U also contain x. *Hidden hypotheses:* none. *Deviation (restated):* as in V4.50, the local faces are faces of the chart model, and their identification with faces of U itself is proved since (V4.50, gap G.35, closed). The count in (1) of V4.50 already counts the faces through $\psi(x)$, which is the 2021 statement; the germs $\psi^{-1}(G \cap F_i)$ that contain x and lie in U are read off from (4) for $\sigma = \{i\}$.

V4.66 (Theorem 4.3.1 (preprint 4.23) in the 2021 form; `Polyfold.theorem_4_3_1`). Let X be tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, such that (1) $A \cap \{z_1 = \lambda\}$ is compact for every λ with $|\lambda| \leq \varepsilon$ (possibly empty), and (2) A is in general position at every point. Then there is a **measurable** $\Sigma \subseteq \{|\lambda| < \varepsilon\}$ whose complement in $\{|\lambda| < \varepsilon\}$ is Lebesgue-null, consisting of stratified regular values, such that for $\lambda \in \Sigma$ the fibre A_λ is compact, a smooth finite-dimensional submanifold of X in general position at each of its points, with a corner structure, and tame. *Book:* Theorem 4.3.1 (preprint 4.23), p. 193 (preprint p. 165); the 2021 edition drops "non-empty" from (1) and asks Σ to be measurable. *Hidden hypotheses:* none. *Deviation (stronger):* stratified regular values, as in V4.54; the tameness of A and $\varepsilon > 0$ are kept and not used. Σ is that of V4.64 for $S = \{|\lambda| < \varepsilon\}$ with a measurable null hull of its exceptional set removed (Mathlib's `MeasureTheory.toMeasurable`), which keeps every other conclusion; the compactness of A_λ comes from (1). Hypothesis (2) is printed as in the earlier texts and read as general position (Findings, items 101, 151).

V4.67 (Definition 4.4.3 in a local model; `Polyfold.IsRetractVectorField`, `Polyfold.isRetractVectorField_zero`, `Polyfold.IsRetractVectorField.mem_retractTangent`, `Polyfold.IsRetractVectorField.isScSmooth_comp`, `Polyfold.IsRetractVectorField.scDeriv_apply`). For a local model (O, C, E), a map $A : E_1 \to E_0$ is (the principal part of) a vector field on O if $x \mapsto (x, A(x))$ is an sc-smooth map from the retract (O^1, C^1, E^1) to the tangent (TO, TC, TE). Then $(x, A(x)) \in TO$ for $x \in O_1$; for every sc-smooth retraction r onto O the map $\widetilde{A} = A \circ r^1$ is sc-smooth on U^1; and $Dr(x)A(x) = A(x)$ for $x \in O_1$. The zero map is a vector field on every sc-smooth retract. *Book:* Definition 4.4.3, p. 195, new in the 2021 edition (an sc-smooth section of $p : TX \to X^1$, defined on X^1, not on X), and formula (4.8), p. 200. *Hidden hypotheses:* none. *Deviation (restated):* the definition read in the model of a chart, where the text works in Appendix 4.5.

V4.68 (Definition 4.4.3 on an M-polyfold; `Polyfold.MPolyfold.IsScVectorField`, `Polyfold.MPolyfold.zeroVectorField`, `Polyfold.MPolyfold.tangentChartMap_zeroVectorField`, `Polyfold.MPolyfold.continuous_zeroVectorField`, `Polyfold.MPolyfold.isScVectorField_zero`). An sc-smooth vector field on the M-polyfold X is a map $A : X^1 \to TX$, sc-smooth for the M-polyfold structures of X^1 and of the tangent bundle TX (both paracompact), with $p(A(x)) = x$ for every $x \in X^1$. The zero section, which reads $(\psi(x), 0)$ in every chart ψ, is continuous and an sc-smooth vector field. *Book:* Definition 4.4.3, p. 195, new in the 2021 edition. *Hidden hypotheses:* none.

V4.69 (Lemma 4.4.4 in a local model, $i = 1$; `Polyfold.retractTangent_shift`, `Polyfold.IsRetractVectorField.shift`). For an sc-smooth retraction r onto O, the tangent of the retract (O^1, C^1, E^1) is the level-1 part of TO: $T(O^1) = (TO)^1$. If A is a vector field on (O, C, E), its level map $A_1 : E_2 \to E_1$ is a vector field on (O^1, C^1, E^1): $x \mapsto (x, A_1(x))$ is an sc-smooth section of $T(O^1) \to O^2$. *Book:* Lemma 4.4.4, p. 196, new in the 2021 edition, said to be obvious from the identification of $(TX)^i \to (X^1)^i$ with $T(X^i) \to X^{i+1}$. *Hidden hypotheses:* none. *Deviation (narrower):* in a local model and for $i = 1$; larger i follow by iterating the local statement. The lemma on an M-polyfold, with the identification of $(TX)^1$ and $T(X^1)$ of Proposition 2.3.17 (gap G.6, closed), is V4.78, and for every i V4.79.

V4.70 (the second derivative and its symmetry; `Polyfold.scSecondDeriv`, `Polyfold.scSecondDeriv_symm`, `Polyfold.fderivWithin_fderivWithin_eq`, `Polyfold.IsRelOpenInQuadrant.mem_closure_interior_lvlSet`, `Polyfold.scDeriv_tangentMap_inr`). For an sc² map f on $U \subseteq E$, $z \in U_2$ and $a, b \in E_1$, $D^2 f(z)(a, b)$ is the second component of $D(Tf)(z, a)(b, 0)$, the derivative of $z \mapsto Df(z)a$ in the direction b; and $D(Tf)(z, a)(0, l) = (0, Df(z)l)$. If U is relatively open in a partial quadrant, $D^2 f(z)(a, b) = D^2 f(z)(b, a)$. *Book:* §4.5.1, p. 200, new in the 2021 edition ("using the symmetry of the second derivative", for $f = r$). *Hidden hypotheses:* none (r lives on a relatively open subset of a partial quadrant). *Deviation (restated):* the text does not say what $D^2 r(x)(A(x), B(x))$ is when $A(x), B(x)$ lie only in E_1, while r is twice differentiable only as a map $U_2 \to E_0$, in directions of E_2; it is defined through the sc²-structure, the derivative of Tr. The symmetry holds for $a, b \in E_2$ by Mathlib's symmetry of second derivatives within a set (`ContDiffWithinAt.isSymmSndFDerivWithinAt`; points of a partial quadrant lie in the closure of its interior, also on level 2), and for $a, b \in E_1$ by density of E_2 in E_1 and continuity of $D(Tf)$ (Findings, item 154).

V4.71 (§4.5.1, the Lie bracket in a local model; `Polyfold.retractLieBracket`, `Polyfold.retractLieBracket_antisymm`, `Polyfold.IsRetractVectorField.diff_identity`, `Polyfold.retractLieBracket_mem_tangent`). Let r be an sc-smooth retraction onto O and A, B vector fields on (O, C, E). For $x \in E_2$ put $[A, B](x) = D\tilde{A}(x)B_1(x) - D\tilde{B}(x)A_1(x)$, with $\tilde{A} = A \circ r^1$, $\tilde{B} = B \circ r^1$ differentiated as maps on U^1; $[A, B] = -[B, A]$. For $x \in O_2$ and b with $Dr^1(x)b = b$: $D^2 r(x)(A(x), b) + Dr(x)D\tilde{A}(x)b = D\tilde{A}(x)b$. Hence $Dr(x)[A, B](x) = [A, B](x)$ for $x \in O_2$: the bracket is tangent to O. *Book:* §4.5.1, pp. 200–201, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* $DA(x)$ is read as the derivative of $\tilde{A} = A \circ r^1$, the principal part being defined only on O^1; the computation is the text's, with the symmetry of V4.70 (Findings, item 154).

V4.72 (Proposition 4.4.5 (preprint 4.26) in a local model; `Polyfold.proposition_4_4_5_local`, `Polyfold.isScSmooth_scDeriv_apply`, `Polyfold.levelMap_shift_shift_eq`, `Polyfold.levelMap_comp_shift_eq`). For vector fields A, B on (O, C, E) and an sc-smooth retraction r onto O: $(x, [A, B](x)) \in TO$ for every $x \in O_2$, and $x \mapsto (x, [A, B](x))$ is an sc-smooth map from the retract (O^2, C^2, E^2) to (TO, TC, TE), the local form of "$[A, B]$ is an sc-smooth section of $TX|X^2 \to X^2$". *Book:* Proposition 4.4.5 (preprint 4.26), p. 196 (preprint p. 167), with the proof in §§4.5.1–4.5.2, pp. 200–202, new in the 2021 edition (the earlier texts quote it from another paper, as a vector field on X^1). *Hidden hypotheses:* none. *Deviation (narrower):* in a local model. The statement on an M-polyfold is V4.76, with Lemma 4.5.1 (independence of the chart), whose proof the text leaves to the reader, in V4.74, V4.75 and V4.77 (gap G.38, closed); the pull-back bundle $TX|X^2 \to X^2$ is built since, with $[A,B]$ an sc-smooth section of it (V4.76; gap G.92, closed). *Proof:* the second component is $x \mapsto D\tilde{A}(r^2x)(\tilde{B})_1(x) - D\tilde{B}(r^2x)(\tilde{A})_1(x)$ on U^2, sc-smooth as the tangent map of $\tilde{A}$ composed with the sc-smooth pair $(r^2, (\tilde{B})_1)$, and equal to $[A,B] \circ r^2$; the first is V4.71.

V4.73 (the level of the bracket on pp. 200–201; `Polyfold.lieBracket_level_counterexample`, `Polyfold.not_forall_lieBracket_mem_level_one`, `Polyfold.BracketLevel.opS`, `Polyfold.BracketLevel.opT`, `Polyfold.BracketLevel.xPt`, `Polyfold.BracketLevel.bracket_not_mem_level_one`, `Polyfold.isRetractVectorField_scOperator`, `Polyfold.retractLieBracket_scOperator`). There are an sc-Banach space E, the identity retraction of E onto $O = E$ (with $C = E$), vector fields A, B on (O,C,E) and a point $x \in O_2$ with $[A,B](x) \notin E_1$; so the statement "$[A,B](x) \in E_1$ for every local model, all vector fields A, B and all $x \in O_2$" does not hold. Every sc-operator $E^1 \to E$ is a vector field on (E,E,E), and the bracket of two of them is the commutator $S_0T_1 - T_0S_1$. The example: E the weighted sequence space of V1.13 ($|x|_m = \sup_k (k+1)^m|x_k|$), $(Sx)_k = (-1)^k(k+1)x_k$, $(Tx)_k = (k+1)x_{k+1}$, $x_k = (k+1)^{-3}$. *Book:* §4.5.1, pp. 200–201, new in the 2021 edition: $DA(x)B(x) - DB(x)A(x)$ "belongs to the fiber $(T_xO)_1$ whenever $x \in O_2$" (p. 200), and for $x \in O_2$ the bracket "belongs to the level 1" (p. 201). *Hidden hypotheses:* none. *Deviation (not in the book):* both sentences, read as $[A,B](x) \in E_1$, do not hold as we read them: $[S,T]x = ((-1)^k(k+1)(2k+3)x_{k+1})_k$, and for this x the level-1 coordinates have absolute value $P^3 + P^2 \geq 1/4$, $P = (k+1)/(k+2)$, so they do not tend to 0 (here $T_xO = E_0$, so $(T_xO)_1 = E_1$). Read as "a tangent vector at the point x of level 1 of X^1", that is $x \in O_{m+2}$ gives $[A,B](x) \in E_m$, the sentence is V4.72, and Proposition 4.4.5 is not affected. A first example on a weighted ℓ^2 space (sketched in the header of `Polyfold/Chapter4/VectorFields.lean`) is not formalized (Findings, item 156).

V4.74 (Lemma 4.5.1 in a local model, the computation of §4.5.2; `Polyfold.IsRetractVectorField.diff_related`, `Polyfold.IsRetractVectorField.related_shift`, `Polyfold.retractLieBracket_natural`, `Polyfold.levelMap_levelMap_comp_eq`, `Polyfold.IsRetractVectorField.restrict`, `Polyfold.retractLieBracket_restrict`, `Polyfold.retractLieBracket_natural_inter`). Let r, r' be sc-smooth retractions onto O, O', let f be an sc-smooth map from the retract O to the retract O', $g = f \circ r$ and $Df = Dg$, and let A, B be vector fields on (O,C,E) and A', B' on (O',C',F) with $Df(x)A(x) = A'(f(x))$ and $Df(x)B(x) = B'(f(x))$ for every $x \in O_1$. Then for $x \in O_2$: (a) the relation for A differentiated in a direction b with $Dr^1(x)b = b$ reads $D^2g(x)(A(x),b) + Dg(x)D\tilde{A}(x)b = D\tilde{A}'(g_2(x))Dg^1(x)b$, $\tilde{A} = A \circ r^1$, $\tilde{A}' = A' \circ r'^1$; (b) the relation for B holds on level 1, $Dg^1(x)B_1(x) = B'_1(g_2(x))$, and $g_2 = f_2$ on O_2; (c) $Df(x)[A,B](x) = [A',B'](f(x))$. A vector field on O is one on the open part $O \cap W$, W open, with the restricted retraction, and at points of $(O \cap W)_2$ the bracket is the same for both retractions. For f sc-smooth on $O \cap W$ with $Tf(z,A(z)) = (f(z), A'(f(z)))$ and

$Tf(z, B(z)) = (f(z), B'(f(z)))$ on $(O \cap W)_1$, also $Tf(x, [A, B](x)) = (f(x), [A', B'](f(x)))$ on $(O \cap W)_2$. *Book:* Lemma 4.5.1, p. 201, new in the 2021 edition, whose proof is left to the reader, and the computation of §4.5.2, pp. 201–202, for a map φ with $D\varphi \cdot A = A \circ \varphi$ and $D\varphi \cdot B = B \circ \varphi$. *Hidden hypotheses:* the relations on an open set of level 1 (here on all of O_1), since the computation differentiates them at points of O_2. *Deviation (restated):* the computation for an sc-smooth f between two local models with two pairs of vector fields, with $D^2 g$ read as in V4.70 and its symmetry; the form on $O \cap W$ is the shape of a transition map of charts (V4.75). §4.5.2 ends by calling the result "Proposition 4.5.1", a possible misprint for Lemma 4.5.1 (Findings, items 157, 162). *Proof:* the text's: (a) for A with $b = B_1(x)$ and for B with $b = A_1(x)$, both fixed by $Dr^1(x)$ (V4.69), (b) on both sides, the difference, and the symmetry of $D^2 g$.

V4.75 (principal parts in a chart and the independence of the chart; `Polyfold.MPolyfold.tangentChartMap_change`, `Polyfold.MPolyfold.chartRetraction`, `Polyfold.MPolyfold.chartRetraction_retract`, `Polyfold.MPolyfold.principalPart`, `Polyfold.MPolyfold.tangentChartMap_eq_principalPart`, `Polyfold.MPolyfold.IsScVectorField.isRetractVectorField_principalPart`, `Polyfold.MPolyfold.IsScVectorField.principalPart_related`, `Polyfold.MPolyfold.chartLieBracket`, `Polyfold.MPolyfold.IsScVectorField.chartLieBracket_spec`, `Polyfold.MPolyfold.IsScVectorField.chartLieBracket_change`). Let A, B be sc-smooth vector fields on the M-polyfold X (V4.68) and φ, ψ charts of X, the model of each with a chosen sc-smooth retraction. Tangent charts change by $T\psi = T(\psi \circ \varphi^{-1}) \circ T\varphi$ over the overlap (formula (2.9)). The principal part A_φ, given on O_1 by $T\varphi(A(y)) = (\varphi(y), A_\varphi(\varphi(y)))$ for y of level 1 in the domain of φ, is a vector field on the model (O, C, E) (V4.67); the principal parts in φ and ψ are related by the transition map, $T(\psi\varphi^{-1})(z, A_\varphi(z)) = (\psi\varphi^{-1}(z), A_\psi(\psi\varphi^{-1}(z)))$ on level 1 of the overlap. The bracket $[A_\varphi, B_\varphi]$ of the principal parts satisfies Proposition 4.4.5 in the model (V4.72), and the brackets in φ and ψ are related by the tangent map of the transition map on level 2 of the overlap. *Book:* p. 201, new in the 2021 edition: Lemma 4.5.1 shows that the definition of $[A, B]$ does not depend on the local coordinates. *Hidden hypotheses:* none. *Proof:* the overlap is $O \cap W$ for an open W; V4.74 there, with the relations of the principal parts.

V4.76 (Proposition 4.4.5 (preprint 4.26) on an M-polyfold; `Polyfold.MPolyfold.levelLift_levelChart_spec`, `Polyfold.MPolyfold.lieBracketPre`, `Polyfold.MPolyfold.lieBracket`, `Polyfold.MPolyfold.tangentProj_lieBracket`, `Polyfold.MPolyfold.tangentChartMap_lieBracket`, `Polyfold.MPolyfold.tangentChartMap_lieBracket_eq`, `Polyfold.MPolyfold.continuous_lieBracket`, `Polyfold.proposition_4_4_5`). For sc-smooth vector fields A, B on the M-polyfold X, the Lie bracket $[A, B] : X^2 \to TX$, $X^2 = (X^1)^1$, is the tangent vector at x represented by $[A_\varphi, B_\varphi](\varphi(x))$ in a chart φ chosen around x. In every chart ψ around x, $T\psi([A, B](x)) = (\psi(x), [A_\psi, B_\psi](\psi(x)))$, so $[A, B]$ depends neither on the chart nor on the retraction chosen for its model; $p([A, B](x)) = x$; and $[A, B]$ is continuous and sc-smooth from X^2 to TX. *Book:* Proposition 4.4.5 (preprint 4.26), p. 196 (preprint p. 167), with the proof in §§4.5.1–4.5.2, pp. 200–202: the bracket, defined in local coordinates by the usual formula, is an sc-smooth section of $TX|X^2 \to X^2$, the pull-back of $TX \to X^1$ by the inclusion $X^2 \to X^1$ (the preprint and 2017 texts quote it from another paper, as a vector field on X^1). *Hidden hypotheses:*

none. *Deviation (restated):* “an sc-smooth section of $TX|X^2 \to X^2$” is read as an sc-smooth map $X^2 \to TX$ that covers the inclusion $X^2 \to X^1$; (2026-11-12) the pull-back bundle is built as a sub-M-polyfold of $X^2 \times TX$ and $[A, B]$ is an sc-smooth section of it (`Polyfold.proposition_4_4_5_section`; gap G.92, closed). The earlier statement, a vector field on X^1, would through Φ (V4.78) put $[A, B](x)$ in E_1 for $x \in X_2$, which the example of V4.73 excludes; as we read it, it does not hold, and the 2021 statement does. With V4.72, V4.75 and V4.77 this closes gap G.38 (Findings, item 161). *Proof:* in the chart $(\psi^1)^1$ of X^2 and the tangent chart $T\psi$, $[A, B]$ reads $z \mapsto (z, [A_\psi, B_\psi](z))$, sc-smooth by V4.72; the chart does not matter by V4.75.

V4.77 (Lemma 4.5.1 on M-polyfolds; `Polyfold.MPolyfold.liftLevelOneMap`, `Polyfold.MPolyfold.levelTwoMap`, `Polyfold.MPolyfold.levelMap_coord_levelLift`, `Polyfold.lemma_4_5_1`). Let $f : X \to Y$ be an sc-smooth map of M-polyfolds with level maps $f_1 : X^1 \to Y^1$ and $f_2 : X^2 \to Y^2$, A, B sc-smooth vector fields on X and A', B' on Y with $Tf(A(x)) = A'(f_1(x))$ and $Tf(B(x)) = B'(f_1(x))$ for every $x \in X^1$. Then $Tf([A, B](x)) = [A', B'](f_2(x))$ for every $x \in X^2$. *Book:* Lemma 4.5.1, p. 201, new in the 2021 edition, whose proof is left to the reader: for a local sc-diffeomorphism $\varphi : X \to X$ and vector fields with $T\varphi(A(x)) = A(\varphi(x))$ and $T\varphi(B(x)) = B(\varphi(x))$ at points of X_2, $T\varphi([A, B](x)) = [A, B](\varphi(x))$ for $x \in X_2$. *Hidden hypotheses:* the relations on an open set of X^1, which the computation of §4.5.2 differentiates. *Deviation (restated):* more general in the map (any sc-smooth f, between two M-polyfolds, with a pair of vector fields on each side); the relations are assumed on all of X^1, not only at the points of X_2. A local sc-diffeomorphism between open subsets of X is covered by `Polyfold.lemma_4_5_1_open` (2026-11-12), with the open subsets as Part I’s open sub-M-polyfolds, and on sections of $TX|X^2$ by `Polyfold.lemma_4_5_1_section` (gap G.92, closed) (Findings, items 162, 163). *Proof:* in charts φ around x and ψ around $f(x)$, Tf reads $T(\psi \circ f \circ \varphi^{-1})$, the principal parts are related by $\psi f \varphi^{-1}$ on the open part of the model where it is defined, and V4.74 applies there.

V4.78 (Lemma 4.4.4 on an M-polyfold, $i = 1$; `Polyfold.MPolyfold.vectorFieldLevelOne`, `Polyfold.MPolyfold.raiseIndex`, `Polyfold.lemma_4_4_4`, `Polyfold.MPolyfold.tangentShiftEquiv`, `Polyfold.proposition_2_3_17`). For an sc-smooth vector field A on X, let $A_1 : X^2 \to (TX)^1$ be its level-1 map and $\Phi : T(X^1) \to (TX)^1$ the identification of Proposition 2.3.17 (an sc-diffeomorphism, compatible with the projections to X^2). Then $A^{(1)} = \Phi^{-1} \circ A_1$ is an sc-smooth vector field on the M-polyfold X^1, that is, an sc-smooth section of $T(X^1) \to X^2$, and $\Phi(A^{(1)}(x)) = A(x)$ in $(TX)^1$ for every $x \in X^2$. *Book:* Lemma 4.4.4, p. 196, new in the 2021 edition: through the natural identification of $(TX)^i \to (X^1)^i$ with $T(X^i) \to X^{i+1}$ the index of A can be raised by every $i \geq 0$. *Hidden hypotheses:* none. *Deviation (narrower):* $i = 1$; every fixed i follows by applying the lemma to X^1, X^2, ..., since $A^{(1)}$ is again an sc-smooth vector field; the statement for every i is V4.79 (Findings, items 160, 164). The identification $(TX)^i \cong T(X^i)$ is Proposition 2.3.17 (`Polyfold.proposition_2_3_17`, `Polyfold.proposition_2_3_17_two`, and for every i `Polyfold.proposition_2_3_17_all`; gap G.6, closed). *Proof:* A_1 is sc-smooth (the level-1 map of an sc-smooth map), Φ^{-1} is sc-smooth, and Φ is compatible with the projections, so $p \circ A^{(1)}$ is the identity of X^2.

V4.79 (Lemma 4.4.4 on an M-polyfold, every i; `Polyfold.BundledMPolyfold`, `Polyfold.`

`BundledMPolyfold.lift`, `Polyfold.BundledMPolyfold.raiseIndexAll`, `Polyfold.BundledMPolyfold.tangentIncl_raiseIndexAll`, `Polyfold.BundledMPolyfold.vectorFieldScMap`, `Polyfold.BundledMPolyfold.tangentLiftEquiv_raiseIndexAll`, `Polyfold.lemma_4_4_4_all`, `Polyfold.BundledMPolyfold.lemma_4_4_4_all_zero`, `Polyfold.BundledMPolyfold.tangentLiftEquiv`, `Polyfold.BundledMPolyfold.shiftLiftEquiv`, `Polyfold.proposition_2_3_17_all`). Let X be an M-polyfold, X^i its i-th lift ($X^0 = X$, $X^{i+1} = (X^i)^1$), and $\Phi_i : T(X^i) \to (TX)^i$, $J_i : (X^1)^i \to (X^i)^1$ the sc-diffeomorphisms of Proposition 2.3.17 for every i. For an sc-smooth vector field A on X put $A^{(0)} = A$ and $A^{(i+1)} = (A^{(i)})^{(1)}$, the index raise of V4.78 on X^i. Then for every $i \geq 0$: $A^{(i)}$ is an sc-smooth vector field on X^i, a section of $T(X^i) \to X^{i+1}$; it is A, in that its image under the inclusion $T(X^i) \to TX$ at $x \in X^{i+1}$ is $A(x)$; and $\Phi_i \circ A^{(i)} \circ J_i$ is the i-th index raise $(X^1)^i \to (TX)^i$ of $A : X^1 \to TX$. The zero vector field gives a witness for every i. *Book:* Lemma 4.4.4, p. 196, new in the 2021 edition, with the diagram printed before it. *Hidden hypotheses:* none. *Deviation (restated):* the lifts X^i are bundled with their spaces, so that they form one family indexed by i; "the natural identification" is Φ_i and J_i, and "$A^{(i)}$ is A" is read through the inclusions into TX and through the diagram (Findings, item 164). *Proof:* induction on i, with V4.78 on X^i; the two identities are checked after the inclusion into TX, which is injective and commutes with Φ_i (Proposition 2.3.17).

V4.80 (The Whitney sum $\oplus_k TX$, in the local models; `Polyfold.ScBanach.fiberPow`, `Polyfold.ScBanach.whitney`, `Polyfold.retractWhitney`, `Polyfold.ScRetraction.whitney`, `Polyfold.ScRetraction.whitney_retract`, `Polyfold.IsScRetract.whitney`, `Polyfold.retractWhitneyMap`, `Polyfold.IsScSmoothRetractMap.whitney`). `IsScRetract.whitney`: for an sc-retract (O, C, E), $\oplus_k TO = \{(x, h_1, \dots, h_k) : x \in O_1,\ h_i \in T_x O\}$ (`retractWhitney`) is an sc-retract of $E^1 \oplus E^k$ (`ScBanach.whitney`), the image of the retraction $\oplus_k Tr$ (`ScRetraction.whitney`). `IsScSmoothRetractMap.whitney`: $\oplus_k Tf$ (`retractWhitneyMap`) is sc-smooth for f sc-smooth between retracts, so the charts $\oplus_k T\varphi$ have sc-smooth transitions. *Book:* §4.4, p. 195, before Definition 4.4.1, (New in the 2021 edition.) The tangent bundle $TX \to X^1$ is an sc-smooth bundle, and "it is an easy exercise" that the Whitney sum $TX \oplus \cdots \oplus TX \to X^1$ of k copies is an sc-smooth bundle. *Deviation (restated):* The exercise is done in the local models. $\oplus_k TX$ is not assembled into an M-polyfold bundle: forms are read in its charts $\oplus_k T\varphi$ (V4.81). E^k is an iterated product $E \oplus (E \oplus \cdots)$ (`ScBanach.fiberPow`) (Findings, Chapter 4, item 168).

V4.81 (Definition 4.4.1 (preprint 4.25): sc-differential k-forms; `Polyfold.MPolyfold.IsScForm`, `Polyfold.MPolyfold.chartForm`, `Polyfold.MPolyfold.chartForm_eq`, `Polyfold.MPolyfold.chartForm_transition`, `Polyfold.MPolyfold.isScForm_of_cover`, `Polyfold.MPolyfold.scForms`, `Polyfold.MPolyfold.constForm_mem_scForms`, `Polyfold.IsRetractScForm`, `Polyfold.MPolyfold.IsScForm.isRetractScForm`). `MPolyfold.IsScForm P k` : ω is a family of alternating k-linear maps on the tangent spaces $T_x X$, $x \in X^1$, whose representative `chartForm h` on $\oplus_k TO$ ($\omega_\varphi(T\varphi\, \alpha) = \omega(x)(\alpha)$, `chartForm_eq`) is sc-smooth in every chart φ of the maximal atlas. By `isScForm_of_cover` a covering family of charts suffices (chart change: `chartForm_transition`). `MPolyfold.scForms P k` is the vector space $\Omega^k(X)$; witnesses: 0 and the constant 0-forms (`constForm_mem_scForms`). In a local model the notion is `IsRetractScForm`, and `IsScForm.isRetractScForm` says that the representatives are local forms. *Book:* §4.4, Definition 4.4.1 (preprint 4.25), p. 195 (preprint p. 167), An sc-differential k-form on X is an sc-smooth map $\omega : \oplus_k TX \to \mathbb{R}$ which is linear in each argument and skew-symmetric; these forms make up the vector space $\Omega^k(X)$. *Deviation (restated):* "Linear in each argument and skew-symmetric" is built into the type (Mathlib's alternating maps on T_xX, a linear space by V2.19). "Sc-smooth on $\oplus_k TX$" is read in the charts $\oplus_k T\varphi$ (V4.80), for every chart of the maximal atlas; a covering family suffices, so the quantifier over all charts is not vacuous. There is no nonzero witness for $k \geq 1$ (a natural one, df, is not built) (Findings, Chapter 4, item 168).

V4.82 (The directed system (4.4) and Definition 4.4.2: $\Omega^k_\infty(X)$, $\Omega^*_\infty(X)$; `Polyfold.MPolyfold.lowerFiberCLM`, `Polyfold.MPolyfold.formLower`, `Polyfold.MPolyfold.formLower_mem_scForms`, `Polyfold.BundledMPolyfold.forms`, `Polyfold.BundledMPolyfold.formsMap`, `Polyfold.BundledMPolyfold.formsMap_comp`, `Polyfold.BundledMPolyfold.directedSystem_forms`, `Polyfold.BundledMPolyfold.formsInfty`, `Polyfold.BundledMPolyfold.formsInftyOf`, `Polyfold.BundledMPolyfold.formsInftyOf_formsMap`, `Polyfold.BundledMPolyfold.formsInftyGraded`). `formLower_mem_scForms`: the pull-back by $X^1 \to X$ (fibrewise the inclusion `lowerFiberCLM`, the map $T(X^1) \to TX$ of Proposition 2.3.17) maps $\Omega^k(X)$ to $\Omega^k(X^1)$. `BundledMPolyfold.formsMap` is the directed system $\Omega^k(X^i) \to \Omega^k(X^j)$, $i \leq j$ (`formsMap_comp`, `directedSystem_forms`); `formsInfty A k` is its direct limit $\Omega^k_\infty(X)$ (Mathlib's `Module.DirectLimit`), with the classes `formsInftyOf`; `formsInftyGraded A` is $\oplus_k \Omega^k_\infty(X)$. *Book:* §4.4, (4.4) and Definition 4.4.2, p. 195 (preprint p. 167, after Definition 4.25), Pull-back by the inclusions $X^i \to X$ gives the directed system (4.4) $\Omega^k(X) \to \cdots \to \Omega^k(X^i) \to \Omega^k(X^{i+1}) \to \cdots$, whose direct limit $\Omega^k_\infty(X)$ is a real vector space; its elements are the sc-differential k-forms on X_∞, and $\Omega^*_\infty(X) = \oplus_k \Omega^k_\infty(X)$. *Deviation (same):* The lifts X^i are the bundled `BundledMPolyfold.lift`, as in V4.79. The preprint writes Ω^*_∞ both for the limit and for the sum (Findings, Chapter 4, item 124); the 2021 text and the formal version use two names (item 168).

V4.83 (The formula for $d\omega$ (p. 197) with the bracket of §4.5.1; `Polyfold.bookBracket`, `Polyfold.bookBracket_eq_neg_lieBracket`, `Polyfold.invariantFormula`, `Polyfold.correctSign`, `Polyfold.printedSign`, `Polyfold.item158Sign`, `Polyfold.extDeriv_eq_invariantFormula`, `Polyfold.invariantFormula_printedSign_counterexample`). `extDeriv_eq_invariantFormula`: on a real normed space, for a k-form ω and fields $V_0, \dots, V_k$ differentiable at x, $d\omega(V_0, \dots, V_k)$ (Mathlib's `extDeriv`) equals the formula with the text's bracket $[A, B] = DA \cdot B - DB \cdot A$ (`bookBracket`, $=$ −`lieBracket`, `bookBracket_eq_neg_lieBracket`) and the sign $(-1)^{\ell+j+1}$ on the pair $\ell < j$ (`correctSign`). `invariantFormula_printedSign_counterexample`: for $k = 1$, $\omega = dy$ on $\mathbb{R}^2$, $A_0 = (1, 0)$ and $A_1(p) = (0, p_1)$, the formula with the printed sign $(-1)^{\ell+k}$ (`printedSign`) gives 2 at 0, and so does the sign $(-1)^{\ell+j}$ (`item158Sign`), while $d\omega = 0$; with $A_1' = 0$, which equals A_1 at 0, the printed formula gives 0. *Book:* §4.4, p. 197 (preprint p. 167), the formula for $d\omega$; the bracket: Appendix 4.5.1, p. 200, (2021) $d\omega(A_0^{(1)}, \dots, A_k^{(1)})$ is given by "the familiar formula": the sum over j of $(-1)^j D(\omega(A_0, \dots, \hat{A}_j, \dots, A_k)) \cdot A_j^{(1)}$ (printed with the summation index i), plus the sum over $\ell < j$ of $(-1)^{\ell+k}\, \omega([A_\ell, A_j], A_0^{(1)}, \dots, \hat{A}_\ell^{(1)}, \dots, \hat{A}_j^{(1)}, \dots, A_k^{(1)})$, where $[A, B] = DA \cdot B - DB \cdot A$

(p. 200). The preprint (p. 167) and the 2017 text print the sign $(-1)^{i+j}$ and give the bracket only as "the usual formula". *Deviation (narrower):* Narrower: with the text's bracket and the printed sign $(-1)^{\ell+k}$ the formula does not hold as we read it (`invariantFormula_printedSign_counterexample`): its right side is 2 where $d\omega = 0$, and it changes when A_1 is changed away from the point, so it is not tensorial. The classical sign $(-1)^{\ell+j}$, which item 158 proposed, fails on the same example, since the text's bracket is the negative of the classical one. With $(-1)^{\ell+j+1}$ the formula is Mathlib's d (`extDeriv_eq_invariantFormula`); the printed sign is possibly a misprint, or the text may intend another bracket convention. The index i of the first sum is read as j (Findings, Chapter 4, items 158, 167). Read in the classical setting, a real normed space, to which the sc formula reduces in a chart. The formal d on M-polyfolds (V4.84, V4.85) is defined by the local formula, so the sign does not enter it.

V4.84 (The exterior derivative in the local models: the local formula, naturality, independence of the retraction; `Polyfold.ambientD`, `Polyfold.ambientD_apply`, `Polyfold.IsScSmooth.ambientD`, `Polyfold.scDeriv_relation`, `Polyfold.ambientD_whitneyMk`, `Polyfold.IsAmbientScForm`, `Polyfold.IsAmbientScForm.ambientD`, `Polyfold.retractD`, `Polyfold.IsRetractScForm.retractD`, `Polyfold.ambientD_comp_whitneyMap`, `Polyfold.retractD_comp_retractWhitneyMap`, `Polyfold.retractD_eq_of_subset`, `Polyfold.sum_update_symm_eq_zero`). `ambientD U` $\Omega(y, h_0, \dots, h_k) = \sum_j (-1)^j D\Omega(y, h_0, \dots, \widehat{h}_j, \dots, h_k)(h_j, 0)$ (`ambientD_apply`), built from $T\Omega$ and sc-operators. `IsAmbientScForm.ambientD` and `IsRetractScForm.retractD`: d maps sc-forms on U (relatively open in a partial quadrant), and on a retract O (as $d(\omega \circ \oplus_k Tr)$), to sc-forms one level up: sc-smooth (`IsScSmooth.ambientD`) and alternating (`ambientD_whitneyMk`, via `scDeriv_relation`). `ambientD_comp_whitneyMap` and `retractD_comp_retractWhitneyMap`: $d(f^*\Omega) = f^* d\Omega$ for sc-smooth f. `retractD_eq_of_subset`: d does not depend on the retraction. *Book:* §4.4, pp. 196–197 (preprint p. 167), the definition of $d\omega$, (2021) For an sc-differential k-form ω on X^i and sc-smooth vector fields $A_0, \dots, A_k$ on X^i (sections $X^{i+1} \to T(X^i)$), $d\omega$ on $\oplus_{k+1} T(X^{i+1})$ is defined through the maps $D(\omega(A_0, \dots, \widehat{A}_j, \dots, A_k))$ and the brackets $[A_\ell, A_j]$ by the formula of p. 197. *Deviation (restated):* The text defines d through vector fields. Its formula in a chart is used here: on normed spaces it agrees with the invariant formula with the sign $(-1)^{\ell+j+1}$ (V4.83), and it makes d on an M-polyfold well defined (V4.85). Naturality, and with it the independence of the retraction, uses the symmetry of the second derivative (`scSecondDeriv_symm`), as the bracket does in §4.5.2 (Findings, Chapter 4, item 169).

V4.85 (The exterior derivative $d : \Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$ on an M-polyfold; `Polyfold.MPolyfold.chartD`, `Polyfold.MPolyfold.chartD_transition`, `Polyfold.MPolyfold.extD`, `Polyfold.MPolyfold.extD_apply`, `Polyfold.MPolyfold.chartForm_extD`, `Polyfold.MPolyfold.extD_mem_scForms`, `Polyfold.MPolyfold.extD_add`, `Polyfold.MPolyfold.extD_smul`, `Polyfold.MPolyfold.extD`, `Polyfold.BundledMPolyfold.formsD`). `MPolyfold.extD` at $x \in X^2$ is $d(\omega_\varphi)(\varphi(x), T\varphi^1 \cdot)$ on $T_x(X^1)$, with `chartD` the local d of V4.84 in the chart φ. `extD_apply`: the same holds in every chart around x (from `chartD_transition`), so d does not depend on the chart. `extD_mem_scForms`: $d\omega \in \Omega^{k+1}(X^1)$ for $\omega \in \Omega^k(X)$, with representative `chartForm_extD` in the level charts. `extD_add`, `extD_smul`, `extD` : d is linear. On the lifts, `BundledMPolyfold.formsD` is $\Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$. *Book:* §4.4, p. 197 (preprint p. 167),

(2021) The formula defines the exterior derivative as a linear operator $d : \Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$. (The preprint writes $d : \Omega^k(X^{i+1}) \to \Omega^{k+1}(X^i)$.) *Deviation (restated):* d is the local formula in charts (V4.84), not the vector-field formula, which is proved on normed spaces only (V4.83). Domain and target are those of the 2021 text, the direction item 120 proposed for the preprint. On a family that is not an sc-form, `extD` is 0 by convention (Findings, Chapter 4, item 169).

V4.86 ($d^2 = 0$; `Polyfold.sum_removeNth_symm_eq_zero`, `Polyfold.scDeriv_ambientD_whitneyMk`, `Polyfold.ambientD_ambientD`, `Polyfold.retractD_shift_retractD`, `Polyfold.MPolyfold.extD_extD`, `Polyfold.BundledMPolyfold.formsInftyD_comp_formsInftyD`, `Polyfold.BundledMPolyfold.formsInftyGradedD_comp_self`). `extD_extD`: $d(d\omega) = 0$ in $\Omega^{k+2}(X^2)$ for $\omega \in \Omega^k(X)$. In the local models `ambientD_ambientD` and `retractD_shift_retractD`, from the symmetry of the second derivative and `sum_removeNth_symm_eq_zero`. On the limit, `formsInftyD_comp_formsInftyD` and `formsInftyGradedD_comp_self`. *Book:* §4.4, p. 198 (preprint p. 168), $d \circ d = 0$, stated without proof. *Deviation (same):* The text states $d^2 = 0$ without proof (Findings, Chapter 4, item 121); it is proved here in the sc setting, level by level, by the symmetry of the second derivative (item 170).

V4.87 (d commutes with $X^{i+1} \to X^i$; d on $\Omega^*_\infty(X)$; `Polyfold.MPolyfold.extD_formLower`, `Polyfold.BundledMPolyfold.formsD_formsStep`, `Polyfold.BundledMPolyfold.formsD_formsMap`, `Polyfold.BundledMPolyfold.formsInftyD`, `Polyfold.BundledMPolyfold.formsInftyD_of`, `Polyfold.BundledMPolyfold.formsInftyGradedD`, `Polyfold.MPolyfold.formWedge`, `Polyfold.MPolyfold.formWedge_mem_scForms`, `Polyfold.MPolyfold.extD_formWedge`, `Polyfold.BundledMPolyfold.formsInftyWedge`, `Polyfold.BundledMPolyfold.formsInftyD_formsInftyWedge`, `Polyfold.BundledMPolyfold.formsInftyWedge_assoc`, `Polyfold.BundledMPolyfold.formsInftyWedge_comm`, `Polyfold.BundledMPolyfold.formsInftyGAlgebra`, `Polyfold.BundledMPolyfold.formsInftyGraded_of_mul_of_comm`, `Polyfold.BundledMPolyfold.formsInftyGradedD_of_mul_of`). `extD_formLower`: $d(\iota^*\omega) = \iota^*(d\omega)$ for $\iota : X^1 \to X$; `formsD_formsStep` and `formsD_formsMap` give the squares with the maps of (4.4). `formsInftyD A k` : $\Omega^k_\infty(X) \to \Omega^{k+1}_\infty(X)$ is the induced map (`formsInftyD_of`: $d[\omega] = [d\omega]$), and `formsInftyGradedD` the same map on $\oplus_k \Omega^k_\infty(X)$, of degree $+1$. (2026-11-12) The wedge product: `formWedge` is the pointwise $\omega \wedge \tau$, with $\Omega^k(X) \wedge \Omega^l(X) \subseteq \Omega^{k+l}(X)$ (`formWedge_mem_scForms`) and $d(\omega \wedge \tau) = d\omega \wedge \iota^*\tau + (-1)^k \iota^*\omega \wedge d\tau$ (`extD_formWedge`); `formsInftyWedge` is the induced bilinear map on $\Omega^*_\infty(X)$, associative, graded commutative and unital (`formsInftyWedge_assoc`, `formsInftyWedge_comm`, `formsInftyOne`), with $d(a \wedge b) = da \wedge b + (-1)^k a \wedge db$ (`formsInftyD_formsInftyWedge`). $\bigoplus_k \Omega^k_\infty(X)$ is a graded $\mathbb{R}$-algebra (`formsInftyGAlgebra`, Mathlib's `DirectSum.GAlgebra`), graded commutative (`formsInftyGraded_of_mul_of_comm`), on which d is a derivation of degree $+1$ (`formsInftyGradedD_of_mul_of`) with $d^2 = 0$ (`formsInftyGradedD_comp_self`). *Book:* §4.4, pp. 197–198 (preprint p. 168), d commutes with the inclusion $X^{i+1} \to X^i$ of the directed system (the square from $\Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$ to $\Omega^k(X^{i+1}) \to \Omega^{k+1}(X^{i+2})$); passing to the limit gives $d : \Omega^*_\infty(X) \to \Omega^*_\infty(X)$. *Deviation (same):* "Passing to the limit" is Mathlib's `Module.DirectLimit.lift`. The next sentence of the text calls $(\Omega^*_\infty(X), d)$ a graded differential algebra without defining $\wedge$; we take the classical $(k!\,l!)^{-1}$-normalised wedge,

$(a \wedge b)(v) = (k!\,l!)^{-1} \sum_\sigma \operatorname{sign}\sigma\, a(v_{\sigma(1..k)})\, b(v_{\sigma(k+1..k+l)})$, and prove associativity, graded commutativity, the unit and the Leibniz rule, whose sign $(-1)^k$ holds as we read it; the algebra is Mathlib's `DirectSum.GAlgebra` structure on $\bigoplus_k \Omega^k_\infty(X)$ (gap G.80; Findings, Chapter 4, items 170, 177).

V4.88 (Definition 4.4.6 (preprint 4.27): the sc-de Rham cohomology $H^*_{sc}(X)$; `Polyfold.BundledMPolyfold.formsInftyDRange`, `Polyfold.BundledMPolyfold.formsInftyDRange_le_ker`, `Polyfold.BundledMPolyfold.scDeRham`, `Polyfold.BundledMPolyfold.scDeRhamMk`, `Polyfold.BundledMPolyfold.scDeRhamMk_eq_iff`). `BundledMPolyfold.scDeRham A k` is $\ker d / \operatorname{im} d$ in degree k, a real vector space, with `formsInftyDRange` the image $d(\Omega^{k-1}_\infty(X))$ (0 for $k = 0$) and `formsInftyDRange_le_ker`: $\operatorname{im} d \subseteq \ker d$. `scDeRhamMk_eq_iff`: two closed forms have the same class iff they differ by an exact form. *Book:* §4.4, Definition 4.4.6 (preprint 4.27), p. 198 (preprint p. 168), The sc-de Rham cohomology of the M-polyfold X is $H^*_{sc}(X) = \ker(d)/\operatorname{im}(d)$. *Deviation (same):* The text writes $H^*_{sc}(X)$ for the graded space; it is read degree by degree. The relative version $H^*_{dR}(X, \partial X)$ is not formalized (gap G.39) (Findings, Chapter 4, item 170).

V4.89 (The pull-back f^* and $f^* \circ d_Y = d_X \circ f^*$ (p. 198); `Polyfold.MPolyfold.tangentFiberMap`, `Polyfold.MPolyfold.formPullback`, `Polyfold.MPolyfold.chartForm_formPullback`, `Polyfold.MPolyfold.formPullback_mem_scForms`, `Polyfold.MPolyfold.extD_formPullback`, `Polyfold.MPolyfold.formPullback_comp`, `Polyfold.MPolyfold.formPullback_id`, `Polyfold.MPolyfold.formLower_formPullback`, `Polyfold.BundledMPolyfold.formsPullback`, `Polyfold.BundledMPolyfold.formsInftyPullback`, `Polyfold.BundledMPolyfold.formsInftyD_comp_formsInftyPullback`, `Polyfold.BundledMPolyfold.formsInftyPullback_comp`, `Polyfold.BundledMPolyfold.scDeRhamPullback`, `Polyfold.MPolyfold.formPullback_formWedge`, `Polyfold.BundledMPolyfold.formsInftyPullback_formsInftyWedge`, `Polyfold.BundledMPolyfold.formsInftyPullback_formsInftyOne`). `formPullback hf` is $\omega(f(x))(Tf\,\alpha_1, \dots, Tf\,\alpha_k)$, with Tf on the fibres the continuous linear map `tangentFiberMap` (`tangentMapCLM`). In charts it is $\omega_\psi \circ \oplus_k T(\psi \circ f \circ \varphi^{-1})$ (`chartForm_formPullback`), so it maps $\Omega^k(Y)$ to $\Omega^k(X)$ (`formPullback_mem_scForms`). `extD_formPullback`: $d(f^*\omega) = (f^1)^* d\omega$, chartwise from the naturality of the local d. `formPullback_comp`, `formPullback_id`, `formLower_formPullback` (it commutes with (4.4)). On the lifts, `formsPullback` is $(f^i)^* : \Omega^k(Y^i) \to \Omega^k(X^i)$ and `formsInftyPullback` the limit map $\Omega^k_\infty(Y) \to \Omega^k_\infty(X)$, with `formsInftyD_comp_formsInftyPullback` ($d \circ f^* = f^* \circ d$), `formsInftyPullback_comp`, `formsInftyPullback_id`, and `scDeRhamPullback` on H^k_{sc}. f^* is multiplicative and unital, a morphism of graded algebras (`formPullback_formWedge`, `formsInftyPullback_formsInftyWedge`, `formsInftyPullback_formsInftyOne`). *Book:* §4.4, p. 198 (2017 p. 150; preprint p. 168), An sc-smooth $f : X \to Y$ induces $f^* : \Omega^*_\infty(Y) \to \Omega^*_\infty(X)$, on a representative $\omega \in \Omega^k(Y^i)$ by $\omega \circ (Tf \oplus \cdots \oplus Tf)$ with f raised to the index i, and $f^* \circ d_Y = d_X \circ f^*$, the limit of the commutative squares $\Omega^k(Y^i) \to \Omega^k(X^i)$, $\Omega^{k+1}(Y^{i+1}) \to \Omega^{k+1}(X^{i+1})$ (left to the reader). The 2017 text and the preprint have only the first sentence. *Deviation (same):* The text writes Ω^*_∞; the map is given degree by degree, as for d (V4.87). "Raising the index" is `liftLevelOneMap` on X^1 and `BundledMPolyfold.liftMap` on X^i. $f^* \circ d_Y = d_X \circ f^*$ is left to the reader in the text (2021 only); it holds as we read it and is proved here (Findings, Chapter 4, item 173).

V4.90 (Forms on ∂X and the pull-back j^* (pp. 198–199); `Polyfold.MPolyfold.IsAgreeingOver`, `Polyfold.MPolyfold.IsFaceCompatible`, `Polyfold.MPolyfold.bForms`, `Polyfold.MPolyfold.extD_mem_bForms`, `Polyfold.MPolyfold.le_bForms`, `Polyfold.MPolyfold.bFormsInfty`, `Polyfold.MPolyfold.jStar`, `Polyfold.MPolyfold.formsPullback_mem_bForms`, `Polyfold.MPolyfold.jStar_comp_D`, `Polyfold.MPolyfold.isFaceCompatible_extD`, `Polyfold.MPolyfold.bForms_eq_agreeForms`, `Polyfold.MPolyfold.mem_bForms_iff`, `Polyfold.MPolyfold.hasCornerCover_bdryMap`, `Polyfold.MPolyfold.isTangentInjective_bdryMap`, `Polyfold.MPolyfold.CornerChart.exists_tangent`, `Polyfold.BundledMPolyfold.isAgreeingOver_extD`). `bForms hP k i` $= \Omega^k(\partial X^i)$: the sc-forms $\tau \in \Omega^k(bX^i)$ on the tame boundary (Theorem 2.7.13) that agree over B^i (`IsFaceCompatible`: $\tau_p(v) = \tau_{p'}(v')$ when $Bp = Bp'$ and $TB\,v_j = TB\,v'_j$); `mem_bForms_iff`, `bForms_eq_agreeForms`. The definition also lists the agreement of $d\tau$ over B^{i+1}, which follows (`isFaceCompatible_extD`, from the general `BundledMPolyfold.isAgreeingOver_extD` with `isTangentInjective_bdryMap` and `hasCornerCover_bdryMap`). `extD_mem_bForms`: d maps $\Omega^k(\partial X^i)$ to $\Omega^{k+1}(\partial X^{i+1})$. `bFormsInfty` is the direct limit, `jStar` $= B^* : \Omega^k_\infty(X) \to \Omega^k_\infty(\partial X)$ (`formsPullback_mem_bForms`), `jStar_comp_D`: $dj^* = j^*d$. *Book:* §4.4, pp. 198–199 (2017 p. 150; preprint p. 168), For a tame X the inclusion $\partial X \to X$ is sc-smooth on local faces, and local faces and their intersections are tame M-polyfolds, so differential forms on ∂X and ∂X^i make sense; $j : \partial X \to X$ is the inclusion. *Deviation (same):* The text gives no definition of forms on ∂X, which is not an M-polyfold (Findings, Chapter 4, item 122). We read them as forms on the tame boundary bX (Theorem 2.7.13) that coincide on intersections of local faces, i.e. agree over B (reading R-a; item 173). That d preserves this (the text writes $d\tau$) is proved (2026-11-12): if τ agrees over B^i, $d\tau$ agrees over B^{i+1} (`isFaceCompatible_extD`), so `bForms` is the set of agreeing sc-forms (`bForms_eq_agreeForms`). The proof takes, at two faces through a point x, a tame chart with retraction r and coordinates λ_i, λ_j vanishing on the two faces; $r \circ P_i \circ P_j$ maps a neighbourhood in the partial quadrant into both faces, its two lifts to bX have the same composite with B, and $T_xF \cap T_xF'$ is the image of its tangent map (`CornerChart.exists_tangent`); then $d\tau$ is compared by $df^* = f^*d$. No smoothness of x is needed. $dj^* = j^*d$, stated without proof (item 121), is proved.

V4.91 (The relative complex $\Omega^*_\infty(X, \partial X)$, $d \circ d = 0$ and $H^*_{dR}(X, \partial X)$ (p. 199); `Polyfold.MPolyfold.RelForms`, `Polyfold.MPolyfold.relD`, `Polyfold.MPolyfold.relD_comp_relD`, `Polyfold.MPolyfold.relDRange`, `Polyfold.MPolyfold.relDeRham`, `Polyfold.MPolyfold.relDeRhamMk`, `Polyfold.MPolyfold.relDeRhamMk_eq_iff`, `Polyfold.MPolyfold.relDeRhamEquivOfNoBoundary`). `RelForms hP k` is $\Omega^0_\infty(X)$ for $k = 0$ and $\Omega^k_\infty(X) \times \Omega^{k-1}_\infty(\partial X)$ otherwise. `relD` is $d(\omega, \tau) = (d\omega, j^*\omega - d\tau)$ ($\omega \mapsto (d\omega, j^*\omega)$ in degree 0). `relD_comp_relD`: $d \circ d = 0$, from $d^2 = 0$ on X and ∂X and $dj^* = j^*d$. `relDeRham hP k` $= \ker d / \operatorname{im} d$, a real vector space, with `relDeRhamMk_eq_iff`. `relDeRhamEquivOfNoBoundary`: for $\partial X = \emptyset$, $bX = \emptyset$ and $H^k_{dR}(X, \partial X) \cong H^k_{sc}(X)$. *Book:* §4.4, p. 199 (2017 p. 150; preprint p. 168), $\Omega^*_\infty(X, \partial X) = \Omega^*_\infty(X) \oplus \Omega^{*-1}_\infty(\partial X)$ with $d(\omega, \tau) = (d\omega, j^*\omega - d\tau)$; one easily verifies $d \circ d = 0$; the cohomology is $H^*_{dR}(X, \partial X)$. *Deviation (same):* The text writes the graded sum; it is read degree by degree, with $\Omega^{-1} = 0$. The text calls the sum a differential algebra; the wedge product is formalized on $\Omega^*_\infty(X)$ (`formsInftyWedge`, V4.87) but not on the relative complex (G.80). The identification for $\partial X = \emptyset$ is used on p. 279, after the proof of Theorem 6.7.1 (V6.77). Forms on ∂X are those of V4.90 (Findings, Chapter 4, item 173). *Update (2026-11-13):* the relative complex is a differential algebra with the product of the mapping cone of j^*, $(\omega, \tau)(\omega', \tau') = (\omega \wedge \omega', (-1)^k j^*\omega \wedge \tau')$, which the text does not give (our choice): `Polyfold.MPolyfold.relWedge`, `Polyfold.MPolyfold.relD_relWedge` (d is a derivation), `Polyfold.MPolyfold.relWedge_assoc`, `Polyfold.MPolyfold.relCast_relWedge_relOne` (unit on the left; `Polyfold.MPolyfold.relWedge_relOne`: not on the right), `Polyfold.MPolyfold.relGradedNonUnitalRing`, `Polyfold.MPolyfold.relGradedD_of_mul_of` (gap G.80 closed; Findings, item 178).

V4.92 (Theorem 4.4.7 (preprint 4.28) for sc-forms on a finite-dimensional submanifold: Stokes; `Polyfold.MPolyfold.stokes_formsInftyRestrict`, `Polyfold.MPolyfold.scFormRestrict`, `Polyfold.MPolyfold.scFormRestrict_apply_modelDeriv`, `Polyfold.MPolyfold.formsInftyRestrict`, `Polyfold.MPolyfold.contMDiffForm_formsInftyRestrict`, `Polyfold.MPolyfold.extDeriv_formsInftyRestrict`, `Polyfold.MPolyfold.scFormRestrict_formLower`, `Polyfold.MPolyfold.subset_smoothSet_of_hasCornerStructureOfDim`, `Polyfold.exists_hasCornerStructureOfDim_of_tame`, `Polyfold.exists_hasCornerStructureOfDim_solutionSet`, `Polyfold.MPolyfold.theorem_4_4_7_hyp`, `Polyfold.MPolyfold.faceOrientation`, `Polyfold.MPolyfold.isFaceStructured_of_forall_degIndex_eq_zero`, `Polyfold.MPolyfold.hasCornerStructureOfDim_face`, `Polyfold.sum_integral_restrictForm_eq_boundaryIntegral`, `Polyfold.restrictForm`, `Polyfold.IsInducedOrientation`, `Polyfold.inducedOrientation`, `Polyfold.MPolyfold.finite_setOf_isFace`). `MPolyfold.stokes_formsInftyRestrict`: for a compact sub-M-polyfold $S \subseteq X$ with a corner structure of dimension $n+1$ (then $S \subseteq X_\infty$, `subset_smoothSet_of_hasCornerStructureOfDim`), a continuous tangent orientation o of the Mathlib manifold S (`cornerChartedSpace`) and $\omega \in \Omega^n_\infty(X)$, $\int_S (d\omega)|_S = \int_{\partial S} \omega|_S$. The restriction $\omega \mapsto \omega|_S$ (`scFormRestrict` on $\Omega^k(X^i)$, `formsInftyRestrict` on $\Omega^k_\infty(X)$) is the pull-back along $S \to X^i$ (V4.89) read on the corner frames of $T_xS = \mathbb{R}^n$; in a chart of X it is $\omega_\varphi \circ$ `modelDeriv` (`scFormRestrict_apply_modelDeriv`). It is C^∞ (`contMDiffForm_formsInftyRestrict`) and commutes with d (`extDeriv_formsInftyRestrict`) and with the system (4.4) (`scFormRestrict_formLower`). $\int_{\partial S}$ is the boundary integral of the Mathlib manifold S. The corner structures exist for every submanifold in good position of a tame X (`exists_hasCornerStructureOfDim_of_tame`), in particular for the solution sets of Theorem 6.6.18 (`exists_hasCornerStructureOfDim_solutionSet`). (2026-11-12) With the faces: `MPolyfold.theorem_4_4_7_hyp`: for a compact sub-M-polyfold $S \subseteq X$ with a corner structure of dimension $n+1$, face-structured (Definition 2.5.10), a continuous tangent orientation o of the Mathlib manifold S and $\omega \in \Omega^n_\infty(X)$, $\int_S (d\omega)|_S = \sum_F \int_F \omega|_F$, the sum over the finitely many faces F of S (`finite_setOf_isFace`). Each face is a Mathlib manifold modelled on $[0,\infty)^n$ through a corner structure of dimension n (`hasCornerStructureOfDim_face`), oriented by the induced orientation, outward vector first (`faceOrientation`, `IsInducedOrientation`, unique and continuous); $\omega|_F$ is the pull-back of $\omega|_S$ along $F \hookrightarrow S$ (`restrictForm`). The face sum equals the boundary integral of S (`sum_integral_restrictForm_eq_boundaryIntegral`), and `stokes_formsInftyRestrict` gives the rest. *Book:*

§4.4, Theorem 4.4.7 (preprint 4.28), p. 199 (preprint p. 168), For an oriented compact tame smooth n-dimensional submanifold N whose boundary is a union of faces F, and an sc-differential $(n-1)$-form ω, $\int_N d\omega = \sum_F \int_F \omega$. *Deviation (restated):* The text's N is an oriented compact tame smooth submanifold; here S is a compact sub-M-polyfold with a corner structure of dimension $n+1$, which every submanifold in good position of a tame X has, with a continuous tangent orientation (as Theorem 6.6.18 gives the solution sets, V6.61). The boundary being a union of faces is read as S face-structured (Definition 2.5.10): every point lies on exactly $d_S(x)$ faces. This is a reading, used through "a face is a coordinate face near each of its points" and "a point of degeneracy index 1 lies on exactly one face"; whether it can be dropped is not examined. It is satisfiable: it holds without boundary (`isFaceStructured_of_forall_degIndex_eq_zero`; then the sum is empty and the theorem says $\int_S d\omega = 0$) and for the quadrants (`isFaceStructured_quadrant`). The faces are manifolds in their own right and the sum is over them, each with the induced orientation (outward vector first, Findings, Chapter 4, item 171; gap G.40, closed). The restriction of ω to N, "clearly" a classical form in the text (p. 199), is the pull-back along the inclusion (items 172, 174, 177). The 2021 text writes $\int_N$, the preprint and 2017 $\int_M$ (item 123).

V4.93 (Theorem 4.3.1 (preprint 4.23): stratified regularity does not depend on the chart; `Polyfold.StratRegCond`, `Polyfold.StratRegCond.transport`, `Polyfold.stratRegCond_chart_change`, `Polyfold.IsStratRegularPoint.stratRegCond`, `Polyfold.isStratRegularPoint_iff_forall`, `Polyfold.isStratRegularValue_iff_forall`). `StratRegCond` is the chart condition of `IsStratRegularPoint` (every $\mu \in \mathbb{R}^m$ is v_1 for some $v \in T_yQ$ with v_2 in the minimal subspace of C at y_2); it depends only on the tangent space (`StratRegCond.congr`) and is carried by germs of sc-diffeomorphisms F with $(Fw)_1 = w_1 + c$ that preserve the product partial quadrants (`StratRegCond.transport`). Product-chart transitions have first component $w_1 + a - a'$ (`transition_rnChart_prod_fst`), so their derivatives fix the $\mathbb{R}^m$-component (`scDeriv_fst_eq_of_fst_eq`), and the condition passes from one product chart, presentation, data and level-1 lift to any other (`stratRegCond_chart_change`). `IsStratRegularPoint.stratRegCond`: a stratified regular point satisfies the condition in every product chart $(r \mapsto r - a') \times \varphi'$, every data of Definition 4.2.1, every presentation and every lift; `isStratRegularPoint_iff_forall`, `isStratRegularValue_iff_forall`: "some chart" ⇔ "every chart". *Book:* §4.3, proof of Theorem 4.3.1 (preprint 4.23), p. 194 (preprint p. 166), Regular values of the maps $p_\sigma : A \cap F_\sigma^\circ \to \mathbb{R}^m$, intrinsic to A and its strata. *Deviation (restated):* Stratified regularity is read as in V4.53: regularity of p on the stratum through the point, in a product chart (Findings, Chapter 4, items 102, 108). That it is intrinsic to A and its strata, which the proof of Theorem 4.3.1 (preprint 4.23) takes for granted, is proved by the transport argument of V4.39 (gap G.36, closed).

V4.94 (Lemma 4.1.11 (preprint 4.10): the section $u \mapsto (1-p)(u-t(u))$ is sc-Fredholm; `Polyfold.lemma_4_1_11`, `Polyfold.isConjugateToBasicGerm_qOp`, `Polyfold.exists_splitData`, `Polyfold.isScSubspace_splitW`, `Polyfold.qOp`, `Polyfold.LocalBundle.bundle`, `Polyfold.LocalBundle.sec`, `Polyfold.LocalBundle.isScSmoothSection_sec`, `Polyfold.isFilling2021_of_R_eq_self`). `lemma_4_1_11`: in the setting of Proposition 4.1.9 (an sc⁺-retraction t onto $O \ni 0$, $N = T_0O$ in good position, Y a good sc-complement (`hsplit : E.IsScSplitting N Y`), p the projection onto N along Y), the section $u \mapsto (u, (1-p)(u-t(u)))$ of the local strong bundle

$U \lhd Y \to U$ (`LocalBundle.bundle`, `LocalBundle.sec`, one chart with the identity strong bundle retraction) is sc-Fredholm in the sense of 2021 Definition 3.1.16 (`IsScFredholmSection2021`), on all of U. Regularizing: $u = (1-p)(u - tu) + tu + p(u - tu)$. Germ at a smooth e: the chart centred at e (`LocalBundle.centred`), whose retraction is pointwise the identity, so the principal part fills itself (`isFilling2021_of_R_eq_self`); $s(v) = (1-p)(e - t(v+e))$ is the text's s, and $g - s = (1-p)v$ is conjugated to the basic germ $(a+b, w) \mapsto (b, w)$ (`isConjugateToBasicGerm_qOp`), with the splitting $E \cong (T_0 O \oplus B) \oplus W$, B smooth, $W = Y \cap \ker L$ (`exists_splitData`, `isScSubspace_splitW`). *Book:* §4.1, Lemma 4.1.11 (preprint 4.10), pp. 179–181 (preprint pp. 151–153), The section $u \mapsto (u, \mathbf{f}(u))$, $\mathbf{f}(u) = (1-p)(u - t(u))$, of the local strong bundle $U \lhd Y \to U$ is sc-Fredholm. *Deviation (restated):* The text's "good sc-complement" is the hypothesis `hsplit : E.IsScSplitting N Y` together with the project's `IsGoodComplement`, which keeps only the algebraic splitting; it is a hypothesis of the text, not an added one (satisfiable wherever Proposition 4.1.9 applies, with the closed complement chosen there). No witness of the full data of Proposition 4.1.9 is built. In the proof, B must lie in Y_∞ for $E = (T_0 O \oplus B) \oplus W$ to be an sc-splitting; such B exists, since $L(Y \cap E_\infty) = L(Y)$ (`map_le_map_smooth`). The text gets "$C \cap (T_0 O \oplus B)$ is a partial quadrant" from good position; here it comes from the corner coordinates on $T_0 O \oplus B$, which vanish on W. The slips of the printed proof (Findings, item 57) do not enter. Proposition 4.1.9 (preprint 4.9), its only use, is proved directly (V4.20).

V4.95 (Lemma 4.4.8: the Poincaré lemma for sc-forms (p. 199); `Polyfold.lemma_4_4_8`, `Polyfold.lemma_4_4_8_of_local`, `Polyfold.LocalPoincare`, `Polyfold.localPoincare`, `Polyfold.hasPoincare_lift`, `Polyfold.exists_primitive_of_factor`, `Polyfold.MPolyfold.HasPoincare`, `Polyfold.LocalModel.liftEquiv`, `Polyfold.MPolyfold.formRestrict`, `Polyfold.MPolyfold.extD_formRestrict`, `Polyfold.BundledMPolyfold.openSub`, `Polyfold.BundledMPolyfold.formsInftyResOpen`, `Polyfold.BundledMPolyfold.formsInftyResOpenOpen`, `Polyfold.BundledMPolyfold.FormsInftyEqOn`, `Polyfold.BundledMPolyfold.formsInftyEqOn_of_iff`). `lemma_4_4_8 A x`: there are open $V \subseteq U \ni x$ such that for every k every $a \in \Omega^{k+1}_\infty(U)$ with $da = 0$ has $b \in \Omega^k_\infty(V)$ with $db = a|_V$. U, V are Part I's open sub-M-polyfolds (`restrictOpen`, bundled as `BundledMPolyfold.openSub`); forms restrict to open subsets, commuting with d (`formRestrict`, `extD_formRestrict`, `formsInftyResOpen`, `formsInftyResOpenOpen`), and "$[\omega] = [\tau]$ on V" of (4.7) is `FormsInftyEqOn` (`formsInftyEqOn_of_iff`). The proof is the text's: a chart φ and retraction r with $r(B_\delta \cap C) \subseteq B_\varepsilon$ (`exists_poincareChart`), $V(x) = \varphi^{-1}(B_\delta(\varphi x))$, the M-polyfold O^m as the m-th lift (`LocalModel.liftEquiv`), Lemma 4.5.2 on $B_\delta(\varphi x) \cap C$ about a smooth centre on every level (`LocalPoincare`, `hasPoincare_lift`, `localPoincare`), and $d(j^*\tau) = (r \circ j)^*\omega$ on the direct limits (`exists_primitive_of_factor`, `lemma_4_4_8_of_local`). *Book:* §4.4, Lemma 4.4.8, p. 199 (proof §4.5.3, pp. 202–204) (2021), (New in the 2021 edition.) Every $x \in X$ has open neighbourhoods $V(x) \subseteq U(x)$ such that every closed $[\omega] \in \Omega^k_\infty(U(x))$, $k \geq 1$, has $[\tau] \in \Omega^{k-1}_\infty(V(x))$ with $d[\tau] = [\omega]$ on $V(x)$. The proof applies Lemma 4.5.2 in a chart to $B_\delta(x) \cap C$, working with $C^i \subset E^i$ on the levels. *Deviation (restated):* $U(x)$, $V(x)$ are Part I's open sub-M-polyfolds (nonempty, as they contain x); "$d[\tau] = [\omega]$ on $V(x)$" is $db = a|_V$; the text's degrees $k \geq 1$, $k-1$ are $k+1$, k. In the proof Lemma 4.5.2 is applied to the convex set $B_\delta(\varphi(x)) \cap C$, which need not contain 0, and the remark after Lemma 4.5.2 (a centre $(0, w_0)$) does not cover it; Lemma 4.5.2 is therefore used with the homotopy about any smooth centre, which the same proof gives (V4.96; Findings, Chapter 4, item 177).

V4.96 (Lemma 4.5.2: the local Poincaré lemma on a convex relatively open set (p. 202); `Polyfold.hasPoincare_relOpen_of_zero_mem`, `Polyfold.hasPoincare_relOpen`, `Polyfold.lemma_4_5_2`, `Polyfold.homotopyForm`, `Polyfold.isScSmooth_homotopyForm`, `Polyfold.ambientD_homotopyForm`, `Polyfold.retractD_eq_ambientD`, `Polyfold.isScSmooth_intervalIntegral`, `Polyfold.isSck_intervalIntegral`, `Polyfold.scDeriv_intervalIntegral`). `hasPoincare_relOpen_of_zero_mem`: for a local model M with $M.O$ relatively open in the partial quadrant and convex, $0 \in M.O$, and every k, every $\sigma \in \Omega^{k+1}(O)$ with $d\sigma = 0$ has $\tau \in \Omega^k(O)$ with $d\tau = \iota^*\sigma$ on O^1 (`MPolyfold.HasPoincare`); `hasPoincare_relOpen` is the same about any smooth centre $c \in M.O$. The primitive is the homotopy form $\tau(y)(v) = \int_0^1 t^k \, \omega(c + t(y - c))(y - c, v) \, dt$ (`homotopyForm`, `relOpenForm`), sc-smooth by the sc-smoothness of parametric integrals over $[0, 1]$ (`isScSmooth_intervalIntegral`, `isSck_intervalIntegral`, `scDeriv_intervalIntegral`, by sc-induction through the tangent map) (`isScSmooth_homotopyForm`); `ambientD_homotopyForm` and `lemma_4_5_2` give $d\tau = \omega$ in the chart, with the local d the `deltaFun` of the derivative and the identity $d_r = $ the ambient d on a relatively open set (`retractD_eq_ambientD`). *Book:* Appendix 4.5, Lemma 4.5.2, p. 202 (proof §4.5.3, pp. 202–204) (2021), (New in the 2021 edition.) For U a relatively open convex neighbourhood of 0 in $[0, \infty)^n \oplus W$ and a closed sc-form ω on U of degree $k \geq 1$, there is an sc-form τ on U with $d\tau = \omega$ on U^1; τ is given by the homotopy formula $\tau(y)(v) = \int_0^1 t^{k-1} \omega(ty)(y, v) \, dt$. *Deviation (restated):* The text's degrees $k \geq 1$, $k - 1$ are $k + 1$, k; "$d\tau = \omega$ on U^1" is $d\tau = \iota^*\omega$ for $\iota : U^1 \to U$, and "closed" is $d\omega = 0$. The printed case is the centre $c = 0$ (`hasPoincare_relOpen_of_zero_mem`); any smooth centre is allowed (`hasPoincare_relOpen`), which Lemma 4.4.8 needs (V4.95). No hypothesis is added: the hypotheses (convex, relatively open, a smooth centre) are satisfiable, e.g. $D = C = $ the whole space with $c = 0$ (Findings, Chapter 4, item 177).

What is not verified in this chapter

- pp. 195–208, §4.4 and Appendix 4.5, listed here before 2026-11-13 for the wedge product on $\Omega^*_\infty(X, \partial X)$, which the text calls a differential algebra: verified since with the mapping-cone product of j^* (our choice; V4.91; Findings, item 178). Not formalized, and not defined in the 2021 text: the pull-back of forms on ∂X along a general sc-smooth map of tame M-polyfolds; the pull-back to the faces of a finite-dimensional sub-M-polyfold along a map with $d_S = d_X \circ G$, the case $p_1 : \mathbb{R}^1 \times X \to X$ used for Theorem 6.7.1, is formalized (`Polyfold.MPolyfold.faceAlongData`) (gap G.80, closed). Verified: Definitions 4.4.1–4.4.2, the formula for $d\omega$, d on an M-polyfold, $d \circ d = 0$ and Definition 4.4.6 (V4.80–V4.88, gap G.37 closed; Findings, items 167–172), the wedge product and the graded differential algebra $\Omega^*_\infty(X)$ (V4.87), the pull-back f^* with $f^*d = df^*$, multiplicative (V4.89), forms on ∂X and j^*, with the agreement of $d\tau$ proved (V4.90), the relative complex and $H^*_{dR}(X, \partial X)$ (V4.91, gap G.39 closed; item 173), Theorem 4.4.7 with the sum over the faces of a face-structured N (V4.92, gap G.40 closed;

items 174, 177), the Poincaré Lemmas 4.4.8 and 4.5.2 (V4.95, V4.96; item 177), Definition 4.4.3 (V4.67, V4.68), Lemma 4.4.4 (V4.69, V4.78, V4.79), Proposition 4.4.5 (V4.70–V4.72, V4.76, gap G.38 closed) and Lemma 4.5.1 (V4.74, V4.75, V4.77, gap G.92 closed) (Findings, items 152–164).

- preprint p. 159, the open question after Proposition 4.2.3 (preprint 4.15): a question, not a statement. Its last sentence, for a retract without boundary, is V4.32 with $A = O$; that an sc⁺-retract (O, E, E) is never homeomorphic to the "T" is verified (`Polyfold.Question424.question_4_2_4_scPlus`, 2026-11-12); the question for sc-retracts that are not sc⁺ is open in the text (gap G.34).
- p. 176 (preprint p. 148), Remark 4.1.3 (preprint 4.3): why an example of "$X^1 \neq X$" needs a retraction whose first derivative loses a level (one whose level maps are C^1 gives none, by a Baire category argument): not formalized; the example itself is verified (V4.6; Findings, item 141).
- p. 176, the example that the 2021 edition adds to Remark 4.1.3 (preprint 4.3): the space of Subsection 1.4.4, a line with a half-plane attached, made an M-polyfold with $X_\infty = X$, for which 1_0^1 is not sc-smooth by Theorem 4.1.2 (preprint 4.2), as the local dimension of X is not constant. The argument holds as we read it and is not formalized; the formalized example of V4.6 is a different one (gap G.31; Findings, item 159).
- p. 190 (preprint p. 162), Theorem 4.2.11 (4) (preprint 4.22) for the text's F_σ on all of U: verified for its germ at x, where it is the chart's $\psi^{-1}(G \cap \bigcap F_i)$ ((1) with the faces of U is verified, V4.50; gap G.35 closed; Findings, Chapter 4, item 130).
- Definition 4.1.1 (preprint 4.1): the classical manifold structure beyond the C^∞ property of the transition maps, and the local constancy of the dimension (V4.1, V4.2).
- Definitions 4.2.5, 4.2.10 (preprint 4.16, 4.21) and Remark 4.2.2 (preprint 4.14) in T_aX: verified in charts, with chart independence (V4.38, V4.39, V4.44, V4.45), since T_aX carries no linear structure here (gap G.7 of Chapter 2).

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6. Chapter 5 (HWZ §§5.1–5.5 (preprint §§5.1–5.4)): verified except Theorem 5.27 (preprint; not in the 2021 edition) and the Sobolev sc-structure on general bounded domains

Chapter 5 of Part I (pp. 205–236 (preprint pp. 169–199)) is the Fredholm package for M-polyfolds. §5.1 introduces auxiliary norms on the part $Y_{0,1}$ of biregularity $(0, 1)$ of a strong bundle (Definition 5.1.1 (preprint 5.1), Lemma 5.1.2 (preprint 5.2), Proposition 5.1.3 (preprint 5.3)). §5.2 defines compact solution sets, weakly proper and proper sections (Definition 5.2.1 (preprint 5.4)) and proves that the three agree for sc-Fredholm sections (Theorem 5.2.2 (preprint 5.5), proof in §5.2 (preprint Appendix 5.4.1)), that solution sets are locally compact on every level (Theorem 5.2.3 (preprint 5.6)) and compact in X_∞ (Theorem 5.2.4 (preprint 5.7)), and that the shifted sections are sc-Fredholm (Corollary 5.2.5 (preprint 5.8)). §5.3 builds sc⁺-sections from sc-smooth bump functions (Definition 5.3.1 (preprint 5.9), Proposition 5.3.2 (preprint 5.10), Lemma 5.3.3 (preprint 5.11)), introduces the allowable sections $\Gamma_U^{+,1}(P)$ (Remark 5.3.4 (preprint 5.12)), and proves the perturbation theorems: transversality when $\partial X = \emptyset$ (Theorem 5.3.5 (preprint 5.13), Definition 5.3.7 (preprint 5.15)), general position (Definition 5.3.9 (preprint 5.17), Theorem 5.3.10 (preprint 5.18)), good position relative to the boundary (Theorem 5.3.12 (preprint 5.19)), the Morse-type structure of homotopies (Definition 5.3.13 (preprint 5.20), Theorem 5.3.15 (preprint 5.22) with Lemmas 5.3.16, 5.3.17 (preprint 5.23, 5.24)), perturbation into good position (Theorem 5.25 (preprint; not in the 2021 edition), Lemma 5.26 (preprint; not in the 2021 edition)) and cobordisms (Theorem 5.27 (preprint)). §5.5 (preprint Appendix 5.4.2) discusses sc-smooth bump functions and partitions of unity (Definitions 5.5.1–5.5.14 (preprint 5.28–5.41), Theorems 5.5.5–5.5.16 (preprint 5.32–5.43), Corollary 5.5.17 (preprint 5.44)).

What is verified. **Theorems 5.2.2, 5.2.3, 5.2.4, 5.3.5, 5.3.10, 5.3.12, 5.5.7, 5.5.10 (preprint 5.5, 5.6, 5.7, 5.13, 5.18, 5.19, 5.34, 5.37) and Theorem 5.25 (preprint) hold**, with the readings recorded below, and so do Lemmas 5.1.2, 5.3.3, 5.3.16, 5.3.17 (preprint 5.2, 5.11, 5.23, 5.24) and Lemma 5.26 (preprint), Propositions 5.1.3, 5.3.2, 5.5.3, 5.5.8, 5.5.11, 5.5.15 (preprint 5.3, 5.10, 5.30, 5.35, 5.38, 5.42), Remark 5.3.4 (preprint 5.12), Corollary 5.2.5 (preprint 5.8) (for each fixed i) and Corollary 5.5.17 (preprint 5.44) (with no hypothesis since 2026-11-12, V5.63). Every definition of the chapter is formalized: Definitions 5.1.1 (preprint 5.1; V5.3), 5.2.1 (preprint 5.4; V5.7), 5.3.1 and 5.5.2 (preprint 5.9 and 5.29; V5.16), 5.3.7 (preprint 5.15; V5.23), 5.3.9 (preprint 5.17; V5.31), 5.3.13 (preprint 5.20; V5.41), 5.5.1 (preprint 5.28; V5.47), 5.5.4 (preprint 5.31; V5.50), 5.5.6 (preprint 5.33; V5.51), 5.5.9 (preprint 5.36; V5.54), 5.5.12 (preprint 5.39; V5.59), 5.5.14 (preprint 5.41; V5.60). Corollary 5.2.5 (preprint 5.8) is verified on the shifted strong bundle $P^1 : Y^1 \to X^1$, which the text uses without defining it, and by iteration for every fixed i, with the filler condition on level 1 that its "it is a priori clear" hides (V5.13, V5.14). Theorems 5.2.2–5.2.4 (preprint 5.5–5.7), Corollary 5.2.5 (preprint 5.8) and Lemma 5.3.3 (preprint 5.11) need no tameness (V5.10–V5.14, V5.20). Several places of the text needed more than a reading. (a) *Tameness.* The §3.6 implicit function theorems are proved for sections whose germ data lie in a tame strong bundle chart (Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51); Chapter 3, Findings 190–199); Theorems 5.3.5, 5.3.10, 5.3.12 (preprint 5.13, 5.18, 5.19) and Theorem 5.25 (preprint) take f with tame germ data, the reading of the tame strong bundle with which §5.3 opens, and nothing else (Findings 74–78). Theorem 5.3.10 (preprint 5.18) also assumes X tame, which its proof uses through Theorems 4.2.11 and 4.3.1 (preprint 4.22 and 4.23) (V5.38). (b) *"Generic" and "dense".* The perturbations are $s_0 + \sum \lambda_i t_i$ for λ outside a Lebesgue null set of a small ball (V5.29, V5.38), and density is the bound $\rho(s, s') = \sup_x N(s(x) - s'(x)) \leq \varepsilon$ in the metric space of ρ (V5.21, V5.30, V5.39). (c) *Omitted steps proved.* The spanning sections (V5.24), the compactness that keeps the zeros of small perturbations near $\{0\} \times S_{f+s_0}$ (V5.27), the parametrized solution set as a smooth submanifold in the sense of Definition 4.2.1 (preprint 4.13) (V5.32), $T_yS = \ker f'(y)$ and with it the chart independence of Definition 5.3.9 (preprint 5.17) (V5.33, V5.34), the openness of general position (V5.35), and the properties (i)–(ii) of the proof of Theorem 5.3.10 (preprint 5.18) (V5.36). (d) *Repairs.* Definition 5.3.1 (preprint 5.9) is read with $f(x) \neq 0$ (V5.16), which the weak reading gives at the smooth points of an sc-Banach space (V5.16); "$(\Gamma_U^{+,1}(P), \rho)$ is not complete" is false without a hypothesis and is proved in general, by an example (V5.21); the function σ of Proposition 5.5.3 (preprint 5.30) is misprinted (V5.17); the extension by 0 in Theorem 5.5.7 (preprint 5.34) and both directions of Theorem 5.5.10 (preprint 5.37) need smaller charts (V5.52, V5.55, V5.56); Proposition 5.5.11 (preprint 5.38) is not "an immediate consequence of Theorem 5.37" (V5.58). (e) *Theorem 5.3.12 (preprint 5.19)* is proved without applying the implicit function theorem to the parametrized section at the boundary zeros, where it is in general not in good position (V5.40; Finding 88, on paper). (f) *Lemma 5.3.16 (preprint 5.23)* exposes a hidden hypothesis of Theorem 5.3.15 (preprint 5.22), $dt(x) \neq 0$ (V5.42). Not verified: Theorem 5.27 (preprint), Remark 5.3.6 (preprint 5.14) (a comment), and in the Sobolev remark the sc-structure $W^{1+m,4}(\Omega)$ on bounded domains other than $(0,1) \subset \mathbb{R}^1$ (see the list at the end of this section). A fourth addendum (2026-11-02: the shifted strong bundle and Corollary 5.2.5 (preprint 5.8), the metric ρ, weak against pointwise bump functions; Findings, items 140–164) updated V5.13, V5.14, V5.16, V5.21, V5.22 and V5.30 in place. A fifth addendum (2026-11-08; Findings, items 170–179) covers the Springer 2021 edition: Definition 5.2.1 with the new properties weakly proper$_{<\infty}$ and proper$_{<\infty}$, its a priori dependencies and Theorem 5.2.2 for them (V5.64–V5.66), the compact sub-M-polyfold of Theorem 5.3.5 (V5.67), and Definition 5.3.13 with its new condition (3), which supplies the hidden hypothesis of V5.42 (V5.68–V5.70). A sixth addendum (2026-11-12; Findings, items 183–186) adds Theorem 5.3.15 with both conclusions and no added hypothesis, and its proof (V5.71, V5.72; gap G.48 closed), Remark 5.3.8 for an arbitrary sc-Fredholm arc (V5.73, G.44), the dense form of Theorem 5.3.12 (V5.40, G.45), Remark 5.3.14 (V5.74, G.47), Theorem 5.5.5 (V5.75, G.50), the L^p and Sobolev remark with the sc-structure on $(0,1) \subset \mathbb{R}^1$ only (V5.76, narrower; G.51), Theorems 5.5.13 and 5.5.16 proved (V5.59, V5.62; G.52, G.53), Theorem 5.5.10 for every chart (V5.55, now restated) and Corollary 5.5.17 without hypothesis (V5.63, now restated).

The setting. A section of $P : Y \to X$ is a map $s : X \to Y$ with $P \circ s = \mathrm{id}$; its support is the closure of $\{s \neq 0\}$. X and Y are paracompact Hausdorff spaces, as for every M-polyfold of the project. $Y_{0,1}$ is the M-polyfold $Y[1]$ of §2.6 (preprint §2.5), an auxiliary norm is a function on it, and "$N(y) \leq r$" for a point y of Y includes $y \in Y_{0,1}$ (the text's convention $N(y) = \infty$ off $Y_{0,1}$). The linearization $f'(x) : T_xX \to Y_x$ is read in a strong bundle chart, as the operator $\mathrm{Lin}(z) : T_zO \to K_z$ of Chapter 3, because T_xX carries no linear structure here (gap G.7 of Chapter 2); transversality is chart independent (Chapter 3), and general position is shown chart independent at the zeros near a zero in good position (V5.34). In the perturbation theorems $|\lambda|$ is the maximum norm on $\mathbb{R}^m$, and "the solution set is a manifold" is stated for a non-empty solution set (the empty set is trivially one). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 5".

Update (2026-11-13, Part III wave 1): the reflexive auxiliary norms and the reflexive local compactness of §§5.1–5.2 (Definitions 5.1.4, 5.1.6, 5.1.7, 5.2.6, Lemma 5.1.5, Propositions 5.1.8, 5.2.7, Theorem 5.2.8, Remarks 5.2.9, 5.2.10 (2017: Definitions 5.2–5.4, 5.6, Lemma 5.2, Propositions 5.2, 5.3, Theorem 5.4, Remarks 5.1, 5.2); not in the preprint) and §5.4, the ∂-extension theorem (Definition 5.4.1, Theorem 5.4.2 (2017: Definition 5.11, Theorem 5.9)), are verified (V5.77–V5.89), Proposition 5.2.7, Theorem 5.2.8 and the remark after Proposition 5.2.7 in stronger forms; tools V5.90, V5.91. No hypothesis is added: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ is a witness of all hypotheses of Theorem 5.2.8 at once (`Polyfold.TrivialBundle.theorem_5_2_8_witness`), and the zero section of the all-charts Definition 5.4.1 (`Polyfold.StrongBundle.isBoundaryScPlusSection_fiberZero`).

V5.1 (the charts $\Phi[1]$ for every strong bundle chart; `Polyfold.StrongBundle.oneChartOf`, `Polyfold.StrongBundle.isChart_oneChartOf`, `Polyfold.StrongBundle.oneChartOf_compatible`, `Polyfold.StrongBundle.mem_oneSet_iff_of_isChart`). For every strong bundle chart Φ of P compatible with the atlas (not only those of the atlas), a point y in its domain lies in $Y_{0,1}$ iff the fibre component of $\Phi(y)$ lies in F_1; the map $\Phi[1] : P^{-1}(V) \cap Y_{0,1} \to K[1]$ is a chart of the M-polyfold $Y[1]$, and the charts $\Phi[1]$ of a compatible chart and of a chart of the atlas are compatible. *Book:* §5.1, preprint p. 169, and Lemma 5.1.2 (ii) (preprint 5.2), which quantifies over strong bundle isomorphisms $\Phi : Y|V \to K$. *Hidden hypotheses:* none. *Deviation:* none; Chapter 2 built $\Phi[1]$ only for the charts of the atlas (Findings, item 2).

V5.2 (the fibre $(Y_{0,1})_x$ and its chart norm $|\cdot|_1$; `Polyfold.StrongBundle.oneFiber`, `Polyfold.StrongBundle.toOne`, `Polyfold.StrongBundle.zeroOne`, `Polyfold.StrongBundle.liftOneFiber`, `Polyfold.StrongBundle.liftOneFiber_injective`, `Polyfold.StrongBundle.mem_range_liftOneFiber`, `Polyfold.StrongBundle.isClosed_slice`, `Polyfold.StrongBundle.complete_liftOneFiber`, `Polyfold.StrongBundle.exists_norm_liftOne_le_mul`). The points of the fibre $P^{-1}(x)$ that lie in $Y_{0,1}$ form a linear subspace $(Y_{0,1})_x$; its zero is the point 0_x of $Y[1]$. In a strong bundle chart around x the fibre component of $\Phi[1]$ is an injective linear map $(Y_{0,1})_x \to F_1$ onto the slice $\{h \in F_1 \mid (\varphi(x), h) \in K[1]\}$, which is closed in F_1, so $(Y_{0,1})_x$ is complete for $|h|_1$. The norms $|\cdot|_1$ of two charts around x are comparable, with one constant, on the fibres over a neighbourhood of x. *Book:* §5.1, preprint p. 169 ("the fibers Y_x have the structure of Banach spaces"; $|h|_1$ in Lemma 5.1.2 (preprint 5.2)). *Hidden hypotheses:* none. *Deviation (restated):* the text takes the linear and Banach structure of the fibres of $Y_{0,1}$ for granted; it is constructed, and $|h|_1$ depends on the chart up to equivalence (Findings, item 1).

V5.3 (Definition 5.1.1 (preprint 5.1), auxiliary norms; `Polyfold.StrongBundle.IsCompleteFiberNorm`, `Polyfold.StrongBundle.IsAuxNorm`, `Polyfold.StrongBundle.IsAuxNorm.nonneg`, `Polyfold.StrongBundle.IsAuxNorm.smul`). An auxiliary norm is a continuous $N : Y[1] \to \mathbb{R}$ such that (1) on every fibre $(Y_{0,1})_x$, N is a norm ($N(v) = 0 \Rightarrow v = 0$, $N(tv) = |t|N(v)$, the triangle inequality) for which every Cauchy sequence converges, and (2) if $P(w_k) \to x$ in X and $N(w_k) \to 0$, then $w_k \to 0_x$ in $Y[1]$. It is non-negative and homogeneous. *Book:* Definition 5.1.1 (preprint 5.1), p. 205 (preprint p. 169). *Hidden hypotheses:* none. *Deviation (restated):* "complete norm" is read as a norm on $(Y_{0,1})_x$ with complete normed space; the topology of $Y_{0,1}$ is that of $Y[1]$ (Findings, item 1).

V5.4 (Lemma 5.1.2 (preprint 5.2); `Polyfold.lemma_5_1_2`, `Polyfold.StrongBundle.exists_le_mul_norm_liftOne`, `Polyfold.StrongBundle.IsAuxNorm.exists_norm_liftOne_le`, `Polyfold.StrongBundle.IsAuxNorm.exists_bounds`, `Polyfold.lemma_5_1_2_literal`). Let $N : Y[1] \to \mathbb{R}$ be continuous and a complete norm on every fibre. Then N is an auxiliary norm iff every $x \in X$ lies in the domain of a strong bundle chart Φ with an open neighbourhood $V \subseteq \operatorname{dom}\varphi$ of x and constants $0 < a < A$ such that $a\,N(w) \leq |h|_1 \leq A\,N(w)$ for all $w \in Y_{0,1}$ with $P(w) \in V$, $\Phi(w) = (p, h)$. For an auxiliary norm such bounds hold in *every* strong bundle chart around x; in the text's literal form (`lemma_5_1_2_literal`) the bounds hold for all $w \in Y_{0,1}$ over the whole domain of the chart, obtained by restricting the chart to the neighbourhood (V2.57); the upper bound $g(w) \leq C|h|_1$ holds for every g continuous at 0_x and homogeneous on the fibres near x. *Book:* Lemma 5.1.2 (preprint 5.2), p. 205 (preprint p. 169); proof pp. 205–206 (preprint pp. 169–170). *Hidden hypotheses:* the proof gives the bounds near x only; on all of $P^{-1}(V)$ for the domain V of Φ they hold after restricting the chart, which the text does silently. *Deviation:* none for `lemma_5_1_2_literal`; the first form has V a neighbourhood of x on which the bounds hold; the contradiction argument needs no metrizability (Findings, items 2, 3; Chapter 2, items 42, 44). Formerly "restated", before the literal form was proved.

V5.5 (Proposition 5.1.3 (preprint 5.3), existence; `Polyfold.proposition_5_1_3_exists`, `Polyfold.StrongBundle.gluedNorm`, `Polyfold.StrongBundle.continuous_gluedNorm`, `Polyfold.StrongBundle.isCompleteFiberNorm_gluedNorm`, `Polyfold.StrongBundle.gluedNorm_bounds`). Every strong bundle has an auxiliary norm: for the charts Φ_i of the atlas and a partition of unity (ρ_i) subordinate to their domains, $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)_2|_1$ is continuous on $Y[1]$, a complete norm on every fibre, and comparable near each point with the norm $|\cdot|_1$ of any chart there. *Book:* Proposition 5.1.3 (preprint 5.3), p. 206 (preprint p. 171). *Hidden hypotheses:* none. *Deviation:* none; the text's proof is one sentence (Findings, item 4).

V5.6 (Proposition 5.1.3 (preprint 5.3), comparison; `Polyfold.proposition_5_1_3_compare`). For auxiliary norms N_0, N_1 there is a continuous $f : X \to (0, \infty)$ with $f(x)N_0(w) \leq N_1(w) \leq f(x)^{-1}N_0(w)$ for every $w \in Y_{0,1}$ over x. *Book:* Proposition 5.1.3 (preprint 5.3), p. 206 (preprint p. 171). *Hidden hypotheses:* none. *Deviation:* none; the statement's $N_0(h)$ is $N_0(w)$ (Findings, item 5).

V5.7 (Definition 5.2.1 (preprint 5.4); `Polyfold.StrongBundle.AuxLe`, `Polyfold.StrongBundle.IsWeaklyProper`, `Polyfold.StrongBundle.IsProper`, `Polyfold.StrongBundle.IsProper.isWeaklyProper`, `Polyfold.StrongBundle.IsWeaklyProper.isCompact_solutionSet`). A section f has a compact solution set if $f^{-1}(0)$ is compact in X; it is weakly proper if moreover for every auxiliary norm N there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc⁺-section s with support in U and $N(s(y)) \leq 1$ for all y; it is proper if moreover for every N there is an open $U \supseteq f^{-1}(0)$ with $\mathrm{cl}_X\{x \in U \mid N(f(x)) \leq 1\}$ compact, where $N(f(x)) \leq 1$ includes $f(x) \in Y_{0,1}$. A continuous proper section is weakly proper, and a weakly proper section has a compact solution set. *Book:* Definition 5.2.1 (preprint 5.4), p. 209 (preprint p. 171). *Hidden hypotheses:* "proper implies weakly proper" needs f continuous (so that $\{f = s\}$ is closed). *Deviation:* none (Findings, item 26).

V5.8 (Theorem 5.2.2 (preprint 5.5), the local form for basic germs; `Polyfold.IsSc0ContractionGerm.exists_isCompact_bdd`, `Polyfold.IsBasicGerm.exists_isCompact_bdd`). Let f be a basic germ on $\widetilde{C} \oplus W$ with $P \circ f(a, w) = w - B(a, w)$, $\widetilde{C}$ closed. There is an open $O \ni 0$, independent of c, such that for every c the points x of $U_0 \cap O$ with $f(x) \in \mathbb{R}^N \oplus W_1$ and $|f(x)|_1 \leq c$ lie in a compact subset of E_0, namely the image of $\{(a, z) \mid a \in \widetilde{C}, |a|_0 \leq \rho, z \in K_c, |z|_0 \leq r/8\}$ under $(a, z) \mapsto (a, \delta(a, z))$, with K_c a compact set containing the level-1 ball of radius c and δ the parametrized fixed point of $w \mapsto B(a, w) - z$. *Book:* §5.2 (preprint Appendix 5.4.1), preprint pp. 189–192, (82)–(90). *Hidden hypotheses:* $\widetilde{C}$ closed, as in Theorem 3.5.1 (preprint 3.40). *Deviation (restated):* only the level-0 contraction is used, and the neighbourhood does not depend on the bound (Findings, items 23, 25).

V5.9 (Theorem 5.2.2 (preprint 5.5), the local step; `Polyfold.IsStrongBundleMap.exists_fiber_bound_one`, `Polyfold.exists_nbhd_isCompact_closure_bdd`). A strong bundle map Ψ has a uniform $(0,1)$ fibre bound near each point, $|\Psi[1](u,k)_2|_1 \leq M|k|_1$ for $|u-u_0|_0 < \delta$. If (f,x_0) is an sc-Fredholm germ, then in the strong bundle chart Φ of the germ there is an open $V \ni x_0$ such that for every C the closure in X of $\{x \in V \mid \Phi(f(x))_2 \in F_1, |\Phi(f(x))_2|_1 \leq C\}$ is compact. *Book:* §5.2 (preprint Appendix 5.4.1), preprint pp. 189–192 ("without loss of generality $f = h + t$", "it suffices to find in local coordinates"). *Hidden hypotheses:* the uniform fibre bound of the conjugating isomorphism. *Deviation (stronger):* x_0 need not be a zero; subtracting the sc$^+$-part before the conjugation replaces the text's t and its estimates (83), (85), and the closure is pulled back to X because the retract is the fixed point set of the retraction (Findings, items 23, 24).

V5.10 (Theorem 5.2.2 (preprint 5.5); `Polyfold.theorem_5_2_2_preprint`, `Polyfold.theorem_5_2_2_iff`). Let f be an sc-Fredholm section of a strong bundle over an M-polyfold X. If f has a compact solution set, f is proper; and for f the properties proper, weakly proper and having a compact solution set are equivalent. *Book:* Theorem 5.2.2 (preprint 5.5), p. 210 (preprint p. 172); proof §5.2 (preprint Appendix 5.4.1), preprint pp. 189–192. *Hidden hypotheses:* none. *Deviation (stronger):* X need not be tame (the appendix states the theorem without tameness); the finite cover of $f^{-1}(0)$ and the closure of a finite union are made explicit (Findings, items 24, 25).

V5.11 (Theorem 5.2.3 (preprint 5.6); `Polyfold.theorem_5_2_3`). Let f be an sc-Fredholm section and $f(x) = 0$. There are open neighbourhoods $x \in U(i+1) \subseteq U(i)$ of x in X such that for every i the closure in X of $\{y \in U(i) \mid f(y) = 0\}$ is the image of a compact subset of X_i (with its level topology). *Book:* Theorem 5.2.3 (preprint 5.6), p. 213 (preprint p. 172). *Hidden hypotheses:* none. *Deviation:* none; it is Theorem 3.1.13 (preprint 3.12) at a zero, which is a smooth point where (f,x) is an sc-Fredholm germ.

V5.12 (Theorem 5.2.4 (preprint 5.7); `Polyfold.theorem_5_2_4`, `Polyfold.MPolyfold.tendsto_smoothLevel_iff`, `Polyfold.MPolyfold.smoothLevel_topology_eq`, `Polyfold.MPolyfold.metrizableSpace_smoothLevel`). X_∞ carries the initial topology of the inclusions $X_\infty \to X_m$; it is metrizable, and a sequence converges in it iff it converges on every level. If f is an sc-Fredholm section whose solution set S is compact in X, then $S \subseteq X_\infty$ and S is a compact subset of X_∞. *Book:* Theorem 5.2.4 (preprint 5.7), p. 213 (preprint p. 172). *Hidden hypotheses:* the topology of X_∞, which the text does not define. *Deviation (stronger):* X need not be tame. The convergence on every level of a sequence of zeros is Corollary 3.1.15 (preprint 3.13) at its limit, not Theorem 5.2.3 (preprint 5.6) as the text says (Findings, item 20).

V5.13 (Corollary 5.2.5 (preprint 5.8); `Polyfold.corollary_5_2_5`, `Polyfold.corollary_5_2_5_two`, `Polyfold.StrongBundle.solutionSet_sectionShift`, `Polyfold.isCompact_solutionSet_level`). Let f be an sc-Fredholm section of $P : Y \to X$ (Definition 3.1.16 (preprint 3.14), in the text's germ reading of Definition 3.1.11 (preprint 3.11)) whose solution set is compact. Then $f^1 = f|_{X_1}$ is an sc-Fredholm section of the shifted strong bundle $P^1 : Y^1 \to X^1$ (V5.14) and its solution set is compact in X^1; $f^2 = (f^1)^1$ is an sc-Fredholm section of $P^2 = (P^1)^1$ with compact solution set; and for every i the zeros of f in X_i form a compact subset of X_i. *Book:* Corollary 5.2.5 (preprint 5.8), p. 213 (preprint p. 173) (over a tame X, for every i). *Hidden hypotheses:* the bundle P^i, which the text does not define, is the iterated shift $P^i = (P^{i-1})^1$. *Deviation:* none; the first statement holds for every strong bundle, so every fixed i follows by iterating it ($i = 2$ is stated); one statement for a family indexed by i is not formalized, and the base $(X^1)^1$ of P^2 has the points of X_2, and its topology is that of X_2 (V2.93; gap G.6, closed). Tameness is not used; the conclusion is Definition 3.1.16 (preprint 3.14) for f^i, not tame germ data, which would need tame charts of P^1 (Findings, items 140–146). Formerly "narrower": only the compactness half was verified (gap G.41, now closed for each fixed i).

V5.14 (Corollary 5.2.5 (preprint 5.8): the shifted strong bundle, and f^1 sc-Fredholm on it; `Polyfold.StrongBundle.shift`, `Polyfold.StrongBundleChart.shift`, `Polyfold.StrongBundleChart.Compatible.shift`, `Polyfold.StrongBundle.mem_shift_doubleLevel_iff`, `Polyfold.StrongBundle.sectionShift`, `Polyfold.StrongBundle.regularizing_sectionShift`, `Polyfold.ScNbhdGerm.shiftParam`, `Polyfold.IsSc0ContractionGerm.shift`, `Polyfold.IsBasicGerm.shift`, `Polyfold.IsConjugateToBasicGerm.shift`, `Polyfold.IsConjugateToBasicGerm.shift_germ`, `Polyfold.IsFillingGerm.mem_range_toBase_of_fillDefect`, `Polyfold.IsFillingGerm.germ_shift`, `Polyfold.isScFredholmGerm_sectionShift`, `Polyfold.corollary_5_2_5_isScFredholmSection`, `Polyfold.corollary_5_2_5_chart`). *The shifted strong bundle.* $P^1 : Y^1 \to X^1$ is a strong bundle over the M-polyfold X^1 (Definition 2.6.5 (preprint 2.51)): its total space is $Y_{1,1}$, the level 1 of the M-polyfold $Y[0]$ with its level topology; P^1 is the restriction of P; the fibre over x is $P^{-1}(x) \cap Y_{1,1}$, a linear subspace, carrying the topology of F_1; and the atlas consists of the index raises Φ^1 of the strong bundle charts Φ of P (base chart the level chart φ^1, fibre model F^1, retraction R restricted to $U^1 \lhd F^1$), compatible charts having compatible index raises. Its double levels are $(Y^1)_{m,k} = Y_{m+1,k+1}$. *$f^1$ is sc-Fredholm.* For every sc-Fredholm section f (no compactness needed), $f^1 = f|_{X_1}$ is an sc-Fredholm section of P^1: it is sc-smooth, regularizing, and at every smooth point of X^1 it has the data of Definition 3.1.11 (preprint 3.11) in the index raise Φ^1 of the chart Φ of a datum (g, s) of f there. The principal part of f^1 is the level-1 map of that of f; g^1 is a filling germ of it, including condition (3) of Definition 3.1.4 (preprint 3.4) on level 1, which comes from a regularity statement at the centre of every datum of Definition 3.1.11 (preprint 3.11): $v \in \ker Dr(0)$ and $Dk(0)v \in F_m$ imply $v \in E_m$, k the fill defect. *In a chart.* The level shift f^1 of a basic germ f on $\tilde{C} \oplus W$, $\tilde{C}$ a partial quadrant, is a basic germ on $\tilde{C} \oplus W^1$ with B^1 in place of B; if a principal part k is conjugated to a basic germ h by Ψ, then k^1 is conjugated to h^1 by the index raise Ψ^1. Hence for an sc-Fredholm section with compact solution set: the zeros in X_1 form a compact subset of X_1, and at every smooth point x, in the chart Φ centred at x of the germ (f, x) with filling germ g and sc$^+$-germ s (the text's germ reading of Definition 3.1.11 (preprint 3.11)), the level-1 germ $(g - s)^1$ is conjugated to a basic germ (by the germ form of the index raise, which uses $g - s$ only near 0) and s^1 is an sc$^+$-germ on the shifted germ of s. *Book:* Corollary 5.2.5 (preprint 5.8), p. 213 (preprint p. 173) ("it is a priori clear that f^i is sc-Fredholm", as a section of $P^i : Y^i \to X^i$). *Hidden hypotheses:* the bundle $Y^1 \to X^1$, which the text does not define ($Y[i]$ is defined for $i = 0, 1$ only): $Y^1 = Y_{1,1}$, the convention of E^1, forced by "f^i is obtained by raising the index" (an sc-smooth section maps X_1 into $Y_{1,1}$, in general not into $Y_{1,2}$); $\tilde{C}$ a partial quadrant. *Deviation:* none; "it is a priori clear" hides condition (3) of the filling on level 1, which does not follow from level 0 alone (an sc-operator can be bijective on level 0 and not onto on level 1); it is proved for the data of Definition 3.1.11 (preprint 3.11) by reducing to $g(0) = 0$ with Lemma 3.1.20 (preprint 3.17) (subtracting the section $z \mapsto \Gamma(z, -g(0))$, which leaves the fill defect unchanged) and applying Proposition 3.6.5 (preprint 3.52) in sc form. No hypothesis beyond the text's is added; tameness is not used, and the tame form (germ data in tame charts of P^1) is not stated (Findings, items 22, 140–146). Formerly "narrower": the bundle was not built and the filler on level 1 not verified (gap G.41, now closed for each fixed i). In the text's germ reading of Definition 3.1.11 (preprint 3.11) (phase 2; Findings, Chapter 3, item 333) s^1 is an sc⁺-germ on the shifted germ; in the whole-chart reading it was sc⁺ on all of U^1.

V5.15 (ℝ as an M-polyfold; sc-smooth functions; `Polyfold.realPolyfold`, `Polyfold.realChart`, `Polyfold.isChart_realChart`, `Polyfold.MPolyfold.isScSmoothMap_of_local`, `Polyfold.MPolyfold.IsScSmoothMap.real_chart`, `Polyfold.isScSmoothMap_real_of_local`, `Polyfold.isScSmoothMap_real_const`, `Polyfold.isScSmoothMap_real_of_contDiff`, `Polyfold.MPolyfold.IsScSmoothMap.comp_contDiff`). ℝ with the constant sc-structure and the single chart id is an M-polyfold, and an sc-smooth function on X is an sc-smooth map $X \to \mathbb{R}$ (Definition 2.3.8 (preprint 2.14)). Sc-smoothness of a map between M-polyfolds can be checked in one pair of charts around each point (continuity included); an sc-smooth function read in a chart is sc-smooth on the retract; constants are sc-smooth, and $\sigma \circ f$ is sc-smooth for $\sigma : \mathbb{R} \to \mathbb{R}$ of class C^∞. *Book:* Definitions 5.3.1, 5.5.2, 5.5.6 (preprint 5.9, 5.29, 5.33) ("sc-smooth map $f : X \to \mathbb{R}$"). *Hidden hypotheses:* none. *Deviation (not in the book):* the M-polyfold ℝ is presupposed by the text (Findings, item 40).

V5.16 (Definitions 5.3.1 and 5.5.2 (preprint 5.9 and 5.29); `Polyfold.MPolyfold.AdmitsBumpFunctions`, `Polyfold.MPolyfold.AdmitsBumpFunctionsWeak`, `Polyfold.MPolyfold.AdmitsBumpFunctions.weak`, `Polyfold.MPolyfold.HasBumpAt`, `Polyfold.MPolyfold.admitsBumpFunctionsWeak_of_dense`, `Polyfold.MPolyfold.AdmitsBumpFunctionsWeak.hasBumpAt_of_mem_smooth`, `Polyfold.MPolyfold.admitsBumpFunctions_of_hilbert`). X admits sc-smooth bump functions if for every x and every open $U \ni x$ there is an sc-smooth $f : X \to \mathbb{R}$ with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$. Definition 5.5.2 (preprint 5.29) as written ("f not identically zero") is a separate notion, implied by the first. A point x is a *bump point* if every open $U \ni x$ carries an sc-smooth f with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$: the first notion says that every point is a bump point, and the second follows if the bump points are dense. On an sc-Banach space E (as an M-polyfold with the identity chart) the second notion makes every smooth point a bump point; on M-polyfolds built on sc-Hilbert spaces the first notion holds. *Book:* Definition 5.3.1 (preprint 5.9), p. 216 (preprint p. 173); Definition 5.5.2 (preprint 5.29), p. 229 (preprint p. 192). *Hidden hypotheses:* none. *Deviation (restated):* "$f \neq 0$" is read as $f(x) \neq 0$, the reading Proposition 5.5.3's (preprint 5.30) proof uses. Whether the weak reading implies it is open only at non-smooth points: at a smooth point of an sc-Banach space a bump that is non-zero near x is translated to x by a smooth vector, and translation by a non-smooth vector does not preserve the levels; that the weak reading is exactly the density of the bump points is argued, one direction formalized (Findings, items 41, 160–163).

V5.17 (Propositions 5.3.2 and 5.5.3 (preprint 5.10 and 5.30); `Polyfold.proposition_5_5_3`, `Polyfold.proposition_5_3_2`). If X admits sc-smooth bump functions, then for every x and open $U \ni x$ there is an sc-smooth $f : X \to [0,1]$ with $f(x) = 1$, $\operatorname{supp} f \subseteq U$, and $f \equiv 1$ on a neighbourhood of x. *Book:* Proposition 5.3.2 (preprint 5.10), p. 216 (preprint p. 173); Proposition 5.5.3 (preprint 5.30), p. 230 (preprint p. 193). *Hidden hypotheses:* Definition 5.3.1 (preprint 5.9) in the reading of V5.16. *Deviation:* none; the text's σ ("0 for $s \leq 1$ and 1 for $s \geq 1$") is misprinted and $g(x) = 1$ is not given; the proof uses $f = \tau(3g/g(x) - 1)$ with a smooth transition τ (Findings, item 42).

V5.18 (the sc-calculus of sections: $\Gamma^+(P)$, supports, $f \cdot s$ and sums; `Polyfold.sectionSupport`, `Polyfold.sectionSmul`, `Polyfold.sectionSum`, `Polyfold.StrongBundle.scPlusSections`, `Polyfold.StrongBundle.IsScPlusSection.mem_oneSet`, `Polyfold.StrongBundle.isScPlusSection_fiberZero`, `Polyfold.StrongBundle.IsScPlusSection.smul_left`, `Polyfold.StrongBundle.IsScPlusSection.add`, `Polyfold.StrongBundle.isScPlusSection_sectionSum`). $\Gamma^+(P)$ is the set of sc$^+$-sections (Definition 2.6.6 (preprint 2.52)); their values lie in $Y_{0,1}$. The zero section is an sc$^+$-section; for an sc-smooth function f and an sc$^+$-section s, $f \cdot s$ (fibrewise) is an sc$^+$-section; sums and finite sums of sc$^+$-sections are sc$^+$-sections. *Book:* §5.3, preprint p. 173 ($\Gamma^+(P)$); used in the proof of Lemma 5.3.3 (preprint 5.11) ("$s(y) = \beta(y)t(y)$ is a sc$^+$-section") and in the finite combinations $\sum \lambda_i s_i$ of Theorems 5.3.5, 5.3.10 (preprint 5.13, 5.18). *Hidden hypotheses:* none. *Deviation (not in the book):* the text uses these rules without stating them; in a chart $(f \cdot s)^\wedge = (f \circ \varphi^{-1}) \cdot \hat{s}$ and the product rule of the sc-calculus applies (Findings, item 45).

V5.19 (local sc$^+$-sections cut off and extended by 0; `Polyfold.StrongBundle.cutoffSection`, `Polyfold.StrongBundle.isScPlusSection_cutoffSection`). Let Φ be a strong bundle chart of the atlas, $\tau : O \to F_1$ such that $z \mapsto (z, \tau(z))$ is sc-smooth $O \to K[1]$, and $\beta : X \to \mathbb{R}$ sc-smooth with $\operatorname{supp} \beta \subseteq \operatorname{dom} \varphi$. Then $y \mapsto \Phi^{-1}(\varphi(y), \beta(y)\tau(\varphi(y)))$ on $\operatorname{dom} \varphi$, 0_y outside, is an sc$^+$-section. *Book:* proof of Lemma 5.3.3 (preprint 5.11), preprint p. 174 ("if we construct s with support in Q' we can extend it by 0"). *Hidden hypotheses:* none. *Deviation (not in the book):* the extension by 0 is proved (Findings, item 46 (c)).

V5.20 (Lemma 5.3.3 (preprint 5.11); `Polyfold.lemma_5_3_3`). Let X admit sc-smooth bump functions and N be an auxiliary norm. For a smooth point x, a smooth point e of Y with $P(e) = x$, $\varepsilon > 0$ and an open $U \ni x$, there is an sc$^+$-section s with $s(x) = e$, $\operatorname{supp} s \subseteq U$ and $N(s(y)) < N(e) + \varepsilon$ **for all** y. *Book:* Lemma 5.3.3 (preprint 5.11), p. 217 (preprint p. 173); proof preprint p. 174. *Hidden hypotheses:* Definition 5.3.1 (preprint 5.9) in the reading of V5.16. *Deviation (stronger):* the bound holds at every point (the text's bound at x is trivial; the proof gives the uniform one, which §5.3 needs); tameness of P and X, normality of X and a chart centred at x are not needed (Findings, item 46).

V5.21 ($\Gamma_U^{+,1}(P)$ and the metric ρ; `Polyfold.StrongBundle.allowableSections`, `Polyfold.`

`StrongBundle.sectionDistAt`, `Polyfold.StrongBundle.allowableDist`, `Polyfold.StrongBundle.allowableMetric`, `Polyfold.StrongBundle.allowableDist_le_iff`, `Polyfold.TrivialBundle.not_completeSpace_allowableSections`, `Polyfold.StrongBundle.completeSpace_allowableSections_empty`, `Polyfold.TrivialBundle.completeSpace_allowableSections_point`). For an auxiliary norm N and an open U, $\Gamma_U^{+,1}(P)$ is the set of sc$^+$-sections s with $\operatorname{supp} s \subseteq U$ and $N(s(x)) \leq 1$ for all x (the allowable sections). For two sc$^+$-sections $N(s(x)-s'(x))$ is the norm of the difference in the fibre $(Y_{0,1})_x$. The uniform distance $\rho(s,s') = \sup_x N(s(x)-s'(x))$ is finite ($\rho \leq 2$) and, for an auxiliary norm, a metric on $\Gamma_U^{+,1}(P)$; $\rho(s,s') \leq \varepsilon$ iff $N(s(x)-s'(x)) \leq \varepsilon$ for all x. For the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ with $N(x,v) = |v|$ and $U = \mathbb{R}$ the metric space is not complete; for $U = \emptyset$ (any bundle) and over a point with fibre $\mathbb{R}^k$ (any U) it is complete. *Book:* §5.3, preprint p. 174 ("a metric space … not complete"). *Hidden hypotheses:* none for the metric. *Deviation (restated):* the text's "not complete" is false without a hypothesis (the cases $U = \emptyset$ and a point base) and is read as "not complete in general", proved by the example: $s_j(x) = (x, \sqrt{x^2 + 1/(j+1)^2}/(2\sqrt{1+x^2}))$ is a ρ-Cauchy sequence of allowable sections whose pointwise limit has the fibre part $|x|/(2\sqrt{1+x^2})$, not differentiable at 0, while the fibre part of an sc$^+$-section of the trivial bundle is C^1. Non-completeness in the text's setting (U containing a non-isolated point over which $Y_{0,1} \neq 0$) is expected by the same mechanism and not formalized (Findings, items 47, 150–153).

V5.22 (Remark 5.3.4 (preprint 5.12); `Polyfold.remark_5_3_4`, `Polyfold.StrongBundle.IsAuxNorm.auxLe_of_sectionAdd`, `Polyfold.StrongBundle.IsInAllowableCompletion`, `Polyfold.remark_5_3_4_completion`, `Polyfold.remark_5_3_4_of_isScFredholmSection`). Let f have its solution set inside the open U with $\operatorname{cl}\{x \in U \mid N(f(x)) \leq 1\}$ compact. If a section s vanishes outside U and has $N(s(x)) \leq 1$ everywhere — as the uniform limits of allowable sections do — then the solutions of $f + s = 0$ lie in $\{x \in U \mid N(f(x)) \leq 1\}$, and their closure is compact. As stated (`remark_5_3_4_completion`): with the points of the completion represented by sections s with values in $Y_{0,1}$ that are uniform N-limits of allowable sections (`IsInAllowableCompletion`), for f continuous the solution set of $f + s = 0$ is closed, hence compact; for f sc-Fredholm with compact solution set, with the U of Theorem 5.2.2 (preprint 5.5) (`remark_5_3_4_of_isScFredholmSection`). *Book:* Remark 5.3.4 (preprint 5.12), p. 217 (preprint p. 174). *Hidden hypotheses:* none. *Deviation (restated):* the completion of $\Gamma_U^{+,1}$ is represented by sections, uniform N-limits of allowable sections; the metric space itself is V5.21, and that every ρ-Cauchy sequence has such a limit section is not formalized (Findings, items 131, 153). The closedness of the solution set, which the remark does not discuss, is proved by a chart argument: Lemma 5.1.2's (preprint 5.2) bound turns $N(s - s') \leq \varepsilon$ into a level-0 bound in a chart, and $f + s'$ is continuous for allowable s'; "adapted to N" is read as the properness of V5.7 (Findings, items 27, 130, 131).

V5.23 (Definition 5.3.7 (preprint 5.15); `Polyfold.StrongBundle.IsTransversal`, `Polyfold.StrongBundle.IsLinSurjAt.linAt_surjective`). A section g is transversal to the zero section if at every zero x the linearization is onto, in a strong bundle chart around x; it is then onto in every such chart. *Book:* Definition 5.3.7 (preprint 5.15), p. 218 (preprint p. 175). *Hidden hypotheses:*

none. *Deviation (restated):* $g'(x) : T_xX \to Y_x$ is read in charts (gap G.7 of Chapter 2) (Findings, item 70).

V5.24 (spanning sections at a zero; `Polyfold.exists_fin_mem_of_dense_of_closedComplemented`, `Polyfold.exists_spanning_sections`). If R is a closed, complemented subspace of finite codimension of a Banach space K and D a dense subspace, finitely many vectors of D span a complement of R. Hence, if X admits sc-smooth bump functions, N is an auxiliary norm, g is sc-Fredholm and $g(y) = 0$, then for every open $U \ni y$ there are finitely many allowable sc$^+$-sections $s_1, \dots, s_k$ supported in U and a strong bundle chart centred at y in which $\mathrm{Lin}(0)(T_0O) + \mathrm{span}\{\hat{s}_i(0)\} = K_0$. *Book:* proof of Theorem 5.3.5 (preprint 5.13), p. 218 (preprint p. 175) ("we find finitely many allowable sc$^+$-sections such that the range of $(f + s_0)'(x)$ together with the sections span Y_x"). *Hidden hypotheses:* none. *Deviation (not in the book):* the text does not say how; the density of the smooth vectors in K_0 is the point (Findings, item 66).

V5.25 (the parametrized section and its slices; `Polyfold.paramSlice`, `Polyfold.isZeroAt_paramSection_iff`, `Polyfold.isScFredholmSection_paramSlice`, `Polyfold.isScPlusSection_paramSlice`, `Polyfold.sectionSupport_paramSlice_subset`, `Polyfold.auxLe_paramSlice`). For a section g and sections $t_1, \dots, t_m$, the slice $g + \sum \lambda_i t_i$ of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ (Theorem 3.1.21 (preprint 3.18)) is a section of P, and (λ, y) is a zero of F iff y is a zero of the slice. For g sc-Fredholm and t_i sc$^+$ the slices are sc-Fredholm; slices of sc$^+$-sections are sc$^+$ with support in the union of the supports; and if $N(s_0) \leq 1 - \varepsilon$, $N(t_i) \leq 1$ and $m|\lambda| \leq \varepsilon$, then $N(s_0 + \sum \lambda_i t_i) \leq 1$. *Book:* proof of Theorem 5.3.5 (preprint 5.13), preprint p. 176 ("there is a number $\delta_0 > 0$ such that $|\lambda| < \delta_0$ implies $N(\sum \lambda_j t_j(y)) < \varepsilon/2$"). *Hidden hypotheses:* none. *Deviation (not in the book):* bookkeeping the text leaves implicit.

V5.26 ("regular value $\Leftrightarrow$ transversal" and transversality of F at $(0, y)$; `Polyfold.surjective_of_forall_exists_ker_fst`, `Polyfold.forall_exists_ker_fst_of_surjective`, `Polyfold.isLinSurjAt_paramSlice_of_ker_fst`, `Polyfold.isLinSurjAt_paramSection_zero_of_spanning`, `Polyfold.isLinSurjAt_paramSection_zero_of_comp`). For $M(a, u) = Lu + \sum a_i e_i$ onto, the projection $\ker M \to \mathbb{R}^m$ is onto iff L is onto. Consequently, if the linearization of F at a zero (λ, y) is onto and the kernel of $\mathrm{Lin}_F(0)$ in the product chart $(r \mapsto r - \lambda) \times \Phi$ projects onto $\mathbb{R}^m$, the slice $g + \sum \lambda_i t_i$ is transversal at y. Conversely, if the range of $\mathrm{Lin}_g(0)$ and the $\hat{t}_i(0)$ span K_0 in a chart centred at the zero y of g, F is transversal at $(0, y)$, and adding sections keeps this. *Book:* proof of Theorem 5.3.5 (preprint 5.13), pp. 218–219 (preprint pp. 175–176) ("the linearization at $(0, x)$ is surjective"; "it is easily verified that $f + s_0 + s(\lambda_0, \cdot)$ is transversal"). *Hidden hypotheses:* none. *Deviation (not in the book):* only "regular value $\Rightarrow$ transversal" is needed and proved for slices; the text's "$\Leftrightarrow$" for $\tilde{S}$ is not formalized (Findings, items 64, 67).

V5.27 (the openness steps and the omitted compactness step; `Polyfold.isWithoutBoundary_prod_rn`, `Polyfold.isClosed_solutionSet`, `Polyfold.exists_isOpen_isLinSurjAt_zero_near`, `Polyfold.exists_isOpen_goodPosition_of_isLinSurjAt`, `Polyfold.exists_pos_forall_mem_of_isCompact`). If $\partial X = \emptyset$ then $\partial(\mathbb{R}^m \times X) = \emptyset$; the solution set of an sc-smooth section is closed. For $\partial X = \emptyset$ and g with tame germ data: if F is transversal at $(0, x)$, it is transversal at $(0, y)$ for the zeros y of g near x; if F is transversal at every $(0, y)$, $y \in S_g$, there is an open $V \supseteq \{0\} \times S_g$ all of whose zeros of F are in good position in a tame chart. If $Z \subseteq \mathbb{R}^m \times X$ is compact and the open V contains $Z \cap \{\lambda = 0\}$, then V contains $Z \cap \{|\lambda| < \delta\}$ for some $\delta > 0$. *Book:* proof of Theorem 5.3.5 (preprint 5.13), pp. 218–219 (preprint pp. 175–176) (“in view of the interior case of the implicit function theorem, Theorem 3.19”; “$U \subset \{|\lambda| < \delta_0\} \oplus X$”). *Hidden hypotheses:* tame germ data, the reading of the tame strong bundle of §5.3 (Findings, item 75). *Deviation (not in the book):* the passage to $|\lambda| < \delta_0$ is the omitted step, proved with Remark 5.3.4 (preprint 5.12) (Findings, items 63, 68).

V5.28 (the Sard step; `Polyfold.exists_null_isLinSurjAt_paramSlice_near`). Let g have tame germ data, t_i be sc$^+$-sections, and z_0 a zero of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ in good position in a tame chart. There are an open $W \ni z_0$ and a Lebesgue null set $CV \subseteq \mathbb{R}^m$ such that for every zero $(\lambda, y) \in W$ of F with $\lambda \notin CV$ the slice $g + \sum \lambda_i t_i$ is transversal at y. *Book:* proof of Theorem 5.3.5 (preprint 5.13), preprint p. 176 (“taking a regular value λ_0 for the projection $\tilde{S} \to \mathbb{R}^m$”). *Hidden hypotheses:* none. *Deviation (not in the book):* $\tilde{S}$ is not built as a manifold; Sard’s theorem is applied in its stratified form on the model of the regular tangent germ chart of Theorem 3.6.8 (2) (preprint 3.54), which avoids showing that the points over the zeros are interior (Findings, items 64, 65).

V5.29 (Theorem 5.3.5 (preprint 5.13), generic form; `Polyfold.theorem_5_3_5_generic`, `Polyfold.StrongBundle.SolutionSetIsManifold`). Let $\partial X = \emptyset$, X admit sc-smooth bump functions, f be an sc-Fredholm section with tame germ data and compact solution set, and N an auxiliary norm. There is an open $U \supseteq S_f$ such that (1) S_{f+s} is compact for every $s \in \Gamma_U^{+,1}(P)$, and (2) for every sc$^+$-section s_0 with $\operatorname{supp} s_0 \subseteq U$ and $N(s_0(x)) \leq 1 - \varepsilon$ ($\varepsilon > 0$) there are allowable $t_1, \dots, t_m \in \Gamma_U^{+,1}(P)$, $\delta > 0$ and a Lebesgue null set $CV \subseteq \mathbb{R}^m$ such that for every λ with $|\lambda| < \delta$, $\lambda \notin CV$: $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable, $f + s_\lambda$ is transversal, S_{f+s_λ} is compact and, if non-empty, a sub-M-polyfold whose induced structure is tame and that of a manifold without boundary of locally constant dimension, with regular tangent germ data at every zero of dimension the Fredholm index of $(f + s_\lambda)'(y)$. *Book:* Theorem 5.3.5 (preprint 5.13), pp. 217–218 (preprint pp. 174–175); proof pp. 218–219 (preprint pp. 175–176). *Hidden hypotheses:* f with tame germ data (Definition 3.6.4 (preprint 3.51)), the reading of §5.3’s tame strong bundle; the statement’s “$N(f(x)) \leq 1$” is $N(s(x)) \leq 1$ and “(f, x) in good position” is $(f + s, x)$. *Deviation (restated):* the dense set of the text is described as the parameters off a null set of a ball (Findings, items 60, 62, 74, 75).

V5.30 (Theorem 5.3.5 (preprint 5.13), the text’s dense form; `Polyfold.theorem_5_3_5_preprint`, `Polyfold.StrongBundle.allowableDist_le_iff`). Under the hypotheses of V5.29 there is an open $U \supseteq S_f$ such that S_{f+s} is compact for every $s \in \Gamma_U^{+,1}(P)$, and for every allowable s' and $\varepsilon > 0$ there is an allowable s with $N(s(x) - s'(x)) \leq \varepsilon$ for all x such that $f + s$ is transversal and S_{f+s} is compact and, if non-empty, a manifold without boundary as in V5.29. *Book:* Theorem 5.3.5 (preprint 5.13), pp. 217–218 (preprint pp. 174–175). *Hidden hypotheses:* as in V5.29. *Deviation (restated):* density in $(\Gamma_U^{+,1}(P), \rho)$ (V5.21) is the bound $\rho(s, s') \leq \varepsilon$, stated pointwise and equivalent to it; the text’s “$N(s_0(y) + s(\lambda_0, y)) \leq 1 - \varepsilon/2$” is ≤ 1 (Findings, items 60, 62).

V5.31 (Definition 5.3.9 (preprint 5.17), in a tame chart; `Polyfold.StrongBundle.IsGeneralPositionGerm`, `Polyfold.StrongBundleChart.linKer`, `Polyfold.StrongBundle.`

`IsGeneralPositionGerm.isLinSurjAt`). (f, y) with $f(y) = 0$ is in general position if in some tame strong bundle chart around y, at the level-1 point z over $\varphi(y)$, $\mathrm{Lin}(z) : T_zO \to K_z$ is onto, and for every presentation T of the partial quadrant the kernel of $\mathrm{Lin}(z)$ has an sc-complement in T_zO contained in $T_z^RO = T_zO \cap E_z$: an sc-subspace B with $(T_zO)_m = (\ker \mathrm{Lin}(z))_m \oplus B_m$ on every level. General position includes transversality at y. *Book:* Definition 5.3.9 (preprint 5.17), p. 220 (preprint p. 177). *Hidden hypotheses:* the chart is tame (the reduced tangent space of Definition 2.5.8 (preprint 2.38) is that of a tame M-polyfold). *Deviation (restated):* stated in a chart (gap G.7 of Chapter 2); chart independence at the zeros near a zero in good position is V5.34 (Findings, items 71, 76, 82).

V5.32 (the bridge lemma: solution sets as smooth submanifolds; `Polyfold.isScPlus_of_isScSmooth_const`, `Polyfold.MPolyfold.TangentGermChart.exists_scPlusRetraction`, `Polyfold.MPolyfold.TangentGermChart.exists_isScPlusRetractionOn`, `Polyfold.MPolyfold.isSmoothSubmanifold_of_isRegular`, `Polyfold.StrongBundle.isSmoothSubmanifold_solutionSet_inter`). An sc-smooth map on a finite-dimensional space with the constant structure is sc⁺. For regular tangent germ data of Definition 3.2.1 (2) (preprint 3.25), the retraction $\tau = \Gamma \circ \pi \circ r$ of the proof of Theorem 3.2.3 (preprint 3.27) is an sc⁺-retraction, so a set with regular tangent germ data at every point is a smooth finite-dimensional submanifold (Definition 4.2.1 (preprint 4.13)). In particular, if every zero of a section in the open W is in good position in a tame chart, $S_f \cap W$ is a smooth finite-dimensional submanifold. *Book:* proof of Theorem 5.3.10 (preprint 5.18), p. 221 (preprint p. 179) ("hence we can apply Theorem 4.23"). *Hidden hypotheses:* none. *Deviation (not in the book):* the text never verifies Definition 4.2.1 (preprint 4.13) for solution sets (Findings, item 81).

V5.33 ($T_yS = \ker f'(y)$; `Polyfold.tangentSpace_subset_linKer`, `Polyfold.MPolyfold.TangentGermChart.exists_linearIndependent_tangentSpace`, `Polyfold.HasIndepTangent.transport`, `Polyfold.exists_tangentSpace_eq_linKer`). In a strong bundle chart the tangent space of a retract on which the principal part vanishes lies in $\ker \mathrm{Lin}(z)$. For f with tame germ data and (f, x_0) in good position in a tame chart there is an open $V \ni x_0$ whose zeros are all in good position such that at every zero $y \in V$, in every strong bundle chart around y and for every open $G \ni y$ in its domain with $\varphi(G \cap S_f \cap V)$ a retract, the tangent space of that retract at the point over $\varphi(y)$ equals $\ker \mathrm{Lin}(z)$. *Book:* used in the proofs of Theorem 5.3.10 (preprint 5.18) and Theorem 5.25 (preprint) ("T_zS_λ has a sc-complement contained in T_x^RX"); Lemma 5.3.17 (preprint 5.24) ("$T_xS = \ker f'(x)$"). *Hidden hypotheses:* none. *Deviation (not in the book):* "$\supseteq$" is a dimension count, d independent tangent vectors of the bridge retraction carried to every chart (Findings, item 82).

V5.34 (Definition 5.3.9 (preprint 5.17) is Definition 4.2.10 (preprint 4.21) for the solution set; `Polyfold.exists_isGeneralPositionGerm_iff`, `Polyfold.isGoodPositionGermTame_of_isGeneralPositionAt`). For f with tame germ data and (f, x_0) a zero in good position in a tame chart there is an open $V \ni x_0$ such that every zero in V is in good position, $S_f \cap V$ is a smooth finite-dimensional submanifold, and at every zero $y \in V$, (f, y) is in general position (Definition 5.3.9 (preprint 5.17)) iff $S_f \cap V$ is in general position at y (Definition 4.2.10 (preprint 4.21)); both are then independent of the chart. If $f'(y)$ is onto and a smooth submanifold $A \subseteq S_f$ is in general position at y, then (f, y) is in good position in a tame chart. *Book:* Definition 5.3.9 (preprint 5.17), p. 220 (preprint p. 177), and the proofs of Theorem 5.3.10 (preprint 5.18) and Theorem 5.25 (preprint). *Hidden hypotheses:* none. *Deviation (not in the book):* the chart independence of Definition 5.3.9 (preprint 5.17) holds near zeros in good position (Findings, items 82, 85).

V5.35 (openness of general position; `Polyfold.exists_isOpen_isGeneralPositionGerm`, `Polyfold.MPolyfold.IsGeneralPositionAt.congr_nhds`). Definition 4.2.10 (preprint 4.21) depends only on the germ of the submanifold. For X tame, f with tame germ data and (f, x_0) in general position at the zero x_0, there is an open $W \ni x_0$ such that $S_f \cap W$ is a smooth finite-dimensional submanifold and at every zero $y \in W$, (f, y) is in good and in general position and $S_f \cap W$ is in general position at y. *Book:* proof of Theorem 5.3.10 (preprint 5.18), preprint p. 178 ("they will also hold at $(0, y)$ for $y \in S_f$ close to x"; "the properties (i)–(ii) hold for all $(\lambda, y) \in \tilde{S}$"). *Hidden hypotheses:* X tame. *Deviation (not in the book):* used without proof in the text; proved through V5.34 and Theorem 4.2.11 (3) (preprint 4.22) (Findings, item 83).

V5.36 (properties (i)–(ii) of the proof of Theorem 5.3.10 (preprint 5.18); `Polyfold.StrongBundleChart.IsParamGP`, `Polyfold.StrongBundleChart.IsParamGP.comp`, `Polyfold.isGeneralPositionGerm_paramSection_of_isParamGP`, `Polyfold.isParamGP_of_isGeneralPositionGerm`, `Polyfold.exists_sections_isParamGP`). In a chart centred at y, (i) is $\mathrm{Lin}_g(0)(T_0O) + \mathrm{span}\{\hat{t}_i(0)\} = K_0$ and (ii) is: every $w \in T_0O$ is $u + e$ with $e \in E_0$ and $\mathrm{Lin}_g(0)u + \sum a_i\hat{t}_i(0) = 0$ for some a. Adding sections keeps (i)–(ii). For g with tame germ data and a tame chart centred at the zero y, (i)–(ii) hold iff F is in general position at $(0, y)$ (Definition 5.3.9 (preprint 5.17)). If X admits sc-smooth bump functions and N is an auxiliary norm, finitely many allowable sc⁺-sections supported in any given open $U \ni y$ achieve (i)–(ii). *Book:* proof of Theorem 5.3.10 (preprint 5.18), preprint p. 178. *Hidden hypotheses:* none. *Deviation (restated):* the complement L of $T_x^R X$ is spanned by smooth vectors, so that Lemma 5.3.3 (preprint 5.11) applies; (i)–(ii) give good position of F first, and then, with ker $F' = T\tilde{S}$, general position (Findings, item 84).

V5.37 (Theorem 5.3.10 (preprint 5.18) from a family; `Polyfold.theorem_5_3_10_of_family`, `Polyfold.StrongBundle.SolutionSetInGeneralPosition`). Let X be tame, f an sc-Fredholm section with tame germ data and compact solution set, N an auxiliary norm. There is an open $U \supseteq S_f$ such that S_{f+s} is compact for every $s \in \Gamma_U^{+,1}(P)$, and for every sc⁺-section s_0 with supp $s_0 \subseteq U$, $N(s_0) \leq 1 - \varepsilon$, and every family of allowable $t_1, \ldots, t_m$ for which $F(\lambda, y) = f(y) + s_0(y) + \sum \lambda_i t_i(y)$ is in general position at every $(0, x)$, $x \in S_{f+s_0}$, there are $\delta > 0$ and a null set CV such that for $|\lambda| < \delta$, $\lambda \notin CV$, $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable and $f + s_\lambda$ is transversal, in general position at every zero, and S_{f+s_λ} is compact, a smooth finite-dimensional submanifold in general position at each point, whose induced structure has a corner structure and is tame, with $d_{S_{f+s_\lambda}} = d_X$ on it. *Book:* Theorem 5.3.10 (preprint 5.18), p. 220 (preprint p. 177); proof pp. 220–222 (preprint pp. 178–179). *Hidden hypotheses:* as in V5.38. *Deviation (not in the book):* the theorem with the family as a hypothesis; Theorem 4.3.1 (preprint 4.23) is used in its countable-cover form (Findings, item 85).

V5.38 (Theorem 5.3.10 (preprint 5.18), generic form; `Polyfold.theorem_5_3_10_param`). Let X be tame and admit sc-smooth bump functions, f an sc-Fredholm section with tame germ data and compact solution set, N an auxiliary norm. There is an open $U \supseteq S_f$ such that (1) S_{f+s} is compact for every $s \in \Gamma_U^{+,1}(P)$, and (2) for every sc$^+$-section s_0 with $\operatorname{supp} s_0 \subseteq U$ and $N(s_0(x)) \leq 1 - \varepsilon$ there are allowable $t_1, \dots, t_m$, $\delta > 0$ and a Lebesgue null set CV such that for every λ with $|\lambda| < \delta$, $\lambda \notin CV$, $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable and $f + s_\lambda$ has all the conclusions of V5.37. *Book:* Theorem 5.3.10 (preprint 5.18), p. 220 (preprint p. 177); proof pp. 219–222 (preprint pp. 177–179). *Hidden hypotheses:* X tame (the proof uses Theorems 4.2.11 and 4.3.1 (preprint 4.22 and 4.23); the opening of §5.3 places the discussion over a tame strong bundle); f with tame germ data. *Deviation (restated):* the dense set is described as the parameters off a null set, as in V5.29; "replacing $f + s_0$ by f" is not needed (Findings, items 80, 85).

V5.39 (Theorem 5.3.10 (preprint 5.18), the text's dense form; `Polyfold.theorem_5_3_10_dense`). Under the hypotheses of V5.38 there is an open $U \supseteq S_f$ such that S_{f+s} is compact for every $s \in \Gamma_U^{+,1}(P)$, and for every allowable s' and $\varepsilon > 0$ an allowable s with $N(s(x) - s'(x)) \leq \varepsilon$ for all x such that $f + s$ is transversal, $(f + s, x)$ is in general position at every $x \in S_{f+s}$, and S_{f+s} is a compact smooth submanifold in general position with a tame corner structure and $d_{S_{f+s}} = d_X$. *Book:* Theorem 5.3.10 (preprint 5.18), p. 220 (preprint p. 177). *Hidden hypotheses:* as in V5.38. *Deviation (restated):* density as in V5.30 (Findings, item 80).

V5.40 (Theorem 5.3.12 (preprint 5.19), generic form; `Polyfold.theorem_5_3_12_param`, `Polyfold.StrongBundle.IsGoodPositionGermTame.congr_nhds`). Good position depends only on the germ of the section. Let X admit sc-smooth bump functions, f be an sc-Fredholm section with tame germ data and compact solution set, N an auxiliary norm, and (f, x) in good position in a tame chart at every zero $x \in \partial X$. There is an open $U \supseteq S_f$ with S_{f+s} compact for every $s \in \Gamma_U^{+,1}(P)$, and there are allowable $t_1, \dots, t_m$ with supports disjoint from ∂X, $\delta > 0$ and a null set CV such that for $|\lambda| < \delta$, $\lambda \notin CV$, $s_\lambda = \sum \lambda_i t_i$ is allowable, vanishes on the neighbourhood $X \setminus \operatorname{supp} s_\lambda$ of ∂X, $f + s_\lambda$ is transversal and in good position at every zero, and S_{f+s_λ} is compact and, if non-empty, a sub-M-polyfold whose induced structure is tame and a manifold with boundary with corners. *Book:* Theorem 5.3.12 (preprint 5.19), p. 222 (preprint p. 180). *Hidden hypotheses:* none. *Deviation (stronger):* X need not be tame; "arbitrarily small" is $\lambda \to 0$; "(f, x) in good position" is $(f + s, x)$. The text's "as before" would apply the implicit function theorem to F at the boundary zeros, where F is in general not in good position (gap G.46, on paper); the proof uses it only at interior points and keeps $f + s = f$ near the boundary zeros. The dense form is proved too (2026-11-12): within N-distance ε of any allowable s' vanishing near ∂X there is such an s (`Polyfold.theorem_5_3_12_dense`, from `Polyfold.theorem_5_3_12_param_rel`; gap G.45 closed) (Findings, items 87, 88, 89).

V5.41 (Definition 5.3.13 (preprint 5.20); `Polyfold.restrictSection`, `Polyfold.StrongBundle.IsGenCompactHomotopy`, `Polyfold.StrongBundle.IsGenCompactHomotopy.isCompact_solutionSet_restrict`). (f, t) is a generalized compact homotopy if X is tame, f is an sc-Fredholm section with tame germ data and compact solution set, $t : X \to \mathbb{R}$ is sc-smooth with $t(X) = [0, 1]$, and for every $\tau \in [0, 1]$ the fibre $X_\tau = t^{-1}(\tau)$ is a sub-M-polyfold with tame induced structure on which the restriction $f|X_\tau$ — the section $a \mapsto (a, f(a))$ of the pull-back of P along the inclusion — is an sc-Fredholm section with tame germ data. The restrictions then have compact solution sets. *Book:* Definition 5.3.13 (preprint 5.20), pp. 223 ff. (preprint pp. 180–181). *Hidden hypotheses:* none. *Deviation (restated):* $Y|X_\tau$ is the pull-back bundle (Proposition 2.6.8 (preprint 2.54)); the end problems are the restrictions to $t^{-1}(0)$, $t^{-1}(1)$; the identification up to isomorphism (Remark 5.3.14 (preprint 5.21)) is V5.74 (gap G.47 closed; Findings, item 100).

V5.42 (Lemma 5.3.16 (preprint 5.23); `Polyfold.lemma_5_3_16_preprint`, `Polyfold.ne_zero_of_isCompl_ker`, `Polyfold.exists_isCompl_ker_of_ne_zero`). Let $\ell : T \to \mathbb{R}$ be linear and non-zero and Z a complement of $\ker \ell$. For every linear $\tau : S \to \mathbb{R}$ there is a unique linear $b : S \to Z$ with $\ell \circ b = \tau$, and $\ell \circ (b + a) = \tau$ whenever $\ell \circ b = \tau$ and a takes values in $\ker \ell$. A non-zero complement of $\ker \ell$ exists only if $\ell \neq 0$, and then it can be spanned by a vector of any spanning set. *Book:* Lemma 5.3.16 (preprint 5.23), p. 224 (preprint p. 182), and the sentence before it ("since x is a smooth point, we find a one-dimensional smooth linear sc-subspace Z with $T_xX = Z \oplus \ker dt(x)$"). *Hidden hypotheses:* $dt(x) \neq 0$, which Theorem 5.3.15 (preprint 5.22) does not assume. *Deviation (explicit):* linear algebra; the hidden hypothesis is exposed (Findings, item 101).

V5.43 (the product rule at a level-1 point; `Polyfold.scDeriv_smul_apply`, `Polyfold.scDeriv_clm_sub_apply`, `Polyfold.chartLinCutoff`, `Polyfold.isScSmoothMap_chartLinCutoff`). For sc-smooth $a : U \to \mathbb{R}$ and $w : U \to F$, $D(a \cdot w)(z)v = Da(z)v \cdot w(z) + a(z) \cdot Dw(z)v$; $D(\ell(\rho(\cdot) - p))(z) = \ell \circ D\rho(z)$; and $g = \beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ on $\operatorname{dom} \varphi$, 0 outside, is sc-smooth when β is and $\operatorname{supp} \beta \subseteq \operatorname{dom} \varphi$. *Book:* proof of Lemma 5.3.17 (preprint 5.24), p. 225 (preprint pp. 182–183). *Hidden hypotheses:* none. *Deviation (not in the book):* tools for V5.44.

V5.44 (Lemma 5.3.17 (preprint 5.24); `Polyfold.exists_scPlusSection_linAt`, `Polyfold.lemma_5_3_17`). Let X admit sc-smooth bump functions, Φ a strong bundle chart around the smooth point x, z the level-1 point over $\varphi(x)$, and e a smooth point of the fibre over x. For every continuous linear ℓ on the model space and open $U \ni x$ there is an sc$^+$-section s with $s(x) = 0$, $\operatorname{supp} s \subseteq U$ and $\operatorname{Lin}_s(z)v = \ell(v)\Phi(e)_2$ on T_zO. Hence, for a section f, a finite-dimensional $K \subseteq T_zO$ and a linear $b : K \to T_zO$ with $\operatorname{Lin}_f(z)(bv) \in \mathbb{R}\,\Phi(e)_2$ for all v, there is an sc$^+$-section s with $s(x) = 0$, $\operatorname{supp} s \subseteq U$ and $\operatorname{Lin}_f(z)(bv) + \operatorname{Lin}_s(z)v = 0$ for $v \in K$. *Book:* Lemma 5.3.17 (preprint 5.24), p. 225 (preprint pp. 182–183). *Hidden hypotheses:* that $f'(x) \circ b_\tau$ has its image in the line of a smooth point e is a hypothesis (the text's justification, "the image of any vector in T_xS belongs to Y_∞", is not formalized). *Deviation (restated):* $s'(x)$ is read in a chart, T_xS as a finite-dimensional K; the graph of S over T_xS and the case distinction are not needed (Findings, item 102).

V5.45 (Theorem 5.25 (preprint); `Polyfold.theorem_5_25_preprint`). Let X be tame and admit sc-smooth bump functions, f a proper sc-Fredholm section with tame germ data, N an auxiliary norm. There is an open $U \supseteq S_f$ with S_{f+s} compact for every $s \in \Gamma_U^{+,1}(P)$, and an $s \in \Gamma_U^{+,1}(P)$ such that $(f + s, x)$ is in good position in a tame chart at every $x \in S_{f+s}$, and $f + s$ has the conclusions of V5.37; in particular S_{f+s} has a corner structure. *Book:* Theorem 5.25 (preprint), preprint pp. 186–187; proof preprint pp. 187–188. *Hidden hypotheses:* f with tame germ data. *Deviation (restated):* U is the neighbourhood the proof produces (that of Theorem 5.3.10 (preprint 5.18)), not an arbitrary one with the compactness property; the text's "in view of Theorem 3.54 … in general position" is V5.35, and "$x \in S_f$" is $x \in S_{f+s_0}$ (Findings, item 86).

V5.46 (Lemma 5.26 (preprint); `Polyfold.isInGoodPositionIn_of_scComplement_le_reduced`, `Polyfold.StrongBundle.IsGeneralPositionGerm.isGoodPositionGerm`, `Polyfold.StrongBundle.IsGeneralPositionGerm.isGoodPositionGermTame`). In a tame retract, a subspace N of T_zO with an algebraic complement $B \subseteq T_z^R O$ is in good position to the partial cone, with good complement B and $N \cap C_z$ of non-empty interior in N. Hence for an sc-Fredholm section f, a zero y with (f, y) in general position (Definition 5.3.9 (preprint 5.17)) is in good position (Definition 3.6.6 (preprint 3.53)), and in the tame chart of the germ data when f has tame germ data. *Book:* Lemma 5.26 (preprint), preprint p. 187; "clearly general position implies good position", p. 222 (preprint p. 179). *Hidden hypotheses:* none beyond the tame chart of Definition 5.3.9 (preprint 5.17). *Deviation (stronger):* X need not be tame; the interior condition of Definition 3.1.24 (preprint 3.21), which the text omits, is proved (Findings, items 72, 76).

V5.47 (Definition 5.5.1 (preprint 5.28); `Polyfold.MPolyfold.IsScSmoothPartitionOfUnity`, `Polyfold.MPolyfold.AdmitsScPartitionsOfUnity`). X admits sc-smooth partitions of unity if for every open cover $(U_i)_{i\in\iota}$ (with ι in the universe of X) there is a partition of unity (ρ_i) — continuous, $\rho_i \geq 0$, locally finite supports, $\sum \rho_i = 1$ — with $\operatorname{supp} \rho_i \subseteq U_i$ and every ρ_i sc-smooth. *Book:* Definition 5.5.1 (preprint 5.28), p. 229 (preprint p. 192). *Hidden hypotheses:* none. *Deviation (restated):* the index type is restricted to the universe of X, which loses nothing (every cover can be indexed by subsets of X) (Findings, item 110).

V5.48 (sc-smooth functions: algebra, local finiteness, extension by 0; `Polyfold.MPolyfold.IsScSmoothMap.real_add`, `Polyfold.MPolyfold.IsScSmoothMap.real_sub`, `Polyfold.MPolyfold.IsScSmoothMap.real_mul`, `Polyfold.isScSmoothMap_real_finsetSum`, `Polyfold.isScSmoothMap_real_finsetProd`, `Polyfold.isScSmoothMap_real_finsum`, `Polyfold.isScSmoothMap_real_of_eventuallyEq`, `Polyfold.isScSmoothMap_real_of_chart`). Sums, differences, products, finite sums and finite products of sc-smooth functions are sc-smooth; so is a locally finite sum; a function that near every point agrees with an sc-smooth function is sc-smooth; a function supported in the domain of a chart whose representative is sc-smooth on the retract is sc-smooth on X. *Book:* proof of Theorem 5.5.10 (preprint 5.37), pp. 232–234 (preprint pp. 195–197). *Hidden hypotheses:* none. *Deviation (not in the book):* used and not stated (Findings, item 112).

V5.49 (partitions of unity give bump functions; `Polyfold.MPolyfold.AdmitsScPartitionsOfUnity.admitsBumpFunctions`). If X admits sc-smooth partitions of unity, it admits sc-smooth bump functions (Definition 5.3.1 (preprint 5.9) in the reading of V5.16). *Book:* §5.5 (preprint §5.4.2), preprint p. 192 (the open problem whether the converse holds). *Hidden hypotheses:* none. *Deviation (not in the book):* the easy direction, which the text implies; the converse is not addressed (Findings, item 119).

V5.50 (Definition 5.5.4 (preprint 5.31); `Polyfold.HasContDiffBump`, `Polyfold.AdmitsContDiffPartitionsOfUnity`). A Banach space admits a C^k-bump function ($k \in \mathbb{N} \cup \{\infty\}$) if some C^k function $V \to \mathbb{R}$, not identically zero, has bounded support; it admits C^k partitions of unity if every open cover has a subordinate partition of unity of C^k functions. *Book:* Definition 5.5.4 (preprint 5.31), p. 230 (preprint p. 193); partitions of unity of Banach spaces, preprint pp. 192, 197–198. *Hidden hypotheses:* none. *Deviation:* none (Findings, item 114).

V5.51 (Definition 5.5.6 (preprint 5.33); `Polyfold.LocalModel.HasBumpProperty`, `Polyfold.closure_inter_ball_subset`). A local model (O, C, E) has the sc-smooth bump function property if for every $x \in O$ and open $W \ni x$ of E_0 with $\mathrm{cl}_E(O \cap W) \subseteq O$ there is an sc-smooth $f : O \to \mathbb{R}$ with $f(x) \neq 0$ and support (in O) inside W. A retract is closed near each of its points: $O \cap \mathrm{cl}\, B(z, r/2) \subseteq O$ when $B(z, r)$ lies in the open set of the domain of a retraction onto O. *Book:* Definition 5.5.6 (preprint 5.33), p. 231 (preprint p. 194). *Hidden hypotheses:* none. *Deviation (restated):* "$f \neq 0$" is read as $f(x) \neq 0$, as in V5.16 (Findings, items 41, 43).

V5.52 (Theorem 5.5.7 (preprint 5.34); `Polyfold.theorem_5_5_7`). X admits sc-smooth bump functions iff some family of charts of X covering X has local models with the sc-smooth bump function property. *Book:* Theorem 5.5.7 (preprint 5.34), p. 231 (preprint p. 194) ("clearly"). *Hidden hypotheses:* none. *Deviation (explicit):* the extension by 0 of a local bump can fail to be continuous (a chart of the unit ball onto the whole space); the proof takes the local bump in a small ball whose preimage has closure inside the chart domain; "atlas" is read as a covering family of charts (Findings, item 43).

V5.53 (Proposition 5.5.8 (preprint 5.35) and M-polyfolds on sc-Hilbert spaces; `Polyfold.LocalModel.hasBumpProperty_of_quadratic`, `Polyfold.proposition_5_5_8`, `Polyfold.MPolyfold.admitsBumpFunctions_of_hilbert`). If $\kappa\|v\|^2 \leq b(v, v)$ for a continuous bilinear b on E_0 and some $\kappa > 0$ — in particular if E_0 is a Hilbert space, $\|v\|^2 = b(v, v)$ — the local model (O, C, E) has the sc-smooth bump function property. An M-polyfold covered by charts on sc-Hilbert spaces admits sc-smooth bump functions. *Book:* Proposition 5.5.8 (preprint 5.35), p. 231 (preprint p. 194); §5.3, p. 216 (preprint p. 173) ("for example, the case if X is built on sc-Hilbert spaces"). *Hidden hypotheses:* none. *Deviation (restated):* "Hilbert space" is a norm given by a continuous bilinear form, not a second normed structure; translation by a non-smooth point is legitimate because the bump is C^∞ on E_0 (Findings, item 44).

V5.54 (Definition 5.5.9 (preprint 5.36); `Polyfold.LocalModel.HasApproxProperty`). A local model (O, C, E) has the sc-smooth approximation property if for every open W of E_0 with $\mathrm{cl}(O \cap W) \subseteq O$, every continuous $f : O \to [0, 1]$ with support (in O) inside W and every $\varepsilon > 0$ there is an sc-smooth $g : O \to [0, 1]$ with support inside W and $|f - g| \leq \varepsilon$ on O. *Book:* Definition 5.5.9 (preprint 5.36), p. 231 (preprint p. 195). *Hidden hypotheses:* none. *Deviation (restated):* $V = O \cap W$ and the closure is taken in E_0 (C is closed) (Findings, item 110).

V5.55 (Theorem 5.5.10 (preprint 5.37) "(i) ⇒ (ii)"; `Polyfold.hasApproxProperty_restrict`, `Polyfold.MChart.isClosed_inter_preimage`). If $\mathrm{cl}_X(s)$ lies in the domain of a chart φ, then $\varphi^{-1}(A)$ is closed in X for closed $A \subseteq \varphi(s)$. If X admits sc-smooth partitions of unity, every chart restricted to an open s with $\mathrm{cl}_X(s) \subseteq \mathrm{dom}\,\varphi$ has a local model with the sc-smooth approximation property. Since 2026-11-12 every chart has it (`Polyfold.hasApproxProperty_of_isChart`, `Polyfold.theorem_5_5_10_every_chart`): the chart domain is $\{d > 0\}$, the function is cut by sc-smooth Urysohn functions on the shells $\{1/(n+2) \le d \le 1/(n+1)\}$, each piece is approximated with support closed in X, and the pieces are summed with locally finite weights. *Book:* proof of Theorem 5.5.10 (preprint 5.37), pp. 232–233 (preprint pp. 195–196). *Hidden hypotheses:* none. *Deviation (restated):* the text's step "$\mathrm{cl}_X(\varphi^{-1}(V)) \subseteq U$" does not hold as we read it (the chart of the unit ball onto the whole space); the shells avoid it, and the claim for every chart holds (Findings, items 111, 184).

V5.56 (Theorem 5.5.10 (preprint 5.37) "(ii) ⇒ (i)"; `Polyfold.MChart.exists_small_nhds`, `Polyfold.LocalModel.HasApproxProperty.exists_approx`, `Polyfold.exists_isScSmoothMap_urysohn`, `Polyfold.admitsScPartitionsOfUnity_of_approx`). Every point of a chart domain has arbitrarily small open neighbourhoods N with $\mathrm{cl}_X(N) \subseteq \operatorname{dom}\varphi$ and $\mathrm{cl}(\varphi(N)) \subseteq O$; the approximation property transports to continuous functions supported in such N. If a family of charts covering X has local models with the approximation property, then disjoint closed sets of X are separated by sc-smooth Urysohn functions $X \to [0,1]$, and X admits sc-smooth partitions of unity. *Book:* proof of Theorem 5.5.10 (preprint 5.37), pp. 232–234 (preprint pp. 196–197). *Hidden hypotheses:* none. *Deviation (restated):* the refinement "with the same index set" does not exist in general and the closure condition is not arranged in the text; the proof goes through Urysohn functions and Mathlib's bump coverings; metrizability is not used (Findings, item 112).

V5.57 (Theorem 5.5.10 (preprint 5.37); `Polyfold.theorem_5_5_10`). X admits sc-smooth partitions of unity subordinate to any open cover iff some family of charts of X covering X has local models with the sc-smooth approximation property. *Book:* Theorem 5.5.10 (preprint 5.37), p. 232 (preprint p. 195). *Hidden hypotheses:* none. *Deviation (restated):* "atlas" is a covering family of charts; for "(i) ⇒ (ii)" it is the family of V5.55 (Findings, items 111, 112).

V5.58 (Proposition 5.5.11 (preprint 5.38); `Polyfold.proposition_5_5_11`). If E_0 admits C^∞ partitions of unity, every local model (O, C, E) has the sc-smooth approximation property. *Book:* Proposition 5.5.11 (preprint 5.38), p. 234 (preprint p. 197). *Hidden hypotheses:* "the sc-Banach space admits smooth partitions of unity" is read for E_0. *Deviation (explicit):* not "an immediate consequence of Theorem 5.37" (which yields some atlas, and would need partitions of unity on an open subset of C); the argument of "(i) ⇒ (ii)" is repeated in E_0 (Findings, item 113).

V5.59 (Definition 5.5.12 (preprint 5.39) and the statement of Theorem 5.5.13 (preprint 5.40); `Polyfold.cZero`, `Polyfold.mem_cZero_iff`, `Polyfold.norm_cZero`, `Polyfold.IsCoordinatewiseContDiffEmbedding`, `Polyfold.TorunczykStatement`). $c_0(\Gamma)$ is the closed subspace of $\ell^\infty(\Gamma)$ of the functions tending to 0 along the cofinite filter, equivalently those with $\{\gamma \mid |f(\gamma)| > \varepsilon\}$ finite for every $\varepsilon > 0$, with the supremum norm; a homeomorphic embedding into $c_0(\Gamma)$ is coordinate-wise C^k if every coordinate is C^k. Theorem 5.5.13 (preprint 5.40) is recorded as a statement and, since 2026-11-12, proved for every real Banach space and every k (`Polyfold.theorem_5_5_13`; gap G.52 closed). *Book:* Definition 5.5.12 (preprint 5.39), Theorem 5.5.13 (preprint 5.40), p. 234 (preprint p. 198). *Hidden hypotheses:* none. *Deviation (restated):* the text's "max" norm is the supremum (attained for $f \neq 0$) (gap G.52; Findings, items 114, 117).

V5.60 (Definition 5.5.14 (preprint 5.41); `Polyfold.IsWCG`, `Polyfold.IsReflexive`). A Banach space is weakly compactly generated if some set compact in the weak topology has dense linear span; it is reflexive if the canonical map into the bidual is onto. *Book:* Definition 5.5.14 (preprint 5.41), p. 234 (preprint p. 198) (reflexivity is used in Proposition 5.5.15 (preprint 5.42), not defined). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 114).

V5.61 (Proposition 5.5.15 (preprint 5.42); `Polyfold.proposition_5_5_15_reflexive`, `Polyfold.proposition_5_5_15_separable`, `Polyfold.span_closedBall_eq_top`). (i) A reflexive Banach space is WCG; (ii) a separable Banach space is WCG. *Book:* Proposition 5.5.15 (preprint 5.42), p. 235 (preprint p. 198). *Hidden hypotheses:* none. *Deviation:* none; Kakutani's theorem is Mathlib's Banach–Alaoglu argument, and in (ii) a dense sequence of the space, suitably scaled, replaces the text's $\{0\} \cup \{x_n/n\}$ (Findings, item 115).

V5.62 (the statement of Theorem 5.5.16 (preprint 5.43), and Hilbert spaces; `Polyfold.GodefroyTroyanskiWhitfieldZizlerStatement`, `Polyfold.hasContDiffBump_of_hilbert`, `Polyfold.isReflexive_of_hilbert`). Theorem 5.5.16 (preprint 5.43) is recorded as a statement: a WCG space with a C^k-bump function admits C^k partitions of unity. A Banach space whose norm comes from a continuous bilinear form has a C^∞-bump function and, if complete, is reflexive. Since 2026-11-12 Theorem 5.5.16 is proved for every real Banach space and every k (`Polyfold.theorem_5_5_16`), through the Amir–Lindenstrauss projectional resolution of the identity, induction on the density with the separable case as base, and Toruńczyk's theorem for the gluing. *Book:* Theorem 5.5.16 (preprint 5.43), p. 235 (preprint p. 198); proof of Corollary 5.5.17 (preprint 5.44), p. 235 (preprint p. 199) ("a Hilbert space is reflexive"). *Hidden hypotheses:* none. *Deviation (same):* "Banach" is `CompleteSpace`, which the recorded statement leaves out (gap G.53 closed) (Findings, items 116, 117, 184).

V5.63 (Corollary 5.5.17 (preprint 5.44); `Polyfold.corollary_5_5_17`, `Polyfold.MPolyfold.admitsScPartitionsOfUnity_of_hilbert`). If E_0 is a Hilbert space ($\|v\|^2 = b(v,v)$) and Theorem 5.5.16 (preprint 5.43) holds for E_0 and $k = \infty$, the local model (O, C, E) has the sc-smooth approximation property. An M-polyfold covered by charts on such spaces admits sc-smooth partitions of unity. *Book:* Corollary 5.5.17 (preprint 5.44), p. 235 (preprint p. 199). Since 2026-11-12 it holds with no hypothesis (`Polyfold.corollary_5_5_17_unconditional`, `Polyfold.MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional`): E_0 is separable for every sc-Banach space (`Polyfold.ScBanach.separableSpace_zero`), and the separable case of Theorem 5.5.16 is proved (`Polyfold.admitsContDiffPartitionsOfUnity_of_separable`). *Hidden hypotheses:* none. *Deviation (restated):* the proof's "in view of Theorem 5.37" is Proposition 5.5.11 (preprint 5.38), and the bump Theorem 5.5.16 (preprint 5.43) needs is the classical C^∞ one; only the separable case of Theorem 5.5.16 is needed (Findings, items 116, 184).

V5.64 (Definition 5.2.1 (preprint 5.4) in the 2021 form; `Polyfold.StrongBundle.IsWeaklyProperOnU`, `Polyfold.StrongBundle.IsWeaklyProperLtInfty`, `Polyfold.StrongBundle.IsProperLtInfty`, `Polyfold.StrongBundle.isWeaklyProperOnU_iff`, `Polyfold.StrongBundle.IsAuxNorm.auxLe_fiberZero`, `Polyfold.StrongBundle.AuxLe.mono`, `Polyfold.StrongBundle.isZeroAt_of_not_mem_of_support_subset`). Let f be a section of a strong bundle. (2) f is weakly proper if its solution set is compact and for every auxiliary norm N there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section s supported in U with $N(s(y)) \leq 1$ for $y \in U$; (4) f is weakly proper$_{<\infty}$ if the same holds for every sc$^+$-section s supported in U with $N(s(y)) \leq C$ for $y \in U$ and some constant C; (5) f is proper$_{<\infty}$ if its solution set is compact and for every N there is an open $U \supseteq f^{-1}(0)$ such that $\mathrm{cl}_X\{x \in U \mid N(f(x)) \leq C\}$ is compact for every $C > 0$, where $N(f(x)) \leq C$ includes $f(x) \in Y_{0,1}$. The form (2), with the bound on U only, is equivalent to weakly proper in the sense of V5.7 (bound everywhere). *Book:* Definition 5.2.1 (preprint 5.4), p. 209 (preprint p. 171); (4) and (5) are new in the 2021 edition, (1) and (3) are those of V5.7. *Hidden hypotheses:* none. *Deviation (explicit):* "$\sup_{y \in U} N(s(y)) < \infty$" is a constant bound on U, and the convention $N = \infty$ off $Y_{0,1}$ is `AuxLe`, as in V5.7; the bound on U is no weaker than a bound everywhere, since s vanishes off U and $N(0_x) = 0$ (Findings, item 170).

V5.65 (the a priori dependencies after Definition 5.2.1 (preprint 5.4); `Polyfold.StrongBundle.IsProperLtInfty.isProper`, `Polyfold.StrongBundle.IsWeaklyProperLtInfty.isWeaklyProperOnU`, `Polyfold.StrongBundle.IsProper.isWeaklyProperOnU`, `Polyfold.StrongBundle.IsProperLtInfty.isWeaklyProperLtInfty`, `Polyfold.StrongBundle.IsWeaklyProperOnU.isCompact_solutionSet`). For a section f: proper$_{<\infty}$ implies proper, weakly proper$_{<\infty}$ implies weakly proper, and weakly proper implies a compact solution set; if f is continuous, proper implies weakly proper and proper$_{<\infty}$ implies weakly proper$_{<\infty}$. *Book:* the paragraph and diagram after Definition 5.2.1 (preprint 5.4), p. 209; the arrows from (4) and (5) are new in the 2021 edition. *Hidden hypotheses:* f continuous for the two arrows from a properness to a weak properness, so that $\{f = s\}$ is closed, as in V5.7. *Deviation (explicit):* the text gives no argument; a solution of $f = s$ lies in U (outside U, s vanishes, so it would be a zero of f), where $N(f(x)) = N(s(x)) \leq C$ (Findings, item 171).

V5.66 (Theorem 5.2.2 (preprint 5.5) in the 2021 form; `Polyfold.theorem_5_2_2`, `Polyfold.theorem_5_2_2_tfae`). Let f be an sc-Fredholm section of a strong bundle over an M-polyfold X. If f has a compact solution set, f is proper$_{<\infty}$; and for f the five properties of V5.64 — compact solution set, weakly proper, proper, weakly proper$_{<\infty}$ and proper$_{<\infty}$ — are equivalent. *Book:* Theorem 5.2.2 (preprint 5.5), p. 210 (preprint p. 172); proof pp. 210–213. *Hidden hypotheses:* none. *Deviation:* none; as in V5.10, X need not be tame and the finite cover of $f^{-1}(0)$ is explicit. Near a zero, the closure of $\{N(f) \leq C\}$ lies in the compact set of V5.9 for the bound $A \cdot C$, where A compares N with the chart norm, so the neighbourhood does not depend on C (Findings, item 172).

V5.67 (Theorem 5.3.5 (preprint 5.13) in the 2021 form; `Polyfold.theorem_5_3_5`, `Polyfold.isSubMPolyfold_solutionSet_of_manifold`). Under the hypotheses of V5.29 ($\partial X = \emptyset$, sc-smooth bump functions, f sc-Fredholm with tame germ data and compact solution set, N an auxiliary norm) there is an open $U \supseteq S_f$ such that S_{f+s} is compact for every $s \in \Gamma_U^{+,1}(P)$, and for every allowable s' and $\varepsilon > 0$ there is an allowable s with $N(s(x) - s'(x)) \leq \varepsilon$ for all x such that $f + s$ is transversal and S_{f+s} is a compact sub-M-polyfold, which, if non-empty, has the structure of a manifold without boundary as in V5.29. A solution set that is such a manifold whenever it is nonempty is a sub-M-polyfold. *Book:* Theorem 5.3.5 (preprint 5.13), pp. 217–218 (preprint pp. 174–175); the 2021 edition concludes that S_{f+s} is a compact sub-M-polyfold with the structure of a compact manifold without boundary. *Hidden hypotheses:* as in V5.29; the printed bound "$N(f(x)) \leq 1$ for all $x \in X$" is read as $N(s(x)) \leq 1$ (possible misprint; the proof and $\Gamma^{+,1}_U(P)$ bound s). *Deviation (restated):* density is pointwise, as in V5.30; the empty solution set counts as a sub-M-polyfold (Findings, items 173, 174).

V5.68 (Definition 5.3.13 (preprint 5.20) in the 2021 form; `Polyfold.MPolyfold.HasSurjDerivAt`, `Polyfold.StrongBundle.IsCompactFredholmRestriction`, `Polyfold.StrongBundle.IsSubmersiveGenCompactHomotopy`, `Polyfold.StrongBundle.IsGenCompactHomotopy.isSubmersiveGenCompactHomotopy`, `Polyfold.StrongBundle.IsSubmersiveGenCompactHomotopy.isCompact_solutionSet_restrict`, `Polyfold.StrongBundle.isCompactFredholmRestriction_of_eq_empty`, `Polyfold.isScSmoothRetractMap_of_domain_empty`). (f, t) is a generalized compact homotopy in the 2021 sense if X is tame, f is an sc-Fredholm section with tame germ data and compact solution set, $t : X \to \mathbb{R}$ is sc-smooth with values in $[0, 1]$, for $i = 0, 1$ the end $t^{-1}(i)$ is a sub-M-polyfold with tame induced structure over which the restriction of f (a section of the pull-back) has tame germ data, and (3) $dt(x)$ is onto at every smooth x: in some chart φ of the atlas around x, with a retraction R onto its model and a level-1 point z over $\varphi(x)$, every $a \in \mathbb{R}$ is the derivative of $t \circ \varphi^{-1} \circ R$ at z on some tangent vector $DR(z)v \in T_zO$. The data of V5.41 together with (3) give these data; every fibre $t^{-1}(\tau)$ has a compact solution set; and the end conditions hold over an empty end. *Book:* Definition 5.3.13 (preprint 5.20), p. 223 (preprint pp. 180–181); the 2021 edition asks $t : X \to [0, 1]$ without surjectivity, drops the conditions on the fibres X_τ, $0 < \tau < 1$, and adds (3). *Hidden hypotheses:* none. *Deviation (restated):* as in V5.41, $Y|X_i$ is the pull-back and the end problems are the restrictions (Remark 5.3.14 (preprint 5.21) is V5.74, gap G.47 closed); $dt(x)$ is read in a chart, since T_xX carries no linear structure here (gap G.7 of Chapter 2) (Findings, items 175, 176).

V5.69 (Lemma 5.3.16 (preprint 5.23) under the 2021 Definition 5.3.13 (3); `Polyfold.lemma_5_3_16`). If $dt(x)$ is onto in the sense of V5.68, then in a chart around x the derivative ℓ of $t \circ \varphi^{-1} \circ R$ at z, restricted to $T_zO = DR(z)(E_0)$, is non-zero, ker ℓ has a one-dimensional complement Z in T_zO, and for every linear $\tau : S \to \mathbb{R}$ there is a unique linear $b : S \to Z$ with $\ell \circ b = \tau$, while $\ell \circ (b + a) = \tau$ whenever $\ell \circ b = \tau$ and a takes values in ker ℓ. *Book:* Lemma 5.3.16 (preprint 5.23), p. 224 (preprint p. 182), with Definition 5.3.13 (3), p. 223. *Hidden hypotheses:* none: the hypothesis $dt(x) \neq 0$ of V5.42 now follows from (3). *Deviation (explicit):* Z is an algebraic complement, spanned by a vector on which ℓ is non-zero; that it can be chosen smooth, as the proof of Theorem 5.3.15 (preprint 5.22) asks, is not part of the statement (Findings, items 101, 178).

V5.70 (a generalized compact homotopy in the 2021 sense that is not one in the earlier sense; `Polyfold.GenCompactHomotopyWitness.arctanTime`, `Polyfold.GenCompactHomotopyWitness.arctanTime_mem_Ioo`, `Polyfold.GenCompactHomotopyWitness.hasSurjDerivAt_arctanTime`, `Polyfold.GenCompactHomotopyWitness.solutionSet_graphSection_id`, `Polyfold.GenCompactHomotopyWitness.arctanTime_preimage_eq_empty`, `Polyfold.

`GenCompactHomotopyWitness.isSubmersiveGenCompactHomotopy_witness`, `Polyfold.GenCompactHomotopyWitness.not_isGenCompactHomotopy_witness`). On the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$, the graph section of the identity (solution set $\{0\}$) and $t(x) = 1/2 + \arctan(x)/\pi$ satisfy Definition 5.3.13 in the 2021 sense (V5.68): t takes values in $(0, 1)$, so both ends are empty, and $dt(x) = 1/(\pi(1 + x^2)) \neq 0$ at every x. They are not a generalized compact homotopy in the sense of V5.41, since t does not take the value 0. *Book:* Definition 5.3.13 (preprint 5.20), p. 223; not in the book. *Hidden hypotheses:* none. *Deviation (not in the book):* satisfiability of the 2021 definition, and the 2021 data do not imply the earlier ones (Findings, item 177).

V5.71 (Theorem 5.3.15 (preprint 5.22): general position and Morse-type critical points of t; `Polyfold.theorem_5_3_15`, `Polyfold.theorem_5_3_15_generalPosition`, `Polyfold.StrongBundle.IsGeneralPositionAtEnds`, `Polyfold.StrongBundle.IsSubmersiveGenCompactHomotopy.isGeneralPositionGerm_of_mem_boundary`, `Polyfold.isGeneralPositionGerm_of_restrict`, `Polyfold.theorem_5_3_15_generalPosition_witness`, `Polyfold.StrongBundle.hasMorseLocalData`, `Polyfold.StrongBundle.hasFamilyParam`, `Polyfold.StrongBundle.exists_hasRegularFamilyParam`, `Polyfold.StrongBundle.isRegularPointOn_of_extremum`, `Polyfold.scDeriv_eq_zero_of_extremum`, `Polyfold.StrongBundle.exists_fderivWithin_eq_scDeriv_of_mem_linKer`, `Polyfold.MPolyfold.IsSmoothParam`, `Polyfold.MPolyfold.IsMorseOrRegularPoint`, `Polyfold.MPolyfold.HasOnlyMorseCriticalPoints`, `Polyfold.MPolyfold.IsMorseOrRegularPoint.morseClause`, `Polyfold.MPolyfold.IsSmoothParam.fderivWithin_ne_zero_congr`, `Polyfold.MPolyfold.IsSmoothParam.hessian_injective_congr`). `theorem_5_3_15`: X with sc-smooth bump functions, (f, t) a generalized compact homotopy (`IsSubmersiveGenCompactHomotopy`, Definition 5.3.13 in the 2021 form, V5.68), the ends $X_i = t^{-1}(i)$ without boundary and $(f|X_i, x)$ in general position at every zero (`IsGeneralPositionAtEnds`), $\partial X = X_0 \cup X_1$, N an auxiliary norm. Then there is an open $U \supseteq S_f$ with S_{f+s} compact for every $s \in \Gamma^{+,1}_U(P)$, and for every $\varepsilon > 0$ an $s_0 \in \Gamma^{+,1}_U(P)$ with $N(s_0(x)) \leq \varepsilon$, $s_0 = 0$ near ∂X, such that (1) $(f + s_0, x)$ is in general position at every $x \in S_{f+s_0}$ and (2) t has only Morse-type critical points on S_{f+s_0} (`MPolyfold.HasOnlyMorseCriticalPoints`). Conclusion (1) alone, with the same hypotheses, is `theorem_5_3_15_generalPosition`; the general position at ∂X comes from the ends (`isGeneralPositionGerm_of_mem_boundary`, `isGeneralPositionGerm_of_restrict`). The local data of the proof are derived (`StrongBundle.hasMorseLocalData`): smooth parametrizations of the families with μ a submersion (`hasFamilyParam`), small sections making an interior zero a regular point (`exists_hasRegularFamilyParam`), and regular points of $t|S$ at the zeros on ∂X at every degeneracy index (`isRegularPointOn_of_extremum`, from `scDeriv_eq_zero_of_extremum`). "Morse-type" does not depend on the parametrization (`MPolyfold.IsMorseOrRegularPoint.morseClause`). The hypotheses are satisfiable (`theorem_5_3_15_generalPosition_witness`). *Book:* §5.3, Theorem 5.3.15 (preprint 5.22), p. 223 (preprint p. 181) (proof pp. 223–228 (preprint pp. 181–186)), For a generalized compact homotopy with end problems in general position and $\partial X = X_0 \sqcup X_1$, an arbitrarily small $s_0 \in \Gamma^{+,1}_U(P)$ vanishing near ∂X makes $(f + s_0, x)$ in general position at every zero and $t : S_{f+s_0} \to [0, 1]$ a function with only Morse-type critical points. *Deviation (stronger):* No added hypothesis. The bump functions on the X_i are not assumed, and of $\partial X = X_0 \sqcup X_1$

only $\partial X \subseteq X_0 \cup X_1$ is used. The end problems are the restrictions $f|X_i$ in the pull-back bundles (Definition 5.3.13 (2) as read in V5.68; up to isomorphism, Remark 5.3.14 is V5.73). "Morse-type critical points" (not defined in the text) is read through smooth parametrizations of S_{f+s_0}: around every point a parametrization by a subset of $\mathbb{R}^n$ with unique derivatives in which a critical point of t is interior with injective Hessian; the clause then holds in every such parametrization (`IsMorseOrRegularPoint.morseClause`). The comparison with the charts of the Mathlib manifold structure of the solution set is not formalized. The hidden hypothesis $dt(x) \neq 0$ of V5.42 is supplied by the 2021 Definition 5.3.13 (3) (V5.69). At a zero on ∂X, t has an extremum, so $dt(x)$ vanishes on the reduced tangent space at every degeneracy index; with general position $dt(x) \neq 0$ on $\ker f'(x)$ (Findings, Chapter 5, items 101, 178, 183). The proof is V5.72 (gap G.48 closed).

V5.72 (Proof of Theorem 5.3.15 (preprint 5.22): (5.12)–(5.14), the parametrized Sard step and the covering argument; `Polyfold.ae_hasOnlyMorseCriticalPointsOn`, `Polyfold.ae_forall_injective_of_surjective`, `Polyfold.ae_hasOnlyMorseCriticalPointsOn_add_linear`, `Polyfold.HasOnlyMorseCriticalPointsOn`, `Polyfold.lagrangian`, `Polyfold.IsLagrangeNondegenerate`, `Polyfold.fderiv_lagrangian_eq_zero_iff`, `Polyfold.injective_hessian_comp_of_lagrange`, `Polyfold.exists_levelSet_param`, `Polyfold.surjective_fderiv_famGradient`, `Polyfold.ae_isLagrangeNondegenerate_family`, `Polyfold.ae_isMorseOrRegularPoint_famSlice`, `Polyfold.famShift`, `Polyfold.exists_spanning_of_group`, `Polyfold.MPolyfold.IsSmoothParam.fderivWithin_eq_zero_of_multiplier`, `Polyfold.famZero`, `Polyfold.isZeroAt_paramSlice_smul_iff`, `Polyfold.StrongBundle.HasMorseLocalData`, `Polyfold.exists_morse_perturbation_hyp`, `Polyfold.theorem_5_3_15_hyp`, `Polyfold.MPolyfold.TangentGermChart.exists_isSmoothParam`, `Polyfold.StrongBundle.exists_isSmoothParam_solutionSet`, `Polyfold.StrongBundle.exists_isSmoothParam_famZero`). In a chart: `ae_hasOnlyMorseCriticalPointsOn` ($g : \Lambda \times E \to \mathbb{R}$ of class C^2 with $(\lambda, z) \mapsto \partial_z g$ transversal to 0: for almost every λ, $g(\lambda, \cdot)$ has only Morse-type critical points), from `ae_forall_injective_of_surjective` (inverse function theorem and Sard in equal dimensions); `ae_hasOnlyMorseCriticalPointsOn_add_linear` ($h + \ell$ is Morse for almost every $\ell \in E^*$). On the solution manifold: the sections are $a_i\sigma$, so S_λ is a level set in the solution manifold M of $f + \sum \mu_j \sigma_j$ (`famZero`, `isZeroAt_paramSlice_smul_iff`); critical points of t on level sets are handled with Lagrange multipliers (`lagrangian`, `IsLagrangeNondegenerate`, `injective_hessian_comp_of_lagrange`, `exists_levelSet_param`); (5.13)–(5.14) become `surjective_fderiv_famGradient`; the Sard step for the family is `ae_isLagrangeNondegenerate_family`, the local statement `ae_isMorseOrRegularPoint_famSlice`; the transversality persists in the union of the families (`exists_spanning_of_group`); the covering argument is `exists_morse_perturbation_hyp`, and `theorem_5_3_15_hyp` the theorem from the local data, discharged by `StrongBundle.hasMorseLocalData` (V5.71). *Book:* §5.3, proof of Theorem 5.3.15 (preprint 5.22), pp. 223–228 (preprint pp. 181–186), Near a critical point of $t|S$, sections s_i built with Lemma 5.3.17 and a trivialization $\Phi_\lambda : S \to S_\lambda$ of the family $f + \sum \lambda_i s_i$ make $(\lambda, y) \mapsto d(t \circ \Phi_\lambda)(y)$ transversal to the zero section ((5.12)–(5.14)); a parameterized version of Sard's theorem gives, for almost every small λ, only Morse-type critical points near that point; finitely many such neighbourhoods cover the solution set, and "by construction" the union of the families keeps the transversality. *Deviation (explicit):* Lagrange multipliers on level sets of the solution manifold of $f + \sum \mu_j \sigma_j$ replace the trivialization Φ_λ; only first derivatives of f at the point enter. The parameterized Sard theorem follows in a chart from the inverse function theorem and Sard in equal dimensions (Mathlib). The persistence of the transversality in the union of the families of several critical points (p. 227, "by construction") is proved with the functions $a_0 = 1$ and $a_i = \arctan(q_i)/2$ (Findings, Chapter 5, item 183). The sections of Lemma 5.3.17 pass through $-c\, f'(x) w_1$ for a smooth tangent vector w_1 with $dt(x) w_1 \neq 0$: the text's $b(e_i)$ lies in an algebraic complement Z, and for the sc$^+$-section through the value to exist (Lemma 5.3.3) the value must be a smooth point of the fibre. Smooth such w_1 exist by density (`exists_hasRegularFamilyParam`; Findings, Chapter 5, item 183).

V5.73 (Remark 5.3.8 (preprint 5.16): a general sc-Fredholm arc made transversal; `Polyfold.remark_5_3_8_arc`, `Polyfold.remark_5_3_8_relative`, `Polyfold.exists_margin_of_isCompact`, `Polyfold.exists_cobordism`, `Polyfold.theorem_5_3_10_relative_hyp`, `Polyfold.exists_homotopy_cobordism`). `remark_5_3_8_arc`: let $F(t,x) = f_t(x)$ be a section of the pull-back of P to $\mathbb{R}^1 \times X$ with tame germ data, over a tame X with bump functions, in general position at its zeros with $t \notin (a,b)$ ($0 < a$, $b < 1$), with compact zeros over $[0,1]$. Then for every $\varepsilon > 0$ there are $0 < c' < a$, $b < d' < 1$ and an sc$^+$-section s with $N(s) \leq \varepsilon$, $s(t,\cdot) = 0$ for $t \notin (c', d')$, such that $F + s$ is in general position at every zero and its zeros over $[0,1]$ are compact. It is the instance $\tau = t$, $[c,d] = [0,1]$ of `remark_5_3_8_relative` (any continuous time function τ on a tame X), proved through Theorem 5.3.10 relative to a compact part; `exists_margin_of_isCompact` gives the compactness margin. The arcs $f_0 + S(t,\cdot)$ between admissible sections are `exists_cobordism`, `exists_homotopy_cobordism` (V6.76). *Book:* §5.3, Remark 5.3.8 (preprint 5.16), pp. 219–220 (preprint pp. 176–177), For transversal f_0, f_1 joined by a sc-Fredholm arc $(t,x) \mapsto f_t(x)$ with compact solution set, a perturbation s vanishing near $t = 0, 1$ makes the arc transversal, and its solution set is a compact cobordism; "the reader will be able to carry out this construction in more detail". *Deviation (restated):* Reading D2: the text's $[0,1] \times X$ is $\mathbb{R}^1 \times X$; "transversal near $t = 0, 1$" is general position at the zeros with $t \notin (a,b)$, and "s vanishing near $t = 0, 1$" is $s(t,\cdot) = 0$ for $t \notin (c', d')$. The compact solution set of the arc is asked over $[0,1]$. The orientation of the cobordism is gap G.66.

V5.74 (Remark 5.3.14 (preprint 5.21): end problems up to isomorphism, made part of the data; `Polyfold.StrongBundle.IsProblemIso`, `Polyfold.StrongBundle.IsProblemIso.image_solutionSet`, `Polyfold.StrongBundle.EndIso`, `Polyfold.StrongBundle.IsGenCompactHomotopyBetween`, `Polyfold.StrongBundle.IsGenCompactHomotopyBetween.congr_left`, `Polyfold.StrongBundle.IsGenCompactHomotopyBetween.congr_right`, `Polyfold.StrongBundle.IsGenCompactHomotopyBetween.isCompact_solutionSet_left`, `Polyfold.StrongBundle.IsSubmersiveGenCompactHomotopy.isGenCompactHomotopyBetween`). `StrongBundle.IsProblemIso B f B' f'`: a strong bundle isomorphism (h, Φ) with $\Phi \circ f = f' \circ h$ (`.refl`, `.symm`, `.trans`; `.image_solutionSet`: $h(S_f) = S_{f'}$; `.isCompact_solutionSet_iff`). `StrongBundle.EndIso`: the restriction of the problem to a sub-M-polyfold is isomorphic to a given problem. `StrongBundle.IsGenCompactHomotopyBetween B f t B f B f :` a 2021 generalized compact homotopy (`IsSubmersiveGenCompactHomotopy`, V5.68) with `EndIso` at $t^{-1}(0)$ to (B_0, f_0) and at $t^{-1}(1)$ to (B_1, f_1); it is invariant under isomorphism of the end problems (`congr_left`, `congr_right`) and the end solution sets are compact (`isCompact_solutionSet_left`, `isCompact_solutionSet_right`). Every generalized compact homotopy is one between its own restrictions, with identity isomorphisms (`IsSubmersiveGenCompactHomotopy.isGenCompactHomotopyBetween`). *Book:* §5.3, Remark 5.3.14 (preprint 5.21), p. 223 (preprint p. 181), Instead of $(f|X_i, P|Y_i) = (f_i, P_i)$ one should require that the problems are isomorphic and make this part of the data. *Deviation (restated):* "Isomorphic problems" is read as a strong bundle isomorphism carrying f to f'; it is put on the 2021 form of Definition 5.3.13 (V5.68). The notion is satisfiable: every generalized compact homotopy is one between its restrictions (Findings, Chapter 5, item 100).

V5.75 (Theorem 5.5.5 (preprint 5.32): the differentiability of $\|\cdot\|^p$ on L^p (Bonic–Frampton); `Polyfold.BonicFrampton.theorem_5_5_5`, `Polyfold.BonicFrampton.contDiff_normRpow_of_lt`, `Polyfold.BonicFrampton.hasFTaylorSeriesUpTo_normRpow`, `Polyfold.BonicFrampton.theorem_5_5_5_even`, `Polyfold.BonicFrampton.wMul`). `BonicFrampton.theorem_5_5_5 (p) [Fact (1   ENNReal.ofReal p)]`: for $F(f) = \|f\|^p$ on the real $L^p(\mu)$ of any measure space, (1) $p = 2k$ □ F is C^∞; (2) $p = 2k + 1$ □ F is $C^{2k} = C^{p-1}$; (3) $p \notin \mathbb{N}$ □ F is $C^{[p]}$. For even p, $\|\cdot\|^p$ is the diagonal of the continuous p-linear form $\int \prod h_i$ (`wMul`, generalized Hölder; `theorem_5_5_5_even`). For every $p \geq 1$ the forms $T_j(f) = \int g_j(f) \prod h_i$, with g_j the j-th derivative of $|t|^p$ and $g_j(f) \in L^{p/(p-j)}$, are a Taylor series of $\|\cdot\|^p$ up to every integer $n < p$ (`hasFTaylorSeriesUpTo_normRpow`, `contDiff_normRpow_of_lt`). *Book:* §5.5 (preprint §5.4.2), Theorem 5.5.5 (preprint 5.32), p. 230 (preprint p. 193), On L^p, $\|\cdot\|^p$ is C^∞ for p an even integer, C^{p-1} for p odd, and $C^{[p]}$ for p not an integer. *Deviation (same):* none.

V5.76 (The L^p and Sobolev remark after Theorem 5.5.5: $W^{k,p}$ norms, bumps, $W^{1,4}$ and $W^{1,3/2}$; `Polyfold.SobolevRemark.Wsub`, `Polyfold.SobolevRemark.norm_rpow_W`, `Polyfold.SobolevRemark.Dop_zero_injective`, `Polyfold.SobolevRemark.remark_5_5_5_sobolev`, `Polyfold.SobolevRemark.remark_5_5_5_sobolev_bump`, `Polyfold.SobolevRemark.hasContDiffBump_Lp`, `Polyfold.SobolevRemark.theorem_5_5_5_sum`, `Polyfold.LocalModel.hasBumpProperty_of_hasContDiffBump`, `Polyfold.SobolevRemark.remark_5_5_5_W14`, `Polyfold.SobolevRemark.remark_5_5_5_W1_3_2`, `Polyfold.SobolevRemark.sobolevSc`, `Polyfold.SobolevRemark.sobolevSc_hyps_empty`, `Polyfold.SobolevRemark.sobolevSc_admitsBumpFunctions`, `Polyfold.SobolevRemark.example_1_1_3_scPolyfold`, `Polyfold.SobolevCube.isCompactOperator_V`, `Polyfold.SobolevCube.isCompactOperator_inclW_cube`, `Polyfold.SobolevCube.dense_smoothLvl_cube`, `Polyfold.SobolevCube.cubeSc`, `Polyfold.SobolevCube.remark_5_5_5_W14_cube`, `Polyfold.SobolevCube.remark_5_5_5_W14_instance`, `Polyfold.SobolevCube.remark_5_5_5_W1_3_2_cube`). $W^{k,p}(\Omega)$, $\Omega \subseteq \mathbb{R}^n$, $p \geq 1$ (`SobolevRemark.Wsub`): the closed subspace of the ℓ^p-product $\prod_{|\alpha| \leq k} L^p(\Omega)$ cut out by the weak-derivative identities; a Banach space with $\|u\|^p = \sum \|D^\alpha u\|^p_{L^p}$ (`norm_rpow_W`), weak derivatives unique on open Ω (`Dop_zero_injective`). (1) Its norm has the differentiability of Theorem 5.5.5 (`remark_5_5_5_sobolev`). (2) $\beta(\|x\|^p)$ is a bump of L^p and of $W^{k,p}(\Omega)$ in the three cases (`hasContDiffBump_Lp`, `remark_5_5_5_sobolev_bump`). (3) Every sc-Banach space with $E_0 \cong W^{1,4}(\Omega)$ admits sc-smooth bump functions, and so do its local models and M-polyfolds modelled on it (`remark_5_5_5_W14`), through `LocalModel.hasBumpProperty_of_hasContDiffBump`. (4) On $W^{1,3/2}(\Omega)$ the bump $\beta(\|x\|^{3/2})$ is sc^1 (`remark_5_5_5_W1_3_2`). The sc-structure $E_m = W^{1+m,p}(\Omega)$ is `sobolevSc`, which takes the compactness of the inclusions (`hc`) and the density of E_∞ (`hd`) as hypotheses; they are discharged for $\Omega = (0,1) \subset \mathbb{R}^1$ and every $p > 1$ (`SobolevCube.cubeSc`: Rellich `isCompactOperator_inclW_cube` through the compact Volterra operator, density `dense_smoothLvl_cube`), where `remark_5_5_5_W14_cube`, `remark_5_5_5_W14_instance` and `remark_5_5_5_W1_3_2_cube` give the claims on the text's sc-structure; `sobolevSc_hyps_empty` is the trivial instance $\Omega = \emptyset$. *Book:* §5.5 (preprint §5.4.2), pp. 230–231 (preprint pp. 193–194), The usual norms of $W^{k,p}(\Omega)$ have the differentiability of $\|\cdot\|_{L^p}$; $\beta(\|x\|^p)$ is a bump function of L^p; $W^{1,4}(\Omega)$ with $E_m = W^{1+m,4}(\Omega)$ admits sc-smooth bump functions, while for $W^{1,3/2}(\Omega)$ the straightforward one is only sc^1 — "does there exist a sc-smooth bump function on E?" *Deviation (narrower):* The remark presupposes that $E_m = W^{1+m,4}(\Omega)$ is an sc-structure on any bounded domain $\Omega \subseteq \mathbb{R}^n$. `sobolevSc` takes the compactness of the inclusions and the density of E_∞ as hypotheses; they are proved only for the unit interval $(0,1) \subset \mathbb{R}^1$, every $p > 1$ (`cubeSc`). Cubes in $\mathbb{R}^n$ with $n \geq 2$ and other bounded domains (where, as we read it, a regular boundary is presupposed) are not done; hence narrower. The bump statements (3), (4) are stronger than printed: they hold for every sc-structure on $W^{1,4}(\Omega)$ resp. $W^{1,3/2}(\Omega)$ and every Ω, since they use only E_0. "Non-vanishing β" is read as $\beta(0) \neq 0$, as in the Hilbert example on p. 229. The closing question is a question, not a claim (Findings, Chapter 5, items 118, 185).

V5.77 (Definition 5.1.4 (2017: 5.2), reflexive $(0,1)$-fibres; `Polyfold.StrongBundle.HasReflexiveFibers`, `Polyfold.StrongBundle.hasReflexiveFibers_of_atlas`, `Polyfold.IsReflexive`, `Polyfold.TrivialBundle.hasReflexiveFibers_bundle`). `StrongBundle.HasReflexiveFibers B`: every $x \in X$ lies in the domain of a strong bundle chart c compatible with the atlas (`B.IsChart c`) with `IsReflexive (c.F 1)`, the canonical map $F_1 \to F_1^{**}$ onto (the project's §5.5 definition). If all charts of the given atlas have reflexive F_1, the bundle has reflexive fibres (`hasReflexiveFibers_of_atlas`). Witness: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ (`TrivialBundle.hasReflexiveFibers_bundle`). *Book:* §5.1, Definition 5.1.4, p. 207 (2017: Definition 5.2; not in the preprint). *Hidden hypotheses:* none. *Deviation (restated):* "There exists an (equivalent) atlas of reflexive charts" is read as "the reflexive compatible charts cover X"; they cover Y, since chart domains are $P^{-1}(V)$.

V5.78 (Lemma 5.1.5 (2017: 5.2), weak convergence of the fibre parts is chart independent; `Polyfold.lemma_5_1_5`, `Polyfold.lemma_5_1_5_limit`, `Polyfold.StrongBundle.exists_limit_of_weakTendsto`, `Polyfold.StrongBundle.weakTendsto_transfer`, `Polyfold.StrongBundle.weakTendsto_liftOneSnd_of_tendsto`, `Polyfold.StrongBundleChart.mem_K_of_tendsto`, `Polyfold.StrongBundleChart.liftOneSnd`). `lemma_5_1_5`: for compatible charts c, c' with reflexive F_1, F_1', x in both chart domains and $P(w_k) \to x$: $(\exists h,\ h_k \rightharpoonup h) \iff (\exists h',\ h_k' \rightharpoonup h')$, h_k `= c.liftOneSnd (B.oneVal (w k))`. Strong form `lemma_5_1_5_limit`: if $h_k \rightharpoonup h$ (only F_1' reflexive needed), there is one $y \in Y_{0,1}$ over x with Ψ-part h, $y_k \to y$ in Y, and $h_k' \rightharpoonup$ the Ψ'-part of y. *Book:* §5.1, Lemma 5.1.5, p. 207 (2017: Lemma

5.2; not in the preprint). *Hidden hypotheses:* none. *Deviation (same):* The strong form identifies the two limits (one point of $Y_{0,1}$), which the text's proof uses implicitly. Steps made explicit: weak convergence and the compact inclusion $F_1 \to F_0$ give norm convergence in F_0; the limit lies in K; the uniform bound is the local comparison of $|\cdot|_1$ in two charts; bounded and level-0 convergent gives weakly convergent in a reflexive space. Fibre parts are read in the chart only for large k.

V5.79 (Definition 5.1.6 (2017: 5.3), mixed convergence; `Polyfold.StrongBundle.MixedTendsto`, `Polyfold.StrongBundle.MixedTendsto.weakTendsto_of_isChart`, `Polyfold.StrongBundle.MixedTendsto.tendsto_oneVal`). `StrongBundle.MixedTendsto B w y`, exactly this, with a compatible chart. Then $h'_k \rightharpoonup h'$ in every compatible chart around $P(y)$ with reflexive F'_1 (`MixedTendsto.weakTendsto_of_isChart`, chart independence), and $y_k \to y$ in Y (`MixedTendsto.tendsto_oneVal`). *Book:* §5.1, Definition 5.1.6, p. 207 (2017: Definition 5.3; not in the preprint). *Hidden hypotheses:* none. *Deviation (same):* The chart independence ("in view of this lemma we can define") is proved, not assumed.

V5.80 (Definition 5.1.7 (2017: 5.4), reflexive auxiliary norms; `Polyfold.StrongBundle.IsReflexiveAuxNorm`, `Polyfold.StrongBundle.IsReflexiveAuxNorm.le_of_frequently_le`, `Polyfold.StrongBundle.le_liminf_coe_of_eventually`). `StrongBundle.IsReflexiveAuxNorm B N`: `B.IsAuxNorm N` and $(N(y) : \overline{\mathbb{R}}) \leq \liminf_k (N(w_k) : \overline{\mathbb{R}})$ for `B.MixedTendsto w y`. Equivalent forms: `IsReflexiveAuxNorm.le_of_frequently_le` ($N(y_k) \leq r$ infinitely often implies $N(y) \leq r$) and the criterion `le_liminf_coe_of_eventually`. *Book:* §5.1, Definition 5.1.7, p. 208 (2017: Definition 5.4; not in the preprint). *Hidden hypotheses:* none. *Deviation (same):* The text prints $\liminf_{k \to 0}$; presumably $k \to \infty$. The liminf is taken in `EReal`, so that unbounded sequences have liminf ∞.

V5.81 (Proposition 5.1.8 (2017: 5.2), existence and comparison of reflexive auxiliary norms; `Polyfold.proposition_5_1_8_exists`, `Polyfold.proposition_5_1_8_compare`, `Polyfold.proposition_5_1_8`, `Polyfold.StrongBundle.isReflexiveAuxNorm_gluedNorm`, `Polyfold.StrongBundle.le_liminf_gluedNorm`). `proposition_5_1_8_exists` (`B.HasReflexiveFibers` gives some `B.IsReflexiveAuxNorm N`), `proposition_5_1_8_compare`, and `proposition_5_1_8` (both). The norm is the glued norm $\sum_i \rho_i(Pw)\,|\Phi_i[1](w)|_1$ over the reflexive compatible charts (`isReflexiveAuxNorm_gluedNorm`, lower semicontinuity `le_liminf_gluedNorm`). *Book:* §5.1, Proposition 5.1.8, p. 208 (2017: Proposition 5.2; not in the preprint). *Hidden hypotheses:* none. *Deviation (same):* The text gives no proof. Existence: Proposition 5.1.3's glued norm over the reflexive charts is an auxiliary norm by Lemma 5.1.2; near $P(y)$ only finitely many terms matter, in each chart the fibre parts converge weakly, and the norm is weakly lower semicontinuous. Comparison: a reflexive auxiliary norm is an auxiliary norm, then Proposition 5.1.3.

V5.82 (Definition 5.2.6 (2017: 5.6), reflexive local compactness; `Polyfold.StrongBundle.HasReflexiveLocalCompactness`, `Polyfold.TrivialBundle.hasReflexiveLocalCompactness_bundle`). `StrongBundle.HasReflexiveLocalCompactness B f`: for every `N` with `B.IsReflexiveAuxNorm N` and every `x` there is an open `U` `x` with `IsCompact (closure {q U | B.AuxLe N (f q) 1})`. Witness: every section of the trivial bundle (`TrivialBundle.hasReflexiveLocalCompactness_bundle`). *Book:* §5.2, Definition 5.2.6, p. 214 (2017: Definition

5.6; not in the preprint). *Hidden hypotheses:* none. *Deviation (same):* The set prints "$N(f(x)) \leq 1$" inside $\{q \in U(x) \mid ...\}$; presumably $N(f(q))$. $N = \infty$ off $Y_{0,1}$ is `AuxLe`. The standing assumptions (reflexive fibres, sc-smooth, regularizing) are hypotheses of the statements that use the notion.

V5.83 (Proposition 5.2.7 (2017: 5.3), $N(f) > 1$ near points off X_1; `Polyfold.proposition_5_2_7`, `Polyfold.StrongBundle.exists_nhds_not_auxLe`, `Polyfold.StrongBundle.not_auxLe_of_not_mem_levelSet`, `Polyfold.StrongBundle.isClosed_setOf_auxLe`, `Polyfold.StrongBundle.exists_mixedTendsto_subseq`). `proposition_5_2_7`, with the book's hypotheses, both parts. Stronger: (2) holds for every continuous section, with no regularity and no local compactness (`StrongBundle.exists_nhds_not_auxLe`); f regularizing gives $N(f(x)) = \infty$ off X_1 (`not_auxLe_of_not_mem_levelSet`); $\{N \circ f \leq r\}$ is closed (`isClosed_setOf_auxLe`). *Book:* §5.2, Proposition 5.2.7, p. 214 (2017: Proposition 5.3; not in the preprint). *Hidden hypotheses:* none. *Deviation (stronger):* The local compactness hypothesis is not used; (2) needs only continuity of f. The sequence argument uses that X is metrizable (Theorem 2.3.10); the mixed convergent subsequence is `exists_mixedTendsto_subseq` (Lemma 5.1.2 bound in a reflexive chart and weak sequential compactness).

V5.84 (The remark after Proposition 5.2.7, the closure of $\{N(f) \leq 1\}$; `Polyfold.StrongBundle.closure_auxLe_subset`). `StrongBundle.closure_auxLe_subset`: for every U, every continuous section, reflexive fibres and a reflexive N. *Book:* §5.2, after Proposition 5.2.7, p. 214 (not in the preprint). *Hidden hypotheses:* none. *Deviation (stronger):* The printed set is $\mathrm{cl}_X(\{x \in X \mid ...\})$; presumably $x \in U$ (the argument above it is about $x_k \in U$). No compactness assumption is needed.

V5.85 (Theorem 5.2.8 (2017: 5.4), compactness control extends from the boundary; `Polyfold.theorem_5_2_8`, `Polyfold.theorem_5_2_8_of_continuous`, `Polyfold.TrivialBundle.theorem_5_2_8_witness`). `theorem_5_2_8`, with the book's hypotheses (`PX.IsTame`, `B.HasReflexiveFibers`, `B.IsScFredholmSection f`, S compact, `B.HasReflexiveLocalCompactness f`, `B.IsReflexiveAuxNorm N`, $U_\partial = W \cap \partial X$ with W open). `theorem_5_2_8_of_continuous`: the same for every continuous section on every M-polyfold. Witness of all hypotheses at once: `TrivialBundle.theorem_5_2_8_witness` ($\mathbb{R}^n$, the graph section of the identity, $U_\partial = \emptyset$). *Book:* §5.2, Theorem 5.2.8, p. 215 (2017: Theorem 5.4; not in the preprint). *Hidden hypotheses:* none. *Deviation (stronger):* Tameness and the sc-Fredholm property are not used: only continuity of f, compactness of S, local compactness at the given N, and ∂X closed. The text's $\widetilde{U}$ is $U(\bar{x}_1) \cup \cdots \cup U(\bar{x}_\ell) \cup \{N \circ f > 1\}$, open by Proposition 5.2.7 (2).

V5.86 (Remark 5.2.9 (2017: 5.1), Theorem 5.2.8 in words; `Polyfold.theorem_5_2_8`). The statement of `theorem_5_2_8`. *Book:* §5.2, Remark 5.2.9, p. 215 (2017: Remark 5.1; not in the preprint). *Hidden hypotheses:* none. *Deviation (restated):* none.

V5.87 (Remark 5.2.10 (2017: 5.2), use in inductive perturbation arguments). A comment; nothing to formalize. *Book:* §5.2, Remark 5.2.10, p. 216 (2017: Remark 5.2; not in the preprint). *Hidden hypotheses:* none. *Deviation (nothing):* none.

V5.88 (Definition 5.4.1 (2017: 5.11), sc$^+$-sections over ∂X; `Polyfold.StrongBundle.IsBoundaryScPlusSection`, `Polyfold.StrongBundle.isBoundaryScPlusSection_fiberZero`, `Polyfold.StrongBundle.boundarySupport`, `Polyfold.IsScPlusOnFaceNear`). `StrongBundle.IsBoundaryScPlusSection B s`: for $x \in \partial X$, $P(s(x)) = x$ and $s(x) \in W_{0,1}$; and for every tame strong bundle chart (`B.IsTameChart c`), every tame presentation T, every $x \in \partial X$ in the chart and every $i \in I(\varphi(x))$, the local representative restricted to $O \cap V \cap \{t_i = 0\}$ is an sc$^+$-section for some open $V \ni \varphi(x)$, and that face germ is an sc-retract (`IsScPlusOnFaceNear`). The zero section satisfies it for every strong bundle (`isBoundaryScPlusSection_fiberZero`). *Book:* §5.4, Definition 5.4.1, p. 228 (2017: Definition 5.11; not in the preprint). *Hidden hypotheses:* none. *Deviation (restated):* The local faces at x are the germs $O \cap \{t_i = 0\}$, $i \in I(\varphi(x))$, in tame coordinates; "the restriction is sc$^+$" is read chart-wise, in every tame chart and presentation (the all-charts quantifier is satisfiable: the zero section). $s(x) \in W_{0,1}$ is stated outright, so that $N(s(x))$ in Theorem 5.4.2 is defined at every boundary point.

V5.89 (Theorem 5.4.2 (2017: 5.9), the ∂-extension theorem; `Polyfold.theorem_5_4_2`, `Polyfold.StrongBundle.exists_boundary_extension`, `Polyfold.StrongBundle.exists_local_bounded`, `Polyfold.StrongBundle.exists_local_extension`, `Polyfold.StrongBundle.degIndex_ne_zero_of_mem_boundary`, `Polyfold.StrongBundle.isScPlusSection_of_locally_p3b`). `theorem_5_4_2`, with the hypotheses `B.IsTameBundle`, `PX.AdmitsScPartitionsOfUnity`, `B.IsBoundaryScPlusSection s`, `B.IsAuxNorm N`, $\tilde{U}$ open containing `boundarySupport PX P s`, f continuous, $f \geq 0$, `tsupport f   Ũ` and $N(s(x)) < f(x)$ on the support: there is an sc$^+$-section $\tilde{s}$ with $N(\tilde{s}(x)) \leq f(x)$, `sectionSupport P s̃   Ũ` and $\tilde{s} = s$ on ∂X. The proof is complete: the local step `exists_local_extension` (Proposition 12.3.7 in a tame chart, then a bump function), the local bound `exists_local_bounded`, the gluing `exists_boundary_extension`. *Book:* §5.4, Theorem 5.4.2, p. 228 (2017: Theorem 5.9; not in the preprint). *Hidden hypotheses:* none. *Deviation (same):* "Strong bundle over a tame M-polyfold" is the project's `IsTameBundle` (the base is tame and every point lies in a tame strong bundle chart); this forces `X Y : Type`, and the universe-polymorphic core `exists_boundary_extension` takes only the chart hypothesis. supp(s) is the closure of $\{x \in \partial X : s(x) \neq 0_x\}$. The proof does not use that $\tilde{U}$ is open or contains supp(s); both are kept as in the text.

V5.90 (Weak sequential compactness in reflexive spaces (tool); `Polyfold.exists_subseq_weakTendsto`, `Polyfold.exists_subseq_weakTendsto_of_eventually`, `Polyfold.WeakTendsto`, `Polyfold.WeakTendsto.tendsto_of_isCompactOperator`, `Polyfold.weakTendsto_of_bounded_of_tendsto`, `Polyfold.WeakTendsto.exists_bound`, `Polyfold.WeakTendsto.eventually_norm_gt`, `Polyfold.isReflexive_of_finiteDimensional`). In a reflexive real normed space every bounded sequence has a weakly convergent subsequence (`exists_subseq_weakTendsto`; `WeakTendsto u h` means $g(u_k) \to g(h)$ for all $g \in F^*$), through the separable closed span, countably many norming functionals, a diagonal subsequence, Banach–Alaoglu in the bidual and reflexivity. Tools: a compact operator maps weakly convergent sequences to norm convergent ones (`WeakTendsto.tendsto_of_isCompactOperator`), bounded and level-0 convergent gives weakly convergent (`weakTendsto_of_bounded_of_tendsto`), uniform boundedness (`WeakTendsto.exists_bound`), weak lower semicontinuity of the norm

(`WeakTendsto.eventually_norm_gt`), finite-dimensional spaces are reflexive (`isReflexive_of_finiteDimensional`). *Book:* not in the book (used in §§5.1–5.2 and Part III, §12.2). *Hidden hypotheses:* none. *Deviation (extra):* none.

V5.91 (Rescaled and sandwiched (reflexive) auxiliary norms, M-polyfold forms (tool); `Polyfold.StrongBundle.IsAuxNorm.comp_mul`, `Polyfold.StrongBundle.IsReflexiveAuxNorm.comp_mul`, `Polyfold.StrongBundle.exists_reflexiveAuxNorm_sandwich`). `StrongBundle.IsAuxNorm.comp_mul`, `StrongBundle.IsReflexiveAuxNorm.comp_mul`: $(g \circ P) \cdot N$ is a (reflexive) auxiliary norm for continuous $g > 0$. `StrongBundle.exists_reflexiveAuxNorm_sandwich`: with reflexive fibres every auxiliary norm N' has reflexive $N_1 \leq N' \leq N_2$. *Book:* not in the book in this form (M-polyfold forms of Remark 12.2.12 (1) and of the last clause of Theorem 12.2.11, p. 577). *Hidden hypotheses:* none. *Deviation (extra):* none.

What is not verified in this chapter

- p. 213 (preprint p. 173), Corollary 5.2.5 (preprint 5.8) as one statement for all i: it is verified for each fixed i by iterating the shift ($i = 1, 2$ stated; V5.13, V5.14); a family P^i indexed by i and the tame form of the conclusion (the tameness of the index raises of tame charts) are not formalized (Findings, items 145, 146). The identification of the iterated base $(X^1)^1$ with X_2 as a topological space, listed here before, is verified since (V2.93; gap G.6, closed).
- preprint p. 174, the non-completeness of $(\Gamma_U^{+,1}(P), \rho)$ in the text's setting: it is proved for the trivial bundle over $\mathbb{R}$, and it fails for $U = \emptyset$ and over a point (V5.21); a general sufficient condition is not formalized, and neither is the identification of the completion of Remark 5.3.4 (preprint 5.12) with the uniform N-limits of allowable sections (V5.22) (Findings, items 152, 153).
- preprint p. 173 and preprint p. 192, Definitions 5.3.1 and 5.5.2 (preprint 5.9 and 5.29): whether the weak reading gives a bump function non-zero at a non-smooth point (V5.16; Findings, item 163).
- p. 218 (preprint p. 175), Remark 5.3.6 (preprint 5.14): a comment on [21], nothing to formalize (gap G.43). Remark 5.3.8 (preprint 5.16) is verified since 2026-11-12 for an arbitrary sc-Fredholm arc (V5.73, gap G.44 closed), and the dense form of Theorem 5.3.12 (preprint 5.19) too (V5.40, gap G.45 closed).
- preprint p. 180, Finding 88: at a zero $(0, x)$, $x \in \partial X$, where all t_i vanish, the parametrized section is in good position only if ker $f'(x) + E_x \supseteq T_xX$, a general position condition that good position does not imply; checked on paper only. The proof of Theorem 5.3.12 (preprint 5.19) avoids the point (V5.40) (gap G.46).
- p. 223, Theorem 5.3.15 (preprint 5.22): verified since 2026-11-12 (V5.71, V5.72; gap G.48 closed), with "Morse-type" read through smooth parametrizations of the solution set; the comparison with the charts of the Mathlib manifold structure of the solution set is not formalized (Findings, item 183).
- preprint p. 188, Theorem 5.27 (preprint) (cobordism between good position perturbations): its proof is left to the reader, and the theorem is not in the 2021 edition; its general-position

analogue is verified (V5.73), the good-position form would need the step of G.46 at $\{i\} \times \partial X$ (gap G.49; Findings, items 104, 186).

- pp. 230–231 (preprint pp. 193–194), the L^p and Sobolev remark: verified (V5.76; gap G.51) except the sc-structure $E_m = W^{1+m,4}(\Omega)$ itself on bounded domains other than $(0,1) \subset \mathbb{R}^1$: `Polyfold.SobolevRemark.sobolevSc` takes the compactness of the inclusions and the density of E_∞ as hypotheses, discharged for $(0,1)$, every $p > 1$ (`Polyfold.SobolevCube.cubeSc`), not for cubes in $\mathbb{R}^n$, $n \geq 2$, or other bounded domains (Findings, items 118, 185). Theorem 5.5.5 (V5.75, gap G.50) and Theorems 5.5.13, 5.5.16 (V5.59, V5.62; gaps G.52, G.53) are verified since 2026-11-12.
- Definitions 5.3.7 and 5.3.9 (preprint 5.15 and 5.17) in T_xX: verified in charts (V5.23, V5.31), with chart independence for Definition 5.3.9 (preprint 5.17) at the zeros near a zero in good position (V5.34), since T_xX carries no linear structure here (gap G.7 of Chapter 2).
- p. 209, the remark after Definition 5.2.1 (2021) that for sc-smooth sections the five compactness properties are in general not equivalent: no example is formalized (Findings, item 171).
- p. 216, Remark 5.2.10 (2017: 5.2): a comment on inductive perturbation arguments, nothing to formalize (V5.87). Readings of §§5.1–5.2 and §5.4 (2026-11-13): Findings, Chapter 5, items 187–192.

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7. Chapter 6 (HWZ §§6.2–6.8 (preprint §§6.1–6.7)): verified except the global parts of Theorem 6.6.18 (preprint 6.41)

Chapter 6 of Part I (pp. 237–286 (preprint pp. 200–237)) builds the orientation theory of sc-Fredholm sections. §6.2 (preprint §6.1) defines the space $\mathrm{Lin}(f,x)$ of linearizations at a smooth point (Definition 6.2.1 (preprint 6.1)) and the index; Proposition 6.2.2 (preprint 6.2), the stability of the index along paths, was verified with Chapter 3 (V3.98). §6.3 (preprint §6.2) fixes the conventions for determinant lines $\lambda(E) = \Lambda^{\max}E$: the isomorphism $\iota : \lambda(E^*) \to \lambda^*(E)$ (Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4)) and the isomorphism Φ_E of a four-term exact sequence, formula (91) (Lemma 6.3.3 (preprint 6.5), Proposition 6.3.5 (preprint 6.6)). §6.4 (preprint §6.3) defines the determinant $\det(T) = \lambda(\ker T) \otimes \lambda^*(\operatorname{coker} T)$ of a Fredholm operator and its orientations (Definitions 6.4.1, 6.4.2 (preprint 6.7, 6.8)), the good left-projections Π_T (Definition 6.4.3 (preprint 6.9), Lemma 6.10 (preprint; not in the 2021 edition)), the isomorphisms $\gamma_T^P : \det(T) \to \det(PT)$ (Proposition 6.4.10 (preprint Lemma 6.11), Proposition 6.4.11 (preprint 6.12), Corollary 6.4.12 (preprint 6.13)), the stabilization T_φ with formula (94) (Lemma 6.4.14 (preprint 6.14), Proposition 6.4.15 (preprint 6.15)) and the direct sum convention (Definition 6.4.16 (preprint 6.16)). §6.5 (preprint §6.4) builds the local determinant bundles $\mathrm{DET}(T,P,\varepsilon)$ (Lemmas 6.5.1–6.5.3 (preprint 6.17–6.19), Proposition 6.5.4 (preprint 6.20), Lemma 6.5.5 (preprint 6.21)) and the determinant line bundle $\mathrm{DET}(E,F)$ over the Fredholm operators (Definition 6.5.6 (preprint 6.22), Propositions 6.5.7, 6.5.8 (preprint 6.23, 6.24), Lemma 6.5.9 (preprint 6.25)). §6.6 (preprint §6.5) restricts it to $\mathrm{Lin}(f,x)$ (Proposition 6.6.1 (preprint 6.26), Definition 6.6.2 (preprint 6.27)), propagates orientations along sc-smooth paths in small neighbourhoods (Proposition 6.6.3 (preprint 6.28), Lemmas 6.6.4–6.6.6

(preprint 6.29–6.31), Proposition 6.6.7 (preprint 6.32) with Lemmas 6.6.8, 6.6.10 (preprint 6.33, 6.34), Theorem 6.6.11 (preprint 6.35), Definition 6.6.12 (preprint 6.36)), globalizes the propagation to a functor Γ_f on the path category (Theorem 6.6.13 (preprint 6.37)), defines orientability and orientations of sc-Fredholm sections (Definitions 6.6.14–6.6.16 (preprint 6.38–6.40)), and orients the solution set (Theorem 6.6.18 (preprint 6.41)) and the sections $f + s$ over $X \times [0, 1]$ (Remark 6.6.19 (preprint 6.42)). §6.7 (preprint §6.6) states the invariant $\Psi_{(f,o)}$ (Theorem 6.7.1 (preprint 6.43)). Appendix 6.8 (preprint 6.7) proves Lemma 6.3.3 (preprint 6.5) (§6.8.1 (preprint §6.7.1)) and Proposition 6.4.11 (preprint 6.12) (§6.8.2 (preprint §6.7.2)).

What is verified. **Theorems 6.6.11 and 6.6.13 (preprint 6.35 and 6.37) hold**, with the readings recorded below, and so do Propositions 6.3.2, 6.3.5, 6.4.11, 6.4.15, 6.5.4, 6.5.7, 6.5.8, 6.6.1, 6.6.7 (preprint 6.4, 6.6, 6.12, 6.15, 6.20, 6.23, 6.24, 6.26, 6.32), Proposition 6.6.3 (preprint 6.28) with (1′) in place of (1), Lemma 6.3.3, Proposition 6.4.10, Lemma 6.4.14, Lemma 6.5.1, Lemma 6.5.2, Lemma 6.5.5, Lemma 6.5.9, Lemma 6.6.6, Lemma 6.6.8, Lemma 6.6.10 (preprint Lemmas 6.5, 6.11, 6.14, 6.17, 6.18, 6.21, 6.25, 6.31, 6.33, 6.34) and Lemma 6.10 (preprint), Lemma 6.6.4 (preprint 6.29) in the form the text uses afterwards, the kernel statement of Lemma 6.6.5 (preprint 6.30), Corollary 6.4.12 (preprint 6.13), the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), Theorem 6.6.18 (preprint 6.41) in the form of tangent orientations that vary continuously along the paths of zeros in a chart, and Remark 6.6.19 (preprint 6.42) at a point. Every definition of the chapter is formalized: Definitions 6.2.1 (preprint 6.1; V6.1), 6.3.1 (preprint 6.3; V6.3), 6.4.1 (preprint 6.7; V6.8), 6.4.2 (preprint 6.8; V6.9), 6.4.3 (preprint 6.9; V6.10), 6.4.16 (preprint 6.16; V6.19), 6.5.6 (preprint 6.22; V6.25), 6.6.2 (preprint 6.27; V6.31), 6.6.12 (preprint 6.36; V6.50), 6.6.14 (preprint 6.38; V6.58), 6.6.15 (preprint 6.39; V6.59), 6.6.16 (preprint 6.40; V6.60). Several places of the text needed more than a reading. (a) *Lemma 6.10 (preprint).* The projection of the text's proof does not lie below the two given projections, as we read it (an example on $\mathbb{R}^2$); the lemma holds by another construction, without the hypothesis that T is Fredholm (V6.11). The 2017 version drops the lemma and remarks that a common lower bound need not exist; with the order of Definition 6.4.3 (preprint 6.9) it always exists, so the preprint's route to Lemma 6.5.9 (preprint 6.25) is the one formalized. (b) *Signs.* The isomorphism Φ_E of formula (94) does not vary continuously in the topology of DET where $\dim \operatorname{coker} T$ changes parity, as we read it, so the orientations it carries along a family need not be continuous (an example, V6.40); the propagation uses $(-1)^{\dim \operatorname{coker} T}\Phi_E$ at both end points (V6.41). The map $h \mapsto \widetilde{h}$ of Remark 6.6.19 (preprint 6.42) has the same property, and the same sign is used there (V6.62). (c) *Contractibility.* Proposition 6.6.3 (1) (preprint 6.28), the sc-smooth contractibility of the neighbourhood, is not established by its proof as we read it; what Theorem 6.6.11 (preprint 6.35) uses is (1′), two paths with the same end points are homotopic in a larger region where (3) holds, and that is proved (V6.32, V6.48). (d) *The chart-free objects.* T_xX, Y_x, $\operatorname{Lin}(f, x)$ and Or^f_x are made in a chart point and identified between charts by explicit transition maps, and the local propagation is shown to be independent of the chart (V6.53, V6.54). (e) *Omitted proofs.* The orientability of a line bundle over a convex set (V6.29), the general position of Lemma 6.6.4 (preprint 6.29) (V6.33), the "simple computation" of Lemma 6.6.8 (preprint 6.33) (V6.45), the covering argument of Theorem 6.6.13 (preprint 6.37) (V6.56) and the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), stated without proof (V6.59), are proved. Not verified: Theorem 6.7.1 (preprint 6.43) and the notion of an admissible section of §6.7 (preprint §6.6), Proposition 6.6.3 (1) (preprint 6.28), the orientation of the solution set as an abstract manifold and the perturbation paragraph after Theorem 6.6.18 (preprint 6.41), the global part of Remark 6.6.19 (preprint 6.42), the general Lemma 6.5.3 (preprint 6.19), several remarks, and the facts on paths recalled on preprint p. 201 (see the list at the end of this section). A fifth addendum (2026-11-08; Findings, items 110–115) covers the Springer 2021 edition: Lemma 6.4.4 (2017: 6.3) with its new index equalities (V6.65), Proposition 6.4.13 (2017: 6.6) without its last sentence (V6.66, V6.67), and Lemmas 6.6.8 and 6.6.9, the two halves of preprint Lemma 6.33 (V6.68). A sixth addendum (2026-11-08; Findings, items 116–121) adds the last sentence of Proposition 6.4.13, with a closed common complement (V6.69, V6.70; gap G.81, closed), and Theorem 6.6.17, the topology of the orientation space O_f (V6.71–V6.75): exactly one Hausdorff topology with properties (1) and (3), and with (2) when small open sets have connected smooth parts (gap G.90); the remarks after the theorem are gap G.91. A seventh addendum (2026-11-10; Findings, items 128–134) adds Theorem 6.7.1 and the case $\partial X = \emptyset$ (V6.76, V6.77, narrower): proved under two hypotheses that are steps of the text's proof, the orientation of the cobordism S_F (gap G.66) and a relative form on it (gap G.54), with the admissible sections of §6.7 (gap G.55, closed). An eighth addendum (2026-11-12; Findings, items 135–136) adds the remarks on the alternative determinant and on the identification (6.25) (V6.78, V6.79; gaps G.57, G.58 closed), the remark that DET(E, F) is not orientable in a corrected form, which does not hold as we read it for finite-dimensional E, F (V6.80, narrower), and the classical determinant bundle (V6.81) (gap G.60 closed), the remarks after Theorem 6.6.17 (V6.82; gap G.91 closed), property (P) of Lemma 6.6.4 (V6.33, now restated; gap G.63 closed), Lemma 6.6.5 with the fibres, π and L as a smooth vector bundle (V6.34, V6.42, now restated; gap G.62 closed), Remark 6.6.19 as an orientation of $\widehat{f}$ in the sense of Definition 6.6.16 (V6.62, now restated; gap G.66 closed), Proposition 6.6.3 (3), (4), Lemmas 6.6.4–6.6.5 and the path category on $[0, 1]$ and $[0, 1]^n$ themselves (V6.32, V6.33, V6.34, V6.52; gap G.65 in part), and Theorem 6.7.1 without hypotheses on the cobordism: the orientation $e \wedge o_0$ of S_F is $e \wedge o_{S_i}$ on its ends, and the relative form is built on inner ends (V6.76, V6.77, now restated; gaps G.54, G.66 closed). A ninth addendum (2026-11-13; Findings, items 137, 138) verifies Proposition 6.6.3 (1) in its 2021 form, "$U' \subset U$ is sc-smoothly contractible in U to the point x" (V6.32, now restated; gap G.61 closed; the earlier reading, U itself contractible, was the preprint's), (1′) for paths on $[0, 1]$ with sc-smooth homotopies on $[0, 1]^2$, and the bundle L of Lemma 6.6.5 over the manifold with boundary $[0, 1]$ for $n = 1$ (V6.34; gap G.65 in part).

The setting. X and Y are paracompact Hausdorff spaces, as for every M-polyfold of the project, and from §6.6 (preprint §6.5) on X is tame and f is an sc-Fredholm section with tame germ data (Definition 3.6.4 (preprint 3.51)), the reading of Chapter 5. The linearization $(f - s)'(x) : T_xX \to Y_x$ is read in a strong bundle chart, at a *chart point* (c, z) of x, as the operator $T_zO = Dr(z)E \to K_z = \rho(z)F$ on level 0, because T_xX carries no linear structure here (gaps G.6, G.7, G.13); the text's $f - s$ with $s(x) = f(x)$ is $f + s$ with $(f + s)(x) = 0$. The sc$^+$-sections are global, made with sc-smooth bump functions: the text assumes sc-smooth partitions of unity from §6.3 (preprint §6.2) on (preprint p. 203), and these give bump functions (V5.49) (Findings, item 43). Paths, homotopies and families are maps on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ of which only $[0, 1]$, $[0, 1]^2$, $[0, 1]^n$ matters (decision D2, as for Proposition 6.2.2 (preprint 6.2) in V3.98), and $X \times [0, 1]$ is read as $\mathbb{R}^n \times X$ with the pull-back π^*P

of the bundle along the projection. An orientation of a line is a ray (Mathlib's `Module.Ray`), and an orientation of a topological line bundle is a choice of rays given near every point by a continuous nowhere-zero section. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 6".

V6.1 (Definition 6.2.1 (preprint 6.1), the space $\mathrm{Lin}(f, x)$; `Polyfold.ChartPoint`, `Polyfold.ChartPoint.T`, `Polyfold.ChartPoint.K`, `Polyfold.ChartPoint.IsScPlusOp`, `Polyfold.ChartPoint.linSet`, `Polyfold.ChartPoint.mem_linSet_iff`, `Polyfold.ChartPoint.exists_isScPlusOp_linAt_eq`, `Polyfold.ChartPoint.linSet_nonempty`, `Polyfold.ChartPoint.isScPlusOp_sub_of_mem_linSet`, `Polyfold.ChartPoint.isFredholm_isIndexAt_of_mem_linSet`, `Polyfold.ChartPoint.exists_index_linSet`, `Polyfold.ChartPoint.fredholmIndex_of_mem_linSet`, `Polyfold.ChartPoint.convex_linSet`, `Polyfold.MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`). At a chart point $p = (c, z)$ of a smooth point x, $\mathrm{Lin}(f, x)$ is the set of operators $(f + s)'(x) + a : T_zO \to K_z$ for an sc$^+$-section s with $(f + s)(x) = 0$ and an sc$^+$-operator a; for one such s it is $(f + s)'(x)$ plus the sc$^+$-operators. Two such sections give linearizations that differ by an sc$^+$-operator; the members of $\mathrm{Lin}(f, x)$ are Fredholm on level 0, all of the index $\mathrm{ind}(f, x)$ of V3.98; $\mathrm{Lin}(f, x)$ is convex, and it is not empty when P has sc$^+$-sections through the smooth points of its fibres, which bump functions give. *Book:* §6.2 (preprint §6.1), pp. 237–240 (preprint pp. 200–201) (the sections s and the paragraph before the definition), Definition 6.2.1 (preprint 6.1), p. 239 (preprint p. 201), and preprint p. 219 (convexity); the index and Proposition 6.2.2 (preprint 6.2) are V3.98. *Hidden hypotheses:* an sc$^+$-section through $f(x)$; the text constructs a local one in a chart, the project uses global ones (`HasScPlusSectionsThrough`), which exist under the standing bump functions. *Deviation (restated):* made in a chart point; global sections; "sc-Fredholm" is proved on level 0, which is what the determinant lines use, and not on every level (gap G.64). The paragraph before the definition compares $(f - s)'(x)$ with itself where $(f - t)'(x)$ is meant (possible misprint) (Findings, items 40, 41).

V6.2 (the determinant lines $\lambda(E)$, $\lambda^*(E)$ and the maps $\lambda(\Phi)$; `Polyfold.maxPow`, `Polyfold.maxPowDual`, `Polyfold.finrank_maxPow`, `Polyfold.wedge`, `Polyfold.wedge_eq_det_smul`, `Polyfold.wedge_append_basis_left`, `Polyfold.wedge_append_basis_right`, `Polyfold.wedge_append_add_left`, `Polyfold.wedge_append_add_right`, `Polyfold.dualVec`, `Polyfold.dualVec_self`, `Polyfold.eq_dualVec_of_apply_eq_one`, `Polyfold.maxPowMap`, `Polyfold.maxPowMap_wedge`, `Polyfold.maxPowMap_comp`, `Polyfold.maxPowMap_ne_zero_iff`, `Polyfold.maxPowDualMap`). $\lambda(E)$ is the top exterior power of a finite-dimensional real space and $\lambda^*(E)$ its dual, both lines; a change of basis multiplies the wedge by the determinant, also in one block of a two-block wedge, and adding to one block vectors of the span of the other does not change the wedge. For $v \neq 0$ in a line, v^* is the functional with $v^*(v) = 1$. For $\dim E = \dim F$, $\lambda(\Phi)(a_1 \wedge \cdots \wedge a_n) = \Phi a_1 \wedge \cdots \wedge \Phi a_n$, with $\lambda(\Psi\Phi) = \lambda(\Psi)\lambda(\Phi)$, nonzero iff Φ is bijective; $\lambda^*(\Phi)$ is its dual. *Book:* §6.3 (preprint §6.2), pp. 241–243 (preprint pp. 203–204). *Hidden hypotheses:* none. *Deviation:* none.

V6.3 (Definition 6.3.1 (preprint 6.3); `Polyfold.iotaDual`, `Polyfold.iotaDual_wedge_wedge`,

`Polyfold.iotaDual_wedge_dualBasis`, `Polyfold.iotaDual_ne_zero`). $\iota : \lambda(E^*) \to \lambda^*(E)$, $\iota(e_1^* \wedge \cdots \wedge e_n^*)(a_1 \wedge \cdots \wedge a_n) = \det(e_i^*(a_j))$, is a nonzero map of lines, and for a basis and its dual basis $\iota(e_1^* \wedge \cdots \wedge e_n^*) = (e_1 \wedge \cdots \wedge e_n)^*$. *Book:* Definition 6.3.1 (preprint 6.3), p. 242 (preprint p. 204). *Hidden hypotheses:* none. *Deviation:* none; the convention $\lambda(\{0\}^*) = \mathbb{R}$ is the zeroth exterior power, and the text's matrix is the transpose of Mathlib's, with the same determinant.

V6.4 (Proposition 6.3.2 (preprint 6.4); `Polyfold.proposition_6_3_2`). For an isomorphism $\Phi : E \to F$ of finite-dimensional spaces, $\lambda^*(\Phi) \circ \iota = \iota \circ \lambda(\Phi^*)$. *Book:* Proposition 6.3.2 (preprint 6.4), p. 243 (preprint p. 205). *Hidden hypotheses:* none. *Deviation:* none.

V6.5 (four-term exact sequences and the isomorphism Φ_E of (91); `Polyfold.FourTermExact`, `Polyfold.FourTermExact.Choice`, `Polyfold.FourTermExact.Choice.value`, `Polyfold.FourTermExact.canonicalChoice`, `Polyfold.FourTermExact.phi`, `Polyfold.FourTermExact.phi_ne_zero`, `Polyfold.FourTermExact.phiEquiv`). For an exact sequence $0 \to A \to B \to C \to D \to 0$ of finite-dimensional spaces, a *choice* is the data of (91): complements Z of $\alpha(A)$ in B and V of $\beta(B)$ in C, bases of A, D, Z, and the γ-preimages in V of the basis of D; $\Phi_E : \lambda(A) \otimes \lambda^*(D) \to \lambda(B) \otimes \lambda^*(C)$ is the linear map given by (91) for one choice, an isomorphism of lines. *Book:* §6.3 (preprint §6.2), preprint p. 206, formula (91) with its two ordering conventions. *Hidden hypotheses:* none. *Deviation:* none.

V6.6 (Lemma 6.3.3 (preprint 6.5); `Polyfold.FourTermExact.lemma_6_3_3`). Formula (91) holds for every choice: Φ_E, defined from one choice, maps $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$ to the right-hand side of (91) computed with any other choice. *Book:* Lemma 6.3.3 (preprint 6.5), p. 244 (preprint p. 206); proof in §6.8.1 (preprint §6.7.1) (V6.63). *Hidden hypotheses:* none. *Deviation (restated):* "does not depend on the choices" is read as this, the strongest form and the one every later computation of γ_T^P uses (Findings, item 1).

V6.7 (Proposition 6.3.5 (preprint 6.6); `Polyfold.contractDual`, `Polyfold.FourTermExact.psi`, `Polyfold.FourTermExact.proposition_6_3_5`). With $\bar{\iota}(v \otimes b \otimes \varphi) = \varphi(v)\, b$ and $\Psi_E(c \otimes h) = \bar{\iota}(c \otimes \Phi_E(h))$, Ψ_E maps the wedge of the basis of C of (91), tensored with $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, to $\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b_1' \wedge \cdots \wedge b_{m-n}'$. *Book:* (92) and Proposition 6.3.5 (preprint 6.6), p. 246 (preprint p. 207). *Hidden hypotheses:* none. *Deviation (restated):* the wedge in C is taken in the order of (91); the statement exchanges the names of its two blocks (possible misprint) (Findings, item 2).

V6.8 (Definition 6.4.1 (preprint 6.7); `Polyfold.linDet`, `Polyfold.fredholmDet`, `Polyfold.finrank_fredholmDet`, `Polyfold.linDetEquivOfEq`). $\det(T) = \lambda(\ker T) \otimes \lambda^*(F/R(T))$, a line for T Fredholm; equal operators have identified determinants. *Book:* Definition 6.4.1 (preprint 6.7), p. 246 (preprint p. 208). *Hidden hypotheses:* none. *Deviation:* none; the alternative definition $\lambda(\ker T) \otimes \lambda(\mathrm{coker}(T)^*)$ mentioned after it is not formalized (gap G.57).

V6.9 (Definition 6.4.2 (preprint 6.8); `Polyfold.FredholmOrientation`, `Polyfold.nonempty_fredholmOrientation`, `Polyfold.exists_two_fredholmOrientation`). An orientation of T is a ray of $\det(T)$; a Fredholm operator has exactly two. *Book:* Definition 6.4.2 (preprint 6.8), p. 246 (preprint p. 208). *Hidden hypotheses:* none. *Deviation:* none.

V6.10 (Definition 6.4.3 (preprint 6.9) and the facts after it; `Polyfold.IsGoodLeftProj`, `Polyfold.goodLeftProjs`, `Polyfold.ProjLE`, `Polyfold.ProjLE.refl`, `Polyfold.ProjLE.antisymm`, `Polyfold.ProjLE.trans`, `Polyfold.IsGoodLeftProj.isFredholm`, `Polyfold.IsGoodLeftProj.fredholmIndex_eq_zero`, `Polyfold.IsGoodLeftProj.isFredholm_comp`, `Polyfold.IsGoodLeftProj.fredholmIndex_comp`, `Polyfold.IsGoodLeftProj.finiteDimensional_quotient_inf`, `Polyfold.exists_isGoodLeftProj`). A good left-projection for T is a bounded projection P with $\dim F/R(P) < \infty$ and $R(PT) = R(P)$; $P \le Q$ iff $P = PQ = QP$ is a partial order. A good left-projection is Fredholm of index 0, PT is Fredholm with $\operatorname{ind}(PT) = \operatorname{ind}(T)$, $R(P) \cap R(P')$ has finite codimension, and Π_T is not empty for T Fredholm. *Book:* Definition 6.4.3 (preprint 6.9) and the paragraphs after it, preprint p. 208. *Hidden hypotheses:* none; that $\Pi_T \neq \emptyset$ is used from Proposition 6.4.10 (preprint Lemma 6.11) on and is not stated. *Deviation:* none (Findings, item 3).

V6.11 (Lemma 6.10 (preprint); `Polyfold.lemma_6_10_preprint`, `Polyfold.lemma_6_10_text_reading_preprint`). For $P, P' \in \Pi_T$ there is $P'' \in \Pi_T$ with $P'' \le P$ and $P'' \le P'$, for any bounded T. On $\mathbb{R}^2$ with $T = 1$ there are $P, P' \in \Pi_T$ such that no projection with range $R(P) \cap R(P')$ lies below both. *Book:* Lemma 6.10 (preprint), preprint p. 208, proof preprint pp. 208–209. *Hidden hypotheses:* none. *Deviation (stronger):* T need not be Fredholm, and the construction differs from the text's: the text's P'' has range $R(P) \cap R(P')$, and as we read it $P'' = P''P$ does not hold for it in general; the formalized P'' projects onto $R(P) \cap R(P') \cap N$, for a closed finite-codimensional N meeting $\ker P + \ker P'$ only in 0, along a complement containing $\ker P + \ker P'$ (Findings, item 4).

V6.12 (Proposition 6.4.10 (preprint Lemma 6.11); `Polyfold.projSeqJ`, `Polyfold.projSeqPhi`, `Polyfold.projSeqPi`, `Polyfold.proposition_6_4_10`, `Polyfold.projSeq`). For $P \in \Pi_T$ the sequence $0 \to \ker T \to \ker(PT) \to F/R(P) \to \operatorname{coker} T \to 0$, with the inclusion, $x \mapsto Tx + R(P)$ and $y + R(P) \mapsto (I - P)y + R(T)$, is exact. *Book:* Proposition 6.4.10 (preprint Lemma 6.11), p. 249 (preprint p. 209). *Hidden hypotheses:* none. *Deviation:* none.

V6.13 (the isomorphism γ_T^P; `Polyfold.gammaProj`, `Polyfold.gammaProj_tmul`, `Polyfold.gammaProj_wedge_tmul`). $\gamma_T^P : \det(T) \to \det(PT)$ is $\Phi_{E(T,P)}$ followed by the identification of $\lambda^*(F/R(P))$ with $\lambda^*(F/R(PT))$; on wedges it is given by (91), for any choice. *Book:* preprint p. 210. *Hidden hypotheses:* none. *Deviation:* none; "recalling $R(P) = R(PT)$" is read as this identification (Findings, item 5).

V6.14 (Proposition 6.4.11 (preprint 6.12); `Polyfold.proposition_6_4_11`, `Polyfold.IsGoodLeftProj.of_projLE`, `Polyfold.comp_comp_eq_of_projLE`). For $P \le Q$ in Π_T, QT is Fredholm, $P \in \Pi_{QT}$, and $\gamma_{QT}^P \circ \gamma_T^Q = \gamma_T^P$, up to the identification $\det(P(QT)) = \det(PT)$ of the equal operators $PQT = PT$. *Book:* Proposition 6.4.11 (preprint 6.12), p. 251 (preprint p. 211); proof in §6.8.2 (preprint §6.7.2) (V6.64). *Hidden hypotheses:* none. *Deviation:* none.

V6.15 (Corollary 6.4.12 (preprint 6.13); `Polyfold.corollary_6_4_12`). For $P \le S$ and $P \le R$ in Π_T, $(\gamma_{RT}^P)^{-1} \circ \gamma_{ST}^P = \gamma_T^R \circ (\gamma_T^S)^{-1}$, so the left side does not depend on P. *Book:* Corollary 6.4.12 (preprint 6.13), p. 251 (preprint p. 211). *Hidden hypotheses:* none. *Deviation:* none.

V6.16 (the stabilization and Lemma 6.4.14 (preprint 6.14); `Polyfold.stabJ`, `Polyfold.stabPi`, `Polyfold.stabClass`, `Polyfold.lemma_6_4_14`, `Polyfold.stabSeq`, `Polyfold.exists_surjective_coprod`). For T Fredholm and $\varphi : \mathbb{R}^n \to F$ with $T_\varphi(e, r) = Te + \varphi r$ onto, the sequence $0 \to \ker T \to \ker T_\varphi \to \mathbb{R}^n \to F/R(T) \to 0$, with $j(e) = (e, 0)$, $\pi(e, r) = r$ and $[\varphi](r) = \varphi(r) + R(T)$, is exact; a stabilizing φ exists for every Fredholm T. *Book:* pp. 251–256 (preprint pp. 211–212) and Lemma 6.4.14 (preprint 6.14), p. 256 (preprint p. 212). *Hidden hypotheses:* none. *Deviation:* none; the proof writes $j(e) = (0, e)$ for $(e, 0)$ (possible misprint) (Findings, item 8).

V6.17 (formula (94); `Polyfold.stdWedge`, `Polyfold.stabOne`, `Polyfold.stabPsi`, `Polyfold.stabPsi_dualVec`). $\psi : \lambda^*(\mathbb{R}^n) \to \lambda^*(\{0\})$, $(e_1 \wedge \cdots \wedge e_n)^* \mapsto 1^*$ for the standard basis, satisfies $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c_1, \ldots, c_n]^{-1}\, 1^*$ for every basis. *Book:* formula (94), p. 256 (preprint p. 212). *Hidden hypotheses:* none. *Deviation:* none.

V6.18 (Proposition 6.4.15 (preprint 6.15); `Polyfold.stabIota`, `Polyfold.stabIota_bijective`, `Polyfold.stab_card_eq`, `Polyfold.isCompl_range_stabPi_span`, `Polyfold.proposition_6_4_15`). $\iota_\varphi = (1 \otimes \psi) \circ \Phi_E : \det(T) \to \det(T_\varphi)$ is an isomorphism, and for a basis a of $\ker T$, $\bar{c}_1, \ldots, \bar{c}_m$ with $[\varphi](\bar{c}_i)$ a basis of $F/R(T)$ and $\bar{b}_i = (b_i, r_i)$ completing $(a, 0)$ to a basis of $\ker T_\varphi$, $\iota_\varphi(h) = \det(\bar{c}, r)^{-1}\,((a_1, 0) \wedge \cdots \wedge (a_k, 0) \wedge \bar{b}_1 \wedge \cdots \wedge \bar{b}_l) \otimes 1^*$; the r_i and $\bar{c}_j$ form a basis of $\mathbb{R}^n$. *Book:* Proposition 6.4.15 (preprint 6.15), p. 257 (preprint p. 213). *Hidden hypotheses:* none. *Deviation (restated):* the identity $m + l = n$, needed to form $\det(\bar{c}, r)$, is a hypothesis of the statement and is proved to hold always; that the r_i and $\bar{c}_j$ form a basis, which the text notices, is proved; the statement calls ψ “φ” (possible misprint). The identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after the proposition are not formalized (gap G.58) (Findings, items 9, 10).

V6.19 (Definition 6.4.16 (preprint 6.16); `Polyfold.kerInl`, `Polyfold.kerInr`, `Polyfold.cokerInl`, `Polyfold.cokerInr`, `Polyfold.kerProdEquiv`, `Polyfold.cokerProdEquiv`, `Polyfold.finrank_ker_prodMap`, `Polyfold.finrank_coker_prodMap`, `Polyfold.isFredholm_prodMap`, `Polyfold.directSumVec`, `Polyfold.directSumDet`, `Polyfold.directSumVec_eq_smul`, `Polyfold.definition_6_4_16`, `Polyfold.directSumDet_bijective`). $T \oplus S$ is Fredholm, with $\ker(T \oplus S) \cong \ker T \times \ker S$ and $\operatorname{coker}(T \oplus S) \cong \operatorname{coker} T \times \operatorname{coker} S$, and $\det(T) \otimes \det(S) \to \det(T \oplus S)$, $h \otimes h' \mapsto (a \wedge a') \otimes (d \wedge d')^*$, is an isomorphism satisfying the formula for every choice of bases. *Book:* Definition 6.4.16 (preprint 6.16), p. 258 (preprint p. 214). *Hidden hypotheses:* none. *Deviation:* none; “natural” is read as the formula for every choice, as in Lemma 6.3.3 (preprint 6.5) (Findings, item 11).

V6.20 (Lemma 6.5.1 (preprint 6.17); `Polyfold.FredholmOp`, `Polyfold.projRestrict`, `Polyfold.lemma_6_5_1`). For a bounded T, $P \in \Pi_T$ and a topological complement Y of $\ker(PT)$, there is $\varepsilon > 0$ such that for $\|S - T\| < \varepsilon$, $P \in \Pi_S$ and $PS|_Y : Y \to R(P)$ is a topological isomorphism. *Book:* Lemma 6.5.1 (preprint 6.17), p. 258 (preprint p. 214). *Hidden hypotheses:* none. *Deviation (stronger):* the hypothesis that T is Fredholm is not used; a topological complement is an algebraic complement with a continuous projection (Findings, item 20).

V6.21 (Lemma 6.5.2 (preprint 6.18); `Polyfold.LocalDetChart`, `Polyfold.LocalDetChart.`

`ySol`, `Polyfold.LocalDetChart.kerMap`, `Polyfold.LocalDetChart.kerProj`, `Polyfold.LocalDetChart.lemma_6_5_2`). A *local determinant chart* is the data (T, P, Y, ε) of Lemma 6.5.1 (preprint 6.17) with its conclusions. The solution $y(k, S) = -(PS|_Y)^{-1}PSk$ is continuous in (k, S), and $(S, k) \mapsto (S, k + y(k, S))$ is a continuous map onto $\bigcup_{\|S-T\|<\varepsilon}\{S\} \times \ker(PS)$, inverted by the continuous projection onto $\ker(PT)$ along Y. *Book:* Lemma 6.5.2 (preprint 6.18), p. 259 (preprint p. 215). *Hidden hypotheses:* none. *Deviation (restated):* "is a trivial vector bundle" is stated as this trivialization and its inverse (Findings, item 21).

V6.22 (Lemma 6.5.3 (preprint 6.19) in the case used, and Proposition 6.5.4 (preprint 6.20); `Polyfold.LocalDetChart.kerFrame`, `Polyfold.LocalDetChart.frame`, `Polyfold.LocalDetChart.frame_ne_zero`, `Polyfold.proposition_6_5_4`). Over the ball of a local determinant chart, $\omega(S) = (k_1(S) \wedge \cdots \wedge k_n(S)) \otimes (e_1 \wedge \cdots \wedge e_l)^*$, with $k_i(S) = k_i + y(k_i, S)$ and a constant basis e of $F/R(P) = F/R(PS)$, is a nowhere-zero frame of $\mathrm{DET}(T, P, \varepsilon)$, and the frames of two charts with the same P differ by a continuous nowhere-zero factor. *Book:* Lemma 6.5.3 (preprint 6.19) and Proposition 6.5.4 (preprint 6.20), p. 260 (preprint p. 216). *Hidden hypotheses:* none. *Deviation (narrower):* Lemma 6.5.3 (preprint 6.19) (exterior powers, duals, sums and tensor products of topological vector bundles) is not formalized in general (gap G.59); it is applied only to trivial bundles, and Proposition 6.5.4 (preprint 6.20) is proved with the explicit frames; "in a natural way" is read as the relation between frames (Findings, item 22).

V6.23 (Lemma 6.5.5 (preprint 6.21); `Polyfold.LocalDetChart.gammaProj_frame_eq`, `Polyfold.lemma_6_5_5`). For $Q \leq P$, γ^Q_{PS} maps the P-frame at S to a continuous nowhere-zero multiple of the Q-frame, on the common ball of any two local determinant charts. *Book:* Lemma 6.5.5 (preprint 6.21), p. 260 (preprint p. 216), sketch pp. 259–261 (preprint pp. 216–217). *Hidden hypotheses:* none. *Deviation (stronger):* two charts with different centres and radii, the form Lemma 6.5.9 (preprint 6.25) uses; the text's sketch (extend the kernel sections of PS to sections spanning $\ker(QS)$) is the proof (Findings, item 23).

V6.24 ($\mathrm{DET}(E, F)$ and the charts $\gamma_{(T,P,\varepsilon)}$; `Polyfold.DetFiber`, `Polyfold.LocalDetChart.pretriv`, `Polyfold.LocalDetChart.coord`, `Polyfold.exists_localDetChart`, `Polyfold.detPrebundle`). The fibre over a Fredholm S is $\det(S)$; a local determinant chart gives the pretrivialization $(S, h) \mapsto (S, \omega(S)^*(\gamma^P_S h))$ with model fibre $\mathbb{R}$; every Fredholm operator is the centre of one; together they form a Mathlib vector prebundle, whose topology and bundle structure are those of $\mathrm{DET}(E, F)$. *Book:* preprint p. 217. *Hidden hypotheses:* none. *Deviation (restated):* $\gamma_{(T,P,\varepsilon)}$ is read in the frame of Proposition 6.5.4 (preprint 6.20) (Findings, item 24).

V6.25 (Definition 6.5.6 (preprint 6.22); `Polyfold.detBasis`). $\mathscr{B}$ is the collection of sets lying over the ball of some chart whose image in ball × $\mathbb{R}$ is open. *Book:* Definition 6.5.6 (preprint 6.22), p. 261 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none.

V6.26 (Proposition 6.5.7 (preprint 6.23); `Polyfold.proposition_6_5_7`). $\mathscr{B}$ is a basis of the topology of the prebundle, so the topology it generates is that one. *Book:* Proposition 6.5.7 (preprint 6.23), p. 261 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none.

V6.27 (Proposition 6.5.8 (preprint 6.24); `Polyfold.proposition_6_5_8`). $\mathrm{DET}(E, F)$ is a topological vector bundle with model fibre $\mathbb{R}$, and every $\gamma_{(T,P,\varepsilon)}$, read in its frame, is a trivialization. *Book:* Proposition 6.5.8 (preprint 6.24), p. 261 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none.

V6.28 (Lemma 6.5.9 (preprint 6.25); `Polyfold.lemma_6_5_9`, `Polyfold.lemma_6_5_9_local`). The transition between two charts is $(S,t) \mapsto (S, g(S)t)$ with g continuous and nowhere zero on the common ball. *Book:* Lemma 6.5.9 (preprint 6.25), p. 262 (preprint p. 218). *Hidden hypotheses:* none. *Deviation:* none; the proof is the text's, a common lower bound $Q \leq P, P'$ (Lemma 6.10 (preprint), V6.11), Corollary 6.4.12 (preprint 6.13) and Lemma 6.5.5 (preprint 6.21).

V6.29 (orientations of line bundles; `Polyfold.IsLineOrientation`, `Polyfold.LineOrientation`, `Polyfold.LineOrientation.eq_of_apply_eq`, `Polyfold.OrCover`, `Polyfold.isCoveringMap_orCover`, `Polyfold.exists_lineOrientation_apply_eq`, `Polyfold.exists_two_lineOrientation`). For a topological line bundle, two orientations that agree on one fibre agree over a preconnected base; over a simply connected, locally path-connected base every orientation of one fibre extends, and over a nonempty such base there are exactly two orientations; the rays of the fibres form a two-sheeted covering. *Book:* the reason given in Proposition 6.6.1 (preprint 6.26) ("since the base space is contractible", p. 263 (preprint p. 219)) and the opening of §6.6 (preprint §6.5), p. 263 (preprint p. 219). *Hidden hypotheses:* none. *Deviation (not in the book):* the text cites contractibility without the theorem it needs; the proof is the lifting criterion for the orientation double cover, and nonempty convex sets of normed spaces are simply connected and locally path-connected (Findings, item 25).

V6.30 (Proposition 6.6.1 (preprint 6.26); `Polyfold.fredholmIncl`, `Polyfold.DetOn`, `Polyfold.proposition_6_6_1`). Over a nonempty convex set C of Fredholm operators, the restriction $\mathrm{DET}(E,F)|_C$ is a topological line bundle with exactly two orientations. *Book:* Proposition 6.6.1 (preprint 6.26), p. 263 (preprint p. 219), for $C = \mathrm{Lin}(f,x)$. *Hidden hypotheses:* none. *Deviation (stronger):* any nonempty convex C; $\mathrm{Lin}(f,x)$ is one (V6.1) (Findings, item 25).

V6.31 (Definition 6.6.2 (preprint 6.27); `Polyfold.ChartPoint.isFredholm_of_mem_linSet`, `Polyfold.ChartPoint.detLin`, `Polyfold.ChartPoint.Or`, `Polyfold.ChartPoint.exists_two_or`, `Polyfold.ChartPoint.orEquivRay`, `Polyfold.nonempty_equiv_of_two`, `Polyfold.equiv_eq_of_apply_eq_of_two`). At a chart point, $\mathrm{DET}(f,x)$ is the pull-back of $\mathrm{DET}(T_zO, K_z)$ to $\mathrm{Lin}(f,x)$ and Or_x^f is the set of its orientations, with exactly two elements; for each $L \in \mathrm{Lin}(f,x)$, $o \mapsto o(L)$ is a bijection onto the two orientations of $\det(L)$. Between two two-element sets there are bijections, and one value determines a bijection. *Book:* Definition 6.6.2 (preprint 6.27), p. 263 (preprint pp. 219–220). *Hidden hypotheses:* none. *Deviation (restated):* made in a chart point (V6.53 compares charts); the category $\mathcal{O}_f$ is read through its objects, with bijections as morphisms, and $\{o_x, -o_x\}$ as "exactly two elements" (Findings, item 44).

V6.32 (Proposition 6.6.3 (preprint 6.28); `Polyfold.IsScHomotopyRel`, `Polyfold.pt1`, `Polyfold.pt2`, `Polyfold.exists_contDiff_eq_id_near_Icc_mapsTo`, `Polyfold.exists_isScSmoothMap_eq_one_nhdsSet`, `Polyfold.exists_nhds_isCompact_zeros`, `Polyfold.exists_isScPlusSection_pullback_of_mapsTo_of_bump`, `Polyfold.exists_isScPlusSection_pullback_of_family`, `Polyfold.hasScPlusSectionsThrough_pullback_rn_of_bump`,

`Polyfold.proposition_6_6_3`, `Polyfold.MPolyfold.paracompactSpace_prod`, `Polyfold.paracompactSpace_pullbackSpace`). With sc-smooth bump functions, every smooth x has open $U \subseteq W$ in the domain of one chart of the atlas with (1′) any two sc-smooth paths in U with the same end points are sc-smoothly homotopic in W with the end points fixed; (2) $\{y \in \mathrm{cl}(U) \mid f(y) = 0\}$ is compact; (3) for every sc-smooth family $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n) \subseteq W$ there is an sc$^+$-section S of $\pi^*P \to \mathbb{R}^n \times X$ with $(f+S)(q, \Phi(q)) = 0$ for q near $[0,1]^n$, depending on $q \in [0,1]^n$ only through $\Phi(q)$; (4) π^*P has sc$^+$-sections through all smooth fibre points. *Book:* Proposition 6.6.3 (preprint 6.28), p. 263 (preprint p. 220), proof pp. 263–264 (preprint pp. 220–221); in 2021 (as in 2017) (1) reads "$U' \subset U$ is sc-smoothly contractible in U to the point x", in the preprint "U is sc-smoothly contractible". *Hidden hypotheses:* bump functions, for the global sections of (3) and (4) (the text assumes partitions of unity from §6.3 (preprint §6.2) on and works in a chart after "assuming $X = U$"); the paracompactness of $\mathbb{R}^n \times X$ and of π^*P, which holds. *Deviation (restated; narrower before 2026-11-13):* (1′) in place of (1): the contraction $r(ty)$ of the proof contracts $r(V)$ for a convex V, but for $U = r(V')$ it needs $r(V') \subseteq V'$, which is not stated, and as we read it (1) is not established for U (gap G.61); (3) is stated for families, as Theorem 6.6.11 (preprint 6.35) needs it for homotopies; the smoothness of x is not used (Findings, items 45, 46). *Update (2026-11-12):* (3) and (4) are also proved for paths over the M-polyfold $[0,1] \times X$ and for families on $[0,1]^n$ over the M-polyfold with corners $[0,1]^n \times X$ (`Polyfold.proposition_6_6_3_three_interval`, `Polyfold.proposition_6_6_3_four_interval`, `Polyfold.proposition_6_6_3_three_cube`, `Polyfold.proposition_6_6_3_four_cube`; gap G.65 in part); (1) remains gap G.61. *Update (2026-11-13):* (1) is proved in its 2021 form (`Polyfold.proposition_6_6_3_one`, `Polyfold.exists_isScContraction`, `Polyfold.IsScContraction`): an open $U' \ni x$ in U and an sc-smooth H on $[0,1] \times X$ with values in U, $H(0,\cdot) = x$ and $H(1,\cdot) = \mathrm{id}$ on U', $H(t,y) = \varphi^{-1} r(\varphi(x) + t(\varphi(y) - \varphi(x)))$ in a chart; the smoothness of x is used. The sentence above concerns the preprint's (1), which is not formalized, and nothing shows that it fails. (1′) also holds for paths on $[0,1]$ with sc-smooth homotopies on $[0,1]^2$ (`Polyfold.exists_isScHomotopyI`, `Polyfold.proposition_6_6_3_interval`) (gap G.61 closed; Findings, item 137).

V6.33 (Lemma 6.6.4 (preprint 6.29); `Polyfold.isZeroAt_paramSection_zero`, `Polyfold.exists_sections_isGeneralPositionGerm_of_isCompact`, `Polyfold.lemma_6_6_4`). Over a tame X with bump functions, for f with tame germ data, an sc-smooth $\Phi : \mathbb{R}^n \to X$ and an sc$^+$-section S of π^*P with $(f+S)(q, \Phi(q)) = 0$ near $[0,1]^n$, there are finitely many sc$^+$-sections $s_1, \dots, s_k$ of π^*P such that $F(\lambda, (q,y)) = f(y) + S(q,y) + \sum \lambda_i s_i(q,y)$ is in general position at $(0, (q, \Phi(q)))$ for all q near $[0,1]^n$. *Book:* the section F, preprint p. 221, Lemma 6.6.4 (preprint 6.29) and the paragraph after it, preprint p. 222. *Hidden hypotheses:* X tame, bump functions, tame germ data (the standing hypotheses of §6.6 (preprint §6.5) and of Theorem 5.3.10 (preprint 5.18)). *Deviation (restated):* the text gives no proof ("using the results from the transversality theory"); the one formalized is that of Theorem 5.3.10 (preprint 5.18). The conclusion is the general position of F at the points $(t, \varphi(t), 0)$, which the text derives from (P) and which Lemmas 6.6.5, 6.6.6 (preprint 6.30, 6.31) use; (P), the general position of each slice F_t, is derived from it, the converse of the text's order (Findings, item 47). *Update (2026-11-12; gap G.63 closed):* `Polyfold.lemma_6_6_4_propertyP`: the same sections give (P) for every q near $[0,1]^n$; the tangent vectors of the family of zeros lie in ker F' and $\mathbb{R}^n$ has no boundary (`Polyfold.isGeneralPositionGerm_slice`, `Polyfold.isParamGP_slice_of_isParamGP`). On the M-polyfold $[0,1]^n$ itself, boundary and corners included, (P) and the general position of F hold at every q and are equivalent there (`Polyfold.lemma_6_6_4_propertyP_cube`, `Polyfold.isGeneralPositionGerm_cubeSlice_iff`, `Polyfold.lemma_6_6_4_cube`), by a one-sided minimum along reduced directions (`Polyfold.mem_minimalSubspaceWith_of_hasFDerivWithinAt`).

V6.34 (the kernels of Lemma 6.6.5 (preprint 6.30); `Polyfold.StrongBundle.IsSurjLinOfIndex.finrank_linKer`, `Polyfold.exists_local_frame_linKer`, `Polyfold.exists_subspaceFamily_linKer`, `Polyfold.hasFDerivAt_scDeriv_rn`, `Polyfold.fderiv_mem_linKer`). For an sc-Fredholm section with tame germ data and a continuous family y_q of zeros in good position, read in one chart at continuously varying level-1 points, the kernels of the linearizations form a continuous family of subspaces of the model space, of dimension the index; the derivative of an sc-smooth family of zeros $\mathbb{R}^n \to X$, read in a chart, lies in the kernel of the linearization. *Book:* Lemma 6.6.5 (preprint 6.30), p. 265 (preprint p. 222). *Hidden hypotheses:* none beyond the standing ones. *Deviation (restated):* the continuation of orientations uses the continuous family of kernels; the rest of the lemma is proved as well (update below), with decision D2 and the family read in one chart containing $\Phi([0,1]^n)$, the text's "$X = U$" (Findings, items 48, 126). *Update (2026-11-09, 2026-11-12; gap G.62 closed):* near the family the solution set is a manifold with boundary and corners and a submanifold of dimension the index (`Polyfold.lemma_6_6_5_isManifold`, `Polyfold.lemma_6_6_5_isSubmanifoldOfDim`), π is a submersion on a neighbourhood of the family (`Polyfold.lemma_6_6_5_submersion_nhds`), the fibres S_q are manifolds with boundary and corners (`Polyfold.lemma_6_6_5_fibre_isManifold`), and L is a C^∞ vector bundle over an open connected neighbourhood of $[0,1]^n$, with frames $DS(z_q)b_i$ that are C^∞ into every level (`Polyfold.lemma_6_6_5_smoothBundle`, `Polyfold.exists_local_smooth_frame_linKer`). On the M-polyfold $[0,1]^n$ itself the solution set, the fibres and π, the last in the corner sense and at the chart level, are proved at every q (`Polyfold.lemma_6_6_5_isManifold_cube`, `Polyfold.lemma_6_6_5_fibre_isManifold_cube`, `Polyfold.lemma_6_6_5_submersion_cube`). *Update (2026-11-13):* for a path ($n = 1$), L is a smooth vector bundle over Mathlib's manifold with boundary $[0,1]$ itself, the pull-back of `Polyfold.lemma_6_6_5_smoothBundle` along the inclusion (`Polyfold.lemma_6_6_5_smoothBundle_interval`, `Polyfold.SmoothSubspaceFamily.comp`; Findings, item 138).

V6.35 (continuous families of subspaces and their determinant lines; `Polyfold.SubspaceFamily`, `Polyfold.SubspaceFamily.det`, `Polyfold.SubspaceFamily.Frame.pretriv`, `Polyfold.SubspaceFamily.Frame.coordChange_continuousOn`, `Polyfold.SubspaceFamily.comp`, `Polyfold.LineOrientation.comp`, `Polyfold.SubspaceFamily.transport`, `Polyfold.SubspaceFamily.transport_eq`, `Polyfold.SubspaceFamily.transport_injective`, `Polyfold.SubspaceFamily.transport_comp`). A family of n-dimensional subspaces of a normed space, spanned near every point by a continuous frame, has the line bundle $b \mapsto \lambda(L_b)$ with the frames as trivializations; families and orientations restrict along continuous maps, and over $[0,1]$ an orientation of $\lambda(L_0)$ is continued to $\lambda(L_1)$, injectively, and over a simply connected, locally path-connected base the continuation along a path depends only on its end points.

Book: pp. 264–266 (preprint pp. 222–223) (the bundles L, $\lambda(L)$ and "an orientation of the lines … determines by continuation an orientation of all the other lines"), and the bundles $\widetilde{L}$ of pp. 266–273 (preprint pp. 224–227). *Hidden hypotheses:* none. *Deviation (restated):* "vector bundle" is read as a family of subspaces of one model space with continuous local frames.

V6.36 (continuous families of kernels and their images; `Polyfold.exists_subspaceFamily_ker`, `Polyfold.SubspaceFamily.exists_map`, `Polyfold.SubspaceFamily.isLineOrientation_map`, `Polyfold.SubspaceFamily.exists_inf_ker`). The kernels of a norm-continuous family of surjective Fredholm operators over a connected base form a continuous family of subspaces; so do the images of a continuous family under a jointly continuous family of linear maps injective on its members, and orientations are carried along; cutting a continuous family by the kernel of a fixed ℓ onto on every member gives one. *Book:* the continuity statements of pp. 265–272 (preprint pp. 223–226) ("as τ varies, the bundle varies continuously"). *Hidden hypotheses:* none. *Deviation (not in the book):* "varies continuously" is read in the operator norm, and for maps into a model space as joint continuity.

V6.37 (the stabilized section in a product chart; `Polyfold.PullbackChartData.ofProd_scDeriv_r_apply`, `Polyfold.PullbackChartData.ofProd_mem_tangent_iff`, `Polyfold.PullbackChartData.mem_K_ofProd_iff`, `Polyfold.StrongBundle.scDeriv_ofProd_slice_apply`, `Polyfold.StrongBundle.scDeriv_paramSection_apply`, `Polyfold.StrongBundleChart.paramDerivAt`, `Polyfold.StrongBundle.scDeriv_paramSection_ofProd_ofProd`, `Polyfold.sliceProj`, `Polyfold.sliceIncl`, `Polyfold.linKer_inf_ker_sliceProj_eq_map`, `Polyfold.exists_sliceOps`, `Polyfold.mem_linSet_sectionAdd_pullbackSlice`, `Polyfold.surjective_coprod_of_isLinSurjAt`, `Polyfold.exists_sliceKernel`). In the product chart $(\lambda \mapsto \lambda) \times ((q \mapsto q) \times \varphi)$ at $(0,(q,z))$, the linearization of F is $(\lambda,(\sigma,v)) \mapsto D(\hat{g}_q \circ r)(z)v + \sum \lambda_i \hat{s}_i(q,\cdot)(z) + D_q\sigma$, with $g_q = f + S(q,\cdot)$; its kernel cut by $\sigma = 0$ is the image of $\ker(T \oplus \varphi)$, with $T = g_q'(x) \in \mathrm{Lin}(f,x)$ and $\varphi(\lambda) = \sum \lambda_i s_i(q,x)$; and $T \oplus \varphi$ is onto when F is in general position at $(0,(q,\Phi(q)))$. *Book:* F, preprint p. 221, F_t, L_t and (96), pp. 264–266 (preprint pp. 222–223). *Hidden hypotheses:* none. *Deviation (restated):* $\ker F_t'(\varphi(t),0)$ is read as the kernel of the linearization of the whole F cut by $\sigma = 0$ (Findings, item 49).

V6.38 (Lemma 6.6.6 (preprint 6.31); `Polyfold.lemma_6_6_6`). With the operators of V6.37, the sequence (96) $0 \to \ker T \to \ker(T \oplus \varphi) \to \mathbb{R}^k \to K_z/R(T) \to 0$ is the stabilization sequence of T and φ (V6.16), hence exact. *Book:* (96) and Lemma 6.6.6 (preprint 6.31), p. 266 (preprint p. 223). *Hidden hypotheses:* none. *Deviation (restated):* the text has c onto by (P); here it is onto by the general position of F (V6.33) and the derivative of the family of zeros, which absorbs the $\mathbb{R}^n$-directions (`Polyfold.fderiv_mem_linKer`) (Findings, item 49).

V6.39 (continuation and the isomorphisms Φ_{E_t}; `Polyfold.SubspaceFamily.cont`, `Polyfold.SubspaceFamily.cont_eq`, `Polyfold.SubspaceFamily.cont_self`, `Polyfold.SubspaceFamily.cont_trans`, `Polyfold.SubspaceFamily.contEquiv`, `Polyfold.SubspaceFamily.cont_comp`, `Polyfold.SubspaceFamily.cont_eq_transport`, `Polyfold.detEquivOfSurjective`, `Polyfold.stabRayEquiv`, `Polyfold.propagate`, `Polyfold.propagate_self`, `Polyfold.propagate_

`trans`, `Polyfold.propagate_comp`). Over a simply connected, locally path-connected base the continuation of a ray of $\lambda(L_{q_0})$ to $\lambda(L_{q_1})$ is a bijection, compatible with composition and pull-back, and along $[0,1]$ it is the continuation of V6.35. For a stabilization $T \oplus \varphi$ onto and an identification of its kernel with L, the rays of $\det(T)$ and of $\lambda(L)$ correspond (ι_φ of V6.18 and $\det(T \oplus \varphi) \cong \lambda(\ker(T \oplus \varphi))$); Φ_{E_0}, continuation and $\Phi_{E_1}^{-1}$ compose to a bijection of the rays of $\det(T_0)$ and $\det(T_1)$. *Book:* pp. 265–269 (preprint pp. 223–224), after Lemma 6.6.6 (preprint 6.31). *Hidden hypotheses:* none. *Deviation (restated):* over any simply connected, locally path-connected base, so that one statement serves the paths and the squares; the fixed identification $\lambda^*(\mathbb{R}^k) \cong \lambda^*(\{0\})$ of (94) occurs at both ends.

V6.40 (the stabilization sign; `Polyfold.coord_stabIota`, `Polyfold.stabD`, `Polyfold.continuousOn_stabD`, `Polyfold.stabD_ne_zero`, `Polyfold.stabSign`, `Polyfold.isLineOrientation_stabRay`, `Polyfold.not_isLineOrientation_stabRay_uncorrected`). In a local determinant chart with the projection P, the chart coordinate of $\iota_\psi(x) \in \det(S \oplus \psi)$ is $(-1)^{k \dim F/R(P) + \dim \operatorname{coker} S} D(S, \psi)$ times that of x, with D continuous and nowhere zero near the centre. For a continuous orientation o of DET, a norm-continuous family T_q and a continuous stabilization φ_q with $T_q \oplus \varphi_q$ onto, the rays $(-1)^{\dim \operatorname{coker} T_q} \Phi(o(T_q))$ of $\lambda(\ker(T_q \oplus \varphi_q))$ form an orientation; without the sign they need not ($E = F = \mathbb{R}$, $k = 1$, $T_q = q/(1 + |q|)$, $\varphi(r) = r_0$). *Book:* formula (94), p. 256 (preprint p. 212), as it is used in the continuity statements of pp. 265–272 (preprint pp. 223–226). *Hidden hypotheses:* none. *Deviation (restated):* as we read it, the orientations carried by Φ_E of (94) along a continuous family are not continuous where $\dim \operatorname{coker} T$ changes parity; the propagation uses the sign-normalized $(-1)^{\dim \operatorname{coker} T} \Phi_E$, which on each operator carries the text's orientation or its opposite (Findings, item 50).

V6.41 (the push-forward along stabilized kernels; `Polyfold.chartPt`, `Polyfold.ChartPoint.stabT`, `Polyfold.ChartPoint.stabT_mem_linSet`, `Polyfold.ChartPoint.stabPhi`, `Polyfold.ChartPoint.stabPhi_finAppend`, `Polyfold.stabSign_inv`, `Polyfold.toPerm_stabSign_symm`, `Polyfold.isFredholm_stabCoprod`, `Polyfold.IsStabKernels`, `Polyfold.IsStabKernels.comp`, `Polyfold.IsStabKernels.push`, `Polyfold.IsStabKernels.push_congr`, `Polyfold.IsStabKernels.orRay`, `Polyfold.IsStabKernels.push_eq_orRay`, `Polyfold.IsStabKernels.cont_orRay_path`, `Polyfold.IsStabKernels.push_comp`, `Polyfold.IsStabKernels.push_trans`). At a chart point, $T = (f + S)'(x) \in \operatorname{Lin}(f, x)$ and $\varphi(\lambda) = \sum \lambda_i \hat{s}_i(x)$. A family of *stabilized kernels* is a continuous family Λ, over a parameter space, of the subspaces $\ker(T_q \oplus \varphi_q)$ of $E_0 \times \mathbb{R}^k$ at points $y(q)$, read in one chart; its push-forward $\varphi_* : \operatorname{Or}^f_{x_0} \to \operatorname{Or}^f_{x_1}$ between the end points is $o \mapsto o(T_a) \mapsto (-1)^{\dim \operatorname{coker} T_a} \Phi_{E_a}(o(T_a))$, continued to $\lambda(\Lambda_b)$, and carried back in the same way at T_b. It depends only on the data, is compatible with pull-back and with concatenation, and along a path with continuously varying data continuation carries the rays read at one parameter to those read at the next. *Book:* the push-forward $o_1 = \varphi_* o_0$, preprint p. 224. *Hidden hypotheses:* none. *Deviation (restated):* the sign of V6.40 at both end points; for given end-point operators the result differs from the text's by $(-1)^{\dim \operatorname{coker} T_a + \dim \operatorname{coker} T_b}$; the sections enter pointwise, and the continuity in the parameter is that of Λ (Findings, item 50).

V6.42 (Lemma 6.6.5 (preprint 6.30) for the push-forward: the stabilized kernels exist; `Polyfold.`

`exists_continuousOn_lift_chart`, `Polyfold.exists_isStabKernels`). For an sc-smooth $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n)$ in the domain of a chart, an sc⁺-section S with $f + S = 0$ along Φ near $[0,1]^n$ and sc⁺-sections $s_1, \dots, s_k$ as in V6.33, the stabilized kernels of the slices form a continuous family over $[0,1]^n$. *Book:* Lemma 6.6.5 (preprint 6.30), p. 265 (preprint p. 222), and the bundle L after it. *Hidden hypotheses:* none beyond the standing ones. *Deviation (restated):* as V6.34 (gap G.62, closed: L is a C^∞ vector bundle, `Polyfold.lemma_6_6_5_smoothBundle`); the family is the tangent spaces of the solution set (V6.34) cut by $\{dt = 0\}$ (V6.36) and identified with $\ker(T_q \oplus \varphi_q)$ (V6.37).

V6.43 (push-forward data along paths; `Polyfold.pt1_mem_Icc`, `Polyfold.exists_lift_of_smooth`, `Polyfold.chartPtOf`, `Polyfold.isZeroAt_sectionAdd_reparamSection`, `Polyfold.IsStabKernels.push_eq_of_eq`, `Polyfold.IsPush`, `Polyfold.isPush_reparam`, `Polyfold.IsPush.split`, `Polyfold.exists_isPush`). e is a push-forward along an sc-smooth path γ in a chart if there are global sc⁺-sections $S, s_1, \dots, s_k$ of $\pi^*P \to \mathbb{R} \times X$ with $(f + S)(t, \gamma(t)) = 0$ near $[0,1]$ and a family of stabilized kernels of their slices along γ whose push-forward is e. A push-forward along γ is the composite of push-forwards along its two halves, and along every sc-smooth path in a set with property (3) of Proposition 6.6.3 (preprint 6.28) there is one. *Book:* preprint p. 224 (the choices of $s(\cdot, t)$ and $s_1, \dots, s_k$ along the path). *Hidden hypotheses:* none. *Deviation (restated):* global sections of the pull-back in place of sections over $U \times [0,1]$; property (P) is used only to construct a push-forward.

V6.44 (Proposition 6.6.7 (preprint 6.32); `Polyfold.IsStabKernels.congr_data`, `Polyfold.IsStabKernels.push_congr_data`, `Polyfold.IsStabKernels.push_eq_of_prefix`, `Polyfold.isZeroAt_sectionAdd_combo`, `Polyfold.linAt_principal_combo`, `Polyfold.continuous_principal_pullbackSlice`, `Polyfold.proposition_6_6_7`). Two push-forwards along the same sc-smooth path in the domain of a chart agree. *Book:* Proposition 6.6.7 (preprint 6.32), p. 267 (preprint p. 224), proof pp. 266–272 (preprint pp. 224–226). *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) (X tame, bump functions, tame germ data). *Deviation (restated):* "does not depend on the choices" is read as the agreement of any two data; the homotopy of Lemma 6.6.10 (preprint 6.34) and the added sections of Lemma 6.6.8 (preprint 6.33) are combined over one square, $S^\tau = (1 - \tau)S^0 + \tau S^1$ with the two families of sections interpolated and the sections of Lemma 6.6.4 (preprint 6.29) on the square appended, so that Lemma 6.6.8 (preprint 6.33) applies on both edges without reordering (Findings, item 52).

V6.45 (Lemma 6.6.8 (preprint 6.33); `Polyfold.wedgeExt`, `Polyfold.wedgeExt_wedge`, `Polyfold.wedgeExt_congr`, `Polyfold.wedgeExt_bijective`, `Polyfold.appendStab`, `Polyfold.kerAppend`, `Polyfold.detEquiv_stabIota_append`, `Polyfold.SubspaceFamily.exists_continuous_lifts`, `Polyfold.isLineOrientation_wedgeExt`, `Polyfold.kerEquivMap`, `Polyfold.stabRayEquiv_append`, `Polyfold.propagate_append`, `Polyfold.IsStabKernels.push_append`). Appending sc⁺-sections $\bar{s}_1, \dots, \bar{s}_d$ after $s_1, \dots, s_k$ does not change the push-forward: the stabilization isomorphism of (φ, ψ) is that of φ followed by $w \mapsto (w, 0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$, for any $\ell_j \in \ker(T \oplus (\varphi, \psi))$ with ψ-coordinates the standard basis, and this map carries continuous orientations along the family. *Book:* Lemma 6.6.8 (preprint 6.33), p. 267 (preprint p. 224), proof preprint p. 225 ("one verifies by a simple computation"). *Hidden hypotheses:* none. *Deviation (restated):* the homotopy

$\tilde{F}(\tau, \cdot)$ of the text's proof is not needed; with the appended sections last the two isomorphisms agree exactly, with no sign (Findings, item 51).

V6.46 (sections over parameter families; `Polyfold.reparamSection`, `Polyfold.pullbackSlice_reparamSection`, `Polyfold.rnProdMapOp`, `Polyfold.StrongBundle.isScPlusSection_of_local`, `Polyfold.StrongBundle.IsScPlusSection.reparam`, `Polyfold.StrongBundle.isGeneralPositionGerm_paramSection_of_comp`). For g affine, $(q, y) \mapsto S(g(q), y)$ is again an sc⁺-section of the pull-back; sc⁺-sections may be checked in one chart around each point; adding sections keeps the general position of F at $(0, y)$. *Book:* pp. 266–272 (preprint pp. 224–226) (the sections $s(y, \tau, t)$ over a square, and "adding more sc⁺-sections", Lemma 6.6.8 (preprint 6.33), p. 267 (preprint p. 224); the remark that after "adding more sections" the properties still hold is in the proof of Theorem 5.3.10 (preprint 5.18), p. 221 (preprint p. 178)). *Hidden hypotheses:* none. *Deviation:* none.

V6.47 (Lemma 6.6.10 (preprint 6.34); `Polyfold.IsStabKernels.push_square`, `Polyfold.sqEdge`, `Polyfold.sqPath`). For a family of stabilized kernels over the square $[0, 1]^2$ with the end points x_0, x_1 fixed and the end-point operators varying continuously, the push-forwards along the edges $\tau = 0$ and $\tau = 1$ agree. *Book:* Lemma 6.6.10 (preprint 6.34), p. 271 (preprint p. 226). *Hidden hypotheses:* none. *Deviation (restated):* the convex homotopy s^τ of the lemma is part of the square of V6.44; the continuity of the end-point orientations along τ is that of V6.40, with the sign.

V6.48 (Theorem 6.6.11 (preprint 6.35); `Polyfold.push_eq_of_homotopy`, `Polyfold.HasLocalPropagation`, `Polyfold.theorem_6_6_11`). Over a tame X with bump functions, for f with tame germ data, every smooth point x has an open $U \ni x$ and a chart c such that push-forwards exist along the sc-smooth paths in U and depend only on their end points. *Book:* the homotopy argument, pp. 271–273 (preprint pp. 226–227), and Theorem 6.6.11 (preprint 6.35), p. 272 (preprint p. 227). *Hidden hypotheses:* the section f, which the statement does not introduce (an sc-Fredholm section, with tame germ data); bump functions. *Deviation (narrower):* U is not shown to be sc-smoothly contractible (gap G.61); what the proof uses is (1′) of V6.32. Since the homotopy of (1′) meets its edge conditions only on $[0, 1]$ while a push-forward datum needs $f + S = 0$ near $[0, 1]$, the proof passes through the reparametrized paths $\gamma_i \circ \kappa$, with three homotopies whose edges are exact (Findings, item 53). *Update (2026-11-13):* Proposition 6.6.3 (1) is proved in its 2021 form, a smaller U' contractible in U (`Polyfold.proposition_6_6_3_one`, gap G.61 closed); U is not shown to be contractible in itself (the preprint's (1)). For paths on $[0, 1]$ the homotopy of (1′) is an sc-smooth map $[0, 1]^2 \to W$ with exact edge conditions (`Polyfold.exists_isScHomotopyI`), the hypothesis of `Polyfold.orTransportI_eq_of_homotopy`.

V6.49 (sc-smooth paths in a chart and their concatenation; `Polyfold.MPolyfold.isScSmoothMap_of_eventuallyEq`, `Polyfold.pt1_apply`, `Polyfold.contDiff_pt1_comp`, `Polyfold.exists_nhds_scPath`, `Polyfold.concatPath`, `Polyfold.concatPath_of_le`, `Polyfold.concatPath_of_gt`, `Polyfold.isScSmoothMap_concatPath`, `Polyfold.concatPath_zero`, `Polyfold.concatPath_one`, `Polyfold.concatPath_half` , `Polyfold.concatPath_half` , `Polyfold.isScSmoothMap_reparam`, `Polyfold.HasLocalPropagation.eq_trans`). For an open U in the domain of a chart and $x \in U$ there is an open $V \ni x$ in U whose smooth points are pairwise joined by sc-smooth paths in U. The concatenation, $\gamma_1(\kappa(3s))$ for $s \leq 1/2$ and $\gamma_2(\kappa(3s-2))$ for $s > 1/2$ with κ the smooth transition function, is sc-smooth, with the right end points and halves; reparametrizing by a C^∞ map keeps sc-smoothness; push-forwards compose along concatenations. *Book:* the paths after Theorem 6.6.11 (preprint 6.35), p. 272 (preprint p. 227), and the concatenation, preprint p. 228. *Hidden hypotheses:* none. *Deviation (narrower):* paths are joined only locally, inside a chart; the global facts recalled on preprint p. 201 (local path connectedness, sc$^+$-smooth paths within a path component) are not formalized (gap G.56).

V6.50 (Definition 6.6.12 (preprint 6.36); `Polyfold.IsContinuousOrientation`, `Polyfold.exists_two_isContinuousOrientation`, `Polyfold.exists_orientableNbhd`, `Polyfold.IsOrientableNbhd`, `Polyfold.exists_isOrientableNbhd`). An orientable neighbourhood of (f, x) is a pair $V \subseteq U$: U open with the local propagation property in a chart, $V \ni x$ open with any two smooth points joined by an sc-smooth path in U; a continuous orientation is a choice $y \mapsto o_y$ on V_∞ carried into itself by the push-forwards along paths in U; there are exactly two, and every smooth point has an orientable neighbourhood. *Book:* the paragraph after Theorem 6.6.11 (preprint 6.35) and Definition 6.6.12 (preprint 6.36), p. 273 (preprint p. 227). *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5). *Deviation (restated):* the pair $V \subseteq U$ in place of an open contractible U, since Proposition 6.6.3 (preprint 6.28) gives (1′) (gap G.61) (Findings, item 54). *Update (2026-11-13):* the pair $V \subseteq U$ is kept since what the push-forwards use is (1′); Proposition 6.6.3 also gives, in its 2021 form, a U' contractible in U (`Polyfold.proposition_6_6_3_one`; gap G.61 closed).

V6.51 (reparametrization homotopies; `Polyfold.mem_smoothSet_of_isScSmoothMap`, `Polyfold.contDiff_pt2_comp`, `Polyfold.isScHomotopyRel_reparam`, `Polyfold.reparam_mem`, `Polyfold.leftHalf`, `Polyfold.rightHalf`, `Polyfold.halvesReparam`, `Polyfold.concatPath_halves`). For a C^∞ θ with $\theta(0) = 0$, $\theta(1) = 1$, $\gamma \circ \theta$ is sc-smoothly homotopic to γ with the end points fixed, inside every set containing $\gamma([0,1])$ when $\theta([0,1]) \subseteq [0,1]$; the concatenation of the two halves of γ is such a reparametrization. *Book:* the homotopy classes $[\varphi]$, preprint p. 227, and their composition, preprint p. 228. *Hidden hypotheses:* none. *Deviation:* none; these are tools for V6.52 and V6.56.

V6.52 (the path category PX; `Polyfold.smoothTransition_half`, `Polyfold.isScSmoothMap_const_of_mem_smoothSet`, `Polyfold.isScSmoothMap_ite`, `Polyfold.isScSmoothMap_ite `, `Polyfold.concatPath_of_le_two_thirds`, `Polyfold.concatPath_of_one_third_le`, `Polyfold.ScHomotopic`, `Polyfold.ScHomotopic.refl`, `Polyfold.ScHomotopic.symm`, `Polyfold.ScHomotopic.trans`, `Polyfold.isScHomotopyRel_triHomotopy`, `Polyfold.assocReparam`, `Polyfold.ScPath`, `Polyfold.ScPath.source_mem_smoothSet`, `Polyfold.ScPath.target_mem_smoothSet`, `Polyfold.ScPath.const`, `Polyfold.ScPath.concat`, `Polyfold.ScPath.reverse`, `Polyfold.ScPath.concat_congr`, `Polyfold.ScPath.const_concat`, `Polyfold.ScPath.concat_const`, `Polyfold.ScPath.concat_assoc`, `Polyfold.ScPath.concat_reverse`, `Polyfold.ScPath.reverse_concat`, `Polyfold.ScPathClass`, `Polyfold.ScPathClass.comp`, `Polyfold.PathCat`, `Polyfold.PathCat.category`, `Polyfold.PathCat.groupoid`, `Polyfold.PathCat.homMk`, `Polyfold.PathCat.comp_mk`, `Polyfold.PathCat.id_eq_mk`, `Polyfold.PathCat.mk_eq_mk_iff`, `Polyfold.PathCat.exists_eq_mk`, `Polyfold.PathCat.inv_mk`). Sc-smooth homotopy

with the end points fixed is an equivalence relation on the sc-smooth paths between two points; concatenation is well defined on classes, associative, with the classes of the constant paths as units and the reversed paths as inverses; the smooth points with these classes form a category, indeed a groupoid. *Book:* the classes $[\varphi]$, their composition and the category PX, pp. 272–274 (preprint pp. 227–228). *Hidden hypotheses:* none. *Deviation (restated):* the edge conditions of a homotopy hold only on $[0,1]$, so transitivity is proved after reparametrizing by κ; the identities, which the text does not name, are the classes of the constant paths; the composition's second piece $\psi(\beta(s))$ is read as $\psi(\beta(s)-1)$ (possible misprint); Mathlib's $[\varphi] \gg [\psi]$ is the text's $[\psi] * [\varphi]$ (Findings, items 55, 71). *Update (2026-11-12):* the category is also built on $[0,1]$ as in the text, with homotopies sc-smooth on the M-polyfold $[0,1]^2$ and the text's β (`Polyfold.PathCatI`, `Polyfold.PathCatI.groupoid`, `Polyfold.ScPathI.concat_toFun_of_le`, `Polyfold.ScPathI.concat_toFun_of_ge`), and it is equivalent to the category on $\mathbb{R}$ (`Polyfold.PathCatI.toPathCat_isEquivalence`).

V6.53 (transferring orientations between chart points; `Polyfold.conjKerEquiv`, `Polyfold.conjCokerEquiv`, `Polyfold.conjDet`, `Polyfold.conjDet_wedge_tmul`, `Polyfold.conjDet_trans`, `Polyfold.conjDet_congr`, `Polyfold.conjDet_refl`, `Polyfold.IsGoodLeftProj.conj`, `Polyfold.gammaProj_conjDet`, `Polyfold.coord_conjDet_coordInv`, `Polyfold.isLineOrientation_conj`, `Polyfold.LineOrientation.conj`, `Polyfold.LineOrientation.conj_apply`, `Polyfold.LineOrientation.conj_refl`, `Polyfold.LineOrientation.conj_trans`, `Polyfold.LineOrientation.conj_congr`, `Polyfold.ChartPoint.baseDeriv`, `Polyfold.ChartPoint.baseDeriv_self`, `Polyfold.ChartPoint.baseDeriv_trans`, `Polyfold.ChartPoint.isScOperator_baseDeriv`, `Polyfold.ChartPoint.fiberMap`, `Polyfold.ChartPoint.fiberMap_self`, `Polyfold.ChartPoint.fiberMap_trans`, `Polyfold.ChartPoint.isScOperator_fiberMap`, `Polyfold.ChartPoint.transferT`, `Polyfold.ChartPoint.transferK`, `Polyfold.ChartPoint.transferT_self_apply`, `Polyfold.ChartPoint.transferT_trans_apply`, `Polyfold.ChartPoint.scDeriv_principal_baseDeriv`, `Polyfold.ChartPoint.IsScPlusOp.conj`, `Polyfold.ChartPoint.mem_linSet_transfer_iff`, `Polyfold.ChartPoint.orTransfer`, `Polyfold.ChartPoint.orTransfer_apply`, `Polyfold.ChartPoint.orTransfer_self`, `Polyfold.ChartPoint.orTransfer_trans`). For isomorphisms A, Γ with $S'A = \Gamma S$, $\det(S) \cong \det(S')$ by A on kernels and Γ on cokernels, compatible with composition and with the γ^P, and continuous orientations of DET are carried along. For two chart points at x, A is the derivative at z of the base transition and Γ the fibre part of the strong bundle transition, with the cocycle rules; $L \in \mathrm{Lin}_p(f,x)$ iff $\Gamma L A^{-1} \in \mathrm{Lin}_{p'}(f,x)$, and the induced bijection $\mathrm{Or}_x^f(p) \cong \mathrm{Or}_x^f(p')$ satisfies the cocycle rules. *Book:* preprint p. 219 (T_xX, Y_x and $\mathrm{Lin}(f,x)$ as intrinsic objects) and the proof of Theorem 6.6.13 (preprint 6.37), p. 273 (preprint p. 228). *Hidden hypotheses:* none. *Deviation (not in the book):* the text treats these objects as chart-free; the identification between charts is made explicit (Findings, item 70).

V6.54 (the push-forward does not depend on the chart; `Polyfold.stabSign_conj`, `Polyfold.coprod_conj`, `Polyfold.ker_coprod_conj`, `Polyfold.surjective_coprod_conj`, `Polyfold.conj_stab_key`, `Polyfold.stabRayEquiv_conjDet`, `Polyfold.SubspaceFamily.cont_map`, `Polyfold.ChartPoint.fiberMap_principal`, `Polyfold.ChartPoint.conj_of_linAt`, `Polyfold.ChartPoint.conj_of_principal`, `Polyfold.continuous_baseDeriv`, `Polyfold.IsStabKernels.`

`exists_transfer`, `Polyfold.IsStabKernels.orRay_transfer`, `Polyfold.IsPush.transfer`, `Polyfold.HasLocalPropagation.push_compat`). The stabilization isomorphism and the sign are natural under conjugation, and continuation is natural under jointly continuous families of injective linear maps. A family of stabilized kernels in one chart is carried by the derivative of the base transition to one in a second chart, with the same sections; a push-forward in one chart, conjugated by the transfer of V6.53 at the end points, is a push-forward in the other; so the local propagations of two charts agree. *Book:* pp. 264–274 (preprint pp. 222–228) (the push-forward, and the orientable neighbourhoods of the proof of Theorem 6.6.13 (preprint 6.37)). *Hidden hypotheses:* none. *Deviation (not in the book):* the text does not compare charts (Findings, item 70).

V6.55 (Or_x^f in a chosen chart; `Polyfold.orChart`, `Polyfold.isChart_orChart`, `Polyfold.mem_orChart`, `Polyfold.Orx`, `Polyfold.toChartOr`, `Polyfold.toChartOr_orChart`, `Polyfold.toChartOr_trans`, `Polyfold.exists_two_orx`, `Polyfold.restrictOr`). Or_x^f is the set of orientations read in a chosen chart of the atlas at x; it has two elements, and it is identified with the orientations read in any other chart at x by the transfer of V6.53, compatibly with composition. *Book:* Definition 6.6.2 (preprint 6.27) and the objects of Theorem 6.6.13 (preprint 6.37), preprint pp. 220, 228. *Hidden hypotheses:* none. *Deviation (restated):* a chosen chart stands for the text's chart-free Or_x^f (Findings, item 70).

V6.56 (transport along paths from local push-forwards; `Polyfold.LocalTransport`, `Polyfold.LocalTransport.push_eq_refl`, `Polyfold.LocalTransport.push_eq_push`, `Polyfold.LocalTransport.push_congr`, `Polyfold.LocalTransport.transportN`, `Polyfold.LocalTransport.transportN_eq_push`, `Polyfold.LocalTransport.IsFine`, `Polyfold.LocalTransport.transportN_eq_of_isFine`, `Polyfold.LocalTransport.IsLebesgueNumber`, `Polyfold.LocalTransport.exists_isLebesgueNumber`, `Polyfold.LocalTransport.isFine_of_isLebesgueNumber`, `Polyfold.LocalTransport.transport`, `Polyfold.LocalTransport.transport_eq_push`, `Polyfold.LocalTransport.transport_congr`, `Polyfold.LocalTransport.transport_const`, `Polyfold.LocalTransport.transport_halves`, `Polyfold.LocalTransport.transportN_square`, `Polyfold.LocalTransport.exists_isSqLebesgueNumber`, `Polyfold.LocalTransport.transport_eq_of_isScHomotopyRel`, `Polyfold.LocalTransport.transport_reparam`, `Polyfold.LocalTransport.transport_concatPath`, `Polyfold.LocalTransport.transport_unique`, `Polyfold.equiv_paste_squares`, `Polyfold.vSeg`, `Polyfold.hSeg`, `Polyfold.dSeg`). From local data (good open sets covering X_∞ and push-forwards along paths in a good set, compatible with concatenation and independent of the good set) the transport along an sc-smooth path is the composite of the push-forwards along its 2^N dyadic pieces, for any N at which every piece lies in a good set (such N exist by the Lebesgue number lemma). It is the push-forward on a path in a good set, homotopy invariant, compatible with concatenation and reparametrization, the identity on constant paths, and the only family with these properties. *Book:* the proof of Theorem 6.6.13 (preprint 6.37), p. 273 (preprint p. 228) ("using the compactness of $[0,1]$ and a finite covering", "by a covering argument", "it is obvious that Γ_f is unique"). *Hidden hypotheses:* none. *Deviation (restated):* dyadic subdivisions of $[0,1]$ and of $[0,1]^2$ in place of arbitrary finite coverings (Findings, item 72).

V6.57 (Theorem 6.6.13 (preprint 6.37); `Polyfold.IsPropagationNbhd`, `Polyfold.orPush`, `Polyfold.orPush_eq`, `Polyfold.orPush_compat`, `Polyfold.orPush_trans`, `Polyfold.IsStarPropertyOn`, `Polyfold.IsStarProperty`, `Polyfold.orLocalTransport`, `Polyfold.orLocalTransport_transport_eq`, `Polyfold.LocalTransport.restrict`, `Polyfold.LocalTransport.transport_apply_eq_of_local`, `Polyfold.orFunctorMap`, `Polyfold.orFunctorMap_mk`, `Polyfold.orFunctorMap_id`, `Polyfold.orFunctorMap_comp`, `Polyfold.orFunctor`, `Polyfold.orFunctorMap_isStarProperty`, `Polyfold.orFunctorMap_unique_of_cover`, `Polyfold.orFunctorMap_unique`, `Polyfold.theorem_6_6_13_hyp`). Over a tame X with bump functions, for f with tame germ data and P with sc⁺-sections through the smooth fibre points, there is exactly one family of bijections $\Gamma_f([\varphi]) : \mathrm{Or}^f_x \to \mathrm{Or}^f_y$ on the morphisms of PX that takes identities to identities, compositions to composites, and has property (□): for every chart c, every open U with the local propagation property in c and every sc-smooth path φ in U, $\Gamma_f([\varphi])$ is every push-forward along φ in c, read through Or^f of the chosen charts. It is a functor $PX \to$ Type, $x \mapsto \mathrm{Or}^f_x$, and the uniqueness needs (□) only on a family of such open sets covering X_∞. *Book:* Theorem 6.6.13 (preprint 6.37) and its proof, preprint p. 228. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5), and sc⁺-sections through every smooth fibre point, so that every Or^f_x has two elements (implied by bump functions). *Deviation (restated):* (□) is asked for every open set with the local propagation property, not only for sc-smoothly contractible ones (gap G.61); $\mathcal{O}_f$ is read as types and bijections; "uniquely determined" is read for the maps on morphisms (Findings, item 72). *Update (2026-11-13):* two $[0,1]$-paths in a small U with the same end points are joined by such a homotopy (`Polyfold.exists_isScHomotopyI`).

V6.58 (Definition 6.6.14 (preprint 6.38); `Polyfold.scPathSetoid`, `Polyfold.IsOrientableSection`, `Polyfold.isOrientableSection_iff_exists`). f is orientable if $\Gamma_f([\varphi]) = \Gamma_f([\psi])$ for any two morphisms $x \to y$ of PX; equivalently there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for every $[\varphi] : x \to y$. *Book:* Definition 6.6.14 (preprint 6.38), p. 274 (preprint p. 229). *Hidden hypotheses:* none. *Deviation (restated):* "smooth points in the same path component" is read as smooth points joined by an sc-smooth path (Findings, item 73).

V6.59 (Definition 6.6.15 (preprint 6.39) and its equivalence with Definition 6.6.14 (preprint 6.38); `Polyfold.IsOrientableSection'`, `Polyfold.isOrientableSection'_iff_exists`, `Polyfold.isOrientableSection_iff`). f is orientable in the sense of Definition 6.6.15 (preprint 6.39) if there is $y \mapsto o_y$ on X_∞ such that every smooth point has an orientable neighbourhood over which it is a continuous orientation; this holds iff there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi]$, so Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) are equivalent. *Book:* Definition 6.6.15 (preprint 6.39), p. 274 (preprint p. 229) ("an equivalent version"). *Hidden hypotheses:* none. *Deviation (stronger):* the equivalence, stated without proof, is proved; the definition's "$y \in U_\infty$", before any U is introduced, is read as $y \in X_\infty$ (possible misprint) (Findings, items 73, 74).

V6.60 (Definition 6.6.16 (preprint 6.40); `Polyfold.IsSectionOrientation`, `Polyfold.isSectionOrientation_iff`, `Polyfold.exists_isSectionOrientation_iff`, `Polyfold.IsSectionOrientation.apply_eq_iff`, `Polyfold.IsSectionOrientation.exists_two`). An orientation of f is a map $y \mapsto o_y$ on X_∞ whose restriction to every orientable neighbourhood of every smooth point is a continuous orientation; equivalently $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi] : x \to y$. f has an orientation iff it is orientable; two orientations that agree at a point agree on the smooth points joined to it by sc-smooth paths, and there are exactly two on each such class. *Book:* Definition 6.6.16 (preprint 6.40), p. 274 (preprint p. 229). *Hidden hypotheses:* none. *Deviation (restated):* an orientable neighbourhood is the pair of V6.50 (Findings, item 75).

V6.61 (Theorem 6.6.18 (preprint 6.41), the natural orientation of the solution set; `Polyfold.theorem_6_6_18_solutionSet`, `Polyfold.MPolyfold.IsGeneralPositionAt.isGoodPositionAt`, `Polyfold.tangentSpace_solutionSet_eq_linKer`, `Polyfold.ChartPoint.linZero`, `Polyfold.ChartPoint.linZero_mem_linSet`, `Polyfold.ChartPoint.linZero_surjective`, `Polyfold.ChartPoint.map_ker_linZero`, `Polyfold.ChartPoint.tangentOr`, `Polyfold.tangentOrAt`, `Polyfold.stabSign_of_surjective`, `Polyfold.stabRayEquiv_zero`, `Polyfold.ChartPoint.stabRayEquiv_eq_tangentOr`, `Polyfold.IsStabKernels.orRay_eq_tangentOr`, `Polyfold.isStabKernels_zero`, `Polyfold.SubspaceFamily.congrL`, `Polyfold.exists_tangentFamily`, `Polyfold.tangentFamily`, `Polyfold.tangentFamily_L`, `Polyfold.cont_tangentOr`, `Polyfold.isLineOrientation_tangentOr`, `Polyfold.isLineOrientation_tangentOr_of_isSectionOrientation`, `Polyfold.theorem_6_6_18_hyp`). Over a tame X, for a proper sc-Fredholm section f with tame germ data in general position (Definition 5.3.9 (preprint 5.17)) at every zero, the solution set S is compact, a smooth submanifold in general position with the induced structure of a tame manifold with boundary with corners (the conclusions of Theorem 5.3.10 (preprint 5.18)), and $T_xS = \ker f'(x)$, read in a chart. At a zero, $f'(x)$ lies in $\mathrm{Lin}(f, x)$ and is onto, so an element o_x of Or^f_x gives a ray of $\det f'(x) \cong \lambda(\ker f'(x)) = \lambda(T_xS)$, the *tangent orientation*. For an orientation o of f (Definition 6.6.16 (preprint 6.40)) and every sc-smooth path γ of zeros in an orientable neighbourhood $V \subseteq U$, the kernels $\ker f'(\gamma_q)$ form a continuous family of subspaces, and the tangent orientations of o form an orientation of its line bundle. The Fredholm side is the push-forward for the datum $S = 0$, $k = 0$, which along a path of zeros is the continuation between tangent orientations. *Book:* Theorem 6.6.18 (preprint 6.41) and its proof, preprint p. 229; the paragraph after it on S_{f+s}, pp. 274–276 (preprint pp. 229–230). *Hidden hypotheses:* tame germ data (the standing hypothesis of Chapter 5); the orientation o exists iff f is orientable (V6.60). "Proper" is the text's hypothesis (`StrongBundle.IsProper`); the proof uses only the compactness of S, which for sc-Fredholm sections is equivalent to it (V5.10). *Deviation (narrower):* "a natural orientation" is read as the tangent orientations with their continuity along the sc-smooth paths of zeros in a chart, and "the local prolongation of the orientation on S is the Fredholm one" as the identity of continuation and push-forward along such paths. Not formalized: an orientation of S as an abstract manifold (gap G.67), the compatibility of the tangent orientations read in two charts at one point (gap G.68), and the perturbation paragraph after the theorem, with S_{f+s} and $\mathrm{Lin}(f + s, x) = \mathrm{Lin}(f, x)$ (gap G.69) (Findings, items 90–93).

V6.62 (Remark 6.6.19 (preprint 6.42); `Polyfold.extOp`, `Polyfold.isFredholm_extOp`, `Polyfold.finrank_ker_extOp`, `Polyfold.extE`, `Polyfold.range_extOp`, `Polyfold.extCoker`, `Polyfold.extDet`, `Polyfold.extDet_wedge_tmul`, `Polyfold.extDetN`, `Polyfold.extDetN_eq_of_surjective`, `Polyfold.coord_extDet`, `Polyfold.coord_extDetN`, `Polyfold.isLineOrientation_extPull`, `Polyfold.LineOrientation.extPull`, `Polyfold.extConjOp`,

`Polyfold.extConjDet`, `Polyfold.existsUnique_extConjOr`, `Polyfold.ChartPoint.prodPt`, `Polyfold.ChartPoint.mem_prodPt_T_iff`, `Polyfold.ChartPoint.mem_prodPt_K_iff`, `Polyfold.ChartPoint.prodT`, `Polyfold.ChartPoint.prodK`, `Polyfold.ChartPoint.prodLin`, `Polyfold.ChartPoint.IsScPlusOp.prodPt`, `Polyfold.constLift`, `Polyfold.ChartPoint.prodLin_mem_linSet`, `Polyfold.ChartPoint.existsUnique_prodOr`, `Polyfold.ChartPoint.prodOr`, `Polyfold.ChartPoint.prodOr_apply`, `Polyfold.ChartPoint.prodOr_injective`). For Fredholm L, $\tilde{L} = L \oplus 0 : E \times \mathbb{R} \to F$ is Fredholm with $\ker \tilde{L} = \ker L \times \mathbb{R}$ and $\operatorname{coker} \tilde{L} = \operatorname{coker} L$, and $h \mapsto \tilde{h}$, $(a_1 \wedge \cdots \wedge a_n) \otimes d^* \mapsto ((a_1, 0) \wedge \cdots \wedge (a_n, 0) \wedge e) \otimes d^*$ with $e = (0, 1)$, is an isomorphism $\det(L) \cong \det(\tilde{L})$; in local determinant charts its coordinate is $(-1)^{\dim F/R(P) + \dim \operatorname{coker} L}$ times a continuous nowhere-zero function, and $(-1)^{\dim \operatorname{coker} L}\, h \mapsto \tilde{h}$ pulls continuous orientations back along $L \mapsto \tilde{L}$. In charts: for an sc-Fredholm f, an sc$^+$-section S of $\pi^* P \to \mathbb{R} \times X$, $\hat{f} = \tilde{f} + S$ with $\tilde{f}(t, y) = f(y)$, and the product chart point (t, z), the operator $\hat{L}(\sigma, v) = Lv$ lies in $\operatorname{Lin}(\hat{f}, (t, x))$ for every $L \in \operatorname{Lin}(f, x)$, and an orientation o of $\operatorname{DET}(f, x)$ gives exactly one orientation $\hat{o}$ of $\operatorname{DET}(\hat{f}, (t, x))$ with $\hat{o}(\hat{L})$ the image of $o(L)$ for **every** L; $o \mapsto \hat{o}$ is injective. *Book:* Remark 6.6.19 (preprint 6.42), p. 276 (preprint p. 230). *Hidden hypotheses:* x smooth and $\operatorname{Lin}(f, x)$ nonempty (implied by bump functions, V6.1); only the orientation of $\operatorname{DET}(f, x)$ at the one point is used. *Deviation (restated):* at a point, as we read it, the literal $h \mapsto \tilde{h}$ is not continuous in L where the parity of $\dim \operatorname{coker} L$ changes, so the orientation of $\operatorname{DET}(\hat{f}, (t, x))$ it defines could depend on the choice of L; the sign $(-1)^{\dim \operatorname{coker} L}$ (the sign of V6.40) is used, which agrees with the text's map when L is onto. $X \times [0, 1]$ is read as $\mathbb{R} \times X$ (decision D2), with the parameter first, so $\tilde{L}(a, b) = La$ is conjugated by the swap of the factors. That $(t, y) \mapsto e \wedge o_y$ is an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40), built from an orientation of f, is proved since 2026-11-12 (gap G.66 closed): `Polyfold.SectionOrientation.prodFirst` (X tame with bump functions, tame germ data, the hypotheses of `Polyfold.theorem_6_6_18_oriented`); $(t, x) \mapsto e \wedge o_x$ is carried by every push-forward of $\tilde{f}$ (`Polyfold.isSectionOrientation_prodFirstFun`), since a push-forward datum of f lifts to one of $\tilde{f}$ and the lifted push-forward maps $e \wedge o$ to $e \wedge$ (push-forward of o) with no sign (`Polyfold.IsPush.prodLift`), and the chart of X plays no role (`Polyfold.prodOrOf_orTransfer`) (Findings, items 95–97). On a product end the induced orientation of the solution set is the product orientation, the time vector first (`Polyfold.isEndProductOrientation_cobOr`).

V6.63 (Appendix 6.8.1 (preprint 6.7.1), the proof of Lemma 6.3.3 (preprint 6.5); `Polyfold.FourTermExact.Choice.exists_smul_source_value`, `Polyfold.FourTermExact.Choice.wedgeC`). For two choices of (91) the right-hand sides are related as the sources are: a change of the basis of Z multiplies both wedges by its determinant, a change of the bases of A and D multiplies both sides by the same factor, and a change of complements adds to the basis of Z vectors of $\alpha(A)$ and to the γ-preimages vectors of $\beta(B)$, which leaves the wedges unchanged. *Book:* §6.8.1 (preprint §6.7.1), preprint pp. 234–235. *Hidden hypotheses:* none. *Deviation (restated):* the step "independent of the choices of the basis if and only if $\det(\tau) = \det(\varepsilon)$" is read as the homogeneity of (91) in the bases of A and D, τ being the σ just introduced; the corrections p_i lie in $\beta(B)$ (written $\beta(C)$) and are combinations of the β-images, whose names are exchanged with those of the γ-preimages in the last step (possible misprints) (Findings, item 1).

V6.64 (Appendix 6.8.2 (preprint 6.7.2), the proof of Proposition 6.4.11 (preprint 6.12); `Polyfold.mem_span_append_of_ker`, `Polyfold.gammaProj_wedge_tmul`, `Polyfold.comp_comp_eq_of_projLE`). For $P \leq Q$ in Π_T: if $Qg = 0$, then g is a combination of the vectors d_i plus Tx with $x \in \ker(QT)$, so the classes of d_i and Tb_j span what the text calls $\ker \tilde{Q}$; the classes $[T\tilde{b}]$ and $[QT\tilde{b}]$ in $F/R(P)$ differ by elements of that span, which does not change the wedge. *Book:* §6.8.2 (preprint §6.7.2), pp. 280–286 (preprint pp. 235–237). *Hidden hypotheses:* none. *Deviation (restated):* the spanning is proved directly in F; the name α_q in the last part is read as α^P (possible misprint) (Findings, items 6, 7).

V6.65 (Lemma 6.4.4 (2017: 6.3); `Polyfold.projIntertwiner`, `Polyfold.projIntertwiner_self`, `Polyfold.projIntertwiner_mul`, `Polyfold.continuous_projIntertwiner`, `Polyfold.lemma_6_4_4`). Let E, F be Banach spaces, $T_0 : E \to F$ Fredholm and $P_0 \in \Pi_{T_0}$. There is $\varepsilon > 0$ such that for every T and every projection P of F with $\|T - T_0\| < \varepsilon$ and $\|P - P_0\| < \varepsilon$: T is Fredholm, $P \in \Pi_T$, and $\operatorname{ind}(P \circ T) = \operatorname{ind}(T) = \operatorname{ind}(T_0) = \operatorname{ind}(P_0 \circ T_0)$. For projections P_0, P the intertwiner $J(P) = PP_0 + (1-P)(1-P_0)$ has $J(P_0) = 1$ and $J(P)P_0 = PJ(P)$, and depends continuously on P. *Book:* Lemma 6.4.4 (2017: 6.3), p. 247 (2017 p. 189); no preprint counterpart; the index equalities are new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (explicit):* $\operatorname{ind}(P \circ T)$ is the index of $P \circ T$ as a map $E \to F$ (as a map onto $R(P)$ its index would be $\dim \ker(PT)$); E and F are taken in the universe `Type`. The proof is not the text's: $J(P)$ is invertible near P_0 and $P = JP_0J^{-1}$, so $R(P) = JR(P_0)$ has finite codimension, and Lemma 6.5.1 (preprint 6.17) for $J^{-1}T$ near T_0 gives $R(PT) = JR(P_0J^{-1}T) = JR(P_0) = R(P)$ (Findings, items 110, 111).

V6.66 (a common complement; `Polyfold.finrank_sup_span_singleton_of_notMem`, `Polyfold.isCompl_sup_span_singleton`, `Polyfold.exists_isCompl_isCompl_of_finrank_eq_aux`, `Polyfold.exists_isCompl_isCompl_of_finrank_eq`). Two finite-dimensional subspaces K_1, K_2 of the same dimension of a real vector space have a common algebraic complement X. *Book:* the proof of Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), takes a closed subspace X complementing both kernels. *Hidden hypotheses:* none. *Deviation (not in the book):* by induction on $\dim(K_1 + K_2) - \dim K_1$: for $a \in K_2 \setminus K_1$ and $b \in K_1 \setminus K_2$, $a + b$ lies in neither, and a common complement of $K_1 + \mathbb{R}(a+b)$ and $K_2 + \mathbb{R}(a+b)$ plus $\mathbb{R}(a+b)$ is one of K_1 and K_2. That X can be taken closed is V6.69 (Findings, items 112, 116).

V6.67 (Proposition 6.4.13 (2017: 6.6), the algebraic part; `Polyfold.proposition_6_4_13`). Let $T : E \to F$ be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$. There are linear isomorphisms $A_T : \ker(PT) \to \ker(QT)$ and $B_T : F/R(P) \to F/R(Q)$ with $A_T \circ j_P = j_Q$, $\Phi_Q \circ A_T = B_T \circ \Phi_P$ and $\pi_Q \circ B_T = \pi_P$, for the maps of the exact sequences $E(T, P)$ and $E(T, Q)$ of V6.12. They are the text's: $A_T(e) = e - x(e)$, $x(e) \in X$ with $QTx(e) = QTe$, and $B_T(f + H) = \tilde{f} + H$ with $Tx + \tilde{f} = (1-P)f$, $x \in X$, $\tilde{f} \in (1-Q)F$, for a common complement X of the kernels (V6.66), which exists because $\dim \ker(PT) = \dim \ker(QT)$ by the indices. *Book:* Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), proof pp. 253–255; no preprint counterpart; the 2021 edition makes $\Pi_T \cap \mathcal{P}_H \neq \emptyset$ a hypothesis. *Hidden hypotheses:* none. *Deviation (narrower):* the last sentence, that A_T and B_T depend continuously on T, is not part of this item: it is V6.70, with a closed X (V6.69) (gap G.81, closed); here X is algebraic, and F/H appears as $F/R(P)$ and $F/R(Q)$

(Findings, items 112, 113, 117).

V6.68 (Lemmas 6.6.8 and 6.6.9; `Polyfold.lemma_6_6_8`, `Polyfold.lemma_6_6_9`). For stabilized kernels of the data $(f, s_1, \dots, s_k)$ along a family, and of the same data with d sections appended, the push-forwards of orientations between two smooth points agree, both when the appended sections are zero (Lemma 6.6.8: the new parameters μ enter as dummy variables) and when they are arbitrary sections $\bar{s}_1, \dots, \bar{s}_d$ (Lemma 6.6.9). *Book:* Lemma 6.6.8 (preprint 6.33), p. 267 (preprint p. 224), and Lemma 6.6.9, p. 269, new in the 2021 edition as a separate lemma: the 2021 text splits preprint Lemma 6.33 (2017: 6.15), whose statement is that of Lemma 6.6.9 and whose proof begins with the step of Lemma 6.6.8. *Hidden hypotheses:* the stabilized kernels of the appended data are given (`IsStabKernels`), as in V6.45. *Deviation (restated):* both are V6.45 (`Polyfold.IsStabKernels.push_append`), Lemma 6.6.8 for $\bar{s}_j = 0$; the homotopy of the text's proof is not needed (Findings, item 114).

V6.69 (a common closed complement; `Polyfold.isCompl_sup_inf_of_isCompl`, `Polyfold.exists_isClosed_isTopCompl_isTopCompl_of_finrank_eq`). Two finite-dimensional subspaces K_1, K_2 of the same dimension of a real normed space E have a common closed complement X, and $E = K_1 \oplus X = K_2 \oplus X$ topologically (with continuous projections). *Book:* the proof of Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), takes a closed subspace X that is a topological complement of both kernels, without argument. *Hidden hypotheses:* none. *Deviation (not in the book):* the argument. $X = Z + (X_0 \cap (K_1 + K_2))$, where X_0 is the algebraic common complement of V6.66 and Z a closed complement of the finite-dimensional $K_1 + K_2$; it complements K_1 and K_2 (if $X_0 \oplus K' = E$, $K' \subseteq K$ and $K \oplus Z = E$, then $(Z + X_0 \cap K) \oplus K' = E$), it is closed as the sum of a closed and a finite-dimensional subspace, and a closed complement of a finite-dimensional subspace is a topological one (Findings, items 112, 116).

V6.70 (Proposition 6.4.13 (2017: 6.6), the last sentence; `Polyfold.projCod`, `Polyfold.solveOp`, `Polyfold.kerTransition`, `Polyfold.cokerTransition`, `Polyfold.solveOp_mem`, `Polyfold.kerTransition_apply`, `Polyfold.cokerTransition_apply`, `Polyfold.apply_solveOp_of_projRestrict_eq`, `Polyfold.disjoint_ker_comp_of_projRestrict_eq`, `Polyfold.isTopCompl_ker_comp_of_projRestrict_eq`, `Polyfold.diagram_of_solveOp`, `Polyfold.continuousAt_solveOp`, `Polyfold.proposition_6_4_13_continuity`, `Polyfold.proposition_6_4_13_continuity_hyps`). Let E, F be Banach spaces, $T : E \to F$ Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$. For $S : E \to F$ and a subspace X write $\xi_S = (QS|_X)^{-1} \circ Q : F \to X$ (the solution $x \in X$ of $QSx = Qg$), $A_S(e) = e - \xi_S(Se)$ on E and $B_S(f) = (1-P)f - S\xi_S((1-P)f)$ on F. There are a closed subspace $X \subseteq E$ and $\varepsilon > 0$ such that (a) $S \mapsto A_S$ and $S \mapsto B_S$ are continuous on $\|S - T\| < \varepsilon$ in the operator norm, and $(S, e) \mapsto A_S e$, $(S, f) \mapsto B_S f$ are jointly continuous there; (b) for every such S: $P, Q \in \Pi_S$; $E = \ker(PS) \oplus X = \ker(QS) \oplus X$ topologically; $e - A_S e \in X$ and $QSA_S e = 0$ ((6.15), (6.16)); $QB_S f = 0$ and $Sx + B_S f = (1-P)f$ for some $x \in X$ ((6.18), (6.19)); A_S restricts to a linear isomorphism $\ker(PS) \to \ker(QS)$ and B_S induces a linear isomorphism $F/R(P) \to F/R(Q)$, with $A \circ j_P = j_Q$, $\Phi_Q \circ A = B \circ \Phi_P$ and $\pi_Q \circ B = \pi_P$ for the exact sequences $E(S, P)$, $E(S, Q)$ (V6.12). The hypotheses are satisfiable ($T = 1$ on $\mathbb{R}$, $P = Q$). *Book:* Proposition 6.4.13 (2017: 6.6), p. 253 (2017 p. 195), last sentence:

for T' near T with $P, Q \in \Pi_{T'}$ the constructions of $A_{T'}$ and $B_{T'}$ depend continuously on T'; argument pp. 253, 255 (with X fixed, the equation (6.15) for T' has a unique solution depending continuously on T'); no preprint counterpart. *Hidden hypotheses:* none; $P, Q \in \Pi_S$ for S near T follows from Lemma 6.5.1 (preprint 6.17) (V6.20) and is part of the conclusion. *Deviation (explicit):* "depend continuously" is read as the continuity of A_S extended to E and of B_S lifted to F, which determine the maps ker$(PS) \to$ ker(QS) and $F/H \to F/H$ between the varying kernels; the joint continuity says that $(S, e) \mapsto (S, A_S e)$ is continuous on the kernel bundle $\bigcup_S \{S\} \times$ ker(PS) of Remark 6.4.5 (V6.21) with its subspace topology. *Proof:* the text's. X is V6.69 for the kernels of PT and QT, of equal dimension by the indices; by Lemma 6.5.1 (preprint 6.17) for (T, P, X) and (T, Q, X), $PS|_X$ and $QS|_X$ are isomorphisms onto $R(P)$, $R(Q)$ near T, and inversion is continuous at isomorphisms, so $S \mapsto \xi_S$ is continuous; the diagram is the construction of the algebraic part (V6.67) for the solution operator ξ_S (Findings, items 113, 117).

V6.71 (the orientation space and the properties of Theorem 6.6.17; `Polyfold.OrSpace`, `Polyfold.orProj`, `Polyfold.segPath`, `Polyfold.isScSmoothMap_segPath`, `Polyfold.OrPullback`, `Polyfold.IsContSectionOver`, `Polyfold.IsPropagationContinuous`, `Polyfold.HasTwoContSections`, `Polyfold.IsOrientationTopology`). For an sc-Fredholm section f of a strong bundle $P : Y \to X$, O_f is the set of pairs (x, o) with $x \in X_\infty$ and $o \in \mathrm{Or}^f_x$, an orientation of DET(f, x) read in the chosen chart (V6.55), and $\pi_f(x, o) = x$; X_∞ carries the metrizable topology of Theorem 2.3.10 (preprint 2.16) (V2.17). For an sc-smooth path φ (an sc-smooth map $\mathbb{R} \to X$), $\varphi^* O_f$ is the subspace $\{(t, p) : \varphi(t) = \pi_f(p)\}$ of $[0, 1] \times O_f$, and the segment $s \mapsto \varphi(a + s(b - a))$ from $\varphi(a)$ to $\varphi(b)$ is sc-smooth. For a topology on O_f: a continuous section over $U \cap X_\infty$ is a continuous $\sigma : U_\infty \to O_f$ with $\pi_f \circ \sigma = \mathrm{id}$, U_∞ with the topology of X_∞; property (2) asks that every $x \in X_\infty$ have an open $U \ni x$ in X with exactly two continuous sections over $U \cap X_\infty$; property (3) asks that for every sc-smooth path φ, every continuous section $\alpha(t) = (t, (\varphi(t), o_t))$ of $\varphi^* O_f$ and every $t_0 \in [0, 1]$, for all t near t_0 an open U with the local propagation property in some chart contains the segment of φ from t_0 to t, and the push-forward along it in U maps o_{t_0} to o_t; an orientation topology is a Hausdorff topology with (1) π_f a local homeomorphism, (2) and (3). *Book:* pp. 274–275, before Theorem 6.6.17, and the three properties of Theorem 6.6.17, p. 275, new in the 2021 edition (the set O_f appears without a topology in the preprint after Definition 6.27, p. 219, and in the 2017 text after Definition 6.9, p. 205). *Hidden hypotheses:* the topology of X_∞ ("the natural metrizable topology"). *Deviation (explicit):* "o_t is the local propagation of o_{t_0}" is read as the push-forward `orPush` (V6.57) along the segment of φ in a set with the local propagation property, which does not depend on that set (`Polyfold.orPush_compat`); paths are the sc-smooth maps $\mathbb{R} \to X$ of decision D2, restricted to $[0, 1]$ (Findings, item 118).

V6.72 (sc-smooth paths in X_∞; `Polyfold.rnChart_toBase`, `Polyfold.continuousOn_levelLift_comp`, `Polyfold.continuous_smoothMk_comp`, `Polyfold.chartSegPath`, `Polyfold.isScSmoothMap_chartSegPath`, `Polyfold.eventually_exists_scPath_smoothLevel`, `Polyfold.exists_nhds_scPath_of_isOpen`). An sc-smooth map $\Gamma : \mathbb{R}^n \to X$ takes values in X_∞ and is continuous into X_∞, its level-m coordinates in every chart being continuous. For every neighbourhood N of a smooth point x in X_∞, every $y \in X_\infty$ close enough to x in X_∞ is joined to

x by an sc-smooth path all of whose points lie in N: in a chart, $s \mapsto r(u_0 + \kappa(s)(u_1 - u_0))$ with κ the smooth transition function. For an open $W \ni x$ in X there is an open V, $x \in V \subseteq W$, whose smooth points are pairwise joined by sc-smooth paths in W. *Book:* not stated; needed for the uniqueness in Theorem 6.6.17, p. 275, whose proof the text leaves to the reader. *Hidden hypotheses:* none. *Deviation (not in the book):* tools; the second statement makes small neighbourhoods in X_∞ reachable by paths that stay in them, which is how property (3) controls the topology (Findings, item 119).

V6.73 (the topology of O_f, property (1) and Hausdorffness; `Polyfold.IsContOrOn`, `Polyfold.orSec`, `Polyfold.orBasic`, `Polyfold.orTopology`, `Polyfold.IsContOrOn.eventually_iff`, `Polyfold.isOpen_orBasic`, `Polyfold.continuousOn_orSec`, `Polyfold.continuous_orProj`, `Polyfold.eq_orSec_of_mem_orBasic`, `Polyfold.orSec_mem_orBasic`, `Polyfold.exists_isContOrOn`, `Polyfold.orChartHomeo`, `Polyfold.isLocalHomeomorph_orProj`, `Polyfold.t2Space_orTopology`). Let L be local transport data for the sets Or^f_x (good open sets covering X_∞ with push-forwards, V6.56). A continuous orientation over an open U is a family $y \mapsto o_y$ carried along by the transport of L along every sc-smooth path in U. The topology $\mathcal{T}_L$ on O_f: W is open iff $\{y \in U_\infty : (y, o_y) \in W\}$ is open in X_∞ for every continuous orientation o over an open U. For every L: two continuous orientations agree near a smooth point iff they agree at it; the sets $\{(y, o_y) : y \in U_\infty\}$ are open, $y \mapsto (y, o_y)$ is continuous on U_∞ and π_f is continuous; every point of O_f lies on a continuous orientation over an open set inside a good set of L; π_f restricted to $\{(y, o_y) : y \in U_\infty\}$ is a homeomorphism onto U_∞, so (1) π_f is a local homeomorphism; and $\mathcal{T}_L$ is Hausdorff. *Book:* Theorem 6.6.17, p. 275, new in the 2021 edition: "a uniquely defined Hausdorff topology" with (1); the text leaves the proof to the reader, in view of Proposition 6.6.3 and Theorem 6.6.11 (preprint 6.28, 6.35). *Hidden hypotheses:* none (every L). *Deviation (not in the book):* the construction of the topology. Points over different smooth points are separated through π_f (X_∞ is metrizable); the two points over one y lie on two continuous orientations that differ at y, hence near y (Findings, item 118).

V6.74 (property (2) under connected smooth neighbourhoods; `Polyfold.MPolyfold.HasConnectedSmoothNbhds`, `Polyfold.MPolyfold.hasConnectedSmoothNbhds_of_subsingleton`, `Polyfold.rnPolyfold_hasConnectedSmoothNbhds`, `Polyfold.eq_orSec_or_eq_orSec`, `Polyfold.hasTwoContSections_orTopology`). X has connected smooth neighbourhoods if every smooth point x has, inside every open $W \ni x$, an open $U \ni x$ of X with $U \cap X_\infty$ preconnected in X_∞; $\mathbb{R}^n$ and every M-polyfold on a subsingleton have them. If every Or^f_y has exactly two elements and X has connected smooth neighbourhoods, then for every L the topology $\mathcal{T}_L$ has property (2): for such a U inside the domain of a continuous orientation o, the continuous sections over $U \cap X_\infty$ are exactly $y \mapsto (y, o_y)$ and $y \mapsto (y, -o_y)$. *Book:* Theorem 6.6.17 (2), p. 275. *Hidden hypotheses:* connected smooth neighbourhoods (`HasConnectedSmoothNbhds`). *Deviation (narrower):* (2) is proved under this hypothesis only; Part I does not state it, it holds for $\mathbb{R}^n$ and is open in general (gap G.90). The proofs of §7.4 use that $U \cap X_\infty$ is connected for connected open U (`MPolyfold.SmoothConnected`, gap G.85); as we read it, that fact gives the hypothesis, M-polyfolds being locally path connected (Proposition 2.3.7). *Proof:* the set where a continuous section equals (y, o_y) is open, and so is the set where it equals $(y, -o_y)$, its complement; U_∞ is preconnected (Findings, item 120).

V6.75 (Theorem 6.6.17; `Polyfold.isPropagationContinuous_orTopology`, `Polyfold.eventually_eq_orSec_of_continuousOn`, `Polyfold.eq_orTopology_of`, `Polyfold.existsUnique_orientationTopology`, `Polyfold.theorem_6_6_17_hyp`). Let X be a tame M-polyfold with sc-smooth bump functions, $P : Y \to X$ a strong bundle with sc$^+$-sections through every smooth fibre point, f an sc-Fredholm section with tame germ data (the hypotheses of Theorem 6.6.13 (preprint 6.37), V6.57, with its paracompactness instances for $\mathbb{R}^n \times X$, $\mathbb{R}^k \times \mathbb{R}^n \times X$ and their pull-back bundles), and $\mathcal{T}$ the topology of V6.73 for the local transport data of Γ_f. Then $\mathcal{T}$ has property (3); every topology on O_f with (1) and (3) is $\mathcal{T}$; there is exactly one Hausdorff topology on O_f with (1) and (3); and if X has connected smooth neighbourhoods (V6.74), exactly one Hausdorff topology with (1), (2) and (3). *Book:* Theorem 6.6.17, p. 275, new in the 2021 edition (the preprint and the 2017 text define the set O_f, V6.71, but no topology on it): for a strong bundle over a tame X and an sc-Fredholm f there is a uniquely defined Hausdorff topology on O_f characterized by (1)–(3); "the easy proof is left to the reader". *Hidden hypotheses:* bump functions, tame germ data and sc$^+$-sections through the smooth fibre points (those of V6.57); connected smooth neighbourhoods for (2). *Deviation (narrower):* existence with (2) needs `HasConnectedSmoothNbhds` (gap G.90); the uniqueness is stronger than stated, from (1) and (3) alone, without Hausdorffness or (2). The remarks after the theorem (the orientation bundle, orientability iff a global continuous section, the continuous $\mathbb{Z}_2$-action, $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f$) are not formalized (gap G.91). *Proof:* (3): near t_0 a continuous section of $\varphi^* O_f$ stays in one open set $\{(y, o_y)\}$, and o is carried along the segments of φ by the transport, which on a good set is the push-forward. Uniqueness: for a topology with (3), a continuous local section through (y_0, o_{y_0}) equals $y \mapsto (y, o_y)$ near y_0, since y is joined to y_0 by a small sc-smooth path (V6.72), along which (3) and the connectedness of $[0, 1]$ carry the equality; the local homeomorphisms of (1) then match the open sets of both topologies (Findings, items 119, 121).

V6.76 (Theorem 6.7.1 (preprint 6.43): the invariant $\Psi_{(f,o)}$; `Polyfold.theorem_6_7_1`, `Polyfold.cobordismOrientationHyp`, `Polyfold.isEndProductOrientation_cobOr`, `Polyfold.theorem_6_7_1_of_orientation`, `Polyfold.cobordismRelativeFormsInner`, `Polyfold.CobordismRelativeFormsInner`, `Polyfold.psiAt_eq_of_cobordism_of_orientation`, `Polyfold.psiAt_eq_of_admissible_of_orientation`, `Polyfold.psiAt_eq_of_homotopy_of_orientation`, `Polyfold.MPolyfold.exists_closed_form_of_ends`, `Polyfold.MPolyfold.faceRestriction_pullOver`, `Polyfold.ProductEnd.pullAt`, `Polyfold.exists_glued_extension`, `Polyfold.ProductEnd.IsEndProductOrientation.of_subEnd`, `Polyfold.MPolyfold.faceAlongData`, `Polyfold.MPolyfold.isChartFaceForm_faceAlongData`, `Polyfold.cobOr`, `Polyfold.isContinuous_cobOr`, `Polyfold.cobordismOrientation_of_ends`, `Polyfold.theorem_6_7_1_of_indep`, `Polyfold.PsiIndepAdmissible`, `Polyfold.PsiIndepHomotopy`, `Polyfold.psiInvariant_spec`, `Polyfold.psiInvariant_homotopy`, `Polyfold.theorem_6_7_1_witness`, `Polyfold.CobordismOrientation`, `Polyfold.integral_eq_of_cutCobordism`, `Polyfold.psiAt`, `Polyfold.psiForms_relD`, `Polyfold.StrongBundle.IsAdmissible`, `Polyfold.solPartIsOrientedManifold`, `Polyfold.IsProperOrientedHomotopyBetween`). `theorem_6_7_1` (`Chapter6/Theorem671Final2021.lean`): for X tame with sc-smooth bump functions there is Ψ assigning to a proper tame sc-Fredholm section f with an orientation o a linear map $\Psi_{(f,o)} : H^k_{dR}(X, \partial X) \to \mathbb{R}$ (`relDeRham`, V4.91) in every degree k with (1): for every auxiliary norm N there are open $U \supseteq S_f$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \leq C\}$ compact for $C > 0$; for each such U, S_{f+s} is compact for every allowable s, admissible sections exist (`StrongBundle.IsAdmissible`), and for each admissible s, S_{f+s} is the union of its index parts S_k, each a compact oriented Mathlib manifold of dimension k (`SolPartIsOrientedManifold`), with $\Psi([\omega, \tau]) = \int_{S_k} \omega|_S - \int_{\partial S_k} \tau|_{\partial S}$ (`solIntegral`, `solBoundaryIntegral`); and (2): $\Psi_{(f_0,o_0)} = \Psi_{(f_1,o_1)}$ for a proper oriented homotopy (`IsProperOrientedHomotopyBetween`). Ψ is `psiAt` for one chosen admissible section; `psiAt` is well defined on classes by Stokes on S_k and on its faces (`psiForms_relD`). `theorem_6_7_1_of_indep` proves the theorem from (A) `PsiIndepAdmissible` and (B) `PsiIndepHomotopy` with no other hypothesis; `psiAt_eq_of_admissible_of_orientation` and `psiAt_eq_of_homotopy_of_orientation` prove (A) and (B) from the cobordism identity `psiAt_eq_of_cobordism_of_orientation` (Stokes on the index-$(k+1)$ part of S_F with a cut-off in time, `integral_eq_of_cutCobordism`, for the relative form `cobordismRelativeFormsInner`) and the cobordism data `exists_isCobordismData`, `exists_homotopy_cobordism_data`. On the point $\mathbb{R}^0$ with $f = 0$, $\Psi([1]) = \pm 1$ (`theorem_6_7_1_witness`). In `theorem_6_7_1` these two are applied to `cobordismOrientationHyp`: the cobordism S_F carries the orientation $e \wedge o_0$ of F (`cobOr`, `SectionOrientation.prodFirst`), which is $e \wedge o_{S_i}$ on its two ends (`isEndProductOrientation_cobOr`), and the relative form `cobordismRelativeFormsInner`. The earlier form `theorem_6_7_1_of_orientation` (`Chapter6/CobordismRelativeForms2021b.lean`) took the orientation of S_F as a hypothesis. *Book:* §6.7 (preprint §6.6), Theorem 6.7.1 (preprint 6.43), p. 277 (preprint p. 231) (proof pp. 277–279 (preprint pp. 231–233)), Over a tame X with sc-smooth bump functions there is a well-defined map associating with a proper oriented sc-Fredholm section (f, o) a linear map $\Psi_{(f,o)} : H^*_{dR}(X, \partial X) \to \mathbb{R}$ with (1) for an auxiliary norm N and s admissible for (N, U), S_{f+s} is an oriented compact manifold with boundary with corners and $\Psi_{(f,o)}([\omega, \tau]) = \int_{S_{f+s}} \omega - \int_{\partial S_{f+s}} \tau$, the integrals being zero if the dimensions do not agree; and (2) $\Psi_{(f_0,o_0)} = \Psi_{(f_1,o_1)}$ for a proper oriented homotopy. The proof compares two admissible sections through the cobordism S_F of Remark 5.3.8, oriented from o and the standard orientation of $[0, 1]$, and gets (2) by "the same homotopy argument". *Deviation (restated):* (2026-11-12) No hypothesis beyond the text's: the orientation of the cobordism S_F, of product form $e \wedge o_{S_i}$ on its two ends, is proved (`cobordismOrientationHyp`), and the relative form $p_1^*\omega - d\beta$ on S_F is built on the inner ends $(0, 1/16) \times S_0$ and $(15/16, 1) \times S_1$ (`cobordismRelativeFormsInner`), since the cobordism data give the product structure only for $t \leq 1/8$ and $t \geq 7/8$. Before 2026-11-12 the orientation was a hypothesis (`theorem_6_7_1_of_orientation`; Findings, Chapter 6, items 134, 136). The proper oriented homotopy of (2), which the text does not define, is read as in R-d: $f_1 = f_0 + S(1, \cdot)$ for an sc⁺-section S over $\mathbb{R}^1 \times X$ with $S(0, \cdot) = 0$, compact zeros over $t \in [0, 1]$ and the properness clause on that slab, and $o_1 = o_0$ transported (`IsProperOrientedHomotopyBetween`; Findings, Chapter 6, item 130). The neighbourhood U of (1) carries the margin of reading R-b, and (1) holds for every auxiliary norm and every such U; the text defines Ψ from one N (item 128). "Zero if the dimensions do not agree" pairs degree k with the index-k part S_k (R-c, item 129). The text's $[0, 1] \times X$ is read as $\mathbb{R}^1 \times X$ (decision D2), and the decomposition $\partial S_F = S_0 \cup S_1 \cup \tilde{S}$ is replaced by a cut-off in time on the product ends (items 131–133). *Proof:* `psiAt`, for one admissible section, is well defined on classes by Stokes on S_k and on its faces; two admissible sections, or the two ends of a proper oriented homotopy, are compared through the cobordism S_F with a cut-off in time: $\int_{S_F} d(gA) = 0$, and on the product ends $\int_{S_F} d(g_i A) = \int_{S_i} a_i$, for the relative form A built on the inner ends; the orientation $e \wedge o_0$ of S_F is $e \wedge o_{S_i}$ on the ends, the time vector first and with no sign (Findings, Chapter 6, items 128–134, 136).

V6.77 (Theorem 6.7.1 (preprint 6.43) for $\partial X = \emptyset$ (p. 279); `Polyfold.theorem_6_7_1_noBoundary`, `Polyfold.theorem_6_7_1_noBoundary_of_orientation`, `Polyfold.theorem_6_7_1_noBoundary_of_indep`, `Polyfold.psiInvariantNoBoundary`, `Polyfold.psiInvariantNoBoundary_mk`, `Polyfold.psiInvariantNoBoundary_of_ne`, `Polyfold.MPolyfold.relDeRhamEquivOfNoBoundary`). `theorem_6_7_1_noBoundary` (`Chapter6/Theorem671Final2021.lean`): for X tame with sc-smooth bump functions and $\partial X = \emptyset$ (`IsWithoutBoundary`), there is Φ with $\Phi_{(f,o)} : H^k_{sc}(X) \to \mathbb{R}$ in every degree ($\Phi = \Psi$ under $H^k_{dR}(X, \partial X) \cong H^k_{sc}(X)$, `relDeRhamEquivOfNoBoundary`) such that for every N, every U as in (1) and every admissible s: if the index of f is n at every smooth point, then S_{f+s} is compact and equal to S_n, $\Phi([\omega]) = \int_{S_{f+s}} \omega|_S$ in degree n, and $\Phi = 0$ in every degree $k \neq n$; and Φ is invariant under proper oriented homotopies. `theorem_6_7_1_noBoundary_of_indep` is the same from (A) and (B); `theorem_6_7_1_noBoundary_of_orientation` is the earlier form under the orientation of the cobordism. *Book:* §6.7 (preprint §6.6), after the proof of Theorem 6.7.1 (preprint 6.43), p. 279 (preprint p. 233), For $\partial X = \emptyset$, $\Psi_{(f,o)} : H^n_{dR}(X) \to \mathbb{R}$ is $\Psi_{(f,o)}([\omega]) = \int_{S_{f+s}} \omega$ for $[\omega] \in H^n_{dR}(X)$ if n is the index of f, and zero otherwise. *Deviation (restated):* As V6.76: no hypothesis beyond the text's (the orientation of the cobordism is proved, `cobordismOrientationHyp`), and homotopies are read as in R-d. The text's "if n is equal to the index, and zero otherwise" is stated as two clauses for a constant index n: the integral in degree n, and $\Phi = 0$ in the other degrees (R-c). The text writes $H^n_{dR}(X)$; for $\partial X = \emptyset$ it is $H^n_{sc}(X)$ (Definition 4.4.6, V4.88), identified with $H^n_{dR}(X, \partial X)$ by `relDeRhamEquivOfNoBoundary` (V4.91).

V6.78 (The alternative determinant $\lambda(\ker T) \otimes \lambda(\operatorname{coker}(T)^*)$ (p. 246); `Polyfold.iotaDualEquiv`, `Polyfold.altLinDet`, `Polyfold.altLinDetEquiv`, `Polyfold.altLinDetEquiv_tmul_dualBasis`, `Polyfold.altFredholmDet`, `Polyfold.altFredholmDetEquiv`, `Polyfold.finrank_altFredholmDet`). `iotaDualEquiv`: the convention ι of Definition 6.3.1 (preprint 6.3) is an isomorphism $\lambda(A^*) \cong \lambda^*(A)$ for finite-dimensional A, natural by Proposition 6.3.2 (preprint 6.4) (V6.3, V6.4). `altLinDet T` $=$ $\lambda(\ker T) \otimes \lambda((F/R(T))^*)$ and `altLinDetEquiv T` $= \mathrm{id} \otimes \iota : \lambda(\ker T) \otimes \lambda(\operatorname{coker}(T)^*) \cong \det(T)$; it maps $(a_1 \wedge \cdots \wedge a_k) \otimes (d_1^* \wedge \cdots \wedge d_m^*)$ to $(a_1 \wedge \cdots \wedge a_k) \otimes (d_1 \wedge \cdots \wedge d_m)^*$ for a basis d of the cokernel and its dual basis (`altLinDetEquiv_tmul_dualBasis`). For Fredholm T: `altFredholmDet`, `altFredholmDetEquiv`, `finrank_altFredholmDet`. *Book:* §6.4 (preprint §6.3), after Definition 6.4.1 (preprint 6.7), p. 246 (preprint p. 208), Some authors use $\det(T) = \lambda(\ker T) \otimes \lambda(\operatorname{coker}(T)^*)$; the two definitions are naturally isomorphic, given a convention identifying $\lambda^*(A)$ and $\lambda(A^*)$. *Deviation (same):* As in the text, with the convention ι of Definition 6.3.1 (preprint 6.3) as the identification of $\lambda^*(A)$ and $\lambda(A^*)$. The chapter's statements use Definition 6.4.1 (preprint 6.7) (V6.8) (Findings, Chapter 6, item 10).

V6.79 (The identification (6.25) (preprint (95)) and $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ (p. 257); `Polyfold.stdLine`, `Polyfold.stdLine_one`, `Polyfold.stdLine_symm_wedge`, `Polyfold.stdLineDual`, `Polyfold.stdLineDual_dualVec`, `Polyfold.tensorStdLine`, `Polyfold.tensorStdLine_tmul_stdWedge`, `Polyfold.tensorStdLine_tmul_wedge`, `Polyfold.tensorStdLineDual`, `Polyfold.tensorStdLineDual_tmul_dualVec`). `stdLine n` : $\mathbb{R} \cong \lambda(\mathbb{R}^n)$, $1 \mapsto e_1 \wedge \cdots \wedge e_n$ (`stdLine_one`), with inverse (6.25) $c_1 \wedge \cdots \wedge c_n \mapsto \det([c_1, \ldots, c_n])$, the matrix with columns c_j (`stdLine_symm_wedge`). Its transpose `stdLineDual n` : $\lambda^*(\mathbb{R}^n) \cong \mathbb{R}^*$, $(e_1 \wedge \cdots \wedge e_n)^* \mapsto 1^*$ (`stdLineDual_dualVec`), is the isomorphism of Proposition 6.4.15 (preprint 6.15). `tensorStdLine n V` : $V \otimes \lambda(\mathbb{R}^n) \cong V$ and `tensorStdLineDual n V` : $V \otimes \lambda^*(\mathbb{R}^n) \cong V$ send $v \otimes (e_1 \wedge \cdots \wedge e_n)$ and $v \otimes (e_1 \wedge \cdots \wedge e_n)^*$ to v (`tensorStdLine_tmul_stdWedge`, `tensorStdLineDual_tmul_dualVec`), and $v \otimes (c_1 \wedge \cdots \wedge c_n)$ to $\det([c])\, v$ (`tensorStdLine_tmul_wedge`), for every V. *Book:* §6.4 (preprint §6.3), after Proposition 6.4.15 (preprint 6.15), p. 257 (preprint p. 213), Dual to (94), $\mathbb{R} \to \lambda(\mathbb{R}^n)$ maps 1 to the wedge of the standard basis, with inverse (95) $c_1 \wedge \cdots \wedge c_n \mapsto \det([c_1, \ldots, c_n])$; this gives identifications $V \otimes \lambda(\mathbb{R}^n) \to V$ and $V \otimes \lambda^*(\mathbb{R}^n) \to V$ for every finite-dimensional V. *Deviation (stronger):* The text asks V to be finite-dimensional; the identifications hold for every V. The formal statements of the chapter use (94) only (V6.17, V6.18) (Findings, Chapter 6, item 10).

V6.80 (The remark that $\mathrm{DET}(E, F)$ is not orientable (end of §6.5, p. 262); `Polyfold.isEmpty_lineOrientation_det_hilbert`, `Polyfold.isEmpty_lineOrientation_det_of_equiv_hilbert`, `Polyfold.isEmpty_lineOrientation_det_of_equiv_prod`, `Polyfold.isEmpty_lineOrientation_det_of_equiv_ellTwo`, `Polyfold.isEmpty_lineOrientation_det_ltwo`, `Polyfold.nonempty_lineOrientation_det_iff_of_equiv`, `Polyfold.isEmpty_lineOrientation_det_of_equiv`, `Polyfold.isEmpty_lineOrientation_det_ellTwo`, `Polyfold.isEmpty_lineOrientation_det_of_complexStructure`, `Polyfold.isEmpty_lineOrientation_det_of_kernel_loop`, `Polyfold.isEmpty_lineOrientation_det_prod`, `Polyfold.isEmpty_lineOrientation_det_prod_of_equiv`, `Polyfold.continuous_kerDetMk`, `Polyfold.isEmpty_lineOrientation_of_loop`, `Polyfold.exists_two_lineOrientation_det_of_finiteDimensional`, `Polyfold.nonempty_lineOrientation_det_of_finiteDimensional`). Non-orientable is `IsEmpty (LineOrientation (DetFiber (E := E) (F := F)))`: $\mathrm{DET}(E, F)$ over all Fredholm operators has no orientation in the sense of Proposition 6.6.1 (preprint 6.26). `isEmpty_lineOrientation_det_hilbert`: none for $E = F = H$, every infinite-dimensional real Hilbert space H (separable or not); `isEmpty_lineOrientation_det_of_equiv_hilbert`: none when E and F are isomorphic to such an H; `isEmpty_lineOrientation_det_of_equiv_prod`: none when E, F are isomorphic to $H \times H$ for a Banach space H with a surjective $S : H \to H$ with odd-dimensional kernel, e.g. $H \cong \mathbb{R} \times H$; `isEmpty_lineOrientation_det_ellTwo`: $\ell^2 \times \ell^2$. In general, `isEmpty_lineOrientation_det_of_complexStructure`: none whenever E and F carry bounded J with $J^2 = -1$ and there is a surjective $T : E \to F$ with kernel of odd finite dimension; the loop $A_\theta = R'_\theta T R_{-\theta}$, $\theta \in [0, \pi]$, $R_\theta = \cos\theta + \sin\theta J$, consists of surjective operators, a basis of the kernels $R_\theta(\ker T)$ is a continuous section of DET in the chart with $P = 1$ (`continuous_kerDetMk`), and it returns to its negative (`isEmpty_lineOrientation_det_of_kernel_loop`, `isEmpty_lineOrientation_of_loop`). Orientability depends only on the isomorphism classes of E and F (`nonempty_lineOrientation_det_iff_of_equiv`, through `LineOrientation.conj`). For finite-dimensional E, F: `exists_two_lineOrientation_det_of_finiteDimensional`, $\mathrm{DET}(E, F)$ has exactly two orientations. *Book:* §6.5 (preprint §6.4), after Lemma 6.5.9 (preprint 6.25), p. 262 (preprint p. 218), The text remarks, with no hypothesis on E, F and no proof or reference, that the line bundle $\mathrm{DET}(E, F)$ over the Fredholm operators $\mathcal{F}(E, F)$ (with the topology of Proposition 6.5.8 (preprint 6.24)) is known to be non-orientable, though orientable over certain subsets. *Deviation (narrower):* The text gives no hypothesis on E, F. For finite-dimensional E, F the remark does not hold as we read it: every operator is Fredholm, the base $L(E, F)$ is convex, and $\mathrm{DET}(E, F)$ has exactly two orientations (`exists_two_lineOrientation_det_of_finiteDimensional`, on $\mathrm{DET}(E, F)$ itself) (Findings, Chapter 6, items 26, 135). Proved under hypotheses: E, F isomorphic to one infinite-dimensional Hilbert space, or to $H \times H$ with $H \cong \mathbb{R} \times H$, or, in general, complex structures on E and F and a surjective operator with kernel of odd finite dimension. Non-orientability for infinite-dimensional Banach spaces without such structures is not formalized and not claimed. The remark presumably intends infinite-dimensional spaces of the kind in the applications. The orientability "over certain subsets" is Proposition 6.6.1 (preprint 6.26) (V6.30).

V6.81 (The classical determinant line bundle at the opening of §6.6 (p. 262); `Polyfold.derivFredholm`, `Polyfold.ClassicalDet`, `Polyfold.classicalDet_apply`, `Polyfold.classicalDet_vectorBundle`, `Polyfold.classicalDet_two_orientations`). For f of class C^1 on an open $U \subseteq E$ with every $Df(x)$ Fredholm: `derivFredholm`, $x \mapsto Df(x)$ as a continuous map $U \to \mathcal{F}(E, F)$; `ClassicalDet`, the pull-back of $\mathrm{DET}(E, F)$ along it, with fibre $\det(Df(x))$ (`classicalDet_apply`), a topological line bundle (`classicalDet_vectorBundle`). `classicalDet_two_orientations`: for contractible U it has precisely two orientations, and every orientation of one line $\det(Df(x_0))$ extends to exactly one orientation of the bundle. *Book:* §6.6 (preprint §6.5), the opening paragraph, p. 262 (preprint p. 219), For a classical Fredholm map $f : U \to F$ on an open $U \subseteq E$, the determinant line bundle of the $Df(x)$ is a smooth line bundle over U, with precisely two orientations if U is contractible. *Deviation (narrower):* f is of class C^1, which is weaker than the text's "smooth". The bundle is a topological line bundle, the pull-back of $\mathrm{DET}(E, F)$ (V6.28); its smoothness, for which smooth frames of Fredholm charts would be needed, is not formalized (Findings, Chapter 6, item 26).

V6.82 (The remarks after Theorem 6.6.17: the orientation bundle, orientability, $\mathbb{Z}_2$ and $\det_f$ (p. 275); `Polyfold.isSectionOrientation_iff_continuous`, `Polyfold.isOrientableSection_iff_exists_continuous_section`, `Polyfold.continuous_orSec_iff`, `Polyfold.exists_eq_orSec`, `Polyfold.orNegHomeomorph`, `Polyfold.orNeg_ne`, `Polyfold.orMulAction`, `Polyfold.continuous_orUnitsAct`, `Polyfold.detCore`, `Polyfold.detBundleCore`, `Polyfold.detMk_eq_iff`, `Polyfold.localTriv_detMk`). With the topology of Theorem 6.6.17 (`orTopology`, V6.73–V6.75), for Γ_f (`orLocalTransport`): `isSectionOrientation_iff_continuous`: $y \mapsto (y, o_y)$ (`orSec o`) is continuous iff o is an orientation of f (Definition 6.6.16 (preprint 6.40)); `exists_eq_orSec`: every section of π_f has this form; `isOrientableSection_iff_exists_continuous_section`: f is orientable (Definitions 6.6.14–6.6.15 (preprint 6.38–6.39)) iff π_f has a global continuous section. `orNegHomeomorph`: $(x, o_x) \mapsto (x, -o_x)$ is a homeomorphism over π_f without fixed points (`orNeg_ne`), giving a continuous action of $\mathbb{Z}^\times \cong \mathbb{Z}_2$ (`orMulAction`, `continuous_orUnitsAct`). `detCore`, `detBundleCore`: $\det_f$ as a Mathlib `VectorBundleCore` with fibre $\mathbb{R}$ over X_∞, trivialized by the continuous orientations (U, o) with coordinate changes ± 1; its fibre is $\mathbb{R} \otimes_{\mathbb{Z}_2} \mathrm{Or}^f_x$: $r \otimes a = r' \otimes a'$ iff $(r', a') = (\pm r, \pm a)$ (`detMk_eq_iff`, `localTriv_detMk`). The generic statements hold for any local transport with two-element fibres (`continuous_orSec_iff`). *Book:* §6.6, after Theorem 6.6.17, p. 275 (new in the 2021 edition), $\pi_f : O_f \to X_\infty$ with the topology of Theorem 6.6.17 is the orientation bundle of f; f is orientable iff π_f has a global continuous section; $\mathbb{Z}_2$ acts continuously on O_f by $(x, o_x) \mapsto (x, -o_x)$; $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f \to X_\infty$ is the determinant bundle of f. *Deviation (restated):* The hypotheses are those of the existence part of Theorem 6.6.17 (V6.75: tame X, bump functions, tame germ data, sc⁺-sections through points); nothing is added. $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f$ is built as the vector bundle core whose trivializations are the continuous orientations, and its fibres are identified with $\mathbb{R} \otimes_{\mathbb{Z}_2} \mathrm{Or}^f_x$ (Findings, Chapter 6, item 121).

What is not verified in this chapter

- preprint p. 201, the facts recalled before Proposition 6.2.2 (preprint 6.2): an M-polyfold is locally path connected, and two smooth points in one path component are joined by an sc⁺-smooth path. Only the local form inside a chart is formalized (V6.49), and “in the same path component” (Definitions 6.6.2, 6.6.14 (preprint 6.27, 6.38)) is read as “joined by an sc-smooth path” (V6.58) (gap G.56; Findings, item 42).
- preprint p. 201, the members of $\mathrm{Lin}(f, x)$ as sc-Fredholm operators on every level: they are shown Fredholm on level 0, with the index $\mathrm{ind}(f, x)$, which is what the determinant lines use (V6.1) (gap G.64).
- preprint p. 208, the alternative definition $\det(T) = \lambda(\ker T) \otimes \lambda(\mathrm{coker}(T)^*)$ and its identification with Definition 6.4.1 (preprint 6.7). Listed here before 2026-11-12; verified since (V6.78; gap G.57, closed).
- preprint p. 213, the identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after Proposition 6.4.15 (preprint 6.15). Listed here before 2026-11-12; verified since, for every V (V6.79; gap G.58, closed); the chapter’s statements use only (94) (V6.17, V6.18).
- p. 259 (preprint p. 216), Lemma 6.5.3 (preprint 6.19) in general (exterior powers, duals, sums and tensor products of finite-dimensional topological vector bundles); it is applied only to trivial bundles, and Proposition 6.5.4 (preprint 6.20) is proved with explicit frames (V6.22) (gap G.59).
- pp. 261–263 (preprint pp. 218–219), the remark that $\mathrm{DET}(E, F)$ is not orientable, and the classical case at the opening of §6.6 (preprint §6.5). Listed here before 2026-11-12; verified since in a corrected form (V6.80, narrower; V6.81; gap G.60, closed): $\mathrm{DET}(E, F)$ has no orientation for E, F isomorphic to one infinite-dimensional Hilbert space, and under complex structures with a surjective operator of odd-dimensional kernel; for finite-dimensional E, F the remark does not hold as we read it (two orientations; Findings, items 26, 135). Not verified: non-orientability for infinite-dimensional Banach spaces without such structures, and the smoothness of the classical bundle, which is verified as a topological line bundle for C^1 maps (V6.81).

- p. 263 (preprint p. 220), Proposition 6.6.3 (1) (preprint 6.28). Listed here before 2026-11-13 as the sc-smooth contractibility of U, the preprint's wording; the 2021 text (as the 2017 one) prints "$U' \subset U$ is sc-smoothly contractible in U to the point x", verified since (V6.32; gap G.61 closed; Findings, item 137). Not formalized: the preprint's "U contractible" and with it the word "contractible" in the preprint's Theorem 6.6.11, Definition 6.6.12 and Theorem 6.6.13 (□); (1′) is what the proofs use (V6.48, V6.50, V6.57).
- p. 265 (preprint p. 222), Lemma 6.6.5 (preprint 6.30) as a statement about manifolds: the solution set as a manifold with boundary with corners near $(t, \varphi(t), 0)$, the map π and the smooth vector bundle L. Listed here before 2026-11-12; verified since, near the family, with L a C^∞ vector bundle over a neighbourhood of $[0,1]^n$ read in one chart (V6.34, V6.42; gap G.62, closed; Findings, items 48, 126).
- preprint p. 222, property (P) of Lemma 6.6.4 (preprint 6.29) for the slices F_t. Listed here before 2026-11-12; verified since, also on the M-polyfold $[0,1]^n$ at its boundary points (V6.33; gap G.63, closed; Findings, item 47).
- §6.6 (preprint §6.5), paths, homotopies and families sc-smooth on $[0,1]$, $[0,1]^2$, $[0,1]^n$ only, and $X \times [0,1]$ as an M-polyfold with boundary: they are read on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ and $\mathbb{R}^n \times X$ (decision D2), as for Proposition 6.2.2 (preprint 6.2) in Chapter 3; a path that does not extend sc-smoothly to $\mathbb{R}$ is not covered (gap G.65; Findings, item 102). Since 2026-11-12 Proposition 6.6.3 (3), (4) (V6.32), Lemmas 6.6.4–6.6.5 with (P) and π at the boundary points (V6.33, V6.34) and the path category (V6.52) are verified on $[0,1]$, $[0,1]^n$ themselves; still read on $\mathbb{R}$: Proposition 6.6.3 (1′), the smooth bundle L and the stabilized kernels (V6.34, V6.42), and Remark 6.6.19 over $\mathbb{R}^1 \times X$ (V6.62). Since 2026-11-13 (1′) for paths on $[0,1]$ and L over $[0,1]$ for $n = 1$ are verified (V6.32, V6.34; Findings, item 138); still read on $\mathbb{R}$: L over $[0,1]^n$ for $n \geq 2$, the stabilized kernels for families with sections over $[0,1]^n \times X$, and Remark 6.6.19 over $\mathbb{R}^1 \times X$ (gap G.65 in part).
- p. 275 (preprint p. 229), Theorem 6.6.18 (preprint 6.41), the orientation of S as an orientation of the abstract finite-dimensional manifold S (of its tangent bundle in the charts of the induced structure); what is verified is the tangent orientation at each zero and its continuity along the sc-smooth paths of zeros in a chart (V6.61) (gap G.67).
- p. 275 (preprint p. 229), Theorem 6.6.18 (preprint 6.41), the compatibility of the tangent orientations read in two charts at the same zero (V6.61) (gap G.68).
- pp. 274–276 (preprint pp. 229–230), the paragraph after Theorem 6.6.18 (preprint 6.41): for f not generic, a small sc⁺-perturbation s supported near the zero set, the orientation of S_{f+s} from o_f, and $\text{Lin}(f+s, x) = \text{Lin}(f, x)$ for the orientation of $\det((f+s)'(x))$ from $\text{DET}(f, x)$ (gap G.69).
- p. 276 (preprint p. 230), Remark 6.6.19 (preprint 6.42) beyond a point: that $(t, y) \mapsto \hat{o}$ is an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40), built from an orientation of f. Listed here before 2026-11-12; verified since, over $\mathbb{R}^1 \times X$ (V6.62; gap G.66, closed). Its use in Theorem 6.7.1, the value $e \wedge o_{S_i}$ of the orientation of the cobordism S_F on the ends, is verified since 2026-11-12 (`Polyfold.cobordismOrientationHyp`, `Polyfold.isEndProductOrientation_cobOr`; V6.76).

- p. 276 (preprint p. 230), §6.7 (preprint §6.6), the sections "admissible for the pair (N, U)"; the facts recalled there are Theorem 5.3.10 (preprint 5.18) (V5.38, V5.39) and the orientation of S_{f+s} (V6.61) (gap G.55). Listed here before 2026-11-10; verified since, with the neighbourhood U read with a margin (V6.76; gap G.55, closed; Findings, item 128).
- p. 277 (preprint p. 231), Theorem 6.7.1 (preprint 6.43) (the invariant $\Psi_{(f,o)}$ on $H^*_{dR}(X, \partial X)$), with its proof, pp. 277–279 (preprint pp. 231–233), and the case $\partial X = \emptyset$, p. 279 (preprint p. 233). Listed here before 2026-11-12 (proved under steps of the text's proof taken as hypotheses); verified since without hypotheses on the cobordism (`Polyfold.theorem_6_7_1`, `Polyfold.theorem_6_7_1_noBoundary`; V6.76, V6.77, restated; gap G.54 closed): the orientation of S_F is $e \wedge o_{S_i}$ on its ends (`Polyfold.cobordismOrientationHyp`), and the relative form on S_F is built on inner ends (`Polyfold.cobordismRelativeFormsInner`; Findings, item 136). The proper oriented homotopy of (2), which the text does not define, is read through Remark 6.6.19 (Findings, item 130), and the extension $p_1^*\omega \wedge p_2^*dt$ of ω is read as $p_1^*\omega$ (possible misprint; Findings, items 103, 131).
- §§6.2 and §6.6 (preprint §6.1 and §6.5), T_xX, Y_x, $\mathrm{Lin}(f, x)$ and Or_x^f as chart-free objects: they are made in chart points and identified between charts (V6.53–V6.55), since T_xX carries no linear structure here (gaps G.6, G.7, G.13).
- p. 253, Proposition 6.4.13 (2017: 6.6), the last sentence: that A_T and B_T depend continuously on T near a given operator. Listed here when the 2021 additions were first reported; verified since, with a closed common complement (V6.69) and the continuity of A_S on E and B_S on F, also on the kernel bundle (V6.70) (gap G.81, closed; Findings, items 113, 117).
- p. 275, Theorem 6.6.17, property (2) (two continuous sections over $U \cap X_\infty$): verified only for M-polyfolds with connected smooth neighbourhoods (small open U with $U \cap X_\infty$ connected in X_∞; V6.74, V6.75), which holds for $\mathbb{R}^n$; the existence and uniqueness without (2) need no such hypothesis (gap G.90; Findings, item 120).
- p. 275, the remarks after Theorem 6.6.17: π_f as the orientation bundle of f, "f orientable iff π_f has a global continuous section", the continuous $\mathbb{Z}_2$-action on O_f and the determinant bundle $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f$. Listed here before 2026-11-12; verified since (V6.82; gap G.91, closed; Findings, item 121).
- The 2021 changes to items verified earlier: Proposition 6.4.15 (preprint 6.15) with its hypotheses made explicit (V6.18, already with them), Proposition 6.6.3 (preprint 6.28) for a tame X (V6.32, without tameness; (1) in its 2021 form since 2026-11-13, gap G.61 closed), Lemma 6.6.5 (preprint 6.30) with the solution set near $(t, \varphi(t), 0)$ in the points (t, y, λ) (gap G.62, closed since 2026-11-12) (Findings, item 115).

8. Chapter 7 (HWZ, Part II, §§7.1–7.6 (2017 §§7.1–7.5)): verified except the tangent $T\varphi$ of Definition 7.1.4 (2017 Definition 7.4), the remark after Definition 7.1.7 (2017 Definition 7.7) in its 2017 form, and the Hausdorff property of $\mathbf{X}^R$ in the 2017 Proposition 7.9 and Proposition 7.10 (rewritten in 2021 as Proposition 7.2.10 and Theorem 7.2.14, which assume it)

Chapter 7 (pp. 287–356 (2017 pp. 225–256); Part II has no 2014 preprint, so numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) introduces ep-groupoids, the sc-smooth version of étale proper Lie groupoids. §7.1 defines groupoids, isotropy groups and ep-groupoids (Definitions 7.1.1–7.1.3 (2017: 7.1–7.3)), the local sc-diffeomorphism $\hat{\varphi}$ and the tangent $T\varphi$ of a morphism (Definition 7.1.4 (2017: 7.4)), the levels and the degeneracy index of the orbit space (Definition 7.1.5 (2017: 7.5)), tame ep-groupoids (Definition 7.1.6 (2017: 7.6)) and the category $\mathcal{EP}$ (Definition 7.1.7 (2017: 7.7)); it proves the openness of π, the countable neighbourhood bases of $|X|$, the local closure relation and the finiteness of the isotropy groups (Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1)), states the natural representation of G_x (Theorem 7.1.13 (2017: 7.1)), discusses the translation groupoid $G_x \ltimes U(x)$ and the map $\widehat{\Phi}$ (pp. 293–297 (2017 pp. 231–233)), extends the natural representation to finite sets of isomorphic objects (Proposition 7.1.15 (2017: 7.5), Definition 7.1.18 (2017: 7.8)), identifies $|U(x)|$ with $G_x \backslash U(x)$ (Proposition 7.1.19 (2017: 7.6)), and defines local uniformizers and ep-subgroupoids (Definitions 7.1.20, 7.1.21 (2017: 7.9, 7.10), Proposition 7.1.22 (2017: 7.7)). §7.2 treats effective ep-groupoids (Definition 7.2.1 (2017: 7.11), Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Proposition 7.2.4 (2017: 7.8)), the subgroups T_x, the reduced morphism space $\mathbf{X}^R$ and the reduced ep-groupoid X_R (Definitions 7.2.7, 7.2.15 (2017: 7.12, 7.13), Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017)). §7.3 proves that $|X|$ is locally metrizable, regular and Hausdorff (Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2), Definition 7.3.7 (2017: 7.14)) and lifts ep-groupoids to level 1 (Propositions 7.3.9, 7.5.3 (2017: 7.11, 7.12)). §7.5 (2017 §7.4) defines paracompact orbit spaces and sc-smooth partitions of unity (Definitions 7.5.1, 7.5.5 (2017: 7.15, 7.16), Remarks 7.5.2, 7.5.6 (2017: 7.1, 7.2)), proves the existence of invariant partitions of unity (Theorems 7.5.4, 7.5.7 (2017: 7.3, 7.4), Corollary 7.6.6 (2017: 7.3)) and characterizes the closed subsets of $|X|$ (Definition 7.5.8 (2017: 7.17), Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14)). Appendix 7.6 (2017 7.5) proves Theorem 7.1.13 (2017: 7.1) as Theorem 7.6.1 (2017: 7.5) (Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9)) and Theorem 7.5.7 (2017: 7.4) as Theorem 7.6.4 (2017: 7.6) (Lemma 7.6.5 (2017: 7.10)).

What is verified. **Theorems 7.1.13–7.6.4 (2017: 7.1–7.6) hold**, with the readings recorded below, and so do Propositions 7.1.8–7.2.4, 7.3.9–7.5.10 (2017: 7.1–7.8, 7.11–7.14), Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) under the hypothesis that $\mathbf{X}^R$ is Hausdorff and paracompact, Lemmas 7.1.11–7.6.5 (2017: 7.1–7.10), and Corollaries 7.3.6–7.6.6 (2017: 7.1–7.3) (Corollary 7.6.6 (2017: 7.3) with the theorem of Godefroy–Troyanski–Whitfield–Zizler as an explicit hypothesis, as in Part I). Every definition of the chapter is formalized: Definitions 7.1.1 (2017: 7.1; V7.1), 7.1.2 (2017: 7.2; V7.2), 7.1.3 (2017: 7.3; V7.3), 7.1.4 (2017: 7.4; V7.5, $T\varphi$ as a definition only), 7.1.5 (2017: 7.5; V7.7), 7.1.6 (2017: 7.6; V7.8), 7.1.7 (2017: 7.7; V7.9), 7.1.18 (2017: 7.8; V7.26), 7.1.20 (2017: 7.9; V7.28), 7.1.21 (2017: 7.10; V7.29), 7.2.1 (2017: 7.11; V7.33), 7.2.7 (2017: 7.12; V7.38), 7.2.15 (2017: 7.13; V7.41), 7.3.7

(2017: 7.14; V7.49), 7.5.1 (2017: 7.15; V7.53), 7.5.5 (2017: 7.16; V7.56), 7.5.8 (2017: 7.17; V7.59). Several places of the text needed more than a reading. (a) *The remark after Definition 7.1.7 (2017: 7.7).* That sc-smoothness of a functor on objects implies sc-smoothness on morphisms does not hold as we read it (a counterexample on the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$); it holds for functors continuous on morphisms, which is what the text's argument uses (V7.10). (b) *The reduced morphism space.* The proof of Proposition 7.2.10 (2017: 7.9) gives the charts of $\mathbf{X}^R$ and their sc-smooth transitions, but, as we read it, neither Hausdorffness nor paracompactness; both are hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), discharged for effective ep-groupoids (V7.39, V7.40; gap G.71). (c) *Connectedness.* The natural representation of Theorem 7.1.13 (2017: 7.1) is taken on a connected U, which the text chooses on 2017 p. 231 but does not put in the statement; it is what makes Φ a homomorphism, Γ a functor, and Lemma 7.2.2 (2017: 7.2) true, and the local path connectedness of M-polyfolds that supplies it is proved (V7.17, V7.18, V7.34). (d) *Tameness of X^1.* Proposition 7.3.9's (2017: 7.11) "if X is tame so is X^1" needs that the index lift of a tame retraction is tame, which Part I left open; it is proved (V7.52). (e) *Omitted steps.* The uniqueness of natural representations behind "adapting restrictions" (V7.24, V7.35), the conjugation $\varphi T_x \varphi^{-1} = T_y$ needed for $\mathbf{X}^R$ (V7.37), the passage from compact to component compact sets in Corollary 7.3.8 (2017: 7.2) (V7.51), the support bound that makes the sum in the proof of Theorem 7.5.7 (2017: 7.4) locally finite (V7.66, V7.67), and Michael's theorem on unions of closed paracompact sets (V7.50) are proved; Proposition 7.1.15 (2017: 7.5) is proved by a construction without induction (V7.25). Not verified: the properties of $T\varphi$ (Definition 7.1.4 (2017: 7.4)), the set $|X_\infty|$ of the filtration of $|X|$, the tangent facts (3), (4) recalled before Proposition 7.1.22 (2017: 7.7), Hausdorffness and paracompactness of $\mathbf{X}^R$ for a general ep-groupoid, and the open question of 2017 p. 249 (see the list at the end of this section). An addendum (2026-11-12, the Part II wave; Findings, items 9, 95, 107, 111, 130, 131, 139, 156) verifies the 2021 statements as printed: the remark after Definition 7.1.7 with its continuity on morphisms (V7.10), Proposition 7.2.10 (charts and transitions), Theorem 7.2.14 and Proposition 7.2.16 under T_R Hausdorff (V7.39, V7.40), Corollary 7.6.6 without Theorem 5.5.16 as a hypothesis (V7.68), Theorem 7.6.7 and Corollary 7.6.8 for every i as isomorphisms in $\mathcal{EP}$ (V7.83; gap G.82 closed), and Zhou's example of Appendix 7.6.3 itself, on $L^2(\mathbb{R})$ with $E_m = H^{m,\delta_m}(\mathbb{R})$, with its exact non-effective part (V7.82, V7.95; gap G.83 closed); Remark 7.5.6 has nothing to formalize (V7.57). Every gap of the chapter is closed.

The setting. Obj and Mor are types with M-polyfold structures `PObj`, `PMor`, Hausdorff and paracompact, as for every M-polyfold of the project (Definition 2.3.4 (2017: 2.8)). An ep-groupoid is `Polyfold.EpGroupoid`, a `Polyfold.GroupoidStr` (the maps s, t, m, u, ι with the groupoid laws on composable pairs; the composition of a non-composable pair is a junk value, never used) with the conditions (1)–(4) of Definition 7.1.3 (2017: 7.3). Properness is Mathlib's `IsProperMap` of the restriction $t : s^{-1}(\mathrm{cl}_X V) \to X$, equivalent to the sequential form the proofs use (V7.4). "An open neighbourhood $U(x)$ equipped with the natural G_x-action" is a datum `Polyfold.EpGroupoid.NaturalRep` (V7.17), never a hypothesis "for every such $U(x)$". $|X|$ carries the quotient topology. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 7".

V7.1 (Definition 7.1.1 (2017: 7.1), groupoids and the orbit space; `Polyfold.GroupoidStr`,

`Polyfold.GroupoidStr.src`, `Polyfold.GroupoidStr.tgt`, `Polyfold.GroupoidStr.comp`, `Polyfold.GroupoidStr.unit`, `Polyfold.GroupoidStr.inv`, `Polyfold.GroupoidStr.comp_assoc`, `Polyfold.GroupoidStr.comp_unit`, `Polyfold.GroupoidStr.unit_comp`, `Polyfold.GroupoidStr.comp_inv`, `Polyfold.GroupoidStr.inv_comp`, `Polyfold.GroupoidStr.mor`, `Polyfold.GroupoidStr.morOn`, `Polyfold.GroupoidStr.Isomorphic`, `Polyfold.GroupoidStr.orbitSetoid`, `Polyfold.GroupoidStr.OrbitSpace`, `Polyfold.GroupoidStr. `, `Polyfold.GroupoidStr. _eq_ _iff`, `Polyfold.GroupoidStr.saturation`, `Polyfold.GroupoidStr.IsSaturated`, `Polyfold.GroupoidStr.isSaturated_iff`, `Polyfold.GroupoidStr.Ob`, `Polyfold.GroupoidStr.instGroupoid`, `Polyfold.GroupoidStr.orbitSetoid_eq`). A groupoid is a set of objects and a set of invertible morphisms with source, target, an associative composition on composable pairs, units and inverses; $|X|$ is the quotient by $x \sim y$ iff there is a morphism $x \to y$, with projection π. *Book:* Definition 7.1.1 (2017: 7.1) and the paragraph after it, pp. 287–288 (2017 pp. 225–226). *Hidden hypotheses:* none. *Deviation (restated):* a structure of maps on types, the laws required on composable pairs and the composition total with a junk value; the saturation $\pi^{-1}(\pi(A))$ of Definition 7.1.21 (2017: 7.10) is defined here; the bridge to Mathlib's `CategoryTheory.Groupoid` is added so that "full" and "faithful" are Mathlib's (Findings, item 1).

V7.2 (Definition 7.1.2 (2017: 7.2), the isotropy group; `Polyfold.GroupoidStr.Isotropy`, `Polyfold.GroupoidStr.Isotropy.val_mul`, `Polyfold.GroupoidStr.Isotropy.val_one`, `Polyfold.GroupoidStr.Isotropy.val_inv`, `Polyfold.GroupoidStr.isotropyEquivVertexGroup`, `Polyfold.GroupoidStr.isotropyEquivAut`). G_x is the group of morphisms $x \to x$ under composition. *Book:* Definition 7.1.2 (2017: 7.2), p. 288 (2017 p. 226). *Hidden hypotheses:* none. *Deviation (restated):* the multiplication is $gh = g \circ h$, the text's; Mathlib's vertex group multiplies in the other order and is identified with the opposite group (Findings, item 2).

V7.3 (Definition 7.1.3 (2017: 7.3), ep-groupoids; `Polyfold.EpGroupoid`, `Polyfold.EpGroupoid.PObj`, `Polyfold.EpGroupoid.PMor`, `Polyfold.EpGroupoid.isLocalScDiffeo_src`, `Polyfold.EpGroupoid.isLocalScDiffeo_tgt`, `Polyfold.EpGroupoid.isScSmoothMap_unit`, `Polyfold.EpGroupoid.isScSmoothMap_inv`, `Polyfold.EpGroupoid.exists_isProperMap`, `Polyfold.EpGroupoid.isScSmoothComp`, `Polyfold.IsScSmoothComp`, `Polyfold.IsScSmoothCompSheet`, `Polyfold.isScSmoothComp_iff_sheet`, `Polyfold.EpGroupoid.isScSmoothCompSheet`, `Polyfold.EpGroupoid.isScSmoothMap_src`, `Polyfold.EpGroupoid.isScSmoothMap_tgt`, `Polyfold.MPolyfold.isSubMPolyfold_fibredProduct`). An ep-groupoid is a groupoid with M-polyfold structures on objects and morphisms such that (1) s, t are local sc-diffeomorphisms, (2) u and ι are sc-smooth, (3) every x has an open $V(x)$ with $t : s^{-1}(\mathrm{cl}_X V(x)) \to X$ proper, (4) m is sc-smooth on the fibred product $\mathbf{X}\,{}_s{\times}_t\,\mathbf{X}$; all structure maps are then sc-smooth. *Book:* Definition 7.1.3 (2017: 7.3), pp. 288 ff. (2017 pp. 226–227). *Hidden hypotheses:* Obj and Mor Hausdorff and paracompact (the project's M-polyfolds). *Deviation (restated):* (3) existential per point, with Mathlib's proper maps; the universal form "for every open $V \ni x$" fails (the pair groupoid of $\mathbb{Z}$), and every smaller open neighbourhood works. (4) on the sub-M-polyfold of Part I's Proposition 2.5.20 (2017: 2.15), the book's statement of the text's citation "Lemma 2.8 in [38]" (an external reference, not a misprint); the proofs use the equivalent sheet form (Findings, items 3, 4, 103).

V7.4 (properness and its sequential form; `Polyfold.isProperMap_iff_exists_tendsto_subseq`, `Polyfold.exists_tendsto_subseq_of_isProperMap`, `Polyfold.isProperMap_domRestrict_iff`, `Polyfold.isProperMap_domRestrict_iff_of_isClosed`, `Polyfold.MPolyfold.isProperMap_domRestrict_iff`, `Polyfold.MPolyfold.isProperMap_domRestrict_iff_isCompact_preimage`, `Polyfold.isProperMap_domRestrict_closure_mono`, `Polyfold.isClosed_image_preimage_of_isProperMap_domRestrict`, `Polyfold.isClosed_of_isProperMap_domRestrict`, `Polyfold.finite_setOf_of_isProperMap_domRestrict`). For the restriction of t to a closed $s^{-1}(Q)$ between metrizable spaces, properness is equivalent to: every sequence with convergent targets has a convergent subsequence; it passes to smaller closures, and proper restrictions have closed images and finite fibres. *Book:* the notion of Definition 7.1.3 (3) (2017: 7.3), 2017 p. 227, used in sequential form in Proposition 7.1.10 (2017: 7.3), Lemma 7.1.11 (2017: 7.1) (p. 292 (2017 p. 230)), Proposition 7.1.22 (2017: 7.7) (p. 307 (2017 p. 238)), Proposition 7.3.9 (2017: 7.11) (2017 p. 248), Proposition 7.5.9 (2017: 7.13) (p. 342 (2017 p. 251)) and Lemma 7.6.2 (2017: 7.8) (2017 p. 254). *Hidden hypotheses:* metrizability (Theorem 2.3.10 (2017: 2.2)), which M-polyfolds have. *Deviation (not in the book):* the equivalence is used without comment in the text (Findings, item 3).

V7.5 (Definition 7.1.4 (2017: 7.4), the local sc-diffeomorphism $\hat{\varphi}$ and the tangent $T\varphi$; `Polyfold.EpGroupoid.Sheet`, `Polyfold.EpGroupoid.Sheet.hat`, `Polyfold.EpGroupoid.Sheet.isScDiffeoOn_hat`, `Polyfold.EpGroupoid.exists_sheet`, `Polyfold.EpGroupoid.sheetAt`, `Polyfold.EpGroupoid.mem_sheetAt`, `Polyfold.EpGroupoid.hat`, `Polyfold.EpGroupoid.isScDiffeoOn_hat`, `Polyfold.EpGroupoid.hat_src`, `Polyfold.EpGroupoid.tangentMor`). For a morphism $\varphi : x \to y$ there are open $\mathbf{U}(\varphi)$, $U(x)$, $U(y)$ with $s : \mathbf{U}(\varphi) \to U(x)$, $t : \mathbf{U}(\varphi) \to U(y)$ sc-diffeomorphisms, and $\hat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} : U(x) \to U(y)$ is an sc-diffeomorphism with $\hat{\varphi}(x) = y$; $T\varphi := T\hat{\varphi}(x)$. *Book:* 2017 p. 227 (the paragraph before the definition) and Definition 7.1.4 (2017: 7.4), p. 289 (2017 p. 227). *Hidden hypotheses:* none. *Deviation (narrower):* $\hat{\varphi}$ is verified; $T\varphi$ is a definition only, and that it is a bounded linear isomorphism on level 0 for $\varphi \in \mathbf{X}_1$ and an sc-isomorphism for smooth φ is not verified (gaps G.7, G.70). The display reads "$\hat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} \circ U(x) \to U(y)$", $\circ$ for $:$ (possible misprint), and "$T\varphi : T_{s(\varphi)}X \to T_{t(\varphi)}$" lacks the X (possible misprint or extraction artefact) (Findings, items 5, 110).

V7.6 (the levels of an ep-groupoid and the filtration of $|X|$; `Polyfold.EpGroupoid.mem_levelSet_src_iff`, `Polyfold.EpGroupoid.mem_levelSet_tgt_iff`, `Polyfold.MPolyfold.maxLevel`, `Polyfold.MPolyfold.IsLocalScDiffeo.maxLevel_eq`, `Polyfold.MPolyfold.IsLocalScDiffeo.degIndex_eq`, `Polyfold.EpGroupoid.degIndex_src`, `Polyfold.EpGroupoid.degIndex_tgt`, `Polyfold.EpGroupoid.orbitLevel`). s and t preserve the levels, so $ml(x) = ml(\varphi) = ml(y)$ for $\varphi : x \to y$; $d_X(s(\varphi)) = d_{\mathbf{X}}(\varphi) = d_X(t(\varphi))$, so d_X is constant on classes; $|X_k| = \pi(X_k)$. *Book:* pp. 289–291 (2017 pp. 227–228), with Part I's Proposition 2.4.4 (2017: 2.7). *Hidden hypotheses:* none. *Deviation (narrower):* the set $|X_\infty|$ and $|X_\infty| = \bigcap_k |X_k|$ are not stated (gap G.72); the density of X_∞ in every X_k, stated on 2017 p. 227 without proof, is not formalized and is not in Part I for M-polyfolds (gap G.73; Findings, items 6, 112, 116).

V7.7 (Definition 7.1.5 (2017: 7.5), the induced degeneracy index; `Polyfold.EpGroupoid.degIndexOrbit`, `Polyfold.EpGroupoid.degIndexOrbit_`). $d_{|X|}(|x|) = d_X(x)$ is well defined. *Book:* Definition 7.1.5 (2017: 7.5), p. 290 (2017 p. 228). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 6).

V7.8 (Definition 7.1.6 (2017: 7.6), tame ep-groupoids, and the tameness of X; `Polyfold.EpGroupoid.IsTame`, `Polyfold.MPolyfold.IsTame`, `Polyfold.MPolyfold.IsScDiffeoOn.exists_tame_chart`, `Polyfold.MPolyfold.IsLocalScDiffeo.isTame`, `Polyfold.MPolyfold.IsLocalScDiffeo.isTame_iff_of_surjective`). X is tame if the object M-polyfold is tame; then the morphism M-polyfold is tame. *Book:* Definition 7.1.6 (2017: 7.6) and the paragraph after it, 2017 p. 228. *Hidden hypotheses:* none; "tame" is the existence of an equivalent tame atlas (Definition 2.5.7 (2017: 2.19)), satisfiable (`Polyfold.EpGroupoid.isTame_trivial_rn`, `Polyfold.EpGroupoid.Examples.isTame_reflection`); "every chart tame" would be unsatisfiable. *Deviation (restated):* the tame atlas of **X** is the one transported along s (Findings, items 7, 106).

V7.9 (Definition 7.1.7 (2017: 7.7), the category $\mathcal{EP}$; `Polyfold.GroupoidFunctor`, `Polyfold.GroupoidFunctor.mor_inv`, `Polyfold.GroupoidFunctor.id`, `Polyfold.GroupoidFunctor.comp`, `Polyfold.GroupoidFunctor.toFunctor`, `Polyfold.GroupoidFunctor.orbitMap`, `Polyfold.EpFunctor`, `Polyfold.EpFunctor.isScSmoothMap_obj`, `Polyfold.EpFunctor.isScSmoothMap_mor`, `Polyfold.EpFunctor.id`, `Polyfold.EpFunctor.comp`, `Polyfold.EpFunctor.ext`, `Polyfold.EP`). The objects of $\mathcal{EP}$ are the ep-groupoids, its morphisms the functors sc-smooth on objects and on morphisms. *Book:* Definition 7.1.7 (2017: 7.7), p. 290 (2017 p. 228). *Hidden hypotheses:* none. *Deviation (restated):* functors commute with s, t, m, u; the compatibility with ι is derived (Findings, item 8).

V7.10 (the remark after Definition 7.1.7 (2017: 7.7): sc-smoothness on objects and continuity on morphisms suffice; `Polyfold.EpGroupoid.mor_eq_srcInv_obj_src`, `Polyfold.EpGroupoid.isScSmoothMap_mor_of_continuous`, `Polyfold.EpGroupoid.isScSmoothMap_mor_iff_continuous`, `Polyfold.EpFunctor.continuous_mor`, `Polyfold.EpFunctor.ofContinuous`, `Polyfold.EpGroupoid.Examples.remark_definition_7_1_7_counterexample`). A functor sc-smooth on objects and continuous (on level 0) on morphisms is sc-smooth on morphisms, by $f(\gamma) = (s|\mathbf{U}(\psi))^{-1}(f(s(\gamma)))$ on $\mathbf{U}(\varphi)$. *Book:* p. 291 (2017 pp. 228–229); the 2021 text asks the continuity on morphisms. *Hidden hypotheses:* none for the 2021 text. *Deviation (as in the book, 2021):* the argument needs only a sheet $\mathbf{U}(\psi)$ at $\psi = f(\varphi)$ and $\mathbf{U}(\varphi) = f^{-1}(\mathbf{U}(\psi))$. The 2017 sentence, without continuity, does not hold as we read it: a functor of the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$, the identity on objects, is sc-smooth on objects and not continuous on morphisms (Findings, items 9, 104).

V7.11 (the quotient topology and Proposition 7.1.8 (2017: 7.1), openness of π; `Polyfold.EpGroupoid.continuous_` , `Polyfold.EpGroupoid.proposition_7_1_8`, `Polyfold.EpGroupoid.proposition_7_1_8_isOpenQuotientMap`, `Polyfold.EpGroupoid.proposition_7_1_8_map_nhds`). $|X|$ carries the quotient topology and π is open. *Book:* 2017 p. 229 and Proposition 7.1.8 (2017: 7.1), p. 291 (2017 p. 229). *Hidden hypotheses:* none. *Deviation (restated):* proved through the saturation $\pi^{-1}(\pi(U)) = t(s^{-1}(U))$ and the openness of t (Findings, item 10).

V7.12 (Proposition 7.1.9 (2017: 7.2), countable neighbourhood bases; `Polyfold.EpGroupoid.proposition_7_1_9`, `Polyfold.EpGroupoid.proposition_7_1_9_of_hasAntitoneBasis`, `Polyfold.EpGroupoid.proposition_7_1_9_firstCountableTopology`). Every $z \in |X|$ has a monotone countable basis of open neighbourhoods. *Book:* Proposition 7.1.9 (2017: 7.2), p. 291 (2017 p. 229), with Theorem 2.3.10 (2017: 2.2). *Hidden hypotheses:* none. *Deviation (restated):* Mathlib's `HasAntitoneBasis`, radii $1/(k+1)$ (Findings, item 10).

V7.13 (Proposition 7.1.10 (2017: 7.3), the local closure relation; `Polyfold.EpGroupoid.proposition_7_1_10`, `Polyfold.EpGroupoid.proposition_7_1_10_of_isProperMap`). Every x has an open U with $\pi(\mathrm{cl}_X A) = \mathrm{cl}_{|X|}\pi(A)$ for all $A \subseteq U$. *Book:* Proposition 7.1.10 (2017: 7.3), pp. 292 ff. (2017 pp. 229–230). *Hidden hypotheses:* none. *Deviation (stronger):* for every U with the properness property and every $A \subseteq \mathrm{cl}_X U$, proved from the closedness of the proper map instead of sequences (Findings, item 11).

V7.14 (Lemma 7.1.11 (2017: 7.1); `Polyfold.EpGroupoid.lemma_7_1_11`, `Polyfold.EpGroupoid.lemma_7_1_11_swap`, `Polyfold.EpGroupoid.lemma_7_1_11_of_isProperMap`, `Polyfold.EpGroupoid.lemma_7_1_11_swap_of_isProperMap`, `Polyfold.finite_setOf_mem_of_isProperMap_domRestrict`, `Polyfold.IsProperMap.finite_preimage_singleton_of_isLocallyInjective`). If $t : s^{-1}(\mathrm{cl}_X U(x_0)) \to X$ is proper, only finitely many ψ have $s(\psi) = x$ and $t(\psi) \in U(x_0)$. *Book:* Lemma 7.1.11 (2017: 7.1), p. 292 (2017 p. 230). *Hidden hypotheses:* none; the openness of $U(x_0)$ and $x_0 \in U(x_0)$ are not used. *Deviation (restated):* the proof treats the inverses (s and t exchanged), and both forms are proved; finiteness from a compact discrete fibre. Possible misprints in the proof: "$t : V(\psi) \to V(y)$" for $V(x)$, "$s(\psi) \in s^{-1}(\mathrm{cl}_X(U(x_0)))$", and unbalanced parentheses in statement and proof (Findings, items 12, 96).

V7.15 (Proposition 7.1.12 (2017: 7.4), finiteness of the isotropy groups; `Polyfold.EpGroupoid.proposition_7_1_12`, `Polyfold.MPolyfold.finite_isotropy_of_isLocalScDiffeo`). G_x is finite. *Book:* Proposition 7.1.12 (2017: 7.4), p. 293 (2017 p. 230). *Hidden hypotheses:* none. *Deviation:* none; from Lemma 7.1.11 (2017: 7.1) with $x_0 = x$ (Findings, item 13).

V7.16 (the identification $G_x \to G_y$, $g \mapsto \varphi \circ g \circ \varphi^{-1}$; `Polyfold.GroupoidStr.isotropyConj`, `Polyfold.GroupoidStr.isotropyConj_val`, `Polyfold.EpGroupoid.conjIsotropy`, `Polyfold.EpGroupoid.conjIsotropy_val`, `Polyfold.EpGroupoid.nonempty_isotropy_mulEquiv`, `Polyfold.EpGroupoid.nat_card_isotropy_eq`, `Polyfold.EpGroupoid.isotropyConj_eq_isotropyConj_conj`). Isomorphic objects have isomorphic isotropy groups, by conjugation with a chosen $\varphi : x \to y$; there is no canonical identification. *Book:* 2017 p. 230, after Proposition 7.1.12 (2017: 7.4). *Hidden hypotheses:* none. *Deviation (stronger):* two choices of φ differ by an inner automorphism of G_x (equivalently, of G_y) (Findings, item 13).

V7.17 (Theorem 7.1.13 (2017: 7.1), the natural representation of G_x; `Polyfold.EpGroupoid.NaturalRep`, `Polyfold.EpGroupoid.NaturalRep.U`, `Polyfold.EpGroupoid.NaturalRep.isOpen_U`, `Polyfold.EpGroupoid.NaturalRep.isConnected_U`, `Polyfold.EpGroupoid.NaturalRep.mem_U`, `Polyfold.EpGroupoid.NaturalRep.Φ`, `Polyfold.EpGroupoid.NaturalRep.Γ`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_Φ`, `Polyfold.EpGroupoid.NaturalRep.Φ_`

`mul`, `Polyfold.EpGroupoid.NaturalRep.isScSmoothOn_Γ`, `Polyfold.EpGroupoid.NaturalRep.Γ_self`, `Polyfold.EpGroupoid.NaturalRep.src_Γ`, `Polyfold.EpGroupoid.NaturalRep.tgt_Γ`, `Polyfold.EpGroupoid.NaturalRep.existsUnique_Γ`, `Polyfold.EpGroupoid.NaturalRep.Φ_one`, `Polyfold.EpGroupoid.NaturalRep.mapsTo_Φ`, `Polyfold.EpGroupoid.NaturalRep.Φ_inv_Φ`, `Polyfold.EpGroupoid.NaturalRep.bijOn_Γ`, `Polyfold.EpGroupoid.theorem_7_1_13`**)**. For every open $V \ni x$ there are an open $U \subseteq V$ of x, a homomorphism $\Phi : G_x \to \mathrm{Diff}_{sc}(U)$ and an sc-smooth $\Gamma : G_x \times U \to \mathbf{X}$ with (1) $\Gamma(g, x) = g$, (2) $s(\Gamma(g, y)) = y$, $t(\Gamma(g, y)) = \Phi(g)(y)$, (3) Γ a bijection onto the morphisms between points of U. *Book:* Theorem 7.1.13 (2017: 7.1), p. 293 (2017 p. 231) (proved as Theorem 7.6.1 (2017: 7.5), V7.62). *Hidden hypotheses:* U connected, which the text chooses on 2017 p. 231 and the homomorphism property needs. *Deviation (stronger):* U connected; $\Phi(g)$ an sc-diffeomorphism of U with inverse $\Phi(g^{-1})$; Γ sc-smooth on each slice $\{g\} \times U$, which is sc-smoothness on $G_x \times U$ (V7.20) (Findings, item 20).

V7.18 (local path connectedness of M-polyfolds; `Polyfold.ScRetraction.exists_ball_joinedIn`, `Polyfold.IsScRetract.locallyPathConnectedSpace`, `Polyfold.MChart.pathComponentIn_mem_nhds`, `Polyfold.MPolyfold.locallyPathConnectedSpace`, `Polyfold.MPolyfold.locallyConnectedSpace`, `Polyfold.MPolyfold.hasBasis_isOpen_isConnected`, `Polyfold.MPolyfold.isOpen_connectedComponent`, `Polyfold.MPolyfold.isConnected_iff_isPathConnected`, `Polyfold.MPolyfold.locallyPathConnectedSpace_quotient`, `Polyfold.eqOn_sections_of_isPreconnected`**)**. M-polyfolds and all their quotients are locally path connected; open connected sets are path connected; two sections of a local homeomorphism agreeing at a point agree on a connected set. *Book:* used without statement on p. 293 (2017 p. 231) ("the connected open neighborhood $U(x)$"), in Lemma 7.6.3 (2017: 7.9) (p. 345 (2017 p. 254)) and Corollary 7.3.8 (2017: 7.2) (p. 326 (2017 p. 247)). *Hidden hypotheses:* none. *Deviation (not in the book):* the first fact of gap G.56 of Part I, now formalized (Findings, items 21, 114).

V7.19 (the splitting $\mathbf{U}(x) = \bigsqcup_g \mathbf{U}(g)$; `Polyfold.EpGroupoid.NaturalRep.isOpen_image_Γ`, `Polyfold.EpGroupoid.NaturalRep.Γ_mem_image`, `Polyfold.EpGroupoid.NaturalRep.pairwise_disjoint_image_Γ`, `Polyfold.EpGroupoid.NaturalRep.iUnion_image_Γ`, `Polyfold.EpGroupoid.NaturalRep.isOpen_morOn`, `Polyfold.EpGroupoid.NaturalRep.isConnected_image_Γ`, `Polyfold.EpGroupoid.NaturalRep.connectedComponentIn_morOn`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_src_image`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_tgt_image`, `Polyfold.EpGroupoid.NaturalRep.index`, `Polyfold.EpGroupoid.NaturalRep.Γ_index`, `Polyfold.EpGroupoid.lemma_7_6_2`**)**. The morphisms between points of $U(x)$ form the disjoint union of the connected open sets $\mathbf{U}(g) = \Gamma(g, U)$, on which s and t are sc-diffeomorphisms onto $U(x)$, and $\mathbf{U}(x)$ can be put into any open neighbourhood of G_x. *Book:* 2017 p. 231. *Hidden hypotheses:* none. *Deviation:* none; "arbitrarily small" is Lemma 7.6.2 (2017: 7.8) for arbitrary open N_g followed by Theorem 7.1.13 (2017: 7.1) (Findings, item 22).

V7.20 (the M-polyfold $G \times U$; `Polyfold.discretePolyfold`, `Polyfold.discreteChart`, `Polyfold.discreteChart_compatible`, `Polyfold.isScSmoothMap_discrete_iff`, `Polyfold.isScSmoothOn_univ_prod_discrete_iff`, `Polyfold.isScSmoothMap_prod_discrete_iff`, `Polyfold.EpGroupoid.NaturalRep.objPolyfold`, `Polyfold.EpGroupoid.NaturalRep.`

`morPolyfold`, `Polyfold.EpGroupoid.NaturalRep.isScSmoothMap_smul`). $G \times U$ is an M-polyfold, the finite disjoint union of copies of U. *Book:* 2017 p. 232. *Hidden hypotheses:* none. *Deviation (restated):* the product with the discrete M-polyfold on G; maps on it are sc-smooth iff they are on each slice (Findings, item 23).

V7.21 (the translation groupoid $G_x \ltimes U(x)$ is an ep-groupoid; `Polyfold.GroupoidStr.translation`, `Polyfold.EpGroupoid.NaturalRep.translationGroupoid`, `Polyfold.EpGroupoid.NaturalRep.shearHomeomorph`, `Polyfold.EpGroupoid.NaturalRep.translation_isLocalScDiffeo_src`, `Polyfold.EpGroupoid.NaturalRep.translation_isLocalScDiffeo_tgt`, `Polyfold.EpGroupoid.NaturalRep.translation_isScSmoothCompSheet`, `Polyfold.EpGroupoid.NaturalRep.translation_isProperMap`, `Polyfold.EpGroupoid.NaturalRep.translationEpGroupoid`). The category with objects U, morphisms $G \times U$, $s(g, y) = y$, $t(g, y) = \Phi(g)(y)$, $(g, y) \circ (h, z) = (gh, z)$ is an ep-groupoid. *Book:* p. 294 (2017 p. 232) ("using the proof of Theorem 7.1 one verifies"). *Hidden hypotheses:* none. *Deviation (stronger):* built from the data (Φ, Γ) alone, with the properness for every open V (Findings, item 22).

V7.22 (the functor $\Gamma : G_x \ltimes U(x) \to X$ is fully faithful; `Polyfold.EpGroupoid.NaturalRep.Γ_mul`, `Polyfold.EpGroupoid.NaturalRep.Γ_one`, `Polyfold.EpGroupoid.NaturalRep.Γ_inv`, `Polyfold.EpGroupoid.NaturalRep.functor`, `Polyfold.EpGroupoid.NaturalRep.full`, `Polyfold.EpGroupoid.NaturalRep.faithful`, `Polyfold.EpGroupoid.NaturalRep.epFunctor`, `Polyfold.GroupoidStr.fullSub`). Γ is a functor, full and faithful by Theorem 7.1.13 (3) (2017: 7.1), with image the full subcategory on $U(x)$. *Book:* 2017 p. 232. *Hidden hypotheses:* U connected (V7.17), for Γ to be a functor. *Deviation (stronger):* also an ep-functor, `Polyfold.EpGroupoid.NaturalRep.epFunctor` (Findings, item 22).

V7.23 (the map $\hat{\Phi} : G_x \to \mathrm{Diff}_{sc}(\mathbf{U}(x))$; `Polyfold.EpGroupoid.NaturalRep.Φhat`, `Polyfold.EpGroupoid.NaturalRep.Φhat_Γ`, `Polyfold.EpGroupoid.NaturalRep.Φhat_mem_image`, `Polyfold.EpGroupoid.NaturalRep.index_Φhat`, `Polyfold.EpGroupoid.NaturalRep.src_Φhat`, `Polyfold.EpGroupoid.NaturalRep.tgt_Φhat`, `Polyfold.EpGroupoid.NaturalRep.Φ_conj`, `Polyfold.EpGroupoid.NaturalRep.Φhat_one`, `Polyfold.EpGroupoid.NaturalRep.Φhat_mul`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_Φhat`, `Polyfold.EpGroupoid.NaturalRep.image_Φhat`). $\hat{\Phi}(g)(\varphi) = \Gamma(ghg^{-1}, g * s(\varphi))$ for $\varphi \in \mathbf{U}(h)$ is a homomorphism into the sc-diffeomorphisms of $\mathbf{U}(x)$ with $\hat{\Phi}(g)(\mathbf{U}(h)) = \mathbf{U}(ghg^{-1})$ and the two commutative diagrams with s and t. *Book:* pp. 294–297 (2017 pp. 232–233). *Hidden hypotheses:* none. *Deviation:* none; no possible misprint on pp. 293–297 (2017 pp. 231–233) (Findings, item 22).

V7.24 (uniqueness and restriction of natural representations; `Polyfold.EpGroupoid.NaturalRep.Γ_eqOn`, `Polyfold.EpGroupoid.NaturalRep.Φ_eqOn`, `Polyfold.EpGroupoid.NaturalRep.mapsTo_Φ_connectedComponentIn`, `Polyfold.EpGroupoid.NaturalRep.eventually_Φ_eq_of_actsTrivially`, `Polyfold.EpGroupoid.NaturalRep.restrict`, `Polyfold.EpGroupoid.NaturalRep.restrictInter`, `Polyfold.EpGroupoid.exists_naturalRep_restrict`, `Polyfold.EpGroupoid.exists_naturalRep_shrink`). Two natural representations at x agree on the component of x in $U \cap U'$, and a natural representation restricts to invariant connected open neighbourhoods. *Book:* used without statement on pp. 307–311 (2017 pp. 239–241) (“adapting restrictions”, “$U(x)$ … can be chosen arbitrarily small”). *Hidden hypotheses:* none. *Deviation (not in the book):* stated and proved (Findings, item 24).

V7.25 (Proposition 7.1.15 (2017: 7.5), the natural representation of a finite set of isomorphic objects; `Polyfold.EpGroupoid.NaturalRepFinite`, `Polyfold.EpGroupoid.NaturalRepFinite.F`, `Polyfold.EpGroupoid.NaturalRepFinite.isScDiffeoOn_F`, `Polyfold.EpGroupoid.NaturalRepFinite.F_comp`, `Polyfold.EpGroupoid.NaturalRepFinite.F_unit`, `Polyfold.EpGroupoid.NaturalRepFinite.pairwiseDisjoint_U`, `Polyfold.EpGroupoid.NaturalRepFinite.morOn_eq`, `Polyfold.EpGroupoid.NaturalRepFinite.isScDiffeoOn_src`, `Polyfold.EpGroupoid.NaturalRepFinite.isScDiffeoOn_tgt`, `Polyfold.EpGroupoid.NaturalRepFinite.F_eq`, `Polyfold.EpGroupoid.NaturalRepFinite.natural`, `Polyfold.EpGroupoid.NatRepFinite.transportRep`, `Polyfold.EpGroupoid.NatRepFinite.Star`, `Polyfold.EpGroupoid.NatRepFinite.exists_star`, `Polyfold.EpGroupoid.NatRepFinite.emptyNaturalRepFinite`, `Polyfold.EpGroupoid.proposition_7_1_15_2017`, `Polyfold.EpGroupoid.proposition_7_1_15_of_mem_nhdsSet`). For finitely many mutually isomorphic objects Y and a neighbourhood V of Y there is a functor f assigning disjoint connected open $U(y) \subseteq V$ and sc-diffeomorphisms $f(\psi) : U(y) \to U(y')$, such that the morphisms between the $U(y)$ split into components $\mathbf{U}(\psi)$ with (1) the natural action of G_y on $U(y)$, (2) $s, t : \mathbf{U}(\psi) \to U(y)$ sc-diffeomorphisms, (3) $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$. *Book:* Proposition 7.1.15 (2017: 7.5), pp. 297–304 (2017 pp. 233–236). *Hidden hypotheses:* none. *Deviation (restated):* proved by transporting one natural representation along chosen morphisms from a base object, without the text’s induction or shrinking; V a neighbourhood of the set Y; $Y = \emptyset$ allowed. Twelve possible misprints in the proof, one possibly an extraction artefact (Findings, items 25, 26).

V7.26 (Definition 7.1.18 (2017: 7.8), the natural representation of Y; `Polyfold.EpGroupoid.NaturalRepFinite`, `Polyfold.EpGroupoid.NatRepFinite.mono`). The functor of Proposition 7.1.15 (2017: 7.5) with its properties. *Book:* Definition 7.1.18 (2017: 7.8), p. 304 (2017 p. 236). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 25).

V7.27 (Proposition 7.1.19 (2017: 7.6), $|U(x)| \cong G_x \backslash U(x)$; `Polyfold.EpGroupoid.NaturalRep._eq_ _iff_exists_smul`, `Polyfold.EpGroupoid.NaturalRep.orbitQuotientMap`, `Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_orbitQuotientMap`, `Polyfold.EpGroupoid.NaturalRep.range_orbitQuotientMap`, `Polyfold.EpGroupoid.proposition_7_1_19`, `Polyfold.EpGroupoid.proposition_7_1_19_fullyFaithful`, `Polyfold.EpGroupoid.proposition_7_1_19_orbitMap`, `Polyfold.EpGroupoid.translationOrbitHomeomorph`). $\tau : G_x \backslash U(x) \to \pi(U(x))$, $G_x * y \mapsto |y|$, is a homeomorphism; the functor $G_x \ltimes U(x) \to X$ is fully faithful and induces a homeomorphism of orbit spaces. *Book:* Proposition 7.1.19 (2017: 7.6), pp. 304 ff. (2017 pp. 236–237). *Hidden hypotheses:* none. *Deviation (stronger):* τ is an open embedding into $|X|$; the openness of σ is not needed (Findings, item 27).

V7.28 (Definition 7.1.20 (2017: 7.9), local uniformizers, and the claims after it; `Polyfold.EpGroupoid.LocalUniformizer`, `Polyfold.EpGroupoid.LocalUniformizer.rep`, `Polyfold.EpGroupoid.LocalUniformizer.Ψ`, `Polyfold.EpGroupoid.LocalUniformizer.obj_eq`, `Polyfold.EpGroupoid.LocalUniformizer.full`, `Polyfold.EpGroupoid.LocalUniformizer.`

`faithful`, `Polyfold.EpGroupoid.LocalUniformizer.isEmbedding_orbitMap`, `Polyfold.EpGroupoid.LocalUniformizer.range_orbitMap`, `Polyfold.EpGroupoid.NaturalRep.toLocalUniformizer`, `Polyfold.EpGroupoid.exists_localUniformizer`, `Polyfold.EpGroupoid.LocalUniformizer.injective_obj`, `Polyfold.EpGroupoid.LocalUniformizer.injective_mor`,`Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_functor_obj`,`Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_functor_mor`). A local uniformizer is a fully faithful embedding $\Psi_x : G_x \ltimes U(x) \to X$ inducing a homeomorphism onto $\pi(U(x))$; one exists around every object, and it is injective on objects and morphisms. *Book:* Definition 7.1.20 (2017: 7.9) and the two sentences after it, 2017 p. 237. *Hidden hypotheses:* none. *Deviation (restated):* "embedding" is derived, not a field; existence inside every open $V \ni x$. "$\Psi_x \circ G_x \ltimes U(x) \to X$" ($\circ$ for $:$) and "a a local uniformizer" (possible misprints) (Findings, item 28).

V7.29 (Definition 7.1.21 (2017: 7.10), ep-subgroupoids; `Polyfold.EpGroupoid.IsEpSubgroupoid`, `Polyfold.EpGroupoid.IsEpSubgroupoid.isSubMPolyfold`, `Polyfold.EpGroupoid.IsEpSubgroupoid.isSaturated`, `Polyfold.EpGroupoid.isEpSubgroupoid_univ`, `Polyfold.EpGroupoid.isEpSubgroupoid_of_isOpen`, `Polyfold.EpGroupoid.isEpSubgroupoid_saturation`, `Polyfold.EpGroupoid.isEpSubgroupoid_trivial`). An ep-subgroupoid is the full subcategory of a saturated sub-M-polyfold A of X. *Book:* Definition 7.1.21 (2017: 7.10), p. 305 (2017 p. 237). *Hidden hypotheses:* none; open saturated sets and saturations of open sets are witnesses. *Deviation:* none (Findings, items 40, 105).

V7.30 (the facts (1)–(4) on the sub-M-polyfold A; `Polyfold.EpGroupoid.subObjPolyfold`, `Polyfold.EpGroupoid.isScSmoothMap_val_subObj`, `Polyfold.EpGroupoid.isScSmoothOn_retractInto_subObj`, `Polyfold.MPolyfold.subPolyfold`). (1) $A \to X$ is sc-smooth and a homeomorphism onto its image, (2) $i^{-1} \circ r$ is sc-smooth for a retraction onto $A \cap V$, (3), (4) the tangent spaces of A at smooth points. *Book:* 2017 pp. 237–238, from Part I's Proposition 2.3.19 (2017: 2.6). *Hidden hypotheses:* none. *Deviation (narrower):* (1) and (2) are used; the tangent facts (3) and (4) are not restated, as tangent spaces carry no linear structure here (gap G.7) (Findings, item 40).

V7.31 (the morphism set A is a sub-M-polyfold; `Polyfold.EpGroupoid.src_mem_of_tgt_mem_of_isSaturated`, `Polyfold.EpGroupoid.morOn_eq_preimage_src`, `Polyfold.EpGroupoid.morOn_eq_preimage_tgt`, `Polyfold.EpGroupoid.isSubMPolyfold_preimage_src`, `Polyfold.EpGroupoid.isSubMPolyfold_preimage_tgt`, `Polyfold.EpGroupoid.isSubMPolyfold_morOn`, `Polyfold.EpGroupoid.subMorPolyfold`, `Polyfold.EpGroupoid.isScSmoothMap_val_subMor`, `Polyfold.EpGroupoid.isScSmoothOn_retractInto_subMor`). $\mathbf{A} = \{\varphi \mid s(\varphi), t(\varphi) \in A\}$ is a sub-M-polyfold of $\mathbf{X}$, by the retractions $\mathbf{r} = (s|\mathbf{U}(\varphi))^{-1} \circ r \circ s$. *Book:* 2017 p. 238. *Hidden hypotheses:* none. *Deviation (restated):* $\mathbf{A} = s^{-1}(A)$ for saturated A, the preimage of a sub-M-polyfold under a local sc-diffeomorphism. The claim "$t(\mathbf{r}(\psi)) \in U(a)$, so that $\mathbf{r}(\psi) \in \mathbf{U}(a)$" need not hold and is not used (possible misprint or extraction artefact) (Findings, item 41).

V7.32 (Proposition 7.1.22 (2017: 7.7), the full subcategory of a saturated sub-M-polyfold is an ep-groupoid; `Polyfold.EpGroupoid.isLocalScDiffeo_fullSub_src`, `Polyfold.EpGroupoid.isLocalScDiffeo_fullSub_tgt`, `Polyfold.EpGroupoid.isScSmoothMap_`

`fullSub_unit`, `Polyfold.EpGroupoid.isScSmoothMap_fullSub_inv`, `Polyfold.EpGroupoid.exists_isProperMap_fullSub`, `Polyfold.EpGroupoid.isScSmoothComp_fullSub`, `Polyfold.EpGroupoid.instParacompactSpaceMorOn`, `Polyfold.EpGroupoid.subEpGroupoid`, `Polyfold.EpGroupoid.subEpGroupoid_src_val`, `Polyfold.EpGroupoid.subEpGroupoid_comp_val`, `Polyfold.EpGroupoid.proposition_7_1_22`, `Polyfold.EpGroupoid.subEpFunctor`, `Polyfold.EpGroupoid.subEpFunctor_full`, `Polyfold.EpGroupoid.subEpFunctor_faithful`). For A saturated and a sub-M-polyfold, the full subcategory is an ep-groupoid with the restricted structure maps, A and $\mathbf{A}$ sub-M-polyfolds. *Book:* the argument on 2017 p. 238 and Proposition 7.1.22 (2017: 7.7), pp. 307 ff. (2017 pp. 238–239). *Hidden hypotheses:* none; the paracompactness of A and $\mathbf{A}$, which the project's M-polyfolds need, is proved. *Deviation (stronger):* the steps the text leaves out (sc-smoothness into the sub-M-polyfold, the inclusion of fibred products) are proved; any $V(a)$ with the properness property works; the inclusion is a fully faithful ep-functor (Findings, item 42).

V7.33 (Definition 7.2.1 (2017: 7.11), effective ep-groupoids, regular objects, and the sentence after it; `Polyfold.EpGroupoid.IsEffective`, `Polyfold.GroupoidStr.IsRegular`, `Polyfold.GroupoidStr.regularSet`, `Polyfold.GroupoidStr.isRegular_tgt_iff`, `Polyfold.EpGroupoid.isRegular_tgt_of_isRegular_src`, `Polyfold.EpGroupoid.isSaturated_regularSet`). X is effective if every open neighbourhood of every $g \in G_x \setminus \{1_x\}$ contains a ψ with $s(\psi) \neq t(\psi)$; x is regular if $G_x = \{1_x\}$; if $s(\varphi)$ is regular, so is $t(\varphi)$. *Book:* Definition 7.2.1 (2017: 7.11) and the sentences after it, 2017 p. 239. *Hidden hypotheses:* none; satisfiable non-vacuously (`Polyfold.EpGroupoid.Examples.isEffective_reflection_nonvacuous`). *Deviation:* none (Findings, items 50, 101).

V7.34 (Lemma 7.2.2 (2017: 7.2), X_{reg} is open; `Polyfold.EpGroupoid.NaturalRep.subset_regularSet`, `Polyfold.EpGroupoid.lemma_7_2_2`). The regular part is open. *Book:* Lemma 7.2.2 (2017: 7.2), p. 307 (2017 p. 239). *Hidden hypotheses:* $\Gamma(1, a) = 1_a$, i.e. U connected (V7.17). *Deviation (stronger):* $U(x) \subseteq X_{reg}$ for every natural representation at a regular x (Findings, item 51).

V7.35 (Lemma 7.2.3 (2017: 7.3), effectiveness and injective natural representations; `Polyfold.EpGroupoid.NaturalRep.kerΦ`, `Polyfold.EpGroupoid.NaturalRep.injective_Φ_iff`, `Polyfold.EpGroupoid.NaturalRep.kerΦ_eq_bot_iff`, `Polyfold.EpGroupoid.IsEffective.eq_one`, `Polyfold.EpGroupoid.isEffective_iff_forall_naturalRep`, `Polyfold.EpGroupoid.lemma_7_2_3`). X is effective iff every x has, inside every open $V(x)$, a natural representation with Φ injective. *Book:* Lemma 7.2.3 (2017: 7.3), pp. 308–309 (2017 pp. 239–240). *Hidden hypotheses:* none. *Deviation (stronger):* (1) □ (2) for every natural representation, without sequences; "adapting restrictions" read through V7.24; injectivity as trivial kernel (Findings, item 52).

V7.36 (Proposition 7.2.4 (2017: 7.8), X_{reg} is open and dense for effective X; `Polyfold.EpGroupoid.NaturalRep.stabilizerEquivIsotropy`, `Polyfold.EpGroupoid.proposition_7_2_4`, `Polyfold.EpGroupoid.isEffective_of_dense_regularSet`, `Polyfold.EpGroupoid.isEffective_iff_dense_regularSet`). If X is effective, X_{reg} is open and dense. *Book:* Proposition 7.2.4 (2017: 7.8), pp. 309 ff. (2017 pp. 240–241). *Hidden hypotheses:* none. *Deviation (stronger):* the minimum of $\#G$ is taken over the given open set, so $U(x_0)$ is not needed; the converse also holds (Findings, item 53).

V7.37 (the subgroup T_x of elements acting trivially; `Polyfold.EpGroupoid.ActsTrivially`, `Polyfold.EpGroupoid.trivialSet`, `Polyfold.EpGroupoid.one_mem_trivialSet`, `Polyfold.EpGroupoid.NaturalRep.actsTrivially_iff`, `Polyfold.EpGroupoid.mem_trivialSet_of_section`, `Polyfold.EpGroupoid.mem_trivialSet_iff_eventually`, `Polyfold.EpGroupoid.exists_mem_nhds_trivialSet_iff`, `Polyfold.EpGroupoid.exists_naturalRep_trivialSet_iff`, `Polyfold.EpGroupoid.exists_naturalRep_kerΦ_eq`, `Polyfold.EpGroupoid.trivialSubgroup`, `Polyfold.EpGroupoid.trivialSubgroup_normal`, `Polyfold.EpGroupoid.isEffective_iff_trivialSubgroup_eq_bot`, `Polyfold.EpGroupoid.isotropyConj_mem_trivialSet`, `Polyfold.EpGroupoid.map_isotropyConj_trivialSubgroup`, `Polyfold.EpGroupoid.NaturalRep.mem_trivialSet_of_val_eq_Γ`**).** g acts trivially if $\Phi(g) = 1$ for some natural representation; for $U(x)$ small the kernel of Φ is the set T_x of such g, a normal subgroup. *Book:* 2017 p. 241. *Hidden hypotheses:* none. *Deviation (stronger):* there is a neighbourhood N of x with $\ker \Phi = T_x$ for every natural representation inside N; $g \in T_x$ iff $s = t$ near g; $\varphi T_x \varphi^{-1} = T_y$, which Definition 7.2.7 (2017: 7.12) needs and the text does not state. "$\mathrm{Diff}_{sc}(U(x)$" (possible misprint or extraction artefact) (Findings, item 54).

V7.38 (Definition 7.2.7 (2017: 7.12), the non-effective part $\mathbf{X}^T$ and the reduced morphisms $\mathbf{X}^R$; `Polyfold.EpGroupoid.nonEffectivePart`, `Polyfold.EpGroupoid.mem_nonEffectivePart_iff`, `Polyfold.EpGroupoid.nonEffectivePart_eq_interior`, `Polyfold.EpGroupoid.ReducedRel`, `Polyfold.EpGroupoid.reducedRel_iff`, `Polyfold.EpGroupoid.reducedSetoid`, `Polyfold.EpGroupoid.reducedRel_comp_left`, `Polyfold.EpGroupoid.ReducedMor`, `Polyfold.EpGroupoid.reducedMor_eq_quotient`**).** $\mathbf{X}^T = \bigcup_x T_x$; $\varphi \sim \varphi'$ iff $s(\varphi) = s(\varphi') = x$ and $\varphi = \varphi' \circ g$ with $g \in T_x$; $\mathbf{X}^R$ is the set of classes. *Book:* Definition 7.2.7 (2017: 7.12), p. 310 (2017 p. 241). *Hidden hypotheses:* none. *Deviation (restated):* $\sim$ is an equivalence since T_x is a subgroup; $\mathbf{X}^T$ is the interior of $\{s = t\}$; "the subset $\mathbf{X}^R$" is a quotient (possible misprint) (Findings, item 55).

V7.39 (Proposition 7.2.10 (2017: 7.9), the natural charts of $\mathbf{X}^R$; `Polyfold.EpGroupoid.proposition_7_2_10_natural_charts`, `Polyfold.EpGroupoid.reduced`, `Polyfold.EpGroupoid.isQuotientMap_reduced`, `Polyfold.EpGroupoid.isOpenMap_reduced`, `Polyfold.EpGroupoid.injOn_reduced`, `Polyfold.EpGroupoid.reducedChart`, `Polyfold.EpGroupoid.reducedChart_compatible`, `Polyfold.EpGroupoid.reducedCharts`, `Polyfold.EpGroupoid.reducedAtlas`, `Polyfold.EpGroupoid.reducedPolyfold`, `Polyfold.EpGroupoid.isChart_reducedChart`, `Polyfold.EpGroupoid.isScDiffeoOn_reduced`, `Polyfold.EpGroupoid.isLocalScDiffeo_reduced`, `Polyfold.EpGroupoid.proposition_7_2_10`, `Polyfold.EpGroupoid.paracompactSpace_reducedMor`, `Polyfold.EpGroupoid.t2Space_reducedMor_of_isEffective`, `Polyfold.EpGroupoid.paracompactSpace_reducedMor_of_isEffective`, `Polyfold.EpGroupoid.paracompactSpace_reducedMor_of_t3`, `Polyfold.EpGroupoid.Examples.exists_not_t2Space_reducedMor`**).** $\mathbf{X}^R$ with the quotient topology T_R has a natural collection of charts $\psi \circ \sigma \circ s$, which cover it and have sc-smooth transitions; they form `reducedAtlas`. *Book:* Proposition 7.2.10, p. 313 (2017: 7.9, pp. 242–243). *Hidden hypotheses:* none. *Deviation (as in the book, 2021):* the quotient map is open and injective on sheets; the M-polyfold `reducedPolyfold` needs only T_R Hausdorff (paracompactness is an instance, V7.77), which is Proposition 7.2.13 (V7.78). The 2017 Proposition 7.9, a natural M-polyfold structure for every ep-groupoid, does not hold as we read it: for $\mathbb{Z}/2$ acting on the dimension-jumping retract of Part I's Remark 4.2.2 (2017: 4.3), $\mathbf{X}^R$ is not Hausdorff (`Polyfold.EpGroupoid.Examples.exists_not_t2Space_reducedMor`). Four reading points in the 2017 proof (Findings, items 56, 57, 60, 107, 111).

V7.40 (Theorem 7.2.14 and Proposition 7.2.16 (2017: Proposition 7.10), the reduced ep-groupoid X_R; `Polyfold.EpGroupoid.theorem_7_2_14_reduced`, `Polyfold.EpGroupoid.proposition_7_2_16`, `Polyfold.EpGroupoid.reducedSrc`, `Polyfold.EpGroupoid.reducedTgt`, `Polyfold.EpGroupoid.reducedUnit`, `Polyfold.EpGroupoid.reducedInv`, `Polyfold.EpGroupoid.reducedComp`, `Polyfold.EpGroupoid.reducedGroupoidStr`, `Polyfold.EpGroupoid.isLocalScDiffeo_reducedSrc`, `Polyfold.EpGroupoid.isLocalScDiffeo_reducedTgt`, `Polyfold.EpGroupoid.isScSmoothMap_reducedUnit`, `Polyfold.EpGroupoid.isScSmoothMap_reducedInv`, `Polyfold.EpGroupoid.exists_isProperMap_reduced`, `Polyfold.EpGroupoid.isScSmoothCompSheet_reduced`, `Polyfold.EpGroupoid.reducedEpGroupoid`, `Polyfold.EpGroupoid.reductionFunctor`, `Polyfold.EpGroupoid.isEffective_reducedEpGroupoid`, `Polyfold.EpGroupoid.proposition_7_10_2017`, `Polyfold.EpGroupoid.reductionIso`). If T_R is Hausdorff, $(X, \mathbf{X}^R)$ with the induced structure maps is an ep-groupoid X_R, effective, with isotropy groups G_x/T_x, and π is a morphism $X \to X_R$ of $\mathcal{EP}$. *Book:* Theorem 7.2.14, p. 318, Proposition 7.2.16 and the isotropy sentence, p. 319 (2017 p. 243 and Proposition 7.10, p. 244). *Hidden hypotheses:* none (paracompactness of T_R is an instance, V7.77). *Deviation (as in the book, 2021):* (3) via the continuous surjection $\mathbf{X} \to \mathbf{X}^R$; (4) in sheet form, with a step the text leaves implicit; effectiveness from the intrinsic form of T_x; for effective X, $X_R \cong X$. The 2017 Proposition 7.10, for every ep-groupoid, does not hold as we read it, since $\mathbf{X}^R$ need not be Hausdorff (V7.39) (Findings, items 58, 60, 107, 111).

V7.41 (the isotropy groups of X_R and Definition 7.2.15 (2017: 7.13), the effective isotropy group; `Polyfold.EpGroupoid.reducedIsotropyHom`, `Polyfold.EpGroupoid.surjective_reducedIsotropyHom`, `Polyfold.EpGroupoid.ker_reducedIsotropyHom`, `Polyfold.EpGroupoid.quotientTrivialSubgroupEquiv`, `Polyfold.EpGroupoid.EffIsotropy`, `Polyfold.EpGroupoid.normalClosure_trivialSet`, `Polyfold.EpGroupoid.effIsotropyEquiv`, `Polyfold.EpGroupoid.effIsotropyEquivReduced`). The isotropy group of x in X_R is G_x/T_x, called the effective isotropy group G_x^{eff}. *Book:* 2017 p. 244 and Definition 7.2.15 (2017: 7.13), p. 319 (2017 p. 244). *Hidden hypotheses:* the isomorphism with the isotropy of X_R under those of V7.39; G_x/T_x itself needs none. *Deviation:* none; "For and object" (possible misprint) (Findings, item 59).

V7.42 (Theorem 7.3.1 (2017: 7.2), basic topological properties of $|X|$; `Polyfold.EpGroupoid.theorem_7_3_1`, `Polyfold.EpGroupoid.theorem_7_3_1_metrizable`, `Polyfold.metrizableSpace_of_locally`). $|X|$ is locally metrizable, regular and Hausdorff, and metrizable if paracompact. *Book:* Theorem 7.3.1 (2017: 7.2), p. 324 (2017 p. 244). *Hidden hypotheses:* none. *Deviation (restated):* "locally metrizable" as an open metrizable neighbourhood of every point; Smirnov's theorem in Part I's form (Findings, item 70).

V7.43 (Lemma 7.3.2 (2017: 7.4), local metrizability; `Polyfold.EpGroupoid.metrizableSpace_`

`orbitRel_quotient_naturalRep`, `Polyfold.EpGroupoid.lemma_7_3_2_naturalRep`, `Polyfold.EpGroupoid.lemma_7_3_2`). Every point of $|X|$ has a metrizable open neighbourhood, $|U(x)|$ for a natural representation. *Book:* Lemma 7.3.2 (2017: 7.4), pp. 324 ff. (2017 pp. 244–245). *Hidden hypotheses:* none. *Deviation (restated):* $\hat{\rho}$ computed as $\min_h \rho(y, h * y')$ (Findings, item 71).

V7.44 (Lemma 7.3.3 (2017: 7.5), countable neighbourhood bases; `Polyfold.EpGroupoid.lemma_7_3_3`). Every point of $|X|$ has a monotone countable basis of open neighbourhoods. *Book:* Lemma 7.3.3 (2017: 7.5), p. 324 (2017 p. 245). *Hidden hypotheses:* none. *Deviation:* none; it is Proposition 7.1.9 (2017: 7.2) (V7.12) (Findings, item 72).

V7.45 (Lemma 7.3.4 (2017: 7.6), $|X|$ is Hausdorff; `Polyfold.EpGroupoid.t1Space_orbitSpace`, `Polyfold.EpGroupoid.lemma_7_3_4`). *Book:* Lemma 7.3.4 (2017: 7.6), pp. 325 ff. (2017 pp. 245–246). *Hidden hypotheses:* none. *Deviation (restated):* from Lemma 7.3.5 (2017: 7.7) and the closedness of points, instead of the text's direct argument (Findings, item 72).

V7.46 (Lemma 7.3.5 (2017: 7.7), $|X|$ is regular; `Polyfold.EpGroupoid.lemma_7_3_5`). *Book:* Lemma 7.3.5 (2017: 7.7), p. 325 (2017 p. 246). *Hidden hypotheses:* none. *Deviation (restated):* with a closed neighbourhood basis and Proposition 7.1.10 (2017: 7.3) (Findings, item 72).

V7.47 (Corollary 7.3.6 (2017: 7.1), the metrizability criterion; `Polyfold.EpGroupoid.corollary_7_3_6`). If $|X|$ is second countable, it is metrizable. *Book:* Corollary 7.3.6 (2017: 7.1), p. 326 (2017 p. 247). *Hidden hypotheses:* none. *Deviation:* none; Urysohn's theorem (Findings, item 73).

V7.48 (subsets of $|X|$; `Polyfold.EpGroupoid.exists_metrizable_nhds_subtype`, `Polyfold.EpGroupoid.metrizableSpace_of_paracompactSpace_subtype`). A subset of $|X|$ is again locally metrizable, regular and Hausdorff; a paracompact one is metrizable. *Book:* 2017 p. 247, before Definition 7.3.7 (2017: 7.14). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 73).

V7.49 (Definition 7.3.7 (2017: 7.14), component compact sets; `Polyfold.EpGroupoid.IsComponentCompact`). $K \subseteq |X|$ meets every connected component of $|X|$ in a compact set. *Book:* Definition 7.3.7 (2017: 7.14), p. 326 (2017 p. 247). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 74).

V7.50 (Michael's theorem and unions of closed paracompact sets; `Polyfold.lemma_2_7_3`, `Polyfold.michael_tfae`, `Polyfold.proposition_2_7_4`, `Polyfold.paracompactSpace_of_closed_refinement`, `Polyfold.exists_closed_precise_refinement`, `Polyfold.paracompactSpace_iUnion`, `Polyfold.paracompactSpace_iUnion_of_finite`, `Polyfold.paracompactSpace_biUnion_of_finite`, `Polyfold.paracompactSpace_union`, `Polyfold.paracompactSpace_of_isClosed_subset`). A regular space is paracompact iff every open cover has a closed locally finite refinement; a regular space with a locally finite cover by closed paracompact subspaces is paracompact. *Book:* Part I, Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17), used for Corollary 7.3.8 (2017: 7.2) (p. 326 (2017 p. 247)) and on the text's route to Proposition 7.5.3 (2017: 7.12) (p. 340 (2017 p. 249)). *Hidden hypotheses:* none. *Deviation (stronger):* the Hausdorff property assumed by the text is not used; gap G.15 of Part I is now formalized (Findings, items 75, 114).

V7.51 (Corollary 7.3.8 (2017: 7.2), neighbourhoods of component compact sets; `Polyfold.EpGroupoid.locallyPathConnectedSpace_orbitSpace`, `Polyfold.EpGroupoid.exists_isOpen_metrizableSpace_closure`, `Polyfold.EpGroupoid.corollary_7_3_8_of_isCompact`, `Polyfold.EpGroupoid.corollary_7_3_8`**).** A component compact K has an open neighbourhood U with $\mathrm{cl}_{|X|}(U)$ paracompact, hence $\mathrm{cl}(U)$ and U metrizable. *Book:* Corollary 7.3.8 (2017: 7.2), p. 326 (2017 p. 247). *Hidden hypotheses:* none. *Deviation (restated):* the reduction to compact K uses open components (local connectedness of $|X|$, V7.18) and a locally finite union the text does not give; Michael's theorem (V7.50) for the finite union. *"U"* names both $\bigcup \mathrm{cl}\, U(z_i)$ and $\bigcup U(z_i)$ (possible misprint) (Findings, item 74).

V7.52 (Proposition 7.3.9 (2017: 7.11), index lifting; `Polyfold.EpGroupoid.liftSrc`, `Polyfold.EpGroupoid.liftTgt`, `Polyfold.EpGroupoid.liftComp`, `Polyfold.EpGroupoid.comp_mem_levelSet`, `Polyfold.EpGroupoid.liftGroupoidStr`, `Polyfold.MPolyfold.IsLocalScDiffeo.levelOne`, `Polyfold.MPolyfold.IsScSmoothMap.levelOne`, `Polyfold.EpGroupoid.isLocalScDiffeo_liftSrc`, `Polyfold.EpGroupoid.isLocalScDiffeo_liftTgt`, `Polyfold.EpGroupoid.isScSmoothMap_liftUnit`, `Polyfold.EpGroupoid.isScSmoothMap_liftInv`, `Polyfold.EpGroupoid.exists_isProperMap_lift`, `Polyfold.EpGroupoid.isScSmoothCompSheet_lift`, `Polyfold.EpGroupoid.isScSmoothComp_lift`, `Polyfold.EpGroupoid.liftOne`, `Polyfold.MPolyfold.IsTame.levelOne`, `Polyfold.EpGroupoid.isTame_liftOne`, `Polyfold.EpGroupoid.proposition_7_3_9`**).** $X^1 = (X^1, \mathbf{X}^1)$ is an ep-groupoid, tame if X is. *Book:* Proposition 7.3.9 (2017: 7.11), pp. 327–328 (2017 pp. 247–248). *Hidden hypotheses:* none. *Deviation (restated):* (4) in sheet form, avoiding $(\mathbf{X}\,{}_s\!\times_t \mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t \mathbf{X}^1$; (3) with convergence in X_1 for the compact set of the text and without its step (b); the tameness, for which the text gives no argument and which Part I left open (Chapter 5, item 145), is proved. Display (7.5) has an unbalanced parenthesis (possible misprint) (Findings, items 80, 81, 96).

V7.53 (Definition 7.5.1 (2017: 7.15), ep-groupoids with paracompact orbit space, the sentence before Remark 7.5.2 (2017: 7.1), and Remark 7.5.2 (2017: 7.1); `Polyfold.EpGroupoid.HasParacompactOrbitSpace`, `Polyfold.EpGroupoid.remark_7_5_2`, `Polyfold.EpGroupoid.hasParacompactOrbitSpace_of_secondCountableTopology`, `Polyfold.EpGroupoid.hasParacompactOrbitSpace_iff_metrizableSpace`**).** (5): $|X|$ is paracompact, not required to be Hausdorff; if X is second countable, (1)–(4) imply (5). Remark 7.5.2 (2017: 7.1) recalls that [38] required second countability. *Book:* Definition 7.5.1 (2017: 7.15), the paragraph after it, and Remark 7.5.2 (2017: 7.1), p. 340 (2017 p. 249). *Hidden hypotheses:* none. *Deviation (restated):* "X second countable" is read for the object space; Remark 7.5.2 (2017: 7.1) has no formal content; "it is presently not known whether (5) is a consequence of (1)–(4)" is not formalized (Findings, items 82, 113).

V7.54 (Proposition 7.5.3 (2017: 7.12), paracompactness under lifts, and the map $|X^1| \to |X|$; `Polyfold.EpGroupoid.orbitIncl`, `Polyfold.EpGroupoid.continuous_orbitIncl`, `Polyfold.EpGroupoid.orbitIncl_injective`, `Polyfold.EpGroupoid.range_orbitIncl`, `Polyfold.EpGroupoid.NaturalRep.liftMulAction`, `Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_liftOrbitQuotientMap`, `Polyfold.EpGroupoid.NaturalRep.`

`metrizableSpace_preimage_orbitIncl`, `Polyfold.EpGroupoid.metrizableSpace_liftOne_orbitSpace`, `Polyfold.metrizableSpace_of_pullback`, `Polyfold.EpGroupoid.proposition_7_5_3`**).** If $|X|$ is paracompact, so is $|X^1|$. *Book:* Proposition 7.5.3 (2017: 7.12), pp. 340 ff. (2017 pp. 249–250). *Hidden hypotheses:* none. *Deviation (stronger):* $|X^1|$ is metrizable, by Smirnov's theorem pulled back along the continuous injection $|X^1| \to |X|$ onto $\pi(X_1)$; $U(x)_1$ carries the restricted action only (Findings, item 83).

V7.55 (Theorem 7.5.4 (2017: 7.3), continuous partitions of unity; `Polyfold.EpGroupoid.orbitPullback`, `Polyfold.EpGroupoid.orbitPullback_isSubordinate`, `Polyfold.EpGroupoid.exists_isSaturated_nhds_finite_orbitPullback`, `Polyfold.EpGroupoid.theorem_7_5_4`**).** For paracompact $|X|$, every open cover of $|X|$ has a subordinate partition of unity, and every saturated open cover of X an invariant continuous one. *Book:* Theorem 7.5.4 (2017: 7.3), p. 341 (2017 p. 250). *Hidden hypotheses:* $|X|$ Hausdorff, from Lemma 7.3.4 (2017: 7.6), for normality. *Deviation:* none (Findings, item 90).

V7.56 (Definition 7.5.5 (2017: 7.16), ep-groupoids with sc-smooth partitions of unity; `Polyfold.EpGroupoid.HasScPartitionsOfUnity`, `Polyfold.MPolyfold.IsScSmoothPartitionOfUnity`**).** Every saturated open cover of X has a subordinate sc-smooth partition of unity with (1) saturated neighbourhoods meeting finitely many supports, (2) invariance. *Book:* Definition 7.5.5 (2017: 7.16), pp. 342 ff. (2017 pp. 250–251). *Hidden hypotheses:* none. *Deviation (restated):* index types in the universe of X, supports as `tsupport`, (2) as $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ (Findings, item 91).

V7.57 (Remark 7.5.6 (2017: 7.2), on sc-smooth partitions of unity). The remark (the topic is "a zoo" apart from sc-Hilbert spaces) has no formal content: nothing to formalize. *Book:* Remark 7.5.6 (2017: 7.2), p. 342 (2017 p. 251). *Hidden hypotheses:* none. *Deviation:* none; the sc-Hilbert case is Corollary 7.6.6 (V7.68), proved with no hypothesis (`Polyfold.EpGroupoid.corollary_7_6_6_unconditional`, `Polyfold.MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional`) (Findings, item 91).

V7.58 (Theorem 7.5.7 (2017: 7.4), sc-smooth partitions of unity of an ep-groupoid; `Polyfold.EpGroupoid.theorem_7_5_7`**).** If $|X|$ is paracompact and the object M-polyfold admits sc-smooth partitions of unity, X has sc-smooth partitions of unity. *Book:* Theorem 7.5.7 (2017: 7.4), p. 342 (2017 p. 251) (proved as Theorem 7.6.4 (2017: 7.6), V7.65, V7.67). *Hidden hypotheses:* none; satisfiable (`Polyfold.EpGroupoid.theorem_7_5_7_hypotheses_rn`). *Deviation:* none (Findings, items 94, 105).

V7.59 (Definition 7.5.8 (2017: 7.17), proper subsets and the properness property; `Polyfold.EpGroupoid.IsProperSubset`, `Polyfold.EpGroupoid.HasPropernessProperty`, `Polyfold.EpGroupoid.exists_hasPropernessProperty`, `Polyfold.EpGroupoid.HasPropernessProperty.mono`, `Polyfold.EpGroupoid.isProperSubset_iff_isCompact`, `Polyfold.EpGroupoid.isProperSubset_iff_exists_tendsto_subseq`**).** Q is proper if $t : s^{-1}(Q) \to X$ is proper; an open U has the properness property if $\mathrm{cl}_X(U)$ is proper. *Book:* Definition 7.5.8 (2017: 7.17), p. 342 (2017 p. 251). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 92).

V7.60 (Proposition 7.5.9 (2017: 7.13), proper subsets are closed; `Polyfold.EpGroupoid.proposition_7_5_9`, `Polyfold.EpGroupoid.IsProperSubset.isClosed`). *Book:* Proposition 7.5.9 (2017: 7.13), p. 342 (2017 p. 251). *Hidden hypotheses:* none. *Deviation:* none; by the compactness of the preimage (Findings, item 92).

V7.61 (Proposition 7.5.10 (2017: 7.14), closed subsets of $|X|$; `Polyfold.EpGroupoid.isClosed_saturation_of_isProperSubset`, `Polyfold.EpGroupoid.proposition_7_5_10_one`, `Polyfold.EpGroupoid.IsProperSubset.of_subset`, `Polyfold.EpGroupoid.IsProperSubset.union`, `Polyfold.EpGroupoid.isProperSubset_iUnion_of_locallyFinite`, `Polyfold.EpGroupoid.isProperSubset_of_isCompact`, `Polyfold.EpGroupoid.exists_hasPropernessProperty_subset`, `Polyfold.EpGroupoid.proposition_7_5_10_two`). (1) $\pi(Q)$ is closed for proper Q; (2) for paracompact $|X|$, $A \subseteq |X|$ is closed iff $A = \pi(Q)$ for a proper Q. *Book:* Proposition 7.5.10 (2017: 7.14), pp. 343–344 (2017 pp. 251–252). *Hidden hypotheses:* none. *Deviation (stronger):* (2) with only the paracompactness of $|X|$, without Lemma 2.7.3 (2017: 2.7) or regularity; "$\mathrm{cl}_X(U_{x_{z_i}})$" for $\mathrm{cl}_X(U(x_{z_i}))$ (possible misprint) (Findings, item 92).

V7.62 (Theorem 7.6.1 (2017: 7.5), the natural representation, restated in Appendix 7.6.1 (2017 7.5.1); `Polyfold.EpGroupoid.theorem_7_6_1`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets`, `Polyfold.EpGroupoid.NatRepConstruction.exists_sheets`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets.mkRep`, `Polyfold.EpGroupoid.NatRepConstruction.exists_naturalRep_sheets`, `Polyfold.EpGroupoid.NatRepConstruction.restrict`). Theorem 7.1.13 (2017: 7.1) with $\Phi(g) = t_g \circ s_g^{-1}$. *Book:* Theorem 7.6.1 (2017: 7.5) and its proof, pp. 344–346 (2017 pp. 253–254). *Hidden hypotheses:* as V7.17. *Deviation (restated):* `Polyfold.EpGroupoid.theorem_7_6_1` is `Polyfold.EpGroupoid.theorem_7_1_13` (V7.17); the proof starts at $U_0 := V \cap \bigcap_g s(N_g)$ so that $U \subseteq V$; the contractibility of N_g^t, N_g^s is not needed; "$\Phi : G_x \to \mathrm{Diff}_{sc}(U)$" for G_{x_0} (possible misprint) (Findings, item 29).

V7.63 (Lemma 7.6.2 (2017: 7.8); `Polyfold.EpGroupoid.lemma_7_6_2`). There is an open $U_1 \subseteq U_0$ of x_0 such that every morphism between points of U_1 lies in some N_g. *Book:* Lemma 7.6.2 (2017: 7.8), pp. 345 ff. (2017 pp. 253–254). *Hidden hypotheses:* none. *Deviation (stronger):* for any open $N_g \ni g$ and any neighbourhood U_0; "properness assumption of ep-polyfolds" (possible misprint) (Findings, item 30).

V7.64 (Lemma 7.6.3 (2017: 7.9); `Polyfold.EpGroupoid.lemma_7_6_3`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets.unionHat`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets.isConnected_unionHat`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets.eqOn_ _one`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets.eqOn_ _mul`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets.hat_one`, `Polyfold.EpGroupoid.NatRepConstruction.Sheets.hat_mul`). There is an open $U_2 \subseteq U_1$ of x_0 such that $U = \bigcup_g t_g \circ s_g^{-1}(U_2)$ lies in U_1 and is invariant under all $t_g \circ s_g^{-1}$. *Book:* Lemma 7.6.3 (2017: 7.9), p. 345 (2017 p. 254). *Hidden hypotheses:* none. *Deviation (stronger):* U_2, and so U, connected, by V7.18 (Findings, item 31).

V7.65 (Theorem 7.6.4 (2017: 7.6), Theorem 7.5.7 (2017: 7.4) restated in Appendix 7.6.2 (2017 7.5.2); `Polyfold.EpGroupoid.theorem_7_6_4`). *Book:* Theorem 7.6.4 (2017: 7.6), p. 346 (2017 p. 255). *Hid-*

den hypotheses: as V7.58. *Deviation:* none; `Polyfold.EpGroupoid.theorem_7_6_4` is `Polyfold.EpGroupoid.theorem_7_5_7` (V7.58) (Findings, item 95).

V7.66 (Lemma 7.6.5 (2017: 7.10), invariant sc-smooth majorants; `Polyfold.EpGroupoid.lemma_7_6_5`). For $U(x) \subseteq \mathrm{cl}_X U(x) \subseteq V(x)$, $V(x)$ with the natural action and $U(x)$ invariant, every invariant continuous $\beta : X \to [0,1]$ supported in $U(x)$ has an invariant sc-smooth majorant $f \geq 0$ supported in $V(x)$. *Book:* Lemma 7.6.5 (2017: 7.10), p. 346 (2017 p. 255). *Hidden hypotheses:* none. *Deviation (stronger):* $\mathrm{supp} f \subseteq \mathrm{cl}_X U(x)$ added (the proof gives it, the proof of Theorem 7.5.7 (2017: 7.4) needs it); "$x \in U(x)$" omitted; the partition of unity of $V(x)$ is taken for a cover of X (Findings, item 93).

V7.67 (the proof of Theorem 7.5.7 (2017: 7.4); `Polyfold.EpGroupoid.exists_invariant_nhds_of_isOpen`, `Polyfold.EpGroupoid.mem_of_ _mem_image_of_invariant`, `Polyfold.EpGroupoid.saturation_diff_inter_eq_empty`, `Polyfold.EpGroupoid.orbitExtension`, `Polyfold.EpGroupoid.isScSmoothMap_orbitExtension`, `Polyfold.EpGroupoid.exists_isScSmoothMap_orbit_urysohn`, `Polyfold.admitsScPartitionsOfUnity_of_approx`, `Polyfold.EpGroupoid.theorem_7_5_7`). The neighbourhoods $V(x) \subseteq \mathrm{cl}\, V(x) \subseteq W(x)$ with (1)–(4), a locally finite refinement, Lemma 7.6.5 (2017: 7.10), averaging, extension to the saturation, and normalization. *Book:* pp. 346–347 (2017 pp. 255–256). *Hidden hypotheses:* none. *Deviation (restated):* (4) follows from (1); Lemma 7.6.5 (2017: 7.10) is applied with $W(x_i)$ in the role of $V(x)$, where the text reads $V(x_i)$ (possible misprint); local finiteness needs the support bound of V7.66; the saturations are closed by properness; the normalization by Mathlib's bump coverings as in Part I's Theorem 5.5.10 (2017: 5.12); "$t : s^{-1}(\mathrm{cl}_X(V(x)) \to X$" (possible misprint) (Findings, items 94, 96).

V7.68 (Corollary 7.6.6 (2017: 7.3), sc-Hilbert atlases; `Polyfold.EpGroupoid.corollary_7_6_6_unconditional`, `Polyfold.MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional`, `Polyfold.EpGroupoid.corollary_7_6_6_hypotheses_rn_unconditional`, `Polyfold.EpGroupoid.corollary_7_6_6`, `Polyfold.MPolyfold.admitsScPartitionsOfUnity_of_hilbert`,`Polyfold.GodefroyTroyanskiWhitfieldZizlerStatement`,`Polyfold.EpGroupoid.corollary_7_6_6_hypotheses_rn`). If $|X|$ is paracompact and the object M-polyfold has an atlas whose models have Hilbert spaces E_0, every saturated open cover has a subordinate sc-smooth partition of unity of X. *Book:* Corollary 7.6.6 (2017: 7.3), p. 347 (2017 p. 256), with Part I's Proposition 5.5.8 (2017: 5.6), Theorem 5.5.10 (2017: 5.12) and Corollary 5.5.17 (2017: 5.2). *Hidden hypotheses:* none; satisfiable on $\mathbb{R}^1$. *Deviation (as in the book):* the Hilbert atlas existential, one chart per point; "every compatible chart" would be unsatisfiable. Part I's Theorem 5.5.16 (2017: 5.14) is no longer a hypothesis: E_0 is separable and the theorem holds for separable spaces (`Polyfold.godefroyTroyanskiWhitfieldZizlerStatement_of_separable`; gap G.53 closed); the form with it as a hypothesis is kept (Findings, items 95, 106, 114).

The additions of the 2021 edition (V7.69–V7.104)

The Springer edition (2021) changes the remark after Definition 7.1.7 and Proposition 7.1.15, adds Proposition 7.1.14, Lemma 7.1.16 and Corollary 7.1.17, rewrites §7.2 from Definition 7.2.5 on (2017 Proposition 7.10 becomes Theorem 7.2.14, preceded by a Hausdorff criterion for the topology T_R

of $\mathbf{X}^R$), adds §7.4 on the Zhou condition and unique continuation, and adds Appendices 7.6.3 (Zhou's example), 7.6.4 (the reduced groupoid and index lifts) and 7.6.5 (the boundary of a tame ep-groupoid). Numbers and pages below are the 2021 text's; items new in the 2021 edition have no 2017 counterpart. Lean files: `Chapter7/Book2021Basic.lean`, `ReducedCriterion.lean`, `Zhou.lean`, `TameBoundaryEp.lean`, and, for Theorem 7.6.9 in full on top of Part I's Theorem 2.7.13 (2)–(4), `TameBoundaryEpStructure.lean` (V7.99–V7.101), and, for §7.4 on every lift X^i, `ZhouLevels.lean` (V7.102–V7.104).

V7.69 (the remark after Definition 7.1.7, 2021 form; `Polyfold.EpGroupoid.remark_definition_7_1_7`, `Polyfold.EpGroupoid.remark_definition_7_1_7_remarkFunctor`). Let $F : X \to Y$ be a functor between ep-groupoids which is sc-smooth on the object M-polyfolds and continuous on the morphism spaces, $\varphi \in \mathbf{X}$ and $\psi = F(\varphi)$. There are open sets $U(\varphi) \ni \varphi$, $U(\psi) \ni \psi$, $U(s(\varphi))$, $U(s(\psi))$ with $F(U(s(\varphi))) \subseteq U(s(\psi))$ and $F(U(\varphi)) \subseteq U(\psi)$ such that $s : U(\varphi) \to U(s(\varphi))$ and $s : U(\psi) \to U(s(\psi))$ are sc-diffeomorphisms and $F(\gamma) = (s|U(\psi))^{-1}(F(s(\gamma)))$ for $\gamma \in U(\varphi)$; hence F is sc-smooth on the morphism spaces. The functor of the counterexample of V7.10 (on the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$) is sc-smooth on objects and not continuous on morphisms. *Book:* the remark after Definition 7.1.7, p. 291 (2017: after Definition 7.7, p. 228), which in 2021 assumes continuity on morphisms. *Hidden hypotheses:* none. *Deviation (restated):* the neighbourhoods are an existential statement; the 2021 hypothesis is the one under which V7.10 proved the 2017 remark, so the counterexample of V7.10 no longer applies (Findings, items 9, 104, 117).

V7.70 (the data (7.3) and Proposition 7.1.14; `Polyfold.EpGroupoid.IsotropySheetData`, `Polyfold.EpGroupoid.exists_isotropySheetData`, `Polyfold.EpGroupoid.proposition_7_1_14`, `Polyfold.EpGroupoid.proposition_7_1_14_of_naturalRep`). The data (7.3) at an object x are an open $W(x) \ni x$ and, for every $g \in G_x$, open $U^s(g), U^t(g) \ni g$ on which s and t are sc-diffeomorphisms onto $W(x)$, with inverses s_g^{-1}, t_g^{-1}; they exist at every object. Let $U(x)$ be an open neighbourhood of x, $\Phi(g)$ ($g \in G_x$) sc-diffeomorphisms of $U(x)$ with inverse $\Phi(g^{-1})$ and $\Phi(gh) = \Phi(g) \circ \Phi(h)$, and $\Gamma(g, \cdot) : U(x) \to \mathbf{X}$ sc-smooth with $\Gamma(g, x) = g$, $s(\Gamma(g, y)) = y$, $t(\Gamma(g, y)) = \Phi(g)(y)$. Then there is an open Φ-invariant $V(x) \ni x$ with $V(x) \subseteq W(x) \cap U(x)$ such that, for $v \in V(x)$ and $g \in G_x$, $\Gamma(g, v) = s_g^{-1}(v)$, $\Phi(g)(v) = t(s_g^{-1}(v))$ and $\Gamma(g, v) = t_g^{-1}(\Phi(g)(v))$; the same holds for every natural representation. *Book:* (7.3) and Proposition 7.1.14, p. 296, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* property (3) of Theorem 7.1.13 (every morphism between points of $U(x)$ is some $\Gamma(g, y)$) is assumed as printed but not used; the identity $\Gamma(g, v) = t_g^{-1}(\Phi(g)(v))$ of (7.4), from the proof, is added (Findings, item 118).

V7.71 (Proposition 7.1.15 in the 2021 form; `Polyfold.EpGroupoid.proposition_7_1_15`, `Polyfold.EpGroupoid.src_eq_tgt_of_printed_2017`, `Polyfold.EpGroupoid.not_printed_2017_reflection`). For a finite set Y of mutually isomorphic objects and a neighbourhood V of Y there is a natural representation of Y on disjoint connected open $U(y) \subseteq V$ (Definition 7.1.18, `NaturalRepFinite`) such that, for every morphism ψ between points of Y: $\psi \in \mathbf{U}(\psi)$, the connected component of ψ in the set of morphisms between points of $U = \bigcup_y U(y)$; $s : \mathbf{U}(\psi) \to U(s(\psi))$ and $t : \mathbf{U}(\psi) \to U(t(\psi))$ are sc-diffeomorphisms; $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$

on $U(s(\psi))$; and $t(\psi') = f(\psi)(s(\psi'))$ for $\psi' \in \mathbf{U}(\psi)$. The printed 2017 form of (2), $s, t : \mathbf{U}(\psi) \to U(y)$ with one y for both maps, together with the other requirements forces $s(\psi) = t(\psi)$; for $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection and $Y = \{1, -1\}$ the hypotheses hold and a natural representation in the 2021 form exists, but no data of the printed 2017 form. *Book:* Proposition 7.1.15, p. 297 (2017: 7.5, p. 233), with (2) printed as $U(s(\psi))$, $U(t(\psi))$ in 2021. *Hidden hypotheses:* none. *Deviation (same):* none for the 2021 form; the 2017 form of (2) as printed does not hold as we read it, and the structure `NaturalRepFinite` of V7.25 already had the 2021 form (Findings, items 25, 119).

V7.72 (Lemma 7.1.16; `Polyfold.EpGroupoid.NaturalRepFinite.mem_U_of_tendsto`, `Polyfold.EpGroupoid.lemma_7_1_16`). In the construction of V7.71, let $\psi \in \mathbf{Y}$ and $\psi'_k \in \mathbf{U}(\psi)$ converge to $\psi' \in \mathbf{X}$. If $(s(\psi'_k))$ or $(t(\psi'_k))$ converges to a point of $U(s(\psi)) \cup U(t(\psi))$, then $\psi' \in \mathbf{U}(\psi)$, $s(\psi'_k) \to s(\psi') \in U(s(\psi))$ and $t(\psi'_k) \to t(\psi') \in U(t(\psi))$. *Book:* Lemma 7.1.16, p. 300, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none; the step that a limit of points of $U(a)$ lying in $U(b)$ lies in $U(a)$ (the $U(y)$ are open and disjoint) is a separate lemma (Findings, item 120).

V7.73 (Corollary 7.1.17 (4)–(7); `Polyfold.EpGroupoid.corollary_7_1_17_4`, `Polyfold.EpGroupoid.corollary_7_1_17_5`, `Polyfold.EpGroupoid.corollary_7_1_17_6`, `Polyfold.EpGroupoid.corollary_7_1_17_7`). In the construction of V7.71: (4) for $x \in Y$ there is a natural representation (Φ, Γ) of G_x on $U(x)$ with $\Gamma(g, z) = (s|\mathbf{U}(g))^{-1}(z)$ and $f(g)(z) = t(\Gamma(g, z)) = \Phi(g)(z)$; (5) $f(\varphi)$ is equivariant along $\gamma_\varphi(g) = \varphi \circ g \circ \varphi^{-1}$; (6) for composable φ, ψ the multiplication is an sc-diffeomorphism from $\mathbf{U}(\psi)\,{}_s{\times}_t\,\mathbf{U}(\varphi)$, an open subset of the fibred product sub-M-polyfold $\mathbf{X}\,{}_s{\times}_t\,\mathbf{X}$, onto $\mathbf{U}(\psi \circ \varphi)$; (7) the $U(y)$ can be taken with the properness property. *Book:* Corollary 7.1.17, p. 301, new in the 2021 edition ((1)–(3) are those of Proposition 7.1.15). *Hidden hypotheses:* none. *Deviation (restated):* in (6) the domain is read inside Part I's fibred product sub-M-polyfold (Proposition 2.5.20 (2017: 2.15)) (Findings, item 120).

V7.74 (Definition 7.2.5 and Lemma 7.2.6; `Polyfold.EpGroupoid.NaturalRep.kerOn`, `Polyfold.EpGroupoid.NaturalRep.mem_kerOn`, `Polyfold.EpGroupoid.NaturalRep.kerOn_anti`, `Polyfold.EpGroupoid.NaturalRep.kerOn_normal`, `Polyfold.EpGroupoid.NaturalRep.kerOn_le_trivialSubgroup`, `Polyfold.EpGroupoid.definition_7_2_5`, `Polyfold.EpGroupoid.lemma_7_2_6`, `Polyfold.EpGroupoid.trivialSubgroupEquiv`, `Polyfold.EpGroupoid.trivialSubgroupEquiv_apply`). For a natural representation at x and $V \subseteq X$, T_x^V is the subgroup of the g with $\Phi(g) = \mathrm{id}$ on $V \cap U(x)$; it decreases as V grows and is normal for invariant V. For every open $W \ni x$, $T_x^W \subseteq T_x$, and there is an invariant open $V \ni x$ in $U(x)$ with $T_x^W = T_x$ for every open $x \in W \subseteq V$; so T_x, the intrinsic subgroup of V7.37, is the maximal subgroup acting trivially on a small $W(x)$, for every natural representation. For $\varphi : x \to y$, $g \mapsto \varphi g \varphi^{-1}$ is injective on T_x with image T_y, an isomorphism $T_x \cong T_y$. *Book:* Definition 7.2.5 and Lemma 7.2.6, p. 310 (Definition 7.2.5: 2017 p. 241, before Definition 7.12; Lemma 7.2.6 new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (restated):* "maximal subgroup acting trivially on some small $W(x)$" is the pair of statements above (Findings, item 121).

V7.75 (Definition 7.2.7, Remark 7.2.8 and Lemma 7.2.9 in the 2021 form; `Polyfold.EpGroupoid.`

`remark_7_2_8`, `Polyfold.EpGroupoid.isOpenQuotientMap_reduced`, `Polyfold.EpGroupoid.isOpenMap_reduced`, `Polyfold.EpGroupoid.isOpenEmbedding_domRestrict_reduced`, `Polyfold.EpGroupoid.reducedTgt`). The topology T_R of $\mathbf{X}^R$ (the quotient topology) makes $\pi : \mathbf{X} \to \mathbf{X}^R$ continuous and is finer than every topology with this property; π is an open quotient map and an open embedding on every sheet of s. *Book:* Definition 7.2.7, p. 310 (2017: 7.12), Remark 7.2.8, p. 311, new in the 2021 edition, and Lemma 7.2.9, p. 311. *Hidden hypotheses:* none. *Deviation (restated):* "the finest topology for which π is continuous" is the quotient topology. In Definition 7.2.7 the target map is printed "$s([\varphi]) = t_X(\varphi)$" (possible misprint for $t([\varphi])$; `reducedTgt`); Lemma 7.2.9 is V7.39's `isOpenMap_reduced` and `isOpenEmbedding_domRestrict_reduced` (Findings, items 122, 123).

V7.76 (Lemma 7.2.11; `Polyfold.EpGroupoid.finite_reduced_preimage`, `Polyfold.EpGroupoid.IsReducedEvenlyCovered`, `Polyfold.EpGroupoid.exists_preimage_reduced_subset_of_t2Space`, `Polyfold.EpGroupoid.isReducedEvenlyCovered_of_t2Space`, `Polyfold.EpGroupoid.isClosed_nonEffectivePart_of_isReducedEvenlyCovered`, `Polyfold.EpGroupoid.isReducedEvenlyCovered_iff_t2Space`, `Polyfold.EpGroupoid.lemma_7_2_11`, `Polyfold.EpGroupoid.Examples.JumpingRetract.reduced_preimage_one`, `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isReducedEvenlyCovered`). The class $\pi^{-1}([\varphi_0])$ is finite. If T_R is Hausdorff, there are open $U(\varphi) \ni \varphi$ for the members φ of the class of φ_0 and an open $U([\varphi_0]) \ni [\varphi_0]$ such that (1) distinct $U(\varphi)$ have disjoint closures, (2) $\pi(U(\varphi)) = U([\varphi_0])$ and π is an open embedding on $U(\varphi)$, (3) $\pi^{-1}(U([\varphi_0]))$ is covered by the $U(\varphi)$. Conversely, (1)–(3) at every unit force $\mathbf{X}^T$ closed; so (1)–(3) hold at every morphism iff T_R is Hausdorff, and they fail at $(1, o_0)$ for the ep-groupoid of V7.82. *Book:* Lemma 7.2.11, p. 315, new in the 2021 edition, for every ep-groupoid. *Hidden hypotheses:* T_R Hausdorff; the proof of (3) uses the uniqueness of limits in T_R. *Deviation (explicit):* without the hypothesis the lemma does not hold as we read it; "homeomorphism onto $U([\varphi_0])$" is read as (2) (Findings, item 124).

V7.77 (Proposition 7.2.12; `Polyfold.EpGroupoid.paracompactSpace_of_isOpenMap`, `Polyfold.EpGroupoid.exists_isCompact_preimage_reduced_subset`, `Polyfold.EpGroupoid.paracompactSpace_reducedMor`, `Polyfold.EpGroupoid.proposition_7_2_12`). For every ep-groupoid, $(\mathbf{X}^R, T_R)$ is paracompact (every open cover has a locally finite open refinement; Hausdorffness not included). *Book:* Proposition 7.2.12 and its footnote, p. 317, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none in the statement. *Proof:* not the text's, which uses Lemma 7.2.11 (1)–(3) and so, as we read it, T_R Hausdorff: paracompactness descends along an open continuous surjection $p : M \to Q$ if every q has a compact K such that every open $N \supseteq K$ contains $p^{-1}(O)$ for an open $O \ni q$; for π, K is the set of $\varphi \in s^{-1}(\mathrm{cl}\, V)$ with $(s,t)(\varphi) = (s,t)(q)$, by properness (Findings, items 111, 125).

V7.78 (Proposition 7.2.13 and Theorem 7.2.14; `Polyfold.EpGroupoid.proposition_7_2_13`, `Polyfold.EpGroupoid.theorem_7_2_14`, `Polyfold.EpGroupoid.reducedPolyfold`, `Polyfold.EpGroupoid.reducedEpGroupoid`, `Polyfold.EpGroupoid.isEffective_reducedEpGroupoid`). The natural charts (`reducedAtlas`) define an M-polyfold structure on $(\mathbf{X}^R, T_R)$ — some Hausdorff paracompact M-polyfold structure on T_R with the natural atlas exists — iff T_R is Hausdorff. If

T_R is Hausdorff, $(X, \mathbf{X}^R)$ is an ep-groupoid `reducedEpGroupoid` with the groupoid structure of V7.40, the object M-polyfold of X, the natural atlas on $\mathbf{X}^R$, and an ep-functor $X \to X_R$; it is effective (Proposition 7.2.16, V7.40). *Book:* Proposition 7.2.13, p. 318, new in the 2021 edition, and Theorem 7.2.14, p. 318 (2017: Proposition 7.10, p. 244). *Hidden hypotheses:* none. *Deviation (restated):* Proposition 7.2.13 is read as the existence of the structure. Theorem 7.2.14 assumes exactly the hypothesis that V7.39 and V7.40 take, T_R Hausdorff; paracompactness is supplied by V7.77 (gap G.71, changed) (Findings, items 56, 107, 111, 126, 143).

V7.79 (Proposition 7.2.17; `Polyfold.EpGroupoid.reduced_eq_iff_comp_inv`, `Polyfold.EpGroupoid.mem_nonEffectivePart_iff_reduced`, `Polyfold.EpGroupoid.continuous_comp_fibredProduct`, `Polyfold.EpGroupoid.continuousOn_comp_inv`, `Polyfold.EpGroupoid.isClosed_reduced_eq_of_isClosed`, `Polyfold.EpGroupoid.isClosed_nonEffectivePart_of_t2Space`, `Polyfold.EpGroupoid.proposition_7_2_17`, `Polyfold.EpGroupoid.isClosed_src_eq_tgt`, `Polyfold.EpGroupoid.closure_nonEffectivePart_subset`). T_R is Hausdorff iff the non-effective part $\mathbf{X}^T$ is closed in $\mathbf{X}$. Also $[\varphi] = [\psi]$ iff $t(\varphi) = t(\psi)$ and $\psi^{-1} \circ \varphi \in \mathbf{X}^T$, and $\operatorname{cl} \mathbf{X}^T \subseteq \{s = t\}$. *Book:* Proposition 7.2.17, p. 320, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none. *Proof:* the relation $[\varphi] = [\psi]$ is closed in $\mathbf{X} \times \mathbf{X}$ when $\mathbf{X}^T$ is closed, and a space is Hausdorff iff this relation is closed for an open quotient map (Mathlib); properness is not used (Findings, item 127).

V7.80 (Definition 7.2.18 (= Definition 7.4.1), Remarks 7.2.19 and 7.2.20; `Polyfold.EpGroupoid.NaturalRep.HasLUC`, `Polyfold.EpGroupoid.IsLUC`, `Polyfold.EpGroupoid.isLUC_iff_forall_naturalRep`, `Polyfold.EpGroupoid.NaturalRep.hasLUC_of_isClosed`, `Polyfold.EpGroupoid.remark_7_2_20`, `Polyfold.EpGroupoid.Examples.remark_7_2_19`, `Polyfold.EpGroupoid.isLUC_trivial`, `Polyfold.EpGroupoid.isLUC_of_isEffective`, `Polyfold.EpGroupoid.isClosed_range_unit`, `Polyfold.EpGroupoid.nonEffectivePart_eq_range_unit_of_isEffective`, `Polyfold.EpGroupoid.Examples.isLUC_trivialZ3`, `Polyfold.EpGroupoid.Examples.isLUC_reflection`). X has the local unique continuation property (LUC) if every object x has a natural representation $U(x)$ such that $\Phi(g) = \mathrm{id}$ on a nonempty open subset of $U(x)$ forces $\Phi(g) = \mathrm{id}$ on $U(x)$. Under LUC every natural representation has this property, and there are such inside every open $O \ni x$. Not every ep-groupoid has LUC (V7.82). LUC holds for the trivial ep-groupoid of every M-polyfold, for every effective ep-groupoid ($\mathbf{X}^T$ is then the closed set of units), for $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection, and for the non-effective trivial $\mathbb{Z}/3$-action on $\mathbb{R}$. *Book:* Definition 7.2.18, Remarks 7.2.19, 7.2.20, p. 321, new in the 2021 edition; Definition 7.4.1, p. 328, repeats Definition 7.2.18. *Hidden hypotheses:* none. *Deviation (stronger):* that the choice of $U(x)$ does not matter is proved (by V7.81, $\mathbf{X}^T$ is then closed, and a connectedness argument) (Findings, items 128, 144).

V7.81 (Proposition 7.2.21, Theorem 7.2.22 and Proposition 7.2.23; `Polyfold.EpGroupoid.NaturalRep.eventually_Φ_eq_of_Γ_mem`, `Polyfold.EpGroupoid.NaturalRep.Γ_mem_of_eventually`, `Polyfold.EpGroupoid.isClosed_nonEffectivePart_of_isLUC`, `Polyfold.EpGroupoid.proposition_7_2_21`, `Polyfold.EpGroupoid.theorem_7_2_22`, `Polyfold.EpGroupoid.t2Space_reducedMor_of_isLUC`, `Polyfold.EpGroupoid.proposition_7_2_23`,

`Polyfold.EpGroupoid.Examples.nonEffectivePart_trivialZ3`, `Polyfold.EpGroupoid.Examples.t2Space_reducedMor_trivialZ3`**).** X has LUC iff $\mathbf{X}^T$ is closed. Equivalent are: T_R is Hausdorff; X has LUC; $\mathbf{X}^T$ is closed in $\mathbf{X}$. Under LUC, T_R is Hausdorff and `reducedEpGroupoid` is an effective ep-groupoid with the structure of V7.78 and an ep-functor $X \to X_R$. For the trivial $\mathbb{Z}/3$-action, $\mathbf{X}^T = \mathbf{X}$ and T_R is Hausdorff. *Book:* Proposition 7.2.21, p. 321, Theorem 7.2.22 and Proposition 7.2.23, p. 323, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* none. *Proof:* $\Gamma(g, y) \in \mathbf{X}^T$ iff $\Phi(g) = \mathrm{id}$ near y; if $\mathbf{X}^T$ is closed, the set of such y is open and closed in the connected $U(x)$ (Findings, items 60, 128, 129).

V7.82 (Appendix 7.6.3, Zhou's example; `Polyfold.EpGroupoid.Examples.ZhouRetract.appendix_7_6_3`, `Polyfold.EpGroupoid.Examples.ZhouRetract.nonEffectivePart_eq`, `Polyfold.EpGroupoid.Examples.ZhouRetract.closure_nonEffectivePart_diff`, `Polyfold.EpGroupoid.Examples.ZhouRetract.not_t2Space_reducedMor`, `Polyfold.EpGroupoid.Examples.ZhouRetract.not_isLUC`, `Polyfold.EpGroupoid.Examples.ZhouRetract.mem_O_iff'`, `Polyfold.EpGroupoid.Examples.ZhouRetract.Data`, `Polyfold.EpGroupoid.Examples.ZhouRetract.sigmaHomeo`, `Polyfold.EpGroupoid.Examples.ZhouRetract.appendix_7_6_3_sigma`, `Polyfold.ZhouLine.lineSc`, `Polyfold.ZhouLine.zhouRetraction`, `Polyfold.ZhouLine.isLossFamily_zhouFam`, `Polyfold.EpGroupoid.Examples.JumpingRetract.appendix_7_6_3`, `Polyfold.EpGroupoid.Examples.JumpingRetract.not_t2Space_reducedMor`, `Polyfold.EpGroupoid.Examples.JumpingRetract.mem_closure_nonEffectivePart`, `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isClosed_nonEffectivePart`, `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isLUC`, `Polyfold.EpGroupoid.Examples.remark_7_2_19`**).** On $L^2(\mathbb{R})$ with $E_m = H^{m,\delta_m}(\mathbb{R})$, $0 = \delta_0 < \delta_1 < \cdots$, the map $r(t, f) = (t, \pi_t f)$, $\pi_t f = \langle f, \beta(\cdot + e^{1/t})\rangle \beta(\cdot + e^{1/t})$ for $t > 0$ and 0 for $t \leq 0$, is an sc-smooth retraction. For $\mathbb{Z}/2$ acting on its image Y by $(t, f) \mapsto (t, -f)$, $\mathbf{X}^T = (\{1\} \times Y) \cup (\{-1\} \times \{t < 0\})$ and $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T = \{(-1, (0, 0))\}$, so T_R is not Hausdorff and LUC fails; through the homeomorphism $\Sigma \cong Y$ the same sets on $\{0, 1\} \times \Sigma$ as printed. *Book:* Appendix 7.6.3, p. 348, new in the 2021 edition. *Hidden hypotheses:* none; the data (δ, β smooth with compact support and $\int \beta^2 = 1$) are satisfiable (`Data.std`). *Deviation (as in the book):* $\beta \geq 0$ is not needed; $H^{m,\delta}(\mathbb{R})$ is defined by lifting to $\mathbb{R} \times S^1$, with the same norm; the sc-smoothness the text cites from [60] is proved. The analogue on the retract of Part I's Remark 4.2.2 (2017: 4.3) is kept, with its $\mathbf{X}^T$ computed exactly too (Findings, items 107, 129, 130, 156).

V7.83 (Appendix 7.6.4: (7.24), Theorem 7.6.7 and Corollary 7.6.8; `Polyfold.EpGroupoid.theorem_7_6_7_one`, `Polyfold.EpGroupoid.reducedRaise`, `Polyfold.EpGroupoid.isScDiffeoOn_reducedRaise`, `Polyfold.EpGroupoid.raiseReducedFunctor`, `Polyfold.EpGroupoid.raiseReducedFunctorInv`, `Polyfold.EP.reducedOfLUC`, `Polyfold.EP.liftOneIso`, `Polyfold.EP.raiseReducedIso`, `Polyfold.EP.theorem_7_6_7`, `Polyfold.EP.corollary_7_6_8`, `Polyfold.EpGroupoid.Examples.theorem_7_6_7_pointEP`, `Polyfold.EpGroupoid.mem_nonEffectivePart_liftOne_of_levelVal`, `Polyfold.EpGroupoid.nonEffectivePart_liftOne_eq_of_isClosed`, `Polyfold.EpGroupoid.hasConnectedDenseLevelOneReps`, `Polyfold.EpGroupoid.theorem_7_6_7_of_isLUC`, `Polyfold.EpGroupoid.corollary_7_6_8_of_isZhou'`, `Polyfold.EpGroupoid.theorem_7_6_7`, `Polyfold.EpGroupoid.corollary_7_6_8`**).** A morphism of level 1 in $\mathbf{X}^T$ lies in $(\mathbf{X}^1)^T$ ((7.24)), and under LUC for X^1, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$. If T_R is Hausdorff for X and X^1, the identification $(X^1)_R \to (X_R)^1$, the identity on objects and $[\varphi]_1 \mapsto [\varphi]$ on morphisms, is an sc-diffeomorphism and an isomorphism of ep-groupoids; under (SLUC), $(X^i)_R \cong (X_R)^i$ in $\mathcal{EP}$ for every i, and the same under (ZHOU). *Book:* (7.24), Theorem 7.6.7 and Corollary 7.6.8, pp. 348–351, new in the 2021 edition. *Hidden hypotheses:* none (the connectedness and density of $U(x) \cap X_1$ hold for every ep-groupoid, gaps G.73, G.85 closed); satisfiable for the point (`theorem_7_6_7_pointEP`). *Deviation (as in the book):* the proof follows the text's diagram; for $i \geq 2$ the isomorphism is the composite of the text's identifications, and "the identity on objects" is stated for $i = 1$, since for variable i the object types are not definitionally equal; the case i uses LUC for X^{i-1} and X^i (gap G.82 closed) (Findings, item 131).

V7.84 (the index lifts X^i, Proposition 7.4.2 and Remark 7.4.3; `Polyfold.EP.liftOne`, `Polyfold.EP.lift`, `Polyfold.EP.lift_zero`, `Polyfold.EP.lift_succ`, `Polyfold.EP.lift_succ'`, `Polyfold.EpGroupoid.proposition_7_4_2`, `Polyfold.EpGroupoid.remark_7_4_3`). $X^0 = X$ and $X^{i+1} = (X^i)^1$, also $= (X^1)^i$. Under LUC, every natural representation has the property of Definition 7.2.18, in particular every $W(x) \supseteq U(x)$, and there are such inside every open $O \ni x$. *Book:* p. 329 (X^i), Proposition 7.4.2, p. 328, and Remark 7.4.3, p. 329, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the hypotheses $U(x) \subseteq W(x)$ and LUC on $U(x)$ are not needed (Findings, item 132).

V7.85 (Lemma 7.4.4; `Polyfold.EpGroupoid.HasLevelOneConnectedDenseReps`, `Polyfold.EpGroupoid.NaturalRep.eqOn_of_isClosed_liftOne`, `Polyfold.EpGroupoid.isLUC_of_isLUC_liftOne`, `Polyfold.EpGroupoid.lemma_7_4_4`, `Polyfold.EpGroupoid.hasLevelOneConnectedDenseReps_of_subsingleton`, `Polyfold.EpGroupoid.Examples.hasLevelOneConnectedDenseReps_lift_pointEP`, `Polyfold.EpGroupoid.Examples.lemma_7_4_4_pointEP`). If X^i has LUC and, for every $k < i$, every object of X^k has a natural representation $U(x)$ with $U(x) \cap X_1$ connected in X^1 and dense in $U(x)$ (for X^k), then X and every X^k, $k \leq i$, have LUC. The hypotheses hold for all lifts of the point ep-groupoid. *Book:* Lemma 7.4.4, p. 329, new in the 2021 edition. *Hidden hypotheses:* that $U(x) \cap X_1$ is connected in X^1, which the proof states without proof, and dense in $U(x)$ (gaps G.85, G.73). *Deviation (explicit):* the two facts are hypotheses; in the proof, "$s : W(g) \to W(y)$" for $W(h)$ (possible misprint) (Findings, item 133).

V7.86 (the Part I facts used in §7.4; `Polyfold.MPolyfold.DenseSmooth`, `Polyfold.MPolyfold.SmoothConnected`, `Polyfold.MPolyfold.DenseSmoothTangent`, `Polyfold.MPolyfold.TangentDeterminedSmooth`, `Polyfold.MPolyfold.TangentDeterminedSmooth.eq_self`, `Polyfold.MPolyfold.DenseSmooth.dense`, `Polyfold.MPolyfold.DenseSmooth.subset_closure_inter`, `Polyfold.MPolyfold.DenseSmooth.exists_smooth`, `Polyfold.MPolyfold.eqOn_tangentFiber_of_mem_closure`, `Polyfold.MPolyfold.SmoothRegular`, `Polyfold.MPolyfold.smoothRegular_of_subsingleton`, `Polyfold.MPolyfold.smoothRegular_discrete_unit`, `Polyfold.MPolyfold.denseSmoothTangent_discrete_unit`). For an M-polyfold X: `DenseSmooth`, X_∞ is dense in every X_m; `SmoothConnected`, $U \cap X_\infty$ is connected in X_∞ for connected open U; `DenseSmoothTangent`, $(T_xX)_\infty$ is dense in T_xX for smooth x; `TangentDeterminedSmooth`, two sc-smooth maps agreeing near a smooth y in X_∞ have the same tangent on T_yX. `SmoothRegular` bundles the first, the second, the last and the openness of $p : TX \to X^1$. Consequences: X_∞ is dense in X, and, if p is open, a map continuous on an open $W \subseteq TX$ and the identity over a set $A \subseteq X_\infty$ is the identity over its closure in X_∞. All hold for M-polyfolds on a subsingleton with every point smooth. *Book:* used in the proofs of Lemma 7.4.6, Propositions 7.4.8, 7.4.10, 7.4.22 and Lemma 7.4.15, pp. 331–340; the density of X_∞ is stated on p. 289. *Hidden hypotheses:* these Part I facts, not proved in Part I for M-polyfolds. *Deviation (explicit):* explicit hypotheses of the §7.4 results that use them (gaps G.85, G.73, G.8) (Findings, item 134).

V7.87 (the tangent Tg of an isotropy element; `Polyfold.EpGroupoid.NaturalRep.eventually_Φ_eq_Γ`, `Polyfold.EpGroupoid.NaturalRep.isoΓ`, `Polyfold.EpGroupoid.NaturalRep.isoΓ_val`, `Polyfold.EpGroupoid.NaturalRep.isoΓ_mem_trivialSubgroup_iff`, `Polyfold.EpGroupoid.NaturalRep.tangentMor_Γ`, `Polyfold.EpGroupoid.NaturalRep.tangentMor_eq_tΦ`, `Polyfold.EpGroupoid.NaturalRep.tΦ_eq_tΦ`, `Polyfold.EpGroupoid.NaturalRep.tangentMor_isoΓ`). The tangent $T\varphi = Tt \circ (Ts|T_\varphi\mathbf{X})^{-1}$ of Definition 7.1.4 (defined without charts) of the morphism $\Gamma(g, y)$ is the tangent map $T\Phi(g)$ on T_yX; in particular $Tg = T\Phi(g)$ on T_xX, the same for every natural representation. For a fixed point z of $\Phi(g)$, $\Gamma(g, z) \in G_z$, it lies in T_z iff $\Phi(g) = \mathrm{id}$ near z, and it acts near z as g does. *Book:* p. 331 ("the linearization Tg") and Proposition 8.1.16's "by definition $Tg = T\hat{g}(x)$", new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* Tg is a set map on the fibre T_xX; no linear structure is used, since (ZHOU) only asks that Tg be the identity (gap G.7 not needed) (Findings, item 135).

V7.88 (Definitions 7.4.5 and 7.4.7, the groups T_x^∞; `Polyfold.EpGroupoid.IsZhou`, `Polyfold.EpGroupoid.trivialSetInf`, `Polyfold.EpGroupoid.mem_trivialSetInf_iff`, `Polyfold.EpGroupoid.trivialSubgroupInf`, `Polyfold.EpGroupoid.mem_trivialSubgroupInf_iff`, `Polyfold.EpGroupoid.trivialSubgroup_le_trivialSubgroupInf`, `Polyfold.EpGroupoid.IsZhou'`, `Polyfold.EpGroupoid.isZhou_of_trivialSubgroup_eq_top`, `Polyfold.EpGroupoid.isZhou'_of_trivialSubgroup_eq_top`, `Polyfold.EpGroupoid.isZhou_of_subsingleton`, `Polyfold.EpGroupoid.isZhou'_of_subsingleton`, `Polyfold.EpGroupoid.isZhou_trivial`, `Polyfold.EpGroupoid.isZhou'_trivial`, `Polyfold.EpGroupoid.Examples.isZhou_trivialZ3`, `Polyfold.EpGroupoid.Examples.isZhou'_trivialZ3`). (ZHOU): for every smooth object x and $g \in G_x$, if Tg is the identity on T_xX then $g \in T_x$. $T_x^\infty \supseteq T_x$ is the subgroup of the g with $\Phi(g) = \mathrm{id}$ near x in X_∞, for any natural representation. (ZHOU′): if Tg is the identity on $(T_xX)_\infty$ then $g \in T_x^\infty$. Both hold when every $T_x = G_x$, in particular for trivial isotropy (the trivial ep-groupoid of every M-polyfold) and for the trivial $\mathbb{Z}/3$-action. *Book:* Definitions 7.4.5 and 7.4.7 and T_x^∞, p. 331, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* Tg as in V7.87 (Findings, items 135, 144).

V7.89 (Lemma 7.4.6 and Proposition 7.4.8; `Polyfold.EpGroupoid.lemma_7_4_6`, `Polyfold.EpGroupoid.isZhou_of_isZhou'`, `Polyfold.EpGroupoid.proposition_7_4_8`, `Polyfold.EpGroupoid.NaturalRep.mem_tangentU_of_fiber`, `Polyfold.EpGroupoid.NaturalRep.tΦ_eq_self_of_eqOn`, `Polyfold.EpGroupoid.NaturalRep.fix_and_tangent_of_mem_closure`, `Polyfold.EpGroupoid.NaturalRep.eq_self_of_smooth_clopen`, `Polyfold.EpGroupoid.

`NaturalRep.eq_self_of_smooth_tangent`, `Polyfold.eq_of_mem_closure_of_continuousAt`, `Polyfold.EpGroupoid.Examples.proposition_7_4_8_point`). If $(T_xX)_\infty$ is dense in T_xX for smooth x, (ZHOU) implies (ZHOU′). Given the Part I facts `SmoothRegular` of V7.86, (ZHOU′) implies (ZHOU); so with both facts (ZHOU) and (ZHOU′) are equivalent. The hypotheses hold for the point ep-groupoid. *Book:* Lemma 7.4.6, p. 331, and Proposition 7.4.8, p. 332, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (explicit):* the proof of 7.4.8 uses that the smooth points of $U(x)$ near which $\hat{g} = \text{id}$ in X_∞ form an open, nonempty and closed subset of the connected $U(x)_\infty$, instead of sequences; "$z_k \to Y$" for $z_k \to y$ (possible misprint) (Findings, item 136).

V7.90 (Proposition 7.4.10 and Corollary 7.4.11 for X and X^1; `Polyfold.EpGroupoid.proposition_7_4_10`, `Polyfold.EpGroupoid.IsSmoothUC`, `Polyfold.EpGroupoid.isSmoothUC_of_isZhou`, `Polyfold.EpGroupoid.isLUC_of_isSmoothUC`, `Polyfold.EpGroupoid.isLUC_liftOne_of_isSmoothUC`, `Polyfold.EpGroupoid.corollary_7_4_11`, `Polyfold.EpGroupoid.corollary_7_4_11_lift`). Given `SmoothRegular`, (ZHOU) implies LUC, and more: unique continuation through X_∞ (if $\hat{g} = \text{id}$ on the smooth points of a nonempty open subset of X_∞ inside $U(x)$, then $\hat{g} = \text{id}$ on $U(x)$), which gives LUC for X and for X^1. With (ZHOU) and `SmoothRegular` for every X^i, every X^i has LUC. *Book:* Proposition 7.4.10, p. 333, and Corollary 7.4.11, p. 334, new in the 2021 edition (for every X^i, through Proposition 7.4.9). *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (narrower):* this form of Corollary 7.4.11 for all i takes (ZHOU) for every X^i, the conclusion of Proposition 7.4.9, as a hypothesis; with (ZHOU) for X only it is V7.102 (gap G.84, closed); the approximation of tangent vectors at the end of the proof of 7.4.10 is the consequence of the openness of $p : TX \to X^1$ in V7.86 (Findings, items 136, 137).

V7.91 (Definition 7.4.12 and Theorem 7.4.13; `Polyfold.EP.IsSLUC`, `Polyfold.EpGroupoid.theorem_7_4_13`, `Polyfold.EpGroupoid.Examples.pointEP`, `Polyfold.EpGroupoid.Examples.lift_pointEP`, `Polyfold.EpGroupoid.Examples.smoothRegular_lift_pointEP`, `Polyfold.EpGroupoid.Examples.isZhou_lift_pointEP`, `Polyfold.EpGroupoid.Examples.isSLUC_pointEP`, `Polyfold.EpGroupoid.Examples.theorem_7_4_13_pointEP`). (SLUC): every X^i has LUC. If every X^i satisfies (ZHOU) and `SmoothRegular`, X has (SLUC). The point ep-groupoid and all its lifts satisfy (ZHOU), `SmoothRegular` and (SLUC). *Book:* Definition 7.4.12 and Theorem 7.4.13, p. 334, new in the 2021 edition: (ZHOU) for X implies (SLUC). *Hidden hypotheses:* the Part I facts of V7.86 for every X^i (gap G.85). *Deviation (narrower):* (ZHOU) for every X^i is a hypothesis here; Proposition 7.4.9 and Theorem 7.4.13 with (ZHOU) for X only are V7.102 (gap G.84, closed) (Findings, items 137, 148).

V7.92 (Definition 7.4.14, Lemma 7.4.15, Propositions 7.4.16–7.4.18; `Polyfold.EpGroupoid.IsVSLUC`, `Polyfold.EpGroupoid.isSmoothUC_of_isVSLUC`, `Polyfold.EpGroupoid.isVSLUC_of_isSmoothUC`, `Polyfold.EpGroupoid.lemma_7_4_15`, `Polyfold.EpGroupoid.proposition_7_4_16`, `Polyfold.EpGroupoid.proposition_7_4_17`, `Polyfold.EpGroupoid.isVSLUC_of_isZhou`, `Polyfold.EpGroupoid.isSmoothUC_of_subsingleton`, `Polyfold.EpGroupoid.isVSLUC_of_subsingleton`, `Polyfold.EpGroupoid.isLUC_of_subsingleton_mor`, `Polyfold.EpGroupoid.isVSLUC_trivial`, `Polyfold.EpGroupoid.isSmoothUC_trivial`). (VSLUC): every smooth x has a natural representation $U(x)$ and an open $V(x) \ni x$ of X_∞ inside it, invariant under G_x, such that $\hat{g} = \mathrm{id}$ on a nonempty open subset of $V(x)$ forces $\hat{g} = \mathrm{id}$ on $V(x)$. If X_∞ is dense in every X_m and $U \cap X_\infty$ is connected for connected open U: (VSLUC) is equivalent to unique continuation through X_∞, implies LUC for X and for X^1, and (VSLUC) for every X^i gives (SLUC); (ZHOU) with `SmoothRegular` implies (VSLUC). (VSLUC) holds for trivial isotropy. *Book:* Definition 7.4.14 and Lemma 7.4.15, p. 335, Propositions 7.4.16–7.4.18, p. 336, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (narrower):* $V(x)$ is read open in X_∞, inside some $U(x)$ and invariant; here Proposition 7.4.16 for X^1 only and Proposition 7.4.17 with (VSLUC) for every X^i; for every X^i with (VSLUC) for X only they are V7.103, and Proposition 7.4.18 in both directions is V7.104 (gap G.84, closed) (Findings, items 136, 137, 149, 150).

V7.93 (Definitions 7.4.19 and 7.4.21, Lemma 7.4.20; `Polyfold.EpGroupoid.IsLR`, `Polyfold.EpGroupoid.IsLR'`, `Polyfold.EpGroupoid.NaturalRep.exists_Φ_eq_of__eq`, `Polyfold.EpGroupoid.NaturalRep.eq_self_of__eq`, `Polyfold.EpGroupoid.lemma_7_4_20`, `Polyfold.EpGroupoid.isLR_iff_isLR'`). (LR): inside every open $O \ni x$ there is a natural representation $U(x)$ such that every sc-diffeomorphism β of $U(x)$ with $|\beta(y)| = |y|$ is $\hat{g}$ for some $g \in G_x$. (LR′): the same for sc-diffeomorphisms β of $U(x)$ onto an open image with $\beta(x) = x$. If β is an sc-diffeomorphism of $U(x)$ onto $\beta(U(x))$ with $\beta(x) = x$ and $|\beta(y)| = |y|$, then $\beta(U(x)) = U(x)$; hence (LR) and (LR′) are equivalent. *Book:* Definition 7.4.19, Lemma 7.4.20, p. 337, Definition 7.4.21, p. 338, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* "arbitrarily small" is "inside every open $O \ni x$", $|\beta(y)| = |y|$ is equality in $|X|$; Lemma 7.4.20 is proved by the finite closed cover $U(x) = \bigcup_g \{\beta = \hat{g}\}$ and connectedness, instead of paths (Findings, item 138).

V7.94 (Proposition 7.4.22, Definition 7.4.23 and Theorem 7.4.24; `Polyfold.EpGroupoid.NaturalRep.eqOn_Φ_of_tangent`, `Polyfold.EpGroupoid.NaturalRep.exists_eq_Φ_of_isZhou`, `Polyfold.EpGroupoid.proposition_7_4_22`, `Polyfold.EP.IsSLR`, `Polyfold.EpGroupoid.theorem_7_4_24`, `Polyfold.EpGroupoid.Examples.isSLR_pointEP`). Under (ZHOU) and LUC, $\hat{g}$ and $\hat{g}'$ agree on $U(x)$ once they agree at a smooth point together with their tangents there. Given `SmoothRegular`, (ZHOU) implies (LR). (SLR): every X^i has (LR); if every X^i satisfies (ZHOU) and `SmoothRegular`, X has (SLR). The point ep-groupoid has (SLR). *Book:* Proposition 7.4.22, p. 338, Definition 7.4.23, p. 339, Theorem 7.4.24, p. 340, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 (gap G.85). *Deviation (narrower):* Theorem 7.4.24 takes (ZHOU) for every X^i as a hypothesis here; with (ZHOU) for X only it is V7.102 (gap G.84, closed). *Proof:* without passing to the reduced ep-groupoid and without the density of regular points: on the open dense union of the X_∞-interiors of the sets $\{\beta = \hat{g}\}$, two pieces whose closures meet have equal tangents at a common smooth point, hence equal $\hat{g}$ (Findings, item 138).

V7.95 (the example of Appendix 7.6.3 fails (ZHOU); `Polyfold.EpGroupoid.Examples.ZhouRetract.not_isZhou`, `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isZhou_unconditional`, `Polyfold.EpGroupoid.Examples.JumpingRetract.not_isZhou`). Zhou's ep-groupoid of V7.82 does not satisfy (ZHOU): it does not have LUC, and (ZHOU) implies LUC for an M-polyfold, which is always `SmoothRegular` (V7.90). *Book:* p. 331, new in the 2021 edition:

the example of Appendix 7.6.3 does not satisfy the Zhou condition. *Hidden hypotheses:* none (gap G.85 closed). *Deviation (as in the book):* none beyond V7.82; the analogue fails (ZHOU) too (gap G.83 closed) (Findings, items 139, 156).

V7.96 (transport of the tame boundary along index-preserving local homeomorphisms; `Polyfold.TameBoundary.IsDHomeoOn`, `Polyfold.TameBoundary.IsDLocalHomeo`, `Polyfold.TameBoundary.isDHomeoOn_id`, `Polyfold.TameBoundary.IsDHomeoOn.isFaceIn_image`, `Polyfold.TameBoundary.IsDHomeoOn.mem_faceStructuredNbhdsAt_image`, `Polyfold.TameBoundary.IsDHomeoOn.isMor_image`, `Polyfold.TameBoundary.IsDHomeoOn.image_faceExt`, `Polyfold.TameBoundary.BTriple.map`, `Polyfold.TameBoundary.map_cls_eq`, `Polyfold.TameBoundary.IsLiftOf`, `Polyfold.TameBoundary.IsLiftOf.unique`, `Polyfold.TameBoundary.IsLiftOf.comp`, `Polyfold.TameBoundary.isLiftOf_id`, `Polyfold.TameBoundary.bLift`, `Polyfold.TameBoundary.bLift_cls`, `Polyfold.TameBoundary.continuous_bLift`, `Polyfold.TameBoundary.isOpenMap_bLift`, `Polyfold.TameBoundary.isLocalHomeomorph_bLift`, `Polyfold.TameBoundary.isCompact_preimage_bMap`, `Polyfold.TameBoundary.exists_rep_centered`, `Polyfold.MPolyfold.IsScDiffeoOn.isDHomeoOn`, `Polyfold.MPolyfold.IsLocalScDiffeo.isDLocalHomeo`). For a corner index d (the degeneracy index of a tame M-polyfold, or any `IsCornerIndex`): a map f that is a homeomorphism of an open W onto an open set preserving the index carries faces to faces, face-structured neighbourhoods to face-structured neighbourhoods, and admissible triples (y, V, θ) to admissible triples, with a class independent of the chosen W. For an index-preserving local homeomorphism f, $[y, V, \theta] \mapsto [f(y), f(V), f(\theta)]$ is a well defined map bf of the tame boundaries, continuous, open and a local homeomorphism; preimages of compact sets under B are compact. An sc-diffeomorphism between open sets preserves the degeneracy index. *Book:* §7.6.5.2, pp. 353–355, new in the 2021 edition, where the maps (7.25)–(7.28) are defined through sc-diffeomorphisms of sheets. *Hidden hypotheses:* none. *Deviation (extra):* a tool of the formalization: the topological content of those sc-diffeomorphisms (Findings, item 140).

V7.97 ((7.25)–(7.29), the boundary groupoid; `Polyfold.EpGroupoid.isTame_PMor`, `Polyfold.EpGroupoid.cornerObj`, `Polyfold.EpGroupoid.cornerMor`, `Polyfold.EpGroupoid.bObj`, `Polyfold.EpGroupoid.bMor`, `Polyfold.EpGroupoid.isDLocalHomeo_src`, `Polyfold.EpGroupoid.isDLocalHomeo_tgt`, `Polyfold.EpGroupoid.isScDiffeoOn_inv`, `Polyfold.EpGroupoid.isDLocalHomeo_inv`, `Polyfold.EpGroupoid.isDHomeoOn_unit`, `Polyfold.EpGroupoid.isDLocalHomeo_unit`, `Polyfold.EpGroupoid.bSrc`, `Polyfold.EpGroupoid.bTgt`, `Polyfold.EpGroupoid.bInv`, `Polyfold.EpGroupoid.bUnit`, `Polyfold.EpGroupoid.bSrc_cls`, `Polyfold.EpGroupoid.bTgt_cls`, `Polyfold.EpGroupoid.bInv_cls`, `Polyfold.EpGroupoid.bUnit_cls`, `Polyfold.EpGroupoid.eq_of_bMap_eq_of_bSrc_eq`, `Polyfold.EpGroupoid.exists_bComp`, `Polyfold.EpGroupoid.bComp`, `Polyfold.EpGroupoid.bComp_spec`, `Polyfold.EpGroupoid.eventuallyEq_comp_sheet`, `Polyfold.EpGroupoid.bComp_eq`, `Polyfold.EpGroupoid.boundaryGroupoid`, `Polyfold.EpGroupoid.boundaryFunctor`). For a tame ep-groupoid, **X** is tame. On the tame boundaries bX, $b\mathbf{X}$: $s([\varphi, \mathbf{V}, \theta]) = [s(\varphi), s(\mathbf{V}), s(\theta)]$ for a sheet **V** of s, likewise t; $\iota([\varphi, \mathbf{V}, \theta]) = [\varphi^{-1}, \mathbf{V}^{-1}, \theta^{-1}]$; $u([x, V, \theta]) = [1_x, \mathbf{V}, \bar{\theta}]$ with $\bar{\theta}$ the preimage of θ under $s|\mathbf{V}$; and $m(a, b)$ is the unique class over $\psi \circ \varphi$ with source $s(b)$, which is the printed $[\psi \circ \varphi, \mathbf{W}\,{}_s{\circ}_t\,\mathbf{V}, \sigma\,{}_s{\circ}_t\,\theta]$. With these maps $(bX, b\mathbf{X})$ is a groupoid and B a functor $(bX, b\mathbf{X}) \to (X, \mathbf{X})$. *Book:* (7.25)–(7.29), pp. 354–355, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* the sc-diffeomorphisms of the sheets enter through their topological content (V7.96); (7.28) is read through u, an index-preserving homeomorphism onto the open set $u(X)$; off composable pairs m takes a junk value (Findings, item 140).

V7.98 (Theorem 7.6.9, its topological part; `Polyfold.EpGroupoid.metrizableSpace_bObj`, `Polyfold.EpGroupoid.metrizableSpace_bMor`, `Polyfold.EpGroupoid.continuous_bSrc`, `Polyfold.EpGroupoid.continuous_bTgt`, `Polyfold.EpGroupoid.continuous_bInv`, `Polyfold.EpGroupoid.continuous_bUnit`, `Polyfold.EpGroupoid.isLocalHomeomorph_bSrc`, `Polyfold.EpGroupoid.isLocalHomeomorph_bTgt`, `Polyfold.EpGroupoid.continuousOn_bComp`, `Polyfold.EpGroupoid.exists_isProperMap_bTgt`, `Polyfold.EpGroupoid.theorem_7_6_9`, `Polyfold.EpGroupoid.bSrc_trivial`, `Polyfold.EpGroupoid.trivial_quadrant_witness`). For a tame ep-groupoid: (1) bX and $b\mathbf{X}$ are metrizable, every $U(x, V, \theta)$ is open and every $\Phi_{(x,V,\theta)}$ an open embedding, and these properties determine the topologies; (3) both maps B are continuous; (4) a point of index d has exactly d preimages under B, on objects and on morphisms; (5) s, t are local homeomorphisms, u, ι are continuous, m is continuous on the composable pairs, $(bX, b\mathbf{X})$ has the properness property of Definition 7.1.3 (3), and B is a functor. The trivial ep-groupoid of the tame quadrant $[0, \infty)^n \oplus W$, $n \geq 1$, is tame, with bX, $b\mathbf{X}$ nonempty and n classes over the corner. *Book:* Theorem 7.6.9 and the characterization after it, p. 355, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the topological part only; (2), the sc-smoothness in (3), the index d_{bX} in (4) beyond the fibre count, and the sc-smoothness and tameness in (5) depend on 2021 Theorem 2.7.13 (2) and are not covered here (they are V7.99–V7.101, `Chapter7/TameBoundaryEpStructure.lean`); "Φ is a homeomorphism onto θ" holds as we read it only for V centred at x, as for (2.44), and is used as an open embedding (Findings, items 140, 141).

V7.99 (sc-smooth maps of tame boundaries; `Polyfold.MPolyfold.isScSmoothOn_of_bMap`, `Polyfold.MPolyfold.isScSmoothMap_bLift`, `Polyfold.MPolyfold.isLiftOf_bLift_sheet`, `Polyfold.MPolyfold.isLocalScDiffeo_bLift`). Let A, B be tame M-polyfolds with the tame M-polyfold structures of Part I's Theorem 2.7.13 (2) on bA, bB (`Polyfold.MPolyfold.tameBoundaryPolyfold`), and $f : A \to B$ sc-smooth on an open U. A map $g : bA \to bB$, continuous on an open $O \subseteq B^{-1}(U)$ with $B \circ g = f \circ B$ on O, is sc-smooth on O; in the charts $p \mapsto \pi(\varphi(Bp))$ of bA and bB it reads $\pi' \circ (\varphi' \circ f \circ \varphi^{-1}) \circ \iota$. Hence the lift bf, $[y, V, \theta] \mapsto [f(y), f(V), f(\theta)]$ (V7.96), of an sc-smooth index-preserving local homeomorphism is sc-smooth, and the lift of a local sc-diffeomorphism is a local sc-diffeomorphism: over a sheet $f : W \to V$ with inverse σ, bf maps $B^{-1}(W)$ onto $B^{-1}(V)$, with an inverse that covers σ and undoes bf. *Book:* §7.6.5.2, p. 355, new in the 2021 edition: the proof sketch of Theorem 7.6.9, which reads the structure maps in the charts Φ as sc-diffeomorphisms $s : \theta \to s(\theta)$ of faces. *Hidden hypotheses:* none. *Deviation (extra):* a tool of the formalization; it replaces the sketch's sc-diffeomorphism of faces by the computation in the charts of bA, which needs only that f is sc-smooth (Findings, item 146).

V7.100 (Theorem 7.6.9 (2), (3), (5): the boundary ep-groupoid, and B an ep-functor;

`Polyfold.EpGroupoid.paracompactSpace_bObj`, `Polyfold.EpGroupoid.paracompactSpace_bMor`, `Polyfold.EpGroupoid.bPObj`, `Polyfold.EpGroupoid.bPMor`, `Polyfold.EpGroupoid.isLocalScDiffeo_bSrc`,`Polyfold.EpGroupoid.isLocalScDiffeo_bTgt`,`Polyfold.EpGroupoid.isScSmoothMap_bUnit`, `Polyfold.EpGroupoid.isScSmoothMap_bInv`, `Polyfold.EpGroupoid.isScSmoothCompSheet_bComp`, `Polyfold.EpGroupoid.isScSmoothComp_bComp`, `Polyfold.EpGroupoid.boundaryEpGroupoid`, `Polyfold.EpGroupoid.isTame_boundaryEpGroupoid`, `Polyfold.EpGroupoid.boundaryEpFunctor`, `Polyfold.EpGroupoid.phiTot_bSrc`). For a tame ep-groupoid $(X, \mathbf{X})$, give bX and $b\mathbf{X}$ the tame M-polyfold structures of Theorem 2.7.13 (2) for the object and the morphism M-polyfold ($\mathbf{X}$ is tame, V7.97); both are paracompact. Then $s, t : b\mathbf{X} \to bX$ of (7.25) are local sc-diffeomorphisms, u of (7.28) and ι of (7.27) are sc-smooth, and m of (7.29) is sc-smooth on $b\mathbf{X}\,{}_s\!\times_t b\mathbf{X}$; with the properness of V7.98, $(bX, b\mathbf{X})$ is an ep-groupoid (Definition 7.1.3 (2017: 7.3)), the boundary ep-groupoid, and it is tame. Both maps B form an ep-functor $(bX, b\mathbf{X}) \to (X, \mathbf{X})$, a morphism of the category $\mathcal{EP}$ (Definition 7.1.7 (2017: 7.7)). For a representative $(\varphi, \mathbf{V}, \theta)$ with $\mathbf{V}$ in a sheet of s, s maps $U(\varphi, \mathbf{V}, \theta)$ into $U(x, s(\mathbf{V}), s(\theta))$ and $\Phi_{(x,s(\mathbf{V}),s(\theta))} \circ s = s \circ \Phi_{(\varphi,\mathbf{V},\theta)}$ there. *Book:* Theorem 7.6.9 (2), (3), (5), p. 355, new in the 2021 edition, whose verification is left to the reader, and the diagram of its proof sketch. *Hidden hypotheses:* none. *Deviation (stronger):* the text asks B to be sc-smooth; it is also a functor (V7.97), so an ep-functor. Condition (4) of Definition 7.1.3, the sc-smoothness of m, is proved in the sheet form of V7.3: near b, $b' \mapsto m(\sigma(tb'), b')$ covers $\gamma \mapsto m(\sigma_1(t\gamma), \gamma)$ for a sheet σ_1 of s on $\mathbf{X}$, and V7.99 applies (Findings, items 145, 146). *Proof:* s, t, u, ι are lifts of local sc-diffeomorphisms or of sc-smooth index-preserving local homeomorphisms of $(X, \mathbf{X})$ (V7.97), so V7.99 applies; tameness is that of Theorem 2.7.13 (2).

V7.101 (Theorem 7.6.9 in full; `Polyfold.EpGroupoid.theorem_7_6_9_full`, `Polyfold.EpGroupoid.trivial_quadrant_boundaryEp_witness`, `Polyfold.MPolyfold.theorem_2_7_13_structure`). For a tame ep-groupoid $(X, \mathbf{X})$: (1) bX and $b\mathbf{X}$ are metrizable; (2) they carry the tame M-polyfold structures of V7.100, characterized as in Theorem 2.7.13: for every admissible triple with V centred at x, $U(x, V, \theta)$ is open and $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto the face θ with its induced structure, inverse to the homeomorphism of V7.98, and every M-polyfold structure with this property has an atlas equivalent to it, for bX and for $b\mathbf{X}$; (3) both maps B are sc-smooth; (4) $d_X(B(p)) - 1 = d_{bX}(p)$ on bX and $d_{\mathbf{X}}(B(a)) - 1 = d_{b\mathbf{X}}(a)$ on $b\mathbf{X}$; (5) the boundary ep-groupoid is tame, its structure maps are those of (7.25)–(7.29), s, t are local sc-diffeomorphisms, u, ι are sc-smooth and m is sc-smooth on the fibred product; and B is an ep-functor. The trivial ep-groupoid of the tame quadrant $[0, \infty)^n \oplus W$, $n \geq 1$, satisfies the hypothesis, its boundary ep-groupoid is tame with nonempty bX, $b\mathbf{X}$, and the classes over the corner have $d_{bX} = n - 1$. *Book:* Theorem 7.6.9 and the characterization after it, p. 355, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the characterization and its uniqueness are for representatives whose V is centred at x (V7.98); in Part I they are verified since for every representative, with image $\theta \cap V^x$ (V2.104; gap G.78, closed), here not; "natural" in (2) is read through that characterization and its uniqueness. (4) is stated as printed, with the subtraction of natural numbers (B takes values in ∂X, where $d_X \geq 1$); the additive form is `Polyfold.MPolyfold.degIndex_tameBoundaryMap`. The hypotheses of the two uniqueness statements are satisfied by the structures of (2) themselves (Findings, items 145, 147). *Proof:* (1) and the topology are V7.98; (2)–(4) are Theorem 2.7.13 (2)–(4) for the object and the morphism M-polyfold; (5) and the ep-functor are V7.100.

V7.102 (Proposition 7.4.9, and Corollary 7.4.11, Theorems 7.4.13 and 7.4.24 with (ZHOU) for X only; `Polyfold.MPolyfold.continuous_levelDown`, `Polyfold.MPolyfold.shiftSmoothMap`, `Polyfold.MPolyfold.shiftSmoothInv`, `Polyfold.MPolyfold.shiftSmoothHomeomorph`, `Polyfold.EpGroupoid.isotropyLevelVal`, `Polyfold.EpGroupoid.levelVal_Φ_liftOne`, `Polyfold.EpGroupoid.tangentLower_tangentMor_liftOne`, `Polyfold.EpGroupoid.isZhou'_liftOne`, `Polyfold.EP.isZhou'_lift`, `Polyfold.EP.proposition_7_4_9`, `Polyfold.EP.corollary_7_4_11_all`, `Polyfold.EP.theorem_7_4_13_all`, `Polyfold.EP.theorem_7_4_24_all`, `Polyfold.EpGroupoid.Examples.proposition_7_4_9_pointEP`, `Polyfold.EpGroupoid.Examples.theorem_7_4_13_all_pointEP`, `Polyfold.EpGroupoid.Examples.theorem_7_4_24_all_pointEP`). The inclusion $X_{m+1} \to X_m$ is continuous, and $(X^1)_\infty \to X_\infty$ is a homeomorphism. A natural representation of X^1 at x inside $U(x) \cap X_1$ acts by the natural action of X, and the tangent action of $\mathbf{X}^1$ is that of $\mathbf{X}$ under the inclusion $T(X^1) \to TX$. (ZHOU′) for X gives (ZHOU′) for X^1, hence for every X^i. So if X satisfies (ZHOU), $(T_xX)_\infty$ is dense in T_xX at every smooth x, and every X^i satisfies the Part I facts `SmoothRegular` of V7.86: every X^i satisfies (ZHOU) (Proposition 7.4.9) and has LUC (Corollary 7.4.11), and X has (SLUC) (Theorem 7.4.13) and (SLR) (Theorem 7.4.24). The point ep-groupoid satisfies the hypotheses. *Book:* Proposition 7.4.9, p. 333, Corollary 7.4.11 and Theorem 7.4.13, p. 334, Theorem 7.4.24, p. 340, new in the 2021 edition. *Hidden hypotheses:* the Part I facts of V7.86 for every X^i, and $(T_xX)_\infty$ dense in T_xX for X (gap G.85). *Deviation (explicit):* "(ZHOU′) is the same for X and X^i" is proved in the direction the proof uses, from X to X^i, through $(X^1)_\infty \cong X_\infty$, Part I's Proposition 2.3.17 for $i = 1$ and the comparison of natural representations; (ZHOU) gives (ZHOU′) for X by Lemma 7.4.6, and (ZHOU′) gives (ZHOU) for X^i by Proposition 7.4.8 (V7.89). This closes gap G.84 for these statements (Findings, item 148). *Proof:* induction on i; the rest as in V7.90, V7.91, V7.94.

V7.103 (Propositions 7.4.16 and 7.4.17 with (VSLUC) for X only; `Polyfold.MPolyfold.DenseSmooth.shift`, `Polyfold.EP.denseSmooth_lift`, `Polyfold.EpGroupoid.isVSLUC_liftOne`, `Polyfold.EP.isVSLUC_lift`, `Polyfold.EP.proposition_7_4_16_all`, `Polyfold.EP.proposition_7_4_17_all`, `Polyfold.EpGroupoid.Examples.isVSLUC_pointEP`, `Polyfold.EpGroupoid.Examples.proposition_7_4_17_all_pointEP`). (VSLUC) for X gives (VSLUC) for X^1, with $V'(x) = V(x) \cap U'(x)$ under $(X^1)_\infty = X_\infty$, hence for every X^i; the density of X_∞ in every X_m passes from X to every X^i. So if X has (VSLUC), X_∞ is dense in every X_m and every X^i satisfies `SmoothConnected`, every X^i has (VSLUC) and LUC (Proposition 7.4.16) and X has (SLUC) (Proposition 7.4.17). The point ep-groupoid has (VSLUC) and satisfies the hypotheses. *Book:* Propositions 7.4.16 and 7.4.17, p. 336, new in the 2021 edition: (VSLUC) is phrased in terms of $X_\infty = X^i_\infty$. *Hidden hypotheses:* the density of X_∞ for X (gap G.73) and `SmoothConnected` for every X^i (gap G.85), which Lemma 7.4.15 uses. *Deviation (explicit):* "$X_\infty = X^i_\infty$" is the homeomorphism of V7.102; "in particular (LUC)" is Lemma 7.4.15 (V7.92) for every X^i (Findings, item 149).

V7.104 (Proposition 7.4.18, (VSLUC) □ (SLUC); **`Polyfold.MPolyfold.levelDownTo`, `Polyfold.MPolyfold.continuous_levelDownTo`, `Polyfold.MPolyfold.exists_level_nhds_smooth`, `Polyfold.EpGroupoid.isotropyLevelMk`, `Polyfold.EpGroupoid.isotropyLevelMk_levelVal`, `Polyfold.EpGroupoid.isotropyLevelVal_mul`, `Polyfold.EpGroupoid.isotropyLevelVal_inv`, `Polyfold.EpGroupoid.NaturalRep.liftOneΦ`, `Polyfold.EpGroupoid.NaturalRep.levelVal_liftOneΦ`, `Polyfold.EpGroupoid.NaturalRep.mapsTo_liftOneΦ`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_liftOneΦ`, `Polyfold.EpGroupoid.NaturalRep.liftOneRep`, `Polyfold.EpGroupoid.NaturalRep.eqOn_of_liftOneRep`, `Polyfold.EpGroupoid.NaturalRep.LevelConnected`, `Polyfold.EpGroupoid.NaturalRep.levelConnected_liftOneRep`, `Polyfold.EP.eqOn_of_level_smooth`, `Polyfold.EP.isVSLUC_of_isSLUC`, `Polyfold.EP.proposition_7_4_18`, `Polyfold.EpGroupoid.Examples.levelConnected_pointEP`, `Polyfold.EpGroupoid.Examples.proposition_7_4_18_pointEP`).** The sets $O \cap X_\infty$, O open in some X_{N+1}, form a neighbourhood basis of X_∞. A natural representation $U(x)$ of $x \in X_1$ with $U(x) \cap X_1$ connected in X^1 restricts to a natural representation of x in X^1, the isotropy groups of x in X and in X^1 being identified; if every $U(x) \cap X_m$ is connected in X_m, so it is for the restriction. Under LUC for X^{N+1}, if $\widehat{g} = \mathrm{id}$ on the smooth points of a nonempty open subset of $U(x) \cap X_{N+1}$, then $\widehat{g} = \mathrm{id}$ on $U(x)$. Hence, if X_∞ is dense in every X_m and every smooth x has a natural representation with every $U(x) \cap X_m$ connected, (SLUC) implies (VSLUC); with `SmoothConnected` for every X^i as well, (VSLUC) and (SLUC) are equivalent. Both hold for the point ep-groupoid, which satisfies the hypotheses. *Book:* Proposition 7.4.18, p. 336, new in the 2021 edition. *Hidden hypotheses:* the density of X_∞ (gap G.73); the natural representations with every $U(x) \cap X_m$ connected, which the proof takes "by construction" and which are a hypothesis here (gap G.85). *Deviation (explicit):* LUC for X^N is applied to the restriction of $U(x)$ to X^N, built one level at a time; the direction (VSLUC) □ (SLUC) is V7.103 (Findings, item 150). *Proof:* the text's: an open subset of $V(x) = U(x) \cap X_\infty$ contains $W' \cap X_\infty$ for a nonempty open W' of some X_N, $\widehat{g} = \mathrm{id}$ on W' by density and continuity, and LUC for X^N on the restricted $U(x)$ gives $\widehat{g} = \mathrm{id}$ on $U(x)$.

What is not verified in this chapter

- p. 289 (2017 p. 227), Definition 7.1.4 (2017: 7.4), the tangent $T\varphi$ as an sc-isomorphism: that it is well defined, equal to $T\widehat{\varphi}(s(\varphi))$, a bounded linear isomorphism on level 0 for φ on level 1, and an sc-operator for smooth φ; $T\varphi$ is only defined (`Polyfold.EpGroupoid.tangentMor`). Its linearity needs the linear structure of T_xX (gap G.7); deferred to Chapter 8 (V7.5) (gap G.70; Findings, item 110).
- 2017 p. 227, the comparison with the classical properness of $(s,t) : \mathbf{X} \to X \times X$; no formal content.
- 2017 p. 228, the filtration $|X| = |X_0| \supset |X_1| \supset \cdots \supset |X_\infty| = \bigcap_k |X_k|$: $|X_k|$ is defined (`Polyfold.EpGroupoid.orbitLevel`), $|X_\infty|$ and the identity are not stated (V7.6) (gap G.72; Findings, item 112).
- 2017 p. 227, "the set X_∞ is dense in every space X_k": stated without proof; Part I proves it only for sc-Banach spaces, not for M-polyfolds; not formalized (V7.6) (gap G.73; Findings, item 116).

- 2017 pp. 228–229, the remark after Definition 7.1.7 (2017: 7.7) in its 2017 form: it does not hold as we read it. The 2021 text (p. 291) assumes continuity on morphisms, and its form is verified (V7.10, V7.69) (Findings, items 9, 117).
- 2017 pp. 237–238, the tangent facts (3) and (4) recalled before Proposition 7.1.22 (2017: 7.7) (T_aA has an sc-complement in T_aX; $T_aA = Tr(a)T_aX$): not restated, since T_aA and T_aX carry no linear structure here; in a chart they are Part I's `Polyfold.proposition_2_3_19_three`, `Polyfold.proposition_2_3_19_four` (V7.30) (gap G.7; Findings, item 40).
- pp. 313–324 (2017 pp. 242–244), Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) for a general ep-groupoid: that $\mathbf{X}^R$ is Hausdorff and paracompact is a hypothesis, discharged for effective X and the trivial ep-groupoid. Hausdorffness fails in general (`Polyfold.EpGroupoid.Examples.exists_not_t2Space_reducedMor`, formalized after the round-trip audit); not formalized: paracompactness when $\mathbf{X}^R$ is Hausdorff, and whether Hausdorffness with **X** second countable suffices (V7.39, V7.40) (gap G.71; Findings, items 56, 60, 107, 111). *2021 edition:* Proposition 7.2.12 (p. 317) proves T_R paracompact for every ep-groupoid, verified (V7.77), so the question of paracompactness is settled; Theorem 7.2.14 (p. 318) assumes T_R Hausdorff, exactly the hypothesis of V7.39 and V7.40 (V7.78), and Theorem 7.2.22 (p. 323) characterizes it (V7.81); the counterexample is of the kind of Remark 7.2.8 and Appendix 7.6.3 (V7.82) (gap G.71, closed; Findings, items 129, 143). *Update (2026-11-12):* the 2021 Proposition 7.2.10, Theorem 7.2.14 and Proposition 7.2.16 are verified as printed (V7.39, V7.40, same); what remains unverified is only the 2017 form, for every ep-groupoid, which does not hold as we read it.
- p. 340 (2017 p. 249), "It is presently not known whether the property (5) is a consequence of the properties (1)–(4)": an open question of the text, not formalized and not given a gap, since nothing depends on it; Remark 7.5.2 (2017: 7.1) has no formal content (V7.53) (Findings, items 82, 113).
- p. 342 (2017 p. 251), Remark 7.5.6 (2017: 7.2): no formal content, nothing to formalize (V7.57).
- 2017 p. 253, the contractibility of N_g^t, N_g^s in the proof of Theorem 7.6.1 (2017: 7.5): not needed (V7.62) (Findings, item 29).
- Verified since this list was first written (2026-11-12): Corollary 7.6.6 (2017: 7.3), p. 347 (2017 p. 256), without Part I's Theorem 5.5.16 (2017: 5.14) as a hypothesis (V7.68) (gap G.53 closed; Findings, item 95).
- Gaps of Part I taken up: G.15 (Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17)) is now formalized (V7.50); of G.56, the local path connectedness of M-polyfolds is now formalized (V7.18), the sc$^+$-smooth paths in a path component are not; G.53 is closed (`Polyfold.theorem_5_5_16`, 2026-11-12) (Findings, item 114).
- 2017 p. 234, Proposition 7.5 (2) (2017) as printed, $s, t : \mathbf{U}(\psi) \to U(y)$ with one y: it does not hold as we read it; the 2021 form, $U(s(\psi))$ and $U(t(\psi))$, is verified (V7.71) (Findings, item 119).
- p. 315, Lemma 7.2.11 for an ep-groupoid whose T_R is not Hausdorff: it does not hold as we read it; verified with T_R Hausdorff (V7.76) (Findings, item 124).
- pp. 329–340, the Part I facts used in §7.4 (X_∞ dense in every X_m, gap G.73; $U \cap X_\infty$ connected

for connected open U; $p : TX \to X^1$ open, gap G.8; tangent maps determined by X_∞; $(T_xX)_\infty$ dense in T_xX) and the connectedness of $U(x) \cap X_1$ in X^1 that the proof of Lemma 7.4.4 uses: explicit, satisfiable hypotheses (V7.83, V7.85, V7.86), for the results on every X^i taken on every X^i (`SmoothRegular`, `SmoothConnected`; `DenseSmooth` passes from X to every X^i by proof), and, for Proposition 7.4.18, the choice of $U(x)$ with every $U(x) \cap X_m$ connected (`NaturalRep.LevelConnected`, V7.104) (gap G.85; Findings, items 133, 134, 148–150).

- pp. 333–336, §7.4 for the index lifts: in Proposition 7.4.9, "(ZHOU') is the same for X and X^i" is verified from X to X^i, the direction the proof uses (V7.102). Proposition 7.4.9, Corollary 7.4.11, Theorems 7.4.13, 7.4.24 and Propositions 7.4.16–7.4.18 are verified for every X^i (V7.102–V7.104; gap G.84, closed; Findings, items 137, 148–150).
- Verified since this list was first written (2026-11-12): p. 348, Appendix 7.6.3, Zhou's retract itself and the exact non-effective part of its ep-groupoid, and that it fails (ZHOU) (p. 331) (V7.82, V7.95; gap G.83 closed; Findings, items 130, 139, 156); pp. 348–351, Appendix 7.6.4, Theorem 7.6.7 and Corollary 7.6.8, the sc-diffeomorphism $(X^1)_R \cong (X_R)^1$ and every i (V7.83; gap G.82 closed; Findings, item 131). Not formalized: the side remark of Appendix 7.6.3 identifying $H^{m,\delta}(\mathbb{R})$, defined here by lifting to $\mathbb{R} \times S^1$, with the 1-D weak-derivative space (the norm is the same).
- p. 355, after Theorem 7.6.9: the characterization of the structures of bX, $b\mathbf{X}$ for representatives whose V is not centred at x, as formerly in Part I (gap G.78, closed there since by V2.104, with image $\theta \cap V^x$); "Φ is a homeomorphism onto θ" holds as we read it only for V centred at x (V7.98). Theorem 7.6.9 (2)–(5) themselves, which depend on 2021 Theorem 2.7.13 (2), are verified (V7.99–V7.101) (Findings, items 140, 141, 145–147).

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9. Chapter 8 (HWZ, Part II, §§8.1–8.6 (2017 §§8.1–8.5)): verified, with Remark 8.1.10 (2017 Remark 8.1) for $T\varphi$ labelled by φ, Theorem 8.5.3 (5) (2017: 8.5) with s on both arrows, Proposition 8.5.7 (2017: 8.3) for nonempty D and the degeneracy remark after Definition 8.5.1 for tame X, Y

Chapter 8 (pp. 357–400 (2017 pp. 257–289); Part II has no 2014 preprint, so numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) treats bundles over ep-groupoids and covering functors. §8.1 equips the tangent $T(X, \mathbf{X}) = (TX, T\mathbf{X})$ with the structure maps (8.1)–(8.9) and proves that it is an ep-groupoid, tame if X is, with an sc-smooth projection functor to X^1 (Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8)); it describes the morphisms of $T\mathbf{X}$ by the pairs $(T\varphi, a)$ (Remark 8.1.10 (2017: 8.1)), proves that the tangent of an sc-smooth functor is one and that $T : \mathcal{EP} \to \mathcal{EP}$ is a functor (Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3)), defines natural transformations and natural equivalence (Definition 8.1.13 (2017: 8.1)), proves that tangents of natural transformations are natural transformations (Proposition 8.1.14 (2017: 8.1)), and that paracompactness of $|X|$ passes to $|TX|$ (Theorem 8.1.15 (2017: 8.4)). §8.2 defines sc-differential forms on ep-groupoids and the sc-de Rham complex (Definitions 8.2.1, 8.2.2 (2017: 8.2, 8.3)). §8.3 defines strong bundles over ep-groupoids (Definition 8.3.1 (2017: 8.4)) with their functorial picture (pp. 375–377 (2017 pp. 271–272)), strong bundle maps and strong bundle functors (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) and section functors (Definition 8.3.4 (2017: 8.7)). §8.5 (2017 §8.4) defines proper covering functors (Definition 8.5.1 (2017: 8.8)) with the degeneracy remark after it, proves that they compose (Proposition 8.5.2 (2017: 8.2)), describes their local structure (Theorem 8.5.3 (2017: 8.5), Remark 8.5.4 (2017: 8.2)), defines covering functors of finite groupoids and geometric lifts (Definition 8.5.5 (2017: 8.9)), constructs the proper covering functor of a geometric lift (Proposition 8.5.7 (2017: 8.3)), shows that every proper covering functor is locally of this form (Theorem 8.5.8 (2017: 8.6)), and defines proper strong bundle covering functors (Definition 8.5.9 (2017: 8.10)). Appendix 8.6 (2017 8.5) counts the fibres of a proper covering functor (§8.6.1 (2017 §8.5.1), Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11), Proposition 8.6.4 (2017: 8.4)) and describes strong bundle coverings fibrewise (§8.6.2 (2017 §8.5.2)).

What is verified. **Theorems 8.1.1–8.5.8 (2017: 8.1–8.6) hold**, with the readings recorded below, and so do Propositions 8.1.14, 8.5.2, 8.6.4 (2017: 8.1, 8.2, 8.4), Proposition 8.5.7 (2017: 8.3) for nonempty D, Lemmas 8.1.2–8.6.3 (2017: 8.1–8.11) (Lemma 8.1.2 (2017: 8.1) in the sheet form of its proof), Remark 8.1.10 (2017: 8.1) without its linearity claims, and Remark 8.5.4 (2017: 8.2). Every definition of the chapter except Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3) is formalized: Definitions 8.1.13 (2017: 8.1; V8.15), 8.3.1 (2017: 8.4; V8.20), 8.3.2 (2017: 8.5; V8.23), 8.3.3 (2017: 8.6; V8.24), 8.3.4 (2017: 8.7; V8.26), 8.5.1 (2017: 8.8; V8.27), 8.5.5 (2017: 8.9; V8.32), 8.5.9 (2017: 8.10; V8.37). Several places of the text needed more than a reading. (a) *Theorem 8.5.3 (5) (2017: 8.5).* As printed, the right vertical arrow of the square is t and its lower right corner $U(t(F(\psi_0)))$; with t the square says $F(s(\psi)) = t(F(\psi))$ on $\mathbf{U}(\psi_0)$, which does not hold as we read it, since $F(s(\psi)) = s(F(\psi))$ for a functor (checked in Lean on $\mathbb{Z}/2 \ltimes \mathbb{R}$, `Polyfold.CoveringExamples.theorem_8_5_3_five_as_printed`). The square is verified with s on both vertical arrows, together with the companion square with t on both (V8.30). (b) *D nonempty.* Definition 8.5.5 (2017: 8.9) allows $D = \emptyset$, and then the functor of Proposition 8.5.7 (2017: 8.3) is surjective on objects only if $U(x) = \emptyset$; Proposition 8.5.7 (2017: 8.3) is verified for nonempty D, which holds in Theorem 8.5.8 (2017: 8.6) (V8.35). (c) *The degeneracy remark after Definition 8.5.1 (2017: 8.8).* The equality $d_{\mathbf{X}\,{}_s\times_F Y} = \frac{1}{2} d_{\mathbf{X}\times Y}$ uses the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, which Part I has for tame M-polyfolds; it is verified for tame X, Y, the inequality in general (V8.28). (d) *Omitted steps.* The tangents of maps defined on open sets, which the proofs of §8.1 differentiate without a definition (V8.2), the tameness of Tr in Lemma 8.1.9 (2017: 8.8), asserted without argument (V8.11), the claim that $\simeq$ is an equivalence relation (V8.16), the ep-groupoid of the functorial picture, asserted without proof (V8.22), the equivalence of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6) (V8.25), the open embedding $G_x \backslash TU(x) \to |TX|$ in the proof of Theorem 8.1.15 (2017: 8.4) (V8.18), the "one verifies" of Proposition 8.5.7 (2017: 8.3) (V8.35) and of Theorem 8.5.8 (2017: 8.6) (V8.36) are proved. (e) *Other routes.* Theorem 8.1.15 (2017: 8.4) is proved by metrizability, not by the text's closed paracompact cover (V8.18); Theorem 8.5.3 (2017: 8.5) through Definition 8.5.1 (3) (2017: 8.8), not through sheets and natural representations in Y (V8.30); Lemma 8.1.2 (2017: 8.1) and condition (4) of Lemma 8.1.7 (2017: 8.6) in the sheet form of Chapter 7 (V8.4, V8.9). (f) *Readings.* Part I has strong bundle maps only between local models, and the global notion is a reading (V8.19); in Theorem 8.5.8 (2017: 8.6) the text's "geometric lift $A : K \to L$" is read as a geometric lift with its associated functor A (V8.36); in

Definition 8.5.9 (3) (2017: 8.10) the strong bundle structure of $\mathbf{V}\,{}_{s}{\times}_{A}\,W$, which the text does not give, is read as a pull-back (V8.37). Not verified: Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3), the linearity of T_xX and of $T\varphi$, the identification $T(\mathbf{X}\,{}_{s}{\times}_{t}\,\mathbf{X}) = T\mathbf{X}\,{}_{Ts}{\times}_{Tt}\,T\mathbf{X}$, and the M-polyfold structure transported along G in Remark 8.1.10 (2017: 8.1) (see the list at the end of this section). An addendum (2026-11-12; Findings, items 131–133) verifies Lemma 8.1.2 (2017: 8.1) as printed, with $T(\mathbf{X}\,{}_{s}{\times}_{t}\,\mathbf{X}) = T\mathbf{X}\,{}_{Ts}{\times}_{Tt}\,T\mathbf{X}$ and $m_{T\mathbf{X}} = Tm$ (V8.4, same; gap G.75 closed), the M-polyfold structure along G and the triples of Remark 8.1.10 (V8.12, which stays narrower for the bare-map reading), the sc-differential forms of Definitions 8.2.1, 8.2.2 and the Poincaré lemma, Proposition 8.2.3 (V8.53–V8.55, same; gaps G.74, G.87 closed), and Proposition 8.4.1 (3), with $|W[1]|$ the orbit space of $(W[1], \mathbf{X}\,{}_{s}{\times}_{P}\,W[1])$ (V8.47, restated; gap G.86 closed); with the additions of 2026-11-09 (the linearity of $T\varphi$, Propositions 8.1.16, 8.4.3 and Lemma 8.5.10, recorded in the rows), every gap of the chapter is closed.

The setting. Ep-groupoids, sc-smooth functors, natural representations, X^1 and paracompact orbit spaces are those of Chapter 7 (`Polyfold.EpGroupoid`, `Polyfold.EpFunctor`, `Polyfold.EpGroupoid.NaturalRep`, `Polyfold.EpGroupoid.liftOne`, `Polyfold.EpGroupoid.HasParacompactOrbitSpace`). TX and $T\mathbf{X}$ are Part I's tangent M-polyfolds (`Polyfold.MPolyfold.TangentSpace`, `Polyfold.MPolyfold.tangentPolyfold`), which for X in `Type u` live in `Type (max u 1)`; their fibres T_xX carry no linear structure (gap G.7), so no statement of this chapter about the linearity of a tangent map is formalized. Strong bundles over M-polyfolds and their pull-backs are Part I's (`Polyfold.StrongBundle`, `Polyfold.StrongBundle.pullback`). "$f : U \to V$ is an sc-diffeomorphism" between open sets is Part I's `Polyfold.MPolyfold.IsScDiffeoOn`, with an inverse. Part I results are cited by their 2017 numbers. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 8".

V8.1 (the tangent data (8.1)–(8.9); `Polyfold.EpGroupoid.tSrc`, `Polyfold.EpGroupoid.tTgt`, `Polyfold.EpGroupoid.tUnit`, `Polyfold.EpGroupoid.tInv`, `Polyfold.EpGroupoid.tBase`, `Polyfold.EpGroupoid.tComp`, `Polyfold.EpGroupoid.tangent_ext`, `Polyfold.EpGroupoid.exists_tSrc_eq`, `Polyfold.EpGroupoid.src_tBase_eq`, `Polyfold.EpGroupoid.tComp_spec`, `Polyfold.EpGroupoid.tBase_tComp`, `Polyfold.EpGroupoid.tSrc_tComp`, `Polyfold.EpGroupoid.tComp_eq`, `Polyfold.EpGroupoid.tUnit_eq`, `Polyfold.MPolyfold.tangentMap`, `Polyfold.MPolyfold.tangentProj`). The objects of $T(X, \mathbf{X})$ are TX, the morphisms $T\mathbf{X}$; $s_T = Ts$, $t_T = Tt$ (8.3); for $Ts(k) = Tt(h)$ the composite $k \circ h$ is the vector over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$ (8.5), (8.6); $u_T(a) = (1_x, Ts(1_x)^{-1}a)$ (8.7), (8.8); $\iota_T = T\iota$ (8.9). *Book:* §8.1, pp. 357–359 (2017 pp. 257–258). *Hidden hypotheses:* none. *Deviation (restated):* the composition is defined by choice and is total, with a junk value off composable pairs, as Chapter 7's composition; on composable pairs it is unique, since Ts is injective on every fibre $T_\varphi\mathbf{X}$ (`Polyfold.EpGroupoid.tangent_ext`); u_T is Tu, which is (8.8). TX and $T\mathbf{X}$ live in `Type (max u 1)`. (8.5) writes $T_{\varphi\circ\psi}\mathbf{X}$ for $T_{\psi\circ\varphi}\mathbf{X}$ (possible misprint) (Findings, items 1, 2).

V8.2 (tangents of maps sc-smooth on open sets and of local sc-diffeomorphisms; `Polyfold.MPolyfold.tangentPreimage`, `Polyfold.MPolyfold.isOpen_tangentPreimage`, `Polyfold.MPolyfold.tangentMapOn`, `Polyfold.MPolyfold.levelVal_tangentProj_tangentMapOn`,

`Polyfold.MPolyfold.tangentMapOn_eq_tangentMap`, `Polyfold.MPolyfold.tangentMapOn_comp`, `Polyfold.MPolyfold.tangentMap_tangentMapOn`, `Polyfold.MPolyfold.tangentMapOn_isScSmoothOn`, `Polyfold.MPolyfold.IsScDiffeoOn.tangent`, `Polyfold.MPolyfold.IsScDiffeoOn.tangentMap`, `Polyfold.MPolyfold.IsScDiffeoOn.bijOn_tangentFiber`, `Polyfold.MPolyfold.IsLocalScDiffeo.tangentMap`, `Polyfold.MPolyfold.IsLocalScDiffeo.surjective_tangentMap`, `Polyfold.MPolyfold.IsLocalScDiffeo.bijOn_tangentFiber`, `Polyfold.MPolyfold.isScDiffeoOn_tangentMap_subtypeVal`, `Polyfold.MPolyfold.isOpenEmbedding_tangentMap_subtypeVal`, `Polyfold.MPolyfold.nonempty_tangentSpace`). For open U, $TU = p^{-1}(U)$ is open in TX and is the tangent of the open sub-M-polyfold U; a map sc-smooth on U has a tangent on $p^{-1}(U)$, sc-smooth, over the map, with the chain rule, equal to Part I's tangent for a globally sc-smooth map; the tangent of an sc-diffeomorphism $U \to V$ is an sc-diffeomorphism $p^{-1}(U) \to p^{-1}(V)$; the tangent of a local sc-diffeomorphism is one, bijective on every fibre, and surjective if the map is. *Book:* used without statement in (8.5), (8.8) and in the proofs of Lemmas 8.1.2, 8.1.3, 8.1.5, 8.1.7 and 8.1.8 (2017: 8.1, 8.2, 8.4, 8.6 and 8.7) (pp. 358–364 (2017 pp. 258–263)). *Hidden hypotheses:* none. *Deviation (not in the book):* Part I defines the tangent only of globally sc-smooth maps; the tangent of a map defined on an open set, such as $(s|\mathbf{U}(\psi))^{-1}$, is built from the charts here (Findings, item 3).

V8.3 (Theorem 8.1.1 (2017: 8.1), the tangent ep-groupoid; `Polyfold.EpGroupoid.tangentGroupoidStr`, `Polyfold.EpGroupoid.tangentEpGroupoid`, `Polyfold.EpGroupoid.tangentProjFunctor`, `Polyfold.EpGroupoid.theorem_8_1_1`, `Polyfold.EpGroupoid.theorem_8_1_1_tame`). $T(X, \mathbf{X}) = (TX, T\mathbf{X})$ with the structure maps (8.3)–(8.9) is an ep-groupoid, tame if $(X, \mathbf{X})$ is, and $(p, \mathbf{p}) : T(X, \mathbf{X}) \to (X, \mathbf{X})^1$ is an sc-smooth functor. *Book:* Theorem 8.1.1 (2017: 8.1), p. 359 (2017 p. 258), proved on pp. 363–365 (2017 pp. 263–264) from Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8). *Hidden hypotheses:* none. *Deviation (restated):* `Polyfold.EpGroupoid.theorem_8_1_1` records that `Polyfold.EpGroupoid.tangentEpGroupoid` has the M-polyfolds TX, $T\mathbf{X}$ and the text's structure maps, with the base point and source of a composite; the conditions of Definition 7.1.3 (2017: 7.3) are Lemmas 8.1.3–8.1.8 (2017: 8.2–8.7) (V8.5–V8.10), the tameness sentence is `Polyfold.EpGroupoid.theorem_8_1_1_tame` (Lemma 8.1.9 (2017: 8.8), V8.11); $(X, \mathbf{X})^1$ is Chapter 7's `Polyfold.EpGroupoid.liftOne` (Findings, items 2, 9).

V8.4 (Lemma 8.1.2 (2017: 8.1), the composition as a tangent map; `Polyfold.EpGroupoid.lemma_8_1_2`, `Polyfold.EpGroupoid.tComp_eq_tangentMapOn`, `Polyfold.EpGroupoid.tangentMapOn_srcInv_tSrc`, `Polyfold.EpGroupoid.mem_tangentPreimage_tgt`, `Polyfold.EpGroupoid.isScSmoothCompSheet`, `Polyfold.EpGroupoid.compPolyfold`, `Polyfold.EpGroupoid.tangentCompPairs`, `Polyfold.EpGroupoid.range_tangentCompPairs`, `Polyfold.EpGroupoid.isScDiffeoOn_tangentCompPairs`,`Polyfold.EpGroupoid.lemma_8_1_2_tangentMul`,`Polyfold.EpGroupoid.lemma_8_1_2_book`, `Polyfold.EpGroupoid.lemma_8_1_2_compMul`, `Polyfold.MPolyfold.isLocalScDiffeo_of_comp_eq`). For a sheet $s : W \to V$ with inverse σ and composable k, h with k over W: $k = T\sigma(Tt(h))$, $k \circ h = Tg(h)$ for $g(\varphi') = m(\sigma(t\varphi'), \varphi')$, and $k \circ h$ lies over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$. *Book:* Lemma 8.1.2 (2017: 8.1) and its proof, 2017 p. 259 ($(\psi, k) \circ (\varphi, h) = Tm(\psi, \varphi)(k, h)$ on $T(\mathbf{X}\,{}_s\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\times_{Tt} T\mathbf{X}$). *Update (2026-11-12; gap G.75 closed):* as printed. $\mathbf{X}\,{}_s\times_t \mathbf{X}$ is a sub-M-polyfold of $\mathbf{X} \times \mathbf{X}$; $v \mapsto (T\pi_1 v, T\pi_2 v)$

is an sc-diffeomorphism from its tangent onto the sub-M-polyfold $T\mathbf{X}\,{}_{Ts}{\times}_{Tt}\,T\mathbf{X}$ of $T\mathbf{X} \times T\mathbf{X}$; $Tm(v) = T\pi_1(v) \circ T\pi_2(v)$, so $k \circ h = Tm(v)$ for the one v over (k, h), and $m_{T\mathbf{X}} = Tm$. *Hidden hypotheses:* none. *Deviation (same):* none; the sheet form above is the text's proof, $\sigma(t(\varphi'))$ being the text's $F(\varphi') = (s|\mathbf{U}(\psi))^{-1}(t(\varphi'))$. *Proof:* π_2 is a local sc-diffeomorphism, so a tangent vector of the fibred product is determined by its base point and $T\pi_2$, in place of differentiating $s(m(F(\varphi'), \varphi')) = s(\varphi')$ (Findings, item 131). The proof writes $s|\mathbf{U}(\varphi \circ \psi)$ for $s|\mathbf{U}(\psi \circ \varphi)$ (possible misprint) (Findings, items 5, 111).

V8.5 (Lemma 8.1.3 (2017: 8.2), source and target of a composite; `Polyfold.EpGroupoid.lemma_8_1_3`, `Polyfold.EpGroupoid.tTgt_tangentMapOn`). $s_T(\Psi \circ \Phi) = s_T(\Phi)$ and $t_T(\Psi \circ \Phi) = t_T(\Psi)$. *Book:* Lemma 8.1.3 (2017: 8.2) and its proof, pp. 359–361 (2017 pp. 259–260). *Hidden hypotheses:* none. *Deviation:* none; the target is computed as in the text, with the tangent of $t \circ \sigma \circ t$ on $t^{-1}(V)$. The proof has "$==$", "$: \mathbf{U}(\psi \circ \varphi) \to U(x)$" without s, and "$Tt)\varphi)(h)$" (possible misprints) (Findings, items 4, 6).

V8.6 (Lemma 8.1.4 (2017: 8.3), associativity; `Polyfold.EpGroupoid.lemma_8_1_4`). The composition of $T\mathbf{X}$ is associative. *Book:* Lemma 8.1.4 (2017: 8.3), pp. 361 ff. (2017 pp. 260–261). *Hidden hypotheses:* none. *Deviation:* none; by fibre injectivity of Ts (Findings, item 4).

V8.7 (Lemma 8.1.5 (2017: 8.4), units; `Polyfold.EpGroupoid.tSrc_tUnit`, `Polyfold.EpGroupoid.tTgt_tUnit`, `Polyfold.EpGroupoid.tComp_tUnit`, `Polyfold.EpGroupoid.tUnit_tComp`, `Polyfold.EpGroupoid.lemma_8_1_5`). $1_q = (1_x, Ts(1_x)^{-1}q)$ is a morphism $q \to q$ and a two-sided unit. *Book:* Lemma 8.1.5 (2017: 8.4), p. 361 (2017 p. 261). *Hidden hypotheses:* none. *Deviation:* none; $Tt(1_q) = q$ by the chain rule for $t \circ u = \mathrm{id}$ instead of $t \circ (s|\mathbf{U}(1_x))^{-1} = \mathrm{id}$ near x (Findings, items 4, 6).

V8.8 (Lemma 8.1.6 (2017: 8.5), inverses; `Polyfold.EpGroupoid.tSrc_tInv`, `Polyfold.EpGroupoid.tTgt_tInv`, `Polyfold.EpGroupoid.tComp_tInv`, `Polyfold.EpGroupoid.tInv_tComp`, `Polyfold.EpGroupoid.lemma_8_1_6`). $\iota_T(\varphi, k) = (\varphi^{-1}, T\iota(\varphi)k)$ is a two-sided inverse. *Book:* Lemma 8.1.6 (2017: 8.5), pp. 362 ff. (2017 pp. 261–262). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 4).

V8.9 (Lemma 8.1.7 (2017: 8.6), the structure maps; `Polyfold.EpGroupoid.isLocalScDiffeo_tSrc`, `Polyfold.EpGroupoid.isLocalScDiffeo_tTgt`, `Polyfold.EpGroupoid.surjective_tSrc`, `Polyfold.EpGroupoid.surjective_tTgt`, `Polyfold.EpGroupoid.isScSmoothMap_tUnit`, `Polyfold.EpGroupoid.isScSmoothMap_tInv`, `Polyfold.EpGroupoid.isScSmoothCompSheet_tangent`, `Polyfold.EpGroupoid.isScSmoothComp_tangent`, `Polyfold.EpGroupoid.lemma_8_1_7`, `Polyfold.isScSmoothComp_iff_sheet`). s_T, t_T are surjective local sc-diffeomorphisms, u_T and ι_T are sc-smooth, and the multiplication is sc-smooth on $T\mathbf{X}\,{}_{s_T}{\times}_{t_T}\,T\mathbf{X}$. *Book:* Lemma 8.1.7 (2017: 8.6), p. 362 (2017 p. 262). *Hidden hypotheses:* none. *Deviation (restated):* "by construction" is V8.2; the sc-smoothness of the multiplication is proved in Chapter 7's sheet form and from it on the fibred product, not through "$m_{T\mathbf{X}}$ is precisely Tm" (gap G.75) (Findings, item 7).

V8.10 (Lemma 8.1.8 (2017: 8.7), properness; `Polyfold.EpGroupoid.lemma_8_1_8`). Every $a \in TX$ has an open $V' \ni a$ with $t_T : s_T^{-1}(\operatorname{cl} V') \to TX$ proper, $V' = p^{-1}(V)$ for V as in Definition

7.1.3 (3) (2017: 7.3) for X. *Book:* Lemma 8.1.8 (2017: 8.7), pp. 363–364 (2017 pp. 262–263). *Hidden hypotheses:* none. *Deviation:* none; the step “$\varphi_k \to \varphi$ in $\mathbf{X}^1$” is not needed. The proof writes $t(\varphi_k, h_k) = Ts(\varphi_k)(h_k)$ for $Tt(\varphi_k)(h_k)$ (possible misprint) (Findings, item 7).

V8.11 (Lemma 8.1.9 (2017: 8.8) and the tameness of Tr; `Polyfold.ScIso.tangentPres`, `Polyfold.tangentSet_eq_tangentPres`, `Polyfold.degIndex_tangentSet`, `Polyfold.mem_minimalSubspaceWith_tangentPres`, `Polyfold.ScRetraction.scDeriv_tangentRetraction_fst`, `Polyfold.ScRetraction.scDeriv_tangentRetraction_inr`, `Polyfold.ScRetraction.IsTameWith.tangent`, `Polyfold.ScRetraction.IsTame.tangentRetraction`, `Polyfold.IsTameRetract.tangent`, `Polyfold.MPolyfold.IsTame.tangent`, `Polyfold.EpGroupoid.lemma_8_1_9`, `Polyfold.EpGroupoid.isTame_tangentEpGroupoid_trivial_rn`). If $(X, \mathbf{X})$ is tame, so is $(TX, T\mathbf{X})$; the tangent Tr of a tame retraction is tame, and the tangent of a tame M-polyfold is tame. *Book:* Lemma 8.1.9 (2017: 8.8), p. 364 (2017 p. 263). *Hidden hypotheses:* none. *Deviation:* none. “Tr is a tame retraction”, asserted without argument and not in Part I, is proved from Part I’s Definition 2.5.2 (2017: 2.17) for one presentation of TC with the complement $A' = (A \cap E_1) \oplus A$, without the second derivative of *r*, and passed to all presentations by Part I’s Lemma 1.3.4 (2017: 1.1) (`Polyfold.ScRetraction.IsTameWith.isTame`) (Findings, items 8, 104).

V8.12 (Remark 8.1.10 (2017: 8.1), the morphisms of $T\mathbf{X}$ as pairs $(T\varphi, a)$; `Polyfold.EpGroupoid.tangentMor`, `Polyfold.EpGroupoid.tangentMor_tSrc`, `Polyfold.EpGroupoid.tangentMor_comp`, `Polyfold.EpGroupoid.remarkG`, `Polyfold.EpGroupoid.remark_8_1_10`, `Polyfold.EpGroupoid.tangentPairsPolyfold`, `Polyfold.EpGroupoid.remarkGInto`, `Polyfold.EpGroupoid.remark_8_1_10_scDiffeo`, `Polyfold.EpGroupoid.remarkTriple`, `Polyfold.EpGroupoid.remark_8_1_10_triples`, `Polyfold.EpGroupoid.Examples.remark_8_1_10_not_injective`). $t_T \circ (s_T|T_\varphi \mathbf{X})^{-1} = T\varphi$; $T(\psi \circ \varphi) = T\psi \circ T\varphi$ on $T_{s(\varphi)}X$; $G(\varphi, h) = (T\varphi, Ts(h))$ is a bijection onto the pairs with $\varphi \in \mathbf{X}_1$, $a \in T_{s(\varphi)}X$, compatible with the compositions. *Book:* Remark 8.1.10 (2017: 8.1), pp. 364 ff. (2017 pp. 264–265). *Hidden hypotheses:* none. *Deviation (narrower):* $T\varphi$ is Chapter 7’s `Polyfold.EpGroupoid.tangentMor`; the first entry of G is read as $T\varphi$ labelled by φ, since as a bare map G is not injective on $\mathbb{Z}/3 \ltimes \mathbb{R}$ with the trivial action (`Polyfold.EpGroupoid.Examples.remark_8_1_10_not_injective`); the bijectivity of G as printed does not hold as we read it, so the row stays narrower. *Update (2026-11-09):* $T\varphi$ is a bounded linear isomorphism, equal to $T\hat{\varphi}(x)$ (gaps G.7, G.70 closed; V7.5). *Update (2026-11-12):* the pairs form the M-polyfold $\mathbf{X}^1 {}_s\times_p TX$, a sub-M-polyfold of $\mathbf{X}^1 \times TX$, and G is an sc-diffeomorphism $T\mathbf{X} \to \mathbf{X}^1 {}_s\times_p TX$; $(\varphi, h) \mapsto (Ts(h), \varphi, Tt(h))$ is injective onto the triples (a, φ, b) with $b = T\varphi(a)$, compatible with the compositions (Findings, item 131). The remark writes “$T\varphi : T_xX \to T_yY$” for T_yX, and calls $T\varphi$ the linearization “at x” and later “at φ” (possible misprints) (Findings, items 10, 112, 113).

V8.13 (Theorem 8.1.11 (2017: 8.2), the tangent of an sc-smooth functor; `Polyfold.EpFunctor.tangent`, `Polyfold.EpFunctor.theorem_8_1_11`). $(T\Phi, T\Phi)$ is an sc-smooth functor $T(X, \mathbf{X}) \to T(Y, \mathbf{Y})$. *Book:* Theorem 8.1.11 (2017: 8.2) and its proof, pp. 365–367 (2017 pp. 265–266). *Hidden hypotheses:* none. *Deviation:* none; compatibility with sources, targets, units and the composition is stated; the units, not checked in the text, are proved (Findings, items 4, 11).

V8.14 (Theorem 8.1.12 (2017: 8.3), the functor $T : \mathcal{EP} \to \mathcal{EP}$; `Polyfold.EP.tangent`, `Polyfold.`

`tangentFunctor`, `Polyfold.theorem_8_1_12`, `Polyfold.theorem_8_1_12_tame`). T is a functor $\mathcal{EP} \to \mathcal{EP}$, $X \mapsto TX$, $\Phi \mapsto T\Phi$, and TX is tame if X is. *Book:* Theorem 8.1.12 (2017: 8.3), p. 366 (2017 p. 266). *Hidden hypotheses:* none. *Deviation (restated):* a functor `EP.{max u 1}`   `EP.{max u 1}`, since TX lives in `Type  (max u 1)`; the tameness sentence is `Polyfold.theorem_8_1_12_tame` (Findings, items 2, 11).

V8.15 (Definition 8.1.13 (2017: 8.1), natural transformations and natural equivalence; `Polyfold.EpNatTrans`, `Polyfold.EpFunctor.NaturallyEquivalent`). A natural transformation $\tau : F \to G$ is an sc-smooth $\tau : X \to \mathbf{Y}$ with $\tau(x) : F(x) \to G(x)$ and $\tau(x') \circ F(h) = G(h) \circ \tau(x)$; $F \simeq G$ if one exists. *Book:* Definition 8.1.13 (2017: 8.1), p. 367 (2017 p. 266). *Hidden hypotheses:* none. *Deviation:* none (Findings, item 12).

V8.16 (the equivalence relation $\simeq$; `Polyfold.EpNatTrans.refl`, `Polyfold.EpNatTrans.symm`, `Polyfold.EpNatTrans.trans`, `Polyfold.EpNatTrans.isScSmoothMap_comp_app`, `Polyfold.EpFunctor.NaturallyEquivalent.refl`, `Polyfold.EpFunctor.NaturallyEquivalent.symm`, `Polyfold.EpFunctor.NaturallyEquivalent.trans`, `Polyfold.EpFunctor.naturallyEquivalent_equivalence`, `Polyfold.EpNatTrans.whiskerLeft`, `Polyfold.EpNatTrans.whiskerRight`, `Polyfold.EpFunctor.NaturallyEquivalent.comp_left`, `Polyfold.EpFunctor.NaturallyEquivalent.comp_right`). $\simeq$ is an equivalence relation on the sc-smooth functors $X \to Y$. *Book:* 2017 p. 267, without proof. *Hidden hypotheses:* none. *Deviation:* none; the sc-smoothness of $x \mapsto \tau'(x) \circ \tau(x)$ comes from Chapter 7's sheet form of Definition 7.1.3 (4) (2017: 7.3); whiskering, not in the text, is added (Findings, item 12).

V8.17 (Proposition 8.1.14 (2017: 8.1), the tangent of a natural transformation; `Polyfold.EpNatTrans.tangent`, `Polyfold.proposition_8_1_14`). If $\tau : \Phi \to \Psi$ is a natural transformation, $T\tau : TX \to T\mathbf{Y}$ is a natural transformation $T\Phi \to T\Psi$; hence $\Phi \simeq \Psi$ implies $T\Phi \simeq T\Psi$. *Book:* Proposition 8.1.14 (2017: 8.1) and its proof, pp. 367–372 (2017 pp. 267–268). *Hidden hypotheses:* none. *Deviation:* none; naturality by fibre injectivity of Ts instead of differentiating the naturality square. The proof has "$T\tau(x_0(h))$" (possible misprint), and labels the lower arrow of the last diagram $T\tau(x_0)h$ for $T\tau(x_0')h'$ (possible misprint or extraction artefact) (Findings, items 4, 13).

V8.18 (Theorem 8.1.15 (2017: 8.4), the orbit space of TX; `Polyfold.EpGroupoid.tangentOrbitProj`, `Polyfold.EpGroupoid.continuous_tangentOrbitProj`, `Polyfold.EpGroupoid.preimage_tangentOrbitProj_image`, `Polyfold.EpGroupoid.NaturalRep.tangentU`, `Polyfold.EpGroupoid.NaturalRep.tΦ`, `Polyfold.EpGroupoid.NaturalRep.isScDiffeoOn_tΦ`, `Polyfold.EpGroupoid.NaturalRep.tangentMulAction`, `Polyfold.EpGroupoid.NaturalRep.tΓ`, `Polyfold.EpGroupoid.NaturalRep.tangent_ _eq_iff`, `Polyfold.EpGroupoid.NaturalRep.isOpenEmbedding_tangentOrbitQuotientMap`, `Polyfold.EpGroupoid.NaturalRep.range_tangentOrbitQuotientMap`, `Polyfold.EpGroupoid.NaturalRep.image_tangentU_eq_preimage`, `Polyfold.EpGroupoid.NaturalRep.metrizableSpace_image_tangentU`, `Polyfold.EpGroupoid.NaturalRep.metrizableSpace_image_liftU`, `Polyfold.EpGroupoid.metrizableSpace_tangent_orbitSpace`, `Polyfold.EpGroupoid.theorem_8_1_15`, `Polyfold.EpGroupoid.hasParacompactOrbitSpace_tangent_trivial`). If $|X|$ is paracompact, so is $|TX|$. *Book:* Theorem 8.1.15 (2017: 8.4) and its proof, 2017 p. 268. *Hidden hypotheses:* none. *Deviation (stronger):* $|TX|$ is metrizable. The proof differs from the text's: Smirnov's theorem pulled back along $|p| : |TX| \to |X^1|$ (Part I's `Polyfold.metrizableSpace_of_pullback`), as for Proposition 7.5.3 (2017: 7.12), with the metrizable $|TU(x)| = |p|^{-1}(|U(x)_1|)$; the closed paracompact $\widetilde{A}_z$, the regularity of $|TX|$ and Part I's Proposition 2.7.4 (2017: 2.17) are not used. The open embedding $G_x \backslash TU(x) \to |TX|$, left implicit, is proved; the local bundle model $TO \to O^1$ is not used, and the linearity of the action of G_x on $TU(x)$ is not formalized (gap G.7). "parcompact" (possible misprint) (Findings, items 20, 21, 104).

V8.19 (strong bundle maps between strong bundles over M-polyfolds, and the pull-back along s; `Polyfold.StrongBundle.pullback`, `Polyfold.pullbackSpace`, `Polyfold.StrongBundle.paracompactSpace_pullback`, `Polyfold.StrongBundle.pullback_atlas`, `Polyfold.PullbackChartData.exists_mem_atlas`, `Polyfold.StrongBundle.isScSmoothMap_pullbackProj`, `Polyfold.StrongBundle.isScSmoothMap_pullbackProj'`, `Polyfold.StrongBundle.isScSmoothOn_to_pullback_iff`, `Polyfold.StrongBundle.isScSmoothMap_pullbackPair`, `Polyfold.pullbackPair`, `Polyfold.StrongBundle.isScDiffeoOn_pullbackProj'`, `Polyfold.StrongBundle.isLocalScDiffeo_pullbackProj'`, `Polyfold.StrongBundle.IsStrongBundleMap`, `Polyfold.fiberRestrict`, `Polyfold.StrongBundle.IsStrongBundleMap.fiberMap`, `Polyfold.StrongBundle.IsStrongBundleMap.isScSmoothMap_base`, `Polyfold.StrongBundle.IsStrongBundleMap.comp`, `Polyfold.StrongBundle.IsStrongBundleMap.mapsTo_doubleLevel`, `Polyfold.StrongBundle.IsStrongBundleIso`, `Polyfold.StrongBundle.isStrongBundleMap_pullbackProj'`, `Polyfold.MPolyfold.isScSmoothMap_of_forall_charts`). A strong bundle map $\Phi : Y \to Y'$ covering f covers f, is linear on the fibres, sc-smooth on $Y[0]$, maps $Y_{0,1}$ into $Y'_{0,1}$ and is sc-smooth on $Y[1]$; it then preserves the double filtration, has bounded fibre maps and an sc-smooth base map. The pull-back $\mathbf{X}\,{}_s{\times}_P\, W$ is Part I's, with its universal property on [0] and [1], and over a sheet of s the projection to W is an sc-diffeomorphism. *Book:* used in Definitions 8.3.1–8.3.3, 8.5.9 (2017: 8.4–8.6, 8.10) (pp. 373–396 (2017 pp. 270–285)); Part I's Definition 2.6.1 (2017: 2.22) and Proposition 2.6.8 (2017: 2.16). *Hidden hypotheses:* none. *Deviation (not in the book):* Part I defines strong bundle maps only between local models; the global notion is a reading (Findings, items 30, 31).

V8.20 (Definition 8.3.1 (2017: 8.4), strong bundles over ep-groupoids; `Polyfold.EpStrongBundle`, `Polyfold.EpStrongBundle.P_ `, `Polyfold.EpGroupoid.unitPair`, `Polyfold.EpStrongBundle.continuous_ `, `Polyfold.EpStrongBundle.trivial`). A strong bundle (P, μ) over X: a strong bundle $P : W \to X$ and a strong bundle map $\mu : \mathbf{X}\,{}_s{\times}_P\, W \to W$ covering t, with (1) μ a surjective local sc-diffeomorphism, linear on the fibres, (2) $\mu(1_x, w) = w$, (3) $\mu(\psi \circ \gamma, w) = \mu(\psi, \mu(\gamma, w))$. *Book:* Definition 8.3.1 (2017: 8.4), p. 375 (2017 p. 271). *Hidden hypotheses:* none. *Deviation (restated):* on Part I's pull-back strong bundle, with the strong bundle maps of V8.19; the surjectivity in (1) is kept although it follows from (2) (Findings, items 32, 102).

V8.21 (the fibre maps $\mu(\psi, \cdot)$; `Polyfold.EpStrongBundle.fiberMor`, `Polyfold.EpStrongBundle.fiberMor_apply`, `Polyfold.EpStrongBundle. _add`, `Polyfold.EpStrongBundle. _smul`, `Polyfold.EpStrongBundle. _inv_ `, `Polyfold.EpStrongBundle. _ _inv`, `Polyfold.`

`EpStrongBundle.fiberMorEquiv`**).** For $\psi : x \to y$, $\mu(\psi, \cdot) : W_x \to W_y$ is a linear isomorphism. *Book:* 2017 p. 271, after Definition 8.3.1 (2017: 8.4). *Hidden hypotheses:* none. *Deviation (stronger):* a topological linear isomorphism, with inverse $\mu(\psi^{-1}, \cdot)$ (Findings, item 32).

V8.22 (the functorial picture, the ep-groupoid $(W, \mathbf{W})$; `Polyfold.EpStrongBundle.morInv`, `Polyfold.EpStrongBundle.morComp`, `Polyfold.EpStrongBundle.morComp_val`, `Polyfold.EpStrongBundle._morComp`, `Polyfold.EpStrongBundle.toGroupoidStr`, `Polyfold.EpStrongBundle.exists_isProperMap_tgt`, `Polyfold.EpStrongBundle.isScSmoothCompSheet_toGroupoidStr`, `Polyfold.EpStrongBundle.toEpGroupoid`, `Polyfold.EpStrongBundle.projFunctor`, `Polyfold.EpStrongBundle.isStrongBundleMap_src`, `Polyfold.EpStrongBundle.isStrongBundleMap_tgt`, `Polyfold.EpStrongBundle.exists_isomorphic_of_isomorphic`, `Polyfold.EpStrongBundle.surjective_orbitMap_projFunctor`, `Polyfold.EpStrongBundle.continuous_orbitMap_projFunctor`). $(W, \mathbf{W})$ with $\mathbf{W} = \mathbf{X}\,{}_s{\times}_P\, W$, $s_{\mathbf{W}}(\psi, w) = w$, $t_{\mathbf{W}} = \mu$, the units $(1_x, w)$, the inverses $(\psi^{-1}, \mu(\psi, w))$ and the composition $(\psi, w') \circ (\gamma, w) = (\psi \circ \gamma, w)$ is an ep-groupoid; $s_{\mathbf{W}}$, $t_{\mathbf{W}}$ are strong bundle maps covering s, t; (P, π_1) is an sc-smooth functor; $|P| : |W| \to |X|$ is continuous and surjective. *Book:* pp. 375–377 (2017 pp. 271–272). *Hidden hypotheses:* none. *Deviation (restated):* "we can define the ep-groupoid", stated without proof, is proved (properness by sequences, condition (4) in sheet form); from here on W lives in the universe of the objects and morphisms of X (Findings, item 33).

V8.23 (Definition 8.3.2 (2017: 8.5), strong bundle maps over ep-groupoids; `Polyfold.EpStrongBundle.IsMap`, `Polyfold.EpStrongBundle.isMap_id`). A strong bundle map $\Phi : W \to W'$ covering the object map of an sc-smooth functor ϕ, with $\Phi(\mu(\gamma, w)) = \mu'(\phi(\gamma), \Phi(w))$. *Book:* Definition 8.3.2 (2017: 8.5) and the paragraph after it, 2017 p. 273. *Hidden hypotheses:* none. *Deviation:* none (Findings, item 34).

V8.24 (Definition 8.3.3 (2017: 8.6), strong bundle functors; `Polyfold.EpStrongBundle.IsStrongBundleFunctor`, `Polyfold.EpStrongBundle.IsStrongBundleFunctor.projFunctor_comp`, `Polyfold.EpStrongBundle.IsStrongBundleFunctor.mor_val`, `Polyfold.EpStrongBundle.isStrongBundleFunctor_id`). A functor $(W, \mathbf{W}) \to (W', \mathbf{W}')$, linear on the fibres, covering an sc-smooth functor ϕ and inducing strong bundle maps on objects and morphisms, with $P' \circ \Phi = \phi \circ P$. *Book:* Definition 8.3.3 (2017: 8.6), p. 378 (2017 p. 274). *Hidden hypotheses:* none. *Deviation (restated):* "linear on the fibres", "covers ϕ" and "induces strong bundle maps" are read together as two conditions `IsStrongBundleMap`, on objects (for P, P') and on morphisms (for the pull-backs $\mathbf{W} \to \mathbf{X}$, $\mathbf{W}' \to \mathbf{X}'$) (Findings, item 34).

V8.25 (the equivalence of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6); `Polyfold.EpStrongBundle.IsStrongBundleFunctor.isMap`, `Polyfold.EpStrongBundle.liftMor`, `Polyfold.EpStrongBundle.IsMap.liftFunctor`, `Polyfold.EpStrongBundle.IsMap.isStrongBundleFunctor_liftFunctor`, `Polyfold.EpStrongBundle.IsStrongBundleFunctor.eq_liftFunctor`, `Polyfold.EpStrongBundle.isMap_iff_strongBundleFunctor`, `Polyfold.StrongBundle.pullbackProj'One`, `Polyfold.StrongBundle.isScSmoothMap_to_pullback_one`, `Polyfold.isLinearMap_fiberRestrict_pullbackMap`). $\Phi : W \to W'$ is a strong bundle map of Definition 8.3.2 (2017: 8.5) iff it is the object map of a strong bundle functor of Definition 8.3.3 (2017:

8.6), and the functor, $\Phi(\gamma, w) = (\phi(\gamma), \Phi(w))$, is unique. *Book:* pp. 377–379 (2017 pp. 273–274) ("these are identical notions"). *Hidden hypotheses:* none. *Deviation:* none; the $[1]$-smoothness of the lifted morphism map needs the universal property of the pull-back on level $[1]$, proved here (Findings, item 34).

V8.26 (Definition 8.3.4 (2017: 8.7), section functors; `Polyfold.EpStrongBundle.IsFunctorialSection`, `Polyfold.EpStrongBundle.IsScSmoothSectionFunctor`, `Polyfold.EpStrongBundle.IsScPlusSectionFunctor`, `Polyfold.EpStrongBundle.IsScFredholmSectionFunctor`, `Polyfold.EpStrongBundle.sectionFunctors`, `Polyfold.EpStrongBundle.plusSectionFunctors`, `Polyfold.EpStrongBundle.fredholmSectionFunctors`, `Polyfold.EpStrongBundle.fiberZero_mem_sectionFunctors`, `Polyfold.EpStrongBundle.sectionAdd_mem_sectionFunctors`, `Polyfold.EpStrongBundle.sectionSmul_mem_sectionFunctors`, `Polyfold.EpStrongBundle.plusSectionFunctors_subset`, `Polyfold.EpStrongBundle.fiberZero_mem_plusSectionFunctors`, `Polyfold.EpStrongBundle.sectionAdd_mem_plusSectionFunctors`, `Polyfold.EpStrongBundle.sectionSmul_mem_plusSectionFunctors`, `Polyfold.EpStrongBundle.fredholmSectionFunctors_subset`, `Polyfold.StrongBundle.IsScSmoothSection.sectionAdd_of_isScSmoothSection`, `Polyfold.StrongBundle.IsScSmoothSection.sectionSmul_const`, `Polyfold.StrongBundle.isScSmoothSection_fiberZero`). (1) An sc-smooth section functor is an sc-smooth section f with $f(y) = \mu(\varphi, f(x))$ for $\varphi : x \to y$; (2) an sc$^+$-section functor takes values in $W_{0,1}$ and is sc-smooth into $W[1]$; (3) an sc-Fredholm section functor is an sc-Fredholm section; $\Gamma(P, \mu)$ is a vector space and $\Gamma^+(P, \mu) \subseteq \Gamma(P, \mu)$ a subspace. *Book:* Definition 8.3.4 (2017: 8.7), pp. 379 ff. (2017 pp. 274–275). *Hidden hypotheses:* none. *Deviation (restated):* "vector space" is closure under the zero section, sums and constant multiples, with Part I's fibrewise operations; no `Module` structure is built; (3) uses Part I's Definition 3.1.16 (2017: 3.8) in its germ reading. The sum of two sc-smooth sections and constant multiples, which Part I does not state, are proved here in the charts (Findings, items 35, 103).

V8.27 (Definition 8.5.1 (2017: 8.8), proper covering functors; `Polyfold.EpFunctor.IsProperCovering`, `Polyfold.EpFunctor.fibredMap`, `Polyfold.MPolyfold.isSubMPolyfold_fibredProduct`, `Polyfold.EpFunctor.isProperCovering_id`). (1) F is a surjective local sc-diffeomorphism on objects; (2) the fibres $F^{-1}(x)$ are finite and evenly covered; (3) $\varphi \mapsto (F(\varphi), s(\varphi))$ is an sc-diffeomorphism $\mathbf{Y} \to \mathbf{X}\,{}_s\!\times_F Y$. *Book:* Definition 8.5.1 (2017: 8.8), p. 383 (2017 p. 275). *Hidden hypotheses:* none. *Deviation (restated):* in (2) $U(x)$ is open and the $U(y)$ carry local inverses $\sigma(y)$; (3) on Part I's fibred-product sub-M-polyfold of $\mathbf{X} \times Y$, as in Definition 7.1.3 (4) (2017: 7.3) (Findings, items 50, 102).

V8.28 (the degeneracy remark after Definition 8.5.1 (2017: 8.8); `Polyfold.EpFunctor.IsProperCovering.degIndex_fibredMap`, `Polyfold.EpFunctor.IsProperCovering.degIndex_mor`, `Polyfold.EpFunctor.IsProperCovering.degIndex_prod_fibredMap_le`, `Polyfold.EpFunctor.IsProperCovering.degIndex_prod_fibredMap`, `Polyfold.MPolyfold.degIndex_prod_le`, `Polyfold.MPolyfold.isTame_prod`, `Polyfold.proposition_2_4_4`). $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}\,{}_s\!\times_F Y}(F(\varphi), s(\varphi))$, $d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi)) = 2\,d_{\mathbf{Y}}(\varphi)$ and $d_{\mathbf{X}\,{}_s\!\times_F Y} = \frac{1}{2} d_{\mathbf{X}\times Y}$ at

$(F(\varphi), s(\varphi))$. *Book:* pp. 379–385 (2017 pp. 275–276). *Hidden hypotheses:* X and Y tame, for the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, which Part I proves for tame M-polyfolds (Part I's Proposition 2.5.20 (2017: 2.15) states its formula $d = \frac{1}{2} d_{X\times Y}$ for tame X, Y). *Deviation (narrower):* the first equality and $d_{\mathbf{X}}(F(\varphi)) = d_{\mathbf{Y}}(\varphi)$ hold in general; the other two for tame X, Y; in general $d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi)) \leq 2\, d_{\mathbf{Y}}(\varphi)$ (Findings, item 51).

V8.29 (Proposition 8.5.2 (2017: 8.2), composites of proper covering functors; `Polyfold.EpFunctor.IsProperCovering.evenlyCovered_comp`, `Polyfold.EpFunctor.IsProperCovering.isScDiffeo_fibredMap_comp`, `Polyfold.EpFunctor.IsProperCovering.proposition_8_5_2`). If $F : Y \to X$ and $H : Z \to Y$ are proper covering functors, so is $F \circ H$. *Book:* Proposition 8.5.2 (2017: 8.2) and its proof, pp. 384–386 (2017 pp. 276–277). *Hidden hypotheses:* none. *Deviation:* none; (2), "follows easily", is proved by shrinking $U(x)$; (3) through $\tilde{K}$, $\tilde{L}$ as in the text. "fo" for "for" (possible misprint) (Findings, item 52).

V8.30 (Theorem 8.5.3 (2017: 8.5), Proper Covering I; `Polyfold.EpFunctor.ProperCoveringChart`, `Polyfold.EpFunctor.ProperCoveringLocal.Data`, `Polyfold.EpFunctor.ProperCoveringLocal.exists_data`, `Polyfold.EpFunctor.ProperCoveringLocal.Data.toChart`, `Polyfold.EpFunctor.ProperCoveringLocal.isScSmoothOn_of_section`, `Polyfold.EpFunctor.theorem_8_5_3`, `Polyfold.EpFunctor.theorem_8_5_3_of_naturalRep`). For a proper covering functor $F : Y \to X$ and $x \in X$, a connected, sufficiently small $U(x)$ with the natural G_x-action carries sets $U(y)$, $y \in F^{-1}(x)$, and $\mathbf{U}(\psi_0)$, ψ_0 a morphism between points of $F^{-1}(x)$, with (0)–(5). *Book:* Theorem 8.5.3 (2017: 8.5), p. 385 (2017 p. 277), proved on pp. 386–388 (2017 pp. 278–279). *Hidden hypotheses:* none. *Deviation (stronger):* "sufficiently small" is proved in two forms, a chart with $U(x)$ inside every open $V \ni x$, and one open $W \ni x$ that works for every natural representation on a connected $U(x) \subseteq W$; $\mathbf{U}(\varphi)$ for $\varphi \in \mathbf{U}(x)$ is the connected component of $\mathbf{U}(x)$ at φ. (5) as printed, with t on the right and $U(t(F(\psi_0)))$, does not hold as we read it (possible misprint; counterexample `Polyfold.CoveringExamples.theorem_8_5_3_five_as_printed`, the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ at $\psi_0 = (-1, 0)$, for every chart, which exists by `Polyfold.CoveringExamples.theorem_8_5_3_five_as_printed_nonvacuous`); it is formalized with s on both vertical arrows (`comm_src`), with the companion square with t (`comm_tgt`). The proof differs from the text's: Definition 8.5.1 (3) (2017: 8.8) gives the sheets $\mathbf{U}(\psi_0)$ as images of continuous sections of s; the finiteness of $F^{-1}(x)$ and Definition 8.5.1 (1) (2017: 8.8) are not needed. In the proof "$g \in G_y$", "G_y-invariant" and "$F(z') = s(F(\varphi))$" for G_x, G_x-invariant and $t(F(\varphi))$, "admitting admitting", and "morphismsm" in (2) (possible misprints) (Findings, items 53, 54, 55).

V8.31 (Remark 8.5.4 (2017: 8.2); `Polyfold.EpFunctor.remark_8_5_4_a`, `Polyfold.EpFunctor.isotropyMor`, `Polyfold.EpFunctor.remark_8_5_4_b`, `Polyfold.EpFunctor.ProperCoveringChart.connectedComponentIn_U`, `Polyfold.EpFunctor.ProperCoveringChart.connectedComponentIn_sheet`). (a) A morphism between points of $F^{-1}(x)$ is mapped into G_x; (b) everything in Theorem 8.5.3 (2017: 8.5) is determined by $U(x)$. *Book:* Remark 8.5.4 (2017: 8.2), pp. 386–387 (2017 pp. 277–278). *Hidden hypotheses:* none. *Deviation (restated):* (b) as: two charts with the same $U(x)$ have the same Φ and Γ on $U(x)$, the same

$U(y)$ and the same $\mathbf{U}(\psi_0)$, and these are connected components; its remarks on the naturality of Theorem 7.1.13's (2017: 7.1) construction have no further formal content (Findings, item 56).

V8.32 (the category Diff_{sc} **and Definition 8.5.5 (2017: 8.9), covering functors and geometric lifts;** **`Polyfold.MPolyfoldCat`, `Polyfold.ScDiffeo`, `Polyfold.ScDiffeo.id`, `Polyfold.ScDiffeo.comp`, `Polyfold.ScDiffeo.ext`, `Polyfold.coveringMap`, `Polyfold.IsCoveringFunctor`, `Polyfold.GeometricLift`, `Polyfold.IsCoveringFunctor.existsUnique_lift`, `Polyfold.IsCoveringFunctor.lift`, `Polyfold.isCoveringFunctor_of_isEmpty`, `Polyfold.CoveringExamples.isCoveringFunctor_id`).** Diff_{sc} has the M-polyfolds as objects and the sc-diffeomorphisms as morphisms; $f : D \to C$, C with one object, is a covering functor if $\psi \mapsto (f(\psi), s(\psi))$ is a bijection $\mathbf{D} \to C\ {}_s\!\times_f D$; a geometric lift is a pair of functors L_D, L_C into Diff_{sc} with a natural transformation $\tau : L_D \to L_C \circ f$. *Book:* 2017 p. 279 and Definition 8.5.5 (2017: 8.9), pp. 388–389 (2017 pp. 279–280). *Hidden hypotheses:* none. *Deviation (restated):* D a Mathlib groupoid in `Type`, \$C = \$ `SingleObj G` for a group G in `Type`, so $C\ {}_s\!\times_f D = G \times D$; the finiteness of D, its hom-sets and G is carried by the results that use it; Diff_{sc} has bundled M-polyfolds of one universe; $D = \emptyset$ is allowed (Findings, items 57, 100).

V8.33 (the discussion after Definition 8.5.5 (2017: 8.9): $\varphi_{yy'}$**, the actions and equivariance;** **`Polyfold.GeometricLift.Ux`, `Polyfold.GeometricLift. y`, `Polyfold.GeometricLift.isScDiffeoOn_ y`, `Polyfold.GeometricLift. ,` `Polyfold.GeometricLift.isScDiffeoOn_ ,` `Polyfold.GeometricLift. _comp`, `Polyfold.GeometricLift. _id`, `Polyfold.GeometricLift. _inv_ ,` `Polyfold.GeometricLift.smul_def`, `Polyfold.GeometricLift.smul_def'`, `Polyfold.GeometricLift.isScSmoothMap_smul`, `Polyfold.GeometricLift. y_ ,` `Polyfold.GeometricLift. _equivariant`).** $\tau(y) : U(y) \to U(x)$ and $\varphi_{yy'}(\psi) = L_D(\psi)$ are sc-diffeomorphisms, $\varphi_{yy''}(\psi \circ \gamma) = \varphi_{y'y''}(\psi) \circ \varphi_{yy'}(\gamma)$, G_y and G_x act on $U(y)$, $U(x)$, and $\tau(y)(h * z) = f(h) * \tau(y)(z)$. *Book:* pp. 388–390 (2017 pp. 280–281). *Hidden hypotheses:* none. *Deviation:* none; the action of G_x is a left action through `SingleObj G`'s composition (Findings, item 57).

V8.34 (disjoint unions of M-polyfolds; **`Polyfold.paracompactSpace_sigma`, `Polyfold.MChart.pushforward`, `Polyfold.MChart.compatible_of_disjoint`, `Polyfold.MPolyfold.sigmaChart`, `Polyfold.MPolyfold.sigma`, `Polyfold.MPolyfold.isChart_sigmaChart_iff`, `Polyfold.MPolyfold.isScDiffeoOn_sigmaMk`, `Polyfold.MPolyfold.isScSmoothMap_sigma_iff`, `Polyfold.MPolyfold.isLocalScDiffeo_sigma_iff`, `Polyfold.MPolyfold.degIndex_sigma`, `Polyfold.MPolyfold.isTame_sigma_iff`, `Polyfold.MPolyfold.isScDiffeoOn_sigmaProdDistrib`, `Polyfold.isScDiffeoOn_prod_discrete_sigma`).** A disjoint union $\bigsqcup_i X_i$ of M-polyfolds, for any index type, is an M-polyfold whose charts are those of the summands; the summands are open and sc-diffeomorphic to the X_i; sc-smoothness, local sc-diffeomorphisms, the degeneracy index and tameness are summand-wise; products distribute over the union, and $G \times X \cong \bigsqcup_G X$ for discrete G. *Book:* p. 390 (2017 p. 281), "possesses a natural M-polyfold structure", in the proof of Proposition 8.5.7 (2017: 8.3). *Hidden hypotheses:* none. *Deviation (not in the book):* the structure is not described in the text (Findings, item 58).

V8.35 (Proposition 8.5.7 (2017: 8.3), the proper covering functor of a geometric lift; **`Polyfold.`**

`GeometricLift.YObj`, `Polyfold.GeometricLift.YMor`, `Polyfold.GeometricLift.sourceStr`, `Polyfold.GeometricLift.yObjPolyfold`, `Polyfold.GeometricLift.yMorPolyfold`, `Polyfold.GeometricLift.isLocalScDiffeo_ySrc`, `Polyfold.GeometricLift.isLocalScDiffeo_yTgt`, `Polyfold.GeometricLift.isProperMap_yTgt`, `Polyfold.GeometricLift.isScSmoothCompSheet_y`, `Polyfold.geometricLiftSource`, `Polyfold.geometricLiftTarget`, `Polyfold.geometricLiftFunctor`, `Polyfold.geometricLiftFunctor_surjective`, `Polyfold.geometricLiftFunctor_surjective_iff`, `Polyfold.geometricLiftFunctor_isLocalScDiffeo`, `Polyfold.geometricLiftFunctor_finite_fiber`, `Polyfold.geometricLiftFunctor_evenlyCovered`, `Polyfold.GeometricLift.fibredInv`, `Polyfold.GeometricLift.isScSmoothMap_fibredInv`, `Polyfold.geometricLiftFunctor_isScDiffeo_fibredMap`, `Polyfold.proposition_8_5_7`, `Polyfold.EpGroupoid.Examples.actionEpGroupoid`, `Polyfold.CoveringExamples.isProperCovering_reflectionLift`). A covering functor $f : D \to C$ with a geometric lift gives a proper covering functor $F : Y \to X$, $X = G_x \ltimes U(x)$, $Y = \bigsqcup U(y)$, $\mathbf{Y} = \bigsqcup \mathrm{mor}(y, y') \times U(y)$, $F|U(y) = \tau(y)$, $F(\psi, z) = (f(\psi), \tau(y)(z))$. *Book:* Proposition 8.5.7 (2017: 8.3) and its proof, pp. 389–391 (2017 pp. 281–282). *Hidden hypotheses:* D nonempty; for $D = \emptyset$, which Definition 8.5.5 (2017: 8.9) allows, F is surjective on objects only if $U(x) = \emptyset$ (`Polyfold.geometricLiftFunctor_surjective_iff`). *Deviation (narrower):* for nonempty D; the finiteness of Definition 8.5.5 (2017: 8.9) enters as instance arguments; X is Chapter 7's action ep-groupoid, G with the discrete topology and lifted by `ULift`; the "one verifies" (composition sc-smooth, properness, here with $V = Y$) is carried out; the inverse of (8.16) is shown sc-smooth directly. The proof's "$F : X \to Y$" is read as $F : Y \to X$ (possible misprint) (Findings, items 59, 101).

V8.36 (Theorem 8.5.8 (2017: 8.6), Proper Covering II; `Polyfold.GroupoidStr.FullOb`, `Polyfold.EpFunctor.FiberCat`, `Polyfold.EpFunctor.fiberIsotropy`, `Polyfold.EpFunctor.restrictFiber`, `Polyfold.EpFunctor.covering_restrict`, `Polyfold.EpFunctor.fintypeFiberCat`, `Polyfold.EpFunctor.nonempty_fiberCat`, `Polyfold.MPolyfoldCat.ofOpen`, `Polyfold.ScDiffeo.ofOpen`, `Polyfold.EpFunctor.IsFullyFaithfulOpenEmbedding`, `Polyfold.EpFunctor.ProperCoveringChart.geometricLift`, `Polyfold.EpFunctor.ProperCoveringChart.Φ`, `Polyfold.EpFunctor.ProperCoveringChart.Ψ`, `Polyfold.EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Φ`, `Polyfold.EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Ψ`, `Polyfold.EpFunctor.ProperCoveringChart.range_Φ_obj`, `Polyfold.EpFunctor.ProperCoveringChart.range_Φ_mor`, `Polyfold.EpFunctor.ProperCoveringChart.range_Ψ_obj`, `Polyfold.EpFunctor.ProperCoveringChart.range_Ψ_mor`, `Polyfold.EpFunctor.ProperCoveringChart.comp_Ψ`, `Polyfold.EpFunctor.theorem_8_5_8`). For a proper covering functor $F : Y \to X$ and $x \in X$: $f = F|D : D \to C$ is a covering functor (C on x, D on $F^{-1}(x)$), and for $U(x)$ as in Theorem 8.5.3 (2017: 8.5) there are a geometric lift with associated proper covering functor $A : K \to L$ and fully faithful open embeddings $\Phi : L \to X$, $\Psi : K \to Y$ with images the full subcategories on $U(x)$ and $F^{-1}(U(x))$ and $F \circ \Psi = \Phi \circ A$. *Book:* Theorem 8.5.8 (2017: 8.6) and its proof, 2017 pp. 283–284. *Hidden hypotheses:* none. *Deviation (narrower):* for ep-groupoids whose objects and morphisms are in `Type`, since Definition 8.5.5 (2017: 8.9) is stated there; "a sufficiently small $U(x)$ as in Theorem

8.5" is any chart of Theorem 8.5.3 (2017: 8.5) (which exist inside every open $V \ni x$, part (b)); "a geometric lift $A : K \to L$" is a geometric lift with its functor A of Proposition 8.5.7 (2017: 8.3); L_D on the morphism ψ_0 of D is $t \circ (s|\mathbf{U}(\psi_0))^{-1}$ (the text's "$\psi \in \mathbf{U}(\psi_0)$" read as $\psi = \psi_0$); both "one verifies" are carried out. "there exists exists", "$\tau(t)$" for $\tau(y)$, and "$A(\psi, z) = (f(z), F(z))$" for $(f(\psi), F(z))$ (possible misprints) (Findings, items 2, 60).

V8.37 (Definition 8.5.9 (2017: 8.10), proper strong bundle covering functors; `Polyfold.StrongBundle.IsStrongBundleIsoOn`, `Polyfold.StrongBundle.IsStrongBundleIsoOn.fiberEquiv`, `Polyfold.EpFunctor.fibredSnd`, `Polyfold.EpFunctor.isScSmoothMap_fibredSnd`, `Polyfold.EpStrongBundle.fibredLift`, `Polyfold.EpStrongBundle.fibredBundleEquiv`, `Polyfold.EpStrongBundle.pullbackProj_fibredBundleEquiv`, `Polyfold.EpStrongBundle.fibredBundleEquiv_symm_fibredLift`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering`, `Polyfold.EpStrongBundle.isProperStrongBundleCovering_id`, `Polyfold.TrivialBundle.epBundle_isProperStrongBundleCovering`). A strong bundle map A, $Q \circ A = F \circ P$, with (1) A surjective and a local strong bundle isomorphism covering a proper covering functor F; (2) $A : W|U(y) \to V|U(x)$ strong bundle isomorphisms over an even covering; (3) $(\psi, w) \mapsto ((F(\psi), A(w)), w)$ a strong bundle isomorphism $\mathbf{W} \to \mathbf{V}{}_s\times_A W$ covering $\mathbf{Y} \to \mathbf{X}{}_s\times_F Y$. *Book:* Definition 8.5.9 (2017: 8.10), p. 394 (2017 p. 285). *Hidden hypotheses:* none. *Deviation (restated):* A is a strong bundle functor (Definition 8.3.3 (2017: 8.6), V8.24); the finiteness in (2) is that of Definition 8.5.1 (2) (2017: 8.8); the strong bundle structure of $\mathbf{V}{}_s\times_A W$, not given in the text, is read as the pull-back of P along $\mathrm{pr}_2 : \mathbf{X}{}_s\times_F Y \to Y$, with the text's projection $((\varphi, v), w) \mapsto (\varphi, P(w))$ (`Polyfold.EpStrongBundle.fibredBundleEquiv`) (Findings, items 70, 102, 103).

V8.38 (§8.6.1 (2017 §8.5.1), the morphisms over a fibre and the isomorphism classes; `Polyfold.EpFunctor.fiberClass`, `Polyfold.EpFunctor.fiberIndex`, `Polyfold.EpFunctor.fiberClasses`, `Polyfold.EpFunctor.iUnion_fiberClass`, `Polyfold.EpFunctor.pairwiseDisjoint_fiberClass`, `Polyfold.EpFunctor.nat_card_isotropy_eq_of_mem_fiberClass`, `Polyfold.EpFunctor.IsProperCovering.bijective_fibredMap`, `Polyfold.EpFunctor.IsProperCovering.existsUnique_lift`, `Polyfold.EpFunctor.IsProperCovering.morOnFiberEquiv`, `Polyfold.EpFunctor.IsProperCovering.ncard_morOn_fiber`, `Polyfold.EpFunctor.IsProperCovering.finite_morOn_fiber`). $\varphi \mapsto (F(\varphi), s(\varphi))$ is a bijection $\mathrm{mor}(F^{-1}(x)) \to G_x \times F^{-1}(x)$, so $\#\mathrm{mor}(F^{-1}(x)) = \#G_x \cdot \#F^{-1}(x)$; $F^{-1}(x) = \Theta_1 \cup \cdots \cup \Theta_q$, with isomorphic isotropy groups within a class. *Book:* §8.6.1 (2017 §8.5.1), 2017 p. 286. *Hidden hypotheses:* none. *Deviation (restated):* the classes are indexed by their images in $|Y|$, $\Theta_c = F^{-1}(x) \cap \pi^{-1}(c)$; $\#$ is the cardinality of finite sets (Findings, item 71).

V8.39 (Lemma 8.6.1 (2017: 8.9); `Polyfold.EpFunctor.morOn_fiberClass_eq`, `Polyfold.EpFunctor.IsProperCovering.morOnClassEquiv`, `Polyfold.EpFunctor.IsProperCovering.lemma_8_6_1`). $\#G_x \cdot \#\Theta_j = \#\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} = \#G_{y_0} \cdot (\#\Theta_j)^2$ and $\#G_x = \#G_{y_0} \cdot \#\Theta_j$ for $y_0 \in \Theta_j$. *Book:* Lemma 8.6.1 (2017: 8.9) and its proof, pp. 396–398 (2017 pp. 286–287). *Hidden hypotheses:* none. *Deviation:* none; (8.17) writes "$\psi \to (F\psi), s(\psi))$" (possible misprint) (Findings, items 71, 72).

V8.40 (Lemma 8.6.2 (2017: 8.10); `Polyfold.EpFunctor.IsProperCovering.exists_isomorphic_of_isomorphic`, `Polyfold.EpFunctor.IsProperCovering.fiberIndex_eq`, `Polyfold.EpFunctor.IsProperCovering.lemma_8_6_2`, `Polyfold.EpFunctor.IsProperCovering.lemma_8_6_2_equiv`). For isomorphic x, x' the isomorphism classes of $F^{-1}(x)$ and $F^{-1}(x')$ correspond one to one, and isomorphic $y \in F^{-1}(x)$, $y' \in F^{-1}(x')$ lie in corresponding classes for exactly one index. *Book:* Lemma 8.6.2 (2017: 8.10) and the paragraph before it, 2017 p. 287. *Hidden hypotheses:* none. *Deviation (restated):* the correspondence is "the same index $c \in |Y|$"; as a bijection of the sets of classes it is `Polyfold.EpFunctor.IsProperCovering.lemma_8_6_2_equiv` (Findings, items 71, 73).

V8.41 (Lemma 8.6.3 (2017: 8.11); `Polyfold.EpFunctor.IsProperCovering.ncard_fiber_eq_sum`, `Polyfold.EpFunctor.IsProperCovering.lemma_8_6_3`, `Polyfold.EpFunctor.IsProperCovering.ncard_fiber_eq_of_isomorphic`). For isomorphic x, x', $\#\Theta_j = \#\Theta'_j$ and $\#F^{-1}(x) = \#F^{-1}(x')$. *Book:* Lemma 8.6.3 (2017: 8.11) and the paragraph before it, p. 398 (2017 p. 287). *Hidden hypotheses:* none. *Deviation:* none; "the third claim of Lemma 8.10" for that of Lemma 8.6.1 (2017: 8.9), and the summation index i with summands $\#\Theta_j$ (possible misprints) (Findings, item 73).

V8.42 (Proposition 8.6.4 (2017: 8.4), the number of sheets; `Polyfold.EpFunctor.IsProperCovering.one_le_ncard_fiber`, `Polyfold.EpFunctor.IsProperCovering.eventually_ncard_fiber_eq`, `Polyfold.EpFunctor.IsProperCovering.isLocallyConstant_ncard_fiber`, `Polyfold.EpFunctor.IsProperCovering.fiberCardOrbit`, `Polyfold.EpFunctor.IsProperCovering.isLocallyConstant_fiberCardOrbit`, `Polyfold.EpFunctor.IsProperCovering.proposition_8_6_4`, `Polyfold.EpFunctor.IsProperCovering.fiberCardFunctor`, `Polyfold.EpGroupoid.trivial`, `Polyfold.discretePolyfold`). $\#F^{-1}(x)$ is constant on the preimage of every connected component of $|X|$, and $x \mapsto \#F^{-1}(x)$ is an sc-smooth functor $X \to \mathbb{N}$. *Book:* Proposition 8.6.4 (2017: 8.4) and the paragraph before it, 2017 p. 288. *Hidden hypotheses:* none. *Deviation (restated):* $\mathbb{N}$ is Lean's, with 0, as the trivial ep-groupoid of the discrete M-polyfold; the values are ≥ 1; only the local constancy of $\#F^{-1}(x)$ is used, not "the structure of isomorphism classes stays the same" (Findings, item 74).

V8.43 (§8.6.2 (2017 §8.5.2), the structure of strong bundle coverings; `Polyfold.EpStrongBundle.IsProperStrongBundleCovering.mor_val`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering.fiberEquiv`, `Polyfold.EpStrongBundle.IsStrongBundleFunctor. _comm`, `Polyfold.EpStrongBundle.section_8_5_2`). $A(\varphi, w) = (F(\varphi), A_{s(\varphi)}(w))$ with A_y topological linear isomorphisms, and $\mu_X(F(\varphi)) \circ A_{s(\varphi)} = A_{t(\varphi)} \circ \mu_Y(\varphi)$: $y \mapsto A_y$ is a natural transformation $\mu_Y \to \mu_X \circ F$ of functors into Ban. *Book:* §8.6.2 (2017 §8.5.2), 2017 pp. 288–289. *Hidden hypotheses:* none. *Deviation (restated):* the natural transformation is stated as its naturality identity; of Definition 8.5.9 (2017: 8.10) only the strong bundle functor part enters the identity (Findings, item 75).

The additions of the 2021 edition (V8.44–V8.52)

The Springer edition (2021) adds Propositions 8.1.16 and 8.1.17 (the Zhou condition and unique continuation for the tangent TX), the auxiliary norms of Definition 8.3.5, §8.4 (Proposition 8.4.1 on paracompact orbit spaces of strong bundles, the local regularity property of Definition 8.4.2 and Proposition 8.4.3), Example 8.5.6, condition (0) of Definition 8.5.9 and Lemma 8.5.10; Proposition 8.2.3, a Poincaré lemma, belongs to the sc-differential forms of §8.2 (gap G.74). Numbers and pages below are the 2021 text's. Lean files: `Chapter8/Book2021.lean`, `Chapter8/TangentZhou.lean`, and, for Proposition 8.1.17 (2) on every $(TX)^i$, `Chapter8/TangentZhouLevels.lean` with Part I's `Chapter2/TangentLiftAll.lean`.

V8.44 (Proposition 8.1.16, (ZHOU) for TX; `Polyfold.MPolyfold.levelVal_tangentProj_mem_smoothSet`, `Polyfold.EpGroupoid.mem_nonEffectivePart_tangent`, `Polyfold.EpGroupoid.TangentZhouReflect`, `Polyfold.EpGroupoid.proposition_8_1_16`, `Polyfold.EpGroupoid.tangentMor_trivial`, `Polyfold.EpGroupoid.tangentZhouReflect_trivial`, `Polyfold.EpGroupoid.Examples.proposition_8_1_16_point`). A smooth point of TX lies over a smooth point of X; if $s = t$ near the base $p(v)$ of $v \in T\mathbf{X}$, then $v \in (T\mathbf{X})^T$. If X satisfies (ZHOU) (Definition 7.4.5, V7.88) and `TangentZhouReflect` holds — for a smooth $b \in TX$ and $(g,h) \in G_b$, if the tangent of (g,h) in the ep-groupoid TX is the identity on $T_b(TX)$, then Tg is the identity on $T_{p(b)}X$ — then TX satisfies (ZHOU). The hypotheses hold for the trivial ep-groupoid of a point (`TangentZhouReflect` for every trivial ep-groupoid). *Book:* Proposition 8.1.16, p. 369, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the step "$T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$", which the text obtains by differentiating (8.16) and using that $T\tau$ is onto, is the hypothesis `TangentZhouReflect`: the second tangent bundle $T(TX)$ and $T\tau$ are not formalized (gap G.88, open for this step; its Proposition 8.1.17 (2) part is closed, V8.52); as in Chapter 7, tangents of isotropy elements are set maps, no linear structure is used (Findings, item 115).

V8.45 (Proposition 8.1.17, LUC for TX; `Polyfold.EpGroupoid.nonEffectivePart_tangent_eq`, `Polyfold.EpGroupoid.proposition_8_1_17_one`, `Polyfold.EpGroupoid.proposition_8_1_17_two`, `Polyfold.EpGroupoid.isOpenMap_tangentProj_of_subsingleton`, `Polyfold.EpGroupoid.Examples.proposition_8_1_17_one_point`). Assume that every $x \in X_1$ has a natural representation with $U(x) \cap X_1$ connected in X^1 and dense in $U(x)$, and that $p : T\mathbf{X} \to \mathbf{X}^1$ is open. (1) If X^1 has LUC, then $(T\mathbf{X})^T = p^{-1}((\mathbf{X}^1)^T)$ and TX has LUC. (2) If X has (SLUC), TX has LUC. The hypotheses hold for the point ep-groupoid. *Book:* Proposition 8.1.17, p. 371, new in the 2021 edition: (1) as stated; (2) (SLUC) for X gives (SLUC) for TX. *Hidden hypotheses:* the connectedness and density of $U(x) \cap X_1$ (as in Appendix 7.6.4, V7.83; the density is gap G.73), and the openness of p (Part I, gap G.8). *Deviation (narrower):* (1) is proved through Proposition 7.2.21 for TX; (2) here only for $(TX)^0 = TX$; for every $(TX)^i$, through $T(X^i) \cong (TX)^i$ as ep-groupoids, it is V8.52 (gap G.88, closed for Proposition 8.1.17 (2)) (Findings, items 116, 124).

V8.46 (Definition 8.3.5, auxiliary norms; `Polyfold.EpStrongBundle.IsAuxNorm`, `Polyfold.EpStrongBundle.exists_oneVal_eq_`, `Polyfold.EpStrongBundle.exists_isAuxNorm_trivial`, `Polyfold.TrivialBundle.exists_isAuxNorm_epBundle`). For a strong bundle $(P : W \to X, \mu)$ over an ep-groupoid, an auxiliary norm is a continuous $N : W_{0,1} \to \mathbb{R}$

which (1) is a complete norm on every fibre, (2) has $h_k \to 0_x$ in $W_{0,1}$ whenever $P(h_k) \to x$ and $N(h_k) \to 0$ — (1), (2) and continuity are Part I's auxiliary norm — and (3) satisfies $N(h) = N(k)$ whenever $h = \mu(\varphi, k)$, $s(\varphi) = P(k)$. Condition (3) is not vacuous: $\mu(\varphi, k) \in W_{0,1}$ for $k \in W_{0,1}$. Over the trivial ep-groupoid every auxiliary norm of Part I is one, and $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ has one. *Book:* Definition 8.3.5, p. 379, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* values in $[0, \infty)$ follow from (1); in (3), h is given in $W_{0,1}$ and $h = \mu(\varphi, k)$ is an equation in W (Findings, item 117).

V8.47 (Proposition 8.4.1, $|X|$, $|W[0]|$ and $|W[1]|$; `Polyfold.EpStrongBundle. _fiberZero`, `Polyfold.EpStrongBundle.zeroOrbitMap`, `Polyfold.EpStrongBundle.isEmbedding_zeroOrbitMap`, `Polyfold.EpStrongBundle.proposition_8_4_1_two_imp_one`, `Polyfold.EpStrongBundle.proposition_8_4_1_one_iff_two`, `Polyfold.EpStrongBundle.proposition_8_4_1_one_imp_two`, `Polyfold.EpStrongBundle.OrbitSpaceOne`, `Polyfold.EpStrongBundle. One_eq_ One_iff`, `Polyfold.EpStrongBundle.isOpenMap_ One`, `Polyfold.EpStrongBundle.continuous_bundleOneSmul`, `Polyfold.EpStrongBundle.isOpenEmbedding_bundleOneOrbitQuotientMap`, `Polyfold.EpStrongBundle.metrizableSpace_orbitSpaceOne`, `Polyfold.EpStrongBundle.isClosedEmbedding_zeroOrbitMapOne`, `Polyfold.EpStrongBundle.proposition_8_4_1_one_imp_three`, `Polyfold.EpStrongBundle.proposition_8_4_1_three_imp_one`, `Polyfold.EpStrongBundle.proposition_8_4_1`). For a strong bundle over an ep-groupoid, $|X|$, $|W[0]|$ and $|W[1]|$ are paracompact together. $\mu(\varphi, 0_{s(\varphi)}) = 0_{t(\varphi)}$, and the zero section induces embeddings $|0| : |X| \to |W[0]|$ and $|X| \to |W[1]|$ with left inverse $|P|$, the second closed; G_x acts on $W|U(x)$ and on $W[1]|U(x)$ by homeomorphisms, and the quotients embed openly into $|W|$, $|W[1]|$ over $|U(x)|$. *Book:* Proposition 8.4.1, p. 380, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* $W[1]$ is not built as an ep-groupoid; $|W[1]|$ is the orbit space of the groupoid $(W[1], \mathbf{X}\,{}_s\!\times_P W[1])$ with the quotient topology of the M-polyfold $W[1]$, which is what the statement uses (gap G.86 closed, 2026-11-12). *Proof:* by metrizability: $|X|$ is metrizable (Theorem 7.3.1) and Smirnov's theorem is pulled back along $|P|$ and $|P[1]|$, with the quotient metric of Lemma 7.3.2 on $G_x \backslash W|U(x)$; only the continuity of μ (at level one for $W[1]$) is used, not the text's invariant metrics and closed cover; (3) □ (1) is the text's closed embedding (Findings, items 118, 127, 133).

V8.48 (Definition 8.4.2, the local regularity property of a strong bundle; `Polyfold.EpStrongBundle.natLift`, `Polyfold.EpStrongBundle.IsLR`, `Polyfold.EpStrongBundle.isLR_trivial`). The natural lift of the G_x-action to $W|U(x)$ is $g * w = \mu(\Gamma(g, P(w)), w)$. A strong bundle has the local regularity property if inside every open $O \ni x$ there is a natural representation $U(x)$ such that every strong bundle map $\Phi : W \to W$, covering some β, which restricts to a strong bundle isomorphism of $W|U(x)$ and satisfies $|\Phi(w)| = |w|$ for $P(w) \in U(x)$, is $w \mapsto g * w$ on $W|U(x)$ for some $g \in G_x$. Every strong bundle over a trivial ep-groupoid has it. *Book:* Definition 8.4.2, p. 382, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* "strong bundle isomorphism $B : W|U(x) \to W|U(x)$" is read as Part I's `IsStrongBundleIsoOn` for a strong bundle map $W \to W$, as for Definition 8.5.9 (V8.37), and $|B(w)| = |w|$ as equality of orbits in $(W, \mathbf{W})$. Proposition 8.4.3 (effective X with (LR) gives the local regularity property) is not formalized (gap G.86) (Findings, item 119).

V8.49 (the remark after Definition 8.5.5 and Example 8.5.6; `Polyfold.IsCoveringFunctor.injective_map_end`, `Polyfold.Example856D`, `Polyfold.example856Functor`, `Polyfold.example_8_5_6`). For a covering functor $f : D \to C$ and $d \in D$, $f : G_d \to G_x$ is injective. For D with two objects a, b and one morphism between any two objects, $C = \{1_x, \varepsilon\}$ and $f(\varphi) = f(\varphi^{-1}) = \varepsilon$: f is a covering functor, G_a, G_b are trivial, $G_x \cong \mathbb{Z}/2$ is not, and $f : G_d \to G_x$ is injective and not surjective. *Book:* the remark after Definition 8.5.5 and Example 8.5.6, p. 388 (Definition 8.5.5: 2017 Definition 8.9; Example 8.5.6 new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none; C is $\mathbb{Z}^\times$ as a one-object category, $\varepsilon = -1$ (Findings, item 120).

V8.50 (Definition 8.5.9 in the 2021 form; `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering.isProperStrongBundleCovering2021`, `Polyfold.EpStrongBundle.isProperStrongBundleCovering2021_iff`, `Polyfold.EpStrongBundle.isProperStrongBundleCovering2021_id`, `Polyfold.TrivialBundle.epBundle_isProperStrongBundleCovering2021`). For strong bundles (W, μ) over Y and (V, τ) over X and an ep-functor $F : Y \to X$, a map $A : W \to V$ is a proper strong bundle covering functor if it is a strong bundle map with $Q \circ A = F \circ P$ satisfying (0) $\tau(F(\psi), A(w)) = A(\mu(\psi, w))$, and (1)–(3) of V8.37. This holds iff A is the object map of a unique strong bundle functor which is a proper strong bundle covering functor in the sense of V8.37. The identity of every strong bundle over an ep-groupoid is one. *Book:* Definition 8.5.9, p. 394 (2017: 8.10, p. 285), with (0) new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* stated for the object map, (0) being the strong bundle map property of Definition 8.3.2 (2017: 8.5); V8.37 had read 2017's "strong bundle map A" as a strong bundle functor, which contains (0), and the two readings are equivalent (Findings, items 70, 121).

V8.51 (Lemma 8.5.10; `Polyfold.EpStrongBundle.pullback`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bundleB`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_val`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.eq_of_isStrongBundleIsoOn`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bijective_bundleB`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv_eq_of`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleB`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.mapsTo_oneSet_bundleBInv`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleBInv`, `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021.lemma_8_5_10`). For a proper strong bundle covering functor $A : W \to V$ covering $F : Y \to X$ (V8.50), on the pull-back $F^*V = \{(y, v) \mid F(y) = Q(v)\}$ with $\mu_{F^*V}(\varphi, (y, v)) = (t(\varphi), \mu_X(F(\varphi), v))$, the map $B(w) = (P(w), A(w))$ is a strong bundle isomorphism $W \to F^*V$ covering the identity of Y, with inverse B^{-1}, and $B(\mu_Y(\varphi, w)) = \mu_{F^*V}(\varphi, B(w))$ (8.23); $A = \bar{A} \circ B$ with $\bar{A}(y, v) = v$. *Book:*

Lemma 8.5.10, p. 395, new in the 2021 edition. *Hidden hypotheses:* none (the paracompactness of F^*V, a hypothesis of Part I's pull-back, always holds). *Deviation (narrower):* "strong bundle isomorphism over the ep-groupoid Y" is read as a strong bundle isomorphism $W \to F^*V$ together with (8.23); that μ_{F^*V} makes F^*V a strong bundle over the ep-groupoid Y is not proved, only its formula is used (gap G.89). *Proof:* B^{-1} is near every point $(y, v) \mapsto \Psi(v)$ for the inverse Ψ of a local strong bundle isomorphism of Definition 8.5.9 (1) (Findings, item 122).

V8.52 (Proposition 8.1.17 (2) for every $(TX)^i$; `Polyfold.MPolyfold.ScEquiv.toHomeomorph`, `Polyfold.EP.tangentEP`, `Polyfold.EpGroupoid.EpEquiv`, `Polyfold.EpGroupoid.EpEquiv.refl`, `Polyfold.EpGroupoid.EpEquiv.trans`, `Polyfold.EpGroupoid.EpEquiv.shift`, `Polyfold.EpGroupoid.EpEquiv.nonEffectivePart_eq`, `Polyfold.EpGroupoid.EpEquiv.isLUC`, `Polyfold.EpGroupoid.tangentShiftEpEquiv`, `Polyfold.EP.tangentLiftEpEquiv`, `Polyfold.EP.isLUC_tangent_lift`, `Polyfold.EP.proposition_8_1_17_two_all`, `Polyfold.EpGroupoid.hasConnectedDenseLevelOneReps_of_subsingleton`, `Polyfold.EpGroupoid.Examples.proposition_8_1_17_two_all_pointEP`, `Polyfold.BundledMPolyfold.tangentLiftEquiv`, `Polyfold.proposition_2_3_17_all`). An equivalence of ep-groupoids, here, is a pair of sc-diffeomorphisms of the object and the morphism M-polyfolds commuting with source and target; it maps $\mathbf{X}^T$ onto $\mathbf{X}'^T$ and so transports LUC, by Proposition 7.2.21, and it passes to the index raises. Part I's Proposition 2.3.17 for $i = 1$, on objects and morphisms, gives an equivalence $T(X^1) \cong (TX)^1$, and by recursion $T(X^i) \cong (TX)^i$ for every i. So if X has (SLUC), every X^i satisfies the hypothesis of V8.45 (natural representations with $U(x) \cap (X^i)_1$ connected in $(X^i)^1$ and dense in $U(x)$), and every $p : T(\mathbf{X}^i) \to (\mathbf{X}^i)^1$ is open, then Proposition 8.1.17 (1) for X^i gives LUC for $T(X^i)$, hence for $(TX)^i$: TX has (SLUC). The point ep-groupoid satisfies the hypotheses. *Book:* Proposition 8.1.17 (2), p. 371, new in the 2021 edition. *Hidden hypotheses:* those of (1) on every level: the connectedness and density of $U(x) \cap (X^i)_1$ for every X^i (gap G.73) and the openness of every p (Part I, gap G.8). *Deviation (explicit):* "naturally $(TX)^i = T(X^i)$" is used through source and target only; the compatibility of the equivalence with unit, inverse and composition is not formalized, since LUC does not need it. This closes gap G.88 for Proposition 8.1.17 (2); the step of Proposition 8.1.16 stays open (V8.44) (Findings, item 124). *Proof:* the text's.

The Part II wave (2026-11-12): §8.2 (V8.53–V8.55)

Definitions 8.2.1, 8.2.2 and Proposition 8.2.3 are formalized in `Chapter8/GapP2_B_EpForms.lean`, `GapP2_B_EpFormsObj.lean`, `GapP2_B_EpFormsFunctor.lean`, `GapP2_B_EpPoincare.lean` and, since its proof uses the push-forward of §11.2, `Chapter11/GapP2_B2_Poincare.lean`. Forms are Part I's (Definition 4.4.1: families of alternating maps on the tangent spaces of level-one points, sc-smooth in charts), on the object M-polyfold.

V8.53 (Definition 8.2.1 (2017: 8.2), forms on ep-groupoids; `Polyfold.EpGroupoid.IsMorCompatible`, `Polyfold.EpGroupoid.epForms`, `Polyfold.EpGroupoid.isMorCompatible_iff_formPullback`, `Polyfold.EpGroupoid.formLower_mem_epForms`, `Polyfold.EpGroupoid.extD_mem_epForms`, `Polyfold.EpGroupoid.constForm_mem_epForms`, `Polyfold.EP.epForms`, `Polyfold.EP.epFormsMap`, `Polyfold.EP.epFormsInfty`, `Polyfold.EP.epFormsInftyOf_eq_iff`,

`Polyfold.EP.epFormsInftyD`, `Polyfold.EP.epFormsInftyD_comp_epFormsInftyD`). $\Omega^k_{ep}(X)$ is the space of sc-differential k-forms ω of the object M-polyfold with $(T\varphi)^*\omega_y = \omega_x$ for every $\varphi : x \to y$ in $\mathbf{X}_1$, $T\varphi$ the bounded linear isomorphism of V7.5; the condition is equivalent to $s^*\omega = t^*\omega$ on $\mathbf{X}$. A form on X^i is a form on X^{i+1}, $\Omega^k_{ep,\infty}(X)$ is the direct limit over i, two representatives give the same class iff their pull-backs to some X^l agree, $d\omega$ is compatible with morphisms, $d[\omega] = [d\omega]$ and $d \circ d = 0$. Constant 0-forms are in $\Omega^0_{ep}(X)$. *Book:* Definition 8.2.1 (2017: 8.2) and the paragraphs after it, pp. 372–373 (2017 pp. 268–269). *Hidden hypotheses:* none. *Deviation (same):* none; the text's Whitney sum $\oplus^k TX$ is Part I's reading of forms. *Proof:* the compatibility of $d\omega$ by the naturality of d for the global maps s, t, where the text uses $f = t \circ (s|\mathbf{U}(\varphi_0))^{-1}$ locally. The definition writes T_yY for T_yX (Findings, items 110, 132).

V8.54 (Definition 8.2.2 (2017: 8.3), the sc-de Rham complex; `Polyfold.EP.epFormsInftyGraded`, `Polyfold.EP.epFormsInftyGradedD_comp_self`, `Polyfold.EP.epDeRham`, `Polyfold.EP.epDeRhamComplex`, `Polyfold.EP.epFormsInftyToObj`, `Polyfold.EP.epFormsInftyToObj_injective`, `Polyfold.EP.formsInftyD_comp_epFormsInftyToObj`, `Polyfold.EP.epFormsInftyPullback`, `Polyfold.EP.epFormsInftyD_comp_epFormsInftyPullback`, `Polyfold.EP.epFormsInftyPullback_eq_of_natTrans`). $(\Omega^*_{ep,\infty}(X), d)$ is a cochain complex of real vector spaces, a subcomplex of Part I's complex of the object M-polyfold through an injective chain map; sc-smooth functors pull it back, commuting with d, and naturally equivalent functors give the same pull-back. *Book:* Definition 8.2.2 (2017: 8.3), p. 373 (2017 p. 270). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 132).

V8.55 (Proposition 8.2.3, the Poincaré lemma; `Polyfold.EP.proposition_8_2_3`, `Polyfold.EP.proposition_8_2_3_of_isOpen`, `Polyfold.EP.sub`, `Polyfold.EP.subIncl`, `Polyfold.EP.modelFunctor`, `Polyfold.EP.isEquivalence_modelFunctor`, `Polyfold.EpGroupoid.NaturalRep.exists_primitive_translation`, `Polyfold.EP.exists_primitive_of_average`, `Polyfold.EP.proposition_8_2_3_object`). Every object x has saturated open neighbourhoods $V \subseteq U$ such that every closed $a \in \Omega^{k+1}_{ep,\infty}(X|U)$ is db on V for some $b \in \Omega^k_{ep,\infty}(X|V)$; "on V" is the pull-back by the inclusion functor $X|V \to X|U$. The same holds inside any saturated open $U \ni x$ given in advance. *Book:* Proposition 8.2.3 and its proof, pp. 373–374, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* the text's $k \geq 1$ is written $k + 1$. *Proof:* the text's: Lemma 4.4.8 on the object M-polyfold, the average over G_x on the translation groupoid $G_x \ltimes U(x)$ of a natural representation, and the extension $\sigma(y) = \psi^*\tau'(x')$ as the push-forward of §11.2 along the equivalence $G_x \ltimes U(x) \to X|\pi^{-1}(\pi(U(x)))$ (Findings, items 123, 132).

What is not verified in this chapter

- pp. 364 ff. (2017 pp. 264–265), Remark 8.1.10 (2017: 8.1): the bijectivity of G with $T\varphi$ a bare map, which does not hold as we read it ($\mathbb{Z}/3 \ltimes \mathbb{R}$, V8.12, narrower; Findings, item 10). The linearity of $T\varphi$ (since 2026-11-09), the M-polyfold structure along G and the triples (since 2026-11-12) are verified (V8.12; Findings, items 112, 113, 131).
- 2017 p. 268, the proof of Theorem 8.1.15 (2017: 8.4): the sc-isomorphism of $TU(x_z)$ with a local bundle model $TO \to O^1$ and the linearity of the bundle action of G_{x_z}; not used, since

the proof here does not need them (V8.18) (gap G.7; Findings, items 21, 113).

- p. 385 (2017 p. 277), Theorem 8.5.3 (5) (2017: 8.5) as printed, with t on the right arrow: it does not hold as we read it; the square with s and the companion square with t are verified (V8.30) (Findings, item 54).
- p. 390 (2017 p. 281), Proposition 8.5.7 (2017: 8.3) for $D = \emptyset$: the functor is then not surjective on objects unless $U(x) = \emptyset$; the proposition is verified for nonempty D (V8.35) (Findings, item 59).
- pp. 379–385 (2017 pp. 275–276), the degeneracy remark for non-tame X, Y: only $d_{\mathbf{X}\times Y} \leq 2\, d_{\mathbf{Y}}$ (V8.28) (Findings, item 51).
- Witnesses: every hypothesis of the chapter is satisfiable (Findings, item 105), but the proper covering functors constructed have fibres $F^{-1}(x)$ with one point: no proper covering functor with $\#F^{-1}(x) > 1$, and no proper strong bundle covering functor other than identities, is constructed (Findings, items 101, 102).
- Gaps taken up: G.70 in part, the well-definedness and functoriality of $T\varphi$ being now formalized (V8.12); G.7 remains open (Findings, items 112, 114).
- p. 369, Proposition 8.1.16: the step "$T(g, h) = \mathrm{id}$ implies $Tg = \mathrm{id}$" (differentiating (8.16), $T\tau$ onto) is the hypothesis `TangentZhouReflect` (V8.44) (gap G.88, open for this step; Findings, item 115). Proposition 8.1.17 (2) for every $(TX)^i$ is verified (V8.52; G.88 closed for it); of "naturally $(TX)^i = T(X^i)$" only the sc-diffeomorphisms of objects and morphisms commuting with source and target are formalized, not the compatibility with unit, inverse and composition, which (LUC) does not use, nor the linearity on the fibres (gap G.7) (Findings, items 116, 124).
- p. 380, Proposition 8.4.1: the ep-groupoid $W[1]$ itself is not built; (3) is verified with $|W[1]|$ the orbit space of $(W[1], \mathbf{X}\,{}_s\!\times_P W[1])$ (V8.47, restated; gap G.86 closed; Findings, item 133). Proposition 8.4.3 is verified since 2026-11-09 (V8.48; Findings, item 128).
- Verified since this list was first written (2026-11-12): Definitions 8.2.1, 8.2.2 (pp. 372–373 (2017 pp. 268–270); V8.53, V8.54; gap G.74 closed), Lemma 8.1.2 (2017: 8.1) as printed (p. 359 (2017 p. 259); V8.4; gap G.75 closed) and Proposition 8.2.3 (p. 373; V8.55; gap G.87 closed) (Findings, items 110, 111, 123, 131, 132).
- p. 395, Lemma 8.5.10: that F^*V with μ_{F^*V} is a strong bundle over the ep-groupoid Y; only the formula of μ_{F^*V} is used (V8.51) (gap G.89; Findings, item 122).

10. Chapter 9 (HWZ, Part II, §§9.1–9.5 and §9.6.1 (2017 §§9.1–9.5)): verified for §§9.1–9.4 and §9.6.1 with the readings and hypotheses recorded below, unconditionally for ep-groupoids whose object M-polyfold is $\mathbb{R}^m$, except the global part of Proposition 9.1.11, Lemma 9.4.3 and, in Figure 9.1, the orientations of the faces of the faces; for §9.5 the sets of measure zero, the σ-algebras and the isotropy factors are verified, branched integration of sc-forms over sheets covered by countably many graph charts, with "oriented, tame" read as a covering by oriented tame structures, the boundary measures and Stokes' theorem as printed for every $n \geq 1$, and Remark 9.5.20 in part (its M-polyfold, the $\mathbb{Z}_3$-action and the effective isotropy of the boundary points)

Chapter 9 (pp. 401–458 (2017 pp. 291–332); numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) treats branched ep⁺-subgroupoids, the objects that carry the solution sets of perturbed sc-Fredholm problems with symmetries. §9.1 defines them as functors $\Theta : X \to \mathbb{Q}^+$ that are locally weighted sums of indicator functions of properly embedded finite-dimensional submanifolds (Definition 9.1.2 (2017: 9.1)), proves that $|\text{supp}(\Theta)|$ is locally closed (Lemma 9.1.5 (2017: 9.1)), constructs symmetric local branching structures (Lemma 9.1.7 (2017: 9.2)), defines closed, compact, manifold- and orbifold-type Θ (Definitions 9.1.6, 9.1.8 (2017: 9.2, 9.3)), and decomposes Θ by dimension (Propositions 9.1.11 (2017: 9.2) and 9.1.12, the latter new in the 2021 edition). §9.2 introduces the category $\text{Gr}(X)$ of formal sums of tangent spaces and the tangent section T_Θ (Theorem 9.2.1 (2017: 9.1)), good and bad points (Definitions 9.2.2, 9.2.6, Lemmas 9.2.3–9.2.7 (2017: 9.4, 9.5, 9.3–9.5)), the graph representation near a bad point (Proposition 9.2.8, Lemmas 9.2.9–9.2.11 (2017: 9.3, 9.6–9.8)), the tangent $T\Theta$ (Definition 9.2.12 (2017: 9.6)), the position of a submanifold relative to ∂X (Lemma 9.2.13, Remark 9.2.14 (2017: 9.9, 9.5)) and the boundary $\partial\Theta$ (Proposition 9.2.15, Definition 9.2.16, Remark 9.2.17 (2017: 9.4, 9.7, 9.6)). §9.3 defines orientations (the category $\widehat{\text{Gr}}(X)$, Definition 9.3.2 (2017: 9.8), Remark 9.3.3, Lemma 9.3.4, Example 9.3.5 (2017: 9.8, 9.10, 9.1)), tame Θ (Definition 9.3.6 (2017: 9.10), Lemma 9.3.7, new) and boundary orientations ((9.11), Definition 9.3.8, Proposition 9.3.9, Definition 9.3.10 (2017: 9.9, —, 9.11)). §9.4 studies the geometry of local branching structures: essential points, sets of measure zero and the decomposition of an oriented structure (Definitions 9.4.1, 9.4.5, 9.4.7, Lemmas 9.4.2–9.4.8, Propositions 9.4.9, 9.4.10 (2017: Definitions 9.12–9.14, Lemmas 9.11–9.15, Propositions 9.5, 9.6)). §9.5 integrates sc-differential forms over branched ep⁺-subgroupoids and proves Stokes' theorem; verified here are its measure-theoretic part (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16)) and the isotropy and set parts of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 (2017: Lemmas 9.18, 9.19; Lemma 9.5.10 and Proposition 9.5.13 new in the 2021 edition), not the integrals. Appendix §9.6.1 (new in the 2021 edition) proves Proposition 9.1.12 (Lemmas 9.6.1, 9.6.2, 9.6.4, Definition 9.6.3, Proposition 9.6.5); §§9.6.2–9.6.3 discuss open questions and have no formal content.

What is verified. Every definition of §§9.1–9.4, of the measure-theoretic part of §9.5 and of §9.6.1 is formalized: Definitions 9.1.2 (V9.1), 9.1.6 (V9.6), 9.1.8 (V9.8), 9.2.2 (V9.15), 9.2.6 (V9.19), 9.2.12 (V9.23), 9.2.16 (V9.27), 9.3.2 (V9.30), 9.3.6 (V9.34), 9.3.8 (V9.36), 9.3.10 (V9.38), 9.4.1 (V9.39), 9.4.5 (V9.43), 9.4.7 (V9.45), 9.5.1 (V9.49), 9.5.3 (V9.51), 9.5.14 (V9.52) and 9.6.3 (V9.60), with Property (T)

(V9.41). **Theorem 9.2.1 (2017: 9.1) holds** for every branched ep$^+$-subgroupoid of an ep-groupoid whose object M-polyfold is $\mathbb{R}^m$, and in general under the graph-chart hypotheses below (V9.13, V9.14); so do Propositions 9.1.12, 9.2.15, 9.3.9, 9.4.9, 9.4.10 and 9.6.5 in the forms stated below, Lemmas 9.1.5, 9.1.7, 9.2.3, 9.2.5, 9.2.7, 9.2.13 (read at points of ∂X), 9.3.4 (for n-dimensional Θ), 9.3.7, 9.4.2, 9.4.4 (for sheets of one dimension), 9.4.6, 9.4.8, 9.6.1, 9.6.2 and 9.6.4, Remarks 9.1.3 (b), 9.2.4, 9.2.14, 9.2.17 and 9.3.3, and Example 9.3.5 with the corrections recorded in V9.33. In §9.5, Lemma 9.5.2 (2017: 9.16) is verified for structures with sheets of one class (V9.50), the remark after Definition 9.5.3 as the completion by null sets (V9.51), and the parts of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 that do not involve integrals: the orders of the isotropy groups, (9.23) and the decompositions of the preimages (V9.54–V9.57). Several places of the text needed more than a reading. (a) *Proposition 9.1.11 (2017: 9.2).* The uniqueness of the dimensional decomposition, as stated, does not hold as we read it: on the trivial ep-groupoid of $\mathbb{R}^2$ the indicator of the x-axis has two decompositions (`Polyfold.EpGroupoid.not_proposition_9_1_11_rn`); the local construction is verified, the global gluing is not formalized (V9.9). (b) *Lemma 9.2.13 (2017: 9.9).* As printed, (2) □ (1) does not hold as we read it at interior points (`Polyfold.MPolyfold.not_lemma_9_2_13_two_one`); the lemma is verified for $x \in M \cap \partial X$, the case the text uses, together with $d_M(x) = 0 \Leftrightarrow T_xM \subseteq T_x^RX$ at every $x \in M$ (V9.24). (c) *Definition 9.2.16 (2017: 9.7).* (1) and (2) do not characterize $\partial\Theta$ uniquely as we read them (the zero functor satisfies them; `Polyfold.not_unique_isBoundaryCharacterized_halfLine`); uniqueness holds among functors vanishing off $\mathrm{supp}(\Theta)$ (V9.27). (d) *Lemma 9.3.4 (2017: 9.10)* does not hold as we read it without equidimensionality (`Polyfold.lemma_9_3_4_mixed_dim_rn`); it is verified for n-dimensional Θ (V9.32). (e) *Example 9.3.5 (2017: 9.1).* The two orientations also coincide on $\{x < 0\}$, and there are exactly six orientations with weights 1 (four for each of the two structures, two shared), not four; all orientations form an infinite rational family (`Polyfold.example_9_3_5_classification`) (V9.33). (f) *Lemma 9.4.4 (2017: 9.13)* does not hold as we read it for sheets of different dimensions (`Polyfold.lemma_9_4_4_mixed_dim`) (V9.42), and neither does *Lemma 9.5.2 (2017: 9.16)*: two sheets of different dimensions do not induce the same null sets on their overlap (`Polyfold.isNull_transfer_mixed_dim`) (V9.50). (g) *Omitted steps.* The step of Lemma 9.6.4 "similarly as good points" and the independence of the formula on p. 457 from the structure (V9.61, V9.62), the step "the image is M near 0" in the converse of Lemma 9.2.13 (V9.24), and the facts on the TM_i needed for Definition 9.2.12 (V9.23) are proved or, where Part I lacks the tools, made explicit hypotheses. (h) *Other routes.* The "measure zero" arguments of Lemmas 9.6.1 and 9.6.2 are replaced by a topological covering lemma (V9.58, V9.59); cones in Proposition 9.2.8 are replaced by rays and slices (V9.22); Proposition 9.4.10 is proved without Property (T) (V9.48). Not verified: the global part of Proposition 9.1.11, Remark 9.1.4, Proposition 9.1.9 and Remark 9.1.10, Lemma 9.4.3, the identification $T_y\partial M = T_y^RM$ with the outward direction and the continuity of $o_{\partial M}$, the classification of all orientations in Example 9.3.5, and in §9.5 the forms, integrals and measures: Theorems 9.5.4, 9.5.5 and 9.5.15–9.5.18 with the measure identities of Lemmas 9.5.7, 9.5.9–9.5.11 and Proposition 9.5.13. An addendum (2026-11-12; Findings, items 114–118) verifies Remark 9.1.4 (V9.63, same), Proposition 9.1.9 with the text's groupoid $\bigsqcup M_x$ and the independence of the choices up to a generalized isomorphism (V9.64, restated) and Remark 9.1.10 for n-dimensional Θ, without the composition of the orbifold groupoid in Mathlib's language (V9.65, narrower; gaps G.94, G.95 closed), the classification of all orientations in Example 9.3.5, infinitely many with six of integer weights (V9.33; gap G.101 closed), Lemmas 9.4.4, 9.4.6 and Definition 9.5.1, Lemma 9.5.2 for the sheets of every M-polyfold with null sets read in M-polyfold charts (V9.42, V9.44, V9.49, V9.50 explicit; gaps G.105, G.106 closed), and Theorems 9.5.4, 9.5.5, Lemmas 9.5.7, 9.5.9–9.5.11 and Proposition 9.5.13 (1) for an abstract integration of forms over the oriented sheets, which is not constructed (V9.54–V9.57, V9.66, narrower; gap G.107 in part); Theorems 9.5.15–9.5.18 are proved from their local inputs only (gaps G.108–G.110 in part). A second addendum (2026-11-13; Findings, item 119) constructs the integral of an sc-differential n-form over an oriented n-dimensional sheet, $n \geq 1$, in graph charts and over sheets covered by countably many of them, as such an integration, which contains the sheets of every oriented tame structure of a compact n-dimensional Θ, and verifies with it Theorems 9.5.4, 9.5.5 as printed with "oriented, tame" read as a covering by oriented tame structures, derived when the M-polyfold has no boundary (V9.66, explicit; witness: the unit circle), Lemmas 9.5.9–9.5.11 (V9.54–V9.56, explicit), Proposition 9.5.13 (2) at the level of the integrals (V9.57, restated) and (9.22) as an identity of signed measures (V9.69, explicit; gap G.107 closed); the boundary null sets with the $(n-1)$-dimensional Hausdorff measure (V9.52; gap G.108 closed); the stratum $\{d_M = 1\}$ as an oriented submanifold whose orientation is Part I's face orientation, and Figure 9.1 for face-structured sheets (V9.36, V9.38; gap G.103 closed); and Theorems 9.5.15, 9.5.16 (V9.67) and 9.5.17, 9.5.18 (V9.68) for $n \geq 2$ under an added class condition on the boundary sheets, satisfiable with $\partial S \neq \emptyset$ (the closed upper hemisphere), and open near the corners of the sheets (narrower; gaps G.109, G.110 in part). A third addendum (2026-11-13, wave K; Findings, item 120) verifies Theorems 9.5.15–9.5.18 as printed for every $n \geq 1$, with the text's hypotheses only: the class condition is derived for every oriented tame structure, near the corners of a sheet by an estimate in its corner charts and toward the frontier of $U(x)$ by comparing tangent spaces with other structures; for $n = 1$ the 0-forms are integrated over the finitely many oriented boundary points (V9.67, V9.68, restated; gap G.109 closed). Witnesses with $\partial S \neq \emptyset$: the closed upper hemisphere ($n = 2$, also with LUC) and the closed half-circle ($n = 1$). Remark 9.5.20 is verified in part: the M-polyfold $X \subseteq \mathbb{R} \times \mathbb{C}$ with dimension jumps $0 \to 3 \to 0$, $\partial X = \{(\pm 1, 0)\}$, $\mathbb{Z}_3 \ltimes X$ with LUC, and the effective isotropy of a boundary point, of order 3 read in X and 1 read in ∂X; Θ and the integrals are not (gap G.110 in part, for the remark only).

The hypotheses. Part I has no level-0 sc-calculus inside partial quadrants and no linear structure on T_xX (gap G.7), so four facts of the text's §9.2 are taken as hypotheses on a general M-polyfold, each proved for $\mathbb{R}^m$: the graph representation of the M_i near a point (`Polyfold.MPolyfold.HasGraphChartAt`, `Polyfold.MPolyfold.HasGraphChartsAt`, `Polyfold.MPolyfold.HasGraphBallAt`; Lemmas 9.2.9–9.2.11), the local constancy of the dimension (`Polyfold.MPolyfold.HasLocallyConstantDimAt`; Remark 9.1.1), invariance of domain (`Polyfold.MPolyfold.HasDomainInvarianceAt`; Lemma 9.2.7 (2)) and the level-1 continuity of frame fields (`Polyfold.MPolyfold.HasContinuousLevelOneOn`; Lemma 9.3.4). Their witnesses are `Polyfold.MPolyfold.hasGraphChartAt_rn`, `Polyfold.MPolyfold.hasGraphChartsAt_rn`, `Polyfold.hasGraphBallAt_rn`, `Polyfold.MPolyfold.hasLocallyConstantDimAt_rn`, `Polyfold.MPolyfold.hasDomainInvarianceAt_rn` and `Polyfold.hasContinuousLevelOneOn_rn`. On $\mathbb{R}^m$

the results that use them are proved without hypotheses: `Polyfold.EpGroupoid.proposition_9_1_12_rn`, `Polyfold.EpGroupoid.hasIndependentTangentSums_rn`, `Polyfold.EpGroupoid.lemma_9_2_7_2_rn`, `Polyfold.EpGroupoid.lemma_9_3_4_rn`. Definition 9.2.12 keeps, also on $\mathbb{R}^m$, the hypothesis that the TM_i are smooth submanifolds of TX (gap G.99).

The setting. Ep-groupoids, natural representations of the isotropy groups, the orbit space and paracompact orbit spaces are those of Chapter 7 (`Polyfold.EpGroupoid`, `Polyfold.EpGroupoid.NaturalRep`, `Polyfold.EpGroupoid.HasParacompactOrbitSpace`); the tangent ep-groupoid is Chapter 8's (`Polyfold.EpGroupoid.tangentEpGroupoid`). Smooth finite-dimensional submanifolds (images of sc⁺-retractions), the smooth set, the boundary and the degeneracy index are Part I's (`Polyfold.MPolyfold.IsSmoothSubmanifold`, `Polyfold.MPolyfold.smoothSet`, `Polyfold.MPolyfold.boundary`, `Polyfold.MPolyfold.degIndex`). A functor $\Theta : X \to \mathbb{Q}^+$ is a function on objects with $\Theta(s(\varphi)) = \Theta(t(\varphi))$ (`Polyfold.EpGroupoid.IsFunctorial`), with values in $\mathbb{Q}$ and nonnegativity a separate condition. The tangent space T_yM of a submanifold is a set of tangent vectors (`Polyfold.MPolyfold.submanifoldTangent`); linear algebra is done in charts (`Polyfold.MPolyfold.chartTangent`). Witnesses on $\mathbb{R}^m$ are ep-groupoids with object M-polyfold Part I's $\mathbb{R}^m$ (`Polyfold.rnPolyfold`), most often the trivial one (`Polyfold.EpGroupoid.trivial`). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 9".

V9.1 (Definition 9.1.2 (2017: 9.1), branched ep⁺-subgroupoids; `Polyfold.EpGroupoid.IsFunctorial`, `Polyfold.EpGroupoid.supp`, `Polyfold.EpGroupoid.orbitSupp`, `Polyfold.EpGroupoid.LocalBranchingStructure`, `Polyfold.EpGroupoid.IsBranchedEpSubgroupoid`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_zero`, `Polyfold.EpGroupoid.localBranchingStructure_indicator`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_indicator`, `Polyfold.rnPolyfold_isSmoothSubmanifold_singleton`). A branched ep⁺-subgroupoid is a functor $\Theta \geq 0$ whose support lies in the smooth set, whose orbit support $|\mathrm{supp}(\Theta)|$ is paracompact, and which at every $x \in \mathrm{supp}(\Theta)$ has a local branching structure: an open $U(x)$, finitely many smooth submanifolds $M_i \subseteq U(x)$ through x with proper inclusions, weights $\sigma_i > 0$, and $\Theta(y) = \sum_{y \in M_i} \sigma_i$ on $U(x)$. The zero functor and the indicator of a smooth point of $\mathbb{R}^n$ are examples. *Book:* Definition 9.1.2, p. 402 (2017: Definition 9.1, p. 292). *Hidden hypotheses:* none. *Deviation (restated):* Θ takes values in $\mathbb{Q}$ with $\Theta \geq 0$ a field; "the topology induced on $|\mathrm{supp}(\Theta)|$ is paracompact" is a `ParacompactSpace` instance on the subtype; "the inclusion $M_i \to U(x)$ is proper" is `IsProperMap` of the inclusion. The 2017 text's "(ii) and (iii)" in (4) is "(2) and (3)" in 2021 (possible misprint, corrected) (Findings, item 1). *Update (2026-11-12):* by Remark 9.1.4 the paracompactness in (1) holds when $|\mathrm{supp}(\Theta)|$ is component compact (`Polyfold.EpGroupoid.isBranchedEpSubgroupoid_of_isComponentCompact`, V9.63; gap G.94 closed).

V9.2 (Remark 9.1.1 (2017: 9.1) and the last paragraph of Definition 9.1.2, the dimension of a submanifold and n-dimensional Θ; `Polyfold.MPolyfold.HasDimAt`, `Polyfold.MPolyfold.finrank_tangentSpace_chart_change`, `Polyfold.MPolyfold.HasDimAt.finrank_eq`, `Polyfold.MPolyfold.HasDimAt.unique`, `Polyfold.MPolyfold.IsSmoothSubmanifold.exists_hasDimAt`, `Polyfold.MPolyfold.hasDimAt_inter_isOpen_iff`,

`Polyfold.MPolyfold.hasDimAt_image_iff`, `Polyfold.MPolyfold.IsSubmanifoldOfDim`, `Polyfold.EpGroupoid.LocalBranchingStructure.IsOfDim`, `Polyfold.EpGroupoid.LocalBranchingStructure.dim_unique`, `Polyfold.EpGroupoid.IsNDimensional`, `Polyfold.EpGroupoid.isNDimensional_zero`, `Polyfold.MPolyfold.HasLocallyConstantDimAt`, `Polyfold.MPolyfold.hasLocallyConstantDimAt_rn`**).** A submanifold has dimension k at a if the tangent space of its sub-retract at the chart image of a has dimension k; the value does not depend on the data and the chart, is unique, exists at every point, and is invariant under restriction to open sets and under sc-diffeomorphisms. Θ is n-dimensional if at every point of its support some local branching structure has only n-dimensional M_i; the zero functor is n-dimensional for every n. The dimensions of the M_i of one structure through x are determined. *Book:* Remark 9.1.1, p. 402 (2017: Remark 9.1, p. 291); Definition 9.1.2, last paragraph and footnote, p. 402. *Hidden hypotheses:* none. *Deviation (narrower):* the dimension is measured in a chart, since T_aA carries no linear structure (gap G.7). The local constancy of $a \mapsto \dim T_aA$ is not proved in general; it is the hypothesis `HasLocallyConstantDimAt` where it is needed, proved on $\mathbb{R}^m$ from the trace of $Ds(y)$, and elsewhere the dimensions of the M_i enter as labellings $\dim : I \to \mathbb{N}$. The footnote (n-dimensional $\Theta \neq 0$ is not m-dimensional, $m \neq n$) and "locally equidimensional" are not formalized (gap G.93) (Findings, items 2, 3).

V9.3 (branched ep$^+$-subgroupoids of ep-groupoids over $\mathbb{R}^n$; `Polyfold.EpGroupoid.paracompactSpace_orbitSupp_trivial_rn`, `Polyfold.EpGroupoid.localBranchingStructure_sum_rn`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_sum_rn`, `Polyfold.EpGroupoid.isNDimensional_sum_rn`, `Polyfold.EpGroupoid.localBranchingStructure_indicator_rn`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_indicator_rn`, `Polyfold.rnPolyfold_isSmoothSubmanifold_range`, `Polyfold.rnPolyfold_hasDimAt_range`, `Polyfold.graph2`, `Polyfold.rnPolyfold_isSubmanifoldOfDim_graph2`, `Polyfold.EpGroupoid.rnPolyfold_isSubmanifoldOfDim_singleton`**).** On the trivial ep-groupoid of $\mathbb{R}^n$, finite weighted sums of indicators of closed submanifolds through which the sum is taken, and indicators of locally closed submanifolds, are branched ep$^+$-subgroupoids, n-dimensional when all submanifolds are. The image of a smooth idempotent s of $\mathbb{R}^n$ is a closed smooth submanifold of dimension $\operatorname{rank} Ds(a)$; graphs of smooth $\beta : \mathbb{R} \to \mathbb{R}$ in $\mathbb{R}^2$ are closed 1-dimensional submanifolds; points are 0-dimensional. *Book:* not in the book. *Hidden hypotheses:* none. *Deviation (extra):* witnesses for the hypotheses of the chapter (Findings, item 100).

V9.4 (Remark 9.1.3 (b) (2017: 9.2 (b)), branching structures are not unique; `Polyfold.EpGroupoid.Remark913Hyp`, `Polyfold.EpGroupoid.remark913Hyp_expNegInvGlue`, `Polyfold.EpGroupoid.thetaGraphs`, `Polyfold.EpGroupoid.thetaGraphs_eq`, `Polyfold.EpGroupoid.Remark913Hyp.structure `, `Polyfold.EpGroupoid.Remark913Hyp.structure `, `Polyfold.EpGroupoid.Remark913Hyp. _pos`, `Polyfold.EpGroupoid.Remark913Hyp.eq_add_of_thetaGraphs_eq`, `Polyfold.EpGroupoid.remark_9_1_3_b`**).** For smooth $\beta_1, \beta_2 : \mathbb{R} \to \mathbb{R}$ with the sign conditions of the text (satisfiable, from `expNegInvGlue`), $\Theta(p) = \#\{i \mid p \in \operatorname{graph} \beta_i\}$ is a 1-dimensional branched ep$^+$-subgroupoid of the trivial ep-groupoid of $\mathbb{R}^2$, equal to the one given by the graphs of 0 and $\beta_1 + \beta_2$, and no submanifold of the one structure agrees with a submanifold of the other on any neighbourhood of the origin. *Book:* Remark 9.1.3 (b), p. 403 (2017:

Remark 9.2 (b), p. 292). *Hidden hypotheses:* none. *Deviation (restated):* the text gives γ_2 only by $\gamma_2(s) > 0$ for $s \neq 0$; describing the same Θ forces $\gamma_2 = \beta_1 + \beta_2$; "not unique, even close to x" is read as the last conjunct (Findings, item 4).

V9.5 (Lemma 9.1.5 (2017: 9.1), $|\mathrm{supp}(\Theta)|$ is locally closed; `Polyfold.EpGroupoid.preimage__orbitSupp`, `Polyfold.EpGroupoid.LocalBranchingStructure.mem_supp_iff`, `Polyfold.EpGroupoid.LocalBranchingStructure.isClosed_preimage_val_supp`, `Polyfold.EpGroupoid.isLocallyClosed_supp`, `Polyfold.EpGroupoid.lemma_9_1_5`). For a branched ep⁺-subgroupoid Θ, $\pi^{-1}(|\mathrm{supp}(\Theta)|) = \mathrm{supp}(\Theta)$, $\mathrm{supp}(\Theta) \cap U(x) = \bigcup_i M_i$ is closed in $U(x)$, $\mathrm{supp}(\Theta)$ is locally closed in X and $|\mathrm{supp}(\Theta)|$ is locally closed in $|X|$. *Book:* Lemma 9.1.5, p. 403 (2017: Lemma 9.1, p. 293). *Hidden hypotheses:* none. *Deviation (same):* the openness of π (Chapter 7) replaces the text's neighbourhood with the properness property (Findings, item 5).

V9.6 (Definition 9.1.6 (2017: 9.2), closed and compact Θ; `Polyfold.EpGroupoid.IsClosedBranched`, `Polyfold.EpGroupoid.IsCompactBranched`, `Polyfold.EpGroupoid.isCompactBranched_indicator`). Θ is closed if $|\mathrm{supp}(\Theta)|$ is closed in $|X|$, compact if it is compact; the indicator of a point of $\mathbb{R}^n$ is compact. *Book:* Definition 9.1.6, p. 404 (2017: Definition 9.2, p. 293). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 7).

V9.7 (the restriction of a local branching structure, and Lemma 9.1.7 (2017: 9.2), symmetric structures; `Polyfold.EpGroupoid.LocalBranchingStructure.restrict`, `Polyfold.EpGroupoid.LocalBranchingStructure.isClosed_M`, `Polyfold.isProperMap_inclusion_iff`, `Polyfold.EpGroupoid.SymmetricBranchingStructure`, `Polyfold.EpGroupoid.IsFunctorial.apply_Φ`, `Polyfold.EpGroupoid.LocalBranchingStructure.symmetrize`, `Polyfold.EpGroupoid.lemma_9_1_7`, `Polyfold.EpGroupoid.lemma_9_1_7'`). A local branching structure at x restricts to every open $V \subseteq U(x)$ containing x. For a branched ep⁺-subgroupoid and $x \in \mathrm{supp}(\Theta)$ there is, inside every open $V \ni x$, a symmetric local branching structure: the domain of a natural representation (Φ, Γ) of G_x, a G_x-action on the index set with $\Phi(g)(M_i) = M_{g \cdot i}$ and $\sigma_{g \cdot i} = \sigma_i$, and $\Theta(y) = \sum_{y \in M_i} \sigma_i$ there; it is obtained with $J = I \times G_x$, $M_{(i,g)} = \Phi(g)(M_i \cap U)$ and $\sigma_{(i,g)} = \sigma_i / |G_x|$. *Book:* the remark before Lemma 9.1.7, p. 404, and Lemma 9.1.7, p. 405 (2017: Lemma 9.2, p. 294). *Hidden hypotheses:* none. *Deviation (stronger):* the symmetric structure lies inside any given V. The restriction remark is formalized at x itself, not at a point x_0 of $V(x_0) \subseteq U(x)$, where the N_i may miss x_0 (reading). That $\Phi(g)(M_i \cap U)$ is a submanifold properly embedded in U through x, used without statement in the text, is proved (Findings, items 5, 6).

V9.8 (Definition 9.1.8 (2017: 9.3), manifold- and orbifold-type; `Polyfold.EpGroupoid.IsOrbifoldType`, `Polyfold.EpGroupoid.IsManifoldType`, `Polyfold.EpGroupoid.isManifoldType_indicator`). Θ is of orbifold type if it takes only the values 0 and 1, of manifold type if moreover between two objects of $\mathrm{supp}(\Theta)$ there is at most one morphism; the indicator of a point of $\mathbb{R}^n$ is of manifold type. *Book:* Definition 9.1.8, p. 405 (2017: Definition 9.3, p. 294). *Hidden hypotheses:* none. *Deviation (same):* none; Proposition 9.1.9 and Remark 9.1.10, which build on it, were not formalized (gap G.95) (Findings, item 7). *Update (2026-11-12):* Proposition 9.1.9 and Remark 9.1.10 are formalized (V9.64, V9.65; gap G.95 closed; Findings, item

114).

V9.9 (Proposition 9.1.11 (2017: 9.2), the dimensional decomposition; `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPartStructure`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPartStructure_isOfDim`, `Polyfold.EpGroupoid.LocalBranchingStructure.eq_sum_dimPart`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_eq_zero`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_eq_of_isOfDim`, `Polyfold.EpGroupoid.SymmetricBranchingStructure.dim_smul`, `Polyfold.EpGroupoid.SymmetricBranchingStructure.dimPart_Φ`, `Polyfold.EpGroupoid.IsDimDecomposition`, `Polyfold.EpGroupoid.IsLocallyFiniteFamily`, `Polyfold.EpGroupoid.Proposition_9_1_11`, `Polyfold.EpGroupoid.axis2`, `Polyfold.EpGroupoid.decompA`, `Polyfold.EpGroupoid.decompB`, `Polyfold.EpGroupoid.decompA_isDimDecomposition`, `Polyfold.EpGroupoid.decompB_isDimDecomposition`, `Polyfold.EpGroupoid.not_proposition_9_1_11_rn`). For a local branching structure with dimension labelling, $\Theta_k(y) = \sum_{\dim M_i = k,\, y \in M_i} \sigma_i$ is represented on $U(x)$ by the k-dimensional structure $(M_i)_{\dim M_i = k}$, $\Theta = \sum_k \Theta_k$ on $U(x)$, and for a symmetric structure G_x preserves dimension and Θ_k is G_x-invariant. The statement of the proposition, "a unique family (Θ_k) of branched ep$^+$-subgroupoids, Θ_k k-dimensional, with $\Theta = \sum_k \Theta_k$ pointwise, and every such family locally finite", is `Proposition_9_1_11`; on the trivial ep-groupoid of $\mathbb{R}^2$ it does not hold as we read it: for the indicator Θ of the x-axis L, both $\Theta_0 = 1_{\{0\}}$, $\Theta_1 = 1_{L \setminus \{0\}}$ and $\Theta_0 = 0$, $\Theta_1 = 1_L$ are such families. *Book:* Proposition 9.1.11, pp. 407–408 (2017: Proposition 9.2, p. 297). *Hidden hypotheses:* the dimension labellings of V9.2. *Deviation (narrower):* only the local construction is verified; the gluing of the local Θ_k into functors on X (compatibility across morphisms, paracompactness of $|\text{supp}(\Theta_k)|$, local finiteness) and a uniqueness statement under an extra condition (for instance that each Θ_k be represented near every point by a subfamily of a local branching structure of Θ) are not formalized (gap G.96) (Findings, item 8).

V9.10 (Proposition 9.1.12, the decomposition does not depend on the structure; `Polyfold.EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear`, `Polyfold.EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear.exists_naturalRep`, `Polyfold.EpGroupoid.Proposition_9_1_12`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_isOfDim`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_topMultiplicity`, `Polyfold.EpGroupoid.proposition_9_1_12_of_topMultiplicity`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_hasGraphChartAt`, `Polyfold.EpGroupoid.proposition_9_1_12_of_hasGraphChartAt`, `Polyfold.EpGroupoid.proposition_9_1_12_rn`, `Polyfold.proposition_9_1_12_rn_zero`, `Polyfold.proposition_9_1_12_rn_one`, `Polyfold.exists_mixed_rn_one`, `Polyfold.EpGroupoid.Remark913Hyp.exists_mixed_rn_two`). For two local branching structures of Θ at x with dimension labellings there is an open $V \ni x$ in both domains on which their dimensional decompositions agree, and V can be taken to be the domain of a natural representation of G_x. This holds for equidimensional structures without hypothesis, for every M-polyfold with `TopMultiplicity` in every dimension or with `HasGraphChartAt` at every point, and, without hypothesis, for every ep-groupoid whose object M-polyfold is $\mathbb{R}^m$. Structures with sheets of different dimensions exist on $\mathbb{R}^1$ and $\mathbb{R}^2$. *Book:* Proposition 9.1.12, p. 408, proof in §9.6.1, pp. 452–457, new in the 2021 edition (the 2017 proof of Proposition 9.2 calls the uniqueness obvious). *Hidden hypotheses:* the graph representation `HasGraphChartAt` (Lemmas 9.2.9–9.2.11) on a general M-polyfold. *Deviation (explicit):* "open, G_x-invariant, with the usual properties" is any open V in both domains, shrunk to a natural representation by `exists_naturalRep`; the hypothesis is discharged on $\mathbb{R}^m$ (gap G.97 in general) (Findings, items 9, 90–93).

V9.11 (the category $\mathrm{Gr}(X)$, p. 409; `Polyfold.EpGroupoid.tangentMor_unit_of_eq`, `Polyfold.EpGroupoid.tangentMor_injOn`, `Polyfold.EpGroupoid.tangentMor_eq_tangentMapOn_hat`, `Polyfold.EpGroupoid.Gr`, `Polyfold.EpGroupoid.grPush`, `Polyfold.EpGroupoid.grPush_mem`, `Polyfold.EpGroupoid.grPush_comp`, `Polyfold.EpGroupoid.grPush_unit`, `Polyfold.EpGroupoid.GrObj`, `Polyfold.EpGroupoid.SmoothCat`, `Polyfold.EpGroupoid.grProj`). $\mathrm{Gr}(x)$ is the set of finite formal sums $\sum_L \sigma_L L$, $\sigma_L \in \mathbb{Q}$, of subsets L of T_xX; a morphism $\varphi : x \to y$ of X^∞ pushes them forward by the tangent $T\varphi$ of Chapter 7, functorially; the objects (x, a), $a \in \mathrm{Gr}(x)$, and the pairs (a, φ) form a category with the projection functor π to the full subcategory X^∞ of smooth objects. *Book:* §9.2, p. 409 (2017 pp. 297–298). *Hidden hypotheses:* none. *Deviation (restated):* the generators are subsets of T_xX rather than linear subspaces, since T_xX carries no linear structure (gap G.7); in a chart the tangent spaces of submanifolds are linear subspaces (V9.12) (Findings, item 20).

V9.12 (the tangent space T_yM of a submanifold, set form and chart form; `Polyfold.MPolyfold.submanifoldTangent`, `Polyfold.MPolyfold.mem_submanifoldTangent_iff`, `Polyfold.MPolyfold.submanifoldTangent_subset_tangentFiber`, `Polyfold.MPolyfold.submanifoldTangent_inter_isOpen`, `Polyfold.MPolyfold.submanifoldTangent_image`, `Polyfold.MPolyfold.chartTangent`, `Polyfold.MPolyfold.chartTangent_eq`, `Polyfold.MPolyfold.exists_submodule_chartTangent`, `Polyfold.MPolyfold.finrank_chartTangent`, `Polyfold.MPolyfold.chartTangent_chart_change`, `Polyfold.MPolyfold.exists_linearEquiv_chartTangent`, `Polyfold.EpGroupoid.LocalBranchingStructure.exists_submodule_of_mem_support`). T_yM is the set of tangent vectors at y fixed by the tangent of the sc⁺-retraction that presents M; it does not depend on the retraction, is local, and is carried by sc-diffeomorphisms. In a chart its image is a linear subspace of the dimension of M at y, carried linearly by chart changes; the generators T_xM_i of $T_\Theta(x)$ are linear subspaces in charts. *Book:* §9.2, the tangent spaces T_xM_i before Theorem 9.2.1, p. 409 (2017 p. 298), with Part I's Remark 4.2.2 (preprint: 4.14). *Hidden hypotheses:* none. *Deviation (restated):* sets instead of subspaces (gap G.7) (Findings, item 20).

V9.13 (Theorem 9.2.1 (2017: 9.1), the tangent section T_Θ; `Polyfold.EpGroupoid.LocalBranchingStructure.tangentSum`, `Polyfold.EpGroupoid.LocalBranchingStructure.transport`, `Polyfold.EpGroupoid.LocalBranchingStructure.tangentSum_transport`, `Polyfold.EpGroupoid.HasIndependentTangentSums`, `Polyfold.EpGroupoid.tangentValue`, `Polyfold.EpGroupoid.tangentValue_eq`, `Polyfold.EpGroupoid.grPush_tangentValue`, `Polyfold.EpGroupoid.tangentSection`, `Polyfold.EpGroupoid.theorem_9_2_1`). If the sums

$\sum_i \sigma_i\, T_x M_i$ of any two local branching structures of Θ at x agree (`HasIndependentTangentSums`), there is a functor $T_\Theta : X^\infty \to \mathrm{Gr}(X)$ with $\pi \circ T_\Theta = \mathrm{Id}$, $T_\Theta(x) = \sum_i \sigma_i\, T_x M_i$ for every local branching structure at $x \in \mathrm{supp}(\Theta)$, $T_\Theta(x) = 0$ off the support, and $T_\Theta(\varphi) = (T_\Theta(s(\varphi)), \varphi)$. *Book:* Theorem 9.2.1, p. 409 (2017: Theorem 9.1, p. 298), proved on pp. 412–418 (2017 pp. 301–306). *Hidden hypotheses:* the independence, which the text proves through Proposition 9.2.8 (V9.14). *Deviation (explicit):* the functoriality on morphisms uses the transport of structures along $\hat{\varphi}$ and $\Theta \circ \hat{\varphi} = \Theta$ (Findings, items 21, 22).

V9.14 (Theorem 9.2.1 (2017: 9.1), the independence of the local branching structure; `Polyfold.EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isGoodPoint`, `Polyfold.EpGroupoid.hasIndependentTangentSums_of_subset_goodSet`, `Polyfold.EpGroupoid.hasIndependentTangentSums_of_isOfDim`, `Polyfold.EpGroupoid.hasIndependentTangentSums_of_dimPartsAgreeNear`, `Polyfold.EpGroupoid.hasIndependentTangentSums_of_isOfDim_of_hasGraphChartsAt`, `Polyfold.EpGroupoid.hasIndependentTangentSums_of_hasGraphChartAt_of_hasLocallyConstantDimAt`, `Polyfold.EpGroupoid.hasIndependentTangentSums_rn`, `Polyfold.EpGroupoid.Remark913Hyp.tangentSum_structure_eq_structure`, `Polyfold.hasIndependentTangentSums_indicator`). The sums are independent at good points without hypothesis; for equidimensional structures under the slice hypothesis `HasTangentSlicesAt`, which follows from `HasGraphChartsAt`; in general under `HasGraphChartAt` and `HasLocallyConstantDimAt` on the support; and for every Θ on an ep-groupoid whose object M-polyfold is $\mathbb{R}^m$, without hypothesis. In the example of Remark 9.1.3 (b) the two structures have the same tangent sum at the bad point 0. *Book:* Theorem 9.2.1, p. 409 (2017: Theorem 9.1, p. 298), the proof on pp. 412–418 (2017 pp. 301–306). *Hidden hypotheses:* the graph representation and the local constancy of the dimension on a general M-polyfold. *Deviation (explicit):* the text reduces to the equidimensional case by Proposition 9.1.11; the formalization does the same through Proposition 9.1.12 (V9.10) with dimension labellings (gaps G.93, G.97) (Findings, items 21, 25, 100).

V9.15 (Definition 9.2.2 (2017: 9.4), good points; `Polyfold.EpGroupoid.LocalBranchingStructure.MU`, `Polyfold.EpGroupoid.LocalBranchingStructure.IsGoodPoint`, `Polyfold.EpGroupoid.LocalBranchingStructure.isGoodPoint_iff`, `Polyfold.EpGroupoid.LocalBranchingStructure.nbhdAt`, `Polyfold.EpGroupoid.LocalBranchingStructure.eq_iff_of_mem_nbhdAt`). $y \in M_U = \bigcup_i M_i$ is good for a local branching structure if near y all M_i through y coincide; equivalently, $y \in \mathrm{supp}(\Theta)$ and Θ is constant on $\mathrm{supp}(\Theta)$ near y. *Book:* Definition 9.2.2, p. 410 (2017: Definition 9.4, p. 299). *Hidden hypotheses:* none. *Deviation (restated):* stated for every Θ with a local branching structure, not only equidimensional ones; the intrinsic form is the argument of the proof of Lemma 9.2.3 (Findings, item 23).

V9.16 (Lemma 9.2.3 (2017: 9.3), being good does not depend on the structure; `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_2_3`). If y lies in the domains of two local branching structures of Θ, at possibly different points, and is good for one, it is good for the other. *Book:* Lemma 9.2.3, p. 410 (2017: Lemma 9.3, p. 299). *Hidden hypotheses:* none. *Deviation (stronger):*

any two structures whose domains contain y, any Θ; in the proof $Q(z)$ is presumably $Q(y)$ (possible misprint) (Findings, item 23).

V9.17 (Remark 9.2.4 (2017: 9.4); `Polyfold.EpGroupoid.remark_9_2_4_a`, `Polyfold.EpGroupoid.mem_goodSet_iff`, `Polyfold.EpGroupoid.remark_9_2_4_b`). (a) A point good for some structure is good for every structure whose domain contains it. (b) Near a good point y, $\Theta = \Theta(y) \cdot 1_M$ for one submanifold M properly embedded in an open $V \ni y$. *Book:* Remark 9.2.4, p. 411 (2017: Remark 9.4, p. 300). *Hidden hypotheses:* none. *Deviation (restated):* (b) with $M = M_i \cap V(y)$ for any i with $y \in M_i$ (Findings, item 23).

V9.18 (Lemma 9.2.5 (2017: 9.4), good points are preserved by morphisms; `Polyfold.EpGroupoid.Sheet.apply_hat`, `Polyfold.EpGroupoid.src_mem_goodSet_of_tgt`, `Polyfold.EpGroupoid.lemma_9_2_5`). For every morphism φ, $s(\varphi)$ is good iff $t(\varphi)$ is. *Book:* Lemma 9.2.5, p. 411 (2017: Lemma 9.4, p. 300). *Hidden hypotheses:* none. *Deviation (stronger):* for every branched ep⁺-subgroupoid, not only equidimensional ones; the transport of structures is replaced by $\Theta \circ \hat{\varphi} = \Theta$ on the sheet of φ (Findings, item 23).

V9.19 (Definition 9.2.6 (2017: 9.5), the good and the bad set; `Polyfold.EpGroupoid.goodSet`, `Polyfold.EpGroupoid.badSet`, `Polyfold.EpGroupoid.preimage_ _image_goodSet`, `Polyfold.EpGroupoid.preimage_ _image_badSet`). S^g_Θ is the set of points of $\mathrm{supp}(\Theta)$ good for some structure whose domain contains them, $S^b_\Theta = \mathrm{supp}(\Theta) \setminus S^g_\Theta$; both are saturated, so they descend to $|X|$. *Book:* Definition 9.2.6, p. 411 (2017: Definition 9.5, p. 300). *Hidden hypotheses:* none. *Deviation (restated):* the text cites "Remark 9.2.4 (c)" for the descent; Remark 9.2.4 has (a) and (b) only, and Lemma 9.2.5 is what is used (possible misprint) (Findings, item 23).

V9.20 (Lemma 9.2.7 (1) (2017: 9.5 (1)), the good set is open and dense; `Polyfold.EpGroupoid.lemma_9_2_7_1`, `Polyfold.EpGroupoid.dense_goodSet`, `Polyfold.EpGroupoid.isNowhereDense_badSet`, `Polyfold.EpGroupoid.lemma_9_2_7_1_dense`). In $\mathrm{supp}(\Theta)$ the good set is open and dense and the bad set is closed and nowhere dense. *Book:* Lemma 9.2.7 (1), p. 412 (2017: Lemma 9.5 (1), p. 300). *Hidden hypotheses:* none. *Deviation (stronger):* for every branched ep⁺-subgroupoid, without equidimensionality and without (2), from a topological covering argument (Findings, item 24).

V9.21 (Lemma 9.2.7 (2) (2017: 9.5 (2)), good points accumulate on every sheet; `Polyfold.MPolyfold.HasDomainInvarianceAt`, `Polyfold.MPolyfold.hasDomainInvarianceAt_rn`, `Polyfold.EpGroupoid.lemma_9_2_7_2`, `Polyfold.EpGroupoid.lemma_9_2_7_2_rn`, `Polyfold.EpGroupoid.Remark913Hyp.zero_mem_badSet`, `Polyfold.EpGroupoid.Remark913Hyp.exists_seq_goodSet_structure`). For an n-dimensional local branching structure at x and every i there is a sequence of good points in M_i converging to x, under invariance of domain for n-dimensional submanifolds on $U(x)$, which holds on $\mathbb{R}^m$; on $\mathbb{R}^m$ the statement has no hypothesis. In Remark 9.1.3 (b) the origin is bad and is approached by good points of the graph of β_1. *Book:* Lemma 9.2.7 (2), p. 412 (2017: Lemma 9.5 (2), p. 301). *Hidden hypotheses:* invariance of domain (`HasDomainInvarianceAt`) on a general M-polyfold. *Deviation (explicit):* the statement holds at every x, not only at bad points (gap G.98) (Findings, item 24).

V9.22 (Proposition 9.2.8 and Lemmas 9.2.9–9.2.11 (2017: Proposition 9.3, Lemmas 9.6–9.8), the graph representation and tangent slices; `Polyfold.MPolyfold.HasTangentSlicesAt`, `Polyfold.MPolyfold.IsGraphChartAt`, `Polyfold.MPolyfold.HasGraphChartsAt`, `Polyfold.MPolyfold.hasTangentSlicesAt_of_hasGraphChartsAt`, `Polyfold.exists_mem_graph_of_hasStrictFDerivAt`, `Polyfold.MPolyfold.isGraphChartAt_rn`, `Polyfold.MPolyfold.hasGraphChartsAt_rn`, `Polyfold.MPolyfold.hasTangentSlicesAt_rn`, `Polyfold.MPolyfold.HasGraphChartAt`, `Polyfold.MPolyfold.hasGraphChartAt_rn`, `Polyfold.MPolyfold.exists_slice_of_isGraphChartAt`, `Polyfold.MPolyfold.exists_generic_slice`, `Polyfold.MPolyfold.IsGraphBallAt`,`Polyfold.MPolyfold.HasGraphBallAt`,`Polyfold.hasGraphBallAt_rn`, `Polyfold.MPolyfold.hasOrientedGoodPointsAt_of_hasGraphBallAt`). If near x every n-dimensional submanifold through x is, in one chart, a graph over a ray (or ball) in its tangent space (`IsGraphChartAt`, `HasGraphChartAt` for all dimensions at once, `HasGraphBallAt`), then for every class Ξ of sheets with one tangent space L at x and every neighbourhood W of x there is a finite slice of points in W meeting each M_i, $i \in \Xi$, once and missing the other M_i, so that $\sum_{i\in\Xi} \sigma_i = \sum_{y\in F} \Theta(y)$ (`HasTangentSlicesAt`), and the slice can be chosen generic. On $\mathbb{R}^m$ the graph representation holds at every point and in every dimension, by the inverse function theorem. *Book:* Proposition 9.2.8, p. 413 (2017: Proposition 9.3, p. 302), Lemmas 9.2.9–9.2.11, pp. 414–417 (2017: Lemmas 9.6–9.8, pp. 302–305). *Hidden hypotheses:* the graph representation on a general M-polyfold. *Deviation (narrower):* Proposition 9.2.8 is formalized in slice form, not with the text's subsets I^e and sequences of good points; rays and balls replace the cones around h_0; Lemmas 9.2.9–9.2.11 are the hypotheses, proved on $\mathbb{R}^m$ (gap G.97) (Findings, item 25).

V9.23 (Definition 9.2.12 (2017: 9.6), the tangent $T\Theta$; `Polyfold.EpGroupoid.tangentSubmanifold`, `Polyfold.EpGroupoid.tangentTheta`, `Polyfold.EpGroupoid.tangentTheta_eq`, `Polyfold.EpGroupoid.sum_tangent_eq_sum_tangent`, `Polyfold.EpGroupoid.tangentTheta_nonneg`, `Polyfold.EpGroupoid.isFunctorial_tangentTheta`, `Polyfold.EpGroupoid.TangentBranchingHyp`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_tangentTheta`, `Polyfold.EpGroupoid.isClosed_tangentSubmanifold`, `Polyfold.EpGroupoid.paracompactSpace_orbitSupp_tangentTheta`, `Polyfold.EpGroupoid.tangentBranchingHyp_of_isSmoothSubmanifold`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_of_isSmoothSubmanifold`, `Polyfold.EpGroupoid.LocalBranchingStructure.isSubMPolyfold_tangentSubmanifold`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_rn`, `Polyfold.EpGroupoid.tangentBranchingHyp_zero`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_zero`). Under the independence of V9.13, $T\Theta(h) = \sum_{h\in TM_i} \sigma_i$ for h over $U(x)$, and 0 over the complement of the support, is well defined, nonnegative and a functor on the tangent ep-groupoid; it is a branched ep$^+$-subgroupoid of TX if the TM_i are smooth submanifolds of TX (closedness and the paracompactness of $|\mathrm{supp}(T\Theta)|$ for paracompact $|X|$ are proved), and on $\mathbb{R}^m$ the TM_i are sub-M-polyfolds of TX, retracts of Ts. *Book:* Definition 9.2.12, p. 418 (2017: Definition 9.6, p. 306). *Hidden hypotheses:* that the TM_i are smooth submanifolds of TX (the tangent of the sc$^+$-retraction is sc$^+$ on TX), and the paracompactness of $|X|$, neither discussed in the text. *Deviation (explicit):* $h \in TM_i$ is read as $h \in T_{p(h)}M_i$; only the zero functor is a witness of the full hypothesis (gap G.99) (Findings, item 26). *Update (2026-11-12):* our Proposition 9.1.9 does not use the paracompactness of $|X|$ (V9.64).

V9.24 (Lemma 9.2.13 (2017: 9.9), $d_M(x) = 0$ and reduced tangent position; `Polyfold.IsReducedTangentAt`, `Polyfold.isReducedTangentAt_iff_subset_reduced`, `Polyfold.isReducedTangentAt_iff_subset_partialCone`, `Polyfold.MPolyfold.subDegIndex`, `Polyfold.MPolyfold.subDegIndex_le`, `Polyfold.MPolyfold.IsTangentReducedAt`, `Polyfold.MPolyfold.isReducedTangentAt_chart_change`, `Polyfold.MPolyfold.isTangentReducedAt_iff_forall`, `Polyfold.MPolyfold.lemma_9_2_13_chart`, `Polyfold.MPolyfold.lemma_9_2_13`, `Polyfold.MPolyfold.lemma_9_2_13_two_one`, `Polyfold.MPolyfold.lemma_9_2_13_iff`, `Polyfold.MPolyfold.subDegIndex_eq_zero_iff`, `Polyfold.ScRetraction.IsScPlus.exists_injOn_proj`, `Polyfold.ScRetraction.IsScPlus.exists_flat`, `Polyfold.MPolyfold.not_lemma_9_2_13_two_one`). For a smooth submanifold M and $x \in M \cap \partial X$: $d_M(x) = 0$ iff $T_xM \subseteq T_x^RX$ (in every chart, the chart image of T_xM lies in the reduced tangent space). For every $x \in M$: $d_M(x) = 0$ iff $T_xM \subseteq T_x^RX$. As printed, with "$x \in \partial X$" in (1), (2) □ (1) does not hold as we read it at a point p of $\mathbb{R}^n$, which satisfies (2) and lies off $\partial\mathbb{R}^n$. *Book:* Lemma 9.2.13, p. 419 (2017: Lemma 9.9, p. 307). *Hidden hypotheses:* none. *Deviation (stronger):* read with $x \in M \cap \partial X$ as the standing assumption, which is where the text uses it, and proved at every $x \in M$ in the form `subDegIndex_eq_zero_iff`. The converse needs a chart of M without boundary near x: by the inverse function theorem on T_xM and the step that the projection is injective on M near x ("the image is M near 0"), which the text does not spell out; the text's complement Y and map F are not needed (Findings, items 40, 41).

V9.25 (Remark 9.2.14 (2017: 9.5), ∂M; `Polyfold.MPolyfold.remark_9_2_14`, `Polyfold.MPolyfold.remark_9_2_14_mem_boundary`, `Polyfold.MPolyfold.mem_boundary_subPolyfold_iff`, `Polyfold.MPolyfold.remark_9_2_14_not_isTangentReducedAt`, `Polyfold.MPolyfold.remark_9_2_14_iff`, `Polyfold.MPolyfold.remark_9_2_14_not_mem_boundary_iff`). $x \in \partial M$ iff $x \in \partial X$ and $T_xM \nsubseteq T_x^RX$; in particular $\partial M \subseteq \partial X$, and $d_M(x) \geq 1$ at such points. *Book:* Remark 9.2.14, p. 419 (2017: Remark 9.5, p. 307). *Hidden hypotheses:* none. *Deviation (restated):* ∂M is the boundary of the induced M-polyfold structure; the text's "not contained in C_xX" is equivalent to the reduced-tangent form (`isReducedTangentAt_iff_subset_partialQuadrant`) (Findings, item 42).

V9.26 (Proposition 9.2.15 (2017: 9.4), the boundary weights; `Polyfold.MPolyfold.submanifoldBoundary`, `Polyfold.MPolyfold.mem_submanifoldBoundary_iff`, `Polyfold.EpGroupoid.boundaryPart`, `Polyfold.EpGroupoid.boundaryTheta`, `Polyfold.EpGroupoid.boundaryTheta_eq`, `Polyfold.EpGroupoid.proposition_9_2_15_1`, `Polyfold.EpGroupoid.proposition_9_2_15_2`, `Polyfold.EpGroupoid.isFunctorial_boundaryTheta`, `Polyfold.EpGroupoid.proposition_9_2_15_3`, `Polyfold.EpGroupoid.filter_tangentValue_eq`). (1) Under the independence of V9.13, $\sum_{y \in \partial M_i} \sigma_i$ is the same for any two local branching structures whose domains contain y. (2) At y with $d_X(y) = 0$ this sum and the corresponding formal sum of tangent spaces vanish, for every structure. (3) $\partial\Theta$, the part of $T_\Theta(y)$ on tangent spaces not contained in T_y^RX, is a functor, nonnegative, supported in ∂X, and equal to $\sum_{y \in \partial M_i} \sigma_i$ on every structure. *Book:* Proposition 9.2.15, p. 420 (2017: Proposition 9.4, p. 308). *Hidden hypotheses:* the independence of Theorem 9.2.1 for (1) and (3). *Deviation (explicit):* (1) and (2) need neither $x \in \mathrm{supp}(\Theta) \cap \partial X$ nor that Θ be branched, and (2) needs no independence (Findings, item 43).

V9.27 (Definition 9.2.16 (2017: 9.7), the boundary $\partial\Theta$; `Polyfold.EpGroupoid.IsBoundaryCharacterized`, `Polyfold.EpGroupoid.isBoundaryCharacterized_boundaryTheta`, `Polyfold.EpGroupoid.isBoundaryCharacterized_zero`, `Polyfold.EpGroupoid.eq_boundaryTheta`, `Polyfold.not_unique_isBoundaryCharacterized_halfLine`). $\partial\Theta$ satisfies the text's (1) $\mathrm{supp}(\partial\Theta) \subseteq \partial X$ and (2) the local formula at points of $\mathrm{supp}(\partial\Theta)$; a functor that vanishes off $\mathrm{supp}(\Theta)$ and satisfies (2) at the points of $\mathrm{supp}(\Theta)$ equals $\partial\Theta$. As stated, (1) and (2) do not characterize $\partial\Theta$ uniquely as we read them: the zero functor satisfies them for every Θ, and for $\Theta = 1$ on $[0, \infty)$ both 0 and $\partial\Theta \neq 0$ do. *Book:* Definition 9.2.16, p. 421 (2017: Definition 9.7, p. 309). *Hidden hypotheses:* the independence of Theorem 9.2.1. *Deviation (narrower):* "uniquely characterized" is verified with (2) asked on $\mathrm{supp}(\Theta)$. "In view of Proposition 9.2.13" before the definition (2017: "Proposition 9.9") is presumably Proposition 9.2.15, and "branched ep$^+$-groupoid" is "ep$^+$-subgroupoid" (possible misprints) (Findings, items 44, 45). *Update (2026-11-12):* read as "the solution of (1), (2) with the largest support", the characterization is unique: among the functions $F \geq 0$ vanishing off $\mathrm{supp}(\Theta)$ with (1) and (2), $\partial\Theta$ is the one with the largest support (`Polyfold.EpGroupoid.definition_9_2_16_existsUnique`, `Polyfold.EpGroupoid.IsBoundaryCharacterized.eq_boundaryTheta_iff`), and every such F equals $\partial\Theta$ on $U(x)$ for $x \in \mathrm{supp}(F)$ (`Polyfold.EpGroupoid.IsBoundaryCharacterized.eq_boundaryTheta_on`); the independence of Theorem 9.2.1 is a theorem, so there is no hidden hypothesis (Findings, item 44).

V9.28 (Remark 9.2.17 (2017: 9.6); `Polyfold.EpGroupoid.boundaryTheta_le`, `Polyfold.boundaryTheta_halfLineCorner`, `Polyfold.boundaryTheta_halfLine_one`, `Polyfold.EpGroupoid.boundaryTheta_indicator_of_mem`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_one`, `Polyfold.EpGroupoid.boundaryTheta_one`). $\partial\Theta \leq \Theta$; $\partial\Theta$ is in general not the restriction of Θ to ∂X: for the indicator of the corner 0 of $[0, \infty)$, $\Theta(0) = 1$ and $\partial\Theta(0) = 0$, while for $\Theta = 1$ on $[0, \infty)$, $\partial\Theta(0) = 1$. *Book:* Remark 9.2.17, p. 421 (2017: Remark 9.6, p. 310). *Hidden hypotheses:* the independence of Theorem 9.2.1, which the witnesses satisfy. *Deviation (restated):* the forward reference "In Section 9.5, Definition 9.3.6" is to §9.3 (possible misprint, also in 2017) (Findings, items 45, 46).

V9.29 (the category $\widehat{\mathrm{Gr}}(X)$ and oriented tangent spaces, p. 422; `Polyfold.MPolyfold.chartVec`, `Polyfold.MPolyfold.OrTangent`, `Polyfold.MPolyfold.chartFrames`, `Polyfold.MPolyfold.IsChartOrientation`, `Polyfold.MPolyfold.IsOrientedAt`, `Polyfold.MPolyfold.IsOrientedAt.isChartOrientation`, `Polyfold.MPolyfold.isOrientedAt_iff_forall`, `Polyfold.MPolyfold.exists_two_orientations`, `Polyfold.MPolyfold.OrTangent.push`, `Polyfold.MPolyfold.IsOrientedAt.push_tangentMapOn`, `Polyfold.EpGroupoid.isOrientedAt_push_tangentMor`, `Polyfold.EpGroupoid.GrHat`, `Polyfold.EpGroupoid.grHatPush`, `Polyfold.EpGroupoid.GrHatObj`, `Polyfold.EpGroupoid.grHatProj`, `Polyfold.EpGroupoid.grForget`, `Polyfold.EpGroupoid.grForgetFunctor`, `Polyfold.EpGroupoid.grForgetFunctor_comp_grProj`). An oriented tangent space at y is a subset $L \subseteq T_y X$ with a set of positive frames, oriented in a chart and then in every chart; every finite-dimensional tangent space has exactly two orientations; morphisms push oriented tangent spaces forward. $\widehat{\mathrm{Gr}}(X)$ is the category of formal sums of oriented tangent spaces over X^∞, with the forgetful functor to $\mathrm{Gr}(X)$ over X^∞. *Book:* §9.3, p. 422 (2017 p. 310), with Remark 9.3.1 (2017: 9.7), which is historical. *Hidden hypotheses:* none. *Deviation (restated):* frames instead of orientations of vector spaces (gap G.7) (Findings, item 50).

V9.30 (Definition 9.3.2 (2017: 9.8), orientations of Θ; `Polyfold.MPolyfold.SubmanifoldOrientation`, `Polyfold.EpGroupoid.OrientedBranchingStructure`, `Polyfold.EpGroupoid.OrientedBranchingStructure.orientedSum`, `Polyfold.EpGroupoid.OrientedBranchingStructure.grForget_orientedSum`, `Polyfold.EpGroupoid.IsOrientation`, `Polyfold.EpGroupoid.IsOrientable`, `Polyfold.EpGroupoid.IsOrientation.mk'`, `Polyfold.EpGroupoid.orientedSection`, `Polyfold.EpGroupoid.orientedSection_comp_grHatProj`, `Polyfold.EpGroupoid.orientedSection_comp_grForgetFunctor_obj`, `Polyfold.EpGroupoid.isOrientation_zero`, `Polyfold.EpGroupoid.isOrientation_sum_rn`, `Polyfold.EpGroupoid.isOrientation_sum_trivial_rn`). An orientation of Θ is a natural assignment $\widehat{T}_\Theta$ of formal sums of oriented tangent spaces at smooth points, lifting T_Θ, which near every $x \in \mathrm{supp}(\Theta)$ is $\sum_i \sigma_i \, T_y(M_i, o_i)$ for an oriented local branching structure (continuous orientations o_i of the M_i); it defines a section functor of $\widehat{\mathrm{Gr}}(X)$ over X^∞. The zero functor and finite sums of oriented closed submanifolds of $\mathbb{R}^n$ are oriented. *Book:* Definition 9.3.2, p. 423 (2017: Definition 9.8, p. 311). *Hidden hypotheses:* the lift condition uses T_Θ, hence the independence of Theorem 9.2.1, discharged on $\mathbb{R}^m$. *Deviation (restated):* the continuity of o_i is the local constancy of the chart frames (Findings, item 51).

V9.31 (Remark 9.3.3 (2017: 9.8), orientations are not unique; `Polyfold.EpGroupoid.remark_9_3_3`). The indicator of a point of $\mathbb{R}^n$ has infinitely many orientations, pairwise distinct, T_q for $q \in (0,1) \cap \mathbb{Q}$. *Book:* Remark 9.3.3, p. 423 (2017: Remark 9.8, p. 311). *Hidden hypotheses:* none. *Deviation (stronger):* infinitely many, not only more than one (Findings, item 52).

V9.32 (Lemma 9.3.4 (2017: 9.10), orientations are determined by good points; `Polyfold.EpGroupoid.IsOrientationOfDim`, `Polyfold.MPolyfold.HasOrientedGoodPointsAt`, `Polyfold.EpGroupoid.lemma_9_3_4`, `Polyfold.hasOrientedGoodPointsAt_rn_zero`, `Polyfold.hasOrientedGoodPointsAt_rn`, `Polyfold.MPolyfold.HasContinuousLevelOneOn`, `Polyfold.hasContinuousLevelOneOn_rn`, `Polyfold.frameDet`, `Polyfold.orientation_eq_iff_frameDet`, `Polyfold.MPolyfold.SubmanifoldOrientation.exists_isOpen_tangent_eq_iff`, `Polyfold.EpGroupoid.lemma_9_3_4_of_hasGraphBallAt`, `Polyfold.EpGroupoid.lemma_9_3_4_rn`, `Polyfold.mixedTheta`, `Polyfold.lemma_9_3_4_mixed_dim`, `Polyfold.lemma_9_3_4_mixed_dim_rn`). Two orientations of an n-dimensional Θ that agree at the good points agree at every smooth point, under `HasOrientedGoodPointsAt` on the support, which follows from the ball graph representation `HasGraphBallAt` and the level-1 continuity `HasContinuousLevelOneOn`; on $\mathbb{R}^m$ without hypothesis. Without equidimensionality the lemma does not hold as we read it: for $\Theta = 1_{\mathbb{R}\times 0} + 1_{(0,0)}$ on $\mathbb{R}^2$ two orientations agree at all good points and differ at the origin. *Book:* Lemma 9.3.4, p. 423 (2017: Lemma 9.10, p. 311). *Hidden hypotheses:* n-dimensional Θ; on a general

M-polyfold the graph representation and the continuity of frame fields on level 1. *Deviation (narrower):* the proof compares the two structures on one family of slices, applying Proposition 9.2.8 to both families together, and compares frames of different sheets by level-1 continuity; the text takes the good-point sequences from the first structure only, a step that needs an argument it does not give (gaps G.97, G.100) (Findings, item 53). *Update (2026-11-12):* for an n-dimensional Θ the lemma holds for any two orientations, with no hypothesis that they are n-dimensional: every orientation of an n-dimensional Θ has n-dimensional local representations (`Polyfold.EpGroupoid.IsOrientation.isOrientationOfDim`, `Polyfold.EpGroupoid.lemma_9_3_4_of_isNDimensional`), since every local branching structure of Θ has n-dimensional sheets (`Polyfold.EpGroupoid.LocalBranchingStructure.isOfDim_of_isNDimensional`).

V9.33 (Example 9.3.5 (2017: 9.1); `Polyfold.betaPP`, `Polyfold.betaMM`, `Polyfold.betaPM`, `Polyfold.betaMP`, `Polyfold.example935Theta`, `Polyfold.example_9_3_5_count`, `Polyfold.example935Theta_isBranched`, `Polyfold.exampleT1`, `Polyfold.exampleT2`, `Polyfold.example_9_3_5`, `Polyfold.example_9_3_5_rn`, `Polyfold.example_9_3_5_eq_origin`, `Polyfold.example_9_3_5_ne_of_pos`, `Polyfold.example_9_3_5_cannot_coincide`, `Polyfold.example_9_3_5_eq_of_neg`, `Polyfold.orientedLift1_injective`, `Polyfold.exampleT2_ne_orientedLift1`, `Polyfold.exists_goodSet_exampleT1_ne_exampleT2`). For $\beta^{++} = e(s) + e(-s)$ ($e =$ `expNegInvGlue`), $\beta^{--} = -\beta^{++}$ and the two mixed functions, the graphs define a branched ep$^+$-subgroupoid Θ of the trivial ep-groupoid of $\mathbb{R}^2$ by two structures; the orientations $\hat{T}^1$, $\hat{T}^2$ given by $\pm(1, \beta')$ coincide at the origin, differ at $(s, \beta^{++}(s))$ for $s > 0$, and no sign choice for the second structure reproduces $\hat{T}^1$ near the origin. They also coincide wherever $x < 0$, and the four sign choices of the first structure give four different orientations, none equal to $\hat{T}^2$. *Book:* Example 9.3.5, p. 425 (2017: Example 9.1, p. 313). *Hidden hypotheses:* none on $\mathbb{R}^2$. *Deviation (narrower):* "otherwise different" and "four different possibilities" do not hold as we read them (at least five orientations with weights 1); the classification of all orientations of Θ is not formalized (gap G.101) (Findings, item 54). *Update (2026-11-12):* the orientations of Θ are classified (`Polyfold.example_9_3_5_classification`; gap G.101 closed): coefficients $a(s, b) \in [0, 1] \cap \mathbb{Q}$ on the four open arcs, constant along each, with $a(+, \uparrow) + a(+, \downarrow) = a(-, \uparrow) + a(-, \downarrow) = A$ and weights A, $2 - A$ on $\pm(1, 0)$ at the origin. There are infinitely many (`Polyfold.example_9_3_5_infinite`); the sign choices on the two structures give exactly six (`Polyfold.ncard_signOrientations`, `Polyfold.example_9_3_5_six`), and these are the orientations with integer weights (`Polyfold.example_9_3_5_integral`, `Polyfold.example_9_3_5_integral_ncard`). "Four different possibilities" holds for each structure; for Θ it does not hold as we read it, so the row stays narrower (Findings, items 54, 115).

V9.34 (Definition 9.3.6 (2017: 9.10), tame Θ; `Polyfold.EpGroupoid.TameBranchingStructure`, `Polyfold.EpGroupoid.IsTameBranched`, `Polyfold.EpGroupoid.isTameBranched_zero`, `Polyfold.EpGroupoid.isTameBranched_indicator`, `Polyfold.MPolyfold.hasTameInducedStructure_of_degIndex_eq_zero`). Θ is tame if at every point of its support there is a local branching structure on a domain with the natural G_x-action and the properness property all of whose M_i have a tame induced structure; the zero functor and the indicator of a point of $\mathbb{R}^n$ are tame. *Book:* Definition 9.3.6, p. 425 (2017: Definition 9.10, p. 314).

Hidden hypotheses: none. *Deviation (restated):* "an equivalent tame structure" is Part I's tame induced structure (Definition 2.5.18); the consequence "in particular every M_i is a manifold with boundary with corners" is not formalized (gap G.102) (Findings, item 55).

V9.35 (Lemma 9.3.7, equal tangent spaces have equal degeneracy; `Polyfold.MPolyfold.subDegIndex_eq_quotDim_of_hasTameInducedStructure`, `Polyfold.MPolyfold.subDegIndex_eq_of_tangentSpace_eq`, `Polyfold.MPolyfold.lemma_9_3_7`, `Polyfold.MPolyfold.subDegIndex_eq_of_submanifoldTangent_eq`). For submanifolds M_1, M_2 with tame induced structures and $y \in M_1 \cap M_2$, $T_yM_1 = T_yM_2$ implies $d_{M_1}(y) = d_{M_2}(y)$. *Book:* Lemma 9.3.7, p. 426, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* neither "Y tame" nor "of the same dimension" is needed; in the proof "$T_0O \cap C = T_0^RO$" is read through the linearity of the chart image of T_0M (Findings, item 56).

V9.36 (Definition 9.3.8 (2017: 9.9), the boundary orientation of an oriented M; `Polyfold.exists_basis_orientation_iff_det_pos`, `Polyfold.exists_boundary_orientation`, `Polyfold.det_cons_smul_add`, `Polyfold.MPolyfold.boundaryFrames`, `Polyfold.MPolyfold.isOrientedAt_boundaryFrames`, `Polyfold.MPolyfold.boundaryFrames_eq`, `Polyfold.exists_boundaryFrames_graph`). For an oriented (M, o), a hyperplane K of T_yM and $e \notin K$, the frames $(a_1, \dots, a_{n-1})$ of K with $(e, a_1, \dots, a_{n-1})$ positive orient K, and the orientation is unchanged under $e \mapsto ce + k$, $c > 0$, $k \in K$; both orientations occur. *Book:* Definition 9.3.8, p. 427 (2017: Definition 9.9, p. 313). *Hidden hypotheses:* none. *Deviation (explicit since 2026-11-13; narrower before):* $T_y\partial M$ and the outward vector enter as data K, e; that $T_y\partial M = T_y^RM$ and the outward half-space are determined by M (through the cones C_yM) is not formalized (gap G.103) (Findings, item 57). *Update (2026-11-12):* for a tame sheet the stratum $\{d_M = 1\}$ is open and dense in ∂M (`Polyfold.MPolyfold.exists_isOpen_boundaryStratum_eq`, `Polyfold.MPolyfold.submanifoldBoundary_subset_closure_boundaryStratum`); it is not built as a submanifold of its own (gap G.103 in part; Findings, item 110). *Update (2026-11-13):* for a tame sheet the stratum $\{d_M = 1\}$ is a smooth $(n-1)$-dimensional submanifold of X with tangent spaces T_y^RM (`Polyfold.MPolyfold.isSubmanifoldOfDim_stratumOne`, `Polyfold.MPolyfold.submanifoldTangent_stratumOne`), oriented by $o_{\partial M}$ in the sense of Definition 9.3.3 (`Polyfold.MPolyfold.SubmanifoldOrientation.stratumOrientation`); in corner coordinates the convention is "outward normal first" (`Polyfold.MPolyfold.incFrame_mem_boundaryFramesAt_iff`), and for a face-structured sheet Part I's face orientation is $o_{\partial M}$ at the points with $d_M = 1$ (`Polyfold.MPolyfold.faceOrientation_eq_iff_mem_stratumOrientation`). The outward direction is read as "in T_yM, not in C_yM" in a presented chart (gap G.103 closed; Findings, items 110, 119).

V9.37 ((9.11) and Proposition 9.3.9, the boundary orientation does not depend on the structure; `Polyfold.MPolyfold.HasSubDegree`, `Polyfold.MPolyfold.SubmanifoldOrientation.hasSubDegree_tangent_iff`, `Polyfold.EpGroupoid.degreePart`, `Polyfold.EpGroupoid.degreePart_orientedSum`, `Polyfold.EpGroupoid.eq_9_11`, `Polyfold.EpGroupoid.proposition_9_3_9`). For two oriented local branching structures with tame M_i and the same oriented sum at y, the parts of degree $\ell = d_{M_i}(y)$ agree for every ℓ (9.11); hence (9.12) gives the same value for every structure, and vanishes at smooth points off $\mathrm{supp}(\Theta)$. *Book:* (9.11), p. 427, and Proposition 9.3.9, p. 428, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (narrower):* the boundary operation $T_y(M,o) \mapsto T_y(\partial M, o_{\partial M})$ is a function `bd` of the oriented tangent space, which is what "the discussion preceding it" provides, given G.103 (Findings, item 58).

V9.38 (Definition 9.3.10 (2017: 9.11), the oriented boundary $\widehat{T}_{\partial\Theta}$; `Polyfold.EpGroupoid.boundaryLift`). $\widehat{T}_{\partial\Theta}(y) = 0$ off $\mathrm{supp}(\Theta)$ and $\sum_{d_{M_i}(y)=1} \sigma_i \, T_y(\partial M_i, o_{\partial M_i})$ on a structure (9.12). *Book:* Definition 9.3.10, p. 428 (2017: Definition 9.11, p. 314, with (9.12) numbered (9.11) there). *Hidden hypotheses:* none. *Deviation (restated since 2026-11-09; narrower before):* with `bd` as in V9.37; the continuity of $o_{\partial M}$ along ∂M and its extension to the faces (Figure 9.1) are not formalized (gap G.103); Remark 9.3.11 has no formal content (Findings, item 58). *Update (2026-11-12):* the closed faces with their orientations exist in the corner-chart reading for face-structured sheets; their identification with $o_{\partial M}$ is not formalized (gap G.103 in part). Remark 9.3.11 is a comment, with nothing to formalize. *Update (2026-11-13):* for a face-structured tame sheet, read as a manifold with corners, the orientation of the open face $\{d_{M_i} = 1\}$ extends to exactly one continuous orientation of the closed face, Part I's `faceOrientation` (`Polyfold.MPolyfold.figure_9_1_existsUnique_faceOrientation`, through `Polyfold.MPolyfold.faceOrientation_eq_iff_mem_stratumOrientation` and `Polyfold.MPolyfold.dense_faceStratum`). The orientations of the faces of the faces are not formalized (gap G.103 closed; Findings, items 110, 119).

V9.39 (Definition 9.4.1 (2017: 9.12), the relative tangent space and essential points; `Polyfold.chartImage`, `Polyfold.relTangent`, `Polyfold.span_tangentConeAt_chartImage_eq_map`, `Polyfold.relTangent_eq_map`, `Polyfold.relTangent_eq_top_iff`, `Polyfold.essentialSet`, `Polyfold.essentialSet_eq`, `Polyfold.crossLines`, `Polyfold.essentialSet_crossLines`, `Polyfold.nonEssential_crossLines`). For a subset K of a finite-dimensional manifold with corners, $T_z^K M$ is the span of the limits of unit directions $\varphi(z_k)/|\varphi(z_k)|$, $z_k \to z$ in $K \setminus \{z\}$, in a chart centred at z, independent of the chart; $z \in K$ is essential if it is a limit of $z_k \in K$ with $T_{z_k}^K M = T_{z_k} M$. For two crossing lines in $\mathbb{R}^2$ the essential set is $\{0\}$. *Book:* Definition 9.4.1, p. 429 (2017: Definition 9.12, p. 315). *Hidden hypotheses:* none. *Deviation (restated):* on Mathlib manifolds with corners, the span of Mathlib's tangent cone in a chart (Findings, item 70). *Update (2026-11-12):* in M-polyfold charts the condition $T_z^K M = T_z M$ is independent of the chart (`Polyfold.MPolyfold.isRelTangentFull_of_chart`): chart changes are differentiable along a finite-dimensional submanifold for the level-0 norms (`Polyfold.MPolyfold.exists_hasFDerivWithinAt_transition`; Findings, item 116).

V9.40 (Lemma 9.4.2 (2017: 9.11), the essential set is closed; `Polyfold.lemma_9_4_2`, `Polyfold.lemma_9_4_2_open`). K^e is closed in K and $K \setminus K^e$ is open in K. *Book:* Lemma 9.4.2, p. 430 (2017: Lemma 9.11, p. 315). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 70).

V9.41 (Property (T) (2017 p. 316); `Polyfold.EpGroupoid.OrientedBranchingStructure.PropertyT`, `Polyfold.MPolyfold.OrTangent.eq_or_eq_or_eq_of_isOrientedAt`, `Polyfold.MPolyfold.exists_ne_isOrientedAt`, `Polyfold.propertyT_example935B1`). An oriented local branching structure has Property (T) on V if, whenever two sheets have the same tangent space at a point of V and their orientations there are related by ε, they are related by ε at every point of V where their tangent spaces agree; the first structure of Example 9.3.5 has it. *Book:* Property (T), p. 430 (2017 p. 316). *Hidden hypotheses:* none. *Deviation (restated):* Lemma 9.4.3, that (T) holds after shrinking $U(x)$, is not formalized (gap G.104) (Findings, item 71).

V9.42 (Lemma 9.4.4 (2017: 9.13), essential in M_i iff essential in $M_{i'}$; `Polyfold.MPolyfold.SheetGeometry`, `Polyfold.graphSheets`, `Polyfold.graphSheetGeometry`, `Polyfold.lemma_9_4_4_mixed_dim`). For sheets of one dimension the essential points of $K(i, i') = M_i \cap M_{i'}$ do not depend on the reference sheet; this is the field `ess_comm` of a sheet geometry, which exists for smooth graphs in $\mathbb{R}^2$. For sheets of different dimensions the lemma does not hold as we read it: $\mathbb{R} \times 0$ is essential in itself as a subset of $\mathbb{R}$ and has no essential point as a subset of $\mathbb{R}^2$. *Book:* Lemma 9.4.4, p. 431 (2017: Lemma 9.13, p. 317). *Hidden hypotheses:* sheets of one dimension. *Deviation (narrower):* for M-polyfold sheets, which carry no Mathlib manifold structure, Lemmas 9.4.4 and 9.4.6 are the fields of `SheetGeometry` (gap G.105); the note in Definition 9.4.7 ($i \sim_y i'$ implies $T_yM_i = T_yM_{i'}$) is its field `submanifoldTangent_eq` (Findings, item 72). *Update (2026-11-12):* for the sheets of every M-polyfold, read in M-polyfold charts, Lemmas 9.4.4 and 9.4.6 are theorems and `SheetGeometry` is built with no hypothesis (`Polyfold.MPolyfold.lemma_9_4_4_chart`, `Polyfold.MPolyfold.lemma_9_4_6_chart`, `Polyfold.MPolyfold.sheetGeometry`); every local branching structure of an n-dimensional Θ has n-dimensional sheets, so the lemma holds for them (`Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_4_4_of_isNDimensional`; gap G.105 closed). The verdict stays narrower for sheets of different dimensions (Findings, items 72, 113, 116).

V9.43 (Definition 9.4.5 (2017: 9.13), sets of measure zero; `Polyfold.IsMeasureZero`, `Polyfold.isMeasureZero_iUnion`, `Polyfold.isMeasureZero_of_forall_mem`, `Polyfold.not_isMeasureZero_univ`). $K \subseteq M$ has measure zero if its image in every chart has Lebesgue measure zero; measure zero is monotone and closed under countable unions, can be checked on a cover by charts, and $\mathbb{R}^2$ itself does not have it. *Book:* Definition 9.4.5, p. 431 (2017: Definition 9.13, p. 317). *Hidden hypotheses:* none. *Deviation (restated):* for Mathlib manifolds with corners (Findings, item 70). *Update (2026-11-12):* for submanifolds of an M-polyfold the null sets are `Polyfold.MPolyfold.IsHNull` (the n-dimensional Hausdorff measure of the chart image, for the level-0 norm, is 0), checked in one chart near each point (`Polyfold.MPolyfold.isHNull_of_forall_exists_chart`) and equal to Lebesgue measure zero in the smooth charts of a tame sheet (`Polyfold.MPolyfold.isHNull_iff_isMeasureZero_sheet`; Findings, item 116).

V9.44 (Lemma 9.4.6 (2017: 9.14), non-essential points form a null set; `Polyfold.unitDirections`, `Polyfold.span_unitDirections`, `Polyfold.exists_pos_measure_cone_lt`, `Polyfold.addHaar_setOf_span_tangentConeAt_ne_top`, `Polyfold.isMeasureZero_setOf_relTangent_ne_top`, `Polyfold.lemma_9_4_6`). For every subset K of a finite-dimensional manifold, $K \setminus K^e$ has measure zero, by a Lebesgue density argument at points where the relative tangent space is not everything. *Book:* Lemma 9.4.6, p. 431 (2017: Lemma 9.14, p. 317). *Hidden hypotheses:* none. *Deviation (stronger):* any K, not only $K(i, i')$; for M-polyfold sheets it is the field `isNull_diff_` ess of `SheetGeometry` (gap G.105) (Findings, items 70, 72). *Update (2026-11-12):* for every subset K of an n-dimensional submanifold of an M-polyfold, $K \setminus K^e$ is null (`Polyfold.MPolyfold.lemma_9_4_6_chart`; gap G.105 closed).

V9.45 (Definition 9.4.7 (2017: 9.14), y-equivalence; `Polyfold.MPolyfold.SheetGeometry.IsLink`, `Polyfold.MPolyfold.SheetGeometry.YEquiv`, `Polyfold.MPolyfold.SheetGeometry.equivalence_yEquiv`, `Polyfold.MPolyfold.SheetGeometry.ySetoid`, `Polyfold.MPolyfold.SheetGeometry.submanifoldTangent_eq_of_yEquiv`, `Polyfold.MPolyfold.SheetGeometry.numClasses`). $i \sim_y i'$ if a chain of sheets through y links them with y essential in each consecutive intersection; it is an equivalence relation on I_y, y-equivalent sheets have the same tangent space at y, and $\#P_y$ is the number of classes. *Book:* Definition 9.4.7, p. 432 (2017: Definition 9.14, p. 318). *Hidden hypotheses:* none. *Deviation (restated):* none (Findings, item 72).

V9.46 (Lemma 9.4.8 (2017: 9.15), points with two classes form a null set; `Polyfold.MPolyfold.SheetGeometry.IsNullIn`, `Polyfold.MPolyfold.SheetGeometry.lemma_9_4_8`, `Polyfold.MPolyfold.SheetGeometry.MStar`, `Polyfold.MPolyfold.SheetGeometry.isNullIn_diff_MStar`). $\{y \in M_U \mid \#P_y \geq 2\}$ is null in the sheets, and $M_U \setminus M_U^*$ is null. *Book:* Lemma 9.4.8, p. 432 (2017: Lemma 9.15, p. 318). *Hidden hypotheses:* none. *Deviation (restated):* "measure zero" in M_U is null in every sheet through the point; the last step of the proof uses $K(i,j)^{ne}$ inside M_i, while $K(j,j')^{ne}$ is null only in M_j and $M_{j'}$ (reading) (Findings, item 73). *Update (2026-11-12):* for an n-dimensional Θ the sheet geometry is `Polyfold.MPolyfold.sheetGeometry n` (`Polyfold.EpGroupoid.lemma_9_4_8_of_isNDimensional`).

V9.47 (Proposition 9.4.9 (2017: 9.5), the decomposition of M_U^*; `Polyfold.EpGroupoid.LocalBranchingStructure.tangentValue_eq_sum`, `Polyfold.MPolyfold.SheetGeometry.MStarPart`, `Polyfold.EpGroupoid.proposition_9_4_9`). For an oriented local branching structure with Property (T), the independence of Theorem 9.2.1 and sheets in a sheet geometry: $M_U \setminus M_U^*$ is null, M_U^* is the disjoint union of the $M_U^J = \{y \in M_U^* \mid I_y = J\}$, and on M_U^J: $\Theta = \sigma_J$, $T_\Theta = \sigma_J T^J$, $\hat{T}_\Theta = a_J(T^J, o_J^a) + b_J(T^J, o_J^b)$ with $a_J + b_J = \sigma_J$ and $o_J^a = -o_J^b$. *Book:* Proposition 9.4.9, p. 434 (2017: Proposition 9.5, p. 319). *Hidden hypotheses:* the independence of Theorem 9.2.1; the sheet geometry. *Deviation (explicit):* the footnote's "Lemma 9.5.2" is a forward reference (Findings, items 72, 74). *Update (2026-11-12):* for an n-dimensional Θ the sheet geometry is `Polyfold.MPolyfold.sheetGeometry n` and the independence is a theorem; Property (T) remains (`Polyfold.EpGroupoid.proposition_9_4_9_of_isNDimensional`).

V9.48 (Proposition 9.4.10 (2017: 9.6), comparing two oriented structures; `Polyfold.EpGroupoid.proposition_9_4_10`, `Polyfold.EpGroupoid.tangent_eq_iff_of_mem_ess`, `Polyfold.example935B1`, `Polyfold.example935B2`, `Polyfold.example935_two_le_numClasses`, `Polyfold.example935_hypotheses_9_4_10`, `Polyfold.example935_MU_eq`). For two oriented local branching structures on the same $U(x)$ with (a) the same weights and (b) the same oriented sums at every point, $M_U = M_U'$ and there are $M_U^\dagger \subseteq M_U$ with null complement in both families and a finite partition $A_1 \sqcup \cdots \sqcup A_\ell$ of $M_U^\dagger$ with (2)–(5) of the text. The two structures of Example 9.3.5 satisfy (a) and (b), and the origin has two classes. *Book:* Proposition 9.4.10, p. 435 (2017: Proposition 9.6, p. 321). *Hidden hypotheses:* the sheet geometry. *Deviation (stronger):* Property

(T) is not needed (Findings, item 74). *Update (2026-11-12):* for an n-dimensional Θ there is no sheet-geometry hypothesis (`Polyfold.EpGroupoid.proposition_9_4_10_of_isNDimensional`).

V9.49 (Definition 9.5.1 (2017: 9.15), sets of measure zero in $S = |\mathrm{supp}(\Theta)|$; `Polyfold.SheetNullSets`, `Polyfold.SheetNullSets.isNull_union`, `Polyfold.SheetNullSets.isNull_iUnion_countable`, `Polyfold.SheetNullSets.isNull_of_cover`, `Polyfold.MPolyfold.SheetMeasureGeometry`, `Polyfold.MPolyfold.SheetMeasureGeometry.toSheetNullSets`, `Polyfold.EpGroupoid.LocalBranchingStructure.HasNullPreimages`, `Polyfold.EpGroupoid.LocalBranchingStructure.hasNullPreimages_iff_isNullIn`, `Polyfold.EpGroupoid.HasTameSheetsIn`, `Polyfold.EpGroupoid.IsMeasureZeroSupp`, `Polyfold.EpGroupoid.IsMeasureZeroSuppAny`, `Polyfold.EpGroupoid.IsMeasureZeroSuppAll`). For a system of null sets in the sheets of a class $\mathcal{S}$ of submanifolds (a σ-ideal in every sheet, local, the same on the overlap of two sheets of $\mathcal{S}$, a nonempty sheet not null in itself), $A \subseteq |X|$ has measure zero if at every $x \in \mathrm{supp}(\Theta)$ some tame local branching structure (Definition 9.3.6) with sheets in $\mathcal{S}$ has preimages $M_i \cap \pi^{-1}(A)$ null in M_i; the variants ask this of some local branching structure, or of every structure with sheets in $\mathcal{S}$. A `SheetMeasureGeometry` is a §9.4 sheet geometry whose null sets are such a system. *Book:* Definition 9.5.1, p. 437 (2017: Definition 9.15, p. 322). *Hidden hypotheses:* none. *Deviation (restated):* "measure zero in M_i" is the abstract system `SheetNullSets`, satisfiable by Lebesgue null sets of graphs in $\mathbb{R}^2$ and of open subsets of $\mathbb{R}^2$ (V9.53); "a suitable $U(x)$" is a tame structure; compactness and orientation of Θ are not used (gap G.106) (Findings, item 80). *Update (2026-11-12):* with the null sets `Polyfold.MPolyfold.IsHNull` of the sheets of a general M-polyfold, Definition 9.5.1 is `Polyfold.EpGroupoid.IsMeasureZeroH`, and the abstract system is instantiated by `Polyfold.MPolyfold.sheetNullSets` (gap G.106 closed; Findings, items 80, 116).

V9.50 (Lemma 9.5.2 (2017: 9.16), measure zero does not depend on the choice; `Polyfold.EpGroupoid.isMeasureZeroSuppAll_of_any`, `Polyfold.EpGroupoid.isMeasureZeroSuppAny_of_isMeasureZeroSupp`, `Polyfold.EpGroupoid.lemma_9_5_2`, `Polyfold.EpGroupoid.isSigmaIdeal_isMeasureZeroSupp`, `Polyfold.EpGroupoid.isSigmaIdeal_isMeasureZeroSuppAll`, `Polyfold.EpGroupoid.not_isMeasureZeroSuppAny_orbitSupp`, `Polyfold.EpGroupoid.not_isMeasureZeroSupp_orbitSupp`, `Polyfold.isNull_transfer_mixed_dim`). If every point of supp(Θ) has a tame local branching structure with sheets in $\mathcal{S}$, then "null preimages for a choice of tame structures", "for a choice of structures" and "for every structure with sheets in $\mathcal{S}$" agree; the sets of measure zero form a σ-ideal, and $S \neq \emptyset$ does not have measure zero. The axis $\mathbb{R} \times 0$ is null in the sheet $\mathbb{R}^2$ and not null in itself. *Book:* Lemma 9.5.2, p. 437 (2017: Lemma 9.16, p. 322), cited from the literature without proof. *Hidden hypotheses:* the sheets of all structures compared lie in one class $\mathcal{S}$ on which two sheets induce the same null sets on their overlap. *Deviation (explicit since 2026-11-12; narrower before):* this transfer property holds for sheets of one dimension and does not hold across dimensions (`Polyfold.isNull_transfer_mixed_dim`), so the lemma is proved for structures with sheets in one class, as for the n-dimensional structures of an n-dimensional Θ; for every structure of an n-dimensional Θ it would need Proposition 9.1.12 in general (gaps G.97, G.106) (Findings, item 81). *Update (2026-11-12):* for an n-dimensional Θ (the standing assumption of §9.5) every local branching structure has n-dimensional sheets, and with the null sets `IsHNull` the lemma holds as stated: if at every point some local branching structure has null preimages of A, then every local branching structure has (`Polyfold.EpGroupoid.lemma_9_5_2_of_isNDimensional`, `Polyfold.EpGroupoid.lemma_9_5_2_tame`). The verdict is explicit: the one hidden hypothesis is the standing one; across dimensions the transfer still does not hold (`Polyfold.isHNull_mixed_dim`) (gap G.106 closed; Findings, items 81, 116).

V9.51 (Definition 9.5.3 (2017: 9.16) and the remark after it, the σ-algebra $L(S)$; `Polyfold.borelNullSigma`, `Polyfold.IsSigmaIdeal`, `Polyfold.borel_le_borelNullSigma`, `Polyfold.measurableSet_borelNullSigma_of_mem`, `Polyfold.borelNullCompletion`, `Polyfold.measurableSet_borelNullSigma_iff`, `Polyfold.EpGroupoid.measureZeroSets`, `Polyfold.EpGroupoid.lebesgueSigma`, `Polyfold.EpGroupoid.isSigmaIdeal_measureZeroSets`, `Polyfold.EpGroupoid.remark_9_5_3`, `Polyfold.EpGroupoid.measurableSet_lebesgueSigma_of_isMeasureZeroSupp`). $L(S)$ is the smallest σ-algebra on the subspace S containing the Borel sets and the subsets of measure zero; for a σ-ideal $\mathcal{N}$ the σ-algebra generated by the Borel sets and $\mathcal{N}$ is $\{A \mid A \,\Delta\, B \in \mathcal{N} \text{ for a Borel } B\}$, so, under the standing assumption of V9.50, $K \in L(S)$ iff $K \,\Delta\, B$ has measure zero for some Borel B. *Book:* Definition 9.5.3 and the remark after it, p. 438 (2017: Definition 9.16, p. 323). *Hidden hypotheses:* none. *Deviation (restated):* the remark that $L(S)$ is obtained by adding the sets of measure zero is read as the completion $K \,\Delta\, B$ null; the form $B \cup N$ would need Borel null hulls (Findings, item 82). *Update (2026-11-12):* with the null sets `IsHNull`: `Polyfold.EpGroupoid.lebesgueSigmaH`, `Polyfold.EpGroupoid.remark_9_5_3_H`, equal to the earlier σ-algebra for tame Θ (`Polyfold.EpGroupoid.lebesgueSigmaH_eq`).

V9.52 (Definition 9.5.14 (2017: 9.17) and measure zero in ∂S, p. 448; `Polyfold.EpGroupoid.IsMeasureZeroBoundary`, `Polyfold.EpGroupoid.IsMeasureZeroBoundaryAll`, `Polyfold.EpGroupoid.isMeasureZeroBoundaryAll_of`, `Polyfold.EpGroupoid.boundaryMeasureZeroSets`, `Polyfold.EpGroupoid.lebesgueSigmaBoundary`, `Polyfold.EpGroupoid.orbitSupp_boundaryTheta_subset`). For a second null-set system on a class of boundaries ∂M_i, $K \subseteq |X|$ has measure zero in $\partial S = |\mathrm{supp}(\partial\Theta)|$ if at every point of supp($\partial\Theta$) a tame structure has boundary preimages $\partial M_i \cap \pi^{-1}(K)$ null in ∂M_i; then every structure with boundaries in the class has; $\partial S \subseteq S$; $L(\partial S)$ is the σ-algebra generated by the Borel sets of ∂S and these sets. *Book:* p. 448 and Definition 9.5.14, p. 448 (2017: Definition 9.17, p. 327). *Hidden hypotheses:* the independence of Theorem 9.2.1 (through which $\partial\Theta$ is defined). *Deviation (explicit):* $(n-1)$-dimensional measure zero in ∂M_i is an abstract system; its description face by face is not formalized, and no witness with $\partial S \neq \emptyset$ is given (gap G.108) (Findings, item 83). *Update (2026-11-12):* face by face, for face-structured tame sheets (`Polyfold.MPolyfold.IsFaceNull`, a proper σ-ideal), and as a system for $n = 1$ (`Polyfold.SheetNullSets.point`), with the witness $\Theta = 1$ on $[0, \infty)$, $\partial S \neq \emptyset$ (V9.53); for $n \geq 2$ on a general M-polyfold the boundary null sets are not built (gap G.108 in part; Findings, items 83, 117). *Update (2026-11-13):* the $(n-1)$-dimensional null sets of the boundaries are the Hausdorff null sets in the ambient charts (`Polyfold.MPolyfold.IsHNull`), a null-set system on the boundaries of tame sheets with no hypothesis (`Polyfold.MPolyfold.boundaryNullSets`; a nonempty ∂M is not null, `Polyfold.MPolyfold.not_isHNull_submanifoldBoundary`), and Definition

9.5.14 is `Polyfold.EpGroupoid.lebesgueSigmaBoundaryH`, independent of the tame structures (`Polyfold.EpGroupoid.isMeasureZeroBoundary_boundaryNullSets_iff`) (gap G.108 closed; Findings, items 117, 119).

V9.53 (witnesses for §9.5 (not in the book); `Polyfold.isMeasureZero_image_of_local`, `Polyfold.isMeasureZero_of_subsingleton`, `Polyfold.isMeasureZero_rn2_iff`, `Polyfold.exists_tameBranchingStructure_trivial_rn`, `Polyfold.hasTameSheetsIn_trivial_rn`, `Polyfold.injective_ _trivial_rn`, `Polyfold.graphSheetMeasureGeometry`, `Polyfold.graphNull`, `Polyfold.theta935`, `Polyfold.isBranchedEpSubgroupoid_theta935`, `Polyfold.hasTameSheetsIn_theta935`, `Polyfold.zero_mem_supp_theta935`, `Polyfold.isMeasureZeroSupp_theta935_zero`, `Polyfold.not_isMeasureZeroSupp_theta935`, `Polyfold.lebesgueSigma_theta935`, `Polyfold.planeSheets`, `Polyfold.planeNull`, `Polyfold.planeSheet`, `Polyfold.thetaPlane`, `Polyfold.isSmoothSubmanifold_univ_rn2`, `Polyfold.isBranchedEpSubgroupoid_thetaPlane`, `Polyfold.mem_supp_thetaPlane`, `Polyfold.hasTameSheetsIn_thetaPlane`, `Polyfold.exists_mem_lebesgueSigma_not_borel_thetaPlane`). On the trivial ep-groupoid of $\mathbb{R}^n$ every local branching structure restricts to a tame one. The graph sheets of §9.4 carry a `SheetMeasureGeometry`; for the Θ of the two graphs G^{++}, G^{--} of Example 9.3.5 the touching point $|0|$ is a nonempty set of measure zero, S is not null and $S \in L(S)$; for $\Theta = 1$ on $\mathbb{R}^2$ with the Lebesgue null sets there is a member of $L(S)$ that is not Borel. *Book:* not in the book. *Hidden hypotheses:* none. *Deviation (extra):* the hypotheses of V9.49–V9.51 are satisfiable, and $L(S)$ is strictly larger than the Borel σ-algebra (Findings, item 82). *Update (2026-11-12):* for ∂S: $\Theta = 1$ on the half-line has a tame structure at every point (`Polyfold.halfLineTame`), $\partial S \neq \emptyset$ (`Polyfold.halfLine_orbitSupp_boundary_nonempty`), ∂S is not null and $L(\partial S)$ is the Borel σ-algebra (`Polyfold.halfLine_lebesgueSigmaBoundary`).

V9.54 (Lemma 9.5.9 (2017: 9.18), restrictions in the G_x-case: the isotropy and set part; `Polyfold.EpGroupoid.NaturalRep. _Φ`, `Polyfold.EpGroupoid.NaturalRep.mem_stabilizer_iff_Φ`, `Polyfold.EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_isotropy`, `Polyfold.EpGroupoid.NaturalRep.exists_slice`, `Polyfold.EpGroupoid.NaturalRep.mapsTo_Φ_MU`, `Polyfold.EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion`, `Polyfold.EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion_quotient`, `Polyfold.EpGroupoid.NaturalRep.pairwise_disjoint_image_Φ_out`). For a natural representation of G_x on $U(x)$ and $y \in U(x)$ with stabilizer H: $\#(G_x/H) \cdot \#G_y = \#G_x$; there is an open $V \ni y$ inside any given neighbourhood with $\varphi_g(V) = V$ for $g \in H$ and $\varphi_g(V) \cap V = \emptyset$ otherwise; for $K \subseteq |M_U \cap V|$ the preimage $M_U \cap \pi^{-1}(K)$ is the union over coset representatives r of $\varphi_r(M_U \cap V \cap \pi^{-1}(K))$, and the union is disjoint. *Book:* Lemma 9.5.9, p. 442 (2017: Lemma 9.18, p. 326). *Hidden hypotheses:* none. *Deviation (explicit since 2026-11-13; narrower before):* the equality of the measures μ_ω^U, μ_ω^V is not stated (forms and integrals, gap G.107); the representatives R are `Quotient.out` on G_x/H, and the restriction $\widetilde{M}_j = V \cap M_j$ keeps the sheets through y (`restrictAt`) (Findings, item 84). *Update (2026-11-12):* for every integration of forms over the oriented sheets (`Polyfold.EpGroupoid.SheetIntegration`, satisfiable in dimension 0, `Polyfold.EpGroupoid.pointIntegration`) the measure identity $\mu_\omega^U(K) = \mu_\omega^V(K)$ is proved (`Polyfold.EpGroupoid.OrientedTameStructure.lemma_9_5_9`); the integral of a form over a sheet of dimension $n \geq 1$ is not constructed, so the row stays narrower (gap G.107 in part; Findings, items 84, 118). *Update (2026-11-13):* the structure is realized for sc-forms, $n \geq 1$, by the integrals over sheets covered by countably many graph charts (`Polyfold.EpGroupoid.FormEval.msheetIntegration`), which contain the sheets of every oriented tame structure of a compact n-dimensional Θ (`Polyfold.EpGroupoid.FormEval.msheets_of_orientedTameStructure`); the row is explicit (gap G.107 closed; Findings, item 119).

V9.55 (Lemma 9.5.10, the effective orders and (9.23); `Polyfold.EpGroupoid.card_effIsotropy_mul_card_trivialSubgroup`, `Polyfold.EpGroupoid.NaturalRep.kerΦ_eq_trivialSubgroup`, `Polyfold.EpGroupoid.NaturalRep.kerΦ_le_stabilizer`, `Polyfold.EpGroupoid.NaturalRep.stabilizerEquivIsotropy_mem_trivialSubgroup_iff`, `Polyfold.EpGroupoid.NaturalRep.card_trivialSubgroup_eq`, `Polyfold.EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_effIsotropy`). $\#T_x \cdot \#G_x^{\mathrm{eff}} = \#G_x$; under LUC for $U(x)$, T_x is the kernel of the representation, it lies in the stabilizer of every $y \in U(x)$, it corresponds to T_y under $H \cong G_y$, so $\#T_y = \#T_x$, and $\#(G_x/H) \cdot \#G_y^{\mathrm{eff}} = \#G_x^{\mathrm{eff}}$, which is (9.23). *Book:* Lemma 9.5.10, p. 444, and (9.23), p. 445, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (explicit since 2026-11-13; narrower before):* the equality of the effective measures is not stated (gap G.107); LUC is `NaturalRep.HasLUC` of the given $U(x)$ (Definition 7.2.18) (Findings, item 84). *Update (2026-11-12):* for every integration of forms over the oriented sheets (`Polyfold.EpGroupoid.SheetIntegration`, satisfiable in dimension 0, `Polyfold.EpGroupoid.pointIntegration`) the identity of the effective measures under LUC is proved (`Polyfold.EpGroupoid.OrientedTameStructure.lemma_9_5_10`; gap G.107 in part). *Update (2026-11-13):* the structure is realized for sc-forms, $n \geq 1$, by the integrals over sheets covered by countably many graph charts (`Polyfold.EpGroupoid.FormEval.msheetIntegration`), which contain the sheets of every oriented tame structure of a compact n-dimensional Θ (`Polyfold.EpGroupoid.FormEval.msheets_of_orientedTameStructure`); the row is explicit (gap G.107 closed).

V9.56 (Lemma 9.5.11 (2017: 9.19), morphism invariance: the isotropy and set part; `Polyfold.EpGroupoid.card_isotropy_eq_of_mor`, `Polyfold.EpGroupoid.card_effIsotropy_eq_of_mor`, `Polyfold.EpGroupoid.IsFunctorial.eq_of_ _eq`, `Polyfold.EpGroupoid.image_ _MU_eq`, `Polyfold.EpGroupoid.NaturalRep.Φ_hyps`). For a morphism $\varphi : x \to x'$, $\#G_x = \#G_{x'}$ and $\#G_x^{\mathrm{eff}} = \#G_{x'}^{\mathrm{eff}}$; for local branching structures on U, U' and a map $\hat{\varphi}$ of U onto U' with $|\hat{\varphi}(z)| = |z|$, $|M_U| = |M_{U'}|$; the maps φ_g of a natural representation satisfy these hypotheses. *Book:* Lemma 9.5.11, p. 445 (2017: Lemma 9.19, p. 326). *Hidden hypotheses:* none. *Deviation (explicit since 2026-11-13; narrower before):* the equality $\mu_\omega^U(K) = \mu_\omega^{U'}(K)$, "a change of variables", is not stated (gap G.107) (Findings, item 84). *Update (2026-11-12):* for every integration of forms over the oriented sheets (`Polyfold.EpGroupoid.SheetIntegration`, satisfiable in dimension 0, `Polyfold.EpGroupoid.pointIntegration`) both identities are proved (`Polyfold.EpGroupoid.OrientedTameStructure.lemma_9_5_11`); the effective one needs neither LUC nor a smaller U (gap G.107 in part; Findings, item 118). *Update (2026-11-13):* the structure is realized for sc-forms, $n \geq 1$, by the integrals over sheets covered by countably many graph charts (`Polyfold.EpGroupoid.FormEval.msheetIntegration`), which contain the sheets of every oriented tame structure of a compact n-dimensional Θ (`Polyfold.EpGroupoid.FormEval.msheets_of_orientedTameStructure`); the row is explicit (gap G.107 closed).

V9.57 (Proposition 9.5.13, the normalizing factors; `Polyfold.EpGroupoid.card_effIsotropy_eq_of_isEffective`, `Polyfold.EpGroupoid.isFunctorial_reduced_iff`, `Polyfold.EpGroupoid.LocalBranchingStructure.toReduced`, `Polyfold.EpGroupoid.card_effIsotropy_eq_reduced`). For effective X, $\#G_x^{\mathrm{eff}} = \#G_x$; for the reduced ep-groupoid X_R (with Hausdorff paracompact morphism space), functors and local branching structures of X and X_R coincide and $\#G_x^{\mathrm{eff}} = \#G_x^R = \#(G_x^R)^{\mathrm{eff}}$. *Book:* Proposition 9.5.13, p. 446, new in the 2021 edition. *Hidden hypotheses:* the morphism space of X_R Hausdorff and paracompact (instances; Hausdorff under LUC). *Deviation (restated since 2026-11-13; narrower before):* the equality of the integrals in (1) and (2) is not stated (gap G.107); what is proved is the equality of the normalizing factors on which the proof rests (Findings, item 84). *Update (2026-11-12):* for every integration of forms over the oriented sheets (`Polyfold.EpGroupoid.SheetIntegration`, satisfiable in dimension 0, `Polyfold.EpGroupoid.pointIntegration`) (1) is proved (`Polyfold.EpGroupoid.proposition_9_5_13_1`); (2) at the level of the integrals is not (gap G.107 in part). *Update (2026-11-13):* (2) holds at the level of the integrals: under LUC the oriented tame structures, the orientation and the integration over the sheets pass to X_R (`Polyfold.EpGroupoid.NaturalRep.toReduced`, `Polyfold.EpGroupoid.OrientedTameStructure.toReduced`, `Polyfold.EpGroupoid.SheetIntegration.toReduced`), and with $|X| \cong |X_R|$ (`Polyfold.EpGroupoid.orbitSuppHomeomorphReduced`), $\mu^{\mathrm{eff}}_{\hat{\Theta},\omega}(K) = \mu_{\hat{\Theta}',\omega'}(K)$ and $\oint_{\hat{\Theta}} \omega = \int_{\hat{\Theta}'} \omega'$ (`Polyfold.EpGroupoid.proposition_9_5_13_2_measure`, `Polyfold.EpGroupoid.proposition_9_5_13_2`), for every `SheetIntegration`, in particular for sc-forms; witness: the unit circle (`Polyfold.EpGroupoid.circle_proposition_9_5_13_2`). The row is restated (gap G.107 closed; Findings, items 84, 119).

V9.58 (Lemma 9.6.1 and $\dim_\Theta$, the top dimension; `Polyfold.MPolyfold.HasDimAt.le_of_inter_subset`, `Polyfold.MPolyfold.exists_mem_forall_notMem`, `Polyfold.EpGroupoid.LocalBranchingStructure.topDim`, `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_6_1`, `Polyfold.EpGroupoid.localTopDim`, `Polyfold.EpGroupoid.localTopDim_eq`, `Polyfold.EpGroupoid.localTopDim_eq_bot`). Two local branching structures at x with dimension labellings have the same top dimension, which defines $\dim_\Theta(x)$. *Book:* Lemma 9.6.1, p. 453, and $\dim_\Theta$, p. 454, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (stronger):* the "measure zero" argument is replaced by a topological covering lemma: a nonempty open piece of a k-dimensional submanifold is not covered by finitely many submanifolds of smaller dimension; no Lebesgue measure and no Baire argument (Findings, item 90).

V9.59 (Lemma 9.6.2, the union of the top-dimensional sheets; `Polyfold.EpGroupoid.LocalBranchingStructure.topUnion`, `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_6_2`, `Polyfold.EpGroupoid.LocalBranchingStructure.mem_topUnion_iff_of_mem`). On $U \cap U'$ the unions M_U^k of the top-dimensional sheets of two structures agree. *Book:* Lemma 9.6.2, p. 454, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (same):* by the covering lemma of V9.58 (Findings, item 90).

V9.60 (Definition 9.6.3, top-good and top-bad points; `Polyfold.EpGroupoid.LocalBranchingStructure.IsTopGoodPoint`, `Polyfold.EpGroupoid.LocalBranchingStructure.IsTopBadPoint`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_`

`isTopGoodPoint`). $y \in M_U^k$ is top-good if near y all top-dimensional sheets through y coincide; at top-good points the top-dimensional parts Θ_k of two structures agree, without hypothesis. *Book:* Definition 9.6.3, p. 455, new in the 2021 edition. *Hidden hypotheses:* none. *Deviation (restated):* none (Findings, item 91).

V9.61 (Lemma 9.6.4, top-goodness does not depend on the structure; `Polyfold.MPolyfold.TopMultiplicity`, `Polyfold.MPolyfold.topMultiplicity_zero`, `Polyfold.topMultiplicity_rn_of_le`, `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_6_4`, `Polyfold.EpGroupoid.LocalBranchingStructure.isTopBadPoint_iff`, `Polyfold.MPolyfold.TopMultiplicityAt`, `Polyfold.MPolyfold.topMultiplicityAt_of_hasGraphChartAt`, `Polyfold.MPolyfold.topMultiplicity_rn`, `Polyfold.EpGroupoid.LocalBranchingStructure.lemma_9_6_4_of_hasGraphChartAt`). A point in both domains is top-good for one structure iff for the other, under `TopMultiplicity` (the top weight of a second structure at a point equals the sum of the weights of its sheets there), which holds for $k = 0$, for $k \geq n$ on $\mathbb{R}^n$, and wherever `HasGraphChartAt` holds, so on $\mathbb{R}^m$ in all dimensions. *Book:* Lemma 9.6.4, p. 455, new in the 2021 edition. *Hidden hypotheses:* the graph representation on a general M-polyfold. *Deviation (explicit):* the step "similarly as good points" needs that the top weight of the second structure at y is σ, an argument the text does not give; it is `TopMultiplicityAt`, proved from `HasGraphChartAt` by grouping the sheets of both families by tangent space (gap G.97) (Findings, item 91).

V9.62 (Proposition 9.6.5 and the formula on p. 457; `Polyfold.EpGroupoid.LocalBranchingStructure.IsTopSlice`, `Polyfold.EpGroupoid.LocalBranchingStructure.proposition_9_6_5`, `Polyfold.EpGroupoid.LocalBranchingStructure.sum_class_eq_sum_of_isTopSlice`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_sum_slices`, `Polyfold.EpGroupoid.LocalBranchingStructure.sum_class_eq_of_hasGraphChartAt`, `Polyfold.EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_hasGraphChartAt`). Under `HasGraphChartAt` at x, for every class of top-dimensional sheets with one tangent space at x and every neighbourhood W of x there is a finite top slice in W; the top-dimensional part is $\Theta_k(x) = \sum_{\text{classes}} \sum_{y \in \text{slice}} \Theta(y)$, and the class sums agree for any two structures. *Book:* Proposition 9.6.5, p. 456, and the formula on p. 457, new in the 2021 edition. *Hidden hypotheses:* the graph representation on a general M-polyfold. *Deviation (narrower):* slices replace the subsets I^e and the sequences of top-good points; the independence of the right-hand side from the structure, which the text does not show, is proved with one slice for the sheets of both structures. In (1) "$I_k = I^1 \sqcup \cdots \sqcup I^p$" reads as $\Xi = I^1 \sqcup \cdots \sqcup I^p$, the points x_e^k are not introduced, and "This completes the proof of Proposition 9.6.5" refers to Proposition 9.1.12 (possible misprints) (gap G.97) (Findings, items 91, 92).

The Part II wave (2026-11-12): §9.1 and §9.5 (V9.63–V9.66)

V9.63 (Remark 9.1.4: paracompact subsets of $|X|$ are metrizable, and the compactness criterion; `Polyfold.EpGroupoid.remark_9_1_4_metrizable`, `Polyfold.EpGroupoid.remark_9_1_4_exists_nhds`, `Polyfold.EpGroupoid.remark_9_1_4_paracompactSpace`, `Polyfold.EpGroupoid.`

`remark_9_1_4`, `Polyfold.EpGroupoid.isBranchedEpSubgroupoid_of_isComponentCompact`, `Polyfold.EpGroupoid.paracompactSpace_orbitSupp_of_isCompact`). `remark_9_1_4_metrizable`: a paracompact $L \subseteq |X|$ is metrizable (Theorem 7.3.1 and Smirnov's theorem, through `metrizableSpace_of_paracompactSpace_subtype`). `remark_9_1_4_exists_nhds`: a $K \subseteq |X|$ meeting every connected component of $|X|$ in a compact set has an open paracompact, metrizable neighbourhood (Corollary 7.3.8); K itself is metrizable and paracompact (`remark_9_1_4_paracompactSpace`). `remark_9_1_4`: if $|\text{supp}(\Theta)|$ is component compact, it is paracompact, so the paracompactness in Definition 9.1.2 (1) holds automatically (`isBranchedEpSubgroupoid_of_isComponentCompact`); the compact case is `paracompactSpace_orbitSupp_of_isCompact`. *Book:* §9.1, Remark 9.1.4, p. 403 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* As stated. The step from the paracompact neighbourhood to K goes through metrizability, since a subspace of a paracompact space need not be paracompact (gap G.94, closed) (Findings, Chapter 9, items 7, 114).

V9.64 (Proposition 9.1.9 (2017: 9.1), the natural manifold and orbifold structures on $|\text{supp}(\Theta)|$; `Polyfold.EpGroupoid.proposition_9_1_9_one`, `Polyfold.EpGroupoid.proposition_9_1_9_one_unique`, `Polyfold.EpGroupoid.proposition_9_1_9_two`, `Polyfold.EpGroupoid.proposition_9_1_9_two_text`, `Polyfold.EpGroupoid.proposition_9_1_9_two_independent`, `Polyfold.EpGroupoid.textGroupoid`, `Polyfold.EpGroupoid.textOrbitHomeomorph`, `Polyfold.EpGroupoid.OpenCover.isEquivalence_refineFunctor`, `Polyfold.EpGroupoid.nonempty_sheetChoice`, `Polyfold.EpGroupoid.supp_inter_eq_of_isOrbifoldType`, `Polyfold.EpGroupoid.isSmoothSubmanifold_supp_of_isOrbifoldType`, `Polyfold.EpGroupoid.isEpSubgroupoid_supp_of_isOrbifoldType`, `Polyfold.EpGroupoid.orbitSuppHomeomorph`, `Polyfold.EpGroupoid.subsingleton_isotropy_of_isManifoldType`, `Polyfold.EpGroupoid.isDescentMap_orbitSupp`, `Polyfold.MPolyfold.IsDescentMap`, `Polyfold.MPolyfold.descend`, `Polyfold.MPolyfold.isLocalScDiffeo_descend`). (1) `proposition_9_1_9_one`: for Θ of manifold type, $\text{supp}(\Theta)$ is a smooth finite-dimensional submanifold with trivial isotropy (`subsingleton_isotropy_of_isManifoldType`), and $Z = |\text{supp}(\Theta)|$, a subspace of $|X|$, carries an M-polyfold structure for which $a \mapsto |a|$ is a local sc-diffeomorphism, descended with the text's charts Ψ_x (`MPolyfold.descend`, `isLocalScDiffeo_descend`); any two M-polyfold structures on Z with this property agree (`proposition_9_1_9_one_unique`), which is the reading of "natural". (2) `proposition_9_1_9_two_text`: for Θ of orbifold type, $\text{supp}(\Theta) \neq \emptyset$ and every choice c of local branching structures (`SheetChoice`, `nonempty_sheetChoice`), $\text{supp}(\Theta)$ is a smooth submanifold and $\text{supp}(\Theta) \cap U(x) = M_x$ (`supp_inter_eq_of_isOrbifoldType`); the text's groupoid M with objects $\bigsqcup M_x$ and morphisms $\bigsqcup M(x, x')$ is an ep-groupoid, the refinement of Proposition 10.1.3 (`textGroupoid`); the functor $M \to \text{supp}(\Theta)$ to the full ep-subgroupoid is an equivalence (Definition 10.1.1), and $|\beta| : |M| \cong Z$, $|(x, y)| \mapsto |y|$ (`textOrbitHomeomorph`). Two choices c, c' give $M_c \leftarrow M_{c,c'} \to M_{c'}$, both legs equivalences, a generalized isomorphism of $\text{EP}(\mathcal{E}^{-1})$ commuting with the maps to Z (`proposition_9_1_9_two_independent`, through `OpenCover.isEquivalence_refineFunctor`). Also `proposition_9_1_9_two`: the full ep-subgroupoid on $\text{supp}(\Theta)$ has orbit space homeomorphic to Z (`orbitSuppHomeomorph`). *Book:* §9.1, Proposition 9.1.9, p. 405 (2017: Proposition 9.1, p. 294). *Hidden hypotheses:* none. *Deviation (restated):* The paracompactness of $|X|$ is not used: Definition 9.1.2 (1) already makes $|\mathrm{supp}(\Theta)|$ paracompact, hence metrizable (Remark 9.1.4, V9.63). "Natural" in (1) is read as the uniqueness of the M-polyfold structure on Z for which $\mathrm{supp}(\Theta) \to Z$ is a local sc-diffeomorphism; "equivalent orbifold structure" in (2) as a generalized isomorphism of Chapter 10 over Z through the common refinement of two choices. The form with the text's groupoid takes $\mathrm{supp}(\Theta) \neq \emptyset$, which the M-polyfold structure on an open subset needs (for $\Theta = 0$, $Z = \emptyset$) (gap G.95, closed) (Findings, Chapter 9, items 7, 114). *Update (2026-11-13):* the hypotheses, including $\mathrm{supp}(\Theta) \neq \emptyset$, are witnessed together by the x-axis of $\mathbb{R}^2$ (`Polyfold.EpGroupoid.remark_9_1_10_hypotheses_axis`, `Chapter9/GapP2_C2Witness.lean`).

V9.65 (Remark 9.1.10 (2017: 9.3), the classical manifold and orbifold structures when $d_{|X|} = 0$; `Polyfold.EpGroupoid.remark_9_1_10_manifold`, `Polyfold.EpGroupoid.remark_9_1_10_orbifold`, `Polyfold.MPolyfold.HasCornerStructureOfDim.of_isLocalScDiffeo_surjective`, `Polyfold.EpGroupoid.isSubmanifoldOfDim_supp`, `Polyfold.EpGroupoid.hasCornerStructureOfDim_subObj`, `Polyfold.EpGroupoid.remark_9_1_10_orbifold_etale`, `Polyfold.EpGroupoid.remark_9_1_10_manifold_proj`, `Polyfold.MPolyfold.isLocalDiffeomorph_of_isLocalScDiffeo`). For Θ n-dimensional (`IsNDimensional Θ n`) with $d_{|X|} = 0$ on $Z = |\mathrm{supp}(\Theta)|$: `remark_9_1_10_manifold`: in case (1) the M-polyfold structure of V9.64 has a corner structure of dimension n with degeneracy index 0, and Z with its interior charted space is a Mathlib manifold, `IsManifold (  n) ∞`; $\mathrm{supp}(\Theta) \to Z$ is a Mathlib local diffeomorphism (`remark_9_1_10_manifold_proj`). `remark_9_1_10_orbifold`: in case (2), for every choice of structures, the object space $\bigsqcup M_x$ and the morphism space of the text's groupoid M are Mathlib n-manifolds and $|M| \cong Z$; `remark_9_1_10_orbifold_etale`: source, target, unit, inverse and the equivalence $M \to \mathrm{supp}(\Theta)$ on objects are Mathlib local diffeomorphisms (`MPolyfold.isLocalDiffeomorph_of_isLocalScDiffeo`). The x-axis of $\mathbb{R}^2$ satisfies the hypotheses ($n = 1$). The second paragraph is Remark 9.1.4 (V9.63) and V9.64 (paracompactness of $|X|$ not used). *Book:* §9.1, Remark 9.1.10, p. 405 (2017: Remark 9.3, p. 294). *Hidden hypotheses:* Θ is n-dimensional: a Mathlib manifold has one model space, while the text's Z may have components of different dimensions (cf. Proposition 9.1.11). *Deviation (narrower):* Mathlib has no notion of a classical orbifold (proper étale Lie groupoid); case (2) is stated as: object and morphism spaces Mathlib n-manifolds, the structure maps other than the composition Mathlib local diffeomorphisms. The composition $M\,{}_s\!\times_t M \to M$ is sc-smooth, as for every ep-groupoid; it is not stated as Mathlib-smooth, since the fibred product carries no Mathlib structure here (gap G.95, closed) (Findings, Chapter 9, items 7, 114). *Update (2026-11-13):* all hypotheses together (Θ branched, of manifold type, 1-dimensional, $d = 0$ on the support, nonempty support) hold for the x-axis of $\mathbb{R}^2$ (`Polyfold.EpGroupoid.remark_9_1_10_hypotheses_axis`, `Polyfold.EpGroupoid.axis_isManifoldType`, `Polyfold.EpGroupoid.axis_degIndexOrbit_eq_zero`, `Polyfold.EpGroupoid.nonempty_supp_axis`).

V9.66 (Theorems 9.5.4 (2017: 9.2) and 9.5.5, Definition 9.5.6, (9.21) and Lemma 9.5.7, for an integration of forms over the sheets; `Polyfold.EpGroupoid.SheetIntegration`, `Polyfold.EpGroupoid.SheetIntegration.sum_integ_eq_sum_integ`, `Polyfold.EpGroupoid.OrientedTameStructure`, `Polyfold.EpGroupoid.OrientedTameStructure.`

`localMeasure`, `Polyfold.EpGroupoid.OrientedTameStructure.localMeasureEff`, `Polyfold.EpGroupoid.OrientedTameStructure.lemma_9_5_7`, `Polyfold.EpGroupoid.existsUnique_branchedMeasure`, `Polyfold.EpGroupoid.theorem_9_5_4`, `Polyfold.EpGroupoid.theorem_9_5_5`, `Polyfold.EpGroupoid.branchedMeasure`, `Polyfold.EpGroupoid.branchedMeasureEff`, `Polyfold.EpGroupoid.branchedIntegral`, `Polyfold.EpGroupoid.branchedIntegralEff`, `Polyfold.EpGroupoid.branchedMeasure_eq_zero`, `Polyfold.EpGroupoid.pointIntegration`, `Polyfold.existsUnique_linearMap_signedMeasure_of_openCover`, `Polyfold.EpGroupoid.FormEval.theorem_9_5_4_full`, `Polyfold.EpGroupoid.FormEval.theorem_9_5_5_full`, `Polyfold.EpGroupoid.FormEval.multiBranchedIntegral`, `Polyfold.EpGroupoid.FormEval.multiBranchedIntegralEff`, `Polyfold.EpGroupoid.FormEval.msheets_of_orientedTameStructure`, `Polyfold.EpGroupoid.FormEval.msheetIntegration`, `Polyfold.EpGroupoid.circle_theorem_9_5_4_full`, `Polyfold.EpGroupoid.FormEval.theorem_9_5_4_full_of_degIndex_eq_zero`, `Polyfold.EpGroupoid.hloc_of_degIndex_eq_zero`, `Polyfold.EpGroupoid.circle_witness`). For every integration of forms over measurable subsets of the oriented sheets of a class S with null sets G (`SheetIntegration`: linear in ω, countably additive, zero on null sets, equal (opposite) for two sheets on a common set where their oriented tangent spaces agree (are opposite), invariant under the sc-diffeomorphisms whose tangent maps are those of morphisms, sheets Lindelöf): (9.20) for finite families of oriented sheets (`SheetIntegration.sum_integ_eq_sum_integ`); the local measures (9.21) (`localMeasure`, `localMeasureEff`); Lemma 9.5.7 (`lemma_9_5_7`); `theorem_9_5_4`: for Θ branched, compact, with an orientation T and, at every point of the support, an oriented tame structure of dimension n with sheets in S describing T (`hcov`), there is exactly one linear map $\Omega \to$ signed measures on $(S, L(S))$ given on measurable $K \subseteq |U(x)|$ by the local formula with $1/\#G_x$, for all such structures; `theorem_9_5_5`: the same with $1/\#G_x^{\text{eff}}$ and structures with LUC; Definition 9.5.6 (`branchedIntegral`, `branchedIntegralEff`). The gluing is `existsUnique_linearMap_signedMeasure_of_openCover`. The structure is satisfiable in dimension 0 for every ep-groupoid (`pointIntegration`). (2026-11-13) For the integrals of sc-forms over sheets covered by countably many graph charts (`FormEval.msheetIntegration`, $n \geq 1$; gap G.107): `theorem_9_5_4_full`, `theorem_9_5_5_full`. There is exactly one linear map with the local formula for every oriented tame structure describing $\widehat{T}_\Theta$ (no class condition: the sheets of every oriented tame structure of a compact n-dimensional Θ have finite $\int |\omega|$, `msheets_of_orientedTameStructure`), under the printed hypotheses with "oriented, tame" read as `hloc`; Definition 9.5.6: `multiBranchedIntegral`, `multiBranchedIntegralEff`. Without boundary, `hloc` comes from n-dimensionality and the orientation (`theorem_9_5_4_full_of_degIndex_eq_zero`). Witness with nonempty compact support: the unit circle in $\mathbb{R}^2$ (`circle_theorem_9_5_4_full`). *Book:* §9.5, Theorem 9.5.4, p. 438 (2017: Theorem 9.2, p. 323), Theorem 9.5.5, p. 438 (new in the 2021 edition), Definition 9.5.6, (9.21), Lemma 9.5.7, pp. 439–441. *Hidden hypotheses:* the text's "oriented, tame" is read as `hloc`: at every point of the support an oriented tame structure describes $\widehat{T}_\Theta$. It follows from n-dimensionality and the orientation when the M-polyfold has no boundary (`hloc_of_degIndex_eq_zero`, `theorem_9_5_4_full_of_degIndex_eq_zero`, where the theorem is the text's); with boundary, that the oriented local structures can be taken with tame sheets is the reading of "tame". *Deviation (explicit since 2026-11-13; narrower* *before):* The abstract theorems are stated for a `SheetIntegration` with forms in a real vector space Ω; since 2026-11-13 it is realized for $\Omega^n_{ep}(X)$ and $\Omega^n_{ep,\infty}(X)$, $n \geq 1$, by the integrals over sheets covered by countably many graph charts (`FormEval.msheetIntegration`), and the local formula holds for every oriented tame structure describing $\hat{T}_\Theta$ (`theorem_9_5_4_full`, `theorem_9_5_5_full`); the hypotheses hold for the unit circle (`circle_theorem_9_5_4_full`). Dimension 0 is the abstract `pointIntegration` (gap G.107, closed) (Findings, Chapter 9, items 85, 118, 119).

The second Part II wave (2026-11-13): §9.5 (V9.67–V9.69; V9.67, V9.68 as completed by wave K)

V9.67 (Theorems 9.5.15 (2017: 9.3) and 9.5.16, the canonical boundary measures; `Polyfold.EpGroupoid.theorem_9_5_15_msheets`, `Polyfold.EpGroupoid.theorem_9_5_16_msheets`, `Polyfold.EpGroupoid.theorem_9_5_15_hyp`, `Polyfold.EpGroupoid.theorem_9_5_16_hyp`, `Polyfold.EpGroupoid.existsUnique_boundaryMeasure`, `Polyfold.EpGroupoid.existsUnique_familyMeasure`, `Polyfold.EpGroupoid.OrientedTameStructure.boundaryFamily`, `Polyfold.EpGroupoid.OrientedTameStructure.represents_boundaryFamily`, `Polyfold.EpGroupoid.isNaturalOn_boundaryOrientation`, `Polyfold.EpGroupoid.OrientedTameStructure.isMeasIn_boundarySheet`, `Polyfold.EpGroupoid.OrientedSheetFamily.nonemptyPart`, `Polyfold.EpGroupoid.hcov_zero`, `Polyfold.EpGroupoid.hcov_msheets_of_isCompact`, `Polyfold.EpGroupoid.OrientedTameStructure.boundarySheet`, `Polyfold.EpGroupoid.OrientedTameStructure.boundaryOrient`, `Polyfold.EpGroupoid.OrientedTameStructure.boundaryOrientation_eq_sum`, `Polyfold.EpGroupoid.OrientedTameStructure.exists_boundarySheet_mem_sheets`, `Polyfold.EP.hcovB_of_hcovS`, `Polyfold.E8W.e8w_hcov`, `Polyfold.E8W.e8_hemisphere_stokes`, `Polyfold.EP.theorem_9_5_15`, `Polyfold.EP.theorem_9_5_16`, `Polyfold.EP.boundarySheet_mem_msheets_of_loc`, `Polyfold.EP.boundaryMeasure_apply_all`, `Polyfold.MPolyfold.k1_exists_tangent_eq`, `Polyfold.MPolyfold.k1_mem_boundaryStratum_of_tangent_eq`, `Polyfold.EpGroupoid.FormEval.k1_msheets_of_subset_list`, `Polyfold.EP.boundaryMeasure`, `Polyfold.EP.boundaryMeasureEff`, `Polyfold.EP.boundaryMeasure_spec`, `Polyfold.EP.theorem_9_5_15_class`, `Polyfold.EP.theorem_9_5_16_class`, `Polyfold.EP.hcov_of_loc`, `Polyfold.EP.hcov_of_loc_isLUC`, `Polyfold.EP.exists_restrict_boundary_mem_msheets`, `Polyfold.EpGroupoid.OrientedTameStructure.exists_restrict_eq`, `Polyfold.EpGroupoid.LocalBranchingStructure.isCompact_M_inter`, `Polyfold.EP.boundaryStratum_inter_mem_msheets`, `Polyfold.EpGroupoid.FormEval.boundaryStratum_inter_mem_msheets`, `Polyfold.EpGroupoid.FormEval.abs_ev_faceFrame`, `Polyfold.EpGroupoid.integrableOn_facePullback_of_isCompact`, `Polyfold.EpGroupoid.FormEval.exists_chartCover_of_isLindelof`, `Polyfold.E8W.k1_hemisphere_stokes`, `Polyfold.EP.zeroFormEval`, `Polyfold.EP.zeroFormIntegration`, `Polyfold.EP.formsInftyRestrict_zero_eq`, `Polyfold.MPolyfold.exists_isOpen_boundaryStratum_subsingleton`, `Polyfold.EpGroupoid.OrientedTameStructure.exists_boundarySheet_subsingleton`, `Polyfold.EpGroupoid.hcov_dim1_of_loc`, `Polyfold.EpGroupoid.hcovEff_dim1_of_loc`, `Polyfold.EP.theorem_9_5_15_dim1`, `Polyfold.EP.theorem_9_5_16_dim1`, `Polyfold.EP.theorem_9_5_16_dim1_isLUC`, `Polyfold.EpGroupoid.boundaryIntegral_eq_pointInteg`, `Polyfold.EP.boundaryIntegral_`

`eq_zeroFormIntegration`, `Polyfold.EpGroupoid.finPointSheets`, `Polyfold.EpGroupoid.finPointIntegration`, `Polyfold.EpGroupoid.finPointInteg_eq_pointInteg`, `Polyfold.EpGroupoid.submanifoldTangent_finite_eq`, `Polyfold.EP.finZeroFormIntegration`, `Polyfold.EP.finZeroFormIntegration_integ_of_subset`, `Polyfold.EP.boundarySheet_finite_of_loc`, `Polyfold.EP.boundarySheet_mem_finPointSheets_of_loc`, `Polyfold.EP.theorem_9_5_15_dim1_all`, `Polyfold.EP.theorem_9_5_16_dim1_all`, `Polyfold.EP.boundaryBranchedMeasureDim1_apply_all`, `Polyfold.EP.boundaryBranchedMeasureEffDim1_apply_all`, `Polyfold.K2Arc.k2_arc_hypotheses`). `EP.theorem_9_5_15 (hm : 0 < m) hΘ hcpt hT hloc`: for Θ branched and compact with an orientation T and `hloc` (at every point of $\operatorname{supp}\Theta$ an oriented tame structure of dimension $m+1$ describing T: "tame, n-dimensional", the reading of G.107), there is exactly one linear Φ from $\Omega^m_{ep,\infty}(X)$ to the signed measures on $(|\operatorname{supp}\partial\Theta|, L(\partial S))$, $L(\partial S)$ = `lebesgueSigmaBoundaryH`, with $\Phi\,\omega\,K = (1/\#G_x)\sum_{\partial_1 M_i\neq\emptyset}\sigma_i\int_{\partial_1 M_i\cap\pi^{-1}K}\omega|(\partial_1 M_i, o_{\partial M_i})$ for every $x\in\operatorname{supp}(\partial\Theta)$, every oriented tame structure at x describing T, and every measurable K with $|K|\subseteq|U(x)|$. `EP.theorem_9_5_16 (hm) (hLUC : X.IsLUC)` …: the same with $1/\#G_x^{\mathrm{eff}}$. The measures are `EP.boundaryMeasure`, `EP.boundaryMeasureEff` (`EP.boundaryMeasure_apply_all`). $n=1$: `EP.theorem_9_5_15_dim1_all hΘ hcpt hT hloc` and `EP.theorem_9_5_16_dim1_all (hLUC : X.IsLUC)` …, for $\omega\in\Omega^0_{ep,\infty}(X)$, with $\int_{\partial_1 M_i\cap\pi^{-1}K}\omega|(\partial_1 M_i, o) = \sum_p \pm\omega(p)$ over the finitely many boundary points (`EP.finZeroFormIntegration`; the measure `EP.boundaryBranchedMeasureDim1`, `EP.boundaryBranchedMeasureDim1_apply_all`); Part I's boundary integral of a 1-dimensional sheet is this value (`EpGroupoid.boundaryIntegral_eq_pointInteg`). The class condition of the earlier versions holds for every structure (`EP.boundarySheet_mem_msheets_of_loc`; for $n=1$ `EP.boundarySheet_finite_of_loc`), through the corner estimate `FormEval.boundaryStratum_inter_mem_msheets` and the tangent comparison `MPolyfold.k1_exists_tangent_eq`. Earlier forms kept: `theorem_9_5_15_msheets`, `theorem_9_5_16_msheets` (class hypothesis `hcov`, now derived, `EP.hcov_of_loc`), `theorem_9_5_15_hyp`, `theorem_9_5_16_hyp` (any sheet integration with the null sets of m-dimensional measure zero), `EP.theorem_9_5_15_class`, `EP.theorem_9_5_16_class`, `EP.theorem_9_5_15_dim1`, `EP.theorem_9_5_16_dim1_isLUC`. Existence and uniqueness: the boundary sheets form an oriented sheet family described by $\hat{T}_{\partial\Theta}$ (`boundaryFamily`, `represents_boundaryFamily`), and Theorem 9.5.4's machinery is proved for such families (`existsUnique_familyMeasure`). Witnesses with $\partial S\neq\emptyset$: the closed upper hemisphere in $[0,\infty)\times\mathbb{R}^2$ ($n=2$, `E8W.k1_hemisphere_stokes`) and the closed half-circle in $[0,\infty)\times\mathbb{R}$ ($n=1$, `K2Arc.k2_arc_hypotheses`). *Book:* §9.5, Theorem 9.5.15, p. 449 (2017: Theorem 9.3, p. 328), and Theorem 9.5.16, p. 449 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (restated):* No hypothesis beyond the text's. "Oriented, tame, n-dimensional" is read as in Theorem 9.5.4: an oriented tame structure describing $\hat{T}$ at every point (`hloc`, G.107); n-dimensionality follows from it. The class condition of the earlier versions (finite $\int|\omega|$ over the boundary sheets, which the text uses tacitly when it integrates over ∂M_i) is derived for every structure (`EP.boundarySheet_mem_msheets_of_loc`; gap G.109 closed). The sum runs over the nonempty $\partial_1 M_i$: empty sheets contribute nothing in the text. For $n=1$ the 0-forms are integrated over the finite sets $\partial_1 M_i$ of oriented points ($\pm\omega(p)$), the reading of $\int\omega|(\partial M_i, o_{\partial M_i})$ in dimension 0, and the formula holds for every structure (`_dim1_all`). Theorem 9.5.16 assumes that X has LUC, as the text does (Findings, Chapter 9, items 85, 117, 119, 120).

V9.68 (Theorems 9.5.17 (2017: 9.4) and 9.5.18, Stokes' theorem; `Polyfold.EP.theorem_9_5_17_msheets`, `Polyfold.EP.theorem_9_5_18_msheets`, `Polyfold.EP.theorem_9_5_17_hyp`, `Polyfold.EP.theorem_9_5_18_hyp`, `Polyfold.EP.stokes_of_boundaryMeasure`, `Polyfold.EP.boundaryBranchedMeasure`, `Polyfold.EP.boundaryBranchedMeasureEff`, `Polyfold.EP.hcovB_of_hcovS`, `Polyfold.EpGroupoid.IsBoundaryLocal`, `Polyfold.EpGroupoid.boundaryMeasure_univ_eq`, `Polyfold.EpGroupoid.boundaryMeasure_eq_zero_of_integ_eq_zero`, `Polyfold.EP.boundaryIntegral_eq_msInteg`, `Polyfold.E8W.e8_hemisphere_stokes`, `Polyfold.E8W.e8w_stokes_hypotheses`,`Polyfold.EP.theorem_9_5_17_boundary_sum_all`,`Polyfold.EP.theorem_9_5_18_boundary_sum_all`, `Polyfold.EP.stokes_of_partition_all`, `Polyfold.EP.exists_stokes_partition`, `Polyfold.EP.epFormsInftyEval_integ_eq_integral`, `Polyfold.EP.epFormsInftySmul`, `Polyfold.EP.theorem_9_5_17`, `Polyfold.EP.theorem_9_5_18`, `Polyfold.EP.hloc_hasLUC`, `Polyfold.EP.hcovS_of_loc`, `Polyfold.EP.hcovS_of_loc_hasLUC`, `Polyfold.EP.hcovS_of_loc_isLUC`, `Polyfold.E8W.k1_hemisphere_stokes`, `Polyfold.EP.theorem_9_5_17_dim1`, `Polyfold.EP.theorem_9_5_18_dim1`, `Polyfold.EP.boundaryBranchedMeasureDim1`, `Polyfold.EP.boundaryBranchedMeasureEffDim1`, `Polyfold.EP.stokes_of_boundaryMeasure_dim1`, `Polyfold.EpGroupoid.boundaryMeasure_univ_eqI`, `Polyfold.EP.theorem_9_5_18_dim1_isLUC`, `Polyfold.E8W.hasLUC_trivial`, `Polyfold.E8W.e8_hemisphere_stokes_eff`, `Polyfold.E8W.e8w_isLUC`, `Polyfold.E8W.k2_hemisphere_stokes_eff`, `Polyfold.K2Arc.isSubmanifoldOfDim_arc`, `Polyfold.K2Arc.isTameSubMPolyfold_arc`, `Polyfold.K2Arc.boundaryStratum_arc`, `Polyfold.K2Arc.arcOrientation`, `Polyfold.K2Arc.arc_hloc`, `Polyfold.K2Arc.arc_isLUC`, `Polyfold.K2Arc.k2_arc_stokes`, `Polyfold.K2Arc.k2_arc_stokes_eff`, `Polyfold.K2Arc.k2_arc_hypotheses`, `Polyfold.K3.remark_9_5_20_effIsotropy`, `Polyfold.K3. `, `Polyfold.K3. B`, `Polyfold.K3.P`, `Polyfold.K3.boundary_eq`). `EP.theorem_9_5_17 (hm : 0 < m) hPU hΘ hcpt hT hloc` : for X with sc-smooth partitions of unity (`hPU`), Θ branched and compact with an orientation T, `hloc` (at every point of supp Θ an oriented tame structure describing T: "tame, n-dimensional") and $\omega \in \Omega^m_{ep,\infty}(X)$: `tameBranchedIntegral` $(d\omega)$ = `EP.boundaryMeasure` ω (univ), i.e. $\int_{\hat{\Theta}} d\omega = \mu_\omega^{\partial\hat{\Theta}}(\partial S)$ with the measure of Theorem 9.5.15 (V9.67). `EP.theorem_9_5_18 (hm) hPU (hLUC : X.IsLUC) …`: the same with `tameBranchedIntegralEff` and `EP.boundaryMeasureEff`. $n = 1$: `EP.theorem_9_5_17_dim1 hPU hΘ hcpt hT hloc` (`tameBranchedIntegral` $(d\omega)$ = `EP.boundaryBranchedMeasureDim1` ω (univ), for $\omega \in \Omega^0_{ep,\infty}(X)$) and `EP.theorem_9_5_18_dim1_isLUC hPU (hLUC : X.IsLUC) …`. The class hypothesis `hcovS` of the earlier forms (`EP.theorem_9_5_17_msheets`, `EP.theorem_9_5_18_msheets`, `_hyp`, which remain) is derived (`EP.hcovS_of_loc`, `EP.hcovS_of_loc_isLUC`). The boundary-sum form holds for every $n \geq 1$ (`theorem_9_5_17_boundary_sum_all`, `theorem_9_5_18_boundary_sum_all`). Key steps: Part I's boundary integral of a sheet equals the §9.5 integral over $\partial_1 M_i$ with $o_{\partial M_i}$ (`boundaryIntegral_eq_msInteg`; for $n = 1$ `EpGroupoid.boundaryIntegral_eq_pointInteg`); μ^∂ is local (`boundaryMeasure_univ_eq`, `boundaryMeasure_univ_eqI`). Witnesses with $\partial S \neq \emptyset$: the closed upper hemisphere in $[0, \infty) \times \mathbb{R}^2$, $n = 2$ (`E8W.k1_hemisphere_stokes`; with LUC `E8W.k2_hemisphere_stokes_eff`), and the closed half-circle in $[0, \infty) \times \mathbb{R}$, $n = 1$ (`K2Arc.k2_arc_` `stokes`, `K2Arc.k2_arc_stokes_eff`). Remark 9.5.20, in part: `K3.remark_9_5_20_effIsotropy` (the M-polyfold X with $\partial X = \{(\pm 1, 0)\}$, $\mathbb{Z}_3 \ltimes X$ with LUC, $\#G_x^{\text{eff}} = 3$ on M, and at $x \in \partial X$ $\#G_x^{\text{eff}} = 3$ read in X but 1 read in ∂X). *Book:* §9.5, Theorem 9.5.17, p. 449 (2017: Theorem 9.4, p. 328), and Theorem 9.5.18, p. 450 (new in the 2021 edition), with the proof, pp. 450–451. *Hidden hypotheses:* none. *Deviation (restated):* No hypothesis beyond the text's: tameness is read as in Theorem 9.5.4 (an oriented tame structure at each point, `hloc`, G.107), and the class condition `hcovS` of the earlier forms is derived (`EP.hcovS_of_loc`). The integrals of §9.5 are realized on one graph chart (interior sheets) and on countably many graph charts (boundary sheets); for $n = 1$ the boundary measure is the signed sum of the values of ω at the oriented boundary points (`_dim1`). Remark 9.5.20 is formalized in part: the M-polyfold X with $\partial X = \{(\pm 1, 0)\}$, $\mathbb{Z}_3 \ltimes X$ with LUC, and the effective isotropy of a point of ∂X ($\mathbb{Z}_3$ read in X, trivial read in ∂X); Θ, $\oint_{\Theta} df$ and the two boundary integrals are not (gap G.110, in part for the remark only). The M-polyfold structure on X is our choice, one chart into a retract in $\mathbb{R}^2 \oplus c_0 \oplus c_0$ (the text says only that one exists); "∂X is a smooth submanifold" is not shown, only that ∂X is a saturated sub-M-polyfold (Findings, Chapter 9, items 85, 117, 119, 120).

V9.69 ((9.22) (2017: (9.18)), the measure of $f \cdot \omega$; `Polyfold.SignedDensity.setIntegral_eq_jordan`, `Polyfold.SignedDensity.setIntegral_sub_toSignedMeasure`, `Polyfold.SignedDensity.abs_setIntegral_sub_le`, `Polyfold.SignedDensity.eq_setIntegral_of_estimate`, `Polyfold.SignedDensity.eq_setIntegral_of_local_estimate`, `Polyfold.EpGroupoid.FormEval.absInteg_union`, `Polyfold.EpGroupoid.OrientedTameStructure.absSheetSum_union`, `Polyfold.EpGroupoid.eq_setIntegral_of_sheetSum`, `Polyfold.EpGroupoid.branchedMeasure_eq_setIntegral_mul`, `Polyfold.EpGroupoid.branchedMeasureEff_eq_setIntegral_mul`, `Polyfold.EpGroupoid.branchedMeasure_eq_jordan`, `Polyfold.EpGroupoid.branchedMeasure_eq_setIntegral_mul_of_continuous`, `Polyfold.EpGroupoid.branchedMeasureEff_eq_setIntegral_mul_of_continuous`, `Polyfold.EpGroupoid.FormEval.IsMul`, `Polyfold.EpGroupoid.OrientedTameStructure.localMeasure_eq_sum_setIntegral_mul`, `Polyfold.EpGroupoid.FormEval.isMul_smul`). For $\tau = f \cdot \omega$ (`FormEval.IsMul`: $\tau = f\omega$ on the frames of the tangent spaces of the sheets) and $\bar{f} \circ \pi = f$ with $\bar{f}$ bounded and $L(S)$-measurable (in particular continuous): $\mu_\tau^{\hat{\Theta}}(K) = \int_K \bar{f}\, d\mu_\omega^{\hat{\Theta}}$ and $\mu_\tau^{\hat{\Theta},\text{eff}}(K) = \int_K \bar{f}\, d\mu_\omega^{\hat{\Theta},\text{eff}}$ for every $K \in L(S)$ (`branchedMeasure_eq_setIntegral_mul`, `branchedMeasureEff_eq_setIntegral_mul`, `_of_continuous`), the integral against the signed measure being Mathlib's `VectorMeasure.integral`, equal to $\int_K \bar{f}\, d\mu_\omega^+ - \int_K \bar{f}\, d\mu_\omega^-$ (`branchedMeasure_eq_jordan`, `SignedDensity.setIntegral_eq_jordan`). The local form, with f inserted under the sheet integrals of (9.21): `localMeasure_eq_sum_setIntegral_mul`. *Book:* §9.5, (9.22), p. 441 (2017: (9.18), p. 325). *Hidden hypotheses:* $\bar{f}$ bounded and $L(S)$-measurable, weaker than the text's continuity (f sc-smooth, S compact). *Deviation (explicit):* Stated globally on $L(S)$ (the text: on $|U(x)|$; the local form follows from `branchedMeasure_apply`). $\tau = f \cdot \omega$ is `FormEval.IsMul`, read through the evaluation of the forms on the frames of the sheets; the form $f \cdot \omega \in \Omega_{ep}^n(X)$ is built for constant f only (`isMul_smul`), not for a non-constant sc-smooth f (gap G.107) (Findings, Chapter 9, item 119).

What is not verified in this chapter

- p. 402, Remark 9.1.1 on a general M-polyfold: the local constancy of dim $T_m M$ (the hypothesis `HasLocallyConstantDimAt`, proved on $\mathbb{R}^m$); the footnote to Definition 9.1.2 and "locally equidimensional" (V9.2) (gap G.93; Findings, items 2, 3).
- p. 403, Remark 9.1.4: Listed here before 2026-11-12; verified since (V9.63; gap G.94 closed; Findings, items 7, 114).
- p. 405, Proposition 9.1.9 (2017: 9.1) and Remark 9.1.10 (2017: 9.3): Listed here before 2026-11-12; verified since (V9.64, V9.65; gap G.95 closed). Remark 9.1.10 for Θ of several dimensions, and the composition of the orbifold groupoid as a Mathlib-smooth map (Mathlib has no classical orbifolds) (Findings, item 114).
- pp. 407–408, Proposition 9.1.11 (2017: 9.2): the gluing of the local Θ_k into branched ep⁺-subgroupoids and the local finiteness; the uniqueness, which does not hold as we read it, under a repaired condition (V9.9) (gap G.96; Findings, item 8).
- pp. 413–417, Proposition 9.2.8 as printed and Lemmas 9.2.9–9.2.11 (2017: Proposition 9.3, Lemmas 9.6–9.8) on a general M-polyfold: the graph representation is the hypothesis `HasGraphChartAt` (`HasGraphChartsAt`, `HasGraphBallAt`), proved on $\mathbb{R}^m$; on a general M-polyfold Proposition 9.1.12, the independence in Theorem 9.2.1, Lemma 9.3.4, Lemma 9.6.4 and Proposition 9.6.5 rest on it (V9.10, V9.14, V9.22, V9.32, V9.61, V9.62) (gap G.97; Findings, items 25, 100).
- p. 412, Lemma 9.2.7 (2) (2017: 9.5 (2)) on a general M-polyfold: invariance of domain (`HasDomainInvarianceAt`, proved on $\mathbb{R}^m$) (V9.21) (gap G.98; Findings, item 24).
- p. 418, Definition 9.2.12 (2017: 9.6): that the TM_i are smooth submanifolds of TX; also on $\mathbb{R}^m$ (V9.23) (gap G.99; Findings, item 26).
- p. 423, Lemma 9.3.4 (2017: 9.10) on a general M-polyfold: the level-1 continuity of frame fields (`HasContinuousLevelOneOn`, proved on $\mathbb{R}^m$) (V9.32) (gap G.100; Findings, item 53).
- p. 425, Example 9.3.5 (2017: 9.1), the classification of all orientations of Θ: Listed here before 2026-11-12; verified since (V9.33; gap G.101 closed; Findings, items 54, 115).
- p. 426, Definition 9.3.6 (2017: 9.10): that a tame M_i is a manifold with boundary with corners (V9.34) (gap G.102; Findings, item 55).
- pp. 427–429, Definitions 9.3.8 and 9.3.10 (2017: 9.9 and 9.11): $T_y \partial M = T_y^R M$ and the outward direction from the cones $C_y M$, the continuity of $o_{\partial M}$ and Figure 9.1; Remark 9.3.11 is a comment, with nothing to formalize (V9.36–V9.38) (gap G.103; Findings, items 57, 58). Since 2026-11-12 the stratum $\{d_M = 1\}$ is open and dense in ∂M for a tame sheet; verified since 2026-11-13: it is an oriented submanifold, its orientation is Part I's face orientation, and Figure 9.1 for face-structured sheets (V9.36, V9.38; gap G.103 closed; Findings, items 110, 119). Not verified: the orientations of the faces of the faces.
- p. 430, Lemma 9.4.3 (2017: 9.12): Property (T) after shrinking $U(x)$ (V9.41) (gap G.104; Findings, item 71).
- pp. 431–432, Lemmas 9.4.4 and 9.4.6 (2017: 9.13, 9.14) for sheets in an M-polyfold: Listed here before 2026-11-12; verified since in M-polyfold charts (V9.42, V9.44; gap G.105 closed; Findings, items 72, 116). Not compared: the essential points in M-polyfold charts with those

in the corner charts.

- p. 437, Definition 9.5.1 and Lemma 9.5.2 (2017: Definition 9.15, Lemma 9.16) with Lebesgue null sets in the sheets of a general M-polyfold: Listed here before 2026-11-12; verified since (`IsHNull`; V9.49, V9.50; gap G.106 closed; Findings, items 80, 81, 116). Lemma 9.5.2 across dimensions does not hold as we read it (outside the standing assumption of §9.5).
- pp. 438–447, branched integration: Theorems 9.5.4 and 9.5.5 (2017: Theorem 9.2 and new), Definition 9.5.6, (9.21), Lemma 9.5.7, the measure identities of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 (1) are verified since 2026-11-12 for an abstract integration of forms over the oriented sheets, and since 2026-11-13 for the integrals of sc-forms, with (9.22) and Proposition 9.5.13 (2), and a witness with nonempty support, the unit circle (V9.54–V9.57, V9.66, V9.69; gap G.107 closed; Findings, items 84, 85, 118, 119). Not verified: the form $f \cdot \omega$ for a non-constant sc-smooth f ((9.22) reads it through the evaluation on the sheets); dimension 0 is the abstract `pointIntegration`.
- p. 448, $(n-1)$-dimensional measure zero in the boundaries of the M_i: listed here before 2026-11-13; verified since with the Hausdorff null sets in the ambient charts (V9.52, V9.53; gap G.108 closed; Findings, items 83, 117, 119). Not compared: these null sets with the face-by-face ones (`IsFaceNull`).
- p. 449, Theorems 9.5.15 and 9.5.16 (2017: Theorem 9.3 and new), the canonical boundary measures: listed here before 2026-11-13 (wave K); verified since as printed for every $n \geq 1$, with the text's hypotheses only (V9.67; gap G.109 closed; Findings, items 85, 117, 119, 120).
- pp. 449–452, Theorems 9.5.17 and 9.5.18 (2017: Theorem 9.4 and new), Stokes' theorem: verified since 2026-11-13 (wave K) as printed for every $n \geq 1$, with the text's hypotheses only (V9.68); witnesses with $\partial S \neq \emptyset$: the closed upper hemisphere (also with LUC) and the closed half-circle (no corners, no branching). Remark 9.5.20 is verified in part: its M-polyfold X, ∂X, $\mathbb{Z}_3 \ltimes X$ with LUC and the effective isotropy of the boundary points in X and in ∂X. Not verified: $\Theta(t,z) = \delta_0(z)$, $\oint_{\Theta} df$ and the two boundary integrals of the remark, and that ∂X is a smooth submanifold (gap G.110 in part; Findings, items 85, 117, 119, 120). Remarks 9.5.12 (history and open questions) and 9.5.19 (on the partition-of-unity hypothesis) have no formal content.
- pp. 457–458, §§9.6.2–9.6.3: open questions, no formal content.
- Witnesses: every hypothesis of the chapter is satisfiable, and those of §9.2 and §9.6.1 hold on every ep-groupoid over $\mathbb{R}^m$; the hypothesis of Definition 9.2.12 has only the zero functor as a witness, and no witness of Lemma 9.2.13 at a boundary point is constructed (Findings, items 100–102).

11. Chapter 10 (HWZ, Part II, §§10.1–10.5 and Appendix §§10.6.1–10.6.3 (2017 §§10.1–10.5)): verified for equivalences, natural equivalences, the weak fibred product, the localization $\mathcal{EP}(E^{-1})$ with its universal property, strong bundle equivalences and generalized strong bundle maps, Remark 10.1.8 with Zhou's example, and Propositions 10.4.6, 10.4.7 and 10.5.6 with Definition 8.4.2 read for local strong bundle isomorphisms, with the readings recorded below; Theorems 10.3.19 and 10.5.7 do not hold as we read them, and Theorem 10.3.19 holds for an effective target

Chapter 10 (pp. 461–520 (2017 pp. 333–374); numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) treats equivalences of ep-groupoids, the sc-smooth form of equivalences of categories, and prepares their inversion. §10.1 defines equivalences (Definition 10.1.1 (2017: 10.1)), characterizes them (Lemma 10.1.2 (2017: 10.1)), refines an ep-groupoid along an open covering (Proposition 10.1.3 (2017: 10.1)), recalls natural equivalences (Definition 10.1.4 (2017: 10.2)) and shows that they and compositions respect equivalences (Propositions 10.1.5, 10.1.6 (2017: 10.2, 10.3)), that the tangent of an equivalence is one (Theorem 10.1.7 (2017: 10.1)), and, mostly new in the 2021 edition, which properties of Chapter 7 are invariant under equivalences: the local regularity property (Propositions 10.1.9, 10.1.10 (2017: 10.4), Remark 10.1.11 (2017: 10.1)), the non-effective part and the reduced ep-groupoid (Remark 10.1.12, Propositions 10.1.13, 10.1.14 (2017: 10.5), Example 10.1.15 (2017: 10.1)), (LUC), (ZHOU), (SLUC) and (SLR) (Propositions 10.1.16–10.1.19). §10.2 builds the weak fibred product $X \times_Y Z$ (Definition 10.2.1 (2017: 10.3)) and shows that it is an ep-groupoid with an equivalence to Z when F is an equivalence (Theorem 10.2.2 (2017: 10.2), Remark 10.2.3 (2017: 10.2)). §10.3 inverts the equivalences: diagrams $X \overset{F}{\longleftarrow} A \overset{\Phi}{\longrightarrow} Y$ with F an equivalence, their refinements and equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4 (2017: Definitions 10.4–10.6, Lemmas 10.3, 10.4)), the induced maps of orbit spaces (Lemma 10.3.6, Definition 10.3.7 (2017: 10.5, 10.7)), the composition of classes (Theorem 10.3.8 (2017: 10.3), proved in Appendix §10.6.1), generalized maps and isomorphisms (Definition 10.3.9, Theorem 10.3.10 (2017: 10.8, 10.4), the latter proved in Appendix §10.6.2), the category $\mathcal{EP}(E^{-1})$ with the functors $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ and $|\cdot| : \mathcal{EP}(E^{-1}) \to$ TOP (Proposition 10.3.11, Definition 10.3.12, Theorems 10.3.14, 10.3.15 (2017: Proposition 10.6, Definition 10.9, Theorems 10.5, 10.6)), and, mostly new in 2021, the invariance of the properties of Chapter 7 under generalized isomorphisms (Propositions 10.3.16, 10.3.17, Remark 10.3.18 (2017: 10.3), Theorem 10.3.19 (2017: 10.7), Definition 10.3.20, Theorem 10.3.21). §10.4 defines strong bundle equivalences (Definition 10.4.1 (2017: 10.10)), the pull-back and push-forward of section functors and their properties (Propositions 10.4.2–10.4.4 (2017: 10.7–10.9), Theorem 10.4.5 (2017: 10.8)), natural equivalences of strong bundle functors (Definitions 10.4.8, 10.4.10, Remark 10.4.9 (2017: Definitions 10.11, 10.12, Remark 10.4)) and the category $\mathcal{SEP}$ (Definition 10.4.11 (2017: 10.13)). §10.5 repeats the localization for strong bundles: diagrams $W \overset{\Phi}{\longleftarrow} W'' \overset{\Psi}{\longrightarrow} W'$ with Φ a strong bundle equivalence, their classes, the generalized strong bundle maps, with the maps $[d]$ and $|[D]|$, the square $|\mathcal{D}|$ and the homomorphisms γ (pp. 504–506 (2017 pp. 364–366, with Remark 10.5)), the category $\mathcal{SEP}(F^{-1})$ and the generalized strong bundle isomorphisms (Definitions 10.5.1, 10.5.2 (2017: 10.14, 10.15), Remark 10.5.3), the push-forward of section functors along them (Theorem 10.5.4 (2017: 10.9), Remark

10.5.5), and the local regularity property under them (Proposition 10.5.6, new in the 2021 edition, and Theorem 10.5.7 (2017: 10.10), whose local regularity hypothesis is new in the 2021 edition).

What is verified. Every definition of §§10.1, 10.2 and 10.4 is formalized: Definitions 10.1.1 (V10.1), 10.1.4 (V10.4), 10.2.1 (V10.17), 10.4.1 (V10.22), 10.4.8 (V10.28), 10.4.10 (V10.30) and 10.4.11 (V10.31), with the class E of equivalences, the refinement $X_{\mathcal{U}}$ (V10.3) and the pull-back and push-forward of section functors (V10.23). Lemma 10.1.2, Propositions 10.1.3, 10.1.5, 10.1.6, 10.1.13, 10.1.14, 10.1.16, 10.1.17 and 10.1.18, Theorem 10.1.7, Remark 10.1.12 and Remark 10.2.3 hold as stated; Remark 10.1.11 and Example 10.1.15 hold in stronger forms (V10.8, V10.12); Theorem 10.2.2 holds in all its claims, its first claim under the weaker hypothesis that F or G is a local sc-diffeomorphism on objects (V10.18–V10.20); Propositions 10.4.2–10.4.4 hold for all section functors, without sc-smoothness (V10.24–V10.26); Theorem 10.4.5 holds without its Fred part (V10.27); Remark 10.4.9 and the equivalence of Definitions 10.4.8 and 10.4.10 hold for strong bundle functors (V10.29, V10.30). Propositions 10.1.9 and 10.1.19 hold as stated, and Proposition 10.1.10 holds with (LR) on one side only (V10.62, V10.16, V10.63). In §10.3 every definition is formalized (V10.33, V10.35, V10.37, V10.39, V10.41, V10.44, V10.49); the properties (E1)–(E3), Lemmas 10.3.2, 10.3.4, 10.3.6, Theorems 10.3.8, 10.3.10, 10.3.14, 10.3.15, Propositions 10.3.11 and 10.3.17 hold as stated, with the proofs of Appendix §§10.6.1–10.6.2 (V10.32–V10.47); Remark 10.3.18 is verified, its non-uniqueness claim through the example of Remark 10.5.5 (V10.48, V10.60), Proposition 10.3.16 and Theorem 10.3.21 hold as stated (V10.64, V10.50); Theorem 10.3.19 does not hold as we read it, and holds when the target is effective (V10.65, V10.66). In §10.5 the diagrams of strong bundles, their refinements, the maps $[d]$ and $|[D]|$, the square $|\mathcal{D}|$ and the homomorphisms γ are formalized (V10.51–V10.54); Definition 10.5.2 and Remark 10.5.5 hold as stated, the latter with an example the text does not give (V10.56, V10.60); the weak fibred product of strong bundles, which the text leaves to the reader, is constructed, with the strong bundle versions of Theorem 10.2.2, Lemma 10.3.4 and Theorem 10.3.8 (V10.68–V10.70), and Definition 10.5.1 holds, $\mathcal{SEP}(F^{-1})$ being a category with a localization functor (V10.55); Remark 10.5.3 is verified for the base and total spaces and, read as "generalized strong bundle isomorphisms are isomorphisms", as one direction of the strong bundle Theorem 10.3.10 (V10.57, V10.71); Theorem 10.5.4 holds without its Fred part (V10.58, V10.59); Proposition 10.5.6 is reduced to Proposition 10.4.6 (V10.61); and Theorem 10.5.7 does not hold as we read it (V10.67). Several places of the text needed more than a reading. (a) *Proposition 10.1.6 (2017: 10.3).* The argument on objects covers only the points $F(x)$; the transport to every y along a morphism $F(x) \to y$ is supplied (V10.6). (b) *Theorem 10.1.7 (2017: 10.1).* Fullness and faithfulness of TF, left to the reader, are proved (V10.7); Proposition 10.1.5, quoted from the literature, is proved (V10.5). (c) *Propositions 10.1.13 and 10.1.16.* The closedness of $\mathbf{X}^T$ passes along equivalences by a closure argument with a conjugation by a local inverse of t in place of sequences (V10.10, V10.13). (d) *Theorem 10.2.2 (2017: 10.2).* The ep-groupoid structure of $X \times_Y Z$ needs of F only that it is a local sc-diffeomorphism on objects, its properness holds for any F and G, and its M-polyfold structures are read as iterated sub-M-polyfolds of Part I's fibred products (V10.18); the composite in the fullness step is read with the ends it must have (V10.19). (e) *Definition 10.4.1 (2017: 10.10).* $W[1]$ is not an ep-groupoid in the project; clause (1) at level $[1]$ is read as "$|\Phi| : |W[1]| \to |W'[1]|$ is a homeomorphism", the rest of it being a consequence of (2) (V10.22). (f) *Theorem 10.4.5 (2017:*

10.8). The sc-smoothness of $\Phi_* f$, "easily verified" in the text, and the preservation of sc$^+$-sections are proved (V10.27). (g) *Theorem 10.3.8 (2017: 10.3), Appendix §10.6.1.* The functor L between weak fibred products is shown to be an equivalence by two out of three, and the text's identification of the two iterated weak fibred products ("the same space") is a reassociation functor, an equivalence (V10.40). (h) *Theorem 10.3.10 (2017: 10.4), Appendix §10.6.2.* In claim (C2) the text uses that a map Γ' is a local sc-diffeomorphism, which, as we read it, already needs the other half of (C2); the local sc-diffeomorphism property of Φ is proved instead from a local sc-smooth right inverse and local injectivity (V10.42). (i) *Theorem 10.3.14 (2017: 10.5).* With Definition 10.3.1 as stated, it is the composite diagram (10.13) that refines $d_{\Psi\circ\Phi}$, not the reverse; the equality of classes is unchanged (V10.45). (j) *§10.5.* The weak fibred product of strong bundles, which the text uses "following our earlier construction" (p. 504) without constructing it, is the pull-back of W'' along π_Z, so that the squares in the proofs of the composition commute strictly, by the fibrewise injectivity of strong bundle equivalences (V10.68, V10.70); the classes, taken for the generated equivalence relation, are the text's by the transitivity (V10.51, V10.69), and the independence of $[D]_*$ of the representative is proved through the orbit spaces, for any two representatives with the same $|D|$ (V10.58). (k) *Proposition 10.1.9.* That $Q(U(x))$ carries a natural representation of G_y, which the text takes for granted, is proved by restricting one at y (V10.62). (l) *Theorem 10.3.19 (2017: 10.7).* The proof's step on p. 494 uses the families α_a of Proposition 10.1.10 (1) as a natural transformation; on $\mathbb{Z}/3\times\mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action, which has (LR), there is no natural transformation $1 \to \mathrm{Inv}$, and the statement and its strong bundle version, Theorem 10.5.7, do not hold as we read them (V10.65, V10.67); over an effective target the families are unique and glue (V10.66). Possible misprints are recorded in V10.3, V10.6, V10.9, V10.14, V10.15, V10.17, V10.19, V10.24, V10.27, V10.40, V10.42, V10.53 and V10.62; V10.61 records a reading point. Not verified: the subcategory $\mathcal{EP}_{lreg}$, Remark 10.1.8 with its example in Appendix §10.6.3, the agreement of the M-polyfold structure of $X \times_Y Z$ with one induced from the triple products, the universal property of the localization, the Fred part of Theorem 10.4.5, Propositions 10.4.6 and 10.4.7, the converse half of the strong bundle version of Theorem 10.3.10, the symbolic diagram (10.24) made precise with the local expressions q, Q of (10.25), the Fred part of Theorem 10.5.4 and Proposition 10.5.6 itself (gaps G.111–G.127; G.113, G.114, G.115, G.118 and G.119 are closed since, by V10.62, V10.63, V10.16, V10.64 with V10.50, and V10.60, G.123 is closed since by V10.68–V10.71 except the converse half of the strong bundle Theorem 10.3.10, and G.120 and G.127 are settled since by the counterexamples V10.65 and V10.67). Since 2026-11-12 every gap of the chapter is closed: the subcategory $\mathcal{EP}_{lreg}$ (V10.74), Remark 10.1.8 with Appendix §10.6.3 as printed, on the appendix's weighted Sobolev scale with its projections (V10.75), the triple-product structures of Theorem 10.2.2 (V10.18), the universal property of the localization (V10.45), Propositions 10.4.6 and 10.4.7 (V10.72, V10.73) and Proposition 10.5.6 (V10.61), with Definition 8.4.2 read for local strong bundle isomorphisms, (10.24) with generalized vertical maps and the local expressions q, Q of (10.25) (V10.53, V10.54), and the converse half of the strong bundle Theorem 10.3.10 (V10.71) (gaps G.111, G.112, G.116, G.117, G.122, G.124, G.126 closed, and G.123 in full).

The hypotheses. No hypothesis beyond the text's is taken on a general ep-groupoid, apart from two technical ones, each with a witness: the object set is nonempty in Proposition 10.1.3 (the open-subset M-polyfold structure of Part I needs a nonempty set; the trivial covering of any nonempty ep-groupoid and a two-set covering of $\mathbb{Z}/2 \ltimes \mathbb{R}$ are witnesses); the weak fibred product is given its ep-groupoid structure under `Polyfold.WeakFibredProduct.Admissible` (F or G a local sc-diffeomorphism on objects), which every equivalence supplies (`Polyfold.WeakFibredProduct.Admissible.of_isEquivalence_left`, `Polyfold.WeakFibredProduct.Admissible.of_isEquivalence_right`). The reduction of Proposition 10.1.19 took Proposition 10.1.9 for the lifts as a hypothesis (satisfied when the two ep-groupoids agree, `Polyfold.EP.isLR_iff_lift_self`); since Proposition 10.1.9 is proved (V10.62), Proposition 10.1.19 holds without it (V10.16). Theorem 10.3.19 is proved under the added hypothesis that the target is effective (V10.66), without which it does not hold as we read it (V10.65). Proposition 10.1.14 keeps the text's hypothesis that both reduced morphism spaces are Hausdorff. In §10.3 no hypothesis is added; a diagram is a structure one universe above the object and morphism types of the ep-groupoids, so that the classes of diagrams are sets and $\mathcal{EP}(E^{-1})$ is a large category like $\mathcal{EP}$. In §10.5 no hypothesis is added; the classes of diagrams of strong bundles are taken for the equivalence relation generated by having a common refinement, which is that relation itself (V10.51, V10.69). Propositions 10.4.6, 10.4.7 and 10.5.6 (since 2026-11-12) add no hypothesis; they read Definition 8.4.2 as `Polyfold.EpStrongBundle.IsLRLocal`, with local strong bundle isomorphisms of $W|U(x)$ (witness `Polyfold.EpStrongBundle.isLRLocal_trivial`), the reading already used for Theorem 10.5.7 (Findings, items 72, 74), and Proposition 10.4.7 uses (LR) of the target only.

The setting. Ep-groupoids, sc-smooth functors, the orbit spaces with their quotient topologies, the category $\mathcal{EP}$ (with Mathlib's composition order: $f \gg g$ is the text's $g \circ f$, and so is d.comp e for diagrams) and the lifts X^i are those of Chapter 7 (`Polyfold.EpGroupoid`, `Polyfold.EpFunctor`, `Polyfold.EP`, `Polyfold.EP.lift`); (LUC), (ZHOU), (LR), (SLUC) and (SLR) are Chapter 7's `Polyfold.EpGroupoid.IsLUC`, `Polyfold.EpGroupoid.IsZhou`, `Polyfold.EpGroupoid.IsLR`, `Polyfold.EP.IsSLUC`, `Polyfold.EP.IsSLR`, and the reduced ep-groupoid is `Polyfold.EpGroupoid.reducedEpGroupoid`. Natural transformations, natural equivalence and the tangent functor are Chapter 8's (`Polyfold.EpNatTrans`, `Polyfold.EpFunctor.NaturallyEquivalent`, `Polyfold.EpFunctor.tangent`), and so are strong bundles over ep-groupoids, strong bundle functors and local strong bundle isomorphisms (`Polyfold.EpStrongBundle`, `Polyfold.EpStrongBundle.IsStrongBundleFunctor`, `Polyfold.StrongBundle.IsStrongBundleIsoOn`). "Full", "faithful", "essentially surjective" and "equivalence of categories" are Mathlib's notions for the underlying functor of the groupoids. The category $\mathcal{SEP}$ and its class F of strong bundle equivalences are §10.4's (`Polyfold.SEP`, `Polyfold.SEP.equivalences`). Findings referred to below are those of `Exposition/Findings.md`, "Chapter 10".

V10.1 (Definition 10.1.1 (2017: 10.1), equivalences and the class E; `Polyfold.EpFunctor.IsEquivalence`, `Polyfold.EP.equivalences`, `Polyfold.GroupoidFunctor.full_iff`, `Polyfold.GroupoidFunctor.faithful_iff`, `Polyfold.GroupoidFunctor.essSurj_iff`, `Polyfold.EpFunctor.IsEquivalence.essSurj`, `Polyfold.EpFunctor.IsEquivalence.exists_iso`, `Polyfold.EpFunctor.IsEquivalence.exists_mor_eq`, `Polyfold.EpFunctor.IsEquivalence.mor_injective`, `Polyfold.EpFunctor.IsEquivalence.id`, `Polyfold.EpGroupoid.Examples.pairToPoint`, `Polyfold.EpGroupoid.Examples.isEquivalence_pairToPoint`). An equiva-

lence $F : X \to Y$ of ep-groupoids is an sc-smooth functor which is full and faithful (every $\psi : F(x) \to F(x')$ is $F(\varphi)$ for some $\varphi : x \to x'$, and F is injective on each set of morphisms $x \to x'$), induces a homeomorphism $|F| : |X| \to |Y|$ of the orbit spaces, and is a local sc-diffeomorphism on objects. The equivalences form a class E of morphisms of $\mathcal{EP}$ containing the identities and closed under composition. An equivalence is essentially surjective: every y is the target of a morphism from some $F(x)$. The identity and the functor from the pair groupoid of $\mathbb{Z}$ to the point are equivalences. *Book:* Definition 10.1.1, p. 461 (2017: Definition 10.1, p. 333), with the category notions recalled on pp. 461–462. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 1).

V10.2 (Lemma 10.1.2 (2017: 10.1) and the remarks around it; `Polyfold.EpFunctor.lemma_10_1_2`, `Polyfold.EpFunctor.isEquivalence_iff`, `Polyfold.EpFunctor.IsEquivalence.isEquivalence_toFunctor`, `Polyfold.EpFunctor.IsEquivalence.comp`, `Polyfold.GroupoidFunctor.injective_orbitMap_of_full`, `Polyfold.GroupoidFunctor.surjective_orbitMap_iff`, `Polyfold.GroupoidFunctor.bijective_orbitMap_of_isEquivalence`, `Polyfold.EpFunctor.continuous_orbitMap`, `Polyfold.EpFunctor.isOpenMap_orbitMap`, `Polyfold.EpGroupoid.Examples.not_exists_leftInverse_pairToPoint`). An sc-smooth functor which is an equivalence of categories and a local sc-diffeomorphism on objects is an equivalence, and conversely every equivalence is both. For a functor of groupoids, $|F|$ is injective if F is full, surjective if and only if F is essentially surjective, and bijective for an equivalence of categories; $|F|$ is continuous, and open if F is open on objects. The composite of two equivalences is an equivalence, and the equivalence from the pair groupoid of $\mathbb{Z}$ to the point has no left inverse on objects. *Book:* Lemma 10.1.2, p. 462 (2017: Lemma 10.1, p. 334), and the remarks on pp. 461–462. *Hidden hypotheses:* none. *Deviation (stronger):* the lemma is stated with its converse; "not invertible as a functor" gets a witness.

V10.3 (Proposition 10.1.3 (2017: 10.1), the refinement $X_{\mathcal{U}}$; `Polyfold.EpGroupoid.OpenCover`, `Polyfold.EpGroupoid.OpenCover.univ`, `Polyfold.EpGroupoid.OpenCover.objSet`, `Polyfold.EpGroupoid.OpenCover.morSet`, `Polyfold.EpGroupoid.OpenCover.groupoidStr`, `Polyfold.EpGroupoid.OpenCover.objPolyfold`, `Polyfold.EpGroupoid.OpenCover.morPolyfold`, `Polyfold.EpGroupoid.OpenCover.isLocalScDiffeo_src`, `Polyfold.EpGroupoid.OpenCover.isLocalScDiffeo_tgt`, `Polyfold.EpGroupoid.OpenCover.exists_isProperMap`, `Polyfold.EpGroupoid.OpenCover.isScSmoothCompSheet`, `Polyfold.EpGroupoid.OpenCover.epGroupoid`, `Polyfold.EpGroupoid.OpenCover.functor`, `Polyfold.EpGroupoid.OpenCover.proposition_10_1_3`, `Polyfold.EpGroupoid.Examples.reflectionCover`, `Polyfold.EpGroupoid.Examples.isEquivalence_reflectionCover`, `Polyfold.EpGroupoid.Examples.not_injective_reflectionCover_obj`). For a covering of the object M-polyfold of X by open sets U_i, $i \in \iota$, the refinement $X_{\mathcal{U}}$ has as objects the pairs (i, x) with $x \in U_i$, an open subset of $\iota \times X$ with ι discrete, and as morphisms the triples (i, φ, j) with $s(\varphi) \in U_i$, $t(\varphi) \in U_j$, an open subset of $(\iota \times \iota) \times \mathbf{X}$, with $s(i, \varphi, j) = (i, s(\varphi))$, $t(i, \varphi, j) = (j, t(\varphi))$, $(j, \psi, k) \circ (i, \varphi, j) = (i, \psi \circ \varphi, k)$, units $(i, 1_x, i)$ and inverses (j, φ^{-1}, i). If X has an object, $X_{\mathcal{U}}$ is an ep-groupoid and $F : X_{\mathcal{U}} \to X$, $(i, x) \mapsto x$, $(i, \varphi, j) \mapsto \varphi$, is an equivalence. For the covering $\{x > -1\}$, $\{x < 1\}$ of $\mathbb{Z}/2 \ltimes \mathbb{R}$ this F is an equivalence which is not injective on objects. *Book:* Proposition 10.1.3, p. 462 (2017:

Proposition 10.1, p. 334), with the construction of $X_{\mathcal{U}}$ on p. 462. *Hidden hypotheses:* the object set is nonempty (the open-subset M-polyfold structure needs a nonempty set; for an empty X the statement is trivial). *Deviation (explicit):* the covering is indexed by a type, the text's index set being the covering itself; "induced from X and **X**" is read as open subsets of products with a discrete factor; the printed (U, φ, v) in the formula for F is read (U, φ, V) (possible misprint) (Findings, item 2). *Proof:* the properness at (i, x) takes a closed neighbourhood of x inside U_i and a neighbourhood with the properness property of X, and is proved with ultrafilters instead of sequences.

V10.4 (Definition 10.1.4 (2017: 10.2), natural equivalence; `Polyfold.definition_10_1_4`, `Polyfold.EpNatTrans`, `Polyfold.EpFunctor.NaturallyEquivalent`, `Polyfold.EpFunctor.naturallyEquivalent_equivalence`, `Polyfold.EpFunctor.NaturallyEquivalent.comp_left`, `Polyfold.EpFunctor.NaturallyEquivalent.comp_right`, `Polyfold.EpNatTrans.toNatIso`). A natural transformation $\tau : F \to G$ of sc-smooth functors $X \to Y$ is an sc-smooth map $\tau : X \to \mathbf{Y}$ with $\tau(x) : F(x) \to G(x)$ and $\tau(x') \circ F(\varphi) = G(\varphi) \circ \tau(x)$ for $\varphi : x \to x'$; $F \simeq G$ if one exists. $\simeq$ is an equivalence relation, and $F \simeq G$ implies $F \circ \Psi \simeq G \circ \Psi$ and $\Phi \circ F \simeq \Phi \circ G$. A natural transformation is a natural isomorphism of the underlying functors. *Book:* Definition 10.1.4, p. 463 (2017: Definition 10.2, p. 335), and the claims after it, p. 464. *Hidden hypotheses:* none. *Deviation (same):* none; the definition is Definition 8.1.13 (Findings, item 3).

V10.5 (Proposition 10.1.5 (2017: 10.2), natural equivalence preserves equivalences; `Polyfold.EpFunctor.proposition_10_1_5`, `Polyfold.EpFunctor.IsEquivalence.of_natTrans`, `Polyfold.EpNatTrans.continuous_app`, `Polyfold.EpNatTrans.isLocalScDiffeo_app`, `Polyfold.EpNatTrans.isLocalScDiffeo_obj`). If $F \simeq G$, then F is an equivalence if and only if G is. *Book:* Proposition 10.1.5, p. 464 (2017: Proposition 10.2, p. 336), quoted from the literature without proof. *Hidden hypotheses:* none. *Deviation (same):* none. *Proof:* $s \circ \tau = F$ makes τ a local sc-diffeomorphism, so $G = t \circ \tau$ is one on objects; the underlying functors are naturally isomorphic, so G is an equivalence of categories, and Lemma 10.1.2 applies (Findings, item 4).

V10.6 (Proposition 10.1.6 (2017: 10.3), the two-out-of-three property; `Polyfold.EpFunctor.proposition_10_1_6`, `Polyfold.MPolyfold.IsLocalScDiffeoAt`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.comp`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.congr`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.of_comp_left`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.of_comp_right`, `Polyfold.EpFunctor.isLocalScDiffeoAt_mor`, `Polyfold.EpFunctor.isLocalScDiffeo_mor`, `Polyfold.EpFunctor.isLocalScDiffeoAt_obj_tgt`). If $F : X \to Y$ and $\Phi \circ F : X \to Z$ are equivalences and $\Phi : Y \to Z$ is sc-smooth, then Φ is an equivalence. A functor which is a local sc-diffeomorphism on objects at $s(\varphi)$ is one at $t(\varphi)$, and one on morphisms at φ. *Book:* Proposition 10.1.6, p. 464 (2017: Proposition 10.3, p. 336). *Hidden hypotheses:* none. *Deviation (same):* the text's argument on objects gives the local sc-diffeomorphism property of Φ only at the points $F(x)$; the transport along a morphism $F(x) \to y$ is added. The proof writes $|\Phi| : |X| \to |Y|$ for $|\Phi| : |Y| \to |Z|$ (possible misprint) (Findings, item 5). *Proof:* Φ is an equivalence of categories since F and $\Phi \circ F$ are, and Lemma 10.1.2 applies.

V10.7 (Theorem 10.1.7 (2017: 10.1), the tangent of an equivalence; `Polyfold.EpFunctor.`

`theorem_10_1_7`, `Polyfold.EpFunctor.tangent`). If F is an equivalence, its tangent $TF : TX \to TY$ is an equivalence of the tangent ep-groupoids. *Book:* Theorem 10.1.7, p. 465 (2017: Theorem 10.1, p. 337). *Hidden hypotheses:* none. *Deviation (same):* none; fullness and faithfulness of TF, left to the reader, are proved (Findings, item 6). *Proof:* by Lemma 10.1.2; a tangent morphism is determined by its base morphism and its source vector, so faithfulness comes from that of F; fullness lifts the base morphism given by fullness of F and uses the injectivity of TF on tangent fibres; essential surjectivity lifts ψ^{-1} for $\psi : F(x) \to y$ and uses that TF maps T_xX onto $T_{F(x)}Y$.

V10.8 (Remark 10.1.11 (2017: 10.1), α_x cannot be prescribed; `Polyfold.EpGroupoid.Examples.pointReflectionEpGroupoid`, `Polyfold.EpGroupoid.Examples.negFunctor`, `Polyfold.EpGroupoid.Examples.isEquivalence_negFunctor`, `Polyfold.EpGroupoid.Examples.orbitMap_id_eq_negFunctor`, `Polyfold.EpGroupoid.Examples.remark_10_1_11_family`, `Polyfold.EpGroupoid.Examples.remark_10_1_11`, `Polyfold.EpGroupoid.Examples.remark_10_1_11_scSmooth`). On the translation groupoid of $\mathbb{Z}/2$ acting on $\mathbb{R}^2$ by $z \mapsto -z$, with $F = \mathrm{id}$ and $G = -\mathrm{id}$ (both equivalences, $|F| = |G|$), $\alpha_z = (-1, z) : F(z) \to G(z)$ is an sc-smooth family, but no family $\alpha_z : F(z) \to G(z)$ defined near 0 with $\alpha_0 = 1_0$ is continuous at 0, and none is sc-smooth on an open neighbourhood of 0. *Book:* Remark 10.1.11, p. 470 (2017: Remark 10.1, p. 340). *Hidden hypotheses:* none. *Deviation (stronger):* "not in an sc-smooth local family" is proved in the form "not in a family continuous at 0"; that the conclusion of Proposition 10.1.10 (1) holds in the example is added (Findings, item 8).

V10.9 (Remark 10.1.12, isotropy groups under equivalences; `Polyfold.EpFunctor.isotropyHom`, `Polyfold.EpFunctor.isotropyHom_mem_trivialSet_iff`, `Polyfold.EpFunctor.IsEquivalence.bijective_isotropyHom`, `Polyfold.EpFunctor.IsEquivalence.remark_10_1_12`). For a functor which is a local sc-diffeomorphism on objects, $g \mapsto F(g)$, $G_x \to G_{F(x)}$, maps g into $T_{F(x)}$ if and only if $g \in T_x$; for an equivalence it is bijective, and for $\psi : F(x) \to y$ there is a group isomorphism $G_x \to G_y$, $g \mapsto \psi \circ F(g) \circ \psi^{-1}$, mapping T_x onto T_y. *Book:* Remark 10.1.12, p. 472 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the printed $G_x \to Y_y$ is read $G_x \to G_y$ (possible misprint) (Findings, item 9).

V10.10 (Proposition 10.1.13, the reduced morphism spaces; `Polyfold.EpFunctor.mor_mem_nonEffectivePart`, `Polyfold.EpFunctor.mem_nonEffectivePart_of_mor`, `Polyfold.EpFunctor.nonEffectivePart_eq`, `Polyfold.EpGroupoid.continuousAt_comp`, `Polyfold.EpFunctor.IsEquivalence.isClosed_nonEffectivePart_iff`, `Polyfold.EpFunctor.reducedMap`, `Polyfold.EpFunctor.continuous_reducedMap`, `Polyfold.EpFunctor.isOpenMap_reducedMap`, `Polyfold.EpFunctor.proposition_10_1_13`, `Polyfold.EpGroupoid.theorem_7_2_22`). For a functor which is a local sc-diffeomorphism on objects, $\mathbf{X}^T = F^{-1}(\mathbf{Y}^T) \cap \{s = t\}$ and $F_R : \mathbf{X}^R \to \mathbf{Y}^R$, $[\varphi] \mapsto [F(\varphi)]$, is well defined, continuous and open. For an equivalence F, $\mathbf{X}^T$ is closed if and only if $\mathbf{Y}^T$ is, and $\mathbf{X}^R$ is Hausdorff if and only if $\mathbf{Y}^R$ is. *Book:* Proposition 10.1.13, p. 472 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* (1) is proved for any functor which is a local sc-diffeomorphism on objects (Findings, item 10). *Proof:* (2) through Theorem 7.2.22 (T_R Hausdorff if and only if $\mathbf{X}^T$ closed), as in the text; the closedness passes from $\mathbf{X}^T$ to $\mathbf{Y}^T$ by conjugating with a local inverse of t at a morphism $\sigma : F(x) \to y$, with closures in place of sequences.

V10.11 (Proposition 10.1.14 (2017: 10.5), the reduced functor; `Polyfold.EpFunctor.reducedGroupoidFunctor`, `Polyfold.EpFunctor.reduced`, `Polyfold.EpFunctor.reduced_eq_of_mor`, `Polyfold.EpFunctor.proposition_10_1_14`). If F is an equivalence and the reduced morphism spaces $\mathbf{X}^R$, $\mathbf{Y}^R$ are Hausdorff, then $F_R = F$ on objects, $F_R([\varphi]) = [F(\varphi)]$ is a well-defined sc-smooth functor $X_R \to Y_R$ of the reduced ep-groupoids and an equivalence. *Book:* Proposition 10.1.14, p. 473 (2017: Proposition 10.5, p. 341). *Hidden hypotheses:* none. *Deviation (same):* none; by Proposition 10.1.13 (2) one Hausdorff hypothesis would suffice (Findings, item 11).

V10.12 (Example 10.1.15 (2017: 10.1), a functor that does not pass to the reduced groupoids; `Polyfold.EpGroupoid.Examples.example1015X`, `Polyfold.EpGroupoid.Examples.example1015Y`, `Polyfold.EpGroupoid.Examples.example1015Functor`, `Polyfold.EpGroupoid.Examples.example1015X_two_mem_nonEffectivePart`, `Polyfold.EpGroupoid.Examples.example1015Y_two_not_mem_nonEffectivePart`, `Polyfold.EpGroupoid.Examples.example1015X_reduced_one_eq_three`, `Polyfold.EpGroupoid.Examples.example1015Y_reduced_one_ne_three`, `Polyfold.EpGroupoid.Examples.example_10_1_15`). For $X = \mathbb{Z}/4 \ltimes \mathbb{R}$ (1 and 3 acting by $x \mapsto -x$, 2 trivially), $Y = \mathbb{Z}/4 \ltimes \mathbb{R}^2$ (rotations by multiples of 90°) and the sc-smooth functor $\Phi(x) = 0$, $\Phi(i, x) = (i, 0)$: $(2, x) \in \mathbf{X}^T$, $(2, 0) \notin \mathbf{Y}^T$, $[(1, x)] = [(3, x)]$ in $\mathbf{X}^R$ and $[(1, 0)] \neq [(3, 0)]$ in $\mathbf{Y}^R$, so no map $f : \mathbf{X}^R \to \mathbf{Y}^R$ satisfies $f([\varphi]) = [\Phi(\varphi)]$. *Book:* Example 10.1.15, p. 474 (2017: Example 10.1, p. 341). *Hidden hypotheses:* none. *Deviation (stronger):* no Hausdorff hypothesis and no topology on the reduced spaces is involved; no map at all is compatible with Φ (Findings, item 12).

V10.13 (Proposition 10.1.16, (LUC) under equivalences; `Polyfold.EpFunctor.proposition_10_1_16`, `Polyfold.EpFunctor.IsEquivalence.isClosed_nonEffectivePart_iff`, `Polyfold.EpGroupoid.proposition_7_2_21`). If $F : X \to Y$ is an equivalence, X has the local unique continuation property if and only if Y has it. *Book:* Proposition 10.1.16, p. 474 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 10). *Proof:* (LUC) holds if and only if $\mathbf{X}^T$ is closed (Proposition 7.2.21, a part of Theorem 7.2.22), and the closedness passes along equivalences (V10.10).

V10.14 (Proposition 10.1.17, the Zhou condition under equivalences; `Polyfold.EpGroupoid.mem_smoothSet_src_iff`, `Polyfold.EpGroupoid.mem_smoothSet_tgt_iff`, `Polyfold.EpFunctor.tangentMor_mor`, `Polyfold.EpFunctor.IsEquivalence.isZhou_of_isZhou`, `Polyfold.EpFunctor.IsEquivalence.isZhou_of_isZhou_right`, `Polyfold.EpFunctor.proposition_10_1_17`). If $F : X \to Y$ is an equivalence, X satisfies the Zhou condition (for smooth x and $g \in G_x$, $Tg = \mathrm{id}$ on T_xX implies $g \in T_x$) if and only if Y does. *Book:* Proposition 10.1.17, p. 475 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the proof uses the naturality $T(F(\varphi)) \circ TF = TF \circ T\varphi$, the composition rule for T and the bijectivity of TF on tangent fibres instead of the natural representations; the text's $U(F(y))$ and "$x \in U(x)$" in (b) are read $U(F(x))$ and $a \in U(x)$ (possible misprints) (Findings, item 13).

V10.15 (Proposition 10.1.18, (SLUC) under equivalences; `Polyfold.EpFunctor.liftOne`, `Polyfold.EpFunctor.IsEquivalence.liftOne`, `Polyfold.EP.exists_isEquivalence_lift`, `Polyfold.EP.proposition_10_1_18`). The restriction $F^1 : X^1 \to Y^1$ of an sc-smooth functor to level 1 is an sc-smooth functor, and an equivalence if F is; hence for every i there is an equivalence $X^i \to Y^i$, and X satisfies (SLUC) if and only if Y does. *Book:* Proposition 10.1.18, p. 476 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the proof cites Proposition 10.1.9 for the step (LUC) along F^i, which is Proposition 10.1.16 (possible misprint) (Findings, item 14).

V10.16 (Proposition 10.1.19, (SLR) under equivalences; `Polyfold.EP.proposition_10_1_19`, `Polyfold.EP.isSLR_iff_of_isLR_iff`, `Polyfold.EP.isLR_iff_lift_self`). If $F : X \to Y$ is an equivalence, X satisfies (SLR) if and only if Y does. The reduction: if, for every i, every equivalence $X^i \to Y^i$ carries (LR) in both directions, then (SLR) passes along F; Proposition 10.1.9 (V10.62) supplies the hypothesis for the lifts F^i. *Book:* Proposition 10.1.19, p. 477 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the text's proof, Proposition 10.1.9 for the lifts (gap G.115, closed since: V10.16, through V10.62; Findings, items 15, 16).

V10.17 (Definition 10.2.1 (2017: 10.3), the weak fibred product as a groupoid; `Polyfold.WeakFibredProduct.objRight`, `Polyfold.WeakFibredProduct.objSet`, `Polyfold.WeakFibredProduct.Obj`, `Polyfold.WeakFibredProduct.morRight`, `Polyfold.WeakFibredProduct.morSet`, `Polyfold.WeakFibredProduct.Mor`, `Polyfold.WeakFibredProduct.conj`, `Polyfold.WeakFibredProduct.conj_comp_mor`, `Polyfold.WeakFibredProduct.conj_recover`, `Polyfold.WeakFibredProduct.src`, `Polyfold.WeakFibredProduct.tgt`, `Polyfold.WeakFibredProduct.unit`, `Polyfold.WeakFibredProduct.inv`, `Polyfold.WeakFibredProduct.comp`, `Polyfold.WeakFibredProduct.src_eq_tgt_iff`, `Polyfold.WeakFibredProduct.groupoid`). For sc-smooth functors $F : X \to Y$ and $G : Z \to Y$, the objects of $X \times_Y Z$ are the triples (x, ϕ, z) with $\phi : F(x) \to G(z)$, the morphisms the triples (h, ϕ, k) with $\phi : F(s(h)) \to G(s(k))$; $s(h, \phi, k) = (s(h), \phi, s(k))$, $t(h, \phi, k) = (t(h), G(k) \circ \phi \circ F(h)^{-1}, t(k))$, the unit at (x, ϕ, z) is $(1_x, \phi, 1_z)$, the inverse is $(h^{-1}, G(k) \circ \phi \circ F(h)^{-1}, k^{-1})$ and the product of a composable pair is $(h \circ h', \phi', k \circ k')$. This is a groupoid. *Book:* Definition 10.2.1, p. 477 (2017: Definition 10.3, p. 342). *Hidden hypotheses:* none. *Deviation (restated):* objects and morphisms are elements of iterated fibred products, $X\ {}_F\times_s (\mathbf{Y}\ {}_t\times_G Z)$ and $\mathbf{X}\ {}_{F\circ s}\times_s (\mathbf{Y}\ {}_t\times_{G\circ s} \mathbf{Z})$; the multiplication is a total map whose value off composable pairs is not used; the printed inverse $\iota(h, \psi, k) = (h^{-1}, G(k) \circ \phi \circ F(h)^{-1}, k^{-1})$ is read with $\psi = \phi$ (possible misprint) (Findings, item 20).

V10.18 (Theorem 10.2.2 (2017: 10.2), first claim: $X \times_Y Z$ is an ep-groupoid; `Polyfold.WeakFibredProduct.Admissible`, `Polyfold.WeakFibredProduct.Admissible.of_isEquivalence_left`, `Polyfold.WeakFibredProduct.Admissible.of_isEquivalence_right`, `Polyfold.WeakFibredProduct.PA`, `Polyfold.WeakFibredProduct.PB`, `Polyfold.WeakFibredProduct.PObj`, `Polyfold.WeakFibredProduct.PMor`, `Polyfold.WeakFibredProduct.isSubMPolyfold_objSet`, `Polyfold.WeakFibredProduct.isSubMPolyfold_morSet`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_z`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_k`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_x`, `Polyfold.WeakFibredProduct.`

`isLocalScDiffeo_h`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_src`, `Polyfold.WeakFibredProduct.isLocalScDiffeo_tgt`, `Polyfold.WeakFibredProduct.isScSmoothMap_unit`, `Polyfold.WeakFibredProduct.isScSmoothMap_inv`, `Polyfold.WeakFibredProduct.isScSmoothComp`, `Polyfold.WeakFibredProduct.exists_isProperMap`, `Polyfold.WeakFibredProduct.epGroupoid`, `Polyfold.MPolyfold.IsLocalScDiffeo.of_comm`, `Polyfold.MPolyfold.isLocalScDiffeo_snd_fibredProduct`, `Polyfold.MPolyfold.isLocalScDiffeo_fst_fibredProduct`, `Polyfold.EpGroupoid.isScSmoothOn_comp_of`, `Polyfold.EpGroupoid.tendsto_comp`, `Polyfold.WeakFibredProduct.theorem_10_2_2_triple`, `Polyfold.WeakFibredProduct.isSubMPolyfold_objSetTriple`, `Polyfold.WeakFibredProduct.isSubMPolyfold_morSetTriple`, `Polyfold.MPolyfold.isSubMPolyfold_image_of_isScDiffeoOn`). If F or G is a local sc-diffeomorphism on objects, in particular if one of them is an equivalence, the groupoid of Definition 10.2.1 with the object M-polyfold $X\,{}_F\!\times_s (\mathbf{Y}\,{}_t\!\times_G Z)$ and the morphism M-polyfold $\mathbf{X}\,{}_{F\circ s}\!\times_s (\mathbf{Y}\,{}_t\!\times_{G\circ s} \mathbf{Z})$, iterated sub-M-polyfolds of products, is an ep-groupoid. For F a local sc-diffeomorphism on objects, $(x, \phi, z) \mapsto z$ and $(h, \phi, k) \mapsto k$ are local sc-diffeomorphisms ((10.9), (10.10)); for G one, $(x, \phi, z) \mapsto x$ and $(h, \phi, k) \mapsto h$ are. The properness of Definition 7.1.3 (3) holds for any F and G. *Book:* Theorem 10.2.2, p. 477 (2017: Theorem 10.2, p. 343), first claim, and its proof, pp. 478–481 (2017 pp. 343–346). *Hidden hypotheses:* none. *Deviation (stronger):* of F the construction uses only that it is a local sc-diffeomorphism on objects. The text's "sub-M-polyfolds of $X \times \mathbf{Y} \times Z$ and $\mathbf{X} \times \mathbf{Y} \times \mathbf{Z}$" is read as the iterated sub-M-polyfold structures of Part I's fibred products (Proposition 2.5.20); they agree with the structures induced directly from the triple products: L and $\mathbf{L}$ are sub-M-polyfolds of $X \times \mathbf{Y} \times Z$ and $\mathbf{X} \times \mathbf{Y} \times \mathbf{Z}$, sc-diffeomorphic to the iterated structures, by the transitivity of sub-M-polyfolds (`Polyfold.WeakFibredProduct.theorem_10_2_2_triple`, `Polyfold.MPolyfold.isSubMPolyfold_image_of_isScDiffeoOn`; gap G.116 closed); the text's "$t : s^{-1}(U) \to X$ proper" is read with $\mathrm{cl}(U)$, as in Definition 7.1.3 (3) (Findings, items 21, 22). *Proof:* s and t are local sc-diffeomorphisms by the diagrams $\pi_Z \circ s = s_Z \circ \pi_\mathbf{Z}$ and $\pi_Z \circ t = t_Z \circ \pi_\mathbf{Z}$; the properness is the text's sequential argument, with $\phi_n = G(k_n)^{-1} \circ (G(k_n) \circ \phi_n \circ F(h_n)^{-1}) \circ F(h_n)$.

V10.19 (Theorem 10.2.2 (2017: 10.2), the case F an equivalence; `Polyfold.WeakFibredProduct.projLeft`, `Polyfold.WeakFibredProduct.projRight`, `Polyfold.WeakFibredProduct.natTrans`, `Polyfold.WeakFibredProduct.natTrans_app`, `Polyfold.WeakFibredProduct.naturallyEquivalent`, `Polyfold.WeakFibredProduct.isEquivalence_projRight`, `Polyfold.WeakFibredProduct.degIndex_eq_z`, `Polyfold.WeakFibredProduct.theorem_10_2_2`, `Polyfold.EpFunctor.IsEquivalence.degIndex_obj`, `Polyfold.EpFunctor.IsEquivalence.isTame_of_isTame`, `Polyfold.EpFunctor.IsEquivalence.hasParacompactOrbitSpace`, `Polyfold.WeakFibredProduct.Examples.theorem_10_2_2_pairToPoint`). The projections π_X and π_Z are sc-smooth functors and $\sigma(x, \phi, z) = \phi$ is a natural transformation $F \circ \pi_X \to G \circ \pi_Z$ whenever $X \times_Y Z$ is defined. If F is an equivalence and G is sc-smooth, then π_Z is an equivalence, $F \circ \pi_X \simeq G \circ \pi_Z$, $d_{X \times_Y Z}(x, \phi, z) = d_Z(z)$, and if Z is tame, or has a paracompact orbit space, so has $X \times_Y Z$. An equivalence preserves the degeneracy index and pulls back tameness and the paracompactness of the orbit space. The weak fibred product of the equivalence from the pair groupoid of $\mathbb{Z}$ to the point with the identity of the point is a nonempty ep-groupoid whose projection to the point is an equivalence. *Book:* Theorem 10.2.2, pp. 477–478 (2017: Theorem 10.2, p. 343), with the paragraph after it, p. 478, and its proof, pp. 481–482. *Hidden hypotheses:* none. *Deviation (same):* in the fullness step the printed $\phi' \circ G(k) \circ \phi^{-1}$, which does not compose, is read $\phi'^{-1} \circ G(k) \circ \phi : F(x) \to F(x')$ (possible misprint); essential surjectivity is taken from that of F (Findings, item 23). *Proof:* π_Z is full and faithful because the unique morphism over $k : z \to z'$ is (h, ϕ, k) with $F(h) = \phi'^{-1} \circ G(k) \circ \phi$, essentially surjective because every $G(z)$ receives a morphism from some $F(x)$, and Lemma 10.1.2 applies.

V10.20 (Theorem 10.2.2 (2017: 10.2), the cases G an equivalence and both; `Polyfold.WeakFibredProduct.isEquivalence_projLeft`, `Polyfold.WeakFibredProduct.degIndex_eq_x`, `Polyfold.WeakFibredProduct.theorem_10_2_2_right`, `Polyfold.WeakFibredProduct.theorem_10_2_2_both`). If G is an equivalence, π_X is an equivalence, $F \circ \pi_X \simeq G \circ \pi_Z$, $d_{X \times_Y Z}(x, \phi, z) = d_X(x)$, and tameness and a paracompact orbit space pass from X to $X \times_Y Z$; if both are, $d_X(x) = d_{X \times_Y Z}(x, \phi, z) = d_Z(z)$. Since the admissibility is a proposition, the two cases give the same ep-groupoid. *Book:* Theorem 10.2.2, p. 478 (2017: Theorem 10.2, p. 343). *Hidden hypotheses:* none. *Deviation (stronger):* for G an equivalence the text states only that π_X is an equivalence and the degeneracy index; the natural equivalence and the transfer of tameness and paracompactness are added (Findings, item 24).

V10.21 (Remark 10.2.3 (2017: 10.2), two equivalences; `Polyfold.WeakFibredProduct.remark_10_2_3`, `Polyfold.WeakFibredProduct.Examples.remark_10_2_3_negFunctor`). For equivalences $F : X \to Y$ and $G : Z \to Y$, both projections of $X \times_Y Z$ are equivalences and $\tau(x, \phi, z) = \phi$ is a natural transformation $F \circ \pi_X \to G \circ \pi_Z$. For the identity and $-\mathrm{id}$ of $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ the weak fibred product is nonempty and both projections are equivalences. *Book:* Remark 10.2.3, pp. 482–483 (2017: Remark 10.2, p. 347). *Hidden hypotheses:* none. *Deviation (same):* none; the closing sentences on "smaller models" have no formal content (Findings, item 25).

V10.22 (Definition 10.4.1 (2017: 10.10), strong bundle equivalences; `Polyfold.EpStrongBundle.IsStrongBundleEquivalence`, `Polyfold.EpStrongBundle.OneOrbitSpace`, `Polyfold.EpStrongBundle.oneOrbitMap`, `Polyfold.EpStrongBundle.IsStrongBundleFunctor.comp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.covers`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.fiberEquiv`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eq_of_obj_eq`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_obj`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_one`, `Polyfold.EpStrongBundle.isStrongBundleEquivalence_id`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.comp`, `Polyfold.EpStrongBundle.isStrongBundleEquivalence_of_inverse`, `Polyfold.StrongBundleEquivalenceExamples.isStrongBundleEquivalence_pullbackProj'`, `Polyfold.StrongBundleEquivalenceExamples.trivialNeg_obj_ne_id`). A strong bundle equivalence $\Phi : (P : W \to X, \mu) \to (P' : W' \to X', \mu')$ is a strong bundle functor $\Phi : (W, \mathbf{W}) \to (W', \mathbf{W}')$ (linear on fibres, satisfying (10.14)) covering an equivalence $\phi : X \to X'$, with (1) $\Phi : W[0] \to W'[0]$ an equivalence of ep-groupoids and $|\Phi| : |W[1]| \to |W'[1]|$ a homeomorphism, and (2) every point of X, and every morphism of $\mathbf{X}$, having an open neighbourhood over which Φ on objects, respectively on morphisms, is a strong bundle isomorphism covering ϕ. The fibre maps $W_x \to W'_{\phi(x)}$ are then topological linear isomorphisms and $\Phi : W[1] \to W'[1]$ is a local sc-diffeomorphism. The identity and composites of strong bundle equivalences are strong bundle equivalences, and so is every isomorphism of strong bundles over ep-groupoids; the projection $(-\mathrm{id})^*(\mathbb{R}^n \times \mathbb{R}^k) \to \mathbb{R}^n \times \mathbb{R}^k$ over $-\mathrm{id}$ is one, not over the identity for $n \geq 1$. *Book:* Definition 10.4.1, p. 496 (2017: Definition 10.10, p. 357). *Hidden hypotheses:* none. *Deviation (restated):* $W[1]$ is not built as an ep-groupoid in the project; (1) at level [1] is read as the homeomorphism of $|W[1]|$, the quotient of $W_{0,1}$ by the orbit relation of W; the local sc-diffeomorphism property at level [1] follows from (2), and fullness and faithfulness at level [1] are those at level [0] restricted (Findings, item 40).

V10.23 (the pull-back and the push-forward of section functors, (10.15)–(10.17) and (10.22); `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.obj_pullback`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_eq`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pullback`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.preObj`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.preMor`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushforward`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pushforward`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushforward_pullback`, `Polyfold.EpStrongBundle. _injective_snd`, `Polyfold.EpStrongBundle.exists_ _eq_of_orbitMap_eq`, `Polyfold.EpStrongBundle.orbitMap_eq_of_forall_exists`). For a strong bundle equivalence Φ covering ϕ, $\Phi^*(f')(x) = \Phi_x^{-1}(f'(\phi(x)))$ is the unique $w \in W_x$ with $\Phi(w) = f'(\phi(x))$, and $(\Phi_* f)(x') = \mu'(\sigma, \Phi(f(x)))$ for a chosen x and $\sigma : \phi(x) \to x'$. For a section functor f the value does not depend on the choice: $(\Phi_* f)(x') = \mu'(\sigma', \Phi(f(y)))$ for every y and $\sigma' : \phi(y) \to x'$ ((10.22)). Pull-backs and push-forwards of section functors are section functors, and $\Phi^*\Phi_* f = f$, $\Phi_*\Phi^* f' = f'$. $|\Phi| = |\Psi|$ if and only if every $\Phi(w)$ is connected to $\Psi(w)$ by a morphism $(\sigma, \Phi(w))$ with $\mu'(\sigma, \Phi(w)) = \Psi(w)$ ((10.15)). *Book:* §10.4, pp. 496–498 (2017 pp. 357–360). *Hidden hypotheses:* none. *Deviation (restated):* Φ^* and Φ_* are defined on all maps, by the zero of the fibre where the value is not in the right fibre (never for sections), and the choice of x and σ is made once for all (Findings, item 41).

V10.24 (Proposition 10.4.2 (2017: 10.7), pull-backs along $|\Phi| = |\Psi|$; `Polyfold.EpStrongBundle.proposition_10_4_2`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_eq_of_orbitMap_eq`). If Φ and Ψ are strong bundle equivalences $(P, \mu) \to (P', \mu')$ covering ϕ and ψ with $|\Phi| = |\Psi|$, then $\Phi^*(f') = \Psi^*(f')$ for every section functor f' of (P', μ'), sc-smooth or not. *Book:* Proposition 10.4.2, p. 497 (2017: Proposition 10.7, p. 358). *Hidden hypotheses:* none. *Deviation (stronger):* sc-smoothness of f' is not used; the printed "$f' : W' \to X'$" is read $f' : X' \to W'$ (possible misprint) (Findings, item 42). *Proof:* with $g = \Psi^* f'$ and a morphism $(\sigma, \Phi(g(x)))$ with $\mu'(\sigma, \Phi(g(x))) = \Psi(g(x))$, functoriality of f' gives $\mu'(\sigma, \Phi(g(x))) = \mu'(\sigma, \Phi(\Phi^* f'(x)))$; $\mu'(\sigma, \cdot)$

and Φ_x are injective.

V10.25 (Proposition 10.4.3 (2017: 10.8), push-forwards along $|\Phi| = |\Psi|$; `Polyfold.EpStrongBundle.proposition_10_4_3`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq_of_orbitMap_eq`). Under the hypotheses of Proposition 10.4.2, $\Phi_* f = \Psi_* f$ for every section functor f of (P, μ), sc-smooth or not. *Book:* Proposition 10.4.3, p. 499 (2017: Proposition 10.8, p. 360). *Hidden hypotheses:* none. *Deviation (stronger):* sc-smoothness of f is not used (Findings, item 43).

V10.26 (Proposition 10.4.4 (2017: 10.9), functoriality; `Polyfold.EpStrongBundle.proposition_10_4_4`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.comp`). For strong bundle equivalences $\Phi : (P, \mu) \to (P', \mu')$ and $\Psi : (P', \mu') \to (P'', \mu'')$, $(\Psi \circ \Phi)^* = \Phi^* \circ \Psi^*$ on all sections of P'' and $(\Psi \circ \Phi)_* = \Psi_* \circ \Phi_*$ on all section functors of (P, μ). *Book:* Proposition 10.4.4, p. 499 (2017: Proposition 10.9, p. 360). *Hidden hypotheses:* none. *Deviation (stronger):* the pull-back identity holds for all sections and the push-forward identity for all section functors, without sc-smoothness (Findings, item 43).

V10.27 (Theorem 10.4.5 (2017: 10.8), without the Fred part; `Polyfold.EpStrongBundle.theorem_10_4_5`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eqOn_pullback`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pullback`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pullback`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.exists_pushforward_local`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pushforward`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pushforward`, `Polyfold.StrongBundle.isScSmoothOn_to_pullback_one`, `Polyfold.MPolyfold.isScSmoothOn_of_forall_charts`, `Polyfold.StrongBundle.IsStrongBundleIsoOn.restrict`, `Polyfold.StrongBundle.IsStrongBundleIsoOn.comp`, `Polyfold.StrongBundle.IsStrongBundleIsoOn.exists_comp`, `Polyfold.StrongBundleEquivalenceExamples.bijOn_pushforward_example`). For a strong bundle equivalence Φ covering ϕ, $\Phi_* : \Gamma(P, \mu) \to \Gamma(P', \mu')$ and $\Phi^* : \Gamma(P', \mu') \to \Gamma(P, \mu)$ are mutually inverse bijections of the sc-smooth section functors; if Ψ is another strong bundle equivalence with $|\Phi| = |\Psi|$, then $\Phi_* = \Psi_*$ and $\Phi^* = \Psi^*$ on them; and Φ_*, Φ^* map sc^+-section functors to sc^+-section functors. For the projection $(-\mathrm{id})^*(\mathbb{R}^n \times \mathbb{R}^k) \to \mathbb{R}^n \times \mathbb{R}^k$ the push-forward is a bijection of the section functors. *Book:* Theorem 10.4.5, pp. 500–501 (2017: Theorem 10.8, p. 362), with the paragraph before it, p. 500. *Hidden hypotheses:* none. *Deviation (narrower):* the preservation of sc-Fredholm section functors is not formalized (gap G.121); the second display's $\Phi^* : \Gamma(P', \mu') \to \Gamma(P, \mu)$ is read with Γ^+ (possible misprint) (Findings, item 44). *Proof:* the sc-smoothness of $\Phi_* f$, called easily verified in the text, comes from the local form $\Phi_* f = \mu' \circ (\tau, \Phi \circ f \circ g_1 \circ s' \circ \tau)$ near x'_0, with τ a local inverse of t' and g_1 one of ϕ; at level $[1]$ through the universal property of the pull-back on an open set.

V10.28 (Definition 10.4.8 (2017: 10.11), natural equivalence of strong bundle functors; `Polyfold.EpStrongBundle.StrongBundleNatTrans`, `Polyfold.EpStrongBundle.StrongBundleNatEquiv`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.refl`, `Polyfold.StrongBundleEquivalenceExamples.strongBundleNatEquiv_example`). A natural transformation between strong bundle functors $\Phi, \Psi : (W, \mathbf{W}) \to (W', \mathbf{W}')$ covering ϕ, ψ is a map T assigning to each w a morphism $T(w) : \Phi(w) \to \Psi(w)$ of $(W', \mathbf{W}')$ with $T(w') \circ \Phi(h) = \Psi(h) \circ T(w)$ for $h : w \to w'$, together with a natural transformation $\tau : \phi \to \psi$ with $\tau(P(w)) = P'(T(w))$; $\Phi \simeq \Psi$ if one exists. Every strong bundle functor is naturally equivalent to itself. *Book:* Definition 10.4.8, p. 502 (2017: Definition 10.11, p. 362). *Hidden hypotheses:* none. *Deviation (restated):* T is data without smoothness, as in the text, and τ is sc-smooth (Definition 10.1.4) (Findings, item 46).

V10.29 (Remark 10.4.9 (2017: 10.4) and (10.23); `Polyfold.EpStrongBundle.StrongBundleNatTrans.T_val`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.T_eq`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.obj_eq`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.unique`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.unique_functor`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.isStrongBundleMap_T`). For a natural transformation (T, τ) from Φ to Ψ: $T(w) = (\tau(P(w)), \Phi(w))$ and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$; Ψ on objects and T are determined by Φ and τ, and so is Ψ itself among strong bundle functors covering ψ; for a strong bundle functor Φ, $T : W \to \mathbf{X}'\ {}_s{\times}_{P'} W'$ is a strong bundle map covering τ. *Book:* Remark 10.4.9, pp. 502–503 (2017: Remark 10.4, p. 363). *Hidden hypotheses:* that Ψ is a strong bundle functor, for the determination of Ψ as a functor. *Deviation (explicit):* the second line of (10.23), printed with μ, is read with μ' (Findings, item 46).

V10.30 (Definition 10.4.10 (2017: 10.12) and its equivalence with Definition 10.4.8; `Polyfold.EpStrongBundle.IsStrongBundleNatTransMap`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.isStrongBundleNatTransMap`, `Polyfold.EpStrongBundle.IsStrongBundleNatTransMap.toNatTrans`, `Polyfold.EpStrongBundle.StrongBundleNatTrans.orbitMap_eq`, `Polyfold.EpStrongBundle.StrongBundleNatEquiv.orbitMap_eq`). A natural transformation in the sense of Definition 10.4.10 is a strong bundle map $T : W \to \mathbf{X}'\ {}_s{\times}_{P'} W'$ covering a natural transformation $\tau : \phi \to \psi$, with $T(w) = (\tau(P(w)), \Phi(w))$ and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$. For strong bundle functors Φ, Ψ the two definitions agree; $\Phi \simeq \Psi$ implies $|\Phi| = |\Psi|$. *Book:* Definition 10.4.10 and the sentence after it, p. 503 (2017: Definition 10.12, p. 363). *Hidden hypotheses:* that Φ and Ψ are strong bundle functors, for Definition 10.4.10 □ Definition 10.4.8. *Deviation (explicit):* none beyond the hypothesis (Findings, item 46).

V10.31 (Definition 10.4.11 (2017: 10.13), the category $\mathcal{SEP}$; `Polyfold.SEP`, `Polyfold.SEP.Hom`, `Polyfold.SEP.equivalences`, `Polyfold.SEP.toEP`, `Polyfold.SEP.toEP_map_mem_equivalences`, `Polyfold.StrongBundleEquivalenceExamples.negSEPHom`, `Polyfold.StrongBundleEquivalenceExamples.negSEPHom_mem_equivalences`). $\mathcal{SEP}$ is a category whose objects are the strong bundles over ep-groupoids and whose morphisms are the strong bundle functors (ϕ, Φ); the strong bundle equivalences form a class F containing the identities and closed under composition; the forgetful functor $\mathcal{SEP} \to \mathcal{EP}$ maps F into E. The projection $(-\mathrm{id})^*(\mathbb{R}^n \times \mathbb{R}^k) \to \mathbb{R}^n \times \mathbb{R}^k$ is a morphism in F. *Book:* Definition 10.4.11, p. 503 (2017: Definition 10.13, p. 364). *Hidden hypotheses:* none. *Deviation (restated):* a morphism carries both ϕ and Φ, since Φ determines ϕ only on the image of P (Findings, item 47).

V10.32 (the properties (E1)–(E3) of the class E; `Polyfold.EPLocalization.isEquivalence_`

`of_isIso`, `Polyfold.EpFunctor.IsEquivalence.comp`, `Polyfold.EpFunctor.proposition_10_1_5`). Every isomorphism of the category $\mathcal{EP}$ is an equivalence; the composite of two equivalences is an equivalence; a functor naturally equivalent to an equivalence is an equivalence. *Book:* §10.3, (E1)–(E3), p. 483 (2017 p. 347). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 30).

V10.33 (Definition 10.3.1 (2017: 10.4), diagrams and refinements; `Polyfold.EPLocalization.Diagram`, `Polyfold.EPLocalization.Diagram.IsRefinement`, `Polyfold.EPLocalization.Diagram.IsRefinement.refl`, `Polyfold.EPLocalization.Diagram.precomp`, `Polyfold.EPLocalization.Diagram.isRefinement_precomp`, `Polyfold.EPLocalization.Diagram.isRefinement_of_naturallyEquivalent`). A diagram $d : X \xleftarrow{F} A \xrightarrow{\Phi} Y$ consists of an ep-groupoid A, an equivalence $F : A \to X$ and an sc-smooth functor $\Phi : A \to Y$. A diagram $d' : X \xleftarrow{F'} A' \xrightarrow{\Phi'} Y$ is a refinement of d if there is an equivalence $H : A' \to A$ with $F \circ H \simeq F'$ and $\Phi \circ H \simeq \Phi'$. Every diagram refines itself; for an equivalence $H : A' \to A$, $X \xleftarrow{F \circ H} A' \xrightarrow{\Phi \circ H} Y$ refines d; for $G \simeq F$ (an equivalence) and $\Psi \simeq \Phi$, $X \xleftarrow{G} A \xrightarrow{\Psi} Y$ refines d. *Book:* Definition 10.3.1, p. 484 (2017: Definition 10.4, p. 348), with the diagrams of p. 483 and the two examples after the definition. *Hidden hypotheses:* none. *Deviation (restated):* the middle ep-groupoid is an object of $\mathcal{EP}$, so a diagram is a structure one universe above the types of objects and morphisms (Findings, item 30).

V10.34 (Lemma 10.3.2 (2017: 10.3), refinements compose; `Polyfold.EPLocalization.Diagram.lemma_10_3_2`, `Polyfold.EPLocalization.Diagram.IsRefinement.trans`). A refinement of a refinement of d is a refinement of d. *Book:* Lemma 10.3.2, p. 484 (2017: Lemma 10.3, p. 348). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 31).

V10.35 (Definition 10.3.3 (2017: 10.5), common refinements; `Polyfold.EPLocalization.Diagram.IsCommonRefinement`). d'' is a common refinement of (d, d') if it refines both. *Book:* Definition 10.3.3, p. 485 (2017: Definition 10.5, p. 349). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 31).

V10.36 (Lemma 10.3.4 (2017: 10.4), transitivity; `Polyfold.EPLocalization.Diagram.lemma_10_3_4`, `Polyfold.EPLocalization.Diagram.Equiv.trans`, `Polyfold.EPLocalization.Diagram.IsRefinement.equiv`). If (d, d') and (d', d'') have common refinements, so has (d, d''): for W refining d by H and d' by H', and W' refining d' by K and d'' by K', the diagram (10.11) $X \leftarrow W \times_{A'} W' \to Y$, formed with the weak fibred product of H' and K, refines d by $H \circ \pi_W$ and d'' by $K' \circ \pi_{W'}$. A refinement is equivalent to the refined diagram. *Book:* Lemma 10.3.4, p. 485 (2017: Lemma 10.4, p. 349), with (10.11), p. 486. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 31).

V10.37 (Definition 10.3.5 (2017: 10.6), equivalence of diagrams; `Polyfold.EPLocalization.Diagram.Equiv`, `Polyfold.EPLocalization.Diagram.Equiv.refl`, `Polyfold.EPLocalization.Diagram.Equiv.symm`, `Polyfold.EPLocalization.Diagram.setoid`, `Polyfold.EPLocalization.Diagram.equiv_iff`, `Polyfold.EPLocalization.Examples.equiv_pairToPoint_id`). Two diagrams are equivalent if they have a common refinement; this is an equivalence relation on the diagrams $X \to Y$. The diagram pt $\leftarrow$ $\mathbb{Z}$-pair $\to$ pt with both legs the equivalence from the pair groupoid of $\mathbb{Z}$ is equivalent to 1_{pt}. *Book:* Definition 10.3.5, p. 486 (2017: Definition 10.6, p. 350). *Hidden hypotheses:* none. *Deviation (restated):* the equivalence relation is a setoid on the type of diagrams; its quotient lives in the universe of the objects of $\mathcal{EP}$ (Findings, item 31).

V10.38 (Lemma 10.3.6 (2017: 10.5), the map of orbit spaces of a diagram; `Polyfold.EPLocalization.Diagram.orbitMap`, `Polyfold.EPLocalization.Diagram.continuous_orbitMap`, `Polyfold.EPLocalization.Diagram.orbitMap_orbitMap_F`, `Polyfold.EPLocalization.Diagram.lemma_10_3_6`, `Polyfold.EPLocalization.Diagram.Equiv.orbitMap_eq`, `Polyfold.EpFunctor.NaturallyEquivalent.orbitMap_eq`, `Polyfold.EpFunctor.orbitMap_comp`). For a diagram d, $|d| = |\Phi| \circ |F|^{-1} : |X| \to |Y|$ is continuous; naturally equivalent functors induce the same map of orbit spaces; if d' refines d, then $|d| = |d'|$, and equivalent diagrams have the same map. *Book:* Lemma 10.3.6, p. 488 (2017: Lemma 10.5, p. 351), with $|d|$ on p. 487. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 32).

V10.39 (Definition 10.3.7 (2017: 10.7), the map $|[d]|$; `Polyfold.EPLocalization.Diagram.orbitMapQ`, `Polyfold.EPLocalization.Diagram.orbitMapQ_mk`, `Polyfold.EPLocalization.Diagram.isHomeomorph_orbitMap`). $|[d]| : |X| \to |Y|$ is $|\Phi| \circ |F|^{-1}$ for any representative; if Φ is an equivalence too, it is a homeomorphism. *Book:* Definition 10.3.7 and the remark after it, p. 488 (2017: Definition 10.7, p. 351). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 32).

V10.40 (Theorem 10.3.8 (2017: 10.3), the composition of classes, with its proof in Appendix §10.6.1; `Polyfold.EPLocalization.Diagram.theorem_10_3_8`, `Polyfold.EPLocalization.Diagram.id`, `Polyfold.EPLocalization.Diagram.comp`, `Polyfold.EPLocalization.Diagram.admissible`, `Polyfold.EPLocalization.wfp`, `Polyfold.EPLocalization.wfpMap`, `Polyfold.EPLocalization.projLeft_comp_wfpMap`, `Polyfold.EPLocalization.projRight_comp_wfpMap`, `Polyfold.EPLocalization.isEquivalence_wfpMap`, `Polyfold.EPLocalization.assocFunctor`, `Polyfold.EPLocalization.isEquivalence_assocFunctor`, `Polyfold.EPLocalization.Diagram.IsRefinement.comp`, `Polyfold.EPLocalization.Diagram.Equiv.comp`, `Polyfold.EPLocalization.Diagram.isRefinement_assoc`, `Polyfold.EPLocalization.Diagram.equiv_assoc`, `Polyfold.EPLocalization.Diagram.isRefinement_id_comp`, `Polyfold.EPLocalization.Diagram.isRefinement_comp_id`, `Polyfold.EpFunctor.IsEquivalence.of_comp_left`). For $d : X \xleftarrow{F} A \xrightarrow{\Phi} Y$ and $e : Y \xleftarrow{G} B \xrightarrow{\Psi} Z$, the composite $X \xleftarrow{F \circ \pi_A} A \times_Y B \xrightarrow{\Psi \circ \pi_B} Z$, with the weak fibred product of Φ and G, is a diagram whose class depends only on the classes of d and e; the composition of classes is associative, and the classes of $1_X = X \xleftarrow{1} X \xrightarrow{1} X$ are units. If d' refines d by H and e' refines e by K, the composite of d' and e' refines that of d and e by the functor $L(a', \phi, b') = (H(a'), \tau_{B'}(b') \circ \phi \circ \tau_{A'}(a')^{-1}, K(b'))$, an equivalence. If G and $G \circ L$ are equivalences, so is L. *Book:* Theorem 10.3.8, p. 488 (2017: Theorem 10.3, p. 352), proved in Appendix §10.6.1, pp. 509–515. *Hidden hypotheses:* none. *Deviation (restated):* $d.\mathrm{comp}\, e$ is the text's $[e] \circ [d]$; the two iterated weak fibred products of the associativity are related by a reassociation functor, an equivalence, where the text identifies them; L is shown to be an equivalence by two out of three ($\pi_A \circ L = H \circ \pi_{A'}$); in the proof, τ_A in the naturality of

$\tau_{B'}$, "$(\Phi \circ \pi_B) \circ L$", $[1_X] \circ [d]$ and $\tau(a, \phi, a)$ are read $\tau_{B'}$, $(\Psi \circ \pi_B) \circ L$, $[d] \circ [1_X]$ and $\tau(x, \phi, a)$ (possible misprints) (Findings, items 33, 60).

V10.41 (Definition 10.3.9 (2017: 10.8), generalized maps and isomorphisms; `Polyfold.EPLocalization.EPLoc.homMk`, `Polyfold.EPLocalization.EPLoc.IsGeneralizedIso`, `Polyfold.EPLocalization.EPLoc.isGeneralizedIso_iff_isIso`). A generalized map $X \to Y$ is a class $[d]$ of diagrams; it is a generalized isomorphism if some generalized map $g : Y \to X$ composes with it to the units on both sides. The generalized isomorphisms are the isomorphisms of $\mathcal{EP}(E^{-1})$. *Book:* Definition 10.3.9, p. 489 (2017: Definition 10.8, p. 352). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 34).

V10.42 (Theorem 10.3.10 (2017: 10.4), generalized isomorphisms, with its proof in Appendix §10.6.2; `Polyfold.EPLocalization.EPLoc.theorem_10_3_10`, `Polyfold.EPLocalization.Diagram.symm`, `Polyfold.EPLocalization.diagFunctor`, `Polyfold.EPLocalization.isEquivalence_diagFunctor`, `Polyfold.EPLocalization.Diagram.isRefinement_comp_symm`, `Polyfold.EPLocalization.Diagram.isRefinement_id`, `Polyfold.EPLocalization.Diagram.equiv_comp_symm`, `Polyfold.EPLocalization.Diagram.equiv_symm_comp`, `Polyfold.EPLocalization.Diagram.isEquivalence_Φ_of_equiv_id`, `Polyfold.EPLocalization.isEquivalence_toFunctor_of_comp`, `Polyfold.EPLocalization.isLocalScDiffeo_obj_of_comp`, `Polyfold.EPLocalization.Diagram.isEquivalence_Φ_of_equiv`, `Polyfold.EPLocalization.EPLoc.isIso_homMk_iff`, `Polyfold.EPLocalization.EPLoc.isIso_homMk`, `Polyfold.EPLocalization.EPLoc.inv_homMk`, `Polyfold.MPolyfold.isLocalScDiffeoAt_of_injOn`, `Polyfold.MPolyfold.IsLocalScDiffeoAt.exists_injOn`). (1) $[X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ is a generalized isomorphism if and only if Φ is an equivalence; (2) every inverse is then $[Y \xleftarrow{\Phi} A \xrightarrow{F} X]$. An sc-smooth map injective near x with an sc-smooth local right inverse through x is a local sc-diffeomorphism at x. *Book:* Theorem 10.3.10, p. 489 (2017: Theorem 10.4, p. 353), proved in Appendix §10.6.2, pp. 515–519. *Hidden hypotheses:* none. *Deviation (same):* in (C1) the text's $A \times_X B$, $\mathrm{mor}_Y(\Psi(a), \Psi(a'))$ and "x" are read $A \times_Y B$, $\mathrm{mor}_X(\Psi(b), \Psi(b'))$ and "y", and in (C2) $(s|U)^{-1}(b')$ is read $(s|U)^{-1}(\Psi(b'))$ (possible misprints); the step of (C2) that Γ' is a local sc-diffeomorphism does not hold as we read it without the other half of (C2), and the local sc-diffeomorphism property of Φ is proved from a local right inverse obtained from the equivalence $\Phi \circ \pi_{A'}$ and from local injectivity (Findings, items 34, 61).

V10.43 (Proposition 10.3.11 (2017: 10.6), orbit maps compose; `Polyfold.EPLocalization.EPLoc.orbitMap`, `Polyfold.EPLocalization.EPLoc.orbitMap_homMk`, `Polyfold.EPLocalization.EPLoc.continuous_orbitMap`, `Polyfold.EPLocalization.EPLoc.orbitMap_id`, `Polyfold.EPLocalization.Diagram.orbitMap_comp`, `Polyfold.EPLocalization.EPLoc.proposition_10_3_11`). For generalized maps $f : X \to Y$ and $g : Y \to Z$, $|g \circ f| = |g| \circ |f|$; $|f|$ is continuous and $|1_X| = \mathrm{id}$. *Book:* Proposition 10.3.11, p. 490 (2017: Proposition 10.6, p. 353), with the computation before it. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 35).

V10.44 (Definition 10.3.12 (2017: 10.9), the category $\mathcal{EP}(E^{-1})$; `Polyfold.EPLocalization.EPLoc`, `Polyfold.EPLocalization.EPLoc.toEP_of`, `Polyfold.EPLocalization.EPLoc.of_`

`toEP`, `Polyfold.EPLocalization.EPLoc.homMk_comp`, `Polyfold.EPLocalization.EPLoc.id_eq`, `Polyfold.EPLocalization.EPLoc.homMk_eq_homMk_iff`, `Polyfold.EPLocalization.EPLoc.homMk_surjective`). $\mathcal{EP}(E^{-1})$ is a category with the ep-groupoids as objects and the generalized maps as morphisms, composed by Theorem 10.3.8, with units $[1_X]$; $[d] = [d']$ if and only if d and d' are equivalent, and every morphism is some $[d]$. *Book:* Definition 10.3.12, p. 490 (2017: Definition 10.9, p. 354). *Hidden hypotheses:* none. *Deviation (restated):* the objects are a copy of those of $\mathcal{EP}$ (a one-field structure), and $[d] \gg [e] = [d.\mathrm{comp}\, e]$ is the text's $[e] \circ [d]$ (Findings, item 35).

V10.45 (Theorem 10.3.14 (2017: 10.5), the localization functor; `Polyfold.EPLocalization.Diagram.ofFunctor`, `Polyfold.EPLocalization.Diagram.isRefinement_ofFunctor_comp`, `Polyfold.EPLocalization.EPLoc.localizationFunctor`, `Polyfold.EPLocalization.EPLoc.localizationFunctor_obj`, `Polyfold.EPLocalization.EPLoc.localizationFunctor_map`, `Polyfold.EPLocalization.EPLoc.theorem_10_3_14`, `Polyfold.EPLocalization.EPLoc.isInvertedBy_localizationFunctor`, `Polyfold.EPLocalization.EPLoc.inv_localizationFunctor_map`, `Polyfold.EPLocalization.EPLoc.homMk_eq`, `Polyfold.EPLocalization.Examples.pairToPointEP`, `Polyfold.EPLocalization.Examples.not_isIso_pairToPointEP`, `Polyfold.EPLocalization.Examples.isGeneralizedIso_pairToPoint`, `Polyfold.EPLocalization.Examples.inv_localizationFunctor_map_pairToPoint`, `Polyfold.EPLocalization.EPLoc.isLocalization`, `Polyfold.EPLocalization.EPLoc.universal_property`, `Polyfold.EPLocalization.map_eq_of_naturallyEquivalent`, `Polyfold.EPLocalization.secFunctor`). There is a functor $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$, the identity on objects, with $i(\Phi) = [X \xleftarrow{1} X \xrightarrow{\Phi} Y]$; for an equivalence F, $i(F)$ is a generalized isomorphism and every inverse is $[Y \xleftarrow{F} X \xrightarrow{1} X]$. i inverts the class E in the sense of Mathlib, and every generalized map is $i(\Phi) \circ i(F)^{-1}$. The equivalence from the pair groupoid of $\mathbb{Z}$ to the point is not an isomorphism of $\mathcal{EP}$, but its image under i is invertible. *Book:* Theorem 10.3.14, p. 491 (2017: Theorem 10.5, p. 354), with p. 487 for $[d] = i(\Phi) \circ i(F)^{-1}$. *Hidden hypotheses:* none. *Deviation (same):* in the proof, the text calls $d_{\Psi\circ\Phi}$ a refinement of (10.13) by π_X; with Definition 10.3.1 it is (10.13) that refines $d_{\Psi\circ\Phi}$, which is what is proved; the universal property of the localization, mentioned on p. 483, is proved: a functor inverting E identifies naturally equivalent functors, through the section $x \mapsto (x, \tau(x), F'(x))$ of $X \times_Y Y \to X$ (`Polyfold.EPLocalization.map_eq_of_naturallyEquivalent`), so every such functor factors uniquely through i (`Polyfold.EPLocalization.EPLoc.universal_property`) and i is a localization in Mathlib's sense (`Polyfold.EPLocalization.EPLoc.isLocalization`; gap G.117 closed) (Findings, item 36).

V10.46 (Theorem 10.3.15 (2017: 10.6), the orbit space functor; `Polyfold.EPLocalization.EPLoc.orbitFunctor`, `Polyfold.EPLocalization.EPLoc.theorem_10_3_15`). There is a functor $|\cdot| : \mathcal{EP}(E^{-1}) \to \mathrm{TOP}$ taking X to $|X|$ and $[X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ to $|\Phi| \circ |F|^{-1}$. *Book:* Theorem 10.3.15, p. 492 (2017: Theorem 10.6, p. 355). *Hidden hypotheses:* none. *Deviation (same):* TOP is Mathlib's `TopCat` (Findings, item 35).

V10.47 (Proposition 10.3.17, (LUC) under generalized isomorphisms; `Polyfold.EPLocalization.EPLoc.exists_diagram_of_isGeneralizedIso`, `Polyfold.EPLocalization.`

`EPLoc.iff_of_isGeneralizedIso`, `Polyfold.EPLocalization.EPLoc.proposition_10_3_17`). A generalized isomorphism is represented by a diagram whose two legs are equivalences; hence a property of ep-groupoids invariant under equivalences is invariant under generalized isomorphisms, and for a generalized isomorphism $X \to Y$, X has (LUC) if and only if Y has it. *Book:* Proposition 10.3.17, p. 492 (new in the 2021 edition), with the argument before Proposition 10.3.16. *Hidden hypotheses:* none. *Deviation (same):* "if one has it, so has the other" is stated as an equivalence (Findings, item 38).

V10.48 (Remark 10.3.18 (2017: 10.3), equivalence of pairs (X, β); `Polyfold.EPLocalization.OrbitChart`, `Polyfold.EPLocalization.OrbitChart.Rel`, `Polyfold.EPLocalization.OrbitChart.Rel.refl`, `Polyfold.EPLocalization.OrbitChart.Rel.symm`, `Polyfold.EPLocalization.OrbitChart.Rel.trans`, `Polyfold.EPLocalization.OrbitChart.setoid`, `Polyfold.EPLocalization.OrbitChart.rel_pairEP_pointEP`). For a space Z, pairs (X, β) with $\beta : |X| \to Z$ a homeomorphism, with (X', β') equivalent to (X, β) if some generalized isomorphism $f : X \to X'$ has $\beta' \circ |f| = \beta$, form a set with an equivalence relation. The pair groupoid of $\mathbb{Z}$ with the orbit map of its equivalence to the point, and the point with the identity, are equivalent pairs. *Book:* Remark 10.3.18, pp. 492–493 (2017: Remark 10.3, p. 355). *Hidden hypotheses:* none. *Deviation (narrower):* the claim that f is in general not unique is not formalized here (gap G.119, closed since: V10.60, by the base classes of the example of Remark 10.5.5); the metrizability of Z plays no role in the relation and is not assumed (Findings, item 39).

V10.49 (Definition 10.3.20, the subcategories of ep-groupoids with (LR); `Polyfold.EPLocalization.EPreg`, `Polyfold.EPLocalization.EPLocReg`, `Polyfold.EPLocalization.Examples.pointReg`, `Polyfold.EPLocalization.Examples.pointLocReg`). $\mathcal{EP}_{reg}$ and $\mathcal{EP}_{reg}(E^{-1})$ are the full subcategories of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$ on the ep-groupoids with the local regularity property; the point ep-groupoid is an object of both. *Book:* Definition 10.3.20, p. 494 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 39).

V10.50 (Theorem 10.3.21, invariance under generalized isomorphisms; `Polyfold.EPLocalization.EPLoc.theorem_10_3_21`, `Polyfold.EPLocalization.EPLoc.theorem_10_3_21_all`). For a generalized isomorphism $X \to Y$, X has (LR), (LUC), (SLR), (SLUC), respectively (ZHOU), if and only if Y has it. *Book:* Theorem 10.3.21, p. 495 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the proof, left to the reader in the text, applies Propositions 10.1.9, 10.1.16, 10.1.17, 10.1.18 and 10.1.19 to the two legs of a representing diagram; the parts (LR) and (SLR) were not formalized in the first edition of this page (gap G.118, closed since: V10.50, V10.64; Findings, items 16, 38).

V10.51 (the diagrams of strong bundles, their refinements and equivalence; `Polyfold.SBLocalization.SDiagram`, `Polyfold.SBLocalization.SDiagram.IsRefinement`, `Polyfold.SBLocalization.SDiagram.precomp`, `Polyfold.SBLocalization.SDiagram.isRefinement_precomp`, `Polyfold.SBLocalization.SDiagram.isRefinement_of_natEquiv`, `Polyfold.SBLocalization.SDiagram.IsRefinement.refl`, `Polyfold.SBLocalization.SDiagram.IsRefinement.trans`, `Polyfold.SBLocalization.SDiagram.IsCommonRefinement`,

`Polyfold.SBLocalization.SDiagram.Equiv`, `Polyfold.SBLocalization.SDiagram.Equiv.refl`, `Polyfold.SBLocalization.SDiagram.Equiv.symm`, `Polyfold.SBLocalization.SDiagram.IsRefinement.equiv`, `Polyfold.SEP.NatEquiv`, `Polyfold.SEP.NatEquiv.comp_left`, `Polyfold.SBLocalization.GenSBMap`, `Polyfold.SBLocalization.SDiagram.cls`, `Polyfold.SBLocalization.SDiagram.cls_eq_cls_of_equiv`, `Polyfold.SBLocalization.SDiagram.cls_eq_cls_iff`, `Polyfold.StrongBundleLocalizationExamples.foldDiagram`, `Polyfold.StrongBundleLocalizationExamples.isRefinement_foldDiagram`, `Polyfold.StrongBundleLocalizationExamples.cls_foldDiagram`). A diagram $D : W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W'$ between strong bundles $(P : W \to X, \mu)$ and $(P' : W' \to X', \mu')$ over ep-groupoids consists of a strong bundle $(P'' : W'' \to X'', \mu'')$ over an ep-groupoid, a strong bundle equivalence Φ and a strong bundle functor Ψ (morphisms of $\mathscr{SEP}$). D' refines D if a strong bundle equivalence $H : W''' \to W''$ has $\Phi \circ H \simeq \Phi'$ and $\Psi \circ H \simeq \Psi'$ (natural equivalence of strong bundle functors, Definition 10.4.8); every diagram refines itself, composing both legs with a strong bundle equivalence or replacing them by naturally equivalent ones gives a refinement, and a refinement of a refinement is one. Two diagrams are equivalent if they have a common refinement, an equivalence relation (V10.69); a generalized strong bundle map $[D] : W \to W'$ is the class of D, and $[D] = [D']$ iff D and D' have a common refinement. For $W' = \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$, $W' \xleftarrow{P'} (-\mathrm{id})^* W' \xrightarrow{P'} W'$ refines $1_{W'}$, so its class is $[1_{W'}]$. *Book:* §10.5, p. 504 (2017 p. 364), "following our earlier construction". *Hidden hypotheses:* none. *Deviation (same):* none; the classes are formed for the generated equivalence relation, which is having a common refinement by the transitivity, the strong bundle version of Lemma 10.3.4 (V10.69) (gap G.123, closed since; Findings, items 18, 50).

V10.52 ($[D]$ covers $[d]$, and the map $|[D]|$; `Polyfold.SBLocalization.SDiagram.base`, `Polyfold.SBLocalization.SDiagram.total`, `Polyfold.SBLocalization.SDiagram.IsRefinement.base`, `Polyfold.SBLocalization.SDiagram.Equiv.base`, `Polyfold.SBLocalization.SDiagram.Equiv.total`, `Polyfold.SBLocalization.GenSBMap.base`, `Polyfold.SBLocalization.GenSBMap.total`, `Polyfold.SBLocalization.SDiagram.orbitMap`, `Polyfold.SBLocalization.SDiagram.continuous_orbitMap`, `Polyfold.SBLocalization.SDiagram.Equiv.orbitMap_eq`, `Polyfold.SBLocalization.GenSBMap.orbitMap`, `Polyfold.SBLocalization.GenSBMap.orbitMap_eq_total`, `Polyfold.SBLocalization.GenSBMap.continuous_orbitMap`, `Polyfold.SEP.baseEP`, `Polyfold.SEP.totalEP`). A diagram D covers the diagram $d : X \xleftarrow{\phi} X'' \xrightarrow{\psi} X'$ of the underlying ep-groupoids, and refinements and equivalent diagrams cover refinements and equivalent diagrams; so $[D]$ covers a generalized map $[d] : X \to X'$ (Definition 10.3.9), and it induces a generalized map $(W, \mathbf{W}) \to (W', \mathbf{W}')$ of the total-space ep-groupoids as well. $|[D]| = |\Psi| \circ |\Phi|^{-1} : |W| \to |W'|$ is continuous and does not depend on the representative. *Book:* §10.5, p. 504 (2017 pp. 364–365). *Hidden hypotheses:* none. *Deviation (same):* $|\Phi|$ is a homeomorphism by Definition 10.4.1 (1), and $|[D]|$ is the orbit map of the diagram of total spaces (Lemma 10.3.6) (Findings, item 51).

V10.53 (the symbolic diagram (10.24) and the commutative square $|\mathscr{D}|$; `Polyfold.SBLocalization.SDiagram.orbitMap_square`, `Polyfold.SBLocalization.GenSBMap.`

`orbitMap_square`, `Polyfold.SBLocalization.SDiagram.orbitMap_orbitMap_Φ`, `Polyfold.SBLocalization.SDiagram.square_10_24`, `Polyfold.SBLocalization.GenSBMap.square_10_24`, `Polyfold.SEP.projEP`). $|P'| \circ |[D]| = |[d]| \circ |P|$ as maps $|W| \to |X'|$, for diagrams and for their classes. *Book:* §10.5, pp. 504–505 (2017: Remark 10.5, p. 365). *Hidden hypotheses:* none. *Deviation (same):* the symbolic diagram (10.24) itself has no formal content; its version with the generalized maps $i(P) = [W \overset{1}{\leftarrow} W \overset{P}{\longrightarrow} X]$, $i(P')$ as vertical arrows, left to the reader, commutes in $\mathcal{EP}(E^{-1})$ (`Polyfold.SBLocalization.SDiagram.square_10_24`, `Polyfold.SBLocalization.GenSBMap.square_10_24`; gap G.124 closed); the bottom arrow $|[d]|$ is printed $|X| \to |X|$, as are the bottom arrows of (10.24) and of the diagram after Remark 10.5.3, in both editions; we read X' (possible misprint) (Findings, item 51).

V10.54 (the homomorphism $\gamma : G_x \to G'_{x'}$; `Polyfold.EPLocalization.Diagram.isotropyHom`, `Polyfold.EPLocalization.Diagram.isotropyHom_spec`, `Polyfold.EPLocalization.Diagram.bijective_isotropyHom`, `Polyfold.EPLocalization.Diagram.exists_isotropyData`, `Polyfold.EPLocalization.Diagram.exists_germ_q`, `Polyfold.EPLocalization.Diagram.isotropyHom_conj`, `Polyfold.SBLocalization.SDiagram.exists_lift_Q`, `Polyfold.StrongBundle.IsLocalMap`). For a diagram $d : X \overset{\phi}{\leftarrow} X'' \overset{\psi}{\to} X'$ of ep-groupoids and x, x' with $|[d]|(|x|) = |x'|$ there are x'' and morphisms $a : \phi(x'') \to x$, $b : \psi(x'') \to x'$. For such data, γ, the composite of $g \mapsto a^{-1} \circ g \circ a$, of the inverse of $\phi : G_{x''} \to G_{\phi(x'')}$, of $\psi : G_{x''} \to G_{\psi(x'')}$ and of $\ell \mapsto b \circ \ell \circ b^{-1}$, is a group homomorphism with $\gamma(g) = b \circ \psi(k) \circ b^{-1}$ for the k with $\phi(k) = a^{-1} \circ g \circ a$, and it is a group isomorphism when ψ is an equivalence. *Book:* §10.5, p. 505 (2017 p. 365). *Hidden hypotheses:* none. *Deviation (stronger):* stated for every diagram of ep-groupoids (for D it is applied to d), with the bijectivity for ψ an equivalence added; the 2021 first step $a^{-1} \circ g \circ a$ is used (the 2017 text has $a \circ g \circ a^{-1}$); the equivariant germ $q = Q_b \circ \psi \circ Q_a^{-1}$ on a natural representation, sc-smooth with $q(x) = x'$ and $q(g * z) = \gamma(g) * q(z)$, and its lift $Q = H_b \circ \Psi \circ H_a^{-1}$, a local strong bundle map with $P' \circ Q = q \circ P$, (10.25), equivariant on the natural lifts, are stated for each choice of the data, since they depend on the representative (`Polyfold.EPLocalization.Diagram.exists_germ_q`, `Polyfold.SBLocalization.SDiagram.exists_lift_Q`; gap G.124 closed; Findings, item 52).

V10.55 (Definition 10.5.1 (2017: 10.14), the category $\mathcal{SEP}(F^{-1})$; `Polyfold.SBLocalization.SEPLoc`, `Polyfold.SBLocalization.SEPLoc.idHom`, `Polyfold.SBLocalization.SEPLoc.localizationPrefunctor`, `Polyfold.SBLocalization.SEPLoc.localizationPrefunctor_map_id`, `Polyfold.SBLocalization.SDiagram.id`, `Polyfold.SBLocalization.SDiagram.ofHom`, `Polyfold.SBLocalization.SEPLoc.category`, `Polyfold.SBLocalization.SEPLoc.homMk`, `Polyfold.SBLocalization.SEPLoc.homMk_comp`, `Polyfold.SBLocalization.SEPLoc.id_eq`, `Polyfold.SBLocalization.SEPLoc.homMk_eq_homMk_iff`, `Polyfold.SBLocalization.SEPLoc.homMk_surjective`, `Polyfold.SBLocalization.SEPLoc.localizationFunctor`, `Polyfold.SBLocalization.SEPLoc.localizationFunctor_toPrefunctor`, `Polyfold.SBLocalization.SEPLoc.isInvertedBy_localizationFunctor`). The objects of $\mathcal{SEP}(F^{-1})$ are the strong bundles over ep-groupoids and its morphisms $W \to W'$ the generalized strong bundle maps; with the composition of V10.70 and the identities $[1_W]$ they form a category, every morphism is a class $[D]$, and $[D] = [D']$ iff D and D' have a common refinement. $\Psi \mapsto [W \xleftarrow{1} W \xrightarrow{\Psi} W']$, which maps 1_W to $[1_W]$, is a functor $\mathcal{SEP} \to \mathcal{SEP}(F^{-1})$ (the strong bundle analogue of Theorem 10.3.14) that inverts the strong bundle equivalences. *Book:* Definition 10.5.1, p. 506 (2017: Definition 10.14, p. 366). *Hidden hypotheses:* none. *Deviation (explicit):* the composition, which the text does not spell out, is that of V10.70; until it was constructed, $\mathcal{SEP}(F^{-1})$ was formalized as a quiver (gap G.123, closed since; Findings, items 18, 53).

V10.56 (Definition 10.5.2 (2017: 10.15), generalized strong bundle isomorphisms; `Polyfold.SBLocalization.GenSBMap.IsGeneralizedIso`, `Polyfold.SBLocalization.SDiagram.symm`, `Polyfold.SBLocalization.GenSBMap.isGeneralizedIso_symm`, `Polyfold.SBLocalization.GenSBMap.isGeneralizedIso_ofHom`, `Polyfold.StrongBundleLocalizationExamples.negDiagram`, `Polyfold.StrongBundleLocalizationExamples.isGeneralizedIso_negDiagram`). A generalized strong bundle map is a generalized strong bundle isomorphism if it is the class of a diagram $W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W'$ in which both Φ and Ψ are strong bundle equivalences. The reversed diagram of such a representative, and $[W \xleftarrow{1} W \xrightarrow{\Psi} W']$ for a strong bundle equivalence Ψ, give generalized strong bundle isomorphisms; for $W' = \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ and the projection $P' : (-\mathrm{id})^* W' \to W'$ over $-\mathrm{id}$, $[(-\mathrm{id})^* W' \xleftarrow{1} (-\mathrm{id})^* W' \xrightarrow{P'} W']$ is one. *Book:* Definition 10.5.2, p. 506 (2017: Definition 10.15, p. 366). *Hidden hypotheses:* none. *Deviation (same):* none (Findings, items 53, 59).

V10.57 (Remark 10.5.3, the comparison with Theorem 10.3.10; `Polyfold.SBLocalization.GenSBMap.remark_10_5_3`, `Polyfold.SBLocalization.SEPLoc.isIso_of_isGeneralizedIso`). A generalized strong bundle isomorphism $[D] : W \to W'$ covers a generalized isomorphism $[d] : X \to X'$ of the base ep-groupoids, induces a generalized isomorphism of the total-space ep-groupoids, $|[D]|$ is a homeomorphism, and $[D]$ is an isomorphism of $\mathcal{SEP}(F^{-1})$. *Book:* Remark 10.5.3, p. 506 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (restated):* the remark says that a result similar to Theorem 10.3.10 can be proved in the strong bundle case and that it is built into Definition 10.5.2; we read this as "generalized strong bundle isomorphisms are isomorphisms of $\mathcal{SEP}(F^{-1})$", which is proved (V10.71); the converse half of the similar result, that every isomorphism of $\mathcal{SEP}(F^{-1})$ is one, is proved for the base functor only (V10.71); the consequences for the base and the total spaces are proved through Theorem 10.3.10 (gap G.123, closed since except the converse half; Findings, items 19, 53).

V10.58 (the push-forward and pull-back along a diagram, and their independence of the representative; `Polyfold.SBLocalization.SDiagram.pushforward`, `Polyfold.SBLocalization.SDiagram.pullback`, `Polyfold.SBLocalization.SDiagram.pushforward_eq_of_isRefinement`, `Polyfold.EpStrongBundle. _ ,` `Polyfold.EpStrongBundle. _eq_ _of_isFunctorialSection`, `Polyfold.EpStrongBundle.eq_of_forall_ _eq`, `Polyfold.SBLocalization.SDiagram. _pushforward`, `Polyfold.SBLocalization.SDiagram.base_orbitMap_eq_of_orbitMap_eq`, `Polyfold.SBLocalization.SDiagram.pushforward_eq_of_orbitMap_eq`, `Polyfold.SBLocalization.SDiagram.pullback_eq_of_orbitMap_eq`). For a diagram D with both legs strong bundle equivalences, $D_* f = \Psi_* \Phi^* f$ and $D^* g = \Phi_* \Psi^* g$, the push-forward along the reversed diagram. If D' refines D, both with second legs strong bundle equivalences, then $\Psi'_*\Phi'^*f = \Psi_*\Phi^*f$ for every section functor f, by the text's computation through Propositions 10.4.2–10.4.4. A section functor g is determined by the orbits $|g(x')|$ in $|W'|$, and $|(D_*f)(x')| = |[D]|(|f(x)|)$ whenever $|[d]|(|x|) = |x'|$; hence any two such diagrams with the same map $|D|$ have the same push-forward and the same pull-back on section functors. *Book:* §10.5, p. 507 (2017 p. 367), the paragraph before Theorem 10.5.4. *Hidden hypotheses:* none. *Deviation (stronger):* the independence is proved for any two diagrams with the same map of orbit spaces, not only along refinements; the text's own step is proved as printed (Findings, item 54).

V10.59 (Theorem 10.5.4 (2017: 10.9), without the Fred part; `Polyfold.SBLocalization.GenSBMap.rep`, `Polyfold.SBLocalization.GenSBMap.pushforward`, `Polyfold.SBLocalization.GenSBMap.pullback`, `Polyfold.SBLocalization.GenSBMap.pushforward_eq`, `Polyfold.SBLocalization.GenSBMap.pullback_eq`, `Polyfold.SBLocalization.GenSBMap.theorem_10_5_4`, `Polyfold.SBLocalization.SDiagram.invOn_pullback_pushforward`, `Polyfold.SBLocalization.SDiagram.bijOn_pushforward`, `Polyfold.SBLocalization.SDiagram.bijOn_pushforward_plus`, `Polyfold.StrongBundleLocalizationExamples.bijOn_pushforward_negDiagram`, `Polyfold.StrongBundleLocalizationExamples.pushforward_foldDiagram`). For a generalized strong bundle isomorphism $[D] : (P,\mu) \to (P',\mu')$: $[D]_*f = \Psi_*\Phi^*f$ and $[D]^*g = \Phi_*\Psi^*g$ for every representative with both legs strong bundle equivalences; $[D]_* : \Gamma(P,\mu) \to \Gamma(P',\mu')$ and $[D]^*$ are mutually inverse bijections of the sc-smooth section functors; the push-forward along the reversed diagram is $[D]^*$; the same holds for sc$^+$-section functors; and $|[D]| = |[D']|$ implies $[D]_* = [D']_*$ and $[D]^* = [D']^*$. For the example of V10.56 $[D]_*$ is a bijection, and the push-forward along $W' \xleftarrow{P'} (-\mathrm{id})^*W' \xrightarrow{P'} W'$ is the identity of $\Gamma(W')$. *Book:* Theorem 10.5.4, pp. 507–508 (2017: Theorem 10.9, p. 367). *Hidden hypotheses:* none. *Deviation (narrower):* the preservation of sc-Fredholm section functors is not formalized, as for Theorem 10.4.5 (gap G.125); "isomorphism" of the vector spaces Γ is read as mutually inverse bijections, $\Gamma(P,\mu)$ having no vector space structure in the project (Findings, item 55).

V10.60 (Remark 10.5.5, $|[D]| = |[D']|$ does not give $[D] = [D']$; `Polyfold.StrongBundleLocalizationExamples.remark_10_5_5`, `Polyfold.StrongBundleLocalizationExamples.pushforward_inv_eq_pushforward_id`, `Polyfold.StrongBundleLocalizationExamples.z3EpBundle`, `Polyfold.StrongBundleLocalizationExamples.z3InvHom`, `Polyfold.StrongBundleLocalizationExamples.z3InvHom_mem_equivalences`, `Polyfold.StrongBundleLocalizationExamples.orbitMap_ofHom_z3InvHom`, `Polyfold.StrongBundleLocalizationExamples.base_orbitMap_ofHom_z3InvHom`, `Polyfold.EPLocalization.Diagram.IsotropyRel`, `Polyfold.EPLocalization.Diagram.isotropyRel_isotropyHom`, `Polyfold.EPLocalization.Diagram.IsRefinement.isotropyRel_iff`, `Polyfold.EPLocalization.Diagram.Equiv.isotropyRel_iff`, `Polyfold.StrongBundleLocalizationExamples.isotropyRel_id`, `Polyfold.StrongBundleLocalizationExamples.not_isotropyRel_inv`, `Polyfold.SBLocalization.GenSBMap.remark_10_5_3`). For the strong bundle $W = \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (trivial action, $\mu((g,x),w) = w$) and the strong bundle equivalence Inv covering $(g,x) \mapsto (g^{-1},x)$,

$[1_W]$ and $[W \xleftarrow{1} W \xrightarrow{\mathrm{Inv}} W]$ are generalized strong bundle isomorphisms with $|[\mathrm{Inv}]| = |[1_W]|$ and with the same map of base orbit spaces, but $[\mathrm{Inv}] \neq [1_W]$, and already their base classes differ; nevertheless $[\mathrm{Inv}]_* = [1_W]_*$ on section functors. The classes are separated by the relation of the homomorphisms γ (the union of their graphs over all choices), which is invariant under refinement and equivalence of diagrams of ep-groupoids: for $[1]$ the generator of $G_0 = \mathbb{Z}/3$ is related to itself, for $[\mathrm{Inv}]$ only to its inverse. *Book:* Remark 10.5.5, p. 508 (new in the 2021 edition), with the paragraph before it. *Hidden hypotheses:* none. *Deviation (same):* the text gives no example; the positive half is the last assertion of Theorem 10.5.4 (V10.59). The base classes are two distinct generalized isomorphisms of $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (Remark 10.5.3) with the same orbit map, an instance of the non-uniqueness claimed in Remark 10.3.18 (gap G.119, closed since: V10.60); the example has the local regularity property, so Theorems 10.3.19 and 10.5.7, and the 2017 Theorems 10.7 and 10.10, do not hold as we read them (V10.65, V10.67; Findings, items 56, 58).

V10.61 (Proposition 10.5.6, (LR) under generalized strong bundle isomorphisms; `Polyfold.SBLocalization.proposition_10_5_6`, `Polyfold.SBLocalization.GenSBMap.iff_of_isGeneralizedIso`, `Polyfold.EpStrongBundle.proposition_10_4_6`). If $[D] : (P, \mu) \to (P', \mu')$ is a generalized strong bundle isomorphism, one of the two strong bundles has the local regularity property for strong bundles iff the other has, Definition 8.4.2 being read with local strong bundle isomorphisms of $W|U(x)$ (`IsLRLocal`). The proof is the text's: a property of strong bundles invariant under strong bundle equivalences is invariant under generalized strong bundle isomorphisms, and the local regularity property is so invariant by Proposition 10.4.6 (V10.72). *Book:* Proposition 10.5.6, p. 508 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* Definition 8.4.2 is read for local strong bundle isomorphisms, as for Theorem 10.5.7 (Findings, item 72); the text refers to Proposition 10.4.7 for the modification, which we read as Proposition 10.4.6, the statement used (a reading point) (gap G.126 closed; Findings, item 57).

V10.62 (Proposition 10.1.9, (LR) under equivalences; `Polyfold.EpFunctor.proposition_10_1_9`, `Polyfold.EpFunctor.TransportChart`, `Polyfold.EpFunctor.exists_transportChart`, `Polyfold.EpFunctor.TransportChart.conj`, `Polyfold.EpFunctor.TransportChart.bijective_conj`, `Polyfold.EpFunctor.TransportChart.Q_Φ`, `Polyfold.EpFunctor.TransportChart.exists_naturalRep_push`, `Polyfold.EpFunctor.TransportChart.exists_naturalRep_pull`, `Polyfold.EpGroupoid.NaturalRep.HasLR`, `Polyfold.EpGroupoid.isLR_iff_hasLR`, `Polyfold.EpGroupoid.NaturalRep.hasLR_iff_of_equivariant`, `Polyfold.EpGroupoid.NaturalRep.eqOn_Γ`, `Polyfold.EpGroupoid.NaturalRep.eqOn_Φ`, `Polyfold.EquivalenceLRExamples.nonempty_transportChart_pointEP`). If $F : X \to Y$ is an equivalence, X has the local regularity property (Definition 7.4.19) if and only if Y has it. For a morphism $\psi : F(x) \to y$, the composite $Q = \widehat{\psi} \circ F$ is an sc-diffeomorphism of a neighbourhood of x onto one of y, $g \mapsto \psi \circ F(g) \circ \psi^{-1}$ is an isomorphism $G_x \to G_y$ with $Q(g * z) = (\psi \circ F(g) \circ \psi^{-1}) * Q(z)$ ((10.2), (10.3)), natural representations are carried along Q in both directions, and an equivariant sc-diffeomorphism $U(x) \to U(y)$ along a surjection $G_x \to G_y$ carries the condition of (LR) both ways. *Book:* Proposition 10.1.9, pp. 468–470 (new in the 2021 edition), with (10.1)–(10.3). *Hidden* *hypotheses:* none. *Deviation (explicit):* the text takes $U(y) = Q(U(x))$ with the natural action of G_y; that $Q(U(x))$ carries a natural representation is proved by restricting an arbitrary one at y (Theorem 7.1.13), and the converse direction restricts one at x to $Q^{-1}(U(y))$; the isomorphism $\sigma \mapsto \psi \circ \sigma \circ \psi^{-1}$ is printed with target $U(y)$ (p. 469); we read G_y (possible misprint) (gap G.113, closed since: V10.62; Findings, items 7, 16).

V10.63 (Proposition 10.1.10 (2017: 10.4), local families α_z and $|TF| = |TG|$; `Polyfold.EpFunctor.proposition_10_1_10_one`, `Polyfold.EpFunctor.proposition_10_1_10_one_of_isLR_right`, `Polyfold.EpFunctor.proposition_10_1_10_two`, `Polyfold.EquivalenceLRExamples.proposition_10_1_10_hypotheses`). Let $F, G : X \to Y$ be equivalences with $|F| = |G|$, and let X or Y have (LR). (1) Inside every open $O \ni x$ there is a natural representation $U(x)$ with an sc-smooth map $U(x) \to$ **Y**, $z \mapsto \alpha_z : F(z) \to G(z)$. (2) The tangent functors satisfy $|TF| = |TG|$ (stated with (LR) for X). The hypotheses are satisfied by the point ep-groupoid. *Book:* Proposition 10.1.10, p. 470 (2017: Proposition 10.4, p. 340). *Hidden hypotheses:* none. *Deviation (stronger):* the text assumes (LR) for X and Y; the proof uses it for X only, and with (LR) for Y only Proposition 10.1.9 gives it for X; the families α_z are in general not natural (V10.65) (gap G.114, closed since: V10.63; Findings, items 7, 16).

V10.64 (Proposition 10.3.16, (LR) under generalized isomorphisms; `Polyfold.EPLocalization.EPLoc.proposition_10_3_16`, `Polyfold.EPLocalization.EPLoc.iff_of_isGeneralizedIso`). If $f : X \to Y$ is a generalized isomorphism, X has (LR) if and only if Y has it. *Book:* Proposition 10.3.16, p. 492 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* the text's proof, Proposition 10.1.9 (V10.62) for the two legs of a representing diagram (gap G.118, closed since: V10.64, V10.50; Findings, items 16, 38).

V10.65 (Theorem 10.3.19 (2017: 10.7) does not hold as we read it; `Polyfold.EPLocalization.EPLoc.Theorem_10_3_19`, `Polyfold.LocalizationLRCounterexample.not_theorem_10_3_19`, `Polyfold.LocalizationLRCounterexample.theorem_10_3_19_counterexample`, `Polyfold.EpGroupoid.Examples.isLR_trivialZ3`, `Polyfold.LocalizationLRCounterexample.not_naturallyEquivalent_id_inv`, `Polyfold.EpGroupoid.Examples.not_isEffective_trivialZ3`). The statement "if one of X, Y has (LR), generalized isomorphisms $f, f' : X \to Y$ with $|f| = |f'|$ are equal" is false: $X = \mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action has (LR), its orbits being points, and $[X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X]$ and $[1_X]$, $\mathrm{Inv}(g, x) = (g^{-1}, x)$, are generalized isomorphisms with the same orbit map that differ (the base part of V10.60). There is no natural transformation $1 \to \mathrm{Inv}$, although $z \mapsto 1_z : z \to \mathrm{Inv}(z)$ is a family as in Proposition 10.1.10 (1); X is not effective. *Book:* Theorem 10.3.19, pp. 493–494 (2017: Theorem 10.7, p. 355). *Hidden hypotheses:* none. *Deviation (restated):* the statement is formalized as a proposition and its negation is proved; it does not hold as we read it, nor does the 2017 Theorem 10.7, which has no (LR) hypothesis. The step of the proof we cannot follow is on p. 494: the families α_a of Proposition 10.1.10 (1) are used as a natural transformation $H_{\mathcal{U}} \to H'_{\mathcal{U}}$, and in the example no choice of them is natural. The strongest correct version we prove is V10.66 (gap G.120, settled since: V10.65, V10.66; Findings, items 39, 58).

V10.66 (Theorem 10.3.19 for an effective target; `Polyfold.EPLocalization.EPLoc.theorem_10_3_19_of_isEffective`, `Polyfold.EpFunctor.naturallyEquivalent_of_isEffective`, `Polyfold.EpFunctor.eqOn_of_isEffective`, `Polyfold.EpGroupoid.eqOn_unit_of_isEffective`, `Polyfold.EPLocalization.EPLoc.theorem_10_3_19_of_isEffective_hypotheses`, `Polyfold.EquivalenceLREffectiveExamples.naturallyEquivalent_of_isEffective_hypotheses`). If one of X, Y has (LR) and Y is effective, generalized isomorphisms $f, f' : X \to Y$ with $|f| = |f'|$ are equal. On the way: in an effective ep-groupoid a continuous family $v \mapsto k(v) \in G_v$ on an open set of objects is the family of units; over an effective target two continuous families $\alpha_z, \alpha'_z : H(z) \to H'(z)$ on an open set agree when H is a local sc-diffeomorphism on objects; and equivalences $H, H' : X \to Y$ with $|H| = |H'|$, X with (LR) and Y effective are naturally equivalent, by an sc-smooth natural transformation on all of X. The point ep-groupoid satisfies the hypotheses. *Book:* Theorem 10.3.19, pp. 493–494 (2017: Theorem 10.7, p. 355), with the hypothesis that Y is effective added. *Hidden hypotheses:* none; the effectiveness of Y is added. *Deviation (narrower):* the proof is not the text's: for $e = f' \circ f^{-1} : Y \to Y$, represented by $Y \xleftarrow{F} A \xrightarrow{\Phi} Y$ with $|\Phi| = |F|$, the families of Proposition 10.1.10 (1), unique over an effective target, glue to a natural transformation $F \to \Phi$, so $e = 1_Y$; no weak fibred product and no cover $A_{\mathcal{U}}$ are needed (Findings, item 17).

V10.67 (Theorem 10.5.7 (2017: 10.10) does not hold as we read it; `Polyfold.SBLocalization.Theorem_10_5_7`, `Polyfold.LocalizationLRCounterexample.not_theorem_10_5_7`, `Polyfold.LocalizationLRCounterexample.isLR_z3EpBundle`, `Polyfold.StrongBundleLocalizationExamples.remark_10_5_5`). The statement "if both strong bundles have the local regularity property, generalized strong bundle isomorphisms $[D], [D']$ with $|[D]| = |[D']|$ are equal" is false: the strong bundle $\mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial μ has the local regularity property (Definition 8.4.2), its orbits being points, and $[W \xleftarrow{1} W \xrightarrow{\mathrm{Inv}} W]$ and $[1_W]$ of V10.60 have the same orbit map and differ. *Book:* Theorem 10.5.7, p. 508 (2017: Theorem 10.10, p. 367). *Hidden hypotheses:* none. *Deviation (restated):* the statement is formalized as a proposition and its negation is proved; it does not hold as we read it, nor does the 2017 Theorem 10.10, which has no local regularity hypothesis; the text's proof ("immediately from Proposition 10.4.7") meets the step of V10.65 for strong bundles; a version for effective strong bundles is not formalized (gap G.127, settled since: V10.67; Findings, item 58).

V10.68 (Theorem 10.2.2 (2017: 10.2) for strong bundles, the weak fibred product of strong bundles; `Polyfold.SBWeakFibredProduct.wfpSEP`, `Polyfold.SBWeakFibredProduct.projX`, `Polyfold.SBWeakFibredProduct.projZ`, `Polyfold.SBWeakFibredProduct.theorem_10_2_2`, `Polyfold.SBWeakFibredProduct.theorem_10_2_2_right`, `Polyfold.SBWeakFibredProduct.nonempty_wfpSEP`, `Polyfold.EpStrongBundle.pullbackAlong`, `Polyfold.EpStrongBundle.isStrongBundleEquivalence_pullbackProjFunctor`, `Polyfold.EpStrongBundle.transportFunctor`, `Polyfold.EpStrongBundle.invLiftFunctor`, `Polyfold.SEP.pullbackAlong`, `Polyfold.SEP.coneHom`, `Polyfold.SEP.natEquiv_cone`, `Polyfold.StrongBundleWeakFibredProductExamples.theorem_10_2_2_negSEPHom`). For a strong bundle equivalence $\Phi : W \to W'$ covering $\phi : X \to X'$ and a strong bundle functor $\Psi : W'' \to W'$

covering $\psi : X'' \to X'$, the weak fibred product $W \times_{W'} W''$ is the pull-back $\pi_Z^* W''$ over the weak fibred product $X \times_{X'} X''$ of ϕ and ψ, with $\Pi_Z((x,\varphi,z),w'') = w''$ covering π_Z and $\Pi_X((x,\varphi,z),w'') = \Phi_x^{-1}(\mu'(\varphi^{-1},\Psi(w'')))$ covering π_X, where Φ_x is the fibrewise isomorphism of Proposition 10.4.4. Π_Z is a strong bundle equivalence, $\Phi \circ \Pi_X \simeq \Psi \circ \Pi_Z$ (Definition 10.4.8), Π_X is a strong bundle equivalence when Ψ is one, and $W \times_{W'} W''$ is nonempty when W'' is. For Φ the strong bundle equivalence over $-\mathrm{id}$ and $\Psi = 1$ both projections are strong bundle equivalences. *Book:* Theorem 10.2.2, p. 477 (2017: Theorem 10.2, p. 343), in the strong bundle case that §10.5 uses "following our earlier construction", p. 504 (2017 p. 364). *Hidden hypotheses:* none. *Deviation (explicit):* the text states Theorem 10.2.2 for ep-groupoids only and gives no construction for strong bundles; the pull-back of W'' along π_Z makes Π_Z a strong bundle equivalence by construction, and the claims on degeneracy indices, tameness and orbit spaces concern the bases and are V10.18–V10.20 (gap G.123, closed since; Findings, item 18).

V10.69 (Lemma 10.3.4 (2017: 10.4) for strong bundles, transitivity; `Polyfold.SBLocalization.SDiagram.lemma_10_3_4`, `Polyfold.SBLocalization.SDiagram.Equiv.trans`, `Polyfold.SBLocalization.SDiagram.equivalence_equiv`, `Polyfold.SBLocalization.SDiagram.cls_eq_cls_iff`). The equivalence of diagrams of strong bundles is transitive: if w refines D by H and D' by H', and w' refines D' by K and D'' by K', the diagram on the weak fibred product of H' and K (V10.68) refines D by $H \circ \Pi_X$ and D'' by $K' \circ \Pi_Z$. So having a common refinement is an equivalence relation, and $[D] = [D']$ iff D and D' have a common refinement. *Book:* Lemma 10.3.4, p. 485 (2017: Lemma 10.4, p. 349), in the strong bundle case of p. 504 (2017 p. 364). *Hidden hypotheses:* none. *Deviation (explicit):* the strong bundle version, which the text leaves to the reader, is proved as Lemma 10.3.4, with V10.68 for the weak fibred product (gap G.123, closed since; Findings, items 18, 50).

V10.70 (Theorem 10.3.8 (2017: 10.3) for strong bundles, the composition and the category laws; `Polyfold.SBLocalization.SDiagram.compMid`, `Polyfold.SBLocalization.SDiagram.compLeft`, `Polyfold.SBLocalization.SDiagram.compRight`, `Polyfold.SBLocalization.SDiagram.comp`, `Polyfold.SBLocalization.SDiagram.IsRefinement.comp`, `Polyfold.SBLocalization.SDiagram.Equiv.comp`, `Polyfold.SBLocalization.SDiagram.isRefinement_assoc`, `Polyfold.SBLocalization.SDiagram.equiv_assoc`, `Polyfold.SBLocalization.SDiagram.equiv_comp_id`, `Polyfold.SBLocalization.SDiagram.equiv_id_comp`, `Polyfold.SBLocalization.SDiagram.equiv_ofHom_comp`, `Polyfold.SBLocalization.GenSBMap.comp`, `Polyfold.SBLocalization.GenSBMap.cls_comp_cls`, `Polyfold.StrongBundleLocalizationCategoryExamples.homMk_foldDiagram_comp`, `Polyfold.StrongBundleLocalizationCategoryExamples.nonempty_comp_foldDiagram`). For $D : W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W'$ and $E : W' \xleftarrow{\Phi'} V'' \xrightarrow{\Psi'} W'''$, the middle of the composite is the pull-back of W'' over the weak fibred product of ψ and ϕ', with the legs $\Phi \circ \Pi_{W''}$ and $\Psi' \circ \Pi_{V''}$, $\Pi_{W''}$ a strong bundle equivalence and $\Pi_{V''}(p,w) = \Phi_z'^{-1}(\mu'(\varphi,\Psi(w)))$. Refinements of D and E give a refinement of the composite, so $[E] \circ [D]$ depends only on the classes; the composition is associative, $[1]$ is a unit on both sides, and $\Psi \mapsto [W \xleftarrow{1} W \xrightarrow{\Psi} W']$ respects composition. For $F = [W' \xleftarrow{P'} (-\mathrm{id})^* W' \xrightarrow{P'} W']$, $F \circ F = [1_{W'}]$, with a nonempty middle. *Book:* Theorem

10.3.8, p. 488 (2017: Theorem 10.3, p. 352), with Appendix §10.6.1, pp. 509–515, in the strong bundle case that Definition 10.5.1, p. 506 (2017: Definition 10.14, p. 366), uses. *Hidden hypotheses:* none. *Deviation (explicit):* the composition and the category laws, which the text does not spell out, follow §10.3 and Appendix §10.6.1 with V10.68; on the pull-back the squares of the proofs commute strictly, by the fibrewise injectivity of strong bundle equivalences (gap G.123, closed since; Findings, item 18).

V10.71 (Theorem 10.3.10 (2017: 10.4) for strong bundles; `Polyfold.SBLocalization.SEPLoc.theorem_10_3_10`, `Polyfold.SBLocalization.SEPLoc.isIso_of_isGeneralizedIso`, `Polyfold.SBLocalization.SEPLoc.inv_homMk`, `Polyfold.SBLocalization.SDiagram.equiv_comp_symm`, `Polyfold.SBLocalization.SDiagram.equiv_symm_comp`, `Polyfold.SBLocalization.SEPLoc.baseFunctor`, `Polyfold.SBLocalization.SEPLoc.isEquivalence_base_of_isIso`, `Polyfold.SBLocalization.SEPLoc.equivalences_Ψ_of_isIso`, `Polyfold.SBLocalization.SEPLoc.isIso_homMk_iff`, `Polyfold.SBLocalization.SEPLoc.isIso_iff_isGeneralizedIso`, `Polyfold.SEP.equivalences_of_natEquiv_comp`, `Polyfold.SEP.exists_isStrongBundleIsoOn`, `Polyfold.SBLocalization.SDiagram.fibInj_Ψ_of_equiv`, `Polyfold.StrongBundleLocalizationCategoryExamples.isIso_homMk_negDiagram`). $[W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W']$ is an isomorphism of $\mathcal{SEP}(F^{-1})$ if and only if Ψ is a strong bundle equivalence (Definition 10.4.1 in full), and then every inverse of it is $[W' \xleftarrow{\Psi} W'' \xrightarrow{\Phi} W]$; so the isomorphisms of $\mathcal{SEP}(F^{-1})$ are exactly the generalized strong bundle isomorphisms of Definition 10.5.2. For the converse, $[D] \circ [E] = 1$ for an inverse $[E]$ makes Ψ injective on the fibres, $[E] \circ [D] = 1$ makes $\Psi \circ G$ naturally equivalent to a strong bundle equivalence, and these suffice: transports along morphisms and local right inverses at levels 0 and 1 make Ψ a local strong bundle isomorphism on objects and morphisms, an equivalence on total spaces and a homeomorphism on $|W[1]|$. The class over $-\mathrm{id}$ of V10.56 is an isomorphism with that inverse. *Book:* Theorem 10.3.10, p. 489 (2017: Theorem 10.4, p. 353), in the strong bundle case of Remark 10.5.3, p. 506 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none; the proof replaces the claims (C1), (C2) of Appendix §10.6.2, which the text does not restate for strong bundles, by a saturation argument (gap G.123 closed; Findings, item 19).

V10.72 (Proposition 10.4.6, (LR) under strong bundle equivalences; `Polyfold.EpStrongBundle.proposition_10_4_6`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.exists_transport`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.transport_Γ`, `Polyfold.EpStrongBundle.hasLRLocal_iff_of_equivariant`, `Polyfold.EpStrongBundle.HasLRLocal`,`Polyfold.EpStrongBundle.isLRLocal_iff_hasLRLocal`,`Polyfold.StrongBundle.IsLocalIso`, `Polyfold.StrongBundle.IsLocalIso.comp`, `Polyfold.StrongBundle.IsLocalIso.symm`, `Polyfold.StrongBundle.IsLocalIsoOn.exists_isLocalIso`, `Polyfold.EpStrongBundle.isLocalIso_lrAct`, `Polyfold.EpStrongBundle.IsLRLocal`). For strong bundles $(P : V \to X, \mu)$, $(Q : W \to Y, \nu)$ over ep-groupoids and a strong bundle equivalence $\Phi : V \to W$ covering ϕ, V has the local regularity property for strong bundles (Definition 8.4.2, with local strong bundle isomorphisms of $W|U(x)$) iff W has. Local strong bundle isomorphisms over sc-diffeomorphisms of open sets compose, restrict and invert; for $\psi : \phi(x) \to y$ the text's transport $v \mapsto \nu(\psi_{P(v)}, \Phi(v))$ is one of them over the chart Q of a transport chart, equivariant on the natural lifts for $g \mapsto \psi \circ \phi(g) \circ \psi^{-1}$, and conjugation by it carries the property of one natural representation to the other. *Book:* Proposition 10.4.6, p. 501 (new in the 2021 edition); the proof, a strong bundle version of Proposition 10.1.9, is left to the reader. *Hidden hypotheses:* none. *Deviation (same):* Definition 8.4.2 is read for local strong bundle isomorphisms (`IsLRLocal`); with the formal `Polyfold.EpStrongBundle.IsLR`, maps of all of W, the conjugation is only local (Findings, items 72, 74); "$\Phi : W \to W$" is read $\Phi : V \to W$ (Findings, item 57). *Proof:* the text's, with the inverse of a local strong bundle isomorphism constructed (`IsLocalIsoOn.exists_isLocalIso`), which the earlier API lacked (Findings, item 45).

V10.73 (Proposition 10.4.7, natural families for strong bundle equivalences; `Polyfold.EpStrongBundle.proposition_10_4_7`, `Polyfold.EpStrongBundle.proposition_10_4_7'`, `Polyfold.GapP2_GBundleFamilyExamples.proposition_10_4_7_hypotheses`, `Polyfold.SEP.exists_family_of_isLRLocal`). For strong bundle equivalences Φ, Ψ covering ϕ, ψ with $|\Phi| = |\Psi|$, the target bundle having the local regularity property (`IsLRLocal`), every x and every open $O \ni x$ have a natural representation $U(x) \subseteq O$ and an sc-smooth $z \mapsto \alpha_z : \phi(z) \to \psi(z)$ on $U(x)$ with $\nu(\alpha_z, \Phi(v)) = \Psi(v)$ whenever $P(v) = z$. The primed form takes (LR) of both bundles, as the text does; the hypotheses hold for the identity of $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$. *Book:* Proposition 10.4.7, pp. 501–502 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (stronger):* (LR) of the source bundle is not used, and $U(x)$ lies in any given neighbourhood; Definition 8.4.2 is read for local strong bundle isomorphisms (Findings, items 45, 72, 74).

V10.74 (the full subcategory $\mathcal{EP}_{lreg}$; `Polyfold.EP.lreg`, `Polyfold.EP.Lreg`, `Polyfold.EP.fullyFaithful_ _lreg`, `Polyfold.EP.lreg_iff_of_isEquivalence`, `Polyfold.EP.isClosedUnderIsomorphisms_lreg`, `Polyfold.EP.lreg_pointEP`, `Polyfold.EP.Lreg.point`, `Polyfold.EPLocalization.EPLoc.lreg`, `Polyfold.EPLocalization.EPLoc.lreg_A_of_lreg`, `Polyfold.EPLocalization.EPLoc.isClosedUnderIsomorphisms_lreg`). The ep-groupoids with the local regularity property (Definition 7.4.19) form a full subcategory $\mathcal{EP}_{lreg}$ of $\mathcal{EP}$ (Mathlib's full subcategory of an object property; the inclusion is fully faithful). The property passes along equivalences, so it is closed under the isomorphisms of $\mathcal{EP}$ and of $\mathcal{EP}(E^{-1})$, and the middle of a diagram out of an object of $\mathcal{EP}_{lreg}$ has it; the point ep-groupoid is an object. *Book:* §10.1, p. 461 (new in the 2021 edition), the subcategory generated by the ep-groupoids with the local regularity property. *Hidden hypotheses:* none. *Deviation (same):* none; "generated by" is the full subcategory on these objects (gap G.111 closed; Findings, item 7).

V10.75 (Remark 10.1.8 and Appendix §10.6.3, Zhou's example; `Polyfold.EpGroupoid.Examples.ZhouTenSix.remark_10_1_8`, `Polyfold.EpGroupoid.Examples.ZhouTenSix.not_isLR`, `Polyfold.EpGroupoid.Examples.ZhouTenSix.isLossFamily_piAbs`, `Polyfold.EpGroupoid.Examples.ZhouTenSix.piAbs_eq`, `Polyfold.EpGroupoid.Examples.ZhouTenSix.Rt` , `Polyfold.EpGroupoid.Examples.ZhouTenSix.X` , `Polyfold.EpGroupoid.Examples.ZhouTenSix. O`, `Polyfold.EpGroupoid.Examples.ZhouTenSix. O_of_nonpos`, `Polyfold.EpGroupoid.Examples.ZhouTenSix. O_of_pos`, `Polyfold.ZhouLine.lineSc`, `Polyfold.ZhouLine.zhouFam`, `Polyfold.ZhouLine.isLossFamily_zhouFam`, `Polyfold.EpGroupoid.`

`Examples.ZhouRetract.Data`, `Polyfold.IsLossFamily.add`, `Polyfold.IsLossFamily.sub`, `Polyfold.IsLossFamily.comp_neg`, `Polyfold.EpGroupoid.Examples.ZhouSigma.remark_10_1_8`, `Polyfold.EpGroupoid.Examples.ZhouSigma.not_isLR`). For weights $0 = \delta_0 < \delta_1 < \cdots$ and a smooth compactly supported β with $\int \beta^2 = 1$, on $F = \mathbb{R} \oplus L^2(\mathbb{R})$ with $F_m = \mathbb{R} \oplus H^{m,\delta_m}(\mathbb{R})$, $r(s, f) = (s, \pi_{|s|} f)$, with the projections π_t onto $\beta(\cdot + e^{1/t})$ of Subsection 1.4.4 and $\pi_{|s|} = \pi_s + \pi_{-s}$, is an sc-smooth retraction; $\mathbb{Z}/2$ acts on its image Y by $(s, f) \mapsto (s, -f)$; and η, equal to (s, f) for $s \leq 0$ and $(s, -f)$ for $s > 0$, is an sc-diffeomorphism of Y, its own inverse, fixing 0, with $|\eta(y)| = |y|$, which on no neighbourhood of 0 is $y \mapsto g * y$ for one $g \in \mathbb{Z}/2$. Hence $\mathbb{Z}/2 \ltimes Y$ does not have (LR). Such data exist (`Polyfold.EpGroupoid.Examples.ZhouRetract.Data`, with `std`). *Book:* Remark 10.1.8, p. 467, and Appendix §10.6.3, p. 519 (both new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* as printed; $H^{m,\delta}(\mathbb{R})$ is defined by lifting to $\mathbb{R} \times S^1$, with the same norm, and the sc-smoothness of π_t, cited from the literature, is proved (V7.82). The appendix's side remark that Y is homeomorphic to $\mathbb{R}^2 \setminus \{(0, t) \mid t \neq 0\}$ is not formalized; Remark 10.1.8 does not use it. A variant with the moving projections of Remark 4.1.3 and the opposite sign convention for η is kept (`Polyfold.EpGroupoid.Examples.ZhouSigma.remark_10_1_8`) (gap G.112 closed; Findings, item 7).

What is not verified in this chapter

- p. 461, the full subcategory $\mathcal{EP}_{lreg}$ of ep-groupoids with the local regularity property: not formalized, not a numbered item (gap G.111; Findings, item 7); closed since: V10.74, the full subcategory, closed under the isomorphisms of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$.
- p. 467, Remark 10.1.8 and Appendix §10.6.3, p. 519 (Zhou's example of a β with $|\beta(y)| = |y|$ not given by the group): it needs the weighted Sobolev sc-structure and the sc-retraction of the appendix (gap G.112; Findings, item 7); closed since: V10.75, as printed, on the appendix's scale with its projections. Not verified: the appendix's side remark that Y is homeomorphic to $\mathbb{R}^2 \setminus \{(0, t) \mid t \neq 0\}$, which Remark 10.1.8 does not use.
- p. 468, Proposition 10.1.9 (new in the 2021 edition): (LR) under equivalences; it needs a natural representation transported along F and a morphism $F(x) \to y$, which the Chapter 7 API of natural representations does not provide (gap G.113; Findings, item 7); closed since: V10.62, where natural representations are carried along $Q = \widehat{\psi} \circ F$ (Findings, item 16).
- p. 470, Proposition 10.1.10 (2017: 10.4): sc-smooth families $\alpha_z : F(z) \to G(z)$ for $|F| = |G|$, and $|TF| = |TG|$; it assumes (LR) and uses Proposition 10.1.9; the example of Remark 10.1.11 satisfies (1) (V10.8) (gap G.114; Findings, items 7, 8); closed since: V10.63, with (LR) on one side only (Findings, item 16).
- p. 477, Proposition 10.1.19 (new in the 2021 edition) itself: only its reduction to Proposition 10.1.9 is verified (V10.16) (gap G.115; Findings, item 15); closed since: V10.16, through V10.62 (Findings, item 16).
- pp. 478–481, Theorem 10.2.2: the agreement of the iterated sub-M-polyfold structures of $X \times_Y Z$ with structures induced directly from $X \times \mathbf{Y} \times Z$ and $\mathbf{X} \times \mathbf{Y} \times \mathbf{Z}$; nothing depends on it (V10.18) (gap G.116; Findings, item 21); closed since: V10.18, the two structures agree.

- p. 483, the universal property of the localization $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ (not a numbered statement): only the inversion of E is proved (V10.45); the universal property would need that every functor inverting E identifies naturally equivalent sc-smooth functors, through a cylinder ep-groupoid the project does not have (gap G.117; Findings, item 36); closed since: V10.45, with the weak fibred product $X \times_Y Y$ as the cylinder; i is a localization in Mathlib's sense.
- p. 491, Remark 10.3.13 (new in the 2021 edition), X as an atlas on $|X|$: no formal content (Findings, item 37).
- p. 492, Proposition 10.3.16 (new in the 2021 edition), and p. 495, the parts (LR) and (SLR) of Theorem 10.3.21: they rest on Proposition 10.1.9 (G.113); `Polyfold.EPLocalization.EPLoc.iff_of_isGeneralizedIso` reduces them to it, as the text does (gap G.118; Findings, item 38); closed since: V10.64, V10.50 (Findings, item 16).
- pp. 492–493, Remark 10.3.18 (2017: 10.3), the claim that a generalized isomorphism f with $\beta' \circ |f| = \beta$ is in general not unique: no example is formalized (the one behind Remark 10.1.8 is G.112) (gap G.119; Findings, item 39); closed since: V10.60, where the example of Remark 10.5.5 gives two distinct generalized isomorphisms of $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the same orbit map (Findings, item 56).
- p. 493, Theorem 10.3.19 (2017: 10.7): generalized isomorphisms with the same orbit map agree when one side has (LR); it rests on Propositions 10.1.10 and 10.3.16 (gap G.120; Findings, items 39, 58); settled since: V10.65, where the example, which has (LR), shows that the statement does not hold as we read it, and V10.66, where it is proved for an effective target (Findings, item 17).
- pp. 500–501, Theorem 10.4.5 (2017: 10.8), the Fred part: it needs the invariance of sc-Fredholm sections of strong bundles over M-polyfolds under local strong bundle isomorphisms, of which Part I has only chart-level germ statements (V10.27) (gap G.121; Findings, item 44).
- pp. 501–502, Propositions 10.4.6 and 10.4.7 (new in the 2021 edition): the strong bundle versions of Propositions 10.1.9 and 10.1.10, the proof of 10.4.6 left to the reader in the text; the ep-groupoid versions are formalized since (V10.62, V10.63), while the strong bundle versions need the transport $H_{\hat{\psi}}$ as a strong bundle isomorphism at levels 0 and 1 and the inverse of a strong bundle isomorphism on an open set, which the strong bundle API does not provide (gap G.122; Findings, item 45); closed since: V10.72, V10.73, with Definition 8.4.2 read for local strong bundle isomorphisms.
- p. 504 (2017 p. 364), the transitivity of the equivalence of diagrams of strong bundles (the strong bundle version of Lemma 10.3.4), and with it the composition in $\mathcal{SEP}(F^{-1})$ (Definition 10.5.1 (2017: 10.14), p. 506) and the strong bundle version of Theorem 10.3.10 that Remark 10.5.3 (p. 506, new in the 2021 edition) builds into Definition 10.5.2: they need the weak fibred product of strong bundles over ep-groupoids, which the text does not construct and the project does not have; the classes are taken for the generated equivalence relation (V10.51, V10.55, V10.57) (gap G.123; Findings, items 50, 53); closed since: V10.68–V10.71, with the weak fibred product of strong bundles built as a pull-back (Findings, item 18), and since 2026-11-12 also the converse half of the strong bundle Theorem 10.3.10: an isomorphism of $\mathcal{SEP}(F^{-1})$ is represented with both legs strong bundle equivalences, by a saturation argu-

ment in place of strong bundle versions of the claims (C1), (C2) of Appendix §10.6.2 (V10.71; Findings, item 19).
- pp. 505–506 (2017 pp. 365–366), the symbolic diagram (10.24) made precise with generalized vertical maps (left to the reader), and the equivariant germ q with its lift Q of (10.25): local expressions which depend on the representative and of which the text uses no property; γ is formalized (V10.54) (gap G.124; Findings, items 51, 52); closed since: V10.53, V10.54, (10.24) with the vertical maps $i(P)$, $i(P')$, and q, Q for each choice of data.
- pp. 507–508, Theorem 10.5.4 (2017: 10.9), the Fred part: it rests on the Fred part of Theorem 10.4.5 (G.121) (V10.59) (gap G.125; Findings, item 55).
- p. 508, Proposition 10.5.6 (new in the 2021 edition) itself: only its reduction to Proposition 10.4.6 (G.122) is verified (V10.61) (gap G.126; Findings, item 57); closed since: V10.61, with Proposition 10.4.6 (V10.72).
- p. 508, Theorem 10.5.7 (2017: Theorem 10.10, p. 367): generalized strong bundle isomorphisms with $|[D]| = |[D']|$ agree when both strong bundles have the local regularity property; it rests on Proposition 10.4.7 (G.122), and turning the local morphisms α_z into a common refinement needs the weak fibred product of strong bundles; its ep-groupoid analogue, Theorem 10.3.19, is G.120 (gap G.127; Findings, item 58); settled since: V10.67, where the example, which has the local regularity property, shows that the statement does not hold as we read it.
- p. 483, the closing sentences of Remark 10.2.3 (“smaller models”): no formal content.
- Witnesses: every hypothesis taken is satisfiable, and every definition has a non-identity witness (Findings, item 70).

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12. Chapter 11 (HWZ, Part II, §§11.1–11.6 (2017 §§11.1–11.6)): verified for the tangent functor and faces up to equivalence, sc-differential forms up to equivalence, push-forwards and pull-backs of branched ep⁺-subgroupoids with their tangents, boundaries and orientations, auxiliary norms and multisection functors under generalized strong bundle isomorphisms, proper coverings and proper strong bundle coverings up to equivalence with their localizations, and the canonical measures and boundary measures under generalized isomorphisms, with the readings recorded below (the intersection clause of Proposition 11.1.6 does not hold as we read it), except the boundary operations of Theorem 11.3.9 along equivalences; Theorems 11.4.4 and 11.4.5 are verified with the text's hypotheses for every $n \geq 1$

Chapter 11 (pp. 521–566 (2017 pp. 375–416); numbers and pages are those of the book (Springer 2021), with the 2017 text's in parentheses) studies how the notions of Chapters 8 and 9 behave under equivalences and generalized maps. §11.1 extends the tangent functor to $\mathcal{EP}(E^{-1})$ (Theorem 11.1.1 (2017: 11.1)), shows that generalized isomorphisms preserve the degeneracy index (Proposition 11.1.2 (2017: 11.1)), defines faces in the ep-groupoid sense (Definition 11.1.3 (2017: 11.1), Example 11.1.4 (2017: 11.1)) and shows that equivalences and generalized isomorphisms induce bijections of faces (Remark 11.1.5, Proposition 11.1.6, Remark 11.1.7, Proposition 11.1.8 (2017: Remark 11.1, Proposition 11.2, Remark 11.2, Proposition 11.3)). §11.2 carries sc-differential forms along equivalences and generalized maps (Theorem 11.2.1, Lemmas 11.2.2, 11.2.4–11.2.6, Definitions 11.2.3, 11.2.7, Propositions 11.2.8, 11.2.9 (2017: Theorem 11.2, Lemmas 11.1–11.4, Definitions 11.2, 11.3, Propositions 11.4, 11.5)). §11.3 defines the pull-back and the push-forward of functors $\Theta : X \to \mathbb{Q}^+$ along equivalences (Definition 11.3.1, Lemma 11.3.2 (2017: Definition 11.4, Lemma 11.5)) and along generalized isomorphisms (pp. 536–537), shows that they preserve branched ep$^+$-subgroupoids and their properties (Theorem 11.3.3 (2017: 11.3)), and continues with section functors of the bundles of oriented tangent sets, orientations, tangents and boundaries (Lemma 11.3.4 to Theorem 11.3.9 (2017: Lemma 11.6 to Theorem 11.7)). §11.4 transforms the canonical measures and boundary measures (Theorems 11.4.1, 11.4.4 (2017: 11.8, 11.9), Remark 11.4.3 (2017: 11.3), and the effective versions, Theorems 11.4.2 and 11.4.5, new in the 2021 edition). §11.5 carries auxiliary norms and sc$^+$-multisections along generalized strong bundle isomorphisms (Proposition 11.5.1, Lemma 11.5.2, Proposition 11.5.3 (2017: Proposition 11.6, Lemma 11.7, Proposition 11.7)). §11.6 studies proper covering functors up to equivalence: their morphism maps (Lemma 11.6.1, new in the 2021 edition), the fibres as translation groupoids (Lemmas 11.6.2, 11.6.3 (2017: 11.8, 11.9)), functors, equivalences and natural transformations between proper coverings (Definitions 11.6.4, 11.6.5 (2017: 11.6, 11.7)), the weak fibred product (Theorem 11.6.6 (2017: 11.10)), diagrams and their common refinements (Definition 11.6.7, Proposition 11.6.8 (2017: Definition 11.8, Proposition 11.8)), the localization $\mathcal{PC}(E^{-1})$ (pp. 561–563) and, in Remark 11.6.9 (2017: 11.4), proper strong bundle coverings.

What is verified. In §11.1 Theorem 11.1.1, Proposition 11.1.2, Remark 11.1.7 and Definition 11.1.3 are verified as stated, the last with faces recorded by their object sets (V11.1–V11.3, V11.7); Example 11.1.4 holds for $[0,\infty)^2 \oplus W$ with any sc-Banach space W (V11.4); the bijections of faces of Remark 11.1.5, quoted there from the literature, are proved (V11.5); Proposition 11.1.6 is verified for the bijection $[d]_*$, the faces as tame ep-groupoids and the generalized isomorphisms $[d]_F$, with the intersection clause in a corrected form, and Proposition 11.1.8 as stated (V11.6, V11.8). §11.2 is verified as stated, with F_* defined as the inverse of F^* (V11.9–V11.13). In §11.3 Definition 11.3.1, Lemma 11.3.2 in a stronger form, the push-forward and pull-back along generalized maps with their functorial properties, and Theorem 11.3.3, with (6) read as below, are verified (V11.14–V11.19); so are Lemma 11.3.4 with (11.8) and (11.9), Theorem 11.3.5, Definition 11.3.6 with (11.10), and Theorems 11.3.7 and 11.3.8, under Chapter 9's hypothesis that the tangent sums of Theorem 9.2.1 are independent, and Theorem 11.3.9 for a representative under a boundary-transfer hypothesis (V11.20–V11.26). In §11.4 Theorems 11.4.1 and 11.4.2 and Remark 11.4.3 are verified as stated, for the canonical measures of Theorems 9.5.4 and 9.5.5 built from Chapter 9's sheet integrals (V11.27, V11.28), and Theorems 11.4.4 and 11.4.5 for the boundary measures of Theorems 9.5.15 and 9.5.16 under their class hypothesis, for $n \geq 2$ (V11.29). In §11.5 Proposition 11.5.1, Lemma 11.5.2 and Proposition 11.5.3 are verified, the last two for sc$^+$-multisections (V11.30–V11.32; gaps G.132, G.133 closed on 2026-11-09). In §11.6 Lemmas 11.6.1–11.6.3, Definitions 11.6.4, 11.6.5, 11.6.7, Theorem 11.6.6 in all its claims and Proposition 11.6.8 are verified (V11.33–V11.41), and so are the localization $\mathcal{PC}(E^{-1})$ with I (V11.43) and Remark 11.6.9, the strong bundle version (V11.42). Several places of the text needed more than a reading. (a) *Theorem 11.1.1 (2017: 11.1).* The text checks that T respects refinements; that T preserves composition needs a comparison functor $T(A \times_Y B) \to TA \times_{TY} TB$, an equivalence by two out of three (V11.1). (b) *Proposition 11.1.6 (2017: 11.2).* The clause that two faces meet in a face of each does not hold as we read it: two faces of a face structured open subset of $[0, \infty)^3 \oplus W$ meet in two pieces of an edge; what holds, and is proved, is that the intersection is a union of faces of each (V11.6). (c) *Theorem 11.3.3 (2017: 11.3).* The paracompactness of $|\mathrm{supp}(\Theta)|$, which 2021 Definition 9.1.2 (1) asks for, is proved for push-forwards and pull-backs, though the text's proof does not mention it, and the sheets of a local branching structure are first restricted to the domain of the local sc-diffeomorphism (V11.17); the uniqueness part of Proposition 9.1.11 does not hold as we read it (Chapter 9), so (6) is read as "every dimensional decomposition is mapped to one" (V11.18). (d) *Lemma 11.3.2 (2017: 11.5).* The well-definedness of F_* needs only fullness, (2) only essential surjectivity (V11.15). (e) *Definition 11.6.4 (2017: 11.6).* For proper covering functors the fibre condition of (2) follows from the other two conditions (V11.36). (f) *Theorem 11.6.6 (2017: 11.10).* Step (C) is proved through the properness of H on objects instead of sequences (V11.38). (g) *Theorems 11.3.7 and 11.3.8 (2017: 11.5, 11.6).* Their proofs, left to the reader, need Chapter 9's tangent, orientation and boundary notions carried by sc-diffeomorphisms between different M-polyfolds; the independence of the tangent sums, assumed for Θ, is carried to $f_*\Theta$ and $f^*\Theta'$ (V11.23, V11.24). (h) *Theorem 11.3.9 (2017: 11.7).* Chapter 9's boundary orientation depends on a boundary operation taken as data, so the theorem is stated for a representative and boundary operations corresponding under the transfer maps (V11.25). (i) *Proposition 11.6.8 (2017: 11.8).* The displayed chain for the functors ε stops one step short; the last step uses the fourth assumption (V11.41). Reading points on the printed text are recorded in V11.2, V11.14, V11.17, V11.34 and V11.38. Not verified: that the boundary operations of Definition 9.3.8 satisfy the boundary-transfer hypothesis along equivalences, and so a form of Theorem 11.3.9 independent of the representative; Theorems 11.4.4 and 11.4.5 without the class hypothesis near the corners of the sheets, and for $n = 1$ (gap G.131 in part). Conditions (1) and (2) of Definition 8.3.5 in Proposition 11.5.1 and the local section structures of sc⁺-multisections are verified since 2026-11-09 (gaps G.132, G.133 closed). Addendum (2026-11-12): the Part II wave closed gaps G.128, G.129, G.134 and G.135 (faces as ep-groupoids with $[d]_F$ and the intersection clause settled by a counterexample and a corrected form, §11.2 in full, the localization $\mathcal{PC}(E^{-1})$, and Remark 11.6.9 with Theorem 11.6.6 and the diagram calculus for strong bundles), with no hypothesis added; §11.4 remains open. Addendum (2026-11-13): the second integration of the Part II wave formalizes §11.4 (gap G.131 in part): Theorems 11.4.1, 11.4.2 and Remark 11.4.3 as stated, Theorems 11.4.4 and 11.4.5 under the class hypothesis `hcov` inherited from Theorems 9.5.15 and 9.5.16 (Lean), which a closed hemisphere with nonempty boundary satisfies; the text's tacit step that $|f|$ preserves the σ-algebras L and $L(\partial S)$ is proved. Addendum (2026-11-13, wave K): Theorems 11.4.4 and 11.4.5 hold with the text's hypotheses only, for every $n \geq 1$, the class hypothesis being derived in Chapter 9 (gap G.109) and the $n = 1$ boundary measures transported along equivalences, spans and generalized isomorphisms (V11.29, restated; gap G.131 closed).

The hypotheses. Two hypotheses beyond the text's statements are taken, both in §11.3. Chapter 9's `Polyfold.EpGroupoid.HasIndependentTangentSums`, the independence part of Theorem 9.2.1, which Chapter 9's formalization assumes and proves on $\mathbb{R}^n$ (`Polyfold.EpGroupoid.`

`hasIndependentTangentSums_rn`), is assumed for the branched ep$^+$-subgroupoid being moved in Theorems 11.3.5, 11.3.7–11.3.9 and Definition 11.3.6, and proved for its push-forward and pull-back. `Polyfold.MPolyfold.IsBoundaryTransfer`, a pointwise condition on the given transfer maps and boundary operations, is assumed in Theorem 11.3.9; it holds along the identity (`Polyfold.MPolyfold.isBoundaryTransfer_id`). The branched statements of §11.3 take X and Y in one universe; generalized isomorphisms relate objects of one $\mathcal{EP}$ anyway. The tangent functor on $\mathcal{EP}(E^{-1})$ is defined on the ep-groupoids of the universe that Chapter 8's tangent functor uses ($\max(u, 1)$). Several statements need less than the text assumes: Lemma 11.3.2 (V11.15), Proposition 11.1.6, whose bijection of faces holds for every generalized isomorphism and whose other claims need only X face structured (V11.6, V11.8), Theorem 11.6.6 (1) (V11.38), and Theorem 11.2.1 (4), which needs the local regularity property of Y only (V11.9). The sc-differential forms of §11.2 and the localizations of §11.6 take no hypothesis beyond the text's; nor do Theorems 11.4.1, 11.4.2 and Remark 11.4.3 (the independence of the tangent sums holds for every Θ, `Polyfold.EpGroupoid.hasIndependentTangentSums`), while Theorems 11.4.4 and 11.4.5 inherit the class hypothesis `hcov` of Theorems 9.5.15 and 9.5.16 (Lean; nonempty boundary sheets in the class of integrated sheets), proved for the transported subgroupoid and satisfiable with $\partial S \neq \emptyset$ (`Polyfold.E8W.e8w_hcov` with `Polyfold.EP.hcovB_of_hcovS`) (V11.29); Remark 11.6.9 does not define the equivalences of proper strong bundle coverings nor their natural transformations, and both are read (V11.42).

The setting. Ep-groupoids, sc-smooth functors, orbit spaces, the degeneracy index d_X and the induced index $d_{|X|}$ on $|X|$ are those of Chapter 7 (`Polyfold.EpGroupoid`, `Polyfold.EpFunctor`, `Polyfold.EpGroupoid.degIndexOrbit`, `Polyfold.EpGroupoid.IsTame`); the tangent functor, strong bundles over ep-groupoids, auxiliary norms and proper covering functors are Chapter 8's (`Polyfold.EpFunctor.tangent`, `Polyfold.EpStrongBundle.IsAuxNorm`, `Polyfold.EpFunctor.IsProperCovering`); branched ep$^+$-subgroupoids, local branching structures and their properties are Chapter 9's (`Polyfold.EpGroupoid.IsFunctorial`, `Polyfold.EpGroupoid.IsBranchedEpSubgroupoid`, `Polyfold.EpGroupoid.LocalBranchingStructure`, `Polyfold.EpGroupoid.IsDimDecomposition`); equivalences, the weak fibred product, the localization $\mathcal{EP}(E^{-1})$ with its diagrams, refinements and generalized isomorphisms, strong bundle equivalences and generalized strong bundle isomorphisms are Chapter 10's (`Polyfold.EpFunctor.IsEquivalence`, `Polyfold.WeakFibredProduct.Admissible`, `Polyfold.EPLocalization.EPLoc`, `Polyfold.EPLocalization.Diagram.IsRefinement`, `Polyfold.EPLocalization.EPLoc.IsGeneralizedIso`, `Polyfold.SEP.equivalences`, `Polyfold.SBLocalization.SDiagram`, `Polyfold.SBLocalization.GenSBMap.IsGeneralizedIso`). Compositions in $\mathcal{EP}(E^{-1})$ are in Mathlib's order, $f \gg g$ for the text's $g \circ f$. $\mathbb{Q}^+$ is $\mathbb{Q}_{\geq 0}$; functors $X \to \mathbb{Q}$ are maps on objects constant along morphisms. Findings referred to below are those of `Exposition/Findings.md`, "Chapter 11".

V11.1 (Theorem 11.1.1 (2017: 11.1), the tangent functor on $\mathcal{EP}(E^{-1})$; **`Polyfold.EPLocalization.EPLoc.theorem_11_1_1`, `Polyfold.EPLocalization.EPLoc.tangentFunctor`, `Polyfold.EPLocalization.EPLoc.tangentFunctor_obj`, `Polyfold.EPLocalization.EPLoc.tangentFunctor_map_homMk`, `Polyfold.EPLocalization.EPLoc.tangentFunctor_map_`**

`localizationFunctor`, `Polyfold.EPLocalization.EPLoc.isGeneralizedIso_tangent`, `Polyfold.EPLocalization.Diagram.tangent`, `Polyfold.EPLocalization.Diagram.IsRefinement.tangent`, `Polyfold.EPLocalization.Diagram.Equiv.tangent`, `Polyfold.EPLocalization.Diagram.isRefinement_tangent_id`, `Polyfold.EPLocalization.Diagram.isRefinement_tangent_comp`, `Polyfold.EPLocalization.wfpLift`, `Polyfold.EPLocalization.projLeft_comp_wfpLift`, `Polyfold.EPLocalization.projRight_comp_wfpLift`, `Polyfold.EpFunctor.tangent_comp`, `Polyfold.EpFunctor.tangent_id`). The assignment $X \mapsto TX$, $[X \xleftarrow{F} A \xrightarrow{\Phi} Y] \mapsto [TX \xleftarrow{TF} TA \xrightarrow{T\Phi} TY]$ is a functor $T : \mathcal{EP}(E^{-1}) \to \mathcal{EP}(E^{-1})$: a refinement $H : d' \to d$ gives the refinement $TH : Td' \to Td$, equivalent diagrams have equivalent tangent diagrams, T preserves identities and composition, and $T(i(\Phi)) = i(T\Phi)$ for the localization functor $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$. T maps generalized isomorphisms to generalized isomorphisms. For functors $P : C \to X$, $Q : C \to Z$ and a natural transformation $\tau : \Phi \circ P \to G \circ Q$, $c \mapsto (P(c), \tau(c), Q(c))$ is a functor $C \to X \times_Y Z$ with π_X and π_Z composites P and Q. *Book:* Theorem 11.1.1, p. 521 (2017: Theorem 11.1, p. 375), with its proof, pp. 521–522. *Hidden hypotheses:* none. *Deviation (same):* the text checks refinements and leaves "the remaining argument" to the reader; that T preserves composition needs the comparison $T(A \times_Y B) \to TA \times_{TY} TB$, $v \mapsto (T\pi_A(v), T\sigma(v), T\pi_B(v))$, an equivalence by two out of three, through which $T(d \circ e)$ refines $Td \circ Te$; T is defined on the ep-groupoids of Chapter 8's tangent universe (Findings, item 1). *Proof:* TF is an equivalence (Theorem 10.1.7); a natural transformation $\sigma : G \to F \circ H$ gives $T\sigma : TG \to TF \circ TH$ (Proposition 8.1.14).

V11.2 (Proposition 11.1.2 (2017: 11.1), the degeneracy index under generalized isomorphisms; `Polyfold.EPLocalization.EPLoc.proposition_11_1_2`, `Polyfold.EPLocalization.EPLoc.degIndexOrbit_orbitMap`, `Polyfold.EPLocalization.Diagram.degIndexOrbit_orbitMap`, `Polyfold.EpFunctor.IsEquivalence.degIndexOrbit_orbitMap`, `Polyfold.EpGroupoid.degIndexOrbit`). If $f : X \to Y$ is a generalized isomorphism and $|f|(|x|) = |y|$, then $d_X(x_0) = d_Y(y_0)$ for all x_0 with $|x_0| = |x|$ and y_0 with $|y_0| = |y|$, and $d_{|X|}(|x|) = d_{|Y|}(|y|)$. *Book:* Proposition 11.1.2, p. 522 (2017: Proposition 11.1, p. 376), with its proof, pp. 522–523. *Hidden hypotheses:* none. *Deviation (same):* none; the proof cites "Proposition 10.3.10" for Theorem 10.3.10 (possible misprint) (Findings, item 2). *Proof:* a representative with both legs equivalences (Theorem 10.3.10); equivalences are local sc-diffeomorphisms on objects and preserve d.

V11.3 (Definition 11.1.3 (2017: 11.1), faces in the ep-groupoid sense; `Polyfold.EpGroupoid.IsOrbitFace`, `Polyfold.EpGroupoid.faceObj`, `Polyfold.EpGroupoid.faceMor`, `Polyfold.EpGroupoid.IsEpFace`, `Polyfold.EpGroupoid.orbitFacesThrough`, `Polyfold.EpGroupoid.IsEpFaceStructured`, `Polyfold.EpGroupoid.IsWeaklyFaceStructured`, `Polyfold.EpGroupoid.faceMor_eq`, `Polyfold.EpGroupoid.isSaturated_faceObj`, `Polyfold.EpGroupoid.isOrbitFace_iff_isFaceIn`, `Polyfold.EpGroupoid.orbitFacesThrough_eq`, `Polyfold.EpGroupoid.isEpFaceStructured_iff`, `Polyfold.EpGroupoid.faceEpGroupoid`). A face of $|X|$ is the closure in $|X|$ of the connected component, in $\{z \in |X| \mid d_{|X|}(z) = 1\}$, of a point of that set; for $\theta \subseteq |X|$, X_θ is the full subcategory on the objects x with $|x| \in \theta$, recorded by its object set $\pi^{-1}(\theta)$, a saturated set, and its morphisms are those with source in it; a face of X is X_θ for a face θ of $|X|$. X is face structured if it is tame and every $z \in |X|$ lies in exactly $d_{|X|}(z)$ faces of $|X|$ (counted in $\mathbb{N} \cup \{\infty\}$), and weakly face structured if it is tame and its object M-polyfold is face structured (Definition 2.5.10). The faces of $|X|$ are Chapter 2's faces for the index $d_{|X|}$ on the open set $|X|$. *Book:* Definition 11.1.3, p. 523 (2017: Definition 11.1, p. 377). *Hidden hypotheses:* none. *Deviation (restated):* a face of X is recorded by its object set; its sub-M-polyfold and ep-groupoid structure is Proposition 11.1.6 (`Polyfold.EpGroupoid.faceEpGroupoid`, V11.6) (Findings, item 3).

V11.4 (Example 11.1.4 (2017: 11.1); `Polyfold.EpGroupoid.Examples.example_11_1_4`, `Polyfold.EpGroupoid.Examples.swapEpGroupoid`, `Polyfold.EpGroupoid.Examples.isFaceStructured_quadrant`, `Polyfold.EpGroupoid.Examples.isHomeomorph_ _trivial`, `Polyfold.EpGroupoid.Examples.setOf_degIndexOrbit_eq_one_swap`, `Polyfold.EpGroupoid.Examples.isOrbitFace_swap_iff`, `Polyfold.EpGroupoid.Examples.not_isEpFaceStructured_swap`). For $C = [0, \infty)^2 \oplus W$ with any sc-Banach space W: C with only identities is a face structured ep-groupoid; $\mathbb{Z}/2 \ltimes C$, with -1 exchanging the two coordinates, is weakly face structured and not face structured, since $\{d_{|X|} = 1\}$ is connected, so that $|X|$ has exactly one face and $|0|$, with $d_{|X|}(|0|) = 2$, lies in one face only. *Book:* Example 11.1.4, p. 523 (2017: Example 11.1, p. 377). *Hidden hypotheses:* none. *Deviation (stronger):* the text has $W = 0$; the reason for "not face structured", not given in the text, is proved (Findings, item 4).

V11.5 (Remark 11.1.5 (2017: 11.1), the bijections of faces; `Polyfold.EPLocalization.EPLoc.remark_11_1_5`, `Polyfold.EpFunctor.saturation_image_faceObj`, `Polyfold.EpFunctor.preimage_faceObj`, `Polyfold.EpFunctor.IsEquivalence.bijOn_faces`, `Polyfold.EPLocalization.EPLoc.faceMap`, `Polyfold.EPLocalization.EPLoc.faceComap`, `Polyfold.EPLocalization.EPLoc.faceMap_homMk`, `Polyfold.EPLocalization.EPLoc.faceComap_homMk`, `Polyfold.EPLocalization.EPLoc.bijOn_faceMap`, `Polyfold.EPLocalization.EPLoc.isHomeomorph_orbitMap`, `Polyfold.TameBoundary.isDHomeoOn_univ`, `Polyfold.TameBoundary.isFaceIn_univ_image`, `Polyfold.TameBoundary.exists_isFaceIn_univ_preimage`, `Polyfold.TameBoundary.isFaceStructuredOpen_univ_iff`). For an equivalence $\Phi : X \to Y$, $\theta \mapsto |\Phi|(\theta)$ is a bijection of the faces of $|X|$ onto those of $|Y|$ with inverse $\theta' \mapsto |\Phi|^{-1}(\theta')$, and $\Phi_*(X_\theta) = \text{saturation}(\Phi(X_\theta)) = Y_{|\Phi|(\theta)}$, $\Phi^*(Y_{\theta'}) = \Phi^{-1}(Y_{\theta'}) = X_{|\Phi|^{-1}(\theta')}$. For a generalized isomorphism $[d] = [X \overset{\Phi}{\longleftarrow} A \overset{\Psi}{\longrightarrow} Y]$, $[d]_*$ and $[d]^*$ on faces are $\theta \mapsto |[d]|(\theta)$ and $\theta' \mapsto |[d]|^{-1}(\theta')$, equal to $\Psi_* \circ \Phi^*$ and $\Phi_* \circ \Psi^*$, and mutually inverse bijections. The general fact behind it: a homeomorphism preserving the index maps faces to faces and preserves face-structuredness. *Book:* Remark 11.1.5, p. 524 (2017: Remark 11.1, p. 378), quoting [56, Proposition 3.15]. *Hidden hypotheses:* none. *Deviation (same):* the quoted result is proved; the ordered face structures of the remark's first paragraph have no formal content (Findings, item 5).

V11.6 (Proposition 11.1.6 (2017: 11.2), faces up to generalized isomorphisms; `Polyfold.EPLocalization.EPLoc.proposition_11_1_6`, `Polyfold.EPLocalization.EPLoc.isOrbitFace_faceMap`, `Polyfold.EPLocalization.EPLoc.isOrbitFace_faceComap`, `Polyfold.EPLocalization.EPLoc.faceComap_faceMap`, `Polyfold.EPLocalization.EPLoc.faceMap_faceComap`, `Polyfold.EpGroupoid.IsEpFaceStructured.isWeaklyFaceStructured`, `Polyfold.EpGroupoid.IsEpFaceStructured.exists_nhds_eq_face`, `Polyfold.`

`EpGroupoid.faceEpGroupoid`, `Polyfold.EpGroupoid.isTame_faceEpGroupoid`, `Polyfold.EpGroupoid.degIndex_faceEpGroupoid`, `Polyfold.EpGroupoid.proposition_11_1_6_faces`, `Polyfold.EpFunctor.IsEquivalence.restrictSub`, `Polyfold.EPLocalization.faceEP`, `Polyfold.EPLocalization.EPLoc.faceIso`, `Polyfold.EPLocalization.EPLoc.isGeneralizedIso_faceIso`, `Polyfold.EPLocalization.EPLoc.orbitMap_faceIso`, `Polyfold.EPLocalization.EPLoc.proposition_11_1_6_faceIso`, `Polyfold.EpGroupoid.FacesMeetInFaces`, `Polyfold.FaceMeetExample.isEpFaceStructured_XV`, `Polyfold.FaceMeetExample.not_facesMeetInFaces`, `Polyfold.MPolyfold.mem_closure_inter_faces`, `Polyfold.EpGroupoid.IsEpFaceStructured.faceObj_inter_eq_sUnion`, `Polyfold.EpGroupoid.FacesMeetInUnionsOfFaces`, `Polyfold.EpGroupoid.IsEpFaceStructured.facesMeetInUnions`). A generalized isomorphism $[d] : X \to Y$ induces the bijection $[d]_*$ of the faces of $|X|$ onto those of $|Y|$, with inverse $[d]^*$; $[d]_*(X_\theta) = Y_{|[d]|(\theta)}$; and for every θ, $|[d]|$ restricts to a homeomorphism $\theta \to [d]_*\theta$. If X is face structured (Definition 11.1.3), its object M-polyfold is weakly face structured, each X_θ agrees near each of its points with one face of the object M-polyfold, and so X_θ is a closed saturated sub-M-polyfold with a tame induced structure, its morphisms form a sub-M-polyfold with a tame structure, and the full subcategory is a tame ep-groupoid with $d_X = d_{X_\theta} + 1$; $[d]$ restricts to a generalized isomorphism $[d]_F : X_\theta \to Y_{[d]_*\theta}$, independent of the representative, whose orbit map is the restriction of $|[d]|$. For distinct faces $\theta \neq \theta'$ with $X_\theta \cap X_{\theta'} \neq \emptyset$, the intersection is the union of the faces of X_θ it contains, and of the faces of $X_{\theta'}$ it contains. The printed clause, that it is one face of each, is stated (`Polyfold.EpGroupoid.FacesMeetInFaces`) and negated for the trivial ep-groupoid of $([0,\infty)^3 \oplus W) \setminus K$, $K = \{p_0 + p_1 \leq 1/4,\ 1 \leq p_2 \leq 2\}$, which is tame and face structured. *Book:* Proposition 11.1.6, p. 524 (2017: Proposition 11.2, p. 378), with its proof, pp. 524–525. *Hidden hypotheses:* none. *Deviation (narrower):* the clause that two faces meet in a face of each does not hold as we read it: the faces $\{p_0 = 0\}$ and $\{p_1 = 0\}$ of the example meet in the two pieces $p_2 < 1$ and $p_2 > 2$ of the edge, and a face of X_θ is the closure of one connected component of the index-1 set (`Polyfold.FaceMeetExample.not_facesMeetInFaces`); the union form is proved for every face structured X (`Polyfold.EpGroupoid.IsEpFaceStructured.facesMeetInUnions`, from `Polyfold.MPolyfold.mem_closure_inter_faces`: in a tame M-polyfold a point on two distinct faces is a limit of points of index 2 on both). Only X is assumed face structured; Y then is (Remark 11.1.7). The bijection needs no face structure (gap G.128 closed; Findings, items 6, 9).

V11.7 (Remark 11.1.7 (2017: 11.2), face structure under generalized isomorphisms; `Polyfold.EPLocalization.EPLoc.remark_11_1_7`, `Polyfold.EpFunctor.IsEquivalence.isTame_of_isTame_left`, `Polyfold.EpFunctor.IsEquivalence.isTame_iff`, `Polyfold.EpFunctor.IsEquivalence.isTame_of_isTame`). If $f : X \to Y$ is a generalized isomorphism, X is tame and face structured if and only if Y is. If $F : X \to Y$ is an equivalence, X is tame if and only if Y is. *Book:* Remark 11.1.7, p. 525 (2017: Remark 11.2, p. 379). *Hidden hypotheses:* none. *Deviation (same):* the forward transfer of tameness along an equivalence, needed for the remark and not in Chapter 10, is proved through the weak fibred product $X \times_Y Y$ (Findings, item 7).

V11.8 (Proposition 11.1.8 (2017: 11.3), generalized isomorphisms with the same orbit map; `Polyfold.EPLocalization.EPLoc.proposition_11_1_8`, `Polyfold.EPLocalization.EPLoc.`

`faceIso`, `Polyfold.EPLocalization.EPLoc.proposition_11_1_8_faceIso`). For generalized isomorphisms $f, g : X \to Y$ with $|f| = |g|$: $f_* = g_*$ and $f^* = g^*$ on faces, and, for face structured X, the generalized isomorphisms $f_F, g_F : X_\theta \to Y_{f_*\theta}$ of (11.1) have the same orbit map, the restriction of $|f| = |g|$. *Book:* Proposition 11.1.8, p. 526 (2017: Proposition 11.3, p. 380). *Hidden hypotheses:* none. *Deviation (same):* Y need not be assumed face structured; it is, by Remark 11.1.7 (Findings, item 8).

V11.9 (Theorem 11.2.1 (2017: Theorem 11.2), sc-differential forms under generalized maps; `Polyfold.EPLocalization.EPLoc.formsPullback`, `Polyfold.EPLocalization.EPLoc.formsPush`, `Polyfold.EPLocalization.EPLoc.formsPullback_homMk`, `Polyfold.EPLocalization.EPLoc.formsPush_homMk`, `Polyfold.EPLocalization.EPLoc.theorem_11_2_1_one`, `Polyfold.EPLocalization.EPLoc.theorem_11_2_1_two`, `Polyfold.EPLocalization.EPLoc.theorem_11_2_1_three`, `Polyfold.EPLocalization.EPLoc.formsPullback_comp`, `Polyfold.EPLocalization.EPLoc.theorem_11_2_1_four`). $d \circ d = 0$ on $\Omega^*_{ep,\infty}(X)$. A generalized map $f : X \to Y$ (a morphism of $\mathcal{EP}(E^{-1})$) has a pull-back f^* with $d_X \circ f^* = f^* \circ d_Y$, equal to $F_* \circ \Phi^*$ for every diagram $X \xleftarrow{F} A \xrightarrow{\Phi} Y$ representing f. A generalized isomorphism h has a push-forward h_*, equal to $\Phi_* \circ F^*$ for every representing diagram, with $h_* \circ d_X = d_Y \circ h_*$, $h_* \circ h^* = 1$ and $h^* \circ h_* = 1$. If X and Y have the local regularity property and $|h| = |k|$, then $h_* = k_*$ and $h^* = k^*$. *Book:* Theorem 11.2.1, p. 526 (2017: Theorem 11.2, p. 380). *Hidden hypotheses:* none. *Deviation (same):* h_* is defined as the pull-back by the inverse generalized map, which is the text's $G_* \circ F^*$ on every representative (Theorem 10.3.10); beyond the text, $(g \circ f)^* = f^* \circ g^*$, and (4) holds with the local regularity property of Y only (Findings, items 20, 22).

V11.10 (Lemma 11.2.2 and Definition 11.2.3 (2017: Lemma 11.1, Definition 11.2), the push-forward of forms along an equivalence; `Polyfold.EpFunctor.IsEquivalence.bijective_formPullback`, `Polyfold.EpFunctor.IsEquivalence.formPush`, `Polyfold.EpFunctor.IsEquivalence.definition_11_2_3`, `Polyfold.EpFunctor.IsEquivalence.exists_lift_tangent`, `Polyfold.EpFunctor.IsEquivalence.lemma_11_2_2`, `Polyfold.EpFunctor.exists_epForms_formPullback_eq_of_surjective`, `Polyfold.MPolyfold.isScForm_of_formPullback`, `Polyfold.MPolyfold.formDescend`, `Polyfold.MPolyfold.formPullback_formDescend`, `Polyfold.MPolyfold.formPullback_injective_of_isLocalScDiffeo`, `Polyfold.EP.epFormsInftyPush`, `Polyfold.EP.epFormsInftyD_comp_epFormsInftyPush`). For an equivalence $F : X \to Y$ the pull-back $F^* : \Omega^k_{ep}(Y) \to \Omega^k_{ep}(X)$ is a bijection, and F_* is its inverse. For $x \in X_1$, $\psi : F(x) \to y$ in $\mathbf{Y}_1$ and $l_j \in T_xX$, $(F_*\omega)_y(T\psi(TF(x)l_1), \dots, T\psi(TF(x)l_k)) = \omega_x(l_1, \dots, l_k)$; every $h_1, \dots, h_k \in T_yY$ is of this form, and the value does not depend on the choices (Lemma 11.2.2). On $\Omega^*_{ep,\infty}$, $F_*[\omega] = [(F^i)_*\omega]$ and $d_Y \circ F_* = F_* \circ d_X$ ((11.5)). *Book:* Lemma 11.2.2, p. 527 (2017: Lemma 11.1, p. 380); Definition 11.2.3, p. 528 (2017: Definition 11.2, p. 382). *Hidden hypotheses:* none. *Deviation (restated):* the text defines $F_*\omega$ by the formula and proves it well defined and sc-smooth ((11.4)); here F_* is the inverse of F^*, whose bijectivity is proved by descent along $\pi_Y : X \times_Y Y \to Y$, the weak fibred product of F and 1_Y, whose objects are the triples (x, ψ, y) of the definition; the formula and Lemma 11.2.2 are theorems, and sc-smoothness is the text's argument (11.4). Reading point: before Definition 11.2.3 the text writes $F_*\omega \in \Omega_{ep,\infty}(X)$;

we read $\Omega_{ep,\infty}(Y)$ (Findings, items 21, 22).

V11.11 (Lemmas 11.2.4–11.2.6 (2017: Lemmas 11.2–11.4), composition, natural equivalence and refinement; `Polyfold.EP.lemma_11_2_4_two`, `Polyfold.EP.epFormsInftyPush_comp_epFormsInftyPullback`, `Polyfold.EP.epFormsInftyPullback_comp_epFormsInftyPush`, `Polyfold.EP.lemma_11_2_5_push`, `Polyfold.EP.lemma_11_2_6`, `Polyfold.EP.lemma_11_2_6_push`, `Polyfold.EP.epFormsInftyPush_eq_of_comp_eq_id`). Push-forwards and pull-backs of forms along equivalences compose and are mutually inverse; naturally equivalent equivalences have the same push-forwards, and naturally equivalent sc-smooth functors the same pull-backs; refinements of diagrams do not change $F_* \circ G^*$ and $G_* \circ F^*$. *Book:* Lemma 11.2.4, p. 528 (2017: Lemma 11.2, p. 382); Lemma 11.2.5, p. 529 (2017: Lemma 11.3, p. 383); Lemma 11.2.6, p. 531 (2017: Lemma 11.4, p. 385). *Hidden hypotheses:* none. *Deviation (same):* the proofs are formal from $F_* = (F^*)^{-1}$. Reading point: the proof of Lemma 11.2.6 cites Lemmas 11.2.2 and 11.2.3 for what it uses; we read Lemmas 11.2.4 and 11.2.5 (Findings, item 21).

V11.12 (Definition 11.2.7 (2017: Definition 11.3), pull-back and push-forward of forms by generalized maps; `Polyfold.EPLocalization.Diagram.formsPullback`, `Polyfold.EPLocalization.Diagram.formsPush`, `Polyfold.EPLocalization.Diagram.formsPullback_eq_of_equiv`, `Polyfold.EPLocalization.Diagram.formsPush_eq_of_equiv`, `Polyfold.EPLocalization.EPLoc.formsPullback`, `Polyfold.EPLocalization.EPLoc.formsPush`). For $[d] = [X \xleftarrow{F} A \xrightarrow{G} Y]$, $[d]^*\omega' = F_* \circ G^*\omega'$, and for a generalized isomorphism $[d]_*\omega = G_* \circ F^*\omega$; both are independent of the representing diagram. *Book:* Definition 11.2.7, p. 532 (2017: Definition 11.3, p. 386). *Hidden hypotheses:* none. *Deviation (same):* none.

V11.13 (Propositions 11.2.8 and 11.2.9 (2017: Propositions 11.4, 11.5), forms under maps with the same orbit map; `Polyfold.EP.proposition_11_2_8`, `Polyfold.EPLocalization.EPLoc.proposition_11_2_9`, `Polyfold.EpFunctor.HasLocalNatIso`, `Polyfold.EpFunctor.HasLocalNatIso.formPullback_eq`, `Polyfold.EpFunctor.hasLocalNatIso_of_orbitMap_eq`, `Polyfold.EP.epFormsInftyPullback_eq_of_orbitMap_eq`, `Polyfold.EPLocalization.EPLoc.formsPullback_eq_of_orbitMap_eq`). If X and Y have the local regularity property, equivalences $F, G : X \to Y$ with $|F| = |G|$ have $F_* = G_*$ and $F^* = G^*$ on $\Omega^*_{ep,\infty}$, and generalized isomorphisms f, g with $|f| = |g|$ have $f_* = g_*$ and $f^* = g^*$. *Book:* Proposition 11.2.8, p. 532 (2017: Proposition 11.4, p. 386); Proposition 11.2.9, p. 534 (2017: Proposition 11.5, p. 388). *Hidden hypotheses:* none. *Deviation (same):* the text takes from Proposition 10.1.10 a morphism $\psi : F(x) \to G(x)$ with $T\psi \circ TF(x) = TG(x)$; the proof uses the local sc-smooth families $\alpha_z : F(z) \to G(z)$ of Proposition 10.1.10 (1), restricted to the index lifts X^i, so that no local regularity of the lifts is needed; the statements also hold with the local regularity property of one of X, Y only. The hypothesis is satisfiable (`Polyfold.EpGroupoid.Examples.isLR_trivialZ3`, `Polyfold.EP.lreg_pointEP`) (Findings, item 22).

V11.14 (Definition 11.3.1 (2017: 11.4), pull-back and push-forward of functors along a functor; `Polyfold.EpFunctor.pullbackTheta`, `Polyfold.EpFunctor.pushforwardTheta`, `Polyfold.EpFunctor.pullbackTheta_apply`, `Polyfold.EpFunctor.pushforwardTheta_of_`

`not_exists`, `Polyfold.EpFunctor.exists_isomorphic_of_pushforwardTheta_ne_zero`, `Polyfold.EpFunctor.pushforwardTheta_nonneg`). For an sc-smooth functor $F : X \to Y$ and maps $\Theta : X \to \mathbb{Q}$, $\Theta' : Y \to \mathbb{Q}$ on objects, $F^*\Theta' = \Theta' \circ F$ and $(F_*\Theta)(y) = \Theta(x)$ for a chosen x with a morphism $F(x) \to y$, and 0 if there is none; F_* preserves nonnegativity. *Book:* Definition 11.3.1, p. 535 (2017: Definition 11.4, p. 389). *Hidden hypotheses:* none. *Deviation (restated):* both are defined for every sc-smooth functor and every map, the choice made once; for an equivalence a morphism $F(x) \to y$ always exists and the value does not depend on the choice (V11.15). The push-forward is printed as a map $X' \to \mathbb{Q}^+$, also in 2017; we read $Y \to \mathbb{Q}^+$ (possible misprint) (Findings, item 30).

V11.15 (Lemma 11.3.2 (2017: 11.5), in a stronger form; `Polyfold.EpFunctor.pushforwardTheta_eq`, `Polyfold.EpFunctor.pushforwardTheta_obj`, `Polyfold.EpFunctor.pushforwardTheta_eq_of_isomorphic`, `Polyfold.EpFunctor.isFunctorial_pushforwardTheta`, `Polyfold.EpFunctor.isFunctorial_pullbackTheta`, `Polyfold.EpFunctor.pullbackTheta_pushforwardTheta`, `Polyfold.EpFunctor.pushforwardTheta_pullbackTheta`, `Polyfold.EpFunctor.pushforwardTheta_eq_of_naturallyEquivalent`, `Polyfold.EpFunctor.pullbackTheta_eq_of_naturallyEquivalent`, `Polyfold.EpFunctor.pullbackTheta_comp`, `Polyfold.EpFunctor.pushforwardTheta_comp`, `Polyfold.EpFunctor.pushforwardTheta_id`, `Polyfold.EpGroupoid.IsFunctorial.eq_of_isomorphic`, `Polyfold.EpFunctor.isomorphic_obj`, `Polyfold.EpFunctor.isomorphic_of_full`). (1) If F is full and Θ is a functor, then $(F_*\Theta)(y) = \Theta(x)$ for every x with a morphism $F(x) \to y$, $F_*\Theta$ is a functor and $F^*F_*\Theta = \Theta$; (2) if F is essentially surjective and Θ' is a functor, $F_*F^*\Theta' = \Theta'$; (3) naturally equivalent full functors $F \simeq G$ have $F_*\Theta = G_*\Theta$, and naturally equivalent functors have $F^*\Theta' = G^*\Theta'$. Moreover F^* of a functor is a functor, $(G \circ F)^* = F^* \circ G^*$, $(G \circ F)_* = G_* \circ F_*$ for full $F, G, G \circ F$, and $1_*\Theta = \Theta$. *Book:* Lemma 11.3.2, p. 536 (2017: Lemma 11.5, p. 389). *Hidden hypotheses:* none. *Deviation (stronger):* the text assumes equivalences throughout; (1) needs only fullness, (2) only essential surjectivity, (3) fullness for push-forwards and nothing for pull-backs; the composition rules are added (Findings, item 31). *Proof:* for $\psi : F(x) \to y$, $\psi' : F(x') \to y$, fullness lifts $\psi^{-1} \circ \psi'$ to $\sigma : x' \to x$, as in the text.

V11.16 (the push-forward and pull-back along generalized maps, pp. 536–537; `Polyfold.EPLocalization.Diagram.pushTheta`, `Polyfold.EPLocalization.Diagram.pullTheta`, `Polyfold.EPLocalization.Diagram.IsRefinement.isEquivalence_Φ_iff`, `Polyfold.EPLocalization.Diagram.pushTheta_eq_of_isRefinement`, `Polyfold.EPLocalization.Diagram.pullTheta_eq_of_isRefinement`, `Polyfold.EPLocalization.Diagram.pushTheta_eq_of_equiv`, `Polyfold.EPLocalization.Diagram.pullTheta_eq_of_equiv`, `Polyfold.EPLocalization.EPLoc.rep`, `Polyfold.EPLocalization.EPLoc.homMk_rep`, `Polyfold.EPLocalization.EPLoc.equiv_rep_homMk`, `Polyfold.EPLocalization.EPLoc.isEquivalence_rep_Φ`, `Polyfold.EPLocalization.EPLoc.pushTheta`, `Polyfold.EPLocalization.EPLoc.pullTheta`, `Polyfold.EPLocalization.EPLoc.pushTheta_homMk`, `Polyfold.EPLocalization.EPLoc.pullTheta_homMk`, `Polyfold.EPLocalization.EPLoc.isFunctorial_pushTheta`, `Polyfold.EPLocalization.EPLoc.isFunctorial_pullTheta`, `Polyfold.EPLocalization.EPLoc.pushTheta_comp`, `Polyfold.EPLocalization.EPLoc.pullTheta_`

`comp`, `Polyfold.EPLocalization.EPLoc.pushTheta_inv`, `Polyfold.EPLocalization.EPLoc.pullTheta_pushTheta`, `Polyfold.EPLocalization.EPLoc.pushTheta_pullTheta`, `Polyfold.WeakFibredProduct.pushforwardTheta_projRight_pullbackTheta`, `Polyfold.WeakFibredProduct.pushforwardTheta_projLeft_pullbackTheta`**).** For a diagram $d : X \xleftarrow{F} A \xrightarrow{\Phi} Y$, $d_*\Theta = \Phi_* F^*\Theta$ and $d^*\Theta' = F_*\Phi^*\Theta'$; a refinement does not change $d^*\Theta'$, nor $d_*\Theta$ when Φ is an equivalence, so equivalent diagrams have the same pull-backs, and the same push-forwards when Φ is an equivalence. Hence f^* is well defined for every generalized map and f_* for every generalized isomorphism, computed on any representative (with Φ an equivalence for f_*); both map functors to functors; $(f \gg g)^* = f^* \circ g^*$ for all generalized maps, $(f \gg g)_* = g_* \circ f_*$ and $(f^{-1})_* = f^*$ for generalized isomorphisms, and $f^*f_*\Theta = \Theta$, $f_*f^*\Theta' = \Theta'$. In a weak fibred product $X \times_Y Z$, $(\pi_Z)_*(\pi_X)^*\Theta = G^*F_*\Theta$ for an equivalence F, and $(\pi_X)_*(\pi_Z)^*\Theta = F^*G_*\Theta$ for an equivalence G. *Book:* §11.3, pp. 536–537 (2017 pp. 389–390), not a numbered statement. *Hidden hypotheses:* none. *Deviation (stronger):* the text defines f^* for generalized isomorphisms; it is well defined, and contravariantly functorial, for all generalized maps. The composition rules, "easily verified" in the text, go through the base change in the weak fibred product that represents the composite (Theorem 10.3.8); nothing uses Theorem 10.3.19, which does not hold as we read it (Chapter 10, V10.65) (Findings, item 32).

V11.17 (Theorem 11.3.3 (2017: 11.3), the proof's step for an equivalence; `Polyfold.EpFunctor.isBranchedEpSubgroupoid_pullbackTheta`, `Polyfold.EpFunctor.isBranchedEpSubgroupoid_pushforwardTheta`, `Polyfold.EpFunctor.exists_localBranchingStructure_pushforwardTheta`, `Polyfold.EpFunctor.exists_mem_supp_of_mem_supp_pushforwardTheta`, `Polyfold.EpGroupoid.LocalBranchingStructure.mapDiffeo`, `Polyfold.EpGroupoid.LocalBranchingStructure.mapU`, `Polyfold.EpGroupoid.LocalBranchingStructure.mem_image_iff_of_mem_mapU`, `Polyfold.EpGroupoid.LocalBranchingStructure.isOfDim_mapDiffeo`, `Polyfold.EpFunctor.orbitSupp_pullbackTheta`, `Polyfold.EpFunctor.orbitSupp_pushforwardTheta`, `Polyfold.EpFunctor.isClosedBranched_pushforwardTheta`, `Polyfold.EpFunctor.isClosedBranched_pullbackTheta`, `Polyfold.EpFunctor.isCompactBranched_pushforwardTheta`, `Polyfold.EpFunctor.isCompactBranched_pullbackTheta`, `Polyfold.EpFunctor.isOrbifoldType_pushforwardTheta`, `Polyfold.EpFunctor.isOrbifoldType_pullbackTheta`, `Polyfold.EpFunctor.isManifoldType_pushforwardTheta`, `Polyfold.EpFunctor.isManifoldType_pullbackTheta`, `Polyfold.EpFunctor.isNDimensional_pullbackTheta`, `Polyfold.EpFunctor.isNDimensional_pushforwardTheta`, `Polyfold.EpFunctor.exists_pushforwardTheta_eq_apply`, `Polyfold.EpFunctor.isDimDecomposition_pushforwardTheta`, `Polyfold.EpFunctor.isDimDecomposition_pullbackTheta`, `Polyfold.MPolyfold.HasDimAt.of_image`, `Polyfold.MPolyfold.hasDimAt_image_iff_of_isScDiffeoOn`, `Polyfold.MPolyfold.IsSubmanifoldOfDim.image_isScDiffeoOn`, `Polyfold.EpFunctor.TransportChart`**).** For an equivalence $F : X \to Y$: $F^*\Theta'$ and $F_*\Theta$ of branched ep⁺-subgroupoids are branched ep⁺-subgroupoids; $|\mathrm{supp}(F^*\Theta')| = |F|^{-1}(|\mathrm{supp}(\Theta')|)$ and, for full F, $|\mathrm{supp}(F_*\Theta)| = |F|(|\mathrm{supp}(\Theta)|)$; F_* and F^* preserve closedness and compactness (Definition 9.1.6), orbifold-type and manifold-type (Definition 9.1.8), n-dimensionality and dimensional decompositions $\Theta = \sum_k \Theta_k$ into k-dimensional branched ep⁺-subgroupoids. A local branching structure is moved by an sc-diffeomorphism $f : W \to W'$ between open subsets of the object spaces of two ep-groupoids, with $\Theta'(f(u)) = \Theta(u)$ on W, to the sheets $f(M_i \cap W)$ with the same weights; the dimension of a submanifold is invariant under such an f. *Book:* Theorem 11.3.3, proof, pp. 537–538 (2017: Theorem 11.3, pp. 390–391). *Hidden hypotheses:* none. *Deviation (explicit):* the paracompactness of $|\mathrm{supp}|$ required by 2021 Definition 9.1.2 (1) is proved from the formulas for $|\mathrm{supp}|$ and $|F|$ a homeomorphism, which the proof does not mention; the sheets are restricted to the domain of $\widehat{\psi} \circ F$, Chapter 10's transport chart; manifold-type is shown for two morphisms $y \to y'$ directly. In the manifold-type step, $\psi \circ \varphi' \circ \psi^{-1} : F(x) \to F(x)$ for $\psi : F(x) \to y$, $\varphi' : y \to y$ is read $\psi^{-1} \circ \varphi' \circ \psi$ (possible misprint) (Findings, item 33).

V11.18 (Theorem 11.3.3 (2017: 11.3), branched ep$^+$-subgroupoids under generalized isomorphisms; **`Polyfold.EPLocalization.EPLoc.theorem_11_3_3_one`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_3_two`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_3_three`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_3_four`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_3_five`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_3_six`, `Polyfold.EPLocalization.EPLoc.isNDimensional_pushTheta`, `Polyfold.EPLocalization.EPLoc.isNDimensional_pullTheta`, `Polyfold.EPLocalization.EPLoc.orbitSupp_pushTheta`, `Polyfold.EpGroupoid.not_proposition_9_1_11_rn`).** For a generalized isomorphism $f : X \to Y$: (1) $f_*\Theta$ of a branched ep$^+$-subgroupoid Θ of X is one of Y; (2) $f^*\Theta'$ of one of Y is one of X; for functors Θ, Θ', f_* and f^* preserve (3) closedness, (4) compactness, (5) orbifold-type and manifold-type, and (6) n-dimensionality and dimensional decompositions: if $\Theta = \sum_k \Theta_k$ with Θ_k k-dimensional branched ep$^+$-subgroupoids, then $f_*\Theta = \sum_k f_*\Theta_k$ is one for $f_*\Theta$, and likewise for f^*. $|\mathrm{supp}(f_*\Theta)| = |f|(|\mathrm{supp}(\Theta)|)$. *Book:* Theorem 11.3.3, p. 537 (2017: Theorem 11.3, p. 390), with its proof, pp. 537–538. *Hidden hypotheses:* none. *Deviation (restated):* (6) refers to "the dimensional decomposition" of Proposition 9.1.11, whose uniqueness does not hold as we read it (Chapter 9, `Polyfold.EpGroupoid.not_proposition_9_1_11_rn`); (6) is read as "every dimensional decomposition is mapped to one", and n-dimensionality is added; X and Y are in one universe (Findings, item 34). *Proof:* $f_* = \Phi_* F^*$ and $f^* = F_* \Phi^*$ on a representative with both legs equivalences, and V11.17.

V11.19 (witnesses for §11.3; **`Polyfold.EpGroupoid.Examples.reflectionTheta`, `Polyfold.EpGroupoid.Examples.localBranchingStructure_reflectionTheta`, `Polyfold.EpGroupoid.Examples.isBranchedEpSubgroupoid_reflectionTheta`, `Polyfold.EpGroupoid.Examples.isBranchedEpSubgroupoid_pullbackTheta_reflectionCover`, `Polyfold.EpGroupoid.Examples.pullbackTheta_reflectionCover_two_points`, `Polyfold.EpGroupoid.Examples.pushforwardTheta_pullbackTheta_reflectionCover`, `Polyfold.EpGroupoid.Examples.reflectionEP`, `Polyfold.EpGroupoid.Examples.reflectionCoverEP`, `Polyfold.EpGroupoid.Examples.reflectionCoverDiagram`, `Polyfold.EpGroupoid.Examples.isGeneralizedIso_reflectionCoverDiagram`, `Polyfold.EpGroupoid.Examples.pushTheta_reflectionCoverDiagram`).** On $\mathbb{Z}/2 \ltimes \mathbb{R}$ (-1 acting by $x \mapsto -x$), $\Theta_0 = 1_{\{0\}}$ is a branched ep$^+$-subgroupoid, with the local branching structure $U = \mathbb{R}$, $M = \{0\}$, weight 1, at a point with isotropy $\mathbb{Z}/2$. For the equivalence $F : X_{\mathcal{U}} \to \mathbb{Z}/2 \ltimes \mathbb{R}$ of the covering $\{x > -1\}$, $\{x < 1\}$ (Proposition 10.1.3), $F^*\Theta_0$ is a branched ep$^+$-subgroupoid, equal to 1 at two different objects, and

$F_* F^* \Theta_0 = \Theta_0$; for the generalized isomorphism $f = [\mathbb{Z}/2 \ltimes \mathbb{R} \xleftarrow{F} X_{\mathcal{U}} \xrightarrow{1} X_{\mathcal{U}}]$, $f_* \Theta_0 = F^* \Theta_0$, a branched ep⁺-subgroupoid, and $f^* f_* \Theta_0 = \Theta_0$. *Book:* not in the book. *Hidden hypotheses:* none. *Deviation (not in the book):* witnesses that the definitions and Theorem 11.3.3 apply to a non-trivial action groupoid and a functor that is not injective on objects (Findings, item 35).

V11.20 (Lemma 11.3.4 (2017: 11.6) with (11.8) and (11.9), section functors of $\widehat{Gr}$ along equivalences and generalized isomorphisms; `Polyfold.EpGroupoid.IsGrHatSection`, `Polyfold.EpGroupoid.GrHatSections`, `Polyfold.EpGroupoid.IsOrientation.isGrHatSection`, `Polyfold.MPolyfold.fibreInv`, `Polyfold.EpFunctor.pushSection`, `Polyfold.EpFunctor.pullSection`, `Polyfold.EpFunctor.pushSection_eq`, `Polyfold.EpFunctor.isGrHatSection_pushSection`, `Polyfold.EpFunctor.isGrHatSection_pullSection`, `Polyfold.EpFunctor.pullSection_pushSection`, `Polyfold.EpFunctor.pushSection_pullSection`, `Polyfold.EpFunctor.pushSection_eq_of_natTrans`, `Polyfold.EpFunctor.pullSection_eq_of_natTrans`, `Polyfold.EpFunctor.pushSection_comp`, `Polyfold.EpFunctor.pullSection_comp`, `Polyfold.WeakFibredProduct.pushSection_projRight_pullSection`, `Polyfold.WeakFibredProduct.pushSection_projLeft_pullSection`, `Polyfold.EPLocalization.Diagram.pushSection`, `Polyfold.EPLocalization.Diagram.pullSection`, `Polyfold.EPLocalization.Diagram.pushSection_eq_of_equiv`, `Polyfold.EPLocalization.Diagram.pullSection_eq_of_equiv`, `Polyfold.EPLocalization.EPLoc.pushSection`, `Polyfold.EPLocalization.EPLoc.pullSection`, `Polyfold.EPLocalization.EPLoc.pushSection_homMk`, `Polyfold.EPLocalization.EPLoc.pullSection_homMk`, `Polyfold.EPLocalization.EPLoc.pushSection_comp`, `Polyfold.EPLocalization.EPLoc.pullSection_comp`, `Polyfold.EPLocalization.EPLoc.pushSection_inv`, `Polyfold.EPLocalization.EPLoc.lemma_11_3_4`). $\Gamma(\hat{\pi}_X)$ is the set of section functors of $\hat{\pi} : \widehat{Gr}(X) \to X_\infty$, with no continuity required: maps h with $h(x)$ a finite formal sum of oriented tangent sets at x for $x \in X_\infty$, natural along morphisms, recorded as functions on all objects vanishing off X_∞; every orientation is one. For an equivalence $F : X \to Y$, $(F_* h)(y) = T\psi \circ TF(x)\, h(x)$ for a chosen x and $\psi : F(x) \to y$ ((11.8)), the same for every such choice, and $(F^* h')(x) = TF(x)^{-1} h'(F(x))$ ((11.9)). F_*, F^* map section functors to section functors and are mutually inverse; naturally equivalent equivalences give the same F_* and F^*; $(G \circ F)_* = G_* \circ F_*$, $(G \circ F)^* = F^* \circ G^*$; in a weak fibred product of two equivalences, $(\pi_Z)_* (\pi_X)^* h = G^* F_* h$, and symmetrically. For diagrams $d_* = \Phi_* F^*$, $d^* = F_* \Phi^*$, unchanged on equivalent diagrams. Lemma 11.3.4: for a generalized isomorphism f, $f_* : \Gamma(\hat{\pi}_X) \to \Gamma(\hat{\pi}_Y)$ and $f^* : \Gamma(\hat{\pi}_Y) \to \Gamma(\hat{\pi}_X)$ are well defined, $f_* = \Phi_* F^*$ and $f^* = F_* \Phi^*$ for every representative, $(f^{-1})_* = f^*$, and $(g \circ f)_* = g_* \circ f_*$, $(g \circ f)^* = f^* \circ g^*$ for generalized isomorphisms f, g. *Book:* (11.8), p. 538, (11.9) and Lemma 11.3.4, p. 539 (2017: Lemma 11.6, p. 392). *Hidden hypotheses:* none. *Deviation (restated):* section functors are defined on all objects, 0 off X_∞; the formal sums are moved by the image of carriers and frames; the composition laws are stated, as in the text, for generalized isomorphisms (Findings, item 40).

V11.21 (Theorem 11.3.5 (2017: 11.4), orientations under generalized isomorphisms; `Polyfold.EpFunctor.TransportChart.tangentMor_ _tangent_obj`, `Polyfold.MPolyfold.OrTangent.map`, `Polyfold.MPolyfold.IsOrientedAt.map_tangentMapOn`, `Polyfold.MPolyfold.`

`SubmanifoldOrientation.mapDiffeo`, `Polyfold.MPolyfold.SubmanifoldOrientation.mapDiffeo_tangent`, `Polyfold.EpGroupoid.OrientedBranchingStructure.mapDiffeo`, `Polyfold.EpGroupoid.OrientedBranchingStructure.orientedSum_mapDiffeo`, `Polyfold.EpFunctor.isOrientation_pushSection`, `Polyfold.EpFunctor.isOrientation_pullSection`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_5_push`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_5_pull`). For a generalized isomorphism $f : X \to Y$, if $\widehat{T}_\Theta$ is an orientation of a branched ep⁺-subgroupoid Θ of X, then $f_*\widehat{T}_\Theta$ is an orientation of $f_*\Theta$; if $\widehat{T}_{\Theta'}$ is an orientation of Θ' on Y, then $f^*\widehat{T}_{\Theta'}$ is one of $f^*\Theta'$. Both assume the independence of the tangent sums of Theorem 9.2.1 for Θ, resp. Θ'. An orientation of a submanifold and an oriented local branching structure are moved by an sc-diffeomorphism between open subsets of two M-polyfolds, the frames by the tangent map; $T\psi \circ TF = T(\widehat{\psi} \circ F)$ on T_uX. *Book:* Theorem 11.3.5, p. 539 (2017: Theorem 11.4, p. 393), with its proof, pp. 539–540. *Hidden hypotheses:* none beyond Chapter 9's: the independence of the tangent sums in Theorem 9.2.1 (`Polyfold.EpGroupoid.HasIndependentTangentSums`), a hypothesis of Chapter 9's formalization that holds on $\mathbb{R}^n$, is taken for Θ, resp. Θ'. *Deviation (explicit):* the hypothesis is assumed for Θ only and carried to $f_*\Theta$, $f^*\Theta'$ (V11.23); the oriented structure at x is moved by $\widehat{\psi} \circ F$ as in the text (Findings, item 41).

V11.22 (Definition 11.3.6 (2017: 11.5) and (11.10), oriented branched ep⁺-subgroupoids under generalized isomorphisms; `Polyfold.EPLocalization.OrientedBranched`, `Polyfold.EPLocalization.OrientedBranched.push`, `Polyfold.EPLocalization.OrientedBranched.pull`, `Polyfold.EPLocalization.OrientedBranched.push_comp`, `Polyfold.EPLocalization.OrientedBranched.pull_comp`, `Polyfold.EPLocalization.OrientedBranched.pull_push`, `Polyfold.EPLocalization.EPLoc.pushTheta_pullTheta`, `Polyfold.EPLocalization.EPLoc.pushSection_pullSection`). An oriented branched ep⁺-subgroupoid $\widehat{\Theta} = (\Theta, \widehat{T}_\Theta)$ on an object of $\mathcal{EP}$, with the independence of its tangent sums; for a generalized isomorphism f, $f_*\widehat{\Theta} = (f_*\Theta, f_*\widehat{T}_\Theta)$ and $f^*\widehat{\Theta}' = (f^*\Theta', f^*\widehat{T}_{\Theta'})$ are oriented branched ep⁺-subgroupoids; $(g \circ f)_* = g_* \circ f_*$ and $(g \circ f)^* = f^* \circ g^*$ ((11.10)); of the two inverse laws of (11.10), $f^*f_*\widehat{\Theta} = \widehat{\Theta}$ when $\widehat{T}_\Theta$ vanishes off X_∞, and $f_*f^*\widehat{\Theta}' = \widehat{\Theta}'$ for its two components, $f_*f^*\Theta' = \Theta'$ and $f_*f^*h' = h'$ for section functors h'. *Book:* Definition 11.3.6 and (11.10), p. 541 (2017: Definition 11.5, p. 394). *Hidden hypotheses:* the independence of the tangent sums, as in V11.21. *Deviation (restated):* the independence is a field of the structure; the second inverse law is proved componentwise (Findings, item 42).

V11.23 (Theorem 11.3.7 (2017: 11.5), tangents of branched ep⁺-subgroupoids under generalized isomorphisms; `Polyfold.MPolyfold.submanifoldTangent_image_of_isScDiffeoOn`, `Polyfold.MPolyfold.chartVec_tangentMapOn_of_isScDiffeoOn`, `Polyfold.EpGroupoid.LocalBranchingStructure.tangentSum_mapDiffeo`, `Polyfold.EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isScDiffeoOn`, `Polyfold.EpGroupoid.tangentTheta_tangentMapOn`, `Polyfold.EpFunctor.hasIndependentTangentSums_pushforwardTheta`, `Polyfold.EpFunctor.hasIndependentTangentSums_pullbackTheta`, `Polyfold.EpFunctor.tangentValue_pushforwardTheta`, `Polyfold.EpFunctor.tangentValue_pullbackTheta`, `Polyfold.EpFunctor.tangentTheta_pushforwardTheta`, `Polyfold.EpFunctor.tangentTheta_pullbackTheta`, `Polyfold.EPLocalization.`

`EPLoc.hasIndependentTangentSums_pushTheta`, `Polyfold.EPLocalization.EPLoc.hasIndependentTangentSums_pullTheta`, `Polyfold.EPLocalization.EPLoc.tangentFunctor_map_eq`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_7_push`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_7_pull`). For a generalized isomorphism $f : X \to Y$ with tangent Tf (Theorem 11.1.1) and branched ep$^+$-subgroupoids Θ of X and Θ' of Y with independent tangent sums, $(Tf)_*T\Theta = T(f_*\Theta)$ and $(Tf)^*T\Theta' = T(f^*\Theta')$. The independence passes from Θ to $f_*\Theta$ and from Θ' to $f^*\Theta'$; for an equivalence, $T(F_*\Theta)(y)$ is $T\Theta(x)$ moved by $T\psi \circ TF(x)$ for every $\psi : F(x) \to y$; for an sc-diffeomorphism f between open subsets of two M-polyfolds, $T_{f(a)}f(A) = Tf(T_aA)$. *Book:* Theorem 11.3.7, p. 541 (2017: Theorem 11.5, p. 395); the proof is left to the reader. *Hidden hypotheses:* the independence of the tangent sums for Θ, resp. Θ', as in V11.21. *Deviation (explicit):* $(Tf)_*$ is the push-forward of Definition 11.3.1 along the tangent generalized isomorphism; the proof left to the reader is formalized (Findings, item 43).

V11.24 (Theorem 11.3.8 (2017: 11.6), boundaries of branched ep$^+$-subgroupoids under generalized isomorphisms; `Polyfold.MPolyfold.IsTangentReducedAt.image_of_isScDiffeoOn`, `Polyfold.MPolyfold.isTangentReducedAt_image_iff_of_isScDiffeoOn`, `Polyfold.MPolyfold.mem_submanifoldBoundary_image_iff_of_isScDiffeoOn`, `Polyfold.EpGroupoid.boundaryTheta_apply_of_isScDiffeoOn`, `Polyfold.EpFunctor.boundaryTheta_pushforwardTheta`, `Polyfold.EpFunctor.boundaryTheta_pullbackTheta`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_8_push`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_8_pull`). For a generalized isomorphism f and branched ep$^+$-subgroupoids with independent tangent sums, $f_*(\partial\Theta) = \partial(f_*\Theta)$ and $f^*(\partial\Theta') = \partial(f^*\Theta')$ ((11.11)). For an sc-diffeomorphism f between open subsets of two M-polyfolds and a submanifold A, $f(a) \in \partial f(A)$ if and only if $a \in \partial A$. *Book:* Theorem 11.3.8, p. 541 (2017: Theorem 11.6, p. 395), with the sketch of its proof, pp. 541–542. *Hidden hypotheses:* the independence of the tangent sums, as in V11.21. *Deviation (explicit):* "$d_M(x)$ is being preserved" is read as the invariance of ∂M (Definition 9.2.16, through Remark 9.2.14) under sc-diffeomorphisms; the proof left to the reader is formalized (Findings, item 44).

V11.25 (Theorem 11.3.9 (2017: 11.7), oriented boundaries, for a representative and under a boundary-transfer hypothesis; `Polyfold.MPolyfold.IsBoundaryTransfer`, `Polyfold.MPolyfold.isBoundaryTransfer_id`, `Polyfold.EpFunctor.pushSection_boundaryLift`, `Polyfold.EpFunctor.pullSection_boundaryLift`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_9_push`, `Polyfold.EPLocalization.EPLoc.theorem_11_3_9_pull`). A linear map $T_xX \to T_yY$ transfers a boundary operation *bd* at x to *bd*′ at y if it preserves the degree 1 of oriented tangent spaces and $bd'(f_*L) = f_*(bd\,L)$ in degree 1; the identity transfers every boundary operation to itself. For a diagram $X \overset{F}{\longleftarrow} A \overset{\Phi}{\longrightarrow} Y$ with Φ an equivalence and boundary operations *bd*, bd_A, *bd*′ on X, A, Y transferred by $TF(a)^{-1}$ and by $T\psi \circ T\Phi(a)$: $d_*(\partial\Theta) = \partial(d_*\Theta)$ and $d_*\widehat{T}_{\partial\Theta} = \widehat{T}_{\partial(d_*\Theta)}$, the orientation of the boundary being Chapter 9's boundary lift of $\widehat{T}_{\Theta}$; likewise for d^*. *Book:* Theorem 11.3.9, p. 542 (2017: Theorem 11.7, p. 395); the proof is left to the reader. *Hidden hypotheses:* `Polyfold.MPolyfold.IsBoundaryTransfer` for the given maps and boundary operations, pointwise; it holds along the identity (`Polyfold.MPolyfold.isBoundaryTransfer_` `id`); and the independence of the tangent sums. *Deviation (narrower):* in Chapter 9 the boundary operation is data, since the partial cones of submanifolds are not formalized, so the theorem is stated for a representative and boundary operations corresponding under the transfer maps; that the boundary operations of Definition 9.3.8 satisfy the hypothesis along equivalences, and hence a form of the theorem independent of the representative, is not formalized (gap G.130) (Findings, item 45).

V11.26 (witnesses for Lemma 11.3.4 to Theorem 11.3.9; `Polyfold.EpGroupoid.Examples.exampleT1_mem_grHatSections`, `Polyfold.EpGroupoid.Examples.pushSection_id_exampleT1`, `Polyfold.EpGroupoid.Examples.rn2EP`, `Polyfold.EpGroupoid.Examples.example935Oriented`, `Polyfold.EpGroupoid.Examples.isGeneralizedIso_id_rn2EP`, `Polyfold.EpGroupoid.Examples.example935Oriented_push`, `Polyfold.EpGroupoid.Examples.isBoundaryTransfer_id_rn2`). The orientation $\widehat{T}^1$ of Example 9.3.5 on $\mathbb{R}^2$ is a section functor of $\widehat{Gr}$, fixed by the push-forward and the pull-back along the identity equivalence; $(\Theta, \widehat{T}^1)$ of Example 9.3.5 is an oriented branched ep$^+$-subgroupoid in the sense of Definition 11.3.6, pushed forward along the identity generalized isomorphism; the boundary-transfer hypothesis holds along the identity of the trivial ep-groupoid on $\mathbb{R}^2$ for the identity boundary operation. *Book:* not in the book. *Hidden hypotheses:* none. *Deviation (not in the book):* witnesses that the hypotheses of V11.20–V11.25 can hold (Findings, item 46).

V11.27 (Theorems 11.4.1 and 11.4.2 (2017: Theorem 11.8; Theorem 11.4.2 new in the 2021 edition), transformation of the canonical measures; `Polyfold.EPLocalization.EPLoc.theorem_11_4_1_pull`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_1_push`, `Polyfold.EPLocalization.EPLoc.integral_pull`, `Polyfold.EPLocalization.EPLoc.integral_push`, `Polyfold.EPLocalization.Diagram.theorem_11_4_1_pull`, `Polyfold.EPLocalization.Diagram.theorem_11_4_1_push`, `Polyfold.EpFunctor.measure_span_core`, `Polyfold.EpFunctor.theorem_11_4_1_pull_equiv`, `Polyfold.EpFunctor.theorem_11_4_1_push_equiv`, `Polyfold.EpFunctor.integral_pull_equiv`, `Polyfold.EpFunctor.integral_push_equiv`, `Polyfold.EpFunctor.measure_pull_core`, `Polyfold.EpFunctor.measurable_suppMap`, `Polyfold.EpFunctor.measurable_suppInv`, `Polyfold.EpFunctor.isMeasureZeroSupp_image`, `Polyfold.EpFunctor.isMeasureZeroSupp_preimage`, `Polyfold.EpFunctor.hcov_pull`, `Polyfold.EpFunctor.hcov_push`, `Polyfold.EpGroupoid.FormEval.IsPullback`, `Polyfold.EpFunctor.IsEquivalence.epFormEval_isPullback`, `Polyfold.EP.epFormsInftyEval_isPullback`, `Polyfold.MPolyfold.GraphChart.mapDiffeoCross`, `Polyfold.MPolyfold.IsHNull.image_of_isScDiffeoOn_cross`, `Polyfold.MPolyfold.HasTameInducedStructure.image_of_isScDiffeoOn`, `Polyfold.EpGroupoid.OrientedTameStructure.restrictRep`, `Polyfold.EpGroupoid.OrientedTameStructure.move`, `Polyfold.EpGroupoid.OrientedTameStructure.represents_move`, `Polyfold.EpGroupoid.FormEval.integ_image_cross`, `Polyfold.EpGroupoid.OrientedTameStructure.sheetSum_move`, `Polyfold.EpFunctor.TransportChart.evCompatOn_Q`, `Polyfold.EpFunctor.TransportChart.exists_rep_push`, `Polyfold.EpGroupoid.OrientedTameStructure.exists_push_structure`, `Polyfold.EpGroupoid.OrientedTameStructure.pullStructure`, `Polyfold.EpGroupoid.OrientedTameStructure.represents_pullStructure`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_2_pull`,

`Polyfold.EPLocalization.EPLoc.theorem_11_4_2_push`, `Polyfold.EPLocalization.EPLoc.integralEff_pull`, `Polyfold.EPLocalization.EPLoc.integralEff_push`, `Polyfold.EPLocalization.Diagram.theorem_11_4_2_pull`, `Polyfold.EPLocalization.Diagram.theorem_11_4_2_push`, `Polyfold.EpFunctor.measure_span_core_eff`, `Polyfold.EpFunctor.TransportChart.card_effIsotropy_eq`, `Polyfold.EpFunctor.hcovL_pull`, `Polyfold.EpFunctor.hcovL_push`, `Polyfold.EpFunctor.measure_pull_core_eff`, `Polyfold.EpFunctor.theorem_11_4_2_pull_equiv`, `Polyfold.EpFunctor.theorem_11_4_2_push_equiv`, `Polyfold.EpFunctor.integralEff_pull_equiv`, `Polyfold.EpFunctor.integralEff_push_equiv`, `Polyfold.EpGroupoid.hasIndependentTangentSums`, `Polyfold.EpGroupoid.FormEval.hcov_of_loc`). For a generalized isomorphism $f : X \to Y$, oriented, tame, compact, n-dimensional ($n \geq 1$) branched ep⁺-subgroupoids $\hat{\Theta}$ on X and $\hat{\Theta}'$ on Y, forms $\omega \in \Omega^n_{ep,\infty}(X)$, $\omega' \in \Omega^n_{ep,\infty}(Y)$ and $K \in \mathcal{L}(|\mathrm{supp}\, f^*\Theta'|)$ (resp. $\mathcal{L}(|\mathrm{supp}\,\Theta|)$): $|f|(K) \in \mathcal{L}$, $\mu^{f^*\hat{\Theta}'}_{f^*\omega'}(K) = \mu^{\hat{\Theta}'}_{\omega'}(|f|(K))$ and $\mu^{\hat{\Theta}}_{\omega}(K) = \mu^{f_*\hat{\Theta}}_{f_*\omega}(|f|(K))$, for the canonical measures of Theorem 9.5.4 built from Chapter 9's sheet integrals; in particular the integrals agree. Theorem 11.4.2: the same for the effective measures of Theorem 9.5.5 when X and Y have the local unique continuation property. The same holds for a diagram $X \overset{F}{\longleftarrow} A \overset{\Phi}{\longrightarrow} Y$ with Φ an equivalence and for an equivalence, with any evaluation of forms compatible with the pull-back. *Book:* Theorem 11.4.1, p. 543 (2017: Theorem 11.8, p. 396); Theorem 11.4.2, p. 544. *Hidden hypotheses:* none; the independence of the tangent sums taken in §11.3 holds for every Θ (`Polyfold.EpGroupoid.hasIndependentTangentSums`). *Deviation (same):* 'oriented, tame' is read as in Theorem 9.5.4's Lean form, a structure describing $\hat{T}_\Theta$ with sheets in the class of integrated sheets, which `Polyfold.EpGroupoid.FormEval.hcov_of_loc` gives from a plain oriented tame branching structure; f_* is computed on the chosen representative of f; for Theorem 11.4.2 the covering hypothesis asks the property of Definition 7.2.18 of each representation, which LUC gives for all of them. The proof follows the text: structures moved along the local sc-diffeomorphisms of an equivalence, (11.12) = (11.13), $\sharp G_x = \sharp G_y$ (and $\sharp G^{eff}_x = \sharp G^{eff}_y$ under LUC, `Polyfold.EpFunctor.TransportChart.card_effIsotropy_eq`), the uniqueness of the glued measure, then $G_* F^*$ for $X \leftarrow A \to Y$. The text uses without comment that $|f|$ maps $\mathcal{L}$ onto $\mathcal{L}$; it is proved through Lemma 9.5.2 (`Polyfold.EpFunctor.measurable_suppMap`, `Polyfold.EpFunctor.measurable_suppInv`) (gap G.131, closed since 2026-11-13; Findings, item 50).

V11.28 (Remark 11.4.3 (2017: 11.3), equivalent polyfold structures; `Polyfold.EPLocalization.EPLoc.remark_11_4_3`, `Polyfold.EPLocalization.EPLoc.integral_pull`, `Polyfold.EPLocalization.EPLoc.integral_push`). For a generalized isomorphism $f : X \to Y$ and maps $\beta : |X| \to Z$, $\beta' : |Y| \to Z$ with $\beta' \circ |f| = \beta$, and $B \subseteq Z$ with $\beta^{-1}(B) \in \mathcal{L}$: $\beta'^{-1}(B) \in \mathcal{L}$ and $\mu^{f^*\hat{\Theta}'}_{f^*\omega'}(\beta^{-1}B) = \mu^{\hat{\Theta}'}_{\omega'}(\beta'^{-1}B)$; so the signed measure on $S \subseteq Z$ does not depend on the representative, and with $B = Z$ the integrals agree. *Book:* Remark 11.4.3, p. 544 (2017: Remark 11.3, p. 396). *Hidden hypotheses:* none. *Deviation (same):* the text's 'obvious consequence' is stated for the measures on Z; β, β' need not be homeomorphisms for it (gap G.131, closed since 2026-11-13; Findings, item 50).

V11.29 (Theorems 11.4.4 and 11.4.5 (2017: Theorem 11.9), boundary measures; `Polyfold.`

`EPLocalization.EPLoc.theorem_11_4_4_pull`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_4_push`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_5_pull`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_5_push`, `Polyfold.EPLocalization.Diagram.theorem_11_4_4_pull`, `Polyfold.EPLocalization.Diagram.theorem_11_4_4_push`, `Polyfold.EPLocalization.Diagram.theorem_11_4_5_pull`, `Polyfold.EPLocalization.Diagram.theorem_11_4_5_push`, `Polyfold.EPLocalization.Diagram.hcovB_pullTheta`, `Polyfold.EPLocalization.Diagram.hcovB_pushTheta`, `Polyfold.EpFunctor.measure_span_core_bd`, `Polyfold.EpGroupoid.boundaryMeasure`, `Polyfold.EpGroupoid.boundaryMeasureEff`, `Polyfold.EpFunctor.orbitSupp_boundaryTheta_pull`, `Polyfold.EpFunctor.measurable_bsuppMap`, `Polyfold.EpFunctor.measurable_bsuppInv`, `Polyfold.EpFunctor.hcovB_pull`, `Polyfold.EpFunctor.hcovB_push`, `Polyfold.EpFunctor.measure_pull_core_bd`, `Polyfold.EpFunctor.theorem_11_4_4_pull_equiv`, `Polyfold.EpFunctor.theorem_11_4_4_push_equiv`, `Polyfold.EpFunctor.theorem_11_4_5_pull_equiv`, `Polyfold.EpFunctor.theorem_11_4_5_push_equiv`, `Polyfold.EpGroupoid.boundaryStratum_image_cross`, `Polyfold.EpGroupoid.OrientedTameStructure.boundarySheet_move`, `Polyfold.EpGroupoid.OrientedTameStructure.boundaryOrient_move_tangent`, `Polyfold.EpGroupoid.OrientedTameStructure.boundarySheet_move_mem_msheets`, `Polyfold.EpGroupoid.OrientedTameStructure.boundarySheetSum_move`, `Polyfold.EpGroupoid.FormEval.isMeasIn_image_cross`, `Polyfold.EpGroupoid.FormEval.image_mem_msheets_cross`, `Polyfold.EpGroupoid.FormEval.msInteg_image_cross`, `Polyfold.EpGroupoid.FormEval.ChartCover.imageCross`, `Polyfold.EpGroupoid.theorem_9_5_15_msheets`, `Polyfold.EpGroupoid.theorem_9_5_16_msheets`, `Polyfold.EpGroupoid.hcov_msheets_of_isCompact`, `Polyfold.EpGroupoid.hcov_zero`, `Polyfold.EP.hcovB_of_hcovS`, `Polyfold.E8W.e8w_hcov`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_4_pull_loc`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_4_push_loc`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_5_pull_loc`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_5_push_loc`, `Polyfold.EP.boundaryMeasure_eq`, `Polyfold.EP.boundaryMeasureEff_eq`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_4_pull_dim1`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_4_push_dim1`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_5_pull_dim1`, `Polyfold.EPLocalization.EPLoc.theorem_11_4_5_push_dim1`, `Polyfold.EpFunctor.measure_span_core_bd_dim1`,`Polyfold.EpFunctor.theorem_11_4_4_pull_equiv_dim1`,`Polyfold.EpFunctor.theorem_11_4_4_push_equiv_dim1`, `Polyfold.EpFunctor.theorem_11_4_5_pull_equiv_dim1`, `Polyfold.EpFunctor.theorem_11_4_5_push_equiv_dim1`, `Polyfold.EpFunctor.measure_pull_core_bd_dim1`, `Polyfold.EP.zeroFormEval_epFormsInftyPullback`, `Polyfold.EpGroupoid.OrientedTameStructure.boundarySheetSum_move_dim1`). Theorem 11.4.4 for a generalized isomorphism f, with the text's hypotheses (`EPLoc.theorem_11_4_4_pull_loc`, `EPLoc.theorem_11_4_4_push_loc`): $\hat{\Theta}$ (resp. $\hat{\Theta}'$) is branched, compact, oriented and tame (`hloc`: an oriented tame structure of dimension $n = m + 1$ describing $\hat{T}$ at every point of the support), $m > 0$. On $|\mathrm{supp}(\partial\Theta)|$, $|f|(K) \in \mathcal{L}(\partial S)$, $\mu_{\omega}^{\partial\hat{\Theta}}(K) = \mu_{f_*\omega}^{\partial f_*\hat{\Theta}}(|f|(K))$ and $\mu_{f^*\omega'}^{\partial f^*\hat{\Theta}'}(K) = \mu_{\omega'}^{\partial\hat{\Theta}'}(|f|(K))$, for $\omega \in \Omega^m_{ep,\infty}$. Theorem 11.4.5 (`_loc` versions, hypotheses `X.X.IsLUC`, `Y.X.IsLUC`): the same for the effective boundary measures. The measures are those of Theorems 9.5.15 and 9.5.16 (`EP.boundaryMeasure_eq`, `EP.boundaryMeasureEff_eq`). $n = 1$ (`EPLoc.theorem_11_4_4_pull_dim1`, `_push_dim1`, `EPLoc.theorem_11_4_5_pull_dim1`, `_push_dim1`): the same statements for $\omega \in \Omega^0_{ep,\infty}$ and the measures `EP.boundaryBranchedMeasureDim1`, …`EffDim1`, with the same hypotheses in dimension 1; also for equivalences (`EpFunctor.theorem_11_4_4_pull_equiv_dim1`, …). The versions with E7's class hypothesis `hcov` (`EPLoc.theorem_11_4_4_pull`, `_push`, `EPLoc.theorem_11_4_5_pull`, `_push`) remain, as do the forms for a diagram $X \leftarrow A \rightarrow Y$ and for an equivalence with any evaluation of forms compatible with the pull-back; `hcov` passes along equivalences and generalized isomorphisms (`hcovB_pull`, `hcovB_push`, `Diagram.hcovB_pullTheta`, `Diagram.hcovB_pushTheta`), and $|f|$ maps $\mathcal{L}(\partial S)$ onto $\mathcal{L}(\partial S)$ (`measurable_bsuppMap`, `measurable_bsuppInv`). *Book:* §11.4, Theorem 11.4.4, p. 545 (2017: Theorem 11.9, p. 397), and Theorem 11.4.5, p. 546 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (restated):* (1) No hypothesis beyond the text's: E7's class hypothesis `hcov` is derived from tameness (`EP.hcov_of_loc`, gap G.109); it passes to the transported subgroupoids (`hcovB_pull`, `hcovB_push`). Every $n \geq 1$ is covered ($n = 1$ in the `_dim1` theorems, with Ω^0 integrated over oriented points). (2) f_* is computed on the representative `rep f`, as for Theorem 11.4.1; 'oriented, tame' is read as for Theorem 11.4.1, and `HasIndependentTangentSums` holds for every Θ and is not a hypothesis (Findings, Chapter 11, item 50).

V11.30 (Proposition 11.5.1 (2017: 11.6), auxiliary norms under generalized strong bundle isomorphisms, without conditions (1) and (2) of Definition 8.3.5; `Polyfold.SBLocalization.GenSBMap.proposition_11_5_1`, `Polyfold.SBLocalization.GenSBMap.pullNorm`, `Polyfold.SBLocalization.GenSBMap.pushNorm`, `Polyfold.SBLocalization.SDiagram.pullNorm`, `Polyfold.SBLocalization.SDiagram.pushNorm`, `Polyfold.SBLocalization.SDiagram.pullNorm_eq`, `Polyfold.SBLocalization.SDiagram.pullNorm_eq_Ψ`, `Polyfold.SBLocalization.SDiagram.isOneInvariant_pullNorm`, `Polyfold.SBLocalization.SDiagram.pullNorm_eq_lift`, `Polyfold.SBLocalization.SDiagram.continuous_pullNorm`, `Polyfold.SBLocalization.SDiagram.pushNorm_pullNorm`, `Polyfold.SBLocalization.SDiagram.pullNorm_pushNorm`, `Polyfold.SBLocalization.SDiagram.pullNorm_eq_of_orbitMap_eq`, `Polyfold.SBLocalization.SDiagram.pushNorm_eq_of_orbitMap_eq`, `Polyfold.SBLocalization.SDiagram.oneOrbitMap`, `Polyfold.SBLocalization.SDiagram.continuous_oneOrbitMap`, `Polyfold.SBLocalization.SDiagram.oneOrbitMap_oneMap`, `Polyfold.SBLocalization.SDiagram.oneToOrbit_oneOrbitMap`, `Polyfold.SBLocalization.SDiagram.oneOrbitMap_eq_of_orbitMap_eq`, `Polyfold.SBLocalization.oneToOrbit`, `Polyfold.SBLocalization.oneToOrbit_injective`, `Polyfold.SBLocalization.oneMap`, `Polyfold.SBLocalization.continuous_oneMap`, `Polyfold.SBLocalization.oneToOrbit_oneMap`, `Polyfold.SBLocalization.IsOneInvariant`, `Polyfold.SBLocalization.isOneInvariant_const`, `Polyfold.SBLocalization.isOneInvariant_iff`, `Polyfold.EpStrongBundle.IsAuxNorm.isOneInvariant`). For a generalized strong bundle isomorphism $[D] : W \rightarrow W'$ and functions N' on $W'_{0,1}$, N on $W_{0,1}$: $[D]^*N'(e) = N'(e')$ for a representative e' of $|D|(|e|) \in |W'[1]|$, and $[D]_*N$ is the pull-back along the reversed diagram; both are independent of the representative, and for an invariant N', $[D]^*N'(e) = N'(\Psi(e''))$ for every $e'' \in W''_{0,1}$ with $\Phi(e'') \cong e$, the text's formula. If N' (resp. N) is an auxiliary norm, $[D]^*N'$ (resp. $[D]_*N$) is continuous and satisfies condition (3) of Definition 8.3.5; $[D]^*[D]_*N = N$ and $[D]_*[D]^*N' = N'$ for every invariant N, N', in particular for auxiliary norms; if $|[D]| = |[D']|$, then $[D]^*N' = [D']^*N'$ and $[D]_*N = [D']_*N$. Here $|D| : |W[1]| \to |W'[1]|$ is $|\Psi| \circ |\Phi|^{-1}$, continuous, and determined by $|D|$ on $|W|$ because $|W[1]| \to |W|$ is injective. *Book:* Proposition 11.5.1, p. 547 (2017: Proposition 11.6, p. 399). *Hidden hypotheses:* none. *Deviation (narrower):* conditions (1) (a complete norm on each fibre) and (2) (a sequence with $N(h_k) \to 0$ over a convergent sequence of base points converges to the zero section) of Definition 8.3.5 are not shown for $[D]^*N'$ and $[D]_*N$ (gap G.132); the text's choices of e'' and $\Phi(e'') \to e$ are replaced by a representative of $|D|(|e|)$ in $|W'[1]|$ (Findings, item 60).

V11.31 (the pull-back and push-forward of multisection functors along a strong bundle equivalence, and Lemma 11.5.2 (2017: 11.7); `Polyfold.SBLocalization.IsWFunctor`, `Polyfold.SBLocalization.isWFunctor_const`, `Polyfold.SBLocalization.IsWFunctor.eq_of_ _eq`, `Polyfold.SBLocalization.pullMult`, `Polyfold.SBLocalization.IsWFunctor.pullMult`, `Polyfold.SBLocalization.pushMult`, `Polyfold.SBLocalization. _obj_choose`, `Polyfold.SBLocalization.pushMult_eq`, `Polyfold.SBLocalization.lemma_11_5_2`, `Polyfold.SBLocalization.pullMult_pushMult`, `Polyfold.SBLocalization.pushMult_pullMult`). A functor $\Lambda : W \to \alpha$ on the total ep-groupoid of a strong bundle is a map constant along morphisms, $\Lambda(\mu(\varphi, w)) = \Lambda(w)$. For a strong bundle functor Ψ, $\Psi^*\Lambda' = \Lambda' \circ \Psi$ of a functor is a functor; for a strong bundle equivalence Φ, $\Phi_*\Lambda(w') = \Lambda(w)$ for a chosen w and morphism $\Phi(w) \to w'$, equal to $\Lambda(w)$ for every w with $\Phi(w) \cong w'$; $\Phi_*\Lambda$ is a functor (Lemma 11.5.2), and $\Phi^*\Phi_*\Lambda = \Lambda$, $\Phi_*\Phi^*\Lambda' = \Lambda'$. *Book:* §11.5, pp. 547–548 (2017 pp. 399–400), and Lemma 11.5.2, p. 548 (2017: Lemma 11.7, p. 400). *Hidden hypotheses:* none. *Deviation (narrower):* only the functor $\Lambda : W \to \mathbb{Q}^+$ underlying an sc⁺-multisection is treated; its local section structures (2021 Definition 13.2.1) and their transport along Φ are not formalized (gap G.133) (Findings, item 61).

V11.32 (Proposition 11.5.3 (2017: 11.7), multisection functors under generalized strong bundle isomorphisms; `Polyfold.SBLocalization.GenSBMap.proposition_11_5_3`, `Polyfold.SBLocalization.GenSBMap.pushMult`, `Polyfold.SBLocalization.GenSBMap.pullMult`, `Polyfold.SBLocalization.SDiagram.pushMult`, `Polyfold.SBLocalization.SDiagram.pullMult`, `Polyfold.SBLocalization.SDiagram.pushMult_eq`, `Polyfold.SBLocalization.SDiagram.pullMult_eq`, `Polyfold.SBLocalization.SDiagram.isWFunctor_pushMult`, `Polyfold.SBLocalization.SDiagram.isWFunctor_pullMult`, `Polyfold.SBLocalization.SDiagram.pullMult_pushMult`, `Polyfold.SBLocalization.SDiagram.pushMult_pullMult`, `Polyfold.SBLocalization.SDiagram.pushMult_eq_of_orbitMap_eq`, `Polyfold.SBLocalization.SDiagram.pullMult_eq_of_orbitMap_eq`). For a generalized strong bundle isomorphism $[D]$ and functors $\Lambda : W \to \mathbb{Q}_{\geq 0}$, $\Lambda' : W' \to \mathbb{Q}_{\geq 0}$: $[D]_*\Lambda = \Psi_*\Phi^*\Lambda$ and $[D]^*\Lambda' = \Phi_*\Psi^*\Lambda'$ do not depend on the representative $W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W'$ (both legs strong bundle equivalences); they map functors to functors and are mutually inverse; and if $[D']$ is a generalized strong bundle isomorphism with $|[D]| = |[D']|$, the push-forwards and the pull-backs agree. *Book:* Proposition 11.5.3, p. 548 (2017: Proposition 11.7, p. 401). *Hidden hypotheses:* none. *Deviation (narrower):* for the functors underlying sc⁺-multisections, without their local section structures (gap G.133); the text's "generalized strong bundle maps" with $|[D]| = |[D']|$ is read as generalized strong bundle isomorphisms, for which the push-forward is defined (Findings, item 61). *Proof*: $[D]_*\Lambda(w') = \Lambda(w)$ whenever $|D|(|w|) = |w'|$, so everything depends only on $|D|$.

V11.33 (Lemma 11.6.1, the morphism map of a proper covering functor; `Polyfold.EpFunctor.IsProperCovering.lemma_11_6_1`, `Polyfold.EpFunctor.IsProperCovering.lift`, `Polyfold.EpFunctor.IsProperCovering.mor_lift`, `Polyfold.EpFunctor.IsProperCovering.src_lift`, `Polyfold.EpFunctor.IsProperCovering.eq_lift`, `Polyfold.EpFunctor.IsProperCovering.obj_tgt_lift`, `Polyfold.EpFunctor.IsProperCovering.existsUnique_lift_tgt`). For a proper covering functor $F : Y \to X$, F on morphisms is a surjective local sc-diffeomorphism. The inverse $\alpha^{-1}(\varphi, y)$ of $\alpha : \psi \mapsto (F(\psi), s(\psi))$ is the unique ψ with $F(\psi) = \varphi$, $s(\psi) = y$; it has $F(t(\psi)) = t(\varphi)$, and lifts with prescribed target are unique too. *Book:* Lemma 11.6.1, p. 549 (new in the 2021 edition). *Hidden hypotheses:* none. *Deviation (same):* none; the local sc-diffeomorphism half is also Chapter 10's `Polyfold.EpFunctor.isLocalScDiffeo_mor` (Findings, item 70).

V11.34 (Lemma 11.6.2 (2017: 11.8), the fibre as a translation groupoid; `Polyfold.EpFunctor.IsProperCovering.lemma_11_6_2`, `Polyfold.EpFunctor.IsProperCovering.Fiber`, `Polyfold.EpFunctor.IsProperCovering.Fiber.val`, `Polyfold.EpFunctor.IsProperCovering.Fiber.mk`, `Polyfold.EpFunctor.IsProperCovering.fiberSMul`, `Polyfold.EpFunctor.IsProperCovering.fiberSMul_val_of`, `Polyfold.EpFunctor.IsProperCovering.instMulActionFiber`, `Polyfold.EpFunctor.IsProperCovering.smul_val_of`, `Polyfold.EpFunctor.IsProperCovering.FiberFull`, `Polyfold.EpFunctor.IsProperCovering.Translation`, `Polyfold.EpFunctor.IsProperCovering.toTranslation`, `Polyfold.EpFunctor.IsProperCovering.bijective_toTranslation_obj`, `Polyfold.EpFunctor.IsProperCovering.full_toTranslation`, `Polyfold.EpFunctor.IsProperCovering.faithful_toTranslation`). For a proper covering functor $F : Y \to X$ and an object x, $g * y = t(\alpha^{-1}(g, y))$ is an action of the isotropy group G_x on $Y_x = F^{-1}(x)$ ((11.15)); the functor from the full subcategory $Y(x)$ on Y_x to the translation groupoid $G_x \ltimes Y_x$ which is the identity on objects and $\psi \mapsto (F(\psi), s(\psi))$ on morphisms is bijective on objects, full and faithful, hence an isomorphism of categories and an equivalence. *Book:* the action, p. 550, and Lemma 11.6.2, p. 551 (2017: Lemma 11.8, p. 402). *Hidden hypotheses:* none. *Deviation (restated):* "naturally identified" is read as this functor, an isomorphism of categories, as the text says before the lemma; the topology of $Y(x)$ plays no part. In the computation of $\alpha(\psi_2 \circ \psi_1)$ on p. 550 the second factor is printed $(F(\psi_2), s(\psi_1))$; we read $(F(\psi_1), s(\psi_1))$ (possible misprint) (Findings, item 71).

V11.35 (Lemma 11.6.3 (2017: 11.9), transport of fibres along a morphism; `Polyfold.EpFunctor.IsProperCovering.lemma_11_6_3`, `Polyfold.EpFunctor.IsProperCovering.transportFiber`, `Polyfold.EpFunctor.IsProperCovering.transportFiber_val_of`, `Polyfold.EpFunctor.IsProperCovering.transportFiber_inv_transportFiber`, `Polyfold.EpFunctor.IsProperCovering.bijective_transportFiber`, `Polyfold.EpFunctor.IsProperCovering.transportFiber_smul`, `Polyfold.EpFunctor.IsProperCovering.translationMap`, `Polyfold.EpGroupoid.conjIsotropy`). A morphism $\varphi : x \to x'$ of X induces the bijection $B_\varphi : Y_x \to Y_{x'}$, $B_\varphi(y) = t(\psi)$ for $F(\psi) = \varphi$, $s(\psi) = y$, with inverse $B_{\varphi^{-1}}$, equivariant along $\gamma_\varphi(g) = \varphi \circ g \circ \varphi^{-1}$

((11.16)); and $\tilde{\varphi} : G_x \ltimes Y_x \to G_{x'} \ltimes Y_{x'}$, $(g, y) \mapsto (\varphi \circ g \circ \varphi^{-1}, B_\varphi(y))$, is a functor bijective on objects, full and faithful, an equivalence. *Book:* Lemma 11.6.3, p. 551 (2017: Lemma 11.9, p. 403), with its proof, pp. 551–552. *Hidden hypotheses:* none. *Deviation (same):* none (Findings, item 71).

V11.36 (Definition 11.6.4 (2017: 11.6), functors and equivalences between proper coverings; `Polyfold.ProperCoveringHom`, `Polyfold.ProperCoveringHom.ext`, `Polyfold.ProperCoveringHom.mapsTo_fiber`, `Polyfold.ProperCoveringHom.id`, `Polyfold.ProperCoveringHom.comp`, `Polyfold.ProperCoveringHom.IsFiberBijective`, `Polyfold.ProperCoveringHom.IsEquivalence`, `Polyfold.ProperCoveringHom.IsEquivalence.id`, `Polyfold.ProperCoveringHom.IsEquivalence.comp`, `Polyfold.ProperCoveringHom.isFiberBijective_of_isEquivalence`, `Polyfold.ProperCoveringHom.isEquivalence_iff`, `Polyfold.CoveringEquivalence.Examples.pairToPointCovering`, `Polyfold.CoveringEquivalence.Examples.isEquivalence_pairToPointCovering`). (1) An sc-smooth functor $(\Psi, \Phi) : F \to F'$ between functors $F : Y \to X$, $F' : Y' \to X'$ is a pair of sc-smooth functors $\Psi : Y \to Y'$, $\Phi : X \to X'$ with $\Phi \circ F = F' \circ \Psi$ on objects and on morphisms; Ψ then maps Y_x into $Y'_{\Phi(x)}$; identities and composites are such pairs. (2) It is an equivalence if Ψ and Φ are equivalences and Ψ maps Y_x bijectively onto $Y'_{\Phi(x)}$ for every x; identities and composites of equivalences are equivalences. If F and F' are proper covering functors, the fibre condition follows from the other two, so that (Ψ, Φ) is an equivalence if and only if Ψ and Φ are. The pair $(\mathbb{Z}\text{-pair} \to \text{pt}, \mathbb{Z}\text{-pair} \to \text{pt})$ between the identity coverings is an equivalence, though it has no inverse functor. *Book:* Definition 11.6.4, p. 553 (2017: Definition 11.6, p. 404). *Hidden hypotheses:* none. *Deviation (stronger):* the redundancy of the fibre condition for proper coverings is added; the text's remark after the definition treats the fibre bijection as defining a subclass of the functors of (1) (Findings, item 72). *Proof:* injectivity from the faithfulness of Φ and unique lifting along F; surjectivity from the essential surjectivity of Ψ, the fullness of Φ and unique lifting along F'.

V11.37 (Definition 11.6.5 (2017: 11.7), natural transformations between functors of proper coverings; `Polyfold.ProperCoveringHom.NatTrans`, `Polyfold.ProperCoveringHom.NaturallyEquivalent`, `Polyfold.ProperCoveringHom.NatTrans.naturality_F`, `Polyfold.ProperCoveringHom.NatTrans.refl`, `Polyfold.ProperCoveringHom.NatTrans.symm`, `Polyfold.ProperCoveringHom.NatTrans.trans`, `Polyfold.ProperCoveringHom.NatTrans.whiskerLeft`, `Polyfold.ProperCoveringHom.NatTrans.whiskerRight`, `Polyfold.ProperCoveringHom.naturallyEquivalent_equivalence`, `Polyfold.ProperCoveringHom.NaturallyEquivalent.comp_left`, `Polyfold.ProperCoveringHom.NaturallyEquivalent.comp_right`, `Polyfold.CoveringEquivalence.Examples.naturallyEquivalent_pairToPointCovering`). A natural transformation $(\zeta, \xi) \to (\zeta', \xi')$ is a pair of natural transformations $\tau_Y : \zeta \to \zeta'$, $\tau_X : \xi \to \xi'$ of ep-groupoids with $\tau_X(F(y)) = F'(\tau_Y(y))$ for every y ((11.20)); then $\tau_X(x_2) \circ \xi(F(\psi)) = \xi'(F(\psi)) \circ \tau_X(x_1)$ for $\psi : y_1 \to y_2$, $x_i = F(y_i)$ (the diagram after the definition). Natural equivalence, the existence of one, is an equivalence relation compatible with composition on both sides. *Book:* Definition 11.6.5, p. 554 (2017: Definition 11.7, p. 405), and the diagram on p. 555. *Hidden hypotheses:* none. *Deviation (same):* $\tau_X \circ F = F' \circ \tau_Y$ is read objectwise (Findings, item 73).

V11.38 (Theorem 11.6.6 (2017: 11.10) (1), H is a proper covering functor, with steps (A)–(C); `Polyfold.ProperCoveringHom.hFunctor`, `Polyfold.ProperCoveringHom.hObj`, `Polyfold.ProperCoveringHom.hMor`, `Polyfold.ProperCoveringHom.mor_conj`, `Polyfold.ProperCoveringHom.hMor_src`, `Polyfold.ProperCoveringHom.hMor_tgt`, `Polyfold.ProperCoveringHom.surjective_hObj`, `Polyfold.ProperCoveringHom.isLocalScDiffeo_hObj`, `Polyfold.ProperCoveringHom.isScDiffeo_fibredMap_hFunctor`, `Polyfold.ProperCoveringHom.finite_fiber_hObj`, `Polyfold.ProperCoveringHom.isProperMap_hObj`, `Polyfold.ProperCoveringHom.isProperCovering_hFunctor_of`, `Polyfold.MPolyfold.IsScDiffeoOn.restrictTarget`, `Polyfold.MPolyfold.IsLocalScDiffeo.evenlyCovered_of_isProperMap`, `Polyfold.EpFunctor.IsProperCovering.isProperMap`). For proper covering functors $F : Y \to X$, $F' : Y' \to X'$, $F'' : Y'' \to X''$, functors $\Gamma = (\Psi, \Phi) : F \to F'$, $(\zeta, \xi) : F'' \to F'$ and admissible weak fibred products $P = Y \times_{Y'} Y''$, $Q = X \times_{X'} X''$, the maps $H(y, \psi', y'') = (F(y), F'(\psi'), F''(y''))$ and $H(h, \psi', h'') = (F(h), F'(\psi'), F''(h''))$ ((11.18), (11.19)) form a functor $H : P \to Q$; if Ψ and Φ are local sc-diffeomorphisms on objects and Ψ maps each Y_x onto $Y'_{\Phi(x)}$, then H is a proper covering functor: (A) surjective and a local sc-diffeomorphism on objects, (B) its map α an sc-diffeomorphism onto the fibred product, (C) with finite fibres and proper on objects, hence evenly covered. A proper covering functor is a proper map on objects, and a proper local sc-diffeomorphism with a finite fibre is evenly covered near it. *Book:* Theorem 11.6.6 (1), p. 555 (2017: Theorem 11.10, p. 405), and steps (A)–(C) of its proof, pp. 556–560. *Hidden hypotheses:* none. *Deviation (stronger):* (1) is proved under the weaker hypotheses stated; step (C) is proved through properness instead of the text's argument with sequences. The heading of step (A) has the two weak fibred products in the reverse order, "maps $X \times_{X'} X''$ onto $Y \times_{Y'} Y''$"; we read $Y \times_{Y'} Y''$ onto $X \times_{X'} X''$ (possible misprint) (Findings, item 74).

V11.39 (Theorem 11.6.6 (2017: 11.10), the weak fibred product of proper coverings; `Polyfold.ProperCoveringHom.theorem_11_6_6`, `Polyfold.ProperCoveringHom.projLeftHom`, `Polyfold.ProperCoveringHom.projRightHom`, `Polyfold.ProperCoveringHom.natTransProj`, `Polyfold.CoveringEquivalence.Examples.theorem_11_6_6_pairToPoint`, `Polyfold.CoveringEquivalence.Examples.nonempty_fiber_id`, `Polyfold.WeakFibredProduct.isEquivalence_projRight`, `Polyfold.MPolyfold.IsLocalScDiffeo.isTame`). Let F, F', F'' be proper covering functors, $\Gamma = (\Psi, \Phi) : F \to F'$ an equivalence and $(\zeta, \xi) : F'' \to F'$ an sc-smooth functor between them, with P, Q, H as in V11.38. Then (1) H is a proper covering functor; (2) $(\pi_{Y''}, \pi_{X''}) : H \to F''$ is an equivalence; (3) $(\Psi \circ \pi_Y, \Phi \circ \pi_X)$ and $(\zeta \circ \pi_{Y''}, \xi \circ \pi_{X''})$ are naturally equivalent, through $\tau_P(y, \psi', y'') = \psi'$, $\tau_Q(x, \varphi', x'') = \varphi'$; if X'' is tame, P and Q are tame; if (ζ, ξ) is an equivalence, so is $(\pi_Y, \pi_X) : H \to F$. Witness: the equivalence $\mathbb{Z}$-pair → pt of identity coverings and the identity of the identity covering of the point satisfy the hypotheses, with P nonempty. *Book:* Theorem 11.6.6, p. 555 (2017: Theorem 11.10, p. 405), with steps (D)–(F) of its proof, pp. 560–561. *Hidden hypotheses:* none. *Deviation (same):* in (D) the fibre bijection of $\pi_{Y''}$ comes from V11.36 instead of a direct check; the tameness of Y'', "trivial" in the text, is that of a surjective local sc-diffeomorphism onto a tame M-polyfold (Findings, item 74).

V11.40 (the category $\mathscr{PC}$ and Definition 11.6.7 (2017: 11.8), diagrams and common re-

finements; `Polyfold.PC`, `Polyfold.PC.equivalences`, `Polyfold.ProperCoveringHom.comp_assoc`, `Polyfold.PC.fibred`, `Polyfold.PC.fibredFst`, `Polyfold.PC.fibredSnd`, `Polyfold.PC.Diagram`, `Polyfold.PC.Diagram.CommonRefinement`, `Polyfold.PC.Diagram.HasCommonRefinement`, `Polyfold.PC.Examples.pair`, `Polyfold.PC.Examples.point`, `Polyfold.PC.Examples.equivalences_pairToPoint`, `Polyfold.PC.Examples.diagram`, `Polyfold.PC.Examples.fibredPair`, `Polyfold.PC.Examples.isEquivalence_fibredFst`, `Polyfold.PC.Examples.diagram'`). Proper covering functors (in one universe) and the functors of Definition 11.6.4 (1) form a category $\mathcal{PC}$, with composition associative on the nose and a multiplicative class $E_{\mathcal{PC}}$ of equivalences; Theorem 11.6.6 (1) gives the weak fibred product as an object of $\mathcal{PC}$ with its two projections. A diagram $d : F \to F'$ is $F \xleftarrow{\Gamma} A \xrightarrow{\varepsilon} F'$ with Γ an equivalence of proper coverings; a common refinement of d, d' consists of equivalences $H : A'' \to A$, $H' : A'' \to A'$ with $\Gamma \circ H \simeq \Gamma' \circ H'$ and $\varepsilon \circ H \simeq \varepsilon' \circ H'$. Witnesses: pair ← pair → point and pair ← pair $\times_{\text{point}}$ pair → point. *Book:* the category $\mathcal{PC}$ and its diagrams, p. 561, and Definition 11.6.7, p. 561 (2017: Definition 11.8, p. 410). *Hidden hypotheses:* none. *Deviation (same):* all proper coverings of $\mathcal{PC}$ are in one universe, so that weak fibred products stay in $\mathcal{PC}$ (Findings, item 75).

V11.41 (Proposition 11.6.8 (2017: 11.8), common refinements are transitive; `Polyfold.PC.proposition_11_6_8`, `Polyfold.PC.Diagram.hasCommonRefinement_equivalence`, `Polyfold.PC.Examples.hasCommonRefinement_diagram`, `Polyfold.PC.Examples.hasCommonRefinement_diagram_diagram`). If d, d' and d', d'' have common refinements, so have d, d''; having a common refinement is an equivalence relation on diagrams $F \to F'$. The two witness diagrams of V11.40 have a common refinement, built from Theorem 11.6.6 (3). *Book:* Proposition 11.6.8, p. 562 (2017: Proposition 11.8, p. 411), and the sentence before it, p. 561. *Hidden hypotheses:* none. *Deviation (same):* the displayed chain for the ε's ends at $\varepsilon' \circ K \circ \pi_{B''}$; the last step to $\varepsilon'' \circ K' \circ \pi_{B''}$ uses the assumption $\varepsilon' \circ K \simeq \varepsilon'' \circ K'$, as in the chain for the Γ's, and is added (Findings, item 75). *Proof:* the weak fibred product $A'' \times_{A'} B''$ of H' and K with the equivalences $H \circ \pi_{A''}$ and $K' \circ \pi_{B''}$ (Theorem 11.6.6).

V11.42 (Remark 11.6.9 (2017: 11.4), proper strong bundle coverings up to equivalence; `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021`, `Polyfold.SBPC`, `Polyfold.SBPC.Hom`, `Polyfold.SBPC.forget`, `Polyfold.SBPC.equivalences`, `Polyfold.SBPC.Loc`, `Polyfold.SBPC.Q`, `Polyfold.SBPC.isLocalization_Q`, `Polyfold.SBPC.lift`, `Polyfold.SBPC.uniq`, `Polyfold.SBPC.toPCLoc`, `Polyfold.SBPC.toPCLoc_fac`, `Polyfold.SBPC.toSEPLocW`, `Polyfold.SBPC.toSEPLocV`, `Polyfold.SBPC.ofSEP`, `Polyfold.EpStrongBundle.isStrongBundleIso_fibredLift_of_isProperCovering`, `Polyfold.StrongBundle.evenlyCovered_of_isStrongBundleIsoOn`, `Polyfold.EpStrongBundle.isProperStrongBundleCovering2021_of`, `Polyfold.SEP.pullbackCovering`, `Polyfold.SEP.isProperStrongBundleCovering2021_pullbackCovering`, `Polyfold.SEP.isProperStrongBundleCovering2021_pullbackCovering_of_isEquivalence`, `Polyfold.SBPC.equivalences_iff`, `Polyfold.SBPC.NatTrans`, `Polyfold.SBPC.NaturallyEquivalent`, `Polyfold.SBPC.fibred`, `Polyfold.SBPC.fibredFst`, `Polyfold.SBPC.fibredSnd`, `Polyfold.SBPC.fibredNatTrans`, `Polyfold.SBPC.theorem_11_6_6`, `Polyfold.SBPC.fibredLift`, `Polyfold.SBPC.Diagram`, `Polyfold.SBPC.Diagram.CommonRefinement`,

`Polyfold.SBPC.proposition_11_6_8`, `Polyfold.SBPC.Diagram.comp`, `Polyfold.SBPC.Diagram.hasCommonRefinement_comp`, `Polyfold.SBPCLoc`, `Polyfold.SBPCLoc.I`, `Polyfold.SBPCLoc.isIso_I_map`, `Polyfold.SBPCLoc.homMk_eq`, `Polyfold.SBPCLoc.inv_homMk`, `Polyfold.SBPCLoc.fromLoc`, `Polyfold.SBPC.Examples.theorem_11_6_6_ep`). The category SBPC: objects are strong bundle functors A over a functor F satisfying Definition 8.5.9 (proper strong bundle coverings, Chapter 8's `Polyfold.EpStrongBundle.IsProperStrongBundleCovering2021`), morphisms commutative cubes of strong bundle functors; the class E_{SBPC} of morphisms whose two strong bundle components are strong bundle equivalences; the localization $SBPC[E^{-1}]$ with its universal property, and the functors it induces to $\mathscr{PC}(E^{-1})$ and to the localization of strong bundles. Theorem 11.6.6 for SBPC: for $\Lambda \in E_{SBPC}$ the weak fibred product of the strong bundles is a proper strong bundle covering over the weak fibred product of the bases, its second projection lies in E_{SBPC}, and the square commutes up to natural equivalence. $SBPC[E^{-1}]$ is described by diagrams under common refinement (Definition 11.6.7 and Proposition 11.6.8 for SBPC), I inverts E_{SBPC}, every morphism is $I(\Gamma)^{-1} \circ I(\varepsilon)$, and the comparison functor from the universal localization commutes with I. Witness: the identity of $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$. *Book:* Remark 11.6.9, pp. 563–566 (2017: Remark 11.4, pp. 412–413). *Hidden hypotheses:* none. *Deviation (same):* the text leaves the details to the reader and defines neither E_{SBPC} nor natural transformations of SBPC morphisms; they are read as above and as strong bundle natural transformations on both levels whose bases satisfy (11.20). Along the way, conditions (2) and (3) of Definition 8.5.9 follow from (0) and (1) when the base functor is a proper covering (`Polyfold.EpStrongBundle.isProperStrongBundleCovering2021_of`) (gap G.135 closed; Findings, items 76–78).

V11.43 (the localization $\mathscr{PC}(E^{-1})$ and $I : \mathscr{PC} \to \mathscr{PC}(E^{-1})$; `Polyfold.PC.fibredLift`, `Polyfold.PC.isEquivalence_fibredLift`, `Polyfold.PC.IsEquivalence.of_comp_left`, `Polyfold.PC.Diagram.comp`, `Polyfold.PC.Diagram.hasCommonRefinement_comp`, `Polyfold.PC.Diagram.hasCommonRefinement_assoc`, `Polyfold.PCLoc`, `Polyfold.PCLoc.I`, `Polyfold.PCLoc.isIso_I_map`, `Polyfold.PCLoc.homMk_eq`, `Polyfold.PCLoc.isIso_homMk`, `Polyfold.PCLoc.inv_homMk`, `Polyfold.PCLoc.Examples.not_isIso_pairToPoint_isIso_I_map`). Classes of diagrams $F \overset{\Gamma}{\leftarrow} A \overset{\varepsilon}{\rightarrow} F'$ under common refinement are the morphisms of a category $\mathscr{PC}(E^{-1})$, composed through the weak fibred products of Theorem 11.6.6; the class of the composite does not depend on the representatives, and the composition is associative and unital. $I(\varepsilon) = [F \overset{1}{\leftarrow} F \overset{\varepsilon}{\rightarrow} F']$ defines a functor $I : \mathscr{PC} \to \mathscr{PC}(E^{-1})$ mapping equivalences to isomorphisms, every morphism is $I(\Gamma)^{-1} \circ I(\varepsilon)$, and a diagram whose legs are both equivalences is invertible, with inverse $[F' \overset{\varepsilon}{\leftarrow} A \overset{\Gamma}{\rightarrow} F]$. The functor from the pair groupoid of $\mathbb{Z}$ to the point is not an isomorphism of $\mathscr{PC}$, and its image under I is one. *Book:* pp. 561–563 (2017 pp. 410–412), after Proposition 11.6.8. *Hidden hypotheses:* none. *Deviation (restated):* the text leaves the well-definedness of the composition to the reader; it is proved with the common refinement of Definition 11.6.7 directly, through maps into weak fibred products built from natural transformations, which are equivalences by two out of three. The middle of the composite is $A' \times_{F'} A$, the text's $A \times_{F'} A'$ with the factors exchanged, because Theorem 11.6.6 is formalized with the equivalence on the left (gap G.134 closed; Findings, items 76, 78).

What is not verified in this chapter

- pp. 524–525, Proposition 11.1.6 (2017: 11.2): the clause that two faces meet in a face of each does not hold as we read it; a counterexample and the corrected form, a union of faces of each, are proved instead (V11.6) (gap G.128 closed; Findings, items 6, 9).
- p. 524, Remark 11.1.5, the ordered face structures: no formal content (Findings, item 5).
- p. 542, Theorem 11.3.9 (2017: 11.7): that the boundary operations of Definition 9.3.8 satisfy `Polyfold.MPolyfold.IsBoundaryTransfer` along equivalences (it needs sc-diffeomorphisms between different M-polyfolds to preserve tame induced structures, d_M and outward vectors; Chapter 9 takes the boundary operation as data, the partial cones of submanifolds not being formalized), and hence a form of the theorem independent of the representative (V11.25) (gap G.130; Findings, item 45).
- pp. 545–546, Theorems 11.4.4 and 11.4.5: listed here before 2026-11-13 (wave K) without the class hypothesis of Theorems 9.5.15 and 9.5.16 near the corners of the sheets and for $n = 1$; verified since with the text's hypotheses for every $n \geq 1$ (V11.29; gap G.131 closed; Findings, item 50). Theorems 11.4.1, 11.4.2 and Remark 11.4.3 are verified as stated since 2026-11-13 (V11.27, V11.28).
- Readings, not claims of the text: the class E_{SBPC} and the natural transformations of SBPC morphisms of Remark 11.6.9, which the text does not define (V11.42; Findings, item 77).
- Witnesses: every hypothesis taken is satisfiable, and every definition has a witness (Findings, item 80).

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13. Chapter 12 (HWZ, Part III, §§12.1–12.5 (2017 §§12.1–12.5)): begun 2026-11-13 (Part III wave 1, continued in waves 2–7), verified for sc-Fredholm section functors, auxiliary norms and their existence, continuous partitions of unity, the good neighbourhood theorem, the extension from the boundary (Theorem 12.2.7 as printed), reflexive auxiliary norms (Theorem 12.2.13 with the paracompactness of $|X|$ named; Part III wave 3), bump functions and sc⁺-section functors (Theorems 12.3.2 (2) and 12.3.3 with the pointwise reading of bump functions), the local and equivariant extension from the boundary and Theorem 12.3.9, the compactness notions of §12.4 with the reflexive ones, and the orientation bundle with Part I's hypotheses, the last clause of Theorem 12.5.3, the continuity in Proposition 12.5.6 (□) and in Theorem 12.5.7 (2) (Part III wave 5), and Theorem 12.5.7 as printed with the independence of the representative and functoriality for the data of the diagrams (Part III wave 7)

Chapter 12 (pp. 569–598; numbers and pages are those of the book (Springer 2021), with the 2017 text's numbers in parentheses) is the Fredholm theory of Part I carried to strong bundles over ep-groupoids. §12.1 defines sc-Fredholm section functors (Definition 12.1.1 (2017: 12.1)). §12.2 defines auxiliary norms (Definition 12.2.1 (2017: 12.2)) and proves their existence (Theorem 12.2.2 (2017: 12.1)), with continuous partitions of unity (Definitions 12.2.3, 12.2.4, Proposition 12.2.5 (2017: Definitions 12.3, 12.4, Proposition 12.1)), the good neighbourhood theorem (Theorem 12.2.6 (2017: 12.2)) and the extension from the boundary (Theorem 12.2.7 (2017: 12.3)). §12.3 treats bump functions and sc$^+$-section functors (Definition 12.3.1, Theorems 12.3.2, 12.3.3, Remarks 12.3.4, 12.3.5 (2017: Definition 12.7, Theorems 12.6, 12.7, Remarks 12.3, 12.4)) and the extension of sc$^+$-sections from the boundary (Definition 12.3.6, Proposition 12.3.7, Corollary 12.3.8 (2017: Definition 12.8, Proposition 12.2, Corollary 12.1)). §12.4 compares compactness notions (Definition 12.4.1, Theorem 12.4.2, Lemma 12.4.4 (2017: Definition 12.9, Theorem 12.9, Lemma 12.1)). §12.5 builds the orientation bundle (Lemma 12.5.1, Definition 12.5.2, Theorem 12.5.3, Definition 12.5.4, Proposition 12.5.6 (2017: Lemma 12.2, Definition 12.11, Theorem 12.11, Definition 12.12, Proposition 12.4)). Part III wave 2 added the reflexive auxiliary norms (Definition 12.2.8 to Theorem 12.2.13 (2017: Definition 12.5 to Theorem 12.5)), Theorem 12.3.9 (2017: 12.8), Definition 12.4.3, Proposition 12.4.5, Theorem 12.4.6 (2017: Definition 12.10, Proposition 12.3, Theorem 12.10) and Theorem 12.5.7 (2017: 12.12).

What is verified. §12.1 and Definition 12.2.1 are verified as stated (V12.1–V12.3). Theorem 12.2.2, Definitions 12.2.3, 12.2.4, Proposition 12.2.5 and Theorem 12.2.6 are verified as stated (V12.4–V12.8); Theorem 12.2.7 is verified as printed (V12.9; gap G.136 closed in wave 2). Definition 12.3.1 is verified as printed, with the pointwise variant kept (V12.10); Theorem 12.3.2 (1) is proved in each consistent pairing of Definition 12.3.1 with Part I's Definition 5.3.1 (V12.11); Theorem 12.3.2 (2) and Theorem 12.3.3 are verified with the pointwise reading of bump functions (V12.12, V12.13); the checkable claims of Remarks 12.3.4 and 12.3.5 are proved (V12.14, V12.15). Definition 12.3.6 is verified at chart level and on ep-groupoids (V12.16; gap G.137 closed in wave 2), Proposition 12.3.7 in a stronger form (V12.17) and Corollary 12.3.8 with the isotropy action made explicit (V12.18). §12.4 up to Lemma 12.4.4 is verified as stated (V12.19–V12.21). In §12.5 Lemma 12.5.1, Definitions 12.5.2, 12.5.4 are verified with Part I's hypothesis of sc$^+$-sections through fibre points (V12.22, V12.23, V12.25); Theorem 12.5.3 is verified with Part I's hypotheses of Theorem 6.6.17, its last clause in Part III wave 5 (V12.24; gap G.138 closed, gap G.140); Proposition 12.5.6 is verified, (□) with the continuity of the lifted section in Part III wave 5 (V12.26; gap G.139 closed). Part III wave 2: the reflexive fibres are restated for ep-groupoids, Definitions 12.2.8, 12.2.10 and Theorem 12.2.11 are verified as stated and Remarks 12.2.9, 12.2.12 in stronger forms (V12.28–V12.33); Theorem 12.2.13 is verified with the paracompactness of $|X|$ named, the local step being proved in Part III wave 3 (V12.34; gap G.147 closed); Theorem 12.3.9 as stated (V12.35); Definition 12.4.3 as stated, Proposition 12.4.5 and Theorem 12.4.6 in stronger forms (V12.36–V12.38); Theorem 12.5.7 as a bijection of orientation spaces with a unique invariant push-forward, for one representative (V12.39; gap G.158); tools V12.40–V12.44. Part III wave 5: Theorem 12.5.7 (2) with the continuity of o' (V12.39); tools V12.45, V12.46. Part III waves 6–7: Theorem 12.5.7 (1) as a homeomorphism and (2) as printed, the independence of the representative and functoriality for the data of the diagrams (V12.39; gap G.158 closed, gap G.192); tools V12.47 (a Part I fact), V12.48.

The hypotheses. Theorem 12.2.7 has none: the hypothesis `Polyfold.EpStrongBundle.HasLocalExtensions` of the global part `theorem_12_2_7_hyp` is proved for tame X (`Polyfold.EpStrongBundle.hasLocalExtensions_of_isTame`, Part III wave 2). The pointwise reading of Definition 12.3.1, `Polyfold.EpGroupoid.AdmitsBumpFunctionsPointwise`, in Theorems 12.3.2 (2) and 12.3.3 (witness `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`). In §12.5, `Polyfold.StrongBundle.HasScPlusSectionsThrough` (a consequence of bump functions, `Polyfold.MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`), and, in Theorem 12.5.3, the hypotheses of Part I's Theorem 6.6.17 (bump functions on the object M-polyfold and tame germ data), inherited from Part I (gap G.140). The all-charts definitions of the boundary sections have the zero section as witness (`Polyfold.isBoundaryScPlusSectionOn_zero`, `Polyfold.proposition_12_3_7_nonvacuous`), and the isotropy action of Corollary 12.3.8 the trivial action (`Polyfold.LocalBundleAction.trivial`). Part III wave 2: `Polyfold.EpStrongBundle.HasReflexiveLocalExtensions` in Theorem 12.2.13 (witnesses `Polyfold.EpStrongBundle.theorem_12_2_13_hyp_witness`, `Polyfold.TrivialBundle.epBundle_reflexive_witness`); in Theorem 12.5.7 the standing `HasScPlusSectionsThrough` of §12.5 for the three bundles and the push-forward relation (witness `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward_witness`). Theorem 12.3.9 adds no hypothesis; all of its hypotheses hold at once for the trivial bundle (`Polyfold.EpStrongBundle.theorem_12_3_9_hypotheses_rn`). Part III wave 3: `Polyfold.EpStrongBundle.HasReflexiveLocalExtensions` is proved for tame X (`Polyfold.EpStrongBundle.hasReflexiveLocalExtensions_of_isTame`); `Polyfold.EpStrongBundle.theorem_12_2_13` names `h : X.HasParacompactOrbitSpace` (witness `Polyfold.TrivialBundle.theorem_12_2_13_witness`; Findings, item 19). The step lemmas of `Chapter12/P3W30_*` carry only proof data and the standing `hsec`. Part III wave 5 adds no hypothesis: `Polyfold.EpStrongBundle.theorem_12_5_3_push` and `Polyfold.EpStrongBundle.proposition_12_5_6_mpr_hyp` take Part I's hypotheses of Theorem 6.6.17 and the standing `hsec`; the paracompactness instances are discharged inside (Findings, items 21–23). Part III waves 6–7 add no hypothesis: Theorem 12.5.7 (1), (2) take Part I's hypotheses for X, X' and V and the standing `hsec`; the independence of the representative and functoriality take none (Findings, items 24, 25).

The setting. Ep-groupoids, natural representations and orbit spaces are those of Chapter 7; strong bundles over ep-groupoids, section functors and auxiliary norms those of Chapter 8 (`Polyfold.EpStrongBundle`, `Polyfold.EpStrongBundle.IsAuxNorm`, `Polyfold.EpStrongBundle.IsScFredholmSectionFunctor`); sc⁺-sections, linearizations, the determinant bundle and the orientation space O_f with its topology are Part I's (Chapters 5 and 6). Tame local strong bundle models are those of Chapter 2.

V12.1 (Definition 12.1.1 (2017: 12.1), sc-Fredholm section functors; `Polyfold.EpStrongBundle.definition_12_1_1`). `EpStrongBundle.definition_12_1_1`: `E.IsScFredholmSectionFunctor f   E.IsScSmoothSectionFunctor f   B.IsScFredholmSection f` (`Iff.rfl`, Definition 8.3.4 (3)). *Book:* §12.1, Definition 12.1.1, p. 569 (2017: Definition 12.1). *Hidden hypotheses:* none. *Deviation (same):* The project's existing definition, unfolded.

V12.2 (The remark after Definition 12.1.1: regularizing on objects and morphisms, germs; `Polyfold.EpStrongBundle.remark_12_1_1_regularizing_obj`, `Polyfold.EpStrongBundle.remark_12_1_1_regularizing_mor`, `Polyfold.EpStrongBundle.remark_12_1_1_germ`). Objects: $x \in X_m$ and $f(x) \in W_{m,m+1}$ give $x \in X_{m+1}$ (`remark_12_1_1_regularizing_obj`). Morphisms: $\varphi \in \mathbf{X}_m$ and $f(\varphi) = (\varphi, f(s\varphi)) \in (\mathbf{X}\,{}_s\!\times_P W)_{m,m+1}$ give $\varphi \in \mathbf{X}_{m+1}$ (`remark_12_1_1_regularizing_mor`, through the pull-back projection, a strong bundle map). Germs:

`IsScFredholmGerm` at each smooth x (`remark_12_1_1_germ`). *Book:* §12.1, after Definition 12.1.1, p. 569 (2017: after Definition 12.1). *Hidden hypotheses:* none. *Deviation (explicit):* "$f(\varphi)$" is read as the induced section of the pull-back bundle $\mathbf{X}\,{}_s\!\times_P W \to \mathbf{X}$.

V12.3 (Definition 12.2.1 (2017: 12.2), auxiliary norms; `Polyfold.EpStrongBundle.IsAuxNorm`). The existing `EpStrongBundle.IsAuxNorm` (Definition 8.3.5): `B.IsAuxNorm N` (Definition 5.1.1: continuity, (1), (2)) and invariance under μ (3). *Book:* §12.2, Definition 12.2.1, p. 570 (2017: Definition 12.2). *Hidden hypotheses:* none. *Deviation (same):* The text itself recalls Definition 8.3.5; no new declaration. Non-negativity follows from (1).

V12.4 (Theorem 12.2.2 (2017: 12.1), existence of auxiliary norms; `Polyfold.EpStrongBundle.theorem_12_2_2`, `Polyfold.SEP.theorem_12_2_2`, `Polyfold.EpStrongBundle.avgNorm`, `Polyfold.EpStrongBundle.isLocalAuxNorm_avgNorm`, `Polyfold.EpStrongBundle.avgNorm_actOne`, `Polyfold.EpStrongBundle.transNorm`, `Polyfold.EpStrongBundle.isLocalAuxNorm_transNorm`, `Polyfold.EpStrongBundle.transNorm_ `, `Polyfold.EpStrongBundle.glueNorm`, `Polyfold.EpStrongBundle.isAuxNorm_glueNorm`). `EpStrongBundle.theorem_12_2_2 (h : X.HasParacompactOrbitSpace) :  N, E.IsAuxNorm N`, and the form `SEP.theorem_12_2_2`. The proof follows the text's three steps: average a local norm over G_x on $U(x)$ (`avgNorm`), transport it to the saturation (`transNorm`, invariant under all morphisms by `transNorm_ `), glue with a partition of unity of $|X|$ (`isAuxNorm_glueNorm`). *Book:* §12.2, Theorem 12.2.2, p. 570 (proof pp. 572–574) (2017: Theorem 12.1). *Hidden hypotheses:* none. *Deviation (same):* The local norm is a Part I auxiliary norm (Proposition 5.1.3) instead of a chart norm, and the transport through a chosen morphism uses the invariance under G_x, so Theorem 12.2.6 (5) is not needed. Completeness of the glued fibre norms comes from `StrongBundle.isAuxNorm_of_tendsto` (V12.29).

V12.5 (Definition 12.2.3 (2017: 12.3) and the remark after it, continuous partitions of unity; `Polyfold.EpGroupoid.IsContPartitionOfUnity`, `Polyfold.EpGroupoid.isContPartitionOfUnity_comp_ `, `Polyfold.EpGroupoid.IsContPartitionOfUnity.descend`, `Polyfold.EpGroupoid.isSubordinate_comp_ `). `EpGroupoid.IsContPartitionOfUnity `, with fields `continuous`, `mem_Icc`, `map_mor`, `locallyFinite` and `sum_eq_one` (as `finsum`). Both directions of the remark: `isContPartitionOfUnity_comp_ ` (Mathlib `PartitionOfUnity` on the orbit space composed with π) and `IsContPartitionOfUnity.descend` (a `PartitionOfUnity` on $|X|$ with $\beta_i'(\pi y) = \beta_i(y)$). *Book:* §12.2, Definition 12.2.3, pp. 570–571 (2017: Definition 12.3). *Hidden hypotheses:* none. *Deviation (same):* The supports are closures in X. The paracompactness of $|X|$ in the preamble is not needed for the definition. The descent uses that π is open (Proposition 7.1.8).

V12.6 (Definition 12.2.4 (2017: 12.4), subordinate partitions of unity; `Polyfold.EpGroupoid.IsContPartitionOfUnity.IsSubordinate`). `IsContPartitionOfUnity.IsSubordinate X U:  i, tsupport ( i)  U i`. *Book:* §12.2, Definition 12.2.4, p. 571 (2017: Definition 12.4). *Hidden hypotheses:* none. *Deviation (same):* none.

V12.7 (Proposition 12.2.5 (2017: 12.1), subordinate continuous partitions of unity exist;

`Polyfold.EpGroupoid.proposition_12_2_5`). `EpGroupoid.proposition_12_2_5`: for `X.HasParacompactOrbitSpace` and saturated open U_i covering X there is a continuous partition of unity subordinate to (U_i). *Book:* §12.2, Proposition 12.2.5, p. 571 (2017: Proposition 12.1). *Hidden hypotheses:* none. *Deviation (same):* The text's proof: a partition of unity of $|X|$ subordinate to $(\pi(U_\lambda))$, composed with π; $|X|$ is normal by Theorem 7.3.1.

V12.8 (Theorem 12.2.6 (2017: 12.2), the good neighbourhood theorem; `Polyfold.EpGroupoid.theorem_12_2_6`, `Polyfold.EpGroupoid.exists_subseq_of_isProperMap`, `Polyfold.EpGroupoid.NaturalRep.image_Φ_eq`). `EpGroupoid.theorem_12_2_6 B x`: there are U', U'' and a natural representation `R : X.NaturalRep x` with `R.U = U''`, U' open, $x \in U'$, and (1)–(5): (2) $\Phi(g)(U') = U'$ for $g \in G_x$; (3) U'' lies in the base domain of one strong bundle chart compatible with the atlas; (4) with sequences and a strictly monotone reindexing. *Book:* §12.2, Theorem 12.2.6, pp. 571–572 (2017: Theorem 12.2). *Hidden hypotheses:* none. *Deviation (same):* The text's proof: properness of t on $s^{-1}(\overline{V_0})$, two shrinkings, and (5) from the uniqueness in Theorem 7.1.13 (3).

V12.9 (Theorem 12.2.7 (2017: 12.3), extension of auxiliary norms from the boundary; `Polyfold.EpStrongBundle.theorem_12_2_7`, `Polyfold.EpStrongBundle.hasLocalExtensions_of_isTame`, `Polyfold.StrongBundle.exists_localExtension_of_isTame`, `Polyfold.StrongBundle.isLocalAuxNorm_retractNorm`, `Polyfold.StrongBundle.retractNorm_eq`, `Polyfold.StrongBundle.retractNorm`, `Polyfold.MPolyfold.exists_retraction_boundary`, `Polyfold.boundaryShift`, `Polyfold.minCoord`, `Polyfold.EpStrongBundle.theorem_12_2_7_hyp`, `Polyfold.EpStrongBundle.theorem_12_2_7_hyp_witness`, `Polyfold.EpStrongBundle.IsAuxNormOn`, `Polyfold.EpStrongBundle.IsAuxNorm.isAuxNormOn`, `Polyfold.EpStrongBundle.HasLocalExtensions`, `Polyfold.EpStrongBundle.hasLocalExtensions_of_isAuxNorm`, `Polyfold.EpGroupoid.tgt_mem_boundary_iff`, `Polyfold.EpGroupoid.isOpen_compl_boundary`, `Polyfold.StrongBundle.IsLocalAuxNorm.mono`). `EpStrongBundle.theorem_12_2_7 (hX : X.IsTame) (h : X.HasParacompactOrbitSpace) (hN : E.IsAuxNormOn X N)`: there is $\tilde{N}$ with `E.IsAuxNorm Ñ` and $\tilde{N} = N$ over ∂X. It is `theorem_12_2_7_hyp` (the global part) with its hypothesis `HasLocalExtensions X B N` discharged by `hasLocalExtensions_of_isTame hX hN`. The local step is the Part I theorem `StrongBundle.exists_localExtension_of_isTame (hP : PX.IsTame)`: for N continuous on $W_{0,1}|\partial X$, a seminorm on the fibres over ∂X and with condition (2) for sequences in ∂X, every $x_0 \in \partial X$ has an open $V \ni x_0$ and a local auxiliary norm M on V with $M = N$ over $V \cap \partial X$. `IsAuxNormOn A N` is Definition 12.2.1 over A. *Book:* §12.2, Theorem 12.2.7, pp. 574–576 (2017: Theorem 12.3). *Hidden hypotheses:* none. *Deviation (same):* Hypotheses as printed (tame X, paracompact $|X|$, N an auxiliary norm over ∂X in the reading `IsAuxNormOn`). Readings of the local step: the retraction $r \circ q$ is built in a tame chart ψ of X (`MPolyfold.exists_retraction_boundary`: $\rho = \psi^{-1} r q \psi$, continuous on an open $V \ni x_0$, values in ∂X by tameness, identity on $V \cap \partial X$), with $\tau =$ min over the coordinates of a presentation of C (`minCoord`); the fibre formula is evaluated in an arbitrary strong bundle chart of the atlas around x_0, through $\Phi[1]$ (`retractNorm`), with $A(u)$ the fibre part of $R[1]$ (`chartFibMap`). The text's single local model with a tame base retraction is not needed; only the continuity of ρ and its two properties enter. Continuity of N' on $W_{0,1}$ uses the joint continuity of $(u, h) \mapsto A(u)h$

on $U \oplus F_1$ (level 0 of the sc-smooth R on $(U \lhd F)[1]$). Condition (2) for N': $N(Tw_k) \to 0$ gives $Tw_k \to 0_{\rho(y)}$ by (2) for N (applied to a tail of the sequence), hence $A(u_k)h_k \to 0$ in F_1, and $\|h_k - A(u_k)h_k\| \to 0$ gives $h_k \to 0$. The invariance of N is used only in the averaging step of `theorem_12_2_7_hyp`, which stays as the global part (`theorem_12_2_7_hyp`, with the witness `theorem_12_2_7_hyp_witness` of its hypothesis). Tame X is satisfiable (`isTame_trivial`).

V12.10 (Definition 12.3.1 (2017: 12.7), sc-smooth bump functions on ep-groupoids; `Polyfold.EpGroupoid.AdmitsBumpFunctions`, `Polyfold.EpGroupoid.AdmitsBumpFunctionsPointwise`, `Polyfold.EpGroupoid.AdmitsBumpFunctionsPointwise.weak`, `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`). `EpGroupoid.AdmitsBumpFunctions`, the printed form (`f  0`, `tsupport f  U`). The variant with $f(x) \neq 0$, the project's reading of Part I's Definition 5.3.1, is `AdmitsBumpFunctionsPointwise`; it implies the printed form (`AdmitsBumpFunctionsPointwise.weak`). Witness: the trivial ep-groupoid on $\mathbb{R}^n$ (`admitsBumpFunctionsPointwise_trivial_rn`). *Book:* §12.3, Definition 12.3.1, p. 579 (2017: Definition 12.7). *Hidden hypotheses:* none. *Deviation (same):* Part I's Definition 5.3.1 is read pointwise in this project (Findings, Chapter 5, items 41, 160–163); both readings are kept here.

V12.11 (Theorem 12.3.2 (1) (2017: 12.6 (1)), bump functions on X and on the object M-polyfold; `Polyfold.EpGroupoid.theorem_12_3_2_1`, `Polyfold.EpGroupoid.theorem_12_3_2_1_weak`, `Polyfold.EpGroupoid.exists_objBump_of_epBump`, `Polyfold.EpGroupoid.exists_epBump_of_objBump`, `Polyfold.EpGroupoid.exists_natAverage`, `Polyfold.EpGroupoid.exists_invariant_extension`). `EpGroupoid.theorem_12_3_2_1`: `X.AdmitsBumpFunctionsPointwise  X.PObj.AdmitsBumpFunctions` (pointwise on both sides). `theorem_12_3_2_1_weak`: `X.AdmitsBumpFunctions  X.PObj.AdmitsBumpFunctionsWeak` ("not identically 0" on both sides). No hypothesis. *Book:* §12.3, Theorem 12.3.2 (1), p. 579 (2017: Theorem 12.6 (1)). *Hidden hypotheses:* none. *Deviation (restated):* The printed statement pairs Definition 12.3.1 ("not identically 0") with Part I's Definition 5.3.1 (read pointwise); its (□) direction would then be Part I's open weak-to-pointwise question (Findings, Chapter 5, item 163), so the theorem is proved in each consistent reading. The text averages a bump function f over G_x, and that average can cancel to 0; we average f^2 (presumably intended, since Proposition 5.5.3's functions are ≥ 0). (□) uses the properness of cl V for closedness of the saturation.

V12.12 (Theorem 12.3.2 (2) (2017: 12.6 (2)), invariant bump functions with $f(x) = 1$; `Polyfold.EpGroupoid.theorem_12_3_2_2`, `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`). `EpGroupoid.theorem_12_3_2_2`: from `X.AdmitsBumpFunctionsPointwise`, an sc-smooth f with values in $[0, 1]$, $f(x) = 1$, `tsupport f  U` and $f(s\varphi) = f(t\varphi)$. Witness of the hypothesis: `admitsBumpFunctionsPointwise_trivial_rn`. *Book:* §12.3, Theorem 12.3.2 (2), p. 579 (2017: Theorem 12.6 (2)). *Hidden hypotheses:* the pointwise reading of Definition 12.3.1 ($f(x) \neq 0$), the project's reading of Part I; witness `admitsBumpFunctionsPointwise_trivial_rn`. *Deviation (narrower):* From the printed "not identically 0" reading the result would need Part I's open weak-to-pointwise question (Findings, Chapter 5, item 163). The proof is the text's: Proposition 5.5.3, then average, then extend.

V12.13 (Theorem 12.3.3 (2017: 12.7), sc⁺-section functors through invariant points; `Polyfold.`

`EpStrongBundle.theorem_12_3_3`, `Polyfold.EpStrongBundle.exists_equivariant_average`, `Polyfold.EpStrongBundle.natTerm`, `Polyfold.EpStrongBundle.natTerm_tgt`, `Polyfold.EpStrongBundle.exists_isScSmoothOn_one_secAct`, `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`). `EpStrongBundle.theorem_12_3_3`, with `X.AdmitsBumpFunctionsPointwise`, `X.HasParacompactOrbitSpace`, `E.IsAuxNorm N`, x smooth, e smooth with $P(e) = x$ and $\mu(g, e) = e$ for $g \in G_x$, O open with $\pi(x) \in O$, $\varepsilon > 0$: $e \in W_{0,1}$ and there is σ with `E.IsScPlusSectionFunctor` , support in $\pi^{-1}(O)$, $\sigma(x) = e$ and $N(\sigma(y)) \leq N(e) + \varepsilon$. *Book:* §12.3, Theorem 12.3.3, p. 581 (2017: Theorem 12.7). *Hidden hypotheses:* the pointwise reading of Definition 12.3.1, as in V12.12; witness `admitsBumpFunctionsPointwise_trivial_rn`. *Deviation (narrower):* Paracompactness is kept but unused: the text needs it only for the existence of N, which is a hypothesis. The local section through e is Part I's Lemma 5.3.3 (cut-off and bound included); averaging $\sigma'' = |G_x|^{-1} \sum_k h_k$ (`exists_equivariant_average`); the extension uses the morphisms (V12.15).

V12.14 (Remark 12.3.4 (2017: 12.3), functoriality and transversality compete; `Polyfold.EpStrongBundle.remark_12_3_4`). `EpStrongBundle.remark_12_3_4`: for every functorial section σ and $g \in G_x$, $\mu(g, \sigma(x)) = \sigma(x)$, so the invariance hypothesis of Theorem 12.3.3 is necessary. *Book:* §12.3, Remark 12.3.4, p. 584 (2017: Remark 12.3). *Hidden hypotheses:* none. *Deviation (explicit):* Only the checkable claim is formalized; the rest of the remark is commentary.

V12.15 (Remark 12.3.5 (2017: 12.4), extension of equivariant local sc⁺-sections; `Polyfold.EpStrongBundle.remark_12_3_5`, `Polyfold.EpStrongBundle.natExtension`, `Polyfold.EpStrongBundle.natExtension_eq`, `Polyfold.EpStrongBundle.natExtension_isFunctorial`, `Polyfold.EpStrongBundle.natExtension_eq_zero`). `EpStrongBundle.remark_12_3_5`: for `R : NaturalRep x`, `closure V   R.U` proper, σ sc⁺, equivariant on the morphisms of $U(x)$, with support in `closure V`: `natExtension E R` is an sc⁺-section functor equal to σ on $U(x)$, supported in the saturation of the support of σ, and every bound $N(\sigma) \leq b$ passes to it. *Book:* §12.3, Remark 12.3.5, p. 584 (2017: Remark 12.4). *Hidden hypotheses:* none. *Deviation (explicit):* The section σ plays the role of $f \cdot s$ (already cut off); "controlled" becomes the bound b.

V12.16 (Definition 12.3.6 (2017: 12.8), sc⁺-sections over the boundary: the local form; `Polyfold.retractBoundary`, `Polyfold.IsScPlusOnFaceNear`, `Polyfold.IsBoundaryScPlusSectionOn`, `Polyfold.mem_retractBoundary_iff`, `Polyfold.isBoundaryScPlusSectionOn_zero`, `Polyfold.StrongBundle.IsBoundaryScPlusSection`, `Polyfold.EpStrongBundle.IsBoundaryScPlusSection`, `Polyfold.EpStrongBundle.IsBoundaryScPlusSectionFunctor`, `Polyfold.EpStrongBundle.isBoundaryScPlusSectionFunctor_fiberZero`, `Polyfold.EpGroupoid.tgt_mem_boundary_iff_p3x`). In a tame local strong bundle model $K = R(U \lhd F) \to O$ with presentation T: `IsBoundaryScPlusSectionOn T C O K s D` (for $x \in D$ and $i \in I(x)$, `IsScPlusOnFaceNear T C O K s x i`); `retractBoundary E C O` is ∂O, with `mem_retractBoundary_iff`. The ep-groupoid form is `EpStrongBundle.IsBoundaryScPlusSection` (Definition 5.4.1 on the object M-polyfold; it does not involve μ), and the functor version used by Theorem 12.3.9 is `IsBoundaryScPlusSectionFunctor` (also $s(t\varphi) = \mu(\varphi, s(s\varphi))$ for $s(\varphi) \in \partial X$; then $t(\varphi) \in \partial X$, `EpGroupoid.tgt_mem_` `boundary_iff_p3x`). Witnesses: the zero section (`isBoundaryScPlusSectionOn_zero`, `isBoundaryScPlusSectionFunctor_fiberZero`). *Book:* §12.3, Definition 12.3.6, p. 584 (2017: Definition 12.8). *Hidden hypotheses:* none. *Deviation (restated):* The chart-level form is the content used by Proposition 12.3.7 and Corollary 12.3.8. The ep-groupoid form is the M-polyfold form on objects; the local faces are read chart-wise, as in Definition 5.4.1 (V5.88). The functor condition "on $W|\partial X$" is read on morphisms between boundary points.

V12.17 (Proposition 12.3.7 (2017: 12.2), local extension from the boundary; `Polyfold.proposition_12_3_7`, `Polyfold.proposition_12_3_7_of_faces_at`, `Polyfold.proposition_12_3_7_nonvacuous`, `Polyfold.boundaryExtension`, `Polyfold.boundaryExtension_toBase_snd`, `Polyfold.boundaryExtension_fst`, `Polyfold.boundaryExtension_mem`, `Polyfold.exists_isScSmooth_faceTerm`, `Polyfold.sum_powerset_erase_empty_alternating`, `Polyfold.cornerKill_insert_of_coord_eq_zero`). `proposition_12_3_7` (book form): for V_0 open, $x_0 \in \partial O \cap V_0$ and the face condition on $\partial O \cap V_0$, an open $V \ni x_0$ with $O \cap V$ a retract and $\tilde{s}$ sc$^+$ on $O \cap V$ equal to s on $\partial O \cap V$. Stronger: `proposition_12_3_7_of_faces_at` needs the face condition only at x_0, with the explicit extension `boundaryExtension`, which agrees with the text's formula (`boundaryExtension_toBase_snd`). Witness with $\partial O \neq \emptyset$: `proposition_12_3_7_nonvacuous`. *Book:* §12.3, Proposition 12.3.7, p. 585 (2017: Proposition 12.2). *Hidden hypotheses:* none. *Deviation (stronger):* Only the face germs at x_0 itself are used; $E = \mathbb{R}^n \oplus H$ is replaced by an arbitrary tame presentation T; $(t,h)_I = r(t_I, h)$ is `base RB (cornerKill T I 0 z)`; the cancellation is `sum_powerset_erase_empty_alternating`.

V12.18 (Corollary 12.3.8 (2017: 12.1), equivariant local extension from the boundary; `Polyfold.corollary_12_3_8`, `Polyfold.LocalBundleAction`, `Polyfold.LocalBundleAction.trivial`, `Polyfold.LocalBundleAction.average`, `Polyfold.mem_strongShiftOne_sum_smul`). `corollary_12_3_8` for a finite group G with `A : LocalBundleAction G C O K` (an action by sc-smooth maps of O with an sc-smooth, fibre-linear lift to $K[1]$ covering it, unit and composition laws), with `hfix` (G fixes x_0), $\partial O \cap V_0$ invariant, the face condition and equivariance of s there: an open invariant $V \ni x_0$ and $\widehat{s}$ (`LocalBundleAction.average` of the extension of Proposition 12.3.7), sc$^+$ on $O \cap V$, equivariant, equal to s on $\partial O \cap V$. Witness: `LocalBundleAction.trivial`. *Book:* §12.3, Corollary 12.3.8, p. 586 (2017: Corollary 12.1). *Hidden hypotheses:* none. *Deviation (explicit):* "G_{x_0} acts with a lift to K" is the structure `LocalBundleAction`; that the isotropy group fixes x_0 is the explicit `hfix`; "invariant" is read as equivariant, $\widehat{s}(gx) = g * \widehat{s}(x)$.

V12.19 (Definition 12.4.1 (2017: 12.9), compact solution set, proper, proper$_{<\infty}$; `Polyfold.EpStrongBundle.HasCompactSolutionSetEp`, `Polyfold.EpStrongBundle.IsProperFunctor`, `Polyfold.EpStrongBundle.IsProperLtInftyFunctor`). `EpStrongBundle.HasCompactSolutionSetEp`, `IsProperFunctor`, `IsProperLtInftyFunctor`, with Chapter 8's auxiliary norms and Part I's `AuxLe` for $N(f(x)) \leq C$. *Book:* §12.4, Definition 12.4.1, p. 587 (2017: Definition 12.9). *Hidden hypotheses:* none. *Deviation (same):* none.

V12.20 (Theorem 12.4.2 (2017: 12.9), the three compactness notions agree; `Polyfold.EpStrongBundle.theorem_12_4_2`, `Polyfold.EpStrongBundle.isProperLtInftyFunctor_of_hasCompactSolutionSetEp`, `Polyfold.EpStrongBundle.IsProperFunctor.`

`hasCompactSolutionSetEp`, `Polyfold.EpStrongBundle.IsProperLtInftyFunctor.isProperFunctor`, `Polyfold.EpStrongBundle.auxLe_tgt_of_auxLe_src`, `Polyfold.exists_nhds_isCompact_closure_auxLe`). `EpStrongBundle.theorem_12_4_2 (hf : E.IsScFredholmSectionFunctor f) (_hpara : X.HasParacompactOrbitSpace): List.TFAE [IsProperFunctor, HasCompactSolutionSetEp, IsProperLtInftyFunctor]`. *Book:* §12.4, Theorem 12.4.2, pp. 587–588 (2017: Theorem 12.9). *Hidden hypotheses:* none. *Deviation (same):* Paracompactness is kept but unused, since (1)–(3) quantify over all auxiliary norms. The proof's labels "(3)□ (2), (2)□ (1), (1)□ (3)" are presumably (3)□ (1), (1)□ (2), (2)□ (3), matching their arguments. The shrinking to natural neighbourhoods is not needed: the closure lies in a finite union of compact sets. The local step is Part I's Theorem 5.2.2.

V12.21 (Lemma 12.4.4 (2017: 12.1), closures in $|X|$ and in X; `Polyfold.EpGroupoid.lemma_12_4_4_1`, `Polyfold.EpGroupoid.lemma_12_4_4_2`, `Polyfold.EpGroupoid.isClosed_image_ _of_isSaturated`). `EpGroupoid.lemma_12_4_4_1 (hA : X.IsSaturated A) : closure (  '' A) =  '' closure A`; `lemma_12_4_4_2 (hA : IsCompact (closure A))`, the same conclusion; `isClosed_image_ _of_isSaturated`. *Book:* §12.4, Lemma 12.4.4, p. 589 (2017: Lemma 12.1). *Hidden hypotheses:* none. *Deviation (same):* Through π open (`IsOpenMap.preimage_closure_eq_closure_preimage`).

V12.22 (Lemma 12.5.1 (2017: 12.2), morphisms act on the linearizations; `Polyfold.EpStrongBundle.lemma_12_5_1_one`, `Polyfold.EpStrongBundle.lemma_12_5_1_two`, `Polyfold.EpStrongBundle.lemma_12_5_1_three_unit`, `Polyfold.EpStrongBundle.lemma_12_5_1_three_comp`, `Polyfold.EpStrongBundle.linMor`, `Polyfold.EpStrongBundle.morTx`, `Polyfold.EpStrongBundle.morKx`, `Polyfold.EpStrongBundle.morTx_comp`, `Polyfold.EpStrongBundle.orMor`, `Polyfold.EpStrongBundle.orMor_comp`, `Polyfold.EpStrongBundle.orMor_unit`, `Polyfold.EpStrongBundle.orMor_inv`, `Polyfold.StrongBundle.LocalIso.pullPt`, `Polyfold.StrongBundle.LocalIso.linSet_pullPt`, `Polyfold.EpStrongBundle.morIso`, `Polyfold.EpStrongBundle.eqOn_morIso`, `Polyfold.EpStrongBundle.germAgree_morIso_comp`). `lemma_12_5_1_one` (membership), `lemma_12_5_1_two` (`Set.BijOn (linMor …) (Lin x) (Lin x )`), `lemma_12_5_1_three_unit`, `lemma_12_5_1_three_comp`; on orientations `orMor` with `orMor_comp`, `orMor_unit`, `orMor_inv`. Lin is Part I's, read in chart points; $T\varphi$ and $\mu(\varphi, \cdot)$ are the chart identifications `morTx`, `morKx`, built from the local strong bundle isomorphism `morIso` of the sheet of φ. *Book:* §12.5, Lemma 12.5.1, p. 592 (2017: Lemma 12.2). *Hidden hypotheses:* `hsec : B.HasScPlusSectionsThrough` (sc$^+$-sections through smooth fibre points), Part I's standing hypothesis for $\mathrm{Lin}(f, x) \neq \emptyset$; it follows from bump functions (`MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`). *Deviation (explicit):* Lin is read in chart points, Part I's convention. "(1) and (3) give (2)" is proved directly, by conjugation with isomorphisms. The text calls the lemma obvious; the proof pushes $(f - s_0)'(x)$ by $\widehat{\Phi}$ to $(f - \widehat{\Phi}_* s_0)'(x')$ (`linSet_pullPt`, using $f = \Psi \circ f \circ \phi$ by functoriality) and gets functoriality from the germ identity of the local inverses (`germAgree_morIso_comp`).

V12.23 (Definition 12.5.2 (2017: 12.11), the orientation bundle O_f; `Polyfold.EpStrongBundle.definition_12_5_2`, `Polyfold.orProj_two`, `Polyfold.exists_other_or`, `Polyfold.`

`EpStrongBundle.OrRel`, `Polyfold.EpStrongBundle.orRel_equivalence`, `Polyfold.EpStrongBundle.orbitProj`, `Polyfold.EpGroupoid.SmoothRel`, `Polyfold.EpGroupoid.smoothRel_equivalence`). `EpStrongBundle.definition_12_5_2 := OrSpace f hF.2`, Part I's orientation space of f on the object M-polyfold, with `orProj`; exactly two points over each x (`orProj_two`, with `hsec`); the topology is Part I's, characterized in V12.24. The action of morphisms as a relation: `OrRel` (an equivalence, `orRel_equivalence`), the orbit map `orbitProj : |O_f| → |X_∞|`. *Book:* §12.5, Definition 12.5.2, p. 593 (2017: Definition 12.11). *Hidden hypotheses:* none. *Deviation (restated):* O_f and its topology are Part I's (germs continuous = Part I's `IsContOrOn`, via path transport); `orProj_two` needs `hsec` (as in V12.22). The relations `OrRel`, `SmoothRel` and the orbit spaces used in Proposition 12.5.6 are not defined in the text.

V12.24 (Theorem 12.5.3 (2017: 12.11), the topology of O_f; `Polyfold.EpStrongBundle.theorem_12_5_3_hyp`, `Polyfold.EpStrongBundle.theorem_12_5_3_push`, `Polyfold.EpStrongBundle.theorem_12_5_3_push_hyp`, `Polyfold.EpStrongBundle.pullOrx_morIso_tgt`, `Polyfold.EpStrongBundle.eqOn_morIso_of_functor`, `Polyfold.EpStrongBundle.pullOrx_morIso_eq_orMor`, `Polyfold.EpStrongBundle.germEq_morIso_sheet`, `Polyfold.StrongBundle.LocalIso.isContOrOn_pullOrx`, `Polyfold.StrongBundle.LocalIso.orxVia_comm`, `Polyfold.StrongBundle.LocalIso.orxVia_comm_local`, `Polyfold.StrongBundle.LocalIso.orxVia`, `Polyfold.StrongBundle.LocalIso.pullOrx`). `EpStrongBundle.theorem_12_5_3_hyp hX hbump hfT`: exactly one topology on O_f with `IsOrientationTopology` (Hausdorff, `orProj` a local homeomorphism, exactly two continuous sections near each point, propagation), and every fibre has two points; through Part I's `theorem_6_6_17` and `orProj_two`.

The last sentence (Part III wave 5): `EpStrongBundle.theorem_12_5_3_push hX hbump hfT hF hx hx' hs hs'`: for a morphism $\sigma : x \to x'$ between smooth points and a continuous orientation over an open $V \ni x$ (`IsContOrOn L V o`, Part I's transport L), there is an open $V' \ni x'$ over which the section carried by the local isomorphism of the sheet of σ (`(E.morIso  ).pullOrx … o`) is a continuous orientation, with value $\sigma_* o_x$ at x' (`orMor`). `theorem_12_5_3_push_hyp` is the form with Part I's instance arguments and the explicit open set $t(\text{sheet}) \cap \phi^{-1}(V)$; `pullOrx_morIso_eq_orMor`: the value of the carried section at y' is $\sigma'_* o_{s(\sigma')}$ for the sheet morphism σ' through y'. General form: `StrongBundle.LocalIso.isContOrOn_pullOrx` (a local strong bundle isomorphism between bundles over two M-polyfolds carries continuous orientations), from `LocalIso.orxVia_comm_local` and `orxVia_comm` (the transport commutes with the isomorphism along every sc-smooth path in its domain, by the dyadic induction `scPath_induction_or`). *Book:* §12.5, Theorem 12.5.3, p. 594 (2017: Theorem 12.11). *Hidden hypotheses:* `hbump : X.PObj.AdmitsBumpFunctions` and the tame germ data `hfT : B.IsScFredholmSectionTame f`, the hypotheses of Part I's Theorem 6.6.17, inherited (gap G.140); the text assumes a tame X only. *Deviation (narrower):* The last sentence is `theorem_12_5_3_push`, with Part I's hypotheses only (gap G.138 closed in Part III wave 5); "continuous local section" is read as Part I's `IsContOrOn` over an open set, whose section $y \mapsto (y, o_y)$ is continuous for Part I's topology (`continuousOn_orSec`); $\widehat{\Phi}$ is read as `E.morIso  `, the local isomorphism of the sheet of σ. The fibre bijection of a morphism and its functoriality are `orMor` (V12.22).

V12.25 (Definition 12.5.4 (2017: 12.12), orientations of sc-Fredholm functors; `Polyfold.`

`EpStrongBundle.definition_12_5_4`). `EpStrongBundle.definition_12_5_4 t o`: `Continuous[t] (orSec o)` and `orMor  (o x) = o x` for every morphism $\varphi : x \to x'$ between smooth points, for a topology `t` on O_f (Part I's, by V12.24). *Book:* §12.5, Definition 12.5.4, p. 594 (2017: Definition 12.12). *Hidden hypotheses:* none. *Deviation (same):* With Part I's hypotheses the topology is the unique one of `theorem_12_5_3_hyp`. "Smooth morphism" is read as a morphism with smooth source and target (equivalent by `mem_smoothSet_src_iff`).

V12.26 (Proposition 12.5.6 (2017: 12.4), orientability through $|\sigma|$; `Polyfold.EpStrongBundle.proposition_12_5_6_mp`, `Polyfold.EpStrongBundle.proposition_12_5_6_mpr_invariant`, `Polyfold.EpStrongBundle.IsTwoToOne`, `Polyfold.EpStrongBundle.proposition_12_5_6_mpr_hyp`, `Polyfold.EpStrongBundle.proposition_12_5_6_mpr_hyp'`, `Polyfold.EpStrongBundle.exists_nhds_orMor_eq_hyp`, `Polyfold.EpStrongBundle.isOpen_orSat_hyp`, `Polyfold.EpStrongBundle.orMor_isotropy_of_isTwoToOne`, `Polyfold.EpStrongBundle.eq_of_quot_eq_of_isTwoToOne`). `EpStrongBundle.proposition_12_5_6_mp`: an orientation (`definition_12_5_4 t o`) gives `IsTwoToOne` and a continuous section of `orbitProj`. `proposition_12_5_6_mpr_invariant`: from `IsTwoToOne` and any section s of $|\sigma|$, a section o of σ with $\varphi_* o_x = o_{x'}$ for all morphisms, lifting s. Orbit spaces: quotients by `SmoothRel` and `OrRel`.

(□) with continuity (Part III wave 5): `EpStrongBundle.proposition_12_5_6_mpr_hyp hX hbump hfT hF h  hs`: with Part I's hypotheses, `IsTwoToOne` and a continuous section s of $|\sigma|$ (quotient topologies on $|O_f|$ and $|X_\infty|$, Part I's `orTopology` on O_f), there is o with `definition_12_5_4 hF hsec (orTopology …) o` (continuous and invariant) and $|(x, o_x)| = s(|x|)$. `proposition_12_5_6_mpr_hyp'` is the form with instance arguments. Steps: `eq_of_quot_eq_of_isTwoToOne`, `orMor_isotropy_of_isTwoToOne` (G_x fixes the orientations at x), `exists_nhds_orMor_eq_hyp` (near x_0 every morphism maps a continuous $\hat{o}$ to itself; natural representation, finiteness of G_{x_0}, the last sentence of Theorem 12.5.3), `isOpen_orSat_hyp`. *Book:* §12.5, Proposition 12.5.6, p. 595 (2017: Proposition 12.4). *Hidden hypotheses:* `hbump`, the tame germ data `hfT` (Part I's, gap G.140, as in V12.24), and `hsec` as in V12.22 (taken as `hbump.hasScPlusSectionsThrough` in `proposition_12_5_6_mpr_hyp`). *Deviation (narrower):* (□) is complete; (□) with continuity is `proposition_12_5_6_mpr_hyp`, for Part I's topology, which by V12.24 is the unique orientation topology (gaps G.138, G.139 closed in Part III wave 5). The orbit spaces are quotients defined here.

V12.27 (Complete fibre norms are automatic (tool); `Polyfold.StrongBundle.isAuxNorm_of_tendsto`, `Polyfold.StrongBundle.exists_norm_liftOne_le_of_tendsto`, `Polyfold.StrongBundle.IsFiberSeminorm`, `Polyfold.StrongBundle.TendstoZeroOne`). `StrongBundle.isAuxNorm_of_tendsto`: a continuous fibrewise seminorm with property (2) of Definition 5.1.1 is an auxiliary norm, since near each point it is squeezed between multiples of the chart norm $|\cdot|_1$. *Book:* not in the book (used for Definition 5.1.1 (1) and Definition 12.2.1 (1)). *Hidden hypotheses:* none. *Deviation (extra):* none.

V12.28 (Reflexive 1-fibres for strong bundles over ep-groupoids (before Definition 12.2.8); `Polyfold.EpStrongBundle.HasReflexiveFibers`). `EpStrongBundle.HasReflexiveFibers E` is `B.HasReflexiveFibers`, Part I's Definition 5.1.4 (V5.77) for the object bundle $W \to X$. *Book:* §12.2, before Definition 12.2.8, p. 576 (2017: before Definition 12.5). *Hidden hypotheses:* none. *Deviation* *(restated):* The atlas condition is read as in Part I (V5.77): the reflexive compatible charts cover X.

V12.29 (Definition 12.2.8 (2017: 12.5), mixed convergence; `Polyfold.EpStrongBundle.MixedTendsto`, `Polyfold.EpStrongBundle.definition_12_2_8`). `EpStrongBundle.MixedTendsto E w y` is `B.MixedTendsto w y` (Part I's Definition 5.1.6, V5.79); `definition_12_2_8` unfolds it (`Iff.rfl`). *Book:* §12.2, Definition 12.2.8, p. 576 (2017: Definition 12.5). *Hidden hypotheses:* none. *Deviation (same):* None.

V12.30 (Remark 12.2.9 (2017: 12.1), mixed convergence is chart independent and preserved by morphisms; `Polyfold.EpStrongBundle.remark_12_2_9`, `Polyfold.EpStrongBundle.mixedTendsto_actOne`, `Polyfold.EpStrongBundle.exists_mixedTendsto_`, `Polyfold.StrongBundle.mixedTendsto_of_bounded`, `Polyfold.StrongBundle.MixedTendsto.exists_eventually_le`, `Polyfold.StrongBundle.exists_le_mul_auxNorm`). `EpStrongBundle.remark_12_2_9`: for $w_k \to^m w$ (and reflexive fibres): (i) weak convergence in every compatible chart around $P(w)$ with reflexive F_1; (ii) $w_k \to w$ in W; (iii) for every sc-smooth local section α of s on an open $S \ni P(w)$, $\mu(\alpha(Pw_k), w_k) \to^m \mu(\alpha(Pw), w)$ (`mixedTendsto_actOne`; pointwise form at a morphism `exists_mixedTendsto_`). *Book:* §12.2, Remark 12.2.9, pp. 576–577 (2017: Remark 12.1). *Hidden hypotheses:* none. *Deviation (stronger):* (i) is the remark (Part I's `MixedTendsto.weakTendsto_of_isChart`). (iii) is added: the same argument for the fibre maps of μ instead of a transition map (level-0 continuity of μ, a local bound $N_0 \circ \mu(\alpha, \cdot) \leq CN_0$ for an auxiliary norm, `exists_le_mul_auxNorm`, then "bounded and level-0 convergent implies mixed convergent", `mixedTendsto_of_bounded`). It keeps the lower semicontinuity through the transports in the proofs of Theorems 12.2.11 and 12.2.13.

V12.31 (Definition 12.2.10 (2017: 12.6), reflexive auxiliary norms on ep-groupoids; `Polyfold.EpStrongBundle.IsReflexiveAuxNorm`, `Polyfold.EpStrongBundle.IsReflexiveAuxNorm.toStrong`, `Polyfold.EpStrongBundle.isReflexiveAuxNorm_of_mixedLscAt`). `EpStrongBundle.IsReflexiveAuxNorm E N`: a structure extending `E.IsAuxNorm N` (Definition 12.2.1 = 8.3.5, continuity and (0)–(2)) with the field `le_liminf` (the liminf in `EReal`). `IsReflexiveAuxNorm.toStrong` gives Part I's Definition 5.1.7; `isReflexiveAuxNorm_of_mixedLscAt`. *Book:* §12.2, Definition 12.2.10, p. 577 (2017: Definition 12.6). *Hidden hypotheses:* none. *Deviation (same):* The printed $\liminf_{k\to 0}$ is presumably $k \to \infty$, as in Part I (V5.80).

V12.32 (Theorem 12.2.11 (2017: 12.4), existence of reflexive auxiliary norms; `Polyfold.EpStrongBundle.theorem_12_2_11`, `Polyfold.EpStrongBundle.theorem_12_2_11_exists`, `Polyfold.EpStrongBundle.theorem_12_2_11_mul`, `Polyfold.EpStrongBundle.theorem_12_2_11_sandwich`, `Polyfold.EpStrongBundle.mixedLscAt_avgNorm`, `Polyfold.EpStrongBundle.mixedLscAt_transNorm`, `Polyfold.EpStrongBundle.mixedLscAt_glueNorm`, `Polyfold.EpStrongBundle.isReflexiveAuxNorm_glueNorm`). `EpStrongBundle.theorem_12_2_11 E h hB` (the three clauses), with the parts `theorem_12_2_11_exists`, `theorem_12_2_11_mul`, `theorem_12_2_11_sandwich`. The hypotheses are exactly `X.HasParacompactOrbitSpace` and `E.HasReflexiveFibers`. *Book:* §12.2, Theorem 12.2.11, p. 577 (proof p. 578) (2017: Theorem 12.4). *Hidden hypotheses:* none. *Deviation (same):* The proof is the text's: average over G_x on a natural-representation domain, transport to the saturation, glue with a partition of unity of

$|X|$ (wave 1's `avgNorm`, `transNorm`, `glueNorm`). The local norm is a Part I reflexive auxiliary norm (Proposition 5.1.8) instead of $\|\mathrm{pr}_2 \circ \Psi(g * h)\|_{F_1}$. The lower semicontinuity is kept by the average (`mixedLscAt_avgNorm`), the transport (`mixedLscAt_transNorm`, through `mixedTendsto_actOne`) and the gluing (`mixedLscAt_glueNorm`: finitely many terms near $|P(w)|$, and terms with $\beta_\lambda(|P(w)|) = 0$ are ≥ 0). The sandwich: $N \leq (f \circ P)N'$ and $N' \leq (g \circ P)N$ by Remark 12.2.12 (2), then $N_1 = (f \circ P)^{-1}N$, $N_2 = (g \circ P)N$ by Remark 12.2.12 (1).

V12.33 (Remark 12.2.12 (2017: 12.2), scaling, comparison and cut-off of reflexive auxiliary norms; `Polyfold.EpStrongBundle.remark_12_2_12_1`, `Polyfold.EpStrongBundle.remark_12_2_12_2`, `Polyfold.EpStrongBundle.remark_12_2_12_3`, `Polyfold.StrongBundle.MixedLscAt`, `Polyfold.StrongBundle.MixedLscAt.mul_comp`). `remark_12_2_12_1` (as printed); `remark_12_2_12_2` (with $|X|$ paracompact: a continuous $f > 0$ with $f(s\varphi) = f(t\varphi)$ and $N_1 \leq (f \circ P)N_2$); `remark_12_2_12_3` (U open, $N \geq 0$ and mixed-lsc over U, $f \geq 0$ continuous: the liminf inequality for $\tilde{N} = \mathbf{1}_U f N$). *Book:* §12.2, Remark 12.2.12, pp. 577–578 (2017: Remark 12.2). *Hidden hypotheses:* none. *Deviation (stronger):* (1) and (2) as printed; (2) uses the paracompactness of $|X|$, which the remark's context has (local comparison by the bounds of Lemma 5.1.2, made invariant, glued by a partition of unity of $|X|$). (3) holds without the saturation of U, the functoriality of f and $\operatorname{supp} f \subset U$, which serve the continuity and functoriality of $\tilde{N}$, not the estimate: off U, $\tilde{N}(w) = 0 \leq \tilde{N}(w_k)$.

V12.34 (Theorem 12.2.13 (2017: 12.5), extension of reflexive auxiliary norms from the boundary; `Polyfold.EpStrongBundle.theorem_12_2_13`, `Polyfold.EpStrongBundle.hasReflexiveLocalExtensions_of_isTame`, `Polyfold.StrongBundle.exists_reflexiveLocalExtension_of_isTame`, `Polyfold.StrongBundle.mixedLscAt_retractNorm`, `Polyfold.StrongBundle.weakTendsto_chartFibMap`, `Polyfold.StrongBundle.exists_norm_chartFibMap_le`, `Polyfold.StrongBundle.toBase_chartFibMap`, `Polyfold.WeakTendsto.sub`, `Polyfold.TrivialBundle.theorem_12_2_13_witness`, `Polyfold.EpStrongBundle.theorem_12_2_13_hyp`, `Polyfold.EpStrongBundle.IsReflexiveAuxNormOn`, `Polyfold.EpStrongBundle.HasReflexiveLocalExtensions`, `Polyfold.EpStrongBundle.HasReflexiveLocalExtensions.hasLocalExtensions`, `Polyfold.EpStrongBundle.hasReflexiveLocalExtensions_of_isReflexiveAuxNorm`, `Polyfold.EpStrongBundle.theorem_12_2_13_hyp_witness`, `Polyfold.TrivialBundle.epBundle_reflexive_witness`). `EpStrongBundle.theorem_12_2_13 (hX : X.IsTame) (h : X.HasParacompactOrbitSpace) (hB : E.HasReflexiveFibers) (hN : E.IsReflexiveAuxNormOn X.PObj.boundary N) :  Ñ, E.IsReflexiveAuxNorm Ñ   w, P (B.oneVal w)  X.PObj.boundary → Ñ w = N w`. It is `theorem_12_2_13_hyp` with `hloc` discharged by `hasReflexiveLocalExtensions_of_isTame hX hB hN`. The local step is the Part I theorem `StrongBundle.exists_reflexiveLocalExtension_of_isTame (hP : PX.IsTame) (hB : B.HasReflexiveFibers)`. Suppose N is continuous over ∂X, a fibre seminorm over ∂X, satisfies condition (2) for sequences in ∂X, and is lsc along mixed convergent sequences in ∂X. Then every $x_0 \in \partial X$ has an open $V \ni x_0$ and a local auxiliary norm M on V such that: M is lsc along mixed convergent sequences (`MixedLscAt`) at every point over V; $M = N$ over $V \cap \partial X$. M is wave 2's `retractNorm` (E), taken in a strong bundle chart around x_0 with reflexive F_1. Such a chart is supplied by `HasReflexiveFibers`.

The global part is `theorem_12_2_13_hyp` (with `hloc : HasReflexiveLocalExtensions X B N`), now discharged; its witness `theorem_12_2_13_hyp_witness` stays. *Book:* §12.2, Theorem 12.2.13, p. 578 (2017: Theorem 12.5). *Hidden hypotheses:* none. *Deviation (explicit):* Hypotheses are the text's: reflexive 1-fibres, tame X, and N reflexive over ∂X in the reading `IsReflexiveAuxNormOn`. `h : X.HasParacompactOrbitSpace` is not printed in the statement. The proof says "partitions of unity, as in Theorem 12.2.11", and T12.2.11 assumes paracompact $|X|$. T12.2.7, whose construction is reused, states it too. The statement only names it. The "inspection" is `StrongBundle.mixedLscAt_retractNorm`. Let $w_k \to^m w$ with $P(w) \in V$, and write $u = \varphi(\rho(P(\cdot)))$ and $A(u) =$ `chartFibMap`. *Second term.* $h_k \rightharpoonup h$ in the reflexive F_1 of the chart, and $A(u_k)h_k \rightharpoonup A(u)h$ (`weakTendsto_chartFibMap`). The second fact holds because the sequence is bounded, by the local uniform bound `exists_norm_chartFibMap_le` from the joint continuity of $R[1]$, and converges in F_0, by the level-0 continuity of R and the compact $F_1 \to F_0$. So $\|h_k - A(u_k)h_k\|$ is weakly lsc. *First term.* The points $\Phi[1]^{-1}(u_k, A(u_k)h_k)$ lie over $\rho(\cdot) \in \partial X$ and are mixed convergent in the same chart. So the term is lsc by the lsc of N over ∂X. Tameness enters only through `MPolyfold.exists_retraction_boundary` (E). Witness for all hypotheses together: `TrivialBundle.theorem_12_2_13_witness n k`. It uses the tame trivial ep-groupoid of $\mathbb{R}^n$, paracompact $|X|$, the reflexive fibres of $\mathbb{R}^n \times \mathbb{R}^k$, and a reflexive auxiliary norm over ∂X.

V12.35 (Theorem 12.3.9 (2017: 12.8), extension of sc⁺-section functors from the boundary; `Polyfold.EpStrongBundle.theorem_12_3_9`, `Polyfold.EpStrongBundle.theorem_12_3_9_hypotheses_rn`, `Polyfold.EpStrongBundle.exists_extension_of_tameCharts`, `Polyfold.EpStrongBundle.exists_extension_bounded_of_tameCharts`, `Polyfold.EpStrongBundle.exists_local_functor_extension`, `Polyfold.EpStrongBundle.exists_equivariant_average_p3x`, `Polyfold.EpStrongBundle.isScPlusSectionFunctor_glueSection`, `Polyfold.EpStrongBundle.isFunctorialSection_glueSection`, `Polyfold.EpStrongBundle.isFunctorialSection_sectionSmul_p3x`, `Polyfold.EpStrongBundle.isScPlusSectionFunctor_fiberZero_p3x`, `Polyfold.glueSection`, `Polyfold.glueSection_eq_of`, `Polyfold.sectionSupport_glueSection_subset`, `Polyfold.isScPlusSection_glueSection`, `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_of_hasScPartitionsOfUnity`, `Polyfold.EpGroupoid.isSaturated_compl_boundary_p3x`). `EpStrongBundle.theorem_12_3_9 E (hB : B.IsTameBundle) (_hpara : X.HasParacompactOrbitSpace) (hpu : X.HasScPartitionsOfUnity) (hs : E.IsBoundaryScPlusSectionFunctor s) (hU : IsOpen U) (hUsat : X.IsSaturated U) (hbU : X  U)`: (1) there is $\tilde{s}$ with `E.IsScPlusSectionFunctor s̃`, `sectionSupport P s̃  U` and $\tilde{s} = s$ on ∂X; (2) for every `E.IsAuxNorm N`, $\delta > 0$ and $N(s(x)) < \delta$ on ∂X, the same with $N(\tilde{s}(x)) \leq \delta$ for all x. The cores `exists_extension_of_tameCharts`, `exists_extension_bounded_of_tameCharts` assume only that every object lies in a tame strong bundle chart. Witness of all hypotheses: `theorem_12_3_9_hypotheses_rn`. *Book:* §12.3, Theorem 12.3.9, pp. 586–587 (2017: Theorem 12.8). *Hidden hypotheses:* none. *Deviation (same):* Readings: "strong bundle over the tame ep-groupoid" is `B.IsTameBundle` for the object strong bundle, as in Theorem 5.4.2; the partitions of unity are Definition 7.5.5; "supported in U" is $\mathrm{cl}\{\tilde{s} \neq 0\} \subseteq U$. Paracompactness of $|X|$ is kept and unused, since the partitions of unity are assumed. The proof is the text's, with two differences: the local step averages Part I's local extension (Proposition 12.3.7 in a tame chart, cut off) over G_x in the natural representation, which is Corollary 12.3.8's formula, and extends to the saturation with `remark_12_3_5`; the average still extends s because s is functorial on ∂X. For the bound, the cover $\{V, X \setminus \partial X\}$ with $V = \{N(\tilde{s}) < \delta\}$ gives β with $\operatorname{supp} \beta \subseteq V$ and $\beta = 1$ on ∂X; the text's second neighbourhood V' is not needed. Bump functions on objects come from the ep partitions of unity (`admitsBumpFunctionsPointwise_of_hasScPartitionsOfUnity`, then `theorem_12_3_2_1`), which the text uses tacitly ("Using Theorem 12.3.2 (2)"); no bump hypothesis is added.

V12.36 (Definition 12.4.3 (2017: 12.10), the ep-groupoid reflexive local compactness property; `Polyfold.EpStrongBundle.HasReflexiveLocalCompactnessEp`, `Polyfold.EpStrongBundle.hasReflexiveLocalCompactnessEp_of_strong`). `EpStrongBundle.HasReflexiveLocalCompactnessEp E f`: for every `N` with `E.IsReflexiveAuxNorm N` and every `x` there is an open `U  x` with `IsCompact (closure {y  U | B.AuxLe N (f y) 1})`. `hasReflexiveLocalCompactnessEp_of_strong`: Part I's Definition 5.2.6 on the object bundle implies it. *Book:* §12.4, Definition 12.4.3, pp. 588–589 (2017: Definition 12.10). *Hidden hypotheses:* none. *Deviation (same):* The standing assumptions are hypotheses of the statements that use the definition. $N(f(y)) \leq 1$ is `AuxLe` ($N = \infty$ off $W_{0,1}$).

V12.37 (Proposition 12.4.5 (2017: 12.3), three forms of reflexive local compactness; `Polyfold.EpStrongBundle.proposition_12_4_5`, `Polyfold.EpStrongBundle.proposition_12_4_5_of_functorial`, `Polyfold.EpStrongBundle.isSaturated_setOf_auxLe`, `Polyfold.EpGroupoid.isCompact_closure_of_isProperMap`, `Polyfold.EpGroupoid.exists_lift_tendsto`, `Polyfold.EpGroupoid.isSaturated_inter`). `EpStrongBundle.proposition_12_4_5 (_hB) (hf : E.IsScSmoothSectionFunctor f) (_hreg)`: `List.TFAE [(1), (2), (3)]`; `proposition_12_4_5_of_functorial` for every section functor ($f(t\varphi) = \mu(\varphi, f(s\varphi))$). *Book:* §12.4, Proposition 12.4.5, p. 590 (2017: Proposition 12.3). *Hidden hypotheses:* none. *Deviation (stronger):* Reflexive fibres, sc-smoothness and the regularizing property are not used. (2) □ (3) by Lemma 12.4.4 (1) ($\{y \in \tilde{U} \mid N(f(y)) \leq 1\}$ is saturated, `isSaturated_setOf_auxLe`). (1) □ (3): $\tilde{U}$ is the saturation of $U(x)$, then Lemma 12.4.4 (2). (3) □ (1), "easily obtained" in the text, needs an argument: with $V_0 \ni x$ such that $t : s^{-1}(\mathrm{cl} V_0) \to X$ is proper, $U = \tilde{U} \cap V_0$; a set inside V_0 whose orbit set has compact closure has compact closure in X (`EpGroupoid.isCompact_closure_of_isProperMap`: orbit subsequence, lift of convergent sequences of $|X|$, `exists_lift_tendsto`, connecting morphisms, properness).

V12.38 (Theorem 12.4.6 (2017: 12.10), compactness control extended from the boundary; `Polyfold.EpStrongBundle.theorem_12_4_6`, `Polyfold.EpStrongBundle.theorem_12_4_6_of_continuous`). `EpStrongBundle.theorem_12_4_6` with the book's hypotheses (`_hT : X.IsTame`, `_hpara`, `hB`, `hN : E.IsReflexiveAuxNorm N`, `E.IsScFredholmSectionFunctor f`, `E.HasCompactSolutionSetEp f`, `E.HasReflexiveLocalCompactnessEp f`, $U_\partial = W' \cap \partial X$ with W' open, `X.IsSaturated Ub`, $S_f \cap \partial X \subseteq U_\partial$, the compactness). `theorem_12_4_6_of_continuous`: for every continuous section functor, with local compactness only at the given N. *Book:* §12.4, Theorem 12.4.6, pp. 590–592 (2017: Theorem 12.10). *Hidden hypotheses:* none. *Deviation (stronger):*

Tameness, paracompactness and the Fredholm property beyond continuity are not used. “Open in ∂X” is read as $W' \cap \partial X$. The proof is the text’s, in $|X|$: finitely many $|U(\bar{x}_i)|$ cover $\mathrm{cl}|U_\partial \cap \{N \circ f \leq 1\}|$; $\{N \circ f > 1\}$ is open (Proposition 5.2.7 (2)) and saturated; finitely many interior $|U(x_i) \setminus \partial X|$ cover the rest of $|S_f|$; U is the preimage of the union cut by $|W'| \cup |X \setminus \partial X|$, so $U \cap \partial X = U_\partial$.

V12.39 (Theorem 12.5.7 (2017: 12.12), naturality of the orientation bundle O_f; `Polyfold.theorem_12_5_7_one_hyp`, `Polyfold.theorem_12_5_7_two_invariant`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_bijective`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orbitProj_orbitOrMap`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_bijective`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullOr`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushOr`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pullOr`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pushOr`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushOr_obj`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pushOr_unique`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward_witness`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orEqv`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orEqv_apply`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orMor_orEqv`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.locIso`, `Polyfold.theorem_12_5_7_two_hyp`, `Polyfold.theorem_12_5_7_two_cont`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isContOrOn_pullOr_hyp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isContOrOn_pushOr_hyp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.pullOr_eq_pullOrx`, `Polyfold.StrongBundle.LocalIso.isContOrOn_of_pullOrx`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orEqv_eq_orxVia`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.germEq_locIso`, `Polyfold.theorem_12_5_7_one_homeo_hyp`, `Polyfold.theorem_12_5_7_one_homeo`, `Polyfold.theorem_12_5_7_two_book_hyp`, `Polyfold.theorem_12_5_7_two_book`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orbitOrHomeo`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitHomeo`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.continuous_orSpaceMap`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isOpenMap_orSpaceMap`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.continuous_smoothMap`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.isOpenMap_smoothMap`, `Polyfold.EpStrongBundle.isOpen_orSat`, `Polyfold.EpStrongBundle.isOpenMap_quotMk_or`, `Polyfold.EpGroupoid.isOpenMap_quotMk_smooth`, `Polyfold.MPolyfold.IsScDiffeoOn.smoothFwd`, `Polyfold.MPolyfold.IsScDiffeoOn.smoothBwd`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orEqv_comp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eT_comp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eK_comp`, `Polyfold.`

`EpStrongBundle.IsStrongBundleEquivalence.pullback_comp_of`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orSpaceMap_comp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_comp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_comp`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.compLocSet`, `Polyfold.StrongBundle.LocalIso.transferT_mpullPt` , `Polyfold.StrongBundle.LocalIso.transferK_mpullPt` , `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orEqv_nat`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.germ_nat`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eT_nat`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.eK_nat`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_nat`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_nat`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_refine`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_refine`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_refinement`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_refinement_base`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_rep_indep`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_rep_indep_base`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_functorial`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_functorial_base`, `Polyfold.isContOrOn_univ_iff_continuous`). For a representative $h_1 : V \to W$ over ϕ_1, $h_2 : V \to W'$ over ϕ_2 (two `IsStrongBundleEquivalence`s) with `hpb : f'' = h .pullback f`, `hpb : f'' = h . pullback f'` and sc-Fredholm `hF hF' hF''`: `theorem_12_5_7_one_hyp h h hF hF' hF'' hpb hpb hsec hsec' hsec''` gives bijections `F : |O_f|   |O_{f'}|` and `G : |X_∞|   |X'_∞|` with $|\sigma'| \circ F = G \circ |\sigma|$, $F \circ |\Phi_1|_* = |\Phi_2|_*$, $G \circ |\phi_1| = |\phi_2|$. `theorem_12_5_7_two_invariant … ho`: for o with `E.IsOrInvariant` there is exactly one invariant o' of σ' with $\mathrm{orEqv}_2\, o'_{\phi_2 v} = \mathrm{orEqv}_1\, o_{\phi_1 v}$ for every smooth $v \in V$. Steps: `orEqv a : Or^{f'}_{ a}   Or^f_a` (conjugation by the linearizations of the local strong bundle isomorphism at a), its naturality `orMor_orEqv`, `orbitOrMap` (bijective), `pullOr`/`pushOr`, `pushOr_unique`.

(2) with continuity (Part III wave 5): `theorem_12_5_7_two_hyp h h hF hF' hF'' hpb hpb hsec hsec' hsec'' hX hbump hfT hX' hbump' hfT' hX'' hbump'' hfT'' ho hoc`: for an invariant o with `IsContOrOn L univ o`, the unique invariant o' of `theorem_12_5_7_two_invariant` satisfies `IsContOrOn L' univ o'`; `theorem_12_5_7_two_cont` is the same without instance arguments. For one equivalence: `IsStrongBundleEquivalence.isContOrOn_pullOr_hyp` (`pullOr` keeps continuity), `isContOrOn_pushOr_hyp` (`pushOr` of an invariant continuous o is continuous on all of X', near $\phi(a)$ by `LocalIso.isContOrOn_of_pullOrx`, near any x' by a morphism and the last sentence of Theorem 12.5.3), `orEqv_eq_orxVia` (`orEqv b` is the orientation map of the local isomorphism at any a with $b \in U(a)$), `LocalIso.morVia_germ` (it depends only on the germ).

Part III wave 7 (`Chapter12/P3W7W3_*`). (1) as printed: `theorem_12_5_7_one_homeo_hyp h h hF hF' hF'' hpb hpb hsec hsec' hsec'' hX hbump hfT hX' hbump' hfT' hX'' hbump''` `hfT''`: with Part I's topology on O_f, $O_{f'}$ (`orTopology`), the quotient topologies on $|O_f|$, $|O_{f'}|$, $|X_\infty|$ and the topology of Theorem 2.3.10 on X_∞, there are homeomorphisms $F : |O_f| \simeq_t |O_{f'}|$ and $G : |X_\infty| \simeq_t |X'_\infty|$ with $|\sigma'| \circ F = G \circ |\sigma|$, $F \circ |\Phi_1|_* = |\Phi_2|_*$, $G \circ |\phi_1| = |\phi_2|$ (`theorem_12_5_7_one_homeo` without instance arguments). For one equivalence: `orbitOrHomeo`, `smoothOrbitHomeo`, from `continuous_orSpaceMap`, `isOpenMap_orSpaceMap`, `continuous_smoothMap`, `isOpenMap_smoothMap` (the local sc-diffeomorphisms `smoothFwd`/`smoothBwd` and the continuity of sc-smooth maps on X_∞, V12.47), `isOpen_orSat`, `isOpenMap_quotMk_or`. (2) as printed: `theorem_12_5_7_two_book_hyp … ho`, for o with `E.definition_12_5_4 hF hsec (orTopology …) o` there is exactly one o' with `E'.definition_12_5_4 … o'` corresponding to o over V; the continuity of Definition 12.5.4 is passed to and from `IsContOrOn L univ` by Part I's `continuous_orSec_iff` (V12.48). Functoriality: `orEqv_comp` ($\mathrm{orEqv}_{\Psi \circ \Phi}(a) = \mathrm{orEqv}_\Phi(a) \circ \mathrm{orEqv}_\Psi(\phi a)$, from `eT_comp`, `eK_comp` and the three-term transfer `transferT_mpullPt`), `orbitOrMap_comp`, and `theorem_12_5_7_functorial` (with `_base`): for $W \leftarrow V_1 \rightarrow W'$, $W' \leftarrow V_2 \rightarrow W''$ and a middle V_3 with equivalences to V_1, V_2 and a strong bundle natural transformation (Definition 10.4.8) as in Chapter 10's weak fibred product, the map of the composite is the composite of the maps. Independence of the representative: `orEqv_nat` (invariance under a natural equivalence $T : \Phi \simeq \Psi$, through `germ_nat`, `eT_nat`, `eK_nat`), `orbitOrMap_nat`, `theorem_12_5_7_refinement`, and `theorem_12_5_7_rep_indep` (with `_base`): two diagrams with a common refinement (Chapter 10's `SDiagram.Equiv`) give the same maps of $|O_f|$ and $|X_\infty|$. *Book:* §12.5, Theorem 12.5.7, pp. 595–597 (2017: Theorem 12.12). *Hidden hypotheses:* `hsec`, `hsec'`, `hsec''` (`HasScPlusSectionsThrough` for W, W', V), the standing hypothesis of §12.5 (V12.22), with the witnesses `MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough` and `StrongBundle.hasScPlusSectionsThrough_of_chart`; f' is any section with the same pull-back f'', and the push-forward is one (`pullback_pushforward_witness`); a representative is $h_1 = h_2 = \mathrm{id}$ (`isStrongBundleEquivalence_id`). *Deviation (narrower):* (1) and (2) hold as printed under Part I's hypotheses on X, X' and V (tame, bump functions, tame germ data, as in `theorem_12_5_3_hyp`; gap G.140) and the standing `hsec` (Part III wave 7; gap G.158 closed). The independence of the representative and the functoriality (left to the reader in the text) are proved for the data of the diagrams (a common refinement; a composite through equivalences with a natural equivalence), with no further hypotheses; since the underlying functions agree, so do the homeomorphisms of (1). The same statements for Chapter 10's `SDiagram` and `GenSBMap` objects are gap G.192.

V12.40 (Pointwise lower semicontinuity along mixed convergence and its algebra (tool); `Polyfold.StrongBundle.MixedLscAt`, `Polyfold.StrongBundle.MixedLscAt.le_liminf`, `Polyfold.StrongBundle.MixedLscAt.add`, `Polyfold.StrongBundle.MixedLscAt.const_mul`, `Polyfold.StrongBundle.mixedLscAt_finset_sum`, `Polyfold.EpStrongBundle.mixedLscAt_comp_actOne`). `StrongBundle.MixedLscAt B M y` (for $w_k \rightarrow^m y$ and $\varepsilon > 0$, eventually $M(y) - \varepsilon < M(w_k)$), with `.le_liminf`, `.add`, `.const_mul`, `.mul_comp`, `.congr`, `mixedLscAt_finset_sum`, `mixedLscAt_const`, `mixedLscAt_of_nonpos`, `isReflexiveAuxNorm_of_mixedLscAt`, `IsReflexiveAuxNorm.mixedLscAt`; ep-groupoid form `mixedLscAt_comp_actOne`. *Book:* not in the book (used for Definition 12.2.10 (3) and Theorems 12.2.11, 12.2.13). *Hidden hypotheses:* none. *Deviation (extra):* None.

V12.41 (The retraction onto the boundary near a boundary point of a tame M-polyfold (proof of Theorem 12.2.7); `Polyfold.MPolyfold.exists_retraction_boundary`, `Polyfold.boundaryShift`, `Polyfold.minCoord`, `Polyfold.one_le_degIndexWith_iff`, `Polyfold.coord_boundaryShift`, `Polyfold.boundaryShift_eq_self`, `Polyfold.continuous_boundaryShift`). `MPolyfold.exists_retraction_boundary (hP : P.IsTame) (hx : x  P.boundary)`: there are an open $V \ni x_0$ and ρ, continuous on V, with values in ∂X and $\rho = \mathrm{id}$ on $V \cap \partial X$. Coordinates: `minCoord`, `boundaryShift` (q) with `coord_boundaryShift`, `mem_quadrant_boundaryShift`, `exists_coord_boundaryShift_eq_zero`, `boundaryShift_eq_self`, `continuous_boundaryShift`. *Book:* §12.2, proof of Theorem 12.2.7, p. 575 (2017: Theorem 12.3). *Hidden hypotheses:* none. *Deviation (extra):* The text assumes $x_0 = (0, h_0)$; here x_0 is any boundary point (if some coordinates of x_0 are positive, q still maps into ∂C and fixes ∂C). $d_X = d_C$ in the tame chart by `degIndex_eq_chartDegIndex_of_isTameWith`.

V12.42 (The local norm N' for an arbitrary continuous retraction (proof of Theorem 12.2.7); `Polyfold.StrongBundle.isLocalAuxNorm_retractNorm`, `Polyfold.StrongBundle.retractNorm_eq`, `Polyfold.StrongBundle.retractNorm`, `Polyfold.StrongBundle.retractTransport`, `Polyfold.StrongBundle.chartFibMap`, `Polyfold.StrongBundle.isLinearMap_chartFibMap`, `Polyfold.MChart.tendsto_of_tendsto_apply`). `StrongBundle.isLocalAuxNorm_retractNorm hc y (hV : IsOpen V) (hVc : V  c..source) (h : ContinuousOn  V) (h A : MapsTo  V A) (h c) hNc hNs hN2 : B.IsLocalAuxNorm V (retractNorm hc y  N)` for any $A \subseteq X$ and N an auxiliary norm over A; `retractNorm_eq`: $N' = N$ over $V \cap A$ where $\rho = \mathrm{id}$. Fibre transport `retractTransport` (linear), $A(u) =$ `chartFibMap`. *Book:* §12.2, proof of Theorem 12.2.7, p. 575 (2017: Theorem 12.3). *Hidden hypotheses:* none. *Deviation (extra):* The formula of the text, read through $\Phi[1]$ in any strong bundle chart of the atlas, with $A(u)$ the fibre part of $R[1]$; continuity of N' uses the joint continuity of $(u, h) \mapsto A(u)h$ on $U \oplus F_1$.

V12.43 (The invariance clause of Definition 12.5.4 (tool); `Polyfold.EpStrongBundle.IsOrInvariant`, `Polyfold.EpStrongBundle.definition_12_5_4_iff`). `EpStrongBundle.IsOrInvariant E hF hsec o` (the morphism clause alone) and `definition_12_5_4_iff : definition_12_5_4 t o  Continuous (orSec o)  IsOrInvariant o` (`Iff.rfl`). *Book:* §12.5, Definition 12.5.4, p. 595 (2017: Definition 12.12). *Hidden hypotheses:* none. *Deviation (extra):* None; this names the second conjunct of V12.25.

V12.44 (Orientation transport through a local strong bundle isomorphism between different bundles (tool); `Polyfold.StrongBundle.LocalIso.mpullPt`, `Polyfold.StrongBundle.LocalIso.linSet_mpullPt`, `Polyfold.StrongBundle.LocalIso.morVia`, `Polyfold.StrongBundle.LocalIso.morVia_eq`, `Polyfold.StrongBundle.LocalIso.GermAgree`, `Polyfold.StrongBundle.LocalIso.transferT_mpullPt`, `Polyfold.StrongBundle.LocalIso.mpullPt_eq_pullPt`). `StrongBundle.LocalIso.mpullPt`, `linSet_mpullPt`, `transferT_mpullPt`, `transferK_mpullPt`, `morPull`, `morVia`, `morVia_apply`, `morVia_eq` (independent of the auxiliary chart point); two-step composition `GermAgree`, `transferT_mpullPt`, `transferK_mpullPt`; `mpullPt_eq_pullPt` (`rfl`) relates it to wave 1's `pullPt`. *Book:* §12.5, proof of Theorem 12.5.7, pp. 595–597 (2017: Theorem 12.12). *Hidden hypotheses:* none. *Deviation (extra):* None; it generalizes wave 1′s `pullPt` / `orVia` (from a bundle to itself) to maps between different bundles.

V12.45 (Continuity of an orientation is a local property (tool); `Polyfold.isContOrOn_of_local_or`, `Polyfold.IsContOrOn.mono_or`, `Polyfold.IsContOrOn.congr_or`, `Polyfold.scPath_induction_or`, `Polyfold.smoothTransPath_or`, `Polyfold.isScSmoothMap_smoothTransPath_or`, `Polyfold.smoothTransPath_or_mem`). `isContOrOn_of_local_or (hU : IsOpen U) (h :  x  U, x  smoothSet →  V, x  V  IsContOrOn L V o) : IsContOrOn L U o`, for any `LocalTransport` L; `IsContOrOn.mono_or`, `IsContOrOn.congr_or`. `scPath_induction_or`: a property of sc-smooth paths that holds on short paths and passes from the two halves to the whole path holds for every path (dyadic induction with a Lebesgue number); `smoothTransPath_or`: the reparametrization by `Real.smoothTransition`. *Book:* §12.5, used in the proofs of Theorems 12.5.3 and 12.5.7, pp. 594–597 (2017: Theorems 12.11, 12.12). *Hidden hypotheses:* none. *Deviation (extra):* None.

V12.46 (The orientation map of a local isomorphism depends only on its germ (tool); `Polyfold.StrongBundle.LocalIso.GermEq`, `Polyfold.StrongBundle.LocalIso.morVia_germ`, `Polyfold.StrongBundle.LocalIso.mmorT_germ`, `Polyfold.StrongBundle.LocalIso.mmorK_germ`, `Polyfold.EpGroupoid.Sheet.tgtInv_eq_near_or`). `StrongBundle.LocalIso.GermEq L L  b`: the base and bundle maps agree on an open $N \ni b$. `LocalIso.morVia_germ` (from `mmorT_germ`, `mmorK_germ`); `EpGroupoid.Sheet.tgtInv_eq_near_or`: two sheets through σ have the same $(t|)^{-1}$ near $t(\sigma)$. Companion of V12.44. *Book:* §12.5, implicit in the proofs of Theorems 12.5.3 and 12.5.7, pp. 594–597 (2017: Theorems 12.11, 12.12). *Hidden hypotheses:* none. *Deviation (extra):* None.

V12.47 (sc-smooth maps are continuous on the levels and on X_∞ (Part I tool, used for Theorem 12.5.7 (1)); `Polyfold.MPolyfold.continuousAt_levelLift`, `Polyfold.MPolyfold.IsScSmoothOn.continuousOn_level_comp`, `Polyfold.MPolyfold.IsScSmoothOn.continuousOn_smoothLevel`, `Polyfold.MPolyfold.IsScSmoothMap.continuous_smoothLevel`). `MPolyfold.IsScSmoothOn.continuousOn_level_comp`: if u is continuous into X_m and $v = f \circ u$ into Y_m, then v is continuous. `MPolyfold.IsScSmoothOn.continuousOn_smoothLevel`: the map $X_\infty \cap U \ \to \ Y_\infty$ is continuous; `MPolyfold.IsScSmoothMap.continuous_smoothLevel`; `MPolyfold.continuousAt_levelLift` (a chart is continuous on level m). A new Part I fact (Part III wave 7, `Chapter5/P3W7W3_SmoothLevelCont.lean`). *Book:* not in the book (used in §2.3 after Definition 2.3.8 and Theorem 2.3.10, and in §12.5). *Hidden hypotheses:* none. *Deviation (extra):* none.

V12.48 (Continuity of an orientation: Definition 12.5.4 and Part I's `IsContOrOn` agree (tool); `Polyfold.isOpen_setOf_orSec_eq`, `Polyfold.isContOrOn_univ_of_continuous_orSec`, `Polyfold.isContOrOn_univ_iff_continuous`). `isContOrOn_univ_of_continuous_orSec`: for any `LocalTransport L` with two orientations everywhere, on an M-polyfold with `HasConnectedSmoothNbhds`, continuity of `orSec  o` for `orTopology  L` gives `IsContOrOn L univ  o` (a clopen argument with `isOpen_setOf_orSec_eq` and `isContOrOn_of_local_or`). **`isContOrOn_univ_iff_continuous hX hbump hfT hsec ht`**: with Part I's hypotheses, `Continuous[t] (orSec  o)    IsContOrOn univ  o` for every topology `t` with `IsOrientationTopology`. *Book:* not in the book (Definition 12.5.4, p. 594, against Part I's continuity of orientations along paths). *Hidden hypotheses:* none. *Deviation (extra):* Neither hypothesis of `isContOrOn_univ_of_continuous_orSec` restricts: two orientations at every point hold under `hsec` (`exists_two_orx`), and every M-polyfold has connected smooth neighbourhoods (`MPolyfold.hasConnectedSmoothNbhds`).

What is not verified in this chapter

- p. 579, Theorem 12.3.2 (2) and p. 581, Theorem 12.3.3 (2017: 12.6 (2), 12.7): from the printed reading of Definition 12.3.1 they would need Part I's open question whether a bump function not identically 0 gives one non-zero at a given point (V12.12, V12.13; Findings, Chapter 5, item 163; Chapter 12, item 3).
- p. 594, Theorem 12.5.3 (2017: 12.11) for a tame X only, without Part I's hypotheses of Theorem 6.6.17 (gap G.140; Findings, item 10).
- p. 595, Remark 12.5.5 (2017: 12.5): prose on orientations of solution sets; no row.
- pp. 595–597, Theorem 12.5.7 (2017: 12.12): the restatement for Chapter 10's `SDiagram`/`GenSBMap` objects (V12.39; gap G.192; Findings, item 24).

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14. Chapter 13 (HWZ, Part III, §§13.1–13.5 (2017 §§13.1–13.5)): begun 2026-11-13 (Part III wave 1, continued in waves 2–5), verified for good systems of neighbourhoods (Theorem 13.1.7 as printed), the natural bi-action, the existence of good systems and Proposition 13.1.9; for the support, pointwise norm, symmetric structures and the decorated algebra of sc$^+$-multisections; for structurable sc$^+$-multisections with Proposition 13.3.19; for the pull-back and push-forward along equivalences and coverings (Theorems 13.4.1, 13.4.3, 13.4.4) and commensurable sums; and for atomic sc$^+$-multisections, Theorem 13.5.3 given a good system, Theorem 13.5.7, and Proposition 13.4.2

Chapter 13 (pp. 599–640; §§13.1–13.5 are pp. 599–635; numbers and pages are those of the book (Springer 2021), with the 2017 text's numbers in parentheses) is the theory of sc$^+$-multisections. §13.1 constructs good systems of neighbourhoods (Lemma 13.1.1, Definition 13.1.2, Proposition 13.1.3, Definitions 13.1.4, 13.1.5, Lemma 13.1.6, Theorems 13.1.7, 13.1.8, Proposition 13.1.9 (2017: Lemma 13.1, Definition 13.1, Proposition 13.1, Definitions 13.2, 13.3, Lemma 13.2, Theorems 13.1, 13.2, Proposition 13.2)). §13.2 defines sc$^+$-multisections (Definition 13.2.1 (2017: 13.4), formalized in Chapter 11), their support and pointwise norm (Definitions 13.2.2–13.2.4 (2017: 13.5–13.7)), symmetric structures (Definition 13.2.5, Lemma 13.2.6 (2017: Definition 13.8, Lemma 13.3)), decorated multisections (Definitions 13.2.7, 13.2.8 (2017: 13.9, 13.10)) and their algebra (Definitions 13.2.9, 13.2.10, Lemma 13.2.11, Definition 13.2.12, Lemma 13.2.13 (2017: Definitions 13.11, 13.12, Lemma 13.4, Definition 13.13, Lemma 13.5)). §13.3 (2017: §13.3) defines index correspondences and structurable sc$^+$-multisections (Definitions 13.3.1–13.3.17, Corollary 13.3.18, Proposition 13.3.19 (2017: Definitions 13.14–13.21, Corollary 13.1, Proposition 13.5)); §13.4 treats equivalences and coverings (Theorems 13.4.1, 13.4.3, 13.4.4, Proposition 13.4.2, Definition 13.4.5 (2017: Theorems 13.3–13.5, Proposition 13.6, Definition 13.22)); §13.5 constructs atomic sc$^+$-multisections (Definition 13.5.1, Theorems 13.5.3–13.5.7 (2017: Definition 13.23, Theorems 13.6–13.10)). §§13.2–13.3 and §13.5 were added in Part III wave 2, §13.4, Proposition 13.3.19 and Theorem 13.5.7 in waves 3–4, Proposition 13.4.2 in wave 5.

What is verified. In §13.1 Lemma 13.1.1 holds without the invariance of V (V13.1); Definition 13.1.2, Proposition 13.1.3, Definitions 13.1.4, 13.1.5 and Lemma 13.1.6 are verified with the reading of "$U(x)$ with the natural G_x-action" as an open invariant subset of the domain of a natural representation (V13.2–V13.6); Theorem 13.1.7 is verified as printed, with properness relative to Q, and in a stronger form (V13.7; gap G.141 closed in wave 2), Theorem 13.1.8 as stated (V13.8) and Proposition 13.1.9 in a stronger form (V13.9). In §13.2 Definitions 13.2.2–13.2.5, 13.2.12 and Lemma 13.2.6 are verified (V13.10–V13.14, V13.18); Definitions 13.2.9, 13.2.10 and Lemmas 13.2.11, 13.2.13 are verified at the level of sc$^+$-multisection functors and, in Part III wave 2, of decorations (V13.15–V13.17, V13.19), with Definitions 13.2.7, 13.2.8 (V13.22, V13.23) and the remark that inverses do not exist (V13.24) (gaps G.142, G.143 closed). Part III wave 2, §13.3: Definitions 13.3.1, 13.3.4, 13.3.6, 13.3.8, 13.3.10, 13.3.11, 13.3.17, Example 13.3.3, Remarks 13.3.5, 13.3.7, Lemma 13.3.14 and Corollary 13.3.18 are verified as stated, Definition 13.3.2 in a stronger form, Propositions 13.3.12, 13.3.15 restated, Remark 13.3.13 in its positive half (V13.25–V13.43); Proposition 13.3.19 is restated, for a nonempty family or given a good system (V13.50; gap G.148 closed in wave 3). §13.5: Definition 13.5.1 restated (V13.44); Theorem 13.5.3 as an equivalence with the existence of a good system, narrower since Chapter 13 has no standing paracompactness (V13.46; gap G.149 closed in wave 3); Theorems 13.5.4–13.5.6 with bump functions in the pointwise reading, Theorem 13.5.5 with $I \neq \emptyset$ (V13.47–V13.49); tools V13.21. Part III waves 3–4: Theorem 13.4.1 in a stronger form, Theorems 13.4.3, 13.4.4 as stated (V13.51–V13.53); Definition 13.4.5 (1) restated, (2) and the locally finite sum as stated (V13.54–V13.56); the reading of Definition 13.5.1 on a natural neighbourhood (V13.57); Theorem 13.5.7 in a stronger form (V13.58); the structurable clause of Theorem 13.5.5 given a good system (V13.48). Part III wave 5: Proposition 13.4.2 restated, with the relative data `Polyfold.SBLocalization.StructuredOn` (V13.59; gap G.163 closed).

The hypotheses. Theorem 13.1.7 has none: the printed hypothesis, a good system of the full subcategory on Q (`Polyfold.EpGroupoid.IsGoodIn`, witness `Polyfold.EpGroupoid.GoodSystem.isGoodIn_univ`), implies the project's `Polyfold.EpGroupoid.IsGoodOn` (witness `Polyfold.EpGroupoid.GoodSystem.isGoodOn`; every M-polyfold as a trivial ep-groupoid has a good system, `Polyfold.EpGroupoid.nonempty_goodSystem_trivial`). Natural neighbourhoods with proper closure exist inside every open set (`Polyfold.EpGroupoid.exists_natNbhd_isProperSubset`). §13.2 adds no hypothesis beyond the paracompactness of $|X|$ for the existence of decorations, which the text takes from Theorem 13.1.8 (witness `Polyfold.EpGroupoid.hasParacompactOrbitSpace_trivial`), and $c \neq 0$ in the example without inverse (`Polyfold.SBLocalization.NoInverse.exists_no_inverse`). §13.5: the existence of a good system in Theorems 13.5.3, 13.5.5 and Proposition 13.3.19 (`Nonempty A.X.GoodSystem`, necessary; witness `Polyfold.SBLocalization.nonempty_goodSystem_epSEP_zero`; from paracompactness by Theorem 13.1.8) (the earlier hypothesis `hcrit` of wave 2, witness `Polyfold.SBLocalization.theorem_13_5_3_hyp_witness`, is no longer needed), $I \neq \emptyset$ in Theorem 13.5.5 (witness `Polyfold.SBLocalization.SymmetricFamily.ofSeed`), and bump functions in the pointwise reading in Theorems 13.5.4, 13.5.6 (witness `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`). Structurable and atomic sc⁺-multisections exist (`Polyfold.SBLocalization.isStructurable_zeroMultisection`, `Polyfold.SBLocalization.isAtomic_zeroMultisection`). §13.4 adds no hypothesis, Proposition 13.4.2 included (its data are inhabited, `Polyfold.SBLocalization.StructuredTuple.toStructuredOn`). Readings: Findings, items 16–21.

The setting. Ep-groupoids, natural representations, proper subsets and orbit spaces are Chapter 7′s; strong bundles over ep-groupoids, auxiliary norms and the localization of strong bundles are Chapter 8′s and Chapter 10′s; sc⁺-multisection functors and local section structures are Chapter 11′s (`Polyfold.SBLocalization.IsScPlusMultisection`, `Polyfold.SBLocalization.IsLocalRepOn`).

V13.1 (Lemma 13.1.1 (2017: 13.1), closures in X and in $|X|$; `Polyfold.EpGroupoid.lemma_13_1_1`, `Polyfold.EpGroupoid.lemma_13_1_1_book`, `Polyfold.EpGroupoid.exists_natNbhd_isProperSubset`). `EpGroupoid.lemma_13_1_1`: for `N : X.NatNbhd x` with `X.IsProperSubset (closure N.U)` and any $V \subseteq N.U$, $\overline{V} \subseteq N.U \iff \overline{\pi(V)} \subseteq \pi(N.U)$. `lemma_13_1_1_book` adds the invariance hypothesis. *Book:* §13.1, Lemma 13.1.1, p. 599 (2017: Lemma 13.1). *Hidden hypotheses:* none. *Deviation (stronger):* The G_x-invariance of V is not used. (□) uses only properness (Proposition 7.1.10); (□) uses only the natural action (pigeonhole over the finite G_x). "$U(x)$ equipped with the natural G_x-action" is read as a `NatNbhd`: an open invariant subset of the domain of a natural representation, which need not be connected (Findings, Chapter 13, item 1).

V13.2 (Definition 13.1.2 (2017: 13.1), good systems of neighbourhoods; `Polyfold.EpGroupoid.NatNbhd`, `Polyfold.EpGroupoid.morBetween`, `Polyfold.EpGroupoid.IsDiffeoComponent`, `Polyfold.EpGroupoid.isDiffeoComponent_iff_injOn`, `Polyfold.EpGroupoid.GoodSystem`, `Polyfold.EpGroupoid.nonempty_goodSystem_trivial`). `EpGroupoid.GoodSystem`, with fields `nbhd :   x, X.NatNbhd x` (2), `isProperSubset_closure` (1) and `isDiffeoComponent` (3): for $\varphi \in \mathcal{U}(x, x')$, the component of φ in `morBetween U(x) U(x )` is an `IsDiffeoComponent` (s, t sc-diffeomorphisms onto their images, which are open automatically). For open components (3) is injectivity of s and t (`isDiffeoComponent_iff_injOn`). Witness: every M-polyfold, as a trivial ep-groupoid, has a good system (`nonempty_goodSystem_trivial`). *Book:* §13.1, Definition 13.1.2, p. 600 (2017: Definition 13.1). *Hidden hypotheses:* none. *Deviation (restated):* Condition (2) is read through `NatNbhd` (Findings, Chapter 13, item 1); components are indexed by a point in them.

V13.3 (Proposition 13.1.3 (2017: 13.1), invariant shrinkings of good systems; `Polyfold.EpGroupoid.proposition_13_1_3`, `Polyfold.EpGroupoid.GoodSystem.restrict`, `Polyfold.EpGroupoid.GoodSystem.exists_refinement_subset`, `Polyfold.EpGroupoid.NatNbhd.invariantCore`, `Polyfold.EpGroupoid.GoodSystem.isDiffeoComponent_of_subset`). `EpGroupoid.proposition_13_1_3`: for open $W(x) \ni x$ inside $U(x)$, invariant under the Φ_g of $U(x)$, there is a good system with $U'(x) = W(x)$ and the same natural representations (`GoodSystem.restrict`). `GoodSystem.exists_refinement_subset`: an invariant refinement inside $U(x) \cap V(x)$ (`NatNbhd.invariantCore`). *Book:* §13.1, Proposition 13.1.3, p. 600 (2017: Proposition 13.1). *Hidden hypotheses:* none. *Deviation (same):* Holds as printed with the `NatNbhd` reading; with "natural action on a connected domain", $\widetilde{U}(x)$ could fail to be the domain of a natural representation. (1) and (3) pass to subsets: closed subsets of proper sets are proper, and components of a smaller set lie in components of the larger one.

V13.4 (Definition 13.1.4 (2017: 13.2), refinements and subordinate good systems; `Polyfold.EpGroupoid.GoodSystem.IsRefinement`, `Polyfold.EpGroupoid.GoodSystem.IsSubordinate`, `Polyfold.EpGroupoid.GoodSystem.isRefinement_restrict`). `GoodSystem.IsRefinement   :=  x, .U x  .U x;` `GoodSystem.IsSubordinate  O :=  x,  i, .U x  O i;` `isRefinement_restrict`. *Book:* §13.1, Definition 13.1.4, p. 600 (2017: Definition 13.2). *Hidden hypotheses:* none. *Deviation (same):* none.

V13.5 (Definition 13.1.5 (2017: 13.3), the natural bi-action; `Polyfold.EpGroupoid.biact`, `Polyfold.EpGroupoid.src_biact`, `Polyfold.EpGroupoid.tgt_biact`, `Polyfold.EpGroupoid.biact_mem`, `Polyfold.EpGroupoid.biact_biact`, `Polyfold.EpGroupoid.isScSmoothOn_biact`, `Polyfold.EpGroupoid.biactMulAction`, `Polyfold.EpGroupoid.GoodSystem.biact`). `EpGroupoid.biact N N' g'   g` $= \Gamma'(g', t\varphi) \circ \varphi \circ \Gamma(g, \Phi_{g^{-1}}(s\varphi))$, with source $\Phi_{g^{-1}}(s\varphi)$ (`src_biact`), target $\Phi'_{g'}(t\varphi)$ (`tgt_biact`), values in $\mathcal{U}(x, x')$ (`biact_mem`), the action laws (`biact_biact`) and sc-smoothness of each map (`isScSmoothOn_biact`); packaged as a `MulAction` (`biactMulAction`). *Book:* §13.1, Definition 13.1.5, p. 601 (2017: Definition 13.3). *Hidden hypotheses:* none. *Deviation (restated):* The book names the map; the formula is the one used in the book's proofs. Convention: `biact g   g` is the book's $g' * \varphi * g$, with source $g^{-1} * s(\varphi)$.

V13.6 (Lemma 13.1.6 (2017: 13.2), with the discussion before it; `Polyfold.EpGroupoid.GoodSystem.image_biact_comp`, `Polyfold.EpGroupoid.GoodSystem.image_inv_comp`, `Polyfold.EpGroupoid.GoodSystem.components`, `Polyfold.EpGroupoid.GoodSystem.lemma_13_1_6_one`, `Polyfold.EpGroupoid.GoodSystem.lemma_13_1_6_two`, `Polyfold.EpGroupoid.GoodSystem.lemma_13_1_6_three_src`, `Polyfold.EpGroupoid.GoodSystem.lemma_13_1_6_three_tgt`, `Polyfold.EpGroupoid.GoodSystem.eq_one_of_biact_left_mem`, `Polyfold.EpGroupoid.GoodSystem.eq_one_of_biact_right_mem`, `Polyfold.EpGroupoid.GoodSystem.exists_biact_left_of_src`, `Polyfold.EpGroupoid.GoodSystem.exists_biact_right_of_tgt`). `GoodSystem.image_biact_comp` (components go to components; `image_inv_comp` for inversion; the components partition $\mathcal{U}(x, x')$). `lemma_13_1_6_one`, `lemma_13_1_6_two` from the pointwise forms `eq_one_of_biact_left_mem`, `eq_one_of_biact_right_mem` (one $\psi \in \Sigma$ with $g' * \psi \in \Sigma$ forces $g' = 1$); `lemma_13_1_6_three_src`, `_tgt` for the components of any two morphisms. *Book:* §13.1, Lemma 13.1.6, p. 602 (2017: Lemma 13.2), and the discussion before it. *Hidden hypotheses:* none. *Deviation (stronger):* (1) and (2) need only one $\psi \in \Sigma$ with $g' * \psi \in \Sigma$; (3) has no membership hypotheses (for $\varphi \notin \mathcal{U}(x, x')$ the component is empty).

V13.7 (Theorem 13.1.7 (2017: 13.1), extension of good partial systems; `Polyfold.EpGroupoid.theorem_13_1_7_book`, `Polyfold.EpGroupoid.IsGoodIn`, `Polyfold.EpGroupoid.IsProperSubsetIn`, `Polyfold.EpGroupoid.subSrc`, `Polyfold.EpGroupoid.subTgt`, `Polyfold.EpGroupoid.isGoodIn_iff`, `Polyfold.EpGroupoid.IsGoodIn.isGoodOn`, `Polyfold.EpGroupoid.IsGoodIn.closure_subset`, `Polyfold.EpGroupoid.closure_subset_of_isProperSubsetIn`, `Polyfold.EpGroupoid.isProperSubset_of_isProperSubsetIn`, `Polyfold.EpGroupoid.isProperSubsetIn_of_isProperSubset`, `Polyfold.EpGroupoid.GoodSystem.isGoodIn_univ`,

`Polyfold.EpGroupoid.theorem_13_1_7`, `Polyfold.EpGroupoid.IsTransport`, `Polyfold.EpGroupoid.injOn_comp_of_transport`, `Polyfold.EpGroupoid.exists_nhds_injOn_of_isProperSubset`, `Polyfold.EpGroupoid.exists_transport_of_isGoodOn`, `Polyfold.EpGroupoid.IsGoodOn`). Printed form: `EpGroupoid.theorem_13_1_7_book (hQ : IsOpen Q) (h Q : X.  '' Q = univ) N (hN : X.IsGoodIn N Q) W hW hyW :    : X.GoodSystem, (  q   Q,  .nbhd q = N q)     y   Q,  .U y   W y`. `IsGoodIn N Q`: for $q \in Q$, $U(q) \subseteq Q$; (1) read in the full subcategory on Q (`X.IsProperSubsetIn Q (closure (↑ ¹' U(q) : Set Q))`: $t : s^{-1}(\mathrm{cl}_Q U(q)) \cap \mathrm{Mor}(Q) \to Q$ proper, through `subSrc`, `subTgt` on `X.morOn Q`); (3) for $q, q' \in Q$. `isGoodIn_iff (hQ) (h Q) : X.IsGoodIn N Q   X.IsGoodOn N Q    q   Q, closure (N q).U   Q`. Stronger form: `EpGroupoid.theorem_13_1_7` for any Q with $\pi(Q) = |X|$ and `X.IsGoodOn N Q` (properness in X): there is a good system with ` .nbhd q = N q` for $q \in Q$ and $U(y) \subseteq W(y)$ for $y \notin Q$. Witnesses: `GoodSystem.isGoodOn`, `GoodSystem.isGoodIn_univ`. *Book:* §13.1, Theorem 13.1.7, p. 602 (2017: Theorem 13.1). *Hidden hypotheses:* none. *Deviation (stronger):* `theorem_13_1_7_book` is the printed statement. The project's `theorem_13_1_7` takes the weaker `IsGoodOn N Q` (properness in X; implied by the printed hypothesis, `IsGoodIn.isGoodOn`) and needs neither Q open nor $\mathrm{cl}_X U(q) \subseteq Q$, so the formalization is stronger. Proof of Q-proper □ X-proper (`isProperSubset_of_isProperSubsetIn`), by ultrafilters: a local section σ of s through a morphism $\chi : z \to q'' \in Q$ (it exists since $|Q| = |X|$) with $t \circ \sigma \in Q$ first shows $\mathrm{cl}_X U \subseteq Q$, then moves φ to $\sigma(t\varphi) \circ \varphi$ in $\mathrm{Mor}(Q)$ and back by $\chi^{-1} \circ \rho$. No sub-ep-groupoid structure is needed: the full subcategory is described by its morphism set and its source and target maps. Reading kept from wave 1: conditions (2), (3) are the same in X and in Q (a natural representation of the ep-groupoid Q is one of X). The natural action and $\mathcal{U}(q, q')$ are the same in Q and in X. The proof of `theorem_13_1_7` does not use the book's smallness condition on the closures; it uses transports carrying $U(y)$ into $U(q_y)$.

V13.8 (Theorem 13.1.8 (2017: 13.2), existence of good systems; `Polyfold.EpGroupoid.theorem_13_1_8`). `EpGroupoid.theorem_13_1_8 (hX : X.HasParacompactOrbitSpace) W hW hxW :  : X.GoodSystem,   x,  .U x   W x`. *Book:* §13.1, Theorem 13.1.8, p. 604 (2017: Theorem 13.2). *Hidden hypotheses:* none. *Deviation (same):* The book's "for all $\in X$" is read as "for all $x \in X$". The proof uses paracompactness of $|X|$ through a precise refinement and the shrinking lemma; metrizability is not used.

V13.9 (Proposition 13.1.9 (2017: 13.2), with the stabilisers and D_Σ before it; `Polyfold.EpGroupoid.GoodSystem.stabSrc`, `Polyfold.EpGroupoid.GoodSystem.stabTgt`, `Polyfold.EpGroupoid.GoodSystem.D`, `Polyfold.EpGroupoid.GoodSystem.isScDiffeoOn_D`, `Polyfold.EpGroupoid.GoodSystem.proposition_13_1_9`, `Polyfold.EpGroupoid.GoodSystem.proposition_13_1_9_D`, `Polyfold.EpGroupoid.GoodSystem.partner`, `Polyfold.EpGroupoid.GoodSystem.partnerHom`, `Polyfold.EpGroupoid.GoodSystem.srcInv`). `GoodSystem.stabSrc`, `stabTgt` (subgroups), `GoodSystem.D` with `isScDiffeoOn_D`. `GoodSystem.proposition_13_1_9 (h  :      .mor x x )`: `f : stabSrc  * stabTgt` with $\mathrm{srcInv}(\Phi_g z) = \mathrm{biact}(fg)(\mathrm{srcInv}\, z)\, g^{-1}$ on $s(\Sigma)$, and every map with this property equals f; `proposition_13_1_9_D`: $\Phi'_{g'(g)} \circ D_\Sigma = D_\Sigma \circ \Phi_g$ on $s(\Sigma)$. *Book:* §13.1, Proposition 13.1.9, p. 607 (2017: Proposition 13.2), and the discussion before it, p. 606. *Hidden hypotheses:* none. *Deviation (stronger):* Uniqueness holds among all maps, not only homomorphisms. The proof goes through Lemma 13.1.6 and the bi-action laws, not through smoothness of Θ_g.

V13.10 (Definition 13.2.2 (2017: 13.5), domain support and support; `Polyfold.SBLocalization.supp`, `Polyfold.SBLocalization.suppAt`, `Polyfold.SBLocalization.domSupp`, `Polyfold.SBLocalization.IsLocalRepOn.suppAt_eq`, `Polyfold.SBLocalization.IsScPlusMultisection.finite_suppAt`, `Polyfold.SBLocalization.IsScPlusMultisection.sum_suppAt`, `Polyfold.SBLocalization.IsScPlusMultisection.suppAt_nonempty`, `Polyfold.SBLocalization.domSupp_eq_closure_image`). `SBLocalization.domSupp Λ = closure {x |   w, P w = x   w   0_x   0 < Λ w}`, `supp`, `suppAt`; the claims: `IsLocalRepOn.suppAt_eq`, `IsScPlusMultisection.finite_suppAt`, `sum_suppAt`, `suppAt_nonempty`, and `domSupp_eq_closure_image` (the first relation after Definition 13.2.3). *Book:* §13.2, Definition 13.2.2, p. 608 (2017: Definition 13.5). *Hidden hypotheses:* none. *Deviation (same):* At functor level: Λ is an `IsScPlusMultisection` (Chapter 11's reading of Definition 13.2.1); the decorated objects $[\Lambda, U, S]$ are V13.22, V13.23 (gap G.142 closed).

V13.11 (Definition 13.2.3 (2017: 13.6), the pointwise norm $N(\Lambda)$; `Polyfold.SBLocalization.normW`, `Polyfold.SBLocalization.pointwiseNorm`, `Polyfold.SBLocalization.IsScPlusMultisection.suppAt_subset_oneSet`, `Polyfold.SBLocalization.IsScPlusMultisection.isGreatest_pointwiseNorm`, `Polyfold.SBLocalization.IsScPlusMultisection.domSupp_eq_closure_pointwiseNorm`). `SBLocalization.pointwiseNorm N Λ x = sSup (normW N '' suppAt Λ x)`, with the supremum attained (`isGreatest_pointwiseNorm`); both relations for every auxiliary norm (`domSupp_eq_closure_pointwiseNorm`). *Book:* §13.2, Definition 13.2.3, p. 608 (2017: Definition 13.6). *Hidden hypotheses:* none. *Deviation (restated):* The maximum is written as a supremum with an attainment lemma; attainment uses that sc⁺-section values lie in $W_{0,1}$.

V13.12 (Definition 13.2.4 (2017: 13.7), trivial on V; `Polyfold.SBLocalization.IsTrivialOn`, `Polyfold.SBLocalization.IsScPlusMultisection.apply_fiberZero_eq_one_iff`, `Polyfold.SBLocalization.IsScPlusMultisection.isLeast_domSupp`). `SBLocalization.IsTrivialOn Λ V :=   x   V, Λ (0_x) = 1`; the "hence" clause is `apply_fiberZero_eq_one_iff` (an iff); the remark is `isLeast_domSupp`. *Book:* §13.2, Definition 13.2.4, p. 609 (2017: Definition 13.7). *Hidden hypotheses:* none. *Deviation (same):* none.

V13.13 (Definition 13.2.5 (2017: 13.8), symmetric sc⁺-section structures; `Polyfold.SBLocalization.SymmetricStructure`, `Polyfold.SBLocalization.prodIsotropyAction`). `SBLocalization.SymmetricStructure A Λ x`, with a natural representation `R` ($U = R.U$), the properness property, a finite index type `I` with a `MulAction` of the isotropy group, sections `s`, sc⁺-regularity, equivariance and the representation of Λ. *Book:* §13.2, Definition 13.2.5, p. 610 (2017: Definition 13.8). *Hidden hypotheses:* none. *Deviation (restated):* "G_x-invariant with the natural action" is read as "$U(x)$ is the domain of a natural representation", Chapter 11's reading of Definition 13.2.1, which includes connectedness; properness is the existing `HasPropernessProperty`; weights are in $\mathbb{Q}_{\geq 0}$.

V13.14 (Lemma 13.2.6 (2017: 13.3), symmetric structures exist locally; `Polyfold.SBLocalization.transportSection`, `Polyfold.SBLocalization.isScPlusSectionOn_transportSection`, `Polyfold.SBLocalization.IsLocalRepOn.exists_equal_weights`, `Polyfold.SBLocalization.lemma_13_2_6`, `Polyfold.SBLocalization.extendZero`, `Polyfold.SBLocalization.IsLocalRepOn.extendZero`). `SBLocalization.lemma_13_2_6 (hΛ : IsScPlusMultisection A Λ) (x) : Nonempty (SymmetricStructure A Λ x)`, by the text's construction: shrink a local representation to a natural representation with the properness property, equalise the weights (`IsLocalRepOn.exists_equal_weights`), and transport by G_x (`transportSection`, index set $I \times G_x$). *Book:* §13.2, Lemma 13.2.6, p. 610 (2017: Lemma 13.3). *Hidden hypotheses:* none. *Deviation (same):* Before transporting, the sections are extended by 0 off U, so that sections are global functions.

V13.15 (Definition 13.2.9 (2017: 13.11), the sum $\Lambda' \oplus \Lambda''$; `Polyfold.SBLocalization.addAt`, `Polyfold.SBLocalization.multisectionAdd`, `Polyfold.SBLocalization.multisectionAdd_eq_conv`, `Polyfold.SBLocalization.multisectionAdd_eq_sum_of_isLocalRepOn`, `Polyfold.SBLocalization.definition_13_2_9`, `Polyfold.SBLocalization.multisectionAdd_comm`, `Polyfold.SBLocalization.multisectionAdd_assoc`, `Polyfold.SBLocalization.convAt`, `Polyfold.SBLocalization.conv`, `Polyfold.SBLocalization.isLocalRepOn_conv`, `Polyfold.SBLocalization.isScPlusMultisection_conv`, `Polyfold.SBLocalization.IsScPlusSectionOn.sectionAdd`, `Polyfold.EpGroupoid.NatNbhd.Le`, `Polyfold.EpGroupoid.NatNbhd.exists_le_of_finite`, `Polyfold.SBLocalization.fibAct`, `Polyfold.SBLocalization.fibAct_sectionAdd`, `Polyfold.SBLocalization.SymmetricStructureOn.restrict`, `Polyfold.SBLocalization.SymmetricStructureOn.add`, `Polyfold.SBLocalization.Decoration.exists_common`, `Polyfold.SBLocalization.Decoration.add`, `Polyfold.SBLocalization.Decoration.add_I`, `Polyfold.SBLocalization.Decoration.add_s_apply`, `Polyfold.SBLocalization.Decoration.add_U_subset`, `Polyfold.SBLocalization.Decoration.add_rel`, `Polyfold.SBLocalization.Decoration.add_comm_rel`, `Polyfold.SBLocalization.Decoration.add_assoc_rel`, `Polyfold.SBLocalization.definition_13_2_9_decorated`, `Polyfold.SBLocalization.definition_13_2_9_decorated_equiv`, `Polyfold.SBLocalization.DecoratedMultisection.add`, `Polyfold.SBLocalization.DecoratedMultisection.forget_add`, `Polyfold.SBLocalization.DecoratedMultisection.add_comm`, `Polyfold.SBLocalization.DecoratedMultisection.add_assoc`). `SBLocalization.multisectionAdd`, (1) verbatim, is an sc$^+$-multisection (`definition_13_2_9`); local form `multisectionAdd_eq_sum_of_isLocalRepOn`; at functor level `multisectionAdd_comm`, `multisectionAdd_assoc`; the n-fold sum `conv`. Decorated: `Decoration.add` with `add_I` ($I_x = I'_x \times I''_x$), `add_s_apply` ($s_{(i,i')}(y) = s'_i(y) + s''_{i'}(y)$), `add_U_subset` ($U(x) \subseteq U'(x) \cap U''(x)$), `definition_13_2_9_decorated`; the sum respects Definition 13.2.8 (`Decoration.add_rel`), so it is defined on decorated sc$^+$-multisections (`DecoratedMultisection.add`, `forget_add`), commutative and associative there (`DecoratedMultisection.add_comm`, `add_assoc`). *Book:* §13.2, Definition 13.2.9, p. 611 (2017: Definition 13.11). *Hidden hypotheses:* none. *Deviation (restated):* (1) and the closure under $\oplus$ are as printed. $U(x) = U'(x) \cap U''(x)$ is read as an open neighbourhood of x inside the intersection with the natural action of $U'(x)$, on which the actions of $U'(x)$ and

$U''(x)$ agree (`NatNbhd.exists_le_of_finite`): the intersection itself need not carry one natural action for which both structures are symmetric. The class does not depend on the choice, since Definition 13.2.8 allows shrinking. The local section structures of the decoration are symmetric with the diagonal action $g(i, i') = (g(i), g(i'))$.

V13.16 (Definition 13.2.10 (2017: 13.12), scalar multiples $a\Lambda$; `Polyfold.SBLocalization.smulAt`, `Polyfold.SBLocalization.multisectionSmul`, `Polyfold.SBLocalization.multisectionSmul_zero`, `Polyfold.SBLocalization.multisectionSmul_eq_sum_of_isLocalRepOn`, `Polyfold.SBLocalization.definition_13_2_10`, `Polyfold.SBLocalization.IsScPlusSectionOn.sectionSmul`, `Polyfold.SBLocalization.SymmetricStructureOn.smulSection`, `Polyfold.SBLocalization.SymmetricStructureOn.smulSection_apply`, `Polyfold.SBLocalization.fibAct_sectionSmul`, `Polyfold.SBLocalization.SymmetricStructureOn.smul`, `Polyfold.SBLocalization.Decoration.smul`, `Polyfold.SBLocalization.definition_13_2_10_decorated`, `Polyfold.SBLocalization.DecoratedMultisection.smul`). `SBLocalization.multisectionSmul a Λ`, exactly the printed cases; 0Λ is the zero multisection (`multisectionSmul_zero`); $a\Lambda$ is an sc⁺-multisection (`definition_13_2_10`); local sections as_i with the same weights (`multisectionSmul_eq_sum_of_isLocalRepOn`). Decorated: `Decoration.smul` ($(\mathcal{U}, (as_i^x))$, the same index sets), `definition_13_2_10_decorated`, and on classes `DecoratedMultisection.smul a` (well defined by `Decoration.smulFun_rel`). *Book:* §13.2, Definition 13.2.10, p. 611 (2017: Definition 13.12). *Hidden hypotheses:* none. *Deviation (same):* As printed. For $a = 0$ the sections $0 \cdot s_i$ are the zero section and the counting formula gives Λ_0.

V13.17 (Lemma 13.2.11 (2017: 13.4), $\beta \odot \Lambda$; `Polyfold.SBLocalization.multisectionSmulFun`, `Polyfold.SBLocalization.lemma_13_2_11`, `Polyfold.SBLocalization.SymmetricStructureOn.smulFun`, `Polyfold.SBLocalization.Decoration.smulFun`, `Polyfold.SBLocalization.Decoration.smulFun_rel`, `Polyfold.SBLocalization.lemma_13_2_11_decorated`). `SBLocalization.lemma_13_2_11`, with `h :    ,   (tgt  ) =   (src  )` and β sc-smooth on the object M-polyfold (the book's “sc-smooth functor”): `IsScPlusMultisection A (multisectionSmulFun   Λ)`. Decorated: `Decoration.smulFun` (sections βs_i^x, the same good system and index sets), `lemma_13_2_11_decorated`, `Decoration.smulFun_rel` (compatible with Definition 13.2.8). *Book:* §13.2, Lemma 13.2.11, p. 612 (2017: Lemma 13.4). *Hidden hypotheses:* none. *Deviation (same):* “Sc-smooth functor” is read as invariance along morphisms plus sc-smoothness on objects.

V13.18 (Definition 13.2.12 (2017: 13.13), locally finite families; `Polyfold.SBLocalization.nontrivIdx`, `Polyfold.SBLocalization.nontrivIdx_mono`, `Polyfold.SBLocalization.IsLocallyFiniteFamily`, `Polyfold.SBLocalization.IsLocallyFiniteFamily.exists_stable`). `SBLocalization.IsLocallyFiniteFamily Λ :=   x,   O, IsOpen O   IsSaturated O   x   O   (nontrivIdx Λ O).Finite`; the stability remark is `IsLocallyFiniteFamily.exists_stable` (minimising the cardinality), for open saturated $O' \ni x$. *Book:* §13.2, Definition 13.2.12, p. 612 (2017: Definition 13.13). *Hidden hypotheses:* none. *Deviation (same):* none.

V13.19 (Lemma 13.2.13 (2017: 13.5), locally finite sums; `Polyfold.SBLocalization.lfSum`,

`Polyfold.SBLocalization.lfSum_eq_conv`, `Polyfold.SBLocalization.lfSum_eq_conv_of_mem`, `Polyfold.SBLocalization.trivial_of_not_mem_nontrivIdx`, `Polyfold.SBLocalization.lemma_13_2_13`, `Polyfold.SBLocalization.SymmetricStructureOn.sumSection`, `Polyfold.SBLocalization.SymmetricStructureOn.sumSection_apply`, `Polyfold.SBLocalization.fibAct_sectionSum`, `Polyfold.SBLocalization.SymmetricStructureOn.pi`, `Polyfold.SBLocalization.SymmetricStructureOn.congr`, `Polyfold.SBLocalization.exists_lfData`, `Polyfold.SBLocalization.lfSumStructure`, `Polyfold.SBLocalization.Decoration.lfSum`, `Polyfold.SBLocalization.lemma_13_2_13_decorated`). `SBLocalization.lfSum Λ w =   v : J → W_{P w}, if   v = w then   j, Λ j (v j) else 0`; equal to `conv` for finite J and near x to the finite convolution over J_x (`lfSum_eq_conv_of_mem`); `lemma_13_2_13`: for sc⁺-multisections in a locally finite family, `IsScPlusMultisection A (lfSum Λ)`. Decorated: `Decoration.lfSum hΛ hlf D  `, with $U(x) \subseteq O(x) \cap \bigcap_{j \in J_x} U_j(x)$ (with the natural action of $U_0(x)$, agreeing with those of $U_j(x)$), index set $\prod_{j \in J_x} I^j_x$ and sections $\sum_{j \in J_x} s^{j,x}_{i_j}$ (`lfSumStructure`, through `SymmetricStructureOn.pi`); `lemma_13_2_13_decorated`. *Book:* §13.2, Lemma 13.2.13, p. 612 (2017: Lemma 13.5). *Hidden hypotheses:* none. *Deviation (restated):* The text does not write out $\oplus$ over infinite J; it is read through finitely supported families (indices off J_x contribute the zero vector with weight 1), which agrees with Definition 13.2.9 (1) for finite J. The intersection is read as for Definition 13.2.9 (V13.15). When no Λ_j is non-trivial near x ($J_x = \emptyset$) the text leaves the natural action on $O(x)$ open: it is taken from a given good system $\mathcal{U}_0$ (an argument of `Decoration.lfSum`; any $\mathcal{U}_j$ when $J \neq \emptyset$, or Theorem 13.1.8 for paracompact $|X|$).

V13.20 (Tools for good systems; `Polyfold.MPolyfold.IsLocalScDiffeo.isScDiffeoOn_invFunOn`, `Polyfold.EpGroupoid.continuousOn_comp_of_continuousOn`, `Polyfold.EpGroupoid.exists_natNbhd_isProperSubset`, `Polyfold.EpGroupoid.exists_isOpen_injOn_src_tgt`). `MPolyfold.IsLocalScDiffeo.isScDiffeoOn_invFunOn`: a local sc-diffeomorphism injective on an open S is an sc-diffeomorphism $S \to f(S)$ with inverse `invFunOn`; `exists_natNbhd_isProperSubset`: neighbourhoods with the natural action and proper closure inside any open $W \ni x$; `exists_isOpen_injOn_src_tgt`. *Book:* not in the book (used in §13.1). *Hidden hypotheses:* none. *Deviation (extra):* none.

V13.21 (Properness by ultrafilters, and local sections of s (tools for Theorem 13.1.7); `Polyfold.EpGroupoid.isProperMap_iff_ultrafilter_of_isInducing`, `Polyfold.EpGroupoid.isProperSubset_iff_ultrafilter`, `Polyfold.EpGroupoid.isProperSubsetIn_iff_ultrafilter`, `Polyfold.EpGroupoid.exists_localSection`, `Polyfold.EpGroupoid.exists_mor_tgt_mem`). `isProperMap_iff_ultrafilter_of_isInducing` (proper maps on subspaces via ultrafilters on the ambient space), `isProperSubset_iff_ultrafilter` (Definition 7.5.8 by ultrafilters), `isProperSubsetIn_iff_ultrafilter`, `exists_localSection` (a local section of s through χ with targets in an open set), `exists_mor_tgt_mem`. *Book:* not in the book (proof of Theorem 13.1.7, p. 602). *Hidden hypotheses:* none. *Deviation (extra):* None.

V13.22 (Definition 13.2.7 (2017: 13.9), sc⁺-multisections with a system of symmetric structures; `Polyfold.SBLocalization.SymmetricStructureOn`, `Polyfold.SBLocalization.SymmetricStructureOn.P_s`, `Polyfold.SBLocalization.SymmetricStructureOn.`

`apply_fib`, `Polyfold.SBLocalization.SymmetricStructureOn.rep_of_apply_fib`, `Polyfold.SBLocalization.SymmetricStructureOn.card_pos`, `Polyfold.SBLocalization.SymmetricStructureOn.s_eq_fiberZero`, `Polyfold.SBLocalization.SymmetricStructure.toOn`, `Polyfold.SBLocalization.exists_symmetricStructureOn`, `Polyfold.SBLocalization.Decoration`, `Polyfold.SBLocalization.DecoratedTuple`, `Polyfold.SBLocalization.exists_decoration`). `structure Decoration A Λ` with `  : A.X.GoodSystem` and `S :   x, SymmetricStructureOn A Λ ( .nbhd x)`; `structure DecoratedTuple A` with `Λ`, `isScPlusMultisection`, `dec : Decoration A Λ`. `SymmetricStructureOn A Λ N` is Definition 13.2.5 on the neighbourhood N of x with its natural G_x-action: a finite I with a G_x-action, sc⁺-sections s_i on $N.U$ with $s_{g(i)}(g * y) = \mu(\Gamma(g, y), s_i(y))$ and $\Lambda(w) = |I|^{-1}\#\{i \mid s_i(P(w)) = w\}$ over $N.U$. Witnesses: `exists_decoration` (for paracompact $|X|$, from Lemma 13.2.6 and Theorem 13.1.8), `exists_symmetricStructureOn`. *Book:* §13.2, Definition 13.2.7, p. 610 (2017: Definition 13.9). *Hidden hypotheses:* none. *Deviation (same):* The good system carries a natural representation at every point (wave 1's `NatNbhd`), so "symmetric on $U(x)$" uses the action of $U(x)$ itself. The properness clause of Definition 13.2.5 is part of the good system (Definition 13.1.2 (1)). `exists_decoration` assumes $|X|$ paracompact, as the text takes good systems from Theorem 13.1.8 (witness `hasParacompactOrbitSpace_trivial`).

V13.23 (Definition 13.2.8 (2017: 13.10), equivalence of systems; decorated sc⁺-multisections; `Polyfold.EpGroupoid.GoodSystem.ofSubset`, `Polyfold.EpGroupoid.GoodSystem.exists_subset_inter`, `Polyfold.SBLocalization.Decoration.Rel`, `Polyfold.SBLocalization.Decoration.Rel.refl`, `Polyfold.SBLocalization.Decoration.Rel.symm`, `Polyfold.SBLocalization.Decoration.Rel.trans`, `Polyfold.SBLocalization.DecoratedTuple.Equiv`, `Polyfold.SBLocalization.DecoratedTuple.equivalence`, `Polyfold.SBLocalization.DecoratedTuple.setoid`, `Polyfold.SBLocalization.DecoratedMultisection`, `Polyfold.SBLocalization.DecoratedMultisection.forget`, `Polyfold.SBLocalization.DecoratedMultisection.isScPlusMultisection_forget`). `Decoration.Rel D D'` (conditions (1)–(2), for decorations of possibly different functors), `Rel.refl`, `Rel.symm`, `Rel.trans`; `DecoratedTuple.Equiv T T' := T.Λ = T'.Λ  T.dec.Rel T'.dec`, `DecoratedTuple.equivalence`; `DecoratedMultisection A` (the quotient), `DecoratedMultisection.forget`, `isScPlusMultisection_forget`. *Book:* §13.2, Definition 13.2.8, p. 611 (2017: Definition 13.10). *Hidden hypotheses:* none. *Deviation (same):* Transitivity uses a common refinement of two good systems (`GoodSystem.exists_subset_inter`, from wave 1's `exists_refinement_subset`).

V13.24 (The remark after Definition 13.2.9 (2017: after 13.11): in general there are no inverses; `Polyfold.SBLocalization.zeroMultisection_fib`, `Polyfold.SBLocalization.multisectionAdd_zeroMultisection`, `Polyfold.SBLocalization.NoInverse.twoValued`, `Polyfold.SBLocalization.NoInverse.isLocalRepOn_twoValued`, `Polyfold.SBLocalization.NoInverse.isScPlusMultisection_twoValued`, `Polyfold.SBLocalization.NoInverse.add_le_one_of_ne`, `Polyfold.SBLocalization.NoInverse.remark_13_2_9_no_inverse`, `Polyfold.SBLocalization.NoInverse.exists_no_inverse`). `multisectionAdd_zeroMultisection`: the zero multisection Λ_0 is neutral, $\Lambda \oplus \Lambda_0 = \Lambda$. `NoInverse.remark_13_2_9_no_inverse`: on the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid, for $c \neq 0$ the sc⁺-multisection `twoValued` c (weight ½ on the zero section and ½ on the constant section c) satisfies twoValued $c \oplus \Lambda' \neq \Lambda_0$ for every sc⁺-multisection Λ'. `NoInverse.exists_no_inverse` ($k \geq 1$): such Λ exists. *Book:* §13.2, after Definition 13.2.9, p. 611 (2017: after Definition 13.11). *Hidden hypotheses:* none. *Deviation (same):* "Inverse" is read with respect to the neutral element Λ_0. Over $x = 0$, $(\Lambda \oplus \Lambda')(0_x) = \frac{1}{2}(\Lambda'(0) + \Lambda'(-c)) \leq \frac{1}{2}$ (`add_le_one_of_ne`), while $\Lambda_0(0_x) = 1$.

V13.25 (The diagrams (13.5) and Definition 13.3.1 (2017: 13.14), correspondences $J(x, x')$; `Polyfold.SBLocalization.IndexDiagram`, `Polyfold.SBLocalization.IndexDiagram.Equiv`, `Polyfold.SBLocalization.IndexDiagram.equivalence`, `Polyfold.SBLocalization.IndexDiagram.setoid`, `Polyfold.SBLocalization.IndexCorr`, `Polyfold.SBLocalization.IndexCorr.mk`, `Polyfold.SBLocalization.IndexCorr.mk_eq_mk`, `Polyfold.SBLocalization.IndexCorr.ind`). `structure IndexDiagram` (fields `I`, `Fintype I`, `a`, `b`, `surjective_a`, `surjective_b`, `uniform_a`, `uniform_b`, with `Nat.card` of the fibres); `IndexDiagram.Equiv`, `IndexDiagram.equivalence`, `IndexDiagram.setoid`; `IndexCorr` (the quotient, $J(x, x')$ for $\alpha = I_x$, $\beta = I_{x'}$), `IndexCorr.mk`, `mk_eq_mk`, `ind`. *Book:* §13.3, (13.5) and Definition 13.3.1, p. 613 (2017: Definition 13.14). *Hidden hypotheses:* none. *Deviation (same):* I is finite (the text's diagrams are between finite index sets).

V13.26 (Definition 13.3.2 (2017: 13.15), the pseudo-inverse and the bi-action on $J(x, x')$; `Polyfold.SBLocalization.IndexCorr.swap`, `Polyfold.SBLocalization.IndexCorr.swap_swap`, `Polyfold.SBLocalization.IndexCorr.swapEquiv`, `Polyfold.SBLocalization.IndexCorr.biact`, `Polyfold.SBLocalization.IndexCorr.biact_one_one`, `Polyfold.SBLocalization.IndexCorr.biact_biact`, `Polyfold.SBLocalization.IndexDiagram.swap`, `Polyfold.SBLocalization.IndexDiagram.map`, `Polyfold.SBLocalization.IndexCorr.map`, `Polyfold.SBLocalization.IndexCorr.map_refl`, `Polyfold.SBLocalization.IndexCorr.map_map`). `IndexCorr.swap`, `swap_swap`, `swapEquiv` (a bijection); `IndexCorr.biact g'   g` for actions of groups G, G' on I_x, $I_{x'}$, with `biact_one_one` and `biact_biact` ($g' {*} (h' {*} \tau {*} h) {*} g = (g'h') {*} \tau {*} (hg)$, an action of $G_{x'} \times G_x^{op}$). *Book:* §13.3, Definition 13.3.2, p. 614 (2017: Definition 13.15). *Hidden hypotheses:* none. *Deviation (stronger):* The text defines the maps; the action laws and the bijectivity of $\sharp$ are proved.

V13.27 (Example 13.3.3 (2017: 13.1), the bi-action is trivial when $I_x = \{*\}$; `Polyfold.SBLocalization.IndexCorr.example_13_3_3`). `IndexCorr.example_13_3_3`: for any group actions, `biact g'   g =`   for every   `: IndexCorr PUnit` . *Book:* §13.3, Example 13.3.3, p. 614 (2017: Example 13.1). *Hidden hypotheses:* none. *Deviation (same):* c is built fibrewise (`Equiv.ofFiberEquiv`) from the equal fibre cardinalities.

V13.28 (Definition 13.3.4 (2017: 13.16), correspondences between local symmetric structures; `Polyfold.SBLocalization.Decoration.Compat`, `Polyfold.SBLocalization.Decoration.IsCorrespondence`). `Decoration.IsCorrespondence D x x'` with fields `locallyConstant` (for $\varphi \in \mathcal{U}(x, x')$, eventually $\tau\psi = \tau\varphi$ near φ), `finite` ($\tau(\mathcal{U}(x, x'))$ finite), `biact` (with wave 1's `GoodSystem.biact`) and `compat` (`(   ).Sat` of `Decoration.Compat`, condition (3) for every representative). *Book:* §13.3, Definition 13.3.4, p. 614 (2017: Definition 13.16). *Hidden hypotheses:* none. *Deviation (same):* $\mu(\psi, \cdot)$ is `fibAct`   ($= \mu$ on the fibre over $s(\psi)$). The action of G_x on I_x is that of the symmetric structure.

V13.29 (Remark 13.3.5 (2017: 13.1), condition (3) does not depend on the representative; `Polyfold.SBLocalization.IndexDiagram.Sat`, `Polyfold.SBLocalization.IndexDiagram.sat_iff_of_equiv`, `Polyfold.SBLocalization.IndexCorr.Sat`, `Polyfold.SBLocalization.IndexCorr.sat_mk`, `Polyfold.SBLocalization.IndexCorr.sat_iff`, `Polyfold.SBLocalization.IndexCorr.sat_map`, `Polyfold.SBLocalization.IndexCorr.sat_swap`, `Polyfold.SBLocalization.IndexCorr.sat_mono`, `Polyfold.SBLocalization.IndexCorr.sat_of_forall`, `Polyfold.SBLocalization.IndexCorr.sat_prod`). For any relation R on $I_x \times I_{x'}$: `IndexDiagram.sat_iff_of_equiv` ("$R(a(k), b(k))$ for all k" is invariant under equivalence), hence `IndexCorr.Sat R` is defined on classes, and `IndexCorr.sat_iff` (τ satisfies R iff every representative does). *Book:* §13.3, Remark 13.3.5, p. 615 (2017: Remark 13.1). *Hidden hypotheses:* none. *Deviation (same):* Stated for all conditions of the form $R(a(k), b(k))$, which includes (3).

V13.30 (Definition 13.3.6 (2017: 13.17), structurable sc⁺-multisections; `Polyfold.SBLocalization.IndexCorr.ofEquiv`, `Polyfold.SBLocalization.IndexCorr.ofPerm`, `Polyfold.SBLocalization.Decoration.Tau`, `Polyfold.SBLocalization.IsStructurable`, `Polyfold.SBLocalization.DecoratedMultisection.IsStructurable`, `Polyfold.SBLocalization.isStructurable_iff`, `Polyfold.SBLocalization.Decoration.zeroTau`, `Polyfold.SBLocalization.isStructurable_zeroMultisection`). `structure Decoration.Tau D` (fields , `isCorrespondence`, `diag` with `IndexCorr.ofPerm g`, `swap` for $\varphi \in \mathcal{U}(x', x)$); `DecoratedMultisection.IsStructurable` (some representative carries a `Tau`); for functors `IsStructurable A Λ := IsScPlusMultisection A Λ    D, Nonempty D.Tau`, with `isStructurable_iff` (a functor is structurable iff it is `forget` of a structurable decorated class). Witness: `isStructurable_zeroMultisection` (the zero multisection, $\tau \equiv [\{*\} \leftarrow \{*\} \rightarrow \{*\}]$, for paracompact $|X|$). *Book:* §13.3, Definition 13.3.6, p. 615 (2017: Definition 13.17). *Hidden hypotheses:* none. *Deviation (same):* $\tau_{x,x'}$ is a function on all morphisms; only its values on $\mathcal{U}(x, x')$ enter.

V13.31 (Remark 13.3.7 (2017: 13.2), the structured tuple $(\Lambda, \mathcal{U}, S, \tau)$; `Polyfold.SBLocalization.StructuredTuple`). `structure StructuredTuple A extends DecoratedTuple A with tau : dec.Tau`. *Book:* §13.3, Remark 13.3.7, p. 616 (2017: Remark 13.2). *Hidden hypotheses:* none. *Deviation (same):* None.

V13.32 (Definition 13.3.8 (2017: 13.18), equivalence of structured tuples; `Polyfold.SBLocalization.StructuredTuple.Equiv`, `Polyfold.SBLocalization.StructuredTuple.equivalence`, `Polyfold.SBLocalization.StructuredTuple.setoid`, `Polyfold.SBLocalization.IndexCorr.map_cast_rfl`, `Polyfold.SBLocalization.IndexCorr.map_cast_symm`, `Polyfold.SBLocalization.IndexCorr.map_cast_trans`). `StructuredTuple.Equiv` (with `hI :  x, (T.dec.S x).I = (T'.dec.S x).I` and `' = .map (Equiv.cast (hI x)) (Equiv.cast (hI x'))` on $\mathcal{U}''(x, x')$), `StructuredTuple.equivalence`, `StructuredTuple.setoid`. *Book:* §13.3, Definition 13.3.8, p. 616 (2017: Definition 13.18). *Hidden hypotheses:* none. *Deviation (same):* (3) is an equality of types and (4) compares after transport along it. As printed, the definition puts no condition on the sections s^x, s'^x (presumably they are meant to agree on $U''(x)$, as in Definition 13.2.8); it is formalized as printed, and nothing below uses the sections of an equivalent tuple.

V13.33 (Remark 13.3.9 (2017: 13.3), another possible equivalence; `Polyfold.SBLocalization.IndexCorr.map`). Nothing to formalize: a comment on the choice of definition. The relabelling it uses is `IndexCorr.map`. *Book:* §13.3, Remark 13.3.9, p. 616 (2017: Remark 13.3). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V13.34 (Definition 13.3.10 (2017: 13.19), structured sc$^+$-multisections; `Polyfold.SBLocalization.StructuredMultisection`, `Polyfold.SBLocalization.StructuredMultisection.forget`). `StructuredMultisection A := Quotient (StructuredTuple.setoid A)`, `StructuredMultisection.forget`. *Book:* §13.3, Definition 13.3.10, p. 616 (2017: Definition 13.19). *Hidden hypotheses:* none. *Deviation (same):* None.

V13.35 (Definition 13.3.11 (2017: 13.20), $\Gamma^+_m(P)$, $\Gamma^+_s(P)$, $\Gamma^+_{st}(P)$; `Polyfold.SBLocalization.Γm`, `Polyfold.SBLocalization.Γs`, `Polyfold.SBLocalization.Γst`, `Polyfold.SBLocalization.Γs_subset_Γm`, `Polyfold.SBLocalization.StructuredMultisection.forget_mem_Γs`, `Polyfold.SBLocalization.StructuredMultisection.range_forget`). `Γm A`, `Γs A` (sets of functors), `Γst A := StructuredMultisection A`, `Γs_subset_Γm`, `StructuredMultisection.forget_mem_Γs`, `StructuredMultisection.range_forget` (`range forget = Γs A`). *Book:* §13.3, Definition 13.3.11, p. 617 (2017: Definition 13.20). *Hidden hypotheses:* none. *Deviation (same):* None.

V13.36 (Proposition 13.3.12 (2017: 13.3), sums of structurable sc$^+$-multisections; `Polyfold.SBLocalization.IndexDiagram.prod`, `Polyfold.SBLocalization.IndexDiagram.card_fiber_prodMap`, `Polyfold.SBLocalization.IndexCorr.prod`, `Polyfold.SBLocalization.IndexCorr.prod_swap`, `Polyfold.SBLocalization.IndexCorr.prod_biact`, `Polyfold.SBLocalization.IndexCorr.prod_ofPerm`, `Polyfold.EpGroupoid.biact_congr_of_le`, `Polyfold.EpGroupoid.GoodSystem.biact_congr_of_le`, `Polyfold.SBLocalization.Decoration.restrictTo`, `Polyfold.SBLocalization.Decoration.Tau.restrict`, `Polyfold.SBLocalization.Decoration.Tau.prod`, `Polyfold.SBLocalization.Decoration.Tau.add`, `Polyfold.SBLocalization.StructuredTuple.add`, `Polyfold.SBLocalization.StructuredTuple.add_Λ`, `Polyfold.SBLocalization.proposition_13_3_12`). `proposition_13_3_12 : IsStructurable A Λ' → IsStructurable A Λ'' → IsStructurable A (multisectionAdd Λ' Λ'')`; structured: `Decoration.Tau.add` (a `Tau` for `D'.add D''`), `StructuredTuple.add`, `StructuredTuple.add_Λ`; on a common good system `Decoration.Tau.prod` with `IndexCorr.prod` ($[a_1, b_1] \times [a_2, b_2] = [a_1 \times a_2, b_1 \times b_2]$, `prod_swap`, `prod_biact`, `prod_ofPerm`). *Book:* §13.3, Proposition 13.3.12, p. 617 (2017: Proposition 13.3). *Hidden hypotheses:* none. *Deviation (restated):* "We may assume $\mathcal{U}_1 = \mathcal{U}_2$" is realized by restricting both structures to the common neighbourhoods of the decorated sum (same actions; `Decoration.restrictTo`, `Decoration.Tau.restrict`, with `GoodSystem.biact_congr_of_le`: the bi-action does not change). Compatibility (3) is the text's computation (`fibAct_sectionAdd`).

V13.37 (Remark 13.3.13 (2017: 13.4), $\Lambda_1 \oplus \Lambda_2$ versus $\Lambda_2 \oplus \Lambda_1$; `Polyfold.SBLocalization.`

`StructuredTuple.add_Λ`, `Polyfold.SBLocalization.multisectionAdd_comm`, `Polyfold.SBLocalization.DecoratedMultisection.add_comm`). The agreement of the functors: `StructuredTuple.add_Λ` with `multisectionAdd_comm`; on decorated classes `DecoratedMultisection.add_comm`. The difference of the structured classes is not expressible: Lean cannot prove two product types unequal, so "$I_x^1 \times I_x^2 \neq I_x^2 \times I_x^1$" has no Lean statement. *Book:* §13.3, Remark 13.3.13, p. 618 (2017: Remark 13.4). *Hidden hypotheses:* none. *Deviation (narrower):* Only the positive half is formalized; the negative half is about the equality of types, which has no Lean statement.

V13.38 (Lemma 13.3.14 (2017: 13.6), $r\Lambda$ and $\beta\Lambda$ are structurable; `Polyfold.SBLocalization.Decoration.Tau.smulFun`, `Polyfold.SBLocalization.Decoration.Tau.smul`, `Polyfold.SBLocalization.lemma_13_3_14`, `Polyfold.SBLocalization.lemma_13_3_14_fun`). `lemma_13_3_14 (h : IsStructurable A Λ) (a :  ) : IsStructurable A (multisectionSmul a Λ)`, `lemma_13_3_14_fun` (for β as in Lemma 13.2.11); the same τ serves (`Decoration.Tau.smul`, `Decoration.Tau.smulFun`). *Book:* §13.3, Lemma 13.3.14, p. 618 (2017: Lemma 13.6). *Hidden hypotheses:* none. *Deviation (same):* Compatibility uses $\beta(t(\psi)) = \beta(s(\psi))$ (`fibAct_sectionSmul`). Includes $r = 0$.

V13.39 (T_Λ and the construction before Proposition 13.3.15 (2017: before 13.4); `Polyfold.SBLocalization.trivialSet`, `Polyfold.SBLocalization.isOpen_trivialSet`, `Polyfold.SBLocalization.isTrivialOn_trivialSet`, `Polyfold.SBLocalization.apply_fiberZero_tgt`, `Polyfold.SBLocalization.TightIdx`, `Polyfold.SBLocalization.TightIdx.eT`, `Polyfold.SBLocalization.TightIdx.eN`, `Polyfold.SBLocalization.SymmetricStructureOn.tight`, `Polyfold.SBLocalization.Decoration.tightNbhd`, `Polyfold.SBLocalization.Decoration.tight`, `Polyfold.SBLocalization.Decoration.tight `, `Polyfold.SBLocalization.Decoration.Tau.tight`, `Polyfold.SBLocalization.IndexCorr.toUnit`, `Polyfold.SBLocalization.IndexCorr.fromUnit`). `trivialSet Λ`, `isOpen_trivialSet`, `isTrivialOn_trivialSet`; `TightIdx p I` ($= \{* \mid p\} \sqcup \{i \in I \mid \neg p\}$) with `TightIdx.eT : {*}  I'_x` ($x \in T_\Lambda$), `TightIdx.eN : I_x  I'_x` ($x \notin T_\Lambda$); `SymmetricStructureOn.tight`, `Decoration.tightNbhd`, `Decoration.tight`, `Decoration.tight ` (`tight _TT`, `_TN`, `_NT`, `_NN`), `Decoration.Tau.tight` (the new tuple is structured). *Book:* §13.3, before Proposition 13.3.15, pp. 618–619 (2017: before Proposition 13.4). *Hidden hypotheses:* none. *Deviation (stronger):* The text does not check (3) of Definition 13.3.4 for the new τ'; it holds because all sections vanish at points y with $\Lambda(0_y) = 1$ (`SymmetricStructureOn.s_eq_fiberZero`) and $\Lambda(0_{t(\psi)}) = \Lambda(0_{s(\psi)})$ (`apply_fiberZero_tgt`). $I'_x = \{*\}$ and $I'_x = I_x$ are bijections (`eT`, `eN`), not equalities of types.

V13.40 (Proposition 13.3.15 (2017: 13.4), the tight version of a structured tuple; `Polyfold.SBLocalization.proposition_13_3_15`, `Polyfold.SBLocalization.Decoration.isTight_tight`). `proposition_13_3_15 hΛ D t`: there are `D'` and `t' : D'.Tau` with `D'.IsTight t'` ((1), (4), (5)); for $x \in T_\Lambda$, $U'(x) \subseteq T_\Lambda$ carries the action of $U(x)$; and equivariant bijections $e_x : I_x \simeq I'_x$ for $x \notin T_\Lambda$ with `D'. .nbhd x = D. .nbhd x`, $s'_{e_x(i)} = s_i$, and $\tau'_{x,x'} = \tau_{x,x'}$ relabelled by $e_x, e_{x'}$ ((2), (3)). *Book:* §13.3, Proposition 13.3.15, p. 619 (2017: Proposition 13.4). *Hidden hypotheses:* none. *Deviation (restated):* Equalities of index sets are bijections (as for T_Λ, V13.39); (3) holds for all morphisms.

V13.41 (Remark 13.3.16 (2017: 13.5), the choice of $U'(x)$ for $x \in T_\Lambda$; `Polyfold.SBLocalization.Decoration.exists_tightNbhd`). Prose. The construction takes any natural neighbourhood inside $U(x) \cap T_\Lambda$ with the same action (`Decoration.exists_tightNbhd`). *Book:* §13.3, Remark 13.3.16, p. 619 (2017: Remark 13.5). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V13.42 (Definition 13.3.17 (2017: 13.21), tightly structured sc$^+$-multisections; `Polyfold.SBLocalization.Decoration.IsTight`, `Polyfold.SBLocalization.StructuredMultisection.IsTight`). `structure Decoration.IsTight D t` (fields `unique`, `zero`, `_TN`, `_NT`, `_TT`, on $\varphi \in \mathcal{U}(x, x')$), `StructuredMultisection.IsTight`. *Book:* §13.3, Definition 13.3.17, p. 620 (2017: Definition 13.21). *Hidden hypotheses:* none. *Deviation (same):* $I_x = \{*\}$ is a bijection $\{*\} \simeq I_x$; (2)–(3) are stated for every such bijection (all are equal).

V13.43 (Corollary 13.3.18 (2017: 13.1), tightly structured versions exist; `Polyfold.SBLocalization.corollary_13_3_18`, `Polyfold.SBLocalization.corollary_13_3_18_structured`). `corollary_13_3_18 : IsStructurable A Λ →  D t, D.IsTight t`; `corollary_13_3_18_structured`: every $\Lambda \in \Gamma_s^+$ is `forget` of a tightly structured class. *Book:* §13.3, Corollary 13.3.18, p. 620 (2017: Corollary 13.1). *Hidden hypotheses:* none. *Deviation (same):* None.

V13.44 (Definition 13.5.1 (2017: 13.23), atomic sc$^+$-multisection functors; `Polyfold.SBLocalization.AtomicData`, `Polyfold.SBLocalization.AtomicData.count`, `Polyfold.SBLocalization.AtomicData.Reach`, `Polyfold.SBLocalization.AtomicData.IsRelated`, `Polyfold.SBLocalization.IsAtomic`, `Polyfold.SBLocalization.AtomicData.count__eq`, `Polyfold.SBLocalization.AtomicData.reach_iff_mem_saturation`, `Polyfold.SBLocalization.AtomicData.IsRelated.eq_mult`, `Polyfold.SBLocalization.AtomicData.IsRelated.symmetricStructure`, `Polyfold.SBLocalization.AtomicData.IsRelated.nonempty`, `Polyfold.SBLocalization.AtomicData.zero`, `Polyfold.SBLocalization.isAtomic_zeroMultisection`). `structure AtomicData A` holds (1)–(2): `x`, `R : NaturalRep x` with $U(x) =$ `R.U`, `hasPropernessProperty`, an open G_x-invariant `V` with $x \in V$ and $\overline{V} \subseteq U(x)$, `I` with `Fintype` and `MulAction (Isotropy x)`, `s` with `isScPlusSectionOn` on $U(x)$, `equivariant` and `vanish` on $U(x) \setminus V$. `AtomicData.IsRelated Λ` is (3). `IsAtomic A Λ := IsScPlusMultisection A Λ  D, D.IsRelated Λ`. `IsRelated.symmetricStructure` (the data is a symmetric structure of Definition 13.2.5), `IsRelated.nonempty` ($I_x \neq \emptyset$), `count_ _eq` (the formula in (3) does not depend on φ), `IsRelated.eq_mult` (Λ is determined by the data). Witness: `isAtomic_zeroMultisection`. *Book:* §13.5, Definition 13.5.1, p. 629 (2017: Definition 13.23). *Hidden hypotheses:* none. *Deviation (restated):* "G_x-invariant $U(x)$ carrying a symmetric structure" is read as in Definition 13.2.5 (wave 1): $U(x)$ is the domain of a natural representation; $V(x)$ is invariant for the natural action and contains x. The data is bundled as a structure; Λ is an sc$^+$-multisection, as the text presupposes. In (3) the count is taken at $P(w') = t(\varphi)$ (Findings, Chapter 13, item 13).

V13.45 (Remark 13.5.2 (2017: 13.6), atomic perturbations and bump functions; `Polyfold.SBLocalization.theorem_13_5_6_2`). Nothing to formalize beyond Theorems 13.5.4–13.5.6, which use only `AdmitsBumpFunctionsPointwise` and no partition of unity. The "finitely many suffice" claim belongs to Theorem 15.3.7. *Book:* §13.5, Remark 13.5.2, p. 630 (2017: Remark 13.6). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V13.46 (Theorem 13.5.3 (2017: 13.6), atomic sc⁺-multisections are structurable; Polyfold.SBLocalization.IndexDiagram.card_fst_fiber, Polyfold.SBLocalization.IndexDiagram.card_snd_fiber, Polyfold.SBLocalization.IndexDiagram.full, Polyfold.SBLocalization.IndexCorr.full, Polyfold.SBLocalization.IndexCorr.swap_full, Polyfold.SBLocalization.IndexCorr.biact_full, Polyfold.SBLocalization.IndexCorr.biact_ofEquiv, Polyfold.SBLocalization.IndexCorr.swap_ofEquiv, Polyfold.SBLocalization.IndexCorr.sat_ofEquiv, Polyfold.SBLocalization.eventually_eq_of_existsUnique, Polyfold.SBLocalization.eq_of_mem_connectedComponentIn, Polyfold.SBLocalization.AtomicStructData.fibAct_comp, Polyfold.SBLocalization.AtomicStructData.fibAct_unit, Polyfold.SBLocalization.AtomicStructData.fibAct_inv_fibAct, Polyfold.SBLocalization.AtomicStructData.fibAct_fibAct_inv, Polyfold.SBLocalization.AtomicStructData.transportSection_eq_fibAct, Polyfold.SBLocalization.AtomicStructData.comp_inv_comp, Polyfold.SBLocalization.AtomicStructData, Polyfold.SBLocalization.exists_atomicStructData, Polyfold.SBLocalization.AtomicStructData.Z, Polyfold.SBLocalization.AtomicStructData.isClosed_Z, Polyfold.SBLocalization.AtomicStructData.mem_Z_iff_of_mor, Polyfold.SBLocalization.AtomicStructData., Polyfold.SBLocalization.AtomicStructData.Dom, Polyfold.SBLocalization.AtomicStructData.conj, Polyfold.SBLocalization.AtomicStructData.existsUnique_rel, Polyfold.SBLocalization.AtomicStructData.H, Polyfold.SBLocalization.AtomicStructData.H_spec, Polyfold.SBLocalization.AtomicStructData.H_unique, Polyfold.SBLocalization.AtomicStructData.H_comp, Polyfold.SBLocalization.AtomicStructData.H_unit, Polyfold.SBLocalization.AtomicStructData.H_inv, Polyfold.SBLocalization.AtomicStructData.H_eventually, Polyfold.SBLocalization.AtomicStructData., Polyfold.SBLocalization.AtomicStructData.W, Polyfold.SBLocalization.AtomicStructData., Polyfold.SBLocalization.AtomicStructData.W_cond, Polyfold.SBLocalization.AtomicStructData.sec, Polyfold.SBLocalization.AtomicStructData.sec_compat, Polyfold.SBLocalization.AtomicStructData.sec_eq_zero, Polyfold.SBLocalization.AtomicStructData.S, Polyfold.SBLocalization.AtomicStructData.dec, Polyfold.SBLocalization.AtomicStructData.corr, Polyfold.SBLocalization.AtomicStructData.H_biact, Polyfold.SBLocalization.AtomicStructData.isCorrespondence, Polyfold.SBLocalization.AtomicStructData.tau, Polyfold.SBLocalization.AtomicStructData.isStructurable, Polyfold.SBLocalization.theorem_13_5_3_of_goodSystem, Polyfold.SBLocalization.theorem_13_5_3, Polyfold.SBLocalization.theorem_13_5_3_of_paracompact, Polyfold.SBLocalization.AtomicData.IsLocallyTransported, Polyfold.SBLocalization.theorem_13_5_3_local, Polyfold.SBLocalization.theorem_13_5_3_hyp, Polyfold.SBLocalization.theorem_13_5_3_hyp_witness, Polyfold.SBLocalization.AtomicData.exists_isLocalRepOn_transport). `theorem_13_5_3 (hΛ : IsAtomic A Λ) : IsStructurable A Λ  Nonempty A.X.GoodSystem.` `theorem_13_5_3_of_goodSystem (h  : Nonempty A.X.GoodSystem) (hΛ :`

`IsAtomic A Λ) : IsStructurable A Λ.` `theorem_13_5_3_of_paracompact (hX : A.X.HasParacompactOrbitSpace)`. `IsStructurable` is Definition 13.3.6. There is no `hcrit`. The proof follows the text, with $Z = \pi^{-1}\pi(\overline{V(x_0)})$ (closed, saturated): `AtomicStructData`, `exists_atomicStructData`: for $y \in Z$, an open $O(y) \ni y$ and τ_y with $t\tau_y = \mathrm{id}$ and $s\tau_y \in U(x_0)$ (`exists_isLocalRepOn_transport`). `H y y'    G_{x }` is the unique h with $\Gamma(h, s\tau_y(s\psi)) = \tau_{y'}(t\psi)^{-1} \circ \psi \circ \tau_y(s\psi)$. It is functorial (`H_comp`, `H_unit`, `H_inv`). It is locally constant (`H_eventually`, from the finiteness of G_{x_0} and Hausdorffness). $\gamma_y(g) = H(\Gamma(g, y))$ is a homomorphism ( ). The good system □ the given one has the same natural actions. On it $H(\Gamma(g, z)) = \gamma_y(g)$ for $z \in U(y)$, $y \in Z$, and $U(y) \cap Z = \emptyset$ for $y \notin Z$. `S y` has index set I_{x_0} at every y, with sections pulled back by τ_y on Z and zero sections off Z. It is equivariant through γ_y (`sec_compat`), and the counting formula comes from D13.5.1 (3). `corr y y'  :` $[I \xleftarrow{id} I \xrightarrow{H(\varphi)} I]$ when y, y' are both in Z or both outside it ($H = 1$ outside); the full $[I \leftarrow I \times I \to I]$ otherwise. `tau : dec.Tau` checks D13.3.4 (1)–(3) and D13.3.6 (1)–(2). *Book:* §13.5, Theorem 13.5.3, p. 630 (2017: Theorem 13.6). *Hidden hypotheses:* none. *Deviation (narrower):* Definition 13.3.6 includes a good system, so a structurable Λ gives one. The form shows the hypothesis `Nonempty A.X.GoodSystem` is exactly what is needed. The text takes it from Theorem 13.1.8 ($|X|$ paracompact): `theorem_13_5_3_of_paracompact`. The index set is I_{x_0} everywhere, with zero sections where Λ is trivial. The text uses $\{*\}$ there. Between Z and its complement the correspondences are the full ones. These are structures in the sense of D13.3.6, so this is a different choice of witness, not a different statement. Hidden hypotheses: `Nonempty A.X.GoodSystem`; it is necessary (the form) and holds for paracompact $|X|$ (`theorem_13_5_3_of_paracompact`, from Theorem 13.1.8). Witness: `nonempty_goodSystem_epSEP_zero` ($\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$). Theorem 13.5.3 as printed assumes no paracompactness, and Chapter 13 has no standing assumption that supplies it; the proof constructs the neighbourhoods itself. Theorem 13.1.8 is where paracompactness enters. The earlier hypothesis form `theorem_13_5_3_hyp` (with `hcrit`) and its witness stay as supporting declarations; `hcrit` is no longer needed (gap G.149 closed).

V13.47 (Theorem 13.5.4 (2017: 13.7), Construction I; `Polyfold.SBLocalization.SymmetricFamily`, `Polyfold.SBLocalization.SymmetricFamily.ofSeed`, `Polyfold.SBLocalization.SymmetricFamily.ofSeed_s`, `Polyfold.SBLocalization.SymmetricFamily.ofSeed_vanish`, `Polyfold.SBLocalization.exists_cutoff`, `Polyfold.SBLocalization.theorem_13_5_4_1`, `Polyfold.SBLocalization.theorem_13_5_4_2`). `theorem_13_5_4_1 hbump hx R _hprop hVo hxV hVinv hVcl hw hwx :   F : SymmetricFamily A R, (  i y, y   V → F.s i y = 0)    i , F.s i  x = w;` `theorem_13_5_4_2 … (hs : IsScPlusSectionOn A s 0)`: the same with $s_{i_0} = s$ near x. `SymmetricFamily A R`: global sc⁺-sections of $W \to X$ with support in `R.U`, a G_x-action on I, equivariance on $U(x)$. Proof: $I = G_x$ acting by left multiplication, s_g the `natTerm` translates of a seed s_0 (`SymmetricFamily.ofSeed`); (1) s_0 from Part I's `lemma_5_3_3` with an auxiliary norm of `proposition_5_1_3_exists`; (2) $s_0 = \beta s$ with β from `proposition_5_5_3` (`exists_cutoff`). *Book:* §13.5, Theorem 13.5.4, p. 632 (2017: Theorem 13.7). *Hidden hypotheses:* bump functions in the pointwise reading `AdmitsBumpFunctionsPointwise`, as in Theorem 12.3.3 (V12.13); witness `EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`.

Deviation (narrower): Otherwise stronger: global sc⁺-sections that vanish off V, not only on $U \setminus V$. In (2), "$s_{i_0} = s$ near x" needs $\beta \equiv 1$ near x, which Proposition 5.5.3 provides. `_hprop`, and in (2) `_hx`, are kept as in the text and unused.

V13.48 (Theorem 13.5.5 (2017: 13.8), Construction II; `Polyfold.SBLocalization.SymmetricFamily.eq_zero_iff`, `Polyfold.SBLocalization.SymmetricFamily.exists_vanishing_nbhd`, `Polyfold.SBLocalization.AtomicData.ofFamily`, `Polyfold.SBLocalization.theorem_13_5_5`, `Polyfold.SBLocalization.AtomicData.mult`, `Polyfold.SBLocalization.AtomicData.isRelated_mult`, `Polyfold.SBLocalization.AtomicData.isWFunctor_mult`, `Polyfold.SBLocalization.AtomicData.domSupp_mult_subset`, `Polyfold.SBLocalization.AtomicData.isScPlusMultisection_mult`, `Polyfold.SBLocalization.AtomicData.isAtomic_mult`, `Polyfold.SBLocalization.theorem_13_5_5_structurable_hyp`, `Polyfold.SBLocalization.theorem_13_5_5_structurable`). `theorem_13_5_5 _hbump _hx R hprop (F : SymmetricFamily A R) [Nonempty F.I] : ! Λ, IsAtomic A Λ ( w, P w saturation R.U → Λ w = zeroMultisection A w) w, P w R.U → Λ w = |I| ¹ * #{i | s i (P w) = w}`. The $V(x)$ of the proof comes from `SymmetricFamily.exists_vanishing_nbhd`; the multisection is `AtomicData.mult` (`isScPlusMultisection_mult`: near points joined to $U(x)$ the sections transported by a local inverse of t, `exists_isLocalRepOn_transport`; elsewhere the zero section). "Structurable" is `theorem_13_5_5_structurable` (given a good system), from `theorem_13_5_3_of_goodSystem`, as in the text; the earlier hypothesis form is `theorem_13_5_5_structurable_hyp`. *Book:* §13.5, Theorem 13.5.5, p. 633 (2017: Theorem 13.8). *Hidden hypotheses:* $I \neq \emptyset$ (`[Nonempty F.I]`), implicit in the text (with $I = \emptyset$ the printed formula gives $\Lambda = 0$ over $U(x)$, which is not a multisection; for atomic Λ it is automatic, `IsRelated.nonempty`); witness `SymmetricFamily.ofSeed` ($I = G_x$). The structurable clause assumes that a good system exists, as Theorem 13.5.3 does (V13.46; paracompact $|X|$ via Theorem 13.1.8); gap G.149 is closed. *Deviation (narrower):* Existence and uniqueness are as printed. "Supported in the saturation of $U(x)$" is read as: Λ is trivial over points not isomorphic to a point of $U(x)$. The first-case formula of the proof is read with $w' = \mu(\varphi, w)$, as in Definition 13.5.1 (3). Bump functions and the smoothness of x are not used.

V13.49 (Theorem 13.5.6 (2017: 13.9), atomic sc⁺-multisections through a given vector or section; `Polyfold.SBLocalization.normW_`, `Polyfold.SBLocalization.AtomicData.pointwiseNorm_mult_le`, `Polyfold.SBLocalization.AtomicData.isLocalRepOn_mult`, `Polyfold.SBLocalization.theorem_13_5_6_1`, `Polyfold.SBLocalization.theorem_13_5_6_2`, `Polyfold.SBLocalization.theorem_13_5_6`). `theorem_13_5_6_1 _hpara hbump _hN hw _hw0 hwx R : Λ, IsAtomic A Λ domSupp Λ saturation R.U g, 0 < Λ ( (g, w));` `theorem_13_5_6_2 _hpara hbump hN R hs h : Λ, IsAtomic A Λ domSupp Λ saturation R.U ( y, pointwiseNorm N Λ y normW N (s x) + ) O k s' , IsOpen O x O IsLocalRepOn A Λ O s' j, 0 < j y O, s' j y = s y`; `theorem_13_5_6` combines both, with $x = P(w)$. N is `A.E.IsAuxNorm` (μ-invariant); the bound uses `normW_` and `AtomicData.pointwiseNorm_mult_le`. *Book:* §13.5, Theorem 13.5.6, p. 634 (2017: Theorem 13.9). *Hidden hypotheses:* bump functions in the pointwise reading `AdmitsBumpFunctionsPointwise` (as V12.13); witness `EpGroupoid.`

admitsBumpFunctionsPointwise_trivial_rn. *Deviation (narrower):* Otherwise as printed. $\pi^{-1}(U(x))$ is the saturation. "Contains s with positive weight" is read as a local representation (Definition 13.2.1) on an open $O \ni x$, one of whose sections equals s on O. $|X|$ paracompact is kept and unused, since the text uses it only for the existence of N, a hypothesis here; N and $w \neq 0$ are unused in (1). $P(w)$ smooth is derived from w smooth.

V13.50 (Proposition 13.3.19 (2017: 13.5), locally finite sums of structurable sc$^+$-multisections; Polyfold.SBLocalization.finite_pi_of_subsingleton, Polyfold.SBLocalization.IndexDiagram.card_fiber_pi, Polyfold.SBLocalization.IndexDiagram.pi, Polyfold.SBLocalization.IndexCorr.out, Polyfold.SBLocalization.IndexCorr.mk_out, Polyfold.SBLocalization.IndexCorr.subsingleton_out_I, Polyfold.SBLocalization.IndexCorr.pi, Polyfold.SBLocalization.IndexCorr.pi_eq_mk, Polyfold.SBLocalization.IndexCorr.pi_congr, Polyfold.SBLocalization.IndexCorr.sat_pi, Polyfold.SBLocalization.IndexCorr.swap_pi, Polyfold.SBLocalization.IndexCorr.biact_pi, Polyfold.SBLocalization.IndexCorr.pi_ofPerm, Polyfold.SBLocalization.SymmetricStructureOn.reindex, Polyfold.SBLocalization.SymmetricStructureOn.reindex_s, Polyfold.SBLocalization.fibAct_sum, Polyfold.SBLocalization.SymmetricStructureOn.tightRel, Polyfold.SBLocalization.Decoration.IsAdapted, Polyfold.SBLocalization.Decoration.tightOn, Polyfold.SBLocalization.Decoration.tightOn , Polyfold.SBLocalization.Decoration.tightOn _TT, Polyfold.SBLocalization.Decoration.tightOn _TN, Polyfold.SBLocalization.Decoration.tightOn _NT, Polyfold.SBLocalization.Decoration.tightOn _NN, Polyfold.SBLocalization.Decoration.Tau.tightOn, Polyfold.SBLocalization.Decoration.isTight_tightOn, Polyfold.SBLocalization.subset_trivialSet_of_not_mem_nontrivIdx, Polyfold.SBLocalization.tightIdx_elim_zero, Polyfold.SBLocalization.LFStructData, Polyfold.SBLocalization.LFStructData.O, Polyfold.SBLocalization.LFStructData.F, Polyfold.SBLocalization.LFStructData.V, Polyfold.SBLocalization.LFStructData.M, Polyfold.SBLocalization.LFStructData. , Polyfold.SBLocalization.LFStructData.isAdapted, Polyfold.SBLocalization.LFStructData.Dt, Polyfold.SBLocalization.LFStructData.tt, Polyfold.SBLocalization.LFStructData.K, Polyfold.SBLocalization.LFStructData.finite_K, Polyfold.SBLocalization.LFStructData.Sx, Polyfold.SBLocalization.LFStructData.Dsum, Polyfold.SBLocalization.LFStructData.Sx_s_apply, Polyfold.SBLocalization.LFStructData. , Polyfold.SBLocalization.LFStructData.isCorrespondence, Polyfold.SBLocalization.LFStructData.tau, Polyfold.SBLocalization.LFStructData.isStructurable, Polyfold.SBLocalization.proposition_13_3_19_of_goodSystem, Polyfold.SBLocalization.proposition_13_3_19, Polyfold.SBLocalization.proposition_13_3_19_of_paracompact, Polyfold.SBLocalization. _ne_fiberZero_lf, Polyfold.SBLocalization.isLocallyFiniteFamily_of_locallyFinite, Polyfold.SBLocalization.proposition_13_3_19_locallyFinite, Polyfold.SBLocalization.proposition_13_3_19_locallyFinite_of_paracompact). proposition_13_3_19 [Nonempty L] (h :   l, IsStructurable A (Λ l)) (hlf : IsLocallyFiniteFamily Λ) : IsStructurable A (lfSum Λ). proposition_13_3_19_of_goodSystem (h : Nonempty A.X.GoodSystem) covers any family, including the empty one. proposition_13_3_19_of_ `paracompact (hX : A.X.HasParacompactOrbitSpace)`. The proof is the text's: `LFStructData` holds the data, with the common good system (`isAdapted`). Tight versions over a given good system: `Decoration.Tau.tightOn`, `isTight_tightOn` (file 2). $K_x = \prod_{l \in L} K_x^l$, with `finite_K`. The sum structure `Sx`, relabelled from M/`lfSumStructure`. $\tau = \prod_{l \in L} \tau^l$ via `IndexCorr.pi`, which is well defined because all but finitely many factors are $[\{*\} \leftarrow \{*\} \rightarrow \{*\}]$ (file 1). `tau : Dsum.Tau` checks Definitions 13.3.4 and 13.3.6 for it. This is the step left to the reader.

With the hypothesis as printed (the domain supports form a locally finite family of subsets of X): `isLocallyFiniteFamily_of_locallyFinite (hΛ : l, IsWFunctor A (Λ l)) (h : LocallyFinite fun l => domSupp (Λ l)) : IsLocallyFiniteFamily Λ`. `proposition_13_3_19_locallyFinite [Nonempty L] (h : l, IsStructurable A (Λ l)) (hlf : LocallyFinite fun l => domSupp (Λ l)) : IsStructurable A (lfSum Λ)`; `proposition_13_3_19_locallyFinite_of_paracompact (hX : A.X.HasParacompactOrbitSpace)` (any family). Proof: for `x`, an open `V` `x` meeting finitely many `domSupp (Λ l)`; `O =` ^{1}`( (V))` is open (Proposition 7.1.8) and saturated. A `w` `O` over `y` `O` with `Λ_l(w) > 0` is moved by `: y → v`, `v` `V`, to `( , w)` `O` over `v` (`_ne_fiberZero_lf`) with the same weight, so `v` `domSupp (Λ l)` `V`. Saturation of `domSupp` is not needed. *Book:* §13.3, Proposition 13.3.19, p. 620 (2017: Proposition 13.5). *Hidden hypotheses:* none. *Deviation (restated):* L lives in the bundle's universe (`L : Type u`). "Locally finite domain supports" is read as `IsLocallyFiniteFamily` (Definition 13.2.12): every point has an open saturated neighbourhood over which only finitely many members are non-trivial. For the empty family the sum is the trivial multisection. Its structurability needs a good system, which Definition 13.3.6 requires. The `[Nonempty L]` form takes the good system from a member. The printed hypothesis, Mathlib's `LocallyFinite` of the family of domain supports, implies `IsLocallyFiniteFamily` (`isLocallyFiniteFamily_of_locallyFinite`); `proposition_13_3_19_locallyFinite` is the proposition in that form. Saturation of `domSupp` is not needed.

V13.51 (Theorem 13.4.1 (2017: 13.3), the pull-back along a strong bundle equivalence is structurable; **`Polyfold.EpGroupoid.isOpen_setOf_eq_of_src`, `Polyfold.SBLocalization.PullMap`, `Polyfold.SBLocalization.PullMap.ofEquivalence`, `Polyfold.SBLocalization.PullMap.ofCovering`, `Polyfold.SBLocalization.PullMap.covers`, `Polyfold.SBLocalization.PullMap.fiberEquiv`, `Polyfold.SBLocalization.PullMap.eq_of_obj_eq`, `Polyfold.SBLocalization.PullMap.obj_fibAct`, `Polyfold.SBLocalization.PullMap.pullSec`, `Polyfold.SBLocalization.PullMap.pullSec_eq_iff`, `Polyfold.SBLocalization.PullMap.isScPlusSectionOn_pullSec`, `Polyfold.SBLocalization.PullMap.pull`, `Polyfold.SBLocalization.PullMap.isWFunctor_pull`, `Polyfold.SBLocalization.PullMap.pull_ofEquivalence`, `Polyfold.SBLocalization.PullMap.isScPlusMultisection_pull`, `Polyfold.SBLocalization.PullMap.IsPullNbhd`, `Polyfold.SBLocalization.PullMap.exists_isPullNbhd`, `Polyfold.SBLocalization.PullMap.isDiffeoComponent_of_isPull`, `Polyfold.SBLocalization.PullMap.IsPullNbhd.mor_biact`, `Polyfold.SBLocalization.PullMap.PullData`, `Polyfold.SBLocalization.PullMap.exists_pullData`, `Polyfold.SBLocalization.PullMap.somePullData`, `Polyfold.SBLocalization.PullMap.pullStructure`, `Polyfold.SBLocalization.PullMap.pullDecoration`, `Polyfold.SBLocalization.PullMap.biact_`**

`compHom`, `Polyfold.SBLocalization.PullMap.ofPerm_compHom`, `Polyfold.SBLocalization.PullMap.pullTau`, `Polyfold.SBLocalization.PullMap.pullTuple`, `Polyfold.SBLocalization.PullMap.isStructurable_pull`, `Polyfold.SBLocalization.PullMap.pullTuple_equiv`, `Polyfold.SBLocalization.PullMap.pullStructured`, `Polyfold.SBLocalization.PullMap.pullStructured_mk`, `Polyfold.SBLocalization.PullMap.forget_pullStructured`, `Polyfold.SBLocalization.theorem_13_4_1`, `Polyfold.SBLocalization.theorem_13_4_1_structured`). `theorem_13_4_1 (hF : SEP.equivalences F) (h : IsStructurable A Λ') : IsStructurable C (pullMult F Λ')`. `theorem_13_4_1_structured hF`: there is `pull : StructuredMultisection A → StructuredMultisection C` with `(pull L).forget = pullMult F L.forget` and, for every structured tuple `T'` and every adapted good system `P : PullData T'.dec.`, `pull ⟦T'⟧ = ⟦pullTuple T' P⟧`, `I_x = I'_{F(x)}`, $\tau_{x,y}(\varphi) = \tau'_{F(x),F(y)}(F(\varphi))$. Uniform machinery `PullMap C A` (a map of strong bundles that is a local sc-diffeomorphism on objects, fibrewise a linear homeomorphism, faithful): `PullMap.ofEquivalence`, `PullMap.ofCovering`; adapted neighbourhoods `IsPullNbhd` (F injective on $U(x)$, $F(U(x)) \subset U'(F(x))$, $F(\Gamma(g,z)) = \Gamma'(F(g),F(z))$), `exists_isPullNbhd` (inside any open set, with properness), `PullData`, `exists_pullData`; sections `pullSec` ($\Phi^{-1} \circ s' \circ F$), `isScPlusSectionOn_pullSec`; `pullStructure`, `pullDecoration`, `pullTau`, `pullTuple`, `isStructurable_pull`; canonicity `pullTuple_equiv`, `pullStructured`, `pullStructured_mk`, `forget_pullStructured`. *Book:* §13.4, Theorem 13.4.1, pp. 621–622 (2017: Theorem 13.3). *Hidden hypotheses:* none. *Deviation (stronger):* The canonical structure is shown independent of the representative of $[\Lambda', \mathcal{U}', S', \tau']$ and of the adapted good system (Definition 13.3.8 equivalence), which the text asserts without proof. The text takes $F(U(x)) = U'(F(x))$ "without loss of generality"; the Lean proof keeps $\mathcal{U}'$ and uses only $F(U(x)) \subset U'(F(x))$ with $F|_{U(x)}$ injective (condition (3) of good systems from faithfulness). No paracompactness and no added hypothesis.

V13.52 (Theorem 13.4.3 (2017: 13.4), the push-forward along a strong bundle equivalence is structurable; `Polyfold.SBLocalization.PushChart.fibAct_comp`, `Polyfold.SBLocalization.PushChart.fibAct_unit`, `Polyfold.SBLocalization.PushChart.fibAct_inv_fibAct`, `Polyfold.SBLocalization.PushChart.fibAct_fibAct_inv`, `Polyfold.SBLocalization.PushChart. _fibAct`, `Polyfold.SBLocalization.PushChart`, `Polyfold.SBLocalization.pushChart`, `Polyfold.SBLocalization.conjEquiv`, `Polyfold.SBLocalization.IsPushNbhd`, `Polyfold.SBLocalization.IsPushNbhd.Φ_eq`, `Polyfold.SBLocalization.IsPushNbhd.Qinv_Φ`, `Polyfold.SBLocalization.IsPushNbhd.Γ_comp_ ,` `Polyfold.SBLocalization.exists_isPushNbhd`, `Polyfold.SBLocalization.pushSec`, `Polyfold.SBLocalization.IsPushNbhd.pushSec_eq`, `Polyfold.SBLocalization.IsPushNbhd.isScPlusSectionOn_pushSec`, `Polyfold.SBLocalization.pushStructure`, `Polyfold.SBLocalization.PushChart.continuousAt_of_mor`, `Polyfold.SBLocalization.PushChart.mapsTo_connectedComponentIn`, `Polyfold.SBLocalization.PushChart.liftMor`, `Polyfold.SBLocalization.PushChart.comp_liftMor`, `Polyfold.SBLocalization.PushChart.liftMor_eq_of_comp`, `Polyfold.SBLocalization.PushChart.continuousOn_liftMor`, `Polyfold.SBLocalization.PushChart.injOn_liftMor`, `Polyfold.SBLocalization.PushChart.liftMor_comp`, `Polyfold.SBLocalization.PushChart.liftMor_inv`, `Polyfold.SBLocalization.PushChart.liftMor_Γ`, `Polyfold.SBLocalization.`

`PushChart.liftMor_biact`, `Polyfold.SBLocalization.PushChart.pushSec_compat`, `Polyfold.SBLocalization.PushChart.isDiffeoComponent_push`, `Polyfold.SBLocalization.pushNbhd`, `Polyfold.SBLocalization.pushGood`, `Polyfold.SBLocalization.pushDecoration`, `Polyfold.SBLocalization.pushTau`, `Polyfold.SBLocalization.pushTuple`, `Polyfold.SBLocalization.theorem_13_4_3`, `Polyfold.SBLocalization.theorem_13_4_3_structured`). `theorem_13_4_3 hF (h : IsStructurable C Λ) : IsStructurable A (pushMult F hF Λ)`; `theorem_13_4_3_structured hF T`: `pushTuple hF T` has $\Lambda = \Phi_* \Lambda$, $I'_y = I_{x_{y'}}$, $\tau'_{y,y'}(\varphi') = \tau_{x_y, x_{y'}}(\mathrm{liftMor}\, \varphi')$. Charts: `PushChart F y` / `pushChart hF y` (a point x_y, $\sigma_y : F(x_y) \to y$ and Chapter 10's `TransportChart` $Q = t \circ (s|U(\sigma_y))^{-1} \circ F$ with $\psi_u : F(u) \to Q(u)$); `conjEquiv` ($G_{x_y} \cong G'_y$). `IsPushNbhd T N N'` ($U'(y) \subset Q(U(x_y))$, $\Gamma'(\psi F(g) \psi^{-1}, z) = \psi_{g*u} \circ F(\Gamma(g,u)) \circ \psi_u^{-1}$), `exists_isPushNbhd` (with properness), `IsPushNbhd.Γ_comp_ `, `Qinv_Φ`. `pushSec T s` ($z \mapsto \mu'(\psi_u, \Phi(s(u)))$), `IsPushNbhd.isScPlusSectionOn_pushSec`, `pushStructure`. `PushChart.liftMor T T'  '` (the φ with $\psi'_{u'} \circ F(\varphi) = \varphi' \circ \psi_u$): `comp_liftMor`, `liftMor_eq_of_comp`, `continuousOn_liftMor` (via `continuousAt_of_mor`: F faithful and a local sc-diffeomorphism), `injOn_liftMor`, `liftMor_comp`, `liftMor_inv`, `liftMor_Γ`, `liftMor_biact`, `pushSec_compat`, `isDiffeoComponent_push`. `pushGood`, `pushDecoration`, `pushTau`, `pushTuple`. *Book:* §13.4, Theorem 13.4.3, pp. 625–626 (2017: Theorem 13.4). *Hidden hypotheses:* none. *Deviation (same):* Proposition 13.4.2 is not used: every $y \in X'$ (in $F(X)$ or not) gets a chart along a morphism $\sigma_y : F(x_y) \to y$, which combines the text's two steps ($F|_{U(x_y)}$ and the move along ψ in 13.4.2). No canonicity claim in the statement and none proved (the charts and neighbourhoods are chosen).

V13.53 (Theorem 13.4.4 (2017: 13.5), the pull-back along a proper strong bundle covering is structurable; `Polyfold.SBLocalization.PullMap.ofCovering`, `Polyfold.SBLocalization.theorem_13_4_4`, `Polyfold.SBLocalization.theorem_13_4_4_structured`). `theorem_13_4_4 (hΦ : C.E.IsProperStrongBundleCovering2021 A.E F Φ) (h : IsStructurable A Λ) : IsStructurable C (Λ   Φ)`; `theorem_13_4_4_structured hΦ` (as for 13.4.1: canonical pull with $I_y = I_{F(y)}$, $\tau_{y,y'}(\varphi) = \tau_{F(y),F(y')}(F(\varphi))$, independent of choices). Same `PullMap` machinery through `PullMap.ofCovering`. *Book:* §13.4, Theorem 13.4.4, pp. 626–628 (2017: Theorem 13.5). *Hidden hypotheses:* none. *Deviation (same):* The text's disjoint decomposition $F^{-1}(U(x)) = \bigsqcup V(z)$ is not needed: adapted $V(y)$ with F injective on $V(y)$ and $F(V(y)) \subset U(F(y))$ suffice (local sc-diffeomorphism, faithfulness, fibrewise iso of Φ). The canonicity statement is extra.

V13.54 (Definition 13.4.5 (1) (2017: 13.22 (1)), commensurable structured sc⁺-multisections; `Polyfold.SBLocalization.CommData`, `Polyfold.SBLocalization.CommData.smul_eq_of_diag`, `Polyfold.SBLocalization.IndexCorr.ofEquiv_injective`, `Polyfold.SBLocalization.CommData.Compat`, `Polyfold.SBLocalization.CommData.Sections`, `Polyfold.SBLocalization.CommData.Over`, `Polyfold.SBLocalization.CommData.Over.toTuple`, `Polyfold.SBLocalization.CommData.Over.toStructured`, `Polyfold.SBLocalization.CommData.ofTuple`, `Polyfold.SBLocalization.CommData.Over.ofTuple`, `Polyfold.SBLocalization.CommData.Over.ofTuple_toTuple`, `Polyfold.SBLocalization.Commensurable`, `Polyfold.SBLocalization.Commensurable.refl`, `Polyfold.SBLocalization.Commensurable.symm`).

`CommData A`: the choice c: a `GoodSystem` $\mathcal{U}$, index types I_x (finite, non-empty) with G_x-actions, and maps $\tau_{x,x'}$ with the conditions of D13.3.4 (1)–(2) and D13.3.6 (1)–(2) that do not involve sections (`locallyConstant`, `finite`, `biact`, `diag`, `swap`). `CommData.Sections c` (sections s_i^x on $U(x)$: sc⁺, equivariant, compatible with τ — D13.3.4 (3)) and `CommData.Over c Λ` (plus the counting formula for Λ on every $U(x)$). `Over.toTuple`, `Over.toStructured`: the structured tuple and class. `CommData.ofTuple T`, `Over.ofTuple T`, `Over.ofTuple_toTuple` (rfl): every structured tuple is of this form. `Commensurable L L := c Λ Λ (o : c.Over Λ) (o : c.Over Λ), o .toStructured = L o .toStructured = L;` `Commensurable.refl`, `Commensurable.symm`. `CommData.smul_eq_of_diag` (with `IndexCorr.ofEquiv_injective`): the G_x-action on I_x is determined by τ (D13.3.6 (1) at $y = x$), so taking the actions into c adds nothing to "$\tau_1 = \tau_2$". *Book:* §13.4, Definition 13.4.5 (1), p. 628 (2017: Definition 13.22 (1)). *Hidden hypotheses:* none. *Deviation (restated):* c carries the G_x-actions explicitly (determined by τ, `CommData.smul_eq_of_diag`) and `nonempty` ($I_x \neq \emptyset$; it holds for the data of every structured sc⁺-multisection, `SymmetricStructureOn.card_pos`, and is needed for the empty family in the lf sum). "We can take representatives" is an existential over representatives of both classes, as printed. The paracompactness of $|X|$ is not needed for the definition and is not assumed.

V13.55 (Definition 13.4.5 (2) (2017: 13.22 (2)), the sum of two commensurable structured sc⁺-multisections; `Polyfold.SBLocalization.IndexCorr.sat_iInter`, `Polyfold.SBLocalization.IndexCorr.card_filter_comp_eq`, `Polyfold.SBLocalization.IndexCorr.count_eq_of_sat`, `Polyfold.SBLocalization.IsScPlusSectionOn.of_locally`, `Polyfold.SBLocalization.CommData.countAt`, `Polyfold.SBLocalization.CommData.sumMult`, `Polyfold.SBLocalization.CommData.Sections.P_s`, `Polyfold.SBLocalization.CommData.Sections.compat_self`, `Polyfold.SBLocalization.CommData.Sections.countAt_fibAct`, `Polyfold.SBLocalization.CommData.Sections.fibAct_unit`, `Polyfold.SBLocalization.CommData.Sections.unit_mem_mor`, `Polyfold.SBLocalization.CommData.Sections.countAt_eq`, `Polyfold.SBLocalization.CommData.Sections.isWFunctor_sumMult`, `Polyfold.SBLocalization.CommData.Sections.countAt_eq_sum`, `Polyfold.SBLocalization.CommData.Sections.card_ne_zero`, `Polyfold.SBLocalization.CommData.Sections.isScPlusMultisection_sumMult`, `Polyfold.SBLocalization.CommData.Sections.toOver`, `Polyfold.SBLocalization.CommData.Over.eq_sumMult`, `Polyfold.SBLocalization.CommData.Over.isScPlusMultisection`, `Polyfold.SBLocalization.CommData.Over.sym`, `Polyfold.SBLocalization.CommData.Over.toDecoration`, `Polyfold.SBLocalization.CommData.Over.tau`, `Polyfold.SBLocalization.CommData.Over.s_eq_fiberZero`, `Polyfold.SBLocalization.fibAct_finsum`, `Polyfold.SBLocalization.CommData.isLocallyFiniteFamily_of_finite`, `Polyfold.SBLocalization.CommData.pairFam`, `Polyfold.SBLocalization.CommData.pairOver`, `Polyfold.SBLocalization.CommData.commSum`, `Polyfold.SBLocalization.CommData.Over.commSum`, `Polyfold.SBLocalization.CommData.definition_13_4_5`). `CommData.commSum o o` (the functor), `CommData.Over.commSum o o : c.Over (commSum o o)` (the data $(\mathcal{U}, S, \tau)$ with the sums; `.toStructured` is the class), and `CommData.definition_13_4_5 o o : IsScPlusMultisection A (commSum o o)` ∧ (on $U(x)$ the sections are $s_i^{1,x} + s_i^{2,x}$, fibrewise sum) ∧ (the counting formula at every x with $P(w) \in U(x)$). Ingredients: `IndexCorr.count_` `eq_of_sat` (count transport: along a correspondence the normalized counts $|I_x|^{-1}\#\{i \mid s_i = v\}$ and $|I_{x'}|^{-1}\#\{j \mid s'_j = \mu(\varphi, v)\}$ agree), `Sections.countAt_eq` (the count does not depend on x), `Sections.isWFunctor_sumMult`, `Sections.isScPlusMultisection_sumMult` (the count of any `Sections` is an sc$^+$-multisection functor), `fibAct_finsum` ($\mu(\psi, \cdot)$ is additive on fibres), `IsScPlusSectionOn.of_locally` (being sc$^+$ on U is local). *Book:* §13.4, Definition 13.4.5 (2), p. 628 (2017: Definition 13.22 (2)). *Hidden hypotheses:* none. *Deviation (same):* The sum is defined through the general locally finite sum (V13.56) over `Bool`; the stated section formula recovers the binary $s^1 + s^2$. The well-definedness, which the text calls evident, is proved.

V13.56 (Definition 13.4.5, the locally finite commensurable sum; `Polyfold.SBLocalization.CommData.sumFib`, `Polyfold.SBLocalization.CommData.sumSec`, `Polyfold.SBLocalization.CommData.P_sumSec`, `Polyfold.SBLocalization.CommData.exists_finite_support`, `Polyfold.SBLocalization.CommData.support_subset`, `Polyfold.SBLocalization.CommData.sumSec_eq_sectionSum`, `Polyfold.SBLocalization.CommData.lfSections`, `Polyfold.SBLocalization.CommData.lfSum`, `Polyfold.SBLocalization.CommData.Over.lfSum`, `Polyfold.SBLocalization.CommData.definition_13_4_5_lf`). `CommData.lfSum o hlf` (the functor), `CommData.Over.lfSum o hlf : c.Over (lfSum o hlf)` (data with $s_i^x = \sum_j^f s_i^{j,x}$, `CommData.lfSections`), and `CommData.definition_13_4_5_lf o hlf: IsScPlusMultisection A (lfSum o hlf)` ∧ (on $U(x)$ the sections are the fibrewise `finsum` of the $s_i^{j,x}$) ∧ (the counting formula). Hypotheses: `o :   j, c.Over (Λ j)`, `hlf : IsLocallyFiniteFamily Λ` (D13.2.12). Helpers `sumFib`, `sumSec`, `P_sumSec`, `exists_finite_support`, `support_subset`, `sumSec_eq_sectionSum` (locally the sum is the finite `sectionSum` of `extendZeros`, hence sc$^+$). *Book:* §13.4, after Definition 13.4.5 (2), p. 629 (2017: after Definition 13.22). *Hidden hypotheses:* none. *Deviation (same):* "Locally finite" is `IsLocallyFiniteFamily` (open saturated O on which only finitely many Λ_j are non-trivial); "pairwise commensurable" is read as "over one common c", which is what the text's $[\Lambda_j, \mathcal{U}, S_j, \tau]$ writes.

V13.57 (Definition 13.5.1 (2017: 13.23), atomic on a neighbourhood with the natural action; `Polyfold.SBLocalization.AtomicDataOn`, `Polyfold.SBLocalization.AtomicDataOn.count`, `Polyfold.SBLocalization.AtomicDataOn.Reach`, `Polyfold.SBLocalization.AtomicDataOn.IsRelated`, `Polyfold.SBLocalization.IsAtomicOn`, `Polyfold.SBLocalization.AtomicData.toOn`, `Polyfold.SBLocalization.IsAtomic.isAtomicOn`). `AtomicDataOn A` (fields `x`, `N : NatNbhd x`, `isProperSubset_closure`, `V`, `isOpen_V`, `mem_V`, `mapsTo_V`, `closure_V_subset`, `I`, `s`, `isScPlusSectionOn`, `equivariant`, `vanish`), `AtomicDataOn.count`, `.Reach`, `.IsRelated` (condition (3)), `IsAtomicOn A Λ`. `AtomicData.toOn`, `IsAtomic.isAtomicOn`: `IsAtomic` implies it. *Book:* §13.5, Definition 13.5.1, p. 629 (2017: Definition 13.23). *Hidden hypotheses:* none. *Deviation (extra):* "G_x-invariant $U(x)$" is read as a `NatNbhd` (open, with the natural action), the reading of the good systems in §13.1. `IsAtomic` asks $U(x)$ to be the (connected) domain of a natural representation, which is stronger; the good-system sets $U(x_j)$ in T13.5.7 need not be such domains.

V13.58 (Theorem 13.5.7 (2017: 13.10), a structurable sc$^+$-multisection is a locally finite sum of atomic ones; `Polyfold.SBLocalization.CommData.smulFunOver`, `Polyfold.SBLocalization.multisectionSmulFun_of_eq_zero`, `Polyfold.SBLocalization.exists_`

orbit_nbhd, Polyfold.SBLocalization.theorem_13_5_7, Polyfold.SBLocalization.theorem_13_5_7_book). `theorem_13_5_7 (hpu : A.X.HasScPartitionsOfUnity) (T : StructuredTuple A)`: there are `Λf : A.Obj → A.W →  0, o :  j, (CommData.ofTuple T).Over (Λf j)`, `hlf : IsLocallyFiniteFamily Λf` with `j, IsAtomicOn A (Λf j)`, `CommData.lfSum o hlf = T.Λ`, the sum's sections equal T's on every $U(x)$, and `(CommData.Over.lfSum o hlf).toStructured = ⟦T⟧`. `theorem_13_5_7_book (_hpara : HasParacompactOrbitSpace) (hpu) (hΛ : IsStructurable A Λ)`: some T with `T.Λ = Λ` and such a family. Proof as in the text: $O_x \ni |x|$ open with $\mathrm{cl} O_x \subseteq |U(x)|$ (`exists_orbit_nbhd`, regularity of $|X|$, L7.3.5), a partition of unity (β_x) subordinate to $\pi^{-1}(O_x)$, $\Lambda_j = \beta_j \odot \Lambda$ with data $\beta_j s_i^x$ (`CommData.smulFunOver`, from `smulFun`/`Tau.smulFun`), atomic at $x_j = j$ with $V = U(j) \cap \pi^{-1}(O_j)$ (closure inside $U(j)$ by `lemma_13_1_1`), $\sum_j \beta_j = 1$. Helper `multisectionSmulFun_of_eq_zero`. *Book:* §13.5, Theorem 13.5.7, p. 635 (2017: Theorem 13.10). *Hidden hypotheses:* none. *Deviation (stronger):* For every structured lift T (the text: "a structured lift"), with $J = X$; the class equality is literal (`Quotient.sound`), and the sections agree on every $U(x)$. Atomic is `IsAtomicOn` (V13.57). With `IsAtomic` the statement would need connected natural-rep domains in the good system, which `GoodSystem` does not give. Paracompactness of $|X|$ is not used (`_hpara` is kept in `_book` only); regularity of $|X|$ (L7.3.5) suffices for the shrinking, and `HasScPartitionsOfUnity` for (β_j).

V13.59 (Proposition 13.4.2 (2017: 13.6), extension of a structured sc⁺-multisection from V with $|V| = |X|$; Polyfold.SBLocalization.IsWFunctorOn, Polyfold.SBLocalization.IsScPlusMultisectionOn, Polyfold.SBLocalization.StructuredOn, Polyfold.SBLocalization.StructuredTuple.toStructuredOn, Polyfold.SBLocalization.extendMult, Polyfold.SBLocalization.isWFunctor_extendMult, Polyfold.SBLocalization.extendMult_eq, Polyfold.SBLocalization.eq_extendMult, Polyfold.SBLocalization.unitChart, Polyfold.SBLocalization.isPushNbhd_unit, Polyfold.SBLocalization.ExtChart, Polyfold.SBLocalization.isScPlusMultisection_extendMult, Polyfold.SBLocalization.extTuple, Polyfold.SBLocalization.proposition_13_4_2, Polyfold.SBLocalization.proposition_13_4_2_structurable). `proposition_13_4_2 (hV : IsOpen V) (h V :  '' V = univ) (L : StructuredOn A V)`: (1) `IsScPlusMultisection A (extendMult V L.Λ)`, it agrees with Λ over V, and every functor Λ' on W agreeing with Λ over V equals it (`eq_extendMult`); (2) for every family of open $O(x) \ni x$ there is a `StructuredTuple A` with $\Lambda = \bar{\Lambda}$, $U(q)$ the given ones for $q \in V$, $U(x) \subset O(x)$ for $x \notin V$, and I_q (with its G_q-action), the sections on $U(q)$ and $\tau_{q,q'}$ on $\mathcal{U}(q, q')$ equal to the given ones (via casts of the index types). `proposition_13_4_2_structurable`: $\bar{\Lambda}$ is structurable. The relative data: `StructuredOn A V` (Definitions 13.3.4, 13.3.6 with all indices in V, the good-system condition read as `IsGoodIn` for V), `IsScPlusMultisectionOn A V`, `IsWFunctorOn`. The proof uses the push machinery of §13.4 for $F = \mathrm{id}$: the unit transport chart at $q \in V$ (`unitChart`, `isPushNbhd_unit`), push charts inside $O(x)$ for $x \notin V$, and τ through `liftMor`, which covers the text's cases (i)–(iii). *Book:* §13.4, Proposition 13.4.2, pp. 623–625 (2017: Proposition 13.6). *Hidden hypotheses:* none. *Deviation (restated):* The project has no strong bundle over the open sub-ep-groupoid V; "Λ structured on $W|V$" is read as the relative data `StructuredOn A V` (Definitions 13.3.4, 13.3.6 with all indices in V, the good-system condition as `IsGoodIn` for V, as in `theorem_13_1_7_book`), and "Λ an sc⁺-multisection on $W|V$" as `IsScPlusMultisectionOn`

`A V` (functor along morphisms inside V, local structures on natural neighbourhoods inside V). "Canonical" is read as uniqueness among functors on W extending Λ, which is proved. No added hypothesis; the data are inhabited (`StructuredTuple.toStructuredOn`, $V = X$, e.g. the zero multisection). Gap G.163 closed.

What is not verified in this chapter

- p. 618, Remark 13.3.13 (2017: 13.4), the negative half: a statement about an equality of types, with no Lean statement (V13.37; Findings, item 12).
- p. 630, Theorem 13.5.3 (2017: 13.6) without a good system: as printed it assumes no paracompactness of $|X|$; the existence of a good system is necessary for structurability (V13.46; Findings, item 17).

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15. Chapter 14 (HWZ, Part III, §§14.1–14.6 (2017 §§14.1–14.6)): begun 2026-11-13 (Part III waves 3–7), verified for good neighbourhoods and good systems for ∂X (Theorem 14.1.6 with the chapter's standing assumptions), the tameness of the faces $\bar{\theta}_j$, sc⁺-multisections and structured sc⁺-multisections over ∂X, the local extension result, Proposition 14.1.8 in a stronger form, the good structured version, the extension of the correspondences, z-loops, the relations $\sim_z$ and $\approx_z$ with Proposition 14.4.6 as printed, coherent neighbourhood systems and the local extensions of Proposition 14.4.14 for the objects of §14.3, Lemma 14.5.1 with (14.53), Proposition 14.5.4, the structurability criterion (Theorem 14.5.5), Theorem 14.1.9 with tame strong bundle charts and the reading $I_x \neq \emptyset$, and §14.6 for agreeing representatives, with sc⁺ on the corners, except the multisection form of Lemma 14.5.2, §14.6 from a class-level compatible assignment and the natural representation of a face

Chapter 14 (pp. 637–682; numbers and pages are those of the book (Springer 2021), with the 2017 text's numbers in parentheses) extends a structurable sc⁺-multisection from ∂X to X with control of its size and support. §14.1 (2017: §14.1) defines good neighbourhoods of boundary points and good systems for ∂X, sc⁺-multisections over ∂X and their structured versions, and states the main result (Proposition 14.1.1, Definitions 14.1.2, 14.1.3, Proposition 14.1.4, Definition 14.1.5, Theorem 14.1.6, Definition 14.1.7, Proposition 14.1.8, Theorem 14.1.9 (2017: Definitions 14.1–14.4, Theorem 14.1, Proposition 14.1, Theorem 14.2)). §14.2 builds a good structured version of Λ (Lemmas 14.2.1, 14.2.2, Proposition 14.2.3 (2017: Lemma 14.1, Proposition 14.2)). §14.3 extends the correspondences (Lemmas 14.3.1, 14.3.3, 14.3.6, 14.3.7, Definition 14.3.2, Propositions 14.3.4, 14.3.8 (2017: Lemmas 14.2–14.5, Propositions 14.3, 14.4)). §14.4 introduces z-loops, the relations $\sim_z$, $\approx_z$ and coherent neighbourhood systems, and constructs the local extensions (Definitions 14.4.2–14.4.4, 14.4.8, 14.4.11, Lemmas 14.4.5, 14.4.7, 14.4.9, 14.4.13, Propositions 14.4.6, 14.4.10, 14.4.14 (2017: Definitions 14.5–14.9, Lemmas 14.6–14.9, Propositions 14.5–14.7)). §14.5 assembles the extension (Lemmas 14.5.1, 14.5.2, Proposition 14.5.4, Theorem 14.5.5 (2017: Lemmas 14.10, 14.11, Proposition 14.8, Theorem 14.3)); §14.6 assembles face-wise structured multisections (Definitions 14.6.1, 14.6.2, Theorem 14.6.3 (2017: Definitions 14.10, 14.11, Theorem 14.4)). The chapter was begun in Part III waves 3 and 4 and continued in waves 5–7.

What is verified. Proposition 14.1.1 is verified as stated, Definitions 14.1.2, 14.1.3, 14.1.5, 14.1.7 restated (V14.1–V14.3, V14.5, V14.7); Proposition 14.1.4 as stated with the standing assumptions read as `hB`, `hpu` (V14.4); Theorem 14.1.6 with tameness and the paracompactness of $|X|$ (V14.6). Theorem 14.1.9 is verified with tame strong bundle charts and the reading $I_x \neq \emptyset$ of Definition 14.1.3 (`ChargesBoundary`, necessary) added, from the data of §§14.3–14.5, which are built (Part III wave 6), with a witness with $\partial X \neq \emptyset$ (Part III wave 7); the size control, Proposition 14.5.4 and the assembly are proved (V14.8, V14.37, V14.38, V14.46; gaps G.170, G.171 closed, gaps G.172, G.186). §14.2 is verified as stated (V14.10–V14.12). §14.3: Lemma 14.3.1 in a stronger form, Definition 14.3.2 restated, Lemma 14.3.3 and Proposition 14.3.4 as stated, Lemma 14.3.6 with $|\partial X|$ closed named, Lemma 14.3.7 in a stronger form, Proposition 14.3.8 abstract and instantiated (V14.13–V14.20). §14.4: Definitions 14.4.2–14.4.4, 14.4.8, 14.4.11, Lemma 14.4.7 and Proposition 14.4.14 restated, Lemma 14.4.5 with `diag` named (V14.22–V14.24, V14.26, V14.27, V14.30, V14.33); Proposition 14.4.6, Lemma 14.4.9 and Proposition 14.4.10 as stated, (2) of Proposition 14.4.6 in Part III wave 5, also in a stronger form (V14.25, V14.28, V14.29; gap G.168 closed), the setting of §14.4 from the objects of §14.3 (V14.39; gap G.169 closed), Lemma 14.4.13 with (2) for $z' \in \mathrm{cl}_X(V_a)$ (V14.32). §14.5: Lemma 14.5.1 and Theorem 14.5.5 in stronger forms (V14.34, V14.35). Remarks 14.1.10, 14.3.5, 14.4.1, 14.4.12 are prose (V14.9, V14.17, V14.21, V14.31). Part III wave 5: the tameness of the $\bar{\theta}_j$ as stated (V14.36; gap G.165 closed); Proposition 14.1.8 for a representative with tame middle, with the transport of the face condition named, in Part III wave 6 in a stronger form at class level (V14.44; gap G.164 closed); §14.6: the preamble and Definitions 14.6.1, 14.6.2 restated, Theorem 14.6.3 for agreeing representatives (V14.40–V14.43; gaps G.173, G.184, G.185). Part III waves 6–7: (14.53) for the $\hat{V}_a$ themselves (V14.45); the face-wise $U_F(x)$ good up to equivalence (V14.47; gap G.185); sc$^+$ on the corners (V14.41; gap G.184 closed); the extension part of Theorem 14.6.3 in the proved form of Theorem 14.1.9 (V14.43).

The hypotheses. Tameness and the paracompactness of $|X|$, the chapter's standing assumptions (witnesses `Polyfold.EpGroupoid.trivial_quadrant_witness`, `Polyfold.EpGroupoid.hasParacompactOrbitSpace_trivial`); `hB`, `hpu` in Proposition 14.1.4 and the later theorems (witness `Polyfold.SBLocalization.proposition_14_1_4_hypotheses_rn`); the fields `diag`, `swap` of the z-loop setting (witness `Polyfold.EpGroupoid.ZSetup.trivialSetup`); `HasThetaLocalEq`, Proposition 14.4.6 (2) in the proved form, now a theorem (`Polyfold.SBLocalization.hasThetaLocalEq_of_limit`, `Polyfold.SBLocalization.ZConcrete.hasThetaLocalEq`); the extension statement of Theorem 14.1.9, proved from `Polyfold.SBLocalization.AssemblyHyp` (witness for $\partial X = \emptyset$ only, `Polyfold.SBLocalization.assemblyHyp_of_boundary_eq_empty`); since Part III wave 6 the formal statement without `ChargesBoundary` holds only for $\partial X = \emptyset$ (`Polyfold.SBLocalization.not_theorem_14_1_9_statement_a6`, V14.8), and the forms resting on it are vacuous with boundary. The other hypotheses of Theorem 14.1.9 hold together (`Polyfold.SBLocalization.theorem_14_1_9_statement_hyp_witness`). Part III wave 5: `SuppInSat` in Proposition 14.5.4 (witness `Polyfold.SBLocalization.TransportInput.suppInSat_of_zero`; Lemma 14.5.1 gives it); `ZLimitSetting` and `CompatNbhd` in the abstract form (from the objects of §14.3, `Polyfold.SBLocalization.ZConcrete.toLimit`, `Polyfold.SBLocalization.ZConcrete.compatNbhd`); `CompatibleFaceData` in Theorem 14.6.3 (witness `Polyfold.SBLocalization.CompatibleFaceData.zero`); `FacePull_hyp` and `FacePush_hyp` in Proposition 14.1.8, theorems since Part III wave 6 (`Polyfold.SBLocalization.PullMap.facePull_hyp`, `Polyfold.SBLocalization.facePush_hyp_of_equivalence`). Part III waves 6–7: in Theorem 14.1.9 `hB` (tame strong bundle charts at every object) and `ChargesBoundary Λ`, both with the witness `Polyfold.HalfSpaceBundle.theorem_14_1_9_constSec` ($\partial X \neq \emptyset$, nonzero fibres, nonempty domain support, trivial isotropy; all hypotheses together `Polyfold.HalfSpaceBundle.theorem_14_1_9_charged_hypotheses_halfSEP`); in §14.6 the text's standing `IsEpFaceStructured` and `HasParacompactOrbitSpace`. Readings: Findings, items 1–16.

The setting. Ep-groupoids, natural representations, tameness and the degeneracy index are Chapter 7's; strong bundles over ep-groupoids and auxiliary norms Chapters 8 and 12's; sc^+-multisections, good systems, decorations, correspondences and structurability Chapter 13's (`Polyfold.SBLocalization.IsStructurable`); faces and the boundary calculus are Part I's (Chapter 2).

V14.1 (Proposition 14.1.1 (2017: before Definition 14.1), good open neighbourhoods of boundary points; `Polyfold.EpGroupoid.HasBoundaryFaces`, `Polyfold.EpGroupoid.HasBoundaryFaces.inter_boundary_eq`, `Polyfold.EpGroupoid.exists_isGoodBoundaryNbhd_le`, `Polyfold.EpGroupoid.proposition_14_1_1`, `Polyfold.TameBoundary.isPreconnected_faceAt`, `Polyfold.TameBoundary.isCenteredAt_connectedComponentIn`, `Polyfold.TameBoundary.connectedComponentIn_inter_deg_of_mem_cc`). `proposition_14_1_1 (hX : X.IsTame) (hx : x   X) (hW : IsOpen W) (hxW : x   W)` gives a `V : X.NatNbhd x` with $V \subseteq W$, (1), (3), and some θ with `HasBoundaryFaces x V.U`   and $V \cap \partial X = \bigcup_i V \cap \overline{\theta_i}$. (2) is the `NatNbhd` structure. The relative form `exists_isGoodBoundaryNbhd_le` keeps a given natural action ($V \leq N$). *Book:* §14.1, Proposition 14.1.1, p. 638 (2017: the discussion before Definition 14.1). *Hidden hypotheses:* none. *Deviation (same):* "Arbitrarily small" is read as "inside any open $W \ni x$". $\bar{\theta}_i$ is the closure in V, that is $V \cap \operatorname{cl} \theta_i$. The proof uses Part I's abstract face calculus (`Chapter2/TameBoundary`: `reach`, `faceAt`, centred face-structured sets) instead of the text's chart argument.

V14.2 (Definition 14.1.2 (2017: 14.1), good open neighbourhoods; `Polyfold.EpGroupoid.IsGoodBoundaryNbhd`, `Polyfold.EpGroupoid.NatNbhd.bdry`, `Polyfold.EpGroupoid.NatNbhd.bdry_subset`, `Polyfold.EpGroupoid.NatNbhd.bdry_subset_boundary`, `Polyfold.EpGroupoid.NatNbhd.degIndex_Φ`, `Polyfold.EpGroupoid.NatNbhd.mapsTo_Φ_bdry`, `Polyfold.EpGroupoid.NatNbhd.bdry_mono`, `Polyfold.EpGroupoid.IsGoodBoundaryNbhd.bdry_eq`, `Polyfold.EpGroupoid.IsGoodBoundaryNbhd.mem_boundary`, `Polyfold.EpGroupoid.exists_isGoodBoundaryNbhd`). `IsGoodBoundaryNbhd V` for `V : X.NatNbhd x`, with fields `isConnected`, `isConnected_bdry`, `isProperSubset_closure` and `hasFaces`. `NatNbhd.bdry V = V.U   X` is $U(x)$. The remarks: `IsGoodBoundaryNbhd.bdry_eq` ($U(x) = \bigcup \bar{\theta}_i$), `NatNbhd.mapsTo_Φ_bdry` (invariance), `IsGoodBoundaryNbhd.mem_boundary` and `NatNbhd.degIndex_Φ`. Witness: `exists_isGoodBoundaryNbhd` (tame X). *Book:* §14.1, Definition 14.1.2, p. 641 (2017: Definition 14.1). *Hidden hypotheses:* none. *Deviation (restated):* "Admits the natural G_x-action" is `NatNbhd`. The remark that the $\bar{\theta}_j$ are tame M-polyfolds is V14.36 (Part III wave 5; gap G.165 closed).

V14.3 (Definition 14.1.3 (2017: 14.2), sc$^+$-multisections over ∂X; `Polyfold.SBLocalization.IsFaceScPlusSectionOn`, `Polyfold.SBLocalization.IsFaceScPlusSectionOn.mono`, `Polyfold.SBLocalization.isFaceScPlusSectionOn_of_isBoundaryScPlusSection`, `Polyfold.SBLocalization.isFaceScPlusSectionOn_fiberZero`, `Polyfold.SBLocalization.BoundarySymmetricStructure`, `Polyfold.SBLocalization.BoundarySymmetricStructure.P_s`, `Polyfold.SBLocalization.BoundarySymmetricStructure.restrict`, `Polyfold.SBLocalization.IsBoundaryWFunctor`, `Polyfold.SBLocalization.IsBoundaryScPlusMultisection`, `Polyfold.SBLocalization.zeroBoundaryStructure`, `Polyfold.SBLocalization.isBoundaryScPlusMultisection_zero`, `Polyfold.SBLocalization.ChargesBoundary`, `Polyfold.SBLocalization.BoundarySymmetricStructure.nonempty_I_a6`, `Polyfold.SBLocalization.BoundaryStructuredTuple.nonempty_I_a6`, `Polyfold.SBLocalization.chargesBoundary_of_nonempty_a6`, `Polyfold.SBLocalization.chargesBoundary_zeroMultisection_a6`, `Polyfold.SBLocalization.chargesBoundary_of_structurable_eq_a6`, `Polyfold.SBLocalization.IndexCorr.eq_of_isEmpty_a6`, `Polyfold.SBLocalization.emptyBoundaryStructure_a6`, `Polyfold.SBLocalization.emptyTau_a6`, `Polyfold.SBLocalization.isBoundaryStructurable_zeroFun_a6`, `Polyfold.SBLocalization.boundaryDomSupp_zeroFun_a6`, `Polyfold.SBLocalization.isFaceScPlusSectionOn_of_isScPlusSectionOn_w1`). `IsFaceScPlusSectionOn A s U` is (1): Part I's Definition 5.4.1 with ∂X replaced by U, that is, in every tame strong bundle chart and at every point of U, sc$^+$ on every local face germ. `BoundarySymmetricStructure A Λ V` has fields `I`, `fintype`, `mulAction`, `s`, `isFaceScPlus` on `V.bdry`, `equivariant` on `V.bdry`, and `rep` for $P(w) \in$ `V.bdry`. `IsBoundaryWFunctor` and `IsBoundaryScPlusMultisection A Λ` (functor on $W|\partial X$, plus a good `V` with a structure at every boundary point). Witness: `isBoundaryScPlusMultisection_zero` (tame X).

Part III wave 6 (`Chapter14/P3W6A6_EmptyIndex.lean`): the formal structure admits $I_x = \emptyset$; then (3) reads $\Lambda(w) = 0^{-1} \cdot 0 = 0$ in `0`, and $\Lambda \equiv 0$ is structurable over ∂X with empty index sets (`isBoundaryStructurable_zeroFun_a6`, `emptyTau_a6`, `IndexCorr.eq_of_isEmpty_a6`). `ChargesBoundary Λ`: every $x \in \partial X$ has w over x with $\Lambda(w) \neq 0$; for a tuple it is the same as all $I_x \neq \emptyset$ (`BoundaryStructuredTuple.nonempty_I_a6`, `chargesBoundary_of_nonempty_a6`); nonvacuity `chargesBoundary_zeroMultisection_a6`; a structurable Λ' on X charges every fibre (`chargesBoundary_of_structurable_eq_a6`). Part III wave 7: (1) holds for every sc$^+$-section on X (`isFaceScPlusSectionOn_of_isScPlusSectionOn_w1`, any SEP, through `corner_retraction`). *Book:* §14.1, Definition 14.1.3, p. 642 (2017: Definition 14.2). *Hidden hypotheses:* none. *Deviation (restated):* "$s|\bar{\theta}_j$ is sc$^+$" is read through the local faces: in a tame chart, the germs of the $\bar{\theta}_j$ at a point of $U(x)$ are the local faces there. "A functor on $W|\partial X$" is invariance along morphisms with source in ∂X. Only the values over ∂X enter. The text's standing hypothesis "$|X|$ paracompact" is not used. The text's factor $1/|I_x|$ in (3) presumes $I_x \neq \emptyset$; the formal `BoundarySymmetricStructure` does not require it. Where it matters (Theorem 14.1.9 and the statements built on it, V14.8, V14.43, V15.32, V15.34, V16.71) the reading $I_x \neq \emptyset$ enters as `ChargesBoundary`, which is necessary for the extension (`theorem_14_1_9_iff_a6`). The formal statements without it are vacuous when $\partial X \neq \emptyset$ (V14.8).

V14.4 (Proposition 14.1.4 (2017: Remark 14.2), the local extension result; `Polyfold.StrongBundle.exists_local_extension_of_face`, `Polyfold.SBLocalization.exists_scPlusSection_extension_of_face`, `Polyfold.SBLocalization.proposition_14_1_4_of_closure`, `Polyfold.SBLocalization.proposition_14_1_4`, `Polyfold.SBLocalization.IsScPlusSectionOn.of_locally_p3w4`, `Polyfold.SBLocalization.exists_scSmooth_cutoff_p3w4`, `Polyfold.SBLocalization.exists_scPlusSectionOn_extension_of_face`, `Polyfold.SBLocalization.proposition_14_1_4_full`, `Polyfold.SBLocalization.proposition_14_1_4_hypotheses_rn`, `Polyfold.SBLocalization.fibAct_comp_p3w3`, `Polyfold.SBLocalization.fibAct_unit_p3w3`, `Polyfold.SBLocalization.avgTerm`, `Polyfold.SBLocalization.avgTerm_apply`, `Polyfold.SBLocalization.isScPlusSectionOn_avgTerm`, `Polyfold.SBLocalization.P_avgTerm`, `Polyfold.SBLocalization.avgTerm_equivariant`, `Polyfold.SBLocalization.avgSection`, `Polyfold.SBLocalization.isSection_sumSection_p3w3`, `Polyfold.SBLocalization.isScPlusSectionOn_avgSection`, `Polyfold.SBLocalization.avgSection_apply`, `Polyfold.SBLocalization.avgSection_equivariant`, `Polyfold.SBLocalization.avgSection_eq_of_equivariant`, `Polyfold.SBLocalization.proposition_14_1_4_hyp`, `Polyfold.SBLocalization.proposition_14_1_4_hyp_witness`). `proposition_14_1_4_full hB hpu S`: for every `S : BoundarySymmetricStructure A Λ V` there are $\bar{s}_i$ with `IsScPlusSectionOn A (s̄ i) V.U`, $\bar{s}_i = s_i$ on `V.bdry`, and $\bar{s}_{g(i)}(g * y) = \mu(\Gamma(g, y), \bar{s}_i(y))$ for $y \in V(x)$. Hypotheses: `hB` (every object lies in a tame strong bundle chart) and `hpu : A.X.PObj.AdmitsScPartitionsOfUnity`. The non-equivariant step is `exists_scPlusSectionOn_extension_of_face` (face-wise sc$^+$ on $V \cap \partial X$ □ an sc$^+$-section on V equal to s on $V \cap \partial X$, any open V). Also: `proposition_14_1_4_of_closure` (on any $V' \leq V(x)$ with cl $V' \subseteq V(x)$; a global sc$^+$-section of P there) and `proposition_14_1_4` (a good $V'(x) \leq V(x)$ with cl $V' \subseteq V(x)$, via Lemma 14.2.1).

`proposition_14_1_4_hyp S s h h b`. Given sc$^+$-sections s_i^0 on $V(x)$ with $s_i^0 = s_i$ on $U(x)$, there are sc$^+$ $\bar{s}_i$ on $V(x)$ with $\bar{s}_i = s_i$ on $U(x)$ and $\bar{s}_{g(i)}(g * y) = \mu(\Gamma(g, y), \bar{s}_i(y))$ for all $y \in V(x)$. The sections are $\bar{s}_j(y) = |G_x|^{-1} \sum_g \mu(\Gamma(g^{-1}, g * y), s^0_{g(j)}(g * y))$ (`avgSection`). `avgSection_equivariant` and `avgSection_eq_of_equivariant` give the two properties. Witness of the hypothesis: `proposition_14_1_4_hyp_witness`. *Book:* §14.1, Proposition 14.1.4, p. 642 (2017: Remark 14.2). *Hidden hypotheses:* none. *Deviation (same):* "A strong bundle over a tame ep-groupoid" provides the tame strong bundle charts (`hB`), as in the cores of Theorem 12.3.9; "admitting sc-smooth partitions of unity" is Definition 5.5.1 on the object M-polyfold, as in `theorem_14_1_9_statement`. The paracompact orbit space and the goodness of $V(x)$ are not used ($V(x)$ any `NatNbhd`). The "partition of unity argument" on the open set $V(x)$: the local extensions (Part I's local step with local hypotheses, `StrongBundle.exists_local_extension_of_face`) are glued by a partition of unity of X on each $G_n = \{\text{dist}(\cdot, X \setminus V) > 1/(n+1)\}$ (the object space is metrizable, `MPolyfold.metrizableSpace`), and the G_n-extensions are joined by sc-smooth cut-offs χ_n ($= 1$ on cl G_n, supported in G_{n+1}) with weights $\chi_n - \chi_{n-1}$, locally finite on V. A single partition of unity of X cannot do it: it must cover the frontier of $V(x)$, where the face data may be unbounded. We average over all of G_x (`avgSection`) instead of the stabiliser and orbit transport. The text calls the $\bar{s}_i$ "sc$^+$-multisections"; presumably sc$^+$-sections are meant. The averaging step (`proposition_14_1_4_hyp`, `avgSection`) takes the non-equivariant extensions as data and averages over all of G_x instead of the stabiliser followed by orbit transport; this gives the same symmetry and needs no orbit representatives. `proposition_14_1_4_full` supplies those extensions on $V(x)$ (`exists_scPlusSectionOn_extension_of_face`). The text calls the $\bar{s}_i$ "sc$^+$-multisections"; presumably sc$^+$-sections are meant. Witness for `hB`, `hpu`: `proposition_14_1_4_hypotheses_rn`.

V14.5 (Definition 14.1.5 (2017: 14.3), good systems of neighbourhoods for ∂X; `Polyfold.EpGroupoid.BoundaryGoodSystem`, `Polyfold.EpGroupoid.BoundaryGoodSystem.U`, `Polyfold.EpGroupoid.BoundaryGoodSystem.mor`, `Polyfold.EpGroupoid.BoundaryGoodSystem.mem_U`, `Polyfold.EpGroupoid.BoundaryGoodSystem.U_subset_boundary`, `Polyfold.EpGroupoid.BoundaryGoodSystem.mor_subset`, `Polyfold.EpGroupoid.BoundaryGoodSystem.isDiffeoComponent_of_subset`, `Polyfold.EpGroupoid.BoundaryGoodSystem.exists_refinement`). `BoundaryGoodSystem` has fields `nbhd : x : X, X.NatNbhd x`, `isGood`, and `isDiffeoComponent` (`IsDiffeoComponent` on the components of `morBetween (nbhd x).U (nbhd x').U`). `U x = (nbhd x).bdry` and `mor x x' = morBetween (U x) (U x')`. `isDiffeoComponent_of_subset` (condition (2) is inherited by open subsets). `exists_refinement` (tame X; inside open $W(x)$, with the same actions). *Book:* §14.1, Definition 14.1.5, p. 643 (2017: Definition 14.3). *Hidden hypotheses:* none. *Deviation (restated):* One $V(x)$ is fixed for each x (the text says "there exist $V(x)$, $V(x')$" for each pair). This is how the text uses it: $V(x)$ is chosen once in the proof of Theorem 14.1.6. The $V(x)$ carry their natural actions as data.

V14.6 (Theorem 14.1.6 (2017: 14.1), existence of good systems for ∂X; `Polyfold.EpGroupoid.theorem_14_1_6_hyp`). `theorem_14_1_6_hyp (hX : X.IsTame) (hpara : X.HasParacompactOrbitSpace) O hO hxO`. For $O(x)$ open in X, there is a good system with $V(x) \subseteq O(x)$, so $U(x) \subseteq O(x) \cap \partial X$. *Book:* §14.1, Theorem 14.1.6, p. 643 (2017: Theorem 14.1). *Hidden hypotheses:* none. *Deviation (narrower):* We assume X tame and $|X|$ paracompact instead of $|\partial X|$ paracompact. The text's proof applies Theorem 13.1.8 to X (which needs $|X|$ paracompact) and Proposition 14.1.1 (which needs X tame); the chapter's standing assumptions include both. Sets open in ∂X are written $O(x) \cap \partial X$ with $O(x)$ open in X. Hidden hypotheses: `hX : X.IsTame`, `hpara : X.HasParacompactOrbitSpace` in place of the printed "$|\partial X|$ paracompact"; both are standing assumptions of the chapter (Theorem 14.1.9). Witness: the tame trivial ep-groupoid of $\mathbb{R}^n$ (`proposition_14_1_4_hypotheses_rn`).

V14.7 (Definition 14.1.7 (2017: 14.4), structured sc$^+$-multisections over ∂X; `Polyfold.SBLocalization.BoundaryDecoration`, `Polyfold.SBLocalization.BoundaryDecoration.Compat`, `Polyfold.SBLocalization.BoundaryDecoration.Tau`, `Polyfold.SBLocalization.BoundaryStructuredTuple`, `Polyfold.SBLocalization.IsBoundaryStructurable`, `Polyfold.SBLocalization.BoundaryStructuredTuple.Equiv`, `Polyfold.SBLocalization.BoundaryStructuredTuple.equivalence`, `Polyfold.SBLocalization.BoundaryStructuredTuple.setoid`, `Polyfold.SBLocalization.BoundaryStructuredTuple.equiv_iff_goodSystem`, `Polyfold.SBLocalization.BoundaryStructuredMultisection`,

`Polyfold.SBLocalization.BoundaryStructuredMultisection.forget`, `Polyfold.SBLocalization.BoundaryStructuredMultisection.forget_mk`, `Polyfold.SBLocalization.isBoundaryStructurable_iff`, `Polyfold.SBLocalization.BoundaryDecoration.zeroTau`, `Polyfold.SBLocalization.isBoundaryStructurable_zero`). `BoundaryDecoration A Λ` has fields and `S`. `BoundaryDecoration.Tau D` has fields , `locallyConstant` (in `[ . mor x x']`), `finite`, `swap`, `diag`, `biact` (`biact` with the actions of the `V(x)`) and `compat` (`IndexCorr.Sat`). `BoundaryStructuredTuple`. `Equiv`: $\Lambda = \Lambda'$ over ∂X; open $O(x) \ni x$ with $O(x) \cap \partial X \subseteq U(x) \cap U'(x)$; `hI : I_x = I'_x`; $\tau' = \tau$ (cast) on `morBetween (O x  X) (O x'  X)`. `equivalence`, `setoid` and `BoundaryStructuredMultisection` (with `forget` and `forget_mk`). `equiv_iff_goodSystem` (tame X): the same relation with a good system $\mathcal{U}'' \subseteq \mathcal{U} \cap \mathcal{U}'$, the literal analogue of Definition 13.3.8. `IsBoundaryStructurable`, `isBoundaryStructurable_iff`. Witness: `isBoundaryStructurable_zero` (tame, $|X|$ paracompact). *Book:* §14.1, Definition 14.1.7, p. 644 (2017: Definition 14.4). *Hidden hypotheses:* none. *Deviation (restated):* The equivalence is the text's "similar to the X-case". `equiv_iff_goodSystem` shows the open-set form agrees with the good-system form for tame X. The open-set form is used for transitivity without tameness. (4) is read pointwise on $\mathcal{U}(x, x')$, as printed. Λ is a function on all of W; only its values over ∂X enter.

V14.8 (Theorem 14.1.9 (2017: 14.2), the ∂-extension theorem; `Polyfold.SBLocalization.theorem_14_1_9_charged_a6`, `Polyfold.SBLocalization.theorem_14_1_9_charged_statement`, `Polyfold.SBLocalization.sizeControl_a6`, `Polyfold.SBLocalization.theorem_14_1_9_iff_a6`, `Polyfold.SBLocalization.theorem_14_1_9_statement_of_boundary_eq_empty_a6`, `Polyfold.SBLocalization.boundary_eq_empty_of_assemblyHyp_a6`, `Polyfold.SBLocalization.not_theorem_14_1_9_statement_a6`, `Polyfold.HalfSpaceBundle.halfSEP`, `Polyfold.HalfSpaceBundle.isTame_halfSEP`, `Polyfold.HalfSpaceBundle.hasParacompactOrbitSpace_halfSEP`, `Polyfold.HalfSpaceBundle.admitsScPartitionsOfUnity_halfSEP`, `Polyfold.HalfSpaceBundle.hB_halfSEP`, `Polyfold.HalfSpaceBundle.zero_mem_boundary_halfSEP`, `Polyfold.HalfSpaceBundle.mem_boundary_halfSEP_iff`, `Polyfold.HalfSpaceBundle.nontrivial_fibre_halfSEP`, `Polyfold.HalfSpaceBundle.theorem_14_1_9_charged_hypotheses_halfSEP`, `Polyfold.HalfSpaceBundle.theorem_14_1_9_halfSEP`, `Polyfold.HalfSpaceBundle.epBundleW1`, `Polyfold.HalfSpaceBundle.constSec`, `Polyfold.HalfSpaceBundle.isScPlusSectionOn_constSec`, `Polyfold.HalfSpaceBundle.constMulti`, `Polyfold.HalfSpaceBundle.constStructure`, `Polyfold.HalfSpaceBundle.constTau`, `Polyfold.HalfSpaceBundle.isBoundaryStructurable_constMulti`, `Polyfold.HalfSpaceBundle.chargesBoundary_constMulti`, `Polyfold.HalfSpaceBundle.boundaryDomSupp_constMulti_nonempty`, `Polyfold.HalfSpaceBundle.pointwiseNorm_constMulti`, `Polyfold.HalfSpaceBundle.continuous_normW_constSec`, `Polyfold.HalfSpaceBundle.theorem_14_1_9_constSec`, `Polyfold.SBLocalization.boundaryDomSupp`, `Polyfold.SBLocalization.boundaryDomSupp_zero`, `Polyfold.SBLocalization.theorem_14_1_9_statement`, `Polyfold.SBLocalization.theorem_14_1_9_statement_hyp_witness`, `Polyfold.SBLocalization.theorem_14_1_9_extension_statement`, `Polyfold.SBLocalization.theorem_14_1_9_hyp`, `Polyfold.SBLocalization.theorem_14_1_9_extension_statement_of_boundary_eq_empty`, `Polyfold.`

`SBLocalization.isOpen_pointwiseNorm_lt`, `Polyfold.SBLocalization.pointwiseNorm_tgt`, `Polyfold.SBLocalization.pointwiseNorm_src_le_tgt`, `Polyfold.SBLocalization.pointwiseNorm_nonneg`, `Polyfold.SBLocalization.IsScPlusSectionOn.continuousOn_normW`, `Polyfold.SBLocalization.normW_le_of_multisectionSmul_pos`, `Polyfold.SBLocalization.pointwiseNorm_smulFun_le`, `Polyfold.SBLocalization.multisectionSmulFun_of_eq_one`, `Polyfold.SBLocalization.domSupp_zeroMultisection_gl`, `Polyfold.SBLocalization.isSaturated_boundary_gl`, `Polyfold.SBLocalization.isSaturated_closure_gl`, `Polyfold.SBLocalization.isSaturated_union_gl`, `Polyfold.SBLocalization.AssemblyHyp`, `Polyfold.SBLocalization.theorem_14_1_9_extension_statement_of_assembly`, `Polyfold.SBLocalization.theorem_14_1_9_of_assembly`, `Polyfold.SBLocalization.assemblyHyp_of_boundary_eq_empty`, `Polyfold.SBLocalization.emptyZSetup`, `Polyfold.SBLocalization.emptyTransportInput`, `Polyfold.SBLocalization.emptyAssemblyInput`, `Polyfold.SBLocalization.fibAct_eq_fiberZero_iff_t9`, `Polyfold.SBLocalization.fibAct_comp_inv_t9`,`Polyfold.SBLocalization.CriterionData.hasLocalSectionStructure_t9`,`Polyfold.SBLocalization.AssemblyInput.mult`, `Polyfold.SBLocalization.AssemblyInput.mult_eq`, `Polyfold.SBLocalization.AssemblyInput.isWFunctor_mult`, `Polyfold.SBLocalization.AssemblyInput.crit`, `Polyfold.SBLocalization.AssemblyInput.crit_nonempty`, `Polyfold.SBLocalization.AssemblyInput.isStructurable_mult`, `Polyfold.SBLocalization.AssemblyInput.mult_boundary`, `Polyfold.SBLocalization.AssemblyInput.domSupp_mult_subset`, `Polyfold.SBLocalization.AssemblyInput.isSaturated_boundaryDomSupp`, `Polyfold.SBLocalization.AssemblyInput.exists_cutoff`, `Polyfold.SBLocalization.AssemblyInput.extension`, `Polyfold.SBLocalization.AssemblyInput.ssec_boundary`, `Polyfold.SBLocalization.AssemblyInput.s0_eq_zero`, `Polyfold.SBLocalization.AssemblyInput.mult_of_not_mem_C`, `Polyfold.SBLocalization.AssemblyInput.critV`, `Polyfold.SBLocalization.AssemblyInput.crit` , `Polyfold.SBLocalization.AssemblyInput.isOpen_critV`). `theorem_14_1_9_charged_a6 hX hpara hpu hB Λ hΛ hch N hN Ũ hUo hUs hsub f hf hf0 hfinv htsupp hlt` (Part III wave 6, `Chapter14/P3W6A6_*`): `theorem_14_1_9_charged_statement A`, the book's statement with two added hypotheses, tame strong bundle charts at every object (`hB`) and `ChargesBoundary Λ` (V14.3), proved with no further hypothesis. The route: Proposition 14.2.3, the objects of §14.3 (`exists_zConcrete_of_hasP4`), Theorem 7.5.7, Lemma 14.5.1 with (14.53) (V14.45), the instance `exists_assemblyInput_a6` (V14.46), the assembly and the size control (`sizeControl_a6`). `theorem_14_1_9_iff_a6`: for structurable Λ over ∂X such an extension exists iff `ChargesBoundary Λ`. `not_theorem_14_1_9_statement_a6 hX hpara hpu (hx : x   X)`: the statement without `ChargesBoundary` does not hold as we read the formal definition when $\partial X \neq \emptyset$ ($\Lambda \equiv 0$ with empty index sets, $\widetilde{U} = \emptyset$, $f = 0$); it holds when $\partial X = \emptyset$ (`theorem_14_1_9_statement_of_boundary_eq_empty_a6`), and `AssemblyHyp` holds only then (`boundary_eq_empty_of_assemblyHyp_a6`). Witness with boundary (Part III wave 7, `Chapter14/P3W7W1_*`): on `HalfSpaceBundle.halfSEP m k`, the trivial bundle $H \times \mathbb{R}^k \to H$, $H = [0, \infty) \times \mathbb{R}^m$ (tame, paracompact orbit space, partitions of unity, `hB_halfSEP`, $\partial H = \{x_1 = 0\}$), `theorem_14_1_9_charged_hypotheses_halfSEP` gives all hypotheses at once and **`theorem_14_1_9_constSec hv`** applies the theorem to the constant multisection Λ_v, $v \neq 0$

(structurable over ∂H with $I_x = \{*\}$, charging, nonempty domain support), with $\tilde{U} = X$ and $f(x) = N(x, v) + 1$.

The earlier forms: `theorem_14_1_9_statement (A : SEP) : Prop`, with the hypotheses in the order: `A.X.IsTame`, `A.X.HasParacompactOrbitSpace`, `A.X.PObj.AdmitsScPartitionsOfUnity`, `IsBoundaryStructurable A Λ`, `A.E.IsAuxNorm N`, `IsOpen Ũ`, `A.X.IsSaturated Ũ`, `boundaryDomSupp Λ   Ũ`, `Continuous f`, `0   f`, `f   tgt = f   src`, `tsupport f   Ũ`, and `pointwiseNorm N Λ x < f x` on `boundaryDomSupp Λ`. The conclusion is `  Λ', IsStructurable A Λ'   (  x, pointwiseNorm N Λ' x   f x)   domSupp Λ'   Ũ     w, P w   X → Λ' w = Λ w`. `boundaryDomSupp Λ` is the closure of $\{x \in \partial X \mid \exists w \neq 0_x,\ P(w) = x,\ \Lambda(w) > 0\}$, with `boundaryDomSupp_zero`. `theorem_14_1_9_statement_hyp_witness`: $\Lambda = 0$, $\tilde{U} = \emptyset$, $f = 0$ satisfy the hypotheses.

`theorem_14_1_9_hyp (hext : theorem_14_1_9_extension_statement A) : theorem_14_1_9_statement A`, where `theorem_14_1_9_statement` is statement. The hypothesis `theorem_14_1_9_extension_statement A` (a `Prop`) is the output of §§14.3–14.5. For every structurable Λ over ∂X and saturated open $\tilde{U} \supseteq$ dom-suppΛ there is a structurable Λ'' on X with dom-supp$\Lambda'' \subseteq \tilde{U}$ and $\Lambda''|\partial X = \Lambda$. Witness: `theorem_14_1_9_extension_statement_of_boundary_eq_empty` (it holds when $\partial X = \emptyset$, with $\Lambda'' = 0$).

The assembly (Part III wave 5, `Chapter14/P3W5T9_Assembly.lean`): `theorem_14_1_9_extension_statement_of_assembly (h : AssemblyHyp A) : theorem_14_1_9_extension_statement A` and `theorem_14_1_9_of_assembly (h : AssemblyHyp A) : theorem_14_1_9_statement A`. `AssemblyHyp A`: every structurable boundary Λ has an `AssemblyInput A Λ`, the data of §§14.3–14.5 (the z-loop setting S, the transport input T with Lemma 14.5.1 (1) `suppInSat`, local finiteness and $\sum_a \beta_a = 1$, the index sets, $\hat{\tau}$ finite and bi-equivariant, the representation on $U^0(r(a))$ and the functor). From these: Λ' is `mult` , computed in any chart (`mult_eq`) and a functor (`isWFunctor_mult`); the criterion data `crit` over the index type $A \sqcup X$ ($\hat{V}_a$ with the summed sections `ssec` for `inl  a`; for `inr  x` the invariant core of $X \setminus C$ in a natural neighbourhood of x with the zero section; the text's correspondences `toUnit`, its `swap`, `ofEquiv refl`); `CriterionData.hasLocalSectionStructure_t9` and Theorem 14.5.5 give `isStructurable_mult`; `mult_boundary` ($\Lambda' = \Lambda$ over ∂X where the cut-off is 1), `domSupp_mult_subset`, `exists_cutoff` (Urysohn on $|X|$), `AssemblyInput.extension`. Witness of `AssemblyHyp`: `assemblyHyp_of_boundary_eq_empty` ($\partial X = \emptyset$; `emptyAssemblyInput`). *Book:* §14.1, Theorem 14.1.9, p. 646; proof §14.5, pp. 677–678 (2017: Theorem 14.2). *Hidden hypotheses:* none. *Deviation (narrower):* Added hypotheses: tame strong bundle charts at every object (`hB`, the chapter's standing assumption that W is a strong bundle over the tame X, as Proposition 14.1.4 reads it) and `ChargesBoundary Λ`, the reading $I_x \neq \emptyset$ of the factor $1/|I_x|$ in Definition 14.1.3 (V14.3); it is necessary (`theorem_14_1_9_iff_a6`). Both have the witness `theorem_14_1_9_constSec` with $\partial X \neq \emptyset$, nonzero fibres and nonempty domain support; the ep-groupoid there is trivial (no isotropy), so it shows joint satisfiability, not the isotropy part of the proof. Whether `hB` follows from tameness, and the form of Definition 14.1.3 with nonempty index sets built in, are gap G.186. The proof is the text's: §14.2, §14.3, §14.4 (z-loops, coherent systems, Proposition 14.4.14 for the objects of §14.3), Lemma 14.5.1 with (14.53) for the $\widehat{V}_a$ themselves (no shrinking of the cover is needed), Proposition 14.5.4 and the assembly, Theorem 14.5.5, the size control (gaps G.170, G.171 closed). The earlier forms rest on `theorem_14_1_9_statement` without `ChargesBoundary` and are vacuous when $\partial X \neq \emptyset$: `theorem_14_1_9_hyp` (from the extension statement) and `theorem_14_1_9_of_assembly` (from `AssemblyHyp`, which holds only for $\partial X = \emptyset$). They stay as intermediate steps. "X admits sc-smooth partitions of unity" is read as the M-polyfold of objects (Definition 5.5.1, as printed: "X, viewed as an M-polyfold"). "Supported in $\tilde{U}$" is read as `tsupport f  Ũ`. N is real-valued on $W_{0,1}$ (the project's Definition 8.3.5). The size control applies the extension with $\tilde{U}' = \{f > 0\}$, which is saturated open, contains dom-suppΛ and lies in $\tilde{U}$. $G = \{N(\Lambda'') < f\} \cup (X \setminus \text{dom-supp}\Lambda'')$ is a saturated open set containing ∂X. σ comes from `exists_isScSmoothMap_orbit_urysohn`, with $\sigma = 1$ on ∂X and $\sigma = 0$ off G. $\Lambda' = \sigma \odot \Lambda''$ is structurable by `lemma_13_3_14_fun`. The text writes $h(|x|) = N(\Lambda)(x) < f(x)$ for $x \in \partial X$; the theorem's hypothesis gives this on dom-suppΛ, and we read it there. In the assembly the size-control cut-off σ is folded into the sections before Theorem 14.5.5 is applied (the text multiplies Λ' afterwards), and the points away from the closed saturated set C off which every transported section vanishes are covered by invariant cores of natural neighbourhoods (the text uses $|x| \notin \bigcup |\widehat{V}_b|$). Everything else is the text's construction.

V14.9 (Remark 14.1.10 (2017: 14.3), the proof of Theorem 14.1.9 is involved; `Polyfold.SBLocalization.proposition_14_1_4_full`). Nothing to formalize: a comment on the proof. The section-functor case it mentions is Proposition 14.1.4 on $V(x)$ (`proposition_14_1_4_full`, V14.4). *Book:* §14.1, Remark 14.1.10, p. 646 (2017: Remark 14.3). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V14.10 (Lemma 14.2.1 (2017: before Lemma 14.1), shrinking good neighbourhoods; `Polyfold.EpGroupoid.lemma_14_2_1`, `Polyfold.EpGroupoid.lemma_14_2_1_system`). `lemma_14_2_1 (hX : X.IsTame) (hx) V (hV : IsGoodBoundaryNbhd V )` gives $V \leq V_0$, good, with $\text{cl}\, V \subseteq V_0$, $\text{cl}\,(V \cap \partial X) \subseteq V_0 \cap \partial X$, $\text{cl}(\pi(V \cap \partial X)) \subseteq \pi(V_0 \cap \partial X)$. `lemma_14_2_1_system` does this for a whole good system. *Book:* §14.2, Lemma 14.2.1, p. 648 (2017: the text before Lemma 14.1). *Hidden hypotheses:* none. *Deviation (same):* $|\partial X|$ is closed in $|X|$, so the closure is taken in $|X|$. The faces, connectedness and invariance are those of Definition 14.1.2. The orbit closure is obtained from Proposition 7.1.10 (properness of V_0).

V14.11 (Lemma 14.2.2 (2017: 14.1); `Polyfold.SBLocalization.closure_mor_subset`, `Polyfold.SBLocalization.lemma_14_2_2`). `lemma_14_2_2`. For good systems $\mathcal{U}, \mathcal{U}^0$ with $\text{cl}_X U(x) \subseteq U^0(x)$ and any τ locally constant on $\mathcal{U}^0(x, x')$ (in its relative topology), $d \neq d'$ implies $\text{cl}(\tau^{-1}(d) \cap \mathcal{U}(x, x')) \cap \text{cl}(\tau^{-1}(d') \cap \mathcal{U}(x, x')) = \emptyset$. `closure_mor_subset` gives $\text{cl}\, \mathcal{U}(x, x') \subseteq \mathcal{U}^0(x, x')$. *Book:* §14.2, Lemma 14.2.2, p. 648 (2017: Lemma 14.1). *Hidden hypotheses:* none. *Deviation (same):* The text defines $\mathcal{U}(x, x')$ with $t(\varphi) \in U(x)$; presumably $U(x')$ is meant, as in (P3).

V14.12 (Proposition 14.2.3 (2017: 14.2), a good structured version of Λ; `Polyfold.SBLocalization.BoundaryDecoration.restrictTo`, `Polyfold.SBLocalization.BoundaryDecoration.mor_mono`, `Polyfold.SBLocalization.BoundaryDecoration.Tau.restrict`, `Polyfold.SBLocalization.BoundaryStructuredTuple.restrict`, `Polyfold.SBLocalization.BoundaryStructuredTuple.equiv_restrict`, `Polyfold.SBLocalization.

`BoundaryStructuredTuple.HasP4`, `Polyfold.SBLocalization.BoundaryStructuredTuple.HasP4.restrict`, `Polyfold.SBLocalization.BoundaryStructuredTuple.proposition_14_2_3`, `Polyfold.SBLocalization.BoundaryStructuredTuple.proposition_14_2_3_class`). `BoundaryStructuredTuple.HasP4` and `HasP4.restrict` (persistence). `proposition_14_2_3 (hX : A.X.IsTame) T`: an equivalent T' with the same Λ, natural actions restricted ($V'(x) \leq V(x)$) and (P4). `proposition_14_2_3_class` (every class has a representative with (P4)). Restriction: `BoundaryDecoration.restrictTo`, `BoundaryDecoration.Tau.restrict`, `BoundaryStructuredTuple.restrict`, `equiv_restrict`. *Book:* §14.2, Proposition 14.2.3, p. 649 (2017: Proposition 14.2). *Hidden hypotheses:* none. *Deviation (same):* (P1)–(P3) are the fields of a tuple. The text's "under the assumption of Theorem 14.1.9" is used only for "tame".

V14.13 (Lemma 14.3.1 (2017: 14.2), the open sets $\tilde{O}_{x,x',[d]}$; `Polyfold.P3W4S3.exists_open_disjoint_closure`, `Polyfold.SBLocalization.IndexCorr.swap_biact_p3w4`, `Polyfold.EpGroupoid.inv_biact_p3w4`, `Polyfold.EpGroupoid.biact_congr_rep_p3w4`, `Polyfold.SBLocalization.boundaryMor`, `Polyfold.EpGroupoid.BoundaryGoodSystem.mor_subset_boundaryMor_p3w4`, `Polyfold.EpGroupoid.BoundaryGoodSystem.biact_mem_mor_p3w4`, `Polyfold.EpGroupoid.BoundaryGoodSystem.inv_mem_mor_p3w4`, `Polyfold.SBLocalization.P5Nbhds`, `Polyfold.SBLocalization.P5Nbhds.ofGood`, `Polyfold.SBLocalization.P5Nbhds.mor`, `Polyfold.SBLocalization.P5Nbhds.isOpen_mor`, `Polyfold.SBLocalization.P5Nbhds.biact`, `Polyfold.SBLocalization.P5Nbhds.biact_eq`, `Polyfold.SBLocalization.P5Nbhds.biact_mem`, `Polyfold.SBLocalization.P5Nbhds.mor_inter_boundaryMor`, `Polyfold.SBLocalization.P5Nbhds.mor_subset`, `Polyfold.SBLocalization.CorrPre`, `Polyfold.SBLocalization.CorrPre.exists_of_hasP4`, `Polyfold.SBLocalization.CorrPre.biInv`, `Polyfold.SBLocalization.CorrPre.biInv_biact_mem_iff`, `Polyfold.SBLocalization.CorrPre.symm`, `Polyfold.SBLocalization.CorrPre.symm_inv_mem_iff`, `Polyfold.SBLocalization.CorrPre.symm_biact_mem_iff`, `Polyfold.SBLocalization.CorrExtension`, `Polyfold.SBLocalization.lemma_14_3_1`, `Polyfold.SBLocalization.CorrExtension.disjoint`, `Polyfold.SBLocalization.CorrExtension.eq_of_mem`, `Polyfold.SBLocalization.CorrExtension.inter_boundaryMor`, `Polyfold.SBLocalization.CorrExtension.image_biact`, `Polyfold.SBLocalization.CorrExtension.iUnion_inter_boundaryMor`, `Polyfold.SBLocalization.lemma_14_3_1_book`). `lemma_14_3_1 (hT : T.HasP4) (P : P5Nbhds T) : Nonempty (CorrExtension T P)`. A `CorrExtension` has fields: `O x x' d`, open, for all $x, x' \in \partial X$ and $[d] \in J(x,x')$; `fiber_subset`: $\tau^{0-1}(d) \cap U^0(x,x') \subseteq \tilde{O}_d$; `subset_mor`, which is (1); `disjoint_closure`, which is (2); `biact_mem_iff`: $h * \varphi * g \in \tilde{O}_{h*d*g} \iff \varphi \in \tilde{O}_d$ on $\tilde{U}'(x,x')$, which is (3); `inv_mem_iff`: $\varphi^{-1} \in \tilde{O}_{x',x,d^\sharp} \iff \varphi \in \tilde{O}_{x,x',d}$. (4) is `iUnion_inter_boundaryMor`, and (14.20) is `inter_boundaryMor`. `image_biact` gives (3) as printed (set equality). `lemma_14_3_1_book` is the printed statement (1)–(4) with an existential $\tilde{O}$. *Book:* §14.3, Lemma 14.3.1, p. 650 (2017: Lemma 14.2). *Hidden hypotheses:* none. *Deviation (stronger):* Besides (1)–(4) the sets are symmetric under inversion (`inv_mem_iff`). The text does not state this. It gives $\hat{\tau}_{x',x}(\varphi^{-1}) = \hat{\tau}_{x,x'}(\varphi)^\sharp$, Definition 14.1.7 (a) for $\hat{\tau}$, which `ZSetup.swap` (Lemma 14.4.5, §14.4) needs. The proof is the text's: separate the closures (P4) in the normal space **X** (`P3W4S3.exists_open_disjoint_closure`), and intersect with $\tilde{U}'(x,x')$; intersect over the bi-action (written

$\bigcap_{h,g}(h * \cdot * g)^{-1}(\tilde{O}'_{h*[d]*g})$, which is the text's $\bigcap h * \tilde{O}'_{[d']} * g$ over $h * [d'] * g = [d]$); as an added step, intersect with the inverted sets of (x', x). This keeps (1)–(4). Only finitely many $[d]$ have nonempty $\tau^{0-1}([d])$ (`Tau.finite`). For the others $\tilde{O}_{[d]} = \emptyset$. Reading of (P5)(i): "admits the natural G_x-action" is read as the action of the good system $\mathcal{U}^0$ (the same representation). The bi-action in (3) is then that of Definition 14.1.7 (c).

V14.14 (Definition 14.3.2 (2017: before Lemma 14.3), $\hat{O}_{x,x'}$ and $[d]_{\varphi}$; `Polyfold.P3W4S3.connectedComponentIn_iUnion_subset`, `Polyfold.SBLocalization.CorrExtension.Ot`, `Polyfold.SBLocalization.CorrExtension.Ohat`, `Polyfold.SBLocalization.CorrExtension.tauHat`, `Polyfold.SBLocalization.CorrExtension.OhatD`, `Polyfold.SBLocalization.CorrExtension.tauHat_eq`, `Polyfold.SBLocalization.CorrExtension.mem_O_tauHat`, `Polyfold.SBLocalization.CorrExtension.isOpen_Ot`, `Polyfold.SBLocalization.CorrExtension.Ot_subset_mor`, `Polyfold.SBLocalization.CorrExtension.mor_subset_Ot`, `Polyfold.SBLocalization.CorrExtension.component_subset`, `Polyfold.SBLocalization.CorrExtension.tauHat_eq_of_mem_component`, `Polyfold.SBLocalization.CorrExtension.tauHat_eq_tau_of_mem_component_Ot`, `Polyfold.SBLocalization.CorrExtension.Ohat_subset_Ot`, `Polyfold.SBLocalization.CorrExtension.mor_subset_Ohat`, `Polyfold.SBLocalization.CorrExtension.Ohat_subset_mor`, `Polyfold.SBLocalization.CorrExtension.Ohat_inter_boundaryMor`, `Polyfold.SBLocalization.CorrExtension.component_subset_Ohat`, `Polyfold.SBLocalization.CorrExtension.isOpen_Ohat`, `Polyfold.SBLocalization.CorrExtension.OhatD_eq`). `CorrExtension.Ot` is $\tilde{O}_{x,x'} = \bigcup_d \tilde{O}_{x,x',d}$. `CorrExtension.Ohat` is $\{\varphi \in \tilde{O} \mid \exists \psi \in \mathrm{cc}_{\tilde{O}}(\varphi) \cap \partial X\}$. `CorrExtension.tauHat` is $\hat{\tau}(\varphi)$, the d with $\varphi \in \tilde{O}_{x,x',d}$. It is unique (`tauHat_eq`). It equals $\tau^0(\psi)$ for every $\psi \in \partial X$ in the component of φ (`tauHat_eq_tau_of_mem_component`, also in $\tilde{O}$: `tauHat_eq_tau_of_mem_component_Ot`). `CorrExtension.OhatD` is $\hat{O}_{x,x',[d]}$, and `OhatD_eq` gives $\hat{O}_{[d]} = \hat{O} \cap \tilde{O}_{[d]}$. (14.23): `isOpen_Ohat`; `mor_subset_Ohat`, `Ohat_subset_Ot`, `Ot_subset_mor`, `Ohat_subset_mor`; `Ohat_inter_boundaryMor`. *Book:* §14.3, Definition 14.3.2, p. 651 (2017: the text before Lemma 14.3). *Hidden hypotheses:* none. *Deviation (restated):* $[d]_{\varphi}$ is defined through the clopen decomposition of $\tilde{O}_{x,x'}$ into the $\tilde{O}_{x,x',[d]}$, and is proved equal to the text's definition. The component of φ in $\tilde{O}_{x,x'}$ lies in one $\tilde{O}_{[d]}$ (`component_subset`), and $\tilde{O}_{[d]} \cap \partial X = \tau^{0-1}([d])$. Off $\tilde{O}_{x,x'}$ the value is τ^0; it is irrelevant.

V14.15 (Lemma 14.3.3 (2017: 14.3), $[d]_{\varphi}$ is well defined; `Polyfold.SBLocalization.CorrExtension.lemma_14_3_3`, `Polyfold.SBLocalization.CorrExtension.tauHat_eq_tau_of_mem_component`, `Polyfold.SBLocalization.CorrExtension.exists_boundary_mem_component`). `CorrExtension.lemma_14_3_3`: $\psi, \psi' \in \mathrm{cc}_{\hat{O}_{x,x'}}(\varphi)$, both in ∂X, imply $\tau^0(\psi) = \tau^0(\psi')$. `tauHat_eq_tau_of_mem_component`: $\hat{\tau}(\varphi) = \tau^0(\psi)$ for such ψ. *Book:* §14.3, Lemma 14.3.3, p. 651 (2017: Lemma 14.3). *Hidden hypotheses:* none. *Deviation (same):* The proof is the text's: the $\tilde{O}_{[d]}$ have disjoint closures, so a component lies in one of them.

V14.16 (Proposition 14.3.4 (2017: 14.3), the extended correspondences $\hat{\tau}_{x,x'}$; `Polyfold.SBLocalization.proposition_14_3_4`, `Polyfold.P3W4S3.eq_of_mem_connectedComponentIn`,

`Polyfold.EpGroupoid.eventually_Γ_src_eq_p3w4`, `Polyfold.EpGroupoid.sheetLabel_p3w4`, `Polyfold.EpGroupoid.Γ_sheetLabel_p3w4`, `Polyfold.EpGroupoid.sheetLabel_eq_p3w4`, `Polyfold.EpGroupoid.sheetLabel_locallyConstant_p3w4`, `Polyfold.SBLocalization.CorrExtension.tauHat_eq_tau`, `Polyfold.SBLocalization.CorrExtension.tauHat_locallyConstant`, `Polyfold.SBLocalization.CorrExtension.tauHat_unique`, `Polyfold.SBLocalization.CorrExtension.tauHat_image_subset`, `Polyfold.SBLocalization.CorrExtension.finite_tauHat`, `Polyfold.SBLocalization.CorrExtension.tauHat_biact`, `Polyfold.SBLocalization.CorrExtension.biact_mem_Ot`, `Polyfold.SBLocalization.CorrExtension.biact_mem_Ohat`, `Polyfold.SBLocalization.CorrExtension.tauHat_Γ`, `Polyfold.SBLocalization.CorrExtension.inv_mem_Ohat`, `Polyfold.SBLocalization.CorrExtension.tauHat_inv`, `Polyfold.SBLocalization.CorrExtension.OhatD_inter_boundaryMor`). `proposition_14_3_4 (hT : T.HasP4) (P : P5Nbhds T)` gives $\hat{O}$, $\hat{\tau}$ with: openness; (1); (2); local constancy ($\forall \varphi \in \hat{O}$, $\forall^f \psi \in \mathcal{N}(\varphi)$, $\hat{\tau}\psi = \hat{\tau}\varphi$); $\hat{\tau} = \tau^0$ on $U^0(x, x')$; uniqueness among f locally constant on $\hat{O}$ (in $\mathcal{N}_{\hat{O}}$) with $f = \tau^0$ on $U^0(x, x')$; equivariance; the diagonal identity as printed; $\{\varphi \in \hat{O} \mid \hat{\tau}\varphi = d\} \cap \partial X = \tau^{0-1}(d) \cap U^0(x, x')$. API on any `E : CorrExtension T P`: `biact_mem_Ohat`, `tauHat_locallyConstant`, `tauHat_eq_tau`; `tauHat_unique` (for any $e \circ \hat{\tau}$); `tauHat_biact`, `tauHat_Γ` ($= [I_x \leftarrow I_x \xrightarrow{g} I_x]$), `OhatD_inter_boundaryMor`. Extras: `tauHat_image_subset` and `finite_tauHat`: the values on $\hat{O}$ are among the finitely many values of τ^0; `inv_mem_Ohat`: $\varphi \in \hat{O}_{x,x'} \Rightarrow \varphi^{-1} \in \hat{O}_{x',x}$; `tauHat_inv`: $\hat{\tau}_{x',x}(\varphi^{-1}) = \hat{\tau}_{x,x'}(\varphi)^\sharp$. *Book:* §14.3, Proposition 14.3.4, p. 652 (2017: Proposition 14.3). *Hidden hypotheses:* none. *Deviation (same):* Diagonal property: the text gives no argument. The proof is new: the component of $\Gamma(g, y)$ in $\tilde{O}_{x,x}$ stays in the sheet $\Gamma(g, \cdot)$, because the sheet label is locally constant (`EpGroupoid.sheetLabel_locallyConstant_p3w4`, from s being a local homeomorphism); this component meets ∂X in some $\Gamma(g, z')$, $z' \in U^0(x)$; Definition 14.1.7 (b) gives the value there. Uniqueness uses that every component of $\hat{O}$ meets ∂X (`P3W4S3.eq_of_mem_connectedComponentIn`). The bi-action is the natural one of $\tilde{U}'$ (= that of $\mathcal{U}^0$, (P5) reading).

V14.17 (Remark 14.3.5 (2017: 14.4), $\hat{O}_{x,x'}$ is not of the usual form; `Polyfold.SBLocalization.proposition_14_3_4`). Nothing to formalize: a comment on the construction. The sets and maps it speaks of are `proposition_14_3_4` (V14.16). *Book:* §14.3, Remark 14.3.5, p. 653 (2017: Remark 14.4). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V14.18 (Lemma 14.3.6 (2017: 14.4), the cover $(\tilde{V}_a)_{a \in A}$; `Polyfold.EpGroupoid.lemma_14_3_6`, `Polyfold.EpGroupoid.isClosed_ _image_of_isSaturated`). `EpGroupoid.lemma_14_3_6 (hpara : X.HasParacompactOrbitSpace) (hB : IsClosed (X. '' B)) (N :  x, X.NatNbhd x)`. Conclusion: $\exists A\, V$ with $A \subseteq B$, and for $a \in A$: V_a nonempty, open, $V_a \subseteq \tilde{U}(a)$, $\pi(V_a)$ open, $\overline{\pi(V_a)} \subseteq \pi(\tilde{U}(a))$, and $\Phi_g(V_a) \subseteq V_a$. Also: `LocallyFinite (fun a : A =>  '' V a)` and $\pi(B) \subseteq \bigcup_{a \in A} \pi(V_a)$. `isClosed_ _image_of_isSaturated`: a closed saturated B (as ∂X) has $|B|$ closed. *Book:* §14.3, Lemma 14.3.6, p. 654 (2017: Lemma 14.4). *Hidden hypotheses:* none. *Deviation (explicit):* The text's proof adds $|X \setminus \partial X|$ to the cover, which uses that $|\partial X|$ is closed in $|X|$. This is the hypothesis `hB`. It holds for ∂X (closed, and saturated since d is invariant under morphisms) through `isClosed_ _image_of_isSaturated`. The proof is the text's: a locally finite refinement (`precise_refinement`), a shrinking with closures inside (normality of $|X|$, Hausdorff by Lemma 7.3.4), and $V_a = \tilde{U}(a) \cap \pi^{-1}(W_a)$. A is the set of a with $W_a \neq \emptyset$.

V14.19 (Lemma 14.3.7 (2017: 14.5), the sets $\tilde{O}(y)$ and $o(y)$; `Polyfold.EpGroupoid.lemma_14_3_7`, `Polyfold.EpGroupoid.lemma_14_3_7_stab`, `Polyfold.EpGroupoid.exists_nhds_of_isProperSubset`). `EpGroupoid.lemma_14_3_7 (hB : X.IsSaturated B) (N' : X.NatNbhd a)`. Here B stands for ∂X and $N'.U$ for $\tilde{U}'(a)$. The other hypotheses: `hcl :  b  A, closure (Ut b)  Ut' b`; `hprop :  b  A, X.IsProperSubset (closure (Ut b))`; `hOo`: $\hat{O}_{a,b}$ is open; `hO`: $U'(a,b) = \{\varphi \mid s\varphi \in \tilde{U}'(a) \cap B,\ t\varphi \in \tilde{U}'(b) \cap B\} \subseteq \hat{O}_{a,b}$; `hlf`: $(|\tilde{V}_b|)_{b \in A}$ is locally finite. The conclusion gives $o, \tilde{O}$ with, for $y \in \tilde{U}'(a) \cap B$: $\tilde{O}(y)$ open, $y \in \tilde{O}(y) \subseteq \tilde{U}'(a)$, $o(y)$ finite, $o(y) \subseteq A$, and (1), (2) for all $g \in G_a$, and (3) for every φ with $s(\varphi) \in \tilde{O}(y)$ and $t(\varphi) \in \tilde{U}(b)$. `lemma_14_3_7_stab`: $\tilde{O}(y)$ is invariant under the $g \in G_a$ with $g * y = y$. *Book:* §14.3, Lemma 14.3.7, p. 655 (2017: Lemma 14.5). *Hidden hypotheses:* none. *Deviation (stronger):* (3) is proved without "$\cap\, \tilde{U}(a)$". The properness argument (`exists_nhds_of_isProperSubset`, through $\varphi \mapsto \varphi^{-1}$ and the closed map t on $s^{-1}(\mathrm{cl}\, \tilde{U}(b))$) does not use it. The hypotheses (P5)/(P6)/(14.27)/(14.28) used by the proof are named: $\mathrm{cl}\, \tilde{U}(b) \subseteq \tilde{U}'(b)$, properness, $U'(a,b) \subseteq \hat{O}_{a,b}$ open, and B saturated. "G_y-invariant" is read as invariance under the stabiliser of y in G_a, acting by Φ. Every morphism between points of $\tilde{U}'(a)$ is a $\Gamma(g,\cdot)$, so this is the G_y-action near y. Construction: $o(y)$ is the finite set of b whose $|\tilde{V}_b|$ meets a neighbourhood $W(|y|)$. It depends on $|y|$ only, which gives $o(g * y) = o(y)$. $\tilde{O}(y) = \bigcap_{g \in G_a} \tilde{U}'(a) \cap \Phi_g^{-1}(O_0(g * y))$, which gives $g * \tilde{O}(y) = \tilde{O}(g * y)$.

V14.20 (Proposition 14.3.8 (2017: 14.4), the sets $\hat{V}_a$; `Polyfold.EpGroupoid.isSaturated_boundary_p3w4`, `Polyfold.EpGroupoid.proposition_14_3_8`, `Polyfold.SBLocalization.P5Nbhds.P6Nbhds`, `Polyfold.SBLocalization.P5Nbhds.P6Nbhds.closure_subset`, `Polyfold.SBLocalization.P5Nbhds.P6Nbhds.exists_`, `Polyfold.SBLocalization.P5Nbhds.nbhdObj`, `Polyfold.SBLocalization.P5Nbhds.nbhdObj_of_mem`, `Polyfold.SBLocalization.P5Nbhds.P6Nbhds.nbhdObj`, `Polyfold.SBLocalization.P5Nbhds.P6Nbhds.nbhdObj_of_mem`, `Polyfold.SBLocalization.CorrExtension.OhatObj`, `Polyfold.SBLocalization.CorrExtension.OhatObj_of_mem`, `Polyfold.SBLocalization.CorrExtension.proposition_14_3_8`). `EpGroupoid.proposition_14_3_8` is the abstract form, in the setting of Lemma 14.3.7. Its hypotheses: B saturated; `NatNbhds` $\tilde{U}(a)$, $\tilde{U}'(a)$ with the same representation; $\mathrm{cl}\, \tilde{U}(b) \subseteq \tilde{U}'(b)$, and $\mathrm{cl}\, \tilde{U}(b)$ proper; $\tilde{V}_a \subseteq \tilde{U}(a)$ open and invariant, with $(|\tilde{V}_a|)$ locally finite and covering $|B|$; $\hat{O}_{a,b}$ open with $U'(a,b) \subseteq \hat{O}_{a,b}$. It gives $\hat{V}$ with (1)–(4), (3) including openness of $|\hat{V}_a|$. $\hat{V}_a$ is (14.32), built from `lemma_14_3_7`. `CorrExtension.proposition_14_3_8 E hpara Q` is the instantiated form. For E (Lemma 14.3.1 on (P4), (P5)), (P6) `Q` and $|X|$ paracompact, it gives: $A \subseteq \partial X$; $\tilde{V}$ with the properties of Lemma 14.3.6; $\hat{V}$ with (1)–(4), where $\hat{O}_{a,a'}$ is `E.OhatObj a a'` (= `E.Ohat  a,_  a',_ `). The neighbourhoods are `Q.nbhdObj a` and `P.nbhdObj a` (Obj-indexed through `dite`; `nbhdObj_of_mem`). `EpGroupoid.isSaturated_boundary_p3w4`: ∂X is saturated. *Book:* §14.3, Proposition 14.3.8, p. 656 (2017: Proposition 14.4). *Hidden hypotheses:* none. *Deviation (explicit):* The hypotheses the text's proof uses are named (abstract form): ∂X saturated (proved), the same representation on $\tilde{U}(a) \subseteq \tilde{U}'(a)$ (the (P6) reading), and $U'(a,b) \subseteq \hat{O}_{a,b}$ (Proposition 14.3.4 (1)). The instantiated form discharges all of them from (P4)–(P6) and $|X|$ paracompact.

"Locally finite covering by sets open in $|X|$" is: $|\widehat{V}_a|$ open, `LocallyFinite`, and $|\partial X| \subseteq \bigcup |\widehat{V}_a|$. (4) uses Lemma 14.3.7 (1), (3), as in the text, with $\pi(t\varphi) = \pi(s\varphi)$.

V14.21 (Remark 14.4.1 (2017: 14.5), the implicit relations of the boundary data; `Polyfold.EpGroupoid.ZSetup.ZRel`). Nothing to formalize: it motivates the z-loops and z-arcs of Definitions 14.4.2–14.4.4 (V14.22, V14.23). *Book:* §14.4, Remark 14.4.1, p. 658 (2017: Remark 14.5). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V14.22 (Definitions 14.4.2 and 14.4.3 (2017: 14.5, 14.6), z-loops and z-arcs; `Polyfold.EpGroupoid.ZSetup`, `Polyfold.EpGroupoid.ZSetup.Arc`, `Polyfold.EpGroupoid.ZSetup.Arc.trans`, `Polyfold.EpGroupoid.ZSetup.Arc.symm`, `Polyfold.EpGroupoid.ZSetup.trivialSetup`, `Polyfold.SBLocalization.IndexCorr.Pairs`, `Polyfold.SBLocalization.IndexCorr.pairs_swap`, `Polyfold.SBLocalization.IndexCorr.pairs_ofPerm`). The setting is `EpGroupoid.ZSetup X`: A, `nbhd a` ($\widetilde{U}(a)$ with the natural action), $\widehat{V}_a \subseteq \widetilde{U}(a)$ G_a-invariant, finite I_a with G_a-action, and $\hat{\tau}_{a,a'}$: Mor $\to J(a,a')$. It carries two hypotheses, `diag` ($\hat{\tau}_{a,a}(\Gamma(g,w)) = [\mathrm{id}, g]$ for $w \in \widehat{V}_a$) and `swap` ($\hat{\tau}_{a',a}(\varphi) = \hat{\tau}_{a,a'}(\varphi^{-1})^\sharp$). `ZSetup.Arc S a b i j` is inductive (`nil` at $z \in \widehat{V}_a$ with composite 1_z; `snoc` one step). It says: a chain through the $\widehat{V}$'s with composite φ, together with an arc of indices from i to j. Its pairs are `IndexCorr.Pairs` ("$(i,j) = (\alpha(k), \beta(k))$ for some k", independent of the representative). A z-loop with a z-arc from i to i' is `Arc S a a i i' (1_z)`. Concatenation and reversal: `Arc.trans`, `Arc.symm`. *Book:* §14.4, Definitions 14.4.2, 14.4.3, pp. 659–660 (2017: Definitions 14.5, 14.6). *Hidden hypotheses:* none. *Deviation (restated):* The loop and its arc are encoded together as one inductive predicate, not as two finite sequences. The composite 1_z also fixes $s(\varphi_0) = z$. Length 0 is allowed; it adds nothing. By `diag`, the one-step loop $1_z = \Gamma(1,z)$ relates the same indices (`zRel_iff_unit_step`). `diag` and `swap` are the properties of Definition 13.3.6 (1), (2), carried over to the extensions $\hat{\tau}$. The text uses them, for $\hat{\tau}$, in the proof of Lemma 14.4.5. Witness: `ZSetup.trivialSetup X` on every ep-groupoid ($A = X$, one-point index sets).

V14.23 (Definition 14.4.4 (2017: 14.7) and (14.43), the relation $\sim_z$; `Polyfold.EpGroupoid.ZSetup.ZRel`, `Polyfold.EpGroupoid.ZSetup.zClass`, `Polyfold.EpGroupoid.ZSetup.zRel_equivalence`, `Polyfold.EpGroupoid.ZSetup.ZRel.mem_V`, `Polyfold.EpGroupoid.ZSetup.zRel_iff_unit_step`). `ZSetup.ZRel S a z i i' := S.Arc a a i i' (X.unit z)` and `ZSetup.zClass`. `zRel_equivalence (ha : a  A) (hz : z  V_a) : Equivalence (S.ZRel a z)` proves reflexivity (nil), symmetry (reversed loop, `swap`) and transitivity (concatenation). `ZRel.mem_V`: $i \sim_z i'$ forces $z \in \widehat{V}_a$. *Book:* §14.4, Definition 14.4.4 and (14.43), p. 660 (2017: Definition 14.7). *Hidden hypotheses:* none. *Deviation (restated):* as for D14.4.2–3. The equivalence-relation property, which the text asserts without proof, is proved; symmetry uses `swap`.

V14.24 (Lemma 14.4.5 (2017: 14.6), $\sim_z$ and the G_a-action; `Polyfold.EpGroupoid.ZSetup.lemma_14_4_5`, `Polyfold.EpGroupoid.ZSetup.lemma_14_4_5_class`, `Polyfold.EpGroupoid.ZSetup.zRel_smul`). `ZSetup.lemma_14_4_5 (ha) (hz) g i i' : S.ZRel a z i i'  S.ZRel a (Φ_g z) (g • i) (g • i')` and `ZSetup.lemma_14_4_5_class : S.zClass a (Φ_g z) (g • i) = (g • ·) '' S.zClass a z i`. *Book:* §14.4, Lemma 14.4.5, p. 660 (2017: Lemma 14.6). *Hidden* *hypotheses:* none. *Deviation (explicit):* The text's proof: prepend $\Gamma(g^{-1}, g * z)$ and append $\Gamma(g, z)$. Their correspondences are $[\mathrm{id}, g^{\mp 1}]$; this is `diag`, named as a hypothesis of the setting. The composite is $\Gamma(g, z) \circ \Gamma(g^{-1}, g * z) = \Gamma(1, g * z) = 1_{g*z}$. The reverse implication applies the forward one to g^{-1}.

V14.25 (Proposition 14.4.6 (2017: 14.5), local constancy of $\sim_z$; `Polyfold.EpGroupoid.ZData`, `Polyfold.EpGroupoid.ZData.Arc.eventually_srcInv`, `Polyfold.EpGroupoid.ZData.proposition_14_4_6_1`, `Polyfold.EpGroupoid.ZData.proposition_14_4_6_1_open`, `Polyfold.EpGroupoid.ZData.eventually_zClass_subset`, `Polyfold.EpGroupoid.ZData.trivial`, `Polyfold.EpGroupoid.ZData.trivial_zRel`, `Polyfold.EpGroupoid.tendsto_comp_of_eventually`, `Polyfold.EpGroupoid.Sheet.tendsto_srcInv`, `Polyfold.EpGroupoid.Sheet.eventually_srcInv_eq_unit`, `Polyfold.SBLocalization.ZSections`, `Polyfold.SBLocalization.ZSections.arc_transport`, `Polyfold.SBLocalization.ZSections.zRel_sections_eq`, `Polyfold.SBLocalization.ZSections.proposition_14_4_6_2_of_mem`, `Polyfold.SBLocalization.proposition_14_4_6_2_statement`, `Polyfold.SBLocalization.HasThetaLocalEq`, `Polyfold.SBLocalization.HasThetaLocalEq.proposition_14_4_6_2`, `Polyfold.SBLocalization.ZSections.zero`, `Polyfold.EpGroupoid.eq_of_isPreconnected_z5`, `Polyfold.EpGroupoid.NaturalRepFinite.unit_mem_morOn_z5`, `Polyfold.EpGroupoid.NaturalRepFinite.unit_mem_sheet_z5`, `Polyfold.EpGroupoid.NaturalRepFinite.srcInv_unit_z5`, `Polyfold.EpGroupoid.NaturalRepFinite.eq_of_mem_U_z5`, `Polyfold.EpGroupoid.NaturalRepFinite.eq_of_mem_sheet_z5`, `Polyfold.EpGroupoid.NaturalRepFinite.exists_sheet_z5`, `Polyfold.SBLocalization.ZLimitSetting`, `Polyfold.SBLocalization.ZSections.CompatNbhd`, `Polyfold.SBLocalization.ZLimitRep`, `Polyfold.SBLocalization.ZLimitRep.mem_o_of_mem_W`, `Polyfold.SBLocalization.ZLimitRep.sheet_subset_M`, `Polyfold.SBLocalization.ZLimitRep.tau_eq`, `Polyfold.SBLocalization.ZLimitRep.arc_transport`, `Polyfold.SBLocalization.ZLimitRep.sections_eq`, `Polyfold.SBLocalization.ZLimitSetting.exists_nbhd_near`, `Polyfold.SBLocalization.ZLimitSetting.exists_nbhd_yb`, `Polyfold.SBLocalization.ZLimitSetting.exists_limitRep`, `Polyfold.SBLocalization.sections_eq_of_theta_closure`, `Polyfold.SBLocalization.hasThetaLocalEq_of_limit`, `Polyfold.SBLocalization.proposition_14_4_6_2_of_limit`, `Polyfold.SBLocalization.ZConcrete.proposition_14_4_6_2`, `Polyfold.SBLocalization.ZConcrete.hasThetaLocalEq`). Setting `EpGroupoid.ZData X = ZSetup` plus `isOpen_V` ($\hat{V}_a$ open), `closure_V_subset` ($\mathrm{cl}_X \hat{V}_a \subseteq \tilde{U}(a)$, (14.33)) and `locallyConstant` ($\hat{\tau}_{a,a'}$ locally constant at every φ with $s\varphi \in \hat{V}_a$, $t\varphi \in \hat{V}_{a'}$; P14.3.4 with P14.3.8 (4)). Witness `ZData.trivial X` (every ep-groupoid, $A = X$, nonempty: `trivial_zRel`). (1): `ZData.proposition_14_4_6_1 : S.ZRel a z i i' → z in z , S.ZRel a z i i'`; open-set form `proposition_14_4_6_1_open`; classes `eventually_zClass_subset` ($[i]_{\sim_{z_0}} \subseteq [i]_{\sim_z}$ near z_0). Key lemma `ZData.Arc.eventually_srcInv`: a chain with an arc through φ deforms along any sheet $\mathbf{U} \ni \varphi$: for z near $s(\varphi)$, $(s|\mathbf{U})^{-1}(z)$ carries a chain with the same arc. Boundary data `SBLocalization.ZSections A S` (sections s_i^{0a}, $P \circ s = \mathrm{id}$ and (14.45) on $U(a) = \tilde{U}(a) \cap \partial X$, D14.1.7 (4) for $\hat{\tau}$ along morphisms of $\hat{V}_{a,a'}$ with source in ∂X). `ZSections.arc_transport`: $s_j^b(t\varphi) = \mu(\varphi, s_i^a(s\varphi))$ along chains starting in ∂X; `zRel_sections_eq`: $i \sim_z i'$, $z \in V_a$ □ $s_i^{0a}(z) = s_{i'}^{0a}(z)$. (2) at points of V_a: `ZSections.proposition_14_4_6_`

`2_of_mem`: $i \sim_{z_0} i'$, $z_0 \in V_a$ □ $s^{0a}_i = s^{0a}_{i'}$ on a neighbourhood of z_0 in ∂X. (2) in general: the `Props` `proposition_14_4_6_2_statement D` (as printed) and `HasThetaLocalEq D` (with $z_k \in \hat{V}_a$); `HasThetaLocalEq.proposition_14_4_6_2` (the second implies the first). Not proved (gap G.168).

(2) in general (Part III wave 5, `Chapter14/P3W5Z5_*`): abstract form `proposition_14_4_6_2_of_limit D L hD : proposition_14_4_6_2_statement D` (the printed form), from `hasThetaLocalEq_of_limit D L hD : HasThetaLocalEq D` (the form with $z_k \in \hat{V}_a$), from `sections_eq_of_theta_closure D L hD ha (hz : z   cl V_a   X) (h : (i,i')   Θ_{z }) :   y in  z , y  X → s^{0a}_i y = s^{0a}_{i'} y.` `L : ZLimitSetting S` names the facts of §14.3 the proof uses (open $\hat{O}_{b,c}$ on which $\hat{\tau}_{b,c}$ is locally constant, containing the boundary morphisms from $\tilde{U}(b)$ to $\mathrm{cl}_X\tilde{U}(c)$; $\mathrm{cl}_X\tilde{U}(b)$ proper; $(|\hat{V}_b|)$ locally finite), `hD : D.CompatNbhd` Definition 14.1.7 (4) for $\hat{\tau} = \tau^0$. Concrete form for the objects of §14.3: `ZConcrete.proposition_14_4_6_2 C` and `ZConcrete.hasThetaLocalEq C` for every `C : ZConcrete E Q` (V14.39), with no hypothesis beyond the text's data. The proof is the text's: `ZLimitSetting.exists_limitRep` (finitely many b by local finiteness, $K = \bigcup \mathrm{cl}\,\hat{V}_b$ proper, the finitely many objects of K isomorphic to z_0, Lemma 14.4.7 on them, sheets inside $\hat{O}_{b,c}$), `ZLimitRep.arc_transport` (each step of the z_k-loop moved to y' along its sheet, $\hat{\tau}$ constant on the connected sheets, `ZLimitRep.tau_eq`), `ZLimitRep.sections_eq`. *Book:* §14.4, Proposition 14.4.6, p. 661 (2017: Proposition 14.5). *Hidden hypotheses:* none. *Deviation (same):* (1) is proved as printed. The proof deforms the loop one morphism at a time with the local inverses of s (sheets) and continuity of the multiplication; it does not need Lemma 14.4.7, which the text invokes here. (2) is proved as printed (Part III wave 5; gap G.168 closed), and in a stronger form: for $z_k \in \hat{V}_a$ instead of $z_k \in V_a$ (the form Lemma 14.4.9 (3) uses, `HasThetaLocalEq`), and at every $z_0 \in \mathrm{cl}_X(\hat{V}_a) \cap \partial X$; the neighbourhood is $O(z_0) \cap \partial X$ of Lemma 14.4.7's representation. The z_0-loop of the text is not built separately: each step of the z_k-loop is moved directly along its sheet, and the composite identity is Lemma 14.4.7 (6) (`comp_srcInv_eq`). The facts of §14.3 used are named (`ZLimitSetting`, `CompatNbhd`) in the abstract form; the concrete form discharges them from Propositions 14.3.4, 14.3.8 and (P5)/(P6).

V14.26 (Lemma 14.4.7 (2017: 14.7), finitely many isomorphic objects; `Polyfold.EpGroupoid.lemma_14_4_7`, `Polyfold.EpGroupoid.exists_naturalRepFinite_subset`, `Polyfold.EpGroupoid.NaturalRepFinite.F_isotropy`, `Polyfold.EpGroupoid.NaturalRepFinite.F_equivariant`, `Polyfold.EpGroupoid.NaturalRepFinite.comp_srcInv_eq`, `Polyfold.EpGroupoid.NaturalRepFinite.bijOn_comp`, `Polyfold.EpGroupoid.NaturalRepFinite.isScDiffeoOn_comp`). `EpGroupoid.lemma_14_4_7 Y hY Q hQo hQ` gives `N : X.NaturalRepFinite Y (  y   Y, Q y)`. That is the Part II structure: open, connected, disjoint $U(y)$, F, `sheet`, and (1)–(3) as fields. In addition: $U(y) \subseteq Q(y)$; (4) `NaturalRepFinite.F_isotropy`; (5) `NaturalRepFinite.F_equivariant`, with $\gamma_\varphi =$ `conjIsotropy  `; (6) `NaturalRepFinite.bijOn_comp`: m maps the fibre product bijectively onto $\mathbf{U}(\psi \circ \varphi)$; (6) `NaturalRepFinite.comp_srcInv_eq` and `NaturalRepFinite.isScDiffeoOn_comp`: $m((s|\mathbf{U}(\psi))^{-1}(t\varphi'), \varphi') = (s|\mathbf{U}(\psi \circ \varphi))^{-1}(s\varphi')$, and $(s|\mathbf{U}(\psi\varphi))^{-1} \circ s : \mathbf{U}(\varphi) \to \mathbf{U}(\psi\varphi)$ is an sc-diffeomorphism; (7) $\mathrm{cl}_X U(y)$ proper (`exists_` `naturalRepFinite_subset`). *Book:* §14.4, Lemma 14.4.7, pp. 661–662 (2017: Lemma 14.7). *Hidden hypotheses:* none. *Deviation (restated):* Part II's Proposition 7.1.15 takes one open $V \supseteq Y$. The per-object $Q(x)$ and (7) come from applying it to $\bigcup_y W_y \cap Q(y) \cap P_y$, where the W_y are pairwise disjoint (T2) and the P_y have the properness property. Connectedness of $U(y)$ then puts $U(y)$ inside $W_y \cap Q(y) \cap P_y$. (6): the fibre product is parametrised by its second factor ($(\psi', \varphi') \mapsto \varphi'$, inverse $\varphi' \mapsto ((s|\mathbf{U}(\psi))^{-1}(t\varphi'), \varphi')$). In that chart, m is the sc-diffeomorphism $(s|\mathbf{U}(\psi\varphi))^{-1} \circ s$. No M-polyfold structure on the fibre product itself is used. The key identity is proved by connectedness: the image of $\mathbf{U}(\varphi)$ is connected and contains $\psi \circ \varphi$, so it lies in the component $\mathbf{U}(\psi \circ \varphi)$ (`NatRepFinite.connectedComponentIn_morOn`).

V14.27 (Definition 14.4.8 (2017: 14.8) and (14.44), Θ_z and $\approx_z$; `Polyfold.EpGroupoid.ZData.Theta`, `Polyfold.EpGroupoid.ZData.Approx`, `Polyfold.EpGroupoid.ZData.approxClass`, `Polyfold.EpGroupoid.ZData.theta_iff_seq`, `Polyfold.EpGroupoid.ZData.approx_equivalence`, `Polyfold.EpGroupoid.ZData.approx_symm`, `Polyfold.EpGroupoid.ZData.theta_symm`, `Polyfold.EpGroupoid.ZData.theta_refl`, `Polyfold.EpGroupoid.ZData.theta_subset_closure`, `Polyfold.EpGroupoid.ZData.ZRel.symm'`). `ZData.Theta a z i i' := z   closure {y | S.ZRel a y i i'}`; `ZData.Approx a z := Relation.ReflTransGen (S.Theta a z)`; `ZData.approxClass`. `theta_iff_seq` (Fréchet–Urysohn spaces: the text's sequences); `approx_equivalence` (an equivalence relation), `approx_symm`, `theta_symm`, `theta_refl` ($(i, i) \in \Theta_z$ for $z \in \operatorname{cl} \widehat{V}_a$), `theta_subset_closure`. *Book:* §14.4, Definition 14.4.8 and (14.44), p. 663 (2017: Definition 14.8). *Hidden hypotheses:* none. *Deviation (restated):* the sequences are read as the closure (equal for metrizable object spaces, `theta_iff_seq`). The chain's last index is printed "$i_p = i'$"; $i_k = i'$ is meant. The equivalence-relation property is proved.

V14.28 (Lemma 14.4.9 (2017: 14.8), properties of $\approx_z$; `Polyfold.EpGroupoid.ZData.lemma_14_4_9_1`, `Polyfold.EpGroupoid.ZData.lemma_14_4_9_2`, `Polyfold.EpGroupoid.ZData.lemma_14_4_9_2_class`, `Polyfold.EpGroupoid.ZData.theta_smul`, `Polyfold.EpGroupoid.ZData.lemma_14_4_9_4`, `Polyfold.EpGroupoid.ZData.lemma_14_4_9_4_all`, `Polyfold.SBLocalization.lemma_14_4_9_3_hyp`, `Polyfold.SBLocalization.lemma_14_4_9_3_of_limit`). `ZData.lemma_14_4_9_1`, `ZData.lemma_14_4_9_2`, `ZData.lemma_14_4_9_2_class` ($z \in \operatorname{cl} \widehat{V}_a$), `ZData.lemma_14_4_9_4` and `lemma_14_4_9_4_all` (all i at once), as printed. (3): `SBLocalization.lemma_14_4_9_3_hyp (hP : HasThetaLocalEq D)`.

(3) without condition (Part III wave 5): `lemma_14_4_9_3_of_limit D L hD ha (hz : z   cl V̄_a   X) (h : S.Approx a z i i') :   y in   z, y   X → s^{0a}_i y = s^{0a}_{i'} y` (a chain of `sections_eq_of_theta_closure`); for the objects of §14.3 through `ZConcrete.hasThetaLocalEq` and `lemma_14_4_9_3_hyp`. *Book:* §14.4, Lemma 14.4.9, pp. 663–664 (2017: Lemma 14.8). *Hidden hypotheses:* none. *Deviation (same):* (1), (2), (4) as printed ((2) uses Lemma 14.4.5 and continuity of the natural action on $\widetilde{U}(a)$; (4) the finiteness of I_a). (3) is proved from Proposition 14.4.6 (2) in the form `HasThetaLocalEq`, as the text derives it, and `HasThetaLocalEq` is now a theorem (`hasThetaLocalEq_of_limit`, `ZConcrete.hasThetaLocalEq`; gap G.168 closed); (3) holds at every $z \in \operatorname{cl}_X(\widehat{V}_a) \cap \partial X$.

V14.29 (Proposition 14.4.10 (2017: 14.6), the neighbourhoods $\widetilde{O}(z)$; `Polyfold.SBLocalization.`

`proposition_14_4_10_hyp`, `Polyfold.SBLocalization.mapsTo_Φ_closure_z4`, `Polyfold.SBLocalization.mapsTo_Φ_Vb_z4`, `Polyfold.SBLocalization.Φ_mem_boundary_z4`, `Polyfold.SBLocalization.mem_boundary_of_Φ_z4`, `Polyfold.SBLocalization.tgt_mem_boundary_iff_z4`, `Polyfold.EpGroupoid.NatNbhd.exists_equivariant_nbhds`, `Polyfold.EpGroupoid.NatNbhd.image_Φ_eq`, `Polyfold.EpGroupoid.NatNbhd.isOpen_image_Φ`, `Polyfold.EpGroupoid.NatNbhd.image_Φ_mul`, `Polyfold.EpGroupoid.NatNbhd.continuousAt_Φ`, `Polyfold.SBLocalization.proposition_14_4_10_of_limit`, `Polyfold.SBLocalization.ZConcrete.proposition_14_4_10`). `SBLocalization.proposition_14_4_10_hyp D (hP : HasThetaLocalEq D) ha :   O,   z   closure (S.V a), IsOpen (O z)   z   O z   (1)   (2)   (2')   (3)   (4)`. The equivariant, disjoint choice is the general lemma `EpGroupoid.NatNbhd.exists_equivariant_nbhds` (neighbourhoods with a property preserved by shrinking and by the action, chosen on orbit representatives, made stabiliser-invariant and separated from their translates).

Without the hypothesis (Part III wave 5): `proposition_14_4_10_of_limit D L hD ha` (abstract) and `ZConcrete.proposition_14_4_10 C ha` (the objects of §14.3, the sections of the good structured tuple at a), the conclusion of `proposition_14_4_10_hyp` without `HasThetaLocalEq`. *Book:* §14.4, Proposition 14.4.10, p. 664 (2017: Proposition 14.6). *Hidden hypotheses:* none. *Deviation (same):* (1)–(3) need nothing beyond the setting; (4) comes from Lemma 14.4.9 (3), now proved (gap G.168 closed). The text's "immediate" includes the orbit-representative construction for (2), which is proved here.

V14.30 (Definition 14.4.11 (2017: 14.9), coherent neighbourhood systems; `Polyfold.SBLocalization.ZCoherentNbhdSystem`, `Polyfold.SBLocalization.exists_zCoherentNbhdSystem_hyp`, `Polyfold.SBLocalization.exists_zCoherentNbhdSystem_zero`, `Polyfold.SBLocalization.hasThetaLocalEq_zero`, `Polyfold.SBLocalization.ZSections.toLocalExtensionBase`, `Polyfold.SBLocalization.coherentNbhdSystem_hyp`, `Polyfold.SBLocalization.proposition_14_4_14_of_hyp`, `Polyfold.SBLocalization.LocalExtensionBase`, `Polyfold.SBLocalization.CoherentNbhdSystem`, `Polyfold.SBLocalization.CoherentNbhdSystem.IsEquivariant`, `Polyfold.SBLocalization.LocalExtensionBase.ofZSetup`, `Polyfold.SBLocalization.LocalExtensionBase.zero`, `Polyfold.SBLocalization.CoherentNbhdSystem.zero`, `Polyfold.SBLocalization.exists_coherentNbhdSystem_zero`, `Polyfold.SBLocalization.exists_zCoherentNbhdSystem_of_limit`,`Polyfold.SBLocalization.coherentNbhdSystem_of_limit`,`Polyfold.SBLocalization.ZConcrete.exists_zCoherentNbhdSystem`). `SBLocalization.ZCoherentNbhdSystem D a` (fields `O`, `isOpen`, `mem`, `subset` (1), `equivariant and disjoint` (2), `zClass_subset` (3) for $z \in \widehat{V}_a$ as printed, `sections_eq` (4)). `exists_zCoherentNbhdSystem_hyp D hP ha` (from P14.4.10); witness `exists_zCoherentNbhdSystem_zero`. Hypothesis form: `coherentNbhdSystem_hyp D hP ha hface :   C : CoherentNbhdSystem (D.toLocalExtensionBase ha hface), C.IsEquivariant` (LX reads (3) for $z \in \mathrm{cl}\,\widehat{V}_a$, which P14.4.10 (3) gives).

The abstract form consumed by Proposition 14.4.14: `LocalExtensionBase A a` (the data: `N` = $\widetilde{U}(a)$ with the natural action, `Vh` open invariant $\subseteq$ `N.U`, `I`, `s0` face-wise sc^+ on `N.bdry` with

(14.45), `sim` with Lemma 14.4.5 `sim_smul`, `approx`). `CoherentNbhdSystem D` (fields `O`, `isOpen_O`, `mem_O`, (1) `O_subset`, (3) `sim_approx` for $z \in \operatorname{cl} \widehat{V}_a$, (4) `approx_s0` for $z \in \operatorname{cl}(\widehat{V}_a \cap \partial X)$, $y \in \widetilde{O}(z) \cap U(a)$). (2) is `CoherentNbhdSystem.IsEquivariant`. `LocalExtensionBase.ofZSetup` builds the data from `ZSetup` with $\sim$ = `ZRel` (Lemma 14.4.5 discharged). Witnesses: `LocalExtensionBase.zero`, `CoherentNbhdSystem.zero`, `exists_coherentNbhdSystem_zero` (tame X, every boundary point, $a \in \widehat{V}_a$).

Existence without hypothesis (Part III wave 5): `exists_zCoherentNbhdSystem_of_limit D L hD ha : Nonempty (ZCoherentNbhdSystem D a)`, `coherentNbhdSystem_of_limit D L hD ha hface` (the second form, equivariant), and for the objects of §14.3 `ZConcrete.exists_zCoherentNbhdSystem`. *Book:* §14.4, Definition 14.4.11, p. 664 (2017: Definition 14.9). *Hidden hypotheses:* none. *Deviation (restated):* "open (in X) neighborhoods" includes $z \in \widetilde{O}(z)$. (3) is printed for $z \in \widehat{V}_a$, P14.4.10 (3) for $z \in \operatorname{cl}_X(\widehat{V}_a)$; the structure keeps the printed (3), the existence theorems give the stronger one. The text first calls it a "good neighborhood system" (Remark 14.4.12, p. 665); same object. (3) is printed for $z \in \widehat{V}_a$; Proposition 14.4.10 (3) has $z \in \operatorname{cl}(\widehat{V}_a)$ and the proof of P14.4.14 uses it at points of the closure. We read $z \in \operatorname{cl}(\widehat{V}_a)$, presumably what is meant. The relations $\sim_y$, $\approx_z$ are parameters (Z4 supplies D14.4.8's $\approx_z$); only Lemma 14.4.5 for $\sim$ is required. (2) is a separate predicate since P14.4.14 does not need it. The two structures are one definition read twice: `ZCoherentNbhdSystem` keeps (3) for $z \in \widehat{V}_a$ as printed; `CoherentNbhdSystem` reads (3) for $z \in \operatorname{cl} \widehat{V}_a$, which Proposition 14.4.10 (3) gives and the proof of Proposition 14.4.14 uses. `coherentNbhdSystem_hyp` passes from the first data to the second.

V14.31 (Remark 14.4.12 (2017: 14.6), the meaning of the coherent neighbourhood system; `Polyfold.EpGroupoid.lemma_14_3_6`). Nothing to formalize: a comment. The cover comes from Lemma 14.3.6 with $|X|$ paracompact (`lemma_14_3_6`, V14.18). *Book:* §14.4, Remark 14.4.12, p. 664 (2017: Remark 14.6). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V14.32 (Lemma 14.4.13 (2017: 14.9), the sections over $\widetilde{O}([z])$; `Polyfold.SBLocalization.lemma_14_4_13_1`, `Polyfold.SBLocalization.approxClass_eq_or_disjoint`, `Polyfold.SBLocalization.lemma_14_4_13_2`, `Polyfold.SBLocalization.lemma_14_4_13_3`, `Polyfold.SBLocalization.lemma_14_4_13_classSection`, `Polyfold.SBLocalization.lemma_14_4_13_4`, `Polyfold.SBLocalization.ZCoherentNbhdSystem.disjoint_orbit`). Any `C : ZCoherentNbhdSystem D a`: (1) `lemma_14_4_13_1` ($z' \in \operatorname{cl} \widehat{V}_a$) with `approxClass_eq_or_disjoint`; (2) `lemma_14_4_13_2` for $z' \in \operatorname{cl}_X(V_a)$; the recalled identity is `lemma_14_4_13_classSection` (same hypothesis); (3) `lemma_14_4_13_3` ($z' \in \operatorname{cl} \widehat{V}_a$, every $g \in G_a$); (4) `lemma_14_4_13_4` (every point of $\widetilde{O}(g{*}z') \cap \partial X$ is $g{*}y$, $y \in \widetilde{O}(z') \cap \partial X$, with $s^{0a}_{g(i)}(g{*}y) = \mu(\Gamma(g,y), s^{0a}_i(y))$); disjoint union: `ZCoherentNbhdSystem.disjoint_orbit`. *Book:* §14.4, Lemma 14.4.13, pp. 666–667 (2017: Lemma 14.9). *Hidden hypotheses:* none. *Deviation (narrower):* (2) and the recalled identity are stated for $z' \in \operatorname{cl}_X(V_a)$, $V_a = \widehat{V}_a \cap \partial X$: they are Definition 14.4.11 (4), which covers only those points. A point of $\operatorname{cl}_X(\widehat{V}_a) \cap \partial X$ need not lie in $\operatorname{cl}_X(V_a)$ (a limit of interior points of $\widehat{V}_a$). Presumably $z \in \operatorname{cl}_X(V_a)$ is meant (it is what P14.4.14 uses); no counterexample was attempted. (1), (3), (4) hold for all $z' \in \operatorname{cl} \widehat{V}_a$.

V14.33 (Proposition 14.4.14 (2017: 14.7), the local extensions $\hat{s}^a_i$; `Polyfold.SBLocalization.`

`glueSection_congr_at`, `Polyfold.SBLocalization.CoherentNbhdSystem.eqvGen_s0`, `Polyfold.SBLocalization.CoherentNbhdSystem.exists_local_piece`, `Polyfold.SBLocalization.proposition_14_4_14`, `Polyfold.SBLocalization.proposition_14_4_14_of_hyp`, `Polyfold.SBLocalization.coherentNbhdSystem_hyp`, `Polyfold.SBLocalization.proposition_14_4_14_of_limit`, `Polyfold.SBLocalization.ZConcrete.isFaceScPlus`, `Polyfold.SBLocalization.ZConcrete.proposition_14_4_14`). `proposition_14_4_14 hB hpu C` for `C : CoherentNbhdSystem D`, `D : LocalExtensionBase A a`: there are $\hat{s}_i$ with `IsScPlusSectionOn A (ŝ i) D.Vh`, $\hat{s}_{g(i)}(g * y) = \mu(\Gamma(g, y), \hat{s}_i(y))$ on $\hat{V}_a$, (1) $\hat{s}_i = s^0_i$ on $\hat{V}_a \cap \partial X$, (2) $\sim_y$-compatibility on $\hat{V}_a$. Same `hB`, `hpu` as P14.1.4.

For the data of §§14.3–14.4: `proposition_14_4_14_of_hyp hB hpu D hP ha hface` (`proposition_14_4_14` applied to `coherentNbhdSystem_hyp`), and since Part III wave 5 without `HasThetaLocalEq`: `proposition_14_4_14_of_limit D L hB hpu hD ha hface` (abstract) and `ZConcrete.proposition_14_4_14 C hB hpu ha` (the concrete $\hat{V}_a$, $\hat{\tau}$ and the sections of the good structured tuple, V14.39): there are $\hat{s}_i$, sc⁺ and equivariant on $\hat{V}_a$, with (1) on $\hat{V}_a \cap \partial X$ and (2) on $\hat{V}_a$. `ZConcrete.isFaceScPlus`: the face sc⁺ condition on $U(a)$, from Definition 14.1.3 (1). *Book:* §14.4, Proposition 14.4.14, pp. 667–671 (2017: Proposition 14.7). *Hidden hypotheses:* none. *Deviation (restated):* Stated for abstract data satisfying D14.4.11 (1), (3), (4) and Lemma 14.4.5; the instantiation with the concrete $\hat{V}_a$ (P14.3.8), $\sim$ (`ZRel`, via `ofZSetup`), $\approx$ and $\tilde{O}$ (D14.4.8/P14.4.10) is `ZConcrete.proposition_14_4_14` (V14.39; gaps G.168, G.169 closed). Proof differs: for $z \in \mathrm{cl}(V_a)$, one extension (`exists_scPlusSection_extension_of_face`) of the boundary section of a chosen representative of each $\approx_z$-class (equivalence closure), on G_z with $\mathrm{cl}\, G_z \subseteq \tilde{O}(z)$; for $z \in \mathrm{cl}(\hat{V}_a) \setminus \mathrm{cl}(V_a)$, zero sections on a neighbourhood missing $\mathrm{cl}(V_a)$ (so (4) is needed only as printed); an ordinary partition of unity; then the average over G_a (`avgSection`). The average is equivariant, still extends s^0 (which is equivariant), and stays $\sim$-compatible because $\sim$ is (Lemma 14.4.5). So D14.4.11 (2) and the orbit-by-orbit construction are not needed. $\approx_z$ need not be an equivalence relation. "Symmetric section structure" is read as Definition 13.2.5 (1) (equivariance); the weights and $\hat{\Lambda}_a$ come in §14.5.

V14.34 (Lemma 14.5.1 (2017: 14.10), the functors β_a; `Polyfold.EpGroupoid.lemma_14_5_1`, `Polyfold.EpGroupoid.isSaturated_compl`, `Polyfold.EpGroupoid.isSaturated_preimage_`, `Polyfold.EpGroupoid.isOpen_saturation`, `Polyfold.EpGroupoid.IsSaturated.tgt_mem`). `EpGroupoid.lemma_14_5_1 (hpu : X.HasScPartitionsOfUnity) (hB : IsClosed B) (hBs : X.IsSaturated B) (hAB : A   B) (hV :   a   A, IsOpen (V a)) (hcov : B     a   A, X.saturation (V a))`. It gives $\beta : \mathrm{Obj} \to \mathrm{Obj} \to \mathbb{R}$ with: sc-smooth for $a \in A$, values in $[0, 1]$, $\beta_a(s\varphi) = \beta_a(t\varphi)$; (1) `tsupport (  a)   X.saturation (V a)`; (2) as in the text; (3) `  a   A,   a y = 1` for $y \in B$. *Book:* §14.5, Lemma 14.5.1, p. 671 (2017: Lemma 14.10). *Hidden hypotheses:* none. *Deviation (stronger):* The proof is the text's. The partition of unity is indexed by objects: $U_x = \pi^{-1}\pi(\hat{V}_x)$ for $x \in A$, $U_x = X \setminus B$ otherwise. Tameness is not used. The covering property of the $\hat{V}_a$ (P14.3.8 (3), (14.34)) is the hypothesis `hcov`, and ∂X closed and saturated are named. (14.53) is not part of the lemma; it holds for the $\hat{V}_a$ themselves (V14.45).

V14.35 (Theorem 14.5.5 (2017: 14.3), the structurability criterion; `Polyfold.SBLocalization.`

`CriterionData`, `Polyfold.SBLocalization.CriterionData.exists_transport`, `Polyfold.SBLocalization.CriterionData.Transport`, `Polyfold.SBLocalization.CriterionData.nonempty_transport`, `Polyfold.SBLocalization.CriterionData.Transport.isStructurable`, `Polyfold.SBLocalization.theorem_14_5_5_of_goodSystem`, `Polyfold.SBLocalization.theorem_14_5_5`, `Polyfold.SBLocalization.IndexCorr.biact_compHom_gl`, `Polyfold.SBLocalization.IndexCorr.sat_forall`, `Polyfold.SBLocalization.IndexCorr.ofPerm_compHom_gl`). `theorem_14_5_5 (hX : A.X.HasParacompactOrbitSpace) (hΛ : IsScPlusMultisection A Λ) (D : CriterionData A Λ) : IsStructurable A Λ`. `CriterionData` holds the data (a)–(e). The variant `theorem_14_5_5_of_goodSystem` takes `Nonempty A.X.GoodSystem` in place of paracompactness. *Book:* §14.5, Theorem 14.5.5, pp. 678–680 (2017: Theorem 14.3). *Hidden hypotheses:* none. *Deviation (stronger):* The properness in (a) and the closure condition in (b) are not used, so they are not assumed. "Correspondence" in (e) is read as D13.3.4 (2) together with D13.3.6 (1)–(2): bi-action, compatibility with the sections, `diag` and `swap`. The proof pulls the structures back along local transports τ_y into the V_a. It does not go through Theorem 13.4.3.

V14.36 (The remark after Definition 14.1.2 (2017: after 14.1): the $\bar{\theta}_j$ are tame M-polyfolds; `Polyfold.MPolyfold.isFace_iff_isFaceIn_univ`, `Polyfold.MPolyfold.isFaceStructured_iff_univ`, `Polyfold.EpGroupoid.HasBoundaryFaces.mem_of_mem_closure`, `Polyfold.EpGroupoid.HasBoundaryFaces.eq_faceAt`, `Polyfold.EpGroupoid.HasBoundaryFaces.isFaceIn`, `Polyfold.EpGroupoid.HasBoundaryFaces.isFaceIn_iff`, `Polyfold.EpGroupoid.HasBoundaryFaces.injective_closure`, `Polyfold.EpGroupoid.HasBoundaryFaces.facesThrough_eq`, `Polyfold.EpGroupoid.HasBoundaryFaces.isFaceStructuredOpen`, `Polyfold.EpGroupoid.HasBoundaryFaces.isSubMPolyfold`, `Polyfold.EpGroupoid.HasBoundaryFaces.isFaceStructured_restrictOpen`, `Polyfold.EpGroupoid.HasBoundaryFaces.isFace_restrictOpen`, `Polyfold.EpGroupoid.HasBoundaryFaces.face_isTame`, `Polyfold.EpGroupoid.IsGoodBoundaryNbhd.faces_isSubMPolyfold`). Setting: X tame, V open, `HasBoundaryFaces x V` (Proposition 14.1.1 (4)). `HasBoundaryFaces.isFaceIn_iff`, `injective_closure`, `facesThrough_eq`: the faces of the open set V (Definition 2.7.6) are exactly the $\bar{\theta}_i = V \cap \operatorname{cl}\theta_i$, pairwise distinct. `isFaceStructuredOpen`: V is a face-structured open set. `isSubMPolyfold`: $\bar{\theta}_i \subseteq X$ is a sub-M-polyfold. `isFace_restrictOpen`: $\bar{\theta}_i$ is a face of the tame (`isTame_restrictOpen`), face-structured (`isFaceStructured_restrictOpen`) M-polyfold V. **`HasBoundaryFaces.face_isTame`**: with its sub-M-polyfold structure from Theorem 2.5.17, $\bar{\theta}_i$ is a tame M-polyfold, and $d_X = d_{\bar{\theta}_i} + 1$ on it. `IsGoodBoundaryNbhd.faces_isSubMPolyfold` restates this for good neighbourhoods. Bridge lemmas `MPolyfold.isFace_iff_isFaceIn_univ`, `MPolyfold.isFaceStructured_iff_univ` (Definition 2.5.10 = Definition 2.7.6 for `univ`). *Book:* §14.1, after Definition 14.1.2, p. 641 (2017: after Definition 14.1). *Hidden hypotheses:* none. *Deviation (same):* The structure on $\bar{\theta}_i$ is that of a face of the M-polyfold V (Part I's `restrictOpen`, then `subPolyfold`). `isSubMPolyfold` gives the sub-M-polyfold of X on the same set; that the two structures agree is not stated, though both come from the same local face charts. Gap G.165 closed.

V14.37 (Proposition 14.5.4 (2017: 14.8), the transport of the local extensions; `Polyfold.SBLocalization.IndexCorr.exists_pairs_left_t9`, `Polyfold.SBLocalization.IndexCorr.`

sat_of_pairs_t9, Polyfold.SBLocalization.IndexCorr.rel_of_sat_t9, Polyfold.SBLocalization.sectionSmul_congr_t9, Polyfold.SBLocalization.sectionSmul_of_eq_zero_t9, Polyfold.SBLocalization.src_mem_boundary_t9, Polyfold.SBLocalization.TransportInput.SuppInSat, Polyfold.SBLocalization.TransportInput.bsec, Polyfold.SBLocalization.TransportInput.isSection_bsec, Polyfold.SBLocalization.TransportInput.bsec_eq, Polyfold.SBLocalization.TransportInput.bsec_of_eq_zero, Polyfold.SBLocalization.TransportInput.isScPlusSectionOn_bsec, Polyfold.SBLocalization.TransportInput.fibAct_bsec_eq, Polyfold.SBLocalization.TransportInput.tsec, Polyfold.SBLocalization.TransportInput.tsec_of_arc, Polyfold.SBLocalization.TransportInput.tsec_of_not, Polyfold.SBLocalization.TransportInput.arc_of_pairs, Polyfold.SBLocalization.TransportInput.tsec_of_pairs, Polyfold.SBLocalization.TransportInput.tsec_of_not_reach, Polyfold.SBLocalization.TransportInput.isScPlusSectionOn_tsec, Polyfold.SBLocalization.TransportInput.tsec_equivariant, Polyfold.SBLocalization.TransportInput.tsec_compat, Polyfold.SBLocalization.TransportInput.tsec_boundary, Polyfold.SBLocalization.TransportInput.proposition_14_5_4, Polyfold.SBLocalization.TransportInput.suppInSat_of_zero). `TransportInput.proposition_14_5_4 (h : T.SuppInSat) : proposition_14_5_4_statement T` (the statement of V14.8's setting, exactly). The section is `tsec a b j`, defined through z-arcs (`ZSetup.Arc a b i j `): `tsec a b j (t ) = ( , _a·ŝ (s ))` for every arc (`tsec_of_arc`), and 0 if there is none (`tsec_of_not`). Independence of the arc is `fibAct_bsec_eq`, the text's z-loop argument: two arcs close into a z-loop at a by $\Gamma(g^{-1}, \cdot)$, then Proposition 14.4.14 (2) (`sh_zRel`) and the equivariance of $\hat{s}^a$. sc$^+$ (`isScPlusSectionOn_tsec`): near a target of an arc, a local inverse of t with $\hat{\tau}$ locally constant gives the `transportSection` form; elsewhere the section vanishes off $t(s^{-1}K)$, $K = \mathrm{cl}\{y \in \widetilde{U}(a) \mid \beta_a(y) \neq 0\}$, closed by properness of $\mathrm{cl}\,\widetilde{U}(a)$ and (14.53). `tsec_equivariant`, `tsec_compat`, `tsec_boundary`; `tsec_of_pairs`: the chain form agrees with the text's one-step form. Witness of `SuppInSat`: `suppInSat_of_zero`. *Book:* §14.5, Proposition 14.5.4, pp. 673–677 (2017: Proposition 14.8). *Hidden hypotheses:* none. *Deviation (restated):* One hypothesis is named: `SuppInSat T`, Lemma 14.5.1 (1) ($\beta_a \neq 0$ only on $\pi^{-1}(\pi(\widehat{V}_a))$). It is part of the text's data: the transport input records (14.53) but not this part of Lemma 14.5.1, which is needed for (3) (a point of $\widehat{V}_b \cap \partial X$ outside the saturation of $\widehat{V}_a$ must have $\beta_a = 0$); `lemma_14_5_1` gives it (V14.34). The definition is through chains (arcs) rather than single morphisms; it agrees with the text's one-step form (`tsec_of_pairs`), and the chain form gives equivariance and (4) directly. (2) is read as a closed $K \subseteq \widehat{V}_a$ off which the section vanishes. Gap G.170 closed.

V14.38 (The sum over a in the proof of Theorem 14.1.9 (2017: §14.5), with the identity of Lemma 14.5.2; Polyfold.SBLocalization.AssemblyInput, Polyfold.SBLocalization.AssemblyInput.P_tsec, Polyfold.SBLocalization.AssemblyInput.K, Polyfold.SBLocalization.AssemblyInput.C, Polyfold.SBLocalization.AssemblyInput.isClosed_sat_K, Polyfold.SBLocalization.AssemblyInput.locallyFinite_sat_K, Polyfold.SBLocalization.AssemblyInput.isClosed_C, Polyfold.SBLocalization.AssemblyInput.isSaturated_C, Polyfold.SBLocalization.AssemblyInput.K_subset, Polyfold.

SBLocalization.AssemblyInput.tsec_eq_zero_of_not_mem_C, Polyfold.SBLocalization.AssemblyInput.exists_finite_tsec, Polyfold.SBLocalization.AssemblyInput.sumFib, Polyfold.SBLocalization.AssemblyInput.usec, Polyfold.SBLocalization.AssemblyInput.isSection_usec, Polyfold.SBLocalization.AssemblyInput.support_subset, Polyfold.SBLocalization.AssemblyInput.finite_support, Polyfold.SBLocalization.AssemblyInput.usec_eq_sectionSum, Polyfold.SBLocalization.AssemblyInput.isScPlusSectionOn_usec, Polyfold.SBLocalization.AssemblyInput.usec_fibAct, Polyfold.SBLocalization.AssemblyInput.usec_equivariant, Polyfold.SBLocalization.AssemblyInput.usec_compat, Polyfold.SBLocalization.AssemblyInput.usec_boundary, Polyfold.SBLocalization.AssemblyInput.ssec, Polyfold.SBLocalization.AssemblyInput.isSection_ssec, Polyfold.SBLocalization.AssemblyInput.isScPlusSectionOn_ssec, Polyfold.SBLocalization.AssemblyInput.fibAct_ssec, Polyfold.SBLocalization.AssemblyInput.ssec_equivariant, Polyfold.SBLocalization.AssemblyInput.ssec_compat, Polyfold.SBLocalization.AssemblyInput.usec_eq_zero_of_not_mem_C, Polyfold.SBLocalization.AssemblyInput.ssec_eq_zero_of_). `AssemblyInput A Λ` collects the data of §§14.3–14.5. `usec b j` is the sum as in the fibre vector space, with `isScPlusSectionOn_usec` (locally a finite `sectionSum`), `usec_equivariant`, `usec_compat`, and `usec_boundary : usec b j = s` on $\hat{V}_b \cap \partial X$ (the sum identity of Lemma 14.5.2, with `sum_one`). `ssec b j = · usec b j` folds in the cut-off. `C` is the closed saturated set off which every transported section vanishes (`isClosed_C` from local finiteness and properness, `isSaturated_C`). *Book:* §14.5, proof of Theorem 14.1.9, pp. 676–677 (2017: §14.5, Lemma 14.11). *Hidden hypotheses:* none. *Deviation (same):* None in content. The set C is introduced so that the points outside $\bigcup |\hat{V}_b|$ get a neighbourhood where all sections are 0; the text says this in words. Lemma 14.5.2 enters only through its sum identity, which is proved for the sections (`usec_boundary`); its multisection form (a commensurable family over ∂X) is gap G.172.

V14.39 (The setting of §14.4 from the objects of §14.3, (14.33)–(14.38); Polyfold.SBLocalization.IndexCorr.sat_pairs_z5, Polyfold.SBLocalization.bdryRetr_z5, Polyfold.SBLocalization.bdryRetr_z5_of_mem, Polyfold.SBLocalization.coe_bdryRetr_z5_of_mem, Polyfold.SBLocalization.isoRetr_z5, Polyfold.SBLocalization.isoRetr_z5_val, Polyfold.SBLocalization.rep_isoRetr_z5, Polyfold.SBLocalization.ofPerm_isoRetr_z5, Polyfold.SBLocalization.ZConcrete, Polyfold.SBLocalization.ZConcrete.exists_, Polyfold.SBLocalization.ZConcrete.isProper, Polyfold.SBLocalization.ZConcrete.closure_Vh_subset, Polyfold.SBLocalization.ZConcrete.Vh_subset_P, Polyfold.SBLocalization.ZConcrete.mapsTo_Vh_P, Polyfold.SBLocalization.ZConcrete.morBetween_subset_Ohat, Polyfold.SBLocalization.ZConcrete.mem_bdry, Polyfold.SBLocalization.ZConcrete.toZData, Polyfold.SBLocalization.ZConcrete.toLimit, Polyfold.SBLocalization.ZConcrete.compat_aux, Polyfold.SBLocalization.ZConcrete.toZSections, Polyfold.SBLocalization.ZConcrete.compatNbhd, Polyfold.SBLocalization.exists_zConcrete_of_hasP4, Polyfold.SBLocalization.exists_zConcrete_of_tame). `ZConcrete E Q` (for `E : CorrExtension T P`, `Q : P6Nbhds P`): $x_0 \in \partial X$, $A \subseteq \partial X$, $\tilde{V}_a$, $\hat{V}_a$ with the conclusions of Proposition 14.3.8. `ZConcrete.exists_ E hpara Q x ; exists_` `zConcrete_of_hasP4 hT hpara x` (the chain (P5), Lemma 14.3.1, (P6), Proposition 14.3.8); `exists_zConcrete_of_tame hX hpara x` (tame X, $|X|$ paracompact, $\partial X \neq \emptyset$: the zero tuple and Proposition 14.2.3 for (P4)). `C.toZData : A.X.ZData` ($\tilde{U}(a)$ = `Q.nbhdObj a`, $\hat{V}_a$, $I_a = I_{r(a)}$ of the boundary structure, the action of G_a through `isoRetr_z5`, $\hat{\tau}_{a,a'}$ = `E.tauHat (r a) (r a')`; `diag` from `tauHat_Γ_of_VHat`, `swap` from `tauHat_inv_of_VHat`, local constancy from `tauHat_locallyConstant`, $\mathrm{cl}_X \hat{V}_a \subseteq \tilde{U}(a)$ from Lemmas 13.1.1, 14.3.6 (1)). `C.toLimit : ZLimitSetting C.toZData` ($M_{b,c} = \hat{O}_{b,c}$; (P6) properness; Proposition 14.3.8 (3)). `C.toZSections` (the sections s_i^x of the tuple; (14.45) from Definition 14.1.3 (2); Definition 14.1.7 (4) from `T.tau.compat` and `tauHat_eq_tau`), `C.compatNbhd`. Index transport: r = `bdryRetr_z5 x` ($X \to \partial X$, the identity on ∂X), no casts. *Book:* §§14.3–14.4, (14.33)–(14.38), p. 657 (2017: §§14.3–14.4). *Hidden hypotheses:* none. *Deviation (explicit):* The setting is the text's; the identification of objects with boundary points is named. The base point x_0 only fixes irrelevant values at objects outside ∂X (all of A lies in ∂X). The index types on $x \in \partial X$ are reached through `bdryRetr_z5` and `isoRetr_z5` (a group homomorphism $G_a \to G_{r(a)}$, the identity of morphisms on ∂X). Gap G.169 closed.

V14.40 (The preamble of §14.6 (2017: §14.6): faces as sub-ep-groupoids; sc⁺-sections of $W|F$; `Polyfold.EpGroupoid.IsEpFace.subset_boundary`, `Polyfold.EpGroupoid.exists_isEpFace_mem`, `Polyfold.EpGroupoid.finite_isEpFace_mem`, `Polyfold.EpGroupoid.eventually_mem_of_isEpFace`, `Polyfold.EpGroupoid.exists_isEpFace_germ`, `Polyfold.IsScPlusOnFaceNear.congr_nhds`, `Polyfold.SBLocalization.FaceGermIn`, `Polyfold.SBLocalization.IsFaceRelScPlusSectionOn`, `Polyfold.SBLocalization.IsFaceRelScPlusSectionOn.mono`, `Polyfold.SBLocalization.IsFaceScPlusSectionOn.isFaceRelScPlusSectionOn`, `Polyfold.SBLocalization.isFaceRelScPlusSectionOn_fiberZero`, `Polyfold.SBLocalization.isFaceScPlusSectionOn_of_local`, `Polyfold.EpGroupoid.NaturalRep.eventually_Γ_eq`, `Polyfold.EpGroupoid.NaturalRep.eventually_Φ_eq`, `Polyfold.EpGroupoid.IsEpFace.tgt_mem_iff`, `Polyfold.EpGroupoid.IsEpFace.tgt_mem`, `Polyfold.EpGroupoid.IsEpFace.Φ_mem`). Faces are Chapter 11's ep-faces `IsEpFace` (with `faceEpGroupoid`). `IsEpFace.subset_boundary`, `exists_isEpFace_mem`, `finite_isEpFace_mem`, `eventually_mem_of_isEpFace` (near x, every face through a point contains x), `IsEpFace.tgt_mem_iff` (saturated), `IsEpFace.Φ_mem` (invariance). `exists_isEpFace_germ`: in a tame chart, the local face $\{t_i = 0\}$ near y lies in one face of X. sc⁺ relative to F, `IsFaceRelScPlusSectionOn A F s U`: $P \circ s = \mathrm{id}$, $s \in W_1$ on U, and in every tame strong-bundle chart, on every local face germ of the model lying in F (`FaceGermIn`), the local representative is sc⁺. `isFaceScPlusSectionOn_of_local`: a section that near each y agrees on each face $F \ni y$ with a section sc⁺ relative to F satisfies Definition 14.1.3 (1). *Book:* §14.6, p. 680 (2017: §14.6). *Hidden hypotheses:* none. *Deviation (restated):* The sc⁺-sections of $W|F$ are read through the charts of X on local face germs inside F; those germs are the local models of the sub-M-polyfold F (Theorem 2.5.17). The induced structure on F is therefore not used directly, in the same style as the reading of Definition 14.1.3 (1) (V14.3).

V14.41 (Definition 14.6.1 (2017: 14.10), restriction to $F \cap F'$; `Polyfold.SBLocalization.IndexFamily`, `Polyfold.SBLocalization.RelTuple`, `Polyfold.SBLocalization.RelTuple.U`, `Polyfold.SBLocalization.RelTuple.mor`, `Polyfold.SBLocalization.RelTuple.Equiv`, `Polyfold.SBLocalization.RelTuple.Equiv.`

`refl`, `Polyfold.SBLocalization.RelTuple.Equiv.symm`, `Polyfold.SBLocalization.RelTuple.Equiv.trans`, `Polyfold.SBLocalization.RelTuple.Equiv.restrictTo`, `Polyfold.SBLocalization.RelTuple.Sigma`, `Polyfold.SBLocalization.RelTuple.restrictTo`, `Polyfold.SBLocalization.RelTuple.restrictTo_U`, `Polyfold.SBLocalization.RelTuple.restrictTo_restrictTo_equiv`, `Polyfold.SBLocalization.RelStructured`, `Polyfold.SBLocalization.RelStructured.restrictTo`, `Polyfold.SBLocalization.FaceStructuredTuple`, `Polyfold.SBLocalization.FaceStructuredTuple.restrictFace`, `Polyfold.SBLocalization.FaceStructuredTuple.restrictFace_isFaceRelScPlus`, `Polyfold.SBLocalization.FaceStructuredMultisection`, `Polyfold.SBLocalization.FaceStructuredMultisection.restrictFace`, `Polyfold.SBLocalization.FaceStructuredMultisection.restrictFace'`, `Polyfold.IsScPlusOnFaceNear.isScPlusOnCornerNear`, `Polyfold.SBLocalization.IsCornerRelScPlusSectionOn`, `Polyfold.SBLocalization.IsFaceRelScPlusSectionOn.isCornerRelScPlusSectionOn`, `Polyfold.SBLocalization.FaceStructuredTuple.restrictFace_isCornerRelScPlus`, `Polyfold.SBLocalization.isCornerRelScPlusSectionOn_fiberZero`). `RelTuple A K :` the data of a structured multisection over a set K without regularity (Λ a functor on $W|K$; `nbhd x : NatNbhd x` with t proper on $s^{-1}(\mathrm{cl}(U(x) \cap K))$; sections with $P \circ s = \mathrm{id}$, equivariance and the representation of Λ on $U(x) \cap K$; τ locally constant with finitely many values, with (a)–(c) and (4)). `RelTuple.Equiv` (as for boundary tuples), an equivalence relation, with quotient `RelStructured`. `FaceStructuredTuple A F` adds `isFaceRelScPlus`. Definition 14.6.1: `FaceStructuredTuple.restrictFace T F' : RelTuple A (F  F')`, with `restrictFace_isFaceRelScPlus`; on classes `FaceStructuredMultisection.restrictFace`, `restrictFace'` (well defined, via `RelTuple.Equiv.restrictTo`).

Part III wave 7 (`Chapter14/P3W7W6_CornerScPlus.lean`): `IsCornerRelScPlusSectionOn A F F' s U` (sc$^+$-sections of $W|(F \cap F')$, read in the tame charts of X on the corner germs $\{t_i = t_j = 0\}$, `IsScPlusOnCornerNear`); `IsScPlusOnFaceNear.isScPlusOnCornerNear` (in a tame model, sc$^+$ on the face germ gives sc$^+$ on every corner germ, via `corner_retraction`); `IsFaceRelScPlusSectionOn.isCornerRelScPlusSectionOn`; **`FaceStructuredTuple.restrictFace_isCornerRelScPlus`**: the sections of `T.restrictFace F'` are sc$^+$ relative to the corner $F \cap F'$. *Book:* §14.6, Definition 14.6.1, p. 681 (2017: Definition 14.10). *Hidden hypotheses:* none. *Deviation (restated):* "Good $U_F(x) \subseteq F$" is read as $U(x) \cap F$ for a natural neighbourhood $U(x)$ of X with the properness of Definition 13.1.4 (1); condition (2) of a good system is not part of the data (gap G.185); §14.6 does not use it, and the glued output of Theorem 14.6.3 carries it. On $F \cap F'$ the restriction is sc$^+$ relative to the corner, read through the charts of X as for $W|F$ (`restrictFace_isCornerRelScPlus`; gap G.184 closed in Part III wave 7). A representative whose $U_F(x)$ are good inside the face ep-groupoid exists up to equivalence (V14.47).

V14.42 (Definition 14.6.2 (2017: 14.11), compatible assignments; `Polyfold.SBLocalization.IsCompatibleAssignment`, `Polyfold.SBLocalization.IsCompatibleAssignment.inter`, `Polyfold.SBLocalization.IsCompatibleAssignment.equiv`, `Polyfold.SBLocalization.IsCompatibleAssignment.index_eq`, `Polyfold.SBLocalization.IsCompatibleAssignment.Λ_eq`, `Polyfold.SBLocalization.IndexFamily.unit`, `Polyfold.SBLocalization.`

`FaceStructuredTuple.zero`, `Polyfold.SBLocalization.isCompatibleAssignment_zero`). `IsCompatibleAssignment Λt` for `Λt :  F, IsEpFace F → FaceStructuredMultisection A F`. `IsCompatibleAssignment.equiv` (the representative-level equivalence on $F \cap F'$), `.index_eq` ($I_x^F = I_x^{F'}$), `.Λ_eq` ($\Lambda_F = \Lambda_{F'}$ on $W|(F \cap F')$). Nonvacuity: `isCompatibleAssignment_zero` (the zero multisection on every face). *Book:* §14.6, Definition 14.6.2, p. 681 (2017: Definition 14.11). *Hidden hypotheses:* none. *Deviation (restated):* The equivalence is the analogue of Definition 13.3.8, as for the tuples over ∂X (V14.7). It compares Λ, the index sets and τ on smaller neighbourhoods, but not the sections.

V14.43 (Theorem 14.6.3 (2017: 14.4), assembling the faces; `Polyfold.SBLocalization.CompatibleFaceData`, `Polyfold.SBLocalization.CompatibleFaceData.isCompatibleAssignment`, `Polyfold.SBLocalization.CompatibleFaceData.Λg`, `Polyfold.SBLocalization.CompatibleFaceData.Λg_eq`, `Polyfold.SBLocalization.CompatibleFaceData.faceSel`, `Polyfold.SBLocalization.CompatibleFaceData.faceSel_spec`, `Polyfold.SBLocalization.CompatibleFaceData.sg`, `Polyfold.SBLocalization.CompatibleFaceData. g`, `Polyfold.SBLocalization.CompatibleFaceData.Adapted`, `Polyfold.SBLocalization.CompatibleFaceData.exists_adapted`, `Polyfold.SBLocalization.CompatibleFaceData.sg_eq`, `Polyfold.SBLocalization.CompatibleFaceData.mem_mor_face`, `Polyfold.SBLocalization.CompatibleFaceData. g_eq`, `Polyfold.SBLocalization.CompatibleFaceData.Φ_eq`, `Polyfold.SBLocalization.CompatibleFaceData.S`, `Polyfold.SBLocalization.CompatibleFaceData.dec`, `Polyfold.SBLocalization.CompatibleFaceData.tau`, `Polyfold.SBLocalization.CompatibleFaceData.isBoundaryScPlusMultisection`, `Polyfold.SBLocalization.CompatibleFaceData.glue`, `Polyfold.SBLocalization.CompatibleFaceData.glue_Λ`, `Polyfold.SBLocalization.CompatibleFaceData.glue_U_inter_subset`, `Polyfold.SBLocalization.CompatibleFaceData.glue_s`, `Polyfold.SBLocalization.CompatibleFaceData.glue_ `, `Polyfold.SBLocalization.CompatibleFaceData.theorem_14_6_3_hyp`, `Polyfold.SBLocalization.CompatibleFaceData.theorem_14_6_3_extension_hyp`, `Polyfold.SBLocalization.CompatibleFaceData.zero`, `Polyfold.SBLocalization.CompatibleFaceData.zero_Λg`, `Polyfold.SBLocalization.CompatibleFaceData.theorem_14_6_3_extension_a6`). Input `D : CompatibleFaceData A`: one index family , for each face a `FaceStructuredTuple A F` , and agreement of Λ on $W|(F \cap F')$, of the sections on $U_F(x) \cap U_{F'}(x) \cap F \cap F'$ and of the correspondences on morphisms between those sets. `D.isCompatibleAssignment`: such data is compatible (Definition 14.6.2). `exists_adapted hX hpara`: a boundary good system with faces through points of $V(x)$ containing x, $V(x) \subseteq V_F(x)$ for $F \ni x$, and agreeing natural actions (Theorem 14.1.6, `exists_le_of_finite`, `exists_refinement`). On it `S` ($s_i^x := s_{F,i}^x$ on F, sc⁺ by `isFaceScPlusSectionOn_of_local`) and `tau`; `glue hX hpara : BoundaryStructuredTuple A` with `glue_U_inter_subset`, `glue_s`, `glue_ `. **`theorem_14_6_3_hyp (hX : A.X.IsEpFaceStructured) (hpara : A.X.HasParacompactOrbitSpace)`**: `IsBoundaryStructurable A D.Λg  F hF w, P w  F → D.Λg w = (D.T F hF).Λ w`. `theorem_14_6_3_extension_hyp (h1419 : theorem_14_1_9_statement A) …`, with the hypotheses of Theorem 14.1.9, gives Λ' structurable with $N(\Lambda') \leq f$, dom-supp $\Lambda' \subseteq \tilde{U}$ and $\Lambda' = \Lambda_F$ over every face. Nonvacuity: `CompatibleFaceData.zero`, `zero_`

`Λg`.

Extension part (Part III wave 6): **`CompatibleFaceData.theorem_14_6_3_extension_a6 D hX hpara hpart hB (hch : ChargesBoundary D.Λg) …`**, the extension without the hypothesis `h1419`, from `theorem_14_1_9_charged_a6` (V14.8). `theorem_14_6_3_extension_hyp` took `theorem_14_1_9_statement A`, which holds only for $\partial X = \emptyset$, and is vacuous otherwise. *Book:* §14.6, Theorem 14.6.3, p. 682 (2017: Theorem 14.4). *Hidden hypotheses:* the input is `CompatibleFaceData A`, representatives whose sections and correspondences agree on $F \cap F'$, not a class-level compatible assignment. The text derives $s^x_{F,i} = s^x_{F',i}$ on $U_F(x) \cap U_{F'}(x)$ from compatibility; the equivalence of Definition 14.6.2 (as of Definition 13.3.8) compares Λ, the index sets and τ but not the sections, so the agreement of the representatives is taken as input. Presumably the text means that representatives can be chosen to agree; no counterexample is claimed (gap G.173). Witness `CompatibleFaceData.zero`. *Deviation (narrower):* The extension part is `theorem_14_6_3_extension_a6`, with the two added hypotheses of Theorem 14.1.9 (`hB` and `ChargesBoundary` for the glued Λ; V14.8); it is not instantiated on an example with face data. The standing assumptions are explicit: `IsEpFaceStructured` (tame and face structured; witness `isEpFaceStructured_trivial_restrictOpen`) and `HasParacompactOrbitSpace` (for Theorem 14.1.6).

V14.44 (Proposition 14.1.8 (2017: 14.1), structurability over ∂X under generalized strong bundle isomorphisms; `Polyfold.IsScPlusOnFaceNear.of_restrict_p6`, `Polyfold.SBLocalization.isFaceScPlusSectionOn_iff_singleton`, `Polyfold.SBLocalization.isFaceScPlusSectionOn_of_localIso`, `Polyfold.SBLocalization.PullMap.facePull_hyp`, `Polyfold.SBLocalization.exists_localIso_act_p6`, `Polyfold.SBLocalization.facePush_hyp_of_equivalence`, `Polyfold.SBLocalization.PullMap.isBoundaryStructurable_pull`, `Polyfold.SBLocalization.isBoundaryStructurable_pushMult`, `Polyfold.SBLocalization.proposition_14_1_8`, `Polyfold.SBLocalization.proposition_14_1_8_pullMult`, `Polyfold.SBLocalization.SDiagram.isTame_B_p6`, `Polyfold.SBLocalization.proposition_14_1_8_tame`, `Polyfold.SBLocalization.proposition_14_1_8_pullMult_tame`, `Polyfold.SBLocalization.GenSBMap.proposition_14_1_8`, `Polyfold.SBLocalization.GenSBMap.proposition_14_1_8_pullMult`, `Polyfold.SBLocalization.PullMap.FacePull_hyp`, `Polyfold.SBLocalization.PullMap.facePull_hyp_id`, `Polyfold.SBLocalization.PullMap.facePull_hyp_of_boundary_eq_empty`, `Polyfold.SBLocalization.IsFaceScPlusSectionOn.congr_fm`, `Polyfold.SBLocalization.symm_mem_boundary_fm`, `Polyfold.SBLocalization.PullMap.mem_boundary_iff`, `Polyfold.SBLocalization.PullMap.BdryPullData`, `Polyfold.SBLocalization.PullMap.exists_bdryPullData`, `Polyfold.SBLocalization.PullMap.bdryPullStructure`, `Polyfold.SBLocalization.PullMap.bdryPullTau`, `Polyfold.SBLocalization.PullMap.bdryPullTuple`, `Polyfold.SBLocalization.PullMap.isBoundaryStructurable_pull_hyp`, `Polyfold.SBLocalization.proposition_14_1_8_pull_hyp`, `Polyfold.SBLocalization.FacePush_hyp`, `Polyfold.SBLocalization.facePush_hyp_of_boundary_eq_empty`, `Polyfold.SBLocalization.isBoundaryWFunctor_pushMult_fm`, `Polyfold.SBLocalization.pushMult_eq_bdry_fm`, `Polyfold.SBLocalization.bdryPushGood`, `Polyfold.SBLocalization.bdryPushStructure`, `Polyfold.SBLocalization.`

`bdryPushTau`, `Polyfold.SBLocalization.isBoundaryStructurable_pushMult_hyp`, `Polyfold.SBLocalization.proposition_14_1_8_hyp`, `Polyfold.SBLocalization.proposition_14_1_8_pullMult_hyp`). Part III wave 6 (`Chapter14/P3W6P6_*`), class level: **`GenSBMap.proposition_14_1_8 (hf : f.IsGeneralizedIso) (hA : A.X.IsTame) (hA' : A'.X.IsTame) (h : IsBoundaryStructurable A Λ) : IsBoundaryStructurable A' (pushMult hf Λ)`** and **`GenSBMap.proposition_14_1_8_pullMult hf hA h`** (pull-back; only X tame). For a representative `D : SDiagram A A'` with Ψ an equivalence: `proposition_14_1_8_tame`, `proposition_14_1_8_pullMult_tame` (tame X, X') and `proposition_14_1_8`, `proposition_14_1_8_pullMult` (tame middle, as in the text's proof); the middle is tame when X is (`SDiagram.isTame_B_p6`). The two steps without hypotheses: `PullMap.isBoundaryStructurable_pull` (any `PullMap`) and `isBoundaryStructurable_pushMult` (equivalences). The transport hypotheses of the earlier form are theorems: `PullMap.facePull_hyp (M : PullMap C A) : M.FacePull_hyp` and `facePush_hyp_of_equivalence`, from the key lemma `isFaceScPlusSectionOn_of_localIso` ((1) of Definition 14.1.3 is transported along any local strong bundle isomorphism: a tame chart at x is restricted and transported along L^{-1}, using Part I's `StrongBundle.LocalIso.transportChart`, `isTameChart_transportChart`), with `exists_localIso_act_p6` (the action of a morphism as a local strong bundle isomorphism, with its formula), `IsScPlusOnFaceNear.of_restrict_p6`, `isFaceScPlusSectionOn_iff_singleton`.

The earlier form (Part III wave 5): `proposition_14_1_8_hyp (D : SDiagram A A') (hΨ) (hB : D.B.X.IsTame) (hA' : A'.X.IsTame) (hpull : (PullMap.ofEquivalence D.isEquivalence_Φ).FacePull_hyp) (hpush : FacePush_hyp D.Ψ) (h : IsBoundaryStructurable A Λ) : IsBoundaryStructurable A' (D.pushMult hΨ Λ)`; `proposition_14_1_8_pullMult_hyp` (the same for $[D]^*\Lambda' = \Phi_*\Psi^*\Lambda'$). The two steps: `PullMap.isBoundaryStructurable_pull_hyp` (any `PullMap`, e.g. equivalences and proper coverings; `proposition_14_1_8_pull_hyp` for equivalences) and `isBoundaryStructurable_pushMult_hyp`. On the way: `PullMap.mem_boundary_iff` ($F^{-1}(\partial X') = \partial X$), `isBoundaryWFunctor_pushMult_fm` (Lemma 11.5.2 over the boundary), `pushMult_eq_bdry_fm`, `exists_bdryPullData`, `bdryPushGood` (good systems for the boundary carried along, each neighbourhood good by Proposition 14.1.1), `bdryPullTau`, `bdryPushTau`, `IsFaceScPlusSectionOn.congr_fm` ((1) of Definition 14.1.3 depends only on the values on ∂X near U), `symm_mem_boundary_fm`. *Book:* §14.1, Proposition 14.1.8, p. 645 (2017: Proposition 14.1). *Hidden hypotheses:* none. *Deviation (stronger):* No added hypothesis: the transport of the face condition, the hypotheses of the earlier `proposition_14_1_8_hyp`, is proved (gap G.164 closed in Part III wave 6). The paracompactness of the orbit spaces is not used, hence stronger. "Structurable over ∂X" is `IsBoundaryStructurable` with (1) of Definition 14.1.3 read in every tame strong bundle chart (V14.3). The class-level statement uses the chosen representative of Chapter 10 (`GenSBMap.rep`), as `GenSBMap.pushMult/pullMult` do; the pull-back form needs only X tame.

V14.45 ((14.53) (2017: §14.5): the closure condition of the cut-offs for the $\widehat{V}_a$; `Polyfold.SBLocalization.tgt_mem_of_closure_a6`, `Polyfold.SBLocalization.closure_support_subset_a6`, `Polyfold.SBLocalization.exists_mor_of_ne_zero_a6`, `Polyfold.SBLocalization.ZConcrete.BetaSpec`, `Polyfold.SBLocalization.ZConcrete.exists_`

`betaSpec`). `tgt_mem_of_closure_a6`: for a natural neighbourhood $\tilde{U}(x)$ and invariant $V \subseteq \tilde{U}(x)$, a morphism from V to a point of $\operatorname{cl}\tilde{U}(x)$ ends in V (a limit of morphisms with both ends in $\tilde{U}(x)$, which are $\Gamma(g,\cdot)$; G_x is finite, so one g works, and $\Gamma(g, s(\psi)) = \psi$ is closed near the limit). `closure_support_subset_a6`: hence `tsupport` ^{1} `(V)` gives (14.53). `ZConcrete.exists_betaSpec`: Lemma 14.5.1 on the cover of the objects of §14.3, with $\beta_a := 0$ off A (`ZConcrete.BetaSpec`). *Book:* §14.5, Lemma 14.5.1, p. 671, with (14.53), p. 672 (2017: Lemma 14.10, §14.5). *Hidden hypotheses:* none. *Deviation (same):* (14.53) holds for the $\hat{V}_a$ themselves, by `tgt_mem_of_closure_a6`; a shrinking of the cover, which the earlier gap text expected, is not needed.

V14.46 (The data of §§14.3–14.5 from a structurable boundary multisection (proof of Theorem 14.1.9); `Polyfold.SBLocalization.ZConcrete.shA6`, `Polyfold.SBLocalization.ZConcrete.shA6_spec`, `Polyfold.SBLocalization.ZConcrete.transportInput_a6`, `Polyfold.SBLocalization.ZConcrete.Γ_eq_a6`, `Polyfold.SBLocalization.ZConcrete.Φ_eq_a6`, `Polyfold.SBLocalization.ZConcrete.biact_eq_a6`, `Polyfold.SBLocalization.ZConcrete.biact_ _a6`, `Polyfold.SBLocalization.ZConcrete.assemblyInput_a6`, `Polyfold.SBLocalization.exists_assemblyInput_a6`). `ZConcrete.transportInput_a6 C hB hpu h : TransportInput A C.toZData.toZSetup` (the local extensions `ZConcrete.shA6` from `ZConcrete.proposition_14_4_14`, s^0 from `toZSections`, local constancy from `ZData.locallyConstant`, (14.53) from V14.45). `ZConcrete.assemblyInput_a6 C hB hpu h (hch : ChargesBoundary T.Λ) : AssemblyInput A T.Λ` (`finite_` from `finite_tauHat`, `biact_` through `ZConcrete.biact_eq_a6`, `rep` and `functor` from the tuple, `nonempty_I` from `ChargesBoundary`). **`exists_assemblyInput_a6 hX hpara hpu hB hΛ hch : Nonempty (AssemblyInput A Λ)`**: Proposition 14.2.3, `exists_zConcrete_of_hasP4`, Theorem 7.5.7, Lemma 14.5.1; for $\partial X = \emptyset$ `emptyAssemblyInput`. *Book:* §§14.3–14.5, proof of Theorem 14.1.9, pp. 656–677 (2017: §§14.3–14.5). *Hidden hypotheses:* none. *Deviation (explicit):* `hB` (tame strong bundle charts at every object) is the standing assumption that Proposition 14.4.14 takes (V14.4, V14.33). `ChargesBoundary` is the reading $I_x \neq \emptyset$ (V14.3); `AssemblyHyp` without it holds only for $\partial X = \emptyset$ (`boundary_eq_empty_of_assemblyHyp_a6`). Gap G.171 is closed by this declaration and `theorem_14_1_9_charged_a6` (V14.8).

V14.47 (§14.6 (2017: §14.6): the $U_F(x)$ form a good system for the face F; `Polyfold.EpGroupoid.isProperSubset_face`, `Polyfold.EpGroupoid.isDiffeoComponent_face`, `Polyfold.EpGroupoid.GoodSystem.isProperSubset_face`, `Polyfold.EpGroupoid.GoodSystem.isDiffeoComponent_face`, `Polyfold.SBLocalization.RelTuple.equiv_shrink`, `Polyfold.SBLocalization.RelTuple.exists_equiv_isDiffeoComponent`, `Polyfold.SBLocalization.FaceStructuredTuple.exists_equiv_faceGood`, `Polyfold.SBLocalization.FaceStructuredMultisection.exists_rep_faceGood`). Face level: `EpGroupoid.isProperSubset_face` (properness, (1) of Definition 13.1.4, passes to `faceEpGroupoid hX h`), `EpGroupoid.isDiffeoComponent_face` (the component condition (3)), and the same for every good system of X (`GoodSystem.isProperSubset_face`, `GoodSystem.isDiffeoComponent_face`). Tuples: `RelTuple.shrink` (neighbourhoods replaced by invariant cores, same class, `RelTuple.equiv_shrink`), `RelTuple.exists_equiv_isDiffeoComponent` (with $|X|$ paracompact, through

Theorem 13.1.8). `FaceStructuredTuple.exists_equiv_faceGood hX hpara h T` and on classes `FaceStructuredMultisection.exists_rep_faceGood`: every structured sc⁺-multisection on $W|F$ has a representative whose $U_F(x)$ satisfy (1) and (3) inside the ep-groupoid F and are invariant under the natural G_x-action of X restricted to $U(x) \cap F$. *Book:* §14.6, p. 680 (2017: §14.6). *Hidden hypotheses:* none. *Deviation (restated):* The text takes this for granted for every representative; the formalization gives an equivalent representative with the property. Condition (2) of Definition 13.1.4 is read as the natural action of X restricted to $U(x) \cap F$; a natural representation of the ep-groupoid F itself is not built (gap G.185). Nothing in §14.6 uses it. Standing hypotheses: `IsEpFaceStructured`, `HasParacompactOrbitSpace`.

What is not verified in this chapter

- pp. 642–646, Definition 14.1.3 and Theorem 14.1.9: nonempty index sets in the formal definition, and tame strong bundle charts from the tameness of X (gap G.186; Findings, item 15).
- p. 672, the multisection form of Lemma 14.5.2 and Remark 14.5.3 (2017: Lemma 14.11, Remark 14.7) (gap G.172).
- pp. 681–682, Theorem 14.6.3 (2017: 14.4) from a class-level compatible assignment (gap G.173; Findings, item 12); the natural representation of a face ep-groupoid for good systems on a face (gap G.185).

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16. Chapter 15 (HWZ, Part III, §§15.1–15.4 (2017 §§15.1–15.4)): begun 2026-11-13 (Part III wave 1, continued in waves 2–7), verified for the linearization $T(f, \Lambda)$, its well-definedness and functor property at level-1 points, the sc-Fredholm property of its summands in the chart reading and the separation lemmas (Proposition 15.1.3 for a non-isolated point), for transversality, good position and the local solution sets of §15.2 with Part I's tame germ data, Theorem 15.2.5 also under general position, for the compactness control, admissible perturbations, general position, Remark 15.3.6 and the auxiliary norm on $[0, 1] \times X$ of §15.3, and for Theorems 15.3.7, 15.3.9, Corollary 15.3.10, Theorem 15.4.3 and Corollary 15.4.4 in hypothesis form, with tameness and the ep-groupoid $U(x)$ in §15.2 (Part III wave 5), Theorem 15.3.7 under the text's hypotheses (Part III wave 6), products of strong bundles, sections and auxiliary norms, Theorem 15.4.1 at a chart point and the partitions of unity of $[0, 1] \times X$ (Part III waves 6–7), except products of Fredholm germs and multisections, the relative local families of Theorem 15.3.9, the glued boundary data of Corollary 15.3.10 (3), Theorem 15.4.1 as an orientation and the oriented end identities of Corollary 15.4.4

Chapter 15 (pp. 683–703; §§15.1–15.4 are pp. 683–703; numbers and pages are those of the book (Springer 2021), with the 2017 text's numbers in parentheses) is transversality and the invariants for sc-Fredholm section functors perturbed by sc⁺-multisections. §15.1 defines the linearization $T(f, \Lambda)$ with values in GrF (preamble, Theorem 15.1.1, Remark 15.1.2, Definition 15.1.7 (2017:

Theorem 15.1, Remark 15.1, Definition 15.1)), proved through separating sequences (Proposition 15.1.3, Lemmas 15.1.4, 15.1.5, Remark 15.1.6 (2017: Proposition 15.1, Lemmas 15.1, 15.2; the remark is new in 2021)). §15.2 (2017: §15.2) defines transversality and good position and describes the local solution sets (Definition 15.2.1, Theorem 15.2.2, Remark 15.2.3, Definition 15.2.4, Theorem 15.2.5, Remark 15.2.6 (2017: Definition 15.2, Theorem 15.2, Remark 15.2, Definition 15.3, Theorem 15.3, Remark 15.3)); §15.3 constructs perturbations (Definitions 15.3.1, 15.3.3, 15.3.5, Theorems 15.3.7, 15.3.9, Corollary 15.3.10 (2017: Definitions 15.4–15.6, Theorems 15.4, 15.5, Corollary 15.1)); §15.4 treats orientations and invariants (Theorem 15.4.1, Definition 15.4.2, Theorem 15.4.3, Corollary 15.4.4, Remark 15.4.5 (2017: Theorems 15.6, 15.7, Definition 15.7, Corollary 15.2, Remark 15.8)). §15.2 and part of §15.3 were added in Part III wave 2, the rest of §15.3 and §15.4 in waves 3–4.

What is verified. The well-definedness of $T(f, \Lambda)(x)$, the substance of Theorem 15.1.1, is proved at every level-1 point, independent of the local section structure and of the chart (V15.2), together with Remark 15.1.2 and Definition 15.1.7 (V15.3, V15.8); the functor property is verified at level-1 points in Part III wave 3 (V15.2; gap G.144 closed); the summands are sc-Fredholm in the chart reading, with Part I's tame germ data (V15.9; gap G.145 closed in wave 2). Proposition 15.1.3 is verified for a non-isolated point; at an isolated point no such sequence exists, a degenerate case (V15.4; Findings, item 3). Lemmas 15.1.4, 15.1.5 are verified in a more general topological setting, and Remark 15.1.6 with "slightly" made explicit (V15.5–V15.7). Part III wave 2, §15.2: Definitions 15.2.1, 15.2.4 are restated (V15.11, V15.14); Theorems 15.2.2, 15.2.5 and Remark 15.2.3 are verified with Part I's tame germ data, which §15.2 does not state, and without the tameness of the solution sets (V15.12, V15.13, V15.15; gaps G.150–G.152); tools V15.10. §15.3: the paragraph before Definition 15.3.1, Definition 15.3.3 with its example, Remark 15.3.6 (first sentence) and Corollary 15.3.10 (2) are verified as stated (V15.19, V15.22, V15.23, V15.25, V15.26), Definitions 15.3.1, 15.3.5 restated (V15.20, V15.24); products of ep-groupoids and $[0, 1]^n \times X$ (V15.17, V15.18). Part III waves 3–4: Theorem 15.2.5 also under general position for tame X (V15.15; gap G.151 closed). Remark 15.3.6 after its first sentence, at every support point, with `hft` (V15.27, V15.28; gap G.161 closed). Theorem 15.3.7 from its two analytic steps as hypotheses, given a good system, and its "in particular" (V15.29, V15.30; gap G.162). Theorem 15.3.9 from Theorem 14.1.9 and two steps of its proof (V15.32; gaps G.177–G.179). Corollary 15.3.10 (1), (3) by reduction (V15.33, V15.34; gaps G.180, G.181, G.183). Theorem 15.4.3 with the orientation as data and partitions of unity, Corollary 15.4.4 with the cobordism as a hypothesis (V15.36, V15.37; gaps G.159, G.182). Remarks 15.3.8, 15.3.11, 15.4.5 are prose (V15.31, V15.35, V15.38). Part III wave 5: tameness and the ep-groupoid $U(x)$ in §15.2 (V15.13, V15.15, V15.28, V15.30, V15.43; gaps G.150, G.152 closed); Theorem 15.3.7 from two Props on the object M-polyfold, the parametric step proved (V15.29, V15.39; gap G.162); Theorem 15.3.9 with step (a) proved given tame bundle charts (V15.32, V15.41; gap G.177 closed); Corollary 15.3.10 (1) without added tameness (V15.33, V15.40; gap G.183 closed), (3) in the text's form with witnesses (V15.34; gaps G.180, G.181); the last paragraph of the proof of Corollary 15.3.10 as stated (V15.42). Part III waves 6–7: Theorem 15.3.7 under the text's hypotheses, structurable given a good system (V15.29, V15.44; gap G.162 closed for it); Theorem 15.3.9 with tame bundle charts and nonempty index sets from the relative local families, the Sard and compactness steps proved (V15.32, V15.45; gap G.178); Corollary 15.3.10 (3) from it on the cylinder, with the partitions of unity of $[0,1] \times X$ and the structurability of the glue up to the faces (V15.34, V15.46–V15.48; gap G.181 closed, gaps G.180, G.189, G.190, G.193); Corollary 15.4.4 along the text's argument and its effective version (V15.37, V15.49; gaps G.182, G.191); Theorem 15.4.1 and Definition 15.4.2 at a chart point (V15.50–V15.52; gap G.159); the product strong bundle, its sections and auxiliary norm (V15.53–V15.56; gap G.160).

The hypotheses. One: x_0 is not isolated in Proposition 15.1.3 (`( [ ] x ).NeBot`; witness `Polyfold.Linearization15.proposition_15_1_3_hyp_witness`). The readings (a) and (b) of the proof of Theorem 15.1.1 are recorded in Findings, items 1 and 2. Part III wave 2: `hft` (`Polyfold.StrongBundle.IsScFredholmSectionTame`, Part I's tame germ data) in §15.2, in Theorem 15.1.1 (2) and for $\tilde{f}$ on $[0,1]^n \times X$, inherited from Part I's Theorems 3.1.18 and 3.6.8 (witness `Polyfold.SBLocalization.theorem_15_2_2_hyp_witness`); in the global form of Remark 15.2.3 transversality, $d_X = 0$ on the support and the paracompactness of $|\mathrm{supp}|$ (witness `Polyfold.SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp_witness`). Good position (`Polyfold.SBLocalization.IsGoodPositionAt`) has no separate witness. The all-charts definition of general position has one (`Polyfold.SBLocalization.isInGeneralPosition_witness`). Readings: Findings, items 5–9. Part III waves 3–4: `Nonempty A.X.GoodSystem` in Theorem 15.3.7 (necessary; witness `Polyfold.SBLocalization.nonempty_goodSystem_epSEP_zero`); the analytic steps `HasLocalGPFamilies` and `HasParametricGeneralPosition` (witnesses `Polyfold.SBLocalization.hasLocalGPFamilies_epSEP_zero`, `Polyfold.SBLocalization.hasParametricGeneralPosition_epSEP_zero`; all together `Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_hyp`, on zero fibres); tameness in Theorem 15.2.5 under general position (witness `Polyfold.SBLocalization.theorem_15_2_5_generalPosition_hyp_witness`); `hpar` in Remark 15.3.6 (witness `Polyfold.EpGroupoid.paracompactSpace_orbitSupp_trivial_rn`); the cobordism of Corollary 15.4.4 (witness `Polyfold.SBLocalization.OrientedRegularPerturbation.hasOrientedCobordism_refl`). Part III wave 5: `HasParametricGeneralPosition` is proved (`Polyfold.SBLocalization.hasParametricGeneralPosition`); `HasPointwiseParamOnto`, `HasParamOntoOpen` (witnesses `Polyfold.SBLocalization.hasPointwiseParamOnto_epSEP_zero`, `Polyfold.SBLocalization.hasParamOntoOpen_epSEP_zero`; all together `Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_paramOnto`); `hB` in Theorem 15.3.9 (witness `Polyfold.SBLocalization.proposition_14_1_4_hypotheses_rn`); the hypotheses of Corollary 15.3.10 (3) (witnesses `Polyfold.SBLocalization.corollary_15_3_10_3_boundary_witness`, `Polyfold.SBLocalization.corollary_15_3_10_3_glue_witness`, `Polyfold.SBLocalization.admitsScPartitionsOfUnity_cube_rn`; `hparaC` and tameness are derived). Part III waves 6–7: `HasPointwiseParamOnto` and `HasParamOntoOpen` are proved (`Polyfold.SBLocalization.hasPointwiseParamOnto`, `Polyfold.SBLocalization.hasParamOntoOpen`); `Nonempty A.X.GoodSystem` only for the structurable addendum of Theorem 15.3.7; in Theorem 15.3.9 `ChargesBoundary` (necessary) and `relative_local_families_statement` (witness `Polyfold.SBLocalization.relative_local_families_statement_epSEP_zero`, no boundary and zero fibres: weak); in Corollary 15.3.10 (3) `hBC` (gap G.189) and `CubeLiftFace` (gap G.193), without witnesses; in Corollary 15.4.4 `CylinderOrientationStatement` and `hLUCC`, without a non-degenerate witness (the cobordism only `Polyfold.SBLocalization.OrientedRegularPerturbation.hasOrientedCobordism_refl`). The forms taking the formal Theorem 15.3.9 without `ChargesBoundary` are vacuous with boundary (`Polyfold.SBLocalization.not_theorem_15_3_9_statement_w2`, V15.32, V15.34). Readings: Findings, items 10–24.

The setting. Sc-Fredholm section functors are Chapter 12's (V12.1); sc⁺-multisections and local section structures are Chapter 11's; the tangent fibre T_xX and the linearizations are Part I's, read in charts and transported to the intrinsic fibres.

V15.1 (The preamble of §15.1: GrF(x); **`Polyfold.SBLocalization.GrF`, `Polyfold.Linearization15.GrFModel`).** `SBLocalization.GrF x := (T_xX → [] W_x) →  0` for $x \in X_1$, with $W_x = P^{-1}(x)$ and T_xX the project's tangent fibre; the chart version `Linearization15.GrFModel`. *Book:* §15.1, preamble, p. 683 (2017: §15.1). *Hidden hypotheses:* none. *Deviation (restated):* Formal sums are finitely supported functions, defined at all level-1 points. That the summands are sc-Fredholm is not part of the type; it is proved for the summands that occur (V15.9, gap G.145 closed); φ_* and GrF(X) as a category are not built (gap G.144).

V15.2 (Theorem 15.1.1 (2017: 15.1), the linearization $T(f, \Lambda)$ is well defined; `Polyfold.SBLocalization.theorem_15_1_1_intrinsic`, `Polyfold.SBLocalization.linSumIntr`, `Polyfold.SBLocalization.linSumIntr_chart_independent`, `Polyfold.SBLocalization.toIntr_chart_independent`, `Polyfold.SBLocalization.toIntr`, `Polyfold.SBLocalization.sectionSubOn`, `Polyfold.SBLocalization.linearization_eq`, `Polyfold.SBLocalization.nonempty_localLinData`, `Polyfold.SBLocalization.theorem_15_1_1`, `Polyfold.SBLocalization.linSumChart`, `Polyfold.SBLocalization.IsLocalRepOn.count_eq`, `Polyfold.Linearization15.theorem_15_1_1_model`, `Polyfold.Linearization15.wsum_model_eq`, `Polyfold.Linearization15.continuousOn_scDeriv_coe`, `Polyfold.Linearization15.wsum_pair_eq_of_isLocal`, `Polyfold.SBLocalization.subDeriv_isScFredholm`, `Polyfold.SBLocalization.linearization_summand_isScFredholm`, `Polyfold.SBLocalization.theorem_15_1_1_functor`, `Polyfold.SBLocalization.grfPush`, `Polyfold.SBLocalization.conjOp`, `Polyfold.SBLocalization.conjOp_apply`, `Polyfold.SBLocalization.muLin`, `Polyfold.SBLocalization.muLin_apply`, `Polyfold.SBLocalization.mem_levelSet_one_of_src`, `Polyfold.SBLocalization.transSec`, `Polyfold.SBLocalization.transSet`, `Polyfold.SBLocalization.isOpen_transSet`, `Polyfold.SBLocalization.P_transSec`, `Polyfold.SBLocalization.isScPlusSectionOn_transSec`, `Polyfold.SBLocalization.isWFunctor_morIso_Φ`, `Polyfold.SBLocalization.isLocalRepOn_transSec`, `Polyfold.SBLocalization.principal_transportChart_transSec`, `Polyfold.SBLocalization.chartVec_transport`, `Polyfold.SBLocalization.fiberLin_transport_muLin`, `Polyfold.SBLocalization.fiberLin_transport_mem_range`, `Polyfold.SBLocalization.toIntr_transportChart`, `Polyfold.SBLocalization.linSumIntr_transportChart`). `theorem_15_1_1_intrinsic`: for `A.E.IsScFredholmSectionFunctor f`, two local section structures `IsLocalRepOn A Λ O s` , `IsLocalRepOn A Λ O t` on open sets and compatible charts with domains inside them, `linSumIntr c hx f s` = `linSumIntr c hx f t` at every $x \in X_1$ in both charts ($\sum_{f(x)=s_i(x)} \sigma_i \, \Phi_x^{-1} \circ D((\hat{f} - \hat{s}_i) \circ r)(\varphi x) \circ T\varphi$ in GrF(x)). Chart form `theorem_15_1_1`, model form `theorem_15_1_1_model`. The value: `linearization_eq` (property (2) for any data), `nonempty_localLinData`; property (1): `linearization_eq_zero`.

Functor property: `theorem_15_1_1_functor` (`grfPush`, `toIntr_transportChart`, `linSumIntr_transportChart`): `SBLocalization.theorem_15_1_1_functor (hf : A.E.IsScFredholmSectionFunctor f) (hΛ : IsScPlusMultisection A Λ) (  : A.Mor) (hx : levelVal 1 x = src  ) (hy : levelVal 1 y = tgt  ) : linearization f Λ y = grfPush  (mem_levelSet_one_of_src hx) hx hy (linearization f Λ x)`. Definitions: `muLin` is $\mu(\varphi, \cdot)$ as a linear map $W_x \to W_y$. `conjOp` is $\mu(\varphi, \cdot) \circ L \circ T\varphi^{-1}$, with $T\varphi$ = Chapter 7's `EpGroupoid.tangentMorCLE`. `grfPush` is φ_*, defined as `Finsupp.mapDomain  conjOp`. Ingredients: Local section structures are carried along wave 1's local strong bundle isomorphism `E.morIso`   of the sheet of φ (`transSec`, `isScPlusSectionOn_transSec`, `isLocalRepOn_transSec`). The chart c at x is carried to `(E.morIso  ).transportChart c` at y. In the carried chart $T\varphi$ is the identity on chart vectors (`chartVec_transport`), and the fibre part composes with $\mu(\varphi, \cdot)$ to that of c (`fiberLin_transport_muLin`). Hence `toIntr_transportChart`: the intrinsic operator in the carried chart is `conjOp` of the one in c. Hence `linSumIntr_transportChart`: the formal sums correspond under `grfPush`. *Book:* §15.1, Theorem 15.1.1, pp. 684–687 (2017: Theorem 15.1). *Hidden hypotheses:* none. *Deviation (narrower):* The well-definedness, the substance of the proof, is proved in a stronger form: at all level-1 points, for Λ not necessarily a functor, using only sc-smoothness of f. The functor property $T(f, \Lambda)(t\varphi) = \varphi_* T(f, \Lambda)(s\varphi)$ is proved at all level-1 points (`theorem_15_1_1_functor`, with φ_* = `grfPush`, $T\varphi$ = Chapter 7's `tangentMorCLE`; gap G.144 closed), where the text speaks of smooth points. The row stays narrower for one reason only: the sc-Fredholm property of the summands is proved in the chart reading for f with Part I's tame germ data (`linearization_summand_isScFredholm`, V15.9; gap G.145 closed). Readings (Findings, Chapter 15, items 1, 2): (a) the separation argument runs in the level-1 topology, and the "weak topology" of the limit of derivatives is read as pointwise convergence on E_0 (the derivatives $D(g \circ r)$ of an sc^1 map are continuous only from level 1); (b) in the final display, presumably the sum is over $\{i : f(0) = s_i(0)\}$, since $f(0) = s_i(0)$ need not hold for all i; the proof compares the formal sums of pairs $(s_i(r(y)), D((f - s_i) \circ r)(y))$. "Without loss of generality $X = O$" is the chart change `toIntr_chart_independent`. The functor property: $x \in X_1$ forces $\varphi \in \mathbf{X}_1$ (`mem_levelSet_one_of_src`). Local section structures and charts are carried along the local strong bundle isomorphism `E.morIso`   of the sheet of φ (`isLocalRepOn_transSec`, `LocalIso.transportChart`); in the carried chart $T\varphi$ is the identity on chart vectors and the fibre part composes with $\mu(\varphi, \cdot)$ (`toIntr_transportChart`). The functoriality of $\varphi \mapsto \varphi_*$ itself ($(\psi\varphi)_* = \psi_*\varphi_*$) is not formalized and is not needed for the theorem (about 60–100 lines from Chapter 7).

V15.3 (Remark 15.1.2 (2017: 15.1), (1) follows from (2); `Polyfold.SBLocalization.linearization_eq_zero`, `Polyfold.SBLocalization.remark_15_1_2`, `Polyfold.Linearization15.remark_15_1_2_model`, `Polyfold.Linearization15.mem_support_linSumModel_iff`). `SBLocalization.linearization_eq_zero`: if $\Lambda(f(x)) = 0$ then `linearization f Λ x = 0`; chart forms `remark_15_1_2`, `remark_15_1_2_model`; with positive weights an operator lies in the support iff some i with $s_i(y) = f(y)$ produces it (`mem_support_linSumModel_iff`). *Book:* §15.1, Remark 15.1.2, p. 684 (2017: Remark 15.1). *Hidden* *hypotheses:* none. *Deviation (restated):* In Lean $(f - s_i)'$ is defined for every i and the sum is filtered by $s_i(y) = f(y)$; (1) is derived from positivity of the weights and the count formula.

V15.4 (Proposition 15.1.3 (2017: 15.1), separating sequences; `Polyfold.Linearization15.proposition_15_1_3_hyp`, `Polyfold.Linearization15.proposition_15_1_3_hyp_witness`, `Polyfold.Linearization15.not_proposition_15_1_3_of_isolated`, `Polyfold.Linearization15.not_proposition_15_1_3_point`, `Polyfold.Linearization15.IsSeparatedAt`, `Polyfold.Linearization15.IsLocDich`, `Polyfold.Linearization15.exists_isLocDich_mem`, `Polyfold.Linearization15.exists_isSeparatedAt_subseq`). `Linearization15.proposition_15_1_3_hyp`: in a first countable T_1 space Z, for finitely many maps continuous on an open $V \ni x_0$ with values in a Hausdorff space, and x_0 **not isolated** (`( [ ] x ).NeBot`): $x_k \in V$, $x_k \neq x_0$, $x_k \to x_0$, and one partition $e : I \to \mathrm{Fin}\, p$ with `IsSeparatedAt s e (x k)` for all k. At an isolated point no such sequence exists: `not_proposition_15_1_3_of_isolated`, `not_proposition_15_1_3_point` (a one-point model). Witness: `proposition_15_1_3_hyp_witness` ($\mathbb{R}$ at 0). *Book:* §15.1, Proposition 15.1.3, p. 684 (2017: Proposition 15.1). *Hidden hypotheses:* x_0 is not isolated; witness `proposition_15_1_3_hyp_witness`. *Deviation (narrower):* At an isolated point of the local model (for instance a zero-dimensional X) a sequence $x_k \neq 0$, $x_k \to 0$ cannot exist, so the proposition does not hold as we read it there (`not_proposition_15_1_3_of_isolated`); a degenerate case (Findings, Chapter 15, item 3). Theorem 15.1.1 is unaffected: its core lemma handles isolated points through locality. Otherwise the statement is more general: topological, with no sc-structure.

V15.5 (Lemma 15.1.4 (2017: 15.1), finite closed covers of a ball; `Polyfold.Linearization15.lemma_15_1_4`, `Polyfold.Linearization15.exists_nonempty_interior_of_finite_closed_cover`, `Polyfold.Linearization15.interior_biUnion_eq_empty`). `Linearization15.lemma_15_1_4`, in any pseudo-metric space: for finitely many relatively closed A_i with $\bigcup A_i = B_\varepsilon(y)$, some A_i has nonempty interior; topological core `exists_nonempty_interior_of_finite_closed_cover`. *Book:* §15.1, Lemma 15.1.4, p. 685 (2017: Lemma 15.1). *Hidden hypotheses:* none. *Deviation (stronger):* The proof is by induction on the index set instead of moving z step by step; it is applied on the level-1 topology (reading (a) of V15.2).

V15.6 (Lemma 15.1.5 (2017: 15.2), local dichotomy of finitely many sections; `Polyfold.Linearization15.lemma_15_1_5`, `Polyfold.Linearization15.exists_isOpen_dichotomy`, `Polyfold.Linearization15.exists_fin_partition`). `Linearization15.lemma_15_1_5`: in a pseudo-metric space, for finitely many maps continuous on $B_\varepsilon(y)$ into a Hausdorff space, exactly these conclusions, the partition a surjection $e : I \to \mathrm{Fin}\, p$; topological form `exists_isOpen_dichotomy`. *Book:* §15.1, Lemma 15.1.5, p. 685 (2017: Lemma 15.2). *Hidden hypotheses:* none. *Deviation (stronger):* The maps need only be continuous with values in a Hausdorff space; the sections need not be sc$^+$.

V15.7 (Remark 15.1.6, modifying the sequence; `Polyfold.Linearization15.remark_15_1_6`, `Polyfold.Linearization15.IsSeparatedAt.eventually`). `Linearization15.remark_15_1_6` (metric space): from $x'_k \to x_0$ in V, points $x''_k \in V$ with $d(x'_k, x''_k) < 1/(k+1)$, $x''_k \to x_0$, a subsequence and a partition with `IsSeparatedAt` along it. `IsSeparatedAt.eventually`: separation at

z persists near z. *Book:* §15.1, Remark 15.1.6, p. 686 (not in the 2017 text). *Hidden hypotheses:* none. *Deviation (restated):* "Slightly" is $d < 1/(k+1)$ in the first part and "eventually near z" in the second.

V15.8 (Definition 15.1.7 (2017: 15.1), the linearization of $\Lambda \circ f$; `Polyfold.SBLocalization.linearization`, `Polyfold.SBLocalization.LocalLinData`, `Polyfold.SBLocalization.LocalLinData.sum`, `Polyfold.SBLocalization.linearization_eq`, `Polyfold.SBLocalization.nonempty_localLinData`, `Polyfold.SBLocalization.linearizationIn`, `Polyfold.SBLocalization.linearizationIn_eq`). `SBLocalization.linearization f Λ x : GrF x` for $x \in X_1$, the formal sum of some `LocalLinData A Λ x` (a local section structure and a chart around x inside it), 0 if there is none; independent of the choice (`linearization_eq`), data exist (`nonempty_localLinData`), 0 off the support (`linearization_eq_zero`). Chart reading `linearizationIn`. *Book:* §15.1, Definition 15.1.7, p. 687 (2017: Definition 15.1). *Hidden hypotheses:* none. *Deviation (restated):* none.

V15.9 (Theorem 15.1.1 (2) (2017: 15.1): the summands of $T(f,\Lambda)(x)$ are sc-Fredholm; `Polyfold.SBLocalization.subDeriv_isScFredholm`, `Polyfold.SBLocalization.linearization_summand_isScFredholm`). `subDeriv_isScFredholm`: for `A.B.IsScFredholmSectionTame f`, s sc$^+$ on an open $O \supseteq \operatorname{dom}(c)$, $f(x) = s(x)$ and z over $\varphi(x)$, $D((\widehat{f} - \widehat{s}) \circ r)(z)$ is an sc-operator, T_zO and K_z are sc-subspaces, and the restriction $T_zO \to K_z$ is sc-Fredholm, with every level Fredholm of the level-0 index. `linearization_summand_isScFredholm`: for $x \in X_1$ and any local data, every L in the support of $T(f,\Lambda)(x)$ is `toIntr c hx (subDeriv c f s z)` for some i with $f(x) = s_i(x)$, and that chart operator is sc-Fredholm. *Book:* §15.1, preamble and Theorem 15.1.1 (2), pp. 683–684 (2017: §15.1, Theorem 15.1). *Hidden hypotheses:* `hft : A.B.IsScFredholmSectionTame f` (Part I's tame germ data), inherited from Part I's Theorems 3.1.18 and 3.6.8 (rows V3.94, V3.103, V3.105; with the 2021 untamed notion Theorem 3.6.8 does not hold as we read it, V3.130); the text of §15.1–15.2 does not carry it ("a strong bundle over an ep-groupoid, f an sc-Fredholm section functor"). Witness: `theorem_15_2_2_hyp_witness` (over $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, the graph section of 0 with Λ_0). *Deviation (narrower):* T_xX and W_x carry no sc-structure in the project: "sc-Fredholm $T_xX \to W_x$" is read in the chart, through $T_xX \cong T_zO$ and $W_x \cong K_z$, the identifications that define `toIntr`. Proof: Part I's `StrongBundle.linAt_isScFredholm` for the sc-Fredholm section $(f-s)|_O$ of $W|_O$. `GrF` is not cut down to a subtype; the lemma says what is in the support.

V15.10 (Local sc$^+$-sections and $f - s$ on $W|_O$ (prerequisite for §15.2); `Polyfold.SBLocalization.IsScPlusSectionOn.restrictOpen`, `Polyfold.SBLocalization.subSec`, `Polyfold.SBLocalization.isSection_subSec`, `Polyfold.SBLocalization.subRestrict`, `Polyfold.SBLocalization.isScFredholmSectionTame_subRestrict`, `Polyfold.SBLocalization.subDeriv`, `Polyfold.SBLocalization.ChartOnto`, `Polyfold.SBLocalization.surjective_toIntr_subDeriv_iff`, `Polyfold.SBLocalization.isLinSurjAt_subRestrict`, `Polyfold.SBLocalization.chartOnto_of_isLinSurjAt_subRestrict`, `Polyfold.SBLocalization.exists_solutionSet_of_isGoodPosition`, `Polyfold.SBLocalization.exists_solutionSet_interior`). `IsScPlusSectionOn.restrictOpen` (a local sc$^+$-section on open O is a global sc$^+$-

section of $W|_O$); `subSec f s O` ($f + (-1)s$, s extended by 0), `subRestrict` (its restriction to $W|_O$), `isScFredholmSectionTame_subRestrict` (by Part I's `theorem_3_1_18_tame`); `subDeriv c f s z` $= D((\hat{f} - \hat{s}) \circ r)(z)$, `ChartOnto`, `surjective_toIntr_subDeriv_iff`, `isLinSurjAt_subRestrict`, `chartOnto_of_isLinSurjAt_subRestrict`; `exists_solutionSet_of_isGoodPosition`, `exists_solutionSet_interior` (an open $V \ni x$ with $\{f = s\} \cap V$ a smooth submanifold, from `theorem_3_6_8'`). *Book:* not in the book (used by the proofs of Theorems 15.2.2 and 15.2.5, p. 689). *Hidden hypotheses:* none. *Deviation (extra):* Needs `IsScFredholmSectionTame f`, as Part I's Theorem 3.6.8 (V15.12).

V15.11 (Definition 15.2.1 (2017: 15.2), transversality; `Polyfold.SBLocalization.IsOntoAtLevel`, `Polyfold.SBLocalization.IsOntoAt`, `Polyfold.SBLocalization.IsTransversalAt`, `Polyfold.SBLocalization.IsTransversal`, `Polyfold.SBLocalization.isOntoAtLevel_iff`, `Polyfold.SBLocalization.mem_support_linSumIntr_iff`, `Polyfold.SBLocalization.linearization_ne_zero`, `Polyfold.SBLocalization.mem_smoothSet_of_pos`, `Polyfold.SBLocalization.mem_smoothSet_of_eq`). `IsOntoAtLevel f Λ x` ($\forall L \in \operatorname{supp} T(f,\Lambda)(x)$, L surjective, $x \in X_1$), `IsOntoAt f Λ x` (with $\exists x \in X_1$), `IsTransversalAt f Λ x` ($0 < \Lambda(f(x)) \wedge$ `IsOntoAt`), `IsTransversal f Λ`. `isOntoAtLevel_iff` (any local data): onto iff every `toIntr … (subDeriv c f s z)` with $s_i(x) = f(x)$ is surjective; `linearization_ne_zero` ($T(f,\Lambda)(x) \neq 0$ where $\Lambda(f(x)) > 0$); `mem_smoothSet_of_pos` ($\operatorname{supp}(\Lambda \circ f) \subseteq X_\infty$ under the tame germ data). *Book:* §15.2, Definition 15.2.1, p. 688 (2017: Definition 15.2). *Hidden hypotheses:* none. *Deviation (restated):* "Nonzero" is automatic (`linearization_ne_zero`) and "sc-Fredholm" is automatic under tame germ data (`linearization_summand_isScFredholm`, V15.9), so the definition asks only the surjectivity of the support. $T(f,\Lambda)$ is defined on X_1, so `IsOntoAt` carries $x \in X_1$; under tame germ data this is automatic on the support (`mem_smoothSet_of_pos`).

V15.12 (Theorem 15.2.2 (2017: 15.2), interior local perturbation; `Polyfold.SBLocalization.theorem_15_2_2_hyp`, `Polyfold.SBLocalization.exists_branch_interior`, `Polyfold.SBLocalization.isOpen_sep_ne`, `Polyfold.SBLocalization.IsScPlusSectionOn.continuousOn`, `Polyfold.SBLocalization.theorem_15_2_2_hyp_witness`). `theorem_15_2_2_hyp (hf : A.E.IsScFredholmSectionFunctor f) (hft : A.B.IsScFredholmSectionTame f) (hΛ : IsScPlusMultisection A Λ) (hd : d_X(x) = 0) (htr : IsTransversalAt f Λ x)`: a `NaturalRep` R of G_x, a local section structure `IsLocalRepOn A Λ O s` and a chart c with $R.U \subseteq \operatorname{dom}(c) \subseteq O$ such that, with $M_i = \{y \in R.U \mid f(y) = s_i(y)\}$: (1) $d_X = 0$ on $R.U$; (2) `IsOntoAt` at every $y \in R.U$ with $\Lambda(f(y)) > 0$; (a) `IsSmoothSubmanifold` M_i; (b) proper inclusions; (c) $\{y \in R.U \mid \Lambda(f(y)) > 0\} = \bigcup_i M_i$; (d) the count; (e) `linearization f Λ y = linSumIntr c hy f s` on $\operatorname{dom}(c) \cap X_1$; and only branches with $s_i(x) = f(x)$ meet f in $R.U$. *Book:* §15.2, Theorem 15.2.2, pp. 688–689 (2017: Theorem 15.2). *Hidden hypotheses:* `hft : A.B.IsScFredholmSectionTame f` (Part I's tame germ data), inherited from Part I's Theorems 3.1.18 and 3.6.8 (rows V3.94, V3.103, V3.105; with the 2021 untamed notion Theorem 3.6.8 does not hold as we read it, V3.130); the text of §15.1–15.2 does not carry it ("a strong bundle over an ep-groupoid, f an sc-Fredholm section functor"). Witness: `theorem_15_2_2_hyp_witness` (over $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, the graph section of 0 with Λ_0). *Deviation (narrower):* The index set is all branches of the local structure; branches with $s_i(x) \neq f(x)$ give $M_i = \emptyset$ (the last clause is the text's I). (a) "natural smooth manifold structure" is read as `IsSmoothSubmanifold` (Definition 4.2.1) of X, inside $\{d_X = 0\}$. (e) is read in one chart c (chart independent by `linSumIntr_chart_independent`).

V15.13 (Remark 15.2.3 (2017: 15.2), $\Lambda \circ f$ is locally a branched ep⁺-subgroupoid; `Polyfold.SBLocalization.remark_15_2_3_hyp`, `Polyfold.SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp`, `Polyfold.SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp_witness`, `Polyfold.SBLocalization.remark_15_2_3_tame_hyp`, `Polyfold.SBLocalization.remark_15_2_3_isTameBranched_hyp`, `Polyfold.locallyFinite_image_of_isClosedMap`, `Polyfold.EpGroupoid.NaturalRep.translation_ _eq_ _iff`, `Polyfold.EpGroupoid.NaturalRep.translation_isProperSubset`, `Polyfold.EpGroupoid.NaturalRep.translation_hasPropernessProperty`, `Polyfold.EpGroupoid.NaturalRep.isClosedMap_translation_ `, `Polyfold.EpGroupoid.NaturalRep.finite_fiber_translation_ `, `Polyfold.EpGroupoid.NaturalRep.paracompactSpace_orbitSupp_translation`, `Polyfold.EpGroupoid.NaturalRep.nonempty_tameBranching_translation`, `Polyfold.EpGroupoid.NaturalRep.isTameBranched_translation`, `Polyfold.SBLocalization.remark_15_2_3_restricted_hyp`, `Polyfold.SBLocalization.remark_15_2_3_isTameBranched_hyp_witness`, `Polyfold.SBLocalization.remark_15_2_3_restricted_hyp_witness`). `remark_15_2_3_hyp` (hf, hft, hΛ, $d_X(y) = 0$, `IsTransversalAt f Λ y`): `Nonempty (A.X.LocalBranchingStructure (fun z => (Λ (f z) :  )) y)`, Definition 9.1.2 (2)–(4) with $M_j = \{z \in U \mid f(z) = s_j(z)\}$ over the branches through $f(y)$, weights σ_j. `remark_15_2_3_isBranchedEpSubgroupoid_hyp` (hf, hft, hΛ, `IsTransversal f Λ`, $d_X = 0$ on the support, `ParacompactSpace (A.X.orbitSupp …)`): `A.X.IsBranchedEpSubgroupoid (fun z => (Λ (f z) :  ))`.

Tame, and on the ep-groupoid $U(x)$ (Part III wave 5, `Chapter15/P3W5PR_*`): `remark_15_2_3_tame_hyp hf hft hΛ hd htr : Nonempty (A.X.TameBranchingStructure (fun z => (Λ (f z) :  )) y)` (Definition 9.3.6: a natural representation $U(y)$ with the properness property, the M_j of the branches through $f(y)$, each with a tame induced structure). **`remark_15_2_3_restricted_hyp hf hft hΛ hd htr`**: there is a natural representation R of G_x on $U(x)$ with $d_X = 0$ on $U(x)$ such that on the ep-groupoid $U(x)$ (`R.translationEpGroupoid`, $G_x \ltimes U(x)$, Chapter 7) $z \mapsto \Lambda(f(z))$ is a branched ep⁺-subgroupoid (Definition 9.1.2, including (1): the paracompactness of |supp| is proved) and tame. Global form: `remark_15_2_3_isTameBranched_hyp` (`IsBranchedEpSubgroupoid`   `IsTameBranched` on X). Generic part, for any ep-groupoid and functor $\Theta \geq 0$ with closed support: `NaturalRep.isTameBranched_translation`, from `nonempty_tameBranching_translation`, `paracompactSpace_orbitSupp_translation` (Michael's criterion; the orbit map of $G_x \ltimes U(x)$ is closed with finite fibres, `isClosedMap_translation_ `, `finite_fiber_translation_ `, `locallyFinite_image_of_isClosedMap`), `translation_ _eq_ _iff`, `translation_isProperSubset`, `translation_hasPropernessProperty`. Witnesses `remark_15_2_3_isTameBranched_hyp_witness`, `remark_15_2_3_restricted_hyp_witness`. *Book:* §15.2, Remark 15.2.3, p. 689 (2017: Remark 15.2). *Hidden hypotheses:* `hft : A.B.IsScFredholmSectionTame f` (Part I's tame germ data), inherited from Part I's Theorems 3.1.18 and 3.6.8 (rows V3.94, V3.103, V3.105; with the 2021 untamed notion Theorem 3.6.8 does not hold as we read it, V3.130); the text of §15.1–15.2 does not carry it ("a strong bundle over an ep-groupoid, f an sc-Fredholm section functor"). Witness: `theorem_15_2_2_hyp_witness` (over $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, the graph section of 0 with Λ_0); for the global form also `IsTransversal f Λ`, $d_X = 0$ on the support and the paracompactness of |supp| (Definition 9.1.2 (1), new in 2021, which the remark does not address), with the witness `remark_15_2_3_isBranchedEpSubgroupoid_hyp_witness`. *Deviation (narrower):* "Tame" (Definition 9.3.6) and "on the ep-groupoid $U(x)$" are formalized (gaps G.150, G.152 closed in Part III wave 5). "The ep-groupoid $U(x)$" is read as the translation groupoid $G_x \ltimes U(x)$, which Chapter 7 proves to be an ep-groupoid and which Γ maps fully faithfully onto the full subcategory on $U(x)$ (Theorem 7.1.13 (3); Proposition 7.1.19 identifies the orbit spaces). On it no paracompactness hypothesis is needed. "With support $M_{U(x)}$" is the definition of the support.

V15.14 (Definition 15.2.4 (2017: 15.3), good position at a boundary point; `Polyfold.SBLocalization.IsBranchGoodPosition`, `Polyfold.SBLocalization.IsGoodPositionAt`). `IsBranchGoodPosition f s hO x hxO`: Part I's `IsGoodPositionGermTame` for $(f - s)|_O$ on $W|_O$ at x. `IsGoodPositionAt f Λ x`: $0 < \Lambda(f(x))$, `IsOntoAt f Λ x`, and a local structure `IsLocalRepOn A Λ O s` on an open $O \ni x$ with `IsBranchGoodPosition f (s i) hO x hxO` for every i with $s_i(x) = f(x)$. *Book:* §15.2, Definition 15.2.4, p. 689 (2017: Definition 15.3). *Hidden hypotheses:* none. *Deviation (restated):* (1) and (3) are read per branch as Part I's Definition 3.6.6 in a tame chart (tameness near x, a linearization onto with kernel in good position to the partial quadrant). The condition is asked for one local section structure; the text does not say whether it depends on the structure. $x \in \partial X$ is not required.

V15.15 (Theorem 15.2.5 (2017: 15.3), boundary local perturbation; `Polyfold.SBLocalization.theorem_15_2_5_hyp`, `Polyfold.SBLocalization.exists_branch_goodPosition`, `Polyfold.SBLocalization.theorem_15_2_5_hyp_witness`, `Polyfold.SBLocalization.theorem_15_2_5_hyp_applies`, `Polyfold.StrongBundle.linKer_sup_reduced_eq_of_isGeneralPositionAt`, `Polyfold.SBLocalization.IsReducedOntoAtLevel`, `Polyfold.SBLocalization.IsInGeneralPositionAt`, `Polyfold.SBLocalization.isInGeneralPosition_iff_forall_at`, `Polyfold.SBLocalization.isGeneralPositionGerm_branch_at`, `Polyfold.SBLocalization.isGoodPositionAt_of_isInGeneralPositionAt`, `Polyfold.SBLocalization.exists_reduced_preimage_of_isGeneralPositionGerm`, `Polyfold.SBLocalization.surjective_reduced_toIntr_of_isGeneralPositionGerm`, `Polyfold.SBLocalization.isReducedOntoAtLevel_of_isGeneralPositionGerm`, `Polyfold.SBLocalization.theorem_15_2_5_generalPosition_hyp`, `Polyfold.SBLocalization.theorem_15_2_5_generalPosition_hyp_witness`, `Polyfold.SBLocalization.theorem_15_2_5_tame_hyp`, `Polyfold.SBLocalization.theorem_15_2_5_isTameBranched_hyp`, `Polyfold.SBLocalization.theorem_15_2_5_restricted_hyp`). `theorem_15_2_5_hyp (hf) (hft) (hΛ) (hgp : IsGoodPositionAt f Λ x)`, with the output data of `theorem_15_2_2_hyp`: the M_i are `IsSmoothSubmanifold`; (1) proper inclusions; $M_{U(x)} = \bigcup M_i$; (2) `IsGoodPositionAt f Λ y` (same local structure) at every $y \in R.U$ with $\Lambda(f(y)) > 0$; (3) the count; the formula for `linearization`; only branches through $f(x)$ meet.

Witness of all hypotheses: `theorem_15_2_5_hyp_witness` on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ (zero section, Λ_0, good position everywhere), with `theorem_15_2_5_hyp_applies` (the conclusion there, clause (2) included). The closing sentence (general position): `IsReducedOntoAtLevel f Λ x / IsInGeneralPositionAt f Λ x` (pointwise Definition 15.3.5); `isInGeneralPosition_iff_forall_at hf hΛ` (`IsInGeneralPosition` is the pointwise form on the support). Part I step: `StrongBundle.linKer_sup_reduced_eq_of_isGeneralPositionAt` and `isGoodPositionAt_of_isInGeneralPositionAt hf hft hΛ hpos hgp : IsGoodPositionAt f Λ x` (onto on $T_x^R X$ □ good position at x, Remark 3.6.7 kind, applied to each branch $(f - s_i)|_O$). Converse at nearby points: `isReducedOntoAtLevel_of_isGeneralPositionGerm` (general position germs give onto on $T_y^R X$). `theorem_15_2_5_generalPosition_hyp (hX : A.X.IsTame) hf hft hΛ (hpos : 0 < Λ (f x)) (hgp : IsInGeneralPositionAt f Λ x)`: all clauses of `theorem_15_2_5_hyp`, with (2) as ` y  R.U, 0 < Λ (f y) → IsInGeneralPositionAt f Λ y  IsGoodPositionAt f Λ y`. Proof: `theorem_15_2_5_hyp` at `x`, then Part I's `exists_isOpen_isGeneralPositionGerm` on each branch over `W|O` (tame by `isTame_restrictOpen`) and a shrink of the natural representation.

Tame (Part III wave 5): **`theorem_15_2_5_tame_hyp hf hft (hgp : IsGoodPositionAt f Λ x)`** `: Nonempty (A.X.TameBranchingStructure (fun z => (Λ (f z) :  )) x)`: the natural representation $U(x)$ (inside an open set with the properness property, Definition 7.5.8), $M_j = \{z \in U(x) \mid f(z) = s_j(z)\}$ over the branches through $f(x)$, each `IsSmoothSubmanifold` and `HasTameInducedStructure`, proper inclusions, the count. `theorem_15_2_5_restricted_hyp hf hft hΛ hgp`: on the ep-groupoid $G_x \ltimes U(x)$, $\Lambda \circ f$ is a branched ep$^+$-subgroupoid and tame. `theorem_15_2_5_isTameBranched_hyp hf hft hΛ (hgp :  y, 0 < Λ (f y) → IsGoodPositionAt f Λ y) (hpar)`: `IsBranchedEpSubgroupoid  IsTameBranched` on X. The tame solution sets: V15.43. *Book:* §15.2, Theorem 15.2.5, p. 690 (2017: Theorem 15.3). *Hidden hypotheses:* `hft : A.B.IsScFredholmSectionTame f` (Part I's tame germ data), inherited from Part I's Theorems 3.1.18 and 3.6.8 (rows V3.94, V3.103, V3.105; with the 2021 untamed notion Theorem 3.6.8 does not hold as we read it, V3.130); the text of §15.1–15.2 does not carry it ("a strong bundle over an ep-groupoid, f an sc-Fredholm section functor"). Witness: `theorem_15_2_2_hyp_witness` (over $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, the graph section of 0 with Λ_0). *Deviation (narrower):* "Tame submanifolds" and "tame branched" (Definition 9.3.6) are `theorem_15_2_5_tame_hyp`, `theorem_15_2_5_restricted_hyp` and `theorem_15_2_5_isTameBranched_hyp` (gap G.150 closed in Part III wave 5): each M_j is `IsSmoothSubmanifold` (Definition 4.2.1) with a tame induced structure (Definition 2.5.18). The closing variant with $T_y^R X$ (general position) is `theorem_15_2_5_generalPosition_hyp` (gap G.151 closed), with `hX : A.X.IsTame` and `hft`; the hypothesis is at x only, and the conclusion gives general and good position on $U(x)$. Witness: `theorem_15_2_5_generalPosition_hyp_witness`. The witness of `theorem_15_2_5_hyp` with good position at the support points, missing in the wave-2 list, is `theorem_15_2_5_hyp_witness` (good position from `remark_15_3_6_goodPosition_hyp` and `isInGeneralPosition_epSEP_zero`).

V15.16 (Remark 15.2.6 (2017: 15.3), the formalism for orientations; `Polyfold.SBLocalization.theorem_15_2_2_hyp`, `Polyfold.SBLocalization.isOntoAtLevel_iff`). Prose; §15.4 is gap G.159. The local data it needs are in place: the M_i and the surjective $(f - s_i)'(y)$ of `theorem_15_2_2_hyp` (2) and `isOntoAtLevel_iff`. *Book:* §15.2, Remark 15.2.6, p. 690 (2017: Remark 15.3). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V15.17 (Product ep-groupoids (prerequisite for §15.3); `Polyfold.GroupoidStr.prod`, `Polyfold.MPolyfold.IsScSmoothOn.prodMap`, `Polyfold.MPolyfold.IsScDiffeoOn.prodMap`, `Polyfold.MPolyfold.IsLocalScDiffeo.prodMap`, `Polyfold.isProperMap_domRestrict_prod`, `Polyfold.EpGroupoid.prod_isScSmoothCompSheet`, `Polyfold.EpGroupoid.prod`, `Polyfold.EpGroupoid.prodFst`, `Polyfold.EpGroupoid.prodSnd`, `Polyfold.EpGroupoid.isTame_prod`, `Polyfold.EpGroupoid.degIndex_prod`, `Polyfold.EpGroupoid.prod_ _eq_ _iff`). `GroupoidStr.prod` (componentwise structure maps), `EpGroupoid.prod`, the projections `prodFst`, `prodSnd`; `isTame_prod`, `degIndex_prod` ($d_{X\times X'} = d_X + d_{X'}$), `prod_ _eq_ _iff`; M-polyfold helpers `IsScSmoothOn.prodMap`, `IsScDiffeoOn.prodMap`, `IsLocalScDiffeo.prodMap`, `isProperMap_domRestrict_prod`. *Book:* not in the book as a numbered item (used after Definition 15.3.3, in Remark 15.3.4, Corollaries 15.3.10, 15.4.4). *Hidden hypotheses:* none. *Deviation (extra):* Needs `ParacompactSpace` instances on the product object and morphism spaces, always available from `MPolyfold.paracompactSpace_prod`. The étale, proper and sc-smooth-composition axioms are proved; composition is checked in its local sheet form.

V15.18 ($[0,1]^n \times X$, $\widetilde{W}$ and $\tilde{f}$ (proof of Corollary 15.3.10 (2)); `Polyfold.EpGroupoid.cubeProd`, `Polyfold.EpGroupoid.cubeProj`, `Polyfold.EpGroupoid.cubeIncl`, `Polyfold.EpGroupoid.isTame_cubeProd`, `Polyfold.EpGroupoid.degIndex_cubeProd`, `Polyfold.EpGroupoid.degIndex_cubeProd_one_of_degIndex_eq_zero`, `Polyfold.EpGroupoid.cubeOrbitMap`, `Polyfold.EpGroupoid.isCompact_image_ _cubeProd`, `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_cubeProd`, `Polyfold.SEP.cubeSEP`, `Polyfold.SEP.cubeLiftEp`, `Polyfold.SEP.isScSmoothSectionFunctor_cubeLiftEp`, `Polyfold.SEP.isScFredholmSectionFunctor_cubeLiftEp`, `Polyfold.SEP.solutionSet_cubeLiftEp`, `Polyfold.SEP.hasCompactSolutionSetEp_cubeLiftEp`). `EpGroupoid.cubeProd n` $= [0,1]^n \times X$ (the trivial ep-groupoid of Part I's `cubePolyfold n` times X), `cubeProj`, `cubeIncl q`, `cubeOrbitMap`, `isTame_cubeProd`, `degIndex_cubeProd`, `degIndex_cubeProd_one_of_degIndex_eq_zero` (for $n = 1$ and $d_X \equiv 0$: $\partial\widetilde{X} = \{0,1\} \times X$), `admitsBumpFunctionsPointwise_cubeProd`; `SEP.cubeSEP A n` (the pull-back of W along `cubeProj`), `SEP.cubeLiftEp` ($\tilde{f}$), `isScFredholmSectionFunctor_cubeLiftEp`, `solutionSet_cubeLiftEp` ($S_{\tilde{f}} = [0,1]^n \times S_f$), `hasCompactSolutionSetEp_cubeLiftEp`. *Book:* §15.3, Corollary 15.3.10 (2) and its proof, p. 699 (2017: Corollary 15.1). *Hidden hypotheses:* `hft : A.B.IsScFredholmSectionTame f` for $\tilde{f}$ sc-Fredholm, Part I's tame germ data (as G.140 and G.65 (a)), through Part I's `IsScFredholmSectionTame.paramCube`; witness as for V15.12 (`theorem_15_2_2_hyp_witness`). *Deviation (narrower):* $[0,1]^n$ (`Fin n → unitInterval`) is used so that Part I's `paramCube` applies; the text's $[0,1]$ is $n = 1$. The rest (tameness, boundary, bump functions, compact solution set) is as stated.

V15.19 (The paragraph before Definition 15.3.1 (2017: before 15.4), existence of a controlling pair; `Polyfold.SBLocalization.exists_controlsCompactness`). `exists_controlsCompactness (hf : sc-Fredholm functor) (hS : |S_f| compact) (hN : A.E.IsAuxNorm N) :  U, ControlsCompactness A f N U`, from wave 1's `isProperLtInftyFunctor_of_hasCompactSolutionSetEp` (Theorem 12.4.2 (2) □ (3)). *Book:* §15.3, before Definition 15.3.1, p. 691 (2017: before Definition 15.4). *Hidden hypotheses:* none.

Deviation (same): None.

V15.20 (Definition 15.3.1 (2017: 15.4), (N, U) controls compactness; `Polyfold.SBLocalization.ControlsCompactness`, `Polyfold.SBLocalization.ControlsCompactness.mono`). `ControlsCompactness A f N U := A.E.IsAuxNorm N  IsOpen U  A.X.IsSaturated U  solutionSet A.P f  U  IsCompact (closure (A.X. '' {x  U | A.B.AuxLe N (f x) 1}))`; `ControlsCompactness.mono` (a smaller saturated open $U' \supseteq S_f$). *Book:* §15.3, Definition 15.3.1, p. 691 (2017: Definition 15.4). *Hidden hypotheses:* none. *Deviation (restated):* N is defined on $W_{0,1}$ (Definition 12.2.1 = 8.3.5); "$N(f(x)) \leq 1$" is `AuxLe` ($f(x) \in W_{0,1}$ and $N \leq 1$), the text's $+\infty$ extension on p. 691. Paracompactness of $|X|$ is not part of the definition, which makes it more general; theorems that need it carry it.

V15.21 (Remark 15.3.2 (2017: 15.4), bump functions are needed; `Polyfold.EpGroupoid.admitsBumpFunctionsPointwise_cubeProd`). Prose. The related statement for $[0, 1] \times X$ is `admitsBumpFunctionsPointwise_cubeProd` (V15.18). *Book:* §15.3, Remark 15.3.2, p. 692 (2017: Remark 15.4). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V15.22 (Definition 15.3.3 (2017: 15.5), (N, U)-admissible perturbations; `Polyfold.SBLocalization.IsAdmissible`). `IsAdmissible A f N U Λ := IsScPlusMultisection A Λ  ( x, pointwiseNorm N Λ x  1)  {x |  t  Icc (-1: ) 1, 0 < multisectionSmul t Λ (f x)}  U`. *Book:* §15.3, Definition 15.3.3, p. 692 (2017: Definition 15.5). *Hidden hypotheses:* none. *Deviation (same):* The text's standing data (f sc-Fredholm, $|S_f|$ compact, (N, U) controlling compactness) are hypotheses of the theorems that use the predicate. $t\Lambda$ is `multisectionSmul` (Definition 13.2.10) and $N(\Lambda)$ is `pointwiseNorm` (Definition 13.2.3).

V15.23 (The example after Definition 15.3.3 (2017: after 15.5), domain support in U; `Polyfold.SBLocalization.eq_zero_of_not_mem_domSupp`, `Polyfold.SBLocalization.eq_fiberZero_of_multisectionSmul_pos`, `Polyfold.SBLocalization.isAdmissible_of_domSupp_subset`). `isAdmissible_of_domSupp_subset (hf : IsSection) (hSU : S_f  U) (hΛ) (hN) (hdom : domSupp Λ  U)`; helpers `eq_zero_of_not_mem_domSupp`, `eq_fiberZero_of_multisectionSmul_pos` (off the domain support, $(t\Lambda)(w) > 0$ forces $w = 0$, for every real t). *Book:* §15.3, after Definition 15.3.3, p. 692 (2017: after Definition 15.5). *Hidden hypotheses:* none. *Deviation (same):* None.

V15.24 (Definition 15.3.5 (2017: 15.6), general position; `Polyfold.SBLocalization.IsInGeneralPosition`, `Polyfold.SBLocalization.exists_chart_presentation`, `Polyfold.SBLocalization.isInGeneralPosition_iff_exists`, `Polyfold.SBLocalization.reducedLinearization`, `Polyfold.SBLocalization.isInGeneralPosition_iff_reducedLinearization`, `Polyfold.SBLocalization.mem_levelSet_one_of_pos`, `Polyfold.SBLocalization.isInGeneralPosition_witness`). `IsInGeneralPosition A f Λ`: for all `x : A.X.PObj.Level 1` with `0 < Λ (f x)` and all `L  (linearization f Λ x).support`, `L.domRestrict (reducedTangentFiber h  hx T)` is surjective, for every chart φ around x and presentation T of its partial quadrant. `exists_chart_presentation`; `isInGeneralPosition_iff_exists` (one chart and presentation per point suffices); `reducedLinearization f Λ` `x T` and `isInGeneralPosition_iff_reducedLinearization`; `mem_levelSet_one_of_pos` (supp$(\Lambda \circ f) \subseteq X_1$); witness `isInGeneralPosition_witness n` (on `epSEP n 0`, the zero section with Λ_0, support everything). *Book:* §15.3, Definition 15.3.5, p. 693 (2017: Definition 15.6). *Hidden hypotheses:* none. *Deviation (restated):* The point is taken in `Level 1`, where `linearization` lives; `mem_levelSet_one_of_pos` shows nothing is lost. "Every chart" reads $T_x^R X$ chart-wise (Part I's `reducedTangentFiber`); since that space is chart independent, `isInGeneralPosition_iff_exists` gives the "some chart" form; the all-charts form is satisfiable (`isInGeneralPosition_witness`). Tameness of X is not assumed in the definition; it is needed only for $T_x^R X$ to be the right space.

V15.25 (Remark 15.3.6 (2017: 15.6), first sentence: general position gives surjectivity; `Polyfold.SBLocalization.IsInGeneralPosition.surjective`). `IsInGeneralPosition.surjective`: every operator of the support of `linearization f Λ x` is surjective. *Book:* §15.3, Remark 15.3.6, first sentence, p. 693 (2017: Remark 15.6). *Hidden hypotheses:* none. *Deviation (same):* "Onto" is read operator by operator, as in Definition 15.3.5. The rest of the remark (good position at ∂X, tame branched ep⁺-subgroupoid) is gap G.161.

V15.26 (Corollary 15.3.10 (2) (2017: 15.1 (2)), an auxiliary norm on $[0,1] \times X$ controlling compactness; `Polyfold.StrongBundle.isAuxNorm_max_mul`, `Polyfold.StrongBundle.IsCompleteFiberNorm.congr`, `Polyfold.SEP.cubeAuxNorm`, `Polyfold.SEP.isAuxNorm_cubeAuxNorm`, `Polyfold.SEP.isCompact_closure_image_ _cube`, `Polyfold.SEP.controlsCompactness_cube`, `Polyfold.SEP.interpAuxNorm`, `Polyfold.SEP.isAuxNorm_interpAuxNorm`, `Polyfold.SEP.interpAuxNorm_of_cubeTime_eq_zero`, `Polyfold.SEP.interpAuxNorm_of_cubeTime_eq_one`, `Polyfold.SEP.interpSet`, `Polyfold.SEP.mem_interpSet_zero`, `Polyfold.SEP.mem_interpSet_one`, `Polyfold.SEP.corollary_15_3_10_2`). The proof's step: `SEP.cubeAuxNorm n N` ($\widetilde{N}(q,x,h) = N(h)$, Part I's `pullbackAuxNorm`), `isAuxNorm_cubeAuxNorm` (an auxiliary norm of $(\widetilde{W}, \tilde{\mu})$, invariant under $\tilde{\mu}$), `controlsCompactness_cube` (if (N, U) controls the compactness of f, then $(\widetilde{N}, [0,1]^n \times U)$ controls that of $\tilde{f}$). The statement: `interpAuxNorm A N N` $=$ $\max(\alpha(t)N^0, \beta(t)N^1)$ with $\alpha = \min(1, 3(1-t))$, $\beta = \min(1, 3t)$, `interpSet A U U` $= ([0, 2/3) \times U^0) \cup ((1/3, 1] \times U^1)$, and `corollary_15_3_10_2 (hf) (h : ControlsCompactness A f N U ) (h : … N U ) : ControlsCompactness (A.cubeSEP 1) (cubeLiftEp f hf) (interpAuxNorm A N N ) (interpSet A U U )`, with the restrictions `interpAuxNorm_of_cubeTime_eq_zero`, `_eq_one`, `mem_interpSet_zero`, `mem_interpSet_one`. *Book:* §15.3, Corollary 15.3.10 (2), p. 699, and the first step of its proof, pp. 699–700 (2017: Corollary 15.1 (2)). *Hidden hypotheses:* none. *Deviation (same):* The text does not give $\widetilde{N}$; its proof uses a single N; both forms are proved. The Λ^i play no role in (2). That $\tilde{f}$ is sc-Fredholm carries Part I's tame germ data (V15.18). Parts (1) and (3) of the corollary are gap G.162.

V15.27 (Remark 15.3.6 (2017: 15.6), second sentence: general position gives good position on ∂X; `Polyfold.SBLocalization.exists_reduced_preimage`, `Polyfold.SBLocalization.isGeneralPositionGerm_subRestrict`, `Polyfold.SBLocalization.isGeneralPositionGerm_branch`, `Polyfold.SBLocalization.remark_15_3_6_goodPosition_hyp`). `remark_15_3_6_` `goodPosition_hyp` (hf : `IsScFredholmSectionFunctor f`) (hft : `A.B.IsScFredholmSectionTame f`) (hΛ : `IsScPlusMultisection A Λ`) (hgp : `IsInGeneralPosition A f Λ`) ($\Lambda(f(x)) > 0$): (a) `IsGoodPositionAt f Λ x` (D15.2.4), and (b) for every local sc⁺-section structure `IsLocalRepOn A Λ O s` on an open $O \ni x$ and every i with $s_i(x) = f(x)$, the section $(f - s_i)|_O$ of $W|_O$ is in general position at x in Part I's sense (`IsGeneralPositionGerm`, D5.3.9). The chart step `isGeneralPositionGerm_subRestrict`: in a tame chart c of W centred at x with domain in O, if the intrinsic $(f - s)'(x)$ = `toIntr c hx (subDeriv c f s z)` is onto W_x on $T_x^R X$ (`reducedTangentFiber`, D2.5.8), then $(f - s)|_O$ is in general position at x. Proof: in the transported chart of $W|_O$ the operator is $D((\widehat{f} - \widehat{s}) \circ r)(0)$ (`scDeriv_restrictTo_subRestrict`); onto on $T_0^R O$ gives $\ker \mathrm{Lin}(0) + T_0^R O = T_0 O$; Part I's chart criterion `isGeneralPositionGerm_of_sup_eq`. `isGeneralPositionGerm_branch` (per branch), `exists_reduced_preimage` (the chart reading of "onto on $T_x^R X$"). Good position from general position: Part I's Lemma 5.26 (`IsGeneralPositionGerm.isGoodPositionGermTame`). *Book:* §15.3, Remark 15.3.6, second sentence, p. 693 (2017: Remark 15.6). *Hidden hypotheses:* none. *Deviation (narrower):* Inherited hypothesis `hft` (tame germ data), as in the §15.2 rows and G.140: it supplies the tame chart centred at x, and D15.2.4 is read with it. "Good position" is `IsGoodPositionAt`; "even in general position" is read per branch as Part I's D5.3.9 for $(f - s_i)|_O$, the M-polyfold notion that D15.3.5 extends (the text: "a natural extension of the Definition 5.3.9"). The statement holds at every point of the support, not only on ∂X (stronger in that respect).

V15.28 (Remark 15.3.6 (2017: 15.6), last sentence: $\Lambda \circ f$ is a branched ep⁺-subgroupoid; `Polyfold.SBLocalization.remark_15_3_6_localBranching_hyp`, `Polyfold.SBLocalization.isFunctorial_comp`, `Polyfold.SBLocalization.remark_15_3_6_isBranchedEpSubgroupoid_hyp`, `Polyfold.SBLocalization.remark_15_3_6_isTameBranched_hyp`). `remark_15_3_6_localBranching_hyp` (hf, hft, hΛ, hgp, $\Lambda(f(y)) > 0$) : `Nonempty (A.X.LocalBranchingStructure (fun z => (Λ (f z) :  )) y)` (D9.1.2 (2)–(4)), with $M_j = \{z \in U(y) \mid f(z) = s_j(z)\}$ over the branches through $f(y)$, from `theorem_15_2_5_hyp` at the point of good position given by `remark_15_3_6_goodPosition_hyp` (V15.27). `remark_15_3_6_isBranchedEpSubgroupoid_hyp` (hf, hft, hΛ, hgp, hpar : `ParacompactSpace (A.X.orbitSupp …)`) : `A.X.IsBranchedEpSubgroupoid (fun z => (Λ (f z) :  ))`. Helper `isFunctorial_comp`.

Tame (Part III wave 5): `remark_15_3_6_isTameBranched_hyp hf hft hΛ (hgp : IsInGeneralPosition A f Λ) (hpar) : A.X.IsBranchedEpSubgroupoid (Λ  f)  A.X.IsTameBranched (Λ  f)` (good position at every support point by `remark_15_3_6_goodPosition_hyp`, then `theorem_15_2_5_isTameBranched_hyp`). *Book:* §15.3, Remark 15.3.6, last sentence, p. 693 (2017: Remark 15.6). *Hidden hypotheses:* none. *Deviation (narrower):* `hft` inherited (V15.27). `hpar` is D9.1.2 (1) (new in 2021), not addressed by the remark; it is automatic when |supp| is compact, which is the situation of T15.3.7 (V15.29, V15.30). Unlike interior form `remark_15_2_3_isBranchedEpSubgroupoid_hyp`, no $d_X = 0$ on the support is needed: boundary points are covered through good position. "Tame" (D9.3.6) is `remark_15_3_6_isTameBranched_hyp` (gap G.150 closed in Part III wave 5).

V15.29 (Theorem 15.3.7 (2017: 15.4), general position: existence of the perturbation; Polyfold.SBLocalization.theorem_15_3_7, Polyfold.SBLocalization.theorem_15_3_7_existence, Polyfold.SBLocalization.theorem_15_3_7_existence_statement, Polyfold.SBLocalization.ScPlusGPFamily, Polyfold.SBLocalization.ScPlusGPFamily.mono, Polyfold.SBLocalization.ScPlusGPFamily.zero, Polyfold.SBLocalization.ScPlusGPFamily.add, Polyfold.SBLocalization.ScPlusGPFamily.saturate, Polyfold.SBLocalization.nonempty_scPlusGPFamily_biUnion, Polyfold.SBLocalization.exists_scPlusGPFamily_of_compact, Polyfold.SBLocalization.SeedData.scPlusGPFamily, Polyfold.SBLocalization.exists_scPlusGPFamily_nhds, Polyfold.SBLocalization.theorem_15_3_7_of_goodSystem, Polyfold.SBLocalization.theorem_15_3_7_of_paracompact, Polyfold.SBLocalization.hasPointwiseParamOnto, Polyfold.SBLocalization.hasParamOntoOpen, Polyfold.SBLocalization.exists_tameChart_centred, Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_goodSystem, Polyfold.SBLocalization.IsParamOntoVec, Polyfold.SBLocalization.isParamOntoAt_iff_vec, Polyfold.SBLocalization.muLin_surjective, Polyfold.SBLocalization.tangentMorCLE_mem_reducedTangentFiber_transport, Polyfold.SBLocalization.IsParamOntoVec.of_mor, Polyfold.SBLocalization.transSec_eq_muLin, Polyfold.SBLocalization.paramSec_transSec_eqOn, Polyfold.SBLocalization.transSec_of_eq_tgt, Polyfold.SBLocalization.IsLocalRepOn.congr_eqOn, Polyfold.SBLocalization.exists_linearRep_of_mor, Polyfold.SBLocalization.IsParamOntoAt.of_mor, Polyfold.SBLocalization.IsParamOntoAt.of_mor_inv, Polyfold.SBLocalization.mem_solutionSet_tgt, Polyfold.SBLocalization.src_mem_levelSet_one, Polyfold.SBLocalization.GPFamily.saturate, Polyfold.SBLocalization.hasLocalGPFamilies_of_nhds, Polyfold.SBLocalization.paramSec_eq_paramSlice, Polyfold.SBLocalization.isScPlusSection_paramSec, Polyfold.SBLocalization.sectionSupport_paramSec_subset, Polyfold.SBLocalization.paramSec_fiberZero, Polyfold.SBLocalization.natTerm_paramSec, Polyfold.SBLocalization.SeedData, Polyfold.SBLocalization.SeedData.atom, Polyfold.SBLocalization.SeedData.Λ, Polyfold.SBLocalization.SeedData.atom_R, Polyfold.SBLocalization.SeedData.atom_V, Polyfold.SBLocalization.SeedData.isAtomic_Λ, Polyfold.SBLocalization.SeedData.isScPlusMultisection_Λ, Polyfold.SBLocalization.SeedData.branch, Polyfold.SBLocalization.SeedData.isSection_branch, Polyfold.SBLocalization.SeedData.isLocalRepOn_branch, Polyfold.SBLocalization.SeedData.isLocalRepOn_zero, Polyfold.SBLocalization.SeedData.isLinearFamily, Polyfold.SBLocalization.SeedData.isStructurable_Λ, Polyfold.SBLocalization.SeedData.domSupp_Λ_subset, Polyfold.SBLocalization.SeedData.norm_le, Polyfold.SBLocalization.SeedData.isParamOntoAt, Polyfold.SBLocalization.SeedData.gpFamily, Polyfold.SBLocalization.HasLocalParamSeeds, Polyfold.SBLocalization.hasLocalGPFamilies_of_seeds, Polyfold.SBLocalization.theorem_15_3_7_hyp_seeds, Polyfold.SBLocalization.theorem_15_3_7_of_seeds, Polyfold.SBLocalization.hasLocalParamSeeds_epSEP_zero, Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_seeds, Polyfold.SBLocalization.HasPointwiseParamOnto, Polyfold.

SBLocalization.HasParamOntoOpen, Polyfold.SBLocalization.hasLocalParamSeeds_of, Polyfold.SBLocalization.theorem_15_3_7_of_pointwise, Polyfold.SBLocalization.isParamOntoVec_epSEP_zero, Polyfold.SBLocalization.hasPointwiseParamOnto_epSEP_zero, Polyfold.SBLocalization.hasParamOntoOpen_epSEP_zero, Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_pointwise, Polyfold.SBLocalization.negRestrict, Polyfold.SBLocalization.isSection_negRestrict, Polyfold.SBLocalization.isScPlusSection_negRestrict, Polyfold.SBLocalization.pbLin, Polyfold.SBLocalization.pbLin_restrictSection, Polyfold.SBLocalization.pbLin_injective, Polyfold.SBLocalization.paramSlice_branch_eq, Polyfold.SBLocalization.fiberLin_toIntr_subDeriv, Polyfold.SBLocalization.principal_eq_fiberLin, Polyfold.SBLocalization.principal_restrictTo_negRestrict, Polyfold.SBLocalization.isParamGP_restrictTo_of_onto, Polyfold.SBLocalization.isScPlusSectionOn_of_paramSec, Polyfold.SBLocalization.isGeneralPositionGerm_branchParam, Polyfold.SBLocalization.exists_local_paramGP, Polyfold.SBLocalization.auxLe_of_pos, Polyfold.SBLocalization.exists_pos_auxLe_subset, Polyfold.SBLocalization.hasParametricGeneralPosition, Polyfold.SBLocalization.theorem_15_3_7_hyp_local, Polyfold.SBLocalization.theorem_15_3_7_of_local, Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_local, Polyfold.SBLocalization.theorem_15_3_7_of_paramOnto, Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_paramOnto, Polyfold.SBLocalization.HasLocalGPFamilies, Polyfold.SBLocalization.HasParametricGeneralPosition, Polyfold.SBLocalization.nonempty_gpFamily_biUnion, Polyfold.SBLocalization.exists_gpFamily_of_compact, Polyfold.SBLocalization.exists_norm_lt_of_ae, Polyfold.SBLocalization.sum_abs_lt_one, Polyfold.SBLocalization.isTransversal_of_isInGeneralPosition, Polyfold.SBLocalization.theorem_15_3_7_hyp, Polyfold.SBLocalization.theorem_15_3_7_of_hyp, Polyfold.SBLocalization.nonempty_goodSystem_epSEP_zero, Polyfold.SBLocalization.fiber_eq_epSEP_zero, Polyfold.SBLocalization.hasParametricGeneralPosition_epSEP_zero, Polyfold.SBLocalization.hasLocalGPFamilies_epSEP_zero, Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_hyp, Polyfold.SBLocalization.paramSec, Polyfold.SBLocalization.paramSec_apply, Polyfold.SBLocalization.isSection_paramSec, Polyfold.SBLocalization.paramFib, Polyfold.SBLocalization.paramSec_fin_zero, Polyfold.SBLocalization.isSection_append, Polyfold.SBLocalization.parL, Polyfold.SBLocalization.parR, Polyfold.SBLocalization.paramSec_append, Polyfold.SBLocalization.normW_paramSec_le, Polyfold.SBLocalization.pointwiseNorm_le_of_isLocalRepOn, Polyfold.SBLocalization.pointwiseNorm_le_of_linear, Polyfold.SBLocalization.IsLinearFamily, Polyfold.SBLocalization.IsParamOntoAt, Polyfold.SBLocalization.famAdd, Polyfold.SBLocalization.isLocalRepOn_famAdd, Polyfold.SBLocalization.IsLinearFamily.famAdd, Polyfold.SBLocalization.IsParamOntoAt.famAdd_left, Polyfold.SBLocalization.IsParamOntoAt.famAdd_right, Polyfold.SBLocalization.exists_of_multisectionAdd_pos, Polyfold.SBLocalization.normW_add_le, Polyfold.SBLocalization.pointwiseNorm_multisectionAdd_le, Polyfold.

`SBLocalization.domSupp_multisectionAdd_subset`, `Polyfold.SBLocalization.GPFamily`, `Polyfold.SBLocalization.GPFamily.mono`, `Polyfold.SBLocalization.isLocalRepOn_zeroMultisection`, `Polyfold.SBLocalization.isStructurable_zeroMultisection_of_goodSystem`, `Polyfold.SBLocalization.GPFamily.zero`, `Polyfold.SBLocalization.GPFamily.add`, `Polyfold.SBLocalization.theorem_15_3_7_statement`, `Polyfold.SBLocalization.theorem_15_3_7_statement_epSEP_zero`, `Polyfold.SBLocalization.theorem_15_3_7_of_isInGeneralPosition_zero`, `Polyfold.SBLocalization.pointwiseNorm_zeroMultisection`, `Polyfold.SBLocalization.domSupp_zeroMultisection`, `Polyfold.SBLocalization.isInGeneralPosition_epSEP_zero`). Part III wave 6 (`Chapter15/P3W6G6_*`): **`theorem_15_3_7_existence : theorem_15_3_7_existence_statement A`**: under the text's hypotheses (X tame, `AdmitsBumpFunctionsPointwise`, f an sc-Fredholm section functor with the tame germ data `IsScFredholmSectionTame` inherited from §15.2, compact $|S_f|$, `ControlsCompactness A f N U`, h continuous, constant along morphisms, $0 < h \leq 1$) there is an sc$^+$-multisection Λ with $N(\Lambda)(x) < h(x)$, `domSupp Λ  U`, `IsTransversal f Λ` and `IsInGeneralPosition A f Λ`; no good system and no paracompactness of $|X|$. **`theorem_15_3_7_of_goodSystem (h : Nonempty A.X.GoodSystem) : theorem_15_3_7_statement A`** (the structurable form) and `theorem_15_3_7_of_paracompact` (Theorem 13.1.8 gives the good system); **`theorem_15_3_7`**: the existence and the addendum together. The two Props of the earlier form are proved: `hasPointwiseParamOnto` (Part I's `exists_sections_isParamGP` in a tame chart centred at x for the auxiliary norm $(2/\varepsilon)N$, then the bridge) and `hasParamOntoOpen` (Part I's openness of general position for $F(t, y) = f(y) + \sum_j t_j s_j(y)$ on $\mathbb{R}^\ell \times X$, and back through the bridge at y), through the bridge `isParamOntoVec_iff_isParamGP` (V15.44). Without a good system the local families are sc$^+$ (`ScPlusGPFamily`, sums by Definition 13.2.9, `exists_scPlusGPFamily_of_compact`, `SeedData.scPlusGPFamily`).

The earlier forms: `theorem_15_3_7_hyp` (h□ : `Nonempty A.X.GoodSystem`) (hf : `IsScFredholmSectionFunctor f`) (hS : `HasCompactSolutionSetEp f`) (hcc : `ControlsCompactness A f N U`) ($0 < h \leq 1$) (hloc : `HasLocalGPFamilies A f N U h`) (hsard : `HasParametricGeneralPosition A f N U`) : $\exists \Lambda$, `IsStructurable A Λ ∧` $\forall x$, $N(\Lambda)(x) < h(x)$ `∧ domSupp Λ  U ∧ IsTransversal f Λ ∧ IsInGeneralPosition A f Λ`. `theorem_15_3_7_of_hyp` (h□) (H : under the hypotheses of the theorem — tame, bump functions, f sc-Fredholm with tame germ data, compact $|S_f|$, (N, U) controlling, h a continuous functor into $(0, 1]$ — both `hloc` and `hsard` hold) : `theorem_15_3_7_statement A`. The two hypotheses are the two analytic steps of the proof, stated as Props: `HasLocalGPFamilies A f N U h` (the "Local constructions at $x \in S_f$", interior and boundary): every $x \in S_f$ lies in a saturated open Q carrying a `GPFamily A f N U h Q`: a linear family $t \mapsto \Lambda_t$, $t \in \mathbb{R}^\ell$, of structurable sc$^+$-multisections (`IsLinearFamily`: around every point local representations $(\sum_j t_j s_{ij}, \sigma_i)$ with the same domain, sections and weights for all t), $N(\Lambda_t) \leq |t|_1 h$, dom-supp $\Lambda_t \subseteq U$, and (15.1) at the points of $S_f \cap Q$. `HasParametricGeneralPosition A f N U` ("Achieving global transversality"): for every linear family of sc$^+$-multisections with $N(\Lambda_t) \leq |t|_1$, dom-supp $\Lambda_t \subseteq U$ and (15.1) at every point of S_f, there is $\varepsilon > 0$ with: for almost every t with $\|t\| < \varepsilon$, (f, Λ_t) is in general position. (15.1) is `IsParamOntoAt f Λ x`: in a linear representation around x, for every branch i, every operator

L of $T_{(f,\Lambda_0)}(x)$ and every chart presentation of $T^R_x X$, the map $(v,\tau) \mapsto Lv + \sum_j \tau_j s_{ij}(x)$ is onto W_x from $T^R_x X \times \mathbb{R}^\ell$. Proved (steps (b), (c), (d) and the choice of t in (h) of G.162): sums of families `famAdd`, `GPFamily.add` (structurable by Proposition 13.3.12, linear `IsLinearFamily.famAdd` via `isLocalRepOn_famAdd`/`paramSec_append`, (15.1) `IsParamOntoAt.famAdd_left/right`, norm by `pointwiseNorm_multisectionAdd_le`, domain support by `domSupp_multisectionAdd_subset`), `GPFamily.zero`, `GPFamily.mono`, `nonempty_gpFamily_biUnion`; the finite cover of $|S_f|$ (`exists_gpFamily_of_compact`, with Proposition 7.1.8); the choice of a small t off the null set (`exists_norm_lt_of_ae`: balls have positive Lebesgue measure) with $|t|_1 < 1$ (`sum_abs_lt_one`), so $N(\Lambda_t) \leq |t|_1 h < h$; (3) from (4) (`isTransversal_of_isInGeneralPosition`). Tools for instantiating `hloc` later: `normW_paramSec_le` ($N(\sum_j t_j s_j(y)) \leq \sum_j |t_j| N(s_j(y))$), `pointwiseNorm_le_of_isLocalRepOn`, `pointwiseNorm_le_of_linear` ($N(\Lambda_t)(x) \leq |t|_1 b$ from $N(s_{ij}(x)) \leq b$), `normW_add_le`, `exists_of_multisectionAdd_pos`.

The statement: `theorem_15_3_7_statement A : Prop` (a definition, not proved): `A.X.IsTame → A.X.AdmitsBumpFunctionsPointwise → f, IsScFredholmSectionFunctor f → IsScFredholmSectionTame f → HasCompactSolutionSetEp f → N U, ControlsCompactness A f N U → h, Continuous h → (, h (src) = h (tgt)) → (x, 0 < h x h x 1) → Λ, IsStructurable A Λ (x, pointwiseNorm N Λ x < h x) domSupp Λ U IsTransversal f Λ IsInGeneralPosition A f Λ`. `theorem_15_3_7_statement_epSEP_zero n`: the statement holds on the trivial example $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ (every pair is in general position there; the trivial multisection has all conclusions). `theorem_15_3_7_of_isInGeneralPosition_zero` (paracompact $|X|$, hf, aux norm, $h > 0$, $(f,\Lambda_0)$ in general position): $\Lambda_0$ has all conclusions. Helpers `pointwiseNorm_zeroMultisection`, `domSupp_zeroMultisection`, `isInGeneralPosition_epSEP_zero`.

Part III wave 5 (`Chapter15/P3W5GA_*`, `P3W5GB_*`): **`theorem_15_3_7_of_paramOnto (h : Nonempty A.X.GoodSystem) (H : under the hypotheses of the theorem, HasPointwiseParamOnto A f N  HasParamOntoOpen A f) : theorem_15_3_7_statement A`**, the sharpest form. The two Props are on the object M-polyfold, with no groupoid content: `IsParamOntoVec f x v` is (15.1) at a level-1 point with vectors $v_j \in W_x$ (every operator of $T_{(f,\Lambda_0)}(x)$, every chart presentation of $T^R_x X$; `isParamOntoAt_iff_vec`); `HasPointwiseParamOnto A f N`: at $x \in S_f$ and for $\varepsilon > 0$ there are smooth $w_j \in W_x$ with $N(w_j) < \varepsilon$ and `IsParamOntoVec f x w`; `HasParamOntoOpen A f`: if s_j are sc⁺ and (15.1) holds with $s_j(x)$ at $x \in S_f$, it holds with $s_j(y)$ at every $y \in S_f$ near x. Witnesses `hasPointwiseParamOnto_epSEP_zero`, `hasParamOntoOpen_epSEP_zero`, jointly `theorem_15_3_7_statement_epSEP_zero_of_paramOnto`.

`hsard` is proved: **`hasParametricGeneralPosition hX hf hft hS hcc : HasParametricGeneralPosition A f N U`**, with exactly the hypotheses of the theorem. Steps: Part I's local Sard step without compactness (`StrongBundle.exists_null_isGeneralPositionGerm_paramSlice_near`, V15.39); (15.1) for one branch, read in a tame chart, gives Part I's (i)–(ii) for $F(t,y) = (f-0)|O + \sum_j t_j(-s_{ij})|O$ (`isParamGP_restrictTo_of_onto`), hence general position at $(0,x)$ (`isGeneralPositionGerm_branchParam`), and the slice at t is the branch of Λ_t (`paramSlice_branch_eq`); the local step `exists_local_paramGP`; invariance along morphisms (`isInGeneralPositionAt_of_mor`, V15.39); (15.2) as `exists_pos_auxLe_subset`; the finite cover of $|S_f|$ and the union of the null sets. `theorem_15_3_7_hyp_local`, `theorem_15_3_7_of_local`: the theorem from `HasLocalGPFamilies` alone.

`hloc` is proved from the two Props: `hasLocalGPFamilies_of_seeds` (from `HasLocalParamSeeds A f N`: at $x \in S_f$, for open $O \ni x$ and $\varepsilon > 0$, sc$^+$-sections s_j supported in O with $N(s_j) \leq \varepsilon$ and (15.1) with $s_j(y)$ on S_f near x), and `hasLocalParamSeeds_of` (from the two Props, the bump functions through `EpGroupoid.theorem_12_3_2_1`, Part I's Lemma 5.3.3 and the smoothness of S_f). The groupoid part: `IsParamOntoVec.of_mor` ((15.1) is carried along a morphism by μ_φ, through the functor property of the linearization and the transported chart), `exists_linearRep_of_mor`, `IsParamOntoAt.of_mor`, `mem_solutionSet_tgt` (S_f saturated), `GPFamily.saturate`, `hasLocalGPFamilies_of_nhds`; `SeedData` (a natural representation with the properness property and seeds supported in $V(x)$) with `SeedData.Λ t` the atomic multisection of $\sum_j t_j s_j$, `isStructurable_Λ`, `isLinearFamily`, `norm_le` ($N(\Lambda_t) \leq |t|_1 c$), `domSupp_Λ_subset`, `isParamOntoAt`, `gpFamily`. Forms: `theorem_15_3_7_hyp_seeds`, `theorem_15_3_7_of_seeds`, `theorem_15_3_7_of_pointwise`. *Book:* §15.3, Theorem 15.3.7, pp. 693–698 (2017: Theorem 15.4). *Hidden hypotheses:* none. *Deviation (explicit):* `theorem_15_3_7_existence` has the text's hypotheses. The only hypothesis not printed in the theorem is the tame germ data `hft`, inherited from §15.2 (the Part I hypothesis that the rows of §15.2 carry; gap G.140), through which the proof uses Theorem 15.2.5. Neither a good system nor paracompactness of $|X|$ is used, as Remark 15.3.8 says (V15.31). The addendum "Λ can be taken structurable": structurable (Definition 13.3.6) contains a good system, so `theorem_15_3_7_of_goodSystem` assumes `Nonempty A.X.GoodSystem`; the conclusion implies it, so the hypothesis is necessary for this part. Theorem 13.1.8 gives it for paracompact $|X|$ (`theorem_15_3_7_of_paracompact`); witness `nonempty_goodSystem_epSEP_zero` and `theorem_15_3_7_statement_epSEP_zero_of_goodSystem` on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$. The local constructions use Part I's (i)–(ii) through `isParamOntoVec_iff_isParamGP` ((15.1) with $T_x^R X$ is equivalent to Part I's (i)–(ii) with the full tangent space, read in a tame chart centred at the zero x); the parametric transversality is `hasParametricGeneralPosition`. The earlier forms stay as intermediate steps: `theorem_15_3_7_hyp`, `theorem_15_3_7_of_hyp`, `theorem_15_3_7_of_pointwise`, `theorem_15_3_7_of_local`, and `theorem_15_3_7_of_paramOnto` (a special case of `theorem_15_3_7_of_goodSystem`). Gap G.162 is closed for Theorem 15.3.7. Readings fixed in the statement `theorem_15_3_7_statement`: (3) is `IsTransversal` (Definition 15.2.1); (4) is `IsInGeneralPosition` (Definition 15.3.5), as the proof's first sentence says; "structurable" is Definition 13.3.6; U is open and saturated (`ControlsCompactness`), where the text says "a saturated subset". Readings of the proof: (15.1) with $f'(x)$ read as the operator of $T_{(f,\Lambda_0)}(x)$ (Definition 15.1.7) and the t-derivative of $f - \sum_j t_j s_{ij}$ as $\tau \mapsto -\sum_j \tau_j s_{ij}(x)$ (sign immaterial); the proof's ε_0 is $1/(\ell+1)$ in the sup norm of $\mathbb{R}^\ell$; "a set of measure zero … for t in its complement" is "for almost every t"; Sard "on the intersections of faces" is Part I's Theorem 4.3.1 (local form) per branch on $\mathbb{R}^\ell \times O$; the passage from $V(x)$ to its saturation "by equivariance" is `isInGeneralPositionAt_of_mor`. The supporting inequalities the proof uses silently are proved: `pointwiseNorm_multisectionAdd_le` ($N(\Lambda \oplus \Lambda') \leq N(\Lambda) + N(\Lambda')$) and `domSupp_multisectionAdd_subset`.

V15.30 (Theorem 15.3.7 (2017: 15.4), "in particular": $\Lambda \circ f$ is compact and branched; `Polyfold.SBLocalization.pos_iff_exists_eq`, `Polyfold.SBLocalization.isClosed_supp_comp`, `Polyfold.SBLocalization.isClosed_orbitSupp_comp`, `Polyfold.SBLocalization.supp_comp_subset_controlled`, `Polyfold.SBLocalization.isCompact_orbitSupp_comp`, `Polyfold.SBLocalization.theorem_15_3_7_in_particular_hyp`, `Polyfold.SBLocalization.theorem_15_3_7_in_particular_tame_hyp`). `theorem_15_3_7_in_particular_hyp` (hf) (hft) (hcc : `ControlsCompactness A f N U`) (hΛ) (hN1 : $N(\Lambda) \leq 1$) (hdom : `domSupp Λ  U`) (hgp : `IsInGeneralPosition A f Λ`) : `IsBranchedEpSubgroupoid (Λ  f) IsCompactBranched (Λ  f)`. Ingredients: `isClosed_supp_comp`: $\operatorname{supp}(\Lambda \circ f)$ is closed in X (locally a finite union of $\{f = s_i\}$; helper `pos_iff_exists_eq`); `isClosed_orbitSupp_comp`: $|\operatorname{supp}|$ closed in $|X|$ (saturated, orbit map open, P7.1.8); `supp_comp_subset_controlled`: with $N(\Lambda) \leq 1$ and dom-supp in $U \supseteq S_f$, the support lies in $\{x \in U \mid N(f(x)) \leq 1\}$; `isCompact_orbitSupp_comp`: so $|\operatorname{supp}|$ is compact; compactness gives the paracompactness of D9.1.2 (1), and the local branching comes from R15.3.6.

Tame (Part III wave 5): `theorem_15_3_7_in_particular_tame_hyp hf hft hcc hΛ hN1 hdom hgp : IsBranchedEpSubgroupoid IsCompactBranched IsTameBranched` for $\Lambda \circ f$ (`theorem_15_3_7_in_particular_hyp` with `theorem_15_2_5_isTameBranched_hyp`; paracompactness from compactness). *Book:* §15.3, Theorem 15.3.7, last sentence, p. 694 (2017: Theorem 15.4). *Hidden hypotheses:* none. *Deviation (narrower):* `hft` inherited; "tame" (D9.3.6) is `theorem_15_3_7_in_particular_tame_hyp` (gap G.150 closed in Part III wave 5). Only (1) (with $h \leq 1$), (2) and (4) are used; (3) follows from (4) (`IsInGeneralPosition.surjective`).

V15.31 (Remark 15.3.8 (new in 2021), no paracompactness in Theorem 15.3.7; `Polyfold.SBLocalization.theorem_15_3_7_statement`, `Polyfold.SBLocalization.theorem_15_3_7_hyp`). Nothing to formalize: a comment. `theorem_15_3_7_statement` assumes no paracompactness (V15.29); `theorem_15_3_7_hyp` assumes that a good system exists, which "structurable" (Definition 13.3.6) contains and paracompactness supplies (Theorem 13.1.8). *Book:* §15.3, Remark 15.3.8, p. 694 (new in 2021). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V15.32 (Theorem 15.3.9 (2017: 15.5), the transversal extension theorem; `Polyfold.SBLocalization.theorem_15_3_9_charged_of_local_rel`, `Polyfold.SBLocalization.theorem_15_3_9_charged_statement`, `Polyfold.SBLocalization.chargesBoundary_of_theorem_15_3_9_r6`, `Polyfold.SBLocalization.theorem_15_3_9_charged_of_statement`, `Polyfold.SBLocalization.theorem_15_3_9_charged_hyp`, `Polyfold.SBLocalization.relative_local_families_statement`, `Polyfold.SBLocalization.theorem_15_3_9_charged_statement_epSEP_zero`, `Polyfold.SBLocalization.relative_local_families_statement_epSEP_zero`, `Polyfold.SBLocalization.not_theorem_15_3_9_statement_w2`, `Polyfold.SBLocalization.pointwiseNorm_zeroFun_w2`, `Polyfold.SBLocalization.theorem_15_3_9_of_local_rel`, `Polyfold.SBLocalization.theorem_15_3_9_of_rel_hyp`, `Polyfold.SBLocalization.IsReducedOntoAtPt`, `Polyfold.SBLocalization.isInGeneralPosition_iff_forall_reducedOntoAtPt`, `Polyfold.SBLocalization.IsBoundaryFaceOnto`, `Polyfold.SBLocalization.theorem_15_3_9_statement`, `Polyfold.SBLocalization.lemma_`

`15_3_9_cutoff_statement`, `Polyfold.SBLocalization.theorem_15_3_7_relative_statement`, `Polyfold.SBLocalization.IsInGeneralPosition.isTransversal`, `Polyfold.SBLocalization.theorem_15_3_9_hyp`, `Polyfold.SBLocalization.boundary_epSEP_eq_empty`, `Polyfold.SBLocalization.boundaryDomSupp_epSEP`, `Polyfold.SBLocalization.lemma_15_3_9_cutoff_statement_epSEP_zero`, `Polyfold.SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero`, `Polyfold.SBLocalization.theorem_15_3_9_statement_epSEP_zero`, `Polyfold.SBLocalization.theorem_15_3_9_of_extension_ty`, `Polyfold.SBLocalization.lemma_15_3_9_cutoff_hyp`). Part III wave 6 (`Chapter15/P3W6R6_*`): `theorem_15_3_9_charged_statement A`, the book's statement with tame strong bundle charts `hB` and `ChargesBoundary Λ` added (the hypotheses of Theorem 14.1.9 in the form proved, V14.8). **`theorem_15_3_9_charged_of_local_rel (H : relative_local_families_statement A) : theorem_15_3_9_charged_statement A`**: step (a) `lemma_15_3_9_cutoff_hyp hB` (V15.41), step (b) `theorem_14_1_9_charged_a6`, steps (c)–(d) `theorem_15_3_7_relative_of_local H` with the relative parametric step proved (`hasParametricGeneralPositionRel`, V15.45), step (e) and the "moreover" as before; `theorem_15_3_9_charged_hyp (hrel : theorem_15_3_7_relative_statement A)` is the assembly from the relative statement. `chargesBoundary_of_theorem_15_3_9_r6`: the conclusion forces `ChargesBoundary Λ`. Part III wave 7: `not_theorem_15_3_9_statement_w2`: the statement without `ChargesBoundary` does not hold as we read the formal definition whenever $\partial X \neq \emptyset$ and one admissible f exists ($\Lambda \equiv 0$ with empty index sets).

The earlier forms: `theorem_15_3_9_statement A : Prop`: hypotheses `A.X.IsTame`, `HasParacompactOrbitSpace`, `A.X.PObj.AdmitsScPartitionsOfUnity`, `IsScFredholmSectionFunctor f`, `IsScFredholmSectionTame f`, `HasCompactSolutionSetEp f`, `ControlsCompactness A f N U`, `IsBoundaryStructurable A Λ`, `boundaryDomSupp Λ  U`, $N(\Lambda) \leq 1$ on ∂X, h continuous functor with $0 < h \leq 1$, $N(\Lambda) < h$ on ∂X, `IsBoundaryFaceOnto f Λ`; conclusion $\exists \tilde{\Lambda}$, `IsStructurable` ∧ $\tilde{\Lambda} = \Lambda$ over ∂X ∧ `domSupp  U` ∧ $N(\tilde{\Lambda}) < h$ ∧ `IsAdmissible A f N U Λ̃` ∧ `IsTransversal f Λ̃` ∧ `IsBranchedEpSubgroupoid (Λ̃  f)` ∧ `IsCompactBranched (Λ̃  f)`. `IsBoundaryFaceOnto f Λ`: for every sc$^+$-multisection Λ' on W equal to Λ over ∂X and every smooth $x \in \partial X$ with $\Lambda(f(x)) > 0$, every operator of $T_{(f,\Lambda')}(x)$ is onto on $T_x^R X$ (`IsReducedOntoAtPt`, D15.3.5 at one point; `isInGeneralPosition_iff_forall_reducedOntoAtPt`). **`theorem_15_3_9_hyp`**: `theorem_14_1_9_statement A → lemma_15_3_9_cutoff_statement A → theorem_15_3_7_relative_statement A → theorem_15_3_9_statement A`. The two step-propositions: `lemma_15_3_9_cutoff_statement` (the proof's step (a), p. 698): a continuous functor $h_0 \geq 0$, supported in U, $h_0 < h$, $N(\Lambda) < h_0$ on dom-supp(Λ). `theorem_15_3_7_relative_statement` (steps (c)–(d), "argue as in Theorem 15.3.7"): for structurable Λ_0 with dom-supp in U, $N(\Lambda_0) < h$ and (f, Λ_0) in general position at the points of supp$(\Lambda_0 \circ f) \cap \partial X$, a structurable $\tilde{\Lambda} = \Lambda_0$ over ∂X, dom-supp in U, $N(\tilde{\Lambda}) < h$, transversal and in general position. Proved in the reduction: step (b) (Theorem 14.1.9 with $\tilde{U} = U$, $f = h_0$), the boundary general position of the extension (from `IsBoundaryFaceOnto` at smooth points, `mem_smoothSet_of_pos`), admissibility (`isAdmissible_of_domSupp_subset`), the "moreover" (`theorem_15_3_7_in_particular_hyp`), and `IsInGeneralPosition.isTransversal`. Witnesses on `epSEP n 0` ($\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, $\partial X = \emptyset$): `theorem_15_3_9_statement_epSEP_zero`, `lemma_15_3_`

`9_cutoff_statement_epSEP_zero`, `theorem_15_3_7_relative_statement_epSEP_zero` (helpers `boundary_epSEP_eq_empty`, `boundaryDomSupp_epSEP`).

Part III wave 5: **`theorem_15_3_9_of_extension_ty (hB) (hext : theorem_14_1_9_extension_statement A) (hrel : theorem_15_3_7_relative_statement A) : theorem_15_3_9_statement A`**, with step (a) proved (`lemma_15_3_9_cutoff_hyp hB`, V15.41; `hB`: every object lies in a tame strong bundle chart). `hext` is proved from `AssemblyHyp A` (`theorem_14_1_9_extension_statement_of_assembly`, V14.8). The "moreover" with "tame": `theorem_15_3_7_in_particular_tame_hyp` (V15.30). *Book:* §15.3, Theorem 15.3.9, p. 698 (2017: Theorem 15.5). *Hidden hypotheses:* none. *Deviation (narrower):* Since Part III wave 6 the theorem rests on one proposition, the relative local families `relative_local_families_statement` (gap G.178): step (b) is the proved Theorem 14.1.9 (`theorem_14_1_9_charged_a6`), and steps (c)–(d) have the Sard and compactness parts proved. Added: `hB` (as in Theorem 14.1.9) and `ChargesBoundary Λ`, the reading $I_x \neq \emptyset$ of "sc$^+$-multisection over ∂X", necessary for the conclusion (`chargesBoundary_of_theorem_15_3_9_r6`). The earlier forms that conclude the statement without `ChargesBoundary` (`theorem_15_3_9_hyp`, `theorem_15_3_9_of_extension_ty`, `theorem_15_3_9_of_local_rel`, `theorem_15_3_9_of_rel_hyp`) are vacuous when $\partial X \neq \emptyset$ (`not_theorem_15_3_9_statement_w2`) and stay as intermediate steps. Witnesses (`theorem_15_3_9_charged_statement_epSEP_zero`, `relative_local_families_statement_epSEP_zero`) are on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, with no boundary and zero fibres: weak. Earlier reduction: the theorem follows from the extension statement of Theorem 14.1.9 (V14.8) and steps (a) and (c)–(d) of the text's proof, stated as propositions. Step (a) is proved given tame strong bundle charts (`lemma_15_3_9_cutoff_hyp`, V15.41; gap G.177 closed in Part III wave 5), so `theorem_15_3_9_of_extension_ty` needs the extension statement and steps (c)–(d) (gap G.178). `theorem_15_3_7_hyp` proves T15.3.7 from two hypotheses (V15.29); the relative form needs the same engine with Λ_0 fixed near ∂X. Readings: "surjective with respect to intersections of local faces" = onto on $T_x^R X$ (the smallest intersection of the tangent spaces of the local faces), asked for every sc$^+$ extension of Λ (the restriction to $T_x^R X$ only sees face values; independence of the extension not proved, G.179); "U a saturated subset" read open and saturated; $\Lambda : W|\partial X \to \mathbb{Q}^+$ a function on W of which only the values over ∂X enter; "(U,N)-admissible" over ∂X = dom-supp in U and $N(\Lambda) \leq 1$ on ∂X. Inherited: tame germ data of f (as G.140, §15.2). "Tame" in "tame branched": `theorem_15_3_7_in_particular_tame_hyp` (gap G.150 closed). The conclusion adds general position inside the proof (not in the statement); the statement keeps the text's (3) "onto".

V15.33 (Corollary 15.3.10 (1) (2017: 15.1 (1)), regular perturbations exist; `Polyfold.SBLocalization.IsRegularPerturbation`, `Polyfold.SBLocalization.corollary_15_3_10_1_hyp`, `Polyfold.SBLocalization.corollary_15_3_10_1_ty`, `Polyfold.SBLocalization.theorem_15_3_7_of_paracompact`). `IsRegularPerturbation A f N U Λ := ControlsCompactness IsStructurable domSupp Λ U N(Λ) < 1 IsTransversal f Λ`. **`corollary_15_3_10_1_hyp`** (h157 : `theorem_15_3_7_statement A`) (hd, hpara, hX : tame, hbump : `AdmitsBumpFunctionsPointwise`, hf, hft, hS, hcc): $\exists \Lambda$ regular, in general position, Θ branched and compact (`theorem_15_3_7_in_particular_hyp`).

Without tameness (Part III wave 5): **`corollary_15_3_10_1_ty`** (h157, hd, hpara, hbump, hf, hft, hS, hcc), the same conclusion; tameness of X from $d_X \equiv 0$ by `EpGroupoid.isTame_of_degIndex_eq_zero` (V15.40). *Book:* §15.3, Corollary 15.3.10 (1), p. 699 (2017: Corollary 15.1 (1)). *Hidden hypotheses:* none. *Deviation (narrower):* Reduction to `theorem_15_3_7_statement`, which holds whenever a good system exists (`theorem_15_3_7_of_goodSystem`, V15.29), in particular for paracompact $|X|$, as the corollary assumes (`theorem_15_3_7_of_paracompact hpara` discharges `h157`; the combination is not stated as one declaration). Added (needed by T15.3.7, not in the corollary): bump functions (Remark 15.3.2: the text needs them), tame germ data (inherited). Tameness of X is not added: `corollary_15_3_10_1_ty` derives it from $d_X \equiv 0$ (V15.40; gap G.183 closed in Part III wave 5). Paracompactness is unused given T15.3.7. "Tame" branched: `theorem_15_3_7_in_particular_tame_hyp` (V15.30).

V15.34 (Corollary 15.3.10 (3) (2017: 15.1 (3)), regular perturbations are cobordant; `Polyfold.SBLocalization.corollary_15_3_10_3_charged`, `Polyfold.SBLocalization.chargesBoundary_cubeGlue`, `Polyfold.SBLocalization.exists_ne_zero_of_isScPlusMultisection`, `Polyfold.SBLocalization.not_theorem_15_3_9_statement_cube_w2`, `Polyfold.SBLocalization.corollary_15_3_10_3_pu_w5`, `Polyfold.SBLocalization.corollary_15_3_10_3_glue_w5`, `Polyfold.SBLocalization.corollary_15_3_10_3_struct_w5`, `Polyfold.SBLocalization.isBoundaryStructurable_cubeGlue_w5`, `Polyfold.SBLocalization.isScFredholmSectionTame_cubeLiftEp`, `Polyfold.SBLocalization.mem_boundary_cubeSEP_iff`, `Polyfold.SBLocalization.corollary_15_3_10_3_hyp`, `Polyfold.EpGroupoid.injective_cubeOrbitMap`, `Polyfold.EpGroupoid.surjective_cubeOrbitMap`, `Polyfold.EpGroupoid.isOpenMap_cubeOrbitMap`, `Polyfold.EpGroupoid.cubeOrbitHomeomorph`, `Polyfold.EpGroupoid.hasParacompactOrbitSpace_cubeProd`, `Polyfold.SBLocalization.isReducedOntoAtPt_of_injective`, `Polyfold.SBLocalization.isBoundaryFaceOnto_of_injective`, `Polyfold.SBLocalization.zeroMultisection_eq_one_of_injective`, `Polyfold.SBLocalization.injective_cubeSEP_P`, `Polyfold.SBLocalization.injective_epSEP_zero_P`, `Polyfold.SBLocalization.corollary_15_3_10_3_ty`, `Polyfold.SBLocalization.corollary_15_3_10_3_boundary_witness`, `Polyfold.SBLocalization.admitsScPartitionsOfUnity_cube_rn`, `Polyfold.SBLocalization.cubeGlue`, `Polyfold.SBLocalization.cubeGlue_of_zero`, `Polyfold.SBLocalization.cubeGlue_of_one`, `Polyfold.SBLocalization.P_cubeVal_snd`, `Polyfold.SBLocalization.cubeVal_fiberZero`, `Polyfold.SBLocalization.cubeVal_snd_ne_fiberZero`, `Polyfold.SBLocalization.boundaryDomSupp_cubeGlue_subset`, `Polyfold.SBLocalization.normW_interpAuxNorm_le`, `Polyfold.SBLocalization.pointwiseNorm_cubeGlue_le_zero`, `Polyfold.SBLocalization.pointwiseNorm_cubeGlue_le_one`, `Polyfold.SBLocalization.corollary_15_3_10_3_glue_hyp`, `Polyfold.SBLocalization.corollary_15_3_10_3_glue_witness`). Part III wave 7 (`Chapter15/P3W7W2_ChargedExtension.lean`, `P3W7W5_*`): **`corollary_15_3_10_3_charged h159 hBC hd hpara hPUC hf hft hS hr hr hbs hbonto`**, with `h159 : theorem_15_3_9_charged_statement (A.cubeSEP 1)` (the charged Theorem 15.3.9 on the cylinder, V15.32) and `hBC` (tame strong bundle charts on $[0, 1] \times X$): the conclusion of the earlier forms. The new hypothesis of the charged form is proved: `chargesBoundary_cubeGlue` (the glued data charge $\partial([0, 1] \times X)$, since every sc⁺-multisection charges every fibre, `exists_ne_zero_of_isScPlusMultisection`). Its inputs: `hPUC` from partitions of unity on X (`admitsScPartitionsOfUnity_cubeSEP 1 hPU`, V15.46), `hbs` from `isBoundaryStructurable_cubeGlue_w5` given `CubeLiftFace A` (V15.47). `not_theorem_15_3_9_statement_cube_w2`: under the hypotheses of the corollary with X nonempty, the uncharged statement does not hold on `A.cubeSEP 1`; so `corollary_15_3_10_3_hyp`, `corollary_15_3_10_3_ty`, `corollary_15_3_10_3_glue_hyp`, and `corollary_15_3_10_3_pu_w5`, `corollary_15_3_10_3_glue_w5`, `corollary_15_3_10_3_struct_w5` (which take that statement) hold vacuously for nonempty X.

The earlier forms: **`corollary_15_3_10_3_hyp`** (h159 : `theorem_15_3_9_statement (A.cubeSEP 1)`) (hX, hd : $d_X \equiv 0$, hparaC : paracompact $|[0,1] \times X|$, hPUC : partitions of unity on $[0,1] \times X$, hf, hft, hS, h_i : `ControlsCompactness A f N  U`) and boundary data `Λb` on $\tilde{W}$ with `Λb = Λ` over $\{i\} \times X$, `IsBoundaryStructurable`, `boundaryDomSupp Λb  interpSet A U  U`, $\tilde{N}(\Lambda_b) < 1$ on $\partial\tilde{X}$, `IsBoundaryFaceOnto f̃ Λb`: $\exists\tilde{\Lambda}$, `IsRegularPerturbation (A.cubeSEP 1) f̃ (interpAuxNorm A N  N) (interpSet A U  U) Λ̃`, $\tilde{\Lambda} = \Lambda^i$ over $t = i$, $\tilde{\Theta}$ branched and compact. Helpers: `isScFredholmSectionTame_cubeLiftEp` (tame germ data of $\tilde{f}$, from Part I's `paramCube`), `mem_boundary_cubeSEP_iff` ($\partial([0,1] \times X) = \{0,1\} \times X$ for tame X with $d_X \equiv 0$).

Part III wave 5 (`Chapter15/P3W5TY_Cylinder.lean`, `P3W5TY_CylinderGlue.lean`): `cubeOrbitHomeomorph : [0,1] × |X|  |[0,1] × X|`, `hasParacompactOrbitSpace_cubeProd`. **`corollary_15_3_10_3_ty`**: `corollary_15_3_10_3_hyp` without `hX` (from $d_X \equiv 0$) and `hparaC` (from `hpara`). **`corollary_15_3_10_3_glue_hyp`**, the form of the text: (N^i, U^i, Λ^i) structurable regular perturbations (`IsRegularPerturbation A f N  U  Λ`), Λ_b := `cubeGlue Λ  Λ` (Λ^1 at $t = 1$, Λ^0 otherwise); proved: the values at $t = 0, 1$, `boundaryDomSupp_cubeGlue_subset` (into `interpSet`), $\tilde{N}(\Lambda_b) < 1$ on $\partial\tilde{X}$ (`pointwiseNorm_cubeGlue_le_zero`/`_one`); hypotheses left: `hbs : IsBoundaryStructurable (A.cubeSEP 1) (cubeGlue Λ  Λ)`, `hbonto : IsBoundaryFaceOnto f̃ (cubeGlue Λ  Λ)`, `hPUC`. Witnesses on `epSEP n 0`: `corollary_15_3_10_3_boundary_witness` (tame, $d \equiv 0$, paracompact cube and the six Λ_b hypotheses for $\Lambda_b = \Lambda^i = 0$), `corollary_15_3_10_3_glue_witness` (`hbs`, `hbonto` for $\Lambda^i = 0$), `admitsScPartitionsOfUnity_cube_rn` (`hPUC` for $[0,1] \times \mathbb{R}^n$, Theorem 5.5.10 with product charts, Proposition 5.5.11). Helpers `isReducedOntoAtPt_of_injective`, `isBoundaryFaceOnto_of_injective`, `zeroMultisection_eq_one_of_injective`. *Book:* §15.3, Corollary 15.3.10 (3), pp. 699–700 (2017: Corollary 15.1 (3)). *Hidden hypotheses:* none. *Deviation (narrower):* Reduction to the charged Theorem 15.3.9 on the cylinder (`cubeSEP`, `corollary_15_3_10_2`), itself reduced to the relative local families on the cylinder (gap G.178). Hypotheses not in the text: tame strong bundle charts on $[0,1] \times X$ (`hBC`; gap G.189), the structurability of the glue (`hbs`, proved given `CubeLiftFace A`, gap G.193, V15.47) and its face condition (`hbonto`; gap G.180), and partitions of unity on X (`hPU`, which the text does not state in Corollary 15.3.10; they give those of $[0,1] \times X$, V15.46, gap G.181 closed). Tameness of X and paracompactness of $|[0,1] \times X|$ are derived (`EpGroupoid.isTame_of_degIndex_eq_zero`, `hasParacompactOrbitSpace_cubeProd`). The forms that take the uncharged Theorem 15.3.9 on the cylinder are vacuous for nonempty X (`not_theorem_15_3_9_statement_cube_w2`); they stay as intermediate steps and are not cited as the corollary. Witnesses of the earlier hypotheses (`corollary_15_3_10_3_boundary_witness`, `corollary_15_3_10_3_glue_witness`, `admitsScPartitionsOfUnity_cube_rn`) are on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ with $\Lambda^i = 0$, zero fibres: weak; `CubeLiftFace A` has no witness. The text's proof says only $\tilde{U} = [0,1] \times U$ and $\tilde{f}(t,x) = f(x)$ and does not say how the face condition of the glued data is checked; presumably from the transversality of (f, Λ^i), since $T^R_{(i,x)}\tilde{X} = \{0\} \times T_xX$ when $d_X \equiv 0$. The restriction of $\tilde{\Theta}$ to the ends is V15.48; the last paragraph of the proof, $(N'', U'') = (\max\{N, N'\}, U \cap U')$, is V15.42.

V15.35 (Remark 15.3.11 (2017: 15.7), invariants need orientations; `Polyfold.SBLocalization.theorem_15_4_3_hyp`). Nothing to formalize: a comment. The oriented case is §15.4 (V15.36, V15.37; the canonical orientation is gap G.159). *Book:* §15.3, Remark 15.3.11, p. 699 (2017: Remark 15.7). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V15.36 (Theorem 15.4.3 (2017: 15.7), compactness, branching and Stokes for $\Lambda \circ f$; `Polyfold.SBLocalization.localBranching_of_isGoodPositionAt`, `Polyfold.SBLocalization.theorem_15_4_3_isBranched_hyp`, `Polyfold.SBLocalization.theorem_15_4_3_hyp`, `Polyfold.SBLocalization.theorem_15_4_3_eff_hyp`, `Polyfold.SBLocalization.theorem_15_4_3_dim1_hyp`). **`theorem_15_4_3_isBranched_hyp`** (hf, hft, hcc, hΛ, hdom, hN1, htr : `IsTransversal f Λ`, hgood : `  x   X, 0 < Λ (f x) → IsGoodPositionAt f Λ x`): `IsBranchedEpSubgroupoid Θ` `IsCompactBranched Θ` (interior points: `remark_15_2_3_hyp`; boundary points: `localBranching_of_isGoodPositionAt` from `theorem_15_2_5_hyp`; compactness: `isCompact_orbitSupp_comp`). **`theorem_15_4_3_hyp`** ($0 < m$, hPU : `A.X.HasScPartitionsOfUnity`, the hypotheses of `theorem_15_4_3_isBranched_hyp`, hT : `IsOrientation Θ T`, hloc : oriented tame structures of dimension $m + 1$ representing T): Θ branched, compact, and `tameBranchedIntegral … (d  ) = boundaryMeasure …  univ` for all   : `A.baseEP.epFormsInfty m` (`theorem_9_5_17`). `theorem_15_4_3_eff_hyp` (hLUC, `theorem_9_5_18`); `theorem_15_4_3_dim1_hyp` ($k = 1$, `theorem_9_5_17_dim1`). *Book:* §15.4, Theorem 15.4.3, pp. 702–703 (2017: Theorem 15.7). *Hidden hypotheses:* none. *Deviation (narrower):* The orientation is data (T, `hloc`): T15.4.1 (canonical orientation from o) is not formalized (G.159); "tame" and the dimension k are the `hloc` reading of Theorem 9.5.17. Added: sc-smooth partitions of unity on X (hypothesis of Theorem 9.5.17, inherited); tame germ data of f (inherited); (3) is `IsGoodPositionAt` (D15.2.4 reading). X tame is not used (good position carries tameness near the boundary points).

V15.37 (Corollary 15.4.4 (2017: 15.2), the invariant $\Phi_{(f,o)}$ on de Rham cohomology; `Polyfold.SBLocalization.boundary_cyl_time`, `Polyfold.SBLocalization.boundaryMeasure_cyl_univ`, `Polyfold.SBLocalization.boundaryMeasureEff_cyl_univ`, `Polyfold.EpGroupoid.signedMeasure_univ_eq_cylEnds`, `Polyfold.SBLocalization.IsCylinderBoundaryOf`, `Polyfold.SBLocalization.hasOrientedCobordism_of_cylinder`, `Polyfold.SBLocalization.CylinderOrientationStatement`, `Polyfold.SBLocalization.hasScPartitionsOfUnity_cyl`, `Polyfold.SBLocalization.hasOrientedCobordism_cylinder_hyp`, `Polyfold.SBLocalization.corollary_15_4_4_cylinder_hyp`, `Polyfold.SBLocalization.corollary_15_4_4_cylinder_of_rel`, `Polyfold.SBLocalization.tameBranchedIntegral_zero`, `Polyfold.SBLocalization.signedMeasure_univ_eq_zero_of_boundary`, `Polyfold.SBLocalization.boundaryMeasure_univ_eq_zero`, `Polyfold.SBLocalization.integral_d_eq_zero_of_degIndex_eq_zero`, `Polyfold.SBLocalization.integral_d_eq_zero_of_degIndex_eq_zero_`

`dim1`, `Polyfold.SBLocalization.integral_eq_of_cobordism`, `Polyfold.SBLocalization.OrientedRegularPerturbation`, `Polyfold.SBLocalization.OrientedRegularPerturbation.integral`, `Polyfold.SBLocalization.OrientedRegularPerturbation.HasOrientedCobordism`, `Polyfold.SBLocalization.OrientedRegularPerturbation.hasOrientedCobordism_refl`, `Polyfold.SBLocalization.corollary_15_4_4_statement`, `Polyfold.SBLocalization.corollary_15_4_4_hyp`). Part III wave 7 (`Chapter15/P3W7W2_*`), the text's argument in its order: (a) two regular perturbations; (b) the cylinder perturbation from the charged Theorem 15.3.9 on $[0,1] \times X$ (`corollary_15_3_10_3_charged`, V15.34); the restriction to the ends (V15.48); the splitting of the boundary measure **`boundaryMeasure_cyl_univ`** (for $d_X \equiv 0$, $\mu(\partial S) = \mu(\text{end } 0) + \mu(\text{end } 1)$ for any compact branched oriented tame $\tilde{\Theta}$ on the cylinder; generic `EpGroupoid.signedMeasure_univ_eq_cylEnds`); (d) Stokes and $dF^* = F^*d$. `IsCylinderBoundaryOf P P …`: $\mu^{\partial\tilde{\Theta}}_{pr^*\omega}(\text{end } 1) = \int_{P_1} \omega$ and $\mu^{\partial\tilde{\Theta}}_{pr^*\omega}(\text{end } 0) = -\int_{P_0} \omega$; **`hasOrientedCobordism_of_cylinder`** gives `HasOrientedCobordism P P` with $Y = [0,1] \times X$, $F = pr$. `CylinderOrientationStatement hf P P` (every $\tilde{\Lambda}$ that step (b) can produce has an orientation with oriented tame structures of dimension $m+2$ and the end identities). **`corollary_15_4_4_cylinder_hyp`**: `corollary_15_4_4_statement A f m Can` from h159, hBC, hd, hpara, hPU, hPUC, hf, hft, hS, `hglue` (the glue hypotheses for any two `Can`-perturbations) and `hor` (step (c) for any two); `corollary_15_4_4_cylinder_of_rel` with the relative statement on the cylinder in place of h159.

The earlier form: `OrientedRegularPerturbation A f m` (regular perturbation + orientation T of Θ with oriented tame structures of dimension $m + 1$), `.integral`. **`corollary_15_4_4_statement A f m Can : Prop`**, `Can` the (unformalized) condition "the orientation is the canonical one from (f, o)": (i) $\int_P d\eta = 0$ for every `Can`-perturbation; (ii) closed ω have the same integral over any two `Can`-perturbations. **`corollary_15_4_4_hyp`** (hPU, hd : $d_X \equiv 0$, `Can`, hcob : any two `Can`-perturbations are joined by `HasOrientedCobordism`): the statement, in every dimension $m + 1 \geq 1$. **`integral_eq_of_cobordism`** (the cobordism argument, for any sc-smooth functor $F : Y \to X$ and Y with partitions of unity): if $\int_{\partial\tilde{\Theta}} F^*\omega = I_1(\omega) - I_0(\omega)$ for all ω, then $I_0(\omega) = I_1(\omega)$ for closed ω (Stokes on Y, Ch8/11 $dF^* = F^*d$ `epFormsInftyD_comp_epFormsInftyPullback`, `tameBranchedIntegral_zero`). **`integral_d_eq_zero_of_degIndex_eq_zero`** (dim $\geq$ 2) and `…_dim1`: for $d_X \equiv 0$, $\int_\Theta d\eta = 0$ ($\partial\Theta = 0$; `signedMeasure_univ_eq_zero_of_boundary`, `boundaryMeasure_univ_eq_zero`). `hasOrientedCobordism_refl` (hPU): every oriented regular perturbation is cobordant to itself ($Y = X$, $F = \text{id}$, $\tilde{\Theta} = 0$), so `hcob` is satisfiable with nonempty `Can` (e.g. `Can = (· = P)`). *Book:* §15.4, Corollary 15.4.4, p. 703 (2017: Corollary 15.2). *Hidden hypotheses:* none. *Deviation (narrower):* Formalized: the argument in the text's order, with Theorem 15.3.9 in its charged form; the restriction to the ends; the splitting of the boundary measure over the ends; Stokes and $dF^* = F^*d$. Not formalized: the oriented end identities (`IsCylinderBoundaryOf`, assumed through `CylinderOrientationStatement`; gap G.182), whose parts are the weights on the ends (gap G.190) and the canonical orientation on the cylinder (Theorem 15.4.1, gap G.159), so `Can` stays a parameter; relative general position on the cylinder (gap G.178), `hBC` (gap G.189), the glue hypotheses (gaps G.180, G.193). Witnesses: `hasOrientedCobordism_refl` (every perturbation is cobordant to itself, $\tilde{\Theta} = 0$); no concrete oriented regular perturbation with nonempty support exists in the library, so the cobordism hypotheses have no non-degenerate witness: weak. Φ on H^*_{dR} is read degree-wise in the degree $m+1 = \dim\Theta$ as (i)+(ii). Added: partitions of unity on X (Stokes). Paracompactness of $|X|$ unused. The effective version is V15.49.

V15.38 (Remark 15.4.5 (2017: 15.8), invariants by triangulation). Nothing to formalize: the book does not carry it out. *Book:* §15.4, Remark 15.4.5, p. 703 (2017: Remark 15.8). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V15.39 (General position at a point along morphisms; the local Sard step of Theorem 5.3.10 (tools of the proof of Theorem 15.3.7); `Polyfold.MPolyfold.IsGeneralPositionAt.exists_inter`, `Polyfold.StrongBundle.exists_null_isGeneralPositionGerm_paramSlice_near`, `Polyfold.SBLocalization.isReducedOntoAtLevel_of_mor`, `Polyfold.SBLocalization.isInGeneralPositionAt_of_mor`). `isInGeneralPositionAt_of_mor hf hΛ  : IsInGeneralPositionAt f Λ (src  ) → IsInGeneralPositionAt f Λ (tgt  )` and `isReducedOntoAtLevel_of_mor` (Definition 15.3.5 at one point is invariant under morphisms: the functor property of the linearization, the transported chart, `reducedTangentFiber_eq`). Part I, local form of the Sard step of Theorem 5.3.10, without compactness: `StrongBundle.exists_null_isGeneralPositionGerm_paramSlice_near` (X tame, g tame sc-Fredholm, t_i sc⁺, $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ in general position at a zero z_0: an open $W \ni z_0$ and a null set off which every slice is in general position at its zeros in W), with `MPolyfold.IsGeneralPositionAt.exists_inter`. *Book:* §15.3, proof of Theorem 15.3.7, pp. 696–698 (2017: Theorem 15.4). *Hidden hypotheses:* none. *Deviation (extra):* Part I's `theorem_5_3_10_of_family` is global, under compactness; the local form is what the proof of Theorem 15.3.7 uses near each point of S_f.

V15.40 (An M-polyfold or ep-groupoid with $d_X \equiv 0$ is tame (used by Corollary 15.3.10); `Polyfold.degIndexWith_eq_zero_iff_ty`, `Polyfold.isOpen_setOf_pos_coord_ty`, `Polyfold.degIndex_eq_zero_iff_pos_ty`, `Polyfold.isTameRetract_inter_of_degIndex_eq_zero_ty`, `Polyfold.MPolyfold.isTame_of_degIndex_eq_zero`, `Polyfold.EpGroupoid.isTame_of_degIndex_eq_zero`). `isTameRetract_inter_of_degIndex_eq_zero_ty`: for an sc-smooth retraction r of C and open W with $d_C = 0$ on $r(U) \cap W$, $r(U) \cap W$ is a tame retract (restrict r to $U \cap r^{-1}(W) \cap \{d_C = 0\}$; (2) of Definition 2.5.2 from `ScRetraction.isScSplitting_range_scDeriv`). **`MPolyfold.isTame_of_degIndex_eq_zero`**: $(\forall x,\ d_X(x) = 0) \Rightarrow$ `P.IsTame`; **`EpGroupoid.isTame_of_degIndex_eq_zero`**. *Book:* §15.3, Corollary 15.3.10, p. 699 (2017: Corollary 15.1); Definitions 2.4.1, 2.5.7. *Hidden hypotheses:* none. *Deviation (extra):* None. Gap G.183 closed.

V15.41 (Theorem 15.3.9 (2017: 15.5), proof step (a): the cut-off functor h_0; `Polyfold.SBLocalization.BoundarySymmetricStructure.mem_suppAt_iff_ty`, `Polyfold.SBLocalization.IsBoundaryScPlusMultisection.suppAt_subset_oneSet_ty`, `Polyfold.SBLocalization.normW_image_suppAt_subset_ty`, `Polyfold.SBLocalization.pointwiseNorm_tgt_of_boundary_ty`, `Polyfold.SBLocalization.isSaturated_boundaryDomSupp_ty`, `Polyfold.SBLocalization.boundaryDomSupp_subset_boundary_ty`, `Polyfold.SBLocalization.exists_nhds_pointwiseNorm_lt_ty`, `Polyfold.SBLocalization.lemma_15_3_9_cutoff_hyp`, `Polyfold.SBLocalization.theorem_15_3_9_of_extension_`

ty). **`lemma_15_3_9_cutoff_hyp (hB) : lemma_15_3_9_cutoff_statement A`**, `hB :  x,  c : StrongBundleChart A.X.PObj A.P, A.B.IsTameChart c  x  c. .source`. Ingredients: `BoundarySymmetricStructure.mem_suppAt_iff_ty` (supp $\Lambda(y) = \{s_i(y)\}$ over $U(x) \cap \partial X$), `pointwiseNorm_tgt_of_boundary_ty` ($N(\Lambda)$ a functor on ∂X), `isSaturated_boundaryDomSupp_ty`, **`exists_nhds_pointwiseNorm_lt_ty`** ($N(\Lambda)$ upper semicontinuous along ∂X, through `exists_scPlusSectionOn_extension_of_face`). Construction on $|X|$ (paracompact Hausdorff): g continuous with $0 \leq g < h$ and $g > N(\Lambda)$ on $|\text{dom-supp}\,\Lambda|$ (`exists_continuous_forall_mem_convex_of_local_const`); an Urysohn ψ, 1 on $|\text{dom-supp}\,\Lambda|$ and 0 off V with $\bar{V} \subseteq |U|$; $h_0 = (\psi g) \circ \pi$. *Book:* §15.3, proof of Theorem 15.3.9, p. 698 (2017: Theorem 15.5). *Hidden hypotheses:* none. *Deviation (explicit):* `hB` (every object in a tame strong bundle chart) reads "a strong bundle over a tame ep-groupoid" as including tame bundle charts, as in V14.4; witness `(proposition_14_1_4_hypotheses_rn n k).1`. Without tame bundle charts, Definition 14.1.3 (1) says nothing about the s_i and $N(\Lambda)$ need not be upper semicontinuous along ∂X; the text uses the semicontinuity without comment. Gap G.177 closed.

V15.42 (The last paragraph of the proof of Corollary 15.3.10 (2017: 15.1): $(N, U) = (\max\{N, N'\}, U \cap U')$; `Polyfold.SBLocalization.isAuxNorm_max_ty`, `Polyfold.SBLocalization.controlsCompactness_max`, `Polyfold.SBLocalization.pointwiseNorm_le_max_left`, `Polyfold.SBLocalization.IsRegularPerturbation.of_max_left`, `Polyfold.SBLocalization.IsRegularPerturbation.of_max_right`). `isAuxNorm_max_ty`, **`controlsCompactness_max`**, `pointwiseNorm_le_max_left`, **`IsRegularPerturbation.of_max_left`**, **`IsRegularPerturbation.of_max_right`** (given that (N, U), resp. (N', U'), controls compactness). *Book:* §15.3, proof of Corollary 15.3.10, p. 700 (2017: Corollary 15.1). *Hidden hypotheses:* none. *Deviation (same):* None.

V15.43 (Tame local solution sets (prerequisite of Theorem 15.2.5, "tame submanifolds"); `Polyfold.SBLocalization.exists_tame_solutionSet_of_isBranchGoodPosition`, `Polyfold.SBLocalization.isGoodPositionAt_of_interior`). `exists_tame_solutionSet_of_isBranchGoodPosition hft h0 hs hx0 hfs hgp`: if the branch $(f - s)|_O$ is in good position at x (`IsBranchGoodPosition`, Part I's Definition 3.6.6 on $W|_O$), there is an open $V \ni x$, $V \subseteq O$, with $\{y \in V \mid f(y) = s(y)\}$ a smooth submanifold (Definition 4.2.1) with a tame induced structure (Definition 2.5.18, `HasTameInducedStructure`): Part I's `theorem_3_1_26'` on $W|_O$, carried to X by the sc-diffeomorphism $O \to X$. `isGoodPositionAt_of_interior hf hft hΛ hd htr`: at x with $d_X(x) = 0$, transversality (Definition 15.2.1) gives good position (Definition 15.2.4). *Book:* §15.2, Theorem 15.2.5, p. 690 (2017: Theorem 15.3). *Hidden hypotheses:* none. *Deviation (extra):* The inherited tame germ data `hft`, as in all of §15.2 (V15.13).

V15.44 ((15.1) and Part I's properties (i)–(ii) (the bridge used in the proof of Theorem 15.3.7); `Polyfold.SBLocalization.linAt_principal_eq_subDeriv`, `Polyfold.SBLocalization.mem_support_linearization_zero_iff`, `Polyfold.SBLocalization.levelLift_eq_zero_of_centred`, `Polyfold.SBLocalization.onto_chart_iff`, `Polyfold.SBLocalization.isParamGP_iff_onto_chart`, `Polyfold.SBLocalization.isParamOntoVec_iff_isParamGP`, `Polyfold.SBLocalization.exists_tameChart_centred`). `isParamOntoVec_iff_isParamGP hf hft`

`hsec hc hx h 0 hx0 T hT`: for a chart c of W centred at a zero $x \in X_1$ and a presentation T, `IsParamOntoVec f x (s_j(x))` `c.IsParamGP f s T`. Proof via `onto_chart_iff` ((15.1) read in c: $(v, \tau) \mapsto D((\hat{f} - \hat{0}) \circ r)(0)v + \sum \tau_j \hat{s}_j(0)$ onto K_0 from $(T_0 O \cap T_0^R) \times \mathbb{R}^\ell$) and `isParamGP_iff_onto_chart` ((i)–(ii) □ that reading). Helpers: `linAt_principal_eq_subDeriv`, `mem_support_linearization_zero_iff`, `levelLift_eq_zero_of_centred`, `exists_tameChart_centred` (a tame chart centred at a zero, with a presentation). *Book:* §15.3, proof of Theorem 15.3.7, pp. 694–698 (2017: Theorem 15.4), against Part I's parametric transversality. *Hidden hypotheses:* none. *Deviation (extra):* (15.1) uses $T_x^R X$, Part I's (i)–(ii) the full tangent space with the kernel transversal to $\mathbb{R}^\ell \oplus T_x^R X$; the two are equivalent, as the text presumably takes for granted.

V15.45 (Theorem 15.3.9 (2017: 15.5), proof steps (c)–(d): relative general position; `Polyfold.SBLocalization.IsParamOntoAtRel`, `Polyfold.SBLocalization.isParamOntoAtRel_zero_iff`, `Polyfold.SBLocalization.IsParamOntoAtRel.of_reducedOnto`, `Polyfold.SBLocalization.IsParamOntoAtRel.famAdd_left`, `Polyfold.SBLocalization.IsParamOntoAtRel.famAdd_right`, `Polyfold.SBLocalization.RelGPFamily`, `Polyfold.SBLocalization.RelGPFamily.mono`, `Polyfold.SBLocalization.isLinearFamily_zero`, `Polyfold.SBLocalization.RelGPFamily.zeroOn`, `Polyfold.SBLocalization.RelGPFamily.zero`, `Polyfold.SBLocalization.RelGPFamily.add`, `Polyfold.SBLocalization.eq_zeroMultisection_of_notMem_domSupp`, `Polyfold.SBLocalization.multisectionAdd_eq_left_of_notMem_domSupp`, `Polyfold.SBLocalization.HasLocalGPFamiliesRel`, `Polyfold.SBLocalization.HasParametricGeneralPositionRel`, `Polyfold.SBLocalization.nonempty_relGPFamily_biUnion`, `Polyfold.SBLocalization.exists_relGPFamily_of_compact`, `Polyfold.SBLocalization.isCompact_image_supp_comp`, `Polyfold.SBLocalization.theorem_15_3_7_relative_hyp`, `Polyfold.SBLocalization.theorem_15_3_7_relative_of_hyp`, `Polyfold.SBLocalization.hasLocalGPFamiliesRel_of_boundary`, `Polyfold.SBLocalization.hasLocalGPFamiliesRel_epSEP_zero`, `Polyfold.SBLocalization.hasParametricGeneralPositionRel_epSEP_zero`, `Polyfold.SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero_of_hyp`, `Polyfold.SBLocalization.isLocalRepOn_multisectionAdd`, `Polyfold.SBLocalization.paramSlice_branch_eq_rel`, `Polyfold.SBLocalization.isParamGP_restrictTo_of_onto_rel`, `Polyfold.SBLocalization.isGeneralPositionGerm_branchParam_rel`, `Polyfold.SBLocalization.sectionAdd_paramSec_zero`, `Polyfold.SBLocalization.subSec_eq_of_add`, `Polyfold.SBLocalization.exists_of_multisectionAdd_pos_at`, `Polyfold.SBLocalization.exists_local_paramGP_rel`, `Polyfold.SBLocalization.exists_pos_supp_rel_subset`, `Polyfold.SBLocalization.hasParametricGeneralPositionRel`, `Polyfold.SBLocalization.theorem_15_3_7_relative_of_local`, `Polyfold.SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero_of_local`). Relative (15.1): `IsParamOntoAtRel f Λ Λ x`, with the operators $(f - s_a^0)'(x)$ for every branch s_a^0 of Λ_0 (`.of_reducedOnto`, `.famAdd_left/right`). `RelGPFamily f Λ N U g Q`: structurable, linear, $N(\Lambda_t) \leq |t|_1 g$ with $g = h - N(\Lambda_0)$, domain support in $U \cap \partial X^c$, relative (15.1) on $Q \cap \mathrm{supp}(\Lambda_0 \circ f)$. The two Props `HasLocalGPFamiliesRel A f Λ N U g` and `HasParametricGeneralPositionRel A f Λ N U`; `theorem_15_3_7_relative_hyp`, `theorem_15_3_7_relative_of_hyp`: the relative statement from the two. **Proved:**

`hasParametricGeneralPositionRel hX hf hft hcc hΛ hN hdom` (the local representation of a sum `isLocalRepOn_multisectionAdd`; Part I's general position germ for the branch $f - s_a^0 - \sum_j t_j s_{ij}$, `isGeneralPositionGerm_branchParam_rel`; the local Sard step `exists_local_paramGP_rel`; (15.2) relative to Λ_0, `exists_pos_supp_rel_subset`; a finite cover). `theorem_15_3_7_relative_of_local H`: the relative statement from `HasLocalGPFamiliesRel` alone; `hasLocalGPFamiliesRel_of_boundary` (no support off ∂X). *Book:* §15.3, proof of Theorem 15.3.9, pp. 698–699 (2017: Theorem 15.5). *Hidden hypotheses:* none. *Deviation (narrower):* "Argue as in Theorem 15.3.7" is made precise as relative (15.1) with the branch operators of Λ_0. The Sard and compactness steps are proved; the local families (seeds at interior support points, openness at boundary support points) are assumed (gap G.178). Readings: "supported away from ∂X" is domain support in $U \setminus \partial X$, so $\Lambda' = \Lambda_0$ over ∂X; the family bound is $N(\Lambda_t) \leq |t|_1(h - N(\Lambda_0))$, which gives $N(\Lambda_0 \oplus \Lambda_t) < h$ for $|t|_1 < 1$; relative (15.1) uses every branch of Λ_0, and at boundary support points follows from general position of Λ_0 with $\tau = 0$. `hasParametricGeneralPositionRel` needs only $N(\Lambda_0) \leq 1$ and domain support in U, not the compactness of $|S_f|$. Witnesses of the two Props on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ only (no boundary, zero fibres): weak.

V15.46 (Proof of Corollary 15.3.10 (3): $[0,1] \times X$ admits sc-smooth partitions of unity; `Polyfold.admitsScPartitionsOfUnity_of_urysohn`, `Polyfold.exists_cubeProd_urysohn_w5`, `Polyfold.admitsScPartitionsOfUnity_cubeProd`, `Polyfold.SBLocalization.admitsScPartitionsOfUnity_cubeSEP`, `Polyfold.SBLocalization.corollary_15_3_10_3_pu_w5`, `Polyfold.SBLocalization.corollary_15_3_10_3_glue_w5`). `admitsScPartitionsOfUnity_of_urysohn`: sc-smooth Urysohn functions give sc-smooth partitions of unity (the second half of Theorem 5.5.10). `exists_cubeProd_urysohn_w5`: on $[0,1]^n \times X$, with X admitting partitions of unity, $F(q,y) = \sum_x \rho_x(y)\, g_x(q)$ separates two disjoint closed sets (smooth bumps g_x from the tube lemma, ρ subordinate to the tubes). **`admitsScPartitionsOfUnity_cubeProd`**, **`admitsScPartitionsOfUnity_cubeSEP n hPU`**: the objects of `A.cubeSEP n` admit sc-smooth partitions of unity when X does. `corollary_15_3_10_3_pu_w5`, `corollary_15_3_10_3_glue_w5`: the earlier forms with `hPU` on X in place of `hPUC` (they take the uncharged Theorem 15.3.9 on the cylinder, V15.34). *Book:* §15.3, proof of Corollary 15.3.10 (3), p. 699 (2017: Corollary 15.1). *Hidden hypotheses:* none. *Deviation (explicit):* One hypothesis not stated in Corollary 15.3.10: sc-smooth partitions of unity on X (as for the rest of §15.3, where Theorem 15.3.9 asks for them). Witness `rnPolyfold_admitsScPartitionsOfUnity` ($\mathbb{R}^n$). Gap G.181 is closed.

V15.47 (Proof of Corollary 15.3.10 (3): the glued data are structurable over $\partial([0,1] \times X)$; `Polyfold.SBLocalization.cubeLiftSec`, `Polyfold.SBLocalization.cubeVal_fibAct`, `Polyfold.SBLocalization.cubeIso`, `Polyfold.SBLocalization.cube_Γ_snd`, `Polyfold.SBLocalization.cube_Φ_eq`, `Polyfold.SBLocalization.liftStructure`, `Polyfold.SBLocalization.liftStructure_I`, `Polyfold.SBLocalization.CubeLiftFace`, `Polyfold.SBLocalization.cubeSide`, `Polyfold.SBLocalization.glueO`, `Polyfold.SBLocalization.mor_glue`, `Polyfold.SBLocalization.glueTau`, `Polyfold.SBLocalization.cube_biact_snd`, `Polyfold.SBLocalization.isBoundaryWFunctor_of_sides`, `Polyfold.SBLocalization.isBoundaryStructurable_of_sides`, `Polyfold.SBLocalization.isBoundaryStructurable_`

`cubeGlue_w5`, `Polyfold.SBLocalization.corollary_15_3_10_3_struct_w5`). Lifting (`Chapter15/P3W7W5_CubeLift.lean`): `cubeLiftSec s` ($(q, y) \mapsto ((q, y), s(y))$), `cubeVal_fibAct`, `cubeIso z : G_z →* G_{z }`, **`cube_Γ_snd`** (on a connected natural neighbourhood of z over the domain of N at z_2, $\Gamma_V(g, y)_2 = \Gamma_N(g_2, y_2)$: two continuous lifts agreeing at z), `cube_Φ_eq`, **`liftStructure`** (a Definition 13.2.5 structure of Λ on N gives a Definition 14.1.3 structure of any $\tilde{\Lambda}$ with $\tilde{\Lambda}(w) = \Lambda(w_2)$ over $V \cap \partial$, with $I_z = I_{z_2}$, given (1) for the lifted sections). Gluing (`P3W7W5_GlueStructurable.lean`): `cubeSide`, `glueO`, a good system for ∂ subordinate to `glueO` (Theorem 14.1.6 on the cube), `mor_glue`, `glueTau` ($\tilde{\tau}_{z,z'}(\varphi) = \tau^b_{z_2,z_2'}(\varphi_2)$, (a)–(c) and (4) of Definition 14.1.7 from Definitions 13.3.4, 13.3.6), **`isBoundaryStructurable_of_sides`**, **`isBoundaryStructurable_cubeGlue_w5 hX hd hpara hF h h`**, and `corollary_15_3_10_3_struct_w5` (the earlier form with `hbs` replaced by `hF : CubeLiftFace A`). *Book:* §15.3, proof of Corollary 15.3.10 (3), pp. 699–700 (2017: Corollary 15.1). *Hidden hypotheses:* none. *Deviation (narrower):* Structurability of the glue is proved up to `CubeLiftFace A`: lifted sc$^+$-sections satisfy (1) of Definition 14.1.3 on the faces of $[0, 1] \times X$. It is a theorem in the text's setting but is not proved (gap G.193), and it has no witness. The index sets of the glued structure are those of the Definition 13.2.5 structures, so they are nonempty (`Decoration.nonempty_I`). Only the values of the glue over $t \in \{0, 1\}$ enter. The face condition of the glue is gap G.180.

V15.48 (Proof of Corollary 15.3.10 (3): $\tilde{\Theta}$ restricts to Θ_i on $\{i\} \times X$; `Polyfold.EpGroupoid.cylOrbitTime`, `Polyfold.EpGroupoid.cylOrbitTime_`, `Polyfold.EpGroupoid.continuous_cylOrbitTime`, `Polyfold.EpGroupoid.cylEnd`, `Polyfold.EpGroupoid.isClosed_cylEnd`, `Polyfold.EpGroupoid.measurableSet_cylEnd`, `Polyfold.EpGroupoid.cylOrbitTime_eq_zero_or_one`, `Polyfold.EpGroupoid.cylEndMap`, `Polyfold.EpGroupoid.cylEndMap_`, `Polyfold.EpGroupoid.continuous_cylEndMap`, `Polyfold.EpGroupoid.injective_cylEndMap`, `Polyfold.SBLocalization.theta_cyl_of_time_zero`, `Polyfold.SBLocalization.theta_cyl_of_time_one`, `Polyfold.SBLocalization.theta_cubeIncl_zero`, `Polyfold.SBLocalization.theta_cubeIncl_one`, `Polyfold.SBLocalization.mem_boundary_inter_supp_cyl_iff`, `Polyfold.SBLocalization.cylEndMap_mem_orbitSupp_zero`, `Polyfold.SBLocalization.cylEndMap_mem_orbitSupp_one`). `theta_cubeIncl_zero/one`: $\tilde{\Theta} \circ i_q = \Theta_i$ for the inclusion functor at $q \equiv i$; `theta_cyl_of_time_zero/one` pointwise. **`mem_boundary_inter_supp_cyl_iff`**: for $d_X \equiv 0$, $z \in \partial\tilde{X} \cap \operatorname{supp}\tilde{\Theta}$ iff $t = 0$ and $x \in \operatorname{supp}\Theta_0$, or $t = 1$ and $x \in \operatorname{supp}\Theta_1$. Orbit level: `EpGroupoid.cylOrbitTime` (continuous time on $|[0, 1] \times X|$), `EpGroupoid.cylEnd Θ i` (closed, measurable for $L(\partial S)$), `EpGroupoid.cylEndMap` (continuous, injective), **`cylEndMap_mem_orbitSupp_zero/one`**: the end i of $|\operatorname{supp}\partial\tilde{\Theta}|$ lies over $|\operatorname{supp}\Theta_i|$. *Book:* §15.3, proof of Corollary 15.3.10 (3), p. 700 (2017: Corollary 15.1). *Hidden hypotheses:* none. *Deviation (same):* These are the set and orbit parts of "restricts to Θ_i". The equality of $\partial\tilde{\Theta}$ with Θ_i on the ends as weights is gap G.190.

V15.49 (Corollary 15.4.4 (2017: 15.2), effective version; `Polyfold.SBLocalization.tameBranchedIntegralEff_zero`, `Polyfold.SBLocalization.integralEff_eq_of_cobordism`, `Polyfold.SBLocalization.OrientedRegularPerturbation.integralEff`, `Polyfold.SBLocalization.OrientedRegularPerturbation.HasOrientedCobordismEff`, `Polyfold.SBLocalization.OrientedRegularPerturbation.hasOrientedCobordismEff_refl`,

`Polyfold.SBLocalization.corollary_15_4_4_eff_statement`, `Polyfold.SBLocalization.integralEff_d_eq_zero_of_degIndex_eq_zero`, `Polyfold.SBLocalization.corollary_15_4_4_eff_hyp`, `Polyfold.SBLocalization.IsCylinderBoundaryEffOf`, `Polyfold.SBLocalization.hasOrientedCobordismEff_of_cylinder`, `Polyfold.SBLocalization.CylinderOrientationEffStatement`, `Polyfold.SBLocalization.corollary_15_4_4_eff_cylinder_hyp`). `OrientedRegularPerturbation.integralEff hLUC P`, `HasOrientedCobordismEff`, **`corollary_15_4_4_eff_statement A f m hLUC Can`** (exact forms give 0; closed forms have equal $\oint$). **`corollary_15_4_4_eff_hyp`** (hPU, hLUC, hd, pairwise `HasOrientedCobordismEff`), from `integralEff_eq_of_cobordism` (Theorem 9.5.18 on Y), `integralEff_d_eq_zero_of_degIndex_eq_zero` (Theorem 9.5.18 and its dimension-1 form) and `tameBranchedIntegralEff_zero`; `hasOrientedCobordismEff_refl`. Cylinder forms: `IsCylinderBoundaryEffOf`, `hasOrientedCobordismEff_of_cylinder`, `CylinderOrientationEffStatement`, **`corollary_15_4_4_eff_cylinder_hyp`** (adds `hLUCC`, LUC on $[0,1] \times X$). *Book:* §15.4, Corollary 15.4.4, p. 703 (2017: Corollary 15.2). *Hidden hypotheses:* none. *Deviation (narrower):* Same as V15.37. Added: LUC on $[0,1] \times X$ (`hLUCC`), which presumably follows from LUC on X (gap G.191). Exact forms are handled in every dimension $m + 1 \geq 1$. Witness of the cobordism hypothesis: `hasOrientedCobordismEff_refl` only (weak).

V15.50 (Oriented formal sums $\widehat{Gr}_F(x)$ (2017: before Theorem 15.6); `Polyfold.ChartPoint.OrLin`, `Polyfold.ChartPoint.GrFHat`, `Polyfold.ChartPoint.forgetOr`, `Polyfold.ChartPoint.exists_two_orLin`, `Polyfold.ChartPoint.canonLift`, `Polyfold.ChartPoint.canonLift_unique`, `Polyfold.ChartPoint.canonLift_injective_o`). In a Part I chart point `p: ChartPoint`. `OrLin` is $\Sigma\, L \in \mathrm{Lin}(f, x),\ \mathrm{Ray}(\mathrm{DET}\, L)$ and `ChartPoint.GrFHat` `$= $` `OrLin →   0`; `ChartPoint.forgetOr` is f_F; `ChartPoint.exists_two_orLin`: each L has exactly two orientations; `ChartPoint.canonLift`, `canonLift_unique`, `canonLift_injective_o`. *Book:* §15.4, before Theorem 15.4.1, p. 701 (2017: before Theorem 15.6). *Hidden hypotheses:* none. *Deviation (restated):* Read in a Part I chart point: the operators are $T_zO \to K_z$, and $\mathrm{Lin}(f, x)$ is Part I's `Lin(f, x)`.

V15.51 (Theorem 15.4.1 (2017: 15.6), the canonical oriented lift of T_Θ; `Polyfold.SBLocalization.liftPt`, `Polyfold.SBLocalization.subOp`, `Polyfold.SBLocalization.subOp_mem_linSet`, `Polyfold.SBLocalization.linW`, `Polyfold.SBLocalization.mapDomain_toF_linW`, `Polyfold.SBLocalization.canonLift_spec`, `Polyfold.SBLocalization.theorem_15_4_1_lift`, `Polyfold.SBLocalization.theorem_15_4_1_indep`, `Polyfold.SBLocalization.canonLift_linW_eq_sum`). `subOp_mem_linSet`: $(f - s)'(y) \in \mathrm{Lin}(f, y)$; `mapDomain_toF_linW`: the formal sum `linW` maps to `linSumChart` (Theorem 15.1.1 (2)). **`theorem_15_4_1_lift`**: at the chart point `liftPt c` (level-1 lift), $f_F(\hat{T}) = T(f, \Lambda)(y)$, every summand is oriented by o, and $\hat{T}$ is unique with these two properties; `canonLift_linW_eq_sum` is the formula; **`theorem_15_4_1_indep`**: independence of the local structure (via `linSumChart_eq`); `canonLift_spec` (the generic characterization at any chart point). *Book:* §15.4, Theorem 15.4.1, p. 702 (2017: Theorem 15.6). *Hidden hypotheses:* none. *Deviation (narrower):* Stated in $\widehat{Gr}_F(y)$ at a Part I chart point, for smooth y, not as an orientation `IsOrientation Θ T` in Chapter 9's $\widehat{Gr}(X)$ (gap G.159). The standing hypothesis `HasScPlusSectionsThrough` (V12.22) is used; it follows from bump functions.

V15.52 (Definition 15.4.2 (2017: 15.7), the canonical orientation $\widehat{T}_{((f,o),\Lambda)}$; `Polyfold.SBLocalization.canonicalLift`, `Polyfold.SBLocalization.surjective_subOp_iff`, `Polyfold.SBLocalization.branchOr`, `Polyfold.SBLocalization.branchOr_injective`, `Polyfold.ChartPoint.orKer`, `Polyfold.ChartPoint.orKerEquiv`, `Polyfold.ChartPoint.orKerOrientation`, `Polyfold.ChartPoint.orKerOrientation_injective`). `canonicalLift o` (in the chart point); `surjective_subOp_iff` (onto K_z iff `ChartOnto`); `branchOr`: the kernel orientation in $\Lambda^{\max}N$; `branchOr_injective`: o_y is recovered from it; `ChartPoint.orKer`, `ChartPoint.orKerEquiv`, `ChartPoint.orKerOrientation` (as a Mathlib `Orientation`), `ChartPoint.orKerOrientation_injective`. *Book:* §15.4, Definition 15.4.2, p. 702 (2017: Definition 15.7). *Hidden hypotheses:* none. *Deviation (narrower):* As V15.51. The kernel orientation is given pointwise; its continuity along M_i and the oriented tangent spaces $T_y(M_i, o_i)$ are not formalized (gap G.159).

V15.53 (The product strong bundle $W \times W' \to X \times X'$ (remark after Definition 15.3.3); `Polyfold.bundleSwap`, `Polyfold.prodBundleMap`, `Polyfold.IsStrongBundleMap.prod`, `Polyfold.IsStrongBundleRetraction.prod`, `Polyfold.IsStrongBundleRetract.prod`, `Polyfold.IsScSmoothRetractMap.prodBundleMap`, `Polyfold.prodFiberEquiv`, `Polyfold.prodFiberCLE`, `Polyfold.IsBanachable.prod`, `Polyfold.StrongBundleChart.prod`, `Polyfold.StrongBundleChart.Compatible.prod`, `Polyfold.StrongBundle.prodAtlas`, `Polyfold.StrongBundle.prod`, `Polyfold.StrongBundle.prodOneHomeomorph`, `Polyfold.StrongBundle.isScSmoothMap_prodOneHomeomorph`, `Polyfold.StrongBundle.isScSmoothMap_prodOneHomeomorph_symm`, `Polyfold.StrongBundle.IsStrongBundleMap.prod`). Model level: `bundleSwap` $((a,a'),(b,b')) \mapsto ((a,b),(a',b'))$, `prodBundleMap Φ Φ'`, `IsStrongBundleMap.prod`, `IsStrongBundleRetraction.prod`, `IsStrongBundleRetract.prod`, `IsScSmoothRetractMap.prodBundleMap`. Fibres: `prodFiberEquiv`, `prodFiberCLE`, `IsBanachable.prod`. Charts and atlas: `StrongBundleChart.prod` (retraction $R \times R'$ after the swap), `StrongBundleChart.Compatible.prod` (levels $[0]$ and $[1]$), `StrongBundle.prodAtlas`, **`StrongBundle.prod B B' : StrongBundle (PX.prod PX') (Prod.map P P')`**. Levels: **`StrongBundle.prodOneHomeomorph`** $(W \times W')[1] \simeq_t W[1] \times W'[1]$, sc-smooth both ways. Maps: **`StrongBundle.IsStrongBundleMap.prod`**. *Book:* §15.3, remark after Definition 15.3.3, p. 692 (2017: after Definition 15.5). *Hidden hypotheses:* none. *Deviation (restated):* The text does not describe the structure; the atlas is that of products of charts of the two atlases (as `MPolyfold.prod` does for M-polyfolds). The paracompactness instances on the products, fields of the strong bundle type, are always provided (`MPolyfold.paracompactSpace_prod`, `StrongBundle.paracompactSpace_prod`).

V15.54 (Product sections and products of sc$^+$-sections (remark after Definition 15.3.3); `Polyfold.IsSection.prodMap`, `Polyfold.StrongBundle.IsScSmoothSection.prod`, `Polyfold.StrongBundle.IsScPlusSection.prod`). `IsSection.prodMap`, `StrongBundle.exists_chart_rep` (a chart representative of an sc-smooth section near a point), **`StrongBundle.IsScSmoothSection.prod`**, **`StrongBundle.IsScPlusSection.prod`**. *Book:* §15.3, remark after Definition 15.3.3, p. 692 (2017: after Definition 15.5). *Hidden hypotheses:* none. *Deviation (same):* None (sc-smooth and sc$^+$ read in the product atlas, V15.53).

V15.55 (The product strong bundle over the ep-groupoid $X \times X'$ and the section functor $f \times f'$; `Polyfold.prodPullbackSplit`, `Polyfold.isLocalScDiffeo_prodPullbackSplit`, `Polyfold.EpStrongBundle.prod`, `Polyfold.EpStrongBundle.prod_`, `Polyfold.EpStrongBundle.IsFunctorialSection.prod`, `Polyfold.EpStrongBundle.IsScSmoothSectionFunctor.prod`, `Polyfold.EpStrongBundle.IsScPlusSectionFunctor.prod`). `prodPullbackSplit X X'` (the pull-back $(X \times X')_{1\ s}\times (W \times W')$ as a product of pull-backs, inverse to `prodPullbackJoin`), `isLocalScDiffeo_prodPullbackSplit`; **`EpStrongBundle.prod E E' : EpStrongBundle (X.prod X') (B.prod B')`** (strong bundle map, surjective, local sc-diffeomorphism, unit and composition laws), `EpStrongBundle.prod_` (componentwise); **`EpStrongBundle.IsFunctorialSection.prod`**, **`IsScSmoothSectionFunctor.prod`**, **`IsScPlusSectionFunctor.prod`**. *Book:* §15.3, remark after Definition 15.3.3, p. 692 (2017: after Definition 15.5). *Hidden hypotheses:* none. *Deviation (same):* Paracompactness instances on the product and pull-back spaces, as for `EpGroupoid.prod` (V15.17), produced by `paracompactSpace_pullback`.

V15.56 ($N \times N' = \max(N, N')$ is an auxiliary norm (remark after Definition 15.3.3); `Polyfold.StrongBundle.prodAuxNorm`, `Polyfold.StrongBundle.isAuxNorm_prodAuxNorm`, `Polyfold.EpStrongBundle.isAuxNorm_prodAuxNorm`). `StrongBundle.prodOneFiberFst/Snd/Mk` (the fibre of $(W \times W')_{0,1}$ as a product), **`StrongBundle.prodAuxNorm B B' N N'`**, **`StrongBundle.isAuxNorm_prodAuxNorm`** (Definition 5.1.1), **`EpStrongBundle.isAuxNorm_prodAuxNorm`** (Definition 8.3.5 (2021), with invariance under $\mu \times \mu'$). *Book:* §15.3, remark after Definition 15.3.3, p. 692 (2017: after Definition 15.5). *Hidden hypotheses:* none. *Deviation (same):* None for the auxiliary norm. Products of sc-Fredholm germs, $\Lambda \times \Lambda'$, admissibility and the domain support are gap G.160.

What is not verified in this chapter

- p. 684, Theorem 15.1.1 (2017: 15.1): the functor property $T(f,\Lambda)(t\varphi) = \varphi_* T(f,\Lambda)(s\varphi)$ and GrF(X) as a category (V15.2; gap G.144; Findings, item 4).
- p. 684, Proposition 15.1.3 (2017: 15.1) at an isolated point: does not hold as we read it, a degenerate case not used by Theorem 15.1.1 (V15.4; Findings, item 3).
- pp. 692–693, products of sc-Fredholm germs and of multisections, and Remark 15.3.4 (2017: Remark 15.5) (gap G.160; Findings, item 24).
- pp. 698–699, Theorem 15.3.9 (2017: 15.5): the relative local families of steps (c)–(d), and the independence of the boundary hypothesis of the extension (gaps G.178, G.179; Findings, item 21).
- pp. 699–700, Corollary 15.3.10 (3) (2017: 15.1 (3)): tame strong bundle charts on $[0,1] \times X$, lifted sc$^+$-sections on the faces, the face condition of the glued data, and $\partial\widetilde{\Theta} = \Theta_i$ on the ends as weights (gaps G.180, G.189, G.190, G.193; Findings, item 23).
- pp. 700–703, §15.4: Theorem 15.4.1 and Definition 15.4.2 (2017: Theorem 15.6, Definition 15.7) as an orientation in $\widehat{Gr}(X)$ (gap G.159; Findings, item 22); the oriented end identities of Corollary 15.4.4 (2017: 15.2) and LUC on the cylinder (gaps G.182, G.191; Findings, item 23).

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17. Chapter 16 (HWZ, Part III, §§16.1–16.6 (2017 §§16.1–16.6)): begun 2026-11-13 (Part III wave 1, continued in waves 2–7), verified for polyfold structures and their metrizability, their equivalence and morphisms, polyfolds and sc-smooth maps, the invariance of bump functions, partitions of unity and tameness, the degeneracy index and the boundary; the tangent polyfold, the raised polyfolds, the tangent functor on PS and on effective polyfolds and the de Rham complex with pull-backs; strong polyfold bundles, auxiliary and reflexive auxiliary norms, independent of the model, and sections, the zero section included; branched suborbifolds with the naturality of the structure on the support, their integrals and Stokes' theorem with sc-smooth partitions of unity; sc$^+$-multisections over polyfolds and the Fredholm notions of §16.6, with Theorem 16.6.4 as printed, Theorem 16.5.7 with the readings of Theorem 14.1.9, Theorem 16.6.6 with the integrals well defined and the independence of λ given the cobordisms; with the readings recorded below (the uniqueness sentence after Definition 16.1.2, Remark 16.1.7, Lemma 16.2.1, the composite of Definition 16.1.10 and the uniqueness after Definition 16.3.3 do not hold as we read them, and are verified for effective polyfolds)

Chapter 16 (pp. 705–740; §16.1 is pp. 705–715, §16.2 pp. 715–721, §16.3 pp. 721–729, §16.4 pp. 729–733, §§16.5–16.6 pp. 733–740; numbers and pages are those of the book (Springer 2021), with the 2017 text's numbers in parentheses) defines polyfolds and their structures. §16.1 defines polyfold structures (Definition 16.1.1 (2017: 16.1)), their equivalence (Definition 16.1.2 (2017: 16.2)) and polyfolds (Definition 16.1.3 (2017: 16.3)), shows that bump functions and partitions of unity do not depend on the structure (Proposition 16.1.4, Definition 16.1.5 (2017: Proposition 16.2, Definition 16.4)), discusses generalized isomorphisms over homeomorphisms (Remark 16.1.7 (2017: 16.1)), and defines the degeneracy index, the boundary and tameness (Lemma 16.1.13, Definitions 16.1.14, 16.1.17, Lemma 16.1.19, Definition 16.1.20 (2017: Lemma 16.1, Definitions 16.9, 16.11, Lemma 16.2, Definition 16.12)). §16.2 (2017: §16.2) builds the tangent polyfold and the sc-differential forms (Lemmas 16.2.1, 16.2.2, Definition 16.2.3, Theorem 16.2.4, Definition 16.2.5, Theorem 16.2.6 (2017: Lemmas 16.3, 16.4, Definition 16.14, Theorem 16.1, Definition 16.15, Theorem 16.2)); §16.4 (2017: §16.4) defines branched suborbifolds and proves Stokes' theorem (Definitions 16.4.1, 16.4.2, Proposition 16.4.3, Definitions 16.4.4, 16.4.5, Theorem 16.4.6 (2017: Definitions 16.29, 16.30, Proposition 16.6, Definitions 16.31, 16.34, Theorem 16.5)). §16.3 (2017: §16.3) defines strong polyfold bundles, their auxiliary norms and sections (Definitions 16.3.1–16.3.19, Theorems 16.3.15, 16.3.17 (2017: Definitions 16.16–16.28, Theorems 16.3, 16.4)); §16.5 defines sc$^+$-multisections over polyfolds (Definitions 16.5.1–16.5.6, Theorem 16.5.7 (2017: Definitions 16.35–16.39, Theorem 16.6)); §16.6 the Fredholm theory (Definitions 16.6.1–16.6.5, Theorems 16.6.4, 16.6.6 (2017: Definitions 16.40–16.42, Theorems 16.7, 16.8)). §16.2 and §16.4 were added in Part III wave 2, the rest in waves 3–4.

What is verified. Definitions 16.1.1–16.1.3, 16.1.5, 16.1.14, 16.1.17, 16.1.20 and Lemma 16.1.13 are verified as stated (V16.1, V16.2, V16.4, V16.6, V16.8–V16.10, V16.12); Proposition 16.1.4 with bump functions on the object M-polyfold (V16.5); Lemma 16.1.19 as an iff (V16.11). The sentence after Definition 16.1.2 that the generalized isomorphism is unique, as a consequence of (ZHOU), and Remark 16.1.7 with the sentence on pp. 709–710, do not hold as we read them: the trivial $\mathbb{Z}/3$-

ep-groupoid on $\mathbb{R}$ is a polyfold structure in the sense of Definition 16.1.1, and the identity and the inversion give two generalized isomorphisms with the same orbit map; both statements are proved for an effective target (V16.3, V16.7; Findings, items 1, 2). Part III wave 2: the metrizability remark after Definition 16.1.1 is verified (V16.14; gap G.146 closed). Lemma 16.2.1 does not hold as we read it, for the same pair; its orbit form and its effective form are verified (V16.15; Findings, item 6). The construction of $T(Z, c)$, Lemma 16.2.2, Definitions 16.2.3, 16.2.5 and the remark on p. 720 are verified as stated (V16.16–V16.20); Theorem 16.2.6 for the de Rham complex and, in Part III wave 3, for pull-backs (V16.21). Definitions 16.4.1, 16.4.4, 16.4.5 are verified as stated (V16.22, V16.25, V16.26); Definition 16.4.2 as stated, tameness being invariant (V16.23; gap G.154 closed in wave 3); Proposition 16.4.3 as stated, its naturality in Part III wave 5 (V16.24; gap G.155 closed); the integrals among tame compact representatives (V16.27); Theorem 16.4.6 with sc-smooth partitions of unity (V16.28; gap G.153). Part III waves 3–4: the category PS, Definitions 16.1.6, 16.1.8 as stated (V16.29–V16.31); the composite of Definition 16.1.10 does not hold as we read it in general and is verified for an effective middle polyfold, and the polyfold category of Definition 16.1.11 is that of effective polyfolds (V16.32, V16.33; Findings, item 12); Theorem 16.2.4 on PS and on sc-smooth maps, and in Part III wave 5 as a functor on effective polyfolds (V16.34; gap G.157 closed); the raised polyfolds and τ_Z (V16.35, V16.36; gap G.156 closed). §16.3 as stated, except that the uniqueness after Definition 16.3.3 does not hold as we read it (V16.40; Findings, item 11), Proposition 16.3.16 (2) is read in one model, as stated since Part III wave 7, and Theorem 16.3.17 (2) takes the local step of Theorem 12.2.13, discharged for tame models (V16.37–V16.62; gaps G.166, G.167). §§16.5–16.6: the definitions as stated or restated, Definition 16.5.4 with the structurable pull-back (V16.63–V16.70, V16.72–V16.75); Theorem 16.5.7 from Theorem 14.1.9 in every model, in Part III wave 6 with the readings of Theorem 14.1.9 (V16.71); Theorem 16.6.4 as stated (V16.76); Definition 16.6.5 in one representative, carried along diagrams in Part III wave 6 (V16.77; gap G.176); Theorem 16.6.6 reduced to a model, in Part III wave 6 with the integrals well defined and the independence of λ given the cobordisms (V16.78, V16.80; gaps G.175, G.187, G.188); $n(\lambda)$ as stated, its continuity in Part III wave 5 (V16.67; gap G.174 closed). Part III wave 5: the parenthetical of Definition 16.3.12 in the pull direction, in Part III wave 7 in both directions, with mixed convergence and reflexive auxiliary norms independent of the model (V16.51–V16.53, V16.56, V16.79; gap G.166 closed). Part III wave 7: the zero section (V16.81; gap G.167 closed).

The hypotheses. None beyond the text's statements in §16.1. In §16.4, `hPU : c.AdmitsScPartitionsOfUnity` in Theorem 16.4.6 and the independence of the boundary integrals, which Theorem 9.5.17 assumes and Theorem 16.4.6 does not state (witness `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_hyp_witness`; Findings, item 8). Lemma 16.2.1 for an effective target has the witness `Polyfold.PolyfoldStructure.lemma_16_2_1_of_isEffective_hypotheses`. Every definition has a witness: the polyfold $\mathbb{R}$ with the trivial $\mathbb{Z}/3$-action (`Polyfold.PolyfoldStructure.z3PolyfoldStructure`, `Polyfold.PolyfoldStructure.z3Polyfold` and the `z3Polyfold_*` declarations; V16.13). Part III waves 3–4: the effectiveness of d in Definition 16.1.10 (witness `Polyfold.EffPolyfold.point`); `hTd : d.tangent.IsEffective` in `tangent_comp`, which holds for every effective `d` since Part III wave 5 (`Polyfold.PolyfoldClass.IsEffective.tangent`); `hloc` in Theorem 16.3.17 (2) (witness `Polyfold.`

`StrongPolyfoldBundle.theorem_16_3_17_two_hyp_witness`); `h1419` in Theorem 16.5.7, Theorem 14.1.9 in every model, in the formal form that holds only for $\partial X = \emptyset$ (V14.8), replaced in Part III wave 6 by tame bundle charts in the models and a representative charging the boundary (`Polyfold.StrongPolyfoldBundle.theorem_16_5_7_a6`) (the other hypotheses: `Polyfold.StrongPolyfoldBundle.theorem_16_5_7_hyp_witness`). Part III wave 6: in Theorem 16.6.6 `hPU` (witness `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_hyp_witness`) and the cobordisms `hcob`, `hcobEff` (only the self-cobordism witness `Polyfold.StrongPolyfoldBundle.theorem_16_6_6_hyp_hcob_witness`: weak); the inherited orientation is a parameter. Readings: Findings, items 11–23.

The setting. Ep-groupoids, orbit spaces, the degeneracy index on $|X|$ and tameness are Chapter 7's; generalized isomorphisms and the localization $\mathcal{EP}(E^{-1})$ are Chapter 10's, with (ZHOU) and Theorem 10.3.19 for effective ep-groupoids; the invariance of tameness under equivalences is Chapter 11's.

V16.1 (Definition 16.1.1 (2017: 16.1), polyfold structures; `Polyfold.PolyfoldStructure`, `Polyfold.PolyfoldStructure.toOrbitChart`, `Polyfold.PolyfoldStructure.isLR`, `Polyfold.PolyfoldStructure.z3PolyfoldStructure`). `structure PolyfoldStructure (Z)`, with fields `X`, `paracompact : X.X.HasParacompactOrbitSpace`, `isZhou : X.X.IsZhou` and `: X.X.OrbitSpace  Z`; `toOrbitChart` (the pair of Remark 10.3.18) and `isLR`. Witness: `z3PolyfoldStructure` on $\mathbb{R}$ (the trivial $\mathbb{Z}/3$-action). *Book:* §16.1, Definition 16.1.1, p. 705 (2017: Definition 16.1). *Hidden hypotheses:* none. *Deviation (same):* Z lives in the universe of the object and morphism sets of X. The text's remark that Z is then metrizable is V16.14 (gap G.146 closed). The 2017 definition had neither (ZHOU) nor paracompactness.

V16.2 (Definition 16.1.2 (2017: 16.2), equivalent polyfold structures; `Polyfold.PolyfoldStructure.Equiv`, `Polyfold.PolyfoldStructure.setoid`, `Polyfold.PolyfoldStructure.equiv_iff_rel`, `Polyfold.PolyfoldStructure.orbitMap_eq_of_equiv`). `PolyfoldStructure.Equiv p q :=  f, IsGeneralizedIso f  p. = q.  orbitMap f`; an equivalence relation (`setoid`), equal to Remark 10.3.18's relation (`equiv_iff_rel`); $|f| = \beta^{-1} \circ \alpha$ (`orbitMap_eq_of_equiv`). *Book:* §16.1, Definition 16.1.2, p. 707 (2017: Definition 16.2). *Hidden hypotheses:* none. *Deviation (same):* none.

V16.3 (The uniqueness sentence after Definition 16.1.2; `Polyfold.PolyfoldStructure.Uniqueness_16_1_2`, `Polyfold.PolyfoldStructure.not_uniqueness_16_1_2`, `Polyfold.PolyfoldStructure.uniqueness_16_1_2_counterexample`, `Polyfold.PolyfoldStructure.uniqueness_16_1_2_of_isEffective`). Does not hold as we read it: `PolyfoldStructure.not_uniqueness_16_1_2 : ¬ Uniqueness_16_1_2`. The trivial $\mathbb{Z}/3$-ep-groupoid on $\mathbb{R}$ satisfies (ZHOU) (`isZhou_trivialZ3`) and has paracompact orbit space, so it is a polyfold structure on $\mathbb{R}$ in the sense of Definition 16.1.1; the identity and the generalized isomorphism through the inversion functor both satisfy $\alpha = \alpha \circ |f|$ and differ (`uniqueness_16_1_2_counterexample`). It holds when the target is effective: `uniqueness_16_1_2_of_isEffective` ((ZHOU) gives the local regularity property, then Theorem 10.3.19 for effective ep-groupoids). *Book:* §16.1, after Definition 16.1.2, p. 707 (2017: Definition 16.2). *Hidden hypotheses:* none. *Deviation (narrower):* The sentence does not hold as we read it (Findings, Chapter 16, item 1): the counterexample satisfies every printed hypothesis, (ZHOU), paracompact $|X|$ and β a homeomorphism, all proved for the instance. It is the pair behind `not_theorem_10_3_19` (Chapter 10). The effective form is proved. The 2017 text (Definition 16.2) says "f is necessarily unique", citing the theorem that is now 10.3.19; the 2021 text adds "as a consequence of (ZHOU)".

V16.4 (Definition 16.1.3 (2017: 16.3), polyfolds; `Polyfold.PolyfoldClass`, `Polyfold.PolyfoldStructure.z3Polyfold`). `PolyfoldClass Z := Quotient (PolyfoldStructure.setoid Z)`; witness `z3Polyfold`. *Book:* §16.1, Definition 16.1.3, p. 707 (2017: Definition 16.3). *Hidden hypotheses:* none. *Deviation (same):* A class is a Lean quotient; the book's footnote on classes does not arise.

V16.5 (Proposition 16.1.4 (2017: 16.2), bump functions and partitions of unity are invariant; `Polyfold.PolyfoldStructure.proposition_16_1_4`, `Polyfold.PolyfoldStructure.proposition_16_1_4_two`, `Polyfold.EpFunctor.IsEquivalence.hasScPartitionsOfUnity_iff`, `Polyfold.EpFunctor.IsEquivalence.isScSmoothMap_comp_ _of_comp_obj`, `Polyfold.EpGroupoid.hasScPartitionsOfUnity_iff_orbit`, `Polyfold.EpGroupoid.HasOrbitScPartitionsOfUnity`, `Polyfold.PolyfoldStructure.proposition_16_1_4_one`, `Polyfold.EpFunctor.IsEquivalence.admitsBumpFunctions_iff`, `Polyfold.MPolyfold.IsScDiffeoOn.hasBumpAt`, `Polyfold.EpGroupoid.hasBumpAt_tgt`). `PolyfoldStructure.proposition_16_1_4`: `(p.X.X.PObj.AdmitsBumpFunctions  q.X.X.PObj.AdmitsBumpFunctions)  (p.X.X.HasScPartitionsOfUnity  q.X.X.HasScPartitionsOfUnity)`; parts `proposition_16_1_4_one`, `_two`, through equivalences (`IsEquivalence.admitsBumpFunctions_iff`, `IsEquivalence.hasScPartitionsOfUnity_iff`); the "easy exercise" is `isScSmoothMap_comp_ _of_comp_obj`. *Book:* §16.1, Proposition 16.1.4, p. 708 (2017: Proposition 16.2). *Hidden hypotheses:* none. *Deviation (explicit):* (1) is read for the object M-polyfold (Definition 5.3.1, $f(x) \neq 0$), as the proof uses it; Definition 12.3.1's notion is equivalent by Theorem 12.3.2 (1) (V12.11). (2) is Definition 7.5.5. The proof route differs: local inverses for (1), partitions descended to $|X|$ for (2).

V16.6 (Definition 16.1.5 (2017: 16.4), polyfolds with bump functions and partitions of unity; `Polyfold.PolyfoldClass.AdmitsBumpFunctions`, `Polyfold.PolyfoldClass.definition_16_1_5_one`, `Polyfold.PolyfoldClass.AdmitsScPartitionsOfUnity`, `Polyfold.PolyfoldClass.definition_16_1_5_two`, `Polyfold.PolyfoldStructure.z3Polyfold_admitsBumpFunctions`, `Polyfold.PolyfoldStructure.z3Polyfold_admitsScPartitionsOfUnity`). `PolyfoldClass.AdmitsBumpFunctions`, `PolyfoldClass.AdmitsScPartitionsOfUnity` (some structure in c); `definition_16_1_5_one`, `_two`: equivalent to "every structure in c". Witnesses `z3Polyfold_admitsBumpFunctions`, `z3Polyfold_admitsScPartitionsOfUnity`. *Book:* §16.1, Definition 16.1.5, p. 708 (2017: Definition 16.4). *Hidden hypotheses:* none. *Deviation (same):* Bump functions with the object M-polyfold reading, as in V16.5.

V16.7 (Remark 16.1.7 (2017: 16.1), at most one generalized isomorphism over a homeomorphism; `Polyfold.PolyfoldStructure.Remark_16_1_7`, `Polyfold.PolyfoldStructure.not_remark_16_1_7`, `Polyfold.PolyfoldStructure.remark_16_1_7_of_isEffective`). Does not hold as we read it: `PolyfoldStructure.not_remark_16_1_7 : ¬ Remark_16_1_7` (the pair of V16.3, $h = \mathrm{id}$); the same pair contradicts the sentence on pp. 709–710. Holds for an effective target: `remark_16_1_7_of_isEffective`. *Book:* §16.1, Remark 16.1.7, p. 711 (2017: Remark 16.1), and pp. 709–710. *Hidden hypotheses:* none. *Deviation (narrower):* Does not hold as we read it (Findings, Chapter 16, item 2); the effective form is proved. The "none or many" sentence for continuous h has no content beyond an example and is not formalized.

V16.8 (Lemma 16.1.13 (2017: 16.1), the degeneracy index does not depend on the structure; `Polyfold.PolyfoldStructure.degIndex`, `Polyfold.PolyfoldStructure.degIndex_eq`, `Polyfold.PolyfoldStructure.lemma_16_1_13`). `PolyfoldStructure.degIndex p z := d_ {|X|}(` ^{1} `z)`, equal to $d_X(x)$ for every x with $\beta(|x|) = z$ (`degIndex_eq`); `lemma_16_1_13 (h : Equiv p q) : p.degIndex = q.degIndex`. *Book:* §16.1, Lemma 16.1.13, p. 712 (2017: Lemma 16.1). *Hidden hypotheses:* none. *Deviation (same):* none.

V16.9 (Definition 16.1.14 (2017: 16.9), the degeneracy index of a polyfold; `Polyfold.PolyfoldClass.degIndex`, `Polyfold.PolyfoldClass.definition_16_1_14`, `Polyfold.PolyfoldStructure.z3Polyfold_degIndex`). `PolyfoldClass.degIndex c` (`Quotient.lift`); `definition_16_1_14`: `c.degIndex z = d_X(x)` for every representative and every x over z. Witness `z3Polyfold_degIndex`. *Book:* §16.1, Definition 16.1.14, p. 712 (2017: Definition 16.9). *Hidden hypotheses:* none. *Deviation (same):* none.

V16.10 (Definition 16.1.17 (2017: 16.11), the boundary of a polyfold; `Polyfold.PolyfoldClass.boundary`, `Polyfold.PolyfoldClass.mem_boundary_iff`, `Polyfold.PolyfoldStructure.z3Polyfold_boundary`). `PolyfoldClass.boundary c := {z | 1  c.degIndex z}`; `mem_boundary_iff` (through any representative). Witness `z3Polyfold_boundary` ($= \emptyset$). *Book:* §16.1, Definition 16.1.17, p. 713 (2017: Definition 16.11). *Hidden hypotheses:* none. *Deviation (same):* none.

V16.11 (Lemma 16.1.19 (2017: 16.2), tameness is invariant; `Polyfold.PolyfoldStructure.lemma_16_1_19`). `PolyfoldStructure.lemma_16_1_19 (h : Equiv p q) : p.X.X.IsTame  q.X.X.IsTame` (Chapter 11's `IsEquivalence.isTame_iff`). *Book:* §16.1, Lemma 16.1.19, p. 714 (2017: Lemma 16.2). *Hidden hypotheses:* none. *Deviation (stronger):* An iff, trivially by symmetry.

V16.12 (Definition 16.1.20 (2017: 16.12), tame polyfolds; `Polyfold.PolyfoldClass.IsTame`, `Polyfold.PolyfoldClass.definition_16_1_20`, `Polyfold.PolyfoldStructure.z3Polyfold_isTame`). `PolyfoldClass.IsTame c :=  p, ⟦p⟧ = c  p.X.X.IsTame`; `definition_16_1_20`: equivalent to every representative tame. Witness `z3Polyfold_isTame`. *Book:* §16.1, Definition 16.1.20, p. 714 (2017: Definition 16.12). *Hidden hypotheses:* none. *Deviation (same):* none.

V16.13 (The polyfold $\mathbb{R}$ with nontrivial isotropy (witness); `Polyfold.EpGroupoid.Examples.injective_ _trivialZ3`, `Polyfold.EpGroupoid.Examples.isHomeomorph_ _trivialZ3`, `Polyfold.EpGroupoid.Examples.trivialZ3OrbitHomeomorph`, `Polyfold.EpGroupoid.Examples.hasParacompactOrbitSpace_trivialZ3`, `Polyfold.PartitionOfUnity.comapP3Z`, `Polyfold.EpGroupoid.descendP3Z`, `Polyfold.EpGroupoid.preimage_closure_ `, `Polyfold.MPolyfold.isScSmoothMap_real_of_locally`). The trivial $\mathbb{Z}/3$-ep-groupoid on $\mathbb{R}$: π is injective and a homeomorphism onto $\mathbb{R}$ (`injective_ _trivialZ3`, `isHomeomorph_ _trivialZ3`, `trivialZ3OrbitHomeomorph`), with paracompact orbit space (`hasParacompactOrbitSpace_trivialZ3`); tools `PartitionOfUnity.comapP3Z`, `EpGroupoid.descendP3Z`, `preimage_closure_`, `MPolyfold.isScSmoothMap_real_of_locally`. *Book:* not in the book (witness for §16.1). *Hidden hypotheses:* none. *Deviation (extra):* none.

V16.14 (The remark after Definition 16.1.1 (2017: Proposition 16.1), polyfolds are metrizable; `Polyfold.PolyfoldStructure.metrizableSpace`). `PolyfoldStructure.metrizableSpace (p : PolyfoldStructure Z) : MetrizableSpace Z`, from Theorem 7.3.1's `theorem_7_3_1_metrizable` (paracompact, locally metrizable and Hausdorff orbit space) and β. *Book:* §16.1, after Definition 16.1.1, p. 706 (2017: Proposition 16.1). *Hidden hypotheses:* none. *Deviation (same):* The metrization step is Part I's `metrizableSpace_of_locally` (Chapter 2), so no Smirnov theorem from Mathlib is needed.

V16.15 (Lemma 16.2.1 (2017: 16.3), equal underlying maps give equal $\mathfrak{f}$ and $T\mathfrak{f}$; `Polyfold.PolyfoldStructure.Lemma_16_2_1`, `Polyfold.PolyfoldStructure.not_lemma_16_2_1`, `Polyfold.PolyfoldStructure.lemma_16_2_1_counterexample`, `Polyfold.PolyfoldStructure.lemma_16_2_1_orbit`, `Polyfold.PolyfoldStructure.lemma_16_2_1_of_isEffective`, `Polyfold.PolyfoldStructure.lemma_16_2_1_of_isEffective_hypotheses`, `Polyfold.EPLocalization.EPLoc.tangentHom`, `Polyfold.EPLocalization.EPLoc.orbitMap_tangentHom_eq`). `PolyfoldStructure.Lemma_16_2_1 : Prop` is the printed statement ($\mathfrak{f}, \mathfrak{g}$ generalized isomorphisms, $f \circ \alpha = \beta \circ |\mathfrak{f}|$, $g \circ \alpha = \beta \circ |\mathfrak{g}|$, $f = g$); `T` is `EPLoc.tangentHom` , the class of $[TX \leftarrow TA \rightarrow TY]$. `not_lemma_16_2_1 : ¬ Lemma_16_2_1.{0}` (data `lemma_16_2_1_counterexample`). What holds: `lemma_16_2_1_orbit` (under the printed hypotheses, $|\mathfrak{f}| = |\mathfrak{g}|$ and $|T\mathfrak{f}| = |T\mathfrak{g}|$) and `lemma_16_2_1_of_isEffective` (for Y effective, $\mathfrak{f} = \mathfrak{g}$ and $T\mathfrak{f} = T\mathfrak{g}$), with witness `lemma_16_2_1_of_isEffective_hypotheses` (the point ep-groupoid). *Book:* §16.2, Lemma 16.2.1, p. 716 (2017: Lemma 16.3). *Hidden hypotheses:* Y effective (`lemma_16_2_1_of_isEffective`, witness `lemma_16_2_1_of_isEffective_hypotheses`); without it only the orbit form `lemma_16_2_1_orbit` holds. *Deviation (narrower):* The printed statement does not hold as we read it (`not_lemma_16_2_1`; Findings, Chapter 16, item 6): $Z = W = \mathbb{R}$ with $\mathbb{Z}/3$ acting trivially (ZHOU, $|X|$ paracompact), $f = g = \mathrm{id}$, $\mathfrak{f} = [X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X] \neq [1_X] = \mathfrak{g}$, the same pair as `not_uniqueness_16_1_2` (items 1–2). The proof's last step is Theorem 10.3.19, which needs an effective target (`not_theorem_10_3_19`). Whether $T\mathfrak{f} \neq T\mathfrak{g}$ in the counterexample is not decided; every later use in §16.2 needs only $|T\mathfrak{f}|$ (V16.16).

V16.16 (The construction of $T(Z, c)$, the maps $T\alpha$ and (16.5); `Polyfold.PolyfoldClass.TangentSpace`, `Polyfold.PolyfoldClass.tangentMap`, `Polyfold.PolyfoldClass.tangentMap_eq`, `Polyfold.PolyfoldClass.tangentMap_comp`, `Polyfold.PolyfoldClass.isHomeomorph_tangentMap`, `Polyfold.PolyfoldClass.tangentHomeomorph`). `PolyfoldClass.TangentSpace c := |T X |` with (X_0, α_0) = `c.out`; `tangentMap q hq` $= T\alpha := |T\mathfrak{f}|$ for any $\mathfrak{f} : X \to X_0$ with $\alpha_0 \circ |\mathfrak{f}| = \alpha$, independent of $\mathfrak{f}$ (`tangentMap_eq`); `tangentMap_comp` is (16.5) for every h with $\beta \circ |h| = \alpha$; `isHomeomorph_tangentMap`, `tangentHomeomorph`. *Book:* §16.2, pp. 716–718 (2017: §16.2). *Hidden hypotheses:* none. *Deviation (same):* X_0 is a base point instead of the gluing over all structures. $\mathfrak{f}$ need not be unique (V16.15); the independence of $\mathfrak{f}$ is `EPLoc.orbitMap_tangentHom_` eq: (LR), from ZHOU, gives local families $F(z) \to G(z)$ for $|F| = |G|$ (Proposition 10.1.10 (1)), hence $|TF| = |TG|$ (10.1.10 (2)). The projection τ_Z and the raised polyfolds are gap G.156.

V16.17 (Lemma 16.2.2 (2017: 16.4), the topology of $T(Z, c)$; `Polyfold.PolyfoldClass.lemma_16_2_2`, `Polyfold.PolyfoldClass.tangentSpaceTopology`). `PolyfoldClass.lemma_16_2_2 c`: every `tangentMap q hq` is a homeomorphism for `tangentSpaceTopology c`; a topology for which one $T\alpha$ is a homeomorphism equals it; `MetrizableSpace (TangentSpace c)` (Theorem 8.1.15's `metrizableSpace_tangent_orbitSpace`). *Book:* §16.2, Lemma 16.2.2, p. 718 (2017: Lemma 16.4). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.18 (Definition 16.2.3 (2017: 16.14), the tangent polyfold $(T(Z, c), Tc)$; `Polyfold.PolyfoldClass.tangentStructure`, `Polyfold.PolyfoldClass.tangentStructure_equiv`, `Polyfold.PolyfoldClass.tangent`, `Polyfold.PolyfoldClass.definition_16_2_3`, `Polyfold.PolyfoldStructure.tangent_paracompact`, `Polyfold.PolyfoldStructure.tangent_isZhou`). `PolyfoldClass.tangentStructure q hq : PolyfoldStructure (TangentSpace c)` ($|TX|$ paracompact by Theorem 8.1.15, `tangent_paracompact`; TX satisfies ZHOU by Proposition 8.1.16, `tangent_isZhou`), `tangentStructure_equiv`, `PolyfoldClass.tangent c`, `definition_16_2_3` ($[(TX, T\alpha)] = Tc$ for every $(X, \alpha) \in c$). Instance check: `example : PolyfoldClass (TangentSpace z3Polyfold)`. *Book:* §16.2, Definition 16.2.3, p. 719 (2017: Definition 16.14). *Hidden hypotheses:* none. *Deviation (same):* `TangentSpace c : Type (max u 1)` and `tangentAny : EP.{u} → EP.{max u 1}`, as in Chapter 8. "Equivalent via $T\mathfrak{f}$" is shown with Th for any h with $\beta \circ |h| = \alpha$.

V16.19 (The remark on p. 720: $h_*\omega$ does not depend on h; `Polyfold.EPLocalization.EPLoc.formsPush_eq_of_orbitMap_eq_or`, `Polyfold.EPLocalization.EPLoc.formsPullback_eq_of_orbitMap_eq_or`). Proposition 11.2.9 (`EPLoc.formsPush_eq_of_orbitMap_eq`, Part II) and its forms with (LR) on either side, `EPLoc.formsPush_eq_of_orbitMap_eq_or`, `formsPullback_eq_of_orbitMap_eq_or` (Proposition 10.3.16). *Book:* §16.2, before Definition 16.2.5, p. 720 (2017: before Definition 16.15). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.20 (Definition 16.2.5 (2017: 16.15), sc-differential forms on a polyfold; `Polyfold.PolyfoldClass.FormRep`, `Polyfold.PolyfoldClass.FormRep.Rel`, `Polyfold.PolyfoldClass.ScForm`, `Polyfold.PolyfoldClass.ScForm.mk`, `Polyfold.PolyfoldClass.ScForm.toStructure`, `Polyfold.PolyfoldClass.ScForm.equivStructure`, `Polyfold.PolyfoldClass.ScForm.mk_add`, `Polyfold.PolyfoldClass.ScForm.mk_smul`, `Polyfold.PolyfoldClass.ScForm.equivStructure`). `PolyfoldClass.FormRep`, `FormRep.Rel` (an equivalence relation), `ScForm c k` (the quotient), `ScForm.mk`; `ScForm.toStructure`, `equivStructure` (for every $p \in c$ a bijection to $\Omega^k_{ep,\infty}(X)$); the vector-space structure with `mk_add`, `mk_smul` (the text's formulas for every representative) and `equivStructure` . *Book:* §16.2, Definition 16.2.5, p. 720 (2017: Definition 16.15). *Hidden hypotheses:* none. *Deviation (same):* The vector-space structure is transported from `c.out` and then shown to satisfy the text's formula for every representative. The representatives are related by f^* (`ScForm.toStructure_eq_formsPullback`, in the §16.4 file).

V16.21 (Theorem 16.2.6 (2017: 16.2), the de Rham complex of a polyfold; `Polyfold.`

`PolyfoldClass.theorem_16_2_6_complex`, `Polyfold.PolyfoldClass.ScForm.d`, `Polyfold.PolyfoldClass.ScForm.d_mk`, `Polyfold.PolyfoldClass.ScForm.d_comp_d`, `Polyfold.PolyfoldClass.MapRep.pullback`, `Polyfold.PolyfoldClass.MapRep.pullback_eq_of_equiv`, `Polyfold.PolyfoldClass.ScMap.pullback`, `Polyfold.PolyfoldClass.ScMap.pullback_mk`, `Polyfold.PolyfoldClass.ScMap.d_comp_pullback`, `Polyfold.PolyfoldClass.ScMap.pullback_id`, `Polyfold.PolyfoldClass.ScMap.pullback_mk_comp`, `Polyfold.PolyfoldClass.ScMap.pullback_comp`, `Polyfold.PolyfoldClass.theorem_16_2_6_pullback`). `PolyfoldClass.theorem_16_2_6_complex c k`: $d[(\omega, q)] = [(d\omega, q)]$, $d \circ d = 0$, and $\omega \mapsto [(\omega, q)]$ is the linear isomorphism `equivStructure .symm`; `ScForm.d`, `d_mk`, `d_comp_d`.

The pull-back by sc-smooth maps of polyfolds: `ScMap.pullback F k` (linear). It is well defined on every class, with no effectiveness assumption (`MapRep.pullback_eq_of_equiv`). `pullback_mk`: the text's formula, for every representative. `d_comp_pullback`, `pullback_id`. `pullback_mk_comp`: $(\widehat{g} \circ \widehat{f})^* = \widehat{f}^* \circ \widehat{g}^*$ on representatives. `pullback_comp`: the same for an effective middle polyfold. `PolyfoldClass.theorem_16_2_6_pullback`. *Book:* §16.2, Theorem 16.2.6, p. 721 (2017: Theorem 16.2). *Hidden hypotheses:* none. *Deviation (same):* The pull-back is well defined on every class with no effectiveness assumption (`MapRep.pullback_eq_of_equiv`). The functoriality $(\widehat{g} \circ \widehat{f})^* = \widehat{f}^* \circ \widehat{g}^*$, which the theorem does not state, holds on representatives (`pullback_mk_comp`) and on classes for an effective middle polyfold (`pullback_comp`), where the composition of Definition 16.1.10 is defined (V16.32).

V16.22 (Definition 16.4.1 (2017: 16.29), branched suborbifolds of a polyfold; `Polyfold.PolyfoldClass.BranchedRep`, `Polyfold.PolyfoldClass.BranchedRep.Rel`, `Polyfold.PolyfoldClass.BranchedSuborbifold`, `Polyfold.PolyfoldClass.BranchedSuborbifold.toFun`, `Polyfold.PolyfoldClass.BranchedSuborbifold.definition_16_4_1_wellDefined`, `Polyfold.EPLocalization.EPLoc.pullTheta_eq_of_orbitMap`, `Polyfold.EPLocalization.EPLoc.pullTheta_id_of_isFunctorial`). `PolyfoldClass.BranchedRep`, `BranchedRep.Rel` (an equivalence relation), `BranchedSuborbifold c`; `BranchedSuborbifold.toFun` (θ), `definition_16_4_1_wellDefined` ($\theta(\alpha|x|) = \Theta(x)$ for every representative); key lemma `EPLoc.pullTheta_eq_of_orbitMap` ($(f^*\Theta')(x) = \Theta'(y)$ when $|f||x| = |y|$), `EPLoc.pullTheta_id_of_isFunctorial`. *Book:* §16.4, Definition 16.4.1, p. 730 (2017: Definition 16.29). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.23 (Definition 16.4.2 (2017: 16.30), closed, compact, manifold-type, orbifold-type, tame; `Polyfold.PolyfoldClass.BranchedSuborbifold.HasRepProperty`, `Polyfold.PolyfoldClass.BranchedSuborbifold.hasRepProperty_iff`, `Polyfold.PolyfoldClass.BranchedSuborbifold.IsClosed`, `Polyfold.PolyfoldClass.BranchedSuborbifold.IsCompact`, `Polyfold.PolyfoldClass.BranchedSuborbifold.IsManifoldType`, `Polyfold.PolyfoldClass.BranchedSuborbifold.IsOrbifoldType`, `Polyfold.PolyfoldClass.BranchedSuborbifold.IsTame`, `Polyfold.PolyfoldClass.BranchedSuborbifold.definition_16_4_2_closed`, `Polyfold.PolyfoldClass.BranchedSuborbifold.definition_16_4_2_compact`, `Polyfold.PolyfoldClass.BranchedSuborbifold.definition_16_4_2_manifold`, `Polyfold.PolyfoldClass.BranchedSuborbifold.definition_16_4_2_orbifold`, `Polyfold.`

`PolyfoldClass.BranchedSuborbifold.definition_16_4_2_nDim`, `Polyfold.EpGroupoid.TameBranchingStructure.restrictRep`, `Polyfold.EpGroupoid.TameBranchingStructure.move`, `Polyfold.EpFunctor.isTameBranched_pullbackTheta`, `Polyfold.EpFunctor.isTameBranched_pushforwardTheta`, `Polyfold.EpFunctor.isTameBranched_pullbackTheta_iff`,`Polyfold.EPLocalization.EPLoc.isTameBranched_pullTheta`,`Polyfold.PolyfoldClass.BranchedSuborbifold.definition_16_4_2_tame`, `Polyfold.PolyfoldClass.OrientedRep.Rel.isTameOriented`, `Polyfold.PolyfoldClass.OrientedRep.Rel.isTameOriented_iff`, `Polyfold.PolyfoldClass.OrientedSuborbifold.IsTameOriented`, `Polyfold.PolyfoldClass.OrientedSuborbifold.isTameOriented_iff`). `HasRepProperty P` (some representative has P), `hasRepProperty_iff` (for P stable under pull-backs, some iff all); `IsClosed`, `IsCompact`, `IsManifoldType`, `IsOrbifoldType`, `IsNDimensional`, each with the all-representatives form `definition_16_4_2_closed`, `_compact`, `_manifold`, `_orbifold`, `_nDim` (from Theorem 11.3.3 (3)–(6)); `IsTame` (Definition 9.3.6): some representative is tame, and `definition_16_4_2_tame` gives every representative (G.154 closed).

Tameness is invariant: `TameBranchingStructure.restrictRep` and `move`: D9.3.6 structures restricted, and moved by sc-diffeomorphisms. `EpFunctor.isTameBranched_pullbackTheta`, `isTameBranched_pushforwardTheta`, `isTameBranched_pullbackTheta_iff`, along transport charts as in Theorem 11.4.1. `EPLoc.isTameBranched_pullTheta`. **`BranchedSuborbifold.definition_16_4_2_tame`**: θ is tame iff every representative is. Oriented: `OrientedRep.Rel.isTameOriented(_iff)`, from the structures of Theorem 11.4.1 (`hcov_pullTheta`). `OrientedSuborbifold.IsTameOriented` with `isTameOriented_iff` (all representatives). *Book:* §16.4, Definition 16.4.2, p. 730, and p. 729 (2017: Definition 16.30). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.24 (Proposition 16.4.3 (2017: 16.6), the structure on $\operatorname{supp}\theta$; `Polyfold.PolyfoldClass.BranchedSuborbifold.supp`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_supp`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_homeomorph`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_manifold`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_orbifold`, `Polyfold.PolyfoldClass.BranchedSuborbifold.suppTransition`, `Polyfold.PolyfoldClass.BranchedSuborbifold. _suppTransition`, `Polyfold.PolyfoldClass.BranchedSuborbifold.suppTransition_eq_iff`, `Polyfold.PolyfoldClass.BranchedSuborbifold.suppTransition_self`, `Polyfold.PolyfoldClass.BranchedSuborbifold.suppTransition_trans`, `Polyfold.PolyfoldClass.BranchedSuborbifold.suppTransition_symm`, `Polyfold.PolyfoldClass.BranchedSuborbifold.IsInducedSuppStructure`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality_statement`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality_self`, `Polyfold.PolyfoldClass.BranchedSuborbifold.suppTransition_orbitSuppProj`, `Polyfold.PolyfoldClass.BranchedSuborbifold.isScSmoothOn_suppTransition`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality_all`, `Polyfold.PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality`). `proposition_16_4_3_supp` ($\operatorname{supp}\theta = \alpha(|\operatorname{supp}\Theta|)$), `proposition_16_4_3_homeomorph` ($|\operatorname{supp}\Theta| \simeq_t \operatorname{supp}\theta$), `proposition_16_4_3_manifold` (with Proposition 9.1.9 (1): an M-polyfold structure on $|\mathrm{supp}\,\Theta|$ for which $\mathrm{supp}\,\Theta \to |\mathrm{supp}\,\Theta|$ is a local sc-diffeomorphism), `proposition_16_4_3_orbifold` (Proposition 9.1.9 (2)).

Independence of the representative, for the topology: `suppTransition a a' h : |supp Θ| |supp Θ'|` with `_suppTransition` (α' □ t = α), `suppTransition_self`, `suppTransition_trans`, `suppTransition_symm`; `IsInducedSuppStructure a PZ` (Proposition 9.1.9 (1)); `proposition_16_4_3_naturality_statement c` (Prop: suitable induced structures make the transition an sc-diffeomorphism); `proposition_16_4_3_naturality_self` (the case `a' = a`).

Naturality (Part III wave 5, `Chapter16/P3W5PC_SuppNaturality.lean`): **`proposition_16_4_3_naturality c : proposition_16_4_3_naturality_statement c`**. `suppTransition_orbitSuppProj`: for a generalized isomorphism $[X \overset{F}{\leftarrow} A \overset{\Phi}{\rightarrow} X']$ between the representatives, the transition sends $|Fb|$ to $|\Phi b|$ for $b \in \mathrm{supp}(\Phi^*\Theta')$. `isScSmoothOn_suppTransition`: for any induced structures (Proposition 9.1.9 (1)) the transition is sc-smooth; near $|Fb|$ it is $p' \circ \Phi|_{\mathrm{supp}} \circ (F|_{\mathrm{supp}})^{-1} \circ p^{-1}$, with the restricted equivalences `EpFunctor.IsEquivalence.restrictSub` and the local sc-diffeomorphisms $a \mapsto |a|$. `proposition_16_4_3_naturality_all`: an sc-diffeomorphism for any induced structures, for Θ, Θ' of orbifold type. *Book:* §16.4, Proposition 16.4.3, p. 730 (2017: Proposition 16.6). *Hidden hypotheses:* none. *Deviation (same):* None in content. The structures are the induced structures of Proposition 9.1.9 (1), and `proposition_16_4_3_naturality_all` shows that the choice does not matter. Orbifold type is used; it follows from manifold type. Gap G.155 closed (Part III wave 5).

V16.25 (Definition 16.4.4 (2017: 16.31), oriented branched suborbifolds; `Polyfold.PolyfoldClass.OrientedRep`, `Polyfold.PolyfoldClass.OrientedRep.Rel`, `Polyfold.PolyfoldClass.OrientedSuborbifold`, `Polyfold.PolyfoldClass.OrientedSuborbifold.toBranched`, `Polyfold.EPLocalization.EPLoc.pull_id_of_mem`, `Polyfold.EPLocalization.EPLoc.pullSection_id_of_mem`). `PolyfoldClass.OrientedRep` (fields `p`, `mem`, `A : OrientedBranched p.X`, `smooth`), `OrientedRep.Rel` (an equivalence relation, by `OrientedBranched.pull`, `pull_comp`, `EPLoc.pull_id_of_mem`), `OrientedSuborbifold c`, `OrientedSuborbifold.toBranched`. *Book:* §16.4, Definition 16.4.4, pp. 730–731 (2017: Definition 16.31). *Hidden hypotheses:* none. *Deviation (same):* $\widehat{T}$ is required to vanish off X_∞ (field `smooth`): the text's $\widehat{T}_\Theta$ lives over X_∞, and f^* only reads X_∞ (`EPLoc.pullSection_congr`). Reflexivity needs it ($1^*\widehat{T} = \widehat{T}$ only for such $\widehat{T}$). Every Chapter 11 pull-back satisfies it (`pullSection_mem`), so it is a normalization, not an added hypothesis.

V16.26 (Definition 16.4.5 (2017: 16.34), the boundary $\partial\widehat{\theta}$; `Polyfold.PolyfoldClass.definition_16_4_5_wellDefined`, `Polyfold.PolyfoldClass.OrientedRep.boundaryFun`, `Polyfold.PolyfoldClass.OrientedRep.boundaryFun_eq_of_rel`). `definition_16_4_5_wellDefined` (related representatives, via f, have $f^*\partial\Theta' = \partial\Theta$ and $f^*\widehat{T}_{\partial\Theta'} = \widehat{T}_{\partial\Theta}$); `OrientedRep.boundaryFun` ($\partial\theta = |\partial\Theta| \circ \alpha^{-1}$), `boundaryFun_eq_of_rel` (independent of the representative). *Book:* §16.4, Definition 16.4.5, p. 731 (2017: Definition 16.34). *Hidden hypotheses:* none. *Deviation (same):* $\partial(\Theta, \widehat{T})$ is formalized componentwise (`boundaryTheta`, `boundaryOrientation`), as in Chapter 11, not as an `OrientedBranched`. The compatibility holds without tameness.

V16.27 (The integrals on p. 732 and their independence of the representative; `Polyfold.PolyfoldClass.OrientedRep.IsTameOriented`, `Polyfold.PolyfoldClass.OrientedRep.integral`, `Polyfold.PolyfoldClass.OrientedRep.integralEff`, `Polyfold.PolyfoldClass.OrientedRep.boundaryIntegral`, `Polyfold.PolyfoldClass.OrientedRep.boundaryIntegralEff`, `Polyfold.PolyfoldClass.OrientedRep.integral_eq_of_rel`, `Polyfold.PolyfoldClass.OrientedRep.integralEff_eq_of_rel`, `Polyfold.PolyfoldClass.OrientedRep.boundaryIntegral_eq_of_rel`, `Polyfold.PolyfoldClass.OrientedRep.boundaryIntegralEff_eq_of_rel`, `Polyfold.PolyfoldClass.OrientedRep.boundaryIntegralDim1_eq_of_rel`, `Polyfold.EPLocalization.EPLoc.tameBranchedIntegral_pull`, `Polyfold.EPLocalization.EPLoc.tameBranchedIntegralEff_pull`, `Polyfold.PolyfoldClass.OrientedSuborbifold.integral`, `Polyfold.PolyfoldClass.OrientedSuborbifold.integral_eq`, `Polyfold.PolyfoldClass.OrientedSuborbifold.integralEff`, `Polyfold.PolyfoldClass.OrientedSuborbifold.integralEff_eq`, `Polyfold.PolyfoldClass.OrientedSuborbifold.isCompactBranched_of_mk`). On a representative with `IsTameOriented a n` (Chapter 9's `hloc`): `OrientedRep.integral`, `integralEff` (LUC from ZHOU, Proposition 7.4.10), `boundaryIntegral`, `boundaryIntegralEff`. Independence among tame compact representatives: `integral_eq_of_rel` (Theorem 11.4.1 via `EPLoc.tameBranchedIntegral_pull`), `integralEff_eq_of_rel` (Theorem 11.4.2 via `EPLoc.tameBranchedIntegralEff_pull`), and for the boundary integrals `boundaryIntegral_eq_of_rel`, `boundaryIntegralEff_eq_of_rel`, `boundaryIntegralDim1_eq_of_rel` (with partitions of unity, from Stokes).

On classes: `OrientedSuborbifold.integral` and `integralEff` for tame compact $\hat{\theta}$ (`.toBranched.IsCompact`, `.IsTameOriented n`). `integral_eq` and `integralEff_eq`: the integral is computed on **every** representative. *Book:* §16.4, after Definition 16.4.5, p. 732 (2017: §16.4). *Hidden hypotheses:* the boundary-integral independence lemmas carry `hPU : c.AdmitsScPartitionsOfUnity` (from Stokes, V16.28), with the witness `theorem_16_4_6_hyp_witness`. *Deviation (narrower):* The integrals $\int_{\hat{\theta}}$, $\oint_{\hat{\theta}}$ are defined on classes (`OrientedSuborbifold.integral`, `integralEff`) and equal the integral on every representative (`integral_eq`, `integralEff_eq`); tameness transfers between representatives (`OrientedRep.Rel.isTameOriented`; gap G.154 closed). The boundary integrals on classes are not wrapped; they follow the same pattern from `boundaryIntegral_eq_of_rel` (about 40 lines).

V16.28 (Theorem 16.4.6 (2017: 16.5), Stokes for branched suborbifolds of a polyfold; `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_hyp`, `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_eff_hyp`, `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_dim1_hyp`, `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_eff_dim1_hyp`, `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_both_hyp`, `Polyfold.PolyfoldClass.OrientedRep.theorem_16_4_6_hyp_witness`). `OrientedRep.theorem_16_4_6_hyp (hPU : c.AdmitsScPartitionsOfUnity)` for $n = m + 1 \geq 2$ (from `EP.theorem_9_5_17`); `theorem_16_4_6_eff_hyp` (`EP.theorem_9_5_18`, LUC from ZHOU); `theorem_16_4_6_dim1_hyp`, `theorem_16_4_6_` `eff_dim1_hyp` for $n = 1$ (`theorem_9_5_17_dim1`, `theorem_9_5_18_dim1`); `theorem_16_4_6_both_hyp`. Witness: `theorem_16_4_6_hyp_witness` (= `z3Polyfold_admitsScPartitionsOfUnity`). *Book:* §16.4, Theorem 16.4.6, p. 732 (2017: Theorem 16.5). *Hidden hypotheses:* `hPU : c.AdmitsScPartitionsOfUnity` (Definition 16.1.5 for c; by D16.1.5 (2), `definition_16_1_5_two`, every representative then has them). Witness `theorem_16_4_6_hyp_witness`. Theorem 9.5.17, the ep-groupoid Stokes the text reduces to, assumes "admitting sc-smooth partitions of unity"; Theorem 16.4.6 does not state it, and the standing assumption of Chapter 16 is only ZHOU (p. 705). Removing it is gap G.153. *Deviation (narrower):* Tameness is Chapter 9's (oriented tame structures describing $\widehat{T}$), on the representative used (gap G.154).

V16.29 (The category PS of polyfold structures (2017: §16.1); `Polyfold.PSHom`, `Polyfold.PSHom.id`, `Polyfold.PSHom.comp`, `Polyfold.PSHom.f_eq`, `Polyfold.PSHom.continuous_f`, `Polyfold.PSHom.ext`, `Polyfold.PSHom.IsIdIso`, `Polyfold.PSHom.IsIdIso.inv`, `Polyfold.PSCat`). `PSHom p q` has fields `f`, `gen` and `comm`. `PSHom.id`, `PSHom.comp`, `PSHom.f_eq` ($f = \beta \circ |\mathfrak{f}| \circ \alpha^{-1}$), `continuous_f`, `ext`. `PSCat` with a Mathlib `Category` instance. `PSHom.IsIdIso` (an isomorphism (h, Id)) with `inv`, `comp_inv`, `inv_comp`. *Book:* §16.1, before Definition 16.1.6, p. 709 (2017: §16.1). *Hidden hypotheses:* none. *Deviation (same):* The morphism $\mathfrak{f}$ is a morphism of `EPLoc` (Chapter 10's localization), so composition is in `EPLoc`.

V16.30 (Definition 16.1.6 (2017: 16.5), equivalent morphisms of polyfold structures; `Polyfold.PSHom.Equiv`, `Polyfold.PSHom.Equiv.f_eq`, `Polyfold.PSHom.Equiv.refl`, `Polyfold.PSHom.Equiv.symm`, `Polyfold.PSHom.Equiv.trans`, `Polyfold.PolyfoldClass.MapRep`, `Polyfold.PolyfoldClass.MapRep.setoid`, `Polyfold.PolyfoldClass.ScMap.exists_rep`). `PSHom.Equiv` (literally the square in PS) and `Equiv.f_eq` ($f = f'$). `Equiv.refl/symm/trans`, `equiv_comp_left/right`. `MapRep c d` with `MapRep.setoid`. `ScMap.exists_rep`: every class has a representative between any two given structures. *Book:* §16.1, Definition 16.1.6, p. 710 (2017: Definition 16.5). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.31 (Definition 16.1.8 and Remark 16.1.9 (2017: 16.6, Remark 16.2), sc-smooth maps between polyfolds; `Polyfold.PolyfoldClass.ScMap`, `Polyfold.PolyfoldClass.ScMap.mk`, `Polyfold.PolyfoldClass.ScMap.toFun`, `Polyfold.PolyfoldClass.ScMap.remark_16_1_9`, `Polyfold.PolyfoldClass.ScMap.id`, `Polyfold.PolyfoldClass.ScMap.id_eq`). `PolyfoldClass.ScMap c d` (the quotient) and `ScMap.mk`. `ScMap.toFun`, and `ScMap.remark_16_1_9`: f is continuous and equals $\beta \circ |\mathfrak{f}| \circ \alpha^{-1}$ for every representative. `ScMap.id` with `id_eq`: the class of $([1_X], \mathrm{Id}_Z)$ for every $(X, \alpha) \in c$. *Book:* §16.1, Definition 16.1.8 and Remark 16.1.9, p. 711 (2017: Definition 16.6, Remark 16.2). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.32 (Definition 16.1.10 (2017: 16.7), composition of sc-smooth maps; `Polyfold.PSHom.Definition_16_1_10`, `Polyfold.PSHom.Equiv.comp_of_isEffective`, `Polyfold.PolyfoldClass.MapRep.comp`, `Polyfold.PolyfoldClass.MapRep.comp_equiv`, `Polyfold.PolyfoldClass.ScMap.comp`, `Polyfold.PolyfoldClass.ScMap.comp_mk`, `Polyfold.PolyfoldClass.ScMap.id_comp`, `Polyfold.PolyfoldClass.ScMap.comp_id`, `Polyfold.PolyfoldClass.ScMap.comp_assoc`, `Polyfold.CompositionCounterexample.not_definition_16_1_10`, `Polyfold.CompositionCounterexample.not_exists_comp`, `Polyfold.`

`CompositionCounterexample.not_equiv_comp`, `Polyfold.CompositionCounterexample.a_equiv`, `Polyfold.CompositionCounterexample.not_isEffective_k2`, `Polyfold.CompositionCounterexample.k2Structure`). `PSHom.Definition_16_1_10 : Prop` is the independence claim: equivalent pairs have equivalent composites. `not_definition_16_1_10 : ¬ PSHom.Definition_16_1_10.{0}`. `not_exists_comp`: no composition map on classes agrees with the representatives. Effective form: `PSHom.Equiv.comp_of_isEffective` holds for an effective middle structure. `MapRep.comp_equiv` and `ScMap.comp hd` (with `hd : d.IsEffective`), with `comp_mk`, `id_comp`, `comp_id`, `comp_assoc`. *Book:* §16.1, Definition 16.1.10, p. 711 (2017: Definition 16.7). *Hidden hypotheses:* none. *Deviation (narrower):* The independence of the choices does not hold as we read it (Findings, Chapter 16, item 12; the same cause as items 1, 2, 6 and 11: a non-effective isotropy group). For an effective middle polyfold it holds (`PSHom.Equiv.comp_of_isEffective`), and the composition `ScMap.comp hd` is defined with `hd : d.IsEffective`. **Counterexample.** Take $Z = W = V = \mathbb{R}$ and $(X, \alpha) = (U, \gamma) = [\mathbb{Z}/3 \ltimes \mathbb{R}]$ with the trivial action. Take $(Y, \beta) = [(\mathbb{Z}/3)^2 \ltimes \mathbb{R}]$, also trivial, which satisfies ZHOU and has $|Y| \cong \mathbb{R}$. All maps f, g are Id. Let $\mathfrak{f} = [\iota]$ with $\iota(g) = (g, 1)$, and $\mathfrak{f}' = [\iota] \gg [\sigma]$ with σ the swap, an automorphism over Id. Then $(\mathrm{Id}, \mathfrak{f}) \sim (\mathrm{Id}, \mathfrak{f}')$ (`a_equiv`). Let $\mathfrak{g} = [\mathrm{pr}_1]$. The composites are $[1_X]$ and $[T]$ with $T(g) = 1$. They are not equivalent, because $[T]$ would then be a generalized isomorphism, but T is not faithful (`not_equiv_comp`). The mechanism is that k in D16.1.6 is not determined by Id_W. This is the uniqueness after D16.1.2, which does not hold as we read it (`not_uniqueness_16_1_2`, Z). The middle structure is not effective (`not_isEffective_k2`). For an effective one, Theorem 10.3.19 makes k unique, and the claim holds. The counterexample has exactly the printed hypotheses: polyfold structures (ZHOU, paracompact) and morphisms of PS.

V16.33 (Definition 16.1.11 (2017: 16.8), the polyfold category; `Polyfold.EffPolyfold`, `Polyfold.EffPolyfold.point`, `Polyfold.PolyfoldClass.IsEffective`, `Polyfold.PolyfoldClass.isEffective_iff`, `Polyfold.EpFunctor.IsEquivalence.isEffective_iff`, `Polyfold.EPLoc.isEffective_iff_of_isGeneralizedIso`, `Polyfold.EP.isEffective_iff_trivialSet`). `EffPolyfold` (fields `Z`, `top`, `c`, `isEffective`) with a Mathlib `Category` instance, for **effective** polyfolds. The witness is `EffPolyfold.point`. `PolyfoldClass.IsEffective` with `isEffective_iff`: one structure is effective iff all are. The supporting invariance lemmas are `EpFunctor.IsEquivalence.isEffective_iff` and `EPLoc.isEffective_iff_of_isGeneralizedIso` (Remark 10.1.12). *Book:* §16.1, Definition 16.1.11, p. 711 (2017: Definition 16.8). *Hidden hypotheses:* none. *Deviation (narrower):* The category is built on effective polyfolds. By `not_definition_16_1_10`, the composition of D16.1.10 is not well defined on all polyfolds as we read it. The PS category itself (`PSCat`) has all structures.

V16.34 (Theorem 16.2.4 (2017: 16.1), the tangent functor; `Polyfold.PSCat.tangentFunctor`, `Polyfold.PSHom.ofGen'`, `Polyfold.PolyfoldStructure.tangentSelf`, `Polyfold.PolyfoldClass.tangent_isIdIso`, `Polyfold.PolyfoldClass.MapRep.tangent`, `Polyfold.PolyfoldClass.MapRep.tangent_equiv`, `Polyfold.PolyfoldClass.ScMap.tangent`, `Polyfold.PolyfoldClass.ScMap.tangent_mk`, `Polyfold.PolyfoldClass.ScMap.tangent_toFun`, `Polyfold.PolyfoldClass.ScMap.tangent_id`, `Polyfold.PolyfoldClass.ScMap.tangent_mk_comp`, `Polyfold.PolyfoldClass.ScMap.tangent_comp`, `Polyfold.PolyfoldClass.theorem_`

`16_2_4`, `Polyfold.linearMap_eq_self_of_eqOn_isOpen`, `Polyfold.EpGroupoid.levelVal_tangentProj_mem_smoothSet`, `Polyfold.EpGroupoid.tangent_isotropy_eq_one`, `Polyfold.EpGroupoid.isEffective_tangentEpGroupoid`, `Polyfold.PolyfoldClass.IsEffective.tangent`, `Polyfold.PolyfoldClass.ScMap.tangent_comp_of_isEffective`, `Polyfold.EffPolyfold.tangent`, `Polyfold.EffPolyfold.tangentFunctor`, `Polyfold.theorem_16_2_4_eff`). `PSCat.tangentFunctor : PSCat.{u}  PSCat.{max u 1}` is a Mathlib functor on PS, $(f, \mathfrak{f}) \mapsto (|T\mathfrak{f}|, T\mathfrak{f})$. `ScMap.tangent : ScMap c d → ScMap c.tangent d.tangent` is well defined on classes (`MapRep.tangent_equiv`). `tangent_toFun`: $Tf \circ T\alpha = T\beta \circ |T\mathfrak{f}|$. `tangent_id`: $T1 = 1$. `tangent_mk_comp`: T of a composite of representatives is the composite of the tangents. `tangent_comp`: $T(\widehat{g} \circ \widehat{f}) = T\widehat{g} \circ T\widehat{f}$ when `d` and `d.tangent` are effective. `PolyfoldClass.theorem_16_2_4` collects these.

Part III wave 5 (`Chapter16/P3W5PC_TangentEffective.lean`): `EpGroupoid.isEffective_tangentEpGroupoid (hE : X.IsEffective) (hZ : X.IsZhou) : X.tangentEpGroupoid.IsEffective`; `PolyfoldClass.IsEffective.tangent : c.IsEffective → c.tangent.IsEffective` ((ZHOU) is a field of polyfold structures); `ScMap.tangent_comp_of_isEffective (hd : d.IsEffective)`: $T(\widehat{g} \circ \widehat{f}) = T\widehat{g} \circ T\widehat{f}$; **`EffPolyfold.tangentFunctor : EffPolyfold.{u}  EffPolyfold.{max u 1}`**, a Mathlib functor with `map_id = tangent_id` and `map_comp = tangent_comp_of_isEffective`; `theorem_16_2_4_eff`. Helpers `linearMap_eq_self_of_eqOn_isOpen`, `EpGroupoid.levelVal_tangentProj_mem_smoothSet`, `EpGroupoid.tangent_isotropy_eq_one`. *Book:* §16.2, Theorem 16.2.4, p. 719 (2017: Theorem 16.1). *Hidden hypotheses:* none. *Deviation (narrower):* There is no category of all polyfolds (D16.1.10, V16.32), so the functor lives on effective polyfolds, where composition is defined as we read it: `EffPolyfold.tangentFunctor`, with no hypothesis beyond the text's (gap G.157 closed in Part III wave 5). The proof that T preserves effectiveness goes through a smooth point and uses (ZHOU), Definition 16.1.1 (2): effectiveness is a level-0 statement, while open sets of $T\mathbf{X}$ see only level 1. The hypothesis `d.tangent.IsEffective` of `tangent_comp` now holds for every effective `d` (`tangent_comp_of_isEffective`; witness `EffPolyfold.point.tangent`).

V16.35 (The raised polyfolds $(Z, c)^i = (Z_i, c_i)$; `Polyfold.EpFunctor.liftOne_comp`, `Polyfold.EpFunctor.liftOne_id`, `Polyfold.EpFunctor.orbitIncl_orbitMap_liftOne`, `Polyfold.EPLocalization.Diagram.liftOne`, `Polyfold.EPLocalization.Diagram.IsRefinement.liftOne`, `Polyfold.EPLocalization.Diagram.Equiv.liftOne`, `Polyfold.EPLocalization.Diagram.isRefinement_liftOne_id`, `Polyfold.EPLocalization.Diagram.isRefinement_liftOne_comp`, `Polyfold.EPLocalization.EPLoc.liftOneHom`, `Polyfold.EPLocalization.EPLoc.liftOneHom_id`, `Polyfold.EPLocalization.EPLoc.liftOneHom_comp`, `Polyfold.EPLocalization.EPLoc.isGeneralizedIso_liftOneHom`, `Polyfold.EPLocalization.EPLoc.orbitIncl_orbitMap_liftOneHom`, `Polyfold.EPLocalization.EPLoc.orbitMap_liftOneHom_eq`, `Polyfold.PolyfoldStructure.liftOne_paracompact`, `Polyfold.PolyfoldStructure.liftOne_isZhou`, `Polyfold.PolyfoldClass.RaisedSpace`, `Polyfold.PolyfoldClass.raiseMap`, `Polyfold.PolyfoldClass.raiseMap_eq`, `Polyfold.PolyfoldClass.raiseMap_comp`, `Polyfold.PolyfoldClass.raiseHomeomorph`, `Polyfold.PolyfoldClass.raiseStructure`, `Polyfold.PolyfoldClass.raiseStructure_equiv`, `Polyfold.PolyfoldClass.raise`, `Polyfold.`

PolyfoldClass.raise_eq, Polyfold.PolyfoldClass.raiseIncl, Polyfold.PolyfoldClass.range_raiseIncl, Polyfold.PolyfoldClass.PolyfoldPair, Polyfold.PolyfoldClass.raiseN). Generalized maps: Diagram.liftOne, EPLoc.liftOneHom (liftOneHom_homMk, liftOneHom_id, liftOneHom_comp, isGeneralizedIso_liftOneHom), orbitIncl_orbitMap_liftOneHom (|f^1| = |f| on |X^1| |X|). RaisedSpace c (|X ^{1}|, (X ,   ) = c.out), raiseMap (α^1, raiseMap_eq, raiseMap_comp, a homeomorphism), raiseStructure (ZHOU, paracompact), raise c (the class c₁) with raise_eq. raiseIncl c : RaisedSpace c → Z, injective, continuous, range_raiseIncl : range (raiseIncl c) = '' |X| for every (X,) c; raiseN c i (raiseN_zero, raiseN_succ). *Book:* §16.2, p. 715 (2017: §16.2). *Hidden hypotheses:* none. *Deviation (restated):* $(Z, c)^1$ is a space with an injection into Z with image Z_1 (its topology is that of $|X^1|$, not the subspace topology); mI is not introduced, $Z_i = \alpha(|X_i|)$.

V16.36 (The tangent projection $\tau_Z : T(Z, c) \to (Z, c)^1$; Polyfold.EpFunctor.orbitMap_tangentProj_tangent, Polyfold.EPLocalization.Diagram.orbitMap_tangentProj_tangentAny, Polyfold.EPLocalization.EPLoc.orbitMap_tangentProj_tangentHom, Polyfold.PolyfoldClass.tangentProj, Polyfold.PolyfoldClass.continuous_tangentProj, Polyfold.PolyfoldClass.tangentProj_tangentMap, Polyfold.PolyfoldClass.raiseIncl_tangentProj_tangentMap). PolyfoldClass.tangentProj c : TangentSpace c → RaisedSpace c, continuous_tangentProj, tangentProj_tangentMap (τ_Z □ Tα = α^1 □ |τ_X| for every (X,) c), raiseIncl_tangentProj_tangentMap (the square with values in Z); from EPLoc.orbitMap_tangentProj_tangentHom (τ is natural for generalized maps). *Book:* §16.2, p. 717 (2017: §16.2). *Hidden hypotheses:* none. *Deviation (restated):* the target is (Z, c)^1 as in V16.35 (α^1 in the square; composed with raiseIncl it is the text's α).

V16.37 (Definition 16.3.1 (2017: 16.16), strong polyfold bundle structures; Polyfold.StrongPolyfoldBundleStructure, Polyfold.StrongPolyfoldBundleStructure.p_eq, Polyfold.StrongPolyfoldBundleStructure.p_Γ, Polyfold.StrongPolyfoldBundleStructure.continuous_p, Polyfold.StrongPolyfoldBundleStructure.surjective_p, Polyfold.StrongPolyfoldBundleStructure.z3Structure). structure StrongPolyfoldBundleStructure (p : Y → Z) with fields A : SEP (the strong bundle), isZhou, paracompact (HasParacompactOrbitSpace), isLR : A.E.IsLR (D8.4.2), Γ : |W| Y, : |X| Z, comm : p Γ = |P|. Consequences: p_eq ($p = \gamma \circ |P| \circ \Gamma^{-1}$), p_Γ, continuous_p, surjective_p. *Book:* §16.3, Definition 16.3.1, pp. 721–722 (2017: Definition 16.16). *Hidden hypotheses:* none. *Deviation (same):* “Continuous and surjective” for p is not a field: it follows from the structure (continuous_p, surjective_p). Y, Z live in the universe of the object sets (as for PolyfoldStructure).

V16.38 (Lemma 16.3.2 (new in 2021), Y and Z are metrizable; Polyfold.StrongPolyfoldBundleStructure.lemma_16_3_2). lemma_16_3_2 (S : StrongPolyfoldBundleStructure p) : MetrizableSpace Y MetrizableSpace Z ($|X|$ by Theorem 7.3.1, $|W| = |W[0]|$ via Proposition 8.4.1's paracompactness). *Book:* §16.3, Lemma 16.3.2, p. 722 (new in 2021). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.39 (Definition 16.3.3 (2017: 16.17), equivalence of strong polyfold bun-

dle structures; `Polyfold.SBLocalization.SDiagram.orbitMap_comp'`, `Polyfold.SBLocalization.SDiagram.orbitMap_id'`, `Polyfold.SBLocalization.GenSBMap.orbitMap_comp`, `Polyfold.SBLocalization.GenSBMap.base_comp`, `Polyfold.SBLocalization.GenSBMap.baseOrbitMap_comp`, `Polyfold.SBLocalization.SEPLoc.orbitMap_comp`, `Polyfold.SBLocalization.SEPLoc.orbitMap_id`, `Polyfold.SBLocalization.SEPLoc.baseOrbitMap_comp`, `Polyfold.SBLocalization.SEPLoc.baseOrbitMap_id`, `Polyfold.SBLocalization.SEPLoc.orbitMap_inv_comp`, `Polyfold.SBLocalization.SEPLoc.orbitMap_comp_inv`, `Polyfold.SBLocalization.SEPLoc.baseOrbitMap_comp_inv`, `Polyfold.StrongPolyfoldBundleStructure.Equiv`, `Polyfold.StrongPolyfoldBundleStructure.Equiv.refl`, `Polyfold.StrongPolyfoldBundleStructure.Equiv.symm`, `Polyfold.StrongPolyfoldBundleStructure.Equiv.trans`, `Polyfold.StrongPolyfoldBundleStructure.setoid`, `Polyfold.StrongPolyfoldBundleStructure.equiv_iff`, `Polyfold.StrongPolyfoldBundleStructure.Γ_base`, `Polyfold.StrongPolyfoldBundleStructure.equiv_iff_exists_Γ`). `StrongPolyfoldBundleStructure.Equiv S S' :=  F : GenSBMap S.A S'.A, F.IsGeneralizedIso  '  |F.base| =  Γ'  |F| = Γ`; `Equiv.refl`, `Equiv.symm`, `Equiv.trans`, `setoid p`, `equiv_iff`. The strong bundle form of Proposition 10.3.11 needed for `trans` is new: `SDiagram.orbitMap_comp'`, `GenSBMap.orbitMap_comp`, `base_comp`, `baseOrbitMap_comp`, `SEPLoc.orbitMap_comp`, `orbitMap_id`, `baseOrbitMap_comp`, `baseOrbitMap_id`, `orbitMap_inv_comp`, `orbitMap_comp_inv`, `baseOrbitMap_comp_inv`. Also `Γ_base` (the γ-condition follows from the Γ-condition) and `equiv_iff_exists_Γ`. *Book:* §16.3, Definition 16.3.3, pp. 722–723 (2017: Definition 16.17). *Hidden hypotheses:* none. *Deviation (same):* The text asserts it is an equivalence relation implicitly; proved. The γ-condition is redundant (`Γ_base`).

V16.40 (The remark after Definition 16.3.3: uniqueness of f and F; `Polyfold.StrongPolyfoldBundleStructure.Uniqueness_16_3_3`, `Polyfold.StrongPolyfoldBundleStructure.uniqueness_16_3_3_of_isEffective`, `Polyfold.StrongPolyfoldBundleStructure.z3Structure`, `Polyfold.StrongPolyfoldBundleStructure.z3Bundle`, `Polyfold.StrongPolyfoldBundleStructure.z3_orbitMap_id`, `Polyfold.StrongPolyfoldBundleStructure.z3_baseOrbitMap_id`, `Polyfold.StrongPolyfoldBundleStructure.uniqueness_16_3_3_counterexample`, `Polyfold.StrongPolyfoldBundleStructure.not_uniqueness_16_3_3`, `Polyfold.StrongPolyfoldBundleStructure.not_isEffective_z3Structure`). `not_uniqueness_16_3_3 : ¬ Uniqueness_16_3_3.{0}`, where `Uniqueness_16_3_3` is the claim (for all p, S, S' and generalized strong bundle isomorphisms F, F' satisfying both conditions: `F.base = F'.base  F = F'`). Counterexample `uniqueness_16_3_3_counterexample k`: for `z3Structure k` ($\mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with trivial actions; (ZHOU) `isZhou_trivialZ3`, paracompact `hasParacompactOrbitSpace_trivialZ3`, (LR) `isLR_z3EpBundle`; $\Gamma = \gamma = \mathrm{id}$), the generalized strong bundle isomorphisms $[W \xleftarrow{1} W \xrightarrow{\mathrm{Inv}} W]$ and $[1_W]$ of Remark 10.5.5 both satisfy $\gamma' \circ |f| = \gamma$ and $\Gamma' \circ |F| = \Gamma$, and differ already on the bases. Positive form `uniqueness_16_3_3_of_isEffective (hS' : S'.A.X.IsEffective)`: then `F.base = F'.base  F = F'` (Theorem 10.5.7 for an effective base; (ZHOU) gives (LR) by `proposition_7_4_22_of_isZhou`). `not_isEffective_z3Structure`. *Book:* §16.3, after Definition 16.3.3, p. 723 (2017: after Definition

16.17). *Hidden hypotheses:* none. *Deviation (narrower):* The uniqueness does not hold as we read it (Findings, Chapter 16, item 11; the same cause as items 1, 2, 6 and 12: a non-effective isotropy group). It holds for an effective target base (`uniqueness_16_3_3_of_isEffective`, Theorem 10.5.7 for an effective base; (ZHOU) gives (LR) by `proposition_7_4_22_of_isZhou`). Definitions 16.3.3 and 16.3.4 do not depend on the uniqueness.

V16.41 (The polyfold induced by a strong polyfold bundle; `Polyfold.StrongPolyfoldBundleStructure.toPolyfoldStructure`, `Polyfold.StrongPolyfoldBundleStructure.toPolyfoldStructure_equiv`, `Polyfold.StrongPolyfoldBundle.polyfold`, `Polyfold.StrongPolyfoldBundle.polyfold_mk`). `toPolyfoldStructure S = S.A.baseEP, S.paracompact, S.isZhou, S. ,` `toPolyfoldStructure_equiv` (equivalent bundle structures give equivalent polyfold structures, via `F.base`), `StrongPolyfoldBundle.polyfold` (`Quotient.map`), `polyfold_mk`. *Book:* §16.3, after Definition 16.3.3, p. 723 (2017: after Definition 16.17). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.42 (Definition 16.3.4 (2017: 16.18), strong polyfold bundles; `Polyfold.StrongPolyfoldBundle`, `Polyfold.StrongPolyfoldBundleStructure.z3Bundle`). `abbrev StrongPolyfoldBundle p := Quotient (StrongPolyfoldBundleStructure.setoid p)`; witness `z3Bundle k`. *Book:* §16.3, Definition 16.3.4, p. 723 (2017: Definition 16.18). *Hidden hypotheses:* none. *Deviation (same):* Continuity and surjectivity of p follow from any representative (`continuous_p`, `surjective_p`).

V16.43 (Definition 16.3.5 (2017: 16.19), tame and face-structured strong polyfold bundles; `Polyfold.PolyfoldClass.IsWeaklyFaceStructured`, `Polyfold.PolyfoldClass.IsFaceStructured`, `Polyfold.PolyfoldClass.isFaceStructured_iff`, `Polyfold.StrongPolyfoldBundle.IsTame`, `Polyfold.StrongPolyfoldBundle.IsWeaklyFaceStructured`, `Polyfold.StrongPolyfoldBundle.IsFaceStructured`, `Polyfold.StrongPolyfoldBundle.isTame_iff`, `Polyfold.StrongPolyfoldBundle.z3Bundle_isTame`). `StrongPolyfoldBundle.IsTame c := c.polyfold.IsTame` (D16.1.20), `isTame_iff` (iff every representative's base is tame); `IsWeaklyFaceStructured`, `IsFaceStructured` through `PolyfoldClass.IsWeaklyFaceStructured`, `PolyfoldClass.IsFaceStructured` (some representative weakly face structured / face structured, D11.1.3), `PolyfoldClass.isFaceStructured_iff` (one representative iff all, Remark 11.1.7). Witness `z3Bundle_isTame`. *Book:* §16.3, Definition 16.3.5, p. 723 (2017: Definition 16.19). *Hidden hypotheses:* none. *Deviation (restated):* Definition 16.1.21 (face-structured polyfolds) is not yet in the project; (2), (3) use the representative forms of the Lean column. Invariance is proved for (3); for (2) "some representative" is the reading (no invariance lemma for weak face structure under generalized isomorphisms exists yet). (1) is the text's.

V16.44 (Definition 16.3.6 (2017: 16.20), the category $\mathscr{SPBS}$; `Polyfold.SPBS`, `Polyfold.SPBS.mk'`, `Polyfold.SPBS.Hom`, `Polyfold.SPBS.Hom.p_comm`, `Polyfold.SPBS.Hom.id`, `Polyfold.SPBS.Hom.comp`, `Polyfold.SPBS.category`, `Polyfold.SPBS.comp_` , `Polyfold.SPBS.id_` , `Polyfold.SPBS.Hom.IsInvertible`, `Polyfold.SPBS.Hom.inv`, `Polyfold.SPBS.Hom.comp_inv`, `Polyfold.SPBS.Hom.inv_comp`, `Polyfold.SPBS.isIso_iff_isInvertible`, `Polyfold.StrongPolyfoldBundle.z3_id_isInvertible`). `structure SPBS` (fields `Y Z p S`), `SPBS.mk'`; `SPBS.Hom` (fields `F f` continuous_F continuous_f Γ_comm _comm, $\mathfrak{a} =$ .base); Hom.p_comm (derived); Hom.id, Hom.comp, instance category : Category.{u+1} SPBS.{u}; Hom.IsInvertible, Hom.inv, Hom.comp_inv, Hom.inv_comp, isIso_iff_isInvertible (invertible in the text's sense iff IsIso in the category). Witness z3_id_isInvertible. *Book:* §16.3, Definition 16.3.6, p. 724 (2017: Definition 16.20). *Hidden hypotheses:* none. *Deviation (same):* $p' \circ A = a \circ p$ is not a field: it follows from the two squares (Hom.p_comm). Category laws proved.

V16.45 (The class $\bar{c}$ through isomorphisms $(\mathfrak{A}, \mathrm{Id}_Y)$; Polyfold.SPBS.Hom.f_eq_id, Polyfold.SPBS.exists_isInvertible_iff_equiv, Polyfold.SPBS.inv_F_eq_id). exists_isInvertible_iff_equiv: for S, S' over the same p, an invertible morphism $(\mathfrak{A}, \mathrm{Id}_Y)$ exists iff Equiv S S' (D16.3.3); Hom.f_eq_id ($A = \mathrm{id}$ forces $a = \mathrm{id}$), inv_F_eq_id. *Book:* §16.3, before Definition 16.3.7, p. 725 (2017: before Definition 16.21). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.46 (Definition 16.3.7 (2017: 16.21), equivalence of morphisms in $\mathscr{SPBS}$; Polyfold.SPBS.MorEquiv, Polyfold.SPBS.MorEquiv.F_eq, Polyfold.SPBS.MorEquiv.f_eq, Polyfold.SPBS.MorEquiv.refl, Polyfold.SPBS.MorEquiv.symm, Polyfold.SPBS.MorEquiv.trans). MorEquiv m m (the defining equation in the category, with IsIso of both special morphisms); MorEquiv.F_eq ($A_1 = A_0$), MorEquiv.f_eq ($a_1 = a_0$), MorEquiv.refl/symm/trans. *Book:* §16.3, Definition 16.3.7, p. 725 (2017: Definition 16.21). *Hidden hypotheses:* none. *Deviation (same):* The consequences $A_1 = A_0$, $a_1 = a_0$ are proved; that it is an equivalence relation is proved.

V16.47 (Definition 16.3.8 (2017: 16.22), sc-smooth strong polyfold bundle maps; Polyfold.SPBS.MorRep, Polyfold.SPBS.MorRep.Rel, Polyfold.SPBS.MorRep.setoid, Polyfold.SPBS.StrongPolyfoldBundleMap, Polyfold.SPBS.StrongPolyfoldBundleMap.toFun, Polyfold.SPBS.StrongPolyfoldBundleMap.baseFun, Polyfold.SPBS.StrongPolyfoldBundleMap.continuous_toFun, Polyfold.SPBS.StrongPolyfoldBundleMap.p_comm, Polyfold.SPBS.StrongPolyfoldBundleMap.id, Polyfold.SPBS.StrongPolyfoldBundleMap.toFun_id). structure MorRep c c' (structures $S \in \bar{c}$, $S' \in \bar{c}'$ and a morphism mk' S mk' S'), MorRep.Rel (= MorEquiv), MorRep.setoid, abbrev StrongPolyfoldBundleMap c c'; toFun (A, well defined by MorEquiv.F_eq), baseFun, continuous_toFun, p_comm, id, toFun_id. *Book:* §16.3, Definition 16.3.8, p. 726 (2017: Definition 16.22). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.48 (Definition 16.3.9 (2017: 16.23) and (16.7), auxiliary norms; Polyfold.StrongPolyfoldBundleStructure.oneToY, Polyfold.StrongPolyfoldBundleStructure.Y01, Polyfold.StrongPolyfoldBundleStructure.normOf, Polyfold.StrongPolyfoldBundleStructure.IsAuxNormFor, Polyfold.StrongPolyfoldBundleStructure.normOrbit, Polyfold.StrongPolyfoldBundleStructure.continuous_normOrbit, Polyfold.StrongPolyfoldBundleStructure.isOneInvariant_normOf, Polyfold.StrongPolyfoldBundleStructure.descend, Polyfold.StrongPolyfoldBundleStructure.normOf_descend, Polyfold.StrongPolyfoldBundleStructure.descend_oneToY, Polyfold.StrongPolyfoldBundleStructure.oneToY_pull, Polyfold.StrongPolyfoldBundleStructure.normOf_pull, Polyfold.StrongPolyfoldBundleStructure.isAuxNormFor_of_equiv, Polyfold.StrongPolyfoldBundleStructure.Y01_subset_of_equiv, Polyfold.StrongPolyfoldBundleStructure.Y01_eq_of_equiv, Polyfold.StrongPolyfoldBundle.

`Y01`, `Polyfold.StrongPolyfoldBundle.Y01_eq`, `Polyfold.StrongPolyfoldBundle.IsAuxNorm`, `Polyfold.StrongPolyfoldBundle.definition_16_3_9`, `Polyfold.StrongPolyfoldBundle.IsAuxNorm.nonneg`, `Polyfold.StrongPolyfoldBundle.isAuxNorm_descend`). Per structure: `oneToY e = Γ( (oneVal e))`, `Y01 = range oneToY`, `normOf n e = n(oneToY e)`, `IsAuxNormFor n := S.A.E.IsAuxNorm (S.normOf n)`; `normOrbit`, `continuous_normOrbit` (continuity on $|W[1]|$), `isOneInvariant_normOf`, `descend` (an invariant N read on Y, 0 off $Y_{0,1}$), `normOf_descend`, `descend_oneToY`. (16.7): `oneToY_pull`, `normOf_pull` (`normOf n = GenSBMap.pullNorm hF (S'.normOf n)` when $\Gamma' \circ |F| = \Gamma$), `isAuxNormFor_of_equiv` (Proposition 11.5.1), `Y01_subset_of_equiv`, `Y01_eq_of_equiv` (independence of m). `StrongPolyfoldBundle.Y01`, `Y01_eq`; `IsAuxNorm c n :=  S, ⟦S⟧ = c → S.IsAuxNormFor n`; `definition_16_3_9` (□ □ □ structure), `IsAuxNorm.nonneg`, `isAuxNorm_descend`. *Book:* §16.3, Definition 16.3.9 and (16.7), p. 726 (2017: Definition 16.23). *Hidden hypotheses:* none. *Deviation (restated):* n is a function $Y \to \mathbb{R}$ whose values on $Y_{0,1}$ matter (rather than a function on the subset); continuity is in the topology of $|W[1]|$ transported by Γ and follows from the representing N (`continuous_normOrbit`), nonnegativity from that of N (`IsAuxNorm.nonneg`). The "independent of m" claim and (16.7) are proved.

V16.49 (Proposition 16.3.10 (2017: 16.4), existence of an auxiliary norm; `Polyfold.StrongPolyfoldBundle.proposition_16_3_10`, `Polyfold.StrongPolyfoldBundle.z3Bundle_exists_auxNorm`). `proposition_16_3_10 (c) :  n : Y →  , c.IsAuxNorm n` (Theorem 12.2.2 in a representative, `descend`, then `isAuxNormFor_of_equiv`). Witness `z3Bundle_exists_auxNorm`. *Book:* §16.3, Proposition 16.3.10, p. 726 (2017: Proposition 16.4). *Hidden hypotheses:* none. *Deviation (same):* Paracompactness is part of Definition 16.3.1, so no separate hypothesis appears.

V16.50 (Remark 16.3.11 (2017: 16.4), an auxiliary norm is not a norm; `Polyfold.StrongPolyfoldBundle.IsAuxNorm`). Nothing to formalize: a comment. `n` is a function on Y (V16.48). *Book:* §16.3, Remark 16.3.11, p. 727 (2017: Remark 16.4). *Hidden hypotheses:* none. *Deviation (nothing):* None.

V16.51 (Definition 16.3.12 (2017: 16.24), reflexive 1-fibres; `Polyfold.StrongPolyfoldBundle.HasReflexiveFibers`, `Polyfold.StrongPolyfoldBundle.z3Bundle_hasReflexiveFibers`, `Polyfold.StrongPolyfoldBundle.hasReflexiveFibers_iff_forall`). `HasReflexiveFibers c :=  S, ⟦S⟧ = c  S.A.E.HasReflexiveFibers` (the ep-groupoid notion, V12.28). Witness `z3Bundle_hasReflexiveFibers`. *Book:* §16.3, Definition 16.3.12, p. 727 (2017: Definition 16.24). *Hidden hypotheses:* none. *Deviation (same):* The parenthetical (invariance under equivalence) is proved in both directions (V16.79; gap G.166 closed in Part III wave 7); `hasReflexiveFibers_iff_forall`: $(p, \bar{c})$ has reflexive 1-fibres iff every structure in $\bar{c}$ has.

V16.52 (Definition 16.3.13 (2017: 16.25), reflexive auxiliary norms; `Polyfold.StrongPolyfoldBundle.IsReflexiveAuxNorm`, `Polyfold.StrongPolyfoldBundle.z3Bundle_exists_reflexiveAuxNorm`, `Polyfold.StrongPolyfoldBundle.isReflexiveAuxNorm_iff_forall`). `IsReflexiveAuxNorm c n := c.IsAuxNorm n  S, ⟦S⟧ = c  S.A.E.IsReflexiveAuxNorm (S.normOf n)`. Witness `z3Bundle_exists_reflexiveAuxNorm`. *Book:* §16.3, Definition 16.3.13, p. 727 (2017: Definition 16.25). *Hidden hypotheses:* none. *Deviation (same):*

None. The notion does not depend on the model given reflexive 1-fibres (`isReflexiveAuxNorm_iff_forall`, Part III wave 7).

V16.53 (Definition 16.3.14 (2017: 16.26), mixed convergence for p; `Polyfold.StrongPolyfoldBundle.MixedTendstoIn`, `Polyfold.StrongPolyfoldBundle.MixedTendsto`, `Polyfold.StrongPolyfoldBundle.mixedTendsto_iff_forall`, `Polyfold.StrongPolyfoldBundleStructure.mixedTendstoIn_of_equiv`, `Polyfold.StrongBundle.LocalIso.mixedTendsto`). `MixedTendstoIn S y y` (representatives $w_k, w_0 \in W_{0,1}$ with `oneToY` values y_k, y_0 and `S.A.E.MixedTendsto w w`, D12.2.8); `MixedTendsto c y y :=  S, ⟦S⟧ = c  HasReflexiveFibers ( N reflexive) MixedTendstoIn S y y`. *Book:* §16.3, Definition 16.3.14, p. 727 (2017: Definition 16.26). *Hidden hypotheses:* none. *Deviation (same):* The model is existential, as printed. Mixed convergence does not depend on the model (`mixedTendsto_iff_forall`, through `mixedTendstoIn_of_equiv` and Part I's `StrongBundle.LocalIso.mixedTendsto`; gap G.166 closed in Part III wave 7).

V16.54 (Theorem 16.3.15 (2017: 16.3), reflexive auxiliary norms exist; `Polyfold.StrongPolyfoldBundle.theorem_16_3_15_one`, `Polyfold.StrongPolyfoldBundle.theorem_16_3_15_two`). `theorem_16_3_15_one (hc : c.HasReflexiveFibers) :  n, c.IsReflexiveAuxNorm n`; `theorem_16_3_15_two (hc) (hn : c.IsAuxNorm n) :  n n , c.IsReflexiveAuxNorm n  c.IsReflexiveAuxNorm n  y  c.Y01, n y  n y  n y  n y` (`theorem_12_2_11_exists`, `theorem_12_2_11_sandwich` in the reflexive representative, `descend`, `isAuxNorm_descend`). *Book:* §16.3, Theorem 16.3.15, p. 727 (2017: Theorem 16.3). *Hidden hypotheses:* none. *Deviation (same):* The inequalities are on $Y_{0,1}$, the domain of n.

V16.55 (Proposition 16.3.16 (1) (2017: 16.5 (1)), $n(y_k) \to 0$ implies $y_k \to 0_z$; `Polyfold.StrongPolyfoldBundle.proposition_16_3_16_one`). `proposition_16_3_16_one (hn : c.IsAuxNorm n) (hS : ⟦S⟧ = c) {q :  → |W[1]|} {x}` $(p(\Gamma(q_k)) \to \gamma(|x|), n(\Gamma(q_k)) \to 0)$: `Tendsto q atTop (  |0_x|)` in $|W[1]|$ (Theorem 12.2.2's compactness property via `EpGroupoid.exists_lift_tendsto`). *Book:* §16.3, Proposition 16.3.16 (1), pp. 727–728 (2017: Proposition 16.5). *Hidden hypotheses:* none. *Deviation (stronger):* Holds for every auxiliary norm (reflexivity and reflexive fibres are not used). "$Y_{0,1}$" carries the topology of $|W[1]|$ through Γ; the statement is in $|W[1]|$ of a chosen representative.

V16.56 (Proposition 16.3.16 (2) (2017: 16.5 (2)), lower semicontinuity along mixed convergence; `Polyfold.StrongPolyfoldBundle.proposition_16_3_16_two`, `Polyfold.StrongPolyfoldBundle.mixedTendsto_iff_forall`, `Polyfold.StrongPolyfoldBundle.proposition_16_3_16_two_of_mem`). `proposition_16_3_16_two_of_mem (hN : S.A.E.IsReflexiveAuxNorm (S.normOf n)) (hm : MixedTendstoIn S y y ) : (n y  : EReal)  liminf (fun k => (n (y k) : EReal)) atTop`.

The text's form (Part III wave 7, `Chapter16/P3W7W6_MixedTransfer.lean`): **`proposition_16_3_16_two`** with `hc : c.HasReflexiveFibers`: for $y_k \to^m y$ and a reflexive auxiliary norm n, $n(y) \leq \liminf n(y_k)$, with the models of n and of the convergence independent (`mixedTendsto_iff_forall`, `IsReflexiveAuxNorm.of_mem`). *Book:* §16.3, Proposition 16.3.16 (2), p. 728 (2017:

Proposition 16.5). *Hidden hypotheses:* none. *Deviation (same):* `hc : c.HasReflexiveFibers` is the context of Definition 16.3.13 and Proposition 16.3.16, not an added hypothesis. The single-model form `proposition_16_3_16_two_of_mem` is the step; the invariance of mixed convergence and of reflexive auxiliary norms under generalized strong bundle isomorphisms (the text's remark before Theorem 16.3.17) is proved (gap G.166 closed in Part III wave 7).

V16.57 (Theorem 16.3.17 (1) (2017: 16.4 (1)), extension of auxiliary norms from ∂Z; `Polyfold.StrongPolyfoldBundle.IsAuxNormBoundary`, `Polyfold.StrongPolyfoldBundle.mem_boundary_iff`, `Polyfold.StrongPolyfoldBundle.theorem_16_3_17_one`, `Polyfold.StrongPolyfoldBundle.IsAuxNorm.isAuxNormBoundary`, `Polyfold.StrongPolyfoldBundle.theorem_16_3_17_one_witness`). `IsAuxNormBoundary c n :=  S, ⟦S⟧ = c → S.A.E.IsAuxNormOn  X (S.normOf n)`; `mem_boundary_iff` ($p(\Gamma|e|) \in \partial Z \iff P(e) \in \partial X$); `theorem_16_3_17_one (hc : c.IsTame) (hn : c.IsAuxNormBoundary n) :  ñ, c.IsAuxNorm ñ  y  c.Y01, p y  c.polyfold.boundary → ñ y = n y` (Theorem 12.2.7 in one structure, then `isAuxNormFor_of_equiv`). Witnesses `IsAuxNorm.isAuxNormBoundary`, `theorem_16_3_17_one_witness`. *Book:* §16.3, Theorem 16.3.17 (1), p. 728 (2017: Theorem 16.4). *Hidden hypotheses:* none. *Deviation (same):* "Auxiliary norm of $Y_{0,1}|\partial Z$" is read in the form of Definition 16.3.9 over ∂X (all models).

V16.58 (Theorem 16.3.17 (2) (2017: 16.4 (2)), the reflexive extension; `Polyfold.StrongPolyfoldBundle.theorem_16_3_17_two_hyp`, `Polyfold.StrongPolyfoldBundle.theorem_16_3_17_two_hyp_witness`). `theorem_16_3_17_two_hyp (hn : c.IsAuxNormBoundary n) (hS : ⟦S⟧ = c) (hB : S.A.E.HasReflexiveFibers) (hr : S.A.E.IsReflexiveAuxNormOn  X (S.normOf n)) (hloc : HasReflexiveLocalExtensions S.A.X S.A.B (S.normOf n)) :  ñ, c.IsReflexiveAuxNorm ñ   y  c.Y01, p y  c.polyfold.boundary → ñ y = n y` (`theorem_12_2_13_hyp`). Witness `theorem_16_3_17_two_hyp_witness (hc : c.HasReflexiveFibers)`. *Book:* §16.3, Theorem 16.3.17 (2), p. 728 (2017: Theorem 16.4). *Hidden hypotheses:* none. *Deviation (narrower):* Added `hloc` (the reflexive local step of Theorem 12.2.13, G.147), inherited from `theorem_12_2_13_hyp`. "n reflexive" is read as `hB` + `hr` in one model $S \in \bar{c}$. Tameness is not used. Gap G.147 is closed: for a tame model, `hloc` is `hasReflexiveLocalExtensions_of_isTame` (V12.34); the corollary without `hloc` is not stated.

V16.59 (The zero 0_z of a fibre of p; `Polyfold.SBLocalization. _fiberZero_eq_of_ _eq`, `Polyfold.SBLocalization. _ne_fiberZero`, `Polyfold.SBLocalization. _eq_fiberZero_iff`, `Polyfold.SBLocalization.SDiagram.orbitMap_ _fiberZero`, `Polyfold.SBLocalization.GenSBMap.orbitMap_ _fiberZero`, `Polyfold.StrongPolyfoldBundleStructure.zeroAt`, `Polyfold.StrongPolyfoldBundleStructure.zeroAt_eq`, `Polyfold.StrongPolyfoldBundleStructure.p_zeroAt`, `Polyfold.StrongPolyfoldBundleStructure.zeroAt_eq_of_equiv`, `Polyfold.StrongPolyfoldBundleStructure.Γ_ _eq_zeroAt_iff`, `Polyfold.StrongPolyfoldBundle.zeroAt`, `Polyfold.StrongPolyfoldBundle.zeroAt_eq_of_mem`, `Polyfold.StrongPolyfoldBundle.p_zeroAt`, `Polyfold.StrongPolyfoldBundle.normY`, `Polyfold.StrongPolyfoldBundle.normY_Γ`). `StrongPolyfoldBundleStructure.zeroAt S z = Γ(|0_x|)` for any `x` with `|x| = z` (`zeroAt_eq`), `p_zeroAt`, `zeroAt_eq_of_equiv` (equivalent structures give the same `zeroAt`, via `GenSBMap.orbitMap_ _fiberZero`: generalized strong bundle maps send orbits of zeros to orbits of zeros), `Γ_ _eq_zeroAt_iff`; on the class: `StrongPolyfoldBundle.zeroAt`, `zeroAt_eq_of_mem`, `p_zeroAt`. Also `normY c n` (n extended by 0 off $Y_{0,1}$) and `normY_Γ` (`n(Γ|w|) = N(w)` with `N = normOf n`). *Book:* §16.3, before Definition 16.3.18, p. 728 (2017: before Definition 16.27). *Hidden hypotheses:* none. *Deviation (explicit):* Model independence proved. The zero section as an `ScSection` (gap G.167) is not built; only the function.

V16.60 (The relation of sections before Definition 16.3.18; `Polyfold.StrongPolyfoldBundle.SectionRep`, `Polyfold.StrongPolyfoldBundle.SectionRep.Rel`, `Polyfold.StrongPolyfoldBundle.SectionRep.pushforward_eq_of_Γ`, `Polyfold.StrongPolyfoldBundle.SectionRep.rel_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.rel_of_f_eq`, `Polyfold.StrongPolyfoldBundle.SectionRep.Rel.refl`, `Polyfold.StrongPolyfoldBundle.SectionRep.Rel.symm`, `Polyfold.StrongPolyfoldBundle.SectionRep.Rel.trans`, `Polyfold.StrongPolyfoldBundle.SectionRep.setoid`). `SectionRep c` (fields `S hS F hF f continuous_f isSection comm`), `SectionRep.Rel` (as printed); `pushforward_eq_of_Γ` ($\mathfrak{A}_*F = F'$ follows from $f = f'$ and $\Gamma' \circ |\mathfrak{A}| = \Gamma$: `eq_of_forall_ _eq`, `_pushforward`); `rel_iff` (□ iff $f = f'$ and `Equiv a.S b.S`), `rel_of_f_eq`, `Rel.refl/symm/trans`, `setoid`. *Book:* §16.3, before Definition 16.3.18, pp. 728–729 (2017: before Definition 16.27). *Hidden hypotheses:* none. *Deviation (same):* The equivalence-relation claim is proved. The condition $\mathfrak{A}_*F = F'$ is redundant (`pushforward_eq_of_Γ`), so a class is determined by f (`ScSection.ext_toFun`).

V16.61 (Definition 16.3.18 (2017: 16.27), sc-smooth sections; `Polyfold.StrongPolyfoldBundle.ScSection`, `Polyfold.StrongPolyfoldBundle.ScSection.toFun`, `Polyfold.StrongPolyfoldBundle.ScSection.p_toFun`, `Polyfold.StrongPolyfoldBundle.ScSection.continuous_toFun`, `Polyfold.StrongPolyfoldBundle.ScSection.ext_toFun`). `abbrev ScSection c := Quotient (SectionRep.setoid c)`; `toFun` (the section f, well defined), `p_toFun`, `continuous_toFun`, `ext_toFun`. *Book:* §16.3, Definition 16.3.18, p. 729 (2017: Definition 16.27). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.62 (Definition 16.3.19 (2017: 16.28), sc⁺-sections; `Polyfold.StrongPolyfoldBundle.ScSection.IsScPlus`, `Polyfold.StrongPolyfoldBundle.ScSection.isScPlus_iff`, `Polyfold.StrongPolyfoldBundle.ScPlusSection`). `ScSection.IsScPlus s :=  a, ⟦a⟧ = s  a.F  plusSectionFunctors`, `isScPlus_iff` (one representative iff all, Theorem 10.5.4), `abbrev ScPlusSection c := {s : ScSection c // s.IsScPlus}`. *Book:* §16.3, Definition 16.3.19, p. 729 (2017: Definition 16.28). *Hidden hypotheses:* none. *Deviation (restated):* Realized as the sc-smooth sections whose functor is sc⁺; the continuity of f into $Y_{0,1}$ (topology of $|W[1]|$) is not a separate field: it is that of F into $W_{0,1}$.

V16.63 (The opening of §16.5: structurability under generalized strong bundle isomorphisms; `Polyfold.SBLocalization.SDiagram.isStructurable_pushMult`, `Polyfold.SBLocalization.SDiagram.isStructurable_pullMult`, `Polyfold.SBLocalization.GenSBMap.isStructurable_pushMult`, `Polyfold.SBLocalization.GenSBMap.isStructurable_pullMult`). `SDiagram.isStructurable_pushMult D hΨ h : IsStructurable A' (D.pushMult hΨ Λ)` ($\Psi_*\Phi^*$), `SDiagram.isStructurable_pullMult`, `GenSBMap.isStructurable_pushMult hf h`, `GenSBMap.isStructurable_pullMult hf h`. *Book:* §16.5, opening paragraph, p. 732 (2017: §16.5). *Hidden hypotheses:* none. *Deviation (same):* The interior version. The form over ∂X (Proposition 14.1.8) is gap G.164.

V16.64 (Definition 16.5.1 (2017: 16.35), sc$^+$-multisections of a strong polyfold bundle; `Polyfold.StrongPolyfoldBundleStructure.continuous_` , `Polyfold.StrongPolyfoldBundleStructure.surjective_` , `Polyfold.StrongPolyfoldBundleStructure.isSaturated_preimage_` , `Polyfold.StrongPolyfoldBundleStructure.p_Γ_` , `Polyfold.StrongPolyfoldBundle.pullMult_eq_of_Γ`, `Polyfold.StrongPolyfoldBundle.MultisectionRep`, `Polyfold.StrongPolyfoldBundle.MultisectionRep.Rel`, `Polyfold.StrongPolyfoldBundle.MultisectionRep.rel_iff`, `Polyfold.StrongPolyfoldBundle.MultisectionRep.rel_of_lam_eq`, `Polyfold.StrongPolyfoldBundle.MultisectionRep.setoid`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.toFun`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.toFun_mk`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.ext_toFun`, `Polyfold.StrongPolyfoldBundle.ofFunctor`, `Polyfold.StrongPolyfoldBundle.exists_rep_of_mem`, `Polyfold.StrongPolyfoldBundle.zero`, `Polyfold.StrongPolyfoldBundle.pullMult_z3_independent`). `MultisectionRep c` (fields `S hS Λ isScPlusMultisection lam comm`), `MultisectionRep.Rel` (as printed, $\bar{f}^* =$ `GenSBMap.pullMult`), `rel_iff` (Rel iff `lam = lam'`), `rel_of_lam_eq`, `setoid`, `ScPlusMultisection c` (quotient), `toFun`, `toFun_mk`, `ext_toFun`, `ofFunctor`, `exists_rep_of_mem` (every structure of $\bar{c}$ carries a representative of every λ; Proposition 11.5.3), witness `zero c`. Key lemma `pullMult_eq_of_Γ`: for **every** $\bar{f}$ with $\Gamma' \circ |\bar{f}| = \Gamma$, $\bar{f}^*\Lambda'(w) = \lambda(\Gamma|w|)$. *Book:* §16.5, Definition 16.5.1, p. 733 (2017: Definition 16.35). *Hidden hypotheses:* none. *Deviation (same):* The relation does not depend on the choice of $\bar{f}$, which is not unique for non-effective targets (`uniqueness_16_3_3_counterexample`). Checked on the trivial-action □ /3 example of V16.40: `pullMult_z3_independent` (both $[1_W]$ and $[W \xleftarrow{1} W \xrightarrow{\mathrm{Inv}} W]$ pull every functor back to the same functor). Transitivity and symmetry are by `rel_iff`.

V16.65 (Definition 16.5.2 (2017: 16.36), the domain support; `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.domSupp`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.domSuppText`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.domSuppText_eq_univ`, `Polyfold.StrongPolyfoldBundle.nzSet`, `Polyfold.StrongPolyfoldBundle.isSaturated_nzSet`, `Polyfold.StrongPolyfoldBundle.image_nzSet`). `ScPlusMultisection.domSupp` (with $y \neq 0_z$, as in Definition 13.2.2); printed form `domSuppText`, `domSuppText_eq_univ` (it is all of Z, since weights over a fibre sum to 1). *Book:* §16.5, Definition 16.5.2, p. 733 (2017: Definition 16.36). *Hidden hypotheses:* none. *Deviation (restated):* Presumably $y \neq 0_z$ is meant; the printed set is Z (proved).

V16.66 (Remark 16.5.3 (2017: 16.5), the domain support of a representative; `Polyfold.StrongPolyfoldBundle.remark_16_5_3`). `remark_16_5_3 a : (|domSupp Λ|) = domSupp .` *Book:* §16.5, Remark 16.5.3, p. 733 (2017: Remark 16.5). *Hidden hypotheses:* none. *Deviation (same):*

The text writes Γ; the map is γ (both sets lie in $|X|$, Z). Homeomorphism = restriction of γ.

V16.67 ((16.8), the function $n(\lambda)$; `Polyfold.StrongPolyfoldBundle.image_normY_eq`, `Polyfold.StrongPolyfoldBundle.normLam`, `Polyfold.StrongPolyfoldBundle.normLam_eq`, `Polyfold.SBLocalization.IsLocalRepOn.pointwiseNorm_eq`, `Polyfold.SBLocalization.IsLocalRepOn.univ_nonempty`, `Polyfold.SBLocalization.IsScPlusMultisection.continuous_pointwiseNorm`, `Polyfold.StrongPolyfoldBundle.isQuotientMap_` , `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.continuous_normLam`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.isGreatest_normLam`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.normLam_nonneg`). `normLam c n  z` (`sSup` of `normY`), `image_normY_eq`, `normLam_eq` ($n(\lambda)(\gamma|x|) = N(\Lambda)(x)$, Part I's `pointwiseNorm`).

Continuity (Part III wave 5, `Chapter16/P3W5PC_NormLamContinuous.lean`): `IsLocalRepOn.pointwiseNorm_eq` (in a local section structure $N(\Lambda) = \max_i N(s_i)$), `IsScPlusMultisection.continuous_pointwiseNorm (hΛ) (hN : Continuous N)` (Definition 13.2.3's $N(\Lambda)$ is continuous on X), `isQuotientMap_` , **`ScPlusMultisection.continuous_normLam (s) (hn : c.IsAuxNorm n) : Continuous (c.normLam n s.toFun)`**, `isGreatest_normLam` (the maximum is attained), `normLam_nonneg`. *Book:* §16.5, (16.8), p. 733 (2017: §16.5). *Hidden hypotheses:* none. *Deviation (same):* None. Continuity, the maximum and the values in $[0, \infty)$ are proved as printed (gap G.174 closed in Part III wave 5).

V16.68 (Definition 16.5.4 (2017: 16.37), structurable sc⁺-multisections; `Polyfold.StructurablePullback`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.IsStructurable`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.definition_16_5_4_hyp`, `Polyfold.StrongPolyfoldBundle.zero_isStructurable`, `Polyfold.fnStructurablePullback`, `Polyfold.StrongPolyfoldBundle.ScPlusMultisection.definition_16_5_4`). `ScPlusMultisection.IsStructurable` (some representative, Definition 13.3.6); `definition_16_5_4_hyp (hT : StructurablePullback)` ("therefore all"); `StructurablePullback` = structurability is preserved by `GenSBMap.pullMult` along generalized isomorphisms (Theorems 13.4.1, 13.4.3 for generalized isomorphisms). Witness `zero_isStructurable`.

"One, and therefore all": `definition_16_5_4` (`s.IsStructurable` every representative is structurable), with `StructurablePullback` proved as `fnStructurablePullback` from Theorems 13.4.1 and 13.4.3 for generalized strong bundle isomorphisms (`GenSBMap.isStructurable_pullMult`, V16.63). *Book:* §16.5, Definition 16.5.4, p. 734 (2017: Definition 16.37). *Hidden hypotheses:* none. *Deviation (same):* Text's "$\lambda : Z \to \mathbb{Q}^+$" is presumably $Y \to \mathbb{Q}^+$.

V16.69 (Definition 16.5.5 (2017: 16.38), sc⁺-multisections over ∂Z; `Polyfold.StrongPolyfoldBundle.mem_boundary_iff_obj`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisectionRep`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisectionRep.Rel`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisectionRep.rel_iff`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisectionRep.rel_of_lam_eq`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisectionRep.setoid`, `Polyfold.`

`StrongPolyfoldBundle.BoundaryMultisection`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisection.toFun`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisection.toFun_mk_of_mem`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisection.domSupp`). `BoundaryMultisectionRep c` (comm only over ∂X), `Rel` ($\lambda = \lambda'$ over $p^{-1}(\partial Z)$; $\bar{f}^*$ in orbit form: $\Lambda(w) = \Lambda'(w')$ when $|\bar{f}||w| = |w'|$, $Pw \in \partial X$), `rel_iff`, `rel_of_lam_eq`, `setoid`, `BoundaryMultisection c`, `toFun` (0 off $p^{-1}\partial Z$), `toFun_mk_of_mem`, `BoundaryMultisection.domSupp`, `mem_boundary_iff_obj`. *Book:* §16.5, Definition 16.5.5, p. 734 (2017: Definition 16.38). *Hidden hypotheses:* none. *Deviation (restated):* $\bar{f}^*$ over ∂X has no Lean definition yet (gap G.164); the orbit form is used. As for D16.5.1, the relation does not depend on $\bar{f}$ (`rel_iff`).

V16.70 (Definition 16.5.6 (2017: 16.39), structurable over ∂Z; `Polyfold.StrongPolyfoldBundle.BoundaryMultisection.IsStructurable`, `Polyfold.StrongPolyfoldBundle.boundaryDomSupp_subset`). `BoundaryMultisection.IsStructurable` (`IsBoundaryStructurable`), `boundaryDomSupp_subset`. *Book:* §16.5, Definition 16.5.6, p. 734 (2017: Definition 16.39). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.71 (Theorem 16.5.7 (2017: 16.6), the ∂-extension theorem for polyfolds; `Polyfold.StrongPolyfoldBundle.theorem_16_5_7_a6`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisection.IsStructurableNe`, `Polyfold.StrongPolyfoldBundle.BoundaryMultisection.IsStructurableNe.isStructurable`, `Polyfold.StrongPolyfoldBundle.tsupport_comp_subset_preimage'`, `Polyfold.StrongPolyfoldBundle.theorem_16_5_7_hyp`, `Polyfold.StrongPolyfoldBundle.theorem_16_5_7_hyp_witness`). Part III wave 6 (`Chapter16/P3W6A6_Theorem1657.lean`): **`StrongPolyfoldBundle.theorem_16_5_7_a6 hc hPU hB hn s (hs : s.IsStructurableNe)`** …, the theorem without `h1419`, from `theorem_14_1_9_charged_a6` in a model (V14.8). `BoundaryMultisection.IsStructurableNe`: Definition 16.5.6 with a representative that charges every boundary fibre (`.isStructurable`); `hB`: tame strong bundle charts in the models. The proof is the earlier one.

The earlier form: `theorem_16_5_7_hyp (h1419 :  S, ⟦S⟧ = c → theorem_14_1_9_statement S.A) (hc : c.IsTame) (hPU :  S, ⟦S⟧ = c → S.A.X.PObj.AdmitsScPartitionsOfUnity)` … proved from statement in one model. Witness `theorem_16_5_7_hyp_witness` (the λ, Ũ, f hypotheses are jointly satisfiable on every tame bundle). *Book:* §16.5, Theorem 16.5.7, p. 735 (2017: Theorem 16.6). *Hidden hypotheses:* none. *Deviation (narrower):* Added: the two readings of Theorem 14.1.9 (V14.8), tame strong bundle charts in the models (`hB`) and a representative charging every boundary fibre (`IsStructurableNe`, the reading $I_x \neq \emptyset$). The earlier `theorem_16_5_7_hyp` took `theorem_14_1_9_statement` in every model, which holds only for models without boundary; it is vacuous otherwise. The model witness with boundary of Theorem 14.1.9 (`theorem_14_1_9_constSec`) is not lifted to a polyfold here. `theorem_16_5_7_hyp_witness` shows the hypotheses on λ, $\tilde{U}$, f jointly satisfiable on every tame bundle. Partitions of unity are read in the form Theorem 14.1.9 uses; "support" is read as `tsupport`.

V16.72 (The claim before Definition 16.6.1: sc-Fredholm in one representative, in all; `Polyfold.StrongPolyfoldBundle.ScSection.IsFredholm`, `Polyfold.StrongPolyfoldBundle.`

`ScSection.isFredholm_iff`). `ScSection.IsFredholm` (some representative), `isFredholm_iff` (iff all; Theorem 10.5.4 `theorem_10_5_4_fred`). *Book:* §16.6, before Definition 16.6.1, p. 735 (2017: before Definition 16.40). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.73 (Definition 16.6.1 (2017: 16.40), sc-Fredholm sections; `Polyfold.StrongPolyfoldBundle.FredholmSection`, `Polyfold.StrongPolyfoldBundle.ScSection.exists_rep_of_mem`). `FredholmSection c := {s : ScSection c // s.IsFredholm}`; `ScSection.exists_rep_of_mem` (every structure of $\bar{c}$ carries a representative). *Book:* §16.6, Definition 16.6.1, p. 736 (2017: Definition 16.40). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.74 (Definition 16.6.2 (2017: 16.41), compact solution sets; `Polyfold.StrongPolyfoldBundle.ScSection.solutionSet`, `Polyfold.StrongPolyfoldBundle.ScSection.HasCompactSolutionSet`, `Polyfold.StrongPolyfoldBundle.SectionRep.g`, `Polyfold.StrongPolyfoldBundle.SectionRep.isSection_F`, `Polyfold.StrongPolyfoldBundle.SectionRep.solutionSet_eq`, `Polyfold.StrongPolyfoldBundle.SectionRep.hasCompactSolutionSet_iff`). `ScSection.solutionSet` (via `zeroAt`), `HasCompactSolutionSet`; `SectionRep.solutionSet_eq` ($S_f = \gamma(|S_F|)$), `hasCompactSolutionSet_iff` (iff Definition 12.4.1 (1) in any representative). *Book:* §16.6, Definition 16.6.2, p. 736 (2017: Definition 16.41). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.75 (Definition 16.6.3 (new in 2021), proper and proper$<\infty$ sections; `Polyfold.StrongPolyfoldBundle.ScSection.sublevel`, `Polyfold.StrongPolyfoldBundle.ScSection.ControlsCompactness`, `Polyfold.StrongPolyfoldBundle.ScSection.IsProper`, `Polyfold.StrongPolyfoldBundle.ScSection.IsProperLtInfty`, `Polyfold.StrongPolyfoldBundle.Γ_mem_Y01_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.sublevel_eq`, `Polyfold.StrongPolyfoldBundle.SectionRep.isCompact_closure_g_image_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.isOpen_g_image`, `Polyfold.StrongPolyfoldBundle.SectionRep.preimage_g_image`, `Polyfold.StrongPolyfoldBundle.SectionRep.forall_iff_of_saturated`, `Polyfold.StrongPolyfoldBundle.SectionRep.forall_iff_of_open`, `Polyfold.StrongPolyfoldBundle.SectionRep.solutionSet_subset_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.solutionSet_subset_preimage_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.forall_auxNorm_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.isProper_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.isProperLtInfty_iff`, `Polyfold.StrongPolyfoldBundle.SectionRep.controlsCompactness_iff`). `ScSection.sublevel` ($f(z) \in Y_{0,1}$ and $n(f(z)) \leq C$), `ControlsCompactness`, `IsProper`, `IsProperLtInfty`; in a representative: `Γ_mem_Y01_iff`, `sublevel_eq`, `isCompact_closure_g_image_iff`, `isOpen_g_image`, `preimage_g_image`, `forall_iff_of_saturated`, `forall_iff_of_open`, `solutionSet_subset_iff`, `solutionSet_subset_preimage_iff`, `forall_auxNorm_iff`, `isProper_iff`, `isProperLtInfty_iff` (Definition 12.4.1 (2), (3)), `controlsCompactness_iff` (Definition 15.3.1). *Book:* §16.6, Definition 16.6.3, pp. 736–737 (new in 2021). *Hidden hypotheses:* none. *Deviation (same):* "$n(f(z)) \leq C$" read as Part I's `AuxLe` ($f(z) \in Y_{0,1}$).

V16.76 (Theorem 16.6.4 (2017: 16.7), compact solution set, proper, proper$<\infty$; `Polyfold.StrongPolyfoldBundle.theorem_16_6_4`). `theorem_16_6_4 (s : FredholmSection c) :`

`List.TFAE [s.1.HasCompactSolutionSet, s.1.IsProper, s.1.IsProperLtInfty]`, proved (Theorem 12.4.2 in a representative). *Book:* §16.6, Theorem 16.6.4, p. 737 (2017: Theorem 16.7). *Hidden hypotheses:* none. *Deviation (same):* None.

V16.77 (Definition 16.6.5 (2017: 16.42), orientations of sc-Fredholm sections; `Polyfold.StrongPolyfoldBundle.OrientationRep`, `Polyfold.StrongPolyfoldBundle.FredholmSection.IsOrientable`, `Polyfold.SBLocalization.SDiagram.orientation_transport_hyp`, `Polyfold.SBLocalization.SDiagram.exists_orientation_of_eq`, `Polyfold.StrongPolyfoldBundle.OrientationRep.exists_diagram`, `Polyfold.StrongPolyfoldBundle.OrientationRep.transport_hyp`). `OrientationRep s` (representative, `HasScPlusSectionsThrough`, topology `t` on O_F with `IsOrientationTopology`, `o` with `definition_12_5_4`), `FredholmSection.IsOrientable`.

Transport (Part III wave 6, `Chapter16/P3W606_OrientationTransport.lean`): **`SDiagram.orientation_transport_hyp`**: for $F' = [D]_*F$, an orientation o of F (Definition 12.5.4 for any `IsOrientationTopology`) has exactly one invariant o' compatible over the middle, and o' satisfies Definition 12.5.4 for every `IsOrientationTopology` on $O_{F'}$; `SDiagram.exists_orientation_of_eq`; `OrientationRep.exists_diagram` (a diagram exists for every other representative); **`OrientationRep.transport_hyp`**: an `OrientationRep` on any representative b. *Book:* §16.6, Definition 16.6.5, p. 739 (2017: Definition 16.42). *Hidden hypotheses:* none. *Deviation (narrower):* The orientation is data on one representative; it is carried to every other representative along a diagram D with $[D]_*F = F'$, with continuity in the text's sense, using Part I's hypotheses (tame, bump functions, tame germ data) on W, W' and the middle bundle of D, as in `theorem_12_5_7_two_cont`. Not formalized: the independence of the transported orientation from D, and dropping the hypotheses on the middle bundle (gap G.176). On the trivial-action □ /3 example of V16.40 the two $\bar{f}$ transport identically; no counterexample found.

V16.78 (Theorem 16.6.6 (2017: 16.8), the invariant $\Phi_{(f,o)}$ of a polyfold Fredholm section; `Polyfold.StrongPolyfoldBundle.theorem_16_6_6_hyp`, `Polyfold.StrongPolyfoldBundle.theorem_16_6_6_statement`, `Polyfold.StrongPolyfoldBundle.theorem_16_6_6_hyp_hcob_witness`, `Polyfold.StrongPolyfoldBundle.OrientedPerturbation`, `Polyfold.StrongPolyfoldBundle.OrientedPerturbation.IsAdmissible`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.integral`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.integralEff`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.integral_mk_rep`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.HasCobordism`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.integral_eq_of_hasCobordism`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.hasCobordism_refl`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.HasCobordismEff`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.integralEff_eq_of_hasCobordismEff`, `Polyfold.PolyfoldClass.CompactOrientedSuborbifold.hasCobordismEff_refl`, `Polyfold.PolyfoldClass.OrientedRep.integral_add`, `Polyfold.PolyfoldClass.OrientedRep.integralEff_add`, `Polyfold.PolyfoldClass.OrientedRep.integral_d_eq_zero_of_`

`boundary`, **`Polyfold.PolyfoldClass.OrientedRep.integralEff_d_eq_zero_of_boundary`**, **`Polyfold.PolyfoldClass.OrientedSuborbifold.integral_add`**, **`Polyfold.PolyfoldClass.OrientedSuborbifold.integralEff_add`**, **`Polyfold.PolyfoldClass.OrientedSuborbifold.integral_d_eq_zero`**, **`Polyfold.PolyfoldClass.OrientedSuborbifold.integralEff_d_eq_zero`**, **`Polyfold.PolyfoldClass.OrientedSuborbifold.integral_add_d`**, **`Polyfold.PolyfoldClass.OrientedSuborbifold.integralEff_add_d`**, **`Polyfold.PolyfoldClass.ScForm.toStructure_add`**, **`Polyfold.PolyfoldClass.ScForm.toStructure_zero`**, **`Polyfold.PolyfoldClass.not_mem_boundary_of_eq_empty`**, **`Polyfold.StrongPolyfoldBundle.ScPlusMultisection.IsControlledBy`**, **`Polyfold.StrongPolyfoldBundle.theorem_16_6_6_reduction`**, **`Polyfold.StrongPolyfoldBundle.exists_generalPosition_hyp`**). Part III wave 6 (`Chapter16/P3W6S6_Theorem16_6_6.lean`), on the polyfold: the integrals $\int$, $\oint$ of `CompactOrientedSuborbifold c m` are additive and well defined (`integral_mk_rep`: in every model the polyfold integral is the ep-level integral of Definition 9.5.6); for `c.boundary =` and partitions of unity, $\int d\eta = \oint d\eta = 0$ (`OrientedSuborbifold.integral_d_eq_zero`, `integralEff_d_eq_zero`, from Theorem 16.4.6), hence **`integral_add_d`**, `integralEff_add_d`: the integrals depend only on $[\tau]$. `HasCobordism`, `HasCobordismEff` and **`integral_eq_of_hasCobordism`**, **`integralEff_eq_of_hasCobordismEff`** (Theorem 9.5.18 on Y): cobordant perturbations give equal integrals of closed forms; `hasCobordism_refl`, `hasCobordismEff_refl`. Bundle level: `OrientedPerturbation s m` (a compact oriented suborbifold with $\theta = \lambda \circ f$), `IsAdmissible n U`; **`theorem_16_6_6_statement s n U m IsInherited`** ((i) $\int(\tau + d\eta) = \int \tau$ and $\oint(\tau + d\eta) = \oint \tau$; (ii) for two admissible perturbations with the inherited orientation, equal $\int$ and $\oint$ of closed τ) and **`theorem_16_6_6_hyp`** from `hPU` and `hcob`, `hcobEff` (cobordisms between any two admissible inherited-orientation perturbations); `theorem_16_6_6_hyp_hcob_witness`.

The earlier reduction: `ScPlusMultisection.IsControlledBy` ($n(\lambda) \leq 1$, dom-supp $\subseteq U$); `theorem_16_6_6_reduction` (in every structure $S \in \bar{c}$ the representatives satisfy $\partial X = \emptyset$, Fredholm, `HasCompactSolutionSetEp`, `ControlsCompactness` for `(normOf n,  ¹U)`, $N(\Lambda) \leq 1$, dom-supp$(\Lambda) \subseteq \gamma\pi^{-1}U$: the hypotheses of §§15.3–15.4); `exists_generalPosition_hyp` (from `theorem_15_3_7_statement` in a model: a structurable λ controlled by (n, U) with $n(\lambda) < 1$, representative transversal and in general position). *Book:* §16.6, Theorem 16.6.6, p. 739 (2017: Theorem 16.8). *Hidden hypotheses:* none. *Deviation (narrower):* Proved without the cobordism: well-definedness of the integrals (independent of representative and model) and dependence only on $[\tau]$. Independence of λ rests on `hcob`/`hcobEff` (the cobordism of Corollary 15.4.4, gap G.182) and on `IsInherited`, a parameter for the orientation inherited from (f, o) (Theorem 15.4.1 at the polyfold level with the transport of orientations; gaps G.159, G.175, G.176). The theorem holds for every predicate `IsInherited`. Added: `hPU`, partitions of unity on Z (the hypothesis of Stokes, Theorem 16.4.6; witness `theorem_16_4_6_hyp_witness`). The cobordism hypotheses have only the self-cobordism witness (`theorem_16_6_6_hyp_hcob_witness`, $\widetilde{\Theta} = 0$), and no concrete strong polyfold bundle with an admissible oriented perturbation of nonempty support is built: the statement is nonvacuous only through these witnesses (weak). The text's f oriented, compact solution set and (n, U) enter only through the perturbations and the cobordism. Readings: the degree k is the dimension $m + 1 \geq 1$ of $\widehat{\theta}$ (the case $k = 0$ needs the 0-dimensional branched integral, gap

G.188); “(f, λ) in general position” is read in one model (invariance under generalized isomorphisms is gap G.187); “independent of λ” is equality over any two admissible perturbations with the inherited orientation. The model-level target is `corollary_15_4_4_statement` (V15.37).

V16.79 (Definition 16.3.12 (2017: 16.24), the parenthetical: equivalent structures also have reflexive 1-fibres (pull direction); `Polyfold.StrongBundle.LocalIso.exists_reflexive_chart`, `Polyfold.StrongBundle.hasReflexiveFibers_of_isStrongBundleIsoOn`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.hasReflexiveFibers_of`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.hasReflexiveFibers_push`, `Polyfold.EpStrongBundle.IsStrongBundleEquivalence.hasReflexiveFibers_iff`, `Polyfold.SBLocalization.GenSBMap.hasReflexiveFibers_iff`, `Polyfold.StrongPolyfoldBundleStructure.hasReflexiveFibers_of_equiv`, `Polyfold.StrongPolyfoldBundle.hasReflexiveFibers_iff_forall`). `StrongBundle.LocalIso.exists_reflexive_chart`: for a local strong bundle isomorphism $\Phi : Y|U \to Y'|U'$ covering ϕ and $x \in U$, a compatible chart of Y' at $\phi(x)$ with F_1 reflexive gives one of Y at x (restrict the chart to U', then transport it; the fibre model is unchanged). `StrongBundle.hasReflexiveFibers_of_isStrongBundleIsoOn`: if X is covered by domains of such isomorphisms, reflexive $(0, 1)$-fibres pull back. **`EpStrongBundle.IsStrongBundleEquivalence.hasReflexiveFibers_of`**: for a strong bundle equivalence $\Phi : W \to W'$ (Definition 10.4.1), `B'.HasReflexiveFibers → B.HasReflexiveFibers`.

Push direction (Part III wave 7, `Chapter16/P3W7W6_ReflexivePush.lean`): **`EpStrongBundle.IsStrongBundleEquivalence.hasReflexiveFibers_push`** (for $x' \in X'$ pick x and $\sigma : \phi(x) \to x'$; a reflexive chart at $\phi(x)$ from `exists_reflexive_chart` for `L.symm`, moved to x' by `EpStrongBundle.exists_localIso_mor`), `hasReflexiveFibers_iff`, `SDiagram.hasReflexiveFibers_of`, `GenSBMap.hasReflexiveFibers_iff`, `StrongPolyfoldBundleStructure.hasReflexiveFibers_of_equiv`, **`StrongPolyfoldBundle.hasReflexiveFibers_iff_forall`**. *Book:* §16.3, Definition 16.3.12, p. 727 (2017: Definition 16.24). *Hidden hypotheses:* none. *Deviation (same):* Proved in both directions: along a generalized isomorphism $[W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W']$ reflexive 1-fibres pass between W, W'' and W' (gap G.166 closed in Part III wave 7).

V16.80 (The text before Theorem 16.6.6: $\theta = \lambda \circ f$ is a branched suborbifold; `Polyfold.PolyfoldClass.BranchedRep.rel_of_toFun_eq`, `Polyfold.PolyfoldClass.BranchedSuborbifold.toFun_injective`, `Polyfold.PolyfoldClass.BranchedSuborbifold.toFun_mk`, `Polyfold.StrongPolyfoldBundle.toPolyfoldStructure_mem`, `Polyfold.StrongPolyfoldBundle.perturbedFun`, `Polyfold.StrongPolyfoldBundle.IsPerturbedSuborbifold`, `Polyfold.StrongPolyfoldBundle.lam_f_eq`, `Polyfold.StrongPolyfoldBundle.perturbedRep`, `Polyfold.StrongPolyfoldBundle.exists_functor_rep`, `Polyfold.StrongPolyfoldBundle.perturbedRep_toFun`, `Polyfold.StrongPolyfoldBundle.perturbedRep_toFun_ofFunctor`, `Polyfold.StrongPolyfoldBundle.isPerturbedSuborbifold_iff`, `Polyfold.StrongPolyfoldBundle.isPerturbedSuborbifold_unique`, `Polyfold.StrongPolyfoldBundle.ScSection.IsInGeneralPositionWith`, `Polyfold.`

`StrongPolyfoldBundle.exists_isPerturbedSuborbifold_hyp`). `BranchedRep.rel_of_toFun_eq`, **`BranchedSuborbifold.toFun_injective`**: a branched suborbifold of a polyfold is determined by $\theta : Z \to \mathbb{Q}$. `perturbedFun s t` $= \lambda \circ f$, `IsPerturbedSuborbifold s t` ;`perturbedRep hS F Λ hbr` (the representative $(\Lambda \circ F, (X, \gamma))$ in a model), `exists_functor_rep`, `perturbedRep_toFun` ($|\Lambda \circ F| \circ \gamma^{-1} = \lambda \circ f$ in every model), `isPerturbedSuborbifold_iff`, `isPerturbedSuborbifold_unique`; `ScSection.IsInGeneralPositionWith s t` (general position in some model). **`exists_isPerturbedSuborbifold_hyp`** ($\partial Z = \emptyset$, f sc-Fredholm, (n, U) controlling compactness, λ controlled by (n, U), general position in the model of a, tame germ data): $\exists \theta$ with $\theta = \lambda \circ f$, compact. *Book:* §16.6, p. 739, before Theorem 16.6.6 (2017: before Theorem 16.8). *Hidden hypotheses:* none. *Deviation (explicit):* The branched-suborbifold part is proved: uniqueness and model independence with no extra hypothesis, existence with the inherited tame germ data (gap G.140). "Generic" is read as general position in some model; its invariance under generalized strong bundle isomorphisms is gap G.187. The natural orientation is not built (gaps G.159, G.175).

V16.81 (The zero section of a strong polyfold bundle as an sc⁺-section (§16.3); `Polyfold.StrongPolyfoldBundle.zeroSection`, `Polyfold.StrongPolyfoldBundle.zeroSection_eq_mk`, `Polyfold.StrongPolyfoldBundle.isScPlus_zeroSection`, `Polyfold.StrongPolyfoldBundle.zeroPlusSection`, `Polyfold.StrongPolyfoldBundle.ScSection.eq_zeroSection_iff_F`, `Polyfold.StrongPolyfoldBundle.solutionSet_zeroSection`). `StrongPolyfoldBundle.continuous_zeroAt`, `zeroRepOf S hS : SectionRep c` (F = `fiberZero`), **`zeroSection c : ScSection c`** (`toFun_zeroSection`: it is `c.zeroAt`), `zeroSection_eq_mk` (every structure's zero functor gives the same class), **`isScPlus_zeroSection`**, `zeroPlusSection`, `ScSection.eq_zeroSection_iff_F`, `solutionSet_zeroSection` ($= Z$). *Book:* §16.3, p. 729, Definitions 16.3.18, 16.3.19 (2017: Definitions 16.27, 16.28). *Hidden hypotheses:* none. *Deviation (same):* None.

What is not verified in this chapter

- p. 707, the uniqueness sentence after Definition 16.1.2, and p. 711, Remark 16.1.7 (2017: 16.1), with pp. 709–710: do not hold as we read them; the effective forms are verified (V16.3, V16.7; Findings, items 1, 2).
- p. 711, Remark 16.1.7, the sentence on continuous h (none or many generalized isomorphisms): an example, not formalized.
- p. 716, Lemma 16.2.1 (2017: 16.3): does not hold as we read it; the orbit form and the effective form are verified (V16.15; Findings, item 6).
- p. 732, Theorem 16.4.6 (2017: 16.5) without sc-smooth partitions of unity (gap G.153; Findings, item 8).
- p. 723, the uniqueness after Definition 16.3.3 (2017: 16.17) and p. 710, Definition 16.1.10 (2017: 16.7): do not hold as we read them; the effective forms are verified (V16.32, V16.40; Findings, items 11, 12).
- pp. 733–740, Theorem 16.6.6: the inherited orientation and the cobordism between two perturbations (gap G.175), general position independent of the model (gap G.187), Fredholm index 0 (gap G.188); the independence of a transported orientation of the diagram (gap G.176)

(Findings, items 22, 23).

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18. Sign-off

The statements below are the ones the reader is asked to confirm say what HWZ mean. Each is a summary of the V-item it names; the standing hypotheses of that item are not repeated.

1. (V1.1) An sc-Banach space is a sequence of Banach spaces $E_0 \supset E_1 \supset \cdots$ with compact inclusions and E_∞ dense in every E_m, taken as an indexed family with injective compact inclusions; finite-dimensional spaces carry the constant structure, sums have levels $E_m \oplus F_m$, and E^k has levels E_{k+m}.
2. (V1.2) An sc-operator is a linear map continuous $E_m \to F_m$ on every level, given by its level maps; an sc-isomorphism is one that is a continuous linear equivalence on every level.
3. (V1.3) An sc$^+$-operator is an sc-operator $E \to F^1$; it is compact $E_m \to F_m$ on every level.
4. (V1.4) An sc-subspace is a closed subspace whose smooth points are dense in each $F \cap E_m$; an sc-splitting $E = F \oplus G$ consists of two sc-subspaces with F_m and G_m complementary on every level.
5. (V1.5) A finite-dimensional subspace is an sc-subspace iff it lies in E_∞, and then has an sc-complement.
6. (V1.6) For a closed subspace A of an sc-Banach space E, the quotients E_m/A_m form an sc-Banach space E/A with the quotient map an sc-operator whose level kernels are the A_m.
7. (V1.7) An sc-Fredholm operator is an sc-operator with sc-splittings $E = \ker T \oplus X$, $F = C \oplus \operatorname{range} T$, $\ker T$ and C finite-dimensional, and T bijective $X_m \to \operatorname{range}(T)_m$ on every level; its index $\dim \ker T - \dim C$ is $\dim \ker T_0 - \dim \operatorname{coker} T_0$, and each T_m is Fredholm.
8. (V1.8) An sc-Fredholm operator is regularizing: $Te \in F_m$ implies $e \in E_m$; conversely, an sc-operator that is Fredholm on level 0 and regularizing is sc-Fredholm.
9. (V1.9) The sum of an sc-Fredholm operator and an sc$^+$-operator is sc-Fredholm.
10. (V1.10) A partial quadrant is a set $C = T^{-1}([0,\infty)^n \oplus W)$ for an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$; a tuple (U, C, E) has $U = O \cap C$ with O open.
11. (V1.11) A map is sc^0 on U if it maps U_m into F_m continuously for every m.
12. (V1.12) The tangent TE is $E^1 \oplus E$ and $TU = U^1 \oplus E$, with levels $U_{m+1} \times E_m$.
13. (V1.13) A map is sc^1 on U if it is sc^0, differentiable from U_1 into F_0 at every point of U_1 with a derivative $Df(x)$ bounded on E_0, and $(x,h) \mapsto (f(x), Df(x)h)$ is sc^0 on TU; $Df(x)$ is unique on relatively open subsets of partial quadrants; $x \mapsto Df(x)$ need not be norm continuous (an example on a weighted c_0 scale).
14. (V1.14) f is sc^{k+1} if it is sc^1 with sc^k tangent map, and sc-smooth if sc^k for all k.
15. (V1.15) An sc^0 map on an sc-domain is sc^1 iff on every level $f : U_{m+1} \to F_m$ is differentiable with a derivative extending to a bounded $Df(x) : E_m \to F_m$ that depends jointly continuously on $(x,h) \in U_{m+1} \times E_m$; then $f : U_{m+1} \to F_m$ is C^1, and $Df(x)$ is an sc-operator at smooth points.
16. (V1.16) If f is sc^k on U then $f : U^1 \to F^1$ is sc^k.

17. (V1.17) If f is sc^k then $f : U_{m+k} \to F_m$ is C^k, and $f : U_{m+l} \to F_m$ is C^l for $l \leq k$.
18. (V1.18) If $f : U_{m+l} \to F_m$ is C^{l+1} for all $m \geq 0$ and $0 \leq l \leq k$, then f is sc^{k+1}, as in the text; if this holds for every l, f is sc-smooth.
19. (V1.19) A map into $\mathbb{R}^N$ that is C^{l+1} on U_l for $0 \leq l \leq k$ is sc^{k+1}, as in the text; one that is C^{l+1} on every U_l, or C^∞ on U, is sc-smooth.
20. (V1.20) The composition of sc^1 maps between relatively open subsets of partial quadrants is sc^1 with $T(g \circ f) = Tg \circ Tf$; sc^k and sc-smooth maps compose likewise.
21. (V1.21) The degeneracy index $d_C(x)$ is the number of vanishing half-line coordinates of $T(x)$.
22. (V1.22) d_C does not depend on the presentation T, and neither does the number n of half-line factors.
23. (V1.23) An sc^1 map with sc^1 inverse between relatively open subsets of partial quadrants preserves the degeneracy index at every point.
24. (V1.24) For a compact operator c on a Banach space, $1 + c$ is Fredholm (Riesz–Schauder); a continuous linear map with finite-dimensional kernel and closed range of finite codimension is Fredholm.
25. (V1.25) A compact perturbation of a Fredholm operator between Banach spaces is Fredholm.
26. (V1.26) If $(x, y) \mapsto A(x)y$ is C^k on $s \times Y$ and $\iota : Y' \to Y$ is compact, then $x \mapsto A(x) \circ \iota$ is C^k into $\mathcal{L}(Y', Z)$; for an sc^1 map the level operators $Df(x) : E_m \to F_m$ exist for $x \in U_{m+1}$ and are the derivatives of $f : U_{m+1} \to F_m$.
27. (V1.27) An sc-domain has level sets that are open-in-convex with unique differentiability; relatively open subsets of partial quadrants are sc-domains, and T and the level shift preserve the property.
28. (V1.28) The index dim ker − dim coker is additive under composition, and the Fredholm index between Banach spaces is invariant under small and under compact perturbations.
29. (V1.29) Every level map T_m of an sc-Fredholm operator is Fredholm of index $\text{ind}(T) = \dim K - \dim C$.
30. (V1.30) For T sc-Fredholm and S sc$^+$, $\text{ind}(T + S) = \text{ind}(T)$, on every level.
31. (V1.31) An sc-structure on a finite-dimensional space is sc-isomorphic, through its inclusions, to the constant one; an onto inclusion $E_{m+1} \to E_m$ forces E_m finite-dimensional.
32. (V1.32) $E = A \oplus B$ is an sc-splitting iff some idempotent sc-operator has range A and kernel B.
33. (V1.33) For an sc-subspace A, a class $x + A$ meeting every E_j contains a smooth point: the smooth points of E/A are the classes of smooth points.
34. (V1.34) $(x, h) \mapsto A(x)h$ is jointly continuous iff $x \mapsto A(x)$ is continuous for uniform convergence on compact sets; so Df is compact-open continuous on U_1, and not in general norm continuous (V1.13).
35. (V1.35) Between finite-dimensional spaces, on relatively open subsets of quadrants, sc^1 is C^1 within U (one-sided derivatives at the boundary) and sc-smooth is C^∞.
36. (V1.36) C_1 is a partial quadrant in E^1 and TC one in TE; U_1 and TU are relatively open in them.
37. (V1.37) If $f : U_{m+l} \to F_m$ is C^{l+1} for all m and $l \leq k$, the derivatives $d^i f(x)$ on E_{m+l}

into F_m ($i \le l+1$, $i \le k+1$) are restrictions of one base-level map, continuous in x, and differentiable in x with derivative $d^{i+1}f(x)$ for $i \le l$, $i \le k$.

38. (V1.38) An elementary map $z \mapsto d^i f(\pi_0 z)(\pi_1 z, \dots, \pi_i z)$ with $\pi_0 : X \to E^p$, $\pi_a : X \to E^{n_a}$, into F^m, with $m + i \le p$, $m + i \le n_a + 1$, $i \le k$ and an sc-domain as domain, is sc^s for $i + s \le k + 1$; its tangent map is a pair of elementary maps and sums of them.
39. (V1.39) Sums of sc^k maps on an sc-domain are sc^k; the evaluation of continuous multilinear maps has the product-rule derivative.
40. (V1.40) Two norms on a vector space that induce the same topology are equivalent; two complete norms one of which dominates the other induce the same topology (open mapping theorem).
41. (V1.41) For T sc-Fredholm and S sc⁺, $T + S$ is sc-Fredholm of index $\mathrm{ind}(T)$, and every level map $(T + S)_m$ is Fredholm of that index.
42. (V2.1) An sc-smooth retraction is an sc-smooth $r : U \to U$, U relatively open in a partial quadrant, with $r \circ r = r$; its retract $O = r(U)$ is its fixed-point set, and $Dr(x)$ is a projection for $x \in O_1$.
43. (V2.2) (O, C, E) is a retract if $O = r(U)$ for such an r.
44. (V2.3) Two retractions onto the same O have the same tangent image $Tr(TU) = Tr'(TU')$.
45. (V2.4) $TO = Tr(TU) = \{(r(x), Dr(x)h)\}$, and $T_x O$ is the range of the projection $Dr(x)$.
46. (V2.5) Sc-smoothness of $f \circ r$ and $T(f \circ r)$ on TO do not depend on the retraction r onto O, and $T(f \circ r)$ maps TO into TO'.
47. (V2.6) $f : O \to O'$ is sc-smooth if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth for a (every) retraction onto O; $Tf = T(f \circ r)|_{TO}$.
48. (V2.7) Sc-smooth maps between retracts compose, with $T(g \circ f) = Tg \circ Tf$.
49. (V2.8) (TO, TC, TE) is a retract with retraction Tr, and Tf is sc-smooth between tangents.
50. (V2.9) Open subsets of retracts, and images of idempotent sc-smooth maps on retracts, are retracts.
51. (V2.10) A chart is a homeomorphism of an open set onto a retract; charts are compatible when both transition maps are sc-smooth between the overlap retracts.
52. (V2.11) An atlas is a set of pairwise compatible charts covering X; atlases are equivalent when their union is an atlas, and this is an equivalence relation.
53. (V2.12) An M-polyfold is a Hausdorff paracompact space with an atlas, its charts being all charts compatible with it.
54. (V2.13) f is sc-smooth if it is continuous and all its coordinate representations are sc-smooth between retracts; one pair of charts per point suffices.
55. (V2.14) X_m consists of the points that lie on level m in one (every) chart, with the topology generated by the chart preimages of open subsets of O_m.
56. (V2.15) The inclusions $X_{m+1} \to X_m$ are continuous.
57. (V2.16) The level charts $(V_1, \varphi, (O_1, C_1, E^1))$ make X_1 an M-polyfold.
58. (V2.17) For every M-polyfold, X and every X_m are metrizable, X_m is paracompact, and X_∞ (initial topology of the inclusions) is metrizable; no hypothesis beyond the text's.
59. (V2.18) Tangent tuples (x, φ, h) are equivalent when one is the transport of the other by T of

the transition map; this is an equivalence relation.

60. (V2.19) TX is the set of classes, with $p : TX \to X_1$ and the bijections $T\varphi : TV \to TO$.
61. (V2.20) TX is Hausdorff, metrizable and paracompact, and p is continuous (openness of p not verified).
62. (V2.21) The charts $(TV, T\varphi, (TO, TC, TE))$ make TX an M-polyfold, with transition maps $T(\varphi' \circ \varphi^{-1})$, and $p : TX \to X^1$ is sc-smooth.
63. (V2.22) An sc-smooth $f : X \to Y$ has a well-defined sc-smooth tangent map $Tf : TX \to TY$ covering f, with $T(g \circ f) = Tg \circ Tf$.
64. (V2.23) A sub-M-polyfold is locally the image of an sc-smooth retraction of an open set.
65. (V2.24) A sub-M-polyfold, with the subspace topology, is an M-polyfold for which the inclusion and the maps $i^{-1} \circ r$ are sc-smooth.
66. (V2.25) In a chart, at a smooth a, $T_aO = Dt(a)E \oplus (Dr(a) - Dt(a))E$ is an sc-splitting with $Dt(a)E = T_aA$, and Ts maps T_aO onto T_aA.
67. (V2.26) $d_X(x)$ is the least $d_C(\varphi(x))$ over the charts φ around x.
68. (V2.27) d_X is upper semicontinuous.
69. (V2.28) Sc-diffeomorphisms between open sets preserve d_X.
70. (V2.29) $\partial X = \{d_X \geq 1\}$ is closed; X is without boundary iff $d_X \equiv 0$.
71. (V2.30) On a sub-M-polyfold A, $d_A \leq d_X$.
72. (V2.31) E_x is the closed subspace cut out by the coordinates vanishing at x, of codimension $d_C(x)$; C_x the cone on which they are ≥ 0.
73. (V2.32) $T_x^RO = T_xO \cap E_x$ and $C_xO = T_xO \cap C_x$, defined at every $x \in O_1$.
74. (V2.33) $T_x^RO \subseteq C_xO \subseteq T_xO$, and C_xO is a partial quadrant in T_xO at every smooth point of every sc-smooth retract: the remark's claim that in general it is not is false.
75. (V2.34) For an sc¹ map between quadrants, $Df(x)(E_x) \subseteq E'_{f(x)}$ and $Df(x)(C_x) \subseteq C'_{f(x)}$ at every level-1 point.
76. (V2.35) A germ of sc¹-diffeomorphisms between retracts maps T_x^RO onto T_y^RO' and C_xO onto C_yO', at every level-1 point.
77. (V2.36) A nonempty retract, and a nonempty relatively open subset of a partial quadrant, is an M-polyfold with the chart id.
78. (V2.37) $\dim(T_xO/T_x^RO) \leq d_X(x)$ in every chart of every M-polyfold at points on level 1; the second statement holds without its hypothesis (V2.33).
79. (V2.38) On a relatively open subset of a partial quadrant, the degeneracy index of the M-polyfold equals d_C.
80. (V2.39) C_e and E_e are the cone and the subspace cut out by the coordinates vanishing at e, in a presentation of C.
81. (V2.40) A retraction is tame if it preserves d_C and, at smooth points of O, T_xO has an sc-complement inside E_x (for every presentation of C).
82. (V2.41) For a tame r at a smooth $x \in O$, $(1 - Dr(x))E \subseteq E_x$ is an sc-complement of T_xO; only the second tameness condition is used.
83. (V2.42) For a tame retract, $\dim(T_xO/T_x^RO) = d_O(x)$ at smooth points, where T_x^RO is an sc-subspace and C_xO a partial quadrant in T_xO, and $d_O = d_C$ everywhere; a tame chart of

any M-polyfold realizes d_X.

84. (V2.43) A tame M-polyfold has an equivalent atlas of charts with tame models.
85. (V2.44) A face is the closure of a component of $\{d_X = 1\}$; face-structured means every x lies in exactly $d_X(x)$ faces.
86. (V2.45) Near a point of a tame retract, the retraction maps points of a coordinate hyperplane f_i into the closure of a single component of $\{d_C = 1\} \cap O \cap f_i$.
87. (V2.46) The faces of a tame retract are the components of the sets $f_i \cap O$, distinct for distinct i through a common point.
88. (V2.47) A tame retract is face-structured.
89. (V2.48) In a tame M-polyfold each point has a tame chart realizing d_X whose model is face-structured, and lies in at most $d_X(x)$ global faces.
90. (V2.49) A face of a tame retract is the retract of $r \circ P_i$ on a neighbourhood of the face, hence a sub-M-polyfold, and with its induced structure it is tame with $d_F = d_O - 1$.
91. (V2.50) Every point of a tame M-polyfold lies in exactly $d_X(x)$ local faces, each tame with $d_F(x) = d_X(x) - 1$; the intersection of $\#\sigma$ of them is, near x, a tame sub-M-polyfold A with $d_A = d_X - \#\sigma$.
92. (V2.51) In a tame M-polyfold the interior $F \cap \{d_X = 1\}$ of a face is a component G and a sub-M-polyfold; if X is face-structured, every face is a tame sub-M-polyfold with $d_X = d_F + 1$.
93. (V2.52) $X \times Y$ is an M-polyfold with the product charts, $d_{X\times Y} \leq d_X + d_Y$, with equality and $X \times Y$ tame when X and Y are tame.
94. (V2.53) For a local sc-diffeomorphism f and an sc-smooth g the fibred product is a sub-M-polyfold of $X \times Y$ with $d = d_Y$, tame if Y is; if both are local sc-diffeomorphisms, $d = d_X = d_Y$, and $2d = d_{X\times Y}$ for tame X, Y.
95. (V2.54) A strong bundle map preserves the double filtration, is linear in the fibre and sc-smooth on [0] and [1]; a strong bundle retraction is an idempotent one, covering an sc-smooth retraction.
96. (V2.55) A local strong bundle retract $K = R(U \lhd F)$ and its part $K[1]$ are sc-smooth retracts, with sc-smooth projections to O.
97. (V2.56) Sections of K are sc-smooth into $K[0]$, sc⁺ into $K[1]$; sc⁺ implies sc-smooth.
98. (V2.57) A strong bundle chart is a fibrewise linear homeomorphism $P^{-1}(V) \to K$ over a chart of X; compatibility means sc-smooth transitions on [0] and [1]; equivalence of strong bundle atlases is an equivalence relation, and charts restrict to open subsets of the base.
99. (V2.58) A strong bundle is $P : Y \to X$ with a strong bundle atlas; $Y[0]$ and $Y[1]$ are M-polyfolds, $P[0]$ and $P[1]$ are sc-smooth, and one chart decides whether $y \in Y_{m,k}$; the charts compatible with the atlas depend only on its class.
100. (V2.59) Sections of a strong bundle are sc-smooth into $Y[0]$, sc⁺ in the charts $\Phi[1]$, which is the same as sc-smooth into $Y[1]$.
101. (V2.60) The pull-back f^*Y is the fibre product of P and f.
102. (V2.61) The pull-back of a strong bundle along an sc-smooth map is a strong bundle, with the pulled-back charts and retractions R_g.
103. (V2.62) A paracompact Hausdorff locally metrizable space is metrizable (Smirnov), proved

in a pull-back form.

104. (V2.63) Sc^k-smoothness restricts to smaller domains and is a local property.
105. (V2.64) $Y[1]$ is Hausdorff and paracompact (metrizable), and the inclusion $Y[1] \to Y[0]$ is sc-smooth.
106. (V2.65) For retractions r, s onto O and $f(O) \subseteq O'$: $f \circ r$ is sc-smooth iff $f \circ s$ is, and then $T(f \circ r) = T(f \circ s)$ on TO and $T(f \circ r)$ maps TO into TO'.
107. (V2.66) Every point of a retract has open $V \subseteq W$ with $\mathrm{cl}_O(V) \subseteq W$ and $\mathrm{cl}_O(V)$ complete for $|\cdot|_0$, W arbitrarily small.
108. (V2.67) A locally complete space has, around every point, V, W with $\mathrm{cl}(V) \subseteq W$ and a metric on W for which $\mathrm{cl}(V)$ is complete; it is automatically Hausdorff.
109. (V2.68) If $\mathrm{cl}_X(V) \subseteq W$ then $\mathrm{cl}_W(V) = \mathrm{cl}_X(V)$, and local completeness implies its modified form with $\mathrm{cl}_W(V)$.
110. (V2.69) Sc-smooth retracts are locally complete.
111. (V2.70) Local completeness passes to locally closed subspaces, homeomorphic spaces, open subspaces, and from an open cover with condition (b) to the whole space.
112. (V2.71) Locally complete spaces are first countable and completely regular.
113. (V2.72) Every level U_k, $k \in \mathbb{N} \cup \{\infty\}$, of a connected relatively open U is path-connected, U_∞ with the topology of all the norms.
114. (V2.73) $W = V \cap r^{-1}(V)$ is a relatively open neighbourhood of x with $r(W) = O \cap V \subseteq W$, and retracts are locally path-connected.
115. (V2.74) Retracts are almost locally contractible, by $\Phi(t, y) = r((1-t)y + tx)$.
116. (V2.75) The footnote's paracompactness is Mathlib's, so it changes nothing for M-polyfolds; with "subcovering" instead of "refinement", $\mathbb{R}$ would not be paracompact.
117. (V2.76) In a regular space every open $U \ni x$ contains an open $V \ni x$ with $\mathrm{cl}(V) \subseteq U$, arbitrarily small in a Hausdorff paracompact space.
118. (V2.77) M-polyfolds are locally complete, with arbitrarily small $W(x)$, and locally path-connected.
119. (V2.78) A tame sub-M-polyfold is given locally by a retraction preserving d_X whose image has a tangent complement in the reduced tangent space, in a chart; it is a sub-M-polyfold.
120. (V2.79) Faces of open sets are defined from d_X; the degeneracy index of a tame M-polyfold satisfies the corner axioms from which all of §2.7.4 is proved.
121. (V2.80) Every point of a tame M-polyfold has an open V with $d_X \leq d_X(x)$, every y in exactly $d_X(y)$ faces of V, and a neighbourhood $U(x)$ whose faces in V all contain x.
122. (V2.81) $N_{X,x}$ is nonempty, and N_X and $N_{X,x}$ are closed under nonempty open subsets and intersections as stated; every open subset of a quadrant is face-structured.
123. (V2.82) L_X is a thin category and F_X a functor to sets with injective maps, bijective over a fixed point.
124. (V2.83) Equivalence of admissible triples is an equivalence relation; $B : bX \to X$ has image ∂X and fibres of $d_X(x)$ points.
125. (V2.84) $\Phi_{(x,V,\theta)}$ is injective with image $\theta \cap V^x$, and a bijection onto θ when every face of V contains x; such V are cofinal in $N_{X,x}$.

126. (V2.85) In $[0,\infty)^3$, for $x=(0,0,1)$ and a thin slab $V \in N_{X,x}$ along the segment to $(0,1,0)$, $\Phi_{(x,V,\theta)}$ misses $(0,1,0) \in \theta$.
127. (V2.86) T_{bX}, read with O containing the class, is a topology.
128. (V2.87) The $U(x,V,\theta)$ form a basis of open sets, each Φ is an open embedding into θ, and transition maps are the identity on points of X between open subsets of θ and θ'.
129. (V2.88) bX is Hausdorff and metrizable.
130. (V2.89) T_{bX} is the unique topology making every $U(x,V,\theta)$ open and every Φ open onto its image; B is continuous onto ∂X with fibres of $d_X(x)$ points.
131. (V2.90) The quadrant $[0,\infty)^n \oplus W$ is tame, its corner has n preimages in bX, and it is a tame sub-M-polyfold of itself.
132. (V2.91) $\mathbb{R}$ with the K-topology is Hausdorff and satisfies the modified notion of Remark 2.1.6 but is not locally complete; so the modified notion is strictly weaker.
133. (V2.92) Bundle retractions and bundle maps are defined by their shape $(r(u),\rho(u)v)$, $(a(u),\phi(u)v)$; Tr is a bundle retraction covering r^1, and Tf a bundle map covering f^1.
134. (V2.93) For every m, $(X^1)_m$ is X_{m+1} with its topology and charts, through the identity of points.
135. (V2.94) $p: TX \to X^1$ is sc-smooth, locally the bundle retraction Tr over r^1, with transition maps bundle maps covering the level-1 transition maps of X.
136. (V2.95) $T(X^1)$ and $(TX)^1$ are sc-diffeomorphic through a map commuting with the projections to $(X^1)^1$, independent of the chart and the identity in charts.
137. (V2.96) An sc-diffeomorphism raises to an sc-diffeomorphism of the index lifts; $T(X^2) \cong (TX)^2$ over $(X^1)^2 = (X^2)^1$.
138. (V2.97) For every i, $\Phi_i : T(X^i) \to (TX)^i$ and $J_i : (X^1)^i \to (X^i)^1$ are sc-diffeomorphisms with $J_i \circ p^i \circ \Phi_i = p$, compatible with the inclusions, unique, and natural in X.
139. (V2.98) The diagonal of $[0,\infty)^3 \oplus W$ is a retract sc-diffeomorphic to the half-line, with $d_C = 3$ and $d_\Delta = 1$ at the corner, and one chart onto $[0,\infty) \oplus W$ realizes d_Δ.
140. (V2.99) Condition (3) of Definition 2.5.19 holds in every chart, every presentation and for every sub-M-polyfold retraction once it holds in one chart.
141. (V2.100) A tame sub-M-polyfold of a tame M-polyfold has a tame induced structure.
142. (V2.101) $A = \{(x,y,0)\}$ is a tame sub-M-polyfold, $B = \{(x,x,0)\}$ is a sub-M-polyfold that is not tame as a sub-M-polyfold but has a tame induced structure, for any W in place of $\mathbb{R}$.
143. (V2.102) For tame X, the charts $\pi \circ \varphi \circ B$ make bX a tame M-polyfold, B is sc-smooth and $d_X(B(p)) = d_{bX}(p) + 1$.
144. (V2.103) For centred representatives, Φ is an sc-diffeomorphism of $U(x,V,\theta)$ onto θ, the transition maps are sc-diffeomorphisms, and this characterizes the structure of bX.
145. (V2.104) For every representative, Φ is an sc-diffeomorphism onto the open $\theta \cap V^x$, Proposition 2.7.12 holds as printed, and the characterization holds with this image.
146. (V2.105) For the slab triple of V2.85, Φ is not onto θ.
147. (V2.106) The iterated boundaries b^kX are tame M-polyfolds, B^k is sc-smooth with $d_X \circ B^k = d_{b^kX} + k$, and its image is $\{d_X \geq k\}$.
148. (V2.107) For the quadrant, $n \geq 1$, chart data of bX exist and $d_{bX} = n-1$ over the corner.

149. (V3.1) A germ of neighbourhoods is a nested sequence of relatively open neighbourhoods U_m of 0 in C_m, with tangent germ $(U_{m+1} \times E_m)_m$; a germ is a fixed sequence of sets.
150. (V3.2) sc^0-, sc^1-, sc^k- and sc-smooth germs are the notions of Chapter 1 with $U \cap E_m$ replaced by U_m.
151. (V3.3) Germs restrict, compose (when the inner germ maps U_m into V_m level by level), pair, and sc-operators are sc-smooth germs.
152. (V3.4) Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) hold for germs.
153. (V3.5) An sc-smooth section germ is a section whose principal part composed with r is an sc-smooth germ, on some germ of neighbourhoods inside U, in some centred strong bundle chart compatible with the atlas.
154. (V3.6) Sums of sections have sums of principal parts; sc$^+$-sections have sc-smooth representatives $O \to K[1]$ in every compatible chart.
155. (V3.7) A filling is an sc-smooth germ satisfying (1)–(3) of Definition 3.1.4 (preprint 3.4) near 0, the map in (3) an sc^1-germ (the text's germ reading).
156. (V3.8) For a filler with $g(0) = 0$, $Tf(0) = Dg(0)|_{Dr(0)E}$ is onto iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, with the same index; if $Dg(0)$ is sc-Fredholm, so is $Tf(0)$, and condition (3) holds on every level.
157. (V3.9) A filled version is a filling of the principal part in the chart of Definition 3.1.11 (preprint 3.11), an sc-smooth germ whose levels may shrink with m.
158. (V3.10) A basic germ is an sc-smooth germ with $f(0) = 0$ whose W-component is $w - B(a, w)$ with B an sc^0-contraction germ.
159. (V3.11) For sc^1 sc^0-contraction germs, Lemma 3.1.9 (preprint 3.9) as stated: $|D_2B(x)\zeta|_m \leq \varepsilon|\zeta|_m$ for $x \in U_{m+1}$ small on level m, for every ε and m; and $D_2B(0, 0) = 0$.
160. (V3.12) $D(h + s)(0)$ is sc-Fredholm of index $n - N$ for a basic germ h and an sc$^+$-germ s, and on every level m, $D(h + s)(x)$ is Fredholm of index $n - N$ for $x \in U_{m+1}$ small on level m.
161. (V3.13) An sc-Fredholm germ has a filling germ g and an sc$^+$-germ s, $s(0) = g(0)$, with $g - s$ conjugated to a basic germ near 0 (the text's germ reading).
162. (V3.14) Near a zero of an sc-Fredholm germ, the closures of the zeros in $O(i)$ are compact subsets of X_i.
163. (V3.15) Zeros converging to the base point (or, for sc-Fredholm sections, to any point) converge on every level.
164. (V3.16) An sc-Fredholm section is sc-smooth, regularizing and an sc-Fredholm germ at every smooth point.
165. (V3.17) An sc-Fredholm section plus an sc$^+$-section is sc-Fredholm, for any strong bundle.
166. (V3.18) $s \circ r$ agrees with s on O and R fixes $(r(u), s(r(u)))$.
167. (V3.19) $g + s \circ r$ is a filling germ of $f + s$ when g is one of f, for s an sc-smooth section of K (a germ suffices).
168. (V3.20) Good position: $N \cap C$ has interior in N and an algebraic complement satisfies the ε-condition; the text's sc-complement form is also formalized and holds for smooth N on a partial quadrant.
169. (V3.21) If N is in good position to C, then $N \cap C$ is a partial quadrant in N.

170. (V3.22) The tangent germ property, with Y closed and “sc^∞-germ” read with k-dependent neighbourhoods.
171. (V3.23) The tangent germ property is invariant under sc-diffeomorphisms.
172. (V3.24) Under the regularity conditions (a) δ sc-smooth on V, (b) $\pi(O) \subseteq C$, a set with the tangent germ property is a sub-M-polyfold, tame, with corner charts $\pi\varphi$; without (a), (b) the text’s proof fails in examples.
173. (V3.25) An sc^0-contraction germ has the contraction estimate on small balls on every level, and a working form with explicit regions.
174. (V3.26) The solution germ exists, is sc^0, and is unique in the contraction region.
175. (V3.27) If B is sc^1, the solution germ is sc^1 on the germ shifted by one level.
176. (V3.28) $f^{(1)}$ is a contraction germ of class sc^{min(k, j)} with solution germ $T\delta$; it satisfies Definition 3.3.1 (preprint 3.28).
177. (V3.29) If B is sc^k, the solution germ is sc^j on some shifted germ for every $j \leq k$.
178. (V3.30) If B is sc^k, the solution germ is sc^k on the germ shifted by $2^k - 1$ levels; the text’s hypotheses suffice.
179. (V3.31) If B is sc-smooth, δ is C^k into W_m on a neighbourhood $V_{m,k}$ of 0; δ is an sc-smooth germ on one germ.
180. (V3.32) For a quadrant-like $\widetilde{C}$, any representative of the solution germ of an sc^k (sc-smooth) contraction germ is sc^k (an sc-smooth germ) on a smaller germ.
181. (V3.33) The basic class is Definition 3.1.7 (preprint 3.7).
182. (V3.34) A linear coordinate change makes the index-raised push-forward of a basic germ plus an sc$^+$-germ a basic germ, with $n - N$ unchanged.
183. (V3.35) The solutions near 0 of a contraction germ plus an sc$^+$-germ have compact closures on every level, for closed $\widetilde{C}$.
184. (V3.36) Zeros of $h + s$ near 0 lie on level 1, where the linearization is surjective Fredholm of index $n - N$.
185. (V3.37) Zeros of the push-forward near 0 have their X-component small on level 1.
186. (V3.38) Under Lemma 3.5.5’s (preprint 3.42) regularity, the zeros of g near 0 form the graph of a C^1 map τ over ker $Dg(0) \cap C'$ into a good complement, $\tau(0) = 0$, $D\tau(0) = 0$, and τ is C^k into level k near 0 for every k.
187. (V3.39) ker $Dg(0)$ is the graph of $D\delta(0)$ over ker $DH(0)$.
188. (V3.40) Linear isomorphisms carrying C onto C' carry good complements and good position.
189. (V3.41) ker $DH(0)$ has a good complement Z with $Z \times W$ a good complement of K'.
190. (V3.42) The zeros of $f + s$ near 0 form the graph of a C^1 map σ over ker $D(f + s)(0) \cap C$, $\sigma(0) = 0$, $D\sigma(0) = 0$, C^k into level k near 0 for every k.
191. (V3.43) Definition 3.6.1 (preprint 3.48) is Definition 3.1.4 (preprint 3.4).
192. (V3.44) After shrinking, (1) and (2) of the filler hold on all of $O \cap W$, $U \cap W$.
193. (V3.45) Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51) are Definitions 3.1.11, 3.1.16 (preprint 3.11, 3.14) for a tame strong bundle; the theorems of §3.6 use them with the chart of the germ data tame (V3.103).
194. (V3.46) In a centred chart, $Dg(0)$ and $Tf(x_0)$ are sc-Fredholm with index $n - N$ on every

level, the dimensions of any basic germ to which $g - s$ is conjugated; at a zero of an sc-Fredholm section the linearization is sc-Fredholm in every chart.

195. (V3.47) Good position of the germ: $Tf(0)$ onto and ker $Dg(0)$ in good position to $T_0O \cap C$ in T_0O, in a centred chart; also with the text's sc-complement, the same at zeros of sc-Fredholm sections with tame germ data.
196. (V3.48) Near a zero in good position in a tame chart, the zero set is carried injectively by $\psi\varphi$ onto the graph of a C^1 map σ over ker $D(h+t)(0) \cap C'$, with Y closed, the kernel smooth, and C^k-lifts of σ into level k.
197. (V3.49) Along the graph of τ near 0, Y (closed) stays a good complement of N_v with respect to $C_{v+\tau(v)}$; also for τ on a relatively open V and any presentation of C.
198. (V3.50) Without boundary, a zero set with the regular tangent germ property is a manifold without boundary.
199. (V3.51) A zero set with the regular tangent germ property is a tame sub-M-polyfold and a manifold with boundary with corners.
200. (V3.52) Injectivity of $L(0)$ and the compactness property make $L(b)$ injective for b near 0; in Theorem 3.7.1 (preprint 3.58) both hypotheses follow from its condition (2).
201. (V3.53) On a compact parameter set with injective $L(b)_m$ there is a uniform lower bound c_m.
202. (V3.54) $(v,k) \mapsto L(v)^{-1}k$ is an sc-smooth germ under uniform lower bounds c_m.
203. (V3.55) A good complement has vanishing coordinates in Σ.
204. (V3.56) $d_{N\cap C} = d_C$ off $N \cap W$, and $d_{N\cap C} + \dim(N \cap W) = \dim N$ on it.
205. (V3.57) $d_{N\cap C}(v) = d_C(v + \tau(v))$ for $v \notin N \cap W$ near 0.
206. (V3.58) The quadrant inverse function theorem holds for C^j maps.
207. (V3.59) A C^j map with $\|Df - 1\| \leq \frac{1}{2}$ preserving interior points is a C^j-diffeomorphism onto a relatively open set.
208. (V3.60) Such a map is bi-Lipschitz, injective, and has relatively open image.
209. (V3.61) Such a map is a homeomorphism onto its image.
210. (V3.62) Its inverse has derivative $[Df(g(y))]^{-1}$.
211. (V3.63) The implicit function theorem in partial quadrants, with $f(k+\tau(k)) = 0$ and $k+y \in U$.
212. (V3.64) There is a region on which $B(k,\cdot)$ is a $\frac{1}{2}$-contraction and fixed points are unique.
213. (V3.65) $B(k,\cdot)$ maps $\{|y| \leq \varepsilon|k|\}$ into itself.
214. (V3.66) τ is continuous with $|\tau(k)| \leq \varepsilon|k|/2$, Lipschitz with constant $\varepsilon/(1+\varepsilon) \leq \varepsilon$ on all of V_0.
215. (V3.67) τ is C^j on the whole V_0, with the derivative formula.
216. (V3.68) Under the filler condition at the nearby zeros (a consequence of good position, V3.102), near a zero in good position in a tame chart of an sc-Fredholm section with tame germ data over an M-polyfold without boundary the zeros are smooth points in good position in tame charts and form a manifold without boundary; the forms with the text's hypotheses, the index and the kernel are V3.95, V3.97, V3.99, V3.101.
217. (V3.69) Near a zero in good position in a tame chart whose nearby zeros are in good position in tame charts (or, for tame germ data, under the filler condition of V3.82), the local solution

set consists of smooth points and is a tame sub-M-polyfold equivalent to a d-dimensional manifold with boundary with corners.

218. (V3.70) With good position in a tame chart at every zero ($S \neq \emptyset$), the solution set is a sub-M-polyfold with tame induced structure, a manifold with boundary with corners of locally constant dimension; no hypothesis is added.
219. (V3.71) The image of a tame retract under an sc-diffeomorphism, and an open subset of a tame retract, are tame retracts.
220. (V3.72) A nonempty set with regular data of Definition 3.2.1 (preprint 3.25) of some dimension at every point is a sub-M-polyfold, tame, with an atlas of finite-dimensional corner charts $\pi\varphi$.
221. (V3.73) Two graph descriptions of one set, the first C^1 and the second centred at a point of it with C^k-lifts at 0, have the same dimension, and the first is C^k into every level near that point.
222. (V3.74) Data of Definition 3.2.1 (preprint 3.25) with $\pi(O) \subseteq C$, at a point whose neighbours all have data, can be shrunk to regular data, and the dimension is locally constant.
223. (V3.75) A finite-dimensional subspace in good position has a closed good complement along which the projection maps the partial quadrant into itself.
224. (V3.76) If g is conjugated to a basic germ and s is an sc$^+$-map with $s(0) = 0$, then $(g + s)^1$ is conjugated to a basic germ, by $\Lambda \circ \Phi^1$.
225. (V3.77) Theorems 3.5.6, 3.5.11 (preprint 3.43, 3.47) and the graph description of Theorem 3.6.8 (preprint 3.54) hold with a closed good complement whose projection maps the partial quadrant into itself.
226. (V3.78) At a zero, the linearizations in two strong bundle charts are conjugate by isomorphisms respecting the partial quadrants: surjectivity, Fredholm index and good position of the kernel do not depend on the chart.
227. (V3.79) Surjectivity onto a dense part of K_z and good position of ker $Dg(z) \cap T_zO$, in a chart not centred at the zero y (z in the germ of g), give good position of (f, y) in a tame chart, for tame germ data.
228. (V3.80) In the chart at x_0, at the zeros near x_0, $D(h+t)$ is surjective Fredholm of index $n-N$ and the tangent space N_v of the graph is in good position with complement Y.
229. (V3.81) In the chart at x_0, at the zeros near x_0, ker $Dg(z_0) = D\psi^{-1}(N_v)$ lies in $T_{z_0}O$ and is in good position there; no hypothesis is added.
230. (V3.82) For tame germ data, if the filler in the tame chart at x_0 also satisfies condition (3) of Definition 3.1.4 (preprint 3.4) at its zeros near 0, every zero near x_0 is in good position in a tame chart; the text's argument needs that condition, which holds whenever (f, x_0) is in good position (V3.102).
231. (V3.83) A good complement of $\mathbb{R} \times N$ projects into the lineality space; the tilted kernel of the parameter extension is in good position.
232. (V3.84) At a zero x_0 in good position in a tame chart, the solution set near x_0 has data of Definition 3.2.1 (preprint 3.25) at x_0 in the chart $\psi\varphi$, with $\pi(O) \subseteq C$.
233. (V3.85) If x_0 and the zeros near it are in good position in tame charts, the solution set has

regular data at x_0 of a dimension d that every nearby datum shares; Theorem 3.6.8 (1)–(2) (preprint 3.54) under the filler condition of V3.82.

234. (V3.86) Over an M-polyfold without boundary, with good position in a tame chart at every zero, the solution set is a manifold without boundary of locally constant dimension.
235. (V3.87) With good position in a tame chart at every zero (the text's hypotheses), the solution set is a sub-M-polyfold with tame induced structure, a manifold with boundary with corners of locally constant dimension, the disjoint union of its open and closed parts S_n of dimension n.
236. (V3.88) Given sc-isomorphisms $L(b) = P_Y Df(b, 0)|_X$ with uniform lower bounds and condition (3), $\Phi_*(f - f(0))$ is a basic germ with contraction $x - L(b)^{-1}P_Y(f - f(0))$.
237. (V3.89) Under conditions (1)–(3) of Theorem 3.7.1 (preprint 3.58), in the coordinates of its proof, $f - f(0)$ is conjugated near 0 to a basic germ.
238. (V3.90) A C^j map with $\|Df - 1\| \leq \frac{1}{2}$ on a convex set, whose image has unique differentiability, has a C^j inverse with $Dg = [Df \circ g]^{-1}$.
239. (V3.91) For a strong bundle (tameness not needed), an sc-Fredholm section f and sc⁺-sections s_i, $(r, x) \mapsto f(x) + \sum r_i s_i(x)$ is an sc-Fredholm section of the pull-back bundle over $\mathbb{R}^n \times X$.
240. (V3.92) In the product chart at $(0, x)$ the linearization of the perturbed section is $(\delta r, \xi) \mapsto f'(x)\xi + \sum \delta r_i s_i(x)$ (chart form); at (r_0, x) the first term is that of $f + \sum r_{0,i} s_i$.
241. (V3.93) In the chart at a zero x_0 in good position, at every zero z_0 of the filler near 0 and for every smooth k, $\rho(z_0)k = Dg(z_0)x$ for some $x \in T_{z_0}O$, by a curve of zeros of $g + \mu\rho(r(\cdot))k$ through z_0 from an implicit function theorem at z_0.
242. (V3.94) With the text's hypotheses (tame germ data, good position in a tame chart), every zero near a zero in good position is in good position in a tame chart, and the solution set has there regular data of a dimension that every nearby datum shares; also with good position in the text's sense (sc-complement) in the conclusion.
243. (V3.95) With the text's hypotheses (tame germ data, good position in a tame chart), the local solution set near a zero in good position consists of smooth points in good position in tame charts and is a tame sub-M-polyfold, a d-dimensional manifold with boundary with corners, without boundary if $d_X \equiv 0$.
244. (V3.96) At a zero, the linearization in every chart is sc-Fredholm of the index $n - N$ of any conjugating basic germ on every level, and onto on every level with dim ker $= n - N$ if the germ is in good position.
245. (V3.97) The dimension of the solution set near a zero in good position in a tame chart is the Fredholm index, dim ker $f'(y) =$ ind $f'(y)$, at every zero near it and in every chart; also in Theorems 3.1.26, 3.1.22, 3.6.11 (preprint 3.23, 3.19, 3.57).
246. (V3.98) For tame germ data, near a zero in good position in a tame chart, every zero y has $f'(y)$ onto with ind $f'(y) =$ ind $f'(x_0)$: the index is locally constant along the solution set; for tame germ data, without bump functions, the index is locally constant along the zero set without good position, the index of Definition 6.2.1 (preprint 6.1) with a locally defined sc⁺-section is well defined and exists at every smooth point, and it is constant along every sc-smooth path $\mathbb{R} \to X$ (Proposition 6.2.2 (preprint 6.2)); under bump functions the local

and the global reading of Definition 6.2.1 (preprint 6.1) agree.

247. (V3.99) At the zeros near x_0, $D\psi$ maps ker $f'(y)$, read in the chart at x_0, onto the tangent space of the solution graph.
248. (V3.100) For an sc-Fredholm section with tame germ data and a zero y with $d_X(y) = 0$, (f, y) is in good position in a tame chart if and only if $f'(y)$ is onto.
249. (V3.101) Over an M-polyfold without boundary, Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) hold for tame germ data with the hypothesis "$f'(x)$ surjective", with tame induced structure and the dimension the index.
250. (V3.102) At a point of the retract with ker $Dg \subseteq T_{z_0}O$, $\rho(z_0)F \subseteq Dg(T_{z_0}O)$ implies condition (3) of the filler there (and conversely when Dg is onto); so the filler condition at the nearby zeros holds whenever f has tame germ data and (f, x_0) is in good position in a tame chart.
251. (V3.103) The theorems of §3.6 assume, as Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) do, germ data and good position in a strong bundle chart whose base retraction is tame; compatible charts whose base retraction is the identity of U are tame, so the hypotheses can be met.
252. (V3.104) The product of the constant retraction onto $(1, 0)$ of $[0, \infty) \oplus W$ with any sc-smooth retraction of a nonempty set is not tame: the degeneracy index drops by one under it.
253. (V3.105) If f is an sc-Fredholm section with tame germ data and s an sc⁺-section, $f + s$ has tame germ data, in the same charts.
254. (V3.106) If f has tame germ data, $(r, x) \mapsto f(x) + \sum r_i s_i(x)$ is an sc-Fredholm section of $\pi^* P$ with tame germ data: the product chart of a tame chart is tame.
255. (V3.107) The good position of (f, y) obtained in one chart holds in every chart carrying germ data at y, in particular in the tame one.
256. (V3.108) The 2021 filling is the filling without tameness of r, with $r(0) = 0$; it is the project's filling, and a 2017 filling is a 2021 filling.
257. (V3.109) Over $r(a, b) = (0, b)$ on $[0, \infty) \times \mathbb{R}$, which is not tame, $g(a, b) = a$ fills the zero section: a 2021 filling need not be a 2017 one.
258. (V3.110) The 2021 sc-Fredholm germs and sections are the project's, and the 2017 germs those with tame germ data; the definitions are satisfiable.
259. (V3.111) For a 2021 filling with $\hat{f}(0) = 0$, $T\hat{f}(0)$ and $Dg(0)$ are onto, and Fredholm, together, with the same index.
260. (V3.112) Local compactness and convergence on every level hold for 2021 sc-Fredholm germs.
261. (V3.113) 2021 sc-Fredholm sections are stable under sc$^+$-perturbations and parametrized perturbations.
262. (V3.114) At a zero of a 2021 sc-Fredholm germ, the linearization is Fredholm of index $n - N$.
263. (V3.115) Without a tame chart, good position of ker $T\hat{f}(0)$ in T_0O need not give good position of ker $Dg(0)$ in E, the step of the proof of Theorem 3.6.8 on p. 138.
264. (V3.116) Zeros of a basic germ plus an sc$^+$-germ converging to 0 on level 0 lie, from some index on, on level m and converge there.
265. (V3.117) A pointed $N \cap C$, N in good position, is a quadrant of N, its extreme rays generated

by the preimages of the unit vectors.

266. (V3.118) Vanishing coordinates of points of $N \cap C$ are vanishing coordinates of extreme-ray generators, and good complements vanish there.
267. (V3.119) Either $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = 0$, or N is a line in $\mathbb{R}^{\Gamma^c} \oplus W$ and $\Gamma = \emptyset$; the case $N = 0$ belongs to the first alternative.
268. (V3.120) If $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = 0$, the projection to $\mathbb{R}^{\Gamma}$ carries $(N, N \cap C)$ onto a quadrant; if $0 \neq N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, $(N, N \cap C) \cong (\mathbb{R}, [0, \infty))$.
269. (V3.121) If N is in good position and $N \cap C$ is a closed convex cone, $N \cap C$ is a quadrant of N.
270. (V4.1) A corner chart is a chart onto a relatively open subset of a partial quadrant in a finite-dimensional space with the constant structure; a corner structure is an equivalent atlas of corner charts, of possibly varying dimension, and makes X tame.
271. (V4.2) The transition maps between compatible corner charts are C^{∞} on their relatively open overlaps.
272. (V4.3) An sc-Banach space with finite-dimensional E_0 has all levels equal and is sc-isomorphic to E_0 with the constant structure; all its points are smooth.
273. (V4.4) Corner structures are the atlases of charts onto relatively open subsets of partial quadrants in sc-Banach spaces with finite-dimensional E_0; those of fixed dimension n are the manifolds with corners of Theorem 3.2.3 (preprint 3.27).
274. (V4.5) An M-polyfold, tame or not, has a corner structure iff all its points are smooth, 1_0^1 is sc-smooth, and every point is the centre of a tame chart whose $T_0 O$ is finite-dimensional and in good position.
275. (V4.6) $1_1^0 : X^1 \to X$ is sc-smooth; if $X = X_1$ and 1_0^1 is sc-smooth it is an sc-diffeomorphism; some M-polyfold with $X_0 = X_\infty$ has 1_0^1 not sc-smooth; a finite-dimensional space with the constant structure has $V^1 = V$, and all its retracts are sc⁺-retracts.
276. (V4.7) An sc⁺-map is a map with sc-smooth lift into F^1; it raises levels, is sc-smooth, and is preserved by composition with sc-smooth maps on either side.
277. (V4.8) At a smooth point the derivative of an sc⁺-map is an sc⁺-operator, compact on every level; a compact projection has finite-dimensional range.
278. (V4.9) An sc⁺-retraction is an sc-smooth retraction that is an sc⁺-map (on an open subset of an M-polyfold: with values in X_1 and $1_0^1 \circ s$ sc-smooth).
279. (V4.10) A retraction with the same retract as an sc⁺-retraction is an sc⁺-retraction.
280. (V4.11) An sc⁺-retract is the retract of some, equivalently of every, retraction, which is sc⁺.
281. (V4.12) The retract of an sc⁺-retraction consists of smooth points and its tangent spaces are finite-dimensional.
282. (V4.13) A retract is an sc⁺-retract iff it consists of smooth points and the lift $O \to O^1$ is sc-smooth between retracts.
283. (V4.14) A chart's model is an sc⁺-retract iff its domain is smooth and 1_0^1 is sc-smooth on it; "$X_0 = X_\infty$ and 1_0^1 sc-smooth" iff every chart's model is an sc⁺-retract.
284. (V4.15) C_e is a partial quadrant with $C_e - e = C_e$; near e it agrees with C, and re-centring preserves degeneracy indices, minimal subspaces and the cones C_v.

285. (V4.16) Translation by a smooth e is an sc-diffeomorphism from relatively open subsets of C near e onto relatively open subsets of C_e, and carries retracts (tame retracts) to retracts (tame retracts).
286. (V4.17) Every smooth point of a chart's domain is the centre of a re-centred chart of X, tame if the original is.
287. (V4.18) Re-centring a strong bundle chart in the base gives a strong bundle chart; every smooth point of the base is the centre of one.
288. (V4.19) An sc⁺-retraction is C^1 on every level $k+1$ with operator-norm-continuous derivative, and satisfies the mean value estimate with any constant θ on small balls of C_{k+1}.
289. (V4.20) For an sc⁺-retraction onto $O \ni 0$ with T_0O in good position, a closed good complement Y and the projection p along Y, p is an sc-diffeomorphism from $O \cap W$ onto a relatively open $V \ni 0$ of $T_0O \cap C$, with inverse Γ, $\Gamma(a) - a \in Y$.
290. (V4.21) Near 0, $(1-p)$ contracts differences of points of O on level $k+1$, so two such points with the same projection coincide.
291. (V4.22) For small $a \in N \cap C$ there is a point $u \in O$ with $p(u) = a$, $u - a \in Y$ small on level 1.
292. (V4.23) Projections of points of O near 0 lie in C, and the neighbourhoods W, V and the inverse Γ exist with the required level-1 control.
293. (V4.24) Γ is Lipschitz on level 1, continuous into every level, and differentiable into every level with the frame formula $D\Gamma(a) = Dt(\Gamma(a))|_N(p\,Dt(\Gamma(a))|_N)^{-1}$.
294. (V4.25) Γ is C^∞ into every level, hence sc-smooth on the fixed V; p and Γ are sc-smooth between the retracts.
295. (V4.26) A chart composed with an sc-diffeomorphism between retracts is a chart of the M-polyfold.
296. (V4.27) Corner charts re-centre to tame centred charts whose T_0O is the whole finite-dimensional space, in good position; so (i) implies (ii).
297. (V4.28) Under (ii), $p \circ \psi$ (with a closed good complement) is a corner chart of X at every point; so (ii) implies (i).
298. (V4.29) A smooth finite-dimensional submanifold is locally the image of an sc⁺-retraction on an open set; it is a sub-M-polyfold.
299. (V4.30) In every chart, $\varphi \circ s \circ \varphi^{-1} \circ r$ is an sc⁺-retraction onto the chart image of A, whose points are smooth.
300. (V4.31) In a chart, T_aA is the image of T_aX under the tangent map of s, lies in T_aX, and is finite-dimensional.
301. (V4.32) At points with $d_X = 0$ a submanifold is in good position; if $d_X = 0$ on all of A, A has a corner structure, without tameness of X.
302. (V4.33) $A \subseteq X_\infty$, and all points of A are smooth for its induced structure.
303. (V4.34) A is locally closed in X.
304. (V4.35) $\partial A \subseteq \partial X$.
305. (V4.36) $1_0^1 : A \to A^1$ is sc-smooth for the induced structure.
306. (V4.37) Tangent spaces of retracts are germs, are monotone under inclusion of retracts, and

are mapped into each other by derivatives of maps between retracts.

307. (V4.38) Good position of A at a means, in a chart, that T_zA is in good position to C_z inside T_zO.
308. (V4.39) Good position of A at a does not depend on the chart, the data of Definition 4.2.1 (preprint 4.13) or the presentation.
309. (V4.40) A tame submanifold is one whose induced M-polyfold structure is tame.
310. (V4.41) A smooth finite-dimensional submanifold of a tame M-polyfold, in good position everywhere, has a corner structure and is tame.
311. (V4.42) In a tame centred chart, good position inside T_0O gives good position in E, with good complement $Y \oplus (1-\pi)E$ and constant $\varepsilon/(\|\pi\|+1)$.
312. (V4.43) For tame X, good position at a point holds at all nearby points of A.
313. (V4.44) General position of A at a means, in a chart, an sc-complement of T_zA in T_zO inside T_z^RO, level by level.
314. (V4.45) General position does not depend on the chart, the data or the presentation.
315. (V4.46) At the corner of a centred chart, $N+E_0^{\min} = E_0$ gives onto coordinates, good position with a closed complement in $E_0^{\min}$, and $d_{N\cap C} = d_C$ on $N \cap C$.
316. (V4.47) A finite-dimensional smooth N has in $N + R$ an sc-complement inside the sc-subspace R; $T_yQ + E_0^{\min} = E_0$ gives general position at y.
317. (V4.48) In a tame retract, general position at a smooth point implies good position.
318. (V4.49) In a tame centred chart, general position at the centre gives the graph description with $Y \subseteq E_0^{\min}$, equal degeneracy indices along the graph, and general position along it.
319. (V4.50) Theorem 4.2.11 (1)–(4) (preprint 4.22) holds in a tame chart centred at a point of general position, with F_σ a germ inside U; (1) and (4) for the faces of the chart model, not of U itself.
320. (V4.51) $\Gamma \circ \kappa_\sigma \circ p \circ t$ is an sc⁺-retraction onto $A \cap F_\sigma$ in the chart; its corner model is sc⁺ and in general position in the face model.
321. (V4.52) An sc⁺-retraction in a chart is an sc⁺-retraction on an open subset of X; $A \cap F_\sigma$ is a tame submanifold of the tame F_σ, in general position in it.
322. (V4.53) A stratified regular point is one where the projection maps the tangent space of the stratum through it onto $\mathbb{R}^m$, in some product chart.
323. (V4.54) Theorem 4.3.1 (preprint 4.23): almost every λ with $|\lambda| < \varepsilon$ is a stratified regular value whose fibre is a compact submanifold in general position with a tame corner structure.
324. (V4.55) In a tame chart codim $T_x^RX = d_X(x)$; general position at a gives $d_X(a) \leq \dim T_aA$, and a zero-dimensional submanifold in general position avoids ∂X.
325. (V4.56) Sard: the critical values of a C^∞ map between finite-dimensional spaces form a Haar-null set.
326. (V4.57) The strata of $\{\ell \geq 0\}$ have dimension $\dim V - \#\sigma$, and the stratified critical values of a C^∞ map form a null set.
327. (V4.58) Theorem 3.8.22 (preprint 3.78) for C^∞ data has a C^∞ solution.
328. (V4.59) At a stratified regular point the kernel of Df is in good position to the local partial quadrant, with a closed complement in the stratum.

329. (V4.60) Near a stratified regular point the fibre of f is the image of a C^∞ retraction whose derivative at the point is the identity on ker Df.
330. (V4.61) $e \mapsto (\Gamma(\rho(p(t(c,e)))))_2$ is an sc$^+$-retraction onto the chart image of A_c, whose tangent space contains the vertical tangent vectors of A.
331. (V4.62) General position of A and stratified regularity at z give general position of the fibre at z_2.
332. (V4.63) Every point of A has a neighbourhood and a null set of parameters off which the conclusions of Theorem 4.3.1 (preprint 4.23) hold locally.
333. (V4.64) Countably many such neighbourhoods cover A, so for any parameter set the good parameters have null complement.
334. (V4.65) In the 2021 form of Theorem 4.2.11 (1) (preprint 4.22), each of the $d_X(x)$ coordinate faces of the centred chart has a germ that contains x and lies in U.
335. (V4.66) Theorem 4.3.1 (preprint 4.23) in the 2021 form: with compact, possibly empty, fibres over $|\lambda| \leq \varepsilon$, the good parameters contain a measurable set of full measure in $\{|\lambda| < \varepsilon\}$.
336. (V4.67) In a local model, a vector field is a map $A : E_1 \to E_0$ with $x \mapsto (x, A(x))$ sc-smooth from O^1 to TO; then $Dr(x)A(x) = A(x)$ on O_1, and 0 is one.
337. (V4.68) On an M-polyfold, an sc-smooth vector field is an sc-smooth section $X^1 \to TX$ of the tangent projection; the zero section is one.
338. (V4.69) $T(O^1) = (TO)^1$, and the level map A_1 of a vector field A on O is a vector field on O^1.
339. (V4.70) For an sc^2 map on a relatively open subset of a partial quadrant, $D^2 f(z)(a,b)$, $z \in U_2$, $a, b \in E_1$, is symmetric in a and b.
340. (V4.71) In a local model, $[A,B](x) = D\tilde{A}(x)B(x) - D\tilde{B}(x)A(x)$ satisfies $Dr(x)[A,B](x) = [A,B](x)$ at every $x \in O_2$.
341. (V4.72) In a local model, $x \mapsto (x, [A,B](x))$ is an sc-smooth map from O^2 to TO.
342. (V4.73) For $x \in O_2$ the bracket $[A,B](x)$ need not lie in E_1: two sc-operators on a weighted sequence space.
343. (V4.74) In a local model, an sc-smooth map that relates two pairs of vector fields on level 1 relates their brackets on level 2, also on an open part $O \cap W$ of the retract.
344. (V4.75) The principal parts of an sc-smooth vector field are vector fields on the models, and their brackets in two charts are related by the transition map.
345. (V4.76) On an M-polyfold, $[A,B] : X^2 \to TX$, given in any chart by the bracket of the principal parts, is an sc-smooth section over the inclusion $X^2 \to X^1$, independent of the chart.
346. (V4.77) For an sc-smooth $f : X \to Y$ with $Tf \circ A = A' \circ f_1$ and $Tf \circ B = B' \circ f_1$ on X^1, $Tf \circ [A,B] = [A', B'] \circ f_2$ on X^2.
347. (V4.78) $A^{(1)} = \Phi^{-1} \circ A_1$ is an sc-smooth vector field on X^1, and $\Phi(A^{(1)}(x)) = A(x)$ for $x \in X^2$.
348. (V4.79) For every i, $A^{(i)}$, the index of A raised i times, is an sc-smooth vector field on X^i, and it is A through the identifications of Proposition 2.3.17.
349. (V5.1) Every strong bundle chart compatible with the atlas reads $Y_{0,1}$ by the fibre component

in F_1, and its $\Phi[1]$ is a chart of $Y[1]$.

350. (V5.2) The points of $P^{-1}(x)$ in $Y_{0,1}$ form a linear subspace, complete for the chart norm $|\cdot|_1$; the norms of two charts are locally uniformly comparable.
351. (V5.3) An auxiliary norm is a continuous function on $Y[1]$, a complete norm on every fibre, with $P(w_k) \to x$ and $N(w_k) \to 0$ forcing $w_k \to 0_x$.
352. (V5.4) A continuous fibrewise complete norm is auxiliary iff near every point it is two-sided comparable with $|h|_1$ in some strong bundle chart, also literally on the whole domain of the chart; auxiliary norms are so in every chart.
353. (V5.5) Every strong bundle has an auxiliary norm, glued from the chart norms by a partition of unity.
354. (V5.6) Two auxiliary norms are comparable by a continuous positive function f: $fN_0 \leq N_1 \leq f^{-1}N_0$.
355. (V5.7) Compact solution set, weakly proper and proper are as in the text, with $N = \infty$ off $Y_{0,1}$; for continuous f proper implies weakly proper implies compact solution set.
356. (V5.8) Near 0, the points where a basic germ (on a closed $\tilde{C}$) has level-1 norm $\leq c$ lie in a compact set, on a neighbourhood independent of c.
357. (V5.9) Strong bundle maps have locally uniform $(0,1)$ fibre bounds; at an sc-Fredholm germ the points with $|\Phi(f(x))_2|_1 \leq C$ near x_0 have compact closure, for every C.
358. (V5.10) An sc-Fredholm section with compact solution set is proper, without tameness; proper, weakly proper and compact solution set are equivalent for sc-Fredholm sections.
359. (V5.11) At a zero of an sc-Fredholm section there are nested neighbourhoods $U(i)$ with $\mathrm{cl}\{y \in U(i) \mid f(y) = 0\}$ compact in X_i.
360. (V5.12) With the initial topology of the inclusions, X_∞ is metrizable, and a compact solution set of an sc-Fredholm section is compact in X_∞, without tameness.
361. (V5.13) For an sc-Fredholm section with compact solution set, f^1 is an sc-Fredholm section of P^1 with compact solution set, and so is f^2 of P^2; the solution set is compact in every X_i.
362. (V5.14) $P^1 : Y_{1,1} \to X^1$ with the index raises of the strong bundle charts is a strong bundle, and for every sc-Fredholm section f, f^1 is an sc-Fredholm section of it, the filler condition on level 1 included.
363. (V5.15) $\mathbb{R}$ with the constant structure is an M-polyfold; sc-smoothness of maps is checked in local charts; C^∞ functions of sc-smooth functions are sc-smooth.
364. (V5.16) Admitting sc-smooth bump functions means a bump with $f(x) \neq 0$ inside every neighbourhood of x; Definition 5.5.2 (preprint 5.29) as written is weaker, follows from the density of such points, and gives them at the smooth points of an sc-Banach space.
365. (V5.17) With bump functions, every x has an sc-smooth $f : X \to [0,1]$ supported in a given neighbourhood, equal to 1 near x.
366. (V5.18) The zero section, products of sc-smooth functions and sc⁺-sections, and finite sums of sc⁺-sections are sc⁺-sections; sc⁺-sections take values in $Y_{0,1}$.
367. (V5.19) A local sc⁺-section of a chart of the atlas, cut off by an sc-smooth function supported in the chart domain and extended by 0, is an sc⁺-section.
368. (V5.20) With bump functions and an auxiliary norm, every smooth e over a smooth x is the

value at x of an sc$^+$-section supported in U with $N(s(y)) < N(e) + \varepsilon$ everywhere.

369. (V5.21) $\Gamma^{+,1}_U(P)$ is the set of sc$^+$-sections supported in U with $N \leq 1$; $\rho = \sup_x N(s(x) - s'(x))$ is a metric on it, not complete for the trivial bundle over $\mathbb{R}$, complete for $U = \emptyset$.
370. (V5.22) A section vanishing off an adapted U with $N(s) \leq 1$ has the zeros of $f + s$ in $\{x \in U \mid N(f(x)) \leq 1\}$, with compact closure; for s in the completion (uniform N-limits of allowable sections) the solution set of $f + s$ is closed, hence compact.
371. (V5.23) Transversality to the zero section is surjectivity of the linearization at every zero, in some and then every chart.
372. (V5.24) At a zero of an sc-Fredholm section, finitely many allowable sc$^+$-sections supported in a given U span a complement of the range of the linearization.
373. (V5.25) The slices $g + \sum \lambda_i t_i$ carry the zeros of F, are sc-Fredholm, and stay allowable for $m|\lambda| \leq \varepsilon$ when $N(s_0) \leq 1 - \varepsilon$.
374. (V5.26) A slice over a parameter where $\ker \mathrm{Lin}_F$ projects onto $\mathbb{R}^m$ is transversal; spanning sections make F transversal at $(0, y)$, also after adding sections.
375. (V5.27) $\mathbb{R}^m \times X$ has no boundary if X has none; transversality of F spreads to nearby zeros and gives good position on an open $V \supseteq \{0\} \times S_g$, which contains the zeros over small λ.
376. (V5.28) Near a zero of F in good position, the slices are transversal at their zeros over all λ off a null set.
377. (V5.29) Theorem 5.3.5 (preprint 5.13): for $\partial X = \emptyset$, f with tame germ data, there is U such that allowable perturbations have compact solution sets and $s_0 + \sum \lambda_i t_i$ is transversal with a manifold solution set for small λ off a null set.
378. (V5.30) Theorem 5.3.5 (preprint 5.13), dense form: such good allowable perturbations exist within ρ-distance ε of every allowable s'.
379. (V5.31) General position of (f, y): in a tame chart, $\mathrm{Lin}(z)$ onto and its kernel with an sc-complement inside $T^R_z O$, for every presentation.
380. (V5.32) Regular tangent germ data give sc$^+$-retractions, so solution sets in good position are smooth finite-dimensional submanifolds (Definition 4.2.1 (preprint 4.13)).
381. (V5.33) Near a zero in good position, the tangent space of the solution set equals $\ker \mathrm{Lin}(z)$ in every chart.
382. (V5.34) Near a zero in good position, general position of (f, y) is general position of the solution set at y, and is chart independent.
383. (V5.35) For X tame, general position of (f, x_0) persists at the nearby zeros, with the solution set a submanifold in general position there.
384. (V5.36) Properties (i)–(ii) are general position of F at $(0, y)$, survive added sections, and are achieved by finitely many allowable sections.
385. (V5.37) For X tame, a family with F in general position along $\{0\} \times S_{f+s_0}$ gives, for small λ off a null set, $f + s_\lambda$ with all the conclusions of Theorem 5.3.10 (preprint 5.18).
386. (V5.38) Theorem 5.3.10 (preprint 5.18), generic form, for X tame with bump functions and f with tame germ data.
387. (V5.39) Theorem 5.3.10 (preprint 5.18), dense form: perturbations in general position exist within ρ-distance ε of every allowable s'.

388. (V5.40) Theorem 5.3.12 (preprint 5.19): with good position at the boundary zeros, small $\sum \lambda_i t_i$ vanishing near ∂X give transversality and good position everywhere, a compact manifold with corners.
389. (V5.41) A generalized compact homotopy is a tame X, f with tame germ data and compact solution set, and t onto $[0, 1]$ with tame fibres on which f restricts to sc-Fredholm sections of the pull-backs.
390. (V5.42) Lemma 5.3.16 (preprint 5.23) is linear algebra, and the one-dimensional complement Z of ker $dt(x)$ exists iff $dt(x) \neq 0$.
391. (V5.43) The product rule and the affine chain rule hold at level-1 points; $\beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ is sc-smooth.
392. (V5.44) Lemma 5.3.17 (preprint 5.24): for $\mathrm{Lin}_f(z) \circ b$ with values in the line of a smooth e, an sc$^+$-section vanishing at x with small support cancels it on K.
393. (V5.45) Theorem 5.25 (preprint): for X tame and f proper with tame germ data, some allowable s puts $f + s$ in good position at every zero.
394. (V5.46) Lemma 5.26 (preprint): general position implies good position, with the interior condition, without tameness of X.
395. (V5.47) Admitting sc-smooth partitions of unity: every open cover has a subordinate partition of unity of sc-smooth functions.
396. (V5.48) Sums, products, locally finite sums and chart-supported extensions by 0 of sc-smooth functions are sc-smooth.
397. (V5.49) Sc-smooth partitions of unity give sc-smooth bump functions.
398. (V5.50) C^k-bump functions and C^k partitions of unity of a Banach space are as in the text.
399. (V5.51) The sc-smooth bump function property of a local model, with $f(x) \neq 0$; a retract is closed near each of its points.
400. (V5.52) Theorem 5.5.7 (preprint 5.34): bump functions on X iff a covering family of charts with the local bump property.
401. (V5.53) A local model on a Hilbert E_0 (or a norm dominated by a continuous quadratic form) has the bump property; M-polyfolds on sc-Hilbert spaces admit bump functions.
402. (V5.54) The sc-smooth approximation property of a local model, with $V = O \cap W$ and closures in E_0.
403. (V5.55) With sc-smooth partitions of unity, every chart (in particular every chart restricted to a set with closure in the chart domain) has the approximation property.
404. (V5.56) A covering family of charts with the approximation property gives sc-smooth Urysohn functions and sc-smooth partitions of unity.
405. (V5.57) Theorem 5.5.10 (preprint 5.37): sc-smooth partitions of unity iff a covering family of charts with the approximation property.
406. (V5.58) If E_0 admits C^∞ partitions of unity, every local model on E has the approximation property.
407. (V5.59) $c_0(\Gamma)$ and coordinate-wise C^k embeddings are as in the text; Theorem 5.5.13 (preprint 5.40) holds for every real Banach space.
408. (V5.60) WCG: a weakly compact set with dense span; reflexive: the bidual map is onto.

409. (V5.61) Reflexive and separable Banach spaces are WCG.
410. (V5.62) Theorem 5.5.16 (preprint 5.43) holds for every real Banach space; Hilbert spaces have C^∞-bump functions and are reflexive.
411. (V5.63) Corollary 5.5.17 (preprint 5.44) holds with no hypothesis (E_0 is separable).
412. (V5.64) In the 2021 Definition 5.2.1, weakly proper with the bound $N(s) \leq 1$ on U only is the earlier notion; weakly proper$_{<\infty}$ and proper$_{<\infty}$ are as in the text, with $N = \infty$ off $Y_{0,1}$.
413. (V5.65) Proper$_{<\infty}$ implies proper, weakly proper$_{<\infty}$ implies weakly proper, and for continuous f proper$_{<\infty}$ implies weakly proper$_{<\infty}$.
414. (V5.66) An sc-Fredholm section with compact solution set is proper$_{<\infty}$; for sc-Fredholm sections the five compactness properties are equivalent.
415. (V5.67) In Theorem 5.3.5 (preprint 5.13) the transversal perturbations have solution sets that are compact sub-M-polyfolds.
416. (V5.68) Definition 5.3.13 (preprint 5.20) in the 2021 form, with $dt(x)$ onto read in a chart; the earlier data together with (3) give it.
417. (V5.69) Under the 2021 condition (3), $dt(x) \neq 0$ in a chart, and Lemma 5.3.16 (preprint 5.23) holds with a one-dimensional complement Z of its kernel.
418. (V5.70) On $\mathbb{R}$, the graph section of the identity with $t(x) = 1/2 + \arctan(x)/\pi$ is a generalized compact homotopy in the 2021 sense but not in the earlier one.
419. (V5.77) A strong bundle has reflexive $(0,1)$-fibres when the compatible charts with reflexive F_1 cover X.
420. (V5.78) Weak convergence of the fibre parts in F_1 does not depend on the reflexive chart, and the two limits are the same point of $Y_{0,1}$.
421. (V5.79) Mixed convergence is well defined and implies convergence in Y.
422. (V5.80) A reflexive auxiliary norm is an auxiliary norm with $N(y) \leq \liminf N(y_k)$ along mixed convergent sequences.
423. (V5.81) With reflexive fibres a reflexive auxiliary norm exists, and it is comparable to every auxiliary norm by a continuous $f > 0$.
424. (V5.82) Reflexive local compactness: every point has a neighbourhood U with cl$\{q \in U \mid N(f(q)) \leq 1\}$ compact, for every reflexive N.
425. (V5.83) Points off X_1, and points with $N(f(x)) > 1$, have neighbourhoods on which $N(f) > 1$.
426. (V5.84) The closure of $\{x \in U \mid N(f(x)) \leq 1\}$ lies in $\{x \in \text{cl}(U) \mid N(f(x)) \leq 1\}$.
427. (V5.85) Compactness control on a neighbourhood of $S \cap \partial X$ in ∂X extends to an open $U \supseteq S$ with the same trace on ∂X.
428. (V5.86) Remark 5.2.9 restates Theorem 5.2.8.
429. (V5.88) An sc$^+$-section over ∂X is a section over ∂X with values in $W_{0,1}$ whose restrictions to the local faces are sc$^+$ in every tame chart.
430. (V5.89) An sc$^+$-section over ∂X extends to an sc$^+$-section $\tilde{s}$ with $N(\tilde{s}) \leq f$ and support in $\tilde{U}$.
431. (V5.90) Bounded sequences in a reflexive Banach space have weakly convergent subsequences.
432. (V5.91) $(g \circ P) \cdot N$ is a (reflexive) auxiliary norm for continuous $g > 0$, and every auxiliary

norm lies between two reflexive ones.

433. (V6.1) At a chart point, Lin(f, x) is $(f+s)'(x)$ plus the sc⁺-operators; its members are Fredholm on level 0 of one index, and it is convex and, with sections through $f(x)$, nonempty.
434. (V6.2) $\lambda(E)$ and $\lambda^*(E)$ are lines; wedges change by determinants; $\lambda(\Phi)$ is functorial and nonzero iff Φ is bijective.
435. (V6.3) $\iota : \lambda(E^*) \to \lambda^*(E)$ is given by $\det(e_i^*(a_j))$ and maps the dual-basis wedge to $(e_1 \wedge \cdots \wedge e_n)^*$.
436. (V6.4) $\lambda^*(\Phi) \circ \iota = \iota \circ \lambda(\Phi^*)$ for an isomorphism Φ.
437. (V6.5) Φ_E of a four-term exact sequence is the isomorphism of lines given by (91) for one choice.
438. (V6.6) Formula (91) holds for every choice.
439. (V6.7) Ψ_E maps the wedge of the basis of C, in the order of (91), and $a \otimes d^*$ to the wedge of the basis of B.
440. (V6.8) $\det(T) = \lambda(\ker T) \otimes \lambda^*(\operatorname{coker} T)$ is a line for T Fredholm.
441. (V6.9) An orientation of T is a ray of $\det(T)$; there are exactly two.
442. (V6.10) Good left-projections are as in Definition 6.4.3 (preprint 6.9), $\leq$ is a partial order, the remarks after the definition hold, and $\Pi_T \neq \emptyset$ for T Fredholm.
443. (V6.11) Two good left-projections have a common lower bound in Π_T, for any bounded T; the text's candidate need not be one.
444. (V6.12) The sequence $E(T, P)$ is exact.
445. (V6.13) $\gamma_T^P = \Phi_{E(T,P)}$ is an isomorphism $\det(T) \to \det(PT)$ given on wedges by (91).
446. (V6.14) For $P \leq Q$, $\gamma_{QT}^P \circ \gamma_T^Q = \gamma_T^P$.
447. (V6.15) $(\gamma_{RT}^P)^{-1} \circ \gamma_{ST}^P = \gamma_T^R \circ (\gamma_T^S)^{-1}$ for $P \leq S, R$.
448. (V6.16) The stabilization sequence of T_φ is exact, and stabilizations exist.
449. (V6.17) $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c]^{-1} 1^*$.
450. (V6.18) ι_φ is an isomorphism given by the formula of Proposition 6.4.15 (preprint 6.15), and $m + l = n$ always holds.
451. (V6.19) $T \oplus S$ is Fredholm and $\det(T) \otimes \det(S) \to \det(T \oplus S)$ is the isomorphism given by the formula for every choice.
452. (V6.20) Lemma 6.5.1 (preprint 6.17) holds, without the hypothesis that T is Fredholm.
453. (V6.21) The kernels $\ker(PS)$ near T are trivialized continuously by $k \mapsto k + y(k, S)$, with continuous inverse.
454. (V6.22) The frames $\omega(S)$ are nowhere zero, and the frames of two charts with the same P differ by a continuous nowhere-zero factor.
455. (V6.23) γ_{PS}^Q maps the P-frame to a continuous nowhere-zero multiple of the Q-frame, for any two charts.
456. (V6.24) The charts $\gamma_{(T,P,\varepsilon)}$, read in frames, form a vector prebundle with fibres $\det(S)$.
457. (V6.25) $\mathcal{B}$ is the collection of Definition 6.5.6 (preprint 6.22), read in the frames.
458. (V6.26) $\mathcal{B}$ is a basis of the topology of DET(E, F).
459. (V6.27) DET(E, F) is a topological line bundle, trivialized by the $\gamma_{(T,P,\varepsilon)}$.
460. (V6.28) The transition maps are multiplications by continuous nowhere-zero functions.

461. (V6.29) A line bundle over a nonempty simply connected, locally path-connected base has exactly two orientations.
462. (V6.30) DET(E, F) over a nonempty convex set of Fredholm operators is a line bundle with exactly two orientations.
463. (V6.31) Or_x^f, the orientations of DET(f, x), has exactly two elements, each determined by its value at one L.
464. (V6.32) Proposition 6.6.3 (preprint 6.28) holds with (1) in its 2021 form and with (1′), with global sections over $\mathbb{R}^n \times X$.
465. (V6.33) Finitely many sc⁺-sections put F in general position at all points $(0, (q, \Phi(q)))$, q near $[0, 1]^n$.
466. (V6.34) The kernels of the linearizations along a continuous family of zeros in good position form a continuous family of subspaces.
467. (V6.35) Continuous families of subspaces have determinant line bundles, and orientations are continued along $[0, 1]$.
468. (V6.36) Kernels, images and cuts of continuous families are continuous families, with their orientations.
469. (V6.37) In the product chart the slice kernel is ker$(T \oplus \varphi)$ with $T \in$ Lin(f, x), and $T \oplus \varphi$ is onto.
470. (V6.38) The sequence (96) is exact.
471. (V6.39) Continuation and the stabilization isomorphisms compose to a bijection of the rays at the two ends.
472. (V6.40) With the sign $(-1)^{\dim \operatorname{coker} T}$ the orientations carried by Φ_E along a family are continuous; without it they need not be.
473. (V6.41) The push-forward along stabilized kernels is defined with that sign and is compatible with pull-back and concatenation.
474. (V6.42) With the sections of Lemma 6.6.4 (preprint 6.29) the stabilized kernels along a family form a continuous family.
475. (V6.43) Push-forwards along sc-smooth paths exist in the sets of Proposition 6.6.3 (preprint 6.28) and split along halves.
476. (V6.44) Two push-forwards along the same path agree.
477. (V6.45) Appending sections does not change the push-forward.
478. (V6.46) Affine reparametrizations keep sc⁺-sections, and adding sections keeps general position.
479. (V6.47) Push-forwards along the two edges of a square of stabilized kernels with fixed end points agree.
480. (V6.48) Every smooth point has an open neighbourhood on which push-forwards exist and depend only on the end points.
481. (V6.49) Nearby smooth points are joined by sc-smooth paths in a chart, and concatenations are sc-smooth.
482. (V6.50) An orientable neighbourhood (a pair $V \subseteq U$) has exactly two continuous orientations, and every smooth point has one.

483. (V6.51) Reparametrizations of a path are homotopic to it with the end points fixed.
484. (V6.52) The sc-smooth homotopy classes of paths form a groupoid PX on the smooth points.
485. (V6.53) Two chart points at x identify $\mathrm{Lin}(f, x)$ and Or^f_x, with the cocycle rules.
486. (V6.54) Under this identification the local propagations of two charts agree.
487. (V6.55) Or^f_x read in a chosen chart has two elements and is identified with that of any chart at x.
488. (V6.56) Local push-forwards give a transport along paths, homotopy invariant, functorial, and unique.
489. (V6.57) There is exactly one functor Γ_f on PX with property (□).
490. (V6.58) Orientability in the sense of Definition 6.6.14 (preprint 6.38) is the existence of a family o_y preserved by Γ_f.
491. (V6.59) Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) are equivalent.
492. (V6.60) Orientations of f are the families preserved by Γ_f; they exist iff f is orientable, two on each class.
493. (V6.61) For a proper f in general position at its zeros, S is a compact manifold with boundary with corners, and the tangent orientations o_x of $T_xS = \ker f'(x)$ vary continuously along the sc-smooth paths of zeros in a chart.
494. (V6.62) An orientation of $\mathrm{DET}(f, x)$ gives exactly one orientation of $\mathrm{DET}(\hat{f}, (t, x))$, by $h \mapsto \tilde{h}$ with the sign $(-1)^{\dim \operatorname{coker} L}$, independent of L.
495. (V6.63) The proof of Lemma 6.3.3 (preprint 6.5) in §6.8.1 (preprint §6.7.1) goes through, with homogeneity in the bases of A and D.
496. (V6.64) The proof of Proposition 6.4.11 (preprint 6.12) in §6.8.2 (preprint §6.7.2) goes through, with the spanning proved directly in F.
497. (V6.65) Near (T_0, P_0), T is Fredholm, $P \in \Pi_T$, and $\mathrm{ind}(PT) = \mathrm{ind}(T) = \mathrm{ind}(T_0) = \mathrm{ind}(P_0T_0)$, the indices taken as maps $E \to F$.
498. (V6.66) Two finite-dimensional subspaces of the same dimension have a common algebraic complement.
499. (V6.67) For $P, Q \in \Pi_T$ with the same range, the text's A_T and B_T are isomorphisms making the diagram of $E(T, P)$ and $E(T, Q)$ commute; their continuity in T is V6.70.
500. (V6.68) Appending zero sections (Lemma 6.6.8) or arbitrary sections (Lemma 6.6.9) does not change the push-forward of orientations.
501. (V6.69) Two finite-dimensional subspaces of the same dimension of a normed space have a common closed, topological complement.
502. (V6.70) With such a closed X, A_S (on E) and B_S (lifted to F) depend continuously on S near T, also jointly with the vector, and for every such S they are the text's isomorphisms filling the diagram of $E(S, P)$ and $E(S, Q)$.
503. (V6.71) O_f is the set of (x, o), $x \in X_\infty$, $o \in \mathrm{Or}^f_x$, over X_∞ with the topology of Theorem 2.3.10, and properties (1)–(3) of Theorem 6.6.17 are stated as the text does, (3) with the push-forward along segments of the path.
504. (V6.72) Sc-smooth maps $\mathbb{R}^n \to X$ are continuous into X_∞, and points of X_∞ near x are joined to x by sc-smooth paths inside any given neighbourhood of x in X_∞.

505. (V6.73) The topology defined by the continuous orientations is Hausdorff and makes π_f a local homeomorphism.
506. (V6.74) Over M-polyfolds whose smooth points have small open neighbourhoods with connected smooth part ($\mathbb{R}^n$, for instance), that topology has exactly two continuous sections over such neighbourhoods.
507. (V6.75) Under the hypotheses of Theorem 6.6.13, it has property (3), it is the only topology with (1) and (3), and so the only Hausdorff one with (1)–(3); (2) is proved under the hypothesis of V6.74.
508. (V7.1) A groupoid is a structure of maps with the laws on composable pairs, and $|X|$ is its orbit space with π.
509. (V7.2) The isotropy group G_x has the multiplication $gh = g \circ h$.
510. (V7.3) An ep-groupoid satisfies (1)–(4) of Definition 7.1.3 (2017: 7.3), with (3) existential and (4) on the fibred product of Part I's Proposition 2.5.20 (2017: 2.15), equivalently in sheet form.
511. (V7.4) Properness of $t : s^{-1}(Q) \to X$ is equivalent to the sequential criterion and passes to smaller closures.
512. (V7.5) $\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1}$ is an sc-diffeomorphism with $\widehat{\varphi}(x) = y$; $T\varphi$ is only defined.
513. (V7.6) s, t preserve levels and the degeneracy index, so ml and d_X are constant on classes.
514. (V7.7) $d_{|X|}(|x|) = d_X(x)$ is well defined.
515. (V7.8) If X is tame, so is **X**.
516. (V7.9) The ep-groupoids and the functors sc-smooth on objects and morphisms form the category $\mathcal{EP}$.
517. (V7.10) A functor sc-smooth on objects and continuous on morphisms is sc-smooth on morphisms (2021); without continuity (2017) it need not be.
518. (V7.11) $\pi : X \to |X|$ is open.
519. (V7.12) Every point of $|X|$ has a monotone countable basis of open neighbourhoods.
520. (V7.13) $\pi(\mathrm{cl}_X A) = \mathrm{cl}_{|X|}\pi(A)$ for every $A \subseteq \mathrm{cl}_X U$, U with the properness property.
521. (V7.14) Only finitely many ψ have $s(\psi) = x$ and $t(\psi) \in U(x_0)$, and likewise with s and t exchanged.
522. (V7.15) G_x is finite.
523. (V7.16) $g \mapsto \varphi g \varphi^{-1}$ is an isomorphism $G_x \to G_y$, unique up to an inner automorphism of G_y.
524. (V7.17) Inside every open $V \ni x$ there is a natural representation on a connected open U with (1)–(3) of Theorem 7.1.13 (2017: 7.1).
525. (V7.18) M-polyfolds are locally path connected.
526. (V7.19) $\mathbf{U}(x)$ is the disjoint union of the connected open $\mathbf{U}(g)$, on which s, t are sc-diffeomorphisms onto $U(x)$.
527. (V7.20) $G \times U$ is an M-polyfold on which sc-smoothness is sc-smoothness on each slice.
528. (V7.21) $G_x \ltimes U(x)$ is an ep-groupoid.
529. (V7.22) $\Gamma : G_x \ltimes U(x) \to X$ is a fully faithful functor onto the full subcategory on $U(x)$.
530. (V7.23) $\widehat{\Phi}$ is a homomorphism into the sc-diffeomorphisms of $\mathbf{U}(x)$ with the two commutative diagrams.

531. (V7.24) Two natural representations at x agree on the component of x in $U \cap U'$, and they restrict to invariant connected open neighbourhoods.
532. (V7.25) Every finite set of mutually isomorphic objects has a natural representation on neighbourhoods inside a given neighbourhood.
533. (V7.26) Definition 7.1.18 (2017: 7.8) is the datum of Proposition 7.1.15 (2017: 7.5).
534. (V7.27) $\tau : G_x \backslash U(x) \to \pi(U(x))$ is a homeomorphism, and the functor $G_x \ltimes U(x) \to X$ is fully faithful.
535. (V7.28) Every natural representation is a local uniformizer, local uniformizers exist inside every neighbourhood, and they are injective on objects and morphisms.
536. (V7.29) An ep-subgroupoid is the full subcategory of a saturated sub-M-polyfold.
537. (V7.30) The inclusion $A \to X$ and the maps $i^{-1} \circ r$ are sc-smooth.
538. (V7.31) For saturated A, $\mathbf{A} = s^{-1}(A)$ is a sub-M-polyfold of **X**.
539. (V7.32) The full subcategory of a saturated sub-M-polyfold is an ep-groupoid with the restricted structure maps.
540. (V7.33) Effectiveness and regularity are as in Definition 7.2.1 (2017: 7.11), and the targets of morphisms from regular objects are regular.
541. (V7.34) X_{reg} is open.
542. (V7.35) X is effective iff natural representations with injective Φ exist inside every neighbourhood.
543. (V7.36) For effective X, X_{reg} is open and dense; conversely, density implies effectiveness.
544. (V7.37) T_x is a normal subgroup, the kernel of Φ for small $U(x)$, characterized by $s = t$ near g, and conjugation carries T_x to T_y.
545. (V7.38) $\sim$ is an equivalence relation, and $\mathbf{X}^R$ is the quotient.
546. (V7.39) The charts $\psi \circ \sigma \circ s$ cover $\mathbf{X}^R$ and have sc-smooth transitions, with no hypothesis.
547. (V7.40) If T_R is Hausdorff, X_R is an effective ep-groupoid with isotropy G_x/T_x.
548. (V7.41) The isotropy group of x in X_R is $G_x/T_x = G_x^{\text{eff}}$.
549. (V7.42) $|X|$ is locally metrizable, regular and Hausdorff, and metrizable if paracompact.
550. (V7.43) $|U(x)|$ is metrizable.
551. (V7.44) Every point of $|X|$ has a monotone countable basis of open neighbourhoods.
552. (V7.45) $|X|$ is Hausdorff.
553. (V7.46) $|X|$ is regular.
554. (V7.47) A second countable $|X|$ is metrizable.
555. (V7.48) Subsets of $|X|$ are locally metrizable, and paracompact ones are metrizable.
556. (V7.49) Component compact means compact in every connected component.
557. (V7.50) Michael's characterization of paracompactness and the paracompactness of locally finite unions of closed paracompact sets hold for regular spaces.
558. (V7.51) A component compact K has an open neighbourhood with paracompact metrizable closure.
559. (V7.52) X^1 is an ep-groupoid, tame if X is.
560. (V7.53) Definition 7.5.1 (2017: 7.15) is paracompactness of $|X|$, implied by second countability of X.

561. (V7.54) If $|X|$ is paracompact, $|X^1|$ is metrizable, hence paracompact.
562. (V7.55) For paracompact $|X|$, saturated open covers have invariant continuous partitions of unity.
563. (V7.56) Definition 7.5.5 (2017: 7.16) is the existence of invariant sc-smooth partitions of unity with saturated local finiteness.
564. (V7.57) Remark 7.5.6 (2017: 7.2) has no formal content.
565. (V7.58) Paracompact $|X|$ and sc-smooth partitions of unity on X give sc-smooth partitions of unity of X.
566. (V7.59) Proper subsets and the properness property are as in Definition 7.5.8 (2017: 7.17).
567. (V7.60) Proper subsets are closed.
568. (V7.61) π maps proper subsets to closed sets, and for paracompact $|X|$ every closed set is so obtained.
569. (V7.62) Theorem 7.6.1 (2017: 7.5) is Theorem 7.1.13 (2017: 7.1), proved along Appendix 7.6.1 (2017 7.5.1).
570. (V7.63) Morphisms between points of a small U_1 lie in the given open N_g.
571. (V7.64) $U = \bigcup_g t_g s_g^{-1}(U_2)$ is a connected invariant neighbourhood inside U_1.
572. (V7.65) Theorem 7.6.4 (2017: 7.6) is Theorem 7.5.7 (2017: 7.4).
573. (V7.66) Invariant continuous β supported in $U(x)$ have invariant sc-smooth majorants supported in $\mathrm{cl}_X U(x)$.
574. (V7.67) The proof of Theorem 7.5.7 (2017: 7.4) goes through, with the support bound of Lemma 7.6.5 (2017: 7.10) and Mathlib's bump coverings.
575. (V7.68) With $|X|$ paracompact, a Hilbert atlas gives sc-smooth partitions of unity of X.
576. (V7.69) A functor sc-smooth on objects and continuous on morphisms is sc-smooth on morphisms, with the local formula of the 2021 remark.
577. (V7.70) The data (7.3) exist, and on a small invariant $V(x)$ the natural representation is $\Gamma(g, v) = s_g^{-1}(v)$, $\Phi(g)(v) = t(s_g^{-1}(v))$.
578. (V7.71) Proposition 7.1.15 holds in its 2021 form; the printed 2017 form of (2) would force $s(\psi) = t(\psi)$.
579. (V7.72) Limits of sequences in $\mathbf{U}(\psi)$ whose sources or targets converge in $U(s(\psi)) \cup U(t(\psi))$ stay in $\mathbf{U}(\psi)$.
580. (V7.73) Corollary 7.1.17 (4)–(7): the natural action, equivariance along γ_φ, the multiplication as an sc-diffeomorphism onto $\mathbf{U}(\psi \circ \varphi)$, and the properness property.
581. (V7.74) T_x is the maximal subgroup acting trivially on a small $W(x)$, and conjugation by φ maps T_x onto T_y.
582. (V7.75) T_R is the finest topology making π continuous, and π is an open quotient map.
583. (V7.76) The conclusion of Lemma 7.2.11 holds at every morphism iff T_R is Hausdorff.
584. (V7.77) T_R is paracompact for every ep-groupoid.
585. (V7.78) The natural charts give an M-polyfold structure iff T_R is Hausdorff, and then X_R is an effective ep-groupoid.
586. (V7.79) T_R is Hausdorff iff $\mathbf{X}^T$ is closed in $\mathbf{X}$.
587. (V7.80) LUC is as in Definition 7.2.18, independent of the natural representation, and satisfi-

able, also for non-effective X.

588. (V7.81) T_R Hausdorff, LUC and $\mathbf{X}^T$ closed are equivalent, and under LUC X_R is well defined.
589. (V7.82) In Zhou's example $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$ is one point, T_R is not Hausdorff and LUC fails.
590. (V7.83) Under (SLUC), $(X^i)_R \cong (X_R)^i$ in $\mathcal{EP}$ for every i; for $i = 1$ the identification is an sc-diffeomorphism.
591. (V7.84) $X^{i+1} = (X^i)^1$, and under LUC every natural representation has the LUC property.
592. (V7.85) If X^i has LUC and the level-one sets $U(x) \cap X_1$ are connected and dense, every X^k, $k \leq i$, has LUC.
593. (V7.86) The Part I facts used in §7.4 are explicit hypotheses, satisfiable on a point.
594. (V7.87) Tg, defined without charts, is $T\Phi(g)$ on T_xX for every natural representation.
595. (V7.88) (ZHOU), T_x^∞ and (ZHOU′) are as in Definitions 7.4.5 and 7.4.7, and hold when $T_x = G_x$.
596. (V7.89) With the Part I facts, (ZHOU) and (ZHOU′) are equivalent.
597. (V7.90) With the Part I facts, (ZHOU) gives LUC for X and X^1, and for every X^i if (ZHOU) holds for every X^i.
598. (V7.91) (SLUC) is LUC for every X^i, and (ZHOU) with the Part I facts for every X^i gives (SLUC).
599. (V7.92) (VSLUC) implies LUC for X and X^1, (VSLUC) for every X^i gives (SLUC), and (ZHOU) implies (VSLUC).
600. (V7.93) (LR) and (LR′) are as in Definitions 7.4.19, 7.4.21 and are equivalent by Lemma 7.4.20.
601. (V7.94) With the Part I facts, (ZHOU) implies (LR), and (ZHOU) for every X^i gives (SLR).
602. (V7.95) Zhou's example fails (ZHOU), with no hypothesis.
603. (V7.96) Index-preserving local homeomorphisms induce local homeomorphisms of the tame boundaries.
604. (V7.97) (7.25)–(7.29) define the groupoid $(bX, b\mathbf{X})$, and B is a functor.
605. (V7.98) Theorem 7.6.9 holds in its topological part: metrizable bX, $b\mathbf{X}$, continuous B with d preimages, local homeomorphisms s, t, continuous u, ι, m, properness.
606. (V7.99) A continuous map of tame boundaries that covers an sc-smooth map is sc-smooth; lifts of local sc-diffeomorphisms are local sc-diffeomorphisms.
607. (V7.100) For a tame ep-groupoid, $(bX, b\mathbf{X})$ with (7.25)–(7.29) is a tame ep-groupoid, and B is an ep-functor.
608. (V7.101) Theorem 7.6.9 holds in full: (1)–(5), with the characterization and its uniqueness for centred V.
609. (V7.102) Under (ZHOU) for X and the Part I facts on every level, every X^i satisfies (ZHOU) and has LUC, and X has (SLUC) and (SLR).
610. (V7.103) (VSLUC) for X passes to every X^i; with the Part I facts, every X^i has LUC and X has (SLUC).
611. (V7.104) With the density of X_∞ and level-connected natural representations, (SLUC) implies (VSLUC); so (VSLUC) and (SLUC) are equivalent.
612. (V8.1) The tangent data are Part I's tangent maps of s, t, u, ι and the composition (8.5), unique on composable pairs.

613. (V8.2) Maps sc-smooth on open sets and local sc-diffeomorphisms have tangents with the chain rule, and local sc-diffeomorphisms have local sc-diffeomorphisms as tangents, bijective on the fibres.
614. (V8.3) $T(X, \mathbf{X})$ is an ep-groupoid with the text's structure maps, tame if X is, and $(p, \mathbf{p})$ is an sc-smooth functor into X^1.
615. (V8.4) On a sheet of s, the composition of $T\mathbf{X}$ is the tangent of $\varphi' \mapsto m(\sigma(t\varphi'), \varphi')$; $T(\mathbf{X}{}_s{\times}_t \mathbf{X}) = T\mathbf{X}{}_{Ts}{\times}_{Tt} T\mathbf{X}$ and $m_{T\mathbf{X}} = Tm$.
616. (V8.5) $s_T(\Psi \circ \Phi) = s_T(\Phi)$ and $t_T(\Psi \circ \Phi) = t_T(\Psi)$.
617. (V8.6) The composition of $T\mathbf{X}$ is associative.
618. (V8.7) $1_q = (1_x, Ts(1_x)^{-1}q)$ is a two-sided unit.
619. (V8.8) $(\varphi^{-1}, T\iota(\varphi)k)$ is a two-sided inverse of (φ, k).
620. (V8.9) s_T, t_T are surjective local sc-diffeomorphisms and all structure maps of $T(X, \mathbf{X})$ are sc-smooth.
621. (V8.10) $T(X, \mathbf{X})$ has the properness property, with $V' = p^{-1}(V)$.
622. (V8.11) If X is tame, Tr is a tame retraction and $T(X, \mathbf{X})$ is tame.
623. (V8.12) $T\varphi = t_T \circ (s_T|T_\varphi \mathbf{X})^{-1}$ is functorial and G is an sc-diffeomorphism onto the labelled pairs $\mathbf{X}^1{}_s{\times}_p TX$, compatible with the compositions.
624. (V8.13) The tangent of an sc-smooth functor is an sc-smooth functor.
625. (V8.14) $T : \mathcal{EP} \to \mathcal{EP}$ is a functor preserving tameness.
626. (V8.15) Natural transformations and natural equivalence are as in Definition 8.1.13 (2017: 8.1).
627. (V8.16) Natural equivalence is an equivalence relation.
628. (V8.17) The tangent of a natural transformation is a natural transformation between the tangent functors.
629. (V8.18) If $|X|$ is paracompact, $|TX|$ is metrizable, hence paracompact.
630. (V8.19) Strong bundle maps between strong bundles over M-polyfolds, read globally, preserve the double filtration, and the pull-back along s has its universal property on $[0]$ and $[1]$.
631. (V8.20) A strong bundle over an ep-groupoid is a structure map μ on the pull-back with (1)–(3) of Definition 8.3.1 (2017: 8.4).
632. (V8.21) $\mu(\psi, \cdot) : W_x \to W_y$ is a topological linear isomorphism.
633. (V8.22) $(W, \mathbf{X}{}_s{\times}_P W)$ is an ep-groupoid and (P, π_1) an sc-smooth functor.
634. (V8.23) A strong bundle map over ep-groupoids is compatible with μ as in Definition 8.3.2 (2017: 8.5).
635. (V8.24) A strong bundle functor is a functor of the functorial pictures inducing strong bundle maps on objects and morphisms.
636. (V8.25) Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6) are equivalent, and the strong bundle functor is determined by its object map.
637. (V8.26) Section functors are as in Definition 8.3.4 (2017: 8.7), and Γ, Γ^+ are closed under the linear operations.
638. (V8.27) A proper covering functor satisfies (1)–(3) of Definition 8.5.1 (2017: 8.8), with (3) on

Part I's fibred product.

639. (V8.28) L_F preserves the degeneracy index, and for tame X, Y, $d_{\mathsf{X}\,{}_s\times_F Y} = \frac{1}{2} d_{\mathsf{X}\times Y}$ along its image.
640. (V8.29) Composites of proper covering functors are proper covering functors.
641. (V8.30) Theorem 8.5.3 (2017: 8.5) holds for every natural representation on a small enough connected $U(x)$, with (5) read with s.
642. (V8.31) Morphisms over $F^{-1}(x)$ go to G_x, and the data of Theorem 8.5.3 (2017: 8.5) are determined by $U(x)$.
643. (V8.32) Diff$_{sc}$, covering functors and geometric lifts are as in Definition 8.5.5 (2017: 8.9), with $C =$ `SingleObj G`.
644. (V8.33) The $\varphi_{yy'}$ are functorial and the $\tau(y)$ are equivariant along $f : G_y \to G_x$.
645. (V8.34) Disjoint unions of M-polyfolds are M-polyfolds, with summand-wise sc-smoothness, degeneracy and tameness.
646. (V8.35) For nonempty D, a geometric lift of a covering functor gives a proper covering functor onto $G_x \ltimes U(x)$.
647. (V8.36) For ep-groupoids in `Type`, a proper covering functor is near x and $F^{-1}(x)$ the functor of a geometric lift, up to fully faithful open embeddings.
648. (V8.37) Proper strong bundle covering functors are as in Definition 8.5.9 (2017: 8.10), with (3) on a pull-back.
649. (V8.38) $\#\text{mor}(F^{-1}(x)) = \#G_x \cdot \#F^{-1}(x)$, and the classes of $F^{-1}(x)$ are indexed by their images in $|Y|$.
650. (V8.39) $\#G_x \cdot \#\Theta_j = \#G_{y_0} \cdot (\#\Theta_j)^2$ and $\#G_x = \#G_{y_0} \cdot \#\Theta_j$.
651. (V8.40) For isomorphic x, x' the classes of $F^{-1}(x)$ and $F^{-1}(x')$ correspond, with a unique common index.
652. (V8.41) For isomorphic x, x', corresponding classes have equal cardinality and $\#F^{-1}(x) = \#F^{-1}(x')$.
653. (V8.42) $\#F^{-1}(x)$ is constant on connected components of $|X|$ and an sc-smooth functor $X \to \mathbb{N}$.
654. (V8.43) $\mu_X(F(\varphi)) \circ A_{s(\varphi)} = A_{t(\varphi)} \circ \mu_Y(\varphi)$ for every morphism φ of Y.
655. (V8.44) If X satisfies (ZHOU) and $T(g, h) = \text{id}$ implies $Tg = \text{id}$, then TX satisfies (ZHOU).
656. (V8.45) With $U(x) \cap X_1$ connected and dense and p open, LUC for X^1, or (SLUC) for X, gives LUC for TX.
657. (V8.46) Auxiliary norms over ep-groupoids are Part I's auxiliary norms invariant under μ.
658. (V8.47) $|X|$, $|W[0]|$ and $|W[1]|$ are paracompact together, $|W[1]|$ as an orbit space.
659. (V8.48) The local regularity property of strong bundles is as in Definition 8.4.2, and holds over trivial ep-groupoids.
660. (V8.49) A covering functor is injective on isotropy groups, and Example 8.5.6 shows it need not be surjective.
661. (V8.50) Definition 8.5.9 in the 2021 form is equivalent to the reading of Definition 8.10 (2017) in V8.37.
662. (V8.51) $B(w) = (P(w), A(w))$ is a strong bundle isomorphism $W \to F^*V$ covering the

identity, compatible with the structure maps by (8.23).

663. (V8.52) If X has (SLUC), with the hypotheses of Proposition 8.1.17 (1) on every X^i, then TX has (SLUC), through $T(X^i) \cong (TX)^i$.
664. (V9.1) A branched ep⁺-subgroupoid is a nonnegative functor, supported in the smooth set with paracompact orbit support, locally a weighted sum of indicators of properly embedded smooth submanifolds through the point.
665. (V9.2) The dimension of a submanifold at a point is the dimension of its tangent space in a chart, independent of the chart; Θ is n-dimensional if it has n-dimensional local branching structures.
666. (V9.3) On $\mathbb{R}^n$, finite weighted sums of indicators of closed submanifolds and indicators of locally closed submanifolds are branched ep⁺-subgroupoids.
667. (V9.4) The graphs of β_1, β_2 and those of 0, $\beta_1 + \beta_2$ are two local branching structures of one Θ on $\mathbb{R}^2$ with no common sheet near the origin.
668. (V9.5) $|\mathrm{supp}(\Theta)|$ is locally closed in $|X|$.
669. (V9.6) Θ is closed or compact if $|\mathrm{supp}(\Theta)|$ is closed or compact in $|X|$.
670. (V9.7) At every point of the support there are symmetric local branching structures inside every neighbourhood.
671. (V9.8) Θ is of orbifold type if its values are 0 and 1, of manifold type if moreover there is at most one morphism between two points of its support.
672. (V9.9) The local dimensional decomposition $\Theta = \sum_k \Theta_k$ is constructed; the uniqueness of the global decomposition, as stated, does not hold as we read it.
673. (V9.10) The dimensional decompositions of two local branching structures agree near the point, unconditionally over $\mathbb{R}^m$ and under the graph representation in general.
674. (V9.11) $\mathrm{Gr}(X)$ is the category of formal rational sums of tangent sets at smooth points, with the push-forward by morphisms.
675. (V9.12) T_yM is the set of tangent vectors fixed by the tangent of the retraction; in a chart it is a linear subspace of the dimension of M.
676. (V9.13) If the tangent sums do not depend on the structure, T_Θ is a section functor of $\mathrm{Gr}(X)$ with the text's (1) and (2).
677. (V9.14) The tangent sums do not depend on the structure, unconditionally over $\mathbb{R}^m$ and under the graph representation and local constancy of dimension in general.
678. (V9.15) A point is good if near it all sheets through it coincide, equivalently if Θ is constant on the support near it.
679. (V9.16) Being a good point does not depend on the local branching structure.
680. (V9.17) Good points are good for every structure, and near a good point Θ is a multiple of the indicator of one submanifold.
681. (V9.18) $s(\varphi)$ is good iff $t(\varphi)$ is good.
682. (V9.19) The good and bad sets are saturated subsets of the support.
683. (V9.20) The good set is open and dense in the support, the bad set closed and nowhere dense.
684. (V9.21) Every sheet of an n-dimensional structure at x contains good points converging to x, unconditionally over $\mathbb{R}^m$ and under invariance of domain in general.

685. (V9.22) Under the graph representation, each class of sheets with one tangent space at x has finite slices near x whose Θ-values sum to the weights of the class.
686. (V9.23) $T\Theta$ is a well-defined nonnegative functor on TX, and a branched ep$^+$-subgroupoid if the TM_i are smooth submanifolds of TX and $|X|$ is paracompact.
687. (V9.24) For $x \in M \cap \partial X$, $d_M(x) = 0$ iff $T_xM \subseteq T_x^RX$; this also holds at every $x \in M$, while the printed (2) □ (1) does not hold as we read it at interior points.
688. (V9.25) $x \in \partial M$ iff $x \in \partial X$ and $T_xM \nsubseteq T_x^RX$.
689. (V9.26) The boundary weights do not depend on the structure, vanish where $d_X = 0$, and define a nonnegative functor $\partial\Theta$ supported in ∂X.
690. (V9.27) $\partial\Theta$ satisfies (1) and (2) and is the only functor vanishing off $\mathrm{supp}(\Theta)$ with (2) on $\mathrm{supp}(\Theta)$; (1) and (2) alone do not determine it.
691. (V9.28) $\partial\Theta \leq \Theta$, and $\partial\Theta$ need not be Θ restricted to ∂X.
692. (V9.29) Oriented tangent spaces are tangent sets with a chart-independent class of positive frames, and $\widehat{\mathrm{Gr}}(X)$ is the category of their formal sums.
693. (V9.30) An orientation of Θ is a natural lift of T_Θ that is locally the oriented sum of an oriented local branching structure.
694. (V9.31) The indicator of a point has infinitely many orientations.
695. (V9.32) Two orientations of an n-dimensional Θ that agree at good points agree at all smooth points; without equidimensionality the lemma does not hold as we read it.
696. (V9.33) In Example 9.3.5 the two orientations agree at the origin, differ for $x > 0$, agree for $x < 0$; Θ has infinitely many orientations, a three-parameter rational family, of which six have integer weights.
697. (V9.34) Θ is tame if it has local branching structures with tame induced structures on domains with the natural G_x-action and the properness property.
698. (V9.35) Tame submanifolds with the same tangent space at y have the same degeneracy index at y.
699. (V9.36) The outward-first convention orients a hyperplane of T_yM, independently of the outward vector chosen; for a tame sheet the stratum $\{d_M = 1\}$ is an $(n-1)$-dimensional submanifold oriented by it.
700. (V9.37) The degree-ℓ parts of the oriented sums of two structures agree, so the boundary orientation (9.12) does not depend on the structure.
701. (V9.38) $\hat{T}_{\partial\Theta}$ is 0 off the support and the weighted sum of the boundary orientations of the sheets with $d_{M_i}(y) = 1$; on a face-structured sheet these orientations extend uniquely to the closed faces, as Part I's face orientations.
702. (V9.39) The relative tangent space T_z^KM is the span of limiting unit directions in a chart, independent of the chart, and essential points are limits of points where it is everything.
703. (V9.40) The essential set of K is closed in K.
704. (V9.41) Property (T) says that the orientation relation of two sheets with the same tangent space is the same wherever their tangent spaces agree.
705. (V9.42) For sheets of one dimension, also in M-polyfold charts, essentiality in $M_i \cap M_{i'}$ does not depend on the reference sheet; for different dimensions it does.

706. (V9.43) A set has measure zero if its chart images have Lebesgue measure zero.
707. (V9.44) The non-essential points of any subset of a finite-dimensional manifold, or of an n-dimensional submanifold of an M-polyfold, form a set of measure zero.
708. (V9.45) y-equivalence is an equivalence relation on the sheets through y, and equivalent sheets have the same tangent space.
709. (V9.46) The points of M_U with at least two y-classes form a null set.
710. (V9.47) Off a null set, M_U splits into pieces M_U^J on which Θ, T_Θ and $\hat{T}_\Theta$ have the text's constant form.
711. (V9.48) Two oriented structures with the same weights and oriented sums agree, off a null set, piece by piece as in (1)–(5) of the text, without Property (T).
712. (V9.49) A subset of $S = |\text{supp}(\Theta)|$ has measure zero if, for a system of null sets in the sheets of a class, a tame local branching structure at every point of the support has null preimages.
713. (V9.50) For an n-dimensional Θ, measure zero does not depend on the local branching structures chosen; across dimensions the transfer of null sets between sheets does not hold.
714. (V9.51) $L(S)$ is the σ-algebra generated by the Borel sets and the sets of measure zero, and its members are the sets that differ from a Borel set by a set of measure zero.
715. (V9.52) Measure zero in ∂S and $L(\partial S)$ are defined from the $(n-1)$-dimensional Hausdorff null sets in the boundaries of the sheets and do not depend on the structures chosen.
716. (V9.53) The hypotheses of §9.5 are satisfiable on $\mathbb{R}^2$, with nonempty sets of measure zero and members of $L(S)$ that are not Borel.
717. (V9.54) For a natural representation of G_x and $y \in U(x)$, $\#(G_x/H) \cdot \#G_y = \#G_x$, and preimages decompose disjointly over coset representatives on a slice around y; the measures of sc-forms on U and on V agree.
718. (V9.55) $\#T_x \cdot \#G_x^{\text{eff}} = \#G_x$, and under LUC (9.23) holds and the effective measures agree.
719. (V9.56) The orders of the isotropy groups and of the effective isotropy groups are invariant along morphisms, $|M_U| = |M_{U'}|$, and the measures of sc-forms are invariant.
720. (V9.57) For effective X the effective and full isotropy orders agree, and the effective order on X is the isotropy order on X_R; under LUC the effective integral on X is the integral on X_R.
721. (V9.58) Two structures at x have the same top dimension.
722. (V9.59) The unions of the top-dimensional sheets of two structures agree on the common domain.
723. (V9.60) At top-good points the top-dimensional parts of two structures agree.
724. (V9.61) Top-goodness does not depend on the structure, unconditionally over $\mathbb{R}^m$ and under the graph representation in general.
725. (V9.62) The top-dimensional part at x is the sum of Θ over finite top slices, and the class sums agree for two structures.
726. (V9.63) A paracompact subset of $|X|$ is metrizable, and if $|\text{supp}(\Theta)|$ is component compact it is paracompact.
727. (V9.64) $|\text{supp}(\Theta)|$ carries a unique M-polyfold structure in case (1) and, in case (2), the orbifold structure of the text's groupoid, independent of the choices up to a generalized isomorphism.

728. (V9.65) For n-dimensional Θ with $d_{|X|} = 0$ on the support, these are Mathlib manifolds and local diffeomorphisms, except the composition of the orbifold groupoid, which is sc-smooth.
729. (V9.66) For every integration of forms over the oriented sheets, the branched measures of Theorems 9.5.4, 9.5.5 exist and are unique; for sc-forms and $n \geq 1$ the integration is constructed, and the theorems hold as printed with "oriented, tame" read as a covering by oriented tame structures.
730. (V9.67) For every $n \geq 1$ the boundary measures of Theorems 9.5.15, 9.5.16 exist and are unique, with the local formula for every oriented tame structure, under the text's hypotheses.
731. (V9.68) $\int_{\hat{\Theta}} d\omega = \mu_{\omega}^{\partial\hat{\Theta}}(\partial S)$ for every $n \geq 1$, and the effective form under LUC, under the text's hypotheses; the closed hemisphere and the closed half-circle satisfy them with $\partial S \neq \emptyset$.
732. (V9.69) The measure of $f \cdot \omega$ has density $\bar{f}$ with respect to the measure of ω, globally on $L(S)$, for bounded measurable $\bar{f}$.
733. (V10.1) An equivalence of ep-groupoids is a full and faithful sc-smooth functor, a local sc-diffeomorphism on objects, inducing a homeomorphism of orbit spaces.
734. (V10.2) An sc-smooth functor is an equivalence iff it is an equivalence of categories and a local sc-diffeomorphism on objects.
735. (V10.3) The refinement of a nonempty ep-groupoid along an open covering is an ep-groupoid, and its projection to the ep-groupoid is an equivalence.
736. (V10.4) Natural equivalence of sc-smooth functors is an equivalence relation compatible with composition on both sides.
737. (V10.5) A functor naturally equivalent to an equivalence is an equivalence.
738. (V10.6) If F and $\Phi \circ F$ are equivalences, so is Φ.
739. (V10.7) The tangent of an equivalence is an equivalence.
740. (V10.8) For id and $-$id on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$, no family through 1_0 is even continuous at 0, while $\alpha_z = (-1, z)$ is sc-smooth.
741. (V10.9) For an equivalence and $\psi : F(x) \to y$, $g \mapsto \psi \circ F(g) \circ \psi^{-1}$ is an isomorphism $G_x \to G_y$ mapping T_x onto T_y.
742. (V10.10) For an equivalence, F_R is continuous and open, and one reduced morphism space is Hausdorff iff the other is.
743. (V10.11) An equivalence with Hausdorff reduced morphism spaces descends to an equivalence of the reduced ep-groupoids.
744. (V10.12) The functor of Example 10.1.15 is compatible with no map of the reduced morphism spaces.
745. (V10.13) (LUC) holds for one side of an equivalence iff it holds for the other.
746. (V10.14) The Zhou condition holds for one side of an equivalence iff it holds for the other.
747. (V10.15) (SLUC) holds for one side of an equivalence iff it holds for the other.
748. (V10.16) (SLR) passes along equivalences, through Proposition 10.1.9 for the lifts.
749. (V10.17) The weak fibred product of two sc-smooth functors is a groupoid with the structure maps of Definition 10.2.1.
750. (V10.18) If F or G is a local sc-diffeomorphism on objects, the weak fibred product is an ep-groupoid with the iterated sub-M-polyfold structures.

751. (V10.19) For F an equivalence, π_Z is an equivalence, $F \circ \pi_X \simeq G \circ \pi_Z$, $d = d_Z \circ \pi_Z$, and tameness and paracompactness of the orbit space pass from Z.
752. (V10.20) For G an equivalence, π_X is an equivalence and $d = d_X \circ \pi_X$; for two equivalences, $d_X(x) = d(x, \phi, z) = d_Z(z)$.
753. (V10.21) For two equivalences both projections of the weak fibred product are equivalences, and $\tau(x, \phi, z) = \phi$ is natural.
754. (V10.22) A strong bundle equivalence is a strong bundle functor over an equivalence, an equivalence at level $[0]$, a homeomorphism of $|W[1]|$, and locally a strong bundle isomorphism on objects and morphisms.
755. (V10.23) Push-forward and pull-back of section functors along a strong bundle equivalence are well defined and mutually inverse.
756. (V10.24) $|\Phi| = |\Psi|$ implies $\Phi^* f' = \Psi^* f'$ for every section functor f'.
757. (V10.25) $|\Phi| = |\Psi|$ implies $\Phi_* f = \Psi_* f$ for every section functor f.
758. (V10.26) $(\Psi \circ \Phi)^* = \Phi^* \circ \Psi^*$ and $(\Psi \circ \Phi)_* = \Psi_* \circ \Phi_*$.
759. (V10.27) Φ_* and Φ^* are mutually inverse bijections of sc-smooth section functors, depend only on $|\Phi|$, and preserve sc$^+$-section functors.
760. (V10.28) A natural transformation of strong bundle functors is a natural family $T(w) : \Phi(w) \to \Psi(w)$ over a natural transformation τ of the base functors.
761. (V10.29) $T(w) = (\tau(P(w)), \Phi(w))$ and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$, so T and Ψ are determined by Φ and τ, and T is a strong bundle map.
762. (V10.30) Definitions 10.4.8 and 10.4.10 agree for strong bundle functors, and naturally equivalent strong bundle functors induce the same map of orbit spaces.
763. (V10.31) Strong bundles over ep-groupoids and strong bundle functors form a category with the strong bundle equivalences as a distinguished class.
764. (V10.32) Isomorphisms of $\mathcal{EP}$ and composites of equivalences are equivalences, and so is a functor naturally equivalent to one.
765. (V10.33) A refinement of a diagram $X \leftarrow A \to Y$ is given by an equivalence of the middle ep-groupoids commuting with both legs up to natural equivalence.
766. (V10.34) A refinement of a refinement is a refinement.
767. (V10.35) A common refinement of two diagrams refines both.
768. (V10.36) Having a common refinement is transitive, through the weak fibred product (10.11).
769. (V10.37) Equivalence of diagrams is an equivalence relation.
770. (V10.38) The map $|\Phi| \circ |F|^{-1}$ of orbit spaces is continuous and unchanged under refinement.
771. (V10.39) $|[d]|$ is well defined, and a homeomorphism when both legs are equivalences.
772. (V10.40) The composition of classes of diagrams through weak fibred products is well defined, associative and unital.
773. (V10.41) Generalized isomorphisms are the isomorphisms of $\mathcal{EP}(E^{-1})$.
774. (V10.42) $[X \leftarrow A \xrightarrow{\Phi} Y]$ is a generalized isomorphism iff Φ is an equivalence, with the reversed diagram as inverse.
775. (V10.43) $|g \circ f| = |g| \circ |f|$ for generalized maps.
776. (V10.44) Ep-groupoids and generalized maps form the category $\mathcal{EP}(E^{-1})$.

777. (V10.45) $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ is a functor inverting the equivalences, with $i(F)^{-1} = [Y \xleftarrow{F} X \xrightarrow{1} X]$.
778. (V10.46) The orbit spaces and the maps $|[d]|$ form a functor $\mathcal{EP}(E^{-1}) \to \text{TOP}$.
779. (V10.47) (LUC) is invariant under generalized isomorphisms.
780. (V10.48) Equivalence of pairs (X, β) by generalized isomorphisms is an equivalence relation.
781. (V10.49) The ep-groupoids with (LR) span full subcategories of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$.
782. (V10.50) (LR), (LUC), (SLR), (SLUC) and (ZHOU) are invariant under generalized isomorphisms.
783. (V10.51) Refinement of diagrams of strong bundles is reflexive and transitive, having a common refinement is an equivalence relation, and generalized strong bundle maps are its classes.
784. (V10.52) A generalized strong bundle map covers a generalized map of the base ep-groupoids, and $|[D]| = |\Psi| \circ |\Phi|^{-1}$ is continuous and independent of the representative.
785. (V10.53) $|P'| \circ |[D]| = |[d]| \circ |P|$.
786. (V10.54) A diagram of ep-groupoids and $|[d]|(|x|) = |x'|$ give morphisms a, b and a group homomorphism $\gamma : G_x \to G'_{x'}$, an isomorphism when the second leg is an equivalence.
787. (V10.55) Strong bundles over ep-groupoids and generalized strong bundle maps form a category $\mathcal{SEP}(F^{-1})$, with a functor from $\mathcal{SEP}$ that inverts the strong bundle equivalences.
788. (V10.56) A generalized strong bundle isomorphism is the class of a diagram whose two legs are strong bundle equivalences.
789. (V10.57) A generalized strong bundle isomorphism covers a generalized isomorphism of the base, induces one of the total spaces, has a homeomorphism as its orbit map, and is an isomorphism of $\mathcal{SEP}(F^{-1})$.
790. (V10.58) Push-forward and pull-back of section functors along a diagram of strong bundle equivalences depend only on its map of orbit spaces.
791. (V10.59) A generalized strong bundle isomorphism induces mutually inverse bijections $[D]_*$, $[D]^*$ of the sc-smooth and of the sc$^+$-section functors, which depend only on $|[D]|$.
792. (V10.60) There are generalized strong bundle isomorphisms $[D] \neq [D']$ with $|[D]| = |[D']|$, whose base classes differ too.
793. (V10.61) For a generalized strong bundle isomorphism, one strong bundle has the local regularity property (read with local strong bundle isomorphisms) iff the other has.
794. (V10.62) (LR) passes along equivalences, with natural representations carried along $Q = \widehat{\psi} \circ F$.
795. (V10.63) Equivalences with $|F| = |G|$ have local sc-smooth families $\alpha_z : F(z) \to G(z)$ and $|TF| = |TG|$ when one side has (LR).
796. (V10.64) (LR) is invariant under generalized isomorphisms.
797. (V10.65) On $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action, which has (LR), two generalized isomorphisms with the same orbit map differ, so Theorem 10.3.19 does not hold as we read it.
798. (V10.66) Generalized isomorphisms with the same orbit map are equal when one side has (LR) and the target is effective.
799. (V10.67) The strong bundle of Remark 10.5.5 has the local regularity property, so Theorem

10.5.7 does not hold as we read it.

800. (V10.68) For a strong bundle equivalence Φ and a strong bundle functor Ψ with the same target, the pull-back of the source of Ψ to the weak fibred product of the bases is a weak fibred product of strong bundles: one projection is a strong bundle equivalence, the square commutes up to natural equivalence, and the other projection is a strong bundle equivalence when Ψ is one.
801. (V10.69) Having a common refinement is an equivalence relation on diagrams of strong bundles, and two diagrams have the same class iff they have a common refinement.
802. (V10.70) The composite of diagrams of strong bundles through the weak fibred product is compatible with refinements, associative and unital up to equivalence, and functorial on strong bundle functors.
803. (V10.71) A diagram of strong bundles represents an isomorphism of $\mathcal{SEP}(F^{-1})$ iff its second leg is a strong bundle equivalence, and then the reversed diagram is its inverse.
804. (V11.1) The tangent functor induces a functor $T : \mathcal{EP}(E^{-1}) \to \mathcal{EP}(E^{-1})$, $[X \leftarrow A \to Y] \mapsto [TX \leftarrow TA \to TY]$, extending T on $\mathcal{EP}$.
805. (V11.2) A generalized isomorphism preserves the degeneracy index: $d_X(x_0) = d_Y(y_0)$ whenever $|f|(|x_0|) = |y_0|$.
806. (V11.3) A face of a tame ep-groupoid is the full subcategory over the closure in $|X|$ of a connected component of $\{d_{|X|} = 1\}$; face structured means every z lies in exactly $d_{|X|}(z)$ faces of $|X|$.
807. (V11.4) $[0, \infty)^2 \oplus W$ with only identities is face structured; with $\mathbb{Z}/2$ exchanging the coordinates it is weakly face structured and not face structured.
808. (V11.5) An equivalence induces the bijection $\theta \mapsto |\Phi|(\theta)$ of faces, with $\Phi_*(X_\theta) = \text{saturation}(\Phi(X_\theta))$ and $\Phi^*(Y_{\theta'}) = \Phi^{-1}(Y_{\theta'})$; for generalized isomorphisms $[d]_* = \Psi_* \circ \Phi^*$ and $[d]^* = \Phi_* \circ \Psi^*$ are mutually inverse.
809. (V11.6) A generalized isomorphism induces a bijection of the faces of $|X|$ onto those of $|Y|$ and homeomorphisms $\theta \to [d]_*\theta$; for face structured X each face is a tame ep-groupoid and $[d]$ restricts to generalized isomorphisms $[d]_F$; two faces meet in a union of faces of each, and in general not in one face.
810. (V11.7) Being tame and face structured is invariant under generalized isomorphisms.
811. (V11.8) Generalized isomorphisms with $|f| = |g|$ induce the same bijection of faces, and the generalized isomorphisms f_F, g_F on faces have the same orbit map.
812. (V11.9) $d \circ d = 0$; generalized maps have pull-backs commuting with d; generalized isomorphisms have push-forwards commuting with d and inverse to the pull-backs; under the local regularity property they depend only on the orbit maps.
813. (V11.10) For an equivalence F, F^* on forms is a bijection, and its inverse F_* is given by $(F_*\omega)_y(T\psi(TF(x)l_1), ...) = \omega_x(l_1, ...)$, independently of the choices.
814. (V11.11) Push-forwards and pull-backs of forms along equivalences compose, are mutually inverse, agree for naturally equivalent functors and are unchanged by refinements.
815. (V11.12) $[d]^* = F_* \circ G^*$ and $[d]_* = G_* \circ F^*$ are independent of the representing diagram.
816. (V11.13) Under the local regularity property, equivalences and generalized isomorphisms

with the same orbit map have the same push-forwards and pull-backs of forms.

817. (V11.14) $F^*\Theta' = \Theta' \circ F$ and $(F_*\Theta)(y) = \Theta(x)$ for a chosen morphism $F(x) \to y$.
818. (V11.15) For full F, $F_*\Theta$ is a well-defined functor with $F^*F_*\Theta = \Theta$; for essentially surjective F, $F_*F^*\Theta' = \Theta'$; naturally equivalent functors have the same push-forwards and pull-backs.
819. (V11.16) $f^* = F_*\Phi^*$ is well defined for generalized maps and $f_* = \Phi_*F^*$ for generalized isomorphisms, with $(g \circ f)^* = f^* \circ g^*$, $(g \circ f)_* = g_* \circ f_*$ and $(f^{-1})_* = f^*$.
820. (V11.17) Along an equivalence, push-forwards and pull-backs of branched ep⁺-subgroupoids are branched ep⁺-subgroupoids, with closedness, compactness, type, dimension and dimensional decompositions preserved.
821. (V11.18) Generalized isomorphisms carry branched ep⁺-subgroupoids to branched ep⁺-subgroupoids and preserve properties (3)–(6) of Theorem 11.3.3, (6) for every dimensional decomposition.
822. (V11.19) $1_{\{0\}}$ on $\mathbb{Z}/2 \ltimes \mathbb{R}$ is a branched ep⁺-subgroupoid whose pull-back to a refinement is 1 at two objects.
823. (V11.20) Section functors of $\widehat{Gr}$ are pushed forward by (11.8) and pulled back by (11.9) along equivalences, and along generalized isomorphisms by representatives, independently of the choices.
824. (V11.21) Generalized isomorphisms carry orientations of branched ep⁺-subgroupoids to orientations of their push-forwards and pull-backs.
825. (V11.22) $f_*\hat{\Theta} = (f_*\Theta, f_*\hat{T}_\Theta)$ and $f^*\hat{\Theta}'$ are oriented branched ep⁺-subgroupoids, with $(g \circ f)_* = g_* \circ f_*$ and $(g \circ f)^* = f^* \circ g^*$.
826. (V11.23) $(Tf)_*T\Theta = T(f_*\Theta)$ and $(Tf)^*T\Theta' = T(f^*\Theta')$.
827. (V11.24) $f_*(\partial\Theta) = \partial(f_*\Theta)$ and $f^*(\partial\Theta') = \partial(f^*\Theta')$.
828. (V11.25) For a representative whose transfer maps carry the boundary operations into each other, push-forward and pull-back commute with taking oriented boundaries; the general form is not verified.
829. (V11.26) The orientation of Example 9.3.5 is a section functor and an oriented branched ep⁺-subgroupoid moved along the identity.
830. (V11.27) A generalized isomorphism carries the canonical measures of oriented compact n-dimensional branched ep⁺-subgroupoids, and under LUC the effective ones, by $|f|$, with $|f|$ preserving $\mathcal{L}$: $\mu^{f^*\hat{\Theta}'}_{f^*\omega'} = \mu^{\hat{\Theta}'}_{\omega'} \circ |f|$ and $\mu^{\hat{\Theta}}_{\omega} = \mu^{f_*\hat{\Theta}}_{f_*\omega} \circ |f|$.
831. (V11.28) For $\beta' \circ |f| = \beta$ the measures on Z of corresponding forms over corresponding branched ep⁺-subgroupoids agree, and so do the integrals.
832. (V11.29) The boundary measures, and under LUC the effective ones, transform by $|f|$, with $|f|$ preserving $\mathcal{L}(\partial S)$, for every $n \geq 1$ under the text's hypotheses.
833. (V11.30) Pull-backs and push-forwards of auxiliary norms along generalized strong bundle isomorphisms are well defined, continuous, invariant, mutually inverse and depend only on $|[D]|$; conditions (1) and (2) of Definition 8.3.5 are not verified.
834. (V11.31) The push-forward of a multisection functor along a strong bundle equivalence is a well-defined functor, inverse to the pull-back.

835. (V11.32) Push-forward and pull-back of multisection functors along generalized strong bundle isomorphisms are well defined, mutually inverse and depend only on $|[D]|$.
836. (V11.33) A proper covering functor is a surjective local sc-diffeomorphism on morphisms.
837. (V11.34) G_x acts on $F^{-1}(x)$ by $g * y = t(\alpha^{-1}(g, y))$, and $Y(x)$ is isomorphic to $G_x \ltimes F^{-1}(x)$ by the identity on objects and α on morphisms.
838. (V11.35) A morphism $\varphi : x \to x'$ induces an equivariant bijection B_φ and an isomorphism $G_x \ltimes Y_x \to G_{x'} \ltimes Y_{x'}$.
839. (V11.36) A functor between proper coverings is a commuting pair (Ψ, Φ); it is an equivalence iff Ψ and Φ are equivalences, the fibre condition being automatic.
840. (V11.37) Natural transformations between functors of proper coverings are compatible pairs (τ_Y, τ_X); natural equivalence is an equivalence relation compatible with composition.
841. (V11.38) $H : Y \times_{Y'} Y'' \to X \times_{X'} X''$ is a proper covering functor when Ψ, Φ are local sc-diffeomorphisms on objects and Ψ is onto on fibres.
842. (V11.39) For an equivalence (Ψ, Φ), H is a proper covering, $(\pi_{Y''}, \pi_{X''})$ is an equivalence, the two composites to F' are naturally equivalent, and tameness passes from X'' to P and Q.
843. (V11.40) Proper coverings form a category $\mathcal{PC}$ with a multiplicative class of equivalences, and diagrams between them have common refinements as in Definition 11.6.7.
844. (V11.41) Having a common refinement is an equivalence relation on diagrams of proper coverings.
845. (V11.42) Proper strong bundle coverings form a category with a class of equivalences; Theorem 11.6.6 holds for them, and their localization is described by diagrams under common refinement.
846. (V11.43) Classes of diagrams of proper coverings form a category $\mathcal{PC}(E^{-1})$ with a well-defined composition, and $I : \mathcal{PC} \to \mathcal{PC}(E^{-1})$ maps equivalences to isomorphisms.
847. (V12.1) An sc-Fredholm section functor is an sc-smooth section functor that is sc-Fredholm on the object M-polyfold.
848. (V12.2) An sc-Fredholm section functor is regularizing on objects and on morphisms, and its germs at smooth points are sc-Fredholm germs.
849. (V12.3) An auxiliary norm of a strong bundle over an ep-groupoid is Definition 5.1.1 together with invariance under μ.
850. (V12.4) Every strong bundle over an ep-groupoid with paracompact orbit space has an auxiliary norm.
851. (V12.5) Continuous partitions of unity of X are the partitions of unity of $|X|$ composed with π.
852. (V12.6) A continuous partition of unity is subordinate to a saturated cover when supp $\beta_\lambda \subset U_\lambda$.
853. (V12.7) For paracompact $|X|$ every saturated open cover has a subordinate continuous partition of unity.
854. (V12.8) Every object has nested neighbourhoods $U' \subset U''$ with the natural action, a local strong bundle model, the sequence property and (5).

855. (V12.9) Over a tame ep-groupoid with paracompact orbit space, an auxiliary norm over ∂X extends to an auxiliary norm of W.
856. (V12.10) X admits sc-smooth bump functions: for x and saturated $U \ni x$, an sc-smooth functor $f \neq 0$ supported in U.
857. (V12.11) X admits sc-smooth bump functions if and only if its object M-polyfold does, in each consistent reading of the two definitions.
858. (V12.12) With pointwise bump functions there are invariant sc-smooth $f : X \to [0,1]$ with $f(x) = 1$ supported in a given saturated neighbourhood.
859. (V12.13) A G_x-fixed smooth $e \in W_x$ is the value at x of an sc⁺-section functor supported near $|x|$ with $N(\sigma) \leq N(e) + \varepsilon$.
860. (V12.14) The value of a section functor at x is fixed by G_x.
861. (V12.15) An equivariant local sc⁺-section with support in cl V extends to an sc⁺-section functor with the same bound.
862. (V12.16) Sc⁺-sections over the boundary, in a tame local model and on the object M-polyfold, with a functor version.
863. (V12.17) An sc⁺-section over the boundary extends near $x_0 \in \partial O$ to an sc⁺-section, by the alternating sum over the corner faces.
864. (V12.18) For an isotropy action with a lift to K, an equivariant sc⁺-section over the boundary has an equivariant local extension.
865. (V12.19) Compact solution set, proper and proper$_{<\infty}$ for section functors.
866. (V12.20) For sc-Fredholm section functors, proper, compact solution set and proper$_{<\infty}$ are equivalent.
867. (V12.21) Closure commutes with π for saturated sets and for sets with compact closure.
868. (V12.22) A morphism $\varphi : x \to x'$ maps $\text{Lin}(f, x)$ bijectively onto $\text{Lin}(f, x')$, functorially, and acts on orientations.
869. (V12.23) The orientation space O_f has two points over each smooth x; morphisms act on it by an equivalence relation.
870. (V12.24) The orientation bundle over an ep-groupoid with its topology, with Part I's hypotheses; morphisms keep continuous local orientations continuous.
871. (V12.25) An orientation of f is a continuous section of σ preserved by the morphisms.
872. (V12.26) Orientability is equivalent to a two-to-one orbit map with a continuous section, with the continuity for Part I's topology.
873. (V12.27) A continuous fibrewise seminorm with property (2) of Definition 5.1.1 is an auxiliary norm.
874. (V12.28) Reflexive $(0,1)$-fibres of a strong bundle over an ep-groupoid are those of its object bundle.
875. (V12.29) Mixed convergence in $W_{0,1}$: convergence in W_0 with the F_1-parts weakly convergent.
876. (V12.30) Mixed limits are unique and preserved by the morphism action.
877. (V12.31) A reflexive auxiliary norm is an auxiliary norm that is lower semicontinuous along mixed convergent sequences.

878. (V12.32) Reflexive auxiliary norms exist over an ep-groupoid with reflexive fibres and paracompact orbit space.
879. (V12.33) Reflexive auxiliary norms are comparable, and the averaged and glued norms keep the mixed lower semicontinuity.
880. (V12.34) Over a tame ep-groupoid with paracompact orbit space, a reflexive auxiliary norm over ∂X extends to a reflexive auxiliary norm of W.
881. (V12.35) An sc$^+$-section functor over ∂X extends to an sc$^+$-section functor of W supported near ∂X.
882. (V12.36) Reflexive local compactness of an sc-Fredholm section functor at every point.
883. (V12.37) Points off X_1, and points with $N(f(x)) > 1$, have saturated neighbourhoods on which $N(f) > 1$.
884. (V12.38) With a reflexive auxiliary norm, $\{N(f) \leq 1\}$ near the solution set has compact closure, also relative to ∂X.
885. (V12.39) A generalized strong bundle isomorphism induces homeomorphisms of the orientation and orbit spaces and a unique invariant push-forward, continuous in the book's sense, with Part I's hypotheses; independent of the representative and functorial for the data of the diagrams.
886. (V12.40) Mixed lower semicontinuity of a function on $W_{0,1}$ at a point.
887. (V12.41) Near a boundary point of a tame M-polyfold there is a continuous retraction onto the boundary.
888. (V12.42) The retracted fibre norm is a local auxiliary norm equal to N over the boundary.
889. (V12.43) The invariance clause of Definition 12.5.4 as a predicate.
890. (V12.44) Orientation transport through a local strong bundle isomorphism between different bundles, independent of the auxiliary chart point.
891. (V12.45) Continuity of an orientation is a local property; dyadic induction along sc-smooth paths.
892. (V12.46) The orientation map of a local isomorphism depends only on its germ.
893. (V12.47) sc-smooth maps are continuous on every level and on X_∞ (Part I tool).
894. (V12.48) Definition 12.5.4's continuity of an orientation is Part I's continuity along paths over all of X.
895. (V13.1) For a neighbourhood with the natural action and proper closure, $\mathrm{cl}_X(V) \subset U(x)$ if and only if $\mathrm{cl}_{|X|}|V| \subset |U(x)|$.
896. (V13.2) A good system: natural neighbourhoods with proper closures whose morphism components are sc-diffeomorphisms under s and t.
897. (V13.3) Invariant open shrinkings of a good system are good systems, and invariant refinements exist.
898. (V13.4) Refinements of good systems and good systems subordinate to a cover.
899. (V13.5) The natural bi-action of $G_{x'} \times G_x$ on $\mathcal{U}(x, x')$ is an action by sc-smooth maps.
900. (V13.6) The bi-action permutes the components of $\mathcal{U}(x, x')$ freely on each side, and their images under s and t are equal or disjoint.
901. (V13.7) A good system on an open Q with $|Q| = |X|$, proper in Q, is proper in X and extends

to a good system of X inside given neighbourhoods.

902. (V13.8) For paracompact $|X|$ there is a good system inside any given neighbourhoods.
903. (V13.9) A component Σ determines a unique isomorphism of stabilisers $G_{x,\Sigma} \to G_{x',\Sigma}$ intertwining D_{Σ}.
904. (V13.10) Domain support and support of an sc⁺-multisection; the support is finite over each point with weights summing to 1.
905. (V13.11) The pointwise norm $N(\Lambda)(x)$ is attained, and the domain support is cl$\{N(\Lambda) > 0\}$.
906. (V13.12) Λ is trivial on V when $\Lambda(0_x) = 1$ there; the domain support is the least closed set outside which Λ is trivial.
907. (V13.13) A symmetric sc⁺-section structure at x: a natural representation, sections indexed by a finite G_x-set, equivariant, representing Λ.
908. (V13.14) Every sc⁺-multisection admits a symmetric sc⁺-section structure at every point.
909. (V13.15) The sum of sc⁺-multisections and of decorated sc⁺-multisections; $\oplus$ is associative and commutative.
910. (V13.16) Scalar multiples of sc⁺-multisections and of decorated ones.
911. (V13.17) $\beta\Lambda$ is an sc⁺-multisection for every sc-smooth functor β, also decorated.
912. (V13.18) Locally finite families of sc⁺-multisections, with stable index sets.
913. (V13.19) The sum of a locally finite family of sc⁺-multisections is an sc⁺-multisection, also decorated.
914. (V13.20) Local sc-diffeomorphisms injective on open sets are sc-diffeomorphisms onto their images; natural neighbourhoods with proper closure exist.
915. (V13.21) Ultrafilter tools for properness and closures in a full subcategory.
916. (V13.22) A decoration: a good system with symmetric sc⁺-section structures over it representing Λ.
917. (V13.23) Equivalence of decorations; decorated sc⁺-multisections are its classes.
918. (V13.24) An sc⁺-multisection with no inverse for $\oplus$; the neutral element is Λ_0.
919. (V13.25) Index correspondences $[I \leftarrow K \to I']$ and their composition.
920. (V13.26) Morphisms of index correspondences, and their composition.
921. (V13.27) The examples of index correspondences: bijections, the unit, maps to a point.
922. (V13.28) A structured decoration: correspondences between the index sets over the intersections of a good system.
923. (V13.29) The correspondences of a structured decoration are compatible with composition.
924. (V13.30) A structurable sc⁺-multisection is one with a structured decoration.
925. (V13.31) The zero multisection is structurable.
926. (V13.32) Equivalence of structured decorated tuples, as printed.
927. (V13.33) Remark 13.3.9: prose.
928. (V13.34) Tight structured decorations.
929. (V13.35) Products of structured decorations.
930. (V13.36) Sums of structurable sc⁺-multisections are structurable, through products of correspondences.
931. (V13.37) Remark 13.3.13, the positive half.

932. (V13.38) Every structured decoration has a tight refinement.
933. (V13.39) The tight τ' satisfies conditions (1)–(3) of Definition 13.3.4.
934. (V13.40) Structurable sc⁺-multisections have tight structured decorations.
935. (V13.41) Remark 13.3.16: prose.
936. (V13.42) Scalar multiples and products with sc-smooth functors of structured decorations.
937. (V13.43) Scalar multiples of structurable sc⁺-multisections are structurable.
938. (V13.44) An atomic sc⁺-multisection: transported from a symmetric family at one point to its saturation.
939. (V13.45) Remark 13.5.2: prose.
940. (V13.46) An atomic sc⁺-multisection is structurable if and only if a good system exists; in particular when $|X|$ is paracompact.
941. (V13.47) An sc⁺-section near x can be cut off to a global sc⁺-section vanishing off a neighbourhood.
942. (V13.48) A symmetric family at x with $I \neq \emptyset$ defines a unique atomic sc⁺-multisection supported in the saturation, structurable given a good system.
943. (V13.49) For a smooth w there is an atomic sc⁺-multisection with a section through w and small pointwise norm.
944. (V13.50) A locally finite sum of structurable sc⁺-multisections is structurable.
945. (V13.51) The pull-back of a structurable sc⁺-multisection along a strong bundle equivalence is structurable, canonically on structured classes.
946. (V13.52) The push-forward of a structurable sc⁺-multisection along a strong bundle equivalence is structurable.
947. (V13.53) The pull-back of a structurable sc⁺-multisection along a proper strong bundle covering is structurable.
948. (V13.54) Commensurable structured sc⁺-multisections: the same good system and compatible correspondences.
949. (V13.55) The sum of two commensurable structured sc⁺-multisections is well defined and structured.
950. (V13.56) The sum of a locally finite commensurable family is well defined and structured.
951. (V13.57) Atomic on a natural neighbourhood $U(x)$ (the reading used by Theorem 13.5.7).
952. (V13.58) A structurable sc⁺-multisection is a locally finite sum of atomic ones, for every structured lift.
953. (V13.59) A structured sc⁺-multisection on $W|V$ with $|V| = |X|$ extends uniquely to W, with a structure that agrees on V and has small neighbourhoods off V.
954. (V14.1) Every boundary point has arbitrarily small good open neighbourhoods.
955. (V14.2) Good open neighbourhoods $V(x)$ of boundary points, with the faces θ_j, whose closures are tame.
956. (V14.3) Sc⁺-multisections over ∂X: functors on $W|\partial X$ that are sc⁺ on the local faces; the reading $I_x \neq \emptyset$ as `ChargesBoundary`.
957. (V14.4) A local sc⁺-multisection over $\partial X \cap V(x)$ extends to $V(x)$.
958. (V14.5) Good systems of neighbourhoods for ∂X.

959. (V14.6) For tame X with paracompact $|X|$, good systems for ∂X exist.
960. (V14.7) Structured sc⁺-multisections over ∂X and their equivalence.
961. (V14.8) Theorem 14.1.9 holds with tame strong bundle charts and nonempty index sets (`ChargesBoundary`, necessary) added, with a witness with $\partial X \neq \emptyset$.
962. (V14.9) Remark 14.1.10: prose.
963. (V14.10) Good neighbourhoods of boundary points can be shrunk inside given ones.
964. (V14.11) Lemma 14.2.2, with $t(\varphi) \in U(x')$.
965. (V14.12) A structurable sc⁺-multisection over ∂X has a good structured version.
966. (V14.13) The open sets $\widetilde{O}_{x,x',[d]}$, symmetric under inversion.
967. (V14.14) The sets $\hat{O}_{x,x'}$ and the classes $[d]_\varphi$.
968. (V14.15) The class $[d]_\varphi$ is well defined.
969. (V14.16) The correspondences extend to $\hat{\tau}_{x,x'}$ on $\hat{O}_{x,x'}$, with the diagonal property.
970. (V14.17) Remark 14.3.5: prose.
971. (V14.18) A locally finite cover $(\tilde{V}_a)_{a \in A}$ of $|\partial X|$.
972. (V14.19) The sets $\widetilde{O}(y)$ and $o(y)$ with properties (1)–(6).
973. (V14.20) The sets $\hat{V}_a$ with properties (1)–(4), abstract and instantiated.
974. (V14.21) Remark 14.4.1: prose.
975. (V14.22) z-loops and z-arcs.
976. (V14.23) The relation $\sim_z$ is an equivalence relation.
977. (V14.24) $\sim_z$ is compatible with the G_a-action.
978. (V14.25) The relation $\sim_z$ is locally constant; (2) as printed and for $z_k \in \hat{V}_a$.
979. (V14.26) Finitely many objects isomorphic to a given one in the relevant sets.
980. (V14.27) The sets Θ_z and the relation $\approx_z$.
981. (V14.28) Properties (1)–(4) of $\approx_z$, (3) at every closure point.
982. (V14.29) The neighbourhoods $\widetilde{O}(z)$ with (1)–(4).
983. (V14.30) Coherent neighbourhood systems exist, as printed and with (3) on the closure.
984. (V14.31) Remark 14.4.12: prose.
985. (V14.32) The sections over $\widetilde{O}([z])$ with (1)–(4).
986. (V14.33) Local extensions $\hat{s}_i^a$ exist over a coherent neighbourhood system, for the objects of §14.3.
987. (V14.34) The functors β_a with their properties.
988. (V14.35) The structurability criterion of Theorem 14.5.5, without properness in (a) and the closure in (b).
989. (V14.36) The closures $\bar{\theta}_j$ of the faces of a good neighbourhood are tame M-polyfolds.
990. (V14.37) The local extensions transported along morphisms give sections $\tilde{s}^a_{b,j}$ with (1)–(4), with Lemma 14.5.1 (1) named.
991. (V14.38) The sum over a is sc⁺, equivariant, compatible and equals s_j^{0b} on ∂X.
992. (V14.39) The objects of §14.3 form the setting of §14.4.
993. (V14.40) Faces as sub-ep-groupoids; sc⁺-sections of $W|F$ through local face germs.
994. (V14.41) The restriction of a face-structured multisection to $F \cap F'$, sc⁺ on the corner.
995. (V14.42) Compatible assignments; the index sets agree on $F \cap F'$.

996. (V14.43) Agreeing face-wise representatives glue to a structured multisection over ∂X, which extends by Theorem 14.1.9 in its proved form.
997. (V14.44) Structurability over ∂X is preserved by generalized strong bundle isomorphisms, at class level, with no added hypothesis.
998. (V14.45) (14.53) holds for the $\hat{V}_a$ of Proposition 14.3.8 themselves.
999. (V14.46) The data of §§14.3–14.5 exist for every structurable Λ over ∂X charging the boundary.
1000. (V14.47) Every structured sc⁺-multisection on $W|F$ has a representative whose $U_F(x)$ are good inside the face, up to the natural representation of F.
1001. (V15.1) GrF(x) is the set of finitely supported $\mathbb{Q}_{\geq 0}$-combinations of linear maps $T_xX \to W_x$; the summands that occur are sc-Fredholm (V15.9).
1002. (V15.2) The formal sum $\sum_{f(x)=s_i(x)} \sigma_i(f - s_i)'(x)$ does not depend on the structure or the chart, and $T(f, \Lambda)$ is a functor at level-1 points.
1003. (V15.3) The linearization vanishes off the support of $\Lambda \circ f$.
1004. (V15.4) At a non-isolated point there are $x_k \to x_0$, $x_k \neq x_0$, along which the local sections separate by one partition; at an isolated point there are none.
1005. (V15.5) A ball that is a finite union of relatively closed sets has one of them with nonempty interior.
1006. (V15.6) Finitely many continuous maps on a ball coincide or differ pointwise on a smaller ball, by one partition of the index set.
1007. (V15.7) Any sequence $x'_k \to x_0$ can be modified slightly and thinned to a separating sequence.
1008. (V15.8) The linearization $T(f, \Lambda)(x) \in$ GrF(x) at every level-1 point.
1009. (V15.9) With tame germ data, every summand of $T(f, \Lambda)(x)$ is sc-Fredholm in the chart.
1010. (V15.10) Local sc⁺-sections, $f - s$ on $W|_O$, its chart derivative and local solution sets.
1011. (V15.11) Transversality: every operator of the support of $T(f, \Lambda)(x)$ is onto.
1012. (V15.12) At an interior transversal point, $\{\Lambda \circ f > 0\}$ near x is a finite union of smooth submanifolds with the counting formula.
1013. (V15.13) $\Lambda \circ f$ is locally a tame branched ep⁺-subgroupoid, on the ep-groupoid $G_x \ltimes U(x)$ and globally, with Part I's tame germ data.
1014. (V15.14) Good position at a point, branch by branch as Definition 3.6.6.
1015. (V15.15) In good position at x, $\{\Lambda \circ f > 0\}$ near x is a finite union of tame submanifolds in good position with the counting formula; in general position at x for tame X, general and good position hold near x.
1016. (V15.16) Remark 15.2.6: prose.
1017. (V15.17) The product of two ep-groupoids is an ep-groupoid.
1018. (V15.18) $[0, 1]^n \times X$ is a tame ep-groupoid, and $\tilde{f}$ is sc-Fredholm with compact solution set.
1019. (V15.19) For compact $|S_f|$ and any auxiliary norm there is U with (N, U) controlling compactness.
1020. (V15.20) (N, U) controls compactness: U saturated open around S_f and $\{N(f) \leq 1\} \cap U$ with compact orbit closure.

1021. (V15.21) Remark 15.3.2: prose.
1022. (V15.22) (N, U)-admissible perturbations.
1023. (V15.23) A perturbation with domain support in U and $N(\Lambda) \leq 1$ is admissible.
1024. (V15.24) General position: every operator of the support is onto on $T_x^R X$, in every chart.
1025. (V15.25) General position implies transversality.
1026. (V15.26) An interpolated auxiliary norm on $[0,1] \times W$ controls the compactness of $\tilde{f}$.
1027. (V15.27) General position gives good position at every point of the support.
1028. (V15.28) In general position with paracompact |supp|, $\Lambda \circ f$ is a tame branched ep$^+$-subgroupoid.
1029. (V15.29) Theorem 15.3.7 holds under the text's hypotheses (tame germ data inherited); the structurable addendum given a good system, in particular for paracompact $|X|$.
1030. (V15.30) Under the hypotheses of Theorem 15.3.7, $\Lambda \circ f$ is a compact tame branched ep$^+$-subgroupoid.
1031. (V15.31) Remark 15.3.8: prose.
1032. (V15.32) Theorem 15.3.9 with tame bundle charts and nonempty index sets follows from the relative local families of steps (c)–(d); Sard and compactness proved.
1033. (V15.33) Corollary 15.3.10 (1) follows from Theorem 15.3.7, which holds for paracompact $|X|$.
1034. (V15.34) Corollary 15.3.10 (3) follows from Theorem 15.3.9 in its proved form on the cylinder, with tame cylinder charts and the face data of the glue.
1035. (V15.35) Remark 15.3.11: prose.
1036. (V15.36) With an orientation as data, $\Lambda \circ f$ is compact and branched and satisfies Stokes.
1037. (V15.37) Corollary 15.4.4 follows the text's argument, given the oriented end identities of the cylinder.
1038. (V15.38) Remark 15.4.5: prose.
1039. (V15.39) General position at a point is invariant under morphisms; the local Sard step of Theorem 5.3.10.
1040. (V15.40) An M-polyfold or ep-groupoid with $d_X \equiv 0$ is tame.
1041. (V15.41) The cut-off functor h_0 of the proof of Theorem 15.3.9 exists, given tame bundle charts.
1042. (V15.42) $(\max\{N, N'\}, U \cap U')$ controls compactness, and its regular perturbations are regular for both.
1043. (V15.43) In good position the local solution sets are tame submanifolds; interior transversality gives good position.
1044. (V15.44) (15.1) at a zero is Part I's properties (i)–(ii) in a centred tame chart.
1045. (V15.45) Relative general position: the Sard and compactness steps of Theorem 15.3.9 are proved; the local families are assumed.
1046. (V15.46) $[0,1] \times X$ admits sc-smooth partitions of unity when X does.
1047. (V15.47) The glued data are structurable over $\partial([0,1] \times X)$, given that lifted sc$^+$-sections are sc$^+$ on the faces.
1048. (V15.48) $\tilde{\Theta}$ restricts to Θ_i on the ends, as sets and on orbits.
1049. (V15.49) The effective form of Corollary 15.4.4, given the oriented end identities and LUC on

the cylinder.

1050. (V15.50) Oriented formal sums $\widehat{Gr}_F$ at a chart point, with the forgetful map.
1051. (V15.51) The canonical oriented lift of $T(f,\Lambda)(y)$ exists, has the text's formula, is unique and independent of the local structure, at a chart point.
1052. (V15.52) The canonical orientation pointwise, with the kernel orientations.
1053. (V15.53) The product of two strong bundles is a strong bundle.
1054. (V15.54) Products of sc-smooth and of sc$^+$-sections.
1055. (V15.55) The product strong bundle over $X \times X'$ and the section functor $f \times f'$.
1056. (V15.56) max(N, N') is an auxiliary norm of the product.
1057. (V16.1) A polyfold structure on Z is an ep-groupoid with paracompact orbit space satisfying (ZHOU) and a homeomorphism $|X| \to Z$; Z is then metrizable (V16.14).
1058. (V16.2) Equivalence of polyfold structures through a generalized isomorphism compatible with the homeomorphisms is an equivalence relation.
1059. (V16.3) The generalized isomorphism of an equivalence need not be unique (an example on $\mathbb{R}$ with $\mathbb{Z}/3$ isotropy); it is unique for an effective target.
1060. (V16.4) A polyfold is a space with an equivalence class of polyfold structures.
1061. (V16.5) Equivalent polyfold structures both admit, or both do not admit, sc-smooth bump functions and sc-smooth partitions of unity.
1062. (V16.6) A polyfold admits bump functions (partitions of unity) if one, equivalently every, structure in its class does.
1063. (V16.7) Over a homeomorphism there can be two generalized isomorphisms (the example of V16.3); there is at most one for an effective target.
1064. (V16.8) Equivalent structures give the same degeneracy index on Z.
1065. (V16.9) The degeneracy index d_Z of a polyfold.
1066. (V16.10) The boundary $\partial Z = \{d_Z \geq 1\}$.
1067. (V16.11) Tameness of polyfold structures is invariant under equivalence.
1068. (V16.12) A polyfold is tame if one, equivalently every, structure in its class is tame.
1069. (V16.13) $\mathbb{R}$ with the trivial $\mathbb{Z}/3$-action is a polyfold structure with paracompact orbit space.
1070. (V16.14) The underlying space of a polyfold structure is metrizable.
1071. (V16.15) Equal underlying maps give equal orbit maps $|\mathfrak{f}| = |\mathfrak{g}|$, $|T\mathfrak{f}| = |T\mathfrak{g}|$, and equal $\mathfrak{f}$, $T\mathfrak{f}$ for an effective target; not in general.
1072. (V16.16) The tangent space $T(Z, c)$ and the bijections $T\alpha$, independent of the generalized isomorphism, with (16.5).
1073. (V16.17) $T(Z, c)$ has a unique topology making every $T\alpha$ a homeomorphism, and it is metrizable.
1074. (V16.18) The tangent polyfold $(T(Z, c), Tc)$.
1075. (V16.19) The push-forward of forms does not depend on the generalized isomorphism with the given orbit map.
1076. (V16.20) Sc-differential forms on a polyfold, as classes of forms on its structures, with their vector-space structure.
1077. (V16.21) Sc-differential forms on a polyfold form a de Rham complex, with pull-backs on

classes.

1078. (V16.22) Branched suborbifolds of a polyfold, and the well-defined function θ on Z.
1079. (V16.23) Closed, compact, manifold-type, orbifold-type, n-dimensional and tame branched suborbifolds; all invariant.
1080. (V16.24) $\operatorname{supp}\theta$ is homeomorphic to $|\operatorname{supp}\Theta|$ and carries the structure of Proposition 9.1.9, independent of the representative.
1081. (V16.25) Oriented branched suborbifolds of a polyfold.
1082. (V16.26) The boundary of an oriented branched suborbifold is well defined.
1083. (V16.27) The integrals over tame compact oriented branched suborbifolds are defined on classes.
1084. (V16.28) Stokes' theorem for tame compact oriented branched suborbifolds of a polyfold with sc-smooth partitions of unity.
1085. (V16.29) Polyfold structures with their morphisms form a category PS.
1086. (V16.30) Equivalence of morphisms of polyfold structures.
1087. (V16.31) Sc-smooth maps between polyfolds.
1088. (V16.32) The composite of sc-smooth maps does not depend on the choices for an effective middle polyfold; in general it does not hold as we read it.
1089. (V16.33) Effective polyfolds and sc-smooth maps form a category.
1090. (V16.34) The tangent is a functor on PS, well defined on sc-smooth maps, and a functor on effective polyfolds.
1091. (V16.35) The raised polyfolds $(Z,c)^i$.
1092. (V16.36) The tangent projection $\tau_Z : T(Z,c) \to (Z,c)^1$.
1093. (V16.37) Strong polyfold bundle structures.
1094. (V16.38) Y and Z are metrizable.
1095. (V16.39) Equivalence of strong polyfold bundle structures is an equivalence relation.
1096. (V16.40) The uniqueness of f and F after Definition 16.3.3 does not hold as we read it; it holds for an effective target base.
1097. (V16.41) A strong polyfold bundle induces a polyfold.
1098. (V16.42) Strong polyfold bundles.
1099. (V16.43) Tame and face-structured strong polyfold bundles.
1100. (V16.44) The category $\mathcal{SPBS}$.
1101. (V16.45) The class $\bar{c}$ through isomorphisms over Id_Y.
1102. (V16.46) Equivalence of morphisms in $\mathcal{SPBS}$.
1103. (V16.47) Sc-smooth strong polyfold bundle maps.
1104. (V16.48) Auxiliary norms of a strong polyfold bundle.
1105. (V16.49) Auxiliary norms exist.
1106. (V16.50) Remark 16.3.11: prose.
1107. (V16.51) Reflexive 1-fibres, invariant under equivalence in both directions.
1108. (V16.52) Reflexive auxiliary norms.
1109. (V16.53) Mixed convergence, independent of the model.
1110. (V16.54) Reflexive auxiliary norms exist.

1111. (V16.55) $n(y_k) \to 0$ implies $y_k \to 0_z$, for every auxiliary norm.
1112. (V16.56) Proposition 16.3.16 (2) as stated.
1113. (V16.57) An auxiliary norm over ∂Z extends.
1114. (V16.58) A reflexive auxiliary norm over ∂Z extends, given the local step of Theorem 12.2.13.
1115. (V16.59) The zero 0_z of a fibre, independent of the model.
1116. (V16.60) The relation of sections before Definition 16.3.18.
1117. (V16.61) Sc-smooth sections.
1118. (V16.62) Sc$^+$-sections.
1119. (V16.63) Structurability is preserved by generalized strong bundle isomorphisms (interior form).
1120. (V16.64) Sc$^+$-multisections of a strong polyfold bundle.
1121. (V16.65) The domain support.
1122. (V16.66) The domain support of a representative.
1123. (V16.67) The function $n(\lambda)$ is continuous, with the maximum attained.
1124. (V16.68) Structurable sc$^+$-multisections of a strong polyfold bundle.
1125. (V16.69) Sc$^+$-multisections over ∂Z.
1126. (V16.70) Structurable sc$^+$-multisections over ∂Z.
1127. (V16.71) Theorem 16.5.7 with tame bundle charts in the models and a representative charging the boundary.
1128. (V16.72) Sc-Fredholm in one representative implies sc-Fredholm in all.
1129. (V16.73) Sc-Fredholm sections of a strong polyfold bundle.
1130. (V16.74) Compact solution sets.
1131. (V16.75) Proper and proper$< \infty$ sections.
1132. (V16.76) A compact solution set gives proper and proper$< \infty$.
1133. (V16.77) Orientations of sc-Fredholm sections, carried along a diagram to every representative.
1134. (V16.78) Theorem 16.6.6: the integrals are well defined and depend only on $[\tau]$; independence of λ given the cobordisms.
1135. (V16.79) Reflexive 1-fibres pass along strong bundle equivalences in both directions.
1136. (V16.80) $\lambda \circ f$ is a branched suborbifold, unique and independent of the model, existing under general position.
1137. (V16.81) The zero section of a strong polyfold bundle is an sc$^+$-section.